2.0 SITE CHARACTERISTICS

2.1 GEOGRAPHY AND DEMOGRAPHY

2.1.1 SITE LOCATION AND DESCRIPTION

Figures 2.1.1-1 through 2.1.1-3 show the location of the 3169-acre site within Burke County, Georgia, on the Savannah River at river mile 151.1. Figure 2.1.1-1 indicates the basis for estimated population figures prior to receipt of an operating license. It is considered historical information and will not be updated.

2.1.1.1 Reactor Coordinates

The coordinates of the center of the containment for each of the two units are given below in both latitude and longitude and Universal Transverse Mercator (UTM) coordinates. Latitude and longitude are given to the nearest second and UTM coordinates are given to the nearest 100 m.

Unit	Latitude and Longitude	UTM Coordinates in Zone 17S MG (m)
1	33°08'30" N	N 3,666,900
	81°45'44" W	E 428,900
2	33°08'30" N	N 3,666,900
	81°45'48" W	E 428,800

2.1.1.2 Site Area Map

Drawing CX2D45V002 shows property lines and exclusion boundary lines for the site. Location and orientation of principal plant structures within the site area are shown on drawing CX2D45V003. With the exception of the Georgia Power Company combustion turbine plant, Plant Wilson, there are no commercial, industrial, institutional, recreational, or residential structures within the site area. The nearest point to the exclusion area boundary is the near bank of the Savannah River. Reactor 1 is approximately 3600 ft from the exclusion area boundary, and Reactor 2 is approximately 3900 ft from the exclusion area boundary. A scale that will permit the measurement of distances with reasonable accuracy is shown on drawing CX2D45V002. The Savannah River is adjacent to the site, as shown in figure 2.1.1-2.

2.1.1.3 Boundaries for Establishing Effluent Release Limits

The property lines as shown on drawing CX2D45V002 are the boundaries for determining effluent release limits. Effluent releases will not exceed the limits of 10 CFR 20.106 (a) at the boundary. Information regarding radioactive gaseous and liquid effluents, which will allow identification of structures and release points as well as definition of unrestricted areas within the site boundary that are accessible to members of the public, shall be as shown on drawings CX2D45V002 and AX6DD311. An unrestricted area means an area, to which access is neither limited nor controlled by the licensee, or any area within the site boundary used for residential quarters or for industrial, commercial, institutional, and/or recreational purposes. The site boundary shall be the exclusion boundary line as shown on drawing CX2D45V002. A member of the public means an individual in a controlled area or unrestricted area. However, an individual is not a member of the public during any period in which the individual receives an occupational dose. This category may include persons who use portions of the site for recreational, occupational, or other purposes not associated with the plant.

The definition of unrestricted area used in implementing the Technical Specifications has been expanded over that in 10 CFR 20.1003. The unrestricted area boundary may coincide with the exclusion (fenced) area boundary, as defined in 10 CFR 100.3 (a), but the unrestricted area does not include areas over bodies of water. The concept of unrestricted areas, established at or beyond the site boundary, is utilized in the limiting conditions for operation to keep levels of radioactive materials in liquid and gaseous effluents as low as is reasonably achievable, pursuant to 10 CFR 50.36 (a). The site boundary lines, plant property lines, and the exclusion area lines are shown on drawings CX2D45V002 and CX2D45V003. The property lines are the boundaries for determining effluent release limits.

Access within the property boundary is controlled as discussed in subsection 2.1.2. Detailed discussion of effluent release points is provided in subsection 2.3.5.

2.1.2 EXCLUSION AREA AUTHORITY AND CONTROL

2.1.2.1 Authority

Georgia Power Company (GPC) and the co-owners own the entire plant exclusion area in fee simple including mineral rights. Pursuant to the nuclear operating agreement, Georgia Power Company (GPC), for itself and as agent for the co-owners, has delegated to Southern Nuclear Operating Company, Inc. (SNC) complete authority to regulate any and all access and activity within the entire plant exclusion area. The minimum distance to the exclusion area boundary is discussed in paragraph 2.1.1.2.

2.1.2.2 Control of Activities Unrelated to Plant Operation

There are no areas within the exclusion area in which activities unrelated to plant operation will occur except at the visitors center and at the GPC combustion turbine plant, Plant Wilson.

The exclusion area outside the controlled area fence will be posted and, with the exception of the visitors center, will be closed to persons who have not received permission to enter the property.

The access route to the visitor's center is from River Road along the main plant access road to the road leading to the visitors center. Normally, only a few administrative personnel are located at the visitors center. Because of the remote location of the site, the number of visitors at the center will be minimal. However, persons visiting the center will occupy the center and the area and parking lot immediately adjacent to the center. In the event of emergency conditions at the plant, the emergency plan provides for notification of visitors to the center concerning the proper actions to be taken and evacuation instructions.

The access route to Plant Wilson is along River Road. The emergency plan also provides for notification and evacuation of VEGP personnel at Plant Wilson.

SNC normally will not control passage or use of the Savannah River along the exclusion area boundary. Signs are posted near the river indicating the actions to be taken in the event of emergency conditions at the plant.

2.1.2.3 Arrangements for Traffic Control

SNC has made arrangements with the Burke County Sheriff for control of traffic nearby in the event of an emergency.

The State of Georgia has made arrangements with the U.S. Coast Guard to control traffic on the Savannah River.

2.1.2.4 Abandonment or Relocation of Roads

An improved county road by the name of River Road has been relocated. The relocation of River Road was approved by the Board of County Commissioners of Burke County. Therefore, no further hearings or adjudicatory actions will be necessary.

2.1.3 POPULATION DISTRIBUTION (HISTORICAL)

2.1.3.1 Population Within 10 Miles (HISTORICAL)

Figure 2.1.3-1 shows major population centers within 10 miles of the VEGP. Tables 2.1.3-1 through 2.1.3-16 show population distribution and numbers by 16 compass sectors at 1-mile intervals for the 0- to 10-mile radius. Population projections for the years 1987, 1990, 2000, 2007, 2010, 2020, and 2028 are included in these tables.

Tables 2.1.3-17 and 2.1.3-18 provide population totals for each sector and annular ring, respectively. A total for the 0- to 10-mile enclosed population is also included.

Data sources and methods used to arrive at population distribution and projections are *described below.*

2.1.3.2 Population Between 10 and 50 Miles (HISTORICAL)

Figure 2.1.3-2 and tables 2.1.3-1 through 2.1.3-16 show population distribution and numbers at 10-mile intervals between the 10- and 50-mile radii from the plant reactor. The population projections for 1987, 1990, 2000, 2007, 2010, 2020, and 2028 were derived from 1980 base year data, the census statistic.

Tables 2.1.3-19 and 2.1.3-20 provide population totals for each sector and annular ring, respectively. A total for the 10- to 50-mile enclosed population is also included.

The methodology utilized in the projections is outlined below.

2.1.3.3 Transient Population (HISTORICAL)

Data on transient population are provided in tables 2.1.3-22 through 2.1.3-38. Land uses which draw nonresidents to within 10 miles of the operating units include the VEGP itself (industrial use), Plant Wilson which is adjacent to the VEGP site (industrial use), the Savannah River Plant in South Carolina (industrial use), the Savannah River and adjacent areas (public recreational use), a Georgia Power Company (GPC) recreational facility (private recreational use), and State Highway 125 in South Carolina (transportation). Some variance on a daily basis between weekday and weekend day totals is anticipated, as well as between daytime and nighttime numbers. These variations are detailed in breakdowns of the totals for each sector. Peak totals for weekday and weekday nighttime periods are also shown. The only activity expected to show seasonal variation on a consistent basis is the use of the recreational facility available to GPC and SNC employees. Expected usage in summer versus winter is shown in table 2.1.3-32 and in the summary table 2.1.3-38.

Activity relating to VEGP will consist of employment for VEGP operation, training center functions, visitors center functions, and employment for construction of Unit 2 (1987 only). Most VEGP permanent employees, including SNC and contractor nuclear operations personnel, engineering support personnel, and security personnel, will work on a Monday through Friday 8 am to 5 pm or 7 am to 4 pm schedule. The training center will also operate primarily on a Monday though Friday daytime schedule. However, some nuclear operations and security personnel will be assigned to night or weekend shifts. Most permanent plant workers are expected to live outside the 10-mile radius in the Waynesboro and South Augusta areas.

The visitors center will be open on weekdays and during the day on weekends. Most visitors are expected to come on a prescheduled basis with a school group or as part of a special interest group. Due to the remote location, few individuals or family groups are expected.

The construction work force which will be completing Unit 2 in 1987 is expected to be divided between weekday and weekend shifts. The majority of construction workers will be on a weekday 7 am to 5:30 pm shift (Monday through Thursday). The smaller weekday night shift will operate from 5 pm to 3:30 am (Monday through Thursday). The very small weekend day shift will operate from 7 am to 6:30 pm on Friday, Saturday, and Sunday. The weekend night shift will function from 7 pm to 6:30 am on Friday, Saturday, and Sunday. Some construction workers will live within the 10-mile radius during the week, but most will return to their permanent residences outside the 50-mile radius on weekends. By 2007, only operational, engineering support, and security personnel will be employed.

Plant Wilson is an oil-powered electric generating plant owned by GPC and operated by VEGP personnel.

Savannah River Plant operations in South Carolina have facilities within the VEGP 10-mile radius. These include: 100-C reactor, 100-K reactor, 400-D facility, 100-L reactor, and central shop. Most personnel at these facilities will work on a Monday through Friday weekday schedule. Projected employment at Savannah River Plant facilities for 2007 and 2028 is best represented by the 1987 data according to plant officials. A system is in place to warn and account for transient population in their emergency planning zone. The majority of Savannah River Plant employees live in the Aiken-Augusta corridor.

Additional activities within the area under Savannah River Plant jurisdiction may occur in the following VEGP sectors: N (mile rings 2 through 10), NNE (mile rings 2 through 10), NE (mile rings 1 through 10), ENE (mile rings 2 through 10), E (mile rings 2 through 10), ESE (mile rings 3 through 10), and NNW (mile rings 5 through 8). These activities occur infrequently and at currently unpredictable locations on the Savannah River Plant reservation. No numbers reflecting the people involved are included in tables 2.1.3-22 through 2.1.3-38 but are included here for reference. The U.S. Forest Service monitors timber growth in the area and has approximately 20 employees who work onsite. Contractors are occasionally allowed to conduct logging operations. Approximately 25 to 50 contractor personnel would be involved at any one time. Ecology research teams from the University of Georgia operating out of the Savannah River Ecology Laboratory undertake projects at various locations on the Savannah River Plant site. A maximum of 25 team members may be onsite at any one time. Inspection teams from the South Carolina Department of Health and Environmental Control may have as many as five persons onsite. Savannah River Plant officials also allow deer hunting in limited sectors of the property from October to December on Wednesdays and Saturdays. Hunters may be onsite as early as 4 am but must be offsite by nightfall. The sectors to be hunted are identified at the beginning of each year. A maximum of 150 hunters is allowed on any day.

Recreational activities available to the public at all times within the VEGP 10-mile radius include fishing on the Savannah River and hunting. Fishing activities were surveyed in 1980 by the Georgia Department of Natural Resources at access points above and below the VEGP site. The access points within 10 radial miles of the plant, all of which are on private property, are: Shell Bluff, river mile 162; Hancock Landing, river mile 150; Griffin Landing, river mile 146; Brighams Landing, river mile 143; and Department of Energy Landing, river mile 141.

Fishing activities on sample days for the year were observed only at Hancock and Department of Energy Landings. A total of 21 persons were observed fishing on 6 separate days at these two points, about equally distributed between weekend and weekday. Extrapolation from the sample yields an estimate of 231 fisherman per year within 10 miles of the plant. Since the fishing pressure is approximately evenly distributed over the weekday, this translates into a somewhat greater than 50 percent probability that one or more fisherman would be found on a given day within 10 miles of the plant. Normal plant operations do not affect existing or projected fishing patterns.

Hunting activity on the Georgia side of the river is similarly low. There are no direct data on hunting activity within 10 miles of the site, but surveys conducted by the Georgia Department of Natural Resources indicate that hunting yields in Burke County are among the lowest in the state. There are high hunting license sales in nearby Richmond County (primarily for deer hunting), but the primary hunting areas are reported to be the counties to the northwest of Richmond. The probability of a hunter receiving a license in Richmond County and hunting in one of the counties to the northwest is three to four times greater than the probability of hunting in Burke County. Furthermore, there are no permanent hunting lodges or camps within the 10-mile radius of the plant site.

The Crackerneck Unit of the Sumter National Forest, NNW (mile rings 6, 7, and 8), has a small recreational facility on the South Carolina side of the river. The unit has a primitive campground and trails but no water supply, toilets, or permanent facilities. Usage is primarily as a base for hunting and

fishing, with 1025 user days estimated per year. It is managed by two employees. Discussions are under way to bring the area under Savannah River Plant jurisdiction.

Private recreational facilities are available to GPC and SNC employees at a site in the SW sector (mile ring 3). The facility will not be open to the general public. Year-round usage is expected, but lower attendance is anticipated during winter operating hours (October 16 through April 14) than during summer hours (April 15 through October 15). Since some campsites are available, a small number of persons may be onsite at night. Peak attendance is expected during organized company activities such as baseball tournaments or picnics.

Transient highway traffic within the 10-mile radius is limited to State Highway 125 in South Carolina which passes through the Savannah River Plant. Through-traffic is primarily related to Augusta and the Port of Charleston during the week and recreational usage of South Carolina coastal areas and Savannah on weekends. Access to the Savannah River Plant reservation is by time-stamped travel pass, so that records of the average number of vehicles on weekdays and weekend days are available. Employees also use this route. Estimates were made to eliminate employees with work destinations within the 10-mile radius, since they have been counted in the Savannah River Plant employee figures. Georgia State Highway 23 is within the 10-mile radius. It is used almost exclusively by area residents and may be used by some plant employees. Virtually no transient traffic is expected.

Attendance at churches within the VEGP 10-mile radius is not expected to generate any transient population traffic. Although there are 24 churches within the 10-mile radius, they are small in size and serve residents of the immediate area only.

The only school within the 10-mile radius, Girard Elementary, is scheduled to be closed in 1986.

There are no beaches, federal highways, amusement parks, National Register historic districts or sites, regional shopping malls, or colleges and universities within the 10-mile radius.

The annular ring from 10 to 20 miles of the unit site contains some industrial facilities. Most employees of these facilities live in different sectors and rings from their work location. The NNW-20 segment has several heavy industrial manufacturing plants in the south Augusta area employing approximately 5700 workers. Many of these workers live in the NW-20 and NW-30 sectors. Several segments in this ring in South Carolina contain additional facilities of the Savannah River Plant. The majority of these workers also live in different rings from their work location. Estimated workers in 1987 by segment are: N-20, 2400 employees; NNE-20, 1000 employees; NE-20, 6200 employees; and ENE-20, 500 employees. Socioeconomic base studies completed for the Savannah River Plant show that the majority of plant workers live in the Aiken-Augusta corridor which is comprised of segments N-30 and NNW-30. An additional concentration of industrial employment in this ring is centered in Waynesboro (WSW-20). Most of these workers are estimated to also reside in this segment. Although slight growth in industrial employment at existing and new facilities in the above mentioned segments is anticipated in future years of 2007 and 2028, no significant changes in employment numbers or distribution of facilities is anticipated.

There are no recreational land uses of significance or other sources of daily or seasonal population shifts in the 10- to 20-mile annular ring.

Industrial employment centers in the 20- to 30-mile ring are concentrated in the Augusta area segments NW-30 (3600 employees) and NNW-30 (5368 employees). The majority of these workers live in the same segments, but some live in the adjoining segments of Aiken (N-30) and Columbia County (NW-40).

Additional transient population concentrations are also found in 20- to 30-mile annular rings due to recreational activities on weekends and holidays at several state parks. In Georgia, Magnolia Springs Park near Millen (SSW-30) experiences an estimated daily peak summer (May, June, July, and August) holiday weekend visitor population of 4200. During nonsummer months, the number of daily peak weekend visitors is estimated at 950. On weekdays, visitors are estimated to average approximately 200 during the summer and 75 in the nonsummer months. In South Carolina, two smaller parks also draw some transient visitors. Aiken Park near Windsor (NNE-30) is estimated to draw a maximum of 2000 visitors on a peak summer holiday. In nonsummer months, daily visitors on a peak weekend are estimated to be only 700. Average weekday visitors are estimated to be 100 during the summer and 50 during nonsummer months. Barnwell State Park near Blackville (ENE-30) is smaller, more distant from residential population concentrations, and less subject to seasonal fluctuations. Visitors are estimated to average 700 on a peak holiday and 50 on an average weekday. In future years, some increase in visitors at these parks may occur, although weekend usage is currently near capacity for overnight facilities.

The 30- to 40-mile annular ring contains no significant sources of industrial activity which would result in large transient worker concentrations. The relatively small Rivers Bridge State Park near Ehrhardt, South Carolina (ESE-40), is estimated to attract as many as 700 visitors on a peak holiday and as few as 50 on an average weekday. There is little seasonal fluctuation.

The 40- to 50-mile annular ring has some industrial facilities associated with the cities of Thomson (WNW-50) and Swainsboro (SW-50). The majority of workers live within these segments. Two state parks are located in Georgia within this ring. George L. Smith Park near Twin City (SSW-50) has little seasonal fluctuation in visitors. Peak holiday visitors are estimated to be 500, with 50 visitors on an average weekday. Mistletoe State Park (NW-50) does show seasonal variance in visitors due to its proximity to water at Clarks Hill Reservoir. Visitors are estimated to range from 2200 on a peak summer holiday to 100 on an average summer weekday. In nonsummer months on a peak holiday as many as 800 visitors are estimated to be present. Weekday visitors in nonsummer months are estimated to average 60 persons. This facility is judged to be moderately used at present by the Corps of Engineers. Some increase in usage is expected in the future.

No other activities or attractions produce significant changes in population distribution between segments or in transient population totals on a daily or seasonal basis.

2.1.3.4 Low Population Zone (LPZ) (HISTORICAL)

The LPZ (as defined by 10 CFR 100) for the VEGP is that area falling within a 2-mile radius (figure 2.1.1-3) from the midpoint between the containment buildings. Tables 2.1.3-1 through 2.1.3-16 and table 2.1.3-18 show that the area is expected to remain sparsely populated during the anticipated life of the plant. There is only one road, River Road, within the LPZ, as shown on drawing CX2D45V002. There are no towns, recreational facilities, hospitals, schools, prisons, or beaches within the LPZ or from the LPZ to a radius of 5 miles.

2.1.3.5 Population Center (HISTORICAL)

Augusta is the nearest population center of more than 25,000 people, with a 1980 population of 47,532 people (U.S. census count). The Augusta corporate limit lies approximately 25 miles from the VEGP reactors. The Augusta corporate city limit was selected as the population center boundary because the Savannah River flood plain occupies the immediate area southeast of the corporate limits and creates a sharply defined low population density area from the southeast corporate limits to VEGP.

A list of all communities within 30 miles of the plant with populations greater than 1000 persons are identified in table 2.1.3-21 and figure 2.1.3-5. These also identify the largest city within 50 miles of the plant.

The 1980 population of Richmond County outside Augusta was approximately 134,097 people; the projected 1990 total population of Richmond County, including Augusta, is approximately 203,000. The only defined significant transient population into and out of the Augusta area to the plant site is that portion of the VEGP construction force expected to live in the Augusta vicinity until 1987. The current population distribution indicates high density within the Augusta city limits and the northern portions of Richmond County and much lower density in the rural southern and eastern portions of Richmond County. Projections through 2028 indicate that these distribution and density patterns will remain essentially the same.

2.1.3.6 Population Density (HISTORICAL)

The cumulative resident population for a distance of 50 miles projected for the expected first year of operation (1987) compared to 500 people/mi^2 is shown in figure 2.1.3-3. Projected to the year 2028, this comparison to 1000 people/mi^2 is shown in figure 2.1.3-4.

2.1.3.7 Methodology (HISTORICAL)

The methodology used to determine population distribution is described in appendix 2A.

2.1.3.8 Bibliography (HISTORICAL)

Georgia Office of Planning and Budget, Georgia Population Projections, Atlanta, Georgia.

Georgia State Census Bureau, Georgia Population Statistics, Atlanta, Georgia.

South Carolina State Data Center, Division of Research and Statistical Service, South Carolina Population Projections, Columbia, South Carolina.

U.S. Department of Commerce, Bureau of Census, Advance Reports: 1980 Census of Population and Housing, South Carolina Population Statistics.

U.S. Department of Commerce, Bureau of Census, Illustrative Projections of State Populations by Age, Race, and Sex 1975 to 2000.

U.S. Geological Survey, Branch Division, USGS Quadrangle Maps, Arlington, Virginia.

TABLE 2.1.3-1 (HISTORICAL)

POPULATION BY SEGMENT FOR NORTH SECTOR (0 TO 50 MILES)

Mile (Ring)	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
1	*0*	*0*	*0*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*	*0*
20	*5914*	*6185*	*6815*	*7459*	*7733*	*8779*	*9292*
30	*9462*	*9896*	*10,904*	*11,934*	*12,376*	*14,047*	*15,507*
40	*12,402*	*12,944*	*14,257*	*15,385*	*16,153*	*18,304*	*20,183*
50	*10,944*	*11,196*	*12,278*	*13,223*	*13,627*	*15,130*	*16,419*

TABLE 2.1.3-2 (HISTORICAL)

POPULATION BY SEGMENT FOR NORTH-NORTHEAST SECTOR (0 TO 50 MILES)

Mile (Ring)	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
1	*0*	*0*	*0*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*	*0*
20	*5914*	*6185*	*6815*	*7459*	*7735*	*8779*	*9692*
30	*9462*	*9896*	*10,904*	*11,934*	*12,376*	*14,047*	*15,507*
40	*14,193*	*14,844*	*16,356*	*17,902*	*18,564*	*21,070*	*23,261*
50	*20,134*	*21,222*	*24,196*	*27,216*	*28,511*	*33,765*	*38,818*

TABLE 2.1.3-3 (HISTORICAL)

POPULATION BY SEGMENT FOR NORTHEAST SECTOR (0 TO 50 MILES)

Mile (Ring)	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
1	*0*	*0*	*0*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*	*0*
20	*3698*	*3866*	*4256*	*4653*	*4823*	*5466*	*6034*
30	*5768*	*6031*	*6639*	*7257*	*7523*	*8524*	*9410*
40	*9289*	*9717*	*10,717*	*11,731*	*12,167*	*13,811*	*15,273*
50	*22,327*	*23,417*	*26,060*	*28,626*	*29,824*	*34,134*	*38,032*

TABLE 2.1.3-4 (HISTORICAL)

POPULATION BY SEGMENT FOR EAST-NORTHEAST SECTOR (0 TO 50 MILES)

Mile (Ring)	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
1	*0*	*0*	*0*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*	*0*
20	*2442*	*2552*	*2805*	*3060*	*3170*	*3582*	*3954*
30	*3663*	*3828*	*4208*	*4590*	*4754*	*5373*	*5931*
40	*4774*	*4984*	*5455*	*5936*	*6142*	*6915*	*7612*
50	*12,383*	*12,978*	*14,394*	*15,811*	*16,418*	*18,729*	*20,843*

TABLE 2.1.3-5 (HISTORICAL)

POPULATION BY SEGMENT FOR EAST SECTOR (0 TO 50 MILES)

Mile (Ring)	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
1	*0*	*0*	*0*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*	*0*
20	*2035*	*2108*	*2260*	*2417*	*2485*	*2732*	*2949*
30	*2701*	*2804*	*3025*	*3251*	*3349*	*3709*	*4027*
40	*3963*	*4202*	*4510*	*4831*	*4970*	*5480*	*5924*
50	*4788*	*4954*	*5353*	*5751*	*5922*	*6551*	*7099*

TABLE 2.1.3-6 (HISTORICAL)

POPULATION BY SEGMENT FOR EAST-SOUTHEAST SECTOR (0 TO 50 MILES)

Mile (Ring)	1987	1990	2000	2007	2010	2020	2028
1	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0
6	3	3	3	3	3	3	3
7	0	0	0	0	0	0	0
8	3	3	3	3	3	3	3
9	0	0	0	0	0	0	0
10	5	5	5	5	5	5	5
20	1255	1289	1346	1414	1442	1547	1636
30	2480	2550	2669	2801	2858	3062	3228
40	4712	4880	5216	5575	5728	6293	6783
50	7253	7558	8204	8888	9182	10,275	11,251

TABLE 2.1.3-7 (HISTORICAL)

POPULATION BY SEGMENT FOR SOUTHEAST SECTOR (0 TO 50 MILES)

Mile (Ring)	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
1	*0*	*0*	*0*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*2*	*2*	*3*	*3*	*4*	*4*
4	*3*	*4*	*5*	*6*	*7*	*9*	*11*
5	*28*	*35*	*43*	*50*	*63*	*77*	*95*
6	*31*	*32*	*34*	*36*	*37*	*39*	*41*
7	*35*	*36*	*38*	*40*	*41*	*43*	*45*
8	*6*	*6*	*6*	*6*	*7*	*9*	*11*
9	*18*	*18*	*19*	*20*	*20*	*21*	*22*
10	*27*	*28*	*30*	*32*	*33*	*35*	*37*
20	*1318*	*1333*	*1301*	*1307*	*1309*	*1322*	*1336*
30	*2144*	*2170*	*2042*	*2024*	*2017*	*1999*	*1987*
40	*4304*	*4429*	*4554*	*4774*	*4869*	*5224*	*5542*
50	*7303*	*7609*	*8240*	*8918*	*9209*	*10,294*	*11,267*

TABLE 2.1.3-8 (HISTORICAL)

POPULATION BY SEGMENT FOR SOUTH-SOUTHEAST SECTOR (0 TO 50 MILES)

Mile (Ring)	1987	1990	2000	2007	2010	2020	2028
1	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0
3	0	2	2	3	3	4	4
4	0	2	2	3	3	4	4
5	6	7	8	10	12	15	18
6	64	65	69	72	73	77	81
7	88	90	95	100	102	108	114
8	119	121	128	134	137	145	153
9	88	90	95	100	102	108	114
10	74	75	79	83	85	90	95
20	1176	1190	1171	1182	1186	1206	1224
30	1998	2011	1810	1747	1720	1634	1568
40	2711	2730	2457	2371	2334	2217	2128
50	5619	5801	6064	6357	6529	7087	7617

TABLE 2.1.3-9 (HISTORICAL)

POPULATION BY SEGMENT FOR SOUTH SECTOR (0 TO 50 MILES)

Mile (Ring)	1987	1990	2000	2007	2010	2020	2028
1	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0
3	0	2	2	3	3	4	4
4	0	2	2	3	3	4	4
5	6	7	8	10	12	15	18
6	35	36	38	40	41	43	45
7	59	60	64	67	68	72	76
8	29	30	32	34	35	37	39
9	24	24	25	26	27	29	31
10	35	36	38	40	41	43	45
20	1473	1491	1500	1515	1521	1545	1566
30	2514	2535	2369	2309	2283	2200	2137
40	4117	4212	4386	4574	4639	5109	5493
50	10,080	10,465	12,123	13,336	14,144	16,509	18,722

TABLE 2.1.3-10 (HISTORICAL)

POPULATION BY SEGMENT FOR SOUTH-SOUTHWEST SECTOR (0 TO 50 MILES)

Mile (Ring)	1987	1990	2000	2007	2010	2020	2028
1	0	0	0	0	0	0	0
2	3	4	5	6	7	9	11
3	3	4	5	6	7	9	11
4	3	4	5	6	7	9	11
5	3	4	5	6	7	9	11
6	15	15	16	17	17	18	19
7	6	6	6	7	7	7	8
8	24	24	25	26	27	29	31
9	29	30	32	34	35	37	39
10	56	57	60	63	64	68	72
20	1349	1369	1410	1442	1454	1502	1543
30	2355	2379	2331	2299	2285	2239	2203
40	3626	3685	3744	3796	3824	3919	4011
50	6432	6634	7264	7748	7991	8801	9523

TABLE 2.1.3-11 (HISTORICAL)

POPULATION BY SEGMENT FOR SOUTHWEST SECTOR (0 TO 50 MILES)

Mile (Ring)	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
1	*0*	*0*	*0*	*0*	*0*	*0*	*0*
2	*300*	*3*	*4*	*5*	*6*	*7*	*9*
3	*8*	*8*	*9*	*10*	*10*	*11*	*12*
4	*6*	*7*	*6*	*6*	*6*	*6*	*6*
5	*25*	*26*	*28*	*29*	*30*	*32*	*34*
6	*11*	*11*	*12*	*13*	*14*	*14*	*15*
7	*15*	*15*	*16*	*17*	*18*	*19*	*20*
8	*18*	*18*	*19*	*20*	*20*	*21*	*22*
9	*26*	*27*	*29*	*30*	*31*	*33*	*35*
10	*88*	*90*	*95*	*100*	*102*	*108*	*114*
20	*1411*	*1436*	*1526*	*1593*	*1622*	*1725*	*1811*
30	*2317*	*2351*	*2423*	*2477*	*2501*	*2586*	*2658*
40	*3592*	*3677*	*3934*	*4142*	*4230*	*4550*	*4828*
50	*5386*	*5537*	*5983*	*6351*	*6510*	*7083*	*7582*

TABLE 2.1.3-12 (HISTORICAL)

POPULATION BY SEGMENT FOR WEST-SOUTHWEST SECTOR (0 TO 50 MILES)

Mile (Ring)	1987	1990	2000	2007	2010	2020	2028
1	0	0	0	0	0	0	0
2	0	3	3	4	4	5	5
3	3	4	5	6	7	9	11
4	0	3	3	4	4	5	5
5	14	14	15	16	16	17	18
6	45	46	49	51	52	55	58
7	42	43	46	48	49	52	55
8	50	51	54	57	58	61	64
9	11	11	12	13	14	14	15
10	0	0	0	0	0	0	0
20	1411	1436	1526	1593	1622	1725	1811
30	2314	2354	2501	2613	2660	2829	2974
40	4804	4887	5180	5451	5567	5981	6338
50	5868	5960	6206	6461	6569	6956	7287

TABLE 2.1.3-13 (HISTORICAL)

POPULATION BY SEGMENT FOR WEST SECTOR (0 TO 50 MILES)

Mile (Ring)	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
1	*0*	*0*	*0*	*0*	*0*	*0*	*0*
2	*8*	*8*	*9*	*10*	*10*	*11*	*12*
3	*11*	*11*	*12*	*13*	*13*	*14*	*15*
4	*48*	*8*	*9*	*10*	*10*	*11*	*12*
5	*22*	*22*	*23*	*24*	*25*	*27*	*28*
6	*9*	*9*	*10*	*11*	*11*	*12*	*13*
7	*18*	*18*	*19*	*20*	*20*	*21*	*22*
8	*35*	*36*	*38*	*40*	*41*	*43*	*45*
9	*50*	*51*	*54*	*57*	*58*	*61*	*64*
10	*29*	*30*	*32*	*34*	*35*	*37*	*39*
20	*1411*	*1436*	*1526*	*1593*	*1622*	*1725*	*1811*
30	*2370*	*2411*	*2560*	*2681*	*2733*	*2920*	*3077*
40	*4410*	*4487*	*4756*	*5005*	*5112*	*5496*	*5826*
50	*4641*	*4698*	*4869*	*5059*	*5141*	*5433*	*5684*

TABLE 2.1.3-14 (HISTORICAL)

POPULATION BY SEGMENT FOR WEST-NORTHWEST SECTOR (0 TO 50 MILES)

Mile (Ring)	1987	1990	2000	2007	2010	2020	2028
1	0	0	0	0	0	0	0
2	70	3	4	5	6	7	9
3	258	8	9	10	10	11	12
4	14	14	15	16	16	17	18
5	20	20	21	22	22	23	24
6	3	3	3	3	3	3	3
7	9	9	10	11	11	12	13
8	37	38	42	45	46	51	55
9	37	38	42	45	46	51	55
10	34	35	39	42	43	47	51
20	16,518	17,039	18,794	20,213	20,820	23,067	25,059
30	36,341	37,497	41,392	44,546	45,899	50,900	55,338
40	18,533	19,510	22,043	24,114	25,007	28,418	31,547
50	10,331	10,766	11,892	12,778	13,137	14,564	15,769

TABLE 2.1.3-15 (HISTORICAL)

POPULATION BY SEGMENT FOR NORTHWEST SECTOR (0 TO 50 MILES)

Mile (Ring)	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
1	*0*	*0*	*0*	*0*	*0*	*0*	*0*
2	*136*	*3*	*4*	*5*	*6*	*7*	*9*
3	*6*	*6*	*7*	*8*	*9*	*10*	*12*
4	*43*	*3*	*4*	*5*	*6*	*7*	*9*
5	*86*	*6*	*6*	*7*	*8*	*9*	*10*
6	*0*	*0*	*0*	*0*	*0*	*0*	*0*
7	*56*	*58*	*64*	*69*	*71*	*78*	*84*
8	*7*	*7*	*7*	*8*	*8*	*9*	*9*
9	*20*	*21*	*23*	*25*	*26*	*29*	*31*
10	*7*	*7*	*7*	*8*	*8*	*9*	*9*
20	*33,388*	*34,465*	*38,083*	*41,016*	*42,273*	*46,922*	*51,052*
30	*53,029*	*54,738*	*60,485*	*65,143*	*67,139*	*74,524*	*81,082*
40	*22,581*	*25,013*	*30,215*	*34,658*	*36,563*	*44,179*	*51,546*
50	*26,143*	*28,868*	*34,401*	*39,673*	*41,805*	*50,311*	*58,522*

TABLE 2.1.3-16 (HISTORICAL)

POPULATION BY SEGMENT FOR NORTH-NORTHWEST SECTOR (0 TO 50 MILES)

Mile (Ring)	1987	1990	2000	2007	2010	2020	2028
1	0	0	0	0	0	0	0
2	0	3	4	5	6	7	9
3	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0
7	3	3	3	3	3	3	3
8	3	3	3	3	3	3	3
9	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0
20	19,260	19,930	22,011	23,777	24,534	27,349	29,839
30	32,227	33,331	36,815	39,745	41,001	45,665	49,796
40	9776	9339	10,505	11,601	12,069	13,896	15,596
50	5821	5986	6247	6557	6690	7184	7617

TABLE 2.1.3-17 (HISTORICAL)

POPULATION BY SECTORS
0- TO 10-MILE RADIUS TOTAL

Sector	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
N	*0*	*0*	*0*	*0*	*0*	*0*	*0*
NNE	*0*	*0*	*0*	*0*	*0*	*0*	*0*
NE	*0*	*0*	*0*	*0*	*0*	*0*	*0*
ENE	*0*	*0*	*0*	*0*	*0*	*0*	*0*
E	*0*	*0*	*0*	*0*	*0*	*0*	*0*
ESE	*11*	*11*	*11*	*11*	*11*	*11*	*11*
SE	*148*	*161*	*177*	*193*	*211*	*237*	*266*
SSE	*439*	*452*	*478*	*505*	*517*	*551*	*583*
S	*188*	*197*	*209*	*223*	*230*	*247*	*262*
SSW	*142*	*148*	*159*	*171*	*178*	*195*	*213*
SW	*497*	*205*	*218*	*230*	*237*	*251*	*267*
WSW	*165*	*175*	*187*	*199*	*204*	*218*	*231*
W	*230*	*193*	*206*	*219*	*223*	*237*	*250*
WNW	*482*	*168*	*185*	*199*	*203*	*222*	*240*
NW	*361*	*111*	*122*	*135*	*142*	*158*	*173*
NNW	*6*	*9*	*10*	*11*	*12*	*13*	*15*
Total	*2669*	*1830*	*1962*	*2096*	*2168*	*2340*	*2511*

TABLE 2.1.3-18 (HISTORICAL)

POPULATION BY ANNULAR RINGS
0- TO 10-MILE RADIUS TOTAL

Year	*1 Mile*	*2 Mile*	*3 Mile*	*4 Mile*	*5 Mile*	*6 Mile*	*7 Mile*	*8 Mile*	*9 Mile*	*10 Mile*	*Total*
1987	*0*	*517*	*289*	*117*	*210*	*216*	*331*	*331*	*303*	*355*	*2669*
1990	*0*	*27*	*47*	*47*	*141*	*220*	*338*	*337*	*310*	*363*	*1830*
2000	*0*	*33*	*53*	*51*	*157*	*234*	*361*	*357*	*331*	*385*	*1962*
2007	*0*	*40*	*62*	*59*	*174*	*246*	*382*	*376*	*350*	*407*	*2096*
2010	*0*	*45*	*65*	*62*	*195*	*251*	*390*	*385*	*359*	*416*	*2168*
2020	*0*	*53*	*76*	*72*	*224*	*264*	*415*	*411*	*383*	*442*	*2340*
2028	*0*	*64*	*85*	*80*	*256*	*278*	*440*	*435*	*406*	*467*	*2511*

TABLE 2.1.3-19 (HISTORICAL)

POPULATION BY SECTORS
20- TO 50-MILE RADIUS TOTALS

Sector	*1987*	*1990*	*2000*	*2007*	*2010*	*2020*	*2028*
N	*38,722*	*40,221*	*44,254*	*48,001*	*49,889*	*56,260*	*61,401*
NNE	*49,703*	*52,147*	*58,271*	*64,511*	*67,186*	*77,661*	*87,278*
NE	*41,082*	*43,031*	*47,672*	*52,267*	*54,337*	*61,935*	*68,749*
ENE	*23,262*	*24,342*	*26,862*	*29,397*	*30,484*	*34,599*	*38,340*
E	*13,487*	*14,068*	*15,148*	*16,250*	*16,726*	*18,472*	*19,999*
ESE	*15,700*	*16,277*	*17,435*	*18,678*	*19,210*	*21,177*	*22,898*
SE	*15,069*	*15,541*	*16,137*	*17,023*	*17,404*	*18,839*	*20,132*
SSE	*11,504*	*11,732*	*11,502*	*11,657*	*11,769*	*12,144*	*12,537*
S	*18,184*	*18,703*	*20,378*	*21,734*	*22,587*	*25,363*	*27,918*
SSW	*13,762*	*14,067*	*14,749*	*15,285*	*15,554*	*16,461*	*17,280*
SW	*12,706*	*13,001*	*13,866*	*14,563*	*14,863*	*15,944*	*16,879*
WSW	*14,397*	*14,637*	*15,413*	*16,118*	*16,418*	*17,491*	*18,410*
W	*12,832*	*13,032*	*13,711*	*14,338*	*14,608*	*15,574*	*16,398*
WNW	*81,723*	*84,812*	*94,121*	*101,651*	*104,863*	*116,949*	*127,713*
NW	*135,141*	*143,084*	*163,184*	*180,490*	*187,780*	*215,936*	*242,202*
NNW	*67,084*	*68,586*	*75,578*	*81,680*	*84,294*	*94,094*	*102,848*
Total	*564,358*	*587,281*	*648,281*	*703,643*	*727,972*	*818,899*	*900,982*

TABLE 2.1.3-20 (SHEET 1 OF 2) (HISTORICAL)

POPULATION BY ANNULAR RINGS
20- TO 50-MILE RADIUS TOTALS

Year	*Ring*	*Population*
1987	*20-mile*	*99,973*
	30-mile	*171,145*
	40-mile	*127,787*
	50-mile	*165,453*
	Total	*564,358*
1990	*20-mile*	*103,310*
	30-mile	*176,782*
	40-mile	*133,540*
	50-mile	*173,649*
	Total	*587,281*
2000	*20-mile*	*113,145*
	30-mile	*193,077*
	40-mile	*148,285*
	50-mile	*193,774*
	Total	*648,281*
2007	*20-mile*	*121,693*
	30-mile	*207,351*
	40-mile	*161,846*
	50-mile	*212,753*
	Total	*703,643*
2010	*20-mile*	*125,351*
	30-mile	*213,474*
	40-mile	*167,938*
	50-mile	*221,209*
	Total	*727,972*

TABLE 2.1.3-20 (SHEET 2 OF 2) (HISTORICAL)

Year	*Ring*	*Population*
2020	*20-mile*	*138,973*
	30-mile	*236,258*
	40-mile	*190,862*
	50-mile	*252,806*
	Total	*818,899*
2028	*20-mile*	*150,609*
	30-mile	*256,430*
	40-mile	*211,891*
	50-mile	*282,052*
	Total	*900,982*

TABLE 2.1.3-21 (SHEET 1 OF 2) (HISTORICAL)

COMMUNITIES OF GREATER THAN 1000 PERSONS WITHIN 30 MILES OF VEGP

Community in State of Georgia	*1980 Population*
Augusta	*47,532*
East Boundary	*4,699*
Fort Gordon	*14,069*
Grovetown	*3,384*
Hephzibah	*1,452*
Martinez	*16,472*
Millen	*3,988*
Sardis	*1,180*
South Augusta	*51,072*
Sylvania	*3,352*
Waynesboro	*5,760*
West Augusta	*24,242*
Total	*177,202*

Source: 1980 Census of Population, Characteristics of the Population, Number of Inhabitants, Georgia.

TABLE 2.1.3-21 (SHEET 2 OF 2) (HISTORICAL)

Community in State of South Carolina	*1980 Population*
Aiken	*14,978*
Aiken West	*3,083*
Allendale	*4,400*
Appleton	*2,007*
Bath	*2,242*
Barnwell	*5,572*
Belvedere	*6,859*
Blackville	*2,840*
Clearwater	*3,967*
Denmark	*4,434*
Fairfax	*2,154*
Gloverville	*2,619*
Graniteville	*1,158*
Jackson	*1,771*
Langley	*1,714*
Madison	*1,150*
New Ellenton	*2,628*
North Augusta	*13,593*
Springfield	*2,012*
Warrenville	*1,029*
Williston	*3,173*
Total	*83,383*

Source: 1980 Census of Population, Characteristics of the Population, General Population Characteristics, South Carolina.

TABLE 2.1.3-22 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(NORTH SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	1	3	0	0	3	0
7	4	6	1	3	4	3
8	5	8	1	3	5	4
9	5	8	1	3	5	4
10	5	8	1	3	5	4
Sector total	112	142	28	41	28	18

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	1	3	0	0	3	0
7	4	6	1	3	4	3
8	5	8	1	3	5	4
9	5	8	1	3	5	4
10	5	8	1	3	5	4
Sector total	25	38	4	12	22	15

TABLE 2.1.3-22 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*1*	*3*	*0*	*0*	*3*	*0*
7	*4*	*6*	*1*	*3*	*4*	*3*
8	*5*	*8*	*1*	*3*	*5*	*4*
9	*5*	*8*	*1*	*3*	*5*	*4*
10	*5*	*8*	*1*	*3*	*5*	*4*
Sector total	*25*	*38*	*4*	*12*	*22*	*15*

TABLE 2.1.3-23 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(NORTH-NORTHEAST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	335	462	27	54	32	27
6	10	15	2	4	8	6
7	4	6	1	3	4	3
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	441	592	54	90	50	39

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	335	462	27	54	32	27
6	10	15	2	4	8	6
7	4	6	1	3	4	3
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	354	488	30	61	44	36

TABLE 2.1.3-23 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*335*	*462*	*27*	*54*	*32*	*27*
6	*10*	*15*	*2*	*4*	*8*	*6*
7	*4*	*6*	*1*	*3*	*4*	*3*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*354*	*488*	*30*	*61*	*44*	*36*

TABLE 2.1.3-24 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(NORTHEAST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	27	32	6	4
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	12	16	3	5	11	7
7	0	0	0	0	0	0
8	121	147	26	52	26	26
9	443	473	30	60	30	30
10	800	800	50	50	50	50
Sector total	1468	1545	136	199	123	117

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	3	3	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	12	16	3	5	11	7
7	0	0	0	0	0	0
8	121	147	26	52	26	26
9	443	473	30	60	30	30
10	800	800	50	50	50	50
Sector total	1381	1441	112	170	117	113

TABLE 2.1.3-24 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*3*	*3*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*12*	*16*	*3*	*5*	*11*	*7*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*121*	*147*	*26*	*52*	*26*	*26*
9	*443*	*473*	*30*	*60*	*30*	*30*
10	*800*	*800*	*50*	*50*	*50*	*50*
Sector total	*1381*	*1441*	*112*	*170*	*117*	*113*

TABLE 2.1.3-25 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(EAST-NORTHEAST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	103	127	34	41	13	13
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	15	20	4	6	13	9
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	121	147	26	52	26	26
Sector total	239	294	64	99	52	48

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	14	17	7	10	5	5
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	15	20	4	6	13	9
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	121	147	26	52	26	26
Sector total	150	184	37	68	44	40

TABLE 2.1.3-25 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
			Weekday	*Weekday*		
1	*14*	*17*	*7*	*10*	*5*	*5*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*15*	*20*	*4*	*6*	*13*	*9*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*121*	*147*	*26*	*52*	*26*	*26*
Sector total	*150*	*184*	*37*	*68*	*44*	*40*

TABLE 2.1.3-26 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(EAST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	1010	1089	176	217	132	161
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	10	15	2	3	8	6
9	7	9	2	3	6	3
10	7	9	2	3	6	3
Sector total	1034	1122	182	226	152	173

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	672	730	110	145	105	135
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	10	15	2	3	8	6
9	7	9	2	3	6	3
10	7	9	2	3	6	3
Sector total	696	763	116	154	125	147

TABLE 2.1.3-26 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*672*	*730*	*110*	*145*	*105*	*135*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*10*	*15*	*2*	*3*	*8*	*6*
9	*7*	*9*	*2*	*3*	*6*	*3*
10	*7*	*9*	*2*	*3*	*6*	*3*
Sector total	*696*	*763*	*116*	*154*	*125*	*147*

TABLE 2.1.3-27 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(EAST-SOUTHEAST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	118	141	33	39	11	10
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	118	141	33	39	11	10

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	23	30	7	9	4	4
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	23	30	7	9	4	4

TABLE 2.1.3-27 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*23*	*30*	*7*	*9*	*4*	*4*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*23*	*30*	*7*	*9*	*4*	*4*

TABLE 2.1.3-28 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(SOUTHEAST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	122	259	26	31	26	4
2	41	48	6	11	6	6
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	163	307	32	42	32	10

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	35	155	2	2	20	0
2	52	60	7	14	7	7
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	87	215	9	16	27	7

TABLE 2.1.3-28 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*35*	*155*	*2*	*2*	*20*	*0*
2	*52*	*60*	*7*	*14*	*7*	*7*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*87*	*215*	*9*	*16*	*27*	*7*

TABLE 2.1.3-29 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(SOUTH-SOUTHEAST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	92	109	24	29	6	3

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	5	5	0	0	0	0

TABLE 2.1.3-29 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*5*	*5*	*0*	*0*	*0*	*0*

TABLE 2.1.3-30 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(SOUTH SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	92	109	24	29	6	3

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	5	5	0	0	0	0

TABLE 2.1.3-30 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*5*	*5*	*0*	*0*	*0*	*0*

TABLE 2.1.3-31 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(SOUTH-SOUTHWEST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	92	109	24	29	6	3

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	5	5	0	0	0	0

TABLE 2.1.3-31 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*5*	*5*	*0*	*0*	*0*	*0*

TABLE 2.1.3-32 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(SOUTHWEST SECTOR - 0 TO 10 MILES)[a]

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	75	400	50	100	50	20
	(10)	(400)	(5)	(20)	(15)	(2)
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	167	509	74	129	56	23
	(102)	(509)	(29)	(49)	(21)	(5)

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	75	400	50	100	50	20
	(10)	(400)	(5)	(20)	(15)	(2)
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	80	405	50	100	50	20
	(15)	(405)	(5)	(20)	(15)	(2)

TABLE 2.1.3-32 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	75 (10)	400 (400)	50 (5)	100 (20)	50 (15)	20 (2)
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	80 (15)	405 (405)	50 (5)	100 (20)	50 (15)	20 (2)

a. Numbers in parentheses indicate the transient population expected during winter operating hours at GPC recreational facility (October 16 to April 14).

TABLE 2.1.3-33 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(WEST-SOUTHWEST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	92	109	24	29	6	3

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	5	5	0	0	0	0

TABLE 2.1.3-33 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*5*	*5*	*0*	*0*	*0*	*0*

TABLE 2.1.3-34 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(WEST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	92	109	24	29	6	3

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	5	5	0	0	0	0

TABLE 2.1.3-34 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*5*	*5*	*0*	*0*	*0*	*0*

TABLE 2.1.3-35 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(WEST-NORTHWEST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	92	109	24	29	6	3

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	5	5	0	0	0	0

TABLE 2.1.3-35 (SHEET 2 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(WEST-NORTHWEST SECTOR - 0 TO 10 MILES)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*5*	*5*	*0*	*0*	*0*	*0*

TABLE 2.1.3-36 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(NORTHWEST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	92	109	24	29	6	3

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	5	5	0	0	0	0

TABLE 2.1.3-36 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*0*	*0*	*0*	*0*	*0*	*0*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*5*	*5*	*0*	*0*	*0*	*0*

TABLE 2.1.3-37 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
(NORTH-NORTHWEST SECTOR - 0 TO 10 MILES)

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	92	109	24	29	6	3
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	2	5	0	5	6	4
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	94	114	24	34	12	7

Year 2007

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	5	5	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0
7	2	5	0	5	6	4
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
Sector total	7	10	0	5	6	4

TABLE 2.1.3-37 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Weekday Night Average*	*Weekday Night Peak*	*Weekend Day*	*Weekend Night*
1	*5*	*5*	*0*	*0*	*0*	*0*
2	*0*	*0*	*0*	*0*	*0*	*0*
3	*0*	*0*	*0*	*0*	*0*	*0*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*0*	*0*	*0*	*0*	*0*	*0*
6	*0*	*0*	*0*	*0*	*0*	*0*
7	*2*	*5*	*0*	*5*	*6*	*4*
8	*0*	*0*	*0*	*0*	*0*	*0*
9	*0*	*0*	*0*	*0*	*0*	*0*
10	*0*	*0*	*0*	*0*	*0*	*0*
Sector total	*7*	*10*	*0*	*5*	*6*	*4*

TABLE 2.1.3-38 (SHEET 1 OF 2) (HISTORICAL)

TRANSIENT POPULATION
ANNULAR RINGS AND ENCLOSED POPULATION
(0 TO 10 MILES)[a]

Year 1987

Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	2457	2924	560	679	254	225
2	41	48	6	11	6	6
3	75	400	50	100	50	20
	(10)	(400)	(5)	(20)	(15)	(2)
4	0	0	0	0	0	0
5	335	462	27	54	32	27
6	23	34	5	9	22	13
7	25	37	6	17	27	19
8	136	170	29	58	39	36
9	455	490	33	66	41	37
10	933	964	79	108	87	83
Total enclosed	4480 (4415)	5529 (5529)	795 (750)	1102 (1022)	558 (523)	466 (448)

Year 2007

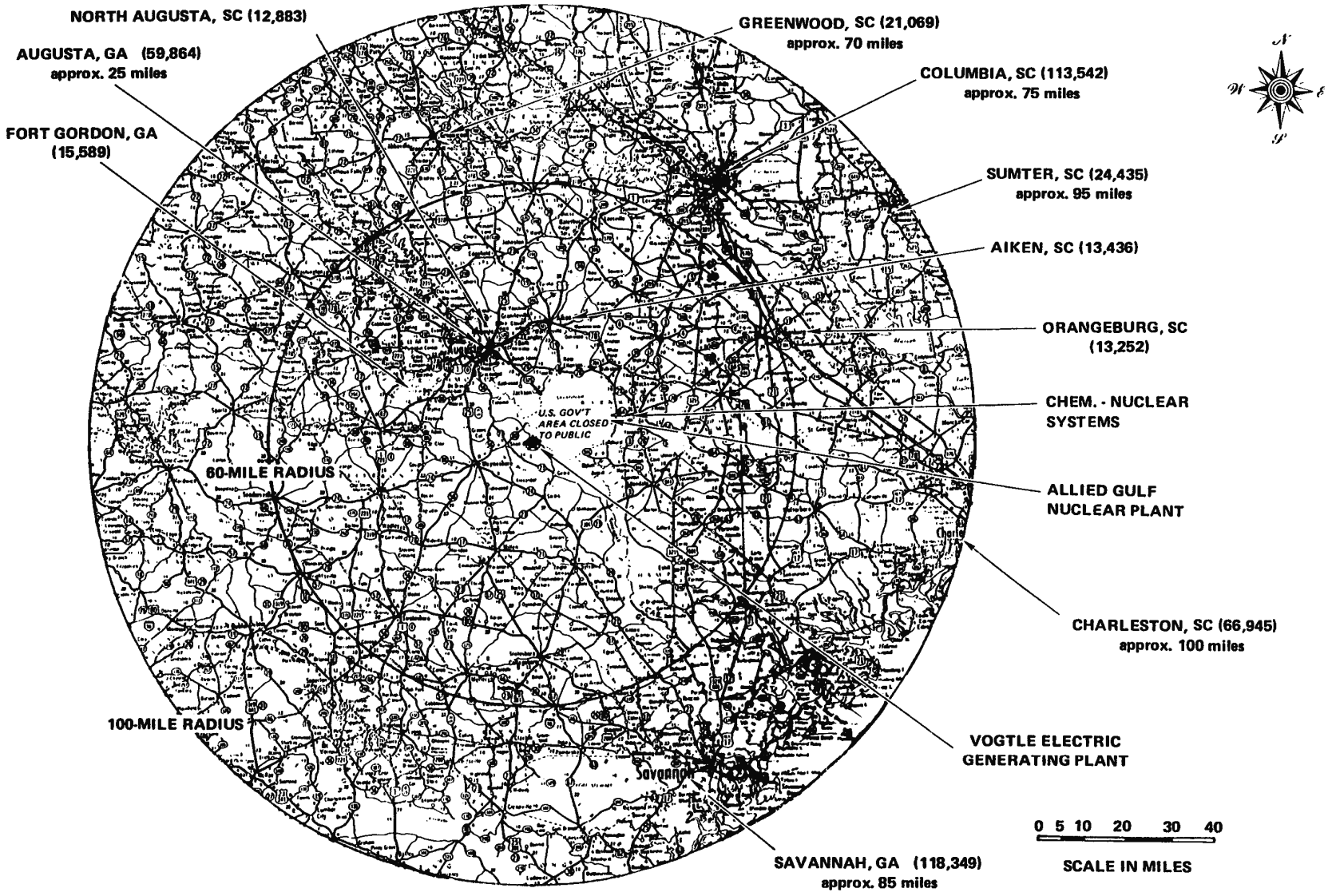
Mile (Ring)	Weekday Average	Weekday Peak	Weekday Night Average	Weekday Night Peak	Weekend Day	Weekend Night
1	804	992	129	169	134	144
2	52	60	7	14	7	7
3	75	400	50	100	50	20
	(10)	(400)	(5)	(20)	(15)	(2)
4	0	0	0	0	0	0
5	335	462	27	54	32	27
6	23	34	5	9	22	13
7	25	37	6	17	27	19
8	136	170	29	58	39	36
9	455	490	33	66	41	37
10	933	964	79	108	87	83
Total enclosed	2838 (2773)	3609 (3609)	365 (320)	595 (515)	439 (404)	386 (368)

TABLE 2.1.3-38 (SHEET 2 OF 2) (HISTORICAL)

Year 2028

Mile (Ring)	*Weekday Average*	*Weekday Peak*	*Night Average*	*Weekday Night Peak*	*Weekday Weekend Day*	*Weekend Night*
1	*804*	*992*	*129*	*169*	*134*	*144*
2	*52*	*60*	*7*	*14*	*7*	*7*
3	*75*	*400*	*50*	*100*	*50*	*20*
	(10)	*(400)*	*(5)*	*(20)*	*(15)*	*(2)*
4	*0*	*0*	*0*	*0*	*0*	*0*
5	*335*	*462*	*27*	*54*	*32*	*27*
6	*23*	*34*	*5*	*9*	*22*	*13*
7	*25*	*37*	*6*	*17*	*27*	*19*
8	*136*	*170*	*29*	*58*	*39*	*36*
9	*455*	*490*	*33*	*66*	*41*	*37*
10	*933*	*964*	*79*	*108*	*87*	*83*
Total enclosed	*2838 (2773)*	*3609 (3609)*	*365 (320)*	*595 (515)*	*439 (404)*	*386 (368)*

a. Numbers in parentheses indicate the transient population expected during winter operating hours of GPC recreational facility (October 16 to April 14).



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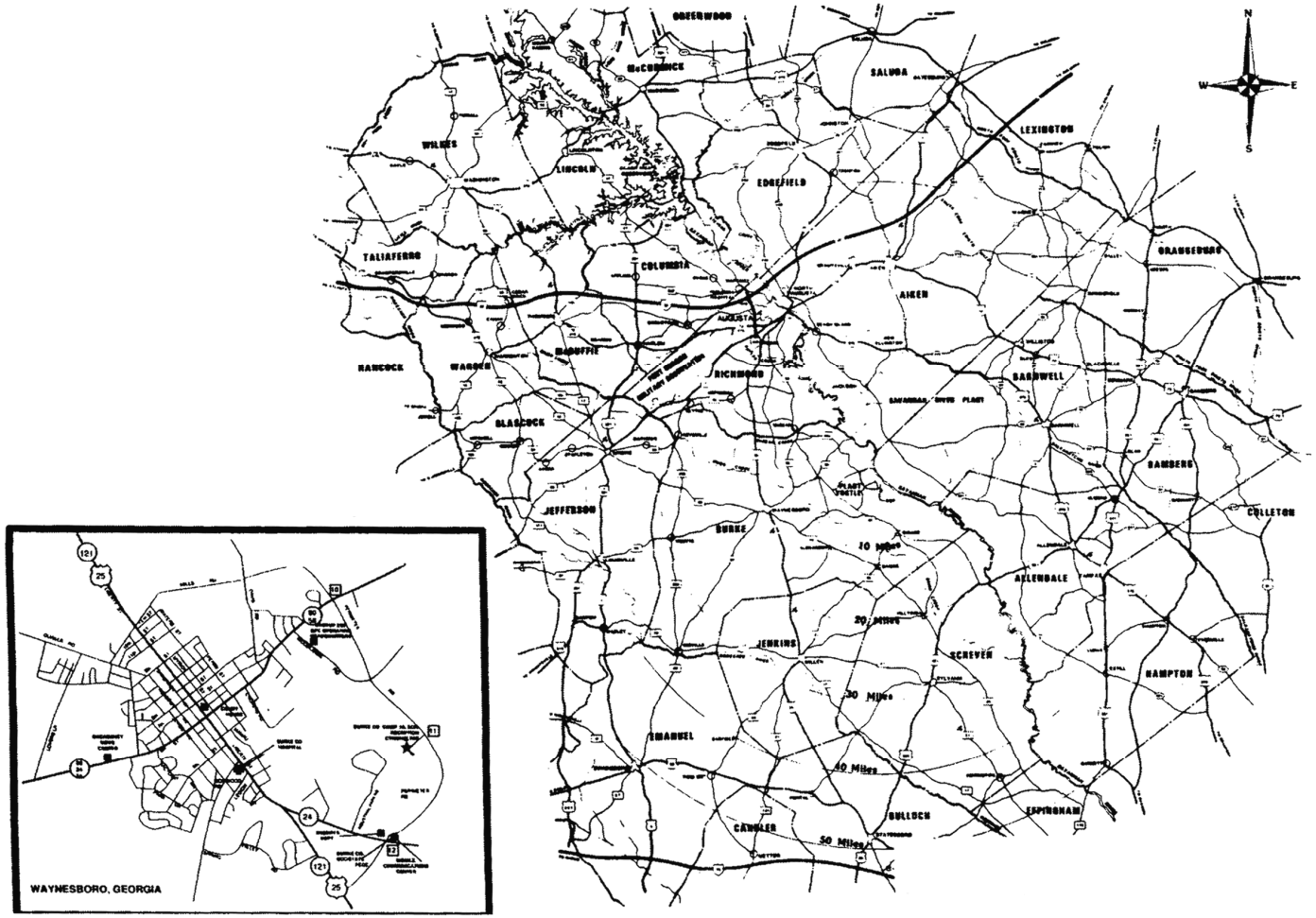
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

POPULATION CENTERS (HISTORICAL)

FIGURE 2.1.1–1



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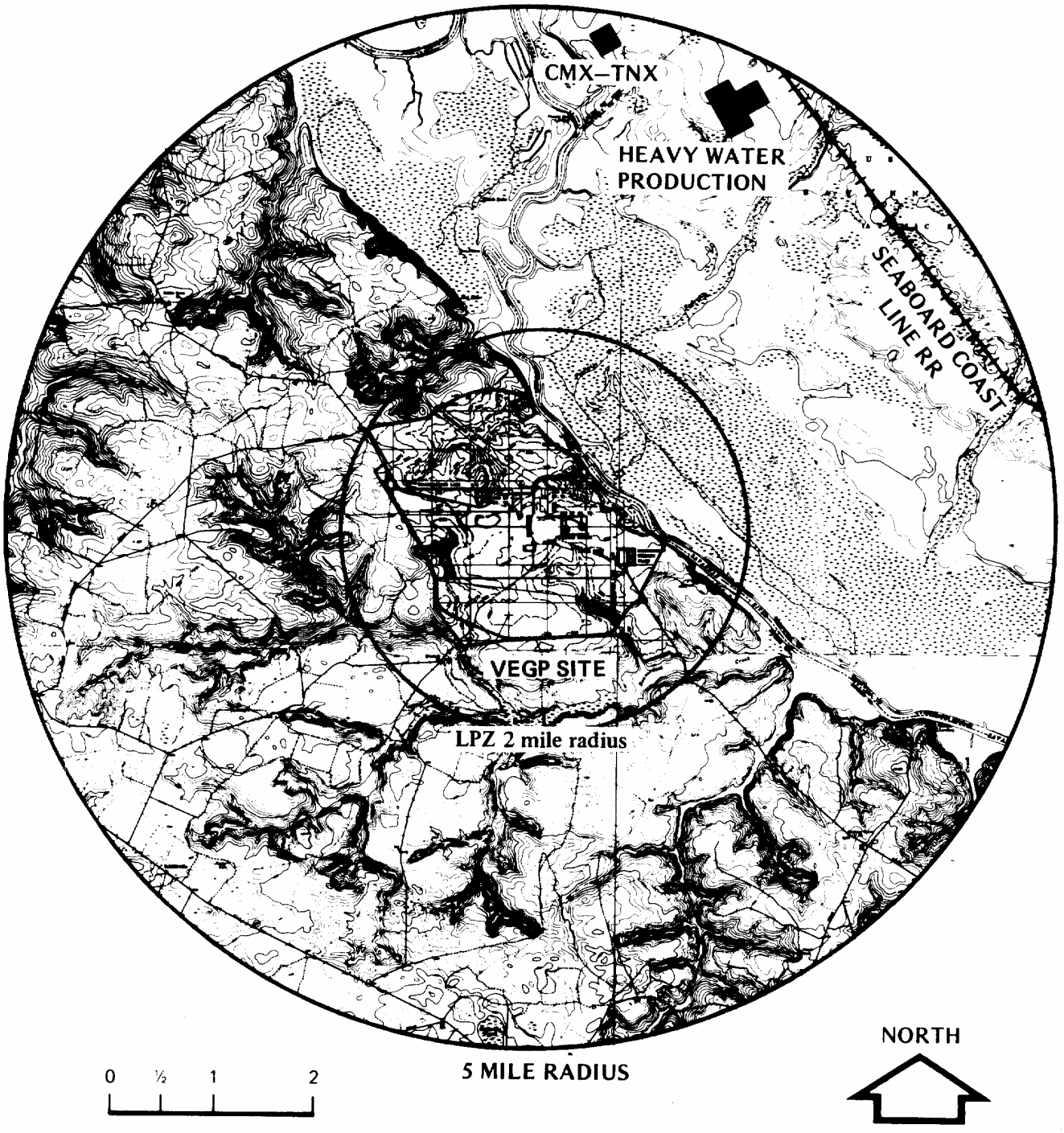
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

AREA MAP

FIGURE 2.1.1–2



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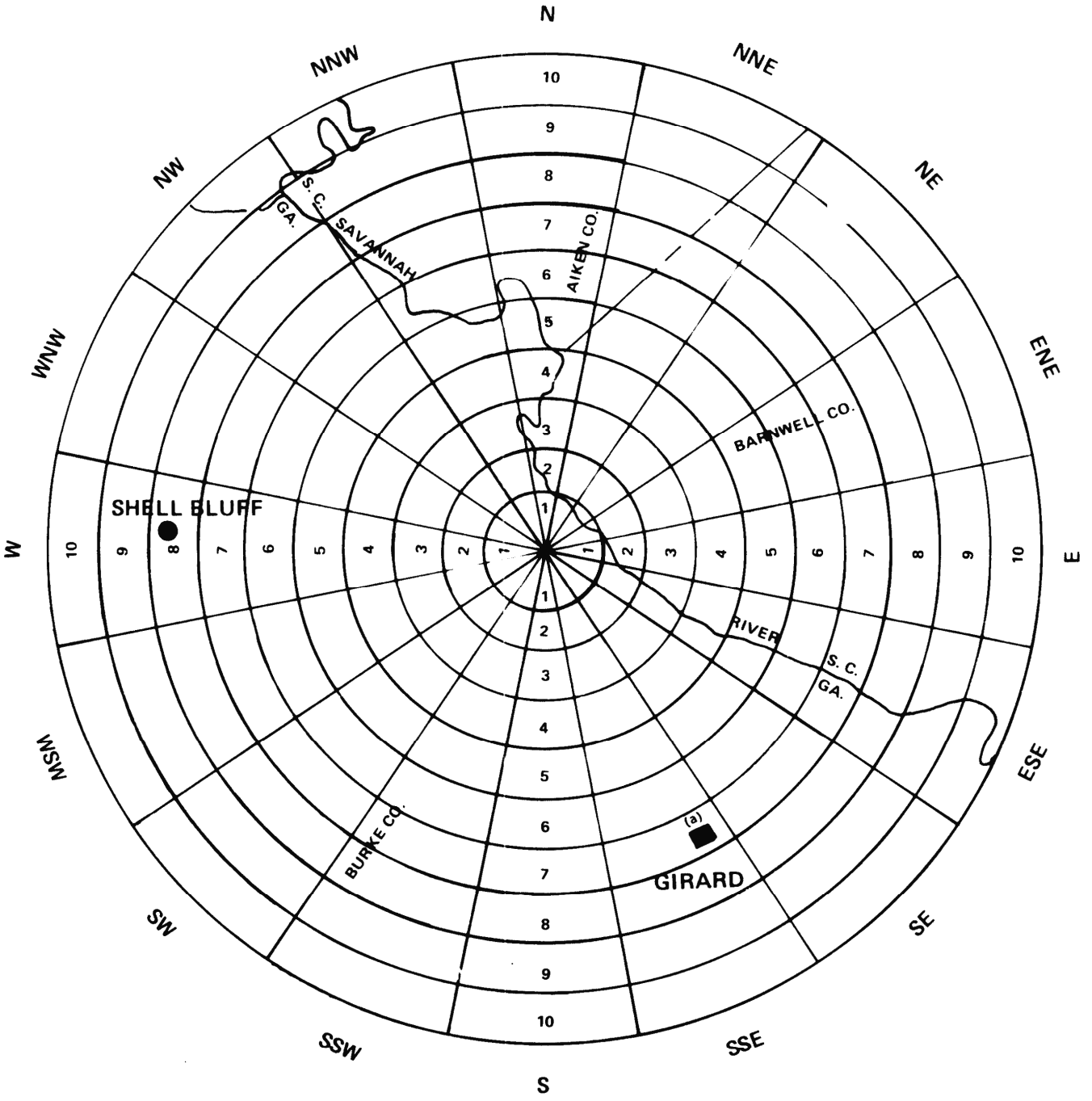
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

AREA MAP

FIGURE 2.1.1–3



a. GIRARD-SEVEN MILES SSE; POPULATION 1980-206, 1985-188

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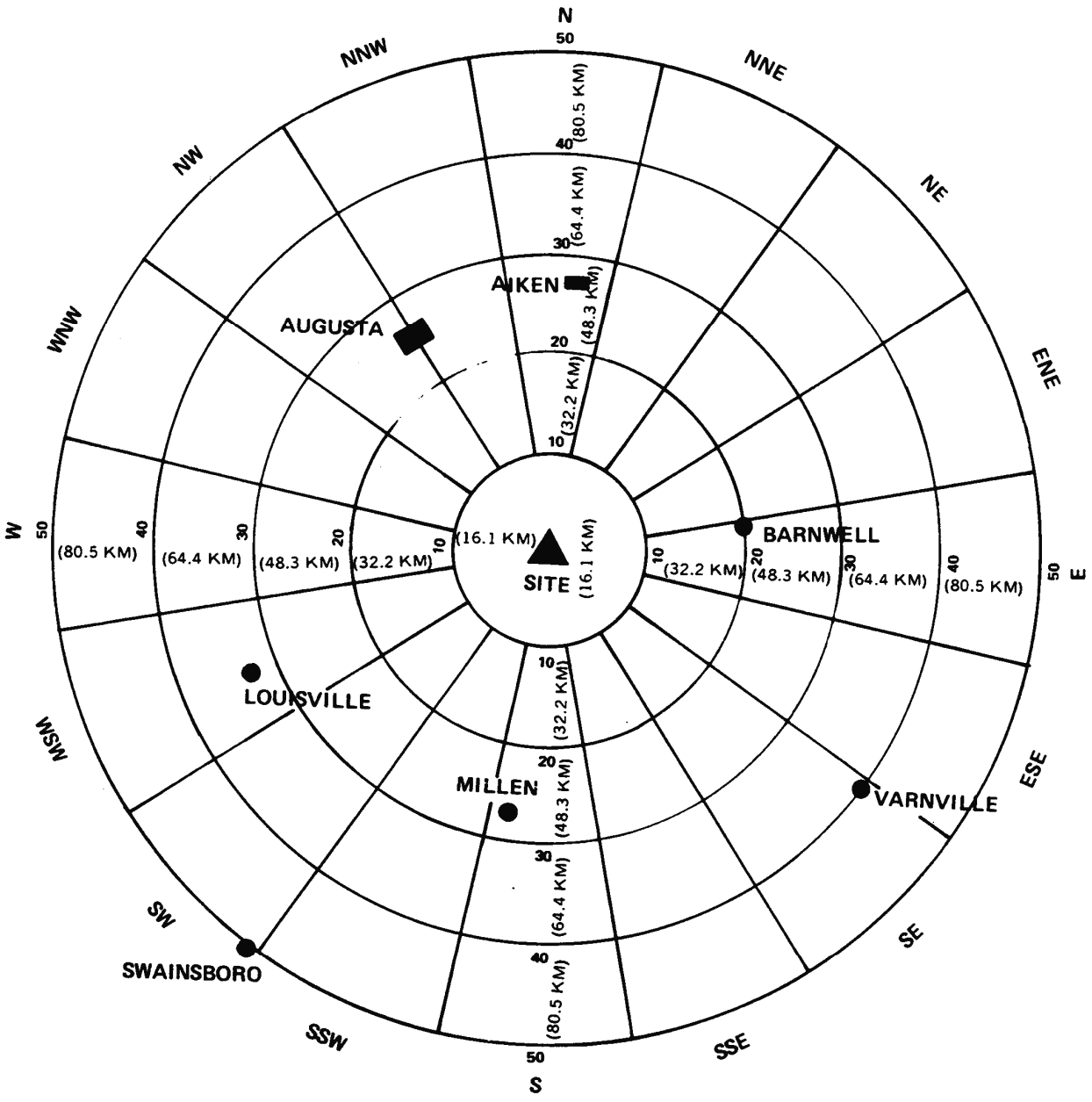
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

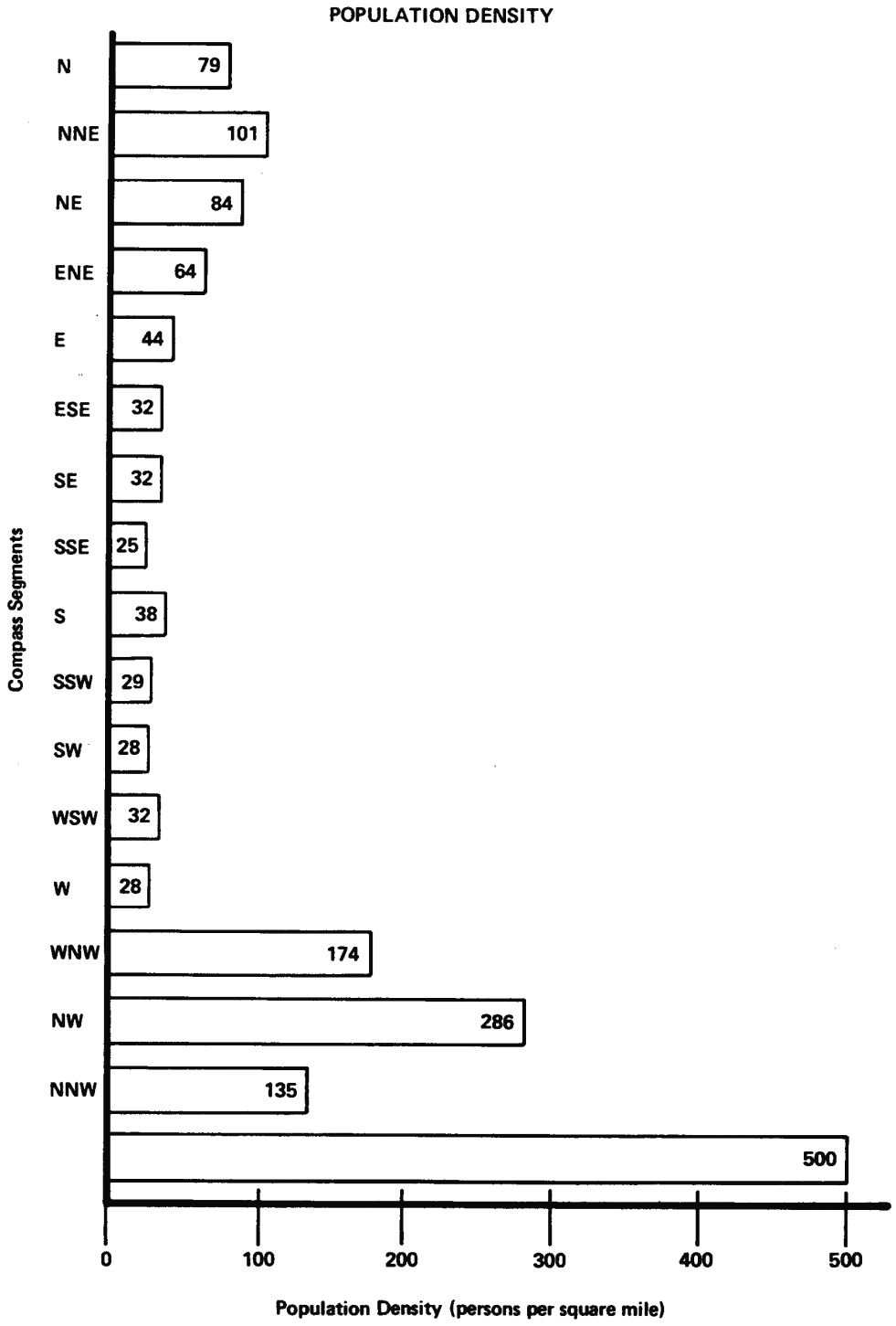
POPULATION DISTRIBUTION WITHIN 10 MILES (HISTORICAL)

FIGURE 2.1.3–1



REV 15 4/09

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	*POPULATION BETWEEN 16.1 AND 80.5 KILOMETERS (10 AND 50 MILES) (HISTORICAL)* *FIGURE 2.1.3–2*



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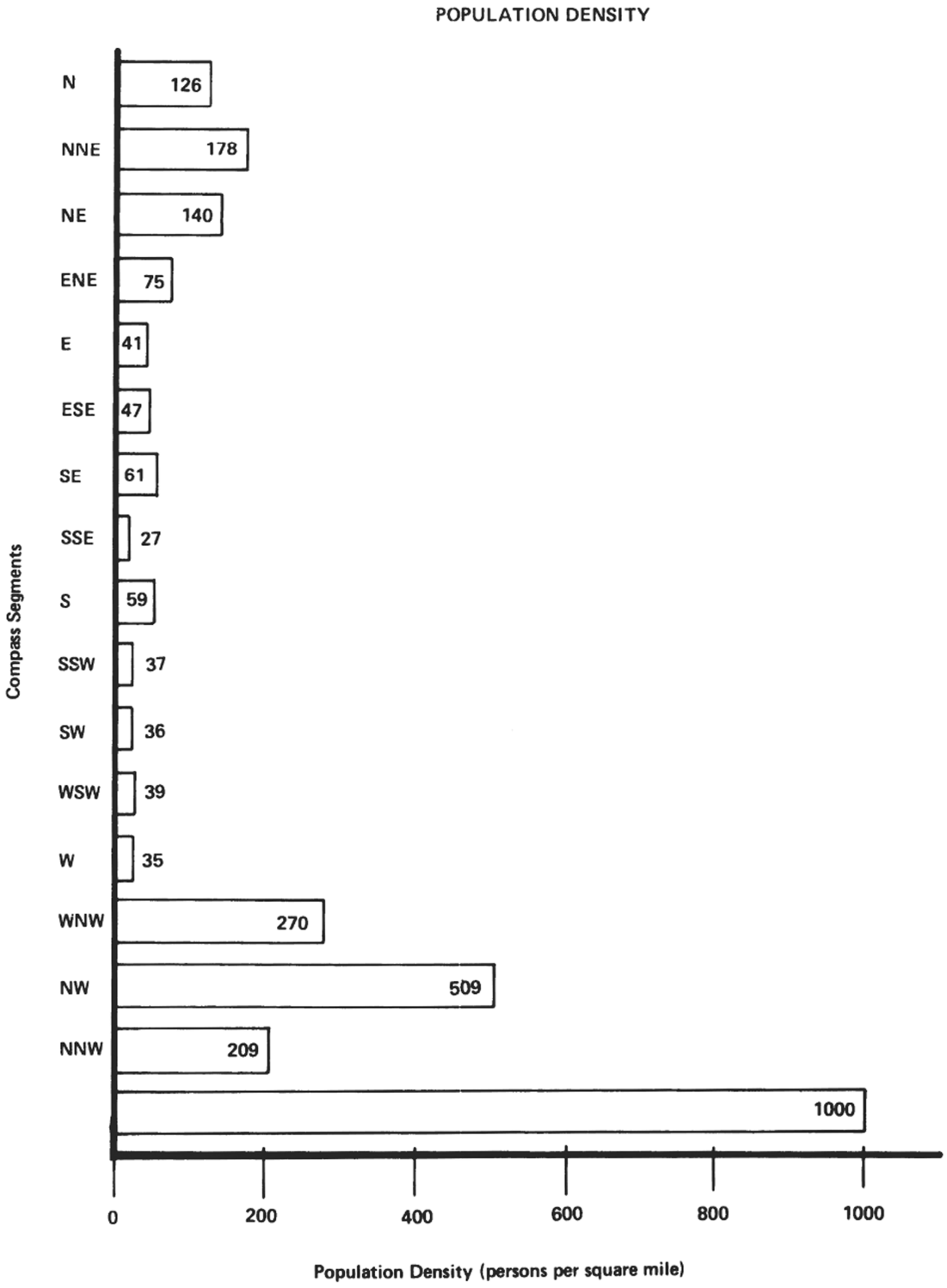
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

1987 POPULATION PROJECTIONS (50–MILE RADIUS) (HISTORICAL)

FIGURE 2.1.3–3



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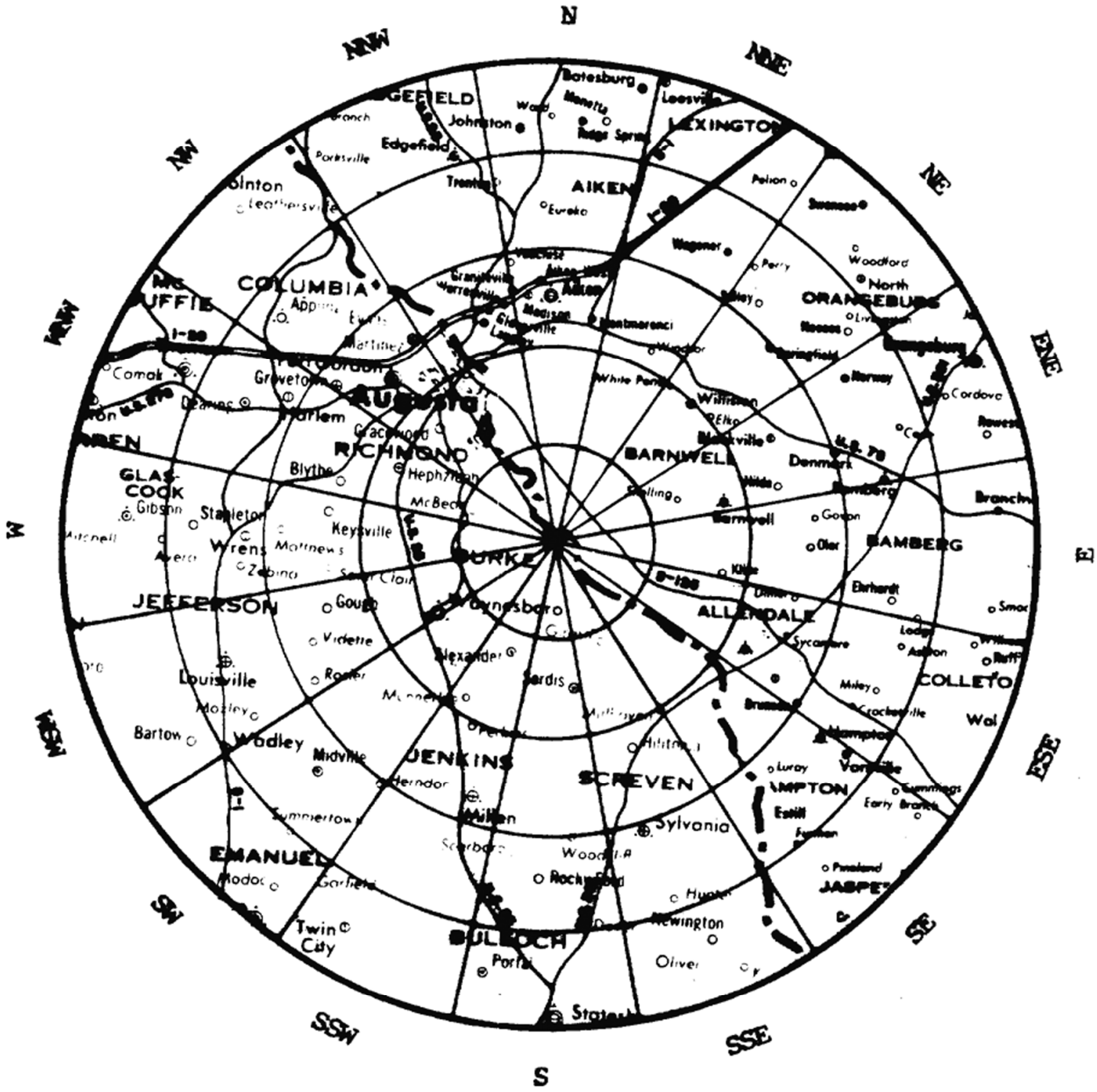
UNIT 1 AND UNIT 2

2028 POPULATION PROJECTIONS
(50–MILE RADIUS) (HISTORICAL)

FIGURE 2.1.3–4


N
NNW
NNE
NW
NE
WNW
ENE
W
E
WSW
ESE
SW
SE
SSW
SSE
S
AIKEN
COLUMBIA
AUGUSTA
RICHMOND
BARNWELL
ORANGEBURG
BAMBERG
ALLENDALE
COLLETON
BURKE
JEFFERSON
JENKINS
SCREVEN
EMANUEL
BULLOCH
JASPER
Louisville
Wadley
Midville
Sylvania
Twin City
Newington
Johnston
Edgefield
Batesburg
Leesville
LEXINGTON
Wagener
Salley
Denmark
Barnwell
Blythe
Waynesboro
Sardis
Millen
Statesboro
Oliver
Hephzibah
Keysville
Wrens
Gough
Alexander
Bartow
Midville
Lyons
Sycamore
Ehrhardt
Olar
Williston
Norway
Cordova
Woodford
North
Perry
Polion
Jackson
Graniteville
Warrenville
Aiken
Eureka
Trenton
Edgefield
Johnston
Batesburg
Leesville
Wagener
Salley
Denmark
Barnwell
Olar
Ehrhardt
Sycamore

2.2 IDENTIFICATION OF POTENTIAL HAZARDS IN SITE VICINITY

2.2.1 LOCATION OF NEARBY INDUSTRIAL, MILITARY, AND TRANSPORTATION FACILITIES

There are two Savannah River Plant facilities, Georgia Power Company's Plant Wilson, a construction site for Units 3&4, one railroad, and one highway with commercial traffic within a 5-mile vicinity of the plant.

Figure 2.2.1-1 shows the location of the major industrial facilities within a 5-mile radius of the VEGP site, including:

- The heavy water production and recovery facility of the Savannah River Plant.
- The TNX-CMX semiworks facilities of the Savannah River Plant.
- Plant Wilson.
- Georgia State Highway 23.
- The Seaboard Coast Line Railroad.
- Construction site for Vogtle Units 3 & 4

Figure 2.2.1-2 shows the location of major industrial facilities, military bases, and pipelines in relation to a 25-mile radius of the site. Highway transportation routes, airports, dock facilities, and railroad facilities are shown in figure 2.2.1-3.

Items illustrated on the maps are described in subsection 2.2.2. There are no military bases within a 25-mile radius of the site. The only major storage facilities within 25 miles of VEGP, other than those at the Savannah River Plant, Allied General Nuclear Services, and Chem-Nuclear Systems, are oil storage tanks associated with the existing combustion turbine-generators for Plant Wilson adjacent to the VEGP site.

2.2.2 DESCRIPTION

2.2.2.1 Facilities

2.2.2.1.1 Savannah River Plant(1)

The Savannah River Plant site (figure 2.2.2-1) borders the Savannah River for approximately 17 miles opposite the VEGP site and is the only industrial facility even partially within a 5-mile radius of the site. The Savannah River Plant occupies an approximately circular area of 300 mi^2 (192,000 acres) in South Carolina, 25 miles southeast of Augusta, Georgia. The Savannah River Plant was built and is operated for the U.S. Department of Energy by E. I. DuPont de Nemours and Company. The site is a closed government reservation, except for through traffic on South Carolina Highway 125 (Savannah River Plant road A) and the Seaboard Coast Line Railroad, and has controlled access for environmental study.

The Savannah River Plant produces nuclear materials, primarily plutonium and tritium, to meet nuclear materials requirements for the Department of Defense. The nuclear products are formed by irradiation of materials in nuclear reactors and recovered by chemical separations

processes. The Savannah River Plant employed a total of 6675 persons as of December 31, 1982.

The Savannah River Plant facilities (figure 2.2.2-1) consist of:

- Four operating nuclear production reactors (P, K, C, and L).
- One nuclear production reactor in standby condition (R).
- A small test reactor in standby condition (U).
- Two separations areas for processing irradiated materials (F and H).
- A heavy water extraction and recovery plant (D).
- A fuel target fabrication facility containing two test reactors (M).
- The Savannah River Laboratory (a process development laboratory to support production operations and containing two test reactors) and administrative facilities (A).
- A variety of nonnuclear facilities necessary for plant operations, including the TNX-CMX semiworks area located near the heavy water production and recovery plant.

The major waste storage areas for radioactive liquids, sludges, and crystallized salts are adjacent to the separations areas and consist of two tank farms linked to the separations areas and to each other by pipelines with secondary containment. In addition, a 195-acre burial ground area located between the F and H separations areas is used for controlled storage of solid radioactive wastes. The reactors, separations areas, and waste storage areas are at least 4 miles from the nearest plant boundary.

2.2.2.1.1.1 Nuclear Production Reactors (100 Area). Four production reactors (P, K, C, and L) of the five originally constructed are currently operating at the Savannah River Plant. The R reactor was shut down and placed in standby condition in 1964. The primary products of the Savannah River Plant reactors are plutonium (Pu^{239}) and tritium (T or H^3). These are produced by absorption of neutrons in uranium (U^{238}) and lithium (Li^6), respectively. The versatility of the reactors has also led to the production of other nuclear materials.

Each production reactor and its auxiliaries are housed in a reactor building that incorporates heavy concrete shielding to protect personnel from radiation. The reactor uses heavy water (D_20) as a neutron moderator and as a recirculating primary coolant to remove heat from the nuclear fission process. Water pumped from the Savannah River or Par Pond, a cooling water impoundment, serves as a secondary coolant.

2.2.2.1.1.2 Chemical Separations Facilities (200 Area). The chemical separations facilities consist of two main operating areas, 200-F and 200-H. Each main area has a large shielded "canyon" building for processing irradiated materials (fuel and/or targets), a waste concentration and storage system, seepage basins, powerhouses, and service facilities. In addition, F area contains the main analytical laboratory, the plutonium metallurgical building, and the plutonium fuel form facility. The H area contains the tritium process buildings, the receiving basin for offsite fuels, and the resin regeneration facility. These areas are the major sources of radioactive liquid and solid wastes that are stored onsite.

In the chemical processing plants the first step is to dissolve the irradiated components which liberate volatile fission products and to generate solutions with high concentrations of radioactivity. The initial separation yields solutions of plutonium, uranium, or neptunium product, and also a high-heat liquid waste, containing the nonvolatile fission products. After the fission products are removed sufficiently from the product solutions, further processing can be done in unshielded areas where the product elements may be converted from solution form to solid. The final steps involve handling powders of both Pu^{238} and Pu^{239} that require additional precautions to avoid airborne releases.

2.2.2.1.1.3 Fuel and Target Fabrication (300 Area). The facilities for fabricating fuel and target elements to be irradiated in Savannah River Plant reactors are located in the 300-M area. Major products are extruded enriched uranium-aluminum alloy fuel and canned depleted uranium metal targets. The only significant radioactive release from these facilities is uranium in aqueous effluents from the canning process. The total amount released since 1955 is about 85,000 lb or 24 Ci of uranium. About 50 percent of the material has settled in plant stream and pond beds adjacent to the area, and none has reached the main stream leading to the river.

2.2.2.1.1.4 Heavy Water Production and Recovery (400 Area). In the 400-D area, heavy water (D_2O) is separated from Savannah River water by a hydrogen sulfide extraction process with final purification by distillation. The most significant release of radioactivity from the 400-D area is tritium from the reprocessing of degraded reactor D_2O coolant that is reprocessed in a separate distillation facility. In 1975, 3000 Ci of tritium were released to the atmosphere and 1600 Ci to Beaver Dam Creek. The major nonradioactive releases from the area in 1975 were about 180,000 lb/year of SO_2 and about 180,000 lb/year of H_2S to the atmosphere from the heavy water extraction plant.

The largest steam power plant (484-D) on the plant site is located in the 400-D area, along with:

- A rework unit (removes H O from D O coolant).
- A drum cleaning facility (where 50-gal drums used for heavy water storage and transport are cleaned).
- An analytical laboratory (for analyses of aqueous and atmospheric releases).
- An extraction plant.
- A distillation plant.

2.2.2.1.1.5 Savannah River Laboratory. The Savannah River Laboratory has primary responsibility for research and development activities at the Savannah River Plant and also has been responsible for small-scale manufacturing activities. Laboratory operations are conducted in the technical area, TNX-CMX semiworks near the river, the thermal effects laboratory on Upper Three Runs, in the Par Pond area, and in the U area.

2.2.2.1.1.6 TNX-CMX Semiworks Area. The TNX-CMX semiworks area is located near the Savannah River adjacent to the heavy water production and recovery plant (400-D area). The purpose of the TNX facilities is to study chemical processing problems and test production-scale equipment. Analyses are conducted with nonradioactive standins or natural uranium. The test equipment includes dissolvers for uranium metal and other nuclear materials,

evaporators, mixer-settlers for solvent extraction, centrifuges, tanks, pumps, large furnaces, vacuum equipment, and distillation equipment. The CMX has nine flow loops and associated equipment for nonnuclear tests of portions of hydraulic systems of Savannah River reactors; only one flow loop is currently in operation.

A concise description of major Savannah River Plant facilities is provided in table 2.2.2-1.

2.2.2.1.2 Transportation Routes

The nearest highway with commercial traffic is Georgia State Highway 23, 5 air miles southwest of the site. The nearest railroad with commercial traffic is approximately 4.5 miles northeast of VEGP. A second railroad lies approximately 12 miles west of VEGP.

2.2.2.1.3 Barnwell Nuclear Fuel Plant

Although not located within 5 miles of the VEGP site, the Barnwell Nuclear Fuel Plant, located in the Barnwell County Industrial Park, is described due to its relative proximity (figure 2.2.2-1) and association with Savannah River Plant. The Barnwell Nuclear Fuel Plant was constructed by Allied-General Nuclear Services for the purpose of reprocessing spent light-water reactor fuel. Such commercial reprocessing has been indefinitely deferred as a matter of U.S. policy. If commercial reprocessing is ever permitted, it is unlikely to occur before the late 1980s. Moreover, the nature of such possible future reprocessing is currently indeterminate, being dependent upon federal decisions related to weapons proliferation concerns. At present the Barnwell Nuclear Fuel Plant is engaged in carrying out only resource and development activities pursuant to a contract with the Department of Energy. However, the federal government has expressed interest in acquiring the Barnwell Nuclear Fuel Plant for use as an away-from-reactor interim storage site for spent fuel.[2]

Table 2.2.2-2 provides a tabular description of Barnwell Nuclear Fuel Plant facilities.

2.2.2.1.4 Chem-Nuclear Systems, Inc.

Chem-Nuclear Systems, Inc., developed, constructed, and operates the largest radioactive waste disposal site in the country near Barnwell, South Carolina. The Chem-Nuclear Systems, Inc., has two licensing agencies and two current licenses to operate the Barnwell Low-Level Waste Disposal Facility, i.e., Radioactive Material License No. 097, issued by the South Carolina Department of Health and Environmental Control, and Radioactive Materials License No. 46-13536-01, issued by the United States Nuclear Regulatory Commission (NRC). The NRC license specifically addresses the possession of special nuclear material. In addition to the two licenses, Chem-Nuclear Systems, Inc., adheres to the established Department of Transportation guidelines of packaging and transportation of hazardous material as stated in Title 49 of the Code of Federal Regulations.

The Chem-Nuclear Systems, Inc., site (figure 2.2.2-2) comprises some 308 acres of which 235 have been deeded to the State of South Carolina as a designated exclusion area. Background radiation averages 0.29 mrem/day. Waste receipts are in the form of solids; no liquids are accepted. The site, now in its 11th year of operation, has disposed of more than 9.5 million ft^3 of waste, having an activity of 2.5 million Ci. Receipts for 1979 exceeded 2.2 million ft^3, having an activity of 314,000 Ci. Isotopes of cobalt and cesium head the list of those received.[3]

Table 2.2.2-3 provides a tabular description of Chem-Nuclear Systems, Inc., facilities.

2.2.2.1.5 Georgia Power Company's Plant Wilson

Plant Wilson is located east-southeast of the power block, approximately 5000 ft along River Road. The existing combustion turbine plant is an electrical peaking power station of Georgia Power Company. The plant consists of six combustion turbines with a total rated capacity of 351.6 MW. The storage capacity of the fuel storage tanks is 9,000,000 gal.

2.2.2.1.6 Major Industries Within 25 Miles

Table 2.2.2-4 describes major industrial facilities of 25 or more employees within a 25-mile radius of the site but greater than 5 miles from the site, with the exception of the previously discussed Barnwell Nuclear Fuel Plant (19 miles away), the Chem-Nuclear Systems, Inc., radioactive waste disposal site (18 miles away), and the facilities of the Savannah River Plant.

2.2.2.1.7 VEGP 3&4 Construction Site

The VEGP 3&4 construction site will be treated as a nearby industrial facility for the purposes of conformance to RG 1.70 and in response to GL 2003-01 and NEI 99-03 on control room habitability and emergency diesel operability. Though part of a shared property, VEGP 3&4 construction site is outside the protected area of VEGP 1&2 and is under control of the construction organization.

Proposed bulk chemicals and gases introduced on the VEGP Units 3&4 construction site are evaluated in accordance with applicable RG 1.70, RG 1.78, RG 1.91, and NUREG/CR-1748 guidance for determination of potential toxic hazards and design basis events for Units 1&2. Bulk chemicals or gases intended to be stored permanently are listed in table 2.2.3-14, Nearby Industrial Facility Chemical Storage or Table 2.2.3-15, Toxic Gas Release Information, as applicable. Temporary bulk chemicals and gases (those used to support construction of Units 3&4 such as anhydrous ammonia, argon, carbon dioxide, nitrogen, etc.) are evaluated for potential hazards with similar rigor as permanent chemicals; however, these are not included in table 2.2.3-14 or 2.2.3-15.

Evaluation of temporary bulk chemicals and gases is necessary to ensure introduction on site does not create a design basis event of VEGP Units 1&2 which may lead to exceeding the criteria established in 10 CFR 100. The impacts of chemical storage on control room habitability and nearest safety-related structures (NSRS) for VEGP 1&2 are considered and, if necessary, a detailed analysis evaluates toxic vapor clouds, tank explosions, flammable/explosive vapor clouds, and/or missiles generated by explosions in accordance with methodology and acceptance criteria consistent with the Units 1&2 licensing bases. Potential overspeed and/or starvation conditions for the Emergency Diesel Generators are also considered for releases, as applicable.

2.2.2.2 Products and Materials

2.2.2.2.1 Savannah River Plant

A description of the products and materials regularly manufactured, stored, used, or transported in relation to those Savannah River Plant facilities within 5 miles of the VEGP site are provided in table 2.2.2-5. The applicable toxicity limits for each hazardous material are provided in table 2.2.2-6.

It should be noted that the amounts of nuclear materials produced by the Savannah River Plant are classified information and that this document represents a best effort at obtaining detailed information for facilities of Savannah River Plant.

2.2.2.2.2 Transportation Routes

2.2.2.2.2.1 Georgia State Highway 23. Segments of Georgia State Highways 23, 80, and 56C are located within a 5-mile radius of the site. The Georgia Department of Transportation does not keep figures on the products and materials carried over these roads.[4] However, major commercial traffic occurs only on State Highway 23, which serves as a major link between Augusta and Savannah. The heaviest truck traffic along State Highway 23 near the site consists primarily of timber and wood products and materials. State Highways 80 and 56C serve primarily as minor transportation routes for local traffic, much of which is associated with development of the VEGP site.

Available statistical data on accidents on these roads (table 2.2.2-7) indicate that the majority of accidents in the vicinity are automobile accidents.

Several access roads to various Savannah River Plant facilities exist within 5 miles of the VEGP site. However, by December 1980 the Savannah River Plant reported that: "Within the past 5 years there have been no transportation accidents on the portion of the SRP that is within the 5-mile radius of the VEGP site."[5]

2.2.2.2.2.2 Seaboard Coast Line Railroad. The only railroad within a 5-mile radius of the site is the Seaboard Coast Line Railroad, which runs through and services the Savannah River Plant approximately 4.5 miles northeast of the site. Data on products and materials carried by the Seaboard Coast Line Railroad are presented in table 2.2.2-8. Hazardous cargo accident data for the Seaboard Coast Line Railroad are provided in table 2.2.2-9.

2.2.2.2.3 Barnwell Nuclear Fuel Plant

A description of the products and materials associated with Barnwell Nuclear Fuel Plant is presented in table 2.2.2-10. Refer to table 2.2.2-6 for applicable toxicity limits.

2.2.2.2.4 Chem-Nuclear Systems, Inc.

A description of the products and materials associated with Chem-Nuclear Systems, Inc., is presented in table 2.2.2-11. Refer to table 2.2.2-6 for applicable toxicity limits.

2.2.2.3 Pipelines

There are three pipelines (figure 2.2.1-2) within 25 miles of VEGP. Pipeline 1, located approximately 21 miles northeast of VEGP, is an 8-in.-diameter line constructed in 1959. It operates at a maximum pressure of 750 psi, is buried 3 ft deep, has 8-in. Rockwell isolation valves at 25-mile intervals, and carries natural gas. It is not used for storage. There are no plans for carrying a different product.[6]

Pipeline 2, located approximately 19 miles southwest of VEGP, has a 14-in.-diameter line constructed in 1954 and a 20-in.-diameter line constructed in 1977. Both lines are buried 3-ft deep, operate at a maximum pressure of 1250 psi, have buried Rockwell isolation valves every 8 to 9 miles, and carry natural gas. They are not used for storage. There are no plans for carrying a different product.[7]

Pipeline 3, located approximately 20 miles northwest of VEGP, has two 16-in.-diameter lines constructed in 1953 and 1957. Both operate at a maximum pressure of 1250 psi, are buried 3 ft deep, have buried Rockwell isolation valves every 8 to 9 miles, and carry natural gas. There are no plans for carrying a different product.(7) Because all of the pipelines identified are well over 5 miles from the VEGP site, the locations at individual pipeline valves are not specified.

2.2.2.4 Waterways

The VEGP is located on the Savannah River. Commercial river traffic is composed of barge-tug tows moving up and down the river channel between Augusta and Savannah. The navigational channel is maintained at a depth of 9 ft by the Corps of Engineers. There are no locks or dams in the vicinity of the plant site. The intake structure is located off the navigational channel at the end of a 400-ft canal.

For the year 1978, a total of 115 tows carried a total freight of 113,900 tons on the Savannah River past the plant site. The freight itself consisted of 102,517 tons of residual fuel oil, 8244 tons of nitrogenous chemical fertilizer, 3099 tons of gasoline, and 40 tons of iron steel and tube.(8)

2.2.2.5 Airports

There are no airports within 15 miles of VEGP. The nearest, the Burke County Airport, approximately 16 air miles WSW (figure 2.2.1-3), has a 3200-ft runway oriented 250° WSW - 70° ENE. The airport uses a nondirectional radio beacon for runway approach. It is used by single-engine private aircraft and by crop-dusting operations. Aircraft operations from May 1979 to May 1980 total 14,401, with about 1600 flights during a peak month.(9) Approach and departure paths extend ENE and WSW approximately 5 miles from the runway. Future plans(10) for the Burke County Airport call for the construction of two hangars 30 x 40 ft each and a 30- x 40-ft terminal building. No runway construction is planned. No increase in aircraft traffic is expected.

Bush Field,(11) located approximately 17 air miles NNW of VEGP, has an 8000-ft runway oriented 170° SSE - 350° NNW and a 6000-ft runway oriented 080° ENE - 260° WSW. It is served by Delta and Atlantis Airlines, with a fiscal year 1980 total of 7781 flights, and air taxi services, with a fiscal year 1980 total of 2026 flights. General aviation of other types flew a total of 56,998 flights during fiscal year 1980. Military planes flew 4921 flights during fiscal year 1980. Total flights into and out of Bush Field during fiscal year 1980 were 71,726.(12) Projections for air traffic at Bush Field up to fiscal year 1990 are given in table 2.2.2-12.(13)

Approach and departure paths at Bush Field are not aligned with VEGP; no regular air traffic patterns for Bush Field extend into air space over VEGP.

Future plans(14) for Bush Field call for lengthening the 8000-ft runway by 500 ft and adding hangar and parking facilities. A new 10,000-ft runway is proposed after 1990. There is an airway, V 185, V 18S, adjacent to VEGP (figure 2.2.1-3). There are no low altitude military training routes, pilot training areas, or approach and departure paths to airports or military facilities in the vicinity of VEGP.(15)

2.2.2.6 Projections of Industrial Growth

Population projections for the three South Carolina counties that lie partially within the 25-mile radius of VEGP indicate an average annual population growth rate for 1980 to 1990 of 1.7 percent.(16) It should be noted that Aiken is a primary growth center, which heavily influences the current (1970 to 1980) percent population increase for Aiken County, the 15.2-

percent increase for Barnwell County, and the 8.6-percent increase for Allendale County. There are no projected major increases to industrial, military, or transportation facilities in South Carolina within a 25-mile radius of VEGP.

Population data for the four Georgia counties that lie partially within a 25-mile radius of VEGP show an average annual growth rate of 0.08 percent for the past census decade. Augusta is the primary growth center and has heavily influenced the decade's 8.8-percent population increase for Richmond County, the 5.5-percent growth for Jenkins County, the 5-percent growth for Burke County, and the 9-percent population increase for Screven County. The only major increase to industry anticipated for the immediate future will be VEGP. The only increase to transportation facilities is the proposed improvement to Bush Field, Augusta Municipal Airport.

2.2.2.7 References

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9. Federal Aviation Administration, Airport Master Record, Burke County, Department of Transportation, 1980.
10. Wainwright Engineering Co., "Airport Layout Plan, Burke County Airport," PGP-A-13-0120-01, Waynesboro, Georgia.
11. Bessom and Pope, consulting engineers, Airport Layout Plan, Bush Field, Augusta, Georgia.
12. Federal Aviation Administration, Air Traffic Activity, Fiscal Year 1980, table 4, Department of Transportation.
13. Bessom and Pope, consulting engineers, Preliminary Airport Layout Plan, Bush Field, Augusta, Georgia.
14. Georgia Department of Transportation, Georgia Aeronautical Chart, 1978-1979.
15. Federal Aviation Administration, Terminal Area Forecast Fiscal Year 1990, Department of Transportation.

16. Lower Savannah Council of Governments, Lower Savannah Overall Economic Development Plan 1981-1985, Aiken, South Carolina.

2.2.3 EVALUATION OF POTENTIAL ACCIDENTS

The accidents considered in this subsection include explosions, delayed ignition of flammable vapor clouds, release of toxic vapors, fires, collisions with the intake structure, and liquid spills.

2.2.3.1 Determination of Design Basis Events

The Standard Review Plan defines design basis events external to the plant as those accidents for which a realistic estimate of the annual probability of exceeding 10 CFR 100 exposure guidelines is in excess of approximately 10^{-7} or for which a conservative estimate of this probability is in excess of approximately 10^{-6}. In lieu of developing specific probabilistic estimates, the accidents considered in this subsection are evaluated deterministically using standardized techniques developed by the NRC and industry. Chemical substances stored at nearby industrial facilities or transported within a 5 mile radius of the plant are identified and evaluated for potential impact on the safe operation of the plant in accordance with established NRC guidelines. Accidents evaluated include explosions, delayed ignition of flammable vapor clouds, release of toxic vapors (from onsite and offsite sources), and fires (including forest fires and fires due to oil or gas pipeline ruptures). Collisions with the intake structure and liquid spills are also evaluated.

A description of data sources, assumptions, and computational methods is presented in the following paragraphs.

2.2.3.1.1 Chemical Substances Stored or Transported Within a 5-Mile Radius of the Plant

As described in subsection 2.2.2, the VEGP site is located along the Savannah River. Commercial traffic on the Savannah River is composed of barge-tug tows moving up and down the river channel between Augusta and Savannah, Georgia. Segments of Georgia State Highways 23, 80, and 56C are located within a 5-mile radius of the VEGP site. However, major commercial traffic occurs only on State Highway 23, which serves as a major link between Augusta and Savannah, Georgia. The only railroad within a 5-mile radius of the VEGP site is the Seaboard Coast Line Railroad, which runs through and services the Savannah River Plant, approximately 4.5 miles northwest of the VEGP site. Chemical substances transported along these major routes within a 5-mile radius of the VEGP site are identified in table 2.2.3-6. Only bulk chemicals or gases transported for use at the permanent Units 3&4 plant are listed on the appropriate tables, as applicable.

As described in subsections 2.2.1 and 2.2.2, the major industrial facilities within a 5-mile radius of the VEGP site include the Savannah River Plant (SRP), Georgia Power Company's combustion turbine plant (Plant Wilson), and the construction site for Units 3 and 4. Chemical substances stored at SRP and Plant Wilson are identified in tables 2.2.3-14 and 2.2.3-17. Only bulk chemicals or gases intended to be stored permanently for use at Units 3&4 are listed on the appropriate tables, as applicable.

2.2.3.1.2 Explosions

Of the chemical substances stored or transported within a 5-mile radius of the VEGP site (tables 2.2.3-6, 2.2.3-14, and 2.2.3-17), only anhydrous ammonia, diesel fuel oil, gasoline, and miscellaneous oils are flammable.(10) The upper and lower limits of flammability for these

substances are depicted in table 2.2.3-12. Accordingly, potential explosions of these substances are evaluated. The remaining chemicals (chlorine, phosphoric acid, sulfuric acid, nitrogen, nitric acid, carbon dioxide, and helium) are nonflammable and are, therefore, excluded from further consideration.[10]

2.2.3.1.2.1 Allowable Distance and Actual Distance. Regulatory Guide 1.91 describes a method for determining the distance between transportation routes and structures beyond which any explosion that might occur on these transportation routes is not likely to have an adverse effect on plant operation or to prevent a safe shutdown. The distance is based on a level of peak positive incident overpressure below which no significant damage would be expected. This pressure is conservatively chosen at 1 psi. A safe distance can be defined by the relationship:

$$R \geq KW^{(1/3)}$$

where

R = distance in feet from an exploding charge of W pounds of TNT.

K = 45, when R is in feet and W is in pounds.

Table 2.2.3-12 shows this allowable distance and compares it with actual transported distance. The method described in Regulatory Guide 1.91 is also used to evaluate stationary flammable sources located within a 5-mile radius of the site. Table 2.2.3-12 shows the allowable distance and compares it with the actual stored distance.

As shown in table 2.2.3-12, the allowable distances for ammonia and miscellaneous oils are much shorter than the actual transported and stored distances. Therefore, according to Regulatory Guide 1.91, an explosion of these substances will not adversely affect safe operation of the plant and can be eliminated from further consideration.

Gasoline and diesel fuel oil were reviewed to determine if these substances are subject to an explosion given the conditions under which they are normally stored or transported. The propensity for an explosion is dependent upon the concentration of the material in the tank vapor space; i.e., the fuel-air mixture. The concentration of the flammable vapor present in the fuel-air mixture was calculated using the following liquid-vapor equilibrium equation:[56]

$$C = [V_{vapor} / (V_{vapor} + V_{air}) \times 100\%$$

where

V_{vapor} = Fuel vapor volume in one mole of vapor

V_{air} = Air volume in one mole of vapor mixture

C = Concentration of fuel in the vapor mixture, percent of volume

For diesel fuel oils, the concentration of flammable material in the vapor space is approximately 0.17 percent, which is below the lower limit of flammability. Similarly, for gasoline the concentration of flammable material in the vapor space is approximately 47 percent which is well above the upper limit of flammability. Accordingly, an explosion of the diesel fuel oil and gasoline stored or transported in the vicinity of the site is not considered a credible event. This conclusion is consistent with that of the National Fire Protection Association,[57] which states that the concentration in the vapor space of tanks storing low vapor pressure liquids, such as kerosene (which is similar to diesel fuel), is normally too lean to burn, while the vapor space concentration in tanks storing high vapor pressure liquids, such as gasoline, is normally too rich to burn.

2.2.3.1.2.2 Deleted.

2.2.3.1.2.3 Missiles Generated in Explosions. A potential hazard to the plant operation is the generation of self-propelled (or rocketing) missiles resulting from an explosion. In a report on rupture of tank cars,[18] a fragment from a liquid propane gas tank car explosion was reported to have been hurled 2,640 ft. The great majority of the fragments generated by exploding tank cars have a range of less than 1,000 ft.[19]

The largest range of a fragment from a flammable gas tank car rupture reported in reference 19 is 4,900 ft (for ethylene oxide tank car rupture). The analytical model shows that, for the worst combination of variables used, the maximum distance traveled by the rocketing fragments is in the range of 4,000 to 11,000 ft (1,220 to 3,350 m).[19]

The distances from the plant site to State Highway 23, to the Seaboard Coast Line Railroad, and to the Savannah River are approximately 7,600 m, 7,250 m, and 1,050 m, respectively. The distances from the plant to SRP and Plant Wilson are approximately 5,000 m and 1,350 m, respectively. Due to the long distances from the highway, railroad, and SRP, it is not credible for missiles generated in explosions to affect any VEGP safety-related structures. As shown in table 2.2.3-12 and discussed in paragraph 2.2.3.1.2.1, explosions of gasoline and diesel fuel oil transported on the Savannah River or stored at Plant Wilson are not credible. As such, explosion generated missiles from these sources are not credible.

There is a total of 2,150 gal of miscellaneous oils stored at Plant Wilson; however, as shown in table 2.2.3-17, no single miscellaneous oil is stored in a quantity larger than 700 gal. This volume represents approximately 3 percent of the 22,675-gal volume associated with the worst case explosion documented in reference 19, which is reported to have hurled fragments 4,900 ft (1,493 m). It is, therefore, reasonable to conclude that an explosion of 700 gal of oil at Plant Wilson would not produce a missile capable of traveling the 1,350 m to the VEGP site.

2.2.3.1.3 Flammable Vapor Clouds (Delayed Ignition)

The necessary condition of ignition requires that volume concentration of a fuel-air mixture be within the limits of flammability[10] as defined below:

$$C_{low} \leq C \leq C_{up}$$

where

C_{low} = the lower limit of volume concentration below which ignition cannot occur.

C_{up} = the upper limit of volume concentration above which ignition cannot occur.

The initial volume concentration of any fuel due to vaporization near the pool area is:

$$C = \frac{Q R_o T_o}{m P_a U A}$$

where

Q = evaporation rate
R_o = gas constant
T_o = ambient air temperature
m = molecular weight
P_a = ambient air pressure
u = wind speed
A = unit area of pool

Due to entrainment, this concentration will be less for distant parts of the plume.

As discussed in paragraph 2.2.3.1.2.1, of the chemical substances stored or transported within a 5-mile radius of the VEGP site, only anhydrous ammonia, diesel fuel oil, gasoline, and miscellaneous oils are flammable. Table 2.2.3-11 gives the fuel-air volume concentrations following a release of these substances and their flammability limits. At windspeeds of 0.25, 5.0, and 10.0 m/sec, the fuel-air concentrations are below the lower limit of flammability, except for ammonia stored at SRP or transported by rail or highway.

For ammonia transported by rail, the ammonia-air concentration is within the flammability range at distances between 6,431 m and 6,758 m from the nearest safety-related structure. The critical distance (given by $kW^{1/3}$ in Regulatory Guide 1.91), which could cause overpressures to safety-related structures, is 686 m.

Therefore, even though an explosion due to an ammonia release from a railroad tank car could occur, it would occur at a distance great enough not to pose an overpressure hazard to VEGP safety-related structures.

The ammonia release on the highway is bounded by the ammonia release on the railway due to the larger quantity transported by rail and the shorter distance between the railway and the plant site.

For ammonia stored at SRP, the ammonia-air concentration is within the flammability range at distances between 4,652 m and 4,791 m from the nearest safety-related structure. The critical distance is 292 m. Therefore, even though an explosion due to an ammonia release from SRP could occur, it would occur at a distance great enough not to pose an overpressure hazard to VEGP safety-related structures.

The zone of combustion for ammonia releases occurs at distances from the VEGP site that are well beyond those traveled by rocketing fragments following tank car explosions (refer to paragraph 2.2.3.1.2.3). Missiles generated from delayed detonations are, therefore, incapable of striking safety-related structures at VEGP.

2.2.3.1.4 Toxic Chemicals

2.2.3.1.4.1 Release of Toxic Chemicals Due to Transportation Accident. There are two classifications of toxic chemical materials that can pose as hazardous chemicals (toxic compressed gases and toxic liquids) being shipped on routes past the VEGP. These chemicals are listed in table 2.2.3-6. Additional chemicals transported to and from the Savannah River Plant are listed in table 2.2.2-5.

Three types of releases are defined for the toxic hazard evaluation methodology:

A. For high vapor pressure liquids such as ammonia and liquid Co_2 the amount of material flashed is estimated. This flash release comprises the "puff" portion of the release for such liquids.

B. The rate of evaporation of high pressure liquids after flashing is also estimated. This release is used as the continuous component of the high vapor pressure liquid source.

C. For normal boiling point liquids, the rate of vaporization is estimated. This evaporative release comprises the entire source for such substances.

Toxic chemicals transported near the VEGP site are analyzed using the same methodology as for onsite toxic hazards. This methodology is described in paragraph 2.2.3.1.4.3. The analysis assumptions of paragraph 2.2.3.1.4.3.1 also apply. For all postulated releases except gasoline,

ammonia, and nitric acid, the average concentration over an 8-h period does not exceed the long-term toxicity limit value. For gasoline and ammonia, it is demonstrated that there is more than 2 min between detection and reaching the short-term toxicity limit value as defined in Regulatory Guide 1.78. Thus, operators have sufficient time to don protective breathing apparatus prior to exposure to incapacitating concentrations of these toxic gases. Nitric acid slightly exceeds the long-term toxicity limit; however, footnote i of table 2.2.3-13 provides justification for acceptability based on the incapacitation criteria.

The results of this analysis are presented in table 2.2.3-13. It shows that the control room will remain habitable for all release scenarios and that only for ammonia and gasoline is operator action required.

2.2.3.1.4.2 Potential Hazard from Major Depots or Storage Areas. The only major depots or storage areas within 5 miles of VEGP are those at the Savannah River Plant and the combustion turbine plant. The chemicals stored at the Savannah River Plant are provided in table 2.2.2-5, and the oils and solvents stored at the combustion turbine plant are provided in table 2.2.3-17. The Savannah River Plant borders the Savannah River for approximately 17 miles opposite the VEGP site. (See subsection 2.2.2.) The combustion turbine plant is located approximately 5000 ft from the VEGP power block.

The chemicals stored at the combustion turbine plant with the exception of the fuel oil No. 2 are stored in small quantities. Due to the fact that these oils and solvents have low volatility and toxicity there is no potential hazard to the control room habitability from these substances.

Fuel oil No. 2 tanks for the combustion turbine plant are located east southeast of the power block, approximately 1350 meters distant. There are three tanks with a capacity of 3,000,000 gal each. These tanks are surrounded by a dike which would prevent the fuel oil from spreading into a large spill area.

The toxic chemical sources at the Savannah River Plant and the combustion turbine plant (table 2.2.3-14) are analyzed using the same methodology as for onsite toxic hazards. This methodology and the assumptions used are described in paragraphs 2.2.3.1.4.3 and 2.2.3.1.4.3.1. The average concentration over an 8-h period does not exceed the long-term toxicity limit except for ammonia. For ammonia, it is calculated that there is more than 2 min from detection to the time that the short-term toxicity limit value is reached. Operators, therefore, have sufficient time to don protective breathing apparatus prior to being exposed to incapacitating concentrations of ammonia.

The results of this analysis are presented in table 2.2.3-15. The control room will remain habitable for all chemicals and only for ammonia is the operator required to take emergency action.

2.2.3.1.4.3 Potential Hazard from Onsite Storage Tanks. The permanent storage facilities on the VEGP site are listed in table 2.2.3-18. The table lists the chemicals, quantities, and their distances from storage to the air intake of the control room.

Several of the chemicals listed in table 2.2.3-18 are excluded from further consideration due to their properties. Those chemicals excluded are:

A. Lube oil because of its relatively low toxicity and nonvolatility in the absence of aerosolization.[38]

B. Oxygen because it does not present a potential hazard for control room habitability.

C. Catalyst (benzoyl peroxide) because the melting point is 108°C; so under ambient conditions it is in the solid state. Also because it is stored in quantities less than 100 pounds.(12)(13)

D. Sodium nitrite because vapor of this liquid is nontoxic.

E. Electrohydraulic control fluid (phosphate esters) because no harmful vapors evolve from this chemical under normal operating temperatures.(39)

F. Liquids stored below ground level because significant spills cannot occur.

G. Seal oil because of its relatively low volatility and nontoxic characteristics.(38)

H. Promotor (dimethyl-p-toluene) because of its low volatility and emission of only NOx vapors when heated to decomposition.(41)

I. Hydrogen as a compressed gas because it is stored in quantities less than 100 lb.

J. Sodium hydroxide since it only poses a threat if it is inhaled in the form of dust or mist in the immediate vicinity of the spill. The control room is sufficiently distant to preclude the inhalation of dust or mist.(45)(46)

K. Sodium hypochlorite because vapor of this liquid is nontoxic.(10)

L. Sodium polyacrylate, sodium tolyltriazole, and zinc salt because these chemicals are nonvolatile.

M. Hydrogen peroxide because vapor of this liquid is nontoxic.(10)

N. Trisodium phosphate because it is a nonvolatile solid and is isolated from the control room by the containment.

O. Hypobromous acid (bromine salt/sodium hypochlorite) because vapor of this liquid is nonvolatile.(58)

The chemicals that are analyzed are divided into three categories: (1) compressed gases; (2) liquified compressed gases and liquids with boiling points below the ambient temperature (low boiling point liquids); and (3) liquids with boiling points above the ambient temperature (normal boiling point liquids). These chemicals can either emit toxic vapors or can be asphyxiates. Liquids can be both pure substances or aqueous solutions. Compressed gases include nitrogen. Liquified compressed gases and low boiling point liquids include Halon 1301, carbon dioxide, hydrogen, propane, and nitrogen. Normal boiling point liquids include methoxypropylamine (30-percent solution), fuel oil no. 2, hydrazine (35-percent solution), sodium hydroxide (50-percent solution), sulfuric acid, unleaded gasoline, and diesel fuel.

The rate at which a chemical is released to the atmosphere depends on the physical properties of the chemical, the geometry of the pool formed by the spilled liquid, meteorological conditions, and the governing equations for the transfer of mass into the vapor phase. Liquids having boiling points above the ambient temperature are governed by mass transfer relationships and constitute continuous releases. Liquids with boiling points below the ambient temperature are governed by heat transfer relationships and have both instantaneous puff releases and continuous releases. Compressed gas spills are entirely puff releases.

The guidelines and methodologies of NUREG-0570(36) are used in the determination of the release rates and the concentration of toxic gases at the control room air intake. This includes the use of heat and mass transfer relationships as well as the relationships for vapor dispersion for both instantaneous puff and continuous releases.

Having the toxic gas concentrations at the control room air intake, and using the parameters of the control room (table 2.2.3-19), the control room toxic gas concentrations are found by solving the following differential equation:

$$\frac{dC_{CR}(t)}{dt} = \lambda_I \chi(t) - \lambda_o C_{CR}(t)$$

where

$C_{CR}(t)$ = control room concentration (gm/m^3).

g_I = control room air inflow rate (s^{-1}).

g_o = control room air outflow rate (s^{-1}).

$\chi(t)$ = control room air intake concentration (gm/m^3).

This inleakage equation is applied to the toxic gas releases. No credit is taken for control room isolation for any chemicals.

2.2.3.1.4.3.1 Analysis Assumptions. Spills are postulated to occur at the site average annual temperature of 17°C and a G Pasquill Stability Category. The centerline of the vapor cloud is assumed to flow directly towards the control room air intake. A range of windspeeds from 0.25 m/s to 10.0 m/s are used in the analysis to determine which windspeeds would result in the maximum control room toxic vapor concentration.

Building wake effects are considered in this analysis as described in NUREG-0570[36] and Slade.[37] The spills are assumed to form circular pools with a uniform initial depth of 1 cm, unless otherwise bounded by dikes or other restrictions. The control room parameters used in this analysis are presented in table 2.2.3-19.

Analysis Results. The results of this analysis, along with toxicity limits and spill parameters, are shown in table 2.2.3-20. For all releases except ammonia and hydrazine, the average concentration over an 8-h period never exceeds the long-term toxicity limit. For the cases where the long-term limit is exceeded, it is shown that there are at least 2 min available between detection and the time that the short-term toxicity limit (as defined in Regulatory Guide 1.78) is reached. This provides control room operators adequate time to don protective breathing apparatus without being exposed to incapacitating concentrations of toxic gases.

This analysis shows that the control room will remain habitable for most release scenarios without any operator action, and that there is sufficient time for control room operators to take emergency action for the remaining release scenarios.

2.2.3.1.5 Fires

In the vicinity of VEGP, the following potential fire hazards exist:

A. Fire due to a transportation accident.

B. Fire due to an oil or gas pipeline rupture accident.

C. Forest fire.

D. Fire due to an accident at industrial storage facilities.

E. Fire due to onsite storage tank spill.

2.2.3.1.5.1 Fire Due to a Transportation Accident. Gasoline, diesel fuel oil, and ammonia are the only flammable substances transported within a 5-mile radius of the VEGP site (table 2.2.3-12). Ammonia is transported along the Seaboard Coast Line Railroad and State Highway 23, which are approximately 7,250 m and 7,600 m from the plant, respectively. The flammability limit of ammonia is in the range of 15.5 to 27 percent by volume.(10) These limits cannot be attained in open spills. Therefore, there is no potential hazard to control room habitability from an ammonia accident with a fire along these transportation routes.

Gasoline and diesel fuel oil are transported in quantities of approximately 5.6×10^6 lb on the Savannah River, which is approximately 1,050 m from the VEGP site. Fuel oil is also transported on State Highway 23; however, due to the longer distance from the plant site to the highway and the relatively small quantities of fuel oil transported along this route (4.21×10^4 lb), a fuel oil accident on the Savannah River is more limiting.

The primary products of combustion emitted from fuel oil and gasoline fires on the Savannah River are CO, CO_2, CH_4, NO_2, SO_2, and SO_3.(30) The toxicity limits in ppm for these constituents are 50 (CO), 5,000 (CO_2), 1.43×10^5 (CH_4), 2 (SO_2 and SO_3), and 3 (NO_2).(10) Using the same methodology as described in paragraph 2.2.3.1.4 for toxic chemical releases, the maximum burning rate, the maximum emission rate, and a wide range of meteorological conditions, it was determined that the resulting concentrations in the unisolated control room will not approach the toxicity limits.

Using the methodology described in NUREG/CR-1748,(28) the heat flux and resultant temperature rise on the VEGP structures due to a barge fire on the Savannah River were also evaluated. The calculated temperature rise is less than the maximum allowable temperature rise. Since the barge fire is limiting (the largest source at the closest approach to the plant site), it is concluded that source fires and vapor cloud fires resulting from delayed ignition (refer to paragraph 2.2.3.1.3) along transportation routes will not cause thermal damage to safety-related structures at VEGP.

2.2.3.1.5.2 Fire Due to Oil or Gas Pipeline Rupture Accident. There are no pipelines within 5 miles of the plant. The nearest pipeline is located approximately 19 miles southwest of the plant. (See paragraph 2.2.2.3.) Thus, a fire from a pipeline is not considered a credible hazard to control room personnel.

2.2.3.1.5.3 Forest Fires. The surrounding plant terrain is characterized by gently rolling hills and is approximately 30-percent farmland, with the remainder primarily wooded areas. The nearest forest has a total area of 3,169 acres and is approximately 1,850 ft from the control room.

The average size of a forest fire in the state of Georgia is approximately 11.4 acres.(53) The rate of spread is conservatively assumed to be 8 ft per minute with a duration of 4 hours.(53)

The toxic chemicals emitted from a forest fire are CO, NO_2, and CH_4.(30) The emission concentrations in the control room air intake were calculated using the infinite line source diffusion equation(54) with the wind direction perpendicular to the line source and blowing directly toward the control room intake, and the Brigg's plume rise equation,(56) which accounts for the buoyancy effect from the heat of the fire. The calculation demonstrated that the pollutant concentrations outside the control room air intake for a variety of windspeeds (from 0.25 to 10 m/sec) and Pasquill stability category G are effectively zero. Therefore, the release of toxic combustion products from an onsite forest fire does not pose a hazard to control room personnel.

Using the methodology described in NUREG/CR-1748,[28] the heat flux and resultant temperature rise on plant structures due to a forest fire were also evaluated. The calculated temperature rise is less than the allowable temperature rise.[28] Therefore, a forest fire will not cause thermal damage to the VEGP safety-related structures.

2.2.3.1.5.4 Fire Due to an Accident at Industrial Storage Facilities. There are three major industrial storage facilities in the vicinity of the plant: The Savannah River Plant (SRP) located at a distance greater than 5,000 m from the VEGP site; Georgia Power Company's combustion turbine plant (Plant Wilson) located approximately 1,350 m from the VEGP site; and the construction site for Units 3&4 located approximately 700 m centerline to centerline. Chemical substances stored at SRP and Plant Wilson are identified in tables 2.2.3-14 and 2.2.3-17.

Of the substances stored at these locations, only ammonia, diesel fuel oil, and miscellaneous oils are flammable. The flammability limit of ammonia is in the range of 15.5 to 27 percent by volume.[10] These limits cannot be attained in open spills. Therefore, there is no potential hazard to the control room habitability from an ammonia accident at SRP.

Approximately 9 million gal of diesel fuel oil is stored at Plant Wilson in three storage tanks (approximately 3 million gal each). Fuel oil is also stored at SRP; however, due to the longer distance from the plant site to SRP and the relatively small quantity of fuel oil stored at SRP (22,500 gal), a fuel oil fire at Plant Wilson is more limiting. Similarly, a diesel fuel oil fire at Plant Wilson will bound fires of miscellaneous oils stored at Plant Wilson (approximately 2,150 gal total).

The primary products of combustion emitted from a diesel fuel oil fire at Plant Wilson are CO, CO_2, CH_4, NO_2, SO_2, and SO_3.[30] The toxicity limits in ppm for these constituents are 50 (CO), 5,000 (CO_2), 1.43×10^5 (CH_4), 2 (SO_2 and SO_3), and 3 (NO_2).[10] Using the same methodology as described in paragraph 2.2.3.1.4 for toxic chemical releases, the maximum burning rate, the maximum emission rate, and a wide range of meteorological conditions, it was determined that the resulting concentrations in the unisolated control room will not approach the toxicity limits.

Using the methodology described in NUREG/CR-1748,[28] the heat flux and resultant temperature rise on the VEGP structures due to a diesel fuel oil fire at Plant Wilson were also evaluated. The calculated temperature rise is less than the maximum allowable temperature rise. Since a fire at Plant Wilson is limiting (the largest source at the closest approach to the plant site), it is concluded that source fires and vapor cloud fires resulting from a delayed ignition (refer to paragraph 2.2.3.1.3) at nearby industrial facilities will not cause thermal damage to safety-related structures at VEGP.

2.2.3.1.5.5 Fire Due to Onsite Storage Tank Spill. There are six gasoline storage tanks located on the Vogtle site. Of these six, four are located underground and no significant spills can occur. The two remaining above-ground tanks are:

A. A 6000-gal gasoline storage tank.

B. A 3000-gal diesel fuel storage tank (evaluated as a gasoline storage tank).

These two tanks are located just outside the primary security fence, near the plant warehouse. The distance from these tanks to the control building is approximately 1200 ft.

The potential hazards involved with these two above-ground tanks are:

A. The shockwave resulting from the ignition and explosion of the tanks' contents.

B. A dangerous thermal environment resulting from a tank fire.

C. A potentially lethal concentration of toxic gases in the control room resulting from a tank fire.

Evaluation of the tank spill, the meteorological conditions, and historical data for outdoor tank spills indicate that an explosion is not probable and is, therefore, not considered a credible hazard to the plant.

The calculated thermal load resulting from a tank fire will produce an insignificant temperature rise at the nearest safety-related structure (NSCW cooling tower). Therefore a tank fire does not create a thermal environment adversely affecting plant safety-related structures.

The primary constituents resulting from either a gasoline or diesel fuel fire are C0, CO_2, CH_4, NO_2, SO_2, and SO_3.[30] The toxicity limits in ppm for these constituents are 50 (CO), 5,000 (CO_2), 1.43 x 10^5 (CH_4), 2 (SO_2 and SO_3), and 3 (NO_2).[10] Using the maximum burning rate, maximum emission rate, and a wide range of meteorological conditions, it was determined that the resulting concentrations of gases, in the unisolated control room will not approach the toxicity limit.

Based on these analyses, the above ground gasoline and diesel fuel storage tanks are not considered to be a credible hazard to the control room.

Three 2000-gal capacity above ground liquid propane tanks are installed at the fire training facility. These tanks provide fuel to fire training aids installed at the facility. The distance from the tanks to the nearest safety-related structure (the NSCW cooling towers) is approximately 3200 ft and the distance to the control room air intake is approximately 3500 ft. The tanks are designed and installed in accordance with ASME Section VIII and the relevant NFPA and insurance standards. The potential hazards associated with these tanks are:

A. The shockwave resulting from an explosion of the tanks' contents or a flammable vapor cloud emanating from a tank rupture.

B. A dangerous thermal environment resulting from a tank or flammable vapor cloud fire.

C. A potentially lethal concentration of toxic gases in the control room resulting from a tank fire.

D. Diesel generator overspeed, starvation, or prevention of a diesel generator start due to a propane tank rupture.

The tanks have been evaluated consistent with the methods described in Regulatory Guide 1.91. Based upon the evaluation the fire training facility propane tanks are located beyond the distance where an exploding tank or an exploding flammable vapor cloud would yield an overpressure of 1 psi on any safety-related structure. As noted in Regulatory Guide 1.91, the effects of blast-generated missiles will be less than those associated with blast overpressure.

The fire training facility propane tanks have also been evaluated for thermal effects of a fire using the methodology described in NUREG/CR-1748. The calculated temperature rise is less than the maximum allowable temperature rise.

Control room habitability following a propane fire at the fire training facility is bounded by the analysis of Plant Wilson discussed in paragraph 2.2.3.1.5.4.

The fire training facility propane tank evaluation also determined that the tanks are located beyond the distance where a propane gas cloud could starve or prevent the start of a diesel generator.

Based upon these evaluations, the propane tanks at the fire training facilicy are located a safe distance from the control room air intake and any safety-related structures. Therefore, the propane tanks are not considered to be a credible hazard.

2.2.3.1.6 Collisions with Intake Structure

The river water intake structure and the river water makeup pumps located inside the structure serve no safety design basis. The pumps provide makeup water to the nonsafety-related circulating water system as described in subsection 10.4.5. A barge or ship colliding with the intake structure would not affect the capability to safely shut down the plant.

2.2.3.1.7 Liquid Spills

Liquid spills during transportation accidents could cause corrosive, cryogenic, or coagulant effects. However, due to the large distances from Georgia State Highway 23 and Seaboard Coast Line Railroad, it is not credible for such spills to affect the safety of the plant. Liquid spills on the Savannah River that could be drawn into the river water intake structure and the circulating water system would not affect the safety of the plant because the circulating water system is nonsafety related. Loss of the circulating water system would not affect the capability to safely shut down the plant.

2.2.3.1.8 Effects of Design Basis Events

As discussed in the previous paragraphs, accidents involving hazardous chemicals transported or stored within a 5-mile radius of the plant site and rupture of onsite storage tanks constitute no hazard to the safe operation of VEGP.

2.2.3.2 References

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TABLE 2.2.2-1

DESCRIPTION OF SAVANNAH RIVER PLANT FACILITIES[a]

Facilities	Major Product(s)/ Primary Function
Savannah River Plant, Aiken, South Carolina, operated by DuPont	Nuclear material production and storage, waste processing, and research and development activities
Production reactors (four active, one in standby)	Production of nuclear materials, primarily plutonium (Pu^{239}) and tritium (T or H^3)
Chemical separation facilities (two)	Chemical processing and high level radioactive waste storage
Fuel and target fabrication facility	Fabrication of elements
Heavy water production and recovery facility[b]	Production of heavy water (D_2O)
Savannah River Laboratory	Research and development
TNX-CMX semiworks area[b]	Production-scale testing

a. Total number of employees as of December 31, 1982, is 6675.

b. Located within a 5-mile radius of VEGP.

TABLE 2.2.2-2

DESCRIPTION OF BARNWELL NUCLEAR FUEL PLANT FACILITY

Facility	Major Product(s)/ Primary Function	Number of Employees
Barnwell Nuclear Fuel Plant	Active only in R and D for DOE Reprocessing of spent light-water reactor fuel (presently and for likely future in abeyance) Possible away-from-reactor interim storage for spent fuel	350

TABLE 2.2.2-3

DESCRIPTION OF CHEM-NUCLEAR SYSTEMS, INC., FACILITY

Facility	Major Product(s)/ Primary Function	Number of Employees
Chem-Nuclear Systems, Inc.	Low level radioactive waste disposal site	185

TABLE 2.2.2-4 (SHEET 1 OF 2)

MAJOR INDUSTRIES (25 OR MORE EMPLOYEES)
WITHIN 25-MILE RADIUS OF VEGP

Location	Facility	Function	Major Product	Number of Employees
Waynesboro, Ga.	Burke Manufacturing Co.	Manufacturer	Men's and boys' jackets	230
	Roy F. Chalker Pub. Co.	Publishers	Books	33
	Keller Aluminum Furniture of Ga.	Manufacture	Aluminum furniture	413
	Kimberly-Clark Corp.	Manufacture	Pine lumber	120
	McKinney Concrete Sers.	Manufacturer	Concrete	45
	Perfection Products Co.	Manufacturer	Room heaters	265
	Samson Manufacturing Co.	Manufacturer	Curtains and draperies	225
	Talley Corbett Box Co.	Manufacturer	Veneer wood chips	30
	Waynesboro Industries	Manufacturer	Food service equipment	117
Millen, Ga.	Brigadier Industries	Manufacturer	Mobile homes	88
	Jockey International	Manufacturer	Aluminum windows and doors	268
	Thompson Co.	Manufacturer	Men's dress slacks	180
Sardis, Ga.	Sardis Manufacturing Co.	Manufacturer	Draperies	150
Barnwell, S.C.	Allied Nuclear Sers.	Manufacturer	Nuclear fuel reprocessing	348
	Chem-Nuclear Systems, Inc.	Waste Disposal	Nuclear waste storage	200
Beach Island, S.C.	Kimberly-Clark Corp.	Manufacturer	Tissue products	842
Aiken, S.C.	E. I. DuPont DeNemours Co., Inc.	Manufacturer	Nuclear materials	5568

TABLE 2.2.2-4 (SHEET 2 OF 2)

Location	Facility	Function	Major Product	Number of Employees
Augusta, Ga.	Abitibi Southern Corp.	Manufacturer	Newsprint	183
	Babcock and Wilcox Co.	Manufacturer	Clay refractories	1171
	Columbia Nitrogen Corp.	Manufacturer	Nitrogen solutions	340
	Con Agra, Inc.	Manufacturer	Animal feed	37
	Continental Forest Industries	Manufacturer	Bleached paperboard	758
	Deerfield Specialty Papers, Inc.	Manufacturer	Specialty papers	94
	Esselte Pendaflex-Dymo	Manufacturer	Office systems and books	204
	Georgia-Carolina Brick and Tile Co.	Manufacturer	Clay products	166
	Graniteville Co.	Manufacturer	Cotton textiles	532
	Homestead Manufacturing Co.	Manufacturer	Draperies	340
	Hydreco Div. General Signal Co.	Manufacturer	Valves and pumps	125
	Ireland Electric Corp.	Manufacturer	Control panels	41
	Kendall Co.	Manufacturer	Surgical dressing	585
	Lily Division	Manufacturer	Paper cups and containers	350
	Mid-South Container Corp.	Manufacturer	Corrugated containers	166
	Modern Roofing and Metal Works, Inc.	Manufacturer	Sheet metal fabrication	65
	Monsanto Co.	Manufacturer	Phosphate compounds	41
	Murray Biscuit Co.	Manufacturer	Cookies	600
	Olin Corp.	Manufacturer	Chemicals	126
	Procter and Gamble Mfg. Co.	Manufacturer	Synthetic detergents	245
	Servomation of Georgia, Inc.	Manufacturer	Vending machinery	78
	Southeastern Newspapers, Inc.	Printer	Newspaper printing	358
	Standard Bag and Synthetics, Inc.	Manufacturer	Bonded synthetics	35
	Sturm Dixie, Inc.	Manufacturer	Pump sleeves	39
	Transco Textile Industries, Ltd.	Manufacturer	Cotton textiles	343

Source: Georgia Manufacturer's Directory: 1980-81, Georgia Department of Industry and Trade, Atlanta, Georgia, 1980.

TABLE 2.2.2-5 (SHEET 1 OF 2)

DESCRIPTION OF PRODUCT AND MATERIALS:
SAVANNAH RIVER PLANT
(FACILITIES WITHIN 5 MILES OF VEGP)

Products or Materials	Status	Annual Amounts	Maximum Quantity At Any Time	Mode of Transport	Frequency of Shipment
400 Area					
Heavy Water Production and Recovery (includes rework, unit, drum cleaning facility, analytical laboratory, extraction plant, and distillation plant)					
Heavy water (D_2O)	Produced	76 tons	330 tons	Truck	3/week
Tritium	Released	3900 Ci	220,000 Ci	NA	NA
Sulfur dioxide (SO_2)	Released	85 tons	NA	NA	NA
Phosphoric acid	Used	1380 lb	460 lb	Truck	3/year
Ammonia	Used present	1.5 tons	2.0 tons	Truck	2/year
Silicone	Not used at present	-	-	-	-
Trisodium phosphate	Used	5000 lb	1000 lb	Truck	5/year
Potassium permangate	Used	220 lb	220 lb	Truck	1/year
Steam and Electric Generating Plant (includes water treatment plant)					
Bituminous coal	Burned	240,000 tons	75,000 tons	Rail	Daily
Sulfur dioxide	Released (continuous boiler emissions)	9600 tons	-	-	-
Chlorine	Used	15 tons	Ten 1-ton cylinders	Truck	Monthly
Trisodium phosphate	Used	10 tons	2 1/2 tons	Truck	6/year
Sulfuric acid	Used	175 tons	270 tons	Rail	3/year
Caustic (NaOH)	Used	290 tons	340 tons	Rail	6/year
Alum	Used	280 tons	100 tons	Truck	8/year

TABLE 2.2.2-5 (SHEET 2 OF 2)

Products or Materials	Status	Annual Amounts	Maximum Quantity At Any Time	Mode of Transport	Frequency of Shipment
TNX-CMX Semiworks Area					
Process waste (kaolin)	Stored (held in retention basin)	-	-	-	-
Steam Boiler					
No. 2 fuel oil	Used	27,000 gal	22,500 gal	Tank truck	1/week

a. About 1550 Ci released to liquid streams, 2350 Ci released to atmosphere.

b. Approximation of quantity resulting from burning of H_2S to SO_2 at top of 400-foot flare stack.

c. Based on sulfur content specification of coal. Actual sulfur content of coal in 1980 was equivalent to 6000 tons of sulfur dioxide.

TABLE 2.2.2-6 (SHEET 1 OF 8)

APPLICABLE TOXICITY LIMITS

Hazardous Substance/ Applicable Facility	Toxicological Data
1. Ammonia, NH^3 RTECS No. B00875000 CAS No. 7664-41-7 Applicable facility: SRP, SRP (HWP)	Colorless gas (or liquid); sharp, intensely irritating odor; lighter than air; easily liquified by pressure; boiling point = -28°F; very soluble in water. Auto-ignition temperature = 1204°F. Combustible. (10-47) At room temperature and atmospheric pressure, a colorless, alkaline gas having a pungent odor. First perceptible odor = 20 ppm. Slight eye irritation at 40 ppm. Fatal almost immediately at 5000 ppm. (11-17) A powerful irritant. Irritant to eyes and mucous membranes of respiratory tract. (3-364) Lowest lethal concentration for humans by inhalation = 3000 ppm for 5 min. Lowest toxic concentration for humans by inhalation = 20 ppm. Aquatic toxicity rating: TLm 96:10-1 ppm. ACGIH threshold limit value = 25 ppm in air. OSHA standard = 50 ppm TWA in air. (1-94)
2. Chlorine, Cl_2 RTECS No. F02100000 CAS No. 7782-50-5 Applicable facility: SRP (HWP)	Green-yellow gas, liquid, or rhombic crystals. Boiling point = -30°F. Extremely irritating to the mucous membranes of the eyes and the respiratory tract. (3-485) Lowest lethal concentration for humans by inhalation = 430 ppm for 30 min. ACGIH TLV = 1 ppm in air. OSHA standard = 1 ppm TWA in air. NIOSH criteria document recommended standard = 0.5 ppm in air for 15 min. (1-403) Odor threshold = 0.3 ppm. (9-21) Level of odor perception = 3.5 ppm. (4-113)

TABLE 2.2.2-6 (SHEET 2 OF 8)

Hazardous Substance/ Applicable Facility	Toxicological Data
3. Coal, bituminous (coal facings, sea coal) RTECS No. GF8300000 Applicable facility: SRP (HWP)	Chiefly amorphous elemental carbon with low percentages of hydrocarbons, complex organic compounds, and inorganic materials. (10-214) Toxic hazard depends upon content of crystalline silicon dioxide. (3-508) Coal dust—an amorphous carbon but sometimes contains hyrocarbons, organic complex salts, and inorganic compounds. Slightly toxic by inhalation. (4-133) ACGIH threshold limit value = 2 mg/m^3 exposure in air (1-441) OSHA exposure limit for respirable coal dust containing less than 5% crystalline silica dioxide = 2.4 mg/m^3; limit for dust containing more than 5% crystalline silica = 10 mg/m^3 (% crystalline silica dioxide + 2). (12-545)
4. Fuel oil (distillate, gas oil, home heating oil No. 2) RTECS No. LS8950000 Applicable facility: BNFP, SRP (TNX-CMX)	A complex mixture of liquid petroleum hydrocarbon products with flash points above 100°F. (1-700) Several grades available. For example: Kerosene--moderate fire hazard. (3-760) Moderately toxic by inhalation and swallowing. (4-263) Toxic oral dose level for rabbit and rat is 20 g/kg. (2-3) Criteria document recommended standard for airborne exposure to refined petroleum solvents to be 100 mg/m^3 time weighted average. (2-3) May contain suspected carcinogens. (3-760)
5. Material Deleted	
6. Manganese dioxide, MnO_2 (manganese oxide, manganese binoxide, manganese black, battery manganese, manganese perioxide) RTECS No. OPO35000 CAS No. 1313-13-9 Applicable facility: SRP, SRP (HWP)	Black crystals or powder; soluble in hydrochloric acid; insoluble in water. Oxidizing agent; may ignite organic materials; moderately toxic. (10-535) Moderate fire hazard by chemical reaction; a powerful oxidizer. Must not be rubbed in contact with easily oxidizable matter. Keep away from heat and flammable materials. (3-787) Lowest lethal dose by intravenous route to rabbits = 45 mg/kg. No OSHA, ACGIH, or NIOSH limits listed. Reported in EPA TSCA inventory (July 1979). (2-28) MnO_2 injected intratracheally into rats, in an attempt to stimulate manganese pneumonitis seen in man, produced characteristic histological changes in the lungs. Inhalation exposures of rabbits to MnO_2 dust 4 h daily for from 3 to 6 months at levels of 10 to 20 mg/m^3 resulted in decreased hemoglobin and erthrocytes in the blood; Mn pneumonitis did not occur, but fibrotic

TABLE 2.2.2-6 (SHEET 3 OF 8)

Hazardous Substance/ Applicable Facility	Toxicological Data
	changes in the lungs resembling those in silicosis were observed. A ceiling exposure limit of 5 mg/m^3 is listed for Mn dust and compounds as Mn. (8-250) OSHA standard for manganese = 5 mg/m^3 as a ceiling for exposure. (12-542)
7. Mercury, Hg (quicksilver) RTECS No. OV4550000 CAS No. 7439-97-6 Applicable facility: SRP	Silvery, extremely heavy liquid. Extremely high surface tension. Insoluble in hydrochloric acid, water, alcohol, and ether; soluble in nitric acid and lipids. High electrical conductivity, noncombustible. Metallic Hg is highly toxic by skin absorption and inhalation of fume or vapor. Tolerance = 0.05 mg/m^3 of air. Absorbed by respiratory and intestinal tract; accidental intake of small amounts is stated to be harmless (Merk Index). FDA permits zero addition to be 20 micrograms of Hg contained in average daily diet. All inorganic compounds of Hg are highly toxic by ingestion, inhalation, and skin absorption. Tolerance = 0.05 mg/m^3 of air. Most organic compounds of Hg are highly toxic. Inorganic Hg can be converted to methyl-mercury (organic) by bacteria in water. Tolerance (alkyl compounds) = 0.01 mg/m^3 of air; all others = 0.05 mg. (10-549) Methylated mercury compounds are especially toxic, because they are readily absorbed into animal tissues, for example, by fish. (13-206) Lowest toxic dose to woman by inhalation = 150 μg/m^3/46D. Lowest lethal concentration to rabbit by inhalation = 29 mg/m^3/30 H. OSHA standard for metallic mercury in air = 1 mg/10 m^3. NIOSH recommends a standard of 0.05 mg(Hg)/m^3 in air for an 8-h time weighted average. Reported in EPA TSCA inventory (July 1979) (2-36) OSHA standard for organic (alkyl) mercury = 0.01 mg/m^3 8-h time weighted average and 0.04 mg/m^3 ceiling. (12-544) Lowest lethal dose to humans by oral route = 1429 mg/kg. Lowest toxic concentration to humans by inhalation = 0.17 mg/m^3 for 40 years for CSNI problems. When heated, Hg emits highly toxic fumes. (3-797)
8. Nitric acid, NHO_3 RTECS No. QU5775000	Lowest published oral human lethal dose = 430 mg/kg. Aquatic toxicity rating: TLm 96: 100-10 ppm. Threshold limit value: air = 2 ppm; OSHA standard air: time weighted average =

TABLE 2.2.2-6 (SHEET 4 OF 8)

Hazardous Substance/ Applicable Facility	Toxicological Data
CAS No. 7697-37-2 Applicable facility: BNFP, SRP	2 ppm. Strong oxidizing agent. (8-301) Corrosive to skin and eyes. (7-Nitric acid)
9. Liquid nitrogen, N_2 RTECS No. QW9720000 CAS No. 7727-37-9 Applicable facility: BNFP	Nontoxic (3-860) Melting point = -209.9°C and boiling point = -195.8°C, can cause severe frostbite on contact. Gaseous nitrogen is simple asphyxiant. Can make confined spaces unsafe due to displacement of oxygen. Colorless, odorless, tasteless gas. (4-369)
10. Nitrogen dioxide, NO_2 (nitrogen peroxide) RTECS NO. QW9800000 CAS No. 10102-44-0 Applicable facility: SRP	A red-brown gas or yellow liquid, melting point = -9.3°C and boiling point = 21°C. Noncombustible. Highly toxic; inhalation may be fatal. Can react with reducing materials. Tolerance = 5 ppm in air. (10-616) Upon inhalation NO_2 affects the essential pulmonary tissues; hemorrhaging in the lungs is common. The only external sign of poisoning is shortness of breath. Pneumonia can develop as a secondary hazard (13-203). Noncombustible but extreme oxidizing agent. May cause fire in contact with clothing and other combustible materials. Lowest toxic concentration to humans by inhalation route = 64 ppm. Lethal concentration or 50% kill to rats = 67 ppm/4 H. (4-370) Lowest lethal concentration to humans by inhalation = 200 ppm/1 M. Lowest toxic concentration to man by inhalation = 90 ppm/40 M. OSHA standard for air = 5 ppm time weighted average. NIOSH recommends 1 ppm ceiling. Reported in EPA TSCA Inventory (July 1979). (2-149) ACGIH lists a threshold limit value of 3 ppm for 8 h average exposure and 5 ppm for a short-term exposure limit (15 min). (8-305) Emergency exposure limits (ppm as NO_2): 5 min = 35 ppm; 15 min = 25 ppm; 30 min = 20 ppm; 60 min = 10 ppm. Exposure for 60 min to 100 ppm, or briefly to 250 ppm, could cause death hours later. (7-Nitrogen dioxide)
11. Perchloroethylene, ClC_3:CCl_2 (tetrachloroethylene, ethylene tetrachloride, carbon dichloride, perk) RTECS NO. KX3850000	Colorless liquid with ether-like odor. Miscible with alcohol, ether, and oils in all proportion. Insoluble in water. Nonflammable. Moderately toxic. Irritant to eyes and skin. Tolerance = 100 ppm in air. (10-660) Vapor may decompose at high temperature such as open flame, red-heated

TABLE 2.2.2-6 (SHEET 5 OF 8)

Hazardous Substance/ Applicable Facility	Toxicological Data
CAS No. 127-18-4 Applicable facility: SRP	materials, with evolution of poisonous gases such as chlorine, carbon monoxide, and phosgene. (4-507) Chloroform-like odor. Moderately toxic via inhalation, oral, subcutaneous, intraperitongal, and dermal routes. Highly toxic via intravenous route. Not corrosive or dangerously reactive but toxic by inhalation, by prolonged or repeated contact with the skin or mucous membranes, or when ingested by mouth. The liquid can cause injuries to the eyes; however, with proper precautions it can be handled safely. The symptoms of acute intoxication from this material are the result of its effects upon the nervous system. (2-892) Lowest toxic concentration for man/human via inhalation route = 96 ppm/7 H, 280 ppm/2 H, 500 ppm/10 M. Lowest lethal concentration for rat by inhalation = 4000 ppm/4 H. Lowest lethal dose for cat and dog by oral route = 4000 mg/kg. Aquatic toxicity rating: TLm 96: 100-10 ppm OSHA standard for air = 100 ppm 8-h time weighted standard of 50 ppm in air and a ceiling of 100 ppm for 15 min. The NCI carcinogenesis bioessay completed with positive results in mouse. Reported in EPA TSCA Inventory (July 1979). (1-669) Odor threshold as low as 5 ppm for some individuals. Potential to produce carcinogenic, mutagenic, or teratogenic effects has not been determined conclusively. NCI is currently conducting a study of the carcinogenic potential. (8-325)
12. Phosphoric acid, H_3PO_4 RTECS NO. TB6300000 CAS No. 7664-38-2 Applicable facility: SRP (HWP)	Colorless, odorless, sparkling liquid or transparent crystalline solid, depending on concentration and temperature. Toxic by ingestion and inhalation. Irritant to skin and eyes. (10-679) Moderately toxic via oral and dermal routes. Used as a general purpose food additive. (3-910) Lethal toxic concentration to humans by inhalation route = 100 mg/m^3. ACGIH TLV for air = 1 mg/m^3 time weighted average. Reported in EPA TSCA Inventory (July 1979). (2-274)
13. Potassium permangate, $KMNO_4$ (permanganic acid, potassium salt) RTECS NO. SD6475000	Dark purple crystal; blue metallic sheen; sweetish, astringent taste; odorless, soluble in water, acetone and methanol; decomposed by alcohol. Oxidizing material. Toxic by ingestion and inhalation; strong irritant to tissue. Dangerous fire

TABLE 2.2.2-6 (SHEET 6 OF 8)

Hazardous Substance/ Applicable Facility	Toxicological Data
CAS No. 7722-64-7 Applicable facility: SRP (HWP)	risk in contact with organic materials. Powerful oxidizing agent. (10-716) May explode in contact with organic materials, especially alcohol, ether glycerin, and combustible gases or with sulfuric acid. Especially dangerous in contact with reducing agents. (4-436) Lowest toxic dose for women by oral route = 2400 μ/kg/D. No published OSHA, NIOSH, or ACGIH limits. Reported in EPA TSCA Inventory (July 1979). Aquatic toxicity rating TLm 96:100-1 ppm. (2-212)
14. Silicone Applicable facility: SRP (HWP)	Silicone is any one of a large group of siloxane polymers based on a structure consisting of alternate silicon and oxygen atoms with various organic radicals attached to the silicon. Silicons are liquids, semisolids, or solids depending on molecular weight and degree of polymerization. Unhalogenated types are combustible and nontoxic. (10-774)
15. Silver nitrate, $AgNO_3$ (lunar caustic, silver nitrate) RTECS NO. VW4725000 CAS No. 7761-88-8 Applicable facility: SRP	Colorless, odorless, sheet-like crystals with a metallic taste. Soluble in water. Has strong oxidizing nature. Its dust irritates skin and respiratory organs and acutely attacks. Orally causes stomachache, diarrhea, dizziness, and convulsion. (4-465) Highly toxic and strong irritant to skin and tissue. (10-777) Lowest lethal dose to man (route unknown) = 29 mg/kg. Lowest lethal dose to dogs by oral route = 20 mg/kg. OSHA standard in air = 10 μg(Ag)/m^3 time weighted average. Reported in EPA TSCA inventory (July 1979). (2-526)
16. Sodium hydroxide, NaOH (caustic soda, soda lye, lye and sodium hydrate) RTECS NO. WB4900000 CAS No. 1310-73-2 Applicable facility: BNFP, SRP	White sodium deliquescent flakes, lumps, or sticks. Absorbs water and carbon dioxide from air. Soluble in water and alcohol. Very corrosive to animal and vegetable tissue. Contact with moisture may generate sufficient heat to ignite combustible materials. Contact with some metals can generate hydrogen gas. Lowest lethal dose to rabbit by oral route = 500 mg/kg. (4-481) Threshold limit value--air = 2 mg/m^3 (ceiling). OSHA standard--air time rated average = 2 mg/m^3. (2-531) Severe irritant to eyes, mucous membranes, and skin. (5-445)

TABLE 2.2.2-6 (SHEET 7 OF 8)

Hazardous Substance/ Applicable Facility	Toxicological Data
17. Sodium nitrate, $NaNO_3$ (sodium niter, cubic niter, natratine chile, salt-peter) RTECS NO. WC5600000 CAS No. 7631-99-4 Applicable facility: SRP	Colorless, transparent, odorless crystals; saline, slightly bitter taste; explodes at 1000°F. Soluble in water and slightly soluble in alcohol. Fire risk near organic materials. Ignites on friction and explodes when shocked. Added to meats, fish, and other food products in concentrations up to 500 ppm. USDA has proposed its elimination. (10-796) Preclude from exposure those individuals with kidney or lung diseases. (4-484) Ignition of mixture of nitrate salt and organic matter is likely to explode. (13-235) Lowest lethal dose for rats by oral route = 200 mg/kg. No OSHA standard for ACGIH TLV listed. Reported in EPA TSCA inventory (July 1979). (2-533)
18. Sulfur dioxide, So_2 RTECS NO. WS4550000 CAS No. 7446-09-5 Applicable facility: SRP (HWP)	Colorless gas or liquid with a pungent odor. Highly irritating via inhalation route to skin, eyes, and mucous membranes. (3-1001) Toxic dose: inhale human, lowest published toxic conclusion = 3 ppm/5 days. ACGIH Threshold limit value--air = 5 ppm. OSHA standard for air: TWA = 5 ppm. NIOSH criteria document recommended standard--air: 0:5 ppm. (2-569) Odor threshold = 0.5 ppm. (9-22)
19. Trisodium phosphate, $NaPO_4$:$12H_2O$ (tribasic trisodium phosphate; TSP) RTECS TC9490000 CAS No. 7601-54-9 Applicable facility: SRP (HWP)	Colorless, water soluble crystals. Nonflammable. Moderately toxic by ingestion; irritant to tissue. (10-798) Lowest lethal dose to rabbits by intravenous route = 1580 mg/kg. (2-274)

TOXICOLOGICAL DATA SOURCES

1. National Institute of Occupational Safety and Health, Registry of Toxic Effects of Chemical Substances, Vol I, U.S. Department of Health and Human Services, 1979.

2. National Institute of Occupational Safety and Health, Registry of Toxic Effects of Chemical Substances, Vol II, U.S. Department of Health and Human Services, 1979.

3. Sax, Irving N., Dangerous Properties of Industrial Materials, 5th edition, 1979.

TABLE 2.2.2-6 (SHEET 8 OF 8)

4. International Technical Information Institute, Toxic and Hazardous Industrial Chemicals Safety Manual, 1980.

5. Proctor, Nick, and Hughes, James, Chemical Hazards of the Workplace, 1978.

6. American Industrial Hygiene Association, Hygienic Guide Series, Vol I.

7. American Industrial Hygiene Association, Hygienic Guide Series, Vol II.

8. American Conference of Governmental Industrial Hygienists, Inc., Documentation of the Threshold Limit Values, 4th edition, 1980.

9. Arthur D. Little, Inc., Odor Thresholds for 53 Commercial Chemicals, part I, research report for the Manufacturing Chemists Association.

10. Hawley, Gessing, revision of The Condensed Chemical Dictionary, 9th edition, Van Nostrand Reinhold Company, 1977.

11. Baker, William, and Mossman, Allen, Matheson Gas Data Book, 5th edition, 1971.

12. U.S. Department of Labor, "General Industry-OSHA Safety and Health Standards (29 CFR 1910)," Publication No. OSHA 2206.

13. Meyer, Eugene, Chemistry of Hazardous Materials, Prentice-Hall, 1977.

TABLE 2.2.2-7 (SHEET 1 OF 2)

BURKE COUNTY, GEORGIA
TRANSPORTATION ACCIDENT DATA,
WITHIN 5 MILES OF THE SITE[a]

	1979	1978	1977	1976	1975
SR 80, ML 25.00 - 32.27					
Accidents	2	1	0	0	0
Disabling injuries	1	0	0	0	0
Fatalities	0	0	0	0	0
Property damage	$1300	$100	0	0	0
Vehicle					
Auto	1	1	0	0	0
Truck	0	0	0	0	0
Other	3	0	0	0	0
SR 23, ML 11.08 - 24.13					
Accidents	8	7	9	6	3
Disabling injuries	7	6	2	1	2
Fatalities	0	1	0	0	1
Property damage	$13,000	$13,100	$7100	$6700	$11,200
Vehicle					
Auto	12	9	13	9	4
Truck	0	0	0	0	0
Other	4	3	0	0	0
SR 56C, ML 5.69 - 6.69					
Accidents	0	0	0	1	0
Disabling injuries	0	0	0	0	0
Fatalities	0	0	0	0	0
Property damage	0	0	0	$300	0

TABLE 2.2.2-7 (SHEET 2 OF 2)

	1979	1978	1977	1976	1975
Vehicle					
Auto	0	0	0	1	0
Truck	0	0	0	0	0
Other	0	0	0	0	0

a. No accident data pertaining specifically to hazardous materials is available.

Source: Georgia Department of Transportation.

TABLE 2.2.2-8

HAZARDOUS RAIL CARGO TRAFFIC ESTIMATES

Cargo Type	Annual Shipments (10^3 tons)	Average Shipment Size (10^3 tons)	Maximum Shipment Size (10^3 tons)
Compressed Gases			
Anhydrous ammonia	90-108	0.72	1.08
Argon	0.0	0.0	0.0
Carbon dioxide	18	0.08	0.16
Chlorine	0.0	0.0	0.0
Helium	0.15	0.03	0.03
Nitrogen	144-162	1.35	2.70
Flammable Liquids			
Gasoline	0.0	0.0	0.0
Diesel oil	0.0	0.0	0.0
Jet fuel	0.0	0.0	0.0
Hydrogen	0.0	0.0	0.0
Flammable Gases			
Propane (liquid) and butane	0.0	0.0	0.0
Liquid nitrogen	0.0	0.0	0.0
Hydrogen	0.0	0.0	0.0
Acetylene	0.0	0.0	0.0
Explosives and Ammunition			
Class A	0.0	0.0	0.0
Class B	0.0	0.0	0.0
Cleaning Products	2.1-2.8	0.07	0.21
Corrosives	0.0	0.0	0.0
Poisons and Pesticides	0.0	0.0	0.0
Solvents	0.0	0.0	0.0
Other Toxic Chemicals	0.0	0.0	0.0

TABLE 2.2.2-9

HAZARDOUS CARGO RAIL ACCIDENT DATA

	1980	1979	1978	1977	1976	1975
Accidents	0.0	0.0	0.0	0.0	0.0	0.0
Hazardous chemical(s) involved	None					Not Applicable
Accident Category						
Collision						
Explosion						
Fire	None					Not Applicable
Flammable vapor cloud release						
Toxic chemical release (gaseous)						
Liquid spill						
Severity						
Disabling injuries						
Fatalities	None					Not Applicable
Property damage						

TABLE 2.2.2-10

DESCRIPTION OF PRODUCTS AND MATERIALS:
BARNWELL NUCLEAR FUEL PLANT
ALLIED-GENERAL NUCLEAR SERVICES

Products or Materials	Status	Annual Amounts	Maximum Quantity at Any Time	Mode of Transport	Frequency of Shipment
Fuel oil	Used	50,000 gal	-	Highway	45/year
Liquid N	Used	50,000 ft^3	-	Highway	1/year
HNO_3	Stored Used	- 35,000 gal	4000/gal -	- Highway	- 1/year
NaOH	Stored Used	- 4000 gal	4000/gal -	- Highway	- 1/year
Natural U	Stored	-	150 MTU	-	-
60CO	Stored	-	200 Ci	-	-
Low level radwaste	Transported offsite	6500 ft^3	-	Highway	30/year
Transported U waste	Transported offsite	100/ft^3	-	Highway	1/year

Source: George T. Stribling, vice president, Marketing and Regulatory Affairs, Allied-General Nuclear Services, Barnwell, South Carolina.

TABLE 2.2.2-11

DESCRIPTION OF PRODUCTS AND MATERIALS:
CHEM-NUCLEAR SYSTEMS, INC.

Products or Materials	Status	Annual Amounts	Maximum Quantity at Any Time	Mode of Transport	Frequency of Shipment
Isotopes Cobalt-60 (by far largest quantity) Cesium-137 Chromium-51 Cobalt-58 Cesium-134 Zinc-65 Zirconium-95 Cerium-141 Praseodymium-143 Strontium-89 Barium-140 Manganese-54 Ruthenium-103 Rhodium-103m Iodine-131 Iron-59 Iron-55 Cerium-144 Hydrogen-3	Stored	1-2 million ft^3	1000 ft^3	Truck, tractor, trailers	5500-6000 shipments/ year; average loads 600 - 1000 ft^3

Source: David Ebenhack, manager, Chem-Nuclear Systems, Barnwell Nuclear Systems.

TABLE 2.2.2-12

TERMINAL AREA FORECAST
FISCAL YEARS 1980-1990 TOTAL FLIGHTS

Year	Flights
1981	97,000
1982	100,000
1983	103,000
1984	106,000
1985	107,000
1986	109,000
1987	111,000
1988	113,000
1989	114,000
1990	115,000

Source: Federal Aviation Administration, Terminal Area Forecast Fiscal Year 1980-Fiscal Year 1990, Department of Transportation.

Table 2.2.3-1 through Table 2.2.3-5

DELETED

TABLE 2.2.3-6 (SHEET 1 OF 2)

CHEMICALS TRANSPORTED ALONG MAJOR TRANSPORTATION ROUTES WITHIN 5 MILES OF VEGP - QUANTITIES AND DISTANCES

Chemical	Mode of Transport	Physical Conditions	Maximum Shipment Size	Minimum Distance from Control Room Air Intake
Chlorine	Truck	Liquefied, compressed gas; ambient conditions	1 ton cylinders	7600 m
Anhydrous ammonia	Truck	Liquefied, compressed gas; 28°F and 250 psi	6 tons	7600 m
Nitrogen (liquid)	Truck	Liquefied compressed gas	6500 gal	7600 m
Phosphoric acid	Truck	Liquid, pure state, ambient conditions	200 lb stainless steel drums	7600 m
Nitric acid	Truck	Liquid, pure state, ambient conditions	5000 gal	7600 m
No. 2 diesel fuel oil	Truck	Liquid, ambient conditions	6000 gal	7600 m
Anhydrous ammonia	Rail	Liquefied compressed gas; 28°F and 80 psi	26 tons	7250 m
Sulfuric Acid	Rail	Liquid, pure state at ambient conditions	13,400 gal	7250 m
Carbon dioxide	Rail	Liquefied compressed gas; 0°C and 300 psi	20 tons	7250 m

TABLE 2.2.3-6 (SHEET 2 OF 2)

Chemical	Mode of Transport	Physical Conditions	Maximum Shipment Size	Minimum Distance from Control Room Air Intake
Helium	Rail	Liquefied compressed gas; ambient temp, 2200 psi	200,000 ft^3	7250 m
Nitrogen (liquid)	Rail	Liquefied compressed gas	14,000 gal	7250 m
No. 2 diesel fuel oil	Barge	Liquid, ambient conditions	5,620,000 lb	1050 m
Gasoline	Barge	Liquid, ambient conditions	5,620,000 lb	1050 m

Table 2.2.3-7 through Table 2.2.3-10

DELETED

TABLE 2.2.3-11 (Sheet 1 of 2)

VOLUME CONCENTRATIONS OF FUEL-AIR MIXTURES
VERSUS FLAMMABILITY LIMIT
(TRANSPORTATION SOURCES)

Substance	Mode of Transport	Fuel-Air Concentration percent[a] Lower	Median	Upper	Lower Flammability Limit Percent[b]
Gasoline	River	0.24	0.13	0.12	1.4
Fuel oil	River	4.08×10^{-4}	2.23×10^{-4}	1.94×10^{-4}	1.3
Ammonia	Rail		[c]		15.5
Ammonia	Highway		[d]		15.5
Fuel oil	Highway	4.64×10^{-4}	2.57×10^{-4}	2.22×10^{-4}	1.3

a. Lower, median, and upper correspond to windspeeds of 0.25, 5.0, and 10.0 m/sec, respectively.

b. All values are from reference 10.

c. The ammonia-air concentration is within the flammability range at distances between 6431 and 6758 meters from the nearest safety-related structure. The critical distance (given by $kW^{1/3}$ in Regulatory Guide 1.91), which could cause overpressures to safety-related structures, is 686 meters. Therefore, even though an explosion due to an ammonia release from a railroad tank car could occur, it would occur at a distance great enough not to pose an overpressure hazard to VEGP safety-related structures.

d. The ammonia release on the highway is bounded by the ammonia release on the railway due to the larger quantity transported by rail and the shorter distance between the railway and the plant site.

TABLE 2.2.3-11 (SHEET 2 OF 2)

(SOURCES STORED OFFSITE)

Substance	Mode of Transport	Fuel-Air Concentration percent[a] Lower	Median	Upper	Lower Flammability Limit Percent[b]
Ammonia	SRP		[c]		15.5
Fuel oil	SRP	4.07×10^{-4}	2.24×10^{-4}	1.94×10^{-4}	1.3
Fuel oil	Plant Wilson	4.16×10^{-4}	2.29×10^{-4}	1.99×10^{-4}	1.3
Misc oils	Plant Wilson	1.83×10^{-4}	2.83×10^{-4}	2.46×10^{-4}	1.3[d]

a. Lower, median, and upper correspond to windspeeds of 0.25, 5.0, and 10.0 m/sec, respectively.

b. All values are from reference 10.

c. The ammonia-air concentration is within the flammability range at distances between 4652 and 4791 meters from the nearest safety-related structure. The critical distance (given by $kW^{1/3}$ in Regulatory Guide 1.91), which could cause overpressures to safety-related structures, is 292 meters. Therefore, even though an explosion due to an ammonia release from SRP could occur, it would occur at a distance great enough not to pose an overpressure hazard to VEGP safety-related structures.

d. Lower limit of flammability is assumed to be the same as that for fuel oil.

TABLE 2.2.3-12 (SHEET 1 OF 2)

REGULATORY GUIDE 1.91 ALLOWABLE DISTANCE AND ACTUAL DISTANCE OF HAZARDOUS CHEMICALS (TRANSPORTED)

Substance	Mode of Transport	Actual Distance (m)	Allowable Distance (m)	Total Mass (lb)	Lower and Upper Flammable Limit (Percent)[a]	Comments
Gasoline	River	1050	N/A	5.62×10^6	1.4 - 7.4	No more consideration[b]
Fuel oil	River	1050	N/A	5.62×10^6	1.3 - 6.0	No more consideration[c]
Ammonia consideration	Rail	7250	686	5.2×10^4	15.5 - 27.0	No more consideration
Ammonia consideration	Highway	7600	420	1.2×10^4	15.5 - 27.0	No more consideration
Fuel oil consideration	Highway	7600	N/A	4.21×10^4	1.3 - 6.0	No more consideration[c]

a. All values are from reference 10.

b. The concentration of flammable material in the tank vapor space is approximately 47 percent, which is well above the upper limit of flammability. Accordingly, an explosion of the gasoline barge on the Savannah River is not considered a credible event. This conclusion is consistent with the National Fire Protection Association,[57] which states that the concentration of vapor in the vapor space of tanks storing high vapor pressure liquids, such as gasoline, is normally too rich to burn.

c. The concentration of flammable material in the tank vapor space is approximately 0.17 percent, which is below the lower limit of flammability. Accordingly, an explosion of these diesel fuel oil tanks is not considered a credible event. This conclusion is consistent with the National Fire Protection Association,[57] which states that the concentration of vapor in the vapor space of tanks storing low vapor pressure liquids, such as kerosene (which is similar to diesel fuel oil), is normally too lean to burn.

TABLE 2.2.3-12 (SHEET 2 OF 2)

REGULATORY GUIDE 1.91 ALLOWABLE DISTANCE AND ACTUAL DISTANCE OF HAZARDOUS CHEMICALS (STORED)

Substance	Mode of Transport	Actual Distance (m)	Allowable Distance (m)	Total Mass (lb)	Lower and Upper Flammable Limit (Percent)[a]	Comments
Ammonia	SRP	>5000	292	4000	15.5 - 27.0	No more consideration
Fuel oil	SRP	>5000	N/A	1.58×10^5	1.3 - 6.0	No more consideration[b]
Fuel oil	Plant Wilson	1350	N/A	2.11×10^7	1.3 - 6.0	No more consideration[b]
Misc oils	Plant Wilson	1350	454	1.51×10^4	1.3 - 6.0	No more consideration

a. All values are from reference 10.

b. The concentration of flammable material in the tank vapor space is approximately 0.17 percent, which is below the lower limit of flammability. Accordingly, an explosion of these diesel fuel oil tanks is not considered a credible event. This conclusion is consistent with the National Fire Protection Association,[57] which states that the concentration of vapor in the vapor space of tanks storing low vapor pressure liquids, such as kerosene (which is similar to diesel fuel oil), is normally too lean to burn.

c. Flammability limits are assumed to be the same as those for fuel oil.

TABLE 2.2.3-13 (SHEET 1 OF 2)

TRANSPORTATION SOURCES - TOXIC GAS RELEASE INFORMATION

Chemical	Toxicity[a] Limit (ppm)	8-h Average[b] Concentration (ppm)	Release[c] Type	Maximum[b,d] Release Rate (gm/s)	Fraction Flashed to Vapor	Vapor[e] Pressure (mm Hg))	Odor[f] Detection (ppm)	Control Room Concentration 2-min After Detection (ppm)[b]
Truck								
Anhydrous ammonia	500	N/A	L	4.996×10^{3}	0.18	N/A	50	69
Nitrogen (liquid)	143,000	0.0	L	N/A	1.0	N/A	N/A	N/A
Phosphoric acid	0.25	3.8×10^{-5}	N	5.0×10^{-3}	N/A	0.0285	N/A	N/A
Nitric acid	2.0	2.85[i]	N	1.601×10^{2}	N/A	10	N/A	N/A
No. 2 fuel oil	300	0.026	N	1.41×10^{1}	N/A	0.408	N/A	N/A
Chlorine	15[g]	N/A	L	N/A[j]	N/A[j]	N/A	N/A[j]	N/A[j]
Rail								
Anhydrous ammonia	500	N/A	L	1.592×10^{4}	0.18	N/A	50	112
Carbon dioxide	5000	41	L	1.187×10^{5}	0.17	N/A	N/A	N/A
Helium	143,000	35	G	N/A	1.0	N/A	N/A	N/A
Nitrogen (liquid)	143,000	0.0	L	N/A	1.0	N/A	N/A	N/A
Sulfuric acid	0.25	0.0037	N	2.6×10^{1}	N/A	0.005	N/A	N/A
Barge								
No. 2 fuel oil	300	44.6	N	3.76×10^{2}	N/A	0.408	N/A	N/A
Gasoline	500[h]	N/A	N	5.22×10^{4}	N/A	403	10	196

TABLE 2.2.3-13 (SHEET 2 OF 2)

a. The 2-min toxicity limit is presented for chlorine, ammonia and gasoline only. The long term (8-h average continuous exposure) toxicity limit is presented for all other materials. All values are from reference 43.

b. At worst case windspeed.

c. N = normal boiling point liquid (boiling point > ambient temperature). Continuous release scenario. L = low boiling point liquid or liquefied compressed gas (boiling point < ambient temperature). Puff release plus continuous release scenario. G = compressed gas release. Puff release scenario.

d. Continuous release rate for normal boiling point liquids. Boiloff rate for low boiling point.

e. Vapor pressure not used in analysis of low boiling point liquids or compressed gases.

f. The odor detection limit is only presented for ammonia and gasoline since the analysis considers control room concentrations 2 min after odor detection.

g. From U.S. NRC Regulatory Guide 1.78, June 1974.

h. The value for fuel oil is used since toxicity limits for gasoline have not been established.

i. The long term toxicity limit is based on continuous exposure for a 40-h work week and results in eye irritation and teeth erosion. An 8-h exposure at levels slightly above this limit will not incapacitate control room operators. Additionally, the 2-min toxicity limit value is never exceeded for nitric acid releases. Furthermore, nitric acid decomposes to nitric oxide and nitrogen dioxide in the presence of air.[44] If all the nitric acid is assumed to be converted to (the more toxic) nitrogen dioxide, the long term 8-h toxicity limit value is never exceeded. Therefore, even though nitric acid exceeds the long term toxicity limit, it will not incapacitate control room operators.

j. The transportation source of chlorine is located greater than 7600 m from the control room air intake. Based on the screening criteria contained in NRC Regulatory Guide 1.78, June 1974, a 1-ton cylinder (largest single container transported) at this distance need not be considered in evaluating control room habitability.

TABLE 2.2.3-14

CHEMICALS ANALYZED THAT ARE AT OFFSITE
STORAGE FACILITIES - QUANTITIES AND DISTANCES

Chemical	Physical Conditions	Maximum Quantity Stored	Distance from Control Room Air Intake
Chlorine at Savannah River Plant (SRP)	Liquefied, compressed gas; ambient conditions	10 1-ton cylinders	>5000 m
Anhydrous ammonia at SRP	Liquefied, compressed gas; ambient conditions	2 tons	>5000 m
Phosphoric acid at SRP	Liquid, pure state; ambient conditions	460 lb	>5000 m
Sulfuric acid at SRP	Liquid, pure state; ambient conditions	270 tons	>5000 m
No. 2 diesel fuel oil at SRP	Liquid, ambient conditions	22,500 gal	>5000 m
No. 2 diesel fuel oil at combustion turbine plant	Liquid, ambient conditions	3,000,000 gal	1350 m

TABLE 2.2.3-15

OFFSITE SOURCES - TOXIC GAS RELEASE INFORMATION

Chemical	Toxicity[a] Limit (ppm)	8-h Average[b] Concentration (ppm)	Release[c] Type	Maximum[b,d] Release Rate (gm/s)	Fraction Flashed to Vapor	Vapor[e] Pressure (mm Hg)	Odor[f] Detection (ppm)	Control Room Concentration 2-min After Detection (ppm)[b]
At Savannah River Plant								
Chlorine	15[g]	N/A	L	N/A[i]	N/A[i]	N/A	N/A[i]	N/A[i]
Anhydrous ammonia	500	N/A	L	2.154×10^{3}	0.18	N/A	50	70
Phosphoric acid	0.25	6.9×10^{-5}	N	3.5×10^{-3}	N/A	0.0285	N/A	N/A
Sulfuric acid	0.25	0.015	N	3.6×10^{-1}	N/A	0.005	N/A	N/A
Diesel fuel oil	300[h]	0.8	N	4.75×10^{1}	N/A	0.408	N/A	N/A
At combustion turbine plant								
Diesel fuel oil	300[h]	4.1	N	1.56×10^{1}	N/A	0.408	N/A	N/A

a. The two-min toxicity limit is presented for chlorine and ammonia only. The long term (8-h average continuous exposure) toxicity limit is presented for all other materials. All values are from reference 43.

b. At worst case windspeed.

c. N = normal boiling point liquid (boiling point > ambient temperature). Continuous release scenario. L = low boiling point liquid or liquefied compressed gas (boiling point < ambient temperature). Puff release plus continuous release scenario. G = compressed gas release. Puff release scenario.

d. Continuous release rate for normal boiling point liquids. Boiloff rate for low boiling point.

e. Vapor pressure not used in analysis of low boiling point liquids or compressed gases.

f. The odor detection limit is only presented for ammonia since the analysis considers control room concentrations 2 min after odor detection.

g. From U.S. NRC Regulatory Guide 1.78, June 1974.

h. The value for gasoline is used since toxicity limits for fuel oil have not been established.

i. The chlorine source at SRP is located greater than 5000 m from the control room air intake. Based on the screening criteria contained in NRC Regulatory Guide 1.78, June 1974, a 1-ton cylinder (largest single container at SRP) at this distance need not be considered in evaluating control room habitability.

TABLE 2.2.3-16

DELETED

TABLE 2.2.3-17

OILS AND SOLVENTS STORED AT THE COMBUSTION TURBINE PLANT

Substance[a]	Quantity (gal)
Fuel oil No. 2	9,000,000 (three 3,000,000-gal tanks)
Gulfcrest 32	700
Gulf diesel Motive 485	700
Gulf Harmony 115	200
Gulf HD SAE 10W	100
Gulf Senate 110	100
Shell 20W Oil	100
Gulf Senate 3200	50
Momar Electraclean	200

a. Brand name or equivalent.

TABLE 2.2.3-18 (SHEET 1 OF 7)

ONSITE CHEMICAL STORAGE

Chemical	State	Quantity	Distance (ft)	Notes
Benzoyl peroxide (catalyst)	40% aqueous solution, ambient conditions	One 6-gal tank	660	(e)
		One 55-gal drum	660	(e)
Carbon dioxide	Liquefied gas, 300 psi, 0°F	One 10-ton tank	310	(a)
	Liquefied gas, 100 psi, ambient temperature	Twelve 150-lb cylinders	160	(b)
Dimethyl-p-toludine (promoter)	Liquid, ambient conditions	One 5-gal tank	660	(c)
		One 55-gal drum	660	(c)
Fuel Oil No. 2	Liquid, ambient conditions	Four 80,000-gal tanks	350	(f)
		Four 1250-gal tanks	310	(b)
		Two 560-gal tanks	720	(b)
		One 100,000-gal tanks	375	(a)
		One 1500-gal tank	1020	(f)

TABLE 2.2.3-18 (SHEET 2 OF 7)

Chemical	State	Quantity	Distance (ft)	Notes
Gasoline	Liquid, ambient conditions	Two 12,000-gal tanks	2675	Unleaded gas[f]
		Two 12,000-gal tanks	2675	Diesel[f]
		One 6000-gal tank	1200	Unleaded gasoline[a]
		One 3000-gal tank	1200	Diesel fuel[b]
Kerosene	Liquid, ambient conditions	One 5000-gal tank	2675	(f)
Halon 1301	Liquefied compressed gas	Two 550-lb capacity cylinders with 376 lb of Halon 1301	Inside control room	These are the largest of the various halon tanks inside the control bldg.[a]
Hydraulic fluid (phosphate esters)	Liquid, ambient conditions	Four 800-gal tanks	240	(c)
		Twelve 20-gal accumulators	240	(c)
Hydrazine	35% aqueous solution ambient conditions	One 6644-gal tank	400	(a)
		Four 250-gal tanks	195	(b)
		Four 800-in.^{3} cylinders	195	(b)
		Four 50-gal tank	>490	(b)

TABLE 2.2.3-18 (SHEET 3 OF 7)

Chemical	State	Quantity	Distance (ft)	Notes
Hydrogen	Compressed gas	130,000-ft^3	930	(a)
	Compressed gas	Twelve 50-lb cylinders	160	(e)
	Compressed gas	8400 ft^3 (Two racks of 16 cylinders each)	850 ft	
Lube oil	Liquid, ambient conditions	Two 30,000-gal tanks	335	(c)
		Two 6400-gal reservoirs	280	(c)
		Twenty-four 55-gal drums	320	(c)
		Four 700-gal reservoirs	120	(c)
		One 300-gal tank	660	(c)
		One 4000-gal tank	2675	(c),(f)
Hydrogen peroxide	Liquid, ambient conditions	One 100-gal tank	200	(d)
Nitrogen	Compressed gas, 2200 psi, 70°F liquefied gas, 200 psi, -320°F	Three 8200-ft^3 vessels	900	(b)
		One 6000-gal tank	900	(a)
		One 2000-gal tank	700	(a)
Oxygen	Compressed gas, 2200 psi, 70°F	Four 8960-ft^3 vessels	890	(d)
Polymer (styrene)	Liquid, ambient conditions	One 6000-gal tank	680	(f)

TABLE 2.2.3-18 (SHEET 4 OF 7)

Chemical	State	Quantity	Distance (ft)	Notes
Seal oil	Liquid, ambient conditions	Two 1360-gal tanks	160	(c)
Sodium hydroxide	50% aqueous solution, ambient conditions	One 20,990-gal tank	580	(a)
		One 175-gal tank	560	(b)
		One 120-gal tank	540	(b)
		One 1130-gal tank	660	(b)
Sulfuric acid	Liquid, ambient conditions	Four 55-gal tanks	530	(b)
		Two 6600-gal tanks	570	(b)
		One 12,850-gal tank	575	(a)
		One 50-gal tank	560	(b)
		Two 200-gal tanks	1490	(b)
		One 15,175-gal tank	1560	(b)
		One 55-gal tank	540	(b)
Sodium polyacrylate	Liquid, ambient conditions	One 5000-gal tank	1380	(c)
		Six 55-gal drums	615	
Sodium tolyltriazole	Liquid, ambient conditions	One 2500-gal tank	1380	(c)
		Six 55-gal drums	615	

TABLE 2.2.3-18 (SHEET 5 OF 7)

Chemical	State	Quantity	Distance (ft)	Notes
Zinc Salt	Liquid, ambient conditions	One 5000-gal tank (g)	1380	(c)
Methoxypropylamine	Liquid, ambient conditions	One 12,789-gal tank Two 250-gal tanks	385 195	(a) (b)
Sodium nitrite	Liquid, ambient conditions	One to ten 55-gal drums (at each of five locations) (h)	75 400	(d)
Sodium hypochlorite	Liquid, ambient conditions	One 500-gal transport tank (i)	>3000 615	(d)
		One 250-gal tank	720	
		Three 6500-gal tanks	1380	
		One 10,000-gal tank	1450	
Trisodium phosphate	Solid, ambient conditions	Three baskets with a total of 220 ft^3 to 260 ft^3	N/A	(j)
Bromine salt (hypobromous acid is a combination of bromine salt and sodium hypochlorite)	Liquid, ambient conditions	One 6800-gal tank (maximum)	1380	(c)
Sodium or ammonium bisulfite	Liquid, ambient conditions	Two 400-gal tanks (one in use)	2025	(a)
Coagulate (polymer, RoQuest 3000)	Liquid, ambient conditions	65 gallons	850 ft	
Anti-scalent (Vitec 3000)	Liquid, ambient conditions	130 gallons	850 ft	
Hydrochloric (muriatic) acid	Liquid, ambient conditions	5 gallons	850 ft	
Propane	Liquified compressed gas			
		Three 2000-gal tanks (fire training facility)	3800 ft	

TABLE 2.2.3-18 (SHEET 6 OF 7)

Chemical	State	Quantity	Distance (ft)	Notes
NALCO 3D TRASAR® 3DT177 (10-30% w/w Phosphoric Acid)	Liquid, ambient conditions	One 6,000-gal tank	1450	(a)
NALCO 3D TRASAR® 3DT190	Liquid, ambient conditions	One 6,000-gal tank	1450	(d)
NALCO ControlBrom® CB70 (10-30% w/w Sodium Bromide)	Liquid, ambient conditions	One 6,000-gal tank	1450	(d)
NALCO STABREX® ST70 (10-30% w/w Sodium Hypochlorite / Sodium Bromide mix: 1-5% w/w Sodium Hydroxide)	Liquid, ambient conditions	One 10,000-gal tank	1450	(b,d)
NALCO 3D TRASAR® 3DT199 (30-60% w/w Sodium Benzotriazole: <1% Sodium Hydroxide)	Liquid, ambient conditions	One 6,000-gal tank	1450	(b,c)

a. Limiting case which results in maximum control room concentration and is analyzed as described in paragraph 2.2.3.1.4.3. Results of analysis are presented in table 2.2.3-20.

b. This case is not limiting. It results in lower control room concentrations.

c. Not analyzed since this material is nonvolatile.

d. Not analyzed since this material is nontoxic.

e. Not analyzed since this material is stored in quantities less than 100 lb.

f. Not analyzed since this material is stored in underground tanks and no significant spill can occur.

g. A small number of 55-gal drums may be stored near the circulating water chemical injection building for emergency use.

h. The 55-gal drums are located at ACCW, CCW, essential chilled water, normal chilled water (all located 75 feet away from the control room) and diesel generator building (400 feet away from the control room).

i. One 500-gal transport tank services the river intake and NSCW systems. The control room is approximately 3000 feet away from the river intake and approximately 615 feet away from the NSCW system.

TABLE 2.2.3-18 (SHEET 7 OF 7)

j. The three trisodium phosphate baskets are located inside containment and isolated from the control room.

k. Deleted.

TABLE 2.2.3-19

PARAMETERS USED IN TOXIC GAS ANALYSIS

Control Room

- Net volume = 168,540 ft^3
- Normal outside air inflow rate = 3000 ft^3

TABLE 2.2.3-20 (SHEET 1 OF 2)

TOXIC GAS RELEASE INFORMATION

Chemical	Toxicity[a] Limit (ppm)	8-h Average[b] Concentration (ppm)	Release[c] Type	Maximum[b,d] Release Rate (gm/s)	Fraction Flashed to Vapor	Vapor[e] Pressure (mm Hg)	Odor[f] Detection (ppm)	Control Room Concentration 2-min After Detection (ppm)[b]
Ammonia (29%)	500[10]	N/A	N	11.627×10^{3}	N/A	474	50	275
Carbon dioxide	5000	4039	L	7.193×10^{4}	0.13	(e)	N/A	N/A
Fuel oil no. 2	300	1.3	N	4.749×10^{-1}	N/A	0.408	N/A	N/A
Halon 1301	70,000[29]	999	L	N/A	1.0	(e)	N/A	N/A
Hydrazine (35%)	30[42]	N/A	N	3.692×10^{1}	N/A	9.2	4	12.9
Hydrogen - liquid[i]	143,000	3201	L	1.18×10^{5}	0.0	(e)	N/A	N/A
Nitrogen - liquid	143,000	5034[h]	L	4.466×10^{5}	0.002	(e)	N/A	N/A
Nitrogen - gas	143,000	2772	G	N/A	1.0	(e)	N/A	N/A
Sulfuric acid	0.25	0.16	N	8.270×10^{-2}	N/A	0.005	N/A	N/A
Gasoline	500[g]	37.91	N	2.1×10^{1}	N/A	620	N/A	N/A
Methoxypropylamine (30%)	5[59]	1.5	N	10.9	N/A	20	<1	N/A
Sodium/ammonium bisulfite	5[j]	0.8	N	6.1	N/A	18	N/A	N/A
Hydrochloric acid (40%)	5 ppm	1.0	N	0.41	N/A	190	N/A	N/A
Propane (fire training facility)[n]	1000	237.6	L	N/A	N/A	(e)	N/A	N/A
NALCO 3D TRASAR® 3DT177 (10-30% w/w Phosphoric Acid)	245	157	N	250	N/A	52	N/A	N/A

a. The 2-min toxicity limit is presented for ammonia and hydrazine only. The long term (8-h average continuous exposure) toxicity limit is presented for all other materials. All values are from reference 43 unless otherwise noted.

b. At worst case windspeed.

c. N = normal boiling point liquid (boiling point > ambient temperature). Continuous release scenario. L = low boiling point liquid or liquefied compressed gas (boiling point < ambient temperature). Puff release plus continuous release scenario. G = compressed gas release. Puff release scenario.

d. Continuous release rate for normal boiling point liquids. Boiloff rate for low boiling point.

e. Vapor pressure not used in analysis of low boiling point liquids or compressed gases.

TABLE 2.2.3-20 (SHEET 2 OF 2)

f. The odor detection limit is only presented for ammonia, gasoline, hydrazine, and methoxypropylamine since the analysis considers control room concentrations 2 min after odor detection.

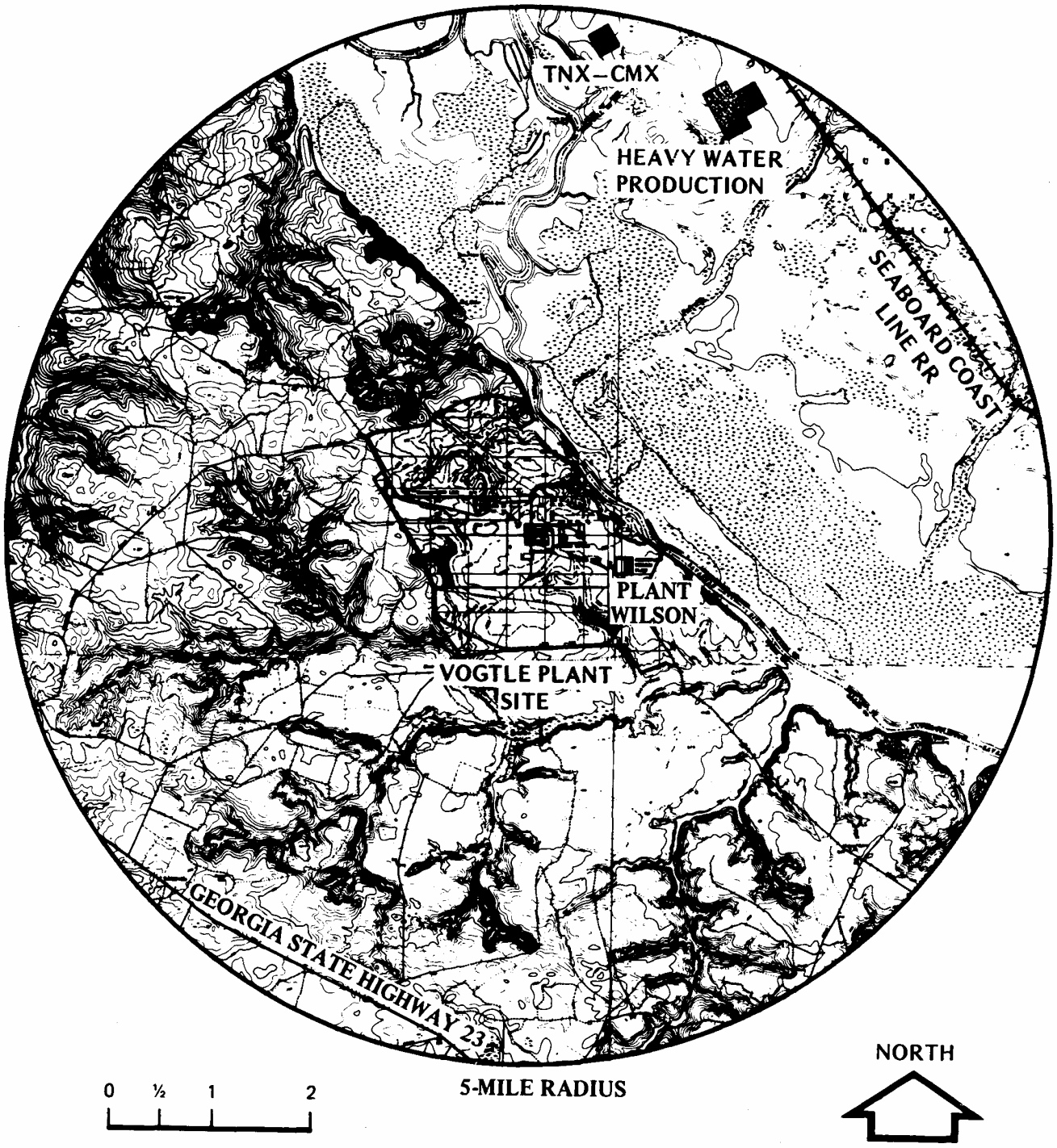
g. The value for fuel oil is used since toxicity limits for gasoline have not been established.

h. Assumes a 9000 gallon liquid tank at 274 meters.

i. The liquid hydrogen tank has been removed and the hydrogen requirements are supplied by a tube trailer. The quantity of hydrogen in the tube trailer is less than the quantity the original installed tank held; therefore, the concentration envelopes the tube trailer quantity.

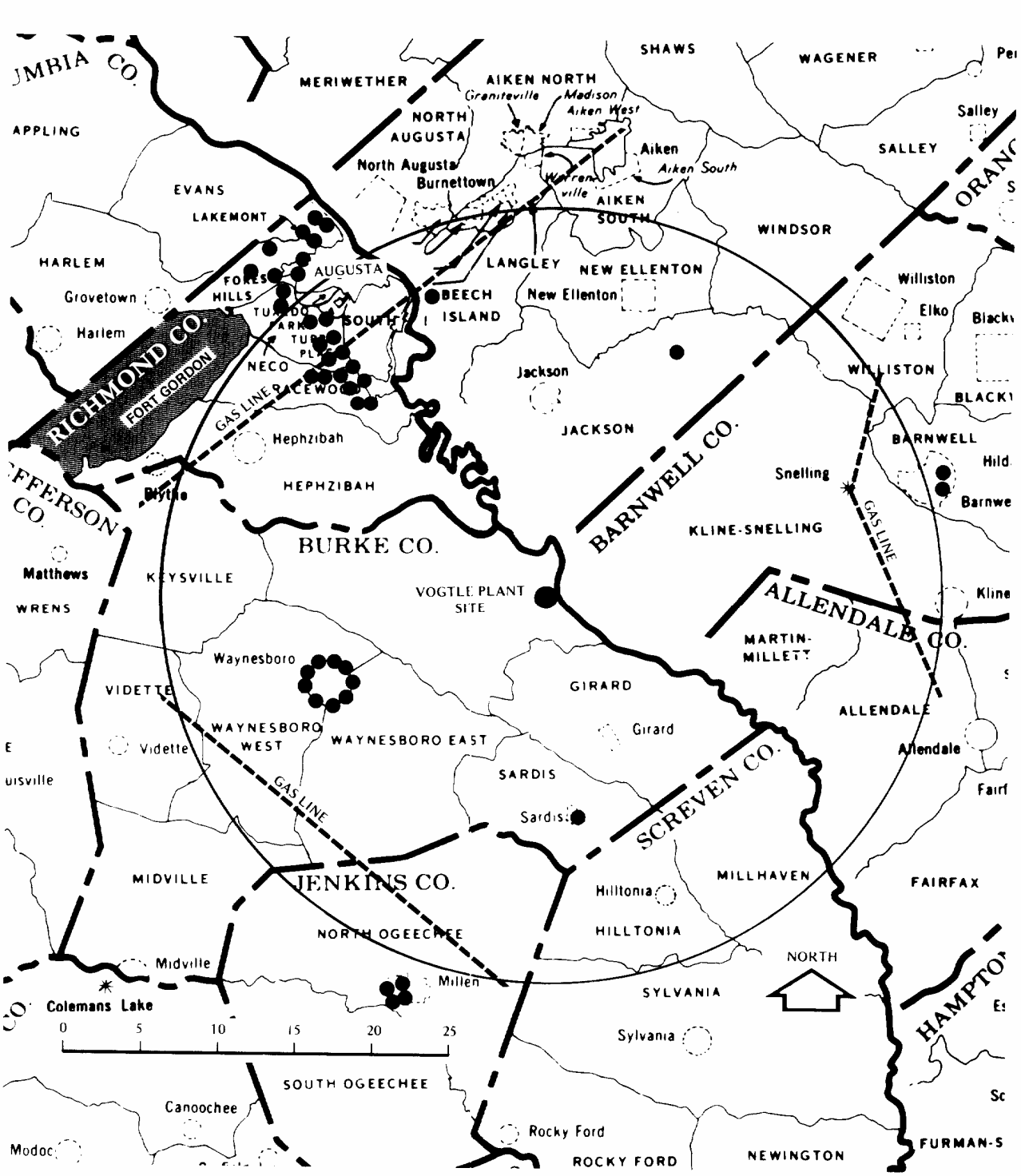
j. See toxic chemicals calculation X6CHH10 for basis used to evaluate toxicity limits.

k. Values evaluated are for a single 8000-gal propane tank located 3500 ft from the control room air intake. This evaluation bounds the installed configuration of three 2000-gal propane tanks located at the fire training facility, 3800 ft from the control room air intake.



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®*	VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	LOCATION OF MAJOR INDUSTRIAL FACILITIES
		FIGURE 2.2.1–1



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	MAJOR INDUSTRIAL FACILITIES, MILITARY BASES, AND PIPELINES WITHIN 25 MILES
	FIGURE 2.2.1–2

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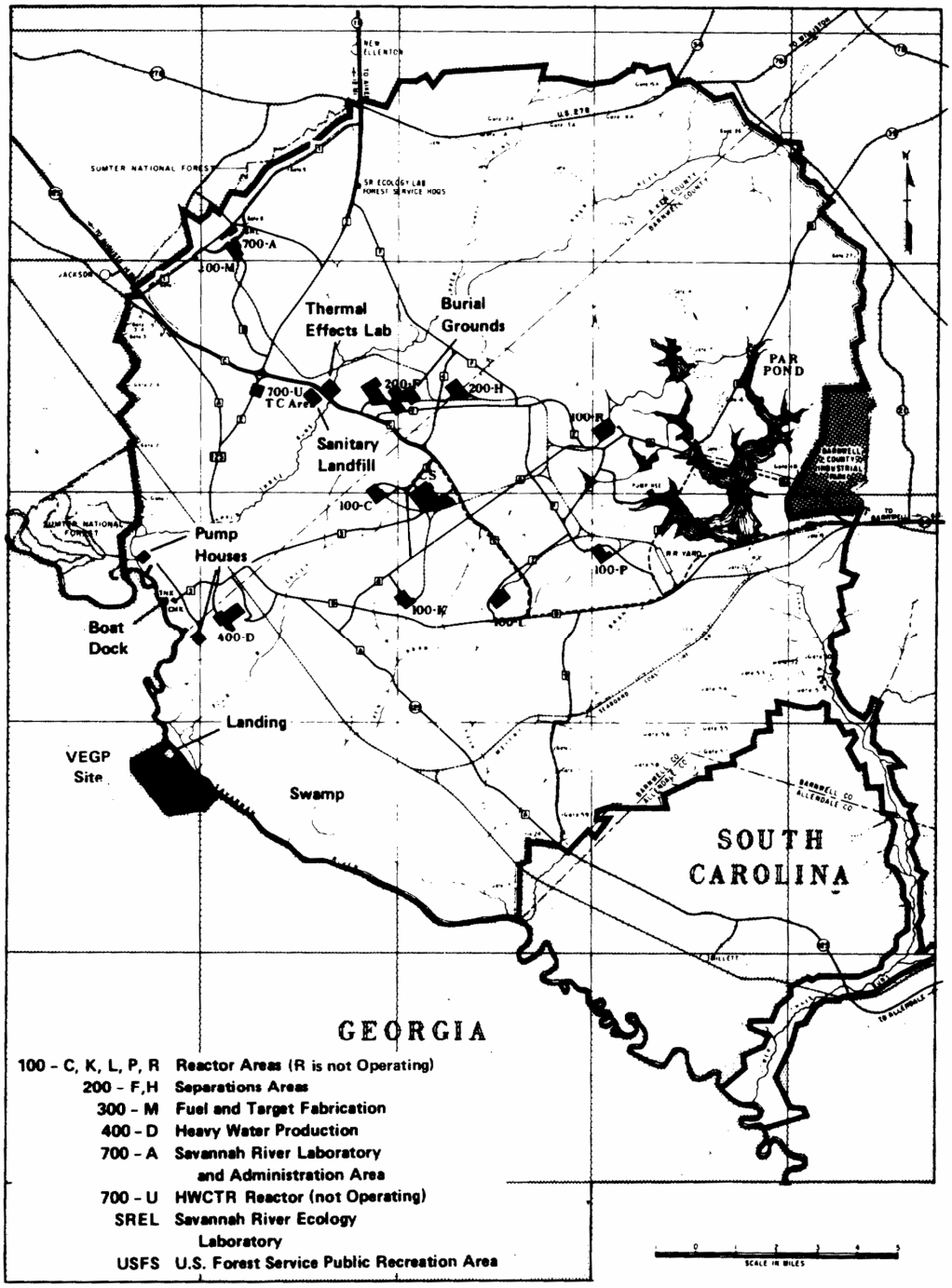
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

HIGHWAY TRANSPORTATION ROUTES, AIRPORTS, DOCK, AND RAILROAD FACILITIES WITHIN 25 MILES

FIGURE 2.2.1–3



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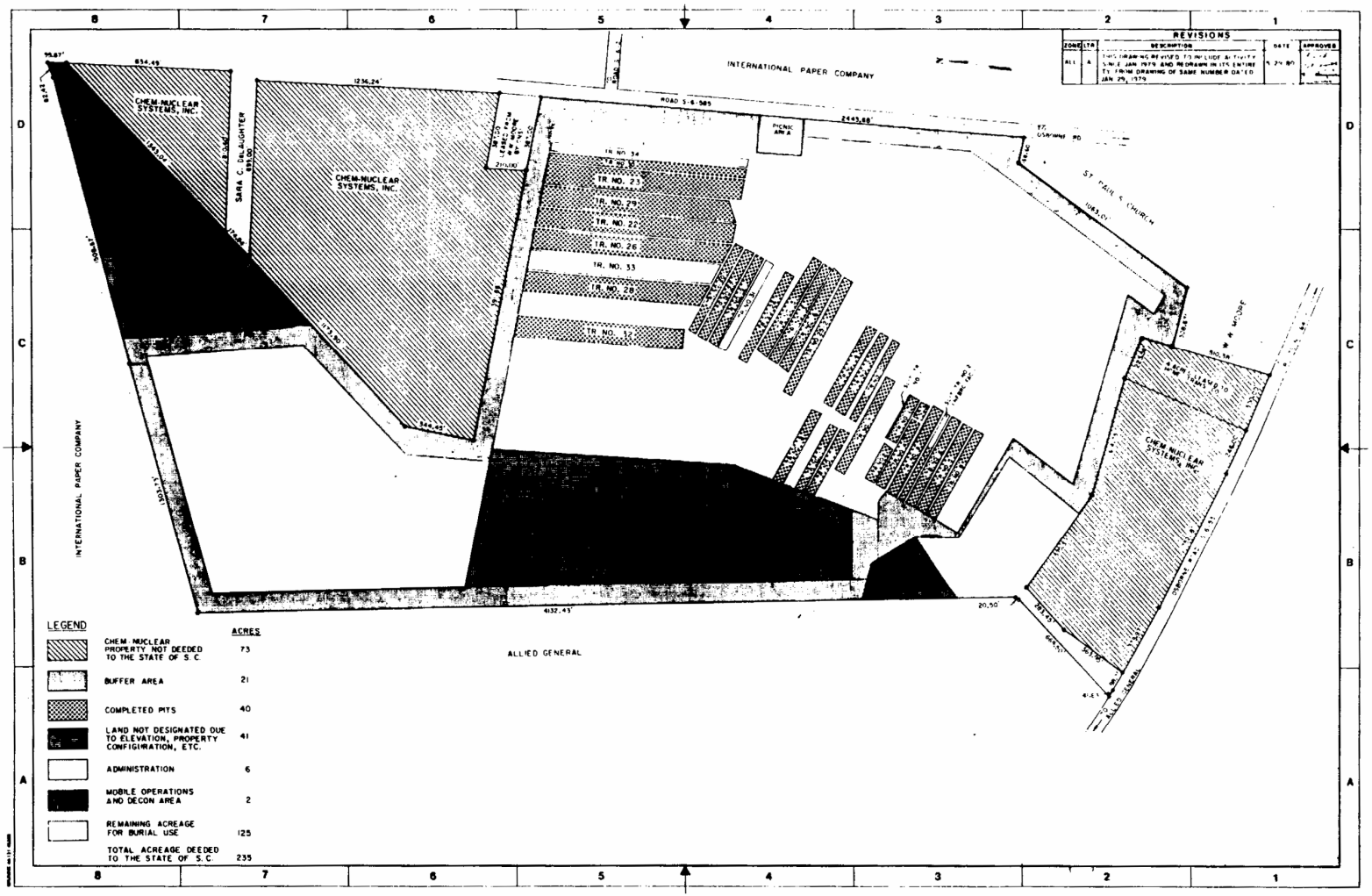
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

SAVANNAH RIVER PLANT SITE MAP

FIGURE 2.2.2–1



REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CHEM–NUCLEAR SYSTEMS, INC.
LAND UTILIZATION PLAN

FIGURE 2.2.2–2

2.3 METEOROLOGY

2.3.1 REGIONAL CLIMATOLOGY

2.3.1.1 General Climate

The general climate in this region [NOAA, 1980(a)[1]] is characterized by mild, short winters, long periods of mild sunny weather in the autumn, somewhat more windy but mild weather in spring, and long, hot summers. Heaviest precipitation in the winter and early spring is associated with low pressure systems moving eastward and northward through the Gulf States and up the Atlantic coast, drawing in moist air from the Gulf of Mexico and the Atlantic Ocean. In summer, heaviest precipitation is due to thunderstorms.

The site is in the area known as the upper Coastal Plain, between the Appalachian Mountains and the Atlantic Ocean, generally 150 to 250 ft above sea level in this region, cut by the valley of the Savannah River, 2 to 5 miles wide near the site.

The Appalachian Mountains to the northwest shelter the region in winter from the cold air masses sweeping down through the continental interior. The cold air that does reach the area is considerably warmed by the descent to the relatively low elevations of the region, as well as by modification due to heating as it passes over the continent. Morning frost conditions are frequent in winter, but daily mean temperatures below freezing are uncommon. Warm, moist air masses from the Gulf of Mexico or the Atlantic Ocean receive little modification; therefore, the area experiences relatively high precipitation in winter and summer.

The site is far enough inland that the strong winds due to occasional tropical storms and hurricanes that affect the area are much reduced. The frequency of strong winds is discussed in paragraph 2.3.1.2.8. Heavy precipitation associated with tropical storms is fairly common in late summer, as discussed in paragraphs 2.3.1.2.1 and 2.3.1.2.5.

The site lies in a region of the United States which experiences relatively low tornado activity; however, the frequency is greater at the site than in the neighboring regions. It is not certain to what extent the relatively greater number of reports in this region (compared to the mountainous areas to the northwest) is due to topographical causes or to the variation of population density and the greater visibility of tornadoes in flatter country.

The area is subject to a relatively high incidence of stagnating anticyclones associated with high air pollution potential, especially in autumn. (See paragraph 2.3.1.2.10.)

2.3.1.2 Regional Meteorological Conditions for Design and Operating Bases (Severe Weather)

2.3.1.2.1 Heavy Precipitation

Unusually heavy precipitation in this area sustained over several hours is associated with the remnants of occasional tropical hurricanes. Although hurricane winds are much reduced in this inland area, they can cause heavy precipitation. For shorter periods, heaviest rates of precipitation are due to thunderstorms.

Maximum rainfall for return periods of 1 to 100 years for duration of 30 min to 10 days is shown in table 2.3.1-1 for the VEGP site region. Estimates for the site region were interpolated from maps [Hershfield (1973),[2] Miller (1964),[3] Frederick (1977)[4]]. For return periods greater than 20 years, frequencies shown in table 2.3.1-1 are estimated by extrapolation using the Gumbel procedure (Gumbel, 1958[5]) which is based on the Fisher-Tippet Type II distribution. Table 2.3.1-2 shows the mean monthly rainfall for Augusta Airport and the annual number of days with 0.01, 0.1, and 0.5 in. of precipitation [NOAA, 1981 (b)[6]].

2.3.1.2.2 Hail

The number of occurrences of heavy hail greater than 3/4 in. in the area is approximately 7 in 13 years, or once in 2 years, for a 1° (latitude and longitude) "square" (Pautz, 1969[7]). (See figure 2.3.1-1.) Isopleths from squares are shown in figure 2.3.1-2. Using cross-checking procedures to eliminate duplication, this figure is considered a reasonably reliable estimate of frequency of occurrence of heavy hail.

2.3.1.2.3 Snow

Snow is infrequent in the site region, and heavy snow is very rare. Loads resulting from snow and rain are discussed in paragraph 2.4.2.3. The greatest 24-h total snowfall on record was 13.7 in. in February 1973. The average annual snowfall at the Augusta Airport is about 1 in. [NOAA, 1981(b)[6]]. Figure 2.3.1-3 shows that the maximum snow load for a 100-year mean recurrence period is about 8 lb/ft^2 (ANSI, 1972[8]). The probable maximum 48-h winter precipitation is 19 in. (U.S. Weather Bureau, 1956[9]).

2.3.1.2.4 Ice Storms

Freezing rain resulting in heavy ice loading is rare in the VEGP site region. Based on a 10-year period (Bennett, 1959[10]) there were 19 days (December, 6 days; January, 10 days; February, 2 days; March, 1 day) with freezing rain in Augusta, Georgia. (See figure 2.3.1-4.) Freezing rain has been reported to last as long as 17 h (National Weather Service, 1960). The extreme radial thickness of ice on utility wires as shown in figure 2.3.1-5 is between 0.50 and 0.74 in.

2.3.1.2.5 Hurricanes

The VEGP is located approximately 100 miles inland from the Atlantic coast; so the effects from hurricanes or tropical depressions are considerably diminished. During the 84-year period from 1886 to 1969, 11 hurricanes and 26 tropical storms passed through a 2.5° (longitude and latitude) box containing the site area (Hope, NOAA SR55, 1971[11]). Figure 2.3.1-6 shows that in the VEGP site region the probability of at least one tropical storm or hurricane per season entering the 2.5°box is about 0.36.

2.3.1.2.6 Thunderstorms

As reported by the observers at the Augusta Airport [NOAA, 1981(b)[6]], the average number of days per year with thunderstorms is 56, with a maximum frequency of 13 days in the month of July and a minimum of 1 day during the months of October through January [NOAA, 1981(b)[6]].

2.3.1.2.7 Lightning

Seasonal and annual frequencies of lightning strikes to the reactor containment building have been estimated based on the frequency of thunderstorms in the VEGP site area and the structure's "attractive area." Since the attractive area of a structure is proportional to the magnitude of the lightning bolt's current, an estimate of the frequency of lightning strikes was determined for lightning bolts having currents of 20, 40, 60, 90, and 135 kA. Table 2.3.1-5 provides a seasonal and annual summary of thunderstorm frequency and frequency of lightning strikes, as well as the various attractive areas.

The reactor containment building is the largest safety-related structure, system, or component on the VEGP site. Strike frequency estimates are provided in table 2.3.1-5 to reflect lightning strikes to the containment building. All other safety-related structures would be subjected to lightning strikes on a less frequent basis than that reported for the containment building.

The lightning strike frequency calculation is based on the technique described by Marshall (1973). The average number of cloud-to-ground lightning strikes is calculated using the following formula:

$$NE = (0.1 + 0.35 \sin \lambda)(0.40 \pm 0.20)$$

where:

NE = number of flashes to earth/km^2/thunderstorm day.

λ = geographical latitude (for the VEGP site, λ =33° 8'30").

To ensure conservatism in the calculation, the term (0.40 $\pm$ 0.20) is assumed to be (0.40 + 0.20), or 0.6. Therefore, for the VEGP site, NE = 0.175 flashes/km^2 thunderstorm day. Multiplying this value by the seasonal and annual number of thunderstorm days at Augusta as taken from NUREG/CR-2252,[19] the number of seasonal and annual cloud-to-ground lightning flashes is determined as shown in table 2.3.1-5.

The attractive area of the containment building is determined following the technique of Marshall:

$$K = \pi R_T^2 = \pi (R_A + R_B)^2$$

where:

K = total attractive area (m^2).

R_T = total radial distance (m).

R_A = radius of containment building (m).

R_B = radial distance to the edge of the attractive zone (m).

The parameter R_B is an increasing multiple of the building height (H) with increasing bolt current. According to Marshall, R_B = 2H, 4H, 6H, 8H, and 10H for bolt currents of 20, 40, 60, 90, and 135 kA, respectively. Given a building height of 55.25 m and a radius of 22.5 m, the attractive area of the containment building is calculated for each of the five bolt currents and presented in table 2.3.1-5.

The number of lightning strikes to the containment building is then calculated by multiplying the number of cloud-to-ground lightning flashes/km^2 by the attractive area of the containment building for each bolt current and then multiplying that product by the frequency of occurrence of each bolt current:

$$NA = A \times K \times F$$

where:

NS = number of lightning strikes to the containment building.

A = average number of cloud-to-ground strikes/km^2.

F = frequency of each bolt current (dimensionless).

Marshall assigns frequencies of 50, 22, 10, 2, and 0.5 percent to bolt currents of 20, 40, 60, 90, and 135 kA, respectively. The results of these calculations are given in table 2.3.1-5.

2.3.1.2.8 Tornado Probability

The probability of a particular point being affected by a tornado is a function of the number of tornadoes occurring on the average in a given area and the average area covered by a tornado.

Based on a 40-year record, the number of tornadoes for the 2° square in which the site is located is between one and two per year (Wolford, 1960[13]). During the period 1955 to 1967, the average number of tornadoes (Pautz, 1969[7]) for the 1° square including the site was about one and one-half per year, corresponding to approximately six per year for the 2° square. (See figures 2.3.1-7 and 2.3.1-8.) This apparent increase is typical for this kind of data and arises in part from increased public awareness of tornadoes and more complete reporting. As even the latter frequency is likely to be an underestimate of the true frequency, a reasonable,

conservative estimate is that the true frequency is twice that reported for the latter period, that is, about three per year for the 1° square.

A typical tornado (Thom, 1963[14]) is about ¼ mile wide, is in contact with the ground for about 10 miles, and covers an area of about 2 ½ mi^2. The 1° square at this latitude has an area of approximately 4000 mi^2. A conservative estimate of the probability of a given point being affected by a tornado in a given year is approximately:

$$\frac{2\ \frac{1}{2} \times 3}{4000}, \text{i.e.,} \frac{1}{500}$$

That is, a given point can be expected to be affected by a tornado once in 500 years, on the average.

Because no large body of water capable of sustaining waterspouts is located near the site, these need not be considered.

2.3.1.2.9 Strong Winds

The frequency of strong winds, 50 knots or greater, has been analyzed for a 2° square (Pautz, 1969[7]) for a 13-year period, 1955 to 1967. (See figure 2.3.1-9.) The total number of windstorms by 1° squares is shown in figure 2.3.1-10. For this area, the frequency is estimated at approximately seven per year per 2° square. Since a considerable proportion of occurrences is likely to be overlooked or unreported, a reasonable, conservative estimate may be twice that number or two occurrences for a 1° square or 4000 mi^2. Extreme winds have been fitted to a statistical distribution, allowing extrapolation to higher windspeeds. At the site, it is estimated that speeds of approximately 105 mph occur once in 100 years (Thom, 1968[15]). The gust factor and vertical wind loading are based on use of a 110-mph windspeed at the 30-ft level (ANSI, 1972[8]). Further discussion of structural loadings is given in section 3.3.

2.3.1.2.10 Probability of High Windspeeds Caused by Tornadoes

The probability of strong winds of a tornado striking a specific site is a function of two factors, the frequency of tornadoes in the site region and the intensity probability (Markee, 1974[16]). The frequency has been estimated in paragraph 2.3.1.2.7; however, there are few actual observations of wind intensity associated with tornadoes.

Data concerning tornado intensity have been compiled during 1971 and 1972. For the contiguous United States, 1612 tornadoes were graded on intensity according to the scale shown in table 2.3.1-3. These tornadoes were categorized by wind groups in table 2.3.1-4, which includes cumulative probability vs intensity. The probabilities in table 2.3.1-4 have been plotted on log probability paper in figure 2.3.1-11 to show the probability of a tornado with a given windspeed.

The design basis tornado (Markee, 1974[16]) has a probability of occurrence of about 10^{-7} per year. For the VEGP site, the tornado wind with a 10^{-7} probability of occurrence has been estimated using the procedure noted below.

The intensity probability, P_s, can be calculated using this relationship:

P_s = tornado strike probability

$P_s P_i \leq 10^{-7}$

$0.002\ P_i \leq 10^{-7}$

$P_i \leq 0.00005$

$P_i \leq 0.005$ percent

From figure 2.3.1-11, the 0.005-percent probable windspeed in a tornado is about 360 mph. Thus, in the site region, the 10^{-7} probable tornado would have a maximum windspeed of about 360 mph, which is considered 290 mph rotational and 70 mph translational (U.S. Nuclear Regulatory Commission, Regulatory Guide 1.76, 1974[17]).

2.3.1.2.11 High Air Pollution Potential

The region is one of high incidence of slow moving anticyclones resulting in high air pollution potential, especially in autumn. This is supported by Korshover (1971)[18] who has reported on the climatology of stagnating anticyclones east of the Rocky Mountains during the period 1936 to 1970. He reports that in the site region there will be, on the average, 13 "stagnation days" per year. A useful indication of the incidence of high air pollution in the site area is given by a recent analysis of high air pollution potential forecasts by the National Environmental Research Center, Environmental Protection Agency, which showed approximately 50 days of forecast high air pollution potential in the VEGP site region in the period August 1, 1960, to April 3, 1970.

2.3.1.2.12 Wet Bulb Temperature (for Ultimate Heat Sink Calculations)

The ultimate heat sink for the VEGP is the nuclear service cooling water tower basins. Heat is transferred to the atmosphere by the cooling tower. A water use analysis has been made to ensure adequate cooling capability. (See subsection 9.2.5 for further information concerning this analysis.)

2.3.1.3 References

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2.3.1.4 Bibliography

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2.3.2 LOCAL METEOROLOGY

2.3.2.1 Normal and Extreme Values of Meteorological Parameters

Augusta Airport is approximately 17 air miles NNW of the plant site, and climatological observations made there can be considered representative of conditions at the site. Ground elevation of Augusta Airport (Bush Field) is 136 ft above sea level; ground elevation at the site is approximately 220 ft. The differences in elevation and in location are of minor significance. Normals, means, and extremes of meteorological data published by the Environmental Data Service of National Oceanic and Atmospheric Administration (NOAA) for the Augusta Airport are given in table 2.3.2-1.

Using 5 years (1959 through 1963) of meteorological data from Augusta Airport, climatological means of temperature, humidity, atmospheric stability, precipitation, and wind have been compiled. Evaluations of atmospheric diffusion conditions are based on records of atmospheric stability, windspeed, and wind direction from a data collection program conducted by Georgia Power Company at the VEGP site in accordance with Regulatory Guide 1.23. In this program, records were taken on an instrumented tower utilizing sensitive instruments at multiple elevations on the tower during a 4-year period from December 1972 through December 1973, from April 1977 through April 1979, and from April 1980 through March 1981. A detailed description of the VEGP program appears in subsection 2.3.3.

2.3.2.1.1 Wind

The mean windspeeds for each month in mph and the most frequent wind direction ("prevailing wind direction") are listed in table 2.3.2-1 (NOAA, 1981(b)). Fastest monthly average winds, 7 to 8 mph, occur in winter; and the slowest, in summer at 5 to 6 mph. Monthly, seasonal, and

annual wind roses are shown in figures 2.3.2-1, 2.3.2-2, and 2.3.2-3, respectively, based on 5 years of Augusta Airport records. Figures 2.3.2-4 through 2.3.2-27 show monthly, seasonal, and annual wind roses for each of 4 years of data for the 33-ft and 150-ft levels from the VEGP site.

The speed of the fastest mile of wind during a 28-year period is 62 mph. Strong winds are associated with thunderstorms, and lower maximum speeds (about 35 mph) are observed in the autumn. This is consistent with the observation that hurricane winds do not extend inland to this distance from the ocean.

The wind roses from the VEGP site agree quite well between levels and for each of the 4 years. Monthly and seasonal data, while comparing well, do show some of the normal year-to-year variation in climatic conditions, particularly during the transitional seasons of spring and fall. The wind roses from the Augusta Airport and the VEGP site also agree closely. There is a little more northerly wind and less westerly wind at Augusta as compared to the VEGP site. This can be attributed to differences in terrain and location of the wind instruments. Also, there are significantly more calms at Augusta, primarily due to the windspeed instrumentation at this station (a National Weather Service Station) having a threshold of about 3 mph, while the VEGP site windspeed instrumentation during the period these data were taken had a threshold speed of about 0.6 mph.

The persistence of windspeed and direction in sectors of 22.5° and 67.5° for periods of 1 to 48 h is given in tables 2.3.2-3 through 2.3.2-10 for each of 4 years from the 33-ft and 150-ft levels, respectively, on the VEGP meteorological tower. Table 2.3.2-11 shows wind persistence based on 5 years of Augusta Airport data.

2.3.2.1.2 Temperature

Monthly averages for the 30-year climatological period 1941 through 1970 (NOAA, 1981(b)) of the daily maximum, daily minimum, and daily mean are listed in table 2.3.2-1. Figure 2.3.2-28 gives the monthly averages and the average of the daily maximum and minimum dry bulb temperature for the 5-year period of Augusta data. Figure 2.3.2-29 provides the dry bulb temperature curves for 3 years of VEGP site data. The normal daily maximum ranges from 58°F in January to 91°F in July. The minimum ranges from 34°F in December and January to 70°F in July. An extreme maximum of 106°F was recorded in July 1952, and a minimum temperature of 3°F was recorded in February 1899.

Based on a 14-year record, the average number of days in a year on which temperatures of 90°F and above occur is 62, as shown in table 2.3.2-1, ranging to approximately two-thirds of the days in July. The temperature remains below freezing all day on the average of 1 day each January. About one-half of the days in December, January, and February have minimum temperatures below freezing.

Figures 2.3.2-30 and 2.3.2-31 show the monthly averages and average of the daily maximum and minimum wet bulb temperatures and dew point temperatures, respectively, from Augusta Airport. Figures 2.3.2-32 and 2.3.2-33 show like data from the VEGP site meteorology for 3 years. Figures 2.3.2-34 through 2.3.2-36 show the monthly averages and average of the daily maximum and minimum dry bulb temperatures, wet bulb temperatures, and dew point temperatures, respectively, from the VEGP site meteorology data for 1980-1981.

The wet bulb, dry bulb, and dew point temperatures from the VEGP site and Augusta Airport show close agreement. The temperatures at the VEGP site are generally a few degrees cooler than at the Augusta Airport. This is probably caused by the influence of the Augusta urban area and local terrain.

2.3.2.1.3 Water Vapor

Normal relative humidities measured four times during the day and based on 14 years of data are given in table 2.3.2-1 (NOAA, 1981(b)). They illustrate the relatively humid climate, with normal afternoon humidities around 45 percent in winter and 50 to 60 percent in summer. Autumn is the least humid period. Figure 2.3.2-37 shows the average monthly relative humidity and the average of daily maxima and minima during the month based on 5 years of Augusta data. Figures 2.3.2-38 and 2.3.2-39 show similar information on relative humidity data average over the 3 years of VEGP site data and a fourth recent year, respectively.

Absolute humidities have been computed based on dry bulb temperature and relative humidity data from 5 years of Augusta data. As shown in figure 2.3.2-40, the average ranges from a maximum of 1.9×10^{-5} g/cm^3 in July to a minimum of 4.6×10^{-6} g/cm^3 in December. Absolute humidity data based on 3 years of VEGP site data shown in figure 2.3.2-41 show a maximum of 1.85×10^{-5} g/cm^3 in September and a minimum of 4.6×10^{-6} g/cm^3 in January. The fourth year of VEGP site data is shown in figure 2.3.2-42, with a maximum of 1.9×10^{-5} g/cm^3 in August and a minimum of 2.7×10^{-6} g/cm^3 in January.

The absolute and relative humidity at Augusta Airport and the VEGP site agree fairly well. The apparent differences in absolute and relative humidity readings between the two sites have not been explained; however, they are probably due to annual variations because measurements are not from the same time period.

2.3.2.1.4 Precipitation

Maximum precipitation amounts recorded at Augusta during intervals of 1 to 24 h, for the period 1902 to 1950 and for the period 1940 to 1950, are listed in tables 2.3.2-12 and 2.3.2-13, respectively. Although only one period of 10 years and one of 50 years are available, averaging the monthly values allows useful rough estimates of the maximum amounts of precipitation to be expected for given time intervals, as shown in the following (Weather Bureau, 1954):

	Maximum Expected Precipitation (in.)	
Duration (h)	From 10-Year Record (in.)	From 50-Year Record (in.)
1	2	2 ½
2	3	3 ½
3	4	4
6	4	4 ½
12	4 1/2	7

	Maximum Expected Precipitation (in.)	
Duration (h)	From 10-Year Record (in.)	From 50-Year Record (in.)
24	4 1/2	10

The last two figures for 50 years are based on one case of sustained heavy rain.

The normal total monthly precipitation, the maximum and minimum observed in a month, and the maximum in 24 h, all for a 28-year period, are listed in table 2.3.2-1 (NOAA, 1981(b)). The maximum precipitation in 24 h recorded in the 28-year period, 6 in., is consistent with the expected maximum for 10- and 50-year periods discussed in paragraph 2.3.1.2.1. Figure 2.3.2-43 gives monthly average and 24-h maximum precipitation. Seasonal and annual precipitation wind roses based on 5 years of Augusta Airport data are shown in figures 2.3.2-44 and 2.3.2-45. Figures 2.3.2-46 through 2.3.2-53 show seasonal and annual precipitation wind roses for each of the 4 years of VEGP site data. During the 1980-81 site year, the precipitation data were considered questionable for extended periods of time, causing the precipitation wind roses to have data in only two of four seasons.

The precipitation wind roses from Augusta Airport and the VEGP site do not show much agreement between sites or from year to year. However, considering the large percentage of data from the northeast and southeast, it is evident that rain patterns are tracking across the Gulf states and/or up the Atlantic coast as generally expected. The differences result from the great variability of rain patterns and storm movements from year to year.

2.3.2.1.5 Fog

Heavy fog, with visibility less than 1/4 mile occurs on roughly 10 percent of days throughout the year, with minimum frequency of about 1 day a month in spring (NOAA, 1981(b)). How representative this figure is for the site depends especially on what proportion of the occurrences of heavy fog at Augusta are due to ground fog, which is largely restricted to valleys. Figure 2.3.2-54 summarizes the visibility data from 5 years of data at the Augusta Airport. The frequency of various visibility conditions taken from the Augusta data is as follows:

Visibility (miles)	Percent Occurrence
< ½	1.21
< 1	2.32
< 2	4.62
< 3	7.87
< 4	12.00

As shown, visibility of less than 1 mile occurs less than 3 percent of the time.

2.3.2.1.6 Atmospheric Stability

Atmospheric stability in the form of joint frequency tables of windspeed and direction from the 33-ft level on the VEGP meteorological tower 33- to 150-ft delta temperature are shown for 3 years of monthly and annual site data in tables 2.3.2-14 and 2.3.2-15. Tables 2.3.2-16 and 2.3.2-17 show monthly and annual joint frequency tables for 3 years of site data using 150-ft windspeed and direction and 33- to 150-ft delta temperature. Results obtained using the STAR computer routine to determine stability are shown in table 2.3.2-18 for data from Augusta, Georgia. Tables 2.3.2-19 and 2.3.2-20 show annual joint frequency tables for the fourth year of VEGP site data from the 33-ft and 150-ft levels.

2.3.2.1.7 Mixing Height

Seasonal and annual mixing heights for the VEGP site region are shown in table 2.3.2-21 (Holzworth, 1972). The maximum morning and afternoon mixing heights are 395 m during the winter and 1800 m during the summer, respectively. A minimum mixing height of 290 m occurs during the fall and a minimum afternoon height of 950 m during the winter.

2.3.2.2 Potential Influence of the Plant and Its Facilities on Local Meteorology

Natural draft cooling towers will be utilized at the VEGP. The elevated release point will result in minimal impact on local meteorology or the plant. The plume rise from the natural draft towers is expected to range from 500 to 1000 ft above the 500-ft towers. Studies of plume effects are described in subsection 5.1.4 of the Operating License Stage Environmental Report (OLSER) where it is determined that visible plumes will extend offsite; however, no increase in ground fog, reduced visibility, or icing due to the operation of these towers is expected. The effect of the plumes generated by the mechanical draft nuclear service cooling water towers is expected to be less than that of the natural draft cooling towers and is discussed further in the OLSER. The topographical features of the plant site are shown on figures 2.3.2-55, 2.3.2-56, and 2.3.2-57, sheets 1 through 4. In general, the terrain is flat with the wide Savannah River valley representing a depression running northwest to southeast approximately 3000 ft from the plant location.

2.3.2.3 Local Meteorological Conditions for Design and Operating Bases

With the exception of short and long term diffusion estimates, discussed in subsections 2.3.4 an 2.3.5, respectively, regional rather than local meteorological and air quality conditions were used for the design and operating bases of the VEGP.

2.3.3 ONSITE METEOROLOGICAL MEASUREMENTS PROGRAMS

The onsite meteorological measurement program commenced operation in April 1972. Instruments for measuring pertinent meteorological parameters were installed on a 45-m tower located in a cleared area SSW at site coordinates N 3260 and E 8040. The tower is not affected by large plant structures since it is more than 1000 ft from the nearest large structure. The base of the tower is at approximately plant grade. The monthly, seasonal, and annual

summaries in this report were taken from hourly values collected using this tower. These data were compiled in accordance with Regulatory Guide 1.23 and were used to determine the joint frequency distributions given in subsection 2.3.2. Four years of onsite data collected at the VEGP site have been prepared for inclusion in this report. These periods are as follows: December 4, 1972, through December 4, 1973; April 4, 1977, through April 4, 1979; and April 1, 1980, through March 31, 1981. A summary of the data recovery percentage from each of the 4 years of data is shown in table 2.3.3-1. A composite data recovery rate of 90 percent or greater was achieved in each of those last 3 years for the pertinent parameters of windspeed (WS) and direction (WD) from the 10-m level and delta temperature (DT) between the 45-m and 10-m level.

The onsite meteorological measurements program and equipment were updated in the first quarter of 1984 in order to meet the intent of NUREG-0654. The updated meteorological data collection center (MDCC) included a 60-m tower located at site coordinates N 3100 and E 7940 with permanent instrumentation at the 10-m and 60-m elevations and a 2-kVA UPS consisting of lead acid batteries, battery charger, and inverter to prevent the loss of meteorological data collection in the event offsite power is interrupted. Temporary instrumentation installed at the 30-m and 45-m levels provided a means to compare data between the 45-m and 60-m towers. This data comparison showed a positive correlation between corresponding parameters allowing the 60-m tower to be used as the primary source of site meteorological data. The 45-m tower was used as a backup for periods of equipment failure on the 60-m tower and consisted of windspeed, wind direction, and ambient temperature instruments at the 10-m elevation. Instrument descriptions are given in table 2.3.2-2. Instrument accuracies given are in conformance with Regulatory Guide 1.23 with the exception of dewpoint and differential temperature.

The onsite meteorological measurements program and equipment were updated in the second quarter of 2015 to replace obsolete equipment and to improve data recovery values. The update replaced the 60-m tower's existing 10-m and 60-m instruments with redundant instruments providing primary and secondary (backup) data. The use of the backup 45-m tower and its instruments was discontinued. Instrument descriptions are given in table 2.3.2-2.

The loop tolerances of the new MET tower windspeed and direction, ambient temperature, vertical temperature difference, relative humidity, and precipitation instrumentation were evaluated and compared to RG 1.23 Rev. 1 requirements to assess the acceptability of the tolerances. Based on review of actual data over a 3-year period, there were less than 2 h per year over this period when the required accuracy for the vertical temperature difference was not met. The remaining loop tolerances all meet or exceed the acceptance criteria of the time averaged accuracies required by RG 1.23 Rev. 1. It is concluded that since the vertical temperature difference inaccuracy over the specified time frame is < 0.02% per yearly period, the accuracy is considered to meet the intent of RG 1.23 Rev. 1.

Data collection for the MDCC consists of redundant data loggers and workstations located in the meteorological tower equipment building. Primary data are transmitted to the Unit 1 IPC and secondary data are transmitted to the Unit 2 IPC at least every 5 s via separate data communication links. The data communication links also transmit MDCC trouble alarms. Unit 1 primary data are transferred to the Unit 2 IPC and the Unit 2 secondary data are transferred to the Unit 1 IPC via Gateways. Each unit's IPC provides primary and secondary data to its respective control room, the TSC, and EOF. The data collected are summarized and edited to provide averages representative of each hour of data.

Major structures nearest to the 60-m tower will be the VEGP Units 3 and 4 reactors and the natural-draft cooling towers, which will be located, respectively, about 4,525 ft (mid-point between the two units) and about 3,025 ft (closest point on the Unit 3 cooling tower) to the north of the 60-m meteorological tower. Regulatory Guide 1.23 indicates that a meteorological tower located at greater than 10-times building-heights horizontal distance downwind will not have adverse building wake effects exerted by the structure. Since the height of the AP1000 units will be about 234 ft above grade, the zone of turbulent flow created by the reactor buildings will be limited to about 2,340 ft (or 10 building heights) downwind. Thus, the reactors will not be expected to adversely affect the measurements taken at the primary tower.

The natural-draft cooling towers for VEGP Units 3 and 4 will be built to a height of 600 ft above plant grade, the tallest structure at the VEGP site. The natural-draft cooling towers are hyperbolically shaped and have a maximum width at the base of the tower, which has an outer diameter of 550 ft. The downwind wake zone for hyperbolically shaped and sloping structures is expected to be smaller than for structures that are square or rectangular and have sharp edges. 40 CFR 51.100(ii)(3) defines good engineering practices (GEP) stack height as that which ensures that emissions from a stack do not result in excessive concentrations of any air pollutant as a result of atmospheric downwash, wakes, or eddy effects created by the source itself, nearby structures, or nearby terrain features. “Nearby structures” is defined in 40 CFR 51.100(ii)(1) as that distance up to five times the lesser of the height or width dimension of a structure. Furthermore, the wake zone area becomes increasingly smaller as the height to width ratio of a structure increases (reference 20). For the natural-draft cooling towers the lesser dimension is the width, which is the base width. Therefore, a conservative method to calculate the outermost boundary of influence exerted by the natural-draft cooling towers is to multiply the maximum width by five. Using this method, with a maximum width of 550 ft at the base of the tower, the downwind wake effect is estimated to extend 2750 ft from the base of the natural-draft cooling towers. The VEGP Unit 3 natural-draft cooling tower is located approximately 3,025 ft north of the meteorological tower. Thus, the meteorological tower is at a distance that will not be affected by the wake zone of the natural-draft cooling towers.

Regulatory Guide 1.23 indicates that ambient temperature and atmospheric moisture measurements should be made to avoid air modification by heat and moisture sources (e.g., ventilation sources, cooling towers, water bodies, large parking lots). An evaluation determined that the 110.000 ft^2 Vogtle Support Center asphalt parking lot, located approximately 800 ft (closest point) northwest of the 60-m tower, and the 1,435,000 ft^2 VEGP Unit 3 and 4 gravel laydown lot, located approximately 650 ft (closest point) east-northeast of the 60-m tower, could slightly increase 10-m ambient temperature sensor measurements in summer months during daytime hours. The system accuracy calculation for the 10-m ambient temperature sensors algebraically added (as a bias) the maximum potential temperature increase calculated for the two lots. Calculated total system accuracy for 10-m ambient temperature is less than the maximum Regulatory Guide 1.23 system accuracy value for ambient temperature (Calculation X5CP02001 Version 1.0).

An assessment was performed to determine the effects of the VEGP Units 3 and 4 natural-draft cooling towers on temperature and moisture measurements at the meteorological tower. A cooling tower plume prediction program was used in the assessment to identify the different categories of plumes which may be expected based on local meteorological data and then examines the potential impact for each of these categories. Only plumes greater than 1000 meters were considered as possibly impacting the meteorological tower instruments. The assessment concluded that, given the very low frequency of winds form the northwest, north-

northwest, and north, the cooling towers will have little to no impact on the measruements collected at the meteorological tower.

The instruments are monitored at least once a week by company personnel. Preventive maintenance is performed by SNC personnel in accordance with the instrument manuals and is intended to maintain 90-percent data recovery.

As discussed in subsection 2.3.2, comparisons of site data with that measured at the Augusta Airport show reasonable agreement. Therefore, it is expected that the site data are representative of longer term conditions.

2.3.4 SHORT TERM DIFFUSION ESTIMATES

Atmospheric diffusion conditions (expressed as values of X/Q) developed for use in evaluating accidents hypothesized in chapter 15 are shown in table 2.3.4-1 for various periods after the accident. This table includes X/Q estimates based on the methods described in Regulatory Guide 1.145. All estimates use vertical temperature difference to determine stability classification. Values in table 2.3.4-1 are based on evaluation of a 3-year period of onsite meteorological data and the most recent year of onsite meteorological data. A description of the site meteorological program is given in subsection 2.3.3.

Methods used to estimate diffusion conditions for evaluating short term accident releases are discussed in paragraph 2.3.4.1, and methods for assessing the consequences of longer term accident releases (up to 30 days) are discussed in paragraph 2.3.4.2.

2.3.4.1 Short Term Releases

The methodology for determining the atmospheric dispersion that exists for short term releases (an hour or so) involves two different calculations as described in Regulatory Guide 1.145. Both methods were used as discussed below.

2.3.4.1.1 Direction-Independent Calculations

The direction-independent approach involves computing X/Q values for each hour of the period of VEGP records used and then counting all of the hours that had X/Q values equal to or greater than a selected value regardless of direction. The number of hours so obtained was then divided by the number of hours in the total period of record to obtain the probability that the selected X/Q value would be equaled or exceeded. The resulting probabilities are independent of wind direction. A plot of cumulative centerline X/Q values as a function of probability of occurrence was constructed using the VEGP hourly data for all 3 years, combined as shown in figure 2.3.4-1. The fourth year of site data is plotted next to the 3 combined years in figure 2.3.4-1, showing a good comparison. Equations 1, 2, and 3 in paragraph 2.3.4.4 were used to compute values of X/Q. The actual distance to the site boundary was used for each 22.5° direction sector. Distances for each sector (given in table 2.3.4-2) represent the minimum distance to the site boundary in a 45° arc centered on the given direction.

2.3.4.1.2 Direction-Dependent Calculations

The direction-dependent calculations outlined in Regulatory Guide 1.145 require the X/Q values to be calculated using the equations given in paragraph 2.3.4.4; however, the results are treated separately for each direction. Figures 2.3.4-2 through 2.3.4-4 show probability distributions for each direction for runs made using each of the 3 years of meteorological data. A single 3-year composite direction-dependent probability distribution was plotted by combining the frequency of occurrence of selected X/Q values for the directions with the highest probabilities (envelopes) at the given X/Q level. The resulting plot is shown in figure 2.3.4-1. The fourth year of data is plotted next to the 3-year composite in figure 2.3.4-1.

2.3.4.1.3 Determining Appropriate Short Term Dispersion

The two cumulative probability distributions shown in figure 2.3.4-1 are then used to determine the appropriate X/Q for the short term accident following the procedure of Regulatory Guide 1.145. The 5 percent probable direction-independent X/Q value read from the distribution in figure 2.3.4-1 is 1.5E-4 s/m^3. For the direction-dependent case, the 0.5 percent probable X/Q is determined to be 1.8×10^{-4} s/m^3 in accordance with Regulatory Guide 1.145. The highest of the two ($X/Q = 1.8 \times 10^{-4}$ s/m^3) is to be used in accident dose calculations and is shown in table 2.3.4-1 for the 1-h case.

A similar procedure was used to determine the short term X/Q value at the low population zone (radius of 2 miles), which is also shown in table 2.3.4-1.

2.3.4.2 Long Term Releases

For releases that occur over a longer period, it is appropriate to incorporate wind direction changes in the model used to estimate concentration at any given point. Using the same 3-year period of data from VEGP, the probability that any particular average diffusion condition (or poorer one) would exist during a selected interval of time (greater than 1 h) was determined. These values are used for 30-day dose calculations at the low population zone (LPZ) as shown in table 2.3.4-1. The values for the fourth year are shown in table 2.3.4-1 and agree closely with the 3-year cumulative values.

The procedure for determining longer term X/Q values is also taken from Regulatory Guide 1.145. The approach involves selection of the highest 0.5 percent direction-dependent and 5 percent direction-independent short term (1 h) X/Q values at the LPZ along with the highest annual average values. These values are plotted on log-log graph paper with X/Q on the ordinate and time on the abscissa. The short term values are plotted at 2 h, and the annual averages (determined as described in subsection 2.3.5) are plotted at 8760 h. The X/Q values for different averaging times are then read from the appropriate locations on the curves. The log-log plots for each direction for each of the 3 years of data are shown in figures 2.3.4-5 through 2.3.4-7. The curve shown in figure 2.3.4-8 is the composite of the highest curves for each of the 4 separate years that were used to determine X/Q values for longer averaging times at the LPZ. No adjustments were made to the annual average X/Q values, since releases are made at ground level and the terrain is relatively uniform.

The diffusion models and assumptions used for determining these values are described in paragraphs 2.3.4.3 and 2.3.4.4.

2.3.4.3 Determination of Stability Classifications

Table 2.3.4-3 gives the temperature difference categories used to classify the VEGP data into Pasquill groups for use in computing σ_y and σ_z in the diffusion equations.

2.3.4.4 Analytical Methods for Dispersion Computations

Plume centerline values of X/Q were estimated using the following relationships described in Regulatory Guide 1.145.

$$\frac{X}{Q} \quad \frac{1}{\bar{u}\left(\pi\,\sigma_y\sigma_z + \frac{A}{2}\right)} \tag{1}$$

$$\frac{X}{Q} \quad \frac{1}{\bar{u}\,(3\pi\,\sigma_y\sigma_z)} \tag{2}$$

$$\frac{X}{Q} \quad \frac{1}{\bar{u}\,\pi\,\Sigma_y\sigma_z} \tag{3}$$

where :

$X/Q =$ relative concentration (s/m^3).

$\pi \quad = 3.14159.$

$\bar{u} \quad =$ windspeed at 33 ft (m/s).

$\sigma_y \quad =$ lateral plume spread coefficient (m).

σ_z = vertical plume spread coefficient (m).

A = smallest vertical plume cross-section area of containment (2368 m^2).

x = distance.

Values of Σ_y were computed as follows:

$$\Sigma_y = M\sigma_y \qquad (X \leq 800\text{ m}) \tag{4}$$

$$y = (M-1)\sigma_y + \sigma_y \qquad (X > 800\text{ m}) \tag{5}$$

Values of M as a function of speed and stability are determined from figure 3 of Regulatory Guide 1.145.

The following procedure is used to determine the appropriate X/Q for each hour of meteorological data. First, the maximum value obtained using equations 1 and 2 is determined. Then the minimum of that value and equation 3 is determined, if the windspeed is less than 6 m/s and the stability is not unstable. This value is then used for all statistical evaluations involving X/Q.

2.3.5 LONG TERM DIFFUSION ESTIMATES

2.3.5.1 Objective

This subsection includes realistic estimates of atmospheric dilution and deposition out to a distance of 50 miles from the plant, based on annual average meteorological conditions. These estimates are used for estimating offsite doses resulting from routine releases of gaseous effluent from the VEGP.

2.3.5.2 Calculations

2.3.5.2.1 General

Calculations are made using 3 years of meteorological data collected from the site meteorological tower. The tower configuration and these data are discussed and summarized in subsection 2.3.3. The 3-year period of data provides a good representation of long term conditions at the site. Comparisons of VEGP site data with meteorological data from the Savannah River Laboratory nearby and with Augusta Airport data show that similar dispersion conditions exist at all sites.

Topography is gently rolling in the site area and should have little effect on wind trajectory. During periods of light winds, local terrain may affect wind trajectory. The most pronounced terrain feature is the river depression. This, however, is a relatively small, wide depression which has little influence. Since the release is at ground level and there are no significant topographic features, no correction factors for recirculation are used.

Atmospheric dispersion models described in this subsection follow those described in Regulatory Guide 1.111. The paragraphs below describe the models used in these evaluations with frequent references to Regulatory Guide 1.111, since most assumptions are identical to those in the guide. These models are used to determine routine (average) X/Q and D/Q values applicable to the site.

2.3.5.2.2 Atmospheric Diffusion Model

$$\overline{(X/Q')}_D = 2.032 \sum_{ij} n_{ij} [N X \bar{u}_i \Sigma_{zj}(x)]^{-1} \exp[-h^2 e/2\sigma^2 zj^{(x)}] \quad (1)$$

Average atmospheric dispersion evaluations are made using the straight line airflow model from Regulatory Guide 1.111, as shown below:

where:

h_e = effective release height.

n_{ij} = length of time (hours of valid data) weather conditions are observed to be at a given wind direction, windspeed class, i, and atmospheric stability class, j.

N = total hours of valid data.

$\bar{u}_i$ = geometrical mean of all speeds in the windspeed class, i, at a height representative of release; calms are one-half the threshold anemometer speed or less; extrapolation to higher levels, if necessary, is done by raising the ratio of the two heights to the n power, where n = 0.25, 0.33, and 0.5 for unstable, neutral, and stable conditions, respectively.

$\sigma_{zj}(x)$ = vertical plume spread without volumetric correction at distance, x, for stability class, j (see figure 1 of Regulatory Guide 1.111) based on vertical temperature difference (ΔT) and Regulatory Guide 1.23 categorization of Pasquill groups by ΔT.

$\Sigma_{zj}(x)$ = vertical plume spread with a volumetric correction for a release within the building wake cavity, at a distance, x, for stability class, j; otherwise $\Sigma_{zj}(x) = \sigma_{zj}(x)$.

$\overline{(X/Q')}_D$ = average effluent concentration, X, normalized by source strength, Q', at distance, x, in a given downwind direction, D.

2.032 = $(2/\pi)^{1/2}$ divided by the width in radians of a 22.5° sector.

In some cases, hourly data are used and the summation over i and j in the above equation is deleted, and the summation is accomplished for all hours at all distances for each direction.

2.3.5.2.3 Source Configuration Considerations

Since the plant vent release point is elevated and there are no buildings which would obstruct the plume in its normal trajectory, equation 1 is used with the height of release defined as follows (from equation 4 of Regulatory Guide 1.111):

$$h_e = h_s + h_{pr} \div h_t - c$$

where:

c = correction for low relative exit velocity (equation 5 of Regulatory Guide 1.111).

h_e = effective release height.

h_{pr} = rise of the plume above the release point based on Briggs. (See below.)

h_s = physical height of the release point. (The elevation of the stack base should be assumed to be zero.)

h_t = maximum terrain height between the release point and the point for which the calculation is made.

Values of h_{pr} are computed as follows for a "jet," since nuclear plant vents have an insignificant amount of buoyancy due to heated discharges:

$$h_{pr} = 1.44D \left(\frac{W_o}{\bar{u}}\right)^{2/3} \left(\frac{x}{D}\right)^{1/3}$$

up to the point where h_{pr} is the minimum of the following two equations:

$$h_{pr_{max}} = 3\left(\frac{W_o}{\bar{u}}\right)D \quad \text{or} \quad h_{pr_{max}} = 1.5\left(\frac{F_m}{\bar{u}}\right)^{1/3} s^{-1/6}$$

where symbols are as before, and:

D	=	stack or vent effective inside diameter (m).
W_O	=	stack or vent exit velocity (m/s).
u	=	windspeed at discharge level (m/s).
F	=	momentum flux (m^4/s^2).
s	=	stability parameter (s^{-2}).

If the plume trajectory from a release point; e.g., turbine building vent, does not remain outside of building wake influences near large structures, all or portions of the plume are considered to be entrapped and brought to ground level in the turbulent wake of the building. The criteria for determining the portion of the plume treated as an elevated or ground release follows from equations 6, 7, and 8 of Regulatory Guide 1.111 and are repeated here for completeness:

If $W_o/u > 5.0$	h_e as calculated above
If $W_o/u \leq 1.0$	$h_e = 0$
If $1 < W_o/u \leq 1.5$	$E_t = 2.58 - 1.58 \left(\frac{W_o}{\bar{u}}\right)$
If $1.5 < W_o/u \leq 5.0$	$E_t = 0.3 - 0.06 \left(\frac{W_o}{\bar{u}}\right)$

The appropriate diffusion estimate is then computed by assuming an elevated release 100 $(1 - E_t)$ percent of the time and by assuming ground release 100 E_t percent of the time. Calculations utilizing this mixed model are referred to as "wake-split" calculations in this report.

A building wake correction is computed for all ground releases near structures in accordance with the following general equation from Regulatory Guide 1.111:

$$\Sigma = \sqrt{\sigma_z{}^2 + \frac{cH^2}{\pi}} \leq 1.73\,\sigma_z$$

where:

Σ = effective dispersion coefficient for use in equation 1 (m).

c = building wake coefficient (c = 0.5).

H = height of tallest structure in the nuclear plant power block (m).

As radioactive effluent in a plume travels downwind, it is subject to several removal mechanisms, including radioactive decay, dry deposition, and wet deposition (during rain). Corrections for radioactive decay are not made in the dispersion estimates reported in this subsection.

Dry deposition which results in depletion of halogen and particulate isotopes from the plume is calculated using figures 2 through 5 in Regulatory Guide 1.111. Depletion factors in these curves are a function of height and distance. Therefore, for sites where elevated releases occur, the terrain must be subtracted from the plume height before entering the curves at the appropriate distance. Each elevated or ground level X/Q is multiplied by the depletion correction factor to estimate the depleted X/Q value.

To determine relative deposition rate as a function of distance and stability, the curves given in figures 6 through 9 of Regulatory Guide 1.111 are used in a computerized table look-up routine. Terrain heights are subtracted before the table look-up is made. Values from the curves are divided by the sector cross-width (arc) at the point of calculation to give units of m^{-2}.

Table 2.3.5-1 lists computer runs made using the diffusion models described above. Since the grazing season is assumed to exist all year, separate runs for the grazing season are not necessary.

A summary of plant vent information for each discharge point is given in tables 2.3.5-2 and 2.3.5-3. Only vents used during routine operation are considered in this evaluation. Inspection of these tables shows that a separate calculation is required to determine diffusion conditions applicable for each vent.

Table 2.3.5-4 summarizes key assumptions utilized in making the model calculations. Table 2.3.5-5 gives terrain elevations for all distances out to 10 miles. Terrain height is conservatively not allowed to decrease with increasing distance nor to decrease below plant grade in accordance with Regulatory Guide 1.111.

A series of four runs is made using different meteorological data bases and vent locations. Resulting X/Q, depleted X/Q, and D/Q values are listed in tables 2.3.5-6 through 2.3.5-9 for each direction sector for 10 distances. Comparisons of the results, using the most recent hourly data with the 3-year joint frequency data, show little difference. Therefore, the 3-year results are used for determining the appropriate dispersion factor for each receptor location in tables 2.3.5-10 and 2.3.5-11. Each table represents model results for one vent location. Table 2.3.5-10 shows the results of calculations made for the plant (wake-split model), and table 2.3.5-11 provides results from the second set of calculations made for all other release points that discharge into the building wake cavity, resulting in a ground level release. Tables 2.3.5-12 and 2.3.5-13 show X/Q, depleted X/Q, and D/Q values for 10 distances out to 50 miles for the fourth year of site meteorological data. These 1-year (1980-1981) tables agree very well with tables 2.3.5-6 through 2.3.5-9 for the 3 composite years. Therefore, the 3-year results shown in tables

2.3.5-10 and 2.3.5-11 for all receptor locations are confirmed by the fourth year of data and should be used for all routine release dose assessments. Isopleths of X/Q, depleted X/Q, and D/Q values based on plant vent releases for the 3-year period of record are shown in figures 2.3.5-1 through 2.3.5-3, respectively, to illustrate annual average dispersion patterns near the site. Figures 2.3.5-4 through 2.3.5-6 show similar information for the fourth recent year.

TABLE 2.3.1-1

RECURRENCE INTERVALS OF RAINFALL FALLING IN PERIODS OF 30 MIN TO 10 DAYS (in.)

Years	30 Min	1 Hour	2 Hours	3 Hours	6 Hours	12 Hours	24 Hours	48 Hours	4 Days	7 Days	10 Days
1	1.3	1.6	1.9	2.1	2.3	2.8	3.2	-	-	-	-
2	1.5	2.0	2.2	2.4	2.8	3.4	3.9	4.4	4.8	5.9	6.3
5	1.7	2.3	2.8	3.0	3.7	4.4	5.0	5.8	6.9	7.7	8.0
10	2.1	2.6	3.2	3.5	4.4	5.0	5.8	6.8	7.8	8.7	9.5
25	2.4	3.0	3.7	4.0	5.0	6.0	6.8	7.9	8.9	10.0	11.6
50	2.6	3.3	4.0	4.4	5.8	6.5	7.5	8.6	9.8	11.5	12.0
100	2.9	4.0	4.5	4.9	6.0	7.0	8.2	9.4	11.5	12.5	13.5

TABLE 2.3.1-2

MEAN NUMBER OF DAYS WITH PRECIPITATION
≥0.01, ≥0.1, ≥0.5 IN.
IN AUGUSTA, GEORGIA

Month	≥0.01 No. of Days	≥0.1 No. of Days	≥0.5 No. of Days
January	10	7	3
February	9	7	3
March	11	8	3
April	8	6	2
May	9	6	3
June	9	6	2
July	12	8	3
August	10	5	2
September	8	7	3
October	6	5	1
November	7	4	1
December	9	5	2
Yearly total	108	74	28

TABLE 2.3.1-3 (SHEET 1 OF 2)

TABLE OF FUJITA-PEARSON TORNADO SCALE[a]

F-Scale Maximum Windspeed				P-Scale Path Length			P-Scale Path Width			
Scale	mph	kt	m/s	Scale	mi	km	Scale	ft	yd	m
0.0	40	35	18	0.0	0.3	0.5	0.0	17	6	5
0.1	43	37	19	0.1	0.4	0.6	0.1	19	6	6
0.2	46	40	21	0.2	0.4	0.6	0.2	21	7	6
0.3	49	43	22	0.3	0.5	0.7	0.3	24	8	7
0.4	52	46	23	0.4	0.5	0.8	0.4	26	9	8
0.5	56	48	25	0.5	0.6	0.9	0.5	30	10	9
0.6	59	51	26	0.6	0.6	1.0	0.6	33	11	10
0.7	63	54	28	0.7	0.7	1.1	0.7	37	13	11
0.8	66	57	30	0.8	0.8	1.3	0.8	42	14	13
0.9	70	60	31	0.9	0.9	1.4	0.9	47	16	14
1.0	73	64	33	1.0	1.0	1.6	1.0	53	18	16
1.1	77	67	34	1.1	1.1	1.8	1.1	59	20	18
1.2	81	70	36	1.2	1.3	2.0	1.2	66	22	20
1.3	84	73	38	1.3	1.4	2.3	1.3	74	25	23
1.4	88	77	40	1.4	1.6	2.6	1.4	84	28	26
1.5	92	80	41	1.5	1.8	2.9	1.5	94	31	29
1.6	96	84	43	1.6	2.0	3.2	1.6	105	35	32
1.7	100	87	45	1.7	2.2	3.6	1.7	118	39	36
1.8	104	91	47	1.8	2.5	4.0	1.8	133	44	40
1.9	109	94	49	1.9	2.8	4.5	1.9	149	50	45
2.0	113	98	50	2.0	3.2	5.1	2.0	167	56	51
2.1	117	102	52	2.1	3.5	5.7	2.1	187	62	57
2.2	121	105	54	2.2	4.0	6.4	2.2	210	70	64
2.3	126	109	56	2.3	4.5	7.2	2.3	235	78	72
2.4	130	113	58	2.4	5.0	8.1	2.4	265	88	81
2.5	135	117	60	2.5	5.6	9.0	2.5	297	99	90
2.6	139	121	62	2.6	6.3	10.2	2.6	333	111	102
2.7	144	125	64	2.7	7.1	11.4	2.7	374	125	114
2.8	148	129	66	2.8	7.9	12.8	2.8	419	140	128
2.9	153	132	68	2.9	8.9	14.3	2.9	470	157	143
3.0	158	137	70	3.0	10.0	16.1	3.0	528	176	161
3.1	162	141	73	3.1	11.2	18.0	3.1	591	197	180
3.2	167	145	75	3.2	12.6	20.3	3.2	665	222	203
3.3	172	149	77	3.3	14.1	22.7	3.3	744	248	227
3.4	177	154	79	3.4	15.9	25.6	3.4	837	279	256
3.5	182	158	81	3.5	17.8	28.6	3.5	940	313	286
3.6	187	162	83	3.6	20.0	32.2	3.6	1054	351	322
3.7	192	167	86	3.7	22.4	36.0	3.7	1183	394	360
3.8	197	171	88	3.8	25.1	40.4	3.8	1326	442	404
3.9	202	175	90	3.9	28.2	45.4	3.9	1489	496	454

TABLE 2.3.1-3 (SHEET 2 OF 2)

F-Scale Maximum Windspeed				P-Scale Path Length			P-Scale Path Width			
Scale	mph	kt	m/s	Scale	mi	km	Scale	ft	yd	m
4.0	207	180	93	4.0	31.6	50.9	4.0	1670	557	509
4.1	212	184	95	4.1	35.5	57.1	4.1	1874	625	571
4.2	218	189	97	4.2	39.8	64.1	4.2	2102	701	641
4.3	223	194	100	4.3	44.7	71.8	4.3	2354	785	718
4.4	228	198	102	4.4	50.1	80.6	4.4	2646	882	806
4.5	233	203	104	4.5	56.2	90.4	4.5	2967	989	904
4.6	238	207	107	4.6	63.1	102.0	4.6	3332	1111	1.0 km
4.7	244	212	109	4.7	70.8	114.0	4.7	3738	1246	1.1
4.8	250	217	112	4.8	79.4	128.0	4.8	4194	1398	1.3
4.9	255	222	114	4.9	89.1	143.0	4.9	4704	1568	1.4
5.0	261	227	117	5.0	100.0	161.0	5.0	1.0 mi	1760	1.6
5.1	267	232	119	5.1	112.0	181.0	5.1	1.1	1971	1.8
5.2	272	236	122	5.2	126.0	203.0	5.2	1.3	2218	2.0
5.3	278	241	124	5.3	141.0	227.0	5.3	1.4	2482	2.3
5.4	284	246	127	5.4	159.0	255.0	5.4	1.6	2798	2.6
5.5	289	251	129	5.5	178.0	286.0	5.5	1.8	3133	2.9
5.6	295	256	132	5.6	200.0	321.0	5.6	2.0	3520	3.2
5.7	301	261	135	5.7	224.0	360.0	5.7	2.2	3912	3.6
5.8	307	267	137	5.8	251.0	404.0	5.8	2.5	4418	4.0
5.9	313	272	140	5.9	282.0	454.0	5.9	2.8	4963	4.5

a. Characteristics of a tornado can be expressed as a combination of Fujita-scale windspeed and Pearson-scale path length and width. This scale permits us to classify tornadoes between two extreme FPP scales, 0.0.0 and 5.5.5.

TABLE 2.3.1-4

TORNADO WIND GROUPS

Windspeed Distribution

Windspeed Classification	No. of Tornadoes	Percent of Total
F5 (<260 mph)	2	0.12
F4 (207 to 260 mph)	34	2.1
F3 (158 to 206 mph)	115	7.2
F2 (113 to 157 mph)	430	26.6
F1 (73 to 112 mph)	710	44.0
F0 (40 to 72 mph)	321	19.9

Cumulative Windspeed Distribution

Windspeed Classification	No. of Tornadoes	Percent of Total
F5 and above (<260 mph)	2	0.12
F4 and above (<206 mph)	36	2.2
F3 and above (<157 mph)	151	9.3
F2 and above (<112 mph)	581	36.0
F1 and above (<74 mph)	1291	80.0
F0 and above (<40 mph)	1612	100.0

TABLE 2.3.1-5

SEASONAL AND ANNUAL FREQUENCY OF LIGHTNING STRIKES
TO THE REACTOR CONTAINMENT BUILDING FOR BOLTS
OF VARIOUS CURRENT MAGNITUDE

			20 kA		40 kA		60 kA		90 kA		135 kA	
Season	Average No. of Thunder-Storm Days (Augusta)[a]	Average No. of Cloud-to-Ground Strikes/km^2	Attractive Area (km^2)	Strikes	Attractive Area (km^2)	Strikes	Attractive Area (km^2)	Strikes	Attractive Area (km^2)	Strikes	Attractive Area (km^2)	Strikes
Winter (Dec-Feb)	4	1	0.056	0.028	0.186	0.041	0.394	0.039	0.679	0.014	1.039	0.005
Spring (Mar-May)	20	4	0.056	0.112	0.186	0.164	0.394	0.158	0.679	0.054	1.039	0.021
Summer (June-Aug)	45	8	0.056	0.224	0.186	0.327	0.394	0.315	0.679	0.109	1.039	0.042
Fall (Sep-Nov)	7	1	0.056	0.028	0.186	0.041	0.394	0.039	0.679	0.014	1.039	0.005
Annual	76	13	0.056	0.364	0.186	0.532	0.394	0.512	0.679	0.177	1.039	0.068

a. From NUREG/CR-2252.

TABLE 2.3.2-1 (SHEET 1 OF 6)

NORMALS, MEANS, AND EXTREMES FOR AUGUSTA, GEORGIA

	Temperatures(°F)							Normal Degree Days Base 65°F –	
	Normal			Extremes					
Month	Daily Max.	Daily Min.	Monthly	Record Highest	Year	Record Lowest	Year	Heating	Cooling
(a)				31		31			
Jan	57.6	34.0	45.8	80	1975	5	1970	601	6
Feb	60.5	36.1	48.3	86	1962	9	1973	475	8
Mar	67.1	42.0	54.6	88	1974	12	1980	346	23
Apr	90.7	76.9	63.8	93	1980	30	1972	90	54
May	89.1	74.2	71.7	99	1964	35	1971	10	218
Jun	87.0	66.7	78.2	105	1952	47	1972	0	396
Jul	90.9	69.9	80.4	107	1980	59	1951	0	477
Aug	90.2	69.8	79.0	104	1968	54	1968	0	453
Sep	89.2	63.2	74.2	101	1957	36	1967	0	279
Oct	77.0	51.2	64.1	97	1954	22	1952	104	76
Nov	67.1	40.2	53.7	90	1961	15	1970	344	5
Dec	68.7	34.1	46.4	82	1967	5	1981	577	0
Year	75.4	51.4	63.4	107	Jul 1980	5	Dec 1981	2547	1995

TABLE 2.3.2-1 (SHEET 2 OF 6)

Precipitation (in.)

	Water Equivalent								Snow, Ice Pellets		
Month	Normal	Max. Monthly	Year	Min. Monthly	Year	Max. (24 h)	Year	Max. Monthly	Year	Max. (24 h)	Year
(a)	--	31	--	31	--	31	--	31	--	31	--
Jan	3.44	8.48	1960	0.75	1981	3.61	1960	1.5	1970	1.5	1970
Feb	3.75	7.67	1961	0.69	1968	3.59	1962	14.0	1973	13.7	1973
Mar	4.67	11.92	1980	0.88	1968	5.31	1967	1.1	1980	1.1	1980
Apr	3.37	8.43	1961	0.60	1970	3.96	1955	0.0	--	0.0	--
May	3.39	9.61	1972	0.48	1951	4.44	1981	0.0	--	0.0	--
Jun	3.66	7.28	1973	1.56	1979	5.08	1981	0.0	--	0.0	--
Jul	5.09	11.43	1967	1.46	1953	3.71	1979	0.0	--	0.0	--
Aug	4.21	9.91	1964	0.65	1980	5.98	1964	0.0	--	0.0	--
Sep	3.26	9.51	1975	0.61	1967	4.93	1969	0.0	--	0.0	--
Oct	2.17	6.90	1959	T	1953	2.67	1964	0.0	--	0.0	--
Nov	2.21	6.18	1957	0.09	1960	2.63	1955	T	1968	T	1968
Dec	3.42	8.65	1981	0.32	1955	3.12	1970	0.9	1958	0.9	1958
Year	42.63	11.92	Mar 1980	T	Oct 1953	5.98	Aug 1964	--	Feb 1973	--	Feb 1973

TABLE 2.3.2-1 (SHEET 3 OF 6)

Relative Humidity percent)					Wind					
							Fastest Mile			
Month	Hour 01	Hour 07	Hour 13	Hour 19	Mean Speed (mph)	Prevailing Direction	Speed (mph)	Direction	Year	Percentage of Possible Sunshine
(Local Time)										
(a)	17	17	17	17	31	13	31	31	--	--
Jan	79	83	54	67	7.1	W	36	23	1978	--
Feb	75	81	48	59	7.7	WNW	40	29	1955	--
Mar	76	83	48	56	8.0	WNW	52	23	1972	--
Apr	79	84	45	55	7.6	SE	39	32	1962	--
May	80	87	50	62	6.5	SE	48	28	1967	--
Jun	87	86	52	63	6.2	SE	62	08	1965	--
Jul	88	88	56	68	5.9	SE	48	33	1970	--
Aug	90	91	57	73	5.5	SE	45	18	1953	--
Sep	91	92	56	77	5.6	NE	35	32	1959	--
Oct	88	90	49	76	5.8	NW	40	18	1977	--
Nov	84	88	49	71	6.2	NW	40	27	1954	--
Dec	81	85	53	70	6.7	NW	34	25	1954	--
Year	84	87	51	66	6.6	SE	62	08	Jun 1965	--

TABLE 2.3.2-1 (SHEET 4 OF 6)

Mean Number of Days

Month	Mean Sky Cover (tenths) Sunrise to Sunset	Sunrise to Sunset			Precipitation (0.1 in. or more)	Snow, Ice Pellets (1.0 in.or more)	Thunderstorms	Heavy Fog Visibility (1/4 mile or less)
		Clear	Partly Cloudy	Cloudy				
(a)	31	31	31	31	31	31	31	31
Jan	6.3	9	6	16	10	*	1	3
Feb	5.8	9	6	13	9	*	2	2
Mar	6.1	9	8	14	11	*	3	2
Apr	5.4	10	9	11	8	0	4	1
May	5.9	5	11	12	9	0	7	1
Jun	5.9	7	12	11	9	0	10	1
Jul	6.3	5	14	12	11	0	13	2
Aug	5.8	7	14	10	10	0	10	3
Sep	5.9	9	9	12	8	0	4	3
Oct	4.6	14	7	10	6	0	1	3
Nov	5.1	12	6	12	7	0	1	3
Dec	5.9	10	6	15	9	*	1	3
Year	5.8	109	108	148	107	*	54	27

TABLE 2.3.2-1 (SHEET 5 OF 6)

	Temperature (°F)				
	Maximum		Minimum		
Month	90° and Above[b]	32° and Below	32° and Below	0° and Below	Average Station Pressure (mb) (el 148 ft msl)
(a)	17	17	17	17	9
Jan	0	*	17	0	1014.6
Feb	0	*	14	0	1014.2
Mar	0	*	6	0	1011.5
Apr	1	0	1	0	1012.0
May	5	0	0	0	1009.5
Jun	13	0	0	0	1010.9
Jul	20	0	0	0	1011.3
Aug	17	0	0	0	1012.5
Sept	8	0	0	0	1011.4
Oct	*	0	1	0	1013.5
Nov	0	0	9	0	1014.6

TABLE 2.3.2-1 (SHEET 6 OF 6)

	Temperature (°F)				
	Maximum		Minimum		
Month	90° and Above[b]	32° and Below	32° and Below	0° and Below	Average Station Pressure (mb) (el 148 ft msl)
Dec	0	0	15	0	1014.6
Year	64	1	60	0	1012.6

a. Length of record, years, through the current year unless otherwise noted, based on January data.

b. 70° and above at Alaskan stations.

* = Less than one half.

T = Trace.

NORMALS - Based on record for the 1941 - 1970 period.

DATE OF AN EXTREME - The most recent in cases of multiple occurrence.

PREVAILING WIND DIRECTION - Record through 1963.

WIND DIRECTION - Numerals indicate tens of degrees clockwise from true north. 00 indicates calm.

FASTEST MILE WIND - Speed is fastest observed 1-min value when the direction is in tens of degrees.

Means and extremes above are from existing and comparable exposures. Annual extremes have been exceeded at other sites in the locality as follows: lowest temperature, 3, in February 1899; maximum monthly precipitation, 14.00, in July 1906; maximum precipitation in 24 h, 7.92, in October 1929.

TABLE 2.3.2-2

VEGP ONSITE WEATHER INSTRUMENTS

Sensed Parameter	Range	System Accuracy	Starting Threshold	Distance Constant	Damping Ratio	Elevation (m)
			60-m Tower Instruments[a]			
Windspeed	0-112 mph (0-50 mps)	≤ ±5%	1.0 mph (0.45 mps)	-	-	10 60
Wind direction	0°-360°	≤ ±5%	1.0 mph (0.45 mps)			10 60
Ambient temperature	-58°F to +122°F (-50°C to +50°C)	≤ ±0.9°F (±0.5°C)	-	-	-	10
Differential temperature	-10°F to ±20°F (-23.3°C to -6.6°C)	≤ ±0.18°F (±0.1°C)	-	-	-	10-60
Relative humidity	0-100%	≤ ±4%	- -	- -	- -	
Dewpoint	40°F to 120°F (40°C to 49°C)	≤ ±2.7°F (±1.5°C)	-	-	-	10
Precipitation	0-100 in. reset daily	±10% of the total accumulated catch	Resolution of 0.01 in. (0.25 mm)	-	-	Tower base
Sigma theta	0°-100°	-	-	-	Sec wind direction	10 60

a. Redundant primary and secondary instruments are installed for each sensed parameter.

TABLE 2.3.2-3 (SHEET 1 OF 4)

1972 TO 1973 WIND PERSISTENCE
VEGP METEOROLOGICAL TOWER - 33-FT LEVEL
22.5° SECTOR WIDTH

Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	84	107	148	147	150	135	118	118	116	126	159	154	157	141	108	80
2	38	39	74	79	75	66	54	42	64	63	86	87	82	77	54	23
4	16	17	31	36	34	25	28	21	28	17	29	35	48	41	23	0
8	2	0	13	8	8	2	3	6	5	2	3	9	11	13	6	0
12	1	0	6	3	0	0	2	1	2	0	1	3	3	5	3	0
18	0	0	2	0	0	0	0	0	0	0	0	0	1	1	1	0
24	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	9	17	25	21	19	20	25	28	31	30	34	38	54	52	43	10
2	4	4	13	10	9	10	17	12	20	13	15	18	30	39	25	3
4	0	1	7	2	1	2	7	5	8	5	4	9	19	23	12	0
8	0	0	3	0	0	1	1	1	0	0	0	0	4	7	3	0
12	0	0	1	0	0	0	0	0	0	0	0	0	2	3	1	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	1	1	1	0	4	5	5	9	6	11	13	23	14	1
2	0	0	0	1	0	0	2	4	3	3	2	3	10	9	7	0
4	0	0	0	0	0	0	0	1	1	2	0	0	5	5	4	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-3 (SHEET 2 OF 4)

22.5° SECTOR WIDTH

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	0	1	0	0	2	4	0	0
2	0	0	0	0	0	0	0	0	0	1	0	0	2	2	0	0
4	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	163	186	194	198	207	211	202	192	197	215	225	225	209	168	177	174
2	88	114	128	137	140	138	127	123	123	139	151	163	151	114	110	104
4	41	64	83	95	99	93	72	75	73	82	93	116	104	84	61	49
8	13	29	39	49	51	41	32	35	32	39	51	53	57	46	31	19
12	7	15	22	31	20	13	16	18	14	19	22	20	27	29	22	10
18	2	7	14	16	9	5	4	4	6	5	4	11	12	15	11	5
24	1	3	7	9	5	3	2	1	1	2	2	4	8	11	4	2
30	1	3	3	4	0	2	1	1	0	1	1	1	7	8	4	1
36	1	0	1	2	0	2	0	0	0	1	0	1	2	5	1	0
48	1	0	0	0	0	1	0	0	0	0	0	0	2	1	0	0

TABLE 2.3.2-3 (SHEET 3 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	30	33	40	48	45	48	48	51	54	60	73	88	80	67	61	57
2	11	17	20	26	25	29	27	36	33	39	36	50	60	50	46	30
4	3	8	8	9	6	12	12	20	22	22	20	30	41	39	29	13
8	0	3	3	4	1	2	7	7	7	5	6	11	19	18	14	3
12	0	2	1	1	1	1	4	3	3	1	2	3	5	11	9	1
18	0	1	1	1	1	1	0	0	0	1	0	0	3	6	8	0
24	0	0	1	1	0	0	0	0	0	0	0	0	2	4	0	0
30	0	0	0	1	0	0	0	0	0	0	0	0	1	1	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	1	1	2	2	1	5	6	10	12	12	22	21	31	30	29	15
2	0	1	2	1	1	2	5	8	7	5	7	14	17	20	16	7
4	0	0	0	1	1	0	1	2	3	3	4	7	10	13	8	4
8	0	0	0	0	0	0	0	1	2	1	1	1	3	4	2	0
12	0	0	0	0	0	0	0	0	1	1	0	0	1	1	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	1	1	1	2	2	2	4	0
2	0	0	0	0	0	0	0	0	1	1	1	2	1	1	2	0
4	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0
8	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-3 (SHEET 4 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 DEGREES)
Speed 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-4 (SHEET 1 OF 4)

1972 TO 1973 WIND PERSISTENCE
VEGP METEOROLOGICAL TOWER - 150-FT LEVEL
22.5° SECTOR WIDTH

Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	121	120	136	176	202	180	173	179	188	171	198	188	191	191	132	81
2	58	56	71	91	104	84	90	84	80	90	113	100	114	103	65	32
4	18	18	35	40	57	38	43	37	32	30	51	41	58	49	24	14
8	2	4	13	9	13	9	7	8	2	7	6	8	19	17	8	1
12	1	0	5	3	3	2	1	2	1	1	0	0	4	3	5	0
18	0	0	2	1	0	0	0	0	0	0	0	0	1	2	2	0
24	0	0	1	1	0	0	0	0	0	0	0	0	1	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	40	41	57	68	88	75	71	77	72	81	102	101	127	123	73	37
2	19	23	34	39	56	39	32	39	37	48	59	60	79	73	42	16
4	5	8	19	18	26	17	18	19	12	17	30	23	40	34	18	4
8	2	2	10	3	6	4	2	2	1	3	2	5	12	14	7	0
12	1	0	4	2	0	1	0	0	0	1	0	0	3	3	4	0
18	0	0	1	0	0	0	0	0	0	0	0	0	1	2	2	0
24	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	8	13	12	8	7	11	12	16	15	14	22	24	50	57	32	2
2	5	3	7	5	1	2	5	9	5	7	10	10	33	36	20	0
4	1	0	5	1	0	1	2	4	2	1	4	5	17	17	7	0
8	0	0	2	0	0	0	0	0	0	1	0	0	4	8	2	0
12	0	0	1	0	0	0	0	0	0	1	0	0	2	2	1	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-4 (SHEET 2 OF 4)

22.5° SECTOR WIDTH

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 22.5 (DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	1	2	1	1	1	1	1	3	3	5	9	18	31	17	0
2	0	0	2	1	0	0	0	1	2	1	2	2	12	16	11	0
4	0	0	0	0	0	0	0	1	1	1	0	0	6	7	4	0
8	0	0	0	0	0	0	0	0	0	1	0	0	0	2	0	0
12	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 to 73120416 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	1	0	2	0	0	4	8	2	0
2	0	0	0	0	0	0	0	0	0	1	0	0	2	3	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	2	2	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 (DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	169	182	212	215	215	243	242	234	249	249	231	232	234	189	213	212
2	102	129	141	149	155	166	168	170	168	184	158	177	171	128	133	137
4	58	79	92	98	108	115	104	115	109	107	107	128	119	92	86	63
8	20	31	44	52	59	61	55	49	48	47	59	66	78	54	41	20
12	11	17	25	33	35	29	31	22	25	27	36	39	40	36	25	14
18	3	10	10	18	20	17	11	6	6	12	18	21	21	27	12	6
24	3	6	7	15	9	8	4	5	2	5	8	12	14	15	5	2
30	2	2	5	7	6	5	3	2	0	3	4	5	5	9	3	1
36	1	2	3	4	3	4	2	0	0	3	2	4	3	8	1	0
48	1	1	1	3	2	2	0	0	0	0	0	0	2	4	0	0

TABLE 2.3.2-4 (SHEET 3 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	72	76	97	124	121	142	143	135	137	141	144	176	161	122	134	111
2	41	51	62	85	84	91	84	87	92	101	101	129	115	86	84	69
4	22	28	40	51	67	56	47	50	58	61	61	80	74	60	46	28
8	8	16	20	25	30	28	15	16	21	19	28	39	45	35	29	15
12	4	10	13	14	13	8	9	6	7	8	13	17	26	28	17	8
18	1	7	6	8	3	3	3	2	1	4	8	10	14	18	8	2
24	1	3	2	2	2	2	2	1	1	2	3	6	10	10	5	0
30	1	1	1	2	1	1	0	1	0	2	2	1	4	9	3	0
36	1	1	1	2	1	0	0	0	0	1	2	1	2	7	1	0
48	0	0	0	1	1	0	0	0	0	0	0	0	2	3	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	20	24	20	18	23	27	28	29	31	36	46	69	62	58	58	40
2	5	12	11	8	7	7	12	15	20	19	21	43	46	43	37	25
4	3	6	5	6	4	2	6	7	10	11	11	27	30	30	22	8
8	0	3	3	2	0	1	2	1	3	2	3	9	16	17	14	2
12	0	2	1	1	0	0	0	1	1	1	1	2	10	13	7	1
18	0	1	1	1	0	0	0	0	0	1	1	0	7	8	3	0
24	0	0	1	1	0	0	0	0	0	0	0	0	2	4	0	0
30	0	0	0	1	0	0	0	0	0	0	0	0	1	2	0	0
38	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 (DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	1	3	3	2	2	2	1	4	6	8	13	25	36	35	34	17
2	0	2	2	2	1	1	1	3	4	4	3	15	21	22	18	11
4	0	0	1	1	1	0	1	2	3	2	3	8	12	16	12	4
8	0	0	0	0	0	0	0	0	1	1	1	1	7	9	5	0
12	0	0	0	0	0	0	0	0	1	1	1	0	2	2	1	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-4 (SHEET 4 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 72120417 TO 73120416 (SECTOR WIDTH = 67.5 (DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	1	1	3	2	2	4	9	10	9	2
2	0	0	0	0	0	0	0	0	1	1	1	2	3	3	3	1
4	0	0	0	0	0	0	0	0	0	0	0	2	3	2	1	0
8	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-5 (SHEET 1 OF 4)

1977 TO 1978 WIND PERSISTENCE
VEGP METEOROLOGICAL TOWER - 33-FT LEVEL
22.5° SECTOR WIDTH

Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	107	136	203	164	143	137	129	124	160	191	213	224	242	196	190	127
2	40	57	84	82	64	67	59	52	75	89	109	109	134	101	85	45
4	13	15	41	33	24	21	16	18	36	32	45	44	55	57	34	6
8	1	7	8	8	5	1	1	3	7	3	7	3	10	12	9	0
12	1	0	4	2	1	0	0	0	0	0	1	1	1	7	4	0
18	0	0	1	1	1	0	0	0	0	0	0	0	0	1	2	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 77040313 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	13	26	38	29	26	12	14	19	22	26	45	46	54	67	50	26
2	4	14	16	18	7	4	2	7	12	13	21	25	30	40	25	7
4	0	5	6	8	2	0	1	1	4	5	4	11	15	20	11	0
8	0	2	1	1	0	0	0	0	0	0	0	0	3	4	3	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	4	1	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 77040313 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	1	3	5	4	2	1	0	2	3	3	5	11	6	8	4	4
2	0	1	1	3	0	0	0	0	1	1	1	2	4	3	3	0
4	0	0	0	1	0	0	0	0	0	1	0	1	2	2	1	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-5 (SHEET 2 OF 4)

22.5° SECTOR WIDTH

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	1	0	1	3	3	1	0	0
2	0	0	0	0	0	0	0	0	0	0	1	2	1	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	236	250	217	246	235	208	220	230	264	298	313	316	305	274	263	262
2	129	148	147	164	149	129	141	155	170	201	198	227	220	198	165	145
4	58	79	86	100	93	73	77	91	95	119	124	146	149	125	99	76
8	23	34	39	46	42	36	29	32	43	53	60	67	69	57	42	31
12	6	18	26	24	25	20	12	14	16	24	24	33	37	34	26	14
18	1	11	12	13	5	2	3	6	3	9	12	12	13	20	15	3
24	0	3	8	7	3	2	0	0	0	4	5	4	7	11	7	2
30	0	2	4	3	1	2	0	0	0	1	3	3	4	8	1	1
36	0	0	4	2	1	2	0	0	0	1	1	0	3	4	0	0
48	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-5 (SHEET 3 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	54	48	50	53	44	38	37	47	54	65	77	97	97	89	97	72
2	27	25	28	32	28	16	12	23	28	37	43	59	70	58	53	30
4	8	14	19	15	13	6	3	8	11	16	26	35	49	45	33	15
8	2	5	5	7	3	1	0	1	3	4	8	12	19	20	12	4
12	0	3	5	3	0	0	0	0	0	1	0	3	5	8	7	2
18	0	1	2	1	0	0	0	0	0	0	0	0	3	5	2	0
24	0	0	1	0	0	0	0	0	0	0	0	0	1	2	0	0
30	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	8	8	9	8	6	3	3	6	7	7	15	16	18	12	11	7
2	1	4	4	4	3	1	1	1	2	3	3	6	8	9	5	2
4	0	0	3	3	1	0	0	0	1	2	2	3	5	6	3	1
8	0	0	0	1	0	0	0	0	0	1	2	1	2	2	2	1
12	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	1	1	1	2	4	5	3	1	0
2	0	0	0	0	0	0	0	0	0	1	1	2	3	2	0	0
4	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0
8	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-5 (SHEET 4 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	0	1	1	1	2	0	0	0
2	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-6 (SHEET 1 OF 4)

1977 TO 1978 WIND PERSISTENCE
VEGP METEOROLOGICAL TOWER - 150-FT LEVEL
22.5° SECTOR WIDTH

Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	152	170	218	223	207	213	174	184	219	259	280	271	309	268	188	134
2	71	75	114	103	99	102	83	77	107	142	153	125	170	145	95	65
4	19	30	57	49	53	40	42	35	40	62	71	52	79	80	37	21
8	6	5	16	10	11	2	8	5	10	8	15	12	20	24	7	2
12	0	1	6	4	2	1	2	3	3	3	4	3	6	11	1	0
18	0	0	3	0	0	1	1	0	0	0	0	1	1	5	0	0
24	0	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	62	83	107	100	101	98	79	96	132	162	183	159	193	187	115	63
2	27	39	59	51	52	51	41	46	65	95	106	84	117	112	56	32
4	5	18	29	29	24	14	16	19	27	45	45	37	55	66	26	11
8	2	5	11	4	6	1	4	5	3	5	7	6	15	21	6	0
12	0	1	5	2	1	1	1	1	0	2	4	2	5	11	1	0
18	0	0	3	0	0	0	1	0	0	0	0	0	1	3	0	0
24	0	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	12	16	28	23	14	19	18	16	25	37	49	44	69	92	46	19
2	5	11	12	9	8	6	5	4	9	16	29	21	43	46	20	8
4	1	4	6	4	1	1	1	3	4	3	11	6	21	24	12	0
8	1	0	2	1	1	0	0	1	0	0	0	0	6	8	1	0
12	0	0	2	0	0	0	0	0	0	0	0	0	2	5	0	0
18	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-6 (SHEET 2 OF 4)

22.5° SECTOR WIDTH

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	1	4	4	0	1	1	1	3	6	4	10	8	20	24	22	5
2	0	3	1	0	1	0	0	1	3	0	4	3	10	18	5	0
4	0	0	0	0	0	0	0	0	2	0	1	1	3	10	1	0
8	0	0	0	0	0	0	0	0	0	0	0	0	1	4	1	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	1	1	0	0	0	0	0	2	0	5	3	4	9	5	2
2	0	0	0	0	0	0	0	0	0	0	1	1	3	6	1	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	253	278	289	265	284	273	286	287	305	306	323	355	305	284	300	270
2	142	175	189	192	188	180	189	187	216	232	231	249	233	205	188	170
4	81	101	122	131	126	121	113	121	135	158	168	169	162	147	128	93
8	39	46	66	73	57	62	53	50	70	83	92	93	95	75	57	40
12	19	25	31	40	31	39	27	30	40	44	52	54	66	49	34	12
18	6	12	17	19	17	15	10	16	12	23	23	27	30	28	19	5
24	1	5	10	10	12	3	3	6	3	15	9	13	18	19	8	1
30	1	2	7	5	9	2	2	3	2	6	8	9	10	16	5	1
36	0	1	3	4	6	2	1	1	0	5	6	8	6	11	4	1
48	0	1	2	4	1	0	1	0	0	1	2	2	3	4	1	0

TABLE 2.3.2-6 (SHEET 3 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	122	132	152	178	176	166	179	187	221	240	245	276	221	180	204	167
2	68	86	97	106	120	113	107	117	156	172	177	192	167	127	132	91
4	45	54	67	88	68	61	58	68	95	118	116	125	120	93	86	54
8	15	21	26	38	21	22	24	27	36	52	61	55	68	56	40	18
12	7	11	13	17	12	10	10	13	18	23	29	35	49	35	27	6
18	3	8	8	9	5	4	2	7	5	10	14	16	21	21	13	1
24	1	3	5	3	3	2	1	0	1	6	6	8	12	17	6	0
30	1	1	5	2	1	2	1	0	1	3	3	6	7	14	5	0
36	0	1	2	2	1	1	1	0	0	1	0	5	4	10	3	0
48	0	1	2	2	0	0	0	0	0	0	0	1	1	2	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	35	40	45	44	46	46	45	46	65	87	86	110	110	107	111	59
2	19	24	21	20	18	17	17	19	28	41	50	71	61	64	63	33
4	8	12	13	11	8	3	6	10	9	20	27	46	44	43	33	19
8	2	5	6	5	2	1	1	1	3	5	9	14	21	21	14	4
12	1	3	3	4	0	0	0	0	0	1	1	7	15	16	9	0
18	0	1	2	2	0	0	0	0	0	0	0	1	7	7	2	0
24	0	0	0	0	0	0	0	0	0	0	0	1	4	6	1	0
30	0	0	0	0	0	0	0	0	0	0	0	1	2	3	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	9	8	8	5	1	2	5	9	11	15	18	32	33	39	37	27
2	3	4	4	3	1	1	2	4	4	7	6	14	21	20	19	7
4	0	1	0	0	0	0	0	2	2	4	3	7	14	12	9	1
8	0	0	0	0	0	0	0	0	0	1	1	1	8	7	6	1
12	0	0	0	0	0	0	0	0	0	0	0	0	2	3	3	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-6 (SHEET 4 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 77040314 TO 78040313 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	3	2	2	1	0	0	0	2	2	6	6	10	13	10	11	7
2	1	1	0	0	0	0	0	0	0	1	1	5	9	7	5	1
4	0	0	0	0	0	0	0	0	0	1	1	1	3	4	3	0
8	0	0	0	0	0	0	0	0	0	0	0	0	1	2	2	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-7 (SHEET 1 OF 4)

1978 TO 1979 WIND PERSISTENCE
VEGP METEOROLOGICAL TOWER - 33-FT LEVEL
22.5º SECTOR WIDTH

Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	140	139	161	158	145	122	125	131	167	157	160	136	148	138	105	100
2	57	69	67	68	78	54	57	59	64	79	74	74	80	66	45	45
4	19	18	37	33	39	19	22	23	30	37	27	35	30	27	22	15
8	2	3	9	8	7	2	2	3	5	5	7	7	8	6	9	0
12	0	0	3	3	1	1	0	1	0	1	0	1	2	2	5	0
18	0	0	1	3	0	0	0	0	0	0	0	0	0	1	3	0
24	0	0	0	1	0	0	0	0	0	0	0	0	0	1	2	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	21	18	31	28	24	10	14	14	22	27	39	34	39	40	32	14
2	7	7	17	10	9	1	5	6	9	15	21	21	18	18	18	5
4	3	3	7	4	4	0	2	3	5	7	10	9	7	8	11	0
8	0	1	0	2	1	0	0	0	0	1	3	2	4	2	6	0
12	0	0	0	2	0	0	0	0	0	0	0	0	0	1	2	0
18	0	0	0	1	0	0	0	0	0	0	0	0	0	1	1	0
24	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	3	3	2	0	0	0	1	1	3	6	9	7	8	9	8	0
2	0	2	0	0	0	0	1	0	2	3	5	4	3	3	6	0
4	0	0	0	0	0	0	0	0	1	2	3	3	2	1	4	0
8	0	0	0	0	0	0	0	0	0	1	0	0	1	1	1	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-7 (SHEET 2 OF 4)

22.5° SECTOR WIDTH

PERSISTENCIES FROM 78040313 to 79040312 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	0	1	0	2	1	2	0	0
2	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 to 79040312 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 to 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	214	238	235	230	210	217	215	237	252	248	228	231	205	201	202	208
2	143	138	129	151	135	129	143	143	167	159	155	161	148	127	114	121
4	70	73	72	92	79	72	76	80	94	89	102	97	94	73	60	66
8	27	32	38	49	40	34	28	35	37	42	46	48	45	38	26	27
12	8	13	15	24	20	17	13	12	16	24	27	22	26	23	13	11
18	2	9	12	13	12	7	3	3	4	9	8	6	9	12	8	5
24	1	6	9	7	5	3	2	2	2	4	2	1	1	7	4	3
30	0	4	7	2	3	2	0	1	1	2	0	1	1	4	4	2
36	0	3	4	1	2	1	0	0	0	0	0	0	1	3	3	1
48	0	0	2	1	1	0	0	0	0	0	0	0	0	1	2	0

TABLE 2.3.2-7 (SHEET 3 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 78040313 to 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	38	55	51	49	39	39	29	38	46	55	62	69	70	65	55	50
2	21	27	33	35	24	18	10	16	23	31	40	43	45	34	28	30
4	9	15	18	20	12	9	2	9	14	18	25	32	30	26	20	16
8	4	3	4	4	4	2	2	3	3	11	12	13	10	15	11	8
12	0	2	3	3	3	0	1	1	3	4	2	1	4	8	7	2
18	0	0	1	2	2	0	1	0	0	1	1	0	1	4	3	1
24	0	0	1	1	1	0	0	0	0	0	0	0	1	1	1	0
30	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0
36	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 to 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	6	7	4	2	0	1	1	4	9	16	18	16	15	14	13	11
2	1	1	3	0	0	1	1	3	4	9	9	9	7	8	7	6
4	0	0	0	0	0	0	0	1	3	6	7	6	5	7	5	4
8	0	0	0	0	0	0	0	0	1	1	3	4	3	3	2	1
12	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 to 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	1	1	3	3	4	2	2	0
2	0	0	0	0	0	0	0	0	0	0	1	2	3	2	1	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-7 (SHEET 4 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 78040313 to 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-8 (SHEET 1 OF 4)

1978 TO 1979 WIND PERSISTENCE
VEGP METEOROLOGICAL TOWER - 150-FT LEVEL
22.5° SECTOR WIDTH

Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	189	167	179	212	217	207	219	195	238	231	239	229	230	208	193	181
2	86	79	84	94	108	96	91	105	116	120	128	108	108	97	99	90
4	29	21	32	46	48	44	32	54	51	50	52	50	51	40	52	39
8	5	1	4	6	14	6	4	11	9	11	14	6	12	6	18	8
12	1	0	3	2	5	0	2	4	2	3	3	1	1	2	7	4
18	0	0	0	0	2	0	1	0	0	0	1	0	0	0	3	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	77	65	82	101	108	104	78	94	120	130	128	121	111	103	104	96
2	39	32	39	52	58	44	33	46	60	65	73	65	63	52	62	47
4	12	6	15	26	22	14	9	22	21	29	34	30	31	21	34	24
8	3	1	3	3	7	2	2	3	3	6	7	4	10	4	15	6
12	1	0	3	2	4	0	1	1	1	1	3	1	1	2	5	3
18	0	0	0	0	2	0	1	0	0	0	1	0	0	0	2	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	29	9	8	26	20	11	7	9	21	30	38	42	39	30	44	31
2	13	5	4	11	10	2	3	4	10	11	20	21	19	17	27	22
4	2	0	1	3	4	1	1	2	4	6	8	10	11	4	13	11
8	0	0	0	1	2	0	0	0	1	1	3	3	5	1	6	3
12	0	0	0	1	1	0	0	0	0	0	0	0	0	0	4	0
18	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-8 (SHEET 2 OF 4)

22.5° SECTOR WIDTH

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	7	1	3	2	2	1	1	0	4	6	10	10	9	11	19	8
2	3	0	1	1	1	0	1	0	2	2	4	7	4	5	10	3
4	0	0	0	0	1	0	0	0	1	1	2	1	2	1	7	0
8	0	0	0	0	1	0	0	0	0	1	0	0	1	1	1	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	1	0	0	0	0	1	5	6	4	1	5	2
2	0	0	0	0	0	0	0	0	0	0	0	2	1	1	0	1
4	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	276	285	260	251	276	297	291	313	314	311	314	305	292	286	268	259
2	179	188	175	168	176	204	194	205	225	202	207	205	203	202	177	176
4	93	104	103	104	113	125	133	141	144	138	140	139	135	138	110	106
8	38	42	44	65	68	63	59	64	75	72	70	75	58	73	57	51
12	26	24	27	33	38	34	37	34	41	47	45	41	37	35	26	33
18	11	4	11	14	19	15	14	10	10	18	15	16	14	15	18	19
24	3	2	4	7	6	7	4	8	6	9	7	6	6	5	9	9
30	3	1	3	6	4	3	0	2	3	6	4	4	2	4	7	3
36	3	0	1	5	2	2	0	1	1	3	2	2	1	3	5	3
48	1	0	1	3	1	0	0	0	0	0	0	0	0	1	3	3

TABLE 2.3.2-8 (SHEET 3 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	136	141	145	145	169	187	190	200	214	202	199	175	169	163	147	144
2	79	84	87	90	107	113	100	119	133	128	128	129	114	112	99	96
4	32	43	46	47	64	51	53	70	81	75	85	84	69	75	63	56
8	19	18	20	21	23	20	16	19	27	37	44	40	35	47	38	31
12	13	10	10	12	11	12	7	9	11	24	24	25	24	23	18	20
18	5	2	5	6	7	5	3	2	4	8	6	11	11	9	13	12
24	3	1	3	6	3	0	1	1	3	4	2	4	3	4	5	4
30	3	1	1	6	3	0	0	0	1	3	1	3	1	3	3	3
36	3	0	1	5	1	0	0	0	0	0	1	0	1	2	3	3
48	1	0	1	2	1	0	0	0	0	0	0	0	0	0	1	2

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	46	40	34	29	34	33	25	34	45	59	72	69	62	72	59	63
2	29	19	17	16	15	14	8	18	23	35	33	40	44	43	43	41
4	15	6	7	8	7	8	4	6	13	19	25	29	24	26	29	25
8	7	1	2	3	3	2	1	2	4	9	13	15	12	16	16	16
12	3	0	1	2	3	1	0	0	2	4	5	3	6	7	9	9
18	1	0	0	2	2	1	0	0	0	0	0	1	2	2	3	3
24	0	0	0	2	2	0	0	0	0	0	0	0	0	2	2	1
30	0	0	0	1	1	0	0	0	0	0	0	0	0	1	1	0
36	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	13	11	5	5	3	3	2	5	9	17	17	18	19	27	27	28
2	7	5	3	3	1	2	0	2	4	9	11	12	16	16	14	15
4	1	0	0	1	1	1	0	1	2	5	5	8	8	14	12	9
8	0	0	0	1	1	1	0	0	1	2	3	2	3	4	4	2
12	0	0	0	1	1	0	0	0	0	0	0	0	0	1	1	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-8 (SHEET 4 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 78040313 TO 79040312 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	2	0	0	1	1	1	0	0	1	6	8	9	8	9	7	6
2	0	0	0	0	0	0	0	0	0	0	4	6	4	2	2	1
4	0	0	0	0	0	0	0	0	0	0	2	3	2	2	2	0
8	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-9 (SHEET 1 OF 4)

1980 TO 1981 WIND PERSISTENCE
VEGP METEOROLOGICAL TOWER - 33-FT LEVEL
22.5° SECTOR WIDTH

START AND END OF PERIOD A0040101 81033124 START AND END OF DATA TO BE ANALYZED
VOGTLE WIND PERSE 33 FT 80-81 2009-A
PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	137	160	180	162	149	146	158	140	178	183	192	240	259	245	176	139
2	71	79	70	79	55	63	70	55	81	75	95	123	153	128	90	58
4	36	34	40	40	22	28	30	24	29	35	39	61	70	48	40	18
8	10	12	14	11	6	6	8	3	6	4	4	24	24	12	14	1
12	1	3	4	2	3	2	1	0	3	1	2	5	6	2	2	0
18	0	1	1	0	0	0	1	0	0	0	0	1	2	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	42	49	58	67	45	42	34	26	29	31	22	50	87	74	62	42
2	24	29	27	30	28	18	16	6	14	14	12	19	55	43	39	18
4	14	14	16	18	8	6	5	2	7	4	2	12	22	15	17	7
8	3	7	6	6	1	0	1	0	2	2	0	2	8	9	6	1
12	1	2	2	1	0	0	0	0	2	0	0	0	2	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	7	10	11	8	5	3	2	2	8	7	2	3	13	7	13	7
2	2	5	5	5	0	1	0	0	3	2	1	0	5	3	7	2
4	0	2	1	1	0	0	0	0	2	1	0	0	2	2	2	0
8	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-9 (SHEET 2 OF 4)

22.5° SECTOR WIDTH

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	1	1	0	1	0	0	0	0	1	1	0	0	2	2	1	2
2	0	1	0	0	0	0	0	0	1	0	0	0	0	0	1	0
4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	240	248	238	234	238	220	221	260	243	288	313	246	289	286	298	260
2	149	152	147	147	153	126	148	155	151	186	192	205	201	199	193	148
4	88	96	94	92	95	80	88	96	98	112	119	140	146	137	117	100
8	41	51	46	44	47	43	39	42	38	44	59	75	89	81	56	47
12	21	29	26	23	23	24	23	22	19	22	29	51	53	51	28	23
18	9	10	19	13	9	9	9	5	5	12	14	25	25	20	11	10
24	5	5	11	9	2	4	1	1	4	4	10	13	13	14	4	4
30	2	2	7	7	0	2	1	0	1	1	5	7	7	4	2	1
36	1	1	5	3	0	1	0	0	1	0	2	5	5	4	0	0
48	0	1	2	0	0	1	0	0	0	0	2	1	3	1	0	0

TABLE 2.3.2-9 (SHEET 3 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	87	95	101	97	104	78	76	67	56	54	78	110	123	119	106	97
2	46	55	61	64	63	43	35	38	31	35	41	76	78	80	68	61
4	29	38	37	31	34	24	17	16	18	18	18	39	53	53	41	36
8	11	17	21	11	10	6	2	5	6	6	6	12	19	28	19	13
12	6	8	13	8	6	1	1	2	2	2	2	5		12	6	5
18	3	3	6	5	1	1	1	0	1	1	0	0		2	1	2
24	2	2	2	4	0	0	0	0	0	0	0	0		1	1	1
30	1	1	2	1	0	0	0	0	0	0	0	0		1	0	0
36	0	1	0	1	0	0	0	0	0	0	0	0		0	0	0
48	0	1	0	0	0	0	0	0	0	0	0	0		0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	22	23	20	18	16	9	5	10	14	16	11	15	20	23	18	23
2	7	8	12	9	6	2	2	3	5	7	4	6	8	12	12	12
4	2	4	6	3	1	0	0	2	3	3	1	2	4	8	4	2
8	1	2	3	1	0	0	0	2	2	1	0	1	2	2	1	1
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	4	2	2	1	1	0	0	1	2	2	1	2	4	3	2	3
2	1	1	1	0	0	0	0	1	1	1	0	0	0	1	1	1
4	0	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-9 (SHEET 4 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-10 (SHEET 1 OF 4)

1980 TO 1981 WIND PERSISTENCE
VEGP METEOROLOGICAL TOWER - 150-FT LEVEL
22.5° SECTOR WIDTH

START AND END OF PERIOD 80040101 81033124 START AND END OF DATA TO BE ANALYZED
VOGTLE WIND PERSE 150 FT 80-81 2009-A
PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	168	152	155	153	167	196	194	199	216	199	196	223	263	258	282	224
2	84	72	65	77	93	90	104	91	94	106	107	107	132	148	180	124
4	39	26	31	25	50	40	45	38	36	43	50	47	62	74	96	57
8	6	6	10	6	12	13	8	5	1	9	3	10	11	22	31	11
12	0	0	4	1	5	3	3	1	0	2	0	3	4	6	11	5
18	0	0	1	0	1	0	1	0	0	0	0	0	1	0	3	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	97	79	69	80	85	91	89	91	92	92	102	106	147	166	196	146
2	56	39	38	47	58	53	51	41	35	53	57	65	79	102	132	85
4	27	17	18	16	26	26	26	14	13	23	22	29	35	56	61	44
8	4	3	6	4	9	6	5	2	0	3	1	6	9	18	20	8
12	0	0	3	1	4	2	2	0	0	0	0	2	4	3	8	4
18	0	0	1	0	1	0	0	0	0	0	0	0	1	0	2	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	25	19	13	19	26	22	14	6	9	19	14	18	25	44	69	51
2	9	6	7	8	10	8	3	3	4	7	7	8	10	24	43	29
4	3	3	4	5	6	5	0	1	2	3	3	2	1	5	18	14
8	0	0	2	1	2	1	0	0	0	2	1	0	1	1	6	5
12	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-10 (SHEET 2 OF 4)

22.5° SECTOR WIDTH

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	2	2	3	4	1	4	2	0	1	4	4	4	3	6	19	14
2	1	0	2	0	1	0	0	0	1	3	1	1	0	3	9	7
4	1	0	2	0	1	0	0	0	1	2	0	0	0	2	3	3
8	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	2	0	1	0	0	0	1	0	1	2	1	1	0	2	2	4
2	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	2
4	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	278	242	242	220	235	256	264	259	286	282	275	324	320	306	286	288
2	174	146	153	145	148	172	171	182	196	198	196	220	212	214	210	204
4	112	94	90	89	91	115	117	114	118	127	137	144	141	158	145	140
8	62	47	39	43	49	63	68	52	62	60	66	69	79	103	98	79
12	27	19	21	28	27	37	39	32	36	30	32	34	51	65	64	54
18	12	10	7	16	14	16	13	13	10	9	14	15	25	35	34	29
24	5	5	3	10	8	8	6	5	4	4	5	6	9	23	22	22
30	2	3	2	7	6	4	4	3	1	2	3	5	8	13	15	8
36	1	2	1	5	6	3	2	3	0	1	1	4	5	8	9	4
48	0	0	1	0	2	2	0	2	0	0	0	0	3	6	6	2

TABLE 2.3.2-10 (SHEET 3 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	177	142	132	120	130	140	150	159	178	169	179	214	262	226	219	204
2	113	95	86	82	84	103	96	94	117	116	129	140	142	159	163	150
4	74	59	50	54	50	59	59	52	69	72	80	86	97	118	105	94
8	36	23	20	25	26	36	29	20	16	18	25	36	54	68	65	50
12	15	7	10	18	14	20	18	8	4	5	11	15	31	40	46	34
18	7	6	4	8	7	8	5	2	0	2	4	6	16	25	28	20
24	5	2	1	5	6	4	2	1	0	1	1	3	5	16	17	14
30	2	2	1	4	5	2	1	1	0	0	0	2	4	8	7	5
36	0	1	1	2	5	2	0	1	0	0	0	2	3	4	4	3
48	0	0	1	0	1	0	0	0	0	0	0	0	1	1	2	1

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	74	47	41	40	45	47	37	26	31	33	36	49	69	46	99	96
2	40	21	17	20	21	19	13	9	12	13	16	22	36	56	63	63
4	19	11	11	15	18	10	6	4	5	9	10	10	11	28	35	31
8	10	3	4	6	6	4	1	0	3	4	4	3	4	16	22	18
12	3	1	2	3	2	2	0	0	0	2	2	0	1	4	7	6
18	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	16	6	7	7	8	5	5	3	4	7	10	10	11	19	26	28
2	9	3	2	3	1	2	1	1	3	3	4	2	6	10	13	14
4	3	3	2	3	1	1	0	1	2	2	2	0	2	4	9	8
8	1	0	0	0	0	0	0	0	2	2	1	0	0	2	3	1
12	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	1
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-10 (SHEET 4 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 80040101 TO 81033124 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	5	3	1	1	0	1	1	2	2	3	4	2	3	4	6	5
2	1	1	0	0	0	0	1	1	1	1	1	0	0	0	3	2
4	1	0	0	0	0	0	0	1	1	1	1	0	0	0	1	1
8	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-11 (SHEET 1 OF 4)

1959 TO 1963 WIND PERSISTENCE
AUGUSTA, GEORGIA
22.5° SECTOR WIDTH

Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	1151	1073	949	1067	1167	1168	1175	1112	1051	1068	1193	1274	1184	1275	1351	1380
2	659	629	686	669	742	722	726	674	602	654	772	832	802	796	840	795
4	319	352	367	405	446	417	392	355	290	379	479	545	537	496	507	412
8	108	151	192	195	172	143	151	126	98	131	190	261	290	255	246	138
12	42	66	114	109	67	47	60	47	38	53	84	111	159	145	118	48
18	15	29	57	47	26	14	18	16	10	9	24	47	92	82	51	11
24	6	13	30	28	12	8	2	5	3	3	5	17	44	47	21	3
30	3	10	21	17	5	3	1	2	0	1	5	7	31	30	9	0
36	2	8	12	8	4	2	1	0	0	1	3	1	23	14	2	0
48	1	4	5	1	0	0	0	0	0	0	1	0	2	5	0	0

PERSISTENCIES FROM 59010101 TO 63123423 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	388	415	439	501	540	540	525	422	369	371	621	774	755	703	690	515
2	207	237	276	301	307	277	280	226	202	246	369	483	506	483	449	271
4	87	119	170	178	169	138	124	98	82	124	204	304	352	304	278	124
8	27	41	83	77	61	32	21	24	19	26	54	112	185	166	120	24
12	7	19	31	31	20	9	5	7	7	8	16	35	92	89	52	7
18	3	9	13	9	4	1	0	0	0	2	1	8	42	42	23	2
24	2	4	5	4	1	0	0	0	0	0	0	3	20	26	7	0
30	1	3	4	2	1	0	0	0	0	0	0	2	14	16	4	0
36	1	3	2	0	0	0	0	0	0	0	0	0	8	8	2	0
48	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0

PERSISTENCIES FROM 59010101 TO 63123423 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	70	80	103	116	100	95	81	69	67	87	159	277	320	301	270	121
2	22	28	47	52	45	31	24	24	23	38	74	134	209	195	163	53
4	6	11	26	25	15	9	4	8	10	15	39	73	146	137	102	13
8	2	2	4	4	3	0	0	2	3	2	4	20	68	66	38	4
12	1	2	2	2	1	0	0	0	1	2	1	4	23	23	9	0
18	0	0	0	1	1	0	0	0	0	0	0	0	5	6	3	0
24	0	0	0	0	0	0	0	0	0	0	0	0	2	3	2	0
30	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-11 (SHEET 2 OF 4)

22.5° SECTOR WIDTH

PERSISTENCIES FROM 59010101 TO 63123423 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	6	8	6	10	5	7	7	11	12	15	32	79	117	128	100	29
2	4	1	0	1	2	4	4	4	1	2	11	41	69	76	54	9
4	1	0	0	0	0	1	1	1	1	1	5	23	49	46	28	2
8	1	0	0	0	0	0	0	0	0	0	0	7	22	16	10	1
12	0	0	0	0	0	0	0	0	0	0	0	2	6	6	2	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 59010101 TO 63123423 (SECTOR WIDTH = 67.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	2	1	1	0	0	1	1	1	0	2	6	18	31	34	22	6
2	1	0	0	0	0	0	0	0	0	0	3	9	17	17	10	1
4	1	0	0	0	0	0	0	0	0	0	1	6	9	10	6	1
8	0	0	0	0	0	0	0	0	0	0	0	1	2	2	2	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Speed GE 5.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	661	525	746	737	644	655	721	590	473	601	703	874	908	836	831	711
2	265	211	360	351	254	290	315	254	198	217	355	440	416	473	365	285
4	79	77	175	170	75	107	110	82	61	70	156	195	176	250	127	83
8	8	12	37	48	12	16	21	18	10	3	23	34	32	84	27	5
12	0	6	11	19	4	4	6	8	1	0	4	9	11	36	6	0
18	0	3	0	8	2	0	1	1	0	0	0	1	3	12	0	0
24	0	1	0	2	0	0	0	0	0	0	0	0	0	5	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-11 (SHEET 3 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 59010101 TO 63123423 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 10.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	158	176	301	285	238	255	239	187	132	198	329	418	538	532	324	204
2	60	85	158	151	99	118	100	79	68	77	145	208	275	324	139	84
4	18	31	62	80	31	38	32	21	17	14	56	92	118	185	47	18
8	0	5	11	25	4	5	5	3	3	0	3	14	20	66	10	0
12	0	4	3	5	1	0	0	1	1	0	0	1	8	30	1	0
18	0	1	0	0	0	0	0	0	0	0	0	0	2	10	0	0
24	0	1	0	0	0	0	0	0	0	0	0	0	0	5	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 59010101 TO 63123423 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 15.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	21	24	46	54	45	25	34	30	17	35	55	105	192	229	86	33
2	7	8	19	25	16	4	5	13	7	7	21	56	94	138	37	14
4	1	2	7	12	5	0	2	1	1	3	6	23	41	81	8	2
8	0	0	1	2	0	0	0	0	1	0	0	1	10	32	2	0
12	0	0	1	1	0	0	0	0	0	0	0	0	2	9	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PERSISTENCIES FROM 59010101 TO 63123423 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 20.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	2	1	5	0	5	0	2	5	4	4	10	21	66	83	26	4
2	1	0	0	0	2	0	1	3	1	0	2	10	36	46	8	3
4	0	0	0	0	0	0	1	0	0	0	0	4	18	28	0	1
8	0	0	0	0	0	0	0	0	0	0	0	0	4	10	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-11 (SHEET 4 OF 4)

67.5° SECTOR WIDTH

PERSISTENCIES FROM 59010101 TO 63123423 (SECTOR WIDTH = 22.5 DEGREES)
Speed GE 25.0 (MPH)

Hours	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
1	0	1	0	0	0	0	1	0	0	0	2	4	15	18	6	1
2	0	0	0	0	0	0	0	0	0	0	0	2	7	10	1	1
4	0	0	0	0	0	0	0	0	0	0	0	1	4	4	0	1
8	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3.2-12

PRECIPITATION DATA FOR AUGUSTA AREA
1902 TO 1950

Month		1 Hour	2 Hours	3 Hours	6 Hours	12 Hours	24 Hours
Jan	Amt	1.10	1.54	1.73	2.61	2.89	2.94
	Date	8/1935	3/1947	3/1947	16/1925	16/1925	16-17/ 1925
Feb	Amt	1.21	1.66	1.80	2.06	2.96	3.17
	Date	28/1916	22/1909	22/1909	22-23/ 1909	25-26/ 1939	7-8/1903
Mar	Amt	1.56	1.60	1.97	2.75	3.67	3.79
	Date	28/1920	28/1920	8/1942	22-23/ 1944	22-23/ 1944	22-23/ 1944
Apr	Amt	1.17	1.41	1.48	2.56	2.96	3.60
	Date	2/1920	25/1946	25/1946	29/1937	29/1937	1-2/1936
May	Amt	2.08	3.05	3.62	3.66	3.66	3.90
	Date	23/1915	23-24/ 1915	23-24/ 1915	23-24/ 1915	23-24/ 1915	23-24/ 1915
Jun	Amt	2.90	3.08	3.09	3.09	4.01	4.17
	Date	18/1911	26/1926	26-27/ 1926	26-27/ 1926	15-16/ 1906	15-16/ 1906
Jul	Amt	2.36	3.36	3.46	3.66	3.68	3.81
	Date	18/1911	6/1950	6/1950	14/1906	14/1906	6-7/1950
Aug	Amt	3.07	3.28	3.46	3.52	3.63	3.93
	Date	14/1903	1/1906	4/1915	4/1915	29/1936	3-4/1915
Sep	Amt	2.47	3.98	4.46	4.72	6.46	9.82
	Date	6/1948	6/1948	6/1948	6/1948	26-27/ 1929	30-1/ 1929
Oct	Amt	2.36	2.78	3.30	5.83	7.79	9.82
	Date	1/1929	1/1929	1/1929	1/1929	1/1929	30-1 1929
Nov	Amt	2.25	2.34	2.34	3.07	4.32	4.33
	Date	2/1903	2/1903	2/1903	23/1942	23/1942	23/1942
Dec	Amt	0.96	1.20	1.56	1.96	2.64	3.99
	Date	15/1941	15/1941	30/1924	2-3/1905	25-26/ 1943	25-26/ 1943
Annual	Amt	3.07	3.98	4.46	5.83	7.79	9.82
	Date	8/14/03	9/6/48	9/6/48	10/1/79	10/1/79	9/30/ 10/1/29

TABLE 2.3.2-13

PRECIPITATION DATA FOR AUGUSTA AREA
1940 TO 1950

Month		1 Hour	2 Hours	3 Hours	6 Hours	12 Hours	24 Hours
Jan	Amt	0.82	1.54	1.73	2.03	2.09	2.36
	Date	3/1947	3/1947	3/1947	3/1947	3/1947	14-15/ 1944
Feb	Amt	0.84	1.23	1.34	1.71	1.86	2.06
	Date	9/1949	9/1949	9/1949	8-9/1949	8-9/1949	8-9/1949
Mar	Amt	1.16	1.44	1.97	2.73	3.67	3.79
	Date	17/1942	8/1942	8/1942	22-23/ 1944	22-23/ 1944	22-23/ 1944
Apr	Amt	0.97	1.41	1.48	1.63	1.99	2.49
	Date	25/1946	25/1946	25/1946	25/1946	16-17/ 1946	25-26/ 1946
May	Amt	1.57	1.67	1.73	2.14	2.14	2.14
	Date	28/1950	25/1948	25/1948	28/1950	28/1950	28/1950
Jun	Amt	1.36	1.45	1.50	2.14	2.32	2.36
	Date	8/1949	8/1949	8/1949	8/1949	12/1947	11-12/ 1947
Jul	Amt	2.14	3.36	3.46	3.67	3.64	3.81
	Date	6/1950	6/1950	6/1950	6-7/1950	6-7/1950	6-7/1950
Aug	Amt	1.83	1.89	1.89	2.17	2.86	3.81
	Date	31/1948	31/1948	31/1948	28/1949	12-13/ 1940	12-13/ 1940
Sep	Amt	2.47	3.98	4.46	4.72	4.80	4.85
	Date	6/1948	6/1948	6/1948	6/1948	6/1948	5-6/1948
Oct	Amt	0.96	1.68	2.12	2.91	3.58	4.12
	Date	25/1946	8/1946	8/1946	8/1946	8/1946	7-8/1946
Nov	Amt	1.39	1.40	1.71	3.07	4.32	4.33
	Date	3/1950	3/1950	23/1942	23/1942	23/1942	23/1942
Dec	Amt	0.96	1.20	1.43	1.68	2.64	3.99
	Date	15/1941	15/1941	25-26/ 1943	25-26/ 1943	25-26/ 1943	25-26/ 1943
Annual	Amt	2.47	3.98	4.46	4.72	4.80	4.85
	Date	9/6/48	9/6/84	9/6/48	9/6/48	9/6/48	9/5-6/48

TABLE 2.3.2-14 (SHEET 1 OF 48)

MONTH OF JANUARY
JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION

REQUEST NUMBER 909-7

FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM 12/73-12/74, 4/77-4/79
SPEED AND DIRECTION 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	3	0	1	1	0	1	0	0	0	0	0	0	0	1	1	2	10	3.5	2.91
3.6 - 7.5	8	0	10	3	3	3	4	5	1	1	0	8	12	11	10	9	98	34.6	5.46
7.6 - 12.5	2	4	5	4	5	4	3	2	0	0	2	7	14	28	22	14	115	41.0	9.71
12.6 - 18.5	0	0	0	4	3	2	0	0	0	0	0	1	5	23	18	0	56	19.8	14.66
18.6 - 24.5	0	0	0	1	0	0	0	0	0	0	0	0	2	0	0	0	3	1.1	19.72
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	13	14	16	13	11	10	7	7	1	1	2	16	33	63	51	25	283	0.0	7.58
PERCENT	4.6	4.9	5.7	4.6	3.9	3.5	2.5	2.5	.4	.4	.7	5.7	11.7	22.3	18.8	8.8	100.0		
AV SPD	5.1	6.1	6.2	10.8	10.1	8.2	7.5	6.2	6.9	4.5	9.9	8.3	9.7	11.0	11.0	8.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

JOINT FREQUENCY TBLES OF WIND SPEED AND DIRECTION

REQUEST NUMBER 909-7

FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 1.0 BUT LESS THAN OR EQUAL TO .9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 100 FT AND 33 FT

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	1	1	0	0	0	0	1	0	0	1	0	1	1	2	4	6.8	3.08
3.6 - 7.5	2	3	2	2	0	1	2	2	1	1	2	4	1	4	3	0	30	50.8	5.91
7.6 - 12.5	1	0	1	0	1	0	1	1	0	0	4	0	2	4	1	0	15	27.1	9.15
12.6 - 18.5	0	0	0	0	0	1	0	0	1	0	0	1	3	1	2	0	9	15.3	13.64
18.6 - 24.5	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	3	3	4	3	1	2	3	3	3	1	6	6	6	9	6	0	59	0.0	6.71
PERCENT	5.1	5.1	6.8	5.1	1.7	3.4	5.1	5.1	5.1	1.7	10.2	15.3	10.2	15.3	10.2	0.0	100.0		
AV SPD	6.2	5.5	6.2	5.1	7.9	9.7	6.6	7.3	7.9	6.6	7.6	6.4	12.3	9.0	9.3	0.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 2 OF 48)

MONTH OF JANUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
3.6 - 7.5	1	2	0	0	1	0	3	1	1	0	0	1	1	1	2	3	17	47.2	5.15
7.6 - 12.5	0	1	1	0	2	0	0	1	1	0	1	3	1	3	0	0	14	38.9	10.10
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	3	6.3	16.06
18.6 - 24.5	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2	5.6	20.18
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	3	1	0	3	0	3	2	2	0	1	4	3	8	2	3	36	0.0	7.24
PERCENT	2.8	8.3	2.8	0.0	8.3	0.0	8.3	5.6	5.6	0.0	2.8	11.1	8.3	22.2	5.6	8.3	100.0		
AV SPD	4.3	7.3	10.9	0.0	10.1	0.0	5.9	7.7	7.9	0.0	7.9	9.0	12.2	13.0	4.1	5.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	5	5	4	1	1	3	2	0	3	0	1	3	3	4	3	5	43	8.5	2.73
3.6 - 7.5	7	12	10	11	15	15	13	12	5	7	6	12	16	14	9	15	179	35.2	5.37
7.6 - 12.5	3	13	13	31	10	9	4	4	8	8	8	16	23	28	14	7	199	39.5	9.43
12.6 - 18.5	0	2	1	2	1	2	0	1	4	13	9	4	11	20	3	0	73	14.5	14.45
18.6 - 24.5	0	0	0	0	0	0	0	0	3	1	0	0	0	5	0	0	9	1.8	20.09
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	.2	25.70
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	15	32	28	45	27	29	19	17	23	29	25	35	53	71	29	27	504	0.0	0.00
PERCENT	3.0	6.3	5.6	6.9	5.4	5.8	3.8	3.4	4.6	5.8	5.0	6.9	10.5	14.1	5.8	5.4	100.0		
AV SPD	5.1	7.3	7.3	9.0	6.9	6.9	6.6	7.0	9.9	11.5	10.8	8.0	9.0	11.1	8.4	5.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 3 OF 48)

MONTH OF JANUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1	1	0	0	0	0	0	0	0	0	0	0	1	0	1	0	4	.7	.50
CALM + - 3.5	5	0	6	5	3	0	2	5	5	4	6	3	8	7	8	5	72	11.8	2.47
3.6 - 7.5	8	6	24	24	9	8	14	9	12	21	15	19	34	51	30	7	291	47.8	5.46
7.6 - 12.5	1	6	6	11	14	8	11	7	8	7	7	8	25	41	22	0	182	29.9	9.54
12.6 - 18.5	0	2	0	0	2	0	0	0	5	4	3	7	2	9	12	0	46	7.6	14.57
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	6	5	0	0	0	11	1.8	20.99
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0	3	.5	28.85
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	15	15	36	40	28	16	27	21	30	36	32	45	75	108	73	12	609	0.0	5.38
PERCENT	2.5	2.5	5.9	6.6	4.6	2.6	4.4	3.4	4.9	5.9	5.3	7.4	2.3	17.7	12.0	2.0	100.0		
AV SPD	4.2	7.8	5.5	6.8	8.0	8.3	7.0	6.6	7.5	7.1	7.1	11.1	7.8	7.9	8.3	4.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	3	1.1	.50
CALM + - 3.5	1	2	1	2	1	0	4	2	6	2	3	3	8	4	9	5	53	20.2	2.49
3.6 - 7.5	5	8	10	10	6	3	6	8	8	3	7	24	21	32	24	11	186	71.0	4.98
7.6 - 12.5	2	1	1	2	0	0	0	0	0	0	0	1	0	4	2	5	18	6.9	9.04
12.6 - 18.5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	.8	14.60
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	9	11	12	14	7	3	10	10	14	6	11	28	30	40	36	21	252	0.0	3.92
PERCENT	3.4	4.2	4.6	5.3	2.7	1.1	3.8	3.8	5.3	2.3	4.2	10.7	11.5	15.3	13.7	8.0	100.0		
AV SPD	6.2	5.5	5.3	5.6	5.0	6.1	3.8	4.5	3.9	5.5	4.2	4.9	4.2	5.2	4.7	6.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 4.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-14 (SHEET 4 OF 48)

MONTH OF JANUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	2	2	0	0	3	1	0	0	0	0	0	1	1	3	1	14	5.8	.50
CALM + - 3.5	7	7	12	4	10	5	2	6	7	1	2	5	5	3	5	3	84	40.6	2.40
3.6 - 7.5	6	11	16	7	7	4	4	1	7	8	9	11	4	6	5	3	109	52.7	4.68
7.6 - 12.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	13	20	30	11	17	12	7	7	14	9	11	16	10	10	13	7	207	0.0	2.40
PERCENT	6.3	9.7	14.5	5.3	8.2	5.8	3.4	3.4	6.8	4.3	5.3	7.7	4.8	4.8	6.3	3.4	100.0		
AV SPD	3.7	3.5	3.8	4.0	3.2	3.4	3.8	2.9	3.9	4.1	4.5	4.4	3.1	3.5	2.7	3.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF JANUARY

SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	1	3	2	0	0	3	1	0	0	0	1	0	3	1	5	1	21	1.1
CALM + - 3.5	21	14	25	14	15	9	10	13	22	7	12	15	24	19	26	20	266	13.6
3.6 - 7.5	37	52	72	57	41	34	46	38	25	41	39	79	89	119	83	48	910	46.4
7.6 - 12.5	9	251	27	48	32	21	19	15	17	15	22	35	65	108	61	26	545	27.9
12.6 - 18.5	1	4	1	6	6	5	0	1	10	18	12	13	21	56	35	0	189	9.6
18.6 - 24.5	0	0	0	0	0	0	0	0	3	1	0	6	8	6	0	0	25	1.3
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	2	2	0	0	0	0	4	.2
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	69	98	127	126	94	72	76	67	87	82	88	150	210	309	210	95	1960	100.0
PERCENT	3.5	5.0	6.5	6.4	4.8	3.7	3.9	3.4	4.4	4.2	4.5	7.7	10.7	15.8	10.7	4.8	100.0	

TABLE 2.3.2-14 (SHEET 5 OF 48)

MONTH OF FEBRUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00	0.00
CALM + - 3.5	1	3	1	2	0	0	0	0	0	0	0	0	0	0	1	3	16	5.4	2.91
3.6 - 7.5	10	9	15	8	8	4	5	1	0	4	3	3	12	5	9	6	101	34.2	5.54
7.6 - 12.5	5	10	11	9	8	6	8	2	1	1	3	12	21	15	18	4	134	45.4	9.53
12.6 - 18.5	1	2	4	4	0	0	0	0	0	1	2	1	4	8	16	1	44	14.9	14.60
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	17	24	31	23	16	10	13	3	1	6	8	19	38	28	44	14	295	0.0	0.00
PERCENT	5.8	8.1	10.5	7.8	5.4	3.4	4.4	1.0	.3	2.0	2.7	6.4	12.9	9.5	14.9	4.7	100.0		
AV SPD	6.8	7.8	7.9	9.2	7.9	8.3	7.8	7.0	11.1	7.5	9.7	8.1	9.0	10.8	10.7	6.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 4

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	4.5	3.40
3.6 - 7.5	4	3	1	2	2	2	1	1	1	0	1	0	4	0	1	0	23	52.3	5.48
7.6 - 12.5	3	1	1	1	0	0	0	0	0	2	1	0	3	0	0	0	12	27.3	9.08
12.6 - 18.5	0	0	1	1	0	0	0	0	0	1	2	0	0	0	2	0	7	15.9	14.03
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	8	5	3	4	2	2	1	1	1	3	4	0	7	0	3	0	44	0.0	6.66
PERCENT	18.2	11.4	6.8	9.1	4.5	4.5	2.3	2.3	2.3	6.8	9.1	0.0	15.9	0.0	6.8	0.0	100.0		
AV SPD	6.6	6.0	8.8	9.3	6.8	6.2	4.4	7.1	6.2	10.8	10.5	0.0	7.5	0.0	10.2	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 6 OF 48)

MONTH OF FEBRUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT
WIND DIRECTION

REQUEST NUMBER 909-7

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	1	4.2	2.60
3.6 - 7.5	0	1	0	1	1	0	1	0	2	0	1	0	1	0	1	1	11	45.4	5.39
7.6 - 12.5	0	3	1	2	0	0	0	1	0	0	0	0	1	1	0	0	9	37.5	9.34
12.6 - 18.5	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	3	12.5	15.38
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	5	1	3	1	0	1	1	2	0	2	1	2	2	1	1	24	0.0	0.00
PERCENT	4.2	20.8	4.2	12.5	4.2	0.0	4.2	4.2	8.3	0.0	8.3	4.2	8.3	8.3	4.2	4.2	100.0		
AV SPD	2.6	10.2	9.4	8.0	6.9	0.0	5.8	10.4	4.3	0.0	4.3	14.1	8.6	12.4	6.4	6.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTION

REQUEST NUMBER 909-7

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	5	3	0	2	2	4	1	1	1	0	2	2	6	3	1	33	9.7	2.60
3.6 - 7.5	16	13	21	9	7	6	8	8	3	2	3	3	7	3	5	5	119	35.0	5.57
7.6 - 12.5	16	9	27	7	6	1	3	2	3	5	2	9	9	16	20	0	135	39.7	9.47
12.6 - 18.5	0	2	8	1	0	0	0	5	2	2	5	5	4	10	7	1	52	15.3	14.27
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.3	18.70
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	32	29	59	17	15	9	15	16	9	10	10	20	22	35	35	7	340	0.0	6.54
PERCENT	9.4	8.5	17.4	5.0	4.4	2.6	4.4	4.7	2.6	2.9	2.9	5.9	6.5	10.3	10.3	2.1	100.0		
AV SPD	7.3	7.0	8.6	7.2	6.9	5.4	6.1	8.7	9.2	8.0	11.2	10.2	9.2	9.5	9.6	5.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 7 OF 48)

MONTH OF FEBRUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	3	2	5	4	0	2	4	2	0	1	2	1	4	6	5	3	44	9.8	2.59
3.6 -7.5	15	9	22	16	24	16	13	17	8	10	11	13	14	18	31	13	250	55.4	5.28
7.6 - 12.5	3	4	9	12	4	5	5	2	6	0	9	0	7	19	32	3	120	25.6	9.10
12.6 - 18.5	0	4	6	1	1	0	0	4	2	1	0	1	4	10	3	0	37	8.2	14.39
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	21	19	42	33	29	23	22	25	16	12	22	15	29	53	71	19	451	0.0	5.63
PERCENT	4.7	4.2	9.3	7.3	6.4	5.1	4.9	5.5	3.5	2.7	4.9	3.3	6.4	11.4	15.7	4.2	100.0		
AV SPD	5.6	7.7	7.4	7.1	6.4	5.7	5.7	7.2	7.7	6.3	7.0	5.5	7.5	8.1	7.4	5.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	8	5	2	2	1	0	4	4	1	2	3	3	1	3	9	6	54	26.6	2.49
3.6 - 7.5	5	8	13	8	4	8	2	2	3	6	15	24	11	12	9	9	139	68.5	5.00
7.6 - 12.5	0	0	2	3	1	1	0	0	0	0	2	0	0	0	0	0	9	4.4	9.85
12.6 - 18.5	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.5	13.20
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	13	14	17	13	6	9	6	6	4	8	20	27	12	15	18	15	203	0.0	4.02
PERCENT	6.4	6.9	8.4	6.4	3.0	4.4	3.0	3.0	2.0	3.9	9.9	13.3	5.9	7.4	8.9	7.4	100		
AV SPD	3.4	5.0	5.6	5.8	6.3	5.6	3.6	3.6	4.8	4.3	5.2	5.1	5.1	4.5	3.6	4.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 4.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 8 OF 48)

MONTH OF FEBRUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	.8	.50
CALM + - 3.5	2	0	0	3	2	3	1	5	3	0	3	1	2	4	1	1	31	26.1	2.53
3.6-7.5	2	2	7	10	2	7	4	2	2	12	10	10	9	5	0	1	85	71.4	4.80
7.6-12.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1.7	9.03
12.6-18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6-24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6-32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	4	4	7	13	4	10	5	7	5	12	14	11	11	9	1	2	119	0.0	0.00
PERCENT	3.4	3.4	5.9	10.9	3.4	8.4	4.2	5.9	4.2	10.1	11.8	9.2	9.2	7.6	.8	1.7	100.0		
AVSPD	3.9	7.3	5.1	4.5	4.2	4.3	4.6	3.0	3.3	4.9	3.9	4.7	4.7	3.8	3.1	4.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 4.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF FEBRUARY
SUMMARY TABLE

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1
CALM + - 3.5	16	16	11	11	5	7	13	12	5	4	9	10	10	19	19	14	181	12.3
3.6-7.5	52	45	79	54	48	43	34	31	19	34	44	53	58	43	66	35	728	49.3
7.6-12.5	27	29	51	34	19	13	16	7	10	8	17	21	41	51	70	7	421	28.5
12.6-18.5	1	10	19	7	1	0	0	9	4	5	9	8	12	29	28	2	144	9.8
18.6-24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.1
24.6-32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	96	100	160	106	73	63	63	59	38	51	80	93	121	142	173	58	1476	100.0
PERCENT	6.5	6.8	10.8	7.2	4.9	4.3	4.3	4.0	2.6	3.5	5.4	6.3	8.2	9.6	11.7	3.9	100.0	

TABLE 2.3.2-14 (SHEET 9 OF 48)

MONTH OF MARCH

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
CALM + - 3.5	0	0	1	1	0	0	0	2	0	0	0	1	1	2	2	0	10	2.2	2.78
3.6 - 7.5	13	12	11	6	7	4	9	5	4	8	12	7	17	23	11	19	169	36.7	5.67
7.6 - 12.5	4	8	11	4	12	12	6	6	10	7	18	23	33	21	11	7	193	42.0	9.59
12.6 - 18.5	0	1	1	0	0	0	5	2	2	4	4	6	21	18	11	1	76	16.5	14.50
18.6 - 24.5	0	0	0	0	0	0	0	0	0	1	0	0	3	7	1	0	12	2.6	19.85
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	17	21	24	11	19	16	20	16	16	20	34	37	75	71	36	27	460	0.0	7.75
PERCENT	3.7	4.6	5.2	2.4	4.1	3.5	4.3	3.5	3.5	4.3	7.4	8.0	16.3	15.4	7.8	5.9	100.0		
AV SPD	6.4	7.6	7.7	6.6	8.5	8.6	9.1	8.2	9.4	10.0	9.1	10.0	10.5	10.7	9.7	7.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 11

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	4	3.8	2.65
3.6 - 7.5	4	5	3	3	0	1	2	2	2	2	1	3	9	3	3	3	46	43.4	5.41
7.6 - 12.5	3	6	10	2	2	3	3	1	0	1	8	4	5	3	0	2	53	50.0	9.10
12.6 - 18.5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	3	2.8	15.01
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	9	11	13	5	2	4	5	3	2	3	9	7	16	5	5	5	106	0.0	6.61
PERCENT	8.5	10.4	12.3	4.7	1.9	3.8	4.7	2.8	1.9	2.8	8.5	6.6	15.1	5.7	5.7	4.7	100.0		
AV SPD	7.0	7.5	8.4	6.6	9.4	8.6	6.4	6.4	4.7	6.7	9.9	8.2	7.6	7.2	6.6	7.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-14 (SHEET 10 OF 48)

MONTH OF MARCH

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2.3	3.50
3.6 -7.5	3	1	1	1	1	0	2	2	0	1	1	0	1	2	1	2	19	43.2	5.83
7.6 - 12.5	0	1	0	0	4	1	0	1	5	2	1	1	2	1	1	0	20	45.5	9.37
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6.8	14.21
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2.3	21.40
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	2	2	1	5	1	2	4	5	4	2	1	3	4	3	2	44	0.0	7.41
PERCENT	6.8	4.5	4.5	2.3	11.4	2.3	4.5	.9.1	11.4	9.1	4.5	2.3	6.8	9.1	6.8	4.5	100.0		
AV SPD	5.7	7.3	4.0	7.2	8.8	7.9	6.6	8.9	9.5	10.8	8.7	8.9	8.5	10.4	10.0	6.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CALM + - 3.5	1	1	2	1	2	4	2	3	3	2	0	1	1	3	1	2	29	5.4	0.00
3.6 -7.5	14	10	13	16	30	23	26	13	17	14	10	13	12	15	8	12	246	46.0	5.65
7.6 - 12.5	7	6	9	12	23	18	14	20	17	8	12	16	20	11	6	5	204	38.1	9.32
12.6 - 18.5	2	1	1	0	1	0	7	7	4	9	2	0	2	5	3	0	44	8.2	14.51
18.6 - 24.5	0	1	0	0	0	0	0	0	2	5	0	0	3	1	0	0	11	2.1	19.93
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.2	25.10
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL		24	18	25	29	56	45	49	43	43	39	24	30	38	35	18	19	535	0.0
PERCENT	4.5	3.4	4.7	5.4	10.5	8.4	9.2	8.0	8.0	7.3	4.5	5.6	7.1	6.5	3.4	3.6	100.0		
AV SPD	7.4	7.0	7.2	7.1	6.7	8.1	8.6	9.0	11.1	8.8	7.9	9.6	8.4	8.8	5.9				

AVERAGE SPEED FOR THIS TABLE EQUALS 8.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-14 (SHEET 11 OF 48)

MONTH OF MARCH

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
CALM + -3.5	3	2	1	3	1	4	2	6	4	7	3	4	2	4	3	4	54	19.7	2.45
3.6 -7.5	6	7	10	17	16	23	21	27	39	55	46	16	18	25	36	7	369	57.5	5.38
7.6 - 12.5	0	0	1	5	11	16	13	16	19	35	23	5	5	18	20	3	190	29.6	9.05
12.6 - 18.5	1	0	0	0	0	0	0	1	6	3	5	0	1	3	2	0	22	3.4	13.94
18.6 - 24.5	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	.2	19.80
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	10	9	12	25	30	43	35	54	72	97	77	24	29	50	62	13	642	0.0	5.63
PERCENT	1.6	1.4	1.9	3.9	4.7	6.7	5.5	8.4	11.2	15.1	12.0	3.7	4.5	7.8	9.7	2.0	100.0		
AV SPD	5.3	5.1	5.8	5.9	6.3	6.9	70	6.6	7.2	7.1	7.0	5.8	6.0	7.3	6.7	5.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	.7	.50
CALM + -3.5	4	1	2	3	1	4	2	6	4	7	3	4	2	4	3	4	54	19.7	2.45
3.6 -7.5	4	3	7	2	10	10	16	15	11	22	15	23	18	14	6	1	178	65.0	4.85
7.6 - 12.5	0	0	1	0	1	7	9	6	6	4	0	0	0	0	3	0	37	13.5	8.81
12.6 - 18.5	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1	0	3	1.1	14.05
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	8	4	11	5	12	21	27	28	21	234	18	27	20	20	13	5	274	0.0	0.00
PERCENT	2.9	1.5	4.0	1.8	4.4	7.7	9.9	10.2	7.7	12.4	6.6	9.9	7.3	7.3	4.7	1.8	100.0		
AV SPD	3.2	4.5	4.9	3.3	5.1	6.0	6.1	5.5	6.1	5.3	4.9	4.6	4.6	4.6	5.9	2.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 5.2
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 12 OF 48)

MONTH OF MARCH

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	3	2.0	.50
CALM + -3.5	1	1	4	1	1	4	3	1	3	2	1	4	4	2	1	2	35	23.3	2.65
3.6 -7.5	1	6	5	4	8	5	3	1	5	13	14	6	9	13	7	0	100	66.7	4.87
7.6 - 12.5	0	0	2	0	0	0	0	2	1	0	1	2	0	0	0	0	8	5.3	8.54
12.6 - 18.5	0	0	0	0	0	1	1	0	0	2	0	0	0	0	0	0	4	2.7	14.49
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	7	12	5	9	10	7	4	9	17	12	12	13	15	8	3	150	0.0	3.66
PERCENT	2.0	4.7	8.0	3.3	6.0	6.7	4.7	2.7	6.0	11.3	10.7	8.0	8.7	10.0	5.3	2.0	100.0		
AV SPD	2.3	5.2	4.8	4.2	4.9	5.3	5.4	5.6	4.9	5.6	5.2	5.0	4.1	4.7	4.9	2.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 4.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF MARCH
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	1	5	.2
CALM + -3.5	10	5	11	9	7	16	8	21	18	15	7	13	14	15	13	11	193	8.7
3.6 -7.5	45	44	50	49	72	66	79	67	78	115	99	68	84	95	72	44	1127	51.0
7.6 - 12.5	14	21	34	23	53	57	45	52	58	57	63	51	65	54	41	17	705	31.9
12.6 - 18.5	4	2	2	0	1	1	13	11	12	20	11	6	25	27	19	1	155	7.0
18.6 - 24.5	0	0	0	0	0	0	0	1	2	6	0	0	6	9	1	0	25	1.1
24.6 - 32.5	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	.0
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	74	72	99	81	133	140	145	152	168	214	180	138	194					
PERCENT	3.3	3.3	4.5	3.7	6.0	6.3	6.6	6.9	7.6	9.7	8.1	6.2	8.8	9.1	6.6	3.3	100.0	

TABLE 2.3.2-14 (SHEET 13 OF 48)

MONTH OF MARCH

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	0	2	0	1	0	2	0	1	0	1	1	3	1	3	2	18	3.6	2.77
3.6 -7.5	8	6	7	13	3	5	12	5	3	4	13	13	22	26	16	13	169	33.4	5.85
7.6 - 12.5	6	2	13	9	7	11	27	10	10	6	13	13	32	26	17	5	207	40.9	9.55
12.6 - 18.5	3	0	0	0	0	0	3	2	3	0	8	16	24	33	11	1	104	20.6	14.63
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	1	1	3	2	1	0	8	1.6	19.78
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	18	8	22	22	11	16	44	17	17	10	36	44	84	88	48	21	506	0.0	
PERCENT	3.6	1.6	4.3	4.3	2.2	3.2	8.7	3.4	3.4	2.0	7.1	8.7	16.6	17.4	9.5	4.2	100.0		
AV SPD	5.3	5.1	5.8	5.9															

AVERAGE SPEED FOR THIS TABLE EQUALS 9.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 7

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	2	5	6.2	2.60
3.6 -7.5	0	0	1	4	2	0	3	1	9	5	2	1	1	0	1	0	30	37.0	5.68
7.6 - 12.5	0	2	0	0	1	3	2	3	2	3	2	3	3	6	1	1	32	39.5	9.71
12.6 - 18.5	0	0	0	0	0	0	2	0	0	3	1	4	4	0	0	0	14	17.3	13.93
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	0	2	1	4	3	4	7	4	11	11	5	8	8	7	3	3	81	0.0	0.00
PERCENT	0.0	2.5	1.2	4.9	3.7	4.9	8.6	5.6	4.9	13.6	6.2	9.9	9.9	8.6	3.7	3.7	100.0		
AV SPD	0.0	9.1	6.3	5.3	6.3	7.4	9.1	6.3	7.0	8.9	9.9	11.4	12.5	8.7	7.0	5.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.7
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 14 OF 48)

MONTH OF APRIL

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN .-9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2.7	3.20
3.6 -7.5	0	1	1	0	0	1	2	3	1	0	4	3	1	0	0	1	18	48.6	5.57
7.6 - 12.5	1	0	0	0	0	0	2	0	2	1	1	0	2	1	1	4	15	40.5	9.68
12.6 - 18.5	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0	3	8.1	13.91
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	1	1	0	0	2	4	3	3	2	6	3	3	2	1	5	37	0.0	6.97
PERCENT	2.7	2.7	2.7	0.0	0.0	5.4	10.8	8.1	8.1	5.4	16	8.1	8.1	5.4	2.7	13.5	100.0		
AV SPD	7.7	6.7	5.7	0.0	0.0	4.5	8.0	5.5	8.0	11.4	7.7	6.3	8.4	12.8	10.2	10.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	.3	.50
CALM + -3.5	2	1	1	2	0	1	2	0	2	4	4	2	0	2	2	0	25	6.7	2.67
3.6 -7.5	2	6	9	6	12	17	8	14	14	22	15	10	9	4	9	3	160	42.7	5.45
7.6 - 12.5	8	2	1	1	5	6	13	9	17	15	14	13	16	22	8	2	152	40.5	9.42
12.6 - 18.5	1	1	0	0	1	0	4	1	2	3	3	3	6	5	6	0	36	9.6	13.89
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	.3	19.40
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	13	10	11	9	18	24	27	14	35	45	37	28	31	33	25	5	375	0.0	6.31
PERCENT	3.5	2.7	2.9	1.4	4.8	6.4	7.2	6.4	9.3	12.0	9.9	7.5	8.3	8.8	6.7	3.3	100.0		
AV SPD	8.4	6.8	5.2	5.3	6.9	6.5	8.3	7.9	7.4	7.2	8.5	8.3	9.4	9.8	8.7	6.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.9
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 3

TABLE 2.3.2-14 (SHEET 15 OF 48)

MONTH OF APRIL

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .-3 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.2	.50
CALM + -3.5	3	0	0	3	4	8	11	7	7	4	6	3	1	5	5	3	70	14.5	2.73
3.6 -7.5	6	5	2	5	22	21	32	35	39	30	17	24	23	13	20	5	299	62.0	5.33
7.6 - 12.5	0	2	1	0	3	1	10	7	11	14	7	6	9	7	14	3	95	19.7	8.85
12.6 - 18.5	0	0	0	0	0	0	0	0	0	6	1	3	1	2	3	0	16	3.3	14.91
18.6 - 24.5	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	.2	18.60
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	9	7	4	8	29	30	53	49	57	55	31	36	34	27	42	11	482	0.0	5.04
PERCENT	1.9	1.5	.8	1.7	6.0	6.2	11.0	10.2	11.8	11.4	6.4	7.5	7.1	5.6	8.7	2.3	100.0		
AV SPD	4.8	6.3	5.9	4.5	5.0	4.5	5.6	5.7	5.8	7.5	6.2	7.0	6.8	6.7	7.3	5.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	3	0	2	1	3	3	5	4	6	6	2	5	4	6	6	8	64	21.4	2.34
3.6 -7.5	9	9	13	13	6	123	8	3	15	20	19	31	26	17	14	3	218	72.9	5.15
7.6 - 12.5	0	0	0	0	0	1	2	1	1	0	0	3	3	3	0	1	15	5.0	9.03
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	.7	14.19
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	12	9	15	14	9	16	15	8	22	26	21	39	33	26	22	12	299	0.0	4.18
PERCENT	4.0	3.0	5.0	4.7	3.0	5.4	5.0	2.7	7.4	8.7	7.0	13.0	11.0	8.7	7.4	4.0	100.0		
AV SPD	4.8	5.7	5.2	4.8	4.0	4.8	5.2	4.6	5.4	4.7	5.0	5.3	5.5	5.0	5.1	3.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 5.0
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 16 OF 48)

MONTH OF APRIL

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION REQUEST NUMBER 909-7
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
CALM + -3.5	4	2	3	3	6	7	4	5	6	2	9	6	6	7	4	3	77	41.6	2.25
3.6 -7.5	4	1	5	3	5	2	2	4	4	6	15	23	12	6	11	3	106	57.3	4.62
7.6 - 12.5	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	2	1.1	9.61
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	8	3	8	6	11	10	6	9	11	8	24	29	18	13	15	6	185	0.0	3.22
PERCENT	4.3	1.6	4.3	3.2	5.9	5.4	3.2	4.9	5.9	4.3	13.0	15.7	9.7	7.0	8.1	3.2	100.0		
AV SPD	3.6	3.5	3.7	3.7	3.2	3.3	3.2	3.5	4.3	4.5	4.3	4.5	3.7	3.7	3.9	3.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF APRIL
SUMMARY OF TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	2	.1
CALM + -3.5	13	3	8	9	14	21	24	15	22	16	22	17	14	22	21	18	260	13.2
3.6 -7.5	29	28	38	44	50	58	67	65	85	87	85	105	94	66	71	28	1000	50.9
7.6 - 12.5	15	8	15	10	16	23	56	30	44	39	37	38	65	65	41	16	518	26.4
12.6 - 18.5	4	1	0	0	1	0	9	3	5	13	14	26	35	41	22	1	175	8.9
18.6 - 24.5	0	0	0	0	0	0	0	0	0	1	2	1	3	2	1	0	10	.5
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL	61	40	62	63	81	102	156	114	156	157	160	187	211	196	156	63	1965	100.0
PERCENT	3.1	2.0	3.2	3.2	4.1	5.2	7.9	5.8	7.9	8.0	8.1	9.5	10.7	10.0	7.9	3.2	100.0	

TABLE 2.3.2-14 (SHEET 17 OF 48)

MONTH OF MAY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	5	2	1	2	1	1	0	0	0	1	0	2	2	2	2	4	25	5.3	2.89
3.6 -7.5	18	16	13	13	11	8	2	2	2	7	6	25	26	21	14	20	204	43.6	5.55
7.6 - 12.5	1	5	7	3	4	2	3	6	8	1	8	21	52	40	20	13	200	42.7	9.28
12.6 - 18.5	0	2	0	0	0	1	2	2	0	1	6	9	11	2	0	1	37	7.9	14.39
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	2	.4	19.74
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	30	25	21	18	16	12	7	10	10	10	21	58	91	65	36	38	468	0.0	6.72
PERCENT	6.4	5.3	4.5	3.8	3.4	2.6	1.5	2.1	2.1	2.1	4.5	12.4	19.4	13.9	7.7	8.1	100.0		
AV SPD	6.1	6.3	6.6	5.8	6.4	6.7	10.6	9.8	8.3	7.2	10.6	8.8	9.0	8.4	7.6	6.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	1	2	0	1	0	0	0	0	1	0	2	1	0	0	8	9.9	2.90
3.6 -7.5	2	3	4	3	3	0	1	3	4	3	7	1	2	3	1	0	40	49.4	5.40
7.6 - 12.5	1	1	1	1	0	1	0	3	3	4	0	4	3	2	1	0	25	30.9	9.45
12.6 - 18.5	0	0	0	1	0	0	1	0	1	0	0	4	1	0	0	0	8	9.9	14.73
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	4	6	7	3	2	2	6	8	7	8	9	8	6	2	0	81	0.0	6.07
PERCENT	3.7	4.9	7.4	8.6	3.7	2.5	2.5	7.4	9.9	8.6	9.9	11.1	9.9	7.4	2.5	0.0	100.0		
AV SPD	7.5	7.1	5.6	5.6	6.2	5.9	6.9	5.9	9.2	8.5	5.9	11.7	8.2	6.5	7.8	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.6
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 18 OF 48)

MONTH OF MAY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	0	1	1	0	0	1	0	0	1	0	2	1	1	0	5	12.5	2.68
3.6 -7.5	1	1	1	4	1	4	1	1	0	3	0	1	0	1	2	0	21	52.5	5.35
7.6 - 12.5	0	1	2	2	0	0	0	0	2	1	0	0	3	0	1	0	12	30.0	9.61
12.6 - 18.5	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	2	5.0	14.20
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	2	3	7	2	4	1	2	3	4	1	2	3	21	4	0	40	0.0	0.00
PERCENT	2.5	5.0	7.5	17.5	5.0	10.0	2.5	5.0	7.5	10.0	2.5	5.0	7.5	2.5	10.0	0.0	100		
AV SPD	4.9	8.7	8.6	5.9	4.2	6.0	7.5	3.9	10.7	6.5	2.7	11.5	10.1	6.2	5.0	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	1	1	4	6	3	4	3	4	0	4	0	0	2	1	2	35	9.4	2.72
3.6 -7.5	9	13	12	19	13	12	13	19	16	10	19	8	4	6	8	6	187	50.4	5.50
7.6 - 12.5	10	8	9	4	3	1	7	16	18	9	8	10	13	3	3	5	127	34.1	9.41
12.6 - 18.5	0	0	0	0	2	0	1	3	7	3	2	0	2	0	2	1	23	6.2	14.23
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	19	22	22	27	24	16	25	41	45	22	33	18	19	11	14	14	372	0.0	6.00
PERCENT	5.1	5.9	5.9	7.3	6.5	4.3	6.7	11.0	12.1	5.9	8.9	4.8	5.1	3.0	3.8	3.8	100.0		
AV SPD	7.4	6.7	7.7	5.6	5.9	5.6	6.8	7.6	8.7	8.2	7.0	8.2	9.2	6.6	7.8	6.8			

AVERAGE SPEED FOR THIS TABLE EQUALS7.3
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 4

TABLE 2.3.2-14 (SHEET 19 OF 48)

MONTH OF MAY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	.2	.50
CALM + -3.5	8	6	2	8	11	17	13	21	12	4	7	8	4	5	1	8	135	25.5	2.55
3.6 -7.5	11	10	18	10	17	19	22	38	37	21	31	31	21	15	15	11	327	61.8	5.06
7.6 - 12.5	2	1	6	2	2	5	2	7	14	2	4	3	0	3	7	0	60	11.3	9.03
12.6 - 18.5	0	0	0	1	0	0	0	0	0	0	1	1	0	0	2	0	5	.9	13.22
18.6 - 24.5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.2	19.80
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	17	26	21	30	41	37	66	63	28	43	43	25	23	25	19	529	0.0	4.18
PERCENT	4.2	3.2	4.9	4.0	5.7	7.8	7.0	12.5	11.9	5.3	8.1	8.1	4.7	4.3	4.7	3.6	100.0		
AV SPD	5.4	4.5	5.8	5.1	4.6	4.3	4.4	4.9	5.6	4.9	5.2	5.4	5.1	5.4	7.6	4.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 5.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	.3	.50
CALM + -3.5	12	6	7	8	7	8	11	7	5	5	8	7	11	8	4	5	119	37.8	2.35
3.6 -7.5	5	6	5	8	4	12	13	1	8	11	16	31	18	26	15	9	188	59.7	4.84
7.6 - 12.5	0	2	0	0	1	0	0	0	1	0	1	0	0	0	1	1	7	2.2	7.98
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	17	14	12	17	12	20	24	8	14	16	25	38	29	34	20	15	315	0.0	3.41
PERCENT	5.4	4.4	3.8	5.4	3.8	6.3	7.6	2.5	4.4	5.1	7.9	12.1	9.2	10.8	6.3	4.8	100.0		
AV SPD	3.5	4.1	3.2	3.6	3.7	4.2	3.6	2.9	4.7	4.3	4.5	4.8	4.1	4.2	4.5	4.5			

AVERAGE SPEED FOR THIS TABLE EQUALS4.1
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 20 OF 48)

MONTH OF MAY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	.6	.50
CALM + -3.5	1	4	6	2	7	3	1	6	5	7	10	7	2	3	9	9	52	48.5	2.40
3.6 -7.5	3	1	4	6	5	4	2	1	4	7	13	14	5	9	3	5	86	50.9	4.72
7.6 - 12.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	4	5	10	8	12	7	3	7	9	14	23	21	7	13	12	14	169	0.0	0.00
PERCENT	2.4	3.0	5.9	4.7	7.1	4.1	1.8	4.1	5.3	8.3	13.6	12.4	4.1	7.7	7.1	8.3	100.0		
AV SPD	4.0	2.6	3.7	3.9	3.5	4.1	4.2	2.4	3.3	4.0	4.1	3.9	4.6	4.3	2.8	3.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

MONTH OF MAY
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	1	0	0	0	0	0	1	0	0	0	1	0	0	3	.2
CALM + -3.5	26	19	18	27	33	33	29	38	26	17	31	24	21	21	18	28	409	20.7
3.6 -7.5	49	50	57	63	54	59	54	65	71	62	92	111	76	81	58	51	1053	53.3
7.6 - 12.5	20	18	25	12	10	9	12	32	46	17	21	38	71	48	33	19	431	21.8
12.6 - 18.5	0	2	0	2	2	1	4	5	9	4	9	15	14	2	4	2	75	3.8
18.6 - 24.5	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	3	.2
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	96	89	100	105	99	102	99	140	152	101	154	189	182	153	113	100	1974	100.0
PERCENT	4.9	4.5	5.1	5.3	5.0	5.2	5.0	7.1	7.7	5.1	7.8	9.6	9.2	7.8	5.7	5.1	100.0	

TABLE 2.3.2-14 (SHEET 21 OF 48)

MONTH OF JUNE

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION REQUEST NUMBER 909-7
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	3	5	4	2	0	1	2	0	0	3	2	1	2	3	0	29	7.1	2.58
3.6 -7.5	19	15	19	13	25	14	9	4	3	3	8	10	25	15	6	5	193	47.3	5.56
7.6 - 12.5	5	3	10	11	9	15	5	5	10	3	7	11	43	23	12	3	175	42.9	9.16
12.6 - 18.5	1	1	0	0	0	0	0	1	0	0	0	3	3	1	0	1	11	2.7	13.72
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	26	22	34	28	36	29	15	12	13	6	18	26	72	41	21	9	408	0.0	6.19
PERCENT	6.4	5.4	8.3	6.9	8.8	7.1	3.7	2.9	3.2	1.5	4.4	6.4	17.6	10.0	5.1	2.2	100.0		
AV SPD	6.3	6.0	6.1	7.0	6.5	7.2	6.6	7.4	9.2	7.0	6.4	7.9	8.4	8.1	7.6	7.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 11

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION REQUEST NUMBER 909-7
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	0	1	0	0	0	2	1	0	0	0	0	0	0	0	0	5	6.2	2.68
3.6 -7.5	4	3	2	3	3	2	3	3	4	4	4	2	5	2	2	1	47	58.0	5.42
7.6 - 12.5	0	2	1	1	2	4	3	2	3	2	3	2	1	0	2	2	29	35.8	8.74
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	5	5	4	4	5	8	7	7	6	6	7	4	6	2	4	3	81	0.0	5.84
PERCENT	6.2	6.2	4.9	4.9	6.2	7.4	8.6	7.4	7.4	8.6	4.9	7.4	7.4	2.5	4.9	3.7	100.0		
AV SPD	4.1	7.2	5.5	6.5	6.5	6.9	5.4	7.0	6.8	6.3	8.6	8.0	5.8	7.1	7.6	8.8			

AVERAGE SPEED FOR THIS TABLE EQUALS6.7
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 22 OF 48)

MONTH OF JUNE

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0	2	9.5	2.71
3.6 -7.5	1	1	2	0	0	0	2	2	0	2	0	0	1	0	0	0	11	52.4	5.98
7.6 - 12.5	0	1	0	0	0	0	0	0	3	0	1	1	1	0	1	0	8	38.1	9.51
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	2	2	0	0	0	2	2	4	2	2	1	2	0	1	0	21	0.0	6.14
PERCENT	4.8	9.5	9.5	0.0	0.0	0.0	9.5	9.5	19.0	9.5	4.8	6.4	9.5	0.0	4.8	0.0	100.0		
AV SPD	6.3	7.7	5.8	0.0	0.0	0.0	6.0	5.6	8.3	6.1	6.2	9.7	8.3	0.0	9.4	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	4	5	2	3	5	3	5	3	2	3	5	3	1	1	2	48	15.1	2.72
3.6 -7.5	3	9	7	9	17	11	14	21	15	15	16	18	11	1	4	5	176	55.5	5.41
7.6 - 12.5	3	2	3	3	10	8	6	9	3	7	10	7	1	3	7	5	87	27.4	8.81
12.6 - 18.5	0	0	0	2	0	0	0	0	0	0	0	1	0	1	0	1	6	1.9	14.37
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	7	15	15	16	30	24	23	35	21	25	19	31	16	5	12	13	317	0.0	5.24
PERCENT	2.2	4.7	4.7	5.0	9.5	7.6	7.3	11.0	6.6	7.9	9.1	9.8	5.0	1.6	3.8	4.1	100.0		
AV SPD	6.7	5.1	5.5	6.6	6.5	6.5	5.8	6.0	5.6	6.9	6.5	6.5	5.8	7.1	7.8	7.1			

AVERAGE SPEED FOR THIS TABLE EQUALS6.3
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-14 (SHEET 23 OF 48)

MONTH OF JUNE

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	.7	.50
CALM + -3.5	4	6	11	5	18	6	22	18	12	11	9	6	4	4	6	2	144	26.7	2.27
3.6 -7.5	7	3	6	12	23	18	27	44	45	28	34	32	31	11	2	8	331	61.4	5.12
7.6 - 12.5	1	1	4	0	1	4	2	7	9	9	9	4	5	1	0	1	58	10.8	8.92
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	2	.4	16.81
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	12	10	22	17	42	28	51	70	66	48	53	43	40	16	8	13	539	0.0	3.78
PERCENT	2.2	1.9	4.1	3.2	7.8	5.2	9.5	13.0	12.2	8.9	9.8	8.0	7.4	3.0	1.5	2.4	100.0		
AV SPD	5.0	3.7	4.5	4.5	4.3	5.4	4.	4.7	5.5	5.5	5.7	5.7	5.5	4.6	2.9	4.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 5.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	.7	.50
CALM + -3.5	8	7	14	5	6	9	13	16	6	6	10	2	7	9	4	4	126	46.2	2.12
3.6 -7.5	5	10	7	9	1	8	13	5	12	13	20	23	11	4	0	0	141	51.6	4.88
7.6 - 12.5	0	0	1	0	0	0	2	0	0	0	0	0	0	0	0	0	3	1.1	9.03
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	.4	14.27
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	13	17	24	14	7	17	28	21	18	19	30	25	18	14	4	4	273	0.0	2.94
PERCENT	4.8	6.2	8.8	5.1	2.6	6.2	10.3	7.7	6.6	7.0	11.0	9.2	6.6	5.1	1.5	1.5	100.0		
AV SPD	3.6	3.9	3.2	3.8	2.7	3.3	4.0	2.7	4.2	4.5	4.4	5.1	4.3	3.9	1.8	2.0			

AVERAGE SPEED FOR THIS TABLE EQUALS3.9
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 24 OF 48)

MONTH OF JUNE

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0.0	0.00
CALM + -3.5	2	3	4	4	3	4	3	3	3	1	1	1	5	4	5	6	52	56.5	2.15
3.6 -7.5	0	1	1	1	1	1	2	3	1	4	5	4	3	3	5	3	38	41.3	4.47
7.6 - 12.5	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	2	2.2	9.47
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	4	5	5	4	5	5	6	4	5	8	5	8	7	10	9	92	0.0	2.80
PERCENT	2.2	4.3	5.4	5.4	4.3	5.4	5.4	6.5	4.3	5.4	8.7	5.4	8.7	7.6	10.9	9.8	100.0		
AV SPD	3.3	2.6	3.1	3.0	2.7	2.5	3.3	3.3	3.0	4.3	5.7	4.4	3.5	3.3	3.2	3.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF JUNE
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	3	0	0	0	0	1	0	0	0	0	0	0	0	2	6	.3
CALM + -3.5	17	23	40	20	32	24	44	45	25	20	27	16	20	20	19	14	406	23.5
3.6 -7.5	39	42	44	47	70	54	70	82	80	69	87	89	87	36	19	22	937	54.1
7.6 - 12.5	9	9	19	15	22	31	17	24	27	21	32	25	51	27	22	11	362	20.9
12.6 - 18.5	1	1	0	2	0	0	0	1	0	1	1	5	4	0	2	20	20	1.2
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	66	75	106	84	124	109	131	153	132	111	147	135	162	85	60	51	1731	100.0
PERCENT	3.8	4.3	6.1	4.9	7.2	6.3	7.6	8.8	7.6	6.4	8.5	7.8	9.4	4.9	3.5	2.9	100.0	

TABLE 2.3.2-14 (SHEET 25 OF 48)

MONTH OF JULY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	1	2	2	0	0	0	0	0	0	1	0	3	1	0	1	12	4.0	2.98
3.6 -7.5	2	8	5	14	7	9	4	0	1	3	13	26	50	21	9	4	176	58.9	5.64
7.6 - 12.5	2	0	2	7	23	9	2	0	0	3	10	19	22	8	2	0	109	2.2	9.47
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	2	.7	13.05
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	5	9	9	23	30	18	6	0	1	6	24	46	75	31	11	5	229	0.0	6.26
PERCENT	1.7	3.0	3.0	7.7	10.0	6.0	2.0	0.0	.3	2.0	8.0	15.4	25.1	10.4	3.7	1.7	100.0		
AV SPD	6.5	5.2	6.0	6.6	8.9	7.6	7.4	0.0	5.8	7.0	7.0	7.2	6.4	6.6	6.2	4.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 5

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	2	3.6	2.35
3.6 -7.5	1	2	2	0	2	2	0	0	3	2	2	4	4	3	0	3	30	53.5	5.29
7.6 - 12.5	0	1	0	0	3	2	2	0	0	2	3	4	3	0	0	0	20	36.4	8.90
12.6 - 18.5	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	1	1.8	13.10
18.6 - 24.5	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	2	3.6	20.73
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	3	3	0	5	4	2	0	3	6	5	5	8	4	0	3	55	0.0	6.15
PERCENT	1.8	5.5	5.5	0.0	9.1	7.3	3.6	0.0	5.5	10.9	9.1	14.5	14.5	7.3	0.0	5.5	100.00		
AV SPD	4.9	6.9	9.8	0.0	7.0	8.0	10.0	0.0	4.6	1.6	7.3	7.7	6.0	4.4	0.0	5.5			

AVERAGE SPEED FOR THIS TABLE EQUALS7.4
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-14 (SHEET 26 OF 48)

MONTH OF JULY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	1	0	0	0	0	0	0	0	0	1	2	0	0	0	4	4.0	2.98
3.6 -7.5	0	0	2	3	3	3	0	1	2	0	0	3	5	0	1	1	24	58.9	5.64
7.6 - 12.5	0	0	0	2	1	1	0	0	1	0	1	2	0	0	1	0	9	2.2	9.47
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.7	13.05
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	0	0	3	5	4	4	0	1	3	0	1	6	7	0	2	1	37	0.0	6.26
PERCENT	0.0	0.0	8.1	10.5	10.8	10.8	0.0	2.7	8.1	0.0	2.7	16.2	18.9	0.0	5.4	2.7	100.0		
AV SPD	0.0	0.0	3.7	6.6	6.8	6.7	0.0	5.7	6.4	0.0	9.3	6.3	4.6	0.0	8.5	4.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	0	2	0	2	4	1	4	2	3	3	0	3	1	0	0	26	9.7	2.60
3.6 -7.5	2	2	14	12	12	17	11	10	12	14	15	16	16	8	3	2	166	62.2	5.35
7.6 - 12.5	1	1	3	5	6	5	7	5	10	10	12	5	0	1	0	1	72	27.0	8.84
12.6 - 18.5	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	3	1.1	16.13
18.6 - 24.5	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0.0	20.73
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	5	3	19	17	20	26	19	19	25	28	30	21	19	10	3	3	267	0.0	5.41
PERCENT	1.9	1.1	7.1	6.4	7.5	9.7	7.1	7.1	9.4	10.5	11.2	7.9	7.1	3.7	1.1	1.1	100.00		
AV SPD	7.2	5.7	6.0	6.8	6.5	5.7	6.7	5.6	7.3	7.1	6.9	6.2	5.1	5.2	4.5	6.3			

AVERAGE SPEED FOR THIS TABLE EQUALS6.4
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-14 (SHEET 27 OF 48)

MONTH OF JULY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO - .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	.0	.50
CALM + -3.5	5	3	6	5	5	11	13	18	11	8	11	17	17	15	5	2	152	3.3	2.49
3.6 -7.5	6	4	11	15	17	20	32	37	62	61	74	39	17	3	6	4	408	23.3	5.14
7.6 - 12.5	3	0	2	5	3	2	6	12	25	17	9	3	3	0	0	0	90	62.5	8.59
12.6 - 18.5	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	13.8	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.3	14.02
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	14	8	20	25	25	33	51	67	98	86	95	59	37	0	11	6	653	0.0	4.27
PERCENT	2.1	1.2	3.1	3.8	3.8	5.1	7.8	10.3	15.0	13.2	4.5	9.0	5.7	2.8	1.7	.9	100.0		
AV SPD	5.1	5.0	5.1	5.5	5.0	4.7	4.9	5.3	6.1	5.8	5.4	4.8	3.9	3.0	4.2	4.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 5.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	2	2	3	6	3	15	9	7	10	11	11	4	1	2	1	87	38.2	2.69
3.6 -7.5	0	3	5	8	8	10	11	17	11	14	15	18	8	5	2	1	138	60.5	4.60
7.6 - 12.5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	2	.9	9.03
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	.4	12.70
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	5	5	7	11	15	13	26	26	18	24	26	30	12	6	5	2	228	0.0	3.64
PERCENT	.9	2.2	3.1	4.8	6.6	5.7	11.4	11.4	7.9	10.5	11.4	13.2	5.3	2.6	2.2	.9	100.0		
AV SPD	3.4	4.0	3.9	4.4	5.1	4.4	3.6	3.9	4.1	4.2	3.9	4.1	4.5	4.5	5.0	7.0			

AVERAGE SPEED FOR THIS TABLE EQUALS4.1
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 28 OF 48)

MONTH OF JULY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	3	1	1	1	0	3	2	2	1	3	2	0	0	0	0	0	12	27.3	2.66
3.6 -7.5	0	1	4	0	2	0	0	0	4	4	2	3	1	1	0	1	30	68.2	4.36
7.6 - 12.5	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	2	4.5	8.38
12.6 - 18.5	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	2	6	1	2	3	2	4	5	7	4	3	1	1	0	2	44	0.0	3.78
PERCENT	6.8	4.5	13.6	2.3	4.5	6.8	4.5	4.5	11.4	15.9	9.1	6.8	2.3	2.3	0.0	4.5	100.0		
AV SPD	2.8	4.1	4.6	2.9	4.8	4.3	4.7	4.0	3.8	3.5	3.6	4.7	4.6	7.2	0.0	6.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 4.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF JULY
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0.0
CALM + -3.5	11	7	14	11	13	18	29	31	21	24	28	29	30	19	7	3	295	38.2
3.6 -7.5	12	20	43	52	52	64	60	67	95	98	121	109	101	41	21	16	972	60.5
7.6 - 12.5	6	2	8	19	36	19	17	17	36	32	35	34	28	9	3	1	304	.9
12.6 - 18.5	1	1	1	0	0	0	0	0	1	0	0	1	0	1	1	3	9	.4
18.6 - 24.5	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0.0
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	30	30	67	82	101	101	106	115	153	157	185	173	159	70	32	22	1583	0.0
PERCENT	1.9	1.9	4.2	5.2	6.4	6.4	6.7	7.3	9.7	9.9	11.7	10.9	10.0	4.4	2.0	1.4	100.0	

TABLE 2.3.2-14 (SHEET 29 OF 48)

MONTH OF AUGUST

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO - 1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.0	0.00
CALM + -3.5	3	4	2	2	0	2	0	2	3	0	2	1	1	2	4	4	32	10.7	2.39
3.6 -7.5	20	16	13	10	13	10	1	0	2	0	11	30	31	12	6	7	182	1.1	5.46
7.6 - 12.5	0	3	10	5	11	11	4	2	0	0	10	21	6	0	0	0	84	8.2	0.00
12.6 - 18.5	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0.0	0.00
TOTAL	23	23	25	17	24	23	5	4	5	0	23	52	38	14	10	12	218	0.0	5.32
PERCENT	7.7	7.7	8.4	5.7	8.1	7.7	1.7	1.3	1.7	0.0	7.7	17.4	12.8	4.7	3.4	4.0	100.0		
AV SPD	5.0	5.5	7.1	6.3	7.5	7.3	8.8	6.1	3.6	0.0	7.4	7.0	5.9	5.2	3.8	4.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	4	1	1	0	3	0	0	1	0	0	0	0	0	0	0	87	7.0	2.37
3.6 -7.5	1	2	7	2	5	10	1	1	0	2	7	5	1	0	0	0	138	64.9	5.41
7.6 - 12.5	0	0	2	1	0	0	3	0	0	0	4	1	0	0	0	0	2	28.1	9.05
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	6	10	4	5	13	4	1	1	2	11	6	1	2	0	0	57	0.0	5.54
PERCENT	3.5	10.5	17.5	7.0	8.8	3.5	7.0	1.8	1.8	3.5	19.3	10.5	1.8	3.5	0.0	0.0	100.0		
AV SPD	4.2	6.6	5.2	6.1	6.7	7.6	7.2	6.1	3.3	7.0	7.1	6.6	4.4	10.3	0.0	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS6.4
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 30 OF 48)

MONTH OF AUGUST

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.0	0.00
CALM + -3.5	1	0	0	2	1	1	0	0	1	0	0	1	0	0	0	2	5	10.9	2.80
3.6 -7.5	0	2	0	1	1	4	0	0	2	3	3	6	4	4	1	0	31	67.4	5.35
7.6 - 12.5	0	0	0	0	1	3	0	0	1	0	1	1	0	0	0	0	9	19.6	8.88
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	2.2	14.70
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	2	3	3	2	8	0	0	4	3	4	7	4	1	4	1	45	0.0	5.31
PERCENT	2.2	4.3	0.0	6.5	4.3	17.4	4.3	0.0	8.7	6.5	8.7	15.2	8.7	2.2	8.7	2.2	100.0		
AV SPD	2.9	4.6	0.0	6.9	6.1	6.8	8.5	0.0	5.2	5.5	6.9	6.9	5.6	5.3	3.9	14.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	.3	.50
CALM + -3.5	2	2	5	3	7	3	4	5	6	8	10	3	7	3	3	3	74	19.8	2.39
3.6 -7.5	6	7	18	17	18	19	19	19	13	24	29	17	3	5	2	2	218	58.4	5.33
7.6 - 12.5	3	2	4	5	7	12	12	7	6	6	10	2	0	1	0	1	78	20.9	8.91
12.6 - 18.5	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0	2	.5	14.40
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	11	11	27	25	32	34	36	31	25	39	49	23	10	9	5	5	373	0.0	4.50
PERCENT	2.9	2.9	7.2	6.7	8.6	9.1	9.7	8.3	6.7	10.5	13.1	6.2	2.7	2.4	1.3	1.6	100.0		
AV SPD	6.0	6.1	5.4	6.0	6.0	6.5	6.8	5.7	5.7	5.7	5.6	4.7	3.2	5.3	3.7	4.7			

AVERAGE SPEED FOR THIS TABLE EQUALS5.7
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-14 (SHEET 31 OF 48)

MONTH OF AUGUST

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	4	.5	.50
CALM + -3.5	7	16	11	11	24	14	33	27	16	21	14	15	13	2	6	3	233	30.1	2.21
3.6 -7.5	14	9	23	29	49	45	46	52	58	75	42	7	4	7	6	4	470	60.6	4.89
7.6 - 12.5	2	0	0	1	1	1	8	10	20	10	6	3	0	0	0	0	65	8.4	8.71
12.6 - 18.5	0	1	0	0	0	0	1	0	0	0	0	0	0	0	1	0	3	.4	14.56
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	23	26	34	41	77	60	88	90	94	107	62	26	18	9	13	7	775	0.0	3.57
PERCENT	3.0	3.4	4.4	5.3	9.9	7.7	11.4	11.6	12.1	13.8	8.0	3.4	2.3	1.2	1.7	.9	100.0		
AV SPD	4.4	3.8	4.1	4.5	4.4	4.7	4.6	4.5	5.7	5.0	4.9	3.8	3.1	3.7	4.5	3.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 4.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 4

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	1	0	0	0	0	0	0	1	0	1	0	0	0	0	3	.8	.50
CALM + -3.5	12	13	15	9	15	15	20	9	11	17	16	10	9	11	6	8	196	54.7	2.17
3.6 -7.5	4	5	12	9	11	17	15	14	22	21	14	3	1	2	4	1	155	43.3	4.35
7.6 - 12.5	0	0	0	0	0	0	0	1	0	0	10	0	1	0	0	0	3	.8	8.33
12.6 - 18.5	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	.3	12.60
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	16	18	28	18	27	32	35	25	33	39	31	13	11	13	10	9	358	0.0	2.71
PERCENT	4.5	5.0	7.8	5.0	7.5	8.9	9.8	7.0	9.2	10.9	8.7	3.6	3.1	3.6	2.8	2.5	100.0		
AV SPD	2.5	2.7	3.1	3.6	3.6	3.5	3.3	4.3	3.9	3.7	3.4	3.0	3.4	2.6	3.1	2.7			

AVERAGE SPEED FOR THIS TABLE EQUALS3.4
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 32 OF 48)

MONTH OF AUGUST

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2.5	.50
CALM + -3.5	5	3	1	3	3	0	2	1	2	4	3	0	2	4	0	3	36	45.1	2.39
3.6 -7.5	4	3	6	1	0	1	0	1	4	8	8	2	0	1	2	1	42	52.5	4.48
7.6 - 12.5	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	9	8	7	4	3	1	2	2	6	12	11	2	2	5	2	4	80	0.0	2.81
PERCENT	11.3	10.0	8.8	5.0	3.8	1.3	2.5	2.5	7.5	15.0	13.8	2.5	2.5	6.3	2.5	5.0	100.0		
AV SPD	4.0	2.9	4.1	3.4	2.2	4.2	2.1	3.0	4.3	3.8	4.0	4.4	2.8	3.3	4.3	3.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF AUGUST
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	2	1	0	0	0	0	1	0	2	1	2	0	0	0	0	0	10	.5
CALM + -3.5	31	38	35	29	49	35	59	44	40	50	45	29	32	22	21	21	21	580	29.2
3.6 -7.5	49	46	79	70	97	97	83	87	101	133	114	70	44	28	22	15	15	1135	57.1
7.6 - 12.5	5	7	16	13	24	28	28	20	27	16	31	28	7	3	0	2	2	255	12.8
12.6 - 18.5	0	1	0	0	0	0	2	1	0	1	0	0	0	0	1	1	7	7	.4
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
TOTAL	85	94	131	112	170	160	172	153	168	202	191	129	84	53	44	39	39	1987	100.00
PERCENT	4.3	4.7	6.6	5.6	8.6	8.1	8.7	7.7	8.5	10.2	9.6	6.5	4.2	2.7	2.2	2.0	2.0	100.0	

TABLE 2.3.2-14 (SHEET 33 OF 48)

MONTH OF SEPTEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.5	0.00
CALM + -3.5	2	3	1	0	0	1	0	2	0	0	1	2	3	3	3	4	25	6.8	2.73
3.6 -7.5	18	15	23	19	20	9	7	5	2	7	11	11	19	15	11	12	204	55.4	5.53
7.6 - 12.5	3	3	18	17	24	6	1	3	5	2	3	19	21	1	3	3	132	35.9	8.99
12.6 - 18.5	0	2	1	2	0	0	0	0	0	0	0	2	0	0	0	0	7	1.9	13.10
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	23	23	43	38	44	16	8	10	7	9	15	34	43	19	17	19	368	0.0	6.01
PERCENT	6.3	6.3	11.7	15.3	12.0	4.3	2.2	2.7	1.9	2.4	4.1	9.2	11.7	5.2	4.6	5.2	100.0		
AV SPD	6.0	6.2	7.5	7.6	8.0	6.7	6.7	6.2	8.3	6.4	5.7	7.9	7.2	5.1	5.7	5.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 7

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM		0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
CALM + -3.5	1	0	2	0	0	1	0	0	1	0	0	1	0	1	0	1	8	11.8	2.69
3.6 -7.5	1	6	4	3	1	1	1	2	2	0	3	2	0	2	4	2	34	50.0	5.35
7.6 - 12.5	0	2	2	6	4	0	0	1	1	2	1	1	0	0	0	2	23	33.8	9.21
12.6 - 18.5	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	3	4.4	13.80
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	8	9	10	5	2	1	3	4	3	4	4	0	3	4	5	68	0.0	5.64
PERCENT	4.4	11.8	13.2	14.7	7.4	2.9	1.5	4.4	5.9	4.4	5.9	5.9	0.0	4.4	5.9	7.4	100.0		
AV SPD	5.2	5.9	6.9	8.9	8.5	4.6	7.3	6.0	6.6	11.2	6.6	6.2	0.0	4.7	4.9	6.8			

AVERAGE SPEED FOR THIS TABLE EQUALS6.9
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 5

TABLE 2.3.2-14 (SHEET 34 OF 48)

MONTH OF SEPTEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	2	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	4	12.5	2.70
3.6 -7.5	1	1	2	0	1	1	0	2	1	0	11	0	2	1	1	3	16	50.0	5.22
7.6 - 12.5	0	0	1	2	1	1	1	0	0	1	0	0	0	0	0	2	2	31.3	9.03
12.6 - 18.5	0	0	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	6.3	13.22
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	1	3	4	2	2	2	2	1	1	2	0	2	1	1	5	32	0.0	5.51
PERCENT	9.4	3.1	9.4	12.5	6.3	6.3	6.3	6.3	3.1	3.1	6.3	0.0	6.3	3.1	3.1	15.6	100.0		
AV SPD	3.6	6.6	7.0	11.3	6.2	5.9	4.9	6.1	5.6	11.4	7.6	0.0	5.9	5.9	6.9	6.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
CALM + -3.5	4	2	5	6	3	3	1	3	1	0	3	1	3	1	3	7	49	13.4	2.63
3.6 -7.5	16	15	27	21	15	23	9	12	7	16	10	2	4	6	7	10	201	54.8	5.36
7.6 - 12.5	5	4	38	18	9	7	0	2	6	10	5	1	1	0	0	0	106	28.9	9.11
12.6 - 18.5	0	0	4	4	2	0	0	0	0	0	0	0	0	0	0	0	10	2.7	14.34
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.3	21.70
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	25	21	74	49	29	33	11	17	14	26	18	4	8	7	10	17	367	0.0	5.36
PERCENT	6.8	5.7	20.2	13.4	7.9	9.0	3.0	4.5	3.8	7.1	4.9	5.9	2.2	1.9	2.7	4.6	100.0		
AV SPD	5.8	5.8	8.0	7.5	7.1	6.1	5.9	5.2	6.7	6.9	6.2	6.2	4.8	4.3	4.4	4.4			

AVERAGE SPEED FOR THIS TABLE EQUALS6.6
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 35 OF 48)

MONTH OF SEPTEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1	0	0	0	2	0	0	0	0	0	0	1	0	0	0	0	4	.6	.50
CALM + -3.5	10	13	11	19	11	15	15	12	8	14	12	11	8	7	9	8	183	27.2	2.29
3.6 -7.5	23	27	68	46	34	30	25	21	42	27	25	24	5	8	14	13	432	64.2	4.89
7.6 - 12.5	0	5	11	9	6	3	5	5	4	2	0	0	0	1	0	1	53	7.9	8.81
12.6 - 18.5	1	0	1	2	0	0	0	0	0	0	0	0	0	0	0	0	1	.1	17.30
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	35	45	90	74	53	48	45	38	54	43	37	37	13	16	23	22	673	0.0	3.69
PERCENT	5.2	6.7	13.4	11.0	7.9	7.1	6.7	5.6	8.0	6.4	5.5	5.5	1.9	2.4	3.4	3.3	100.0		
AV SPD	4.3	4.8	5.6	5.0	4.8	4.3	4.8	5.1	5.2	4.4	4.3	4.1	3.3	3.9	3.9	4.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 4.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	2	2	0	0	0	0	0	0	1	0	0	0	0	0	0	5	1.5	.50
CALM + -3.5	6	12	4	12	16	11	11	8	4	6	9	10	4	7	9	10	139	41.5	2.32
3.6 -7.5	6	13	36	28	23	19	6	7	9	8	7	5	7	3	6	7	190	56.7	4.50
7.6 - 12.5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	.3	8.40
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	12	27	42	40	39	30	17	15	13	15	16	16	11	10	15	17	335	0.0	2.98
PERCENT	3.6	8.1	12.5	11.9	11.6	9.0	5.1	4.5	3.9	4.5	4.8	4.8	3.3	3.0	4.5	5.1	100.0		
AV SPD	3.2	3.6	4.5	4.2	3.8	4.0	3.1	3.3	4.1	3.4	3.3	3.3	3.5	3.0	3.3	3.4			

AVERAGE SPEED FOR THIS TABLE EQUALS3.7
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 36 OF 48)

MONTH OF SEPTEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	3	0	1	1	0	0	2	0	1	0	0	0	0	0	0	0	9	6.7	.50
CALM + -3.5	6	6	3	0	8	1	3	4	1	1	3	10	2	3	7	11	69	51.0	2.20
3.6 -7.5	3	2	1	2	2	4	3	2	4	3	6	9	4	1	5	4	55	41.0	4.26
7.6 - 12.5	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	7.9	8.60
12.6 - 18.5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.7	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	12	9	5	3	10	5	8	6	6	4	9	19	6	4	12	16	134	0.0	2.15
PERCENT	9.0	6.7	3.7	2.2	7.5	3.7	6.0	4.5	4.5	3.0	6.7	14.2	4.5	3.0	9.0	11.9	100.0		
AV SPD	2.4	3.7	2.1	2.6	2.9	4.3	2.4	3.2	3.3	4.8	4.0	3.4	3.5	2.9	2.9	3.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF SEPTEMBER
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	4	2	3	1	2	0	2	0	1	1	0	1	0	0	0	1	18	.9
CALM + -3.5	31	36	26	37	38	32	31	29	15	21	29	38	20	22	31	41	477	24.1
3.6 -7.5	68	79	161	119	96	87	51	51	67	61	62	54	41	36	48	51	1320	57.3
7.6 - 12.5	9	15	70	52	44	17	8	11	16	17	10	22	22	2	3	8	326	18.5
12.6 - 18.5	1	2	6	9	2	0	0	0	0	1	0	0	0	0	0	0	23	1.2
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.1
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	113	134	266	218	182	136	92	91	99	101	101	118	83	60	82	101	1977	100.0
PERCENT	5.7	6.8	13.5	11.0	9.2	6.9	4.7	4.6	5.0	5.1	5.1	6.0	4.2	3.0	4.1	5.1	100.0	

TABLE 2.3.2-14 (SHEET 37 OF 48)

MONTH OF OCTOBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	0	5	0	0	0	1	0	0	0	0	0	1	1	2	2	13	3.1	2.87
3.6 -7.5	14	19	25	14	21	10	2	1	2	3	6	9	11	15	15	15	173	40.6	5.56
7.6 - 12.5	22	26	40	22	20	5	1	0	0	3	4	13	15	10	15	15	210	49.3	9.39
12.6 - 18.5	1	5	1	0	1	0	0	0	0	0	1	1	11	3	4	0	30	7.0	13.81
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	38	50	71	38	42	15	4	1	2	6	11	23	38	29	36	36	426	0.0	7.08
PERCENT	6.9	11.7	16.7	8.5	9.9	3.8	.9	.2	.5	1.4	2.6	5.4	8.0	6.8	8.5	8.5	100.0		
AV SPD	7.9	8.3	8.1	8.5	7.5	7.2	5.4	4.2	4.0	7.4	8.1	8.4	9.5	8.2	8.2	8.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 8

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	0	0	2	0	0	0	0	0	0	0	1	0	0	0	3	5.9	2.88
3.6 -7.5	3	3	3	2	4	1	1	0	0	0	2	1	1	0	1	2	24	47.1	5.32
7.6 - 12.5	0	5	8	2	1	0	2	0	0	0	0	1	1	0	0	2	22	43.1	8.98
12.6 - 18.5	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2	3.9	13.09
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	8	12	5	7	1	3	0	0	0	2	2	3	10	1	4	51	0.0	6.25
PERCENT	5.9	15.7	23.5	9.8	13.7	2.0	5.9	0.0	0.0	0.0	3.9	3.9	5.9	0.0	2.0	7.8	100.0		
AV SPD	7.1	7.6	8.3	8.2	4.7	6.3	7.2	0.0	0.0	0.0	5.0	8.6	7.2	0.0	5.0	8.2			

AVERAGE SPEED FOR THIS TABLE EQUALS7.3
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 38 OF 48)

MONTH OF OCTOBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0	3	11.1	2.26
3.6 -7.5	5	0	3	1	0	1	2	1	2	0	0	0	0	0	1	2	16	59.3	5.39
7.6 - 12.5	0	2	3	0	0	0	0	0	0	1	0	0	1	0	1	0	8	29.3	9.09
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	5	2	6	1	0	1	2	1	0	2	0	0	2	0	3	2	27	0.0	5.21
PERCENT	18.5	7.4	22.2	3.7	0.0	3.7	7.4	3.7	0.0	7.4	0.0	0.0	7.4	0.0	11.1	7.4	100.0		
AV SPD	6.5	9.6	7.0	6.8	0.0	3.7	3.8	4.7	0.0	5.9	0.0	0.0	6.0	0.0	6.3	7.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	4	0	3	0	3	1	1	2	1	1	0	1	3	1	0	2	23	7.9	2.77
3.6 -7.5	14	8	15	20	12	11	5	6	4	5	5	4	5	5	11	9	139	47.6	5.40
7.6 - 12.5	4	21	25	7	13	4	3	1	8	8	6	6	4	5	3	7	125	42.8	9.23
12.6 - 18.5	0	1	2	0	0	0	0	0	1	0	0	0	0	1	0	0	5	1.7	14.20
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	22	30	45	27	28	16	9	9	14	14	11	11	12	12	14	18	292	0.0	6.09
PERCENT	7.5	10.3	15.4	9.2	9.6	5.5	3.1	3.1	4.8	4.8	3.8	3.5	4.1	4.1	4.8	6.2	100.0		
AV SPD	5.7	8.9	8.0	6.9	7.0	6.1	5.9	6.0	8.4	8.0	7.6	7.2	6.3	4.1	5.7	6.3			

AVERAGE SPEED FOR THIS TABLE EQUALS7.2
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 4

TABLE 2.3.2-14 (SHEET 39 OF 48)

MONTH OF OCTOBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	7	9	5	6	8	0	8	4	2	1	7	5	2	2	4	5	79	16.5	2.50
3.6 -7.5	19	22	32	27	22	10	11	16	8	9	17	22	12	14	9	15	265	55.4	5.36
7.6 - 12.5	10	21	15	10	6	3	2	1	1	5	7	8	15	10	8	5	127	26.6	8.97
12.6 - 18.5	1	2	0	0	0	0	0	0	0	0	0	0	1	0	2	0	7	0.0	13.88
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	37	54	52	43	36	17	21	22	11	15	31	35	30	26	23	25	478	0.0	5.00
PERCENT	7.7	11.4	10.9	9.0	7.5	3.6	4.4	4.8	2.3	3.1	6.5	7.3	6.3	5.4	4.8	5.2	100.0		
AV SPD	6.3	6.9	6.4	6.0	5.1	5.3	4.7	5.5	5.3	6.5	5.7	6.1	7.8	6.8	6.8	5.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	2	.5	.50
CALM + -3.5	5	0	9	4	8	2	4	7	5	5	6	7	13	7	5	7	103	26.5	2.55
3.6 -7.5	20	8	42	28	23	12	8	5	9	5	12	11	12	25	16	14	273	70.2	4.83
7.6 - 12.5	1	31	2	7	0	4	0	1	8	1	2	6	4	5	1	3	11	2.8	8.23
12.6 - 18.5	0	1	2	0	0	0	0	0	1	0	0	0	0	1	0	0	5	0.0	14.20
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	26	41	54	32	31	15	12	12	14	11	20	18	25	32	22	24	389	0.0	3.81
PERCENT	6.7	10.5	13.9	8.2	8.0	3.9	3.1	3.1	3.6	2.8	5.1	4.6	6.4	8.2	5.7	6.2	100.0		
AV SPD	4.7	4.6	4.8	4.8	4.5	4.2	4.2	3.5	4.1	4.8	5.1	4.1	3.6	4.4	4.6	4.6			

AVERAGE SPEED FOR THIS TABLE EQUALS4.5
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-14 (SHEET 40 OF 48)

MONTH OF OCTOBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	3	1	1	2	2	0	0	0	1	0	2	1	1	1	2	0	17	4.0	.50
CALM + -3.5	33	20	15	6	7	3	5	4	6	6	14	10	5	10	11	17	168	39.1	2.17
3.6 -7.5	50	28	21	11	23	7	4	1	4	5	22	25	12	6	7	18	244	56.7	4.57
7.6 - 12.5	0	0	0	10	6	0	0	1	0	0	0	8	0	0	0	1	1	.2	7.60
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	86	49	37	19	32	10	9	5	11	11	38	36	18	17	20	32	478	0.0	2.61
PERCENT	20.0	11.4	8.6	4.4	7.4	2.3	2.1	1.2	2.6	2.6	8.8	8.4	4.2	4.0	4.7	7.4	100.0		
AV SPD	3.6	3.6	3.7	3.7	3.9	4.2	3.3	2.5	2.9	3.7	3.6	4.0	3.6	2.9	3.2	3.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

MONTH OF OCTOBER
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	3	1	2	2	2	1	0	0	0	0	2	1	1	1	2	0	19	.9
CALM + -3.5	50	38	37	16	28	10	19	17	14	45	27	23	26	21	23	29	392	18.7
3.6 -7.5	125	111	141	103	105	52	33	30	27	27	64	72	53	65	60	66	1134	54.2
7.6 - 12.5	37	76	93	41	40	12	8	2	9	18	19	28	36	25	28	32	504	24.1
12.6 - 18.5	2	8	4	1	0	0	0	1	1	0	1	0	12	4	0	2	44	2.1
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	217	234	277	163	176	75	60	50	52	59	113	125	128	116	119	129	2093	100.0
PERCENT	10.4	11.2	13.2	7.8	8.4	3.8	2.9	2.4	2.5	2.8	5.4	6.0	6.1	5.5	5.7	7.2	100.0	

TABLE 2.3.2-14 (SHEET 41 OF 48)

MONTH OF NOVEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	2	1	1	1	1	0	2	2	0	0	0	3	3	2	0	0	17	5.6	2.83
3.6 -7.5	10	16	15	4	5	3	4	1	3	2	7	8	18	18	4	4	126	41.6	5.63
7.6 - 12.5	8	9	16	9	5	8	5	3	2	6	14	8	8	13	19	4	141	46.5	9.42
12.6 - 18.5	1	1	2	0	0	0	0	0	0	0	0	0	0	3	4	0	18	5.9	13.58
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	.3	19.70
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	21	27	34	14	11	11	11	6	5	8	22	19	36	36	27	13	313	0.0	6.77
PERCENT	6.9	8.9	11.2	4.6	3.6	3.6	3.6	2.0	1.7	2.6	7.3	6.3	11.9	11.9	8.9	4.3	100.0		
AV SPD	7.1	7.3	8.1	7.7	7.6	8.5	7.0	6.8	6.4	8.8	8.6	6.5	7.9	7.9	9.9	7.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
3.6 -7.5	1	4	3	0	1	0	1	0	0	1	0	2	1	0	1	0	15	42.9	5.30
7.6 - 12.5	1	2	4	0	0	0	2	1	0	0	3	0	2	1	3	0	19	54.3	8.99
12.6 - 18.5	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2.9	13.50
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	6	7	0	1	0	3	1	0	1	3	2	3	1	4	0	35	0.0	6.97
PERCENT	8.6	17.1	20.9	0.0	2.9	0.0	8.6	2.9	0.0	2.9	8.6	5.7	8.6	2.9	11.4	0.0	100.0		
AV SPD	9.4	6.9	7.9	0.0	6.8	0.0	8.4	7.7	0.0	6.1	9.8	4.3	7.8	8.6	7.9	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS7.8
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 42 OF 48)

MONTH OF NOVEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	2	1.1	3.30
3.6 -7.5	0	0	1	0	0	2	1	0	1	2	1	0	0	1	0	1	10	45.5	5.09
7.6 - 12.5	1	1	2	0	0	0	0	0	1	1	0	0	0	3	0	0	10	45.5	8.81
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0.0	13.58
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	9.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0.0	0.00
TOTAL	1	1	4	0	0	2	3	0	2	3	1	0	0	4	0	1	21	0.0	5.94
PERCENT	4.5	4.5	18.2	0.0	0.0	9.1	13.6	0.0	9.1	13.6	4.5	0.0	0.0	18.2	0.0	4.5	100.0		
AV SPD	7.9	7.9	6.3	0.0	0.0	6.1	5.5	0.0	7.5	7.7	7.3	0.0	0.0	7.7	0.0	3.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	5	4	1	0	5	7	3	2	1	4	3	2	5	3	2	2	0	0.0	2.60
3.6 -7.5	8	21	22	21	15	14	8	10	10	15	9	5	19	6	6	4	52	42.2	5.19
7.6 - 12.5	4	3	18	19	3	3	8	8	9	17	6	8	3	3	5	3	184	50.3	9.10
12.6 - 18.5	1	0	0	0	0	0	0	0	0	0	1	0	0	2	2	1	130	32.6	13.57
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	2.7	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	18	29	41	44	23	24	19	20	21	36	19	15	18	14	15	10	366	7.0	5.28
PERCENT	4.9	7.9	11.2	12.0	6.3	6.6	5.2	5.5	5.7	9.8	5.2	4.1	4.9	3.8	4.1	2.7	100.0		
AV SPD	5.8	6.9	7.2	7.3	5.0	5.1	6.2	6.5	7.5	7.2	6.8	7.6	5.1	6.1	7.9	6.4			

AVERAGE SPEED FOR THIS TABLE EQUALS6.5
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 4

TABLE 2.3.2-14 (SHEET 43 OF 48)

MONTH OF NOVEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1	0	2	0	0	2	0	1	0	1	1	0	0	1	1	0	10	1.3	.50
CALM + -3.5	9	11	8	13	12	18	9	21	10	4	3	6	6	7	5	12	154	19.6	2.45
3.6 -7.5	20	27	42	60	43	34	27	21	42	45	22	18	17	26	22	11	477	18.7	5.34
7.6 - 12.5	5	6	16	11	8	11	6	7	2	8	14	2	5	14	11	7	141	???	8.93
12.6 - 18.5	0	0	0	0	0	0	0	0	0	1	0	0	0	3	0	0	3	0.0	13.76
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	35	44	68	84	63	65	42	50	64	59	40	26	28	48	39	30	785	0.0	4.17
PERCENT	4.5	5.6	8.7	10.7	8.0	8.3	5.4	6.4	8.2	7.5	5.1	3.3	3.6	6.1	5.0	3.8	100.0		
AV SPD	5.1	5.6	6.0	5.4	5.2	5.2	5.3	4.6	6.1	6.2	6.7	4.7	5.5	6.5	6.2	499			

AVERAGE SPEED FOR THIS TABLE EQUALS 5.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	1	3	1.1	.50
CALM + -3.5	5	1	4	2	3	8	9	11	1	2	9	1	5	7	3	1	70	26.6	2.37
3.6 -7.5	17	15	15	8	12	11	7	8	11	10	10	3	15	15	18	7	182	69.2	4.75
7.6 - 12.5	0	0	18	0	3	3	8	3	2	1	2	0	0	0	0	0	8	3.0	9.54
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	13.57
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	22	16	19	11	15	17	17	22	14	13	21	4	20	22	21	9	263	0.0	3.52
PERCENT	8.4	6.1	7.2	4.2	5.7	6.5	6.5	8.4	5.3	4.9	8.0	1.5	7.6	8.4	8.0	3.4	100.0		
AV SPD	4.8	4.9	4.7	4.3	4.3	4.2	3.3	4.3	5.1	4.8	4.5	4.2	4.2	4.2	4.6	3.8			

AVERAGE SPEED FOR THIS TABLE EQUALS4.4
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-14 (SHEET 44 OF 48)

MONTH OF NOVEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	3	2	1	1	3	2	0	0	0	0	0	1	1	0	2	0	17	5.7	.50
CALM + -3.5	16	11	8	7	7	6	13	11	10	10	10	9	3	3	6	9	139	46.3	2.19
3.6 -7.5	19	9	6	8	6	4	6	8	12	6	24	11	5	0	7	7	144	48.0	4.56
7.6 - 12.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	38	22	15	16	16	12	20	19	22	18	34	21	9	9	15	16	300	0.0	3.32
PERCENT	12.7	7.3	5.0	5.3	5.3	4.0	6.7	6.3	7.3	5.3	11.3	7.0	3.0	3.0	5.0	5.3	100.0		
AV SPD	3.5	3.0	3.2	3.4	2.9	3.0	3.1	3.6	3.8	3.4	4.0	3.4	3.1	3.9	3.1	3.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF NOVEMBER
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	4	2	3	2	3	4	2	1	0	1	1	1	1	1	3	1	10	1.4
CALM + -3.5	37	28	23	26	28	37	37	47	22	20	25	21	21	22	16	24	434	20.9
3.6 -7.5	75	92	104	101	82	68	54	48	79	81	73	45	66	72	58	38	1131	54.9
7.6 - 12.5	19	21	56	39	16	22	22	22	24	33	39	18	23	33	38	14	439	21.2
12.6 - 18.5	3	2	2	1	0	0	0	0	3	1	0	0	3	0	6	1	32	1.5
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0.0
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	138	145	188	169	129	131	115	118	128	136	140	87	114	136	121	79	2074	100.0
PERCENT	6.7	7.0	9.1	8.1	6.2	6.3	5.5	5.7	6.2	6.6	6.8	4.2	5.5	6.6	5.8	3.8	100.0	

TABLE 2.3.2-14 (SHEET 45 OF 48)

MONTH OF NOVEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO 1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	1	0	0	1	0	0	1	1	0	0	0.0	0.00
CALM + -3.5	1	1	6	0	1	1	0	1	0	2	1	1	1	1	1	2	22	8.2	2.70
3.6 -7.5	10	9	13	5	3	6	9	3	2	3	5	9	3	7	10	3	108	40.4	5.55
7.6 - 12.5	3	9	7	15	8	4	0	1	1	0	1	8	11	26	12	3	110	41.2	9.35
12.6 - 18.5	0	1	1	0	0	0	0	0	3	1	0	3	12	6	8	0	23	8.6	14.90
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	1	4	1.5	18.80
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	14	20	27	20	12	11	9	5	6	6	7	21	26	42	32	9	267	0.0	4.49
PERCENT	5.2	7.5	10.1	7.5	4.5	4.1	3.4	1.9	2.2	2.2	2.6	7.9	9.7	15.7	12.0	3.4	100.0		
AV SPD	6.2	7.5	6.1	8.9	7.9	6.7	6.0	5.8	10.7	6.3	6.1	8.5	7.3	10.0	10.1	7.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 4

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	3	5.2	2.94
3.6 -7.5	3	1	1	9	1	1	2	1	0	0	3	1	5	0	1	2	31	57.4	5.74
7.6 - 12.5	0	0	1	1	4	1	8	3	0	0	3	2	5	0	1	1	19	32.8	9.09
12.6 - 18.5	0	0	0	0	1	0	0	0	0	0	1	2	0	0	1	0	5	8.6	15.35
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	4	1	2	10	6	2	2	1	0	1	7	5	10	1	3	3	58	0.0	6.56
PERCENT	6.9	1.7	3.4	17.2	10.3	3.4	3.4	1.7	0.0	1.7	12.1	8.6	17.2	1.7	5.2	5.2	100.0		
AV SPD	5.6	6.9	8.0	6.4	9.0	7.2	5.0	5.8	0.0	2.8	9.3	12.2	7.3	2.7	9.6	7.6			

AVERAGE SPEED FOR THIS TABLE EQUALS7.7
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-14 (SHEET 46 OF 48)

MONTH OF DECEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + -3.5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3.0	1.10
3.6 -7.5	1	1	0	1	2	1	0	4	2	3	0	1	2	2	0	1	21	63.6	5.51
7.6 - 12.5	0	0	1	1	1	0	1	0	0	2	3	1	1	0	0	0	11	33.3	9.27
12.6 - 18.5	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	8.6	14.90
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.5	18.80
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	1	1	2	3	1	1	4	2	5	3	2	3	2	0	1	37	0.0	5.58
PERCENT	6.1	3.0	3.0	6.1	9.1	3.0	3.0	12.1	6.1	15.2	9.1	6.1	9.1	6.1	0.0	3.0	100.0		
AV SPD	3.3	6.3	7.9	7.5	7.2	4.6	10.7	4.9	6.2	7.1	9.4	8.7	7.1	5.0	0.0	5.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	2	.4	.50
CALM + -3.5	4	6	7	2	2	3	5	10	2	4	5	6	4	4	0	7	71	12.9	2.27
3.6 -7.5	19	16	20	26	17	6	11	15	11	13	16	14	26	13	8	16	247	44.7	5.34
7.6 - 12.5	3	0	17	37	12	6	5	5	7	9	14	25	16	10	13	4	183	33.8	9.24
12.6 - 18.5	0	0	0	3	1	0	0	1	2	2	13	2	0	8	14	2	48	8.7	14.23
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0.0	19.20
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	27	22	44	68	32	2	21	31	22	28	48	48	46	35	35	29	552	0.0	5.27
PERCENT	4.9	4.0	8.0	12.3	5.8	3.4	3.8	5.6	4.0	5.1	8.7	8.7	8.3	6.3	6.3	5.3	100.0		
AV SPD	5.0	4.2	6.6	8.2	7.1	7.2	5.5	5.1	7.5	6.9	9.1	12.2	6.7	8.7	10.6	5.7			

AVERAGE SPEED FOR THIS TABLE EQUALS7.2
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 6

TABLE 2.3.2-14 (SHEET 47 OF 48)

MONTH OF DECEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	2	0	0	1	1	0	0	0	0	0	0	0	4	0.0	0.00
CALM + -3.5	3	8	7	6	3	2	10	10	4	4	4	10	11	5	0	7	94	3.0	1.10
3.6 -7.5	14	13	20	39	33	27	23	28	27	21	33	20	28	32	44	22	424	63.6	5.51
7.6 - 12.5	3	2	2	13	0	0	3	3	2	17	22	21	12	37	26	6	174	33.3	9.27
12.6 - 18.5	0	0	0	2	0	0	0	0	0	2	2	4	0	5	9	1	27	8.6	14.90
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.5	18.80
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	20	23	29	60	38	29	36	42	41	44	61	55	51	79	79	36	723	0.0	5.58
PERCENT	2.8	3.2	4.0	8.3	5.3	4.0	5.0	5.8	5.7	6.1	8.4	7.6	7.1	10.9	10.9	5.0	100.0		
AV SPD	5.1	4.3	5.0	6.3	4.7	5.2	5.0	5.0	6.1	7.2	7.1	7.2	5.7	7.8	7.9	5.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	2	.9	.50
CALM + -3.5	4	2	3	2	6	2	7	2	2	7	3	7	6	4	6	5	70	29.8	2.46
3.6 -7.5	6	7	14	26	11	3	11	7	11	10	17	8	11	13	17	3	157	67.2	4.46
7.6 - 12.5	0	0	0	37	0	0	0	0	7	1	1	0	0	1	1	0	5	2.1	8.56
12.6 - 18.5	0	0	0	3	0	0	0	0	2	2	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	10	9	17	68	17	5	18	10	22	18	21	16	17	18	24	8	200	0.0	3.54
PERCENT	4.3	3.8	7.2	12.3	7.2	2.1	7.7	4.3	4.0	7.7	8.9	6.8	7.2	7.7	10.2	3.4	100.0		
AV SPD	3.3	4.0	4.5	8.2	4.3	4.8	3.9	3.4	7.5	4.5	4.8	3.3	4.0	4.9	4.5	3.4			

AVERAGE SPEED FOR THIS TABLE EQUALS4.3
HOURS ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-14 (SHEET 48 OF 48)

MONTH OF DECEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-7

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	3	1	4	3	4	1	1	1	1	0	1	0	2	3	1	1	26	10.3	0.00
CALM + -3.5	4	11	12	8	11	5	9	10	4	3	1	3	9	14	4	3	108	42.7	1.10
3.6 -7.5	0	5	13	9	5	2	4	28	6	9	33	16	22	5	8	4	118	46.6	5.51
7.6 - 12.5	3	0	0	1	0	0	0	3	0	0	22	0	0	0	0	0	1	.4	9.27
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0.0	14.90
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0.0	18.80
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	9	17	29	21	20	8	14	42	11	11	61	19	33	22	13	8	253	0.0	5.58
PERCENT	3.6	6.7	11.5	8.3	7.9	3.2	5.5	5.8	4.3	4.3	8.4	7.5	13.0	8.7	5.1	3.2	100.0		
AV SPD	2.2	3.0	3.1	3.2	2.7	2.7	2.9	5.0	3.6	4.0	7.1	4.5	3.9	2.6	3.4	2.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF DECEMBER
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	4	1	4	3	6	2	1	2	2	0	1	1	0	3	1	1	74	1.6
CALM + -3.5	18	28	35	19	23	13	31	30	13	21	14	27	6	29	11	24	367	17.4
3.6 -7.5	55	52	81	98	72	46	60	60	59	56	83	69	11	72	88	51	1107	52.2
7.6 - 12.5	9	11	28	68	25	11	9	9	16	29	44	57	0	74	53	14	503	23.7
12.6 - 18.5	0	1	1	5	2	0	0	1	7	5	16	11	0	19	32	3	103	4.9
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	1	0	2	1	1	5	.2
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
32.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	86	93	149	193	128	72	101	102	97	111	158	166	17	199	186	94	2121	100.0
PERCENT	4.1	4.4	7.0	9.1	6.0	3.4	4.8	4.8	4.6	5.2	7.4	7.8	7.2	9.4	8.8	4.4	100.0	

TABLE 2.3.2-15 (SHEET 1 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM 12/73-12/74, 4/77-4/79
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-17

DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	21	18	28	15	6	6	6	11	4	3	10	16	21	18	22	24	229	5.2	2.72
3.6 - 7.5	150	151	169	122	126	85	68	33	25	45	94	159	254	189	121	112	1903	43.4	5.58
7.6 - 12.5	67	82	150	115	136	93	65	40	47	32	93	175	284	210	151	71	1811	41.3	9.33
12.6 - 18.5	7	15	10	10	4	3	10	7	8	7	22	43	82	101	72	7	408	9.3	14.43
18.6 - 24.5	0	0	0	1	0	0	0	0	0	1	2	2	8	11	3	2	30	.7	19.66
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	245	266	357	263	272	187	149	91	84	88	221	395	649	529	369	216	4381	0.0	6.75
PERCENT	5.6	6.1	8.1	6.0	6.2	4.3	3.4	2.1	1.9	2.0	5.0	9.0	14.8	12.1	8.4	4.9	100.0		
AV SPD	6.5	7.0	7.3	7.7	7.8	7.6	7.9	7.7	8.6	7.9	8.3	8.3	8.7	9.3	9.1	7.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 76

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-17

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	6	1	6	4	2	3	2	1	3	1	1	2	5	5	3	3	48	6.2	2.74
3.6 - 7.5	26	37	33	33	24	12	18	16	26	20	34	26	34	17	18	13	387	49.9	5.47
7.6 - 12.5	10	24	31	15	18	15	17	13	8	16	32	22	28	18	9	10	286	36.9	9.13
12.6 - 18.5	2	0	3	4	1	1	3	0	2	6	4	11	9	1	6	0	53	6.8	14.12
18.6 - 24.5	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	2	.3	20.73
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	44	62	74	56	45	31	40	30	39	44	71	61	76	41	36	26	776	0.0	6.29
PERCENT	5.7	8.0	9.5	7.2	5.8	4.0	5.2	3.9	5.0	5.7	9.1	7.9	9.8	5.3	4.6	3.4	100.0		
AV SPD	6.3	6.9	7.2	6.9	7.0	7.3	7.2	6.8	7.0	8.6	8.2	8.8	8.2	7.4	7.7	7.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 16

TABLE 2.3.2-15 (SHEET 2 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-17

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	5	0	3	1	1	2	2	1	2	1	3	1	3	0	4	0	29	7.3	2.61
3.6 - 7.5	13	11	13	13	11	17	15	17	12	14	11	15	18	9	11	15	215	53.9	5.39
7.6 - 12.5	2	10	11	10	10	6	6	3	16	9	10	9	12	9	6	6	135	33.8	9.36
12.6 - 18.5	0	1	0	2	0	0	0	1	1	2	1	2	0	5	1	1	17	4.3	14.54
18.6 - 24.5	0	0	0	1	0	0	0	0	0	0	0	0	1	2	0	0	3	.8	20.57
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	20	22	27	26	22	25	23	22	31	26	25	27	34	25	22	22	399	0.0	5.98
PERCENT	5.0	5.5	6.8	6.5	5.5	6.3	5.8	5.5	7.8	6.5	6.3	6.8	8.5	6.3	5.5	5.5	100.0		
AV SPD	5.2	8.0	6.6	7.4	7.4	6.2	6.5	6.3	7.5	7.8	7.3	7.6	7.4	10.2	6.5	7.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 6

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-17

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1	0	0	0	0	1	0	0	0	1	0	1	0	0	0	0	4	.1	0.50
CALM + - 3.5	29	31	39	24	36	39	32	38	29	29	36	29	34	31	19	33	508	10.9	2.56
3.6 - 7.5	116	132	188	187	183	174	145	159	127	157	153	123	123	86	80	89	2227	47.7	5.41
7.6 - 12.5	67	71	167	149	107	80	83	88	112	112	107	117	106	103	79	40	1588	34.1	9.24
12.6 - 18.5	5	8	16	13	8	2	13	18	24	35	35	15	26	51	37	6	312	6.7	14.28
18.6 - 24.5	0	0	0	0	0	0	0	0	5	6	1	3	3	6	0	0	24	.5	19.95
24.6 - 32.5	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	2	.0	25.40
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	218	242	410	373	334	296	273	303	297	341	333	288	292	277	215	168	4660	0.0	5.73
PERCENT	4.7	5.2	8.8	8.0	7.2	6.4	5.9	6.5	6.4	7.3	7.1	6.2	6.3	5.9	4.6	3.6	100.0		
AV SPD	6.4	6.5	7.3	7.3	6.6	6.2	6.8	6.8	7.8	7.9	7.7	7.5	7.7	8.9	8.4	5.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 32

TABLE 2.3.2-15 (SHEET 3 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-17

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	3	1	4	0	4	2	0	4	1	3	2	2	2	1	2	2	33	.4	.50
CALM + - 3.5	67	76	73	88	102	101	141	154	95	80	84	88	83	69	58	61	1420	19.3	2.42
3.6 - 7.5	149	142	278	300	309	271	293	345	419	403	367	265	224	223	235	120	4343	59.2	5.18
7.6 - 12.5	30	48	73	79	62	59	73	84	134	126	117	64	86	151	140	29	1355	18.5	9.03
12.6 - 18.5	3	10	7	4	3	0	1	6	17	17	13	17	9	29	34	1	171	2.3	14.31
18.6 - 24.5	1	0	0	0	0	0	0	1	0	1	0	6	5	0	0	0	14	.2	20.63
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0	3	.0	28.65
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	253	277	435	471	480	433	508	594	666	630	584	444	409	473	469	213	7339	0.0	4.44
PERCENT	3.4	3.8	5.9	6.4	6.5	5.9	6.9	8.1	9.1	8.6	8.0	6.0	5.6	6.4	6.4	2.9	100.0		
AV SPD	5.1	5.5	5.7	5.7	5.2	5.2	5.1	5.3	6.0	6.1	6.0	6.1	6.1	6.9	7.0	5.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 5.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 11

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-17

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	2	7	2	0	1	1	1	0	2	2	1	1	1	1	1	23	.7	.50
CALM + - 3.5	69	60	65	54	73	63	105	85	59	75	83	70	74	71	66	63	1135	33.1	2.35
3.6 - 7.5	87	118	179	140	120	125	116	93	130	143	167	204	159	168	131	66	2146	62.5	4.79
7.6 - 12.5	3	4	7	5	4	9	13	11	11	7	8	6	4	8	8	11	119	3.5	8.86
12.6 - 18.5	1	1	0	0	0	0	0	1	0	2	0	0	0	2	4	0	11	.3	13.83
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	160	185	258	201	197	198	235	191	200	229	260	281	238	250	210	141	3434	0.0	3.46
PERCENT	4.7	5.4	7.5	5.9	5.7	5.8	6.8	5.6	5.8	6.7	7.6	8.2	6.9	7.3	6.1	4.1	100.0		
AV SPD	4.0	4.3	4.4	4.4	4.2	4.3	4.0	4.0	4.6	4.4	4.4	4.5	4.3	4.4	4.4	4.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 4.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 7

TABLE 2.3.2-15 (SHEET 4 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM _____
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 909-17

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	13	8	10	7	9	6	0	4	1	3	2	2	2	1	2	2	33	.4	.50
CALM + - 3.5	84	69	69	42	65	41	141	154	95	80	84	88	83	69	58	61	1420	19.3	2.26
3.6 - 7.5	94	70	89	62	66	44	36	28	57	82	137	134	86	62	60	50	1157	53.5	4.61
7.6 - 12.5	0	3	3	1	0	1	0	2	2	0	3	2	0	0	0	2	19	.9	8.67
12.6 - 18.5	0	0	0	0	0	1	1	0	0	2	0	0	0	0	0	0	4	.2	14.49
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	191	150	171	112	140	93	88	83	113	124	203	194	136	125	121	119	2163	0.0	2.61
PERCENT	8.8	6.9	7.9	5.2	6.5	4.3	4.1	3.8	5.2	5.7	9.4	9.0	6.3	5.8	5.6	5.5	100.0		
AV SPD	3.5	3.5	3.7	3.6	3.4	3.7	3.4	3.3	3.7	4.3	4.2	4.1	3.8	3.5	3.3	3.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 3.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	17	11	21	9	13	10	6	5	4	6	8	6	8	8	11	7	150	.6
CALM + - 3.5	281	255	283	228	285	255	334	343	243	229	276	262	265	251	225	247	4262	18.4
3.6 - 7.5	635	661	949	857	839	728	691	691	796	864	963	926	898	754	656	465	12373	53.4
7.6 - 12.5	179	242	442	374	337	263	257	241	330	302	370	395	520	499	393	169	5313	22.9
12.6 - 18.5	18	35	36	33	16	7	28	33	52	71	75	88	126	189	154	15	976	4.2
18.6 - 24.5	1	0	1	1	0	0	0	1	5	9	3	11	17	19	3	2	73	.3
24.6 - 32.5	0	0	0	0	0	0	0	0	0	1	2	2	0	0	0	0	5	.0
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	1131	1204	1732	1502	1490	1263	1316	1314	1430	1482	1697	1690	1834	1720	1442	905	23152	100.0
PERCENT	4.9	5.2	7.5	6.5	6.4	5.5	5.7	5.7	6.2	6.4	7.3	7.3	7.9	7.4	6.2	3.9	100.0	

TABLE 2.3.2-16 (SHEET 1 OF 48)

MONTH OF JANUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM 12/73-12/74, 4/77-4/79
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	4	2	0	0	0	1	0	0	0	0	0	0	0	0	1	10	3.6	2.53
3.6 -7.5	10	4	7	2	3	3	4	4	1	1	0	2	6	8	4	4	67	24.3	5.55
7.6 - 12.5	1	5	1	3	8	4	1	1	0	0	2	13	10	17	16	14	96	34.8	9.68
12.6 - 18.5	0	2	2	1	0	2	0	0	0	0	0	1	12	14	14	3	51	18.5	15.14
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	1	7	15	19	0	42	15.2	21.22
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	2	4	4	0	10	3.6	26.84
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	13	15	12	6	11	9	6	5	1	1	2	17	37	58	57	26	276	0.0	8.71
PERCENT	4.7	5.4	4.3	2.2	4.0	3.3	2.2	1.8	.4	.4	.7	6.2	13.4	21.0	20.7	9.4	100.0		
AV SPD	5.6	7.5	8.5	9.3	9.2	5.4	6.2	6.3	5.4	10.4	10.1	14.3	14.9	16.0	8.9				

AVERAGE SPEED FOR THIS TABLE EQUALS 12.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	1	1	0	0	0	1	0	0	2	0	0	0	0	0	5	8.5	2.69
3.6 -7.5	2	0	2	1	0	0	1	2	1	1	4	0	1	0	0	0	15	25.4	5.62
7.6 - 12.5	2	0	2	1	1	1	1	2	0	2	2	1	1	7	0	0	23	39.0	9.20
12.6 - 18.5	0	0	1	0	0	1	0	0	1	0	0	0	3	5	1	0	12	20.3	5.30
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	2	2	0	0	4	6.8	19.99
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	4	0	6	3	1	2	2	5	2	3	8	1	7	14	1	0	59	0.0	7.36
PERCENT	6.8	0.0	10.2	5.1	1.7	3.4	3.4	8.5	3.4	5.1	13.6	1.7	11.9	23.7	1.7	0.0	100.0		
AV SPD	8.3	0.0	7.2	5.9	10.2	12.4	8.7	5.8	9.7	7.7	5.8	9.4	14.0	14.0	15.5	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-16 (SHEET 2 OF 48)

MONTH OF JANUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
3.6 -7.5	3	0	0	0	1	0	1	1	1	0	0	1	1	3	0	1	13	36.1	5.62
7.6 - 12.5	0	2	0	0	1	0	2	1	0	0	1	0	0	2	0	1	10	27.8	9.27
12.6 - 18.5	0	1	0	0	2	0	0	0	0	0	0	3	1	1	0	0	8	22.2	14.10
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	3	8.3	20.61
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	2	5.6	28.34
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	3	0	0	4	0	3	2	1	0	1	4	2	10	1	2	36	0.0	8.61
PERCENT	8.3	8.3	0.0	0.0	11.1	0.0	8.3	5.6	2.8	0.0	2.8	11.1	5.6	27.8	2.8	5.6	100.0		
AV SPD	5.6	12.4	0.0	0.0	10.0	0.0	8.4	7.4	5.9	0.0	7.9	11.6	11.6	14.5	20.9	7.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	1	1	3	0	3	2	0	1	0	0	2	2	2	4	0	23	4.6	2.44
3.6 -7.5	7	6	6	7	5	11	9	7	8	8	2	8	9	5	7	13	118	23.6	5.52
7.6 - 12.5	10	9	16	7	10	16	14	3	4	5	5	9	17	13	5	6	149	29.7	10.00
12.6 - 18.5	2	9	8	19	8	7	1	0	2	12	9	12	19	17	14	2	141	28.1	15.04
18.6 - 24.5	0	1	0	0	1	0	0	2	3	8	5	2	15	13	5	0	53	11.0	20.65
24.6 - 32.5	0	0	0	0	0	0	0	0	3	0	0	0	1	8	1	0	13	2.6	27.47
32.6+	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	2	.4	34.85
TOTAL	21	26	31	36	24	37	26	12	21	33	22	33	63	59	36	21	501	100.0	8.60
PERCENT	4.2	5.2	6.2	7.2	4.8	7.4	5.2	2.4	4.2	6.6	4.4	6.6	12.6	11.8	7.2	4.2	100.0		
AV SPD	8.2	10.6	10.2	11.4	11.3	9.3	8.4	9.6	12.8	13.7	15.7	11.2	13.7	16.1	12.4	7.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =2

TABLE 2.3.2-16 (SHEET 3 OF 48)

MONTH OF JANUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2	.3	.50
CALM + - 3.5	0	1	2	0	1	0	3	1	1	1	1	3	0	3	2	1	20	3.3	2.87
3.6 -7.5	5	0	10	6	6	8	4	3	6	7	7	4	6	7	8	2	87	14.5	5.84
7.6 - 12.5	8	7	17	17	15	10	13	7	9	7	15	15	37	39	34	8	258	43.1	10.04
12.6 - 18.5	0	2	2	14	15	9	8	6	5	9	14	9	30	30	23	4	180	30.1	15.00
18.6 - 24.5	0	2	0	0	0	0	0	1	4	3	1	2	7	6	10	4	40	6.7	20.56
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	1	3	1	0	0	8	1.0	26.59
32.6+	0	0	0	0	0	0	0	0	0	0	2	3	0	1	0	0	6	1.0	37.49
TOTAL	13	13	31	38	37	25	28	18	25	27	41	37	83	87	77	19	599	100.0	9.09
PERCENT	2.2	2.2	5.2	6.3	6.2	4.2	4.7	3.0	4.2	4.5	6.8	6.2	13.9	14.5	12.9	3.2	100.0		
AV SPD	8.5	11.0	8.6	11.1	11.0	10.6	10.2	11.2	11.0	11.4	12.9	13.9	13.4	12.6	12.8	13.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 12.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	.4	.50
CALM + - 3.5	1	0	0	0	0	1	0	0	1	0	0	1	0	0	2	0	0	2.4	2.08
3.6 -7.5	4	1	3	2	1	2	2	4	1	2	2	3	6	3	2	2	40	15 7	5.20
7.6 - 12.5	2	7	4	13	5	4	2	8	14	4	3	10	18	19	22	18	153	60.2	10.11
12.6 - 18.5	0	0	0	5	3	1	0	0	0	0	5	4	5	12	6	11	52	20.5	13.64
18.6 - 24.5	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	2	.8	21.12
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	7	8	7	20	9	8	4	13	16	7	10	18	29	34	33	31	254	0.0	8.02
PERCENT	2.8	3.1	2.8	7.9	3.5	3.1	1.6	3.1	6.3	2.8	3.9	7.1	11.4	13.4	13.0	12.2	100.0		
AV SPD	5.8	9.9	8.4	11.0	10.0	8.4	7.0	7.9	9.2	10.1	11.3	10.0	9.7	11.7	10.9	10.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-16 (SHEET 4 OF 48)

MONTH OF JANUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	2	1.0	.50
CALM + - 3.5	4	0	1	1	2	4	0	0	3	2	0	2	4	1	3	0	27	13.0	2.13
3.6 -7.5	6	9	5	5	7	8	6	4	5	1	1	2	2	5	4	5	75	36.2	5.56
7.6 - 12.5	4	3	2	9	4	11	3	3	7	12	7	7	4	7	4	0	87	42.0	9.70
12.6 - 18.5	0	0	0	0	1	3	0	0	1	2	3	2	1	1	1	1	16	7.7	13.66
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	14	12	8	16	14	26	9	7	16	17	11	13	11	14	12	7	207	0.0	5.14
PERCENT	6.8	5.8	3.9	7.7	6.8	12.6	4.3	3.4	7.7	8.2	5.3	6.3	5.3	5.8	5.8	3.4	100.0		
AV SPD	5.7	6.6	6.4	7.6	7.2	7.9	7.0	8.8	7.7	9.4	11.2	8.8	6.8	8.1	6.6	5.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF JANUARY
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	1	0	2	0	0	0	1	0	0	0	0	0	0	0	1	5	.3
CALM + - 3.5	9	6	7	5	3	8	6	2	6	3	3	8	6	6	11	2	91	4.7
3.6 -7.5	37	20	33	23	23	30	27	25	23	20	16	20	31	31	25	31	415	21.5
7.6 - 12.5	27	33	42	50	44	46	36	25	34	30	35	55	37	104	81	47	776	40.2
12.6 - 18.5	2	14	13	39	29	23	9	6	9	23	31	31	71	80	59	21	460	23.8
18.6 - 24.5	0	3	0	0	1	0	0	3	7	12	6	5	31	38	36	4	146	7.6
24.6 - 32.5	0	0	0	0	0	0	0	0	3	0	1	1	6	15	5	0	31	1.6
32.6+	0	0	0	0	0	0	0	0	0	0	3	3	0	2	0	0	8	.4
TOTAL	75	77	95	119	100	107	78	62	82	88	95	123	232	276	217	106	1932	100.0
PERCENT	3.9	4.0	4.9	8.2	5.2	5.5	4.0	3.2	4.2	4.6	4.9	6.4	12.0	14.3	11.2	5.5	100.0	

TABLE 2.3.2-16 (SHEET 5 OF 48)

MONTH OF FEBRUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	1	2	0	0	0	0	0	0	0	0	0	1	0	0	2	8	2.4	3.18
3.6 -7.5	11	5	10	11	5	3	2	3	2	3	4	2	6	7	5	9	88	26.3	5.69
7.6 - 12.5	12	7	9	8	7	3	9	0	1	0	3	7	9	18	10	6	109	32.6	9.74
12.6 - 18.5	0	11	9	5	2	3	3	0	0	1	4	6	22	14	14	4	98	29.3	14.91
18.6 - 24.5	1	0	0	1	1	0	0	0	0	0	2	1	3	9	9	2	29	8.7	20.52
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	.6	25.29
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	26	24	30	25	15	9	14	3	3	4	13	16	41	48	40	23	334	0.0	8.97
PERCENT	7.8	7.2	9.0	7.5	4.5	2.7	4.2	.9	.9	1.2	3.9	4.8	12.3	14.4	12.0	6.9	100.0		
AV SPD	7.6	11.1	9.6	9.5	10.6	9.8	9.6	5.2	7.8	7.4	12.5	12.0	12.7	13.1	15.0	9.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1.9	2.10
3.6 -7.5	4	3	1	0	2	1	0	1	0	0	0	0	0	1	0	1	14	26.4	5.81
7.6 - 12.5	2	4	1	1	3	0	1	0	1	0	3	1	2	1	0	0	20	37.7	8.96
12.6 - 18.5	0	1	0	0	2	0	0	0	1	0	2	1	5	0	0	1	13	24.5	15.83
18.6 - 24.5	0	0	0	0	1	0	0	0	0	0	1	0	0	0	2	0	4	7.5	20.14
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1.9	26.20
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	6	8	2	1	8	1	1	1	2	0	6	2	7	2	3	3	53	0.0	8.58
PERCENT	11.3	15.1	3.8	1.9	15.1	1.9	1.9	1.9	3.8	0.0	11.3	3.8	13.2	3.8	5.7	5.7	100.0		
AV SPD	7.3	8.7	6.7	9.3	11.7	6.6	7.9	4.8	11.5	0.0	14.4	10.8	13.8	8.7	14.5	15.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-16 (SHEET 6 OF 48)

MONTH OF FEBRUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
3.6 -7.5	0	0	0	1	0	0	0	0	1	0	2	1	0	0	0	0	5	18.5	4.92
7.6 - 12.5	1	1	2	2	0	0	1	0	0	0	0	0	0	1	1	1	10	37.0	9.60
12.6 - 18.5	0	2	1	1	1	0	0	0	0	0	0	0	3	0	1	0	9	33.3	14.74
18.6 - 24.5	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	2	7.4	18.95
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	3.7	25.40
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	3	3	4	1	0	1	0	1	0	2	2	3	2	2	1	27	0.0	9.59
PERCENT	7.4	11.1	11.1	14.8	3.7	0.0	3.7	0.0	3.7	0.0	7.4	7.4	11.1	7.4	7.4	3.7	100.0		
AV SPD	14.3	12.9	11.8	9.5	15.8	0.0	9.4	0.0	4.5	0.0	4.5	12.0	15.2	19.0	12.2	9.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	1	2	0	0	0	3	0	1	0	0	0	2	2	2	2	10	3.8	2.06
3.6 -7.5	8	8	7	3	2	3	3	3	2	1	5	0	4	7	5	5	64	15.3	5.86
7.6 - 12.5	19	21	23	7	11	9	8	5	7	4	11	6	9	10	6	4	161	38.6	9.85
12.6 - 18.5	6	12	30	10	4	3	0	3	1	1	4	5	8	20	18	2	127	30.5	15.03
18.6 - 24.5	1	1	2	0	3	1	0	1	1	2	5	3	5	11	9	1	46	11.0	20.57
24.6 - 32.5	0	0	0	0	2	0	0	0	0	0	0	0	1	0	0	0	3	.7	25.89
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	35	41	64	20	22	16	14	13	12	8	25	14	29	50	40	14	417	0.0	9.10
PERCENT	8.4	9.8	15.3	4.8	5.3	3.8	3.4	3.1	2.9	1.9	6.0	3.4	7.0	12.0	9.6	3.4	100.0		
AV SPD	10.0	10.6	11.9	11.5	14.0	10.8	7.5	11.2	10.1	12.5	12.4	14.8	12.8	14.1	13.7	9.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 12.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 7 OF 48)

MONTH OF FEBRUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	2	2	0	2	3	1	1	1	0	1	0	1	0	1	15	2.6	2.42
3.6 -7.5	8	4	6	7	8	2	9	9	5	4	3	4	4	4	4	0	81	14.0	5.62
7.6 - 12.5	10	12	14	23	24	19	17	18	25	11	10	10	14	27	25	15	280	48.5	9.95
12.6 - 18.5	12	5	10	9	7	3	5	3	2	1	5	8	31	26	27	7	161	27.9	14.76
18.6 - 24.5	0	0	0	1	4	1	0	3	2	0	3	3	6	9	1	1	34	5.9	20.63
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	0	1	3	1	0	8	1.0	25.65
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	30	21	32	42	43	27	34	34	35	17	28	26	56	70	56	24	577	0.0	9.38
PERCENT	5.2	3.0	5.5	7.3	7.5	4.7	5.9	5.9	6.1	2.9	4.9	4.5	9.7	12.1	10.1	4.2	100.0		
AV SPD	10.6	9.9	10.3	10.0	11.3	10.1	8.8	10.1	10.2	8.6	12.2	11.9	14.3	13.8	12.5	11.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	0	0	0	1	0	0	0	1	2	0	1	0	0	1	0	8	3.5	1.91
3.6 -7.5	6	5	2	3	5	2	1	1	1	0	0	2	2	1	2	3	36	15.6	5.96
7.6 - 12.5	4	7	9	14	5	9	7	6	1	3	6	13	15	10	23	6	140	60.6	10.03
12.6 - 18.5	0	0	2	1	2	1	2	1	0	1	13	10	7	3	3	1	47	20.3	14.09
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	12	12	13	18	13	12	10	8	3	8	12	26	24	14	29	10	231	0.0	3.39
PERCENT	5.2	5.2	5.6	7.8	5.6	5.2	4.3	3.5	1.3	3.5	8.2	11.3	10.4	6.1	12.5	4.3	100.0		
AV SPD	6.7	8.6	10.2	9.5	8.8	9.3	10.5	9.5	5.2	6.4	13.1	11.5	11.0	11.1	10.1	9.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 8 OF 48)

MONTH OF FEBRUARY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.7	2.20
3.6 -7.5	0	0	1	0	1	2	2	1	2	4	1	3	4	0	1	2	24	16.4	5.84
7.6 - 12.5	2	0	5	2	7	1	9	9	2	6	10	12	5	6	9	5	90	61.6	10.02
12.6 - 18.5	0	0	0	0	0	3	2	0	0	5	7	6	7	1	0	0	31	21.2	14.18
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	0	7	2	8	6	13	10	4	15	18	21	16	7	10	7	146	0.0	9.28
PERCENT	1.4	0.0	4.8	1.4	5.5	4.1	8.9	6.8	2.7	10.3	12.3	14.0	11.0	4.6	6.8	4.8	100.0		
AV SPD	10.3	0.0	7.8	10.0	9.4	10.3	10.2	10.0	8.5	10.1	12.2	10.8	11.3	11.0	9.8	8.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF FEBRUARY
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
CALM + - 3.5	5	2	7	2	1	2	6	1	3	3	0	2	3	3	4	5	49	2.7
3.6 -7.5	37	23	27	25	23	13	17	18	13	12	15	12	20	20	17	20	312	17.5
7.6 - 12.5	50	52	63	57	57	41	52	39	37	25	49	49	34	73	74	37	810	45.4
12.6 - 18.5	18	31	52	26	18	13	12	7	4	9	35	36	83	64	63	15	486	27.2
18.6 - 24.5	3	1	2	2	9	2	0	4	3	2	11	6	14	29	21	4	115	6.4
24.6 - 32.5	0	0	0	0	2	0	0	0	0	0	1	0	2	4	3	1	13	.7
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	113	109	151	112	110	71	87	69	60	52	111	107	176	193	182	82	1785	100.0
PERCENT	6.3	6.1	8.5	6.3	6.2	4.0	4.9	3.9	3.4	2.9	6.2	6.0	9.9	10.8	10.2	4.6	100.0	

TABLE 2.3.2-16 (SHEET 9 OF 48)

MONTH OF MARCH

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	0	0	1	0	1	1	0	0	0	0	1	0	0	1	6	1.3	1.96
3.6 -7.5	12	9	11	4	5	6	6	4	3	5	7	6	7	8	10	5	108	23.5	5.55
7.6 - 12.5	16	10	5	3	9	12	10	3	9	11	11	14	17	29	14	9	182	39.7	9.74
12.6 - 18.5	1	5	4	1	5	3	3	7	2	4	9	18	18	8	12	2	102	22.2	14.99
18.6 - 24.5	0	0	0	0	0	0	0	1	0	1	3	1	10	10	15	3	44	9.6	20.73
24.6 - 32.5	0	1	0	0	0	0	0	0	0	0	0	0	9	7	0	0	17	3.7	26.98
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	30	25	20	8	20	21	20	16	14	21	30	39	62	62	51	20	459	0.0	9.04
PERCENT	6.5	5.4	4.4	1.7	4.4	4.6	4.4	3.5	3.1	4.5	6.5	8.5	13.5	13.5	11.1	4.4	100.0		
AV SPD	7.9	9.8	8.6	8.2	9.8	9.1	9.4	11.4	10.4	10.7	11.7	12.4	15.1	13.9	13.3	10.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 5

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION REQUEST NUMBER 909.15
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.9	2.30
3.6 -7.5	3	1	2	1	1	3	1	2	0	2	0	2	2	1	1	2	24	22.4	5.11
7.6 - 12.5	4	3	8	2	1	3	3	2	0	4	4	4	4	3	5	4	54	30.5	9.95
12.6 - 18.5	4	3	4	0	1	0	0	0	0	0	5	0	5	1	1	0	24	22.4	14.25
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	1	4	3.7	20.26
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	11	7	14	3	3	6	4	4	0	6	9	6	11	5	8	7	107	0.0	8.60
PERCENT	10.3	6.5	13.1	2.8	2.8	5.6	3.7	3.7	0.0	5.5	8.4	5.6	10.3	7.5	7.5	6.5	100.0		
AV SPD	10.7	11.1	10.7	8.5	9.2	8.2	8.3	6.4	0.0	8.2	12.9	8.6	11.4	14.0	8.8	10.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-16 (SHEET 10 OF 48)

MONTH OF MARCH

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
3.6 -7.5	0	0	1	1	1	1	1	0	1	0	0	0	1	1	1	2	11	26.2	5.71
7.6 - 12.5	2	0	0	1	1	1	0	3	1	3	0	1	1	2	1	1	18	42.9	10.12
12.6 - 18.5	0	1	0	1	1	1	0	1	0	1	3	0	1	0	0	0	10	23.8	14.25
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	2	4.8	19.74
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2.4	29.50
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	1	1	3	3	3	1	4	2	4	3	1	3	4	3	4	42	0.0	9.25
PERCENT	4.8	2.4	2.4	7.1	7.1	7.1	2.4	9.5	4.8	9.5	7.1	2.4	7.1	9.5	7.1	9.5	100.0		
AV SPD	10.8	16.2	4.7	9.8	9.7	10.1	8.1	11.0	7.5	12.7	13.0	11.2	10.4	14.7	12.1	10.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	2	0	1	0	1	3	0	1	1	1	0	0	1	0	0	13	2.4	2.92
3.6 -7.5	6	5	3	8	11	13	17	8	7	7	4	1	4	4	5	4	107	20.1	5.75
7.6 - 12.5	10	10	17	19	31	23	19	11	13	8	13	14	13	9	14	9	233	43.8	9.69
12.6 - 18.5	4	4	3	6	6	7	10	13	16	6	8	9	19	5	5	5	131	24.6	14.82
18.6 - 24.5	0	0	0	0	0	1	2	5	5	12	2	0	0	7	3	2	39	7.3	20.61
24.6 - 32.5	0	0	0	0	0	0	0	0	0	4	0	0	4	1	0	0	9	1.7	26.73
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	22	21	23	34	48	45	51	42	42	38	28	24	40	27	27	20	532	0.0	9.15
PERCENT	4.1	3.9	4.3	6.4	9.0	8.5	9.6	7.9	7.9	7.1	5.3	4.5	7.5	5.1	5.1	3.8	100.0		
AV SPD	9.2	9.2	10.5	9.7	9.6	9.5	9.4	12.9	12.3	15.3	11.5	11.3	13.8	13.2	11.0	11.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-16 (SHEET 11 OF 48)

MONTH OF MARCH

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	1	1	0	1	1	0	0	2	0	2	0	1	0	0	1	10	1.6	1.91
3.6 -7.5	3	5	0	4	3	10	11	14	9	10	12	6	4	4	3	5	103	16.2	5.75
7.6 - 12.5	5	1	7	14	19	16	30	21	37	53	40	13	13	17	33	14	333	52.3	10.05
12.6 - 18.5	1	1	3	1	3	11	7	13	24	29	14	9	3	21	22	6	168	26.4	14.16
18.6 - 24.5	0	0	0	0	0	0	2	0	4	1	2	1	1	1	5	1	16	2.8	19.91
24.6 - 32.5	0	0	0	0	0	0	0	1	0	0	0	0	0	3	1	0	5	.8	27.02
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	9	8	11	19	26	38	50	49	76	93	70	29	22	46	64	27	637	0.0	9.20
PERCENT	1.4	1.3	1.7	3.0	4.1	6.0	7.8	7.7	11.9	14.6	11.0	4.6	3.5	7.2	10.0	4.2	100.0		
AV SPD	8.8	6.9	10.7	9.3	9.9	10.1	10.7	11.3	11.5	10.4	10.4	10.6	13.3	12.4	10.2				

AVERAGE SPEED FOR THIS TABLE EQUALS 10.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	0	0	0	0	0	0	0	2	0	0	0	1	2	1	0	8	8.9	1.69
3.6 -7.5	0	0	2	2	1	3	3	3	3	0	3	4	3	8	3	1	39	14.1	5.81
7.6 - 12.5	2	1	3	5	8	9	12	10	19	24	20	13	14	13	11	5	181	65.6	9.87
12.6 - 18.5	2	0	2	0	2	2	6	3	5	8	5	3	2	4	0	3	45	16.3	14.13
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	.7	22.79
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	.4	26.60
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	6	1	7	7	11	14	28	21	29	30	28	20	20	28	17	9	276	0.0	8.34
PERCENT	2.2	.4	2.5	2.5	4.0	5.1	10.1	7.6	10.5	10.9	10.1	7.2	7.2	10.1	6.3	3.3	100.0		
AV SPD	8.9	12.1	10.5	8.8	10.1	9.7	10.3	9.8	9.9	11.1	10.0	10.1	9.8	9.3	10.2	11.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 12 OF 48)

MONTH OF MARCH

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	.7	0.50
CALM + - 3.5	0	0	0	0	0	1	0	2	2	0	0	0	1	0	1	1	8	5.3	2.33
3.6 -7.5	1	0	0	1	0	3	8	1	6	5	2	1	2	5	4	1	40	26.7	5.51
7.6 - 12.5	0	2	2	2	7	3	2	1	3	4	20	5	7	6	7	4	75	50.0	10.03
12.6 - 18.5	0	0	1	1	1	1	1	1	0	5	5	1	2	7	0	0	26	17.3	13.87
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	2	3	4	8	8	11	5	11	14	28	7	12	18	12	6	150	0.0	6.81
PERCENT	.7	1.3	2.0	2.7	5.3	5.3	7.3	3.3	7.3	9.3	18.7	4.7	8.0	12.0	8.0	4.0	100.0		
AV SPD	3.9	9.1	11.5	9.9	10.7	8.7	7.6	6.0	6.6	9.9	10.4	9.6	9.9	10.6	7.8	7.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF MARCH
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	.0
CALM + - 3.5	5	3	1	1	2	3	4	3	7	1	3	0	4	3	3	3	46	2.1
3.6 -7.5	25	20	19	21	22	39	47	32	29	29	28	20	23	31	27	20	432	19.6
7.6 - 12.5	39	27	42	46	76	67	83	56	32	107	108	64	59	79	85	46	1076	48.8
12.6 - 18.5	12	14	17	10	19	25	27	43	47	51	49	40	50	46	40	16	508	23.0
18.6 - 24.5	0	0	0	0	0	1	4	6	9	14	7	2	11	21	26	8	109	4.9
24.6 - 32.5	0	1	0	0	0	0	0	1	0	4	0	0	13	13	1	0	33	1.5
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	81	65	79	78	119	135	166	141	174	206	198	126	170	193	192	93	2203	100.0
PERCENT	3.7	3.0	3.8	3.5	8.4	6.1	7.8	6.4	7.9	9.4	8.9	5.7	7.7	8.8	8.3	4.2	100.0	

TABLE 2.3.2-16 (SHEET 13 OF 48)

MONTH OF APRIL

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	2	.4	.50
CALM + - 3.5	0	1	0	1	1	0	0	0	1	0	2	1	1	1	2	2	13	2.6	2.70
3.6 -7.5	7	4	5	5	3	2	5	3	6	3	9	9	16	8	6	3	94	19.1	5.69
7.6 - 12.5	5	11	8	6	9	13	20	11	6	8	9	17	25	24	10	3	187	38.0	9.84
12.6 - 18.5	2	1	2	3	4	1	5	5	7	3	12	15	29	29	8	4	130	26.4	15.36
18.6 - 24.5	1	0	0	0	0	0	0	0	0	0	2	9	20	18	5	2	57	11.6	20.76
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	0	4	3	1	0	9	1.8	26.26
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	15	17	15	15	17	16	30	19	20	14	35	51	96	84	32	16	492	0.0	8.80
PERCENT	3.0	3.5	3.0	3.0	3.5	3.3	6.1	3.9	4.1	2.8	7.1	10.4	19.5	17.1	6.5	3.3	100.0		
AV SPD	9.8	8.4	9.2	8.7	9.8	9.3	10.4	10.5	9.9	10.1	11.6	13.0	14.1	14.4	12.2	11.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 12.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 8

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	2	2.6	1.93
3.6 -7.5	0	1	0	1	1	0	1	0	4	4	1	1	0	1	0	1	16	21.1	5.65
7.6 - 12.5	1	1	1	1	1	3	2	3	3	2	1	1	0	3	0	0	30	39.5	9.25
12.6 - 18.5	0	1	0	0	0	0	1	1	0	4	1	4	7	1	1	0	21	27.6	15.44
18.6 - 24.5	0	0	0	0	0	0	0	0	0	1	0	6	2	2	0	0	7	9.2	19.64
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	3	1	2	2	4	4	4	7	18	3	8	9	7	1	2	76	0.0	8.61
PERCENT	1.3	3.9	1.3	2.6	3.6	5.3	5.3	5.3	2.2	23.7	3.9	10.5	11.0	9.2	1.3	2.6	100.0		
AV SPD	11.2	9.4	7.6	7.7	8.3	7.5	10.1	11.2	7.1	10.4	10.8	15.2	17.0	12.0	17.1	4.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-16 (SHEET 14 OF 48)

MONTH OF APRIL

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
3.6 -7.5	0	1	1	0	0	2	1	1	1	1	2	3	0	0	0	1	14	37.8	5.92
7.6 - 12.5	0	0	0	0	0	1	1	3	0	1	2	0	1	1	0	0	10	27.0	9.60
12.6 - 18.5	0	0	0	0	0	0	1	0	0	2	0	0	1	1	1	2	8	21.6	15.69
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	1	5	13.5	19.31
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	0	1	1	0	0	3	3	4	1	4	5	3	2	3	3	4	37	0.0	8.86
PERCENT	0.0	2.7	2.7	0.0	0.0	8.1	8.1	10.8	2.7	10.8	13.5	8.1	5.4	8.1	8.1	10.8	100.0		
AV SPD	0.0	6.6	7.1	0.0	0.0	7.5	8.8	8.6	7.3	12.3	10.5	6.4	13.7	15.3	18.6	13.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	2	0	1	1	1	0	1	1	0	1	0	1	2	1	12	3.3	3.03
3.6 -7.5	4	8	3	8	1	6	3	3	7	9	8	2	5	3	4	2	67	18.5	5.64
7.6 - 12.5	3	1	2	6	7	15	13	16	16	24	14	5	13	4	3	2	145	40.1	9.78
12.6 - 18.5	0	1	0	2	3	4	7	6	5	7	14	0	14	19	9	5	105	28.0	15.14
18.6 - 24.5	1	0	0	0	0	0	0	3	1	0	2	5	4	10	4	2	32	8.8	20.40
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	.3	25.60
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	8	10	7	10	12	28	24	28	32	37	36	25	36	37	22	12	362	0.0	9.23
PERCENT	2.2	2.5	1.9	2.8	3.3	7.2	6.6	7.7	8.8	10.2	9.9	6.9	9.9	10.2	6.1	3.3	100.0		
AV SPD	9.1	7.4	6.2	9.3	9.8	8.9	10.4	12.1	10.3	9.8	12.8	13.1	12.9	15.2	13.0	13.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 4

TABLE 2.3.2-16 (SHEET 15 OF 48)

MONTH OF APRIL

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	2	.4	2.55
3.6 -7.5	3	1	1	3	2	2	9	9	9	8	7	1	4	7	3	0	69	14.8	5.77
7.6 - 12.5	3	4	1	6	8	21	29	31	40	32	11	9	15	15	7	4	237	51.0	10.02
12.6 - 18.5	1	2	1	1	2	0	15	5	10	17	15	15	18	12	22	2	138	29.7	14.13
18.6 - 24.5	0	0	2	0	0	0	0	0	0	3	3	1	1	4	2	0	16	3.4	20.67
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0	0	3	.6	25.57
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	7	7	5	10	13	23	53	45	59	60	38	28	38	38	35	6	465	0.0	9.88
PERCENT	1.5	1.5	1.1	2.2	2.8	4.9	11.4	9.7	12.7	12.9	8.2	6.0	8.2	8.2	7.5	1.3	100.0		
AV SPD	8.9	9.7	12.4	8.5	10.1	9.4	10.6	9.8	10.1	11.5	13.0	13.4	12.0	11.9	13.0	11.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	1	0	0	0	0	1	0	1	0	0	1	0	1	0	0	5	1.7	2.63
3.6 -7.5	3	0	0	1	3	0	1	2	1	4	4	2	3	5	3	5	40	13.5	6.03
7.6 - 12.5	2	3	4	7	15	8	8	7	4	14	20	16	21	12	7	3	151	51.2	10.43
12.6 - 18.5	5	2	2	0	1	4	4	4	8	13	9	14	10	11	4	0	91	30.8	14.39
18.6 - 24.5	0	0	1	0	0	0	0	0	0	0	1	0	1	3	1	1	3	2.7	20.56
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	12	5	7	8	19	12	14	13	14	31	34	33	35	32	15	10	295	0.0	9.93
PERCENT	4.1	2.0	2.4	2.7	6.4	4.1	4.7	4.4	4.7	10.5	11.5	11.2	11.9	10.8	5.1	3.4	100.0		
AV SPD	10.7	11.2	13.4	10.2	9.0	11.4	10.5	10.4	11.8	11.7	11.8	12.0	11.6	12.1	11.1	9.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 16 OF 48)

MONTH OF APRIL

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	1	0	0	1	0	1	1	1	1	0	1	3	1	0	11	5.9	2.34
3.6 -7.5	7	1	2	0	3	4	3	2	4	4	2	2	6	3	3	1	47	25.4	5.17
7.6 - 12.5	2	1	0	2	2	3	5	6	5	3	12	10	12	10	6	3	18	44.3	10.00
12.6 - 18.5	0	0	0	0	3	0	0	0	3	1	15	5	6	7	4	1	45	24.3	14.22
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	9	2	3	2	8	8	8	9	13	9	30	17	25	23	14	5	185	1.0	7.35
PERCENT	4.9	1.1	1.6	1.1	4.3	4.3	4.3	4.9	7.0	4.9	16.2	9.2	13.5	12.4	7.6	2.7	100.0		
AV SPD	6.7	6.4	3.5	9.6	9.3	6.6	8.2	8.5	9.4	7.1	11.7	11.3	9.5	10.3	9.7	10.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF APRIL
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	2	.1
CALM + - 3.5	0	2	3	1	2	3	2	1	4	2	3	4	2	6	6	4	45	2.4
3.6 -7.5	26	16	12	12	13	16	23	20	32	33	30	20	34	27	19	14	347	18.1
7.6 - 12.5	10	21	16	28	43	64	78	77	76	87	69	61	87	69	33	17	842	44.0
12.6 - 18.5	8	7	5	0	13	0	33	21	33	17	68	62	85	80	49	14	538	28.1
18.6 - 24.5	2	0	3	0	0	0	0	3	1	4	9	17	28	38	14	6	125	6.5
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	4	1	4	3	1	0	13	.7
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	52	46	39	47	71	82	136	122	146	173	181	165	241	224	122	55	1912	100.0
PERCENT	2.7	2.4	2.0	2.5	3.7	4.6	7.1	6.4	7.6	9.0	9.5	8.6	12.6	11.7	8.4	2.9	100.0	

TABLE 2.3.2-16 (SHEET 17 OF 48)

MONTH OF MAY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	0	0	2	0	1	0	0	1	1	0	2	0	2	1	0	12	2.8	2.68
3.6 -7.5	19	11	11	8	8	2	2	0	4	4	4	10	20	14	10	6	133	30.6	3.79
7.6 - 12.5	11	6	8	8	11	6	1	5	3	3	11	23	32	25	23	10	186	42.8	9.66
12.6 - 18.5	1	2	3	0	0	0	3	4	1	0	5	8	34	16	9	2	88	20.2	14.68
18.6 - 24.5	0	1	0	0	0	1	0	0	0	0	1	2	6	0	0	0	11	2.5	20.69
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	2	1	0	0	1	5	1.1	27.49
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	33	20	22	18	19	10	6	9	9	8	22	47	93	57	43	19	435	0.0	8.14
PERCENT	7.6	4.6	5.1	4.1	4.4	2.3	1.4	2.1	2.1	1.8	5.1	10.8	21.4	13.1	9.9	4.4	100.0		
AV SPD	7.2	8.2	8.1	7.0	8.2	9.1	11.4	12.1	7.5	7.5	11.7	10.7	12.3	10.1	9.7	9.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 3

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	2	2.9	1.82
3.6 -7.5	4	0	4	3	2	0	0	2	5	2	3	2	2	0	0	0	29	42.0	5.28
7.6 - 12.5	2	2	0	1	1	1	1	0	2	2	2	4	4	3	0	1	26	37.7	9.84
12.6 - 18.5	0	0	2	0	0	0	1	0	3	0	1	1	1	1	0	0	10	14.5	14.89
18.6 - 24.5	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	2	2.9	22.20
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	6	2	7	4	3	3	2	2	10	4	6	8	7	4	0	1	69	0.0	6.91
PERCENT	8.7	2.9	10.1	5.8	4.3	4.3	3.9	2.9	14.5	5.8	8.7	11.6	10.1	5.8	0.0	1.4	100.0		
AV SPD	7.0	10.0	10.8	6.9	8.3	4.5	13.2	4.2	9.3	6.7	7.8	11.1	10.2	11.6	0.0	12.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 18 OF 48)

MONTH OF MAY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	1	0	0	0	0	1	0	1	0	0	0	0	0	3	9.1	2.32
3.6 -7.5	1	0	1	1	3	1	0	0	0	2	0	0	0	0	1	2	12	36.4	5.65
7.6 - 12.5	0	1	2	2	0	1	1	1	1	0	0	1	0	1	0	0	11	33.3	9.47
12.6 - 18.5	0	0	2	0	1	0	0	0	0	0	0	1	1	0	1	0	6	18.2	13.61
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	3.0	22.30
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	1	5	4	4	2	1	1	2	2	1	3	1	1	2	2	33	0.0	6.52
PERCENT	3.0	3.0	15.2	12.1	2.1	8.1	3.0	3.0	6.1	6.1	3.0	9.1	3.0	3.0	6.1	6.1	100.0		
AV SPD	7.3	10.1	10.5	8.5	7.7	8.4	7.8	7.6	7.1	5.8	2.6	14.7	14.2	8.3	9.2	4.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	1	0	2	2	1	0	2	2	0	2	0	1	2	0	1	16	4.8	2.55
3.6 -7.5	4	3	4	10	4	10	9	7	5	4	7	2	3	1	1	5	79	23.7	5.78
7.6 - 12.5	9	5	11	11	6	7	5	18	14	7	13	9	9	6	6	2	141	42.3	9.70
12.6 - 18.5	2	4	11	3	2	2	8	14	11	1	5	2	4	3	4	4	80	24.0	14.77
18.6 - 24.5	0	0	1	0	2	0	0	0	7	1	2	0	2	1	1	0	17	5.1	20.64
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	15	16	27	26	16	20	22	41	39	13	29	13	19	13	12	12	333	0.0	8.18
PERCENT	4.5	4.8	8.1	7.8	4.8	6.0	6.6	12.3	11.7	3.9	8.7	3.9	5.7	3.0	3.6	3.6	100.0		
AV SPD	9.4	9.8	11.8	8.2	9.6	8.1	10.1	10.1	12.7	10.3	10.4	11.3	11.4	10.4	12.0	9.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 3

TABLE 2.3.2-16 (SHEET 19 OF 48)

MONTH OF MAY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	2	1	0	2	0	1	1	0	1	0	0	3	0	0	2	13	2.6	2.60
3.6 -7.5	1	8	9	7	7	9	22	16	7	6	6	2	6	3	5	8	122	24.7	5.41
7.6 - 12.5	7	4	8	11	12	26	16	45	39	20	25	19	12	6	9	9	268	54.3	9.82
12.6 - 18.5	1	2	8	2	4	1	3	6	11	3	7	10	7	9	7	3	84	17.0	14.09
18.6 - 24.5	0	0	0	1	0	0	0	0	0	0	1	0	0	3	1	0	6	1.2	20.03
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	.2	28.10
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	9	16	26	21	25	36	42	68	57	30	39	31	28	21	22	23	494	0.0	8.08
PERCENT	1.8	3.2	5.3	4.3	5.1	7.3	8.5	13.8	11.5	6.1	7.9	6.3	5.7	4.3	4.5	4.7	100.0		
AV SPD	9.8	6.9	9.6	9.2	8.9	8.9	7.9	9.1	10.4	9.4	10.7	11.0	9.3	12.8	10.8	9.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	2	1	0	0	0	0	1	2	0	3	4	0	1	0	0	16	5.4	2.67
3.6 -7.5	2	0	6	3	8	7	10	4	2	2	6	2	1	4	3	5	65	22.1	5.80
7.6 - 12.5	4	4	2	5	4	9	16	7	10	5	10	15	10	25	12	7	145	49.3	9.83
12.6 - 18.5	0	5	3	2	3	1	1	1	3	6	9	10	9	7	4	4	68	23.1	13.79
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	8	11	12	10	15	17	27	13	17	13	28	31	20	37	19	16	294	0.0	7.89
PERCENT	3.7	3.7	4.1	3.4	5.1	5.8	9.2	4.4	5.8	4.4	9.5	10.5	0.8	12.5	6.5	5.4	100.0		
AV SPD	6.8	10.4	7.4	8.0	8.7	8.5	8.8	8.3	9.5	11.1	9.6	10.5	11.0	10.1	10.2	9.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 20 OF 48)

MONTH OF MAY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	.6	.50
CALM + - 3.5	1	1	0	1	0	2	0	2	3	1	2	0	0	0	3	1	17	10.6	2.18
3.6 -7.5	3	1	2	2	1	4	3	1	6	3	2	1	1	3	2	8	43	26.9	5.65
7.6 - 12.5	2	0	3	1	8	3	2	1	3	8	11	5	7	10	5	5	74	46.3	10.14
12.6 - 18.5	1	0	1	0	0	4	0	1	0	3	6	0	4	5	0	0	25	15.6	14.11
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	7	2	6	4	9	13	5	5	12	15	21	6	13	18	10	14	160	0.0	6.04
PERCENT	4.4	1.3	3.8	2.5	5.6	8.1	3.1	3.1	7.5	9.4	13.1	3.8	8.1	11.3	6.3	8.8	100.0		
AV SPD	8.1	4.0	8.8	5.9	9.6	9.3	6.8	6.0	5.1	9.6	10.4	9.9	10.6	10.8	7.5	7.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF MAY
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	.1
CALM + - 3.5	1	6	2	6	4	6	1	6	9	3	8	6	4	5	4	4	79	4.3
3.6 -7.5	34	23	37	34	33	33	46	30	29	23	28	12	33	25	22	34	483	26.6
7.6 - 12.5	15	25	34	39	42	53	42	77	72	45	72	76	74	76	55	34	851	46.8
12.6 - 18.5	5	13	30	7	10	8	16	26	29	13	33	32	60	41	25	13	361	12.9
18.6 - 24.5	0	1	2	1	2	1	0	0	7	1	4	4	8	4	2	0	37	2.0
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	2	1	0	0	2	6	.3
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	79	68	105	87	81	101	105	139	146	85	146	139	181	151	108	87	1818	100.0
PERCENT	4.3	3.7	5.6	4.8	3.0	3.6	5.8	7.0	3.0	4.7	6.0	7.6	10.0	8.3	5.9	4.8	100.0	

TABLE 2.3.2-16 (SHEET 21 OF 48)

MONTH OF JUNE

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	1	1	3	0	1	1	0	0	0	0	1	1	2	0	0	11	2.7	2.76
3.6 -7.5	13	17	7	9	10	7	7	7	3	5	9	11	6	5	1	5	122	29.9	5.53
7.6 - 12.5	7	14	13	17	20	19	8	4	3	7	7	15	31	15	4	6	190	46.6	9.57
12.6 - 18.5	0	1	2	6	2	0	0	2	3	0	4	5	30	16	4	2	77	18.9	14.71
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	2	5	0	1	0	8	2.0	20.30
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	20	33	23	35	32	27	16	13	9	12	20	34	73	30	10	13	408	0.0	7.92
PERCENT	4.9	8.1	5.6	8.6	7.8	6.6	3.9	3.2	2.2	2.9	4.9	8.3	17.0	9.3	10.5	3.2	100.0		
AV SPD	7.0	7.3	8.8	9.2	8.7	8.2	7.4	8.9	10.4	7.6	8.9	9.8	12.4	11.1	11.6	8.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 7

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2.5	2.56
3.6 -7.5	2	1	0	5	2	3	1	2	2	3	1	3	0	0	0	2	37	33.3	5.13
7.6 - 12.5	4	1	1	2	6	4	1	1	3	1	5	3	2	1	2	1	40	49.4	9.31
12.6 - 18.5	1	0	0	0	0	1	1	2	0	1	3	0	2	0	1	0	12	14.8	14.33
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	7	3	2	7	8	8	3	5	7	0	9	6	4	1	3	3	81	0.0	7.25
PERCENT	8.6	3.7	2.5	8.6	9.9	9.9	3.7	6.2	8.6	6.2	11.1	7.4	4.9	1.2	3.7	3.7	100.0		
AV SPD	8.6	6.6	5.9	8.1	8.1	7.9	9.9	9.0	7.9	8.1	11.2	8.1	11.7	10.5	12.0	8.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-16 (SHEET 22 OF 48)

MONTH OF JUNE

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
3.6 -7.5	0	0	1	0	0	1	0	3	0	1	1	0	0	0	0	0	7	33.3	5.21
7.6 - 12.5	1	1	2	0	0	1	0	0	2	1	1	1	0	0	0	1	11	52.4	9.75
12.6 - 18.5	0	0	0	0	0	0	0	0	1	0	0	1	1	0	0	0	3	14.3	14.73
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	1	3	0	0	2	0	3	3	2	2	2	1	0	0	1	21	0.0	7.85
PERCENT	4.8	4.8	14.3	0.0	0.0	9.5	0.0	14.3	14.3	9.5	9.5	9.5	4.8	0.0	0.0	4.8	100.0		
AV SPD	8.5	8.2	8.9	0.0	0.0	7.1	0.0	5.9	11.9	7.2	8.3	12.7	14.6	0.0	0.0	11.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	2	1	2	0	0	0	2	0	2	2	0	0	2	0	0	13	4.1	2.75
3.6 -7.5	1	9	10	4	6	4	14	16	7	9	8	6	5	2	0	4	105	33.1	5.62
7.6 - 12.5	5	5	4	6	22	12	11	15	7	11	18	9	4	2	9	4	142	44.8	9.64
12.6 - 18.5	2	0	1	1	2	5	2	7	0	5	8	5	5	1	4	0	48	15.1	13.79
18.6 - 24.5	0	0	0	2	0	0	0	0	0	1	1	1	1	1	0	0	7	2.2	19.74
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	.6	25.40
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	8	16	16	15	30	21	27	40	14	28	35	21	15	8	13	9	317	0.0	7.54
PERCENT	2.5	8.0	5.0	4.7	9.5	6.6	8.5	12.6	4.4	3.6	11.0	6.6	5.0	2.5	4.1	2.5	100.0		
AV SPD	10.0	6.2	6.6	9.5	9.3	10.0	9.2	8.5	8.1	9.1	2.5	10.3	11.6	8.8	11.1	9.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-16 (SHEET 23 OF 48)

MONTH OF JUNE

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	1	1	1	3	4	1	1	2	0	2	0	0	0	2	1	21	3.9	2.13
3.6 -7.5	6	4	3	7	11	8	21	22	11	9	9	10	3	2	2	4	132	24.5	5.41
7.6 - 12.5	5	1	3	8	18	16	24	60	36	33	28	26	22	5	2	5	292	54.2	9.97
12.6 - 18.5	1	1	5	1	3	3	3	10	10	13	18	8	8	1	2	3	90	16.7	13.91
18.6 - 24.5	0	0	1	0	0	0	0	0	0	0	1	0	0	1	0	0	3	.6	20.39
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	.2	33.90
TOTAL	14	7	13	17	35	31	49	93	59	55	59	44	33	9	8	13	539	100.0	7.08
PERCENT	2.6	1.3	2.4	3.2	6.5	5.8	9.1	17.3	10.9	10.2	10.9	8.2	6.1	1.7	1.5	2.4	100.0		
AV SPD	7.5	6.7	11.3	7.7	8.1	8.1	8.2	9.3	9.7	10.5	11.2	10.0	11.0	10.3	7.5	9.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	2	0	3	4	1	2	0	1	1	0	1	0	1	0	0	16	5.9	2.29
3.6 -7.5	4	5	4	7	5	3	5	8	11	4	4	1	2	5	1	2	71	26.0	5.62
7.6 - 12.5	3	3	6	3	7	10	12	20	10	14	21	6	17	3	2	0	143	52.4	9.70
12.6 - 18.5	1	4	0	0	0	2	0	0	2	6	8	13	2	1	0	1	42	15.4	13.93
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	.4	24.90
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	8	14	10	13	16	16	21	34	24	25	33	21	21	11	3	3	273	0.0	7.30
PERCENT	2.9	5.1	3.7	4.8	5.9	5.9	7.7	12.5	8.0	9.2	12.1	7.7	7.7	4.0	1.1	1.1	100.0		
AV SPD	8.4	8.9	8.6	6.1	6.7	9.2	8.4	8.5	8.4	10.0	10.6	11.5	10.1	5.8	9.5	8.4			

AVERAGE SPEED FOR THIS TABLE EQUALS9.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

TABLE 2.3.2-16 (SHEET 24 OF 48)

MONTH OF JUNE

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	0	0	0	1	0	2	0	0	0	1	0	0	1	1	17	7.6	3.03
3.6 -7.5	0	0	1	0	3	1	5	3	5	1	0	0	2	2	1	2	26	28.3	5.54
7.6 - 12.5	3	1	1	1	1	5	2	5	3	3	4	2	5	4	4	5	49	53.3	9.88
12.6 - 18.5	0	0	0	0	1	0	0	2	0	1	1	2	0	0	1	1	9	9.8	13.48
18.6 - 24.5	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1.1	20.10
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	4	1	2	1	5	7	7	12	9	5	5	5	7	6	7	9	92	0.0	7.26
PERCENT	4.3	1.1	2.2	1.1	5.4	7.6	7.6	13.0	9.8	5.4	5.4	5.4	7.6	6.5	7.6	9.5	100.0		
AV SPD	8.7	8.6	8.6	8.4	8.1	8.9	6.7	8.7	8.9	9.	12.7	9.5	8.2	9.2	8.4	8.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF JUNE
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
CALM + - 3.5	3	7	4	9	7	7	4	5	3	3	4	3	1	5	3	2	70	4.0
3.6 -7.5	26	36	26	32	37	27	53	61	39	32	32	31	18	16	5	19	490	28.3
7.6 - 12.5	28	26	30	37	74	67	58	111	56	70	82	82	81	30	23	22	867	50.1
12.6 - 18.5	5	6	0	8	8	11	8	23	16	26	42	34	48	19	12	7	281	16.2
18.6 - 24.5	0	0	1	2	0	0	0	0	1	1	2	3	6	2	1	0	19	1.1
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	3	.2
32.6+	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	.1
TOTAL	62	75	69	86	126	112	123	200	125	132	163	133	155	73	44	51	1731	100.0
PERCENT	3.0	4.3	4.0	5.1	7.3	6.8	7.1	11.6	7.2	7.0	9.4	7.7	9.0	4.2	2.5	2.9	100.0	

TABLE 2.3.2-16 (SHEET 25 OF 48)

MONTH OF JULY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	1	0	1	0	1	0	0	0	1	1	0	1	1	0	8	2.7	2.81
3.6 -7.5	4	5	7	5	2	3	2	1	2	1	9	22	27	14	5	3	112	37.5	5.69
7.6 - 12.5	2	0	1	6	20	14	4	1	0	2	16	32	36	19	1	0	154	51.5	9.43
12.6 - 18.5	1	0	1	2	10	3	0	0	0	0	1	2	3	2	0	0	25	8.4	14.17
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	8	5	10	13	33	20	7	2	2	3	27	57	66	36	7	3	299	0.0	7.36
PERCENT	2.7	1.7	3.3	4.3	11.0	6.7	2.3	.7	.7	1.0	9.0	19.1	22.1	12.0	2.3	1.0	100.0		
AV SPD	7.5	5.0	6.8	8.8	11.0	9.7	8.5	6.5	7.0	8.1	8.4	8.1	8.4	8.0	5.1	5.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 5

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	2	3.6	3.07
3.6 -7.5	0	0	2	2	1	1	0	1	3	1	1	1	2	2	1	3	21	38.2	5.52
7.6 - 12.5	0	0	0	0	2	4	1	0	0	0	3	6	4	2	1	0	23	41.8	9.73
12.6 - 18.5	0	0	0	0	0	0	0	1	1	1	3	0	0	0	0	0	6	10.9	13.90
18.6 - 24.5	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1.8	19.10
24.6 - 32.5	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	2	3.6	28.37
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	0	1	3	2	3	5	1	2	4	2	8	7	7	5	2	3	55	0.0	7.46
PERCENT	0.0	1.8	5.5	3.6	5.5	9.1	1.8	3.6	7.3	3.6	14.9	12.7	12.7	9.1	3.6	3.5	100.0		
AV SPD	0.0	19.1	12.8	5.5	9.2	9.9	10.6	2.3	8.7	8.6	13.2	6.8	7.1	6.9	5.7	5.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 3

TABLE 2.3.2-16 (SHEET 26 OF 48)

MONTH OF JULY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0	0	3	8.1	3.32
3.6 -7.5	0	0	1	4	1	2	0	1	2	0	1	2	4	0	0	1	19	51.4	5.48
7.6 - 12.5	0	0	1	1	2	1	0	0	0	0	1	2	1	1	1	0	11	29.7	10.22
12.6 - 18.5	0	0	0	0	0	1	0	0	1	0	0	1	0	0	1	0	4	10.8	13.33
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	0	0	3	5	4	4	0	1	3	1	2	5	5	1	2	1	37	0.0	6.44
PERCENT	0.0	0.0	8.1	13.5	10.8	10.8	0.0	2.7	8.1	2.7	5.4	13.5	13.5	2.7	5.4	2.7	100.0		
AV SPD	0.0	0.0	8.0	7.1	8.8	0.0	6.7	8.0	3.5	9.7	9.5	5.9	9.5	11.2	5.9				

AVERAGE SPEED FOR THIS TABLE EQUALS 7.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	1	0	1	0	4	1	1	0	0	0	0	8	3.0	2.88
3.6 -7.5	0	3	9	6	9	9	10	11	7	4	10	10	10	3	2	3	105	39.8	5.57
7.6 - 12.5	0	0	10	5	12	10	9	9	7	10	11	15	11	3	1	0	125	47.3	9.43
12.6 - 18.5	1	1	2	4	0	0	1	1	3	3	4	1	0	1	0	1	23	8.7	14.30
18.6 - 24.5	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	2	.8	22.50
24.6 - 32.5	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	.4	24.50
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	4	21	15	21	25	20	22	18	28	26	27	21	7	3	4	264	0.0	7.21
PERCENT	.8	1.5	8.0	5.7	3.0	9.5	7.6	2.3	8.6	10.8	9.8	10.2	8.0	2.7	1.1	1.5	100.0		
AV SPD	19.4	6.7	8.1	9.4	7.7	8.1	7.9	7.3	9.6	9.4	9.1	8.1	8.0	9.1	5.9	8.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 5

TABLE 2.3.2-16 (SHEET 27 OF 48)

MONTH OF JULY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	1	0	1	0	1	1	3	1	2	3	5	1	1	0	1	22	3.4	2.68
3.6 -7.5	4	4	9	6	8	14	16	19	14	16	20	9	13	14	5	1	172	26.6	5.71
7.6 - 12.5	2	3	10	12	19	17	18	52	54	58	72	33	26	5	1	5	392	60.7	9.81
12.6 - 18.5	0	2	1	1	0	1	1	8	12	11	10	4	3	3	1	1	59	9.1	13.98
18.6 - 24.5	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.2	21.20
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	7	10	21	20	27	33	36	82	81	87	105	56	43	23	7	8	646	0.0	7.82
PERCENT	1.1	1.5	3.3	3.1	4.2	5.1	5.6	12.7	12.5	13.5	16.3	8.7	6.7	3.6	1.1	1.2	100.0		
AV SPD	6.4	8.8	8.5	8.3	8.9	8.0	7.5	9.2	9.9	9.8	9.8	9.0	8.6	7.7	7.5	9.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	.4	.50
CALM + - 3.5	0	1	0	2	0	0	1	0	0	0	1	1	0	0	1	0	7	3.1	1.99
3.6 -7.5	1	1	1	3	2	1	6	9	8	9	8	5	5	1	0	2	52	27.2	5.81
7.6 - 12.5	1	0	2	4	10	10	10	10	11	26	22	19	9	2	0	0	42	72.3	9.63
12.6 - 18.5	0	0	0	1	0	0	0	1	0	5	2	3	2	1	0	0	15	6.6	13.22
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	.4	20.90
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	2	3	10	12	11	17	26	19	40	33	38	17	4	2	2	228	0.0	7.10
PERCENT	.2	.9	1.3	4.4	5.3	4.8	7.5	11.4	8.3	17.5	14.5	12.3	7.5	1.8	.9	.9	100.0		
AV SPD	7.9	5.8	7.7	7.4	9.2	8.8	7.6	8.7	8.1	9.1	9.0	9.5	8.8	10.6	11.0	6.7			

AVERAGE SPEED FOR THIS TABLE EQUALS8.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 28 OF 48)

MONTH OF JULY

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	1	0	1	0	0	0	0	1	0	0	2	5	11.4	2.97
3.6 -7.5	0	0	1	0	2	1	0	0	1	0	1	2	0	0	0	0	8	18.2	5.38
7.6 - 12.5	0	0	0	4	1	2	2	2	3	7	1	0	2	0	0	1	25	56.8	9.93
12.6 - 18.5	0	0	1	0	0	0	1	0	0	1	2	0	0	0	0	0	5	11.4	14.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2.3	18.80
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	0	0	2	4	3	4	3	3	4	8	4	2	3	1	0	3	44	0.0	7.22
PERCENT	0.0	0.0	4.5	9.1	6.8	9.1	6.8	6.8	9.1	18.2	9.1	4.5	6.8	2.3	0.0	6.8	100.0		
AV SPD	0.0	0.0	10.5	9.5	7.9	7.5	12.0	7.0	9.0	10.9	10.4	5.5	7.6	18.8	0.0	5.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

MONTH OF JULY
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	.1
CALM + - 3.5	2	2	2	3	2	3	3	5	1	7	6	8	3	3	8	3	55	3.5
3.6 -7.5	9	13	30	26	25	30	34	42	37	31	50	51	61	34	13	13	499	31.7
7.6 - 12.5	5	3	24	32	66	64	44	88	75	109	126	112	89	32	5	6	872	55.4
12.6 - 18.5	2	3	5	8	10	5	3	11	17	21	22	11	8	7	2	2	137	8.7
18.6 - 24.5	1	1	1	0	0	0	0	0	1	0	0	0	0	1	1	0	6	.4
24.6 - 32.5	0	0	1	0	0	0	0	0	0	1	1	0	0	0	0	0	3	.2
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	19	22	63	60	103	102	84	138	131	169	205	182	152	77	23	24	1573	100.0
PERCENT	1.2	1.4	4.0	4.4	5.5	6.5	5.3	8.8	8.3	10.7	13.0	11.6	10.3	4.9	1.5	1.5	100.0	

TABLE 2.3.2-16 (SHEET 29 OF 48)

MONTH OF AUGUST

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	4	1	2	0	1	1	0	1	2.	0	1	1	0	1	4	1	20	6.7	2.50
3.6 -7.5	18	10	12	8	9	8	1	1	2	1	9	18	20	8	5	6	134	45.0	5.57
7.6 - 12.5	2	7	8	10	16	9	4	0	0	1	13	27	20	4	1	1	123	41.3	9.74
12.6 - 18.5	0	1	4	0	2	5	0	1	0	0	4	3	1	0	0	0	21	7.0	13.68
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	24	19	26	18	28	21	5	3	4	2	27	49	41	13	10	8	298	0.0	7.44
PERCENT	8.1	6.4	8.7	6.0	9.4	7.0	1.7	1.0	1.3	.7	9.1	16.4	13.8	4.4	3.4	2.7	100.0		
AV SPD	5.4	7.5	8.2	8.9	8.7	9.3	10.2	7.6	4.5	7.2	8.9	8.6	7.9	6.0	5.3	5.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 34

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	2	1	0	0	0	0	0	1	0	0	0	0	0	0	5	2.4	3.01
3.6 -7.5	2	2	5	2	2	1	0	0	0	0	3	3	1	0	0	0	21	39.8	5.83
7.6 - 12.5	0	1	4	1	4	1	3	0	1	1	4	3	0	0	0	0	23	43.4	8.50
12.6 - 18.5	0	1	0	0	0	0	0	0	0	0	0	1	0	2	0	0	4	7.5	4.10
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	4	11	4	6	2	3	0	1	2	7	7	1	2	0	0	53	0.0	8.65
PERCENT	5.7	7.5	20.8	7.5	11.3	3.8	5.7	0.0	1.9	3.8	13.2	13.2	1.9	3.8	0.0	0.0	100.0		
AV SPD	4.9	8.8	6.5	6.5	8.0	9.5	10.9	0.0	9.3	6.0	8.5	8.0	6.2	14.2	.0.	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 3

TABLE 2.3.2-16 (SHEET 30 OF 48)

MONTH OF AUGUST

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2.3	3.20
3.6 -7.5	1	0	2	1	2	2	1	0	2	1	3	4	2	0	3	0	24	54.5	5.64
7.6 - 12.5	0	1	0	1	2	5	0	0	2	0	1	3	2	0	0	0	17	38.6	9.42
12.6 - 18.5	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	2.3	13.30
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2.3	21.20
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	1	2	2	4	7	2	0	4	1	4	7	4	0	5	0	44	0.0	6.77
PERCENT	2.3	2.3	4.5	4.5	9.1	15.9	4.5	0.0	9.1	2.3	9.1	15.9	9.1	0.0	11.4	0.0	100.0		
AV SPD	3.8	9.9	4.9	9.3	8.5	8.7	9.4	0.0	6.5	7.2	7.1	7.9	7.6	0.0	8.0	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 3

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	.3	.50
CALM + - 3.5	1	1	5	1	2	2	1	1	2	4	6	3	2	2	1	0	34	9.4	2.95
3.6 -7.5	3	7	11	9	12	15	11	11	19	11	19	14	9	4	4	0	159	43.9	5.45
7.6 - 12.5	4	5	13	6	10	21	13	13	8	8	19	10	2	2	2	1	137	37.8	9.48
12.6 - 18.5	2	2	3	4	1	2	5	3	2	3	1	0	0	0	0	1	29	8.0	14.20
18.6 - 24.5	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	.3	20.90
24.6 - 32.5	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	.3	25.30
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	10	15	32	20	25	40	31	28	31	27	45	28	13	8	7	2	362	0.0	6.11
PERCENT	2.8	4.1	8.5	5.5	6.0	11.0	8.6	7.7	8.6	7.5	12.4	7.7	3.6	2.2	1.9	.6	100.0		
AV SPD	9.4	8.1	7.6	8.7	7.4	8.3	9.1	8.3	7.3	7.8	7.2	6.7	5.5	8.0	5.7	11.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 7.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 15

TABLE 2.3.2-16 (SHEET 31 OF 48)

MONTH OF AUGUST

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	1	3	0	4	4	5	3	4	5	0	2	3	1	2	0	44	5.9	2.44
3.6 -7.5	6	11	18	17	19	27	34	22	34	38	19	9	8	5	10	3	280	37.4	5.58
7.6 - 12.5	5	2	10	16	38	29	42	48	64	72	40	11	4	2	3	3	389	51.9	9.39
12.6 - 18.5	2	0	1	1	2	0	0	4	10	8	3	0	0	0	2	0	33	4.4	14.13
18.6 - 24.5	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	2	.3	20.70
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	.1	25.80
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	15	14	33	39	63	60	81	77	113	123	62	22	15	8	18	6	749	0.0	6.68
PERCENT	2.0	1.9	4.4	5.2	8.4	8.0	10.8	10.3	15.1	16.4	8.4	2.9	2.0	1.1	2.4	.8	100.0		
AV SPD	8.5	6.0	7.1	7.1	8.1	7.3	7.4	8.4	8.8	8.8	8.6	6.8	5.9	6.7	7.5	7.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 16

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2	.6	.50
CALM + - 3.5	2	1	0	0	2	2	2	3	0	1	3	2	3	2	3	1	27	7.6	2.23
3.6 -7.5	6	6	13	6	10	13	15	13	9	16	8	7	5	5	3	4	139	38.9	5.51
7.6 - 12.5	4	0	3	5	10	17	22	13	24	32	21	11	5	1	4	1	178	49.9	9.36
12.6 - 18.5	0	0	0	0	2	2	2	0	2	1	2	0	0	0	0	0	11	3.1	13.52
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	3	16	12	29	34	41	29	35	50	34	20	13	8	10	6	357	0.0	6.24
PERCENT	3.4	2.2	4.5	3.4	8.1	8.5	11.5	8.1	9.8	14.0	9.5	5.6	3.6	2.2	2.8	1.7	100.0		
AV SPD	6.1	4.8	6.2	6.7	8.0	7.6	7.8	7.1	8.1	8.3	8.6	7.6	6.6	5.5	5.9	5.8			

AVERAGE SPEED FOR THIS TABLE EQUALS7.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 12

TABLE 2.3.2-16 (SHEET 32 OF 48)

MONTH OF AUGUST

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	1	1	0	0	0	0	0	1	1	0	1	0	0	0	1	6	7.5	2.16
3.6 -7.5	3	4	0	2	0	5	2	0	2	1	2	1	1	1	1	5	30	37.5	5.55
7.6 - 12.5	0	2	3	0	6	2	0	1	3	11	9	0	0	0	1	1	39	48.8	9.92
12.6 - 18.5	0	0	0	0	1	0	0	0	0	2	2	0	0	0	0	0	5	6.3	13.87
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	7	4	2	7	7	2	1	6	15	13	2	1	1	2	7	80	0.0	6.41
PERCENT	3.8	8.8	5.0	2.5	8.8	8.8	2.5	1.3	7.5	18.8	16.3	2.5	1.3	1.3	2.5	8.8	100.0		
AV SPD	5.6	5.4	7.7	5.6	10.7	7.2	4.1	8.2	5.3	9.9	10.9	4.3	7.1	7.1	6.5	6.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF AUGUST
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	1	0	1	0	0	0	0	0	0	0	1	0	0	0	0	3	.2
CALM + - 3.5	10	5	13	7	9	9	8	8	9	12	10	9	8	6	11	3	137	7.1
3.6 -7.5	39	40	61	45	54	50	54	47	68	68	63	56	46	23	26	18	787	40.5
7.6 - 12.5	15	18	41	39	91	84	84	75	102	125	107	65	33	9	11	7	906	46.6
12.6 - 18.5	4	4	8	5	8	9	8	8	14	14	12	4	1	2	2	1	104	5.4
18.6 - 24.5	0	0	1	0	0	0	1	0	1	0	0	0	0	0	1	0	4	.2
24.6 - 32.5	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	2	.1
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	68	68	124	97	162	171	165	138	194	220	192	135	88	40	52	29	1943	100.0
PERCENT	3.5	3.5	6.4	5.0	8.3	8.8	8.5	7.1	10.0	11.3	9.9	6.9	4.5	2.1	2.7	1.5	100.0	

TABLE 2.3.2-16 (SHEET 33 OF 48)

MONTH OF SEPTEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	3	0	1	1	0	0	1	0	0	0	1	2	7	0	1	1	18	5.0	2.53
3.6 -7.5	11	12	13	17	12	8	4	0	1	5	4	7	12	13	10	9	138	38.5	5.39
7.6 - 12.5	11	5	15	17	23	9	8	1	6	6	8	29	11	5	4	5	163	45.5	9.78
12.6 - 18.5	0	1	7	7	9	2	1	0	2	0	1	4	3	1	0	0	38	10.6	14.61
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.3	19.40
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	25	18	36	42	44	19	14	1	9	11	14	43	33	19	15	15	358	0.0	6.88
PERCENT	7.0	5.0	10.1	11.7	12.3	5.3	3.9	.3	2.5	3.1	3.9	12.0	9.2	5.3	4.2	4.2	100.0		
AV SPD	6.8	6.8	9.2	8.7	10.2	8.6	7.9	11.1	10.7	7.5	8.8	9.9	7.5	7.1	6.3	6.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	1	0	0	1	0	1	1	0	0	0	0	1	0	0	5	7.6	3.01
3.6 -7.5	2	3	1	3	1	0	2	1	1	0	0	2	0	2	0	4	22	33.3	4.79
7.6 - 12.5	1	4	2	2	5	2	1	0	1	2	3	1	0	1	1	1	20	43.9	9.73
12.6 - 18.5	0	0	1	2	3	1	0	0	1	0	0	0	0	0	0	1	9	13.6	14.65
18.6 - 24.5	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1.5	20.80
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	7	5	7	9	4	3	2	5	2	5	2	0	4	1	6	66	0.0	6.57
PERCENT	4.5	10.5	7.6	10.6	13.6	5.1	4.5	3.0	7.6	3.0	7.6	4.5	0.0	5.1	1.5	9.1	100.0		
AV SPD	6.8	7.1	8.8	8.9	11.1	9.0	6.5	5.0	10.7	10.8	10.5	5.3	0.0	6.2	8.4	0.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 3

TABLE 2.3.2-16 (SHEET 34 OF 48)

MONTH OF SEPTEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	1	0	0	0	1	0	1	0	0	0	0	0	0	0	3	10.3	3.04
3.6 -7.5	2	1	0	0	0	0	2	0	1	0	0	0	0	2	1	0	9	31.0	5.28
7.6 - 12.5	1	1	2	1	1	1	0	2	0	0	0	0	0	1	2	0	12	41.4	9.74
12.6 - 18.5	0	0	0	2	0	0	0	0	0	1	1	0	0	0	1	0	5	17.2	15.74
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	3	2	3	3	1	1	2	3	1	2	1	0	0	3	4	0	29	0.0	6.84
PERCENT	10.3	6.9	10.3	10.3	3.4	3.4	6.9	10.3	3.4	6.9	3.4	0.0	0.0	10.3	13.8	0.0	100.0		
AV SPD	6.4	6.5	8.2	15.5	12.2	8.3	5.3	7.8	5.8	9.2	17.4	0.0	0.0	6.9	9.9	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	.3	.50
CALM + - 3.5	3	3	1	2	1	1	2	1	1	1	2	0	0	1	0	2	21	5.9	2.73
3.6 -7.5	19	10	14	17	11	11	12	7	6	7	8	5	2	6	4	5	144	40.2	5.49
7.6 - 12.5	6	6	25	19	18	9	6	5	10	8	3	0	4	5	3	2	129	36.0	9.58
12.6 - 18.5	0	0	16	14	8	1	1	1	1	6	6	0	0	0	0	1	55	15.4	14.73
18.6 - 24.5	1	0	2	2	2	0	0	0	0	0	0	0	0	0	0	0	7	2.0	20.67
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	.3	30.60
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	29	19	58	54	40	22	21	14	18	23	19	6	6	12	7	10	358	0.0	6.69
PERCENT	8.1	5.3	16.2	15.1	11.2	6.1	5.9	3.9	5.0	6.4	5.3	1.7	1.7	3.4	2.0	2.6	100.0		
AV SPD	6.7	6.6	10.8	9.9	10.2	7.4	7.2	6.3	8.2	9.0	9.0	6.5	8.6	6.4	6.7	6.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =4

TABLE 2.3.2-16 (SHEET 35 OF 48)

MONTH OF SEPTEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	1	0	0	1	0	1	1	1	0	0	1	0	0	0	0	6	.9	.50
CALM + - 3.5	5	8	1	4	4	3	2	2	3	5	1	2	2	2	0	1	46	5.9	2.17
3.6 -7.5	11	12	18	25	15	23	23	12	13	12	17	8	5	7	7	7	215	32.5	5.58
7.6 - 12.5	8	10	49	45	37	24	10	24	38	35	15	23	8	6	8	8	348	52.6	9.55
12.6 - 18.5	0	0	10	7	9	4	3	8	1	2	0	0	0	0	0	0	44	6.6	13.94
18.6 - 24.5	0	1	0	1	0	0	0	0	0	0	0	0	0	1	0	0	3	.5	20.45
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	25	32	78	82	66	54	39	47	56	54	33	34	15	16	15	16	662	0.0	5.94
PERCENT	3.8	4.8	11.8	12.4	10.0	8.2	5.9	7.1	8.5	8.2	5.0	5.1	2.3	2.4	2.3	2.4	100.0		
AV SPD	5.9	6.6	9.7	8.7	8.9	7.8	7.1	9.2	8.5	8.5	7.5	7.9	7.2	7.1	7.3	7.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	2	.6	.50
CALM + - 3.5	2	1	2	1	1	2	3	0	1	1	1	0	1	0	1	0	17	5.2	1.85
3.6 -7.5	7	3	11	14	17	8	8	9	6	14	10	4	8	3	6	2	130	39.8	5.54
7.6 - 12.5	4	11	10	22	33	16	14	3	5	14	12	5	4	2	2	7	164	50.2	9.57
12.6 - 18.5	0	0	3	3	4	1	0	0	0	0	1	1	0	0	0	1	14	4.3	13.12
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	13	15	26	40	55	27	25	12	12	30	24	10	14	5	9	10	327	0.0	5.86
PERCENT	4.0	4.6	8.0	12.2	16.8	8.3	7.6	3.7	3.7	9.2	7.3	3.1	4.3	1.5	2.8	3.1	100.0		
AV SPD	6.3	7.9	7.9	8.6	8.9	8.4	7.6	6.6	6.8	6.9	7.7	8.4	5.8	6.7	6.7	9.2			

AVERAGE SPEED FOR THIS TABLE EQUALS7.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 36 OF 48)

MONTH OF SEPTEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	1	0	0	1	1	0	0	0	0	0	0	1	0	0	4	3.1	0.50
CALM + - 3.5	3	1	2	2	2	0	3	3	0	1	3	0	2	1	1	0	24	18.6	1.93
3.6 -7.5	1	3	1	1	3	1	1	2	2	4	4	3	3	2	5	1	37	28.7	5.38
7.6 - 12.5	5	1	1	0	5	4	5	3	2	4	9	4	5	4	4	6	62	48.1	9.42
12.6 - 18.5	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	2	1.6	13.56
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	9	6	5	3	10	6	10	8	4	10	16	7	10	8	10	7	129	0.0	3.79
PERCENT	7.0	4.7	3.9	2.3	7.8	4.7	7.8	6.2	3.1	7.8	12.4	5.4	7.8	6.2	7.8	5.4	100.0		
AV SPD	6.7	7.0	5.0	3.2	7.1	8.0	5.6	6.0	6.9	8.0	7.5	7.1	7.0	6.8	6.5	9.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF SEPTEMBER
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	1	1	0	1	1	2	1	1	2	0	1	1	1	0	0	13	.7
CALM + - 3.5	17	13	9	10	8	7	11	8	6	9	8	4	12	5	3	4	134	6.9
3.6 -7.5	53	44	58	77	59	51	52	31	30	42	43	29	30	35	33	28	695	36.0
7.6 - 12.5	36	38	104	106	122	65	44	38	62	69	52	62	32	24	24	29	907	47.0
12.6 - 18.5	0	2	37	35	33	9	5	9	5	10	9	5	3	1	1	3	167	8.7
18.6 - 24.5	1	1	2	3	2	0	0	0	1	0	0	1	0	1	0	0	12	.6
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	.1
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	107	99	211	231	225	133	114	87	105	132	112	103	78	67	61	64	1929	100.0
PERCENT	5.5	5.1	10.9	12.0	11.7	6.9	5.9	4.5	5.4	6.8	5.8	5.3	4.0	3.5	3.2	3.3	100.0	

TABLE 2.3.2-16 (SHEET 37 OF 48)

MONTH OF OCTOBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	1	2	0	0	0	1	0	0	0	0	1	0	1	0	1	8	1.9	2.55
3.6 -7.5	8	17	8	17	6	5	2	2	2	2	5	7	8	7	9	5	110	26.0	5.53
7.6 - 12.5	20	27	28	19	13	1	0	0	3	4	8	12	14	14	7	13	183	43.3	9.80
12.6 - 18.5	16	8	10	7	4	1	0	0	0	3	9	7	6	9	7	9	96	22.7	14.53
18.6 - 24.5	2	1	0	0	0	0	0	0	0	0	2	2	9	6	3	0	25	5.9	20.86
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0.2	27.30
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	47	54	48	43	23	7	3	2	5	9	24	29	38	37	26	28	423	0.0	8.54
PERCENT	11.1	12.8	11.3	10.2	5.4	1.7	.7	.5	1.2	2.1	5.7	6.9	9.0	8.7	6.1	6.6	100.0		
AV SPD	11.2	9.6	9.7	9.2	9.3	7.1	4.2	5.5	8.4	11.4	11.6	9.9	12.9	12.8	11.2	10.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 5

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	2.0	2.70
3.6 -7.5	1	2	6	1	2	1	0	1	0	0	0	1	1	0	1	1	18	36.0	5.49
7.6 - 12.5	6	3	3	0	2	0	0	1	0	0	0	0	2	0	0	3	20	40.0	9.95
12.6 - 18.5	2	2	2	0	1	0	0	0	0	0	1	1	0	0	0	1	10	20.0	14.78
18.6 - 24.5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2.0	21.20
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	10	7	11	1	6	1	0	2	0	0	1	2	3	0	1	5	50	0.0	7.84
PERCENT	20.0	14.0	22.0	2.0	12.0	2.0	0.0	4.0	0.0	0.0	2.0	4.0	6.0	0.0	2.0	10.0	100.0		
AV SPD	11.7	10.6	8.8	6.9	7.9	7.5	0.0	7.3	0.0	0.0	17.3	8.7	5.0	0.0	4.8	10.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 38 OF 48)

MONTH OF OCTOBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	2	7.4	3.35
3.6 -7.5	2	2	2	0	0	0	3	0	0	0	0	0	0	1	0	0	10	37.0	5.51
7.6 - 12.5	3	2	0	1	0	0	0	1	0	0	0	0	0	1	0	4	12	44.4	9.74
12.6 - 18.5	0	1	0	0	0	0	0	0	0	0	0	1	0	1	0	0	3	11.1	14.26
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	5	5	2	1	0	0	3	1	1	0	1	1	0	3	0	4	27	0.0	7.01
PERCENT	18.5	18.5	7.4	3.7	0.0	0.0	11.1	3.7	3.7	0.0	3.7	3.7	0.0	11.1	0.0	14.6	100.0		
AV SPD	8.3	10.7	4.5	7.5	0.0	0.0	5.7	10.4	3.4	0.0	3.0	12.6	0.0	9.4	0.0	10.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	2	0	2	1	1	1	0	0	0	0	0	0	0	0	0	7	2.4	2.62
3.6 -7.5	4	3	9	10	5	8	3	4	1	1	3	1	4	4	4	13	77	26.9	5.56
7.6 - 12.5	10	13	9	13	7	7	3	3	2	6	8	4	4	4	8	10	111	38.8	9.68
12.6 - 18.5	13	9	6	13	3	3	0	2	11	5	2	2	3	2	1	9	86	30.1	14.79
18.6 - 24.5	0	2	0	0	0	0	0	0	0	0	0	0	1	1	0	0	4	1.4	20.93
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	.3	25.20
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	27	29	26	38	16	19	7	9	14	12	13	7	12	12	13	32	286	0.0	8.40
PERCENT	9.4	10.1	9.1	13.3	5.6	6.0	2.4	3.1	4.9	4.2	4.5	2.4	4.2	4.2	4.5	11.2	100.0		
AV SPD	12.4	11.6	10.4	10.3	8.5	6.3	6.0	6.4	14.0	10.3	9.8	11.2	10.7	10.6	8.5	9.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 39 OF 48)

MONTH OF OCTOBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	3	0	0	2	1	0	1	1	2	0	0	1	0	0	0	1	12	2.5	2.79
3.6 -7.5	10	6	12	10	15	8	2	2	3	6	7	2	8	6	6	6	109	23.0	5.60
7.6 - 12.5	16	17	23	15	12	15	10	8	5	12	8	11	10	6	15	11	194	40.9	10.13
12.6 - 18.5	20	8	21	12	0	2	0	0	2	3	9	13	20	8	12	17	147	31.0	14.58
18.6 - 24.5	1	0	0	0	0	0	1	0	0	0	0	5	1	1	1	2	12	2.5	20.37
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	50	31	56	39	28	25	14	11	12	21	24	32	39	21	34	37	474	0.0	8.84
PERCENT	10.5	6.5	11.8	8.8	5.9	5.3	3.0	2.3	2.5	4.4	5.1	6.8	8.2	4.4	7.2	7.8	100.0		
AV SPD	11.2	10.4	10.6	9.9	7.8	9.0	9.9	9.4	9.0	9.7	10.2	13.3	12.2	11.6	11.4	12.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	1	0	0	1	0	0	0	0	1	0	3	.8	.50
CALM + - 3.5	1	0	0	0	1	1	0	0	0	1	0	0	0	0	0	1	5	1.3	2.38
3.6 -7.5	4	5	7	7	4	5	1	0	3	2	2	1	7	2	4	4	61	15.6	5.92
7.6 - 12.5	19	18	25	24	32	11	7	6	6	8	6	9	20	18	9	11	229	58.7	10.07
12.6 - 18.5	11	10	7	6	5	4	0	1	2	6	9	4	10	10	5	2	92	23.6	13.98
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	35	33	39	37	42	24	9	7	11	18	17	14	37	30	19	18	390	0.0	8.17
PERCENT	9.0	8.5	10.0	9.0	10.8	6.2	2.3	1.9	2.8	4.6	4.4	3.6	9.5	7.7	4.9	4.6	100.0		
AV SPD	11.1	11.0	10.2	10.2	9.6	9.0	8.1	9.5	9.9	9.9	12.5	11.3	10.5	11.2	9.8	9.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 40 OF 48)

MONTH OF OCTOBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	1	0	1	2	1	1	0	0	0	1	0	7	1.6	0.50
CALM + - 3.5	1	1	1	3	2	0	2	1	0	0	2	5	7	0	2	0	27	6.3	2.51
3.6 -7.5	11	8	23	7	8	4	5	2	8	5	4	3	12	6	9	13	128	29.8	5.65
7.6 - 12.5	20	28	22	16	23	7	6	1	6	12	10	11	8	10	5	13	198	46.0	9.64
12.6 - 18.5	14	8	1	4	6	1	0	0	2	14	7	9	2	1	1	0	70	16.3	14.07
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	46	45	47	30	39	13	13	5	18	32	24	28	29	17	18	26	430	0.0	5.86
PERCENT	10.7	10.5	10.9	7.0	9.1	3.0	3.0	1.2	4.2	7.4	5.6	6.5	6.7	4.0	4.2	6.0	100.0		
AV SPD	10.2	9.7	7.6	8.0	9.6	8.2	7.2	4.9	7.3	10.8	10.1	9.5	6.9	8.3	6.3	7.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF OCTOBER
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	1	1	1	2	2	1	0	0	0	2	0	10	.5
CALM + - 3.5	6	4	3	7	6	2	5	2	3	1	3	7	7	1	2	3	62	3.0
3.6 -7.5	40	43	67	52	40	34	16	11	17	16	21	15	40	26	33	42	513	24.7
7.6 - 12.5	94	108	110	88	89	41	26	20	22	42	40	47	58	53	44	65	947	45.5
12.6 - 18.5	76	46	49	42	19	11	0	3	17	31	37	37	41	31	26	38	504	24.2
18.6 - 24.5	4	3	0	0	0	0	1	0	0	0	2	7	11	8	4	2	42	2.0
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	2	.1
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	220	204	229	189	154	89	49	37	61	92	104	113	158	120	111	150	2080	100.0
PERCENT	10.6	9.8	11.0	9.1	7.4	4.3	2.4	1.8	2.9	4.4	5.0	5.4	7.6	5.8	5.3	7.2	100.0	

TABLE 2.3.2-16 (SHEET 41 OF 48)

MONTH OF NOVEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	0	0	1	0	0	0	1	0	0	2	0	0	3	0	0	9	3.2	2.68
3.6 -7.5	4	2	10	7	2	1	3	2	2	2	4	4	7	9	6	4	69	24.6	5.51
7.6 - 12.5	3	6	12	12	4	9	3	5	2	6	3	9	13	12	12	6	117	41.8	10.09
12.6 - 18.5	3	5	6	1	1	3	1	1	0	3	7	5	9	5	13	2	65	23.2	14.62
18.6 - 24.5	0	1	0	0	0	0	0	0	0	0	1	0	5	6	4	2	19	6.8	20.75
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	.4	26.60
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	12	14	28	21	7	13	7	9	4	11	17	18	34	35	36	14	280	0.0	8.52
PERCENT	4.3	5.0	10.0	7.5	2.5	4.6	2.5	3.2	1.4	3.9	6.1	6.4	12.1	12.5	12.9	5.0	100.0		
AV SPD	9.0	11.0	9.4	8.8	9.2	11.7	8.4	9.3	7.5	11.0	11.0	10.6	12.1	10.9	12.8	10.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 3

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
3.6 -7.5	0	2	0	1	0	0	1	0	0	0	0	3	0	0	0	1	8	22.9	5.60
7.6 - 12.5	2	0	2	3	1	0	1	1	0	1	0	1	2	0	3	0	17	48.6	10.24
12.6 - 18.5	0	0	3	0	0	0	0	0	0	0	2	1	0	0	1	1	8	22.9	14.35
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	2	5.7	10.23
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	2	5	4	1	0	2	1	0	1	2	5	2	1	4	3	35	0.0	9.35
PERCENT	5.7	5.7	14.3	11.4	2.9	0.0	5.7	2.9	0.0	2.9	5.7	14.3	5.7	2.9	11.4	8.6	100.0		
AV SPD	11.3	5.3	12.0	8.5	10.2	0.0	8.3	11.5	0.0	11.7	13.7	9.2	11.2	20.9	11.5	13.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 42 OF 48)

MONTH OF NOVEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	2	10.0	3.35
3.6 -7.5	0	0	0	0	0	0	3	0	0	0	1	0	0	0	1	0	5	25.0	5.42
7.6 - 12.5	0	0	1	1	2	0	1	1	1	0	1	0	0	0	0	0	8	40.0	8.89
12.6 - 18.5	0	0	0	0	0	1	0	0	0	1	0	0	0	3	0	0	5	25.0	14.81
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	0	0	1	1	2	1	5	1	1	1	2	0	0	3	2	0	20	3.0	7.26
PERCENT	0.0	0.0	5.0	5.0	10.0	5.0	25.0	5.0	5.0	5.0	10.0	0.0	0.0	15.0	10.0	0.0	100.0		
AV SPD	0.0	0.0	8.7	8.4	8.8	12.8	6.3	8.2	9.9	14.5	7.2	0.0	0.0	15.9	4.0	0.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	3	1	1	1	2	3	1	2	0	0	2	2	2	1	1	2	24	6.7	2.81
3.6 -7.5	6	8	12	7	9	9	11	5	9	5	3	5	4	5	1	2	101	28.1	5.47
7.6 - 12.5	6	6	19	21	14	8	5	8	7	17	10	5	5	11	4	1	147	40.9	9.76
12.6 - 18.5	0	3	14	8	2	5	2	3	6	7	10	4	6	2	6	1	79	22.0	14.46
18.6 - 24.5	2	0	1	1	0	0	1	0	0	0	0	0	0	1	0	1	7	1.9	21.65
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	.3	24.60
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	17	18	47	38	27	25	20	18	22	29	25	16	17	21	12	7	352	0.0	7.50
PERCENT	4.7	5.0	13.1	10.6	7.5	7.0	5.6	5.0	6.1	8.1	7.0	4.5	4.7	5.8	3.3	1.9	100.0		
AV SPD	8.8	8.4	10.7	10.3	7.9	8.5	8.2	8.9	9.1	10.7	11.0	8.9	8.9	10.0	11.6	9.0			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 5

TABLE 2.3.2-16 (SHEET 43 OF 48)

MONTH OF NOVEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	2	.3	.50
CALM + - 3.5	1	2	5	2	5	8	2	4	1	1	3	2	0	2	2	3	43	5.8	2.50
3.6 -7.5	7	9	16	31	22	10	12	15	19	6	6	4	2	9	4	3	175	23.6	5.84
7.6 - 12.5	11	8	43	40	35	39	22	19	22	26	17	19	14	12	12	11	350	47.2	9.89
12.6 - 18.5	7	6	12	6	7	11	9	5	9	12	11	2	14	24	11	7	153	20.6	14.33
18.6 - 24.5	1	1	5	0	0	0	0	0	0	2	0	0	4	2	3	0	18	2.4	19.97
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	27	26	81	79	69	68	45	44	51	47	37	27	35	49	32	24	741	0.0	7.56
PERCENT	3.6	3.5	10.9	10.7	9.3	9.2	6.1	5.9	6.9	6.3	5.0	3.6	4.7	6.6	4.3	3.2	100.0		
AV SPD	10.3	9.3	10.0	8.7	8.9	9.3	9.6	7.9	9.0	11.1	10.1	9.5	12.7	12.0	11.6	9.8			

AVERAGE SPEED FOR THIS TABLE EQUALS9.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	0	1	0	2	0	0	0	0	1	0	0	1	0	1	7	2.8	2.81
3.6 -7.5	2	3	1	0	3	2	3	3	2	0	1	2	1	4	4	2	33	13.4	5.72
7.6 - 12.5	4	10	8	4	7	12	17	8	15	7	13	4	13	10	12	4	148	60.2	10.02
12.6 - 18.5	6	4	3	1	2	6	1	2	2	2	3	1	4	7	12	2	58	23.6	13.74
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	13	17	12	6	12	22	21	13	19	9	18	7	18	22	28	9	246	0.0	9.03
PERCENT	5.3	6.9	4.9	2.4	4.9	8.9	8.5	5.3	7.7	3.7	7.3	2.8	7.3	8.9	11.4	3.7	100.0		
AV SPD	11.4	10.3	10.9	9.1	9.8	10.3	9.8	9.4	9.9	11.1	9.8	8.8	11.0	10.6	11.1	9.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 44 OF 48)

MONTH OF NOVEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	3	0	1	1	1	6	5	1	0	0	1	3	7	2	1	34	12.3	2.34
3.6 -7.5	3	5	3	2	3	8	10	6	3	5	5	1	3	6	6	0	69	25.0	5.37
7.6 - 12.5	6	11	4	5	14	6	9	7	21	10	8	7	11	4	6	6	135	48.9	10.04
12.6 - 18.5	0	1	0	1	2	1	1	1	1	6	5	5	4	7	1	2	38	13.8	13.86
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	11	20	7	9	20	16	26	19	25	21	18	14	21	24	15	9	276	0.0	6.34
PERCENT	4.0	7.2	2.5	3.3	7.2	5.8	9.4	6.9	9.4	7.6	6.5	5.1	7.6	8.7	5.4	3.3	100.0		
AV SPD	7.9	8.3	8.5	8.2	9.4	7.1	7.0	7.1	9.1	10.5	10.4	11.0	9.6	7.9	7.6	9.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

MONTH OF NOVEMBER
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	2	.1
CALM + - 3.5	9	6	6	6	8	14	10	12	2	1	8	5	5	14	6	7	119	6.1
3.6 -7.5	22	29	42	48	39	30	43	31	35	18	20	19	17	33	22	12	460	23.5
7.6 - 12.5	32	41	89	86	77	74	58	49	68	67	52	45	58	49	49	28	922	47.1
12.6 - 18.5	16	19	38	17	14	27	14	12	18	31	38	18	37	48	44	15	406	20.7
18.6 - 24.5	3	2	6	1	0	0	1	0	0	2	1	0	9	10	7	4	46	2.4
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	2	.1
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	82	97	181	158	138	145	126	105	123	119	119	87	127	155	129	66	1957	100.0
PERCENT	4.2	5.0	9.2	8.1	7.1	7.4	6.4	5.4	6.3	6.1	6.1	4.4	5.5	7.0	6.6	3.4	100.0	

TABLE 2.3.2-16 (SHEET 45 OF 48)

MONTH OF DECEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	2	0	0	0	3	0	0	0	0	0	0	2	0	0	0	2	7	2.8	2.92
3.6 -7.5	7	5	4	4	4	3	6	4	2	4	9	4	4	5	3	2	70	28.1	5.27
7.6 - 12.5	5	6	12	13	9	4	3	1	1	2	2	2	12	11	7	6	96	38.6	9.71
12.6 - 18.5	1	3	0	3	3	0	0	0	2	0	2	9	8	13	7	1	52	20.9	14.95
18.6 - 24.5	0	0	0	1	0	0	0	0	2	1	0	2	3	5	4	0	18	7.2	20.75
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	4	2	0	6	2.4	28.77
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	13	14	16	21	19	7	9	5	7	7	13	19	27	38	23	11	249	0.0	8.27
PERCENT	5.2	5.6	6.4	8.4	7.6	2.8	3.6	2.0	2.8	2.8	5.2	7.6	10.8	15.3	9.2	4.4	100.0		
AV SPD	6.9	9.8	8.4	10.3	9.0	9.1	6.7	5.9	12.8	8.4	7.7	11.2	12.0	14.8	15.3	8.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	2	3.8	2.75
3.6 -7.5	0	1	1	1	0	0	1	1	1	0	1	2	1	2	0	1	13	25.0	5.64
7.6 - 12.5	0	1	2	5	4	2	0	0	0	0	1	1	3	3	0	2	24	46.2	9.63
12.6 - 18.5	0	0	0	0	2	0	0	0	0	0	3	3	1	1	0	0	10	19.2	14.99
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1.9	19.20
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	2	3.8	25.79
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	0	2	3	6	6	2	1	1	1	0	7	8	5	6	1	3	52	0.0	8.23
PERCENT	0.0	3.8	5.8	11.5	11.5	3.8	1.9	1.9	1.9	0.0	13.5	15.4	9.6	11.5	1.9	5.8	100.0		
AV SPD	0.0	6.6	9.2	9.1	11.7	8.8	6.6	5.7	7.1	0.0	12.3	11.7	10.5	9.7	19.2	5.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-16 (SHEET 46 OF 48)

MONTH OF DECEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3.1	1.40
3.6 -7.5	1	1	0	1	0	0	1	3	1	2	0	0	0	2	0	0	12	37.5	6.52
7.6 - 12.5	0	0	0	2	2	0	0	0	1	5	0	1	1	1	0	1	14	43.8	9.53
12.6 - 18.5	0	0	0	0	0	0	1	0	0	0	2	0	1	1	0	0	5	15.6	13.07
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2	1	0	3	2	0	2	3	2	7	2	1	2	4	0	1	32	0.0	7.26
PERCENT	6.3	3.1	0.0	9.4	6.3	0.0	6.3	9.4	6.3	21.9	6.3	3.1	6.3	12.5	0.0	3.1	100.0		
AV SPD	3.8	7.4	0.0	9.8	9.8	0.0	9.2	6.9	7.1	8.9	13.1	7.8	11.7	8.8	0.0	10.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	2	2	0	1	2	2	4	3	4	1	2	2	0	2	2	30	5.6	2.45
3.6 -7.5	9	8	17	7	7	3	11	15	6	6	10	9	11	4	3	20	146	27.4	5.37
7.6 - 12.5	8	11	25	34	22	10	4	7	6	6	14	13	22	11	9	2	204	38.3	9.71
12.6 - 18.5	0	0	3	16	9	0	2	1	11	6	17	10	12	13	7	5	112	21.0	14.91
18.6 - 24.5	0	0	0	1	0	0	0	0	1	2	7	5	0	6	13	1	36	6.8	21.44
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	2	0	2	1	0	5	.9	26.66
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	18	21	47	58	39	15	19	27	27	24	49	41	47	36	35	30	533	0.0	7.64
PERCENT	3.4	3.9	8.8	10.9	7.3	2.8	3.6	5.1	5.1	4.5	9.2	7.7	8.8	8.8	6.6	5.6	100.0		
AV SPD	7.2	7.5	8.7	11.3	10.5	8.1	6.9	6.4	10.4	9.5	12.6	11.8	9.6	14.4	15.3	7.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 6

TABLE 2.3.2-16 (SHEET 47 OF 48)

MONTH OF DECEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	1	2	0	1	3	0	0	2	1	0	0	0	1	0	0	12	1.7	2.63
3.6 -7.5	5	6	8	5	6	4	17	9	8	10	3	8	6	4	4	2	105	15.1	5.31
7.6 - 12.5	9	9	15	27	29	23	28	28	11	14	19	22	24	31	23	15	327	47.1	9.88
12.6 - 18.5	1	1	2	8	12	3	2	4	13	31	21	12	25	43	24	13	215	30.9	14.57
18.6 - 24.5	0	0	0	0	0	0	0	0	1	1	3	2	4	19	6	0	36	5.2	20.84
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	16	17	27	40	48	33	47	41	35	57	46	44	59	98	57	30	695	0.0	9.40
PERCENT	2.3	2.4	3.9	5.8	6.9	4.7	6.8	5.9	5.0	8.2	6.6	6.3	8.5	14.1	8.2	4.3	100.0		
AV SPD	8.3	7.8	8.2	10.1	10.4	8.9	8.6	9.5	10.3	12.0	12.8	11.2	12.3	14.3	13.3	11.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 11.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909.15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.4	.50
CALM + - 3.5	1	0	1	0	2	1	0	1	0	0	0	1	0	0	1	0	8	3.5	1.86
3.6 -7.5	3	0	3	2	4	2	1	6	5	3	3	0	3	4	1	2	42	18.5	5.56
7.6 - 12.5	3	1	2	10	20	9	7	6	6	7	13	8	8	8	13	4	125	55.1	9.79
12.6 - 18.5	0	0	0	0	4	1	1	0	7	8	3	0	8	13	3	2	50	22.0	13.91
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0.4	20.50
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	8	1	6	12	30	13	9	13	18	18	19	9	19	26	18	8	227	0.0	7.50
PERCENT	3.5	.4	2.6	5.3	13.2	5.7	4.0	5.7	7.9	7.9	8.4	4.0	8.4	11.5	7.9	3.5	100.0		
AV SPD	6.7	9.8	6.6	8.6	9.6	8.3	9.9	7.1	10.5	11.8	10.0	9.0	10.7	12.0	10.2	9.9			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 2

TABLE 2.3.2-16 (SHEET 48 OF 48)

MONTH OF DECEMBER

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-15

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	3	1	1	1	2	0	3	1	0	0	0	0	2	2	1	17	7.6	2.34
3.6 -7.5	1	2	5	7	5	6	6	9	9	1	4	6	5	3	6	2	77	34.4	5.74
7.6 - 12.5	0	2	5	3	3	9	6	10	4	8	10	2	5	6	10	7	90	40.2	9.87
12.6 - 18.5	0	0	0	0	1	6	1	0	3	5	3	5	9	4	2	1	40	17.9	13.92
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1	7	11	11	10	23	13	22	17	14	17	13	19	15	20	11	224	0.0	6.85
PERCENT	.4	3.1	4.9	4.9	4.5	10.3	5.8	9.8	7.6	6.3	7.6	5.8	8.5	6.7	8.9	4.9	100.0		
AV SPD	5.7	5.0	7.2	6.8	7.6	9.6	8.2	7.4	7.8	11.4	10.7	9.2	10.8	9.1	8.9	8.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

MONTH OF DECEMBER
SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.0
CALM + - 3.5	4	6	6	1	8	8	2	8	6	5	2	6	2	3	5	5	77	3.8
3.6 -7.5	26	23	38	27	26	18	43	47	32	26	30	29	30	24	17	29	465	23.1
7.6 - 12.5	25	30	61	94	89	57	48	52	29	42	59	49	75	71	62	37	880	43.7
12.6 - 18.5	2	4	5	27	31	10	7	5	36	50	51	39	64	88	43	22	484	24.1
18.6 - 24.5	0	0	0	2	0	0	0	0	4	4	10	9	7	31	24	1	92	4.6
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	1	3	0	6	3	0	13	.6
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	58	63	110	151	154	93	100	112	107	127	153	135	178	223	154	94	2012	100.0
PERCENT	2.9	3.1	5.5	7.5	7.7	4.6	5.0	5.6	5.3	6.3	7.6	6.7	8.8	11.1	7.7	4.7	100.0	

TABLE 2.3.2-17 (SHEET 1 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE
PERIOD OF RECORD FROM 12/73-12/74, 4/77-4/79
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-16

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	2	.0	.50
CALM + - 3.5	18	9	11	8	7	3	6	3	4	1	3	11	11	11	9	11	130	3.0	2.62
3.6 -7.5	124	101	105	97	69	49	44	31	30	36	14	102	139	106	74	65	1245	28.9	5.57
7.6 - 12.5	95	104	120	122	149	103	71	32	34	50	30	200	230	193	109	81	1786	41.4	9.72
12.6 - 18.5	25	40	50	36	42	23	16	20	17	14	21	83	175	127	88	29	843	19.6	14.83
18.6 - 24.5	4	3	0	2	1	1	0	1	2	2	1	21	68	69	60	9	254	5.9	20.78
24.6 - 32.5	0	1	0	0	0	0	0	0	0	0	2	2	17	18	10	1	51	1.2	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	266	258	286	265	268	179	137	87	87	103	71	419	641	525	350	196	4311	0.0	8.10
PERCENT	6.2	6.0	6.6	6.1	6.2	4.2	3.2	2.0	2.0	2.4	9.4	9.7		14.9	12.2	8.1	4.5	100.0	
AV SPD	7.9	8.8	8.8	8.9	9.6	9.2	8.8	9.5	9.5	9.3	10.9	10.4	12.1	12.3	12.6	9.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 75

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -1.0 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-16

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	5	2	1	4	0	2	17	1	0	1	1	2	2	1	28	3.7	2.60
3.6 -7.5	20	3	24	21	14	10	8	13	13	13	2	20	10	9	3	16	228	30.2	5.38
7.6 - 12.5	24	0	26	19	31	21	15	10	7	22	4	26	24	24	12	12	329	43.5	9.60
12.6 - 18.5	7	0	13	2	9	3	3	4	1	6	0	12	24	11	5	4	139	18.4	14.80
18.6 - 24.5	1	0	1	0	1	0	0	0	0	1	0	3	4	8	3	2	27	3.6	20.13
24.6 - 32.5	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	1	5	.7	26.85
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	53	3	70	44	56	38	26	29	39	43	6	63	63	54	25	36	756	0.0	7.69
PERCENT	7.0	5.1	9.3	5.8	7.4	5.0	3.4	3.8	5.2	5.7	10.2	8.3	8.3	7.1	3.3	4.8	100.0		
AV SPD	9.1	5.5	9.1	7.5	9.7	8.3	9.3	7.6	8.8	9.0	7.6	10.1	11.9	11.7	11.0	9.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 14

TABLE 2.3.2-17 (SHEET 2 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTIO
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-16

WIND DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	.0	0.00
CALM + - 3.5	1	0	2	1	1	0	1	1	2	2	2	0	0	0	2	0	15	3.9	2.70
3.6 -7.5	10	5	9	9	8	9	13	9	10	7	10	11	8	9	7	7	141	36.6	5.62
7.6 - 12.5	8	9	10	12	11	11	6	12	8	10	7	9	6	11	5	9	144	37.4	9.63
12.6 - 18.5	0	5	3	4	5	3	3	1	2	5	6	7	9	7	5	2	67	17.4	14.39
18.6 - 24.5	1	0	0	0	0	0	0	1	0	0	1	2	0	3	5	2	14	3.6	19.90
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	4	1.0	26.81
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	20	19	24	26	25	23	23	23	22	24	26	29	23	34	24	20	385	0.0	7.56
PERCENT	5.2	4.9	6.2	6.8	6.5	6.0	6.0	6.0	5.7	6.2	6.8	7.5		6.0	8.8	6.2	5.2	100.0	
AV SPD	7.7	10.6	8.2	9.5	9.1	8.7	7.4	8.2	7.5	9.7	9.1	10.1	10.4	12.9	11.1	9.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 6

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-16

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	2	.0	.50
CALM + - 3.5	13	16	15	14	10	16	16	13	12	17	17	11	11	14	12	10	217	4.7	2.64
3.6 -7.5	71	76	105	90	82	101	113	97	84	72	84	63	70	48	40	76	1272	27.5	5.56
7.6 - 12.5	90	95	174	154	170	153	110	114	103	116	137	102	113	80	70	43	1824	39.4	9.70
12.6 - 18.5	32	45	99	100	48	39	39	59	69	62	88	59	90	83	68	36	1016	22.0	14.79
18.6 - 24.5	6	4	6	6	8	2	4	11	19	26	24	16	28	51	35	7	253	5.5	20.72
24.6 - 32.5	0	0	0	0	2	0	0	0	3	6	1	3	7	13	2	1	38	.8	26.68
32.6+	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	2	.0	34.85
TOTAL	212	236	399	364	320	311	282	294	290	300	352	255	319	290	227	173	4624	100.0	7.90
PERCENT	4.6	5.1	8.6	7.9	6.9	6.7	6.1	6.4	6.3	6.5	7.6	5.5	6.9	6.3	4.9	3.7	100.0		
AV SPD	9.3	9.1	10.0	10.2	9.7	8.8	8.6	9.6	10.6	10.8	10.9	10.6	11.5	13.2	12.1	9.2			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.3
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 47

TABLE 2.3.2-17 (SHEET 3 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-16

WIND DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	2	0	1	1	0	1	2	1	0	0	1	1	0	0	0	10	.1	0.50
CALM + - 3.5	16	18	18	17	22	26	19	17	19	17	12	17	10	11	9	12	260	3.6	2.42
3.6 -7.5	69	70	110	128	122	123	180	152	138	132	116	67	69	72	61	41	1650	22.7	5.61
7.6 - 12.5	89	78	200	234	267	255	259	361	380	373	306	216	199	171	172	108	3668	50.4	9.84
12.6 - 18.5	46	30	76	63	64	48	56	72	109	139	127	90	159	177	153	63	1472	20.2	14.40
18.6 - 24.5	2	4	10	3	4	1	3	4	12	10	14	14	24	47	29	8	189	2.6	20.48
24.6 - 32.5	0	0	0	0	0	0	0	1	0	0	4	2	4	7	3	1	22	.3	26.31
32.6+	0	0	0	0	0	0	0	0	0	0	3	3	0	1	0	0	7	.1	36.93
TOTAL	222	202	414	446	480	453	518	609	659	671	582	410	466	486	427	233	7278	100.0	8.03
PERCENT	3.1	2.8	5.7	6.1	6.6	6.2	7.1	8.4	9.1	9.2	8.0	5.6		6.4	6.7	5.9	3.2	100.0	
AV SPD	9.2	8.5	9.6	9.2	9.3	8.8	8.7	9.3	9.8	10.2	10.6	10.7	11.7	12.5	11.9	10.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =24

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE
PERIOD OF RECORD FROM 72120417 TO 79040312
SPEED AND DIRECTION FROM 150FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-16

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1	1	0	1	0	0	1	1	0	2	0	0	2	0	1	0	10	.3	.50
CALM + - 3.5	14	8	15	7	11	10	9	5	9	6	9	12	5	8	10	3	130	3.8	2.15
3.6 -7.5	44	29	105	50	63	51	56	62	52	56	51	33	46	45	32	35	758	22.3	5.64
7.6 - 12.5	52	65	78	116	161	124	141	121	125	160	167	129	154	123	117	66	1899	55.9	9.86
12.6 - 18.5	25	25	22	19	28	25	19	13	31	54	69	63	59	69	37	27	585	17.2	13.92
18.6 - 24.5	0	0	1	0	0	0	0	0	0	1	1	0	1	4	5	1	14	.4	20.95
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	2	.1	25.72
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	136	128	158	193	263	210	226	202	217	279	297	237	267	251	202	132	3398	0.0	7.55
PERCENT	4.0	3.8	8.6	5.7	7.7	6.2	6.7	5.9	6.4	8.2	8.7	7.0	7.9	7.4	5.9	3.9	100.0		
AV SPD	8.7	9.5	9.1	9.0	9.1	9.0	8.8	8.5	9.3	9.7	10.2	10.3	10.1	10.6	10.1	9.6			

AVERAGE SPEED FOR THIS TABLE EQUALS9.5
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 16

TABLE 2.3.2-17 (SHEET 4 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE
PERIOD OF RECORD FROM
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 150 FT

REQUEST NUMBER 909-16

WIND DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	1	1	0	2	1	1	2	1	2	0	1	1	1	1	15	.7	0.50
CALM + - 3.5	12	10	8	9	8	13	11	20	12	6	8	10	19	14	16	8	184	8.7	2.28
3.6 -7.5	36	33	44	27	36	47	51	31	53	34	28	25	41	36	42	40	604	28.5	5.52
7.6 - 12.5	44	51	48	45	81	56	51	49	62	88	111	65	71	67	61	56	1006	47.4	9.86
12.6 - 18.5	15	10	4	6	16	19	6	5	10	46	56	35	35	33	10	6	312	14.7	14.00
18.6 - 24.5	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	2	.1	19.43
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	107	104	105	88	141	137	120	106	140	175	205	135	167	152	130	111	2123	0.0	6.16
PERCENT	5.0	4.9	4.9	4.1	6.6	6.5	5.7	5.0	6.6	8.2	9.7	6.4		7.9	7.2	6.1	5.2	100.0	
AV SPD	8.4	8.2	7.5	7.9	9.0	8.3	7.6	7.6	7.9	10.0	10.6	9.8	9.1	9.3	7.9	8.0			

AVERAGE SPEED FOR THIS TABLE EQUALS8.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 3

SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	1	3	1	3	1	2	3	4	3	4	2	2	5	2	2	1	39	.2
CALM + -3.5	75	62	63	58	60	72	62	61	59	50	58	62	57	60	60	45	964	4.2
3.6 -7.5	374	330	450	422	394	390	465	395	384	350	376	321	383	325	259	280	5898	25.8
7.6 - 12.5	402	422	656	702	870	723	653	699	725	819	851	747	797	669	546	375	10656	46.6
12.6 - 18.5	150	163	267	230	212	160	142	174	245	326	425	349	551	507	366	167	4434	19.4
18.6 - 24.5	14	12	18	11	14	4	7	16	35	40	52	56	125	183	137	29	753	3.3
24.6 - 32.5	0	1	1	0	2	0	0	1	3	6	9	8	28	44	15	4	122	.5
32.6+	0	0	0	0	0	0	0	0	0	0	4	3	0	2	0	0	9	.0
TOTAL	1016	993	1456	1426	1553	1351	1332	1350	1454	1595	1777	1548	1946	1792	1385	901	22875	100.0
PERCENT	4.4	4.3	6.4	6.2	6.8	5.9	5.8	5.9	6.4	7.0	7.8	6.8	8.5	7.8	6.1	3.9	100.0	

TABLE 2.3.2-18 (SHEET 1 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO 1.0
SITE AUGUSTA GA.-VOGTLE
PERIOD OF RECORD FROM 590101010 TO 63123123
SPEED AND DIRECTION FROM 100 FT LEVEL
TEMPERATURE DIFFERENCE BETWEENFT ANDFT

REQUEST NUMBER 909-14

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	28	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	28	7.2	0.50
CALM + - 3.5	14	11	8	8	17	25	12	7	12	9	4	5	10	7	10	6	165	42.3	1.70
3.6 -7.5	14	12	17	20	11	15	10	15	9	14	14	9	8	6	13	10	197	50.5	4.39
7.6 - 12.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	56	23	25	28	28	40	22	22	21	23	18	14	18	13	23	16	390	0.0	1.97
PERCENT	14.4	5.9	6.4	7.2	7.2	10.3	5.6	5.6	5.4	5.9	4.6	3.6		4.6	3.3	5.9	4.1	100.0	
AV SPD	1.8	3.3	3.9	3.8	2.8	2.9	3.1	4.0	3.3	3.5	3.9	3.7	3.4	3.5	3.3	3.4			

AVERAGE SPEED FOR THIS TABLE EQUALS3.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 1.0 BUT LESS THAN OR EQUAL TO 2.0
SITE AUGUSTA GA.-VOGTLE
PERIOD OF RECORD FROM 59010101 TO 63123123
SPEED AND DIRECTION FROM 100 FT LEVEL
TEMPERATURE DIFFERENCE BETWEENFT ANDFT

REQUEST NUMBER 909-14

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	128	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	128	2.8	.50
CALM + - 3.5	258	152	212	187	170	159	254	157	133	64	99	54	65	62	126	135	2287	49.7	2.18
3.6 -7.5	157	108	191	210	144	146	158	85	89	108	141	124	75	80	106	119	2041	44.4	5.27
7.6 - 12.5	7	4	20	15	12	15	9	5	6	10	8	8	5	6	6	8	144	3.1	8.33
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	550	264	423	412	326	320	421	247	228	182	248	186	145	148	238	262	4600	0.0	2.69
PERCENT	12.0	5.7	9.2	9.0	7.1	7.0	9.2	5.4	5.0	4.0	5.4	4.0	3.2	3.2	5.2	5.7	100.0		
AV SPD	2.9	3.8	4.1	4.3	4.0	4.1	3.6	3.6	3.7	4.7	4.5	4.9	4.3	4.3	4.0	4.0			

AVERAGE SPEED FOR THIS TABLE EQUALS3.9
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =0

TABLE 2.3.2-18 (SHEET 2 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.0 BUT LESS THAN OR EQUAL TO 3.0
SITE AUGUSTA GA.-VOGTLE
PERIOD OF RECORD FROM 59010101 TO 63123123
SPEED AND DIRECTION FROM 100 FT LEVEL
TEMPERATURE DIFFERENCE BETWEENFT ANDFT

REQUEST NUMBER 909-14

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	237	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	237	3.7	.50
CALM + - 3.5	163	61	71	55	73	56	159	100	101	46	59	30	48	30	75	79	1206	18.9	1.91
3.6 -7.5	217	178	207	181	156	194	206	129	86	115	146	133	107	106	170	203	2534	39.7	5.13
7.6 - 12.5	114	148	216	231	134	160	139	72	61	95	182	219	179	150	134	116	2350	36.9	8.81
12.6 - 18.5	3	3	3	5	1	2	5	0	0	1	4	1	7	6	2	3	46	.7	14.04
18.6 - 24.5	0	0	1	0	0	0	0	0	0	0	0	0	2	0	0	0	3	0.0	21.18
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	734	390	598	472	364	412	509	301	248	257	391	383	343	292	381	401	6376	0.0	3.41
PERCENT	11.5	6.1	7.8	7.4	5.7	6.5	8.0	4.7	3.9	4.0	6.1	6.0		5.4	4.6	6.0	6.3	100.0	
AV SPD	3.6	6.2	6.6	6.8	6.1	6.4	5.3	5.1	5.0	6.2	6.7	7.3	7.3	7.3	6.1	5.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 6.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 3.0 BUT LESS THAN OR EQUAL TO 4.0
SITE AUGUSTA GA.-VOGTLE
PERIOD OF RECORD FROM 59010101 TO 63123123
SPEED AND DIRECTION FROM 100 FT LEVEL
TEMPERATURE DIFFERENCE BETWEENFT ANDFT

REQUEST NUMBER 909-14

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	49	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	49	.5	.50
CALM + - 3.5	45	19	45	24	25	20	38	32	55	29	25	11	19	15	23	18	443	4.1	2.18
3.6 -7.5	204	119	206	198	166	211	258	218	163	166	223	210	133	113	189	193	2970	27.4	5.38
7.6 - 12.5	126	182	345	386	221	262	280	231	165	186	374	480	467	642	326	183	4856	44.7	9.96
12.6 - 18.5	43	52	94	134	83	53	47	47	43	59	115	237	402	690	139	58	2296	21.2	14.57
18.6 - 24.5	1	0	1	0	5	0	3	2	0	1	4	13	53	19	14	1	200	1.8	20.40
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	2	15	16	2	5	40	.4	26.58
32.6+	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.0	36.00
TOTAL	468	373	691	742	500	546	626	530	426	441	741	953	1089	1578	693	458	10855	100.0	7.19
PERCENT	4.3	3.4	6.4	6.8	4.6	5.0	5.8	4.9	3.9	4.1	6.8	8.8	10.0	14.5	6.4	4.2	100.0		
AV SPD	6.6	8.8	8.9	9.5	8.9	8.4	8.0	8.1	7.7	8.6	9.2	10.4	12.2	13.1	9.9	8.6			

AVERAGE SPEED FOR THIS TABLE EQUALS9.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

TABLE 2.3.2-18 (SHEET 3 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 4.0 BUT LESS THAN OR EQUAL TO 5.0
SITE AUGUSTA GA.-VOGTLE
PERIOD OF RECORD FROM 59010101 TO 63123123
SPEED AND DIRECTION FROM 100 FT LEVEL
TEMPERATURE DIFFERENCE BETWEENFT ANDFT

REQUEST NUMBER 909-14

WIND DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
3.6 -7.5	172	119	198	202	132	167	258	325	204	196	280	255	168	185	267	219	3347	62.8	5.11
7.6 - 12.5	87	126	156	156	76	104	104	73	67	56	101	151	213	320	125	71	1986	37.2	8.75
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	259	245	354	358	208	271	362	398	271	252	381	406	381	505	392	290	5333	0.0	6.05
PERCENT	4.9	4.6	6.6	6.7	3.9	5.1	6.8	7.5	5.1	4.7	7.1	7.6		7.1	9.5	7.4	5.4	100.0	
AV SPD	6.5	7.3	7.1	7.1	6.7	6.7	6.2	5.7	5.9	6.0	6.1	6.6	7.6	7.8	6.5	6.2			

AVERAGE SPEED FOR THIS TABLE EQUALS6.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 5.0 BUT LESS THAN OR EQUAL TO 6.0
SITE AUGUSTA GA.-VOGTLE
PERIOD OF RECORD FROM 59010101 TO 63123123
SPEED AND DIRECTION FROM 100 FT LEVEL
TEMPERATURE DIFFERENCE BETWEENFT ANDFT

REQUEST NUMBER 909-14

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1036	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1036	11.7	.50
CALM + - 3.5	302	101	181	88	180	156	415	333	440	162	351	147	274	170	335	206	3841	43.5	1.82
3.6 -7.5	296	187	305	268	181	164	245	287	213	150	224	285	207	223	417	296	3948	44.7	4.70
7.6 - 12.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	1634	288	486	356	361	320	660	620	653	312	575	432	481	393	752	502	8825	0.0	1.76
PERCENT	18.5	3.3	5.5	4.0	4.1	3.6	7.5	7.0	7.4	3.5	6.5	4.9	5.5	4.5	8.5	5.7	100.0		
AV SPD	1.6	3.9	3.8	4.3	3.5	3.6	3.1	3.4	3.0	3.4	3.2	4.0	3.3	3.7	3.6	3.6			

AVERAGE SPEED FOR THIS TABLE EQUALS3.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =0

TABLE 2.3.2-18 (SHEET 4 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 6.0
SITE AUGUSTA GA.-VOGTLE
PERIOD OF RECORD FROM 59010101 TO 63123123
SPEED AND DIRECTION FROM 100 FT LEVEL
TEMPERATURE DIFFERENCE BETWEENFT ANDFT

REQUEST NUMBER 909-14

WIND DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1759	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1759	23.7	.50
CALM + - 3.5	549	200	291	184	279	214	541	409	474	230	429	266	384	271	565	375	5661	16.3	1.82
3.6 -7.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
7.6 - 12.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
12.6 - 18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	2308	200	291	184	279	214	541	409	474	230	429	266	384	271	565	375	7420	0.0	1.12
PERCENT	31.1	2.7	3.9	2.5	3.8	2.9	7.3	5.5	6.4	3.1	5.8	3.6		5.2	3.7	7.6	5.1	100.0	
AV SPD	.9	2.3	2.3	2.2	2.1	2.2	2.2	2.3	2.1	2.3	2.1	2.2	2.0	2.0	2.1	2.1			

AVERAGE SPEED FOR THIS TABLE EQUALS1.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	3237	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3237	7.4
CALM + - 3.5	1331	544	808	546	744	630	1419	1038	1215	540	967	513	800	555	10	1134	819	31.1
3.6 -7.5	1060	723	1124	1079	790	897	1135	1059	764	749	1028	1016	698	713	1162	1040	15037	34.3
7.6 - 12.5	334	460	737	788	443	541	532	381	299	347	665	858	864	1118	591	378	9336	21.3
12.6 - 18.5	46	55	97	139	84	55	52	47	43	60	119	238	409	696	141	61	2342	5.3
18.6 - 24.5	1	0	2	0	5	0	3	2	0	1	4	13	55	102	14	1	203	.5
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	2	15	16	2	5	40	.1
32.6+	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	.0
TOTAL	6009	1783	2768	2552	2066	2123	3141	2527	2321	1697	2783	2640	2841	3200	3044	2304	43799	100.0
PERCENT	13.7	4.1	6.3	5.8	4.7	4.8	7.2	5.8	5.3	3.9	6.4	6.0	6.5	7.3	6.9	5.3	100.0	

TABLE 2.3.2-19 (SHEET 1 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) LESS THAN OR EQUAL TO -1.0
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	8	11	9	6	0	1	1	2	1	0	2	5	4	4	6	9	63	4.6	2.45
3.6 -7.5	30	12	32	26	19	8	10	4	9	9	17	9	84	76	47	35	470	34.7	5.65
7.6 - 12.5	32	0	29	25	21	22	0	6	2	9	11	0	162	103	63	3	624	46.1	9.56
12.6 - 18.5	8	0	9	6	0	3	0	1	0	0	2	0	61	42	28	6	188	13.9	14.15
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	6	1	3	0	10	.7	20.27
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	.1	25.20
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	78	23	79	63	40	34	28	13	12	18	32	14	317	227	147	85	1355	0.0	7.36
PERCENT	5.8	5.9	5.8	4.6	3.0	2.5	2.1	1.0	.9	1.3	2.4	3.6		23.4	16.8	10.8	6.0	100.0	
AV SPD	7.9	3.3	7.8	7.6	7.9	9.3	8.0	7.1	6.0	7.7	7.4	3.7	9.8	9.2	9.5	7.7			

AVERAGE SPEED FOR THIS TABLE EQUALS8.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 8

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.9
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 33 FT AND 33 FT

REQUEST NUMBER 209.2

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	1	0	2	0	0	0	0	0	0	0	0	0	1	3	0	0	7	3.3	2.71
3.6 -7.5	3	0	10	4	3	8	4	1	2	2	9	6	9	4	3	1	73	34.3	5.54
7.6 - 12.5	6	5	7	6	9	2	7	5	6	1	5	15	14	9	5	4	106	49.8	9.48
12.6 - 18.5	1	2	2	2	0	0	0	0	0	0	0	3	4	1	5	0	25	11.7	14.04
18.6 - 24.5	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	2	.9	21.84
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	11	14	21	12	15	16	11	6	2	3	14	24	28	17	14	5	213	0.0	7.39
PERCENT	5.2	6.6	9.9	5.6	7.0	7.5	5.2	2.8	.9	1.4	6.6	11.3	13.1	8.0	6.6	2.3	100.0		
AV SPD	8.7	9.1	7.9	9.0	10.1	8.7	8.5	8.8	6.4	6.6	7.0	6.1	8.6	7.6	11.4	9.3			

AVERAGE SPEED FOR THIS TABLE EQUALS8.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =2

TABLE 2.3.2-19 (SHEET 2 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.9 BUT LESS THAN OR EQUAL TO -.8
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	1	0	1	1	0	0	0	0	0	0	1	0	0	1	1	6	5.5	2.81
3.6 -7.5	2	3	5	4	0	3	4	2	1	3	5	2	3	2	0	0	39	35.5	5.92
7.6 - 12.5	1	4	7	4	4	4	3	1	2	5	2	5	3	2	1	4	52	47.3	9.50
12.6 - 18.5	1	0	3	3	1	1	0	0	0	1	0	1	1	1	0	0	13	11.8	13.88
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	4	8	15	12	6	8	7	3	3	9	7	9	7	5	2	5	110	0.0	7.27
PERCENT	3.6	7.3	13.6	10.9	5.5	7.3	6.4	2.7	2.7	8.2	6.4	8.2	6.2	4.5	1.8	4.5	100.0		
AV SPD	8.8	7.3	9.1	9.6	9.0	9.3	7.8	6.9	7.1	8.9	6.9	8.4	8.6	10.0	6.0	8.8			

AVERAGE SPEED FOR THIS TABLE EQUALS 8.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN -.8 BUT LESS THAN OR EQUAL TO -.3
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 33 FT AND 33 FT

REQUEST NUMBER 209-2

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	5	4	5	3	4	1	2	4	2	5	2	2	3	3	6	5	56	3.8	2.67
3.6 -7.5	40	33	41	28	28	23	33	31	35	39	28	40	27	22	17	19	484	32.9	5.75
7.6 - 12.5	52	64	88	80	49	62	42	36	34	26	41	53	51	21	28	21	748	50.8	9.52
12.6 - 18.5	7	27	28	27	12	2	4	3	12	9	5	4	7	9	11	2	169	11.5	14.03
18.6 - 24.5	0	1	0	3	0	0	0	1	4	1	0	0	1	0	2	1	14	1.0	20.52
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	104	129	162	141	93	88	81	75	87	80	76	99	89	55	64	48	1471	0.0	7.49
PERCENT	7.1	8.8	11.0	9.6	6.3	6.0	5.5	5.1	5.9	5.4	5.2	6.7	6.1	3.7	4.4	3.3	100.0		
AV SPD	8.2	9.6	9.6	10.1	8.9	8.7	8.3	7.8	9.3	8.0	8.2	8.1	8.9	8.4	9.1	7.6			

AVERAGE SPEED FOR THIS TABLE EQUALS8.8
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 5

TABLE 2.3.2-19 (SHEET 3 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN -.3 BUT LESS THAN OR EQUAL TO .8
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	1	1	0	0	0	0	1	0	0	1	0	0	0	1	0	1	6	.2	.50
CALM + - 3.5	13	16	21	16	12	16	19	26	14	17	14	23	18	11	18	9	263	9.4	2.54
3.6 -7.5	71	56	60	67	61	79	109	123	148	138	153	136	103	72	81	57	1514	54.3	5.39
7.6 - 12.5	56	58	60	65	47	33	40	33	72	47	48	57	49	72	71	49	857	30.8	9.21
12.6 - 18.5	11	17	20	7	7	2	3	0	12	15	1	5	2	7	16	10	135	4.8	14.21
18.6 - 24.5	1	2	1	0	0	0	0	0	3	0	0	0	0	0	2	2	11	.4	19.41
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	153	150	152	155	127	130	172	182	249	218	210	221	172	163	188	128	2780	0.0	5.58
PERCENT	5.5	5.4	5.8	5.6	4.6	4.7	6.2	6.5	8.9	7.8	7.8	7.9		6.2	5.9	6.7	4.6	100.0	
AV SPD	7.6	8.2	8.0	7.3	7.1	6.4	6.2	5.8	7.1	6.7	6.1	6.5	6.4	7.5	7.7	7.8			

AVERAGE SPEED FOR THIS TABLE EQUALS7.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN .8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 33 FT AND 33 FT

REQUEST NUMBER 209-2

WIND DIRECTION

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	3	.2	.50
CALM + - 3.5	6	11	30	12	14	14	13	24	20	17	15	19	17	13	14	11	250	19.8	2.50
3.6 -7.5	30	32	24	32	16	31	53	31	55	64	98	126	94	88	55	23	852	67.5	5.06
7.6 - 12.5	7	4	4	0	2	12	3	6	3	5	10	15	20	10	16	6	129	10.2	8.96
12.6 - 18.5	0	0	0	3	2	0	1	1	0	4	3	5	1	0	3	1	24	1.9	13.64
18.6 - 24.5	0	0	0	3	1	0	0	0	0	1	0	0	0	0	0	0	5	.4	19.43
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	43	48	59	56	35	57	70	63	78	91	126	165	132	111	88	41	1263	0.0	4.34
PERCENT	3.4	3.8	4.7	4.4	2.8	4.5	5.5	5.0	6.2	7.2	10.0	13.1	10.5	8.8	7.0	3.2	100.0		
AV SPD	5.4	4.7	4.0	6.4	5.0	5.4	4.8	4.7	4.8	5.4	5.5	6.8	5.6	5.4	6.2	5.3			

AVERAGE SPEED FOR THIS TABLE EQUALS5.4
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION =0

TABLE 2.3.2-19 (SHEET 4 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100FT) GREATER THAN 2.2
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTIONS

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	2	1	2	1	1	0	0	1	0	1	1	2	0	0	0	0	12	1.5	.50
CALM + - 3.5	22	29	26	15	20	15	6	9	12	7	19	13	15	13	14	13	248	30.7	2.31
3.6 -7.5	8	36	22	7	12	9	12	6	22	31	69	88	55	87	31	8	483	59.9	4.97
7.6 - 12.5	2	2	2	0	1	1	3	4	2	2	7	5	10	7	7	0	55	6.8	9.30
12.6 - 18.5	1	0	1	3	0	0	0	0	0	2	0	0	0	0	1	1	9	1.1	14.44
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	35	68	53	26	34	25	21	20	36	43	96	108	80	87	53	22	807	0.0	3.43
PERCENT	4.3	8.4	6.6	3.2	4.2	3.1	2.6	2.5	4.5	5.3	11.9	13.4		9.9	10.8	6.6	2.7	100.0	
AV SPD	3.6	4.0	3.7	4.6	3.7	3.9	5.0	4.7	4.5	5.2	5.1	5.2	5.4	5.1	4.8	3.8			

AVERAGE SPEED FOR THIS TABLE EQUALS4.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION = 1

SUMMARY TABLE

SPEED (MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	3	3	3	1	1	0	1	2	0	2	1	2	0	1	0	1	21	.3
CALM + - 3.5	55	68	93	53	51	47	41	65	49	46	52	60	58	47	59	48	892	11.1
3.6 -7.5	188	198	194	168	139	159	226	198	272	286	379	429	375	331	234	143	3915	48.9
7.6 - 12.5	156	151	197	186	133	142	114	91	115	95	124	214	309	224	191	119	2571	32.1
12.6 - 18.5	29	58	63	51	25	10	8	5	24	31	11	28	76	60	64	20	563	7.0
18.6 - 24.5	1	4	1	6	1	0	0	1	7	2	0	0	7	1	8	3	42	.5
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	.0
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
TOTAL	426	492	551	465	350	358	390	362	467	462	567	733	825	665	556	334	8005	100.0
PERCENT	5.3	6.1	6.9	5.8	4.4	4.5	4.9	4.5	5.8	5.8	7.1	9.2	10.3	8.3	6.9	4.2	100.0	

TABLE 2.3.2-20 (SHEET 1 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEGF/100FT) LESS THAN OR EQUAL TO-1.0
SITE VOGTLE MET TOWER
PERIOD O FRECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	.1	.50
CALM + - 3.5	7	3	5	7	4	2	2	0	2	1	2	0	5	2	4	9	55	4.0	2.53
3.6 -7.5	33	20	25	27	31	20	17	11	12	8	8	14	29	39	76	39	409	29.7	5.59
7.6 - 12.5	40	32	13	21	23	24	16	25	10	5	4	12	33	99	130	55	542	39.4	9.80
12.6 - 18.5	20	12	7	10	8	6	4	3	2	0	0	1	18	53	114	44	302	21.9	14.74
18.6 - 24.5	2	1	1	2	0	0	0	0	0	0	0	0	1	11	25	14	57	4.1	20.48
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	2	3	3	8	.6	26.80
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2	.1	35.00
TOTAL	102	68	51	67	66	52	39	39	20	14	14	27	80	200	302	107	1376	100.0	7.82
PERCENT	7.4	4.9	3.7	4.9	4.8	3.8	2.8	2.8	1.9	1.0	1.0	2.0		6.3	15.0	25.6	12.1	100.0	
AV SPD	9.2	9.2	8.0	8.5	7.9	8.5	8.4	9.3	7.7	7.5	6.7	7.9	9.3	11.1	11.7	11.3			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.1
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION=5

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN-1.0 BUT LESS THAN OR EQUAL TO-.9
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT
WIND DIRECTION

REQUEST NUMBER 209-1

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	0	1	0	0	1	0	1	0	1	0	0	0	0	0	0	4	1.9	2.64
3.6 -7.5	4	1	4	3	5	6	4	2	7	2	1	7	4	3	8	0	61	28.6	5.59
7.6 - 12.5	1	4	3	6	6	9	9	4	5	4	1	1	10	12	8	6	95	44.6	9.81
12.6 - 18.5	1	3	1	2	6	3	3	1	0	1	0	0	4	5	8	4	42	19.7	14.40
18.6 - 24.5	0	1	0	0	0	1	0	0	0	0	0	0	0	3	2	2	9	4.2	19.92
24.6 - 32.5	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	2	.9	27.72
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	6	9	10	11	17	20	16	8	12	8	2	8	24	23	26	13	213	0.0	8.34
PERCENT	2.8	4.2	4.7	5.2	8.0	9.4	7.5	3.8	5.6	3.8	.9	3.8	11.3	10.0	12.2	6.1	100.0		
AV SPD	7.7	11.7	9.0	9.5	10.4	9.5	9.9	9.3	7.6	8.5	7.8	6.9	10.2	11.5	11.2	14.6			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION=1

TABLE 2.3.2-20 (SHEET 2 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN-.9 BUT LESS THAN OR EQUAL TO-.8
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTIONS

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
CALM + - 3.5	0	1	0	0	0	1	1	0	0	0	1	0	0	0	0	1	5	4.5	2.92
3.6 -7.5	1	0	0	1	4	2	0	4	2	4	2	2	4	0	1	1	28	25.0	6.00
7.6 - 12.5	2	1	4	3	5	3	6	4	3	1	4	2	4	5	2	1	50	44.6	9.76
12.6 - 18.5	2	1	0	1	6	2	3	0	0	1	0	1	3	3	3	1	27	24.1	15.05
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	.9	19.30
24.6 - 32.5	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	.9	25.40
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	5	3	4	5	15	8	10	8	5	7	7	5	12	8	6	4	112	0.0	8.37
PERCENT	4.5	2.7	3.6	4.5	13.4	7.1	8.9	7.1	4.5	6.3	6.3	4.5		10.7	7.1	5.4	3.6	100.0	
AV SPD	12.0	9.1	9.6	10.2	10.7	10.4	10.5	7.2	9.0	10.7	7.6	9.0	10.9	12.7	11.6	9.4			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION=0

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN-.8 BUT LESS THAN OR EQUAL TO-.3
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 150 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

REQUEST NUMBER 209-1

WIND DIRECTION

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	(MPH)
CALM	1	1	2	0	0	0	0	1	0	0	0	0	0	0	0	0	5	.3	.50
CALM + - 3.5	3	5	3	4	6	2	1	2	1	1	4	3	1	4	2	6	48	3.3	2.55
3.6 -7.5	8	16	23	13	28	27	25	19	21	13	24	21	23	16	15	11	308	20.9	5.67
7.6 - 12.5	26	35	54	48	70	55	59	49	42	34	36	44	57	31	30	22	692	47.0	9.90
12.6 - 18.5	16	20	28	23	59	49	23	8	9	19	14	11	18	18	25	22	362	24.6	14.56
18.6 - 24.5	0	0	2	2	3	2	1	0	0	6	5	6	1	3	6	8	45	3.1	20.57
24.6 - 32.5	1	0	1	0	0	0	0	0	1	3	0	0	0	0	1	2	9	.6	27.77
32.6+	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	2	.1	35.19
TOTAL	55	77	113	95	166	155	109	79	75	76	83	86	100	72	79	71	1471	100.0	8.15
PERCENT	3.7	5.2	7.7	6.5	11.3	9.2	7.4	5.4	5.1	5.2	5.6	5.8	6.8	4.9	5.4	4.8	100.0		
AV SPD	10.4	9.7	10.3	10.5	11.1	11.1	10.0	9.1	9.8	11.9	9.9	10.4	10.0	10.6	11.8	12.1			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.6
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION=4

TABLE 2.3.2-20 (SHEET 3 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN-.3 BUT LESS THAN OR EQUAL TO-.8
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTIONS

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	.1	.50
CALM + - 3.5	3	1	0	0	4	2	7	4	5	7	3	4	5	3	3	1	64	2.3	2.33
3.6 -7.5	23	18	15	24	37	35	25	35	72	72	59	37	42	21	27	26	588	21.5	5.72
7.6 - 12.5	48	67	75	50	66	71	90	96	87	122	146	139	107	111	94	63	1432	52.2	9.91
12.6 - 18.5	42	25	31	34	48	37	17	11	10	40	32	22	50	44	67	86	596	21.7	14.26
18.6 - 24.5	4	1	7	1	2	0	0	0	0	9	5	1	1	0	6	13	50	1.8	20.20
24.6 - 32.5	3	1	0	0	0	0	0	0	2	1	0	0	0	0	0	2	9	.3	28.15
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	123	113	154	115	157	145	139	146	176	251	245	203	208	179	197	192	2741	0.0	8.47
PERCENT	4.5	4.1	5.6	4.2	5.7	5.3	5.1	5.3	6.4	9.2	8.9	7.4		7.5	6.5	7.2	7.0	100.0	
AV SPD	11.5	10.5	10.2	10.1	10.4	10.0	9.5	8.9	8.5	9.7	9.8	9.6	10.2	10.5	11.3	12.7			

AVERAGE SPEED FOR THIS TABLE EQUALS 10.2
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION=1

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN. 8 BUT LESS THAN OR EQUAL TO 2.2
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 33 FT AND 33 FT

REQUEST NUMBER 209-2

WIND DIRECTION

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	.1	.50
CALM + - 3.5	0	2	0	4	2	1	6	2	3	1	3	3	2	1	1	2	33	2.6	2.59
3.6 -7.5	13	10	4	11	22	13	20	21	34	29	20	26	26	17	19	22	307	24.5	5.66
7.6 - 12.5	21	10	12	20	14	24	39	40	31	40	40	73	67	78	98	48	667	53.2	9.91
12.6 - 18.5	13	6	2	2	3	4	10	7	5	5	9	18	30	41	50	37	242	19.3	13.90
18.6 - 24.5	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	2	.2	20.08
24.6 - 32.5	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	2	.2	26.54
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	47	34	18	38	41	43	76	70	73	75	79	120	125	137	169	109	1254	0.0	8.15
PERCENT	3.7	2.7	1.4	3.0	3.3	3.4	6.1	5.6	5.8	6.0	6.3	9.6	10.0	10.9	13.5	8.7	100.0		
AV SPD	10.4	9.4	9.2	7.7	7.9	9.0	9.0	8.8	8.3	8.3	9.1	9.7	10.1	10.7	11.0	10.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.7
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION=0

TABLE 2.3.2-20 (SHEET 4 OF 4)

JOINT FREQUENCY TABLES OF WIND SPEED AND DIRECTION
FOR TEMPERATURE DIFFERENCE (DEG F/100 FT) GREATER THAN 2.2
SITE VOGTLE MET TOWER
PERIOD OF RECORD FROM 80040401 TO 81033124
SPEED AND DIRECTION FROM 33 FT LEVEL
TEMPERATURE DIFFERENCE BETWEEN 150 FT AND 33 FT

WIND DIRECTIONS

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT	GEO MEAN SPD (MPH)
CALM	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0	3	.4	.50
CALM + - 3.5	2	3	2	5	2	3	5	3	5	3	2	0	7	1	4	5	52	6.6	2.29
3.6 -7.5	21	12	13	10	8	11	19	16	7	12	10	19	13	12	18	13	214	27.2	5.58
7.6 - 12.5	22	18	13	13	11	6	21	17	15	14	10	32	52	45	52	28	375	47.7	9.79
12.6 - 18.5	1	2	3	5	2	0	1	0	3	2	3	25	29	28	24	14	142	18.1	13.92
18.6 - 24.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
24.6 - 32.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
32.6+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0.00
TOTAL	46	35	31	34	23	20	46	37	30	31	31	77	101	86	98	60	786	0.0	6.80
PERCENT	5.9	4.5	3.9	4.3	2.9	2.5	5.9	4.7	3.8	3.9	3.9	9.8		12.8	10.9	12.5	7.6	100.0	
AV SPD	7.8	7.5	8.3	8.1	7.9	6.3	7.4	7.4	8.0	7.9	8.2	10.3	10.3	10.5	10.1	9.5			

AVERAGE SPEED FOR THIS TABLE EQUALS 9.0
HOURS IN ABOVE TABLE WITH VARIABLE DIRECTION=0

SUMMARY TABLE

SPEED(MPH)	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	TOTAL	PERCENT
CALM	1	1	2	2	0	0	0	2	0	0	0	1	1	0	0	2	12	.2
CALM + - 3.5	15	15	17	26	18	12	22	12	16	14	15	10	20	11	14	24	261	3.3
3.6 -7.5	103	77	104	94	135	114	110	108	155	140	124	126	141	106	164	112	1915	24.1
7.6 - 12.5	160	173	174	161	195	192	240	235	193	220	253	303	336	381	414	223	3853	48.4
12.6 - 18.5	95	69	72	77	132	101	61	30	29	68	58	78	152	192	291	208	1713	21.5
18.6 - 24.5	8	3	10	5	5	4	1	0	0	15	10	7	4	17	40	37	164	2.1
24.6 - 32.5	4	1	2	0	0	0	1	0	3	5	1	0	0	2	4	8	31	.4
32.6+	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	2	4	.1
TOTAL	384	339	381	365	485	423	435	387	397	462	461	526	654	711	927	616	7953	100.0
PERCENT	4.8	4.3	4.8	4.8	6.1	5.3	5.5	4.9	5.0	5.8	5.8	6.6	8.2	8.9	11.7	7.7	100.0	

TABLE 2.3.2-21

SEASONAL AND ANNUAL MORNING AND AFTERNOON MEAN MIXING HEIGHT IN THE VEGP SITE REGION

Season	Morning (m)	Afternoon (m)
Annual	375	1475
Winter	395	950
Spring	380	1700
Summer	390	1800
Fall	290	1400

TABLE 2.3.3-1

VEGP METEOROLOGICAL TOWER SUMMARY OF DATA RECOVERY

Parameter(s)	Percent of Recovery			
	1972-1973	1977-1978	1978-1979	1980-1981
Ambient temperature, 33 ft	87.3	98.8	95.6	96.8
Dewpoint temperature, 33 ft	84.5	99.0	95.3	96.2
Delta temperature, 99-33 ft	86.0	99.3	91.1	98.2
Delta temperature, 150-33 ft	84.2	96.4	96.4	96.1
Windspeed, 33 ft	87.5	98.4	95.3	99.5
Wind direction, 33 ft	88.1	98.9	91.6	96.0
Windspeed, 150 ft	82.1	97.9	94.3	96.4
Wind direction, 150 ft	83.7	99.2	95.0	99.0
Solar radiation	61.5	98.4	96.4	100.0
Rainfall	88.9	99.4	95.1	70.0
All parameters	83.4	98.6	94.6	94.8
Composite				
WS and WD, 33 ft; DT 150-33 ft	79.8	94.8	90.0	92.1
WS and WD, 33 ft; DT 99-33 ft	81.2	97.8	85.5	94.4
WS and WD, 150 ft; DT 150-33 ft	74.4	94.4	92.6	91.6

TABLE 2.3.4-1

SUMMARY OF ACCIDENT X/Q VALVES (s/m^3)

Based on 3 Years of VEGP Site Data

Averaging Time Period after Accident (h)	Site Boundary 0.5 Percent Probable Direction-Dependent	Site Boundary 5 Percent Probable Direction-Independent	Site Boundary 50 Percent Probable Direction-Independent	LPZ 0.5 Percent Probable[a]	LPZ 50 Percent Probable[b]	Worst Condition[c]
1	1.8E-4	1.5E-4	2.8E-5	7.2E-5	1.0E-5	4.7E-4
8	--	--	--	3.3E-5	7.0E-6	1.8E-4
16	--	--	--	2.2E-5	2.0E-6	1.9E-5
72	--	--	--	9.2E-6	1.3E-6	5.0E-6
624	--	--	--	2.7E-6	7.0E-7	1.6E-6

Based on Most Recent Year of VEGP Site Data

1	1.6E-4	1.4E-4	2.7E-5	6.6E-5	1.1E-5	4.7E-4
8	--	--	--	3.8E-5	6.2E-6	1.2E-4
16	--	--	--	6.3E-6	1.8E-6	1.4E-5
72	--	--	--	5.2E-6	1.1E-6	5.0E-6
624	--	--	--	1.6E-6	6.3E-7	1.2E-6

a. Based on log-log interpolation. Five percent probable direction-independent approach gives the same results above.

b. Based on running-mean calculations over appropriate averaging periods using direction-independent approach.

c. Values shown are based on running-mean calculations and are the highest calculated during any of the three 1-year periods of record. The fact that values are lower than reported for the 0.5 percent and 5 percent probable cases for certain averaging periods is due to differences in methodology introduced by the log-log interpolation scheme of Regulatory Guide 1.145. (See figure 2.3.4-8.)

TABLE 2.3.4-2

ASSUMED DISTANCE TO SITE BOUNDARY IN EACH DIRECTION

Direction (from Plant to Site Boundary)	Distance (m)
N	1344
NNE	1097
NE	1097
ENE	1097
E	1369
ESE	1817
SE	1866
SSE	1773
S	1692
SSW	1680
SW	1462
WSW	1462
W	1462
WNW	1649
NW	2240
NNW	1804

TABLE 2.3.4-3

TEMPERATURE DIFFERENCE GROUPS FOR
DETERMINING PASQUILL STABILITY CATEGORIES

Pasquill Category	NRC ΔT Classification (°F/100 ft)[a]
A	$\Delta T < -1.0$
B	$-1.0 \leq \Delta T < -0.9$
C	$-0.9 \leq \Delta T < -0.8$
D	$-0.8 \leq \Delta T < -0.3$
E	$-0.3 \leq \Delta T < 0.8$
F	$0.8 \leq \Delta T < 2.2$
G	$2.2 \leq \Delta T$

a. In conversion from °C/100 m (Regulatory Guide 1.23) to °F/100 ft, values were rounded to nearest tenth of a degree.

TABLE 2.3.5-1

LIST OF COMPUTER RUNS AT VEGP SITE

Run Number	Vent Identification	Data Used	Type of Run	Hourly or Joint Frequency Meteorological Data Used	Season	To Be Used For Evaluating Releases from the Following Vents	Location of Results in the Report
VX-1	Plant vent	1977-78	Wake-split	Hourly	Annual	Used for comparison with VX-3	Table 2.3.5-6
VX-2	All other release points	1977-78	Ground release in building wake	Hourly	Annual	Used for comparison with VX-4	Table 2.3.5-7
VX-3	Plant vent	3-year	Wake-split	Joint frequency	Annual	Plant vent	Tables 2.3.5-8 and 2.3.5-10
VX-4	All other release points	3-year	Ground release in building wake	Joint frequency	Annual	All other vents	Tables 2.3.5-9 and 2.3.5-11
209-10	Plant vent	1980-81	Wake-split	Hourly	Annual	Plant vent	Table 2.3.5-12
209-11	All other release points	1980-81	Ground release in building wake	Hourly	Annual	All other vents	Table 2.3.5-13

TABLE 2.3.5-2

GASEOUS DISCHARGE POINT AT VEGP SITE

System	Vent
Containment preaccess purge exhaust	Plant vent
Turbine building ventilation	Into building wake
Turbine building condenser vacuum exhaust filter system	Into building wake
Steam packing exhauster blower	Into building wake
Fuel handling building normal filter exhaust	Plant vent
Auxiliary building continuous filter exhaust	Plant vent
Laboratory area and hood exhaust	Plant vent
Radwaste processing facility vent	Into building wake

TABLE 2.3.5-3

VENT DESIGN INFORMATION AT VEGP SITE

Vent	Location	Operating Conditions	Height of Release (ft)		Inside Dimensions Exit (in.)	Cross-Sectional Area (ft^2)	Effluent Conditions		
			Above Sea Level	Above Grade[a]			Flowrate (ft^3/min)	Discharge Temperature (°F)	Velocity (ft/min)
Plant vent[b]	Containment Unit 1	Normal power generation	419	200	60 x 120	50	140,600(c)	68-103	2810
		Refueling	419	200	60 x 120	50	150,600	67-103	3010
Plant vent [c]	Containment Unit 2	Normal power generation	419	200	60 x 78	32.5	81,350	71-101	2500
		Refueling	419	200	60 x 78	32.5	91,350	68-103	2810
Turbine building vent[d]	Turbine building Unit 1	Normal power generation	313	94	17.25	1.62	1168	131-193	720
Turbine building vent[d]	Turbine building Unit 2	Normal power generation	313	94	17.25	1.62	1168	131-193	720
Turbine building ventilation-general [e][f]	Turbine building Unit 1	Normal power generation	344	125	97	51.3	1.2 x 10	85-104	2350
Turbine building ventilation-general [e][f]	Turbine building Unit 2	Normal power generator	344	125	97	51.3	1.2 x 10	85-104	2350

a. Nominal building grade elevation is approximately 219 ft.

b. Vent not capped.

c. Unit 1 flowrate includes additional flow from common refueling building.

d. Vent not capped but has 18-in. 180° return elbow.

e. Fan housing is capped with rectangular shroud.

f. Data for each power roof ventilator; nine units total (table 9.4.4-1).

TABLE 2.3.5-4

TABULATION OF INPUT ASSUMPTIONS FOR CALCULATIONS AT VEGP SITE

Parameter	Assumed Value or Characteristic
Height of meteorological instruments for hourly wake split runs	33 ft and 150 ft
Height of meteorological instruments for wake-split runs using joint frequency tables	33 ft extrapolated to vent height
Method for determining stability and diffusion coefficients	Temperature difference using Regulatory Guide 1.23 and Pasquill curves
Calms treatment	Assumed 0.3 mph (threshold for Climet is about 0.6 mph). Assumed to have same direction as measured.
Upper limit for σ_z (m)	1000
Height of tallest structure for computation of Σ effective (m)	55.0
Vent exit conditions	From table 2.3.5-3
Delta temperature correction factor	None
Terrain height	See table 2.3.5-5

TABLE 2.3.5-5

TERRAIN ELEVATION ABOVE PLANT GRADE (m)

Wind Direction from Plant to Receptor

Distance (m)	S	SSW	SW	WSW	W	WNW	NW	NNW	N	NNE	NE	ENE	E	ESE	SE	SSE
500	0.0	4.7	8.7	5.7	1.4	5.8	5.7	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1000	0.0	4.7	16.7	13.4	3.3	10.4	11.8	6.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1500	0.0	4.7	21.7	18.6	7.3	12.2	14.3	7.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2000	0.0	4.7	21.7	18.6	7.3	12.2	14.3	7.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2500	0.0	4.7	21.7	18.6	7.3	12.2	14.3	7.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3000	0.0	4.7	23.7	18.6	7.3	12.2	14.3	7.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3500	0.0	4.7	24.4	18.6	7.3	12.2	16.9	7.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4000	0.0	4.7	24.4	18.6	7.3	12.2	16.9	7.3	0.0	0.0	0.0	0.0	0.0	0.0	4.5	0.0
5000	0.0	4.7	24.7	18.6	7.3	12.2	16.9	7.3	0.0	0.0	0.0	0.0	0.0	0.0	11.1	0.0
6000	0.0	4.7	26.8	18.6	7.3	12.2	16.9	7.3	0.0	0.0	0.0	0.0	0.0	0.0	11.1	0.0
7000	3.6	4.7	26.8	18.6	7.3	12.2	16.9	7.3	0.0	0.0	0.0	7.8	0.0	0.0	11.1	0.0
8000	14.6	4.7	26.8	18.6	7.3	12.2	16.9	7.3	0.0	0.0	21.1	13.9	0.0	0.0	11.8	0.0
9000	14.6	5.1	26.8	18.6	7.3	12.2	16.9	7.3	0.0	0.0	24.4	14.6	0.0	0.0	12.7	7.1
10,000	14.6	6.8	26.8	18.6	7.3	12.2	16.9	7.3	0.0	10.2	24.4	20.2	0.0	0.0	17.1	17.0
12,000	14.6	6.8	34.1	28.9	13.4	12.2	16.9	7.3	0.0	15.9	26.8	20.2	0.0	0.0	17.1	19.5
14,000	14.6	6.8	34.1	28.9	13.4	16.5	19.7	7.3	0.0	15.9	26.8	20.2	0.0	0.0	17.1	19.5
16,000	14.6	6.8	34.1	28.9	13.4	16.5	25.7	7.3	0.0	15.9	26.8	21.7	13.2	0.0	17.1	19.5

TABLE 2.3.5-6 (SHEET 1 OF 2)

ATMOSPHERIC DISPERSION FACTORS FOR VOGTLE
VENT - PLANT VENT SEASON – ANNUAL RUN NO. VX-1

ATMOSPHERIC DISPERSION FACTORS FOR- VENT- SEASON- REQ-

IN DIRECTION SECTOR	RUN TYPE - X/Q SEC/M3 SB	2413	4022	5631	DISTANCE (METERS) 7240	12067	24135	40225	56315	72405
N	1.30E-07	8.00E-08	5.40E-08	4.00E-08	3.10E-08	1.70E-08	8.70E-09	5.20E-09	3.40E-09	2.40E-09
NNE	1.90E-07	9.00E-08	6.00E-08	4.30E-08	3.30E-08	2.20E-08	1.00E-08	5.60E-09	3.80E-09	2.70E-09
NE	2.50E-07	1.10E-07	6.90E-08	5.00E-08	4.00E-08	2.90E-08	1.30E-08	7.50E-09	4.90E-09	3.60E-09
ENE	2.30E-07	9.60E-08	5.90E-08	4.20E-08	3.60E-08	2.20E-08	1.10E-08	5.60E-09	3.80E-09	2.80E-09
E	1.70E-07	9.40E-08	5.70E-08	4.10E-08	3.10E-08	1.70E-08	1.00E-08	5.30E-09	3.50E-09	2.60E-09
ESE	1.20E-07	9.10E-08	5.50E-08	3.90E-08	3.00E-08	1.60E-08	8.00E-09	4.50E-09	3.00E-09	2.20E-09
SE	9.50E-08	7.40E-08	4.70E-08	3.70E-08	2.80E-08	1.60E-08	6.90E-09	3.70E-09	2.50E-09	1.80E-09
SSE	5.80E-08	4.60E-08	3.00E-08	2.20E-08	1.70E-08	1.20E-08	5.00E-09	2.70E-09	1.80E-09	1.40E-09
S	7.10E-08	5.30E-08	3.50E-08	2.60E-08	2.20E-08	1.40E-08	6.40E-09	3.70E-09	2.60E-09	1.90E-09
SSW	7.80E-08	5.80E-08	3.80E-08	2.80E-08	2.20E-08	1.30E-08	6.80E-09	3.80E-09	2.60E-09	2.30E-09
SW	1.80E-07	1.20E-07	8.10E-08	5.80E-08	4.40E-08	2.60E-08	1.10E-08	5.80E-09	3.90E-09	2.80E-09
WSW	1.40E-07	9.80E-08	6.40E-08	4.50E-08	3.40E-08	2.10E-08	8.70E-09	4.60E-09	3.00E-09	2.20E-09
W	1.10E-07	7.30E-08	4.90E-08	3.60E-08	2.70E-08	1.60E-08	7.00E-09	4.00E-09	2.70E-09	2.10E-09
WNW	1.10E-07	8.20E-08	5.40E-08	3.90E-08	3.00E-08	1.70E-08	8.10E-09	4.50E-09	3.60E-09	2.60E-09
NW	8.40E-08	8.00E-08	5.60E-08	4.00E-08	3.10E-08	1.80E-08	8.40E-09	4.90E-09	3.80E-09	2.70E-09
NNW	9.70E-08	7.90E-08	5.30E-08	3.80E-08	3.00E-08	1.60E-08	7.10E-09	4.30E-09	2.90E-09	2.10E-09

IN DIRECTION SECTOR	RUN TYPE - DEPLETED X/Q SEC/M3 SB	2413	4022	5631	DISTANCE (METERS) 7240	12067	24135	40225	56315	72405
N	1.20E-07	7.20E-08	4.90E-08	3.60E-08	2.80E-08	1.50E-08	7.50E-09	3.10E-09	1.70E-09	1.10E-09
NNE	1.70E-07	8.10E-08	5.40E-08	3.90E-08	3.00E-08	1.90E-08	8.50E-09	3.80E-09	2.00E-09	1.30E-09
NE	2.30E-07	9.80E-08	6.10E-08	4.40E-08	3.50E-08	2.60E-08	1.10E-08	5.10E-09	3.00E-09	2.00E-09
ENE	2.10E-07	8.60E-08	5.20E-08	3.70E-08	3.10E-08	2.00E-08	8.80E-09	4.40E-09	2.80E-09	2.00E-09
E	1.50E-07	8.40E-08	5.10E-08	3.60E-08	2.70E-08	1.50E-08	8.00E-09	3.60E-09	2.20E-09	1.40E-09
ESE	1.10E-07	8.10E-08	4.80E-08	3.40E-08	2.60E-08	1.40E-08	6.80E-09	3.50E-09	2.20E-09	1.60E-09
SE	8.50E-08	6.60E-08	4.20E-08	3.30E-08	2.50E-08	1.40E-08	5.90E-09	3.10E-09	2.00E-09	1.40E-09
SSE	5.20E-08	4.20E-08	2.80E-08	2.00E-08	1.50E-08	1.00E-08	4.40E-09	2.40E-09	1.60E-09	1.10E-09
S	6.40E-08	4.80E-08	3.20E-08	2.30E-08	2.00E-08	1.30E-08	5.60E-09	3.20E-09	2.10E-09	1.60E-09
SSW	7.10E-08	5.30E-08	3.50E-08	2.50E-08	2.00E-08	1.20E-08	6.10E-09	3.40E-09	2.30E-09	1.60E-09
SW	1.60E-07	1.10E-07	7.60E-08	5.40E-08	4.10E-08	2.40E-08	9.40E-09	4.70E-09	3.00E-09	2.10E-09
WSW	1.30E-07	9.20E-08	5.90E-08	4.20E-08	3.10E-08	1.90E-08	7.60E-09	3.90E-09	2.50E-09	1.80E-09
W	9.60E-08	6.70E-08	4.50E-08	4.20E-08	2.50E-08	1.50E-08	6.20E-09	3.50E-09	2.30E-09	1.50E-09
WNW	9.80E-08	7.60E-08	5.00E-08	3.60E-08	2.70E-08	1.50E-08	7.10E-09	3.70E-09	1.90E-09	1.30E-09
NW	7.70E-08	7.40E-08	5.10E-08	3.70E-08	2.90E-08	1.60E-08	7.40E-09	3.80E-09	1.90E-09	1.30E-09
NNW	8.80E-08	7.20E-08	4.90E-08	3.50E-08	2.70E-08	1.50E-08	6.30E-09	3.70E-09	2.30E-09	1.70E-09

TABLE 2.3.5-6 (SHEET 2 OF 2)

ATMOSPHERIC DISPERSION FACTORS FOR- VENT- SEASON- REQ-

IN DIRECTION SECTOR	RUN TYPE - DEPOSITION D/Q M-2 SP	2413	4022	5631	DISTANCE (METERS) 7240	12067	24135	40225	56315	72405
N	9.80E-10	4.30E-10	2.00E-10	1.20E-10	7.70E-11	3.40E-11	1.30E-11	1.50E-11	7.30E-12	4.60E-12
NNE	1.40E-09	5.00E-10	2.40E-10	1.40E-10	8.90E-11	4.10E-11	1.50E-11	2.10E-11	8.40E-12	5.10E-12
NE	2.10E-09	6.90E-10	6.10E-10	1.80E-10	1.20E-10	5.40E-11	2.20E-11	2.60E-11	1.30E-11	8.20E-12
ENE	2.80E-09	8.90E-10	4.00E-10	2.30E-10	1.50E-10	6.90E-11	2.20E-11	1.10E-11	7.10E-12	5.40E-12
E	3.10E-09	1.20E-09	5.40E-10	3.10E-10	2.00E-10	9.20E-11	5.40E-11	2.50E-11	1.40E-11	9.30E-12
ESE	1.70E-09	1.10E-09	4.70E-10	2.70E-10	1.70E-10	7.80E-11	2.60E-11	1.30E-11	8.20E-12	6.10E-12
SE	1.30E-09	8.70E-10	3.90E-10	2.20E-10	1.40E-10	6.40E-11	2.10E-11	9.50E-12	6.10E-12	4.50E-12
SSE	7.50E-10	4.80E-10	2.20E-10	1.30E-10	8.40E-11	3.90E-11	1.20E-11	5.50E-12	3.50E-12	2.60E-12
S	7.80E-10	4.70E-10	2.20E-10	1.30E-10	8.60E-11	3.90E-11	1.20E-11	5.60E-12	3.60E-12	2.70E-12
SSW	1.00E-09	6.10E-10	2.80E-10	1.60E-10	1.99E-10	4.70E-11	1.50E-11	6.80E-12	4.30E-12	4.30E-12
SW	2.00E-09	9.50E-10	4.20E-10	2.40E-10	1.60E-10	7.10E-11	2.30E-11	1.10E-11	7.50E-12	5.60E-12
WSW	1.60E-00	7.40E-10	3.30E-10	1.90E-10	1.20E-10	5.50E-11	1.70E-11	8.10E-12	5.40E-12	4.10E-12
W	1.10E-09	5.60E-10	2.60E-10	1.50E-10	9.80E-11	4.40E-11	1.40E-11	6.10E-12	4.10E-12	4.10E-12
WNW	9.00E-10	5.20E-10	2.40E-10	1.40E-10	9.10E-11	4.10E-11	1.30E-11	6.20E-12	6.60E-12	4.20E-12
NW	4.90E-10	4.40E-10	2.00E-10	1.20E-10	7.40E-11	3.30E-11	1.00E-11	7.70E-12	6.20E-12	3.90E-12
NNW	6.70E-10	4.50E-10	2.10E-10	1.30E-10	8.10E-11	3.60E-11	1.10E-11	5.00E-12	3.60E-12	2.80E-12

TABLE 2.3.5-7 (SHEET 1 OF 2)

ATMOSPHERIC DISPERSION FACTORS FOR VOGTLE
VENT - GROUND WAKE SEASON – ANNUAL RUN NO. VX-2

ATMOSPHERIC DISPERSION FACTORS FOR- VENT- SEASON- REQ-

IN DIRECTION SECTOR	RUN TYPE - X/Q SEC/M3 SB	2413	4022	5631	DISTANCE (METERS) 7240	12067	24135	40225	56315	72405
N	1.20E-06	5.10E-07	2.40E-07	1.50E-07	1.10E-07	5.10E-08	1.90E-08	9.60E-09	6.20E-09	4.50E-09
NNE	1.80E-06	6.00E-07	2.90E-07	1.80E-07	1.30E-07	6.00E-08	2.30E-08	1.10E-08	7.40E-09	5.40E-09
NE	2.30E-06	7.40E-07	3.60E-07	2.30E-07	1.60E-07	7.80E-08	3.00E-08	1.50E-08	1.00E-08	7.30E-09
ENE	2.00E-06	6.40E-07	3.10E-07	2.00E-07	1.40E-07	6.80E-08	2.70E-08	1.40E-08	9.00E-09	6.60E-09
E	1.20E-06	5.40E-07	2.60E-07	1.70E-07	1.20E-07	5.70E-08	2.20E-08	1.10E-08	7.40E-09	5.50E-09
ESE	7.90E-07	5.30E-07	2.60E-07	1.60E-07	1.10E-07	5.50E-08	2.10E-08	1.10E-08	7.10E-09	5.20E-09
SE	6.70E-07	4.70E-07	2.30E-07	1.40E-07	1.00E-07	4.90E-08	1.90E-08	9.60E-09	6.30E-09	4.60E-09
SSE	5.80E-07	3.70E-07	1.80E-07	1.10E-07	8.10E-08	3.90E-08	1.50E-08	7.80E-09	5.10E-09	3.70E-09
S	7.60E-07	4.60E-07	2.20E-07	1.40E-07	1.00E-07	4.90E-08	1.90E-08	9.90E-09	6.50E-09	4.80E-09
SSW	7.60E-07	4.50E-07	2.20E-07	1.40E-07	9.90E-08	4.90E-08	1.90E-08	9.80E-09	6.50E-09	4.80E-09
SW	1.20E-06	5.90E-07	2.90E-07	1.80E-07	1.30E-07	6.00E-08	2.30E-08	1.20E-08	7.70E-09	5.60E-09
WSW	9.70E-07	4.80E-07	2.30E-07	1.40E-07	9.90E-08	4.70E-08	1.80E-08	9.10E-09	5.90E-09	4.30E-09
W	1.00E-06	5.00E-07	2.40E-07	1.50E-07	1.10E-07	5.10E-08	2.00E-08	1.00E-08	6.60E-09	4.80E-09
WNW	8.90E-07	5.20E-07	2.50E-07	1.60E-07	1.10E-07	5.30E-08	2.00E-08	1.00E-08	6.70E-09	4.90E-09
NW	5.70E-07	5.10E-07	2.50E-07	1.50E-07	1.10E-07	5.30E-08	2.00E-08	1.00E-08	6.70E-09	4.90E-09
NNW	8.60E-07	5.70E-07	2.70E-07	1.70E-07	1.20E-07	5.70E-08	2.20E-08	1.10E-08	7.20E-09	5.20E-09

IN DIRECTION SECTOR	RUN TYPE - DEPLETED X/Q SEC/M3 SB	2413	4022	5631	DISTANCE (METERS) 7240	12067	24135	40225	56315	72405
N	1.00E-06	4.30E-07	2.00E-07	1.20E-07	8.00E-08	3.50E-08	1.20E-08	5.30E-09	3.10E-09	2.10E-09
NNE	1.60E-06	5.10E-07	2.30E-07	1.40E-07	9.40E-08	4.20E-08	1.40E-08	6.30E-09	3.70E-88	2.50E-09
NE	2.00E-06	6.30E-07	2.90E-07	1.70E-07	1.20E-07	5.40E-08	1.90E-08	8.40E-09	5.00E-09	3.40E-09
ENE	1.80E-06	5.40E-07	2.50E-07	1.50E-07	1.10E-07	4.80E-08	1.70E-08	7.50E-09	4.50E-09	3.00E-09
E	1.10E-06	4.60E-07	2.10E-07	1.30E-07	8.80E-08	4.00E-08	1.40E-08	6.20E-09	3.70E-09	2.50E-09
ESE	6.80E-07	4.50E-07	2.10E-07	1.20E-07	8.60E-08	3.90E-08	1.30E-08	6.00E-09	3.60E-09	2.40E-09
SE	5.80E-07	3.90E-07	1.80E-07	1.10E-07	7.60E-08	3.40E-08	1.20E-08	5.30E-09	3.20E-09	2.10E-09
SSE	5.10E-07	3.20E-07	1.50E-07	8.70E-08	6.40E-08	2.70E-08	9.50E-09	4.30E-09	2.60E-09	1.70E-09
S	6.70E-07	3.90E-07	1.80E-07	1.10E-07	7.60E-08	3.40E-08	1.20E-08	5.40E-09	3.30E-09	2.20E-09
SSW	6.70E-07	3.80E-07	1.80E-07	1.10E-07	7.50E-08	3.40E-08	1.20E-08	5.40E-09	3.20E-09	2.20E-09
SW	1.10E-06	5.00E-07	2.30E-07	1.40E-07	9.40E-08	4.20E-08	1.40E-08	6.40E-09	3.80E-09	2.60E-09
WSW	8.60E-07	4.00E-07	1.80E-07	1.10E-07	7.50E-08	3.30E-08	1.10E-08	5.00E-09	3.00E-09	2.00E-09
W	9.00E-07	4.20E-07	1.90E-07	1.10E-07	8.00E-08	3.60E-08	1.20E-08	5.60E-09	3.30E-09	2.20E-09
WNW	7.80E-07	4.40E-07	2.00E-07	1.20E-07	8.20E-08	3.70E-08	1.30E-08	5.60E-09	3.40E-09	2.30E-09
NW	4.90E-07	4.30E-07	2.00E-07	1.20E-07	8.20E-08	3.70E-08	1.20E-08	5.60E-09	3.30E-09	2.20E-09
NNW	7.50E-07	4.80E-07	2.20E-07	1.80E-07	9.00E-08	4.00E-08	1.40E-08	6.10E-09	3.60E-09	2.40E-09

TABLE 2.3.5-7 (SHEET 2 OF 2)

ATMOSPHERIC DISPERSION FACTORS FOR- VENT- SEASON- REQ-

IN DIRECTION SECTOR	RUN TYPE - X/Q SEC/M3 SB	2413	4022	5631	DISTANCE (METERS) 7240	12067	24135	40225	56315	72405
N	5.10E-09	1.80E-09	7.50E-10	4.20E-10	2.50E-10	1.10E-10	3.30E-11	1.40E-11	7.20E-12	4.60E-12
NNE	7.80E-09	2.00E-09	8.40E-10	4.70E-10	2.80E-10	1.30E-10	3.70E-11	1.50E-11	8.10E-12	5.10E-12
NE	9.60E-09	2.50E-09	1.00E-09	5.70E-10	3.50E-10	1.50E-10	4.50E-11	1.90E-11	9.90E-12	6.30E-12
ENE	9.40E-09	2.40E-09	1.00E-09	5.60E-10	3.40E-10	1.50E-10	4.40E-11	1.80E-11	9.70E-12	6.10E-12
E	7.50E-09	2.70E-09	1.10E-09	6.30E-10	3.80E-10	1.70E-10	5.00E-11	2.10E-11	1.10E-11	6.90E-12
ESE	4.10E-09	2.60E-09	1.10E-09	5.90E-10	3.60E-10	1.60E-10	4.70E-11	1.90E-11	1.00E-11	6.40E-12
SE	3.20E-09	2.10E-09	8.60E-10	4.80E-10	2.90E-10	1.30E-10	3.80E-11	1.60E-11	8.30E-12	5.30E-12
SSE	2.10E-09	1.20E-09	5.10E-10	2.90E-10	1.70E-10	7.70E-11	2.30E-11	9.40E-12	5.00E-12	3.10E-12
S	2.60E-09	1.40E-09	5.60E-10	3.10E-10	1.90E-10	8.40E-11	2.50E-11	1.00E-11	5.40E-12	3.40E-12
SSW	3.10E-09	1.60E-09	6.80E-10	3.80E-10	2.30E-10	1.00E-10	3.00E-11	1.20E-11	6.60E-12	4.20E-12
SW	5.80E-09	2.40E-09	9.70E-10	5.40E-10	3.30E-10	1.50E-10	4.30E-11	1.80E-11	9.40E-12	5.90E-12
WSW	4.80E-09	1.90E-09	8.00E-10	4.40E-10	2.70E-10	1.20E-10	3.50E-11	1.50E-11	7.70E-12	4.90E-12
W	4.10E-09	1.70E-09	6.80E-10	3.40E-10	2.30E-10	1.00E-10	3.00E-11	1.20E-11	6.60E-12	4.20E-12
WNW	3.30E-09	1.70E-09	6.80E-10	3.40E-10	2.30E-10	1.00E-10	3.00E-11	1.20E-11	6.60E-12	4.20E-12
NW	1.70E-09	1.50E-09	6.40E-10	3.50E-10	2.20E-10	9.50E-11	2.80E-11	1.20E-11	6.20E-12	3.90E-12
NNW	2.60E-09	1.60E-09	6.60E-10	3.70E-10	2.20E-10	9.90E-11	2.90E-11	1.20E-11	6.40E-12	4.00E-12

TABLE 2.3.5-8 (SHEET 1 OF 2)

ATMOSPHERIC DISPERSION FACTORS FOR VOGTLE
VENT - PLANT VENT SEASON – ANNUAL RUN NO. VX-3

ATMOSPHERIC DISPERSION FACTORS FOR-VOGTLE GND WAKE VENT-ANNUAL VX-3 3-YR SEASON-S JFT GND REQ-

IN DIRECTION	RUN TYPE - X/Q SEC/M3				DISTANCE (METERS)					
SECTOR	SB	2413	4022	5631	7240	12067	24135	40225	56315	72405
N	1.40E-07	7.80E-08	5.00E-08	3.60E-08	2.80E-08	1.60E-08	8.60E-09	5.80E-09	3.80E-09	2.80E-09
NNE	1.90E-07	8.30E-08	5.20E-08	3.80E-08	2.90E-08	1.90E-08	9.70E-09	5.60E-09	3.90E-09	2.80E-09
NE	2.00E-07	8.60E-08	5.40E-08	3.90E-08	3.10E-08	2.40E-08	1.20E-08	6.80E-09	4.50E-09	3.30E-09
ENE	1.80E-07	7.80E-08	4.80E-08	3.50E-08	3.00E-08	2.00E-08	9.90E-09	5.40E-09	3.60E-09	2.70E-09
E	1.20E-07	7.10E-08	4.40E-08	3.20E-08	2.50E-08	1.40E-08	1.00E-08	5.50E-09	3.70E-09	2.70E-09
ESE	9.40E-08	7.10E-08	4.40E-08	3.10E-08	2.40E-08	1.40E-08	7.60E-09	4.70E-09	3.20E-09	2.30E-09
SE	8.30E-08	6.40E-08	4.10E-08	3.20E-08	2.50E-08	1.50E-08	6.90E-09	3.90E-09	2.60E-09	2.00E-09
SSE	4.80E-08	3.70E-08	2.50E-08	1.80E-08	1.40E-08	1.10E-08	5.20E-09	2.90E-09	2.00E-09	1.50E-09
S	6.80E-08	4.90E-08	3.10E-08	2.30E-08	1.90E-08	1.30E-08	6.20E-09	3.90E-09	2.70E-09	2.10E-09
SSW	7.70E-08	5.50E-08	3.50E-08	2.50E-08	2.00E-08	1.20E-08	6.30E-09	3.60E-09	2.50E-09	2.30E-09
SW	1.70E-07	1.10E-07	7.10E-08	5.10E-08	4.00E-08	2.50E-08	1.10E-08	5.90E-09	4.00E-09	3.00E-09
WSW	1.50E-07	9.60E-08	5.90E-08	4.10E-08	3.10E-08	2.00E-08	8.30E-09	4.50E-09	3.00E-09	2.20E-09
W	1.20E-07	7.70E-08	5.00E-08	3.60E-08	2.80E-08	1.70E-08	7.70E-09	4.60E-09	3.20E-09	2.60E-09
WNW	1.00E-07	7.60E-08	4.90E-08	3.50E-08	2.70E-08	1.50E-08	7.50E-09	4.20E-09	3.60E-09	2.60E-09
NW	8.50E-08	8.00E-08	5.60E-08	4.00E-08	3.10E-08	1.80E-08	8.40E-09	5.00E-09	3.90E-09	2.80E-09
NNW	9.40E-08	7.60E-08	5.10E-08	3.80E-08	2.90E-08	1.60E-08	7.20E-09	4.40E-09	3.00E-09	2.20E-09

IN DIRECTION	RUN TYPE - DEPLETED X/Q SEC/M3				DISTANCE (METERS)					
SECTOR	SB	2413	4022	5631	7240	12067	24135	40225	56315	72405
N	1.20E-07	7.00E-08	4.50E-08	3.20E-08	2.50E-08	1.40E-08	7.50E-09	3.50E-09	1.90E-09	1.30E-09
NNE	1.70E-07	7.40E-08	4.70E-08	3.30E-08	2.60E-08	1.70E-08	8.20E-09	3.80E-09	2.00E-09	1.30E-09
NE	1.80E-07	7.70E-08	4.80E-08	3.40E-08	2.80E-08	2.10E-08	9.70E-09	4.70E-09	2.80E-09	1.90E-09
ENE	1.70E-07	7.00E-08	4.30E-08	3.10E-08	2.70E-08	1.80E-08	8.50E-09	4.40E-09	2.90E-09	2.00E-09
E	1.10E-07	6.30E-08	3.90E-08	2.80E-08	2.20E-08	1.30E-08	8.40E-09	3.90E-09	2.40E-09	1.60E-09
ESE	8.40E-08	6.40E-08	3.90E-08	2.80E-08	2.10E-08	1.20E-08	6.70E-09	3.90E-09	2.50E-09	1.80E-09
SE	7.40E-08	5.70E-08	3.60E-08	2.90E-08	2.20E-08	1.40E-08	6.10E-09	3.40E-09	2.30E-09	1.70E-09
SSE	4.40E-08	3.40E-08	2.20E-08	1.60E-08	1.30E-08	1.00E-08	4.70E-09	2.60E-09	1.80E-09	1.30E-09
S	6.10E-08	4.40E-08	2.80E-08	2.00E-08	1.70E-08	1.20E-08	5.60E-09	3.50E-09	2.40E-09	1.80E-09
SSW	7.00E-08	5.00E-08	3.20E-08	2.30E-08	1.80E-08	1.10E-08	5.70E-09	3.20E-09	2.20E-09	1.60E-09
SW	1.60E-07	1.00E-07	6.50E-08	4.70E-08	3.60E-08	2.20E-08	9.30E-09	4.80E-09	3.10E-09	2.20E-09
WSW	1.40E-07	8.80E-08	5.40E-08	3.70E-08	2.80E-08	1.80E-08	7.20E-09	3.70E-09	2.40E-09	1.70E-09
W	1.10E-07	7.00E-08	4.60E-08	3.30E-08	2.50E-08	1.50E-08	6.80E-09	4.00E-09	2.70E-09	1.80E-09
WNW	9.40E-08	7.00E-08	4.50E-08	3.20E-08	2.50E-08	1.40E-08	6.60E-09	3.50E-09	1.80E-09	1.20E-09
NW	7.80E-08	7.40E-08	5.10E-08	3.70E-08	2.90E-08	1.60E-08	7.50E-09	3.90E-09	2.00E-09	1.30E-09
NNW	8.50E-08	7.00E-08	4.70E-08	3.40E-08	2.70E-08	1.50E-08	6.50E-09	3.80E-09	2.50E-09	1.70E-09

TABLE 2.3.5-8 (SHEET 2 OF 2)

ATMOSPHERIC DISPERSION FACTORS FOR-VOGTLE GND WAKE VENT-ANNUAL VX-3 3-YR SEASON-S JFT GND REQ-

IN DIRECTION	RUN TYPE - DISPOSITION D/QQ M-2				DISTANCE (METERS)					
SECTOR	SB	2413	4022	5631	7240	12067	24135	40225	56315	72405
N	1.30E-09	5.60E-10	2.60E-10	1.50E-10	9.60E-11	4.30E-11	1.50E-11	1.60E-11	7.70E-12	4.90E-12
NNE	1.80E-09	6.00E-10	2.80E-10	1.60E-10	1.00E-10	4.80E-11	1.70E-11	2.00E-11	8.30E-12	5.00E-12
NE	2.30E-09	7.50E-10	3.40E-10	2.00E-10	1.30E-10	5.80E-11	2.20E-11	2.30E-11	1.20E-11	7.50E-12
ENE	2.80E-09	8.70E-10	3.80E-10	2.20E-10	1.50E-10	6.60E-11	2.10E-11	1.00E-11	6.70E-12	5.10E-12
E	2.70E-09	1.10E-09	4.90E-10	2.80E-10	1.80E-10	8.30E-11	4.60E-11	2.20E-11	1.20E-11	8.20E-12
ESE	1.60E-09	1.00E-09	4.30E-10	2.50E-10	1.60E-10	7.30E-11	2.40E-11	1.10E-11	7.40E-12	5.50E-12
SE	1.20E-09	7.70E-10	3.40E-10	2.00E-10	1.30E-10	5.70E-11	1.80E-11	8.30E-12	5.20E-12	3.90E-12
SSE	7.00E-10	4.40E-10	2.00E-10	1.20E-10	7.50E-11	3.50E-11	1.10E-11	5.00E-12	3.20E-12	2.40E-12
S	9.20E-10	5.40E-10	2.40E-10	1.40E-10	9.40E-11	4.30E-11	1.40E-11	6.20E-12	4.00E-12	3.00E-12
SSW	1.10E-09	6.30E-10	2.80E-10	1.60E-10	1.10E-10	4.80E-11	1.50E-11	6.90E-12	4.40E-12	4.30E-12
SW	2.20E-09	9.90E-10	4.40E-10	2.50E-10	1.60E-10	7.30E-11	2.40E-11	1.20E-11	7.70E-12	5.70E-12
WSW	1.80E-09	8.20E-10	3.60E-10	2.10E-10	1.30E-10	6.00E-11	1.90E-11	8.90E-12	5.80E-12	4.40E-12
W	1.50E-09	7.20E-10	3.30E-10	1.90E-10	1.20E-10	5.60E-11	1.70E-11	7.90E-12	5.20E-12	5.10E-12
WNW	1.00E-09	5.90E-10	2.70E-10	1.60E-10	1.00E-10	4.50E-11	1.40E-11	6.80E-12	6.80E-12	4.30E-12
NW	6.30E-10	5.60E-10	2.60E-10	1.50E-10	9.40E-11	4.20E-11	1.30E-11	9.50E-12	7.10E-12	4.50E-12
NNW	7.70E-10	5.10E-10	2.40E-10	1.40E-10	8.90E-11	4.00E-11	1.20E-11	5.60E-12	4.00E-12	3.10E-12

TABLE 2.3.5-9 (SHEET 1 OF 2)

ATMOSPHERIC DISPERSION FACTORS FOR VOGTLE
VENT - GROUND WAKE SEASON – ANNUAL RUN NO. VX-4

AMOSPHERIC DISPERSION FACTORS FOR-VOGTLE GND WAKE VENT-ANNUAL VX-4 3-YR SEASON-S JFT GND REQ-

IN DIRECTION	RUN TYPE - X/Q SEC/M3				DISTANCE (METERS)					
SECTOR	SB	2413	4022	5631	7240	12067	24135	40225	56315	72405
N	1.40E-06	6.00E-07	2.90E-07	1.80E-07	1.30E-07	6.20E-08	2.40E-08	1.20E-08	7.90E-09	5.70E-09
NNE	1.90E-06	6.20E-07	3.00E-07	1.90E-07	1.30E-07	6.30E-08	2.40E-08	1.20E-08	8.00E-09	5.80E-09
NE	2.20E-06	7.20E-07	3.50E-07	2.20E-07	1.60E-07	7.70E-08	3.00E-08	1.50E-08	1.00E-08	7.50E-09
ENE	2.00E-06	6.50E-07	3.20E-07	2.00E-07	1.40E-07	7.00E-08	2.80E-08	1.40E-08	9.30E-09	6.90E-09
E	1.30E-06	5.90E-07	2.90E-07	1.80E-07	1.30E-07	6.30E-08	2.50E-08	1.30E-08	8.40E-09	6.20E-09
ESE	8.80E-07	5.80E-07	2.90E-07	1.80E-07	1.30E-07	6.30E-08	2.50E-08	1.30E-08	8.40E-09	6.20E-09
SE	8.00E-07	5.50E-07	2.70E-07	1.70E-07	1.20E-07	6.00E-08	2.40E-08	1.20E-08	8.00E-09	5.90E-09
SSE	6.60E-07	4.20E-07	2.10E-07	1.30E-07	9.50E-08	4.70E-08	1.90E-08	9.60E-09	6.30E-09	4.70E-09
S	9.90E-07	5.90E-07	2.90E-07	1.90E-07	1.30E-07	6.70E-08	2.70E-08	1.40E-08	9.30E-09	6.80E-09
SSW	9.10E-07	5.40E-07	2.70E-07	1.70E-07	1.20E-07	6.00E-08	2.40E-08	1.20E-08	8.10E-09	6.00E-09
SW	1.50E-06	7.30E-07	3.60E-07	2.30E-07	1.60E-07	7.90E-08	3.10E-08	1.60E-08	1.10E-08	7.80E-09
WSW	1.20E-06	5.80E-07	2.80E-07	1.80E-07	1.30E-07	6.10E-08	2.40E-08	1.20E-08	7.90E-09	5.80E-09
W	1.30E-06	6.50E-07	3.20E-07	2.00E-07	1.40E-07	7.00E-08	2.70E-08	1.40E-08	9.20E-09	6.70E-09
WNW	9.60E-07	5.60E-07	2.70E-07	1.70E-07	1.20E-07	5.80E-08	2.20E-08	1.10E-08	7.40E-09	5.40E-09
NW	6.90E-07	6.20E-07	3.00E-07	1.90E-07	1.30E-07	6.40E-08	2.50E-08	1.20E-08	8.20E-09	6.00E-09
NNW	9.20E-07	6.10E-07	2.90E-07	1.80E-07	1.30E-07	6.10E-08	2.30E-08	1.20E-08	7.60E-09	5.50E-09

IN DIRECTION	RUN TYPE - DEPLETED X/Q SEC/M3				DISTANCE (METERS)					
SECTOR	SB	2413	4022	5631	7240	12067	24135	40225	56315	72405
N	1.20E-06	5.10E-07	2.30E-07	1.40E-07	9.60E-08	4.30E-08	1.50E-08	6.60E-09	4.00E-09	2.70E-09
NNE	1.70E-06	5.30E-07	2.40E-07	1.40E-07	9.90E-08	4.40E-08	1.50E-08	6.70E-09	4.00E-09	2.70E-09
NE	2.00E-06	6.10E-07	2.80E-07	1.70E-07	1.20E-07	5.40E-08	1.90E-08	8.50E-09	5.10E-09	3.40E-09
ENE	1.80E-06	5.50E-07	2.60E-07	1.50E-07	1.10E-07	4.90E-08	1.70E-08	7.80E-09	4.70E-09	3.20E-09
E	1.20E-06	5.00E-07	2.30E-07	1.40E-07	9.70E-08	4.40E-08	1.50E-08	7.00E-09	4.20E-09	2.80E-09
ESE	7.60E-07	5.00E-07	2.30E-07	1.40E-07	9.70E-08	4.40E-08	1.50E-08	7.00E-09	4.20E-09	2.90E-09
SE	6.90E-07	4.70E-07	2.20E-07	1.30E-07	9.10E-08	4.20E-08	1.50E-08	6.60E-09	4.00E-09	2.70E-09
SSE	5.80E-07	3.60E-07	1.70E-07	1.00E-07	7.10E-08	3.30E-08	1.20E-08	5.30E-09	3.20E-09	2.20E-09
S	8.60E-07	5.00E-07	2.30E-07	1.40E-07	1.00E-07	4.70E-08	1.70E-08	7.60E-09	4.60E-09	3.20E-09
SSW	7.90E-07	4.60E-07	2.10E-07	1.30E-07	9.10E-08	4.20E-08	1.50E-08	6.70E-09	4.10E-09	2.80E-09
SW	1.30E-06	6.20E-07	2.90E-07	1.70E-07	1.20E-07	5.50E-08	1.90E-08	8.80E-09	5.30E-09	3.60E-09
WSW	1.10E-06	4.90E-07	2.30E-07	1.40E-07	9.40E-08	4.30E-08	1.50E-08	6.60E-09	4.00E-09	2.70E-09
W	1.20E-06	5.60E-07	2.60E-07	1.50E-07	1.10E-07	4.90E-08	1.70E-08	7.60E-09	4.60E-09	3.10E-09
WNW	8.40E-07	4.70E-07	2.20E-07	1.30E-07	9.00E-08	4.00E-08	1.40E-08	6.20E-09	3.70E-09	2.50E-09
NW	5.90E-07	5.20E-07	2.40E-07	1.40E-07	9.90E-08	4.50E-08	1.50E-08	6.90E-09	4.10E-09	2.80E-09
NNW	8.00E-07	5.20E-07	2.30E-07	1.40E-07	9.60E-08	4.20E-08	1.40E-08	6.40E-09	3.80E-09	2.50E-09

TABLE 2.3.5-9 (SHEET 2 OF 2)

AMOSPHERIC DISPERSION FACTORS FOR-VOGTLE GND WAKE VENT-ANNUAL VX-4 3-YR SEASON-S JFT GND REQ-

IN DIRECTION	RUN TYPE - DEPOSITION D/Q M-2				DISTANCE (METERS)					
SECTOR	SB	2413	4022	5631	7240	12067	24135	40225	56315	72405
N	5.40E-09	1.90E-09	7.90E-10	4.40E-10	2.70E-10	1.20E-10	3.50E-11	1.40E-11	7.70E-12	4.90E-12
NNE	7.70E-09	2.00E-09	8.20E-10	4.60E-10	2.80E-10	1.20E-10	3.60E-11	1.50E-11	8.00E-12	5.00E-12
NE	8.80E-09	2.30E-09	9.40E-10	5.20E-10	3.20E-10	1.40E-10	4.20E-11	1.70E-11	9.10E-12	5.80E-12
ENE	8.80E-09	2.30E-09	9.40E-10	5.20E-10	3.20E-10	1.40E-10	4.10E-11	1.70E-11	9.10E-12	5.70E-12
E	6.80E-09	2.50E-09	1.00E-09	5.70E-10	3.50E-10	1.50E-10	4.50E-11	1.80E-11	9.80E-12	6.20E-12
ESE	3.80E-09	2.30E-09	9.50E-10	5.30E-10	3.20E-10	1.40E-10	4.20E-11	1.70E-11	9.20E-12	5.80E-12
SE	3.00E-09	1.90E-09	8.00E-10	4.50E-10	2.70E-10	1.20E-10	3.50E-11	1.50E-11	7.70E-12	4.90E-12
SSE	2.10E-09	1.20E-09	5.00E-10	2.80E-10	1.70E-10	7.50E-11	2.20E-11	9.10E-12	4.90E-12	3.10E-12
S	2.90E-09	1.50E-09	6.30E-10	3.50E-10	2.10E-10	9.40E-11	2.80E-11	1.10E-11	6.10E-12	3.80E-12
SSW	3.10E-09	1.60E-09	6.70E-10	3.70E-10	2.30E-10	1.00E-10	3.00E-11	1.20E-11	6.50E-12	4.10E-12
SW	5.70E-09	2.30E-09	9.60E-10	5.30E-10	3.30E-10	1.40E-10	4.20E-11	1.70E-11	9.30E-12	5.90E-12
WSW	5.00E-09	2.00E-09	8.30E-10	4.60E-10	2.80E-10	1.20E-10	3.70E-11	1.50E-11	8.10E-12	5.10E-12
W	4.90E-09	2.00E-09	8.30E-10	4.60E-10	2.80E-10	1.20E-10	3.70E-11	1.50E-11	8.00E-12	5.10E-12
WNW	3.40E-09	1.70E-09	7.00E-10	3.90E-10	2.40E-10	1.00E-10	3.10E-11	1.30E-11	6.80E-12	4.30E-12
NW	2.00E-09	1.80E-09	7.30E-10	4.10E-10	2.50E-10	1.10E-10	3.20E-11	1.30E-11	7.10E-12	4.50E-12
NNW	2.90E-09	1.80E-09	7.30E-10	4.10E-10	2.50E-10	1.10E-10	3.20E-11	1.30E-11	7.10E-12	4.50E-12

TABLE 2.3.5-10 (SHEET 1 OF 2)

DIFFUSION AND DEPOSITION ESTIMATES FOR ALL RECEPTOR LOCATIONS

Release Point: Plant Vent/Wake-Split Season: Annual Computer Run ID: VX-3

Direction	Distance to Nearest Milk Cow (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})	Distance to Nearest Meat Animal (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})	Distance to Nearest Milk Goat (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})
N	-	2.5E-08	2.2E-08	8.2E-11	-	2.5E-08	2.2E-08	8.2E-11	-	2.5E-08	2.2E-08	8.2E-11
NNE	-	2.6E-08	2.3E-08	9.0E-11	-	2.6E-00	2.3E-08	9.0E-11	-	2.6E-08	2.3E-08	9.0E-11
NE	-	3.5E-08	3.2E-08	1.1E-10	-	3.5E-08	3.2E-08	1.1E-10	-	3.5E-08	3.2E-08	1.1E-10
ENE	-	2.9E-08	2.6E-08	1.3E-10	-	2.9E-08	2.6E-08	1.3E-10	-	2.9E-08	2.6E-08	1.3E-10
E	-	2.2E-08	2.0E-08	1.6E-10	-	2.2E-08	2.0E-08	1.6E-10	-	2.2E-08	2.0E-08	1.6E-10
ESE	-	2.2E-08	1.9E-08	1.4E-10	-	2.2E-08	1.9E-08	1.4E-10	-	2.2E-08	1.9E-08	1.4E-10
SE	-	2.3E-08	2.0E-08	1.1E-10	6920	2.6E-08	2.3E-08	1.4E-10	-	2.3E-08	2.0E-08	1.1E-10
SSE	-	1.3E-08	1.2E-08	6.4E-11	-	1.3E-08	1.2E-08	6.4E-11	-	1.3E-08	1.2E-08	6.4E-11
S	-	2.0E-08	1.8E-08	8.1E-11	7242	2.0E-08	1.8E-08	9.4E-11	-	2.0E-08	1.8E-08	8.1E-11
SSW	-	1.8E-08	1.6E-08	9.1E-11	7805	1.8E-08	1.6E-08	9.4E-11	-	1.8E-08	1.6E-08	9.1E-11
SW	-	3.6E-08	3.2E-08	1.4E-10	4828	5.9E-08	5.3E-08	3.2E-10	-	3.6E-08	3.2E-08	1.4E-10
WSW	-	2.8E-08	2.5E-08	1.2E-10	3862	6.2E-08	5.6E-08	3.9E-10	-	2.8E-08	2.5E-08	1.2E-10
W	-	2.5E-08	2.3E-08	1.1E-10	5713	3.6E-08	3.2E-08	1.9E-10	-	2.5E-08	2.3E-08	1.1E-10
WNW	-	2.4E-08	2.2E-08	8.7E-11	4184	4.7E-08	4.3E-08	2.5E-10	-	2.4E-08	2.2E-08	8.7E-11
NW	-	2.8E-08	2.6E-08	8.1E-11	6437	3.5E-08	3.2E-08	1.2E-10	-	2.8E-08	2.6E-08	8.1E-11
NNW	-	2.6E-08	2.4E-08	7.6E-11	-	2.6E-08	2.4E-08	7.6E-11	-	2.6E-08	2.4E-08	7.6E-11

TABLE 2.3.5-10 (SHEET 2 OF 2)

Direction	Distance to Nearest Residence (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})	Distance to Nearest Veg. Garden (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})	Nearest Site Boundary (m)	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})
N	-	2.5E-08	2.2E-08	8.2E-11	-	2.5E-08	2.2E-08	8.2E-11	1344	1.4E-07	1.2E-07	1.3E-09
NNE	-	2.6E-08	2.3E-08	9.0E-11	-	2.6E-08	2.3E-08	9.0E-11	1097	1.9E-07	1.7E-07	1.8E-09
NE	-	3.5E-08	3.2E-08	1.1E-10	-	3.5E-08	3.2E-08	1.1E-10	1097	2.0E-07	1.8E-07	2.3E-09
ENE	-	2.9E-08	2.6E-08	1.3E-10	-	2.9E-08	2.6E-08	1.3E-10	1097	1.8E-07	1.7E-07	2.8E-09
E	-	2.2E-08	2.0E-08	1.6E-10	-	2.2E-08	2.0E-08	1.6E-10	1369	1.2E-07	1.1E-07	2.7E-09
ESE	-	2.2E-08	1.9E-08	1.4E-10	-	2.2E-08	1.9E-08	1.4E-10	1817	9.4E-08	8.4E-08	1.6E-09
SE	5150	3.5E-08	3.1E-08	2.3E-10	6920	2.6E-07	2.3E-07	1.4E-10	1866	8.3E-08	7.4E-08	1.2E-09
SSE	-	1.3E-08	1.2E-08	6.4E-11	-	1.3E-08	1.2E-08	6.4E-11	1773	4.8E-08	4.4E-08	7.0E-10
S	7242	1.9E-08	1.7E-08	9.4E-11	7242	2.0E-08	1.8E-08	9.4E-11	1692	6.8E-08	6.1E-08	9.2E-10
SSW	7483	1.9E-08	1.7E-08	1.0E-10	7805	1.8E-08	1.6E-08	9.4E-11	1680	7.7E-08	7.0E-08	1.1E-09
SW	4828	5.9E-08	5.3E-08	3.2E-10	7725	3.7E-08	3.4E-08	1.4E-10	1462	1.7E-07	1.6E-07	1.2E-09
WSW	1931	1.2E-07	1.1E-07	1.1E-09	1931	1.2E-07	1.1E-07	1.1E-09	1462	1.5E-07	1.4E-07	1.8E-09
W	5713	3.6E-08	3.2E-08	1.9E-10	7081	2.8E-08	2.6E-08	1.3E-10	1462	1.2E-07	1.1E-07	1.5E-09
WNW	3701	5.3E-08	4.9E-08	3.1E-10	3701	5.3E-08	4.9E-08	3.1E-10	1649	1.0E-07	9.4E-08	1.0E-09
NW	3701	6.0E-08	5.6E-08	2.9E-10	3701	6.0E-08	5.6E-08	2.9E-10	2240	8.5E-08	7.8E-08	6.3E-10
NNW	-	2.6E-08	2.4E-08	7.6E-11	-	2.6E-08	2.4E-08	7.6E-11	1840	9.4E-08	8.5E-08	7.7E-10

a. Receptor distance greater than 8000 m is indicated by (-); diffusion values given are for 8000 m.

TABLE 2.3.5-11 (SHEET 1 OF 2)

DIFFUSION AND DEPOSITION ESTIMATES FOR ALL RECEPTOR LOCATIONS

Release Point: Assumed Ground Release in Building Wake Season: Annual Computer Run ID: VX-4

Direction	Distance to Nearest Milk Cow (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})	Distance to Nearest Meat Animal (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})	Distance to Nearest Milk Goat (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})
N	-	1.1E-07	8.2E-08	2.4E-10	-	1.1E-07	8.2E-08	2.4E-10	-	1.1E-07	8.2E-08	2.4E-10
NNE	-	1.1E-07	8.4E-08	2.4E-10	-	1.1E-07	8.4E-08	2.4E-10	-	1.1E-07	8.4E-08	2.4E-10
NE	-	1.4E-07	1.0E-07	2.8E-10	-	1.4E-07	1.0E-07	2.8E-10	-	1.4E-07	1.0E-07	2.8E-10
ENE	-	1.2E-07	9.2E-08	2.8E-10	-	1.2E-07	9.2E-08	2.8E-10	-	1.2E-07	9.2E-08	2.8E-10
E	-	1.1E-07	8.2E-08	3.0E-10	-	1.1E-07	8.2E-08	3.0E-10	-	1.1E-07	8.2E-08	3.0E-10
ESE	-	1.1E-07	8.3E-08	2.8E-10	-	1.1E-07	8.3E-08	2.8E-10	-	1.1E-07	8.3E-08	2.8E-10
SE	-	1.1E-07	7.8E-08	2.4E-10	6920	1.3E-07	9.8E-08	2.9E-10	-	1.1E-07	7.8E-08	2.4E-10
SSE	-	8.2E-08	6.1E-08	1.5E-10	-	8.2E-08	6.1E-08	1.5E-10	-	8.2E-08	6.1E-08	1.5E-10
S	-	1.2E-07	8.6E-08	1.9E-10	7242	1.3E-07	1.0E-07	2.1E-10	-	1.2E-08	8.6E-08	1.9E-10
SSW	-	1.0E-07	7.7E-08	2.0E-10	7805	1.1E-07	8.1E-08	2.1E-10	-	1.0E-07	7.7E-08	2.0E-10
SW	-	1.4E-07	1.0E-07	2.9E-10	4828	2.8E-07	2.1E-07	6.9E-10	-	1.4E-07	1.0E-07	2.9E-10
WSW	-	1.1E-07	8.0E-08	2.5E-10	3862	3.0E-07	2.4E-07	9.0E-10	-	1.1E-07	8.0E-08	2.5E-10
W	-	1.2E-07	9.1E-08	2.5E-10	5713	2.0E-07	1.5E-07	4.5E-10	-	1.2E-07	9.1E-08	2.5E-10
WNW	-	1.0E-07	7.6E-08	2.1E-10	4184	2.5E-07	2.0E-07	6.5E-10	-	1.0E-07	7.6E-08	2.1E-10
NW	-	1.1E-07	8.4E-08	2.2E-10	6437	1.6E-07	1.2E-07	3.1E-10	-	1.1E-07	8.4E-08	2.2E-10
NNW	-	1.1E-07	8.1E-08	2.2E-10	-	1.1E-07	8.1E-08	2.2E-10	-	1.1E-07	8.1E-08	2.2E-10

TABLE 2.3.5-11 (SHEET 2 OF 2)

Direction	Distance to Nearest Residence (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})	Distance to Nearest Veg. Garden (m)[a]	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})	Nearest Site Boundary (m)	X/Q (s/m^3)	Depleted X/Q (s/m^3)	D/Q (m^{-2})
N	-	1.1E-07	8.2E-08	2.4E-10	-	1.1E-07	8.2E-08	2.4E-10	1344	1.4E-06	1.2E-06	5.4E-09
NNE	-	1.1E-07	8.4E-08	2.4E-10	-	1.1E-07	8.4E-08	2.4E-10	1097	1.9E-06	1.7E-06	7.7E-09
NE	-	1.4E-07	1.0E-07	2.8E-10	-	1.4E-07	1.0E-07	2.8E-10	1097	2.2E-06	2.0E-06	8.8E-09
ENE	-	1.2E-07	9.2E-08	2.8E-10	-	1.2E-07	9.2E-08	2.8E-10	1097	2.0E-06	1.8E-06	8.8E-09
E	-	1.1E-07	8.2E-08	3.0E-10	-	1.1E-07	8.2E-08	3.0E-10	1369	1.3E.06	1.2E-06	6.8E-09
ESE	-	1.1E-07	8.3E-07	2.8E-10	-	1.1E-07	8.3E-08	2.8E-10	1817	8.8E-06	7.6E-07	3.8E-09
SE	5150	1.9E-07	1.5E-07	5.2E-10	6920	1.3E-07	9.8E-08	2.9E-10	1866	8.0E-07	6.9E-07	3.0E-08
SSE	-	8.2E-08	6.1E-08	1.5E-10	-	8.2E-08	6.1E-08	1.5E-10	1773	6.6E-07	5.8E-07	2.1E-09
S	7242	1.3E-07	1.0E-07	2.1E-10	7242	1.3E-07	1.0E-07	2.1E-10	1692	9.9E-07	8.6E-07	2.9E-09
SSW	7483	1.1E-07	8.6E-08	2.2E-10	7805	1.1E-07	8.1E-08	2.1E-10	1680	9.1E-07	7.9E-07	3.1E-09
SW	4828	2.8E-07	2.1E-07	6.9E-10	7725	1.5E-07	1.1E-07	3.0E-10	1462	1.5E-06	1.3E-06	5.7E-09
WSW	1931	8.1E-07	7.0E-07	2.9E-09	1931	8.1E-07	7.0E-07	2.9E-09	1462	1.2E-06	1.1E-06	5.0E-09
W	5713	2.0E-07	1.5E-07	4.5E-10	7081	1.5E-07	1.1E-07	2.9E-10	1462	1.3E-06	1.2E-06	4.9E-09
WNW	3701	3.0E-07	2.5E-07	8.2E-10	3701	3.0E-07	2.5E-07	3.2E-10	1649	9.6E-07	8.4E-07	3.4E-09
NW	3701	3.3E-07	2.7E-07	8.5E-10	3701	3.3E-07	2.7E-07	8.5E-10	2240	6.9E-07	5.9E-07	2.0E-09
NNW	-	1.1E-07	8.1E-08	2.2E-10	-	1.1E-07	8.1E-08	2.2E-10	1804	9.2E-07	8.0E-07	2.9E-09

a. Receptor distance greater than 8000 m is indicated by (-); diffusion values given are for 8000 m.

TABLE 2.3.5-12 (SHEET 1 OF 2)

ATMOSPHERIC DISPERSION FACTORS 1980-81

VENT - WAKE SPLIT SEASON – ANNUAL RUN NO. 209-10

ATMOSPHERIC DISPERSION FACTORS FOR VENT- SEASON- REQ-

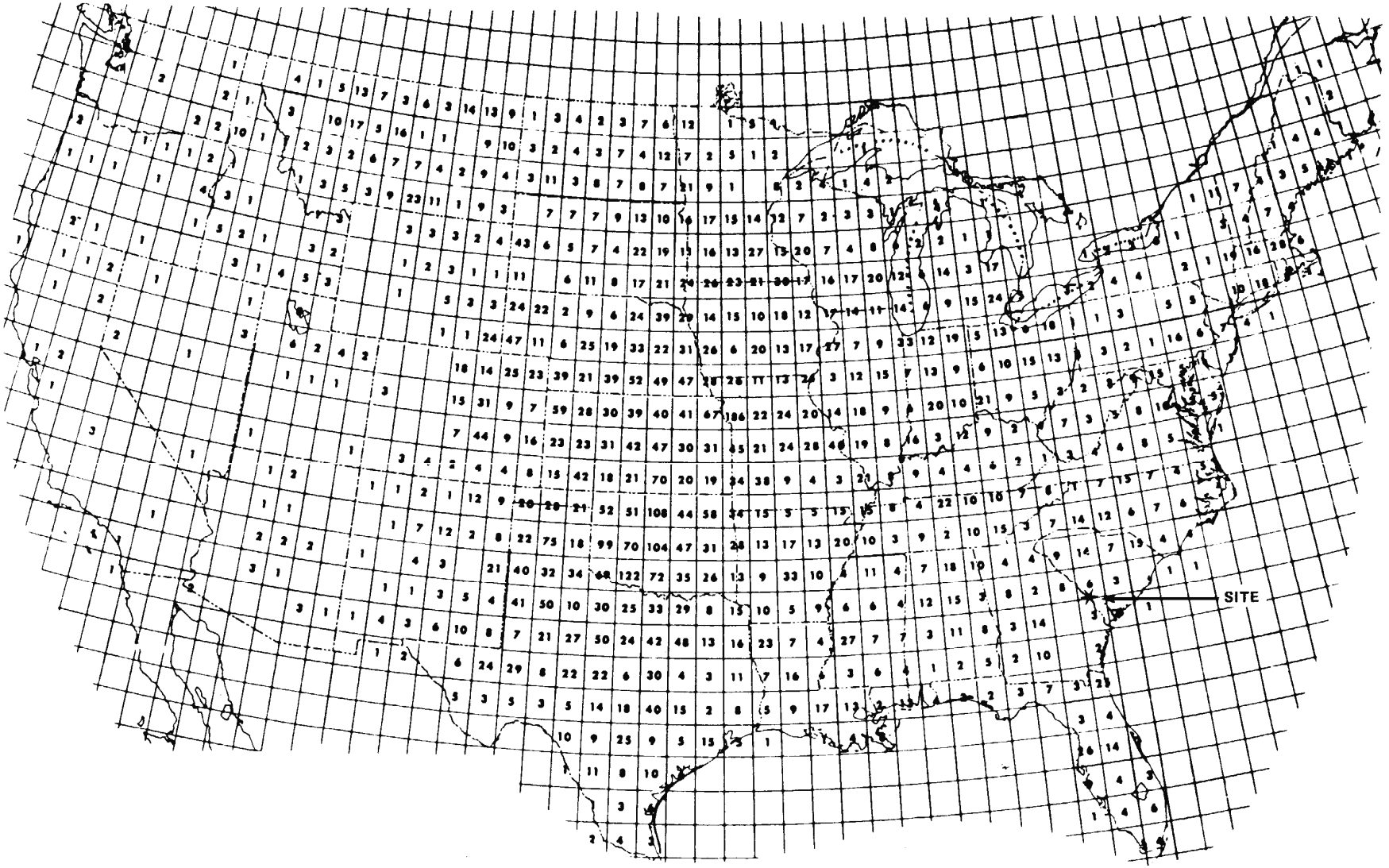
IN DIRECTION	RUN TYPE - X/Q SEC/M3				DISTANCE (METERS)					
SECTOR	SB	2413.5	4022.5	5631.5	7240.5	12067.5	24135.0	40225.0	56315.0	72405.0
N	.905E-07	.634E-07	.378E-07	.262E-07	.209E-07	.122E-07	.530E-08	.308E-08	.208E-08	.154E-08
NNE	.117E-06	.790E-07	.454E-07	.311E-07	.234E-07	.133E-07	.676E-08	.383E-08	.262E-08	.238E-08
NE	.177E-06	.105E-06	.630E-07	.447E-07	.342E-07	.212E-07	.920E-08	.499E-08	.336E-08	.249E-08
ENE	.151E-06	.909E-07	.520E-07	.351E-07	.260E-07	.153E-07	.633E-08	.335E-08	.223E-08	.163E-08
E	.944E-07	.581E-07	.347E-07	.240E-07	.181E-07	.105E-07	.463E-08	.274E-08	.189E-08	.152E-08
ESE	.927E-07	.640E-07	.382E-07	.265E-07	.200E-07	.110E-07	.521E-08	.287E-08	.228E-08	.165E-08
SE	.841E-07	.786E-07	.503E-07	.352E-07	.266E-07	.145E-07	.664E-08	.358E-08	.255E-08	.184E-08
SSE	.883E-07	.689E-07	.439E-07	.313E-07	.239E-07	.134E-07	.580E-08	.348E-08	.235E-08	.172E-08
S	.159E-06	.850E-07	.509E-07	.357E-07	.270E-07	.150E-07	.794E-08	.504E-08	.330E-08	.239E-08
SSW	.206E-06	.853E-07	.514E-07	.362E-07	.275E-07	.178E-07	.878E-08	.499E-08	.351E-08	.257E-08
SW	.252E-06	.990E-07	.589E-07	.417E-07	.332E-07	.246E-07	.119E-07	.704E-08	.467E-08	.345E-08
WSW	.308E-06	.119E-06	.692E-07	.486E-07	.406E-07	.262E-07	.127E-07	.687E-08	.463E-08	.345E-08
W	.185E-06	.961E-07	.563E-07	.397E-07	.303E-07	.174E-07	.120E-07	.625E-08	.416E-08	.308E-08
WNW	.122E-06	.874E-07	.505E-07	.354E-07	.270E-07	.155E-07	.842E-08	.515E-08	.349E-08	.261E-08
NW	.109E-06	.820E-07	.488E-07	.369E-07	.278E-07	.164E-07	.714E-08	.391E-08	.264E-08	.196E-08
NNW	.656E-07	.479E-07	.287E-07	.201E-07	.152E-07	.102E-07	.440E-08	.240E-08	.161E-08	.119E-08

IN DIRECTION	RUN TYPE - DELETED X/Q SEC/M3				DISTANCE (METERS)					
SECTOR	SB	2413.5	4022.5	5631.5	7240.5	12067.5	24135.0	40225.0	56315.0	72405.0
N	.811E-07	.565E-07	.334E-07	.229E-07	.184E-07	.107E-07	.453E-08	.257E-08	.171E-08	.123E-08
NNE	.105E-06	.702E-07	.398E-07	.269E-07	.202E-07	.114E-07	.578E-08	.323E-08	.218E-08	.162E-08
NE	.162E-06	.953E-07	.567E-07	.400E-07	.306E-07	.188E-07	.778E-08	.397E-08	.255E-08	.181E-08
ENE	.138E-06	.823E-07	.465E-07	.310E-07	.230E-07	.133E-07	.526E-08	.264E-08	.169E-08	.119E-08
E	.856E-07	.523E-07	.310E-07	.212E-07	.160E-07	.927E-08	.400E-08	.230E-08	.152E-08	.102E-08
ESE	.841E-07	.578E-07	.342E-07	.235E-07	.177E-07	.964E-08	.445E-08	.227E-08	.115E-08	.765E-09
SE	.762E-07	.712E-07	.456E-07	.317E-07	.239E-07	.129E-07	.552E-08	.265E-08	.128E-08	.852E-09
SSE	.795E-07	.622E-07	.396E-07	.281E-07	.215E-07	.120E-07	.511E-08	.294E-08	.187E-08	.132E-08
S	.142E-06	.750E-07	.446E-07	.310E-07	.235E-07	.129E-07	.668E-08	.295E-08	.166E-08	.111E-08
SSW	.186E-06	.753E-07	.450E-07	.314E-07	.239E-07	.156E-07	.724E-08	.338E-08	.182E-08	.119E-08
SW	.227E-06	.869E-07	.512E-07	.358E-07	.287E-07	.216E-07	.977E-08	.472E-08	.281E-08	.189E-08
WSW	.277E-06	.104E-06	.598E-07	.414E-07	.351E-07	.227E-07	.105E-07	.535E-08	.345E-08	.246E-08
W	.166E-06	.846E-07	.488E-07	.339E-07	.259E-07	.147E-07	.929E-08	.422E-08	.252E-08	.170E-08
WNW	.108E-06	.765E-07	.435E-07	.300E-07	.229E-07	.130E-07	.708E-08	.403E-08	.261E-08	.186E-08
NW	.964E-07	.720E-07	.424E-07	.320E-07	.241E-07	.141E-07	.601E-08	.320E-08	.212E-08	.153E-08
NNW	.585E-07	.425E-07	.252E-07	.175E-07	.133E-07	.896E-08	.378E-08	.200E-08	.132E-08	.951E-09

TABLE 2.3.5-12 (SHEET 2 OF 2)

ATMOSPHERIC DISPERSION FACTORS FOR VENT- SEASON- REQ-

IN DIRECTION SECTOR	RUN TYPE - DEPOSITION D/Q M-2 SB	2413.5	4022.5	5631.5	DISTANCE (METERS) 7240.5	12067.5	24135.0	40225.0	56315.0	72405.0
N	.118E-08	.690E-09	.311E-09	.181E-09	.118E-09	.530E-10	.164E-10	.729E-11	.456E-11	.335E-11
NNE	.146E-08	.840E-09	.375E-09	.216E-09	.138E-09	.617E-10	.190E-10	.830E-11	.507E-11	.497E-11
NE	.233E-08	.106E-08	.461E-09	.262E-09	.165E-09	.758E-10	.247E-10	.118E-10	.735E-11	.526E-11
ENE	.194E-08	.891E-09	.390E-09	.222E-09	.140E-09	.625E-10	.194E-10	.900E-11	.563E-11	.409E-11
E	.118E-08	.562E-09	.253E-09	.147E-09	.933E-10	.419E-10	.128E-10	.570E-11	.374E-11	.345E-11
ESE	.986E-09	.555E-09	.248E-09	.143E-09	.907E-10	.404E-10	.124E-10	.614E-11	.564E-11	.356E-11
SE	.581E-09	.522E-09	.233E-09	.134E-09	.843E-10	.375E-10	.116E-10	.946E-11	.614E-11	.388E-11
SSE	.608E-09	.401E-09	.184E-09	.107E-09	.675E-10	.300E-10	.901E-11	.425E-11	.315E-11	.251E-11
S	.132E-08	.541E-09	.244E-09	.141E-09	.885E-10	.393E-10	.141E-10	.148E-10	.726E-11	.458E-11
SSW	.171E-08	.533E-09	.240E-09	.139E-09	.877E-10	.398E-10	.140E-10	.186E-10	.736E-11	.452E-11
SW	.190E-08	.584E-09	.260E-09	.150E-09	.960E-10	.442E-10	.201E-10	.233E-10	.119E-10	.726E-11
WSW	.319E-08	.959E-09	.421E-09	.243E-09	.157E-09	.705E-10	.232E-10	.116E-10	.781E-11	.602E-11
W	.376E-08	.149E-08	.636E-09	.364E-09	.235E-09	.106E-09	.656E-10	.303E-10	.167E-10	.109E-10
WNW	.179E-08	.112E-08	.476E-09	.272E-09	.175E-09	.780E-10	.260E-10	.129E-10	.850E-11	.637E-11
NW	.141E-08	.934E-09	.402E-09	.232E-09	.148E-09	.661E-10	.211E-10	.951E-11	.598E-11	.441E-11
NNW	.900E-09	.550E-09	.239E-09	.138E-09	.885E-10	.403E-10	.128E-10	.583E-11	.370E-11	.274E-11



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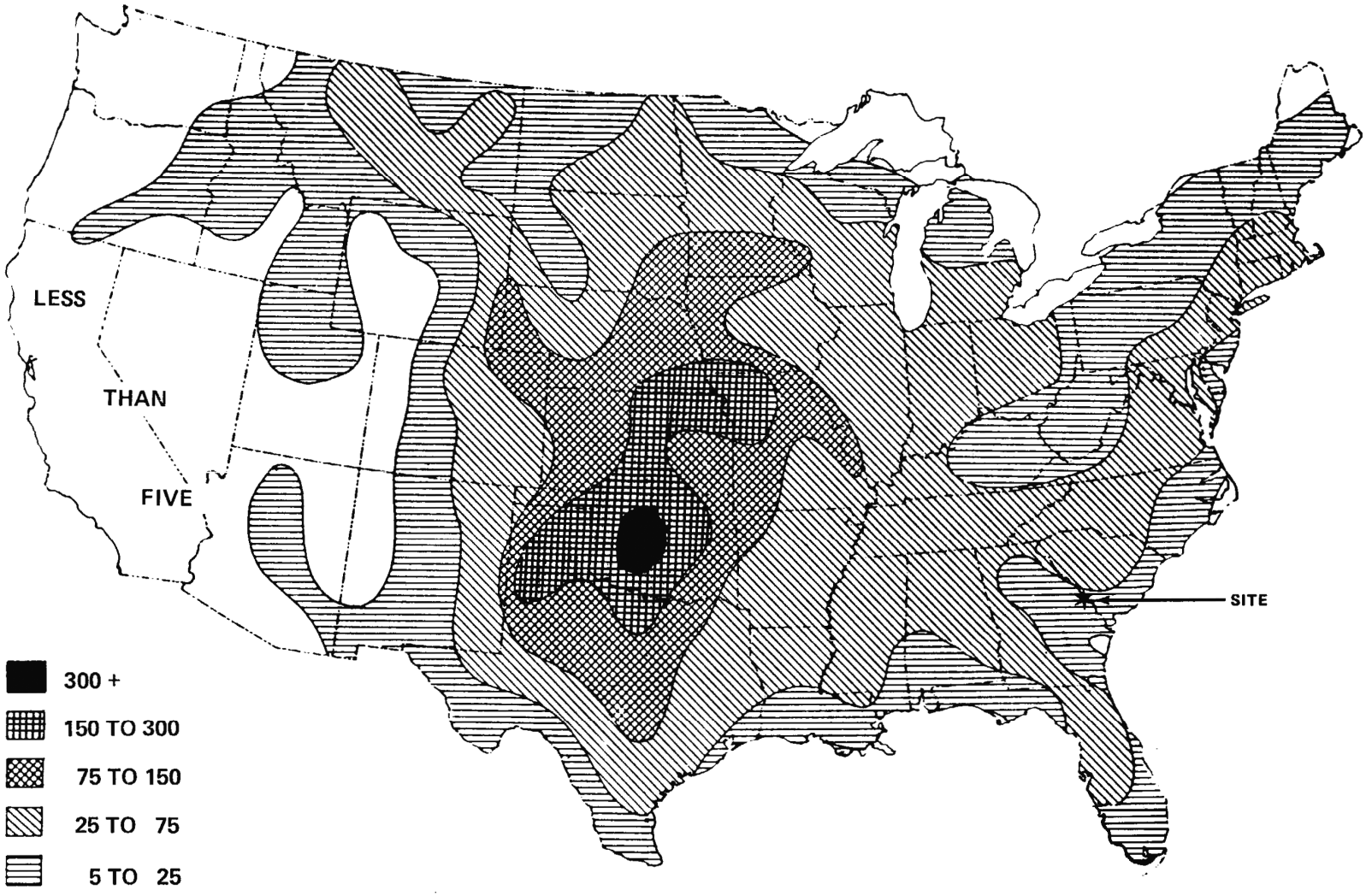
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TOTAL NUMBER OF HAIL REPORTS
3/4 in. AND GREATER
BY 1° SQUARES 1955 TO 1967

FIGURE 2.3.1–1



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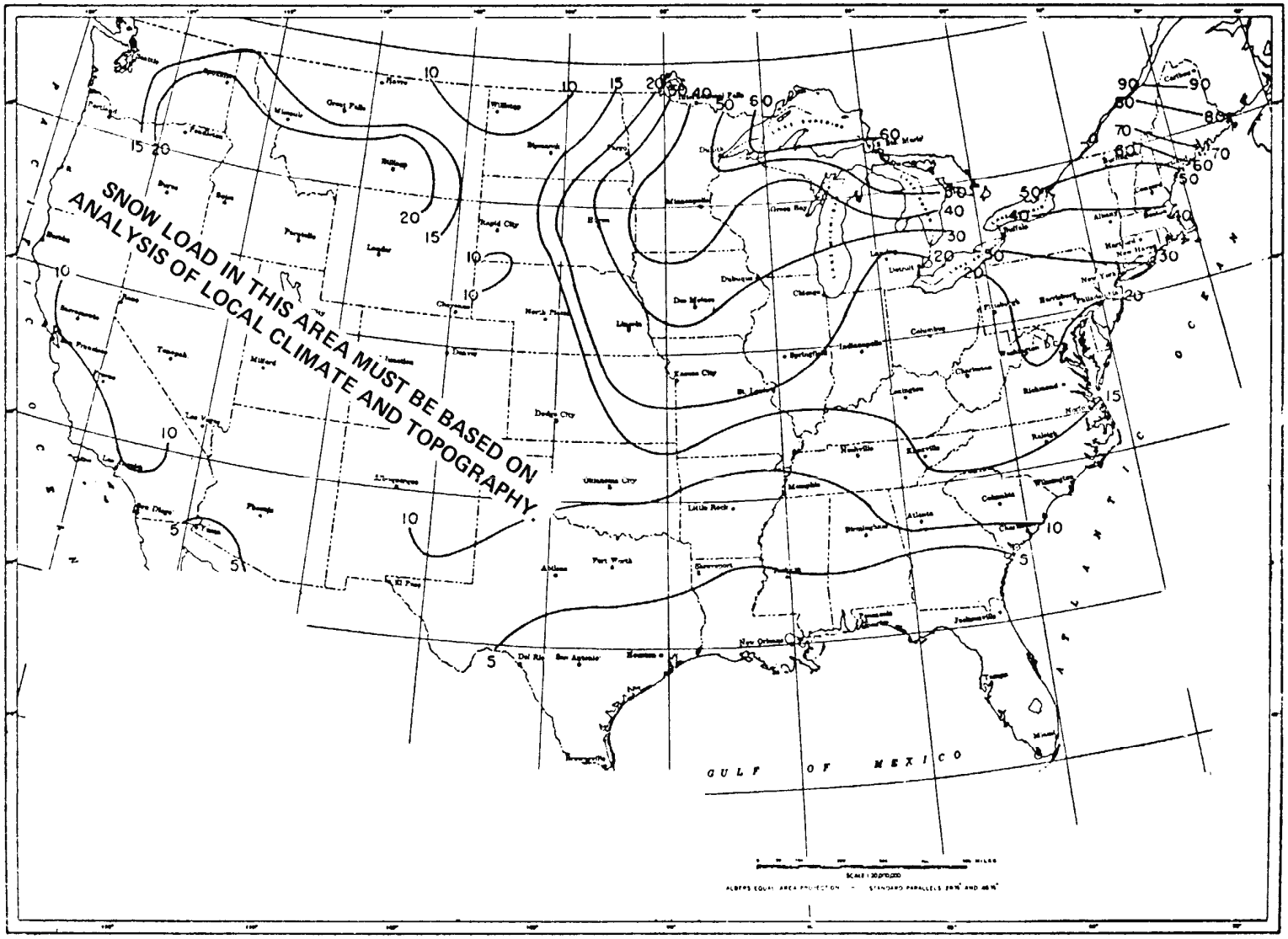
UNIT 1 AND UNIT 2

TOTAL NUMBER OF HAIL REPORTS
3/4 in. AND GREATER
BY 2° SQUARES 1955 TO 1967

FIGURE 2.3.1–2



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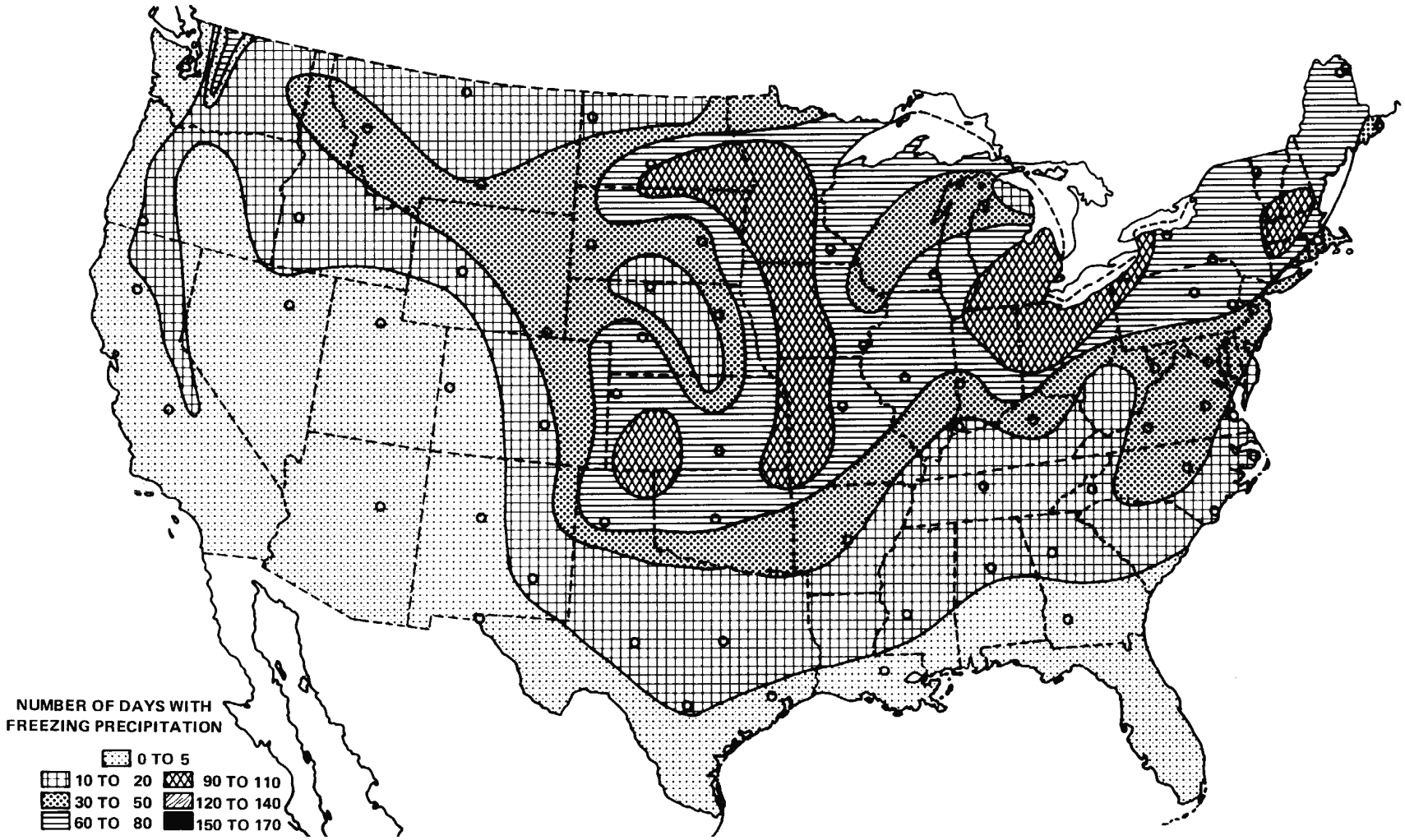
UNIT 1 AND UNIT 2

SNOW LOAD IN lb/ft^2 ON THE GROUND
100–YEAR MEAN RECURRENCE INTERVAL

FIGURE 2.3.1–3



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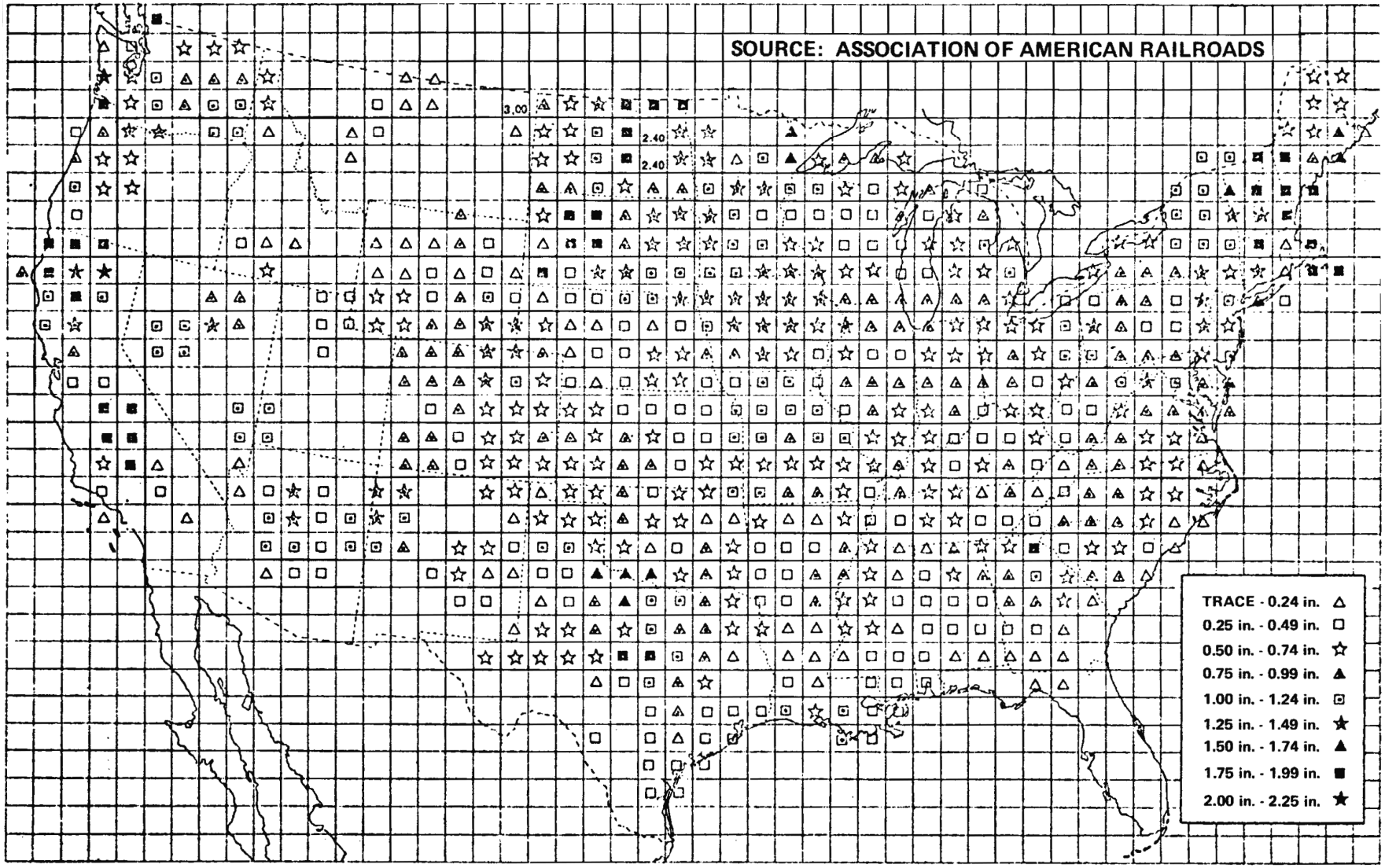
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TOTAL NUMBER DAYS WITH
FREEZING PRECIPITATION 1939 TO 1948

FIGURE 2.3.1–4



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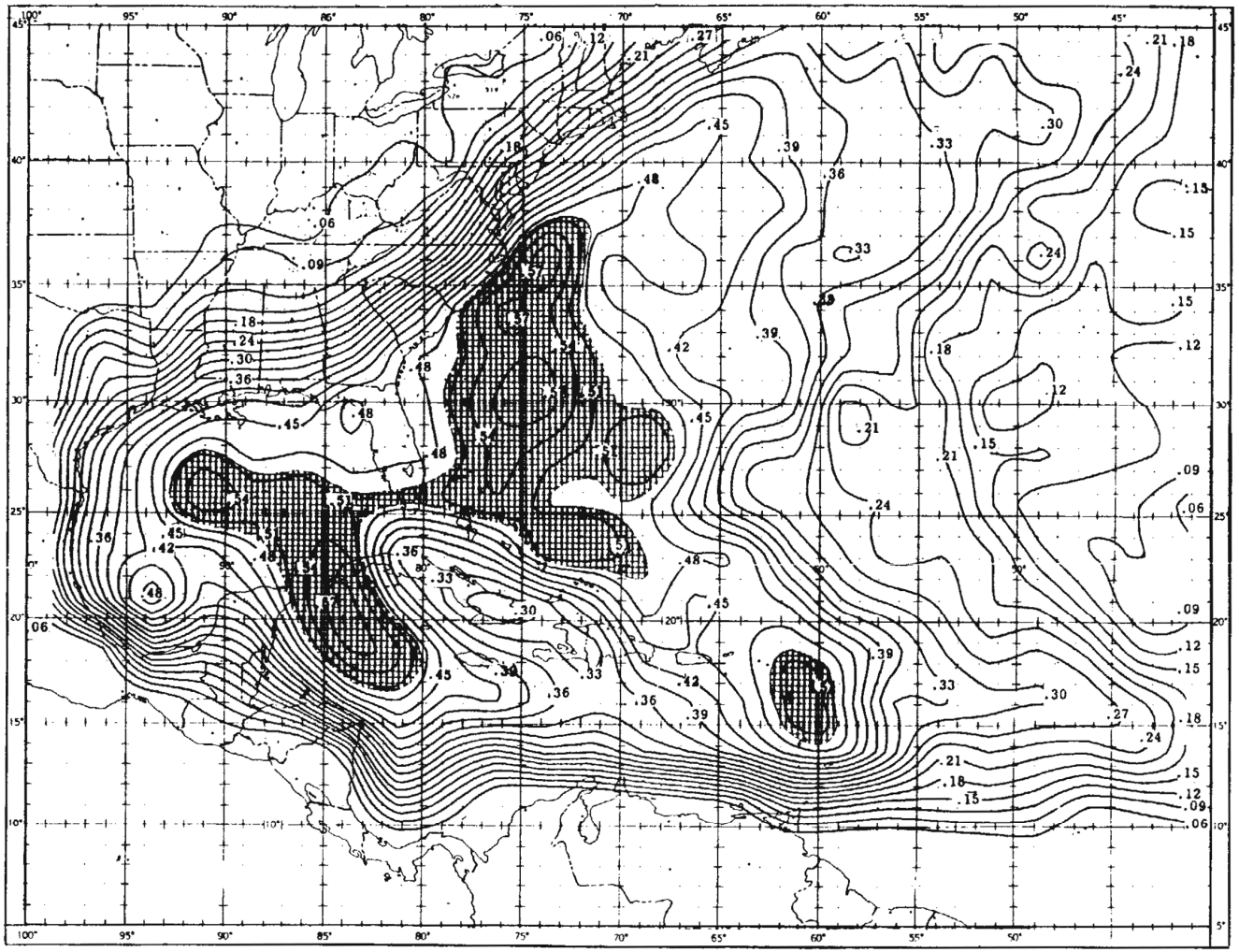
UNIT 1 AND UNIT 2

EXTREME RADIAL THICKNESS OF GLAZE
ON UTILITY WIRES 1928–29 TO 1936–37

FIGURE 2.3.1–5



PROBABILITY OF AT LEAST ONE TROPICAL STORM OR HURRICANE PER SEASON ENTERING UNIT 2½° LATITUDE-LONGITUDE BOXES. THE SHADED AREAS DELINEATE REGIONS WHERE THIS PROBABILITY EXCEEDS 0.50.

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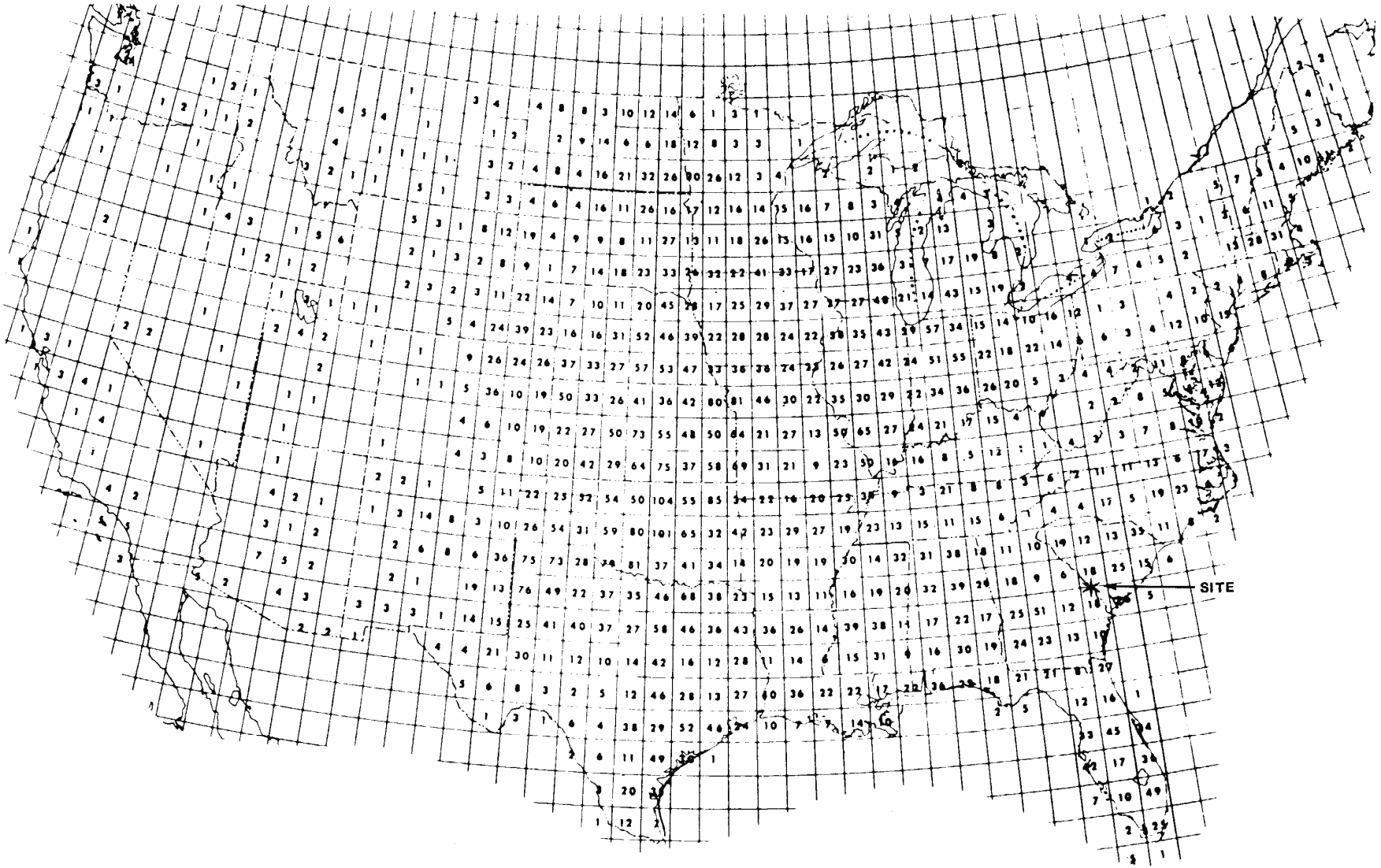
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TROPICAL STORM AND
HURRICANE PROBABILITY

FIGURE 2.3.1–6



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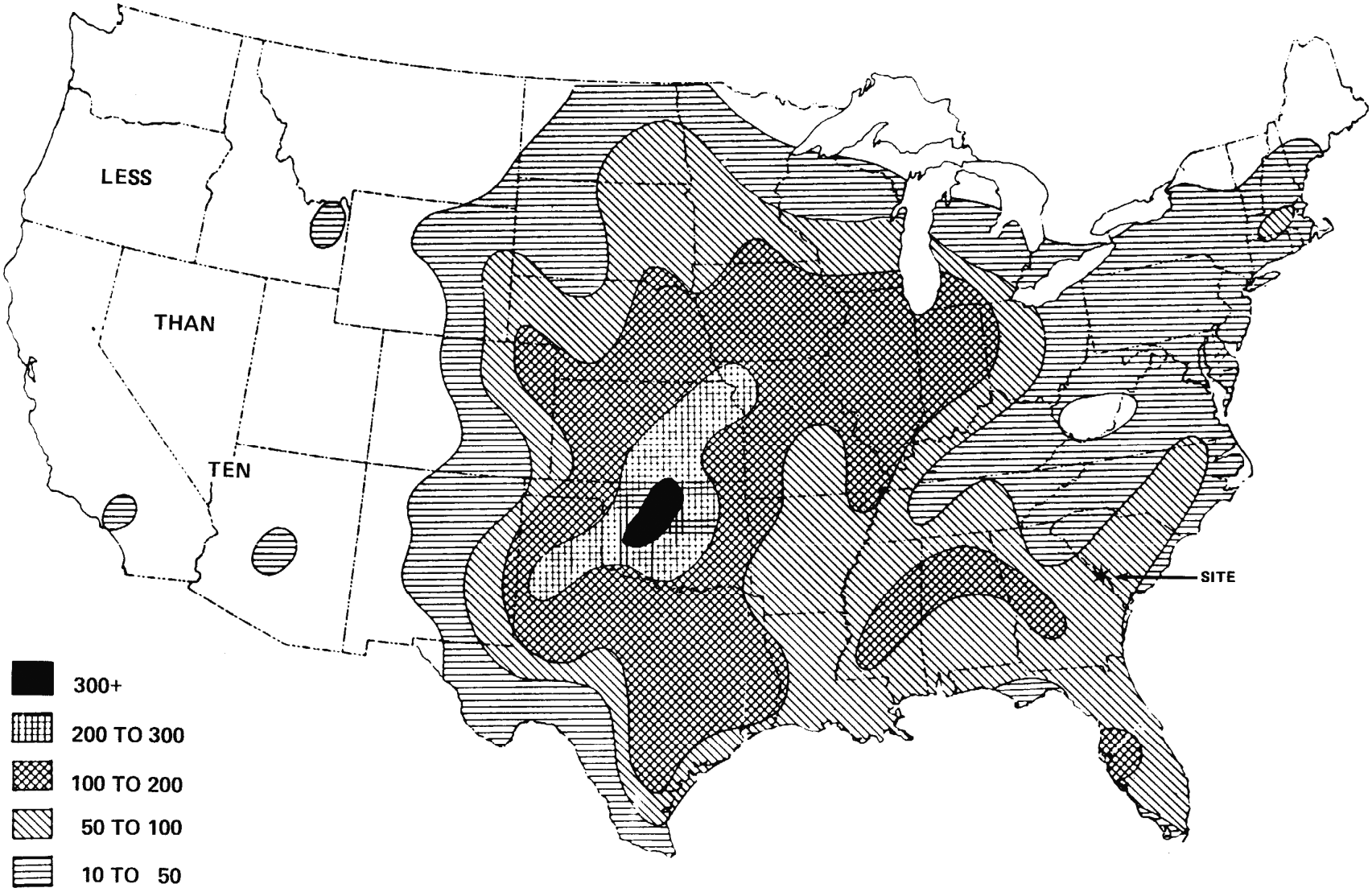
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TOTAL TORNADOES BY 1°SQUARES
1955 TO 1967

FIGURE 2.3.1–7



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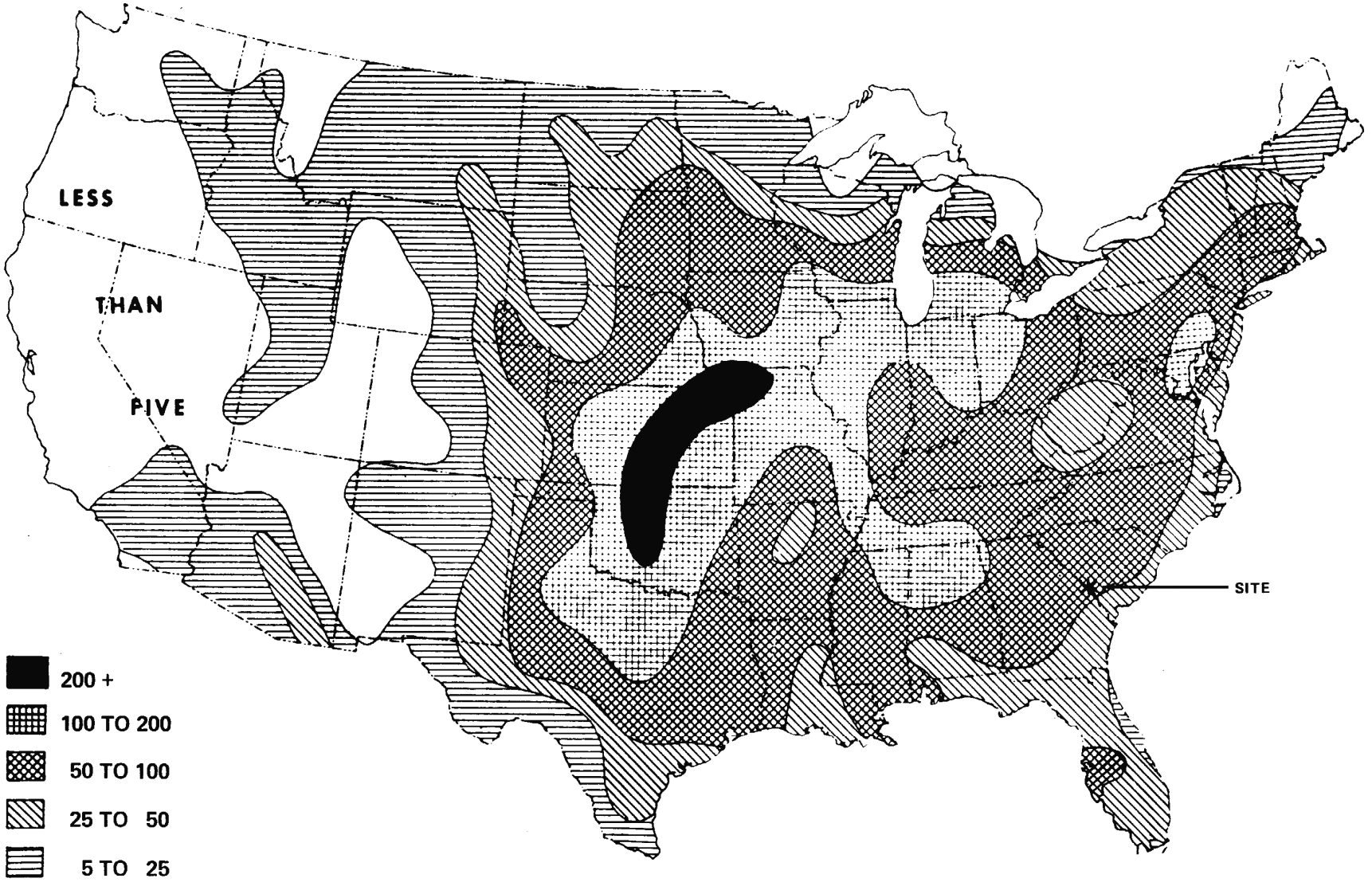
UNIT 1 AND UNIT 2

TOTAL TORNADOES BY 2° SQUARES
1955 TO 1967

FIGURE 2.3.1–8



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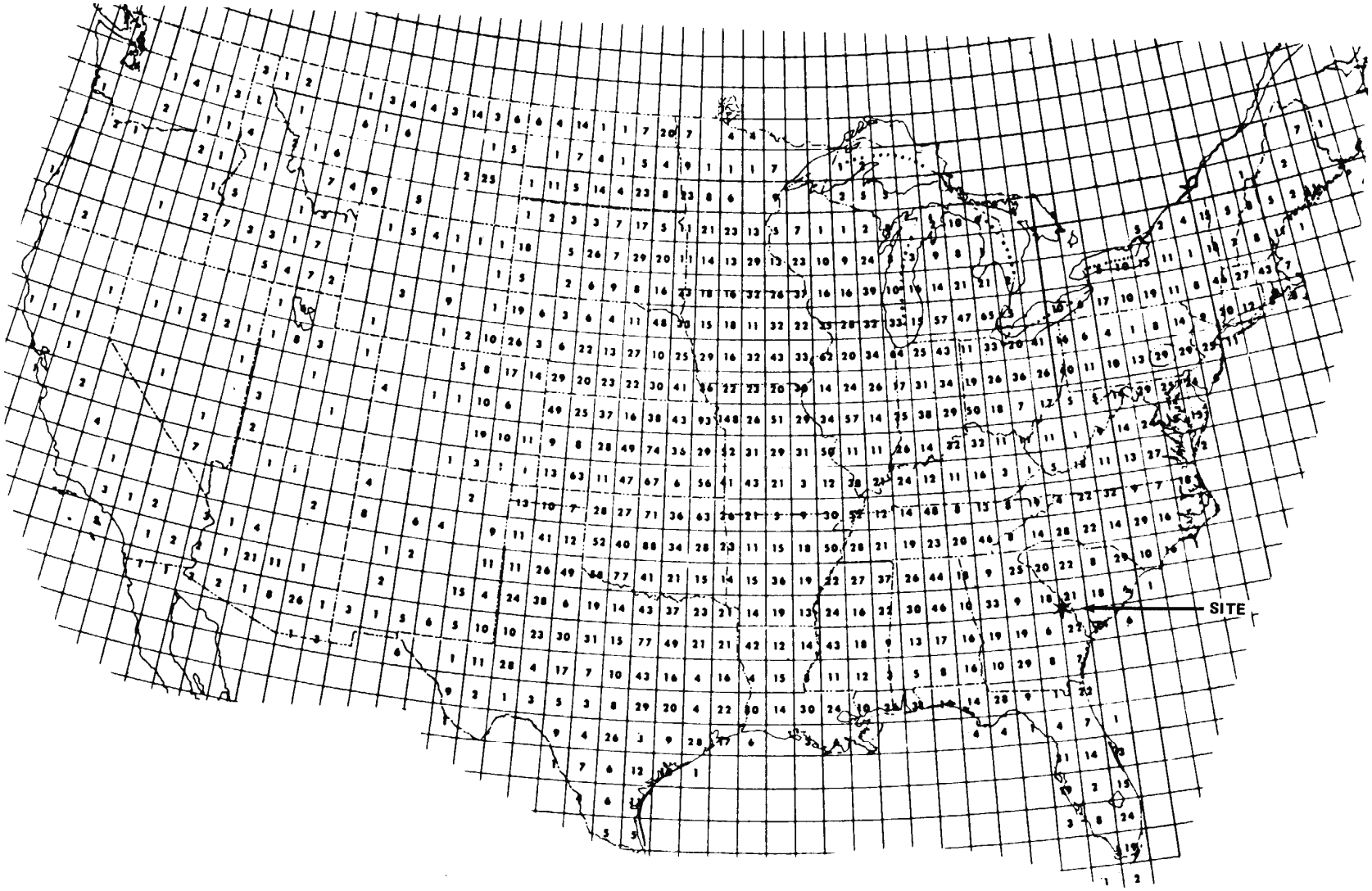
UNIT 1 AND UNIT 2

TOTAL WINDSTORMS
50 KNOTS AND GREATER
BY 2° SQUARES 1955 TO 1967

FIGURE 2.3.1–9



REV 14 10/07

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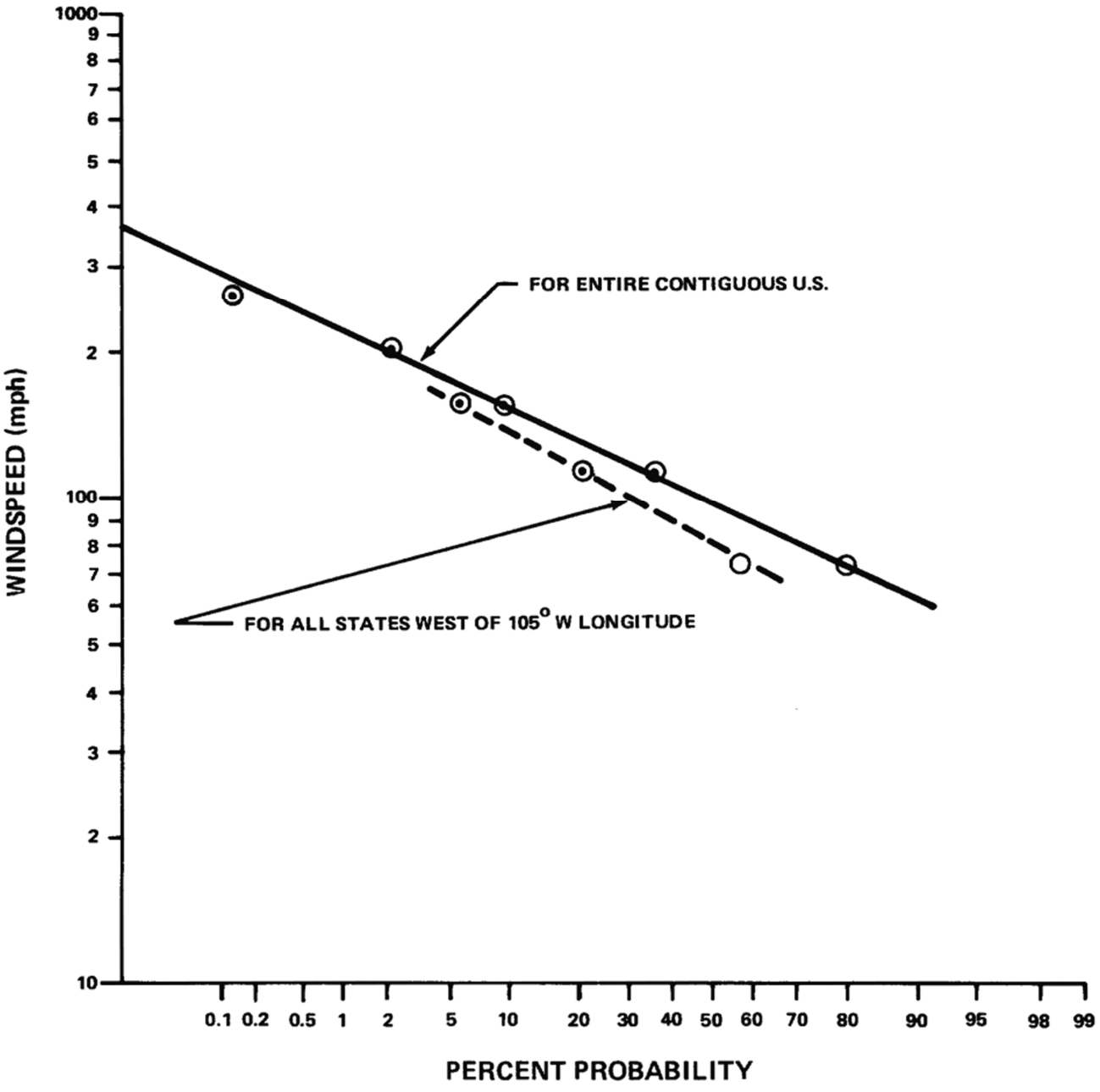
UNIT 1 AND UNIT 2

TOTAL WINDSTORMS
50 KNOTS AND GREATER
BY 1° SQUARES 1955 TO 1967

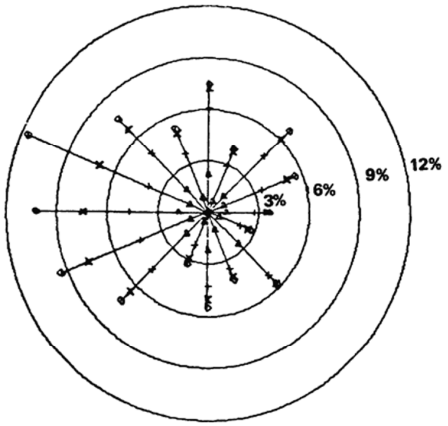
FIGURE 2.3.1–10



REV 14 10/07

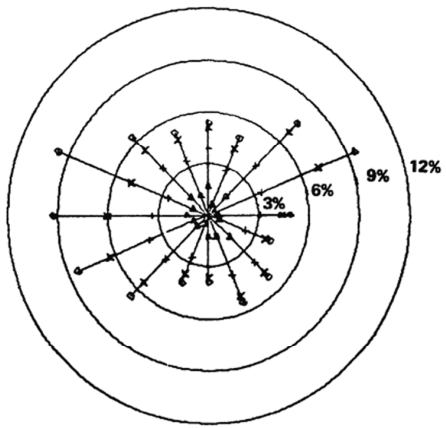
SOUTHERN COMPANY *Energy to Serve Your World®*	VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	TORNADO WIND GROUPS FIGURE 2.3.1–11

JANUARY N



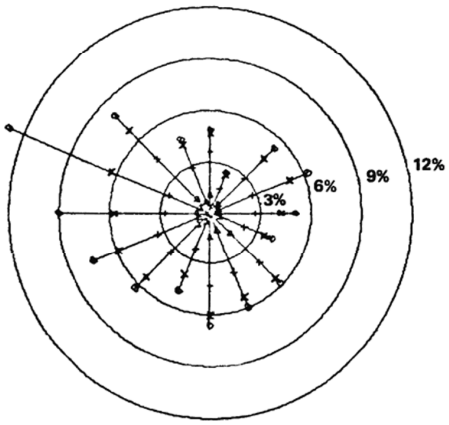
9.65 PERCENT CALMS

FEBRUARY N



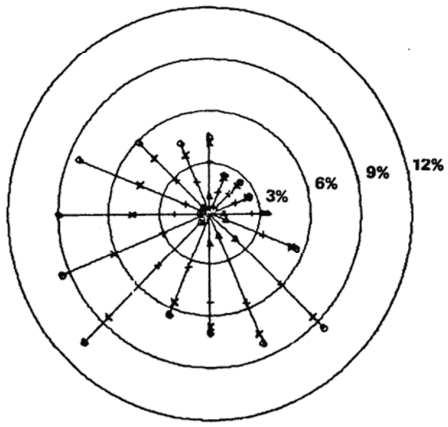
6.95 PERCENT CALMS

MARCH N



5.35 PERCENT CALMS

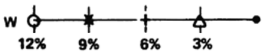
APRIL N



6.01 PERCENT CALMS

NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH [INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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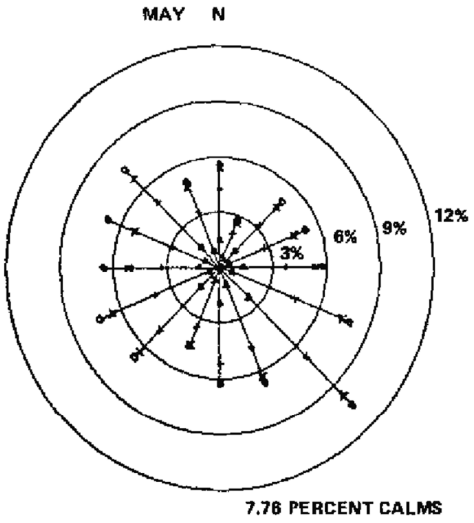
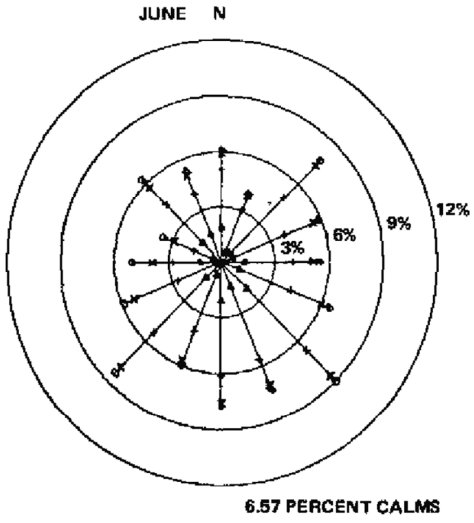
MONTHLY WIND ROSES FROM
AUGUSTA AIRPORT (1959 TO 1963)

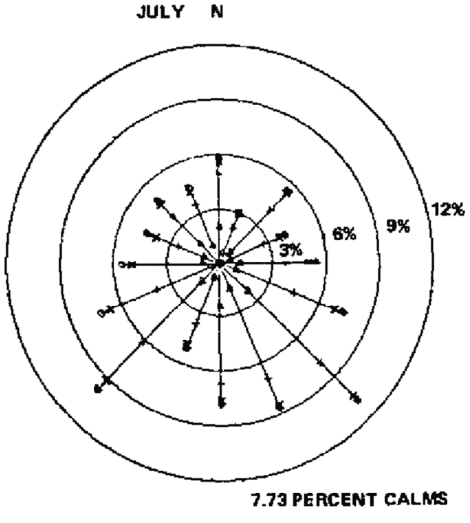
FIGURE 2.3.2–1 (SHEET 1 OF 3)



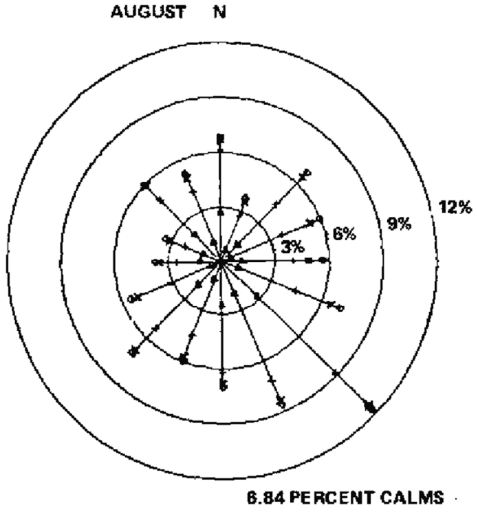
7.76 PERCENT CALMS



6.57 PERCENT CALMS



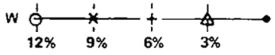
7.73 PERCENT CALMS



6.84 PERCENT CALMS

NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH [INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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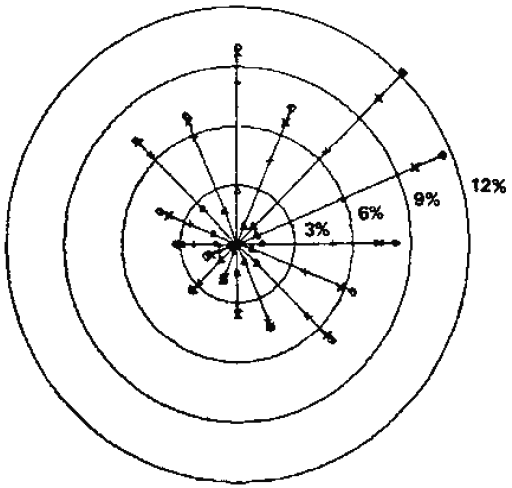
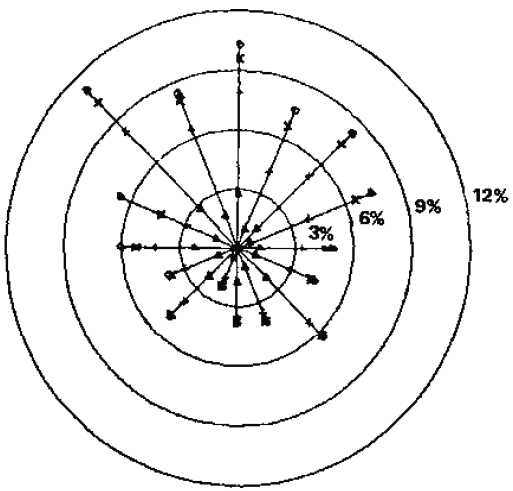
FIGURE 2.3.2–1 (SHEET 2 OF 3)

SEPTEMBER N



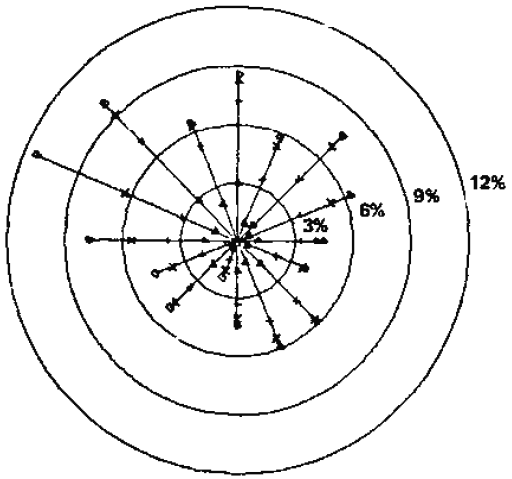
9.79 PERCENT CALMS

OCTOBER N



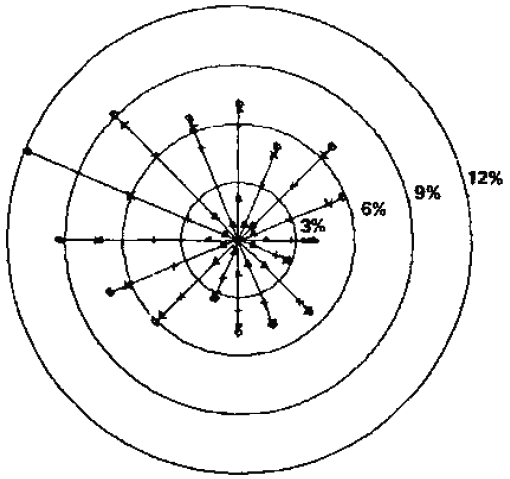
10.65 PERCENT CALMS

NOVEMBER N



10.23 PERCENT CALMS

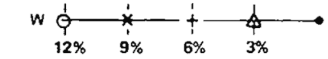
DECEMBER N



8.46 PERCENT CALMS

NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH [INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph

△ WIND SPEED LESS THAN 3.5 mph

\+ WIND SPEED LESS THAN 7.5 mph

X WIND SPEED LESS THAN 12.5 mph

O WIND SPEED LESS THAN 999.0 mph

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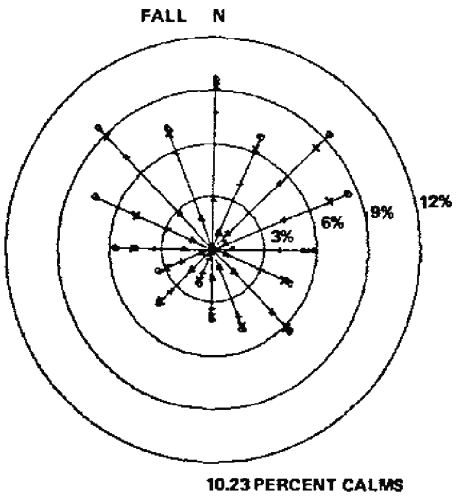
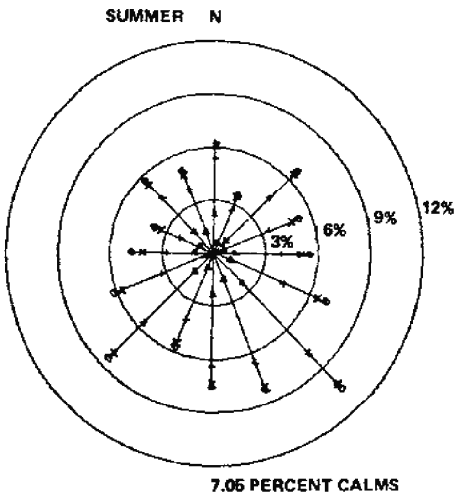
MONTHLY WIND ROSES FROM
AUGUSTA AIRPORT (1959 TO 1963)

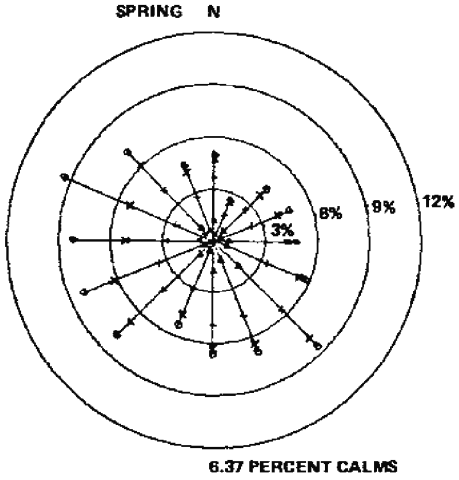
FIGURE 2.3.2–1 (SHEET 3 OF 3)



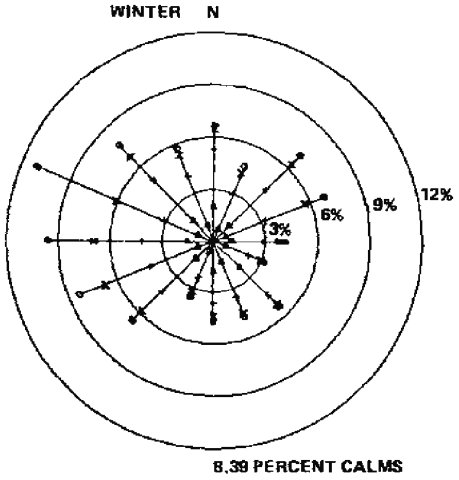
10.23 PERCENT CALMS



7.06 PERCENT CALMS



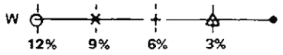
6.37 PERCENT CALMS



8.39 PERCENT CALMS

NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH [INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph

△ WIND SPEED LESS THAN 3.5 mph

\+ WIND SPEED LESS THAN 7.5 mph

X WIND SPEED LESS THAN 12.5 mph

O WIND SPEED LESS THAN 999.0 mph

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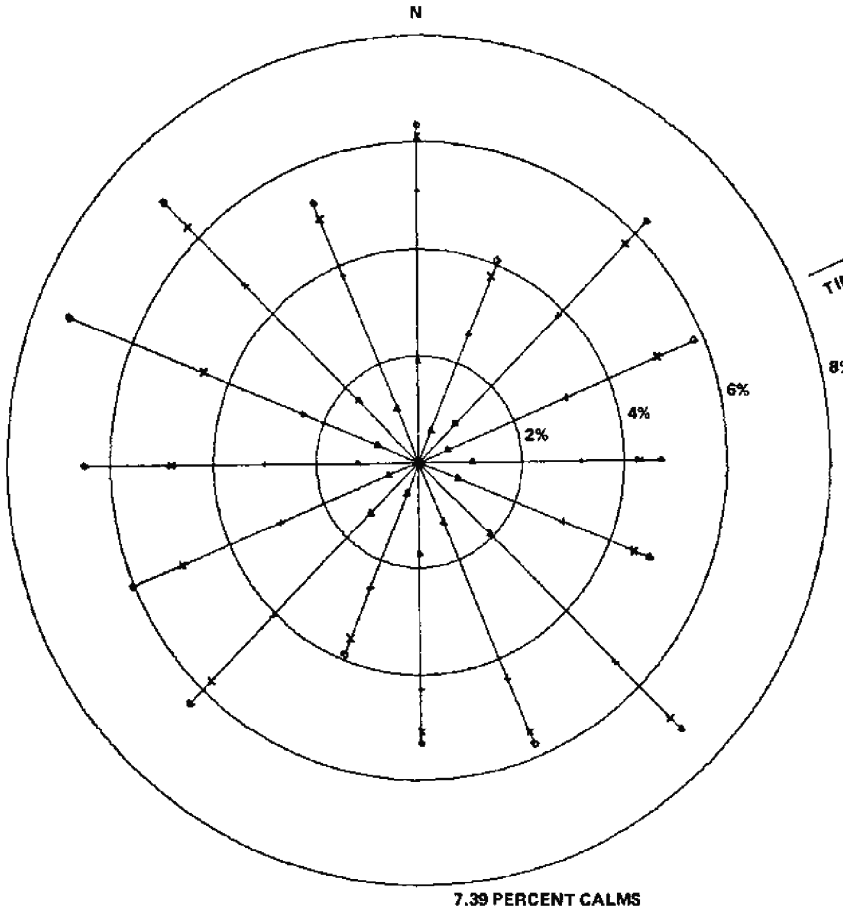
SEASONAL WIND ROSES FOR
AUGUSTA AIRPORT (1959 TO 1963)

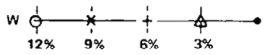
FIGURE 2.3.2–2



NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.

FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH [INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph

△ WIND SPEED LESS THAN 3.5 mph

\+ WIND SPEED LESS THAN 7.5 mph

X WIND SPEED LESS THAN 12.5 mph

O WIND SPEED LESS THAN 999.0 mph

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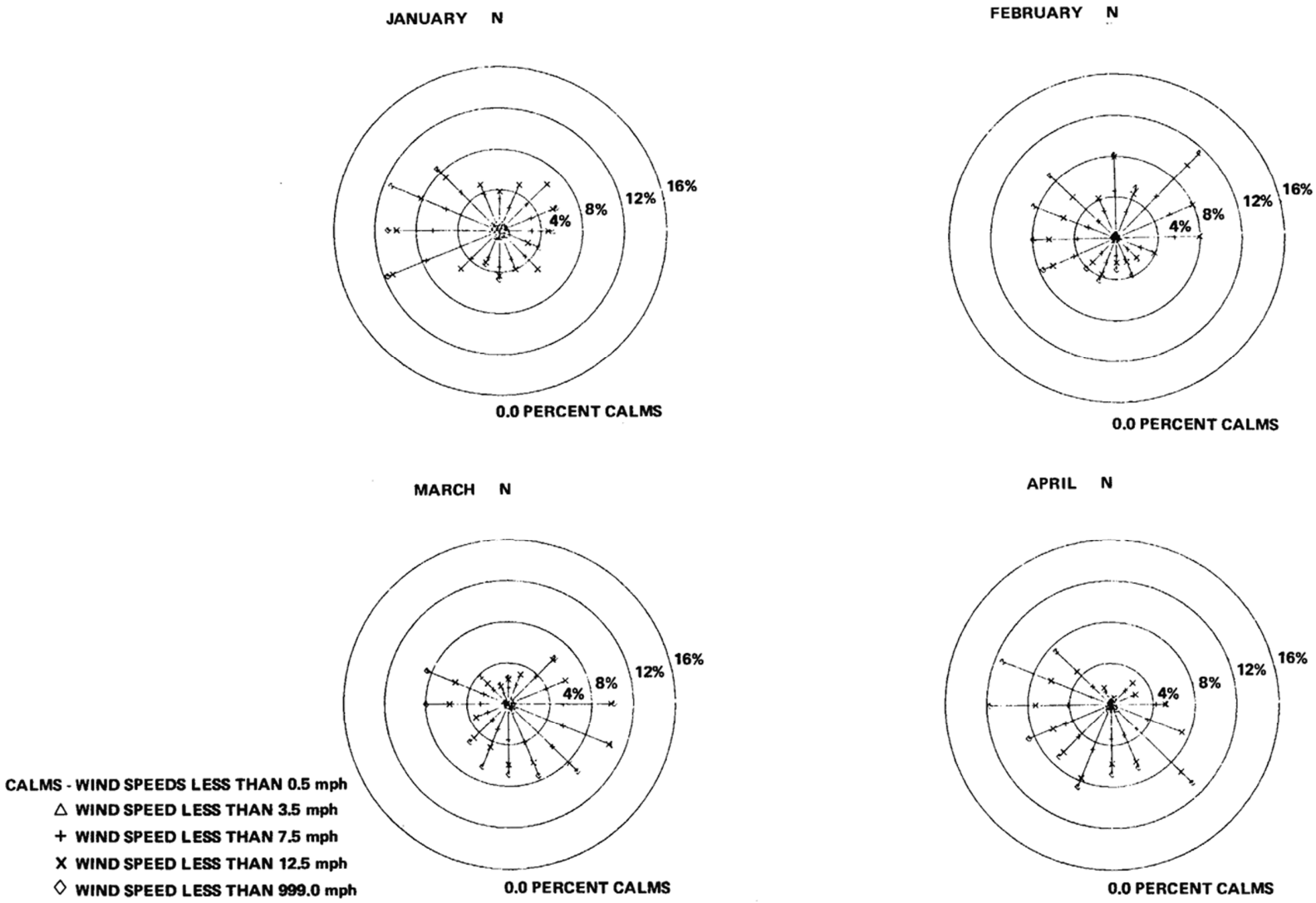
ANNUAL WIND ROSE FOR AUGUSTA AIRPORT 1959 TO 1963

FIGURE 2.3.2–3



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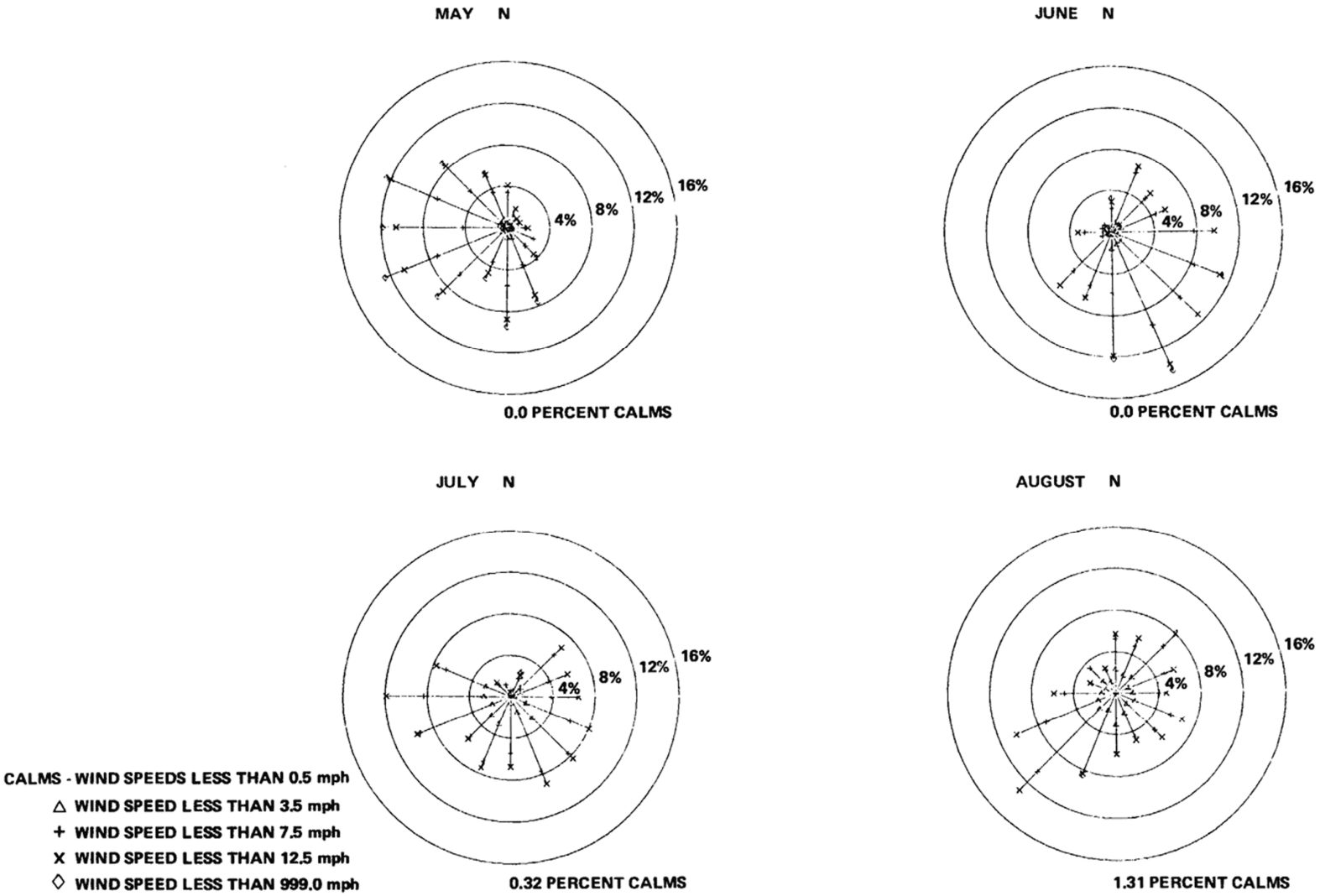
33–FT LEVEL MONTHLY WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–4 (SHEET 1 OF 3)



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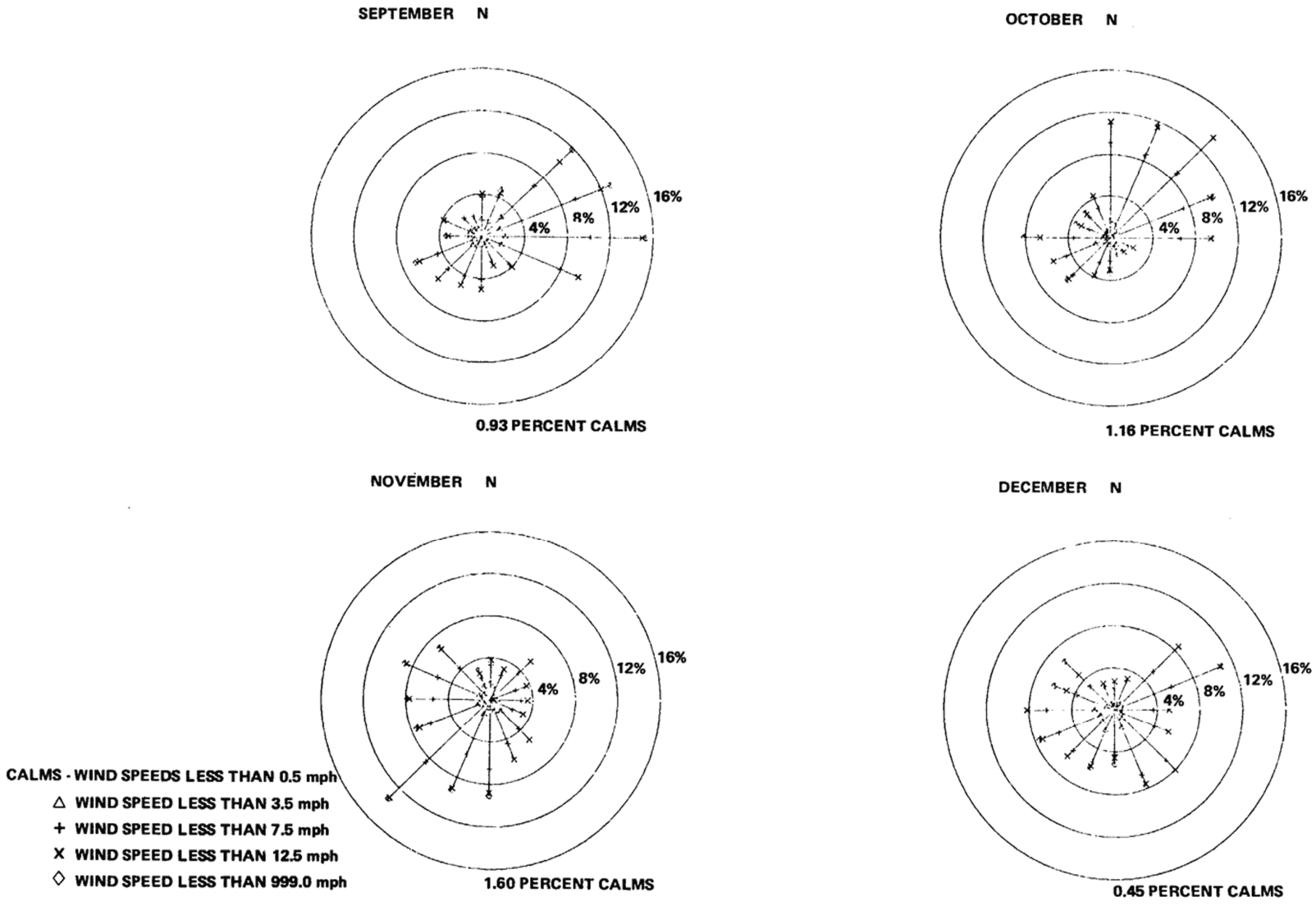
33–FT LEVEL MONTHLY WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–4 (SHEET 2 OF 3)



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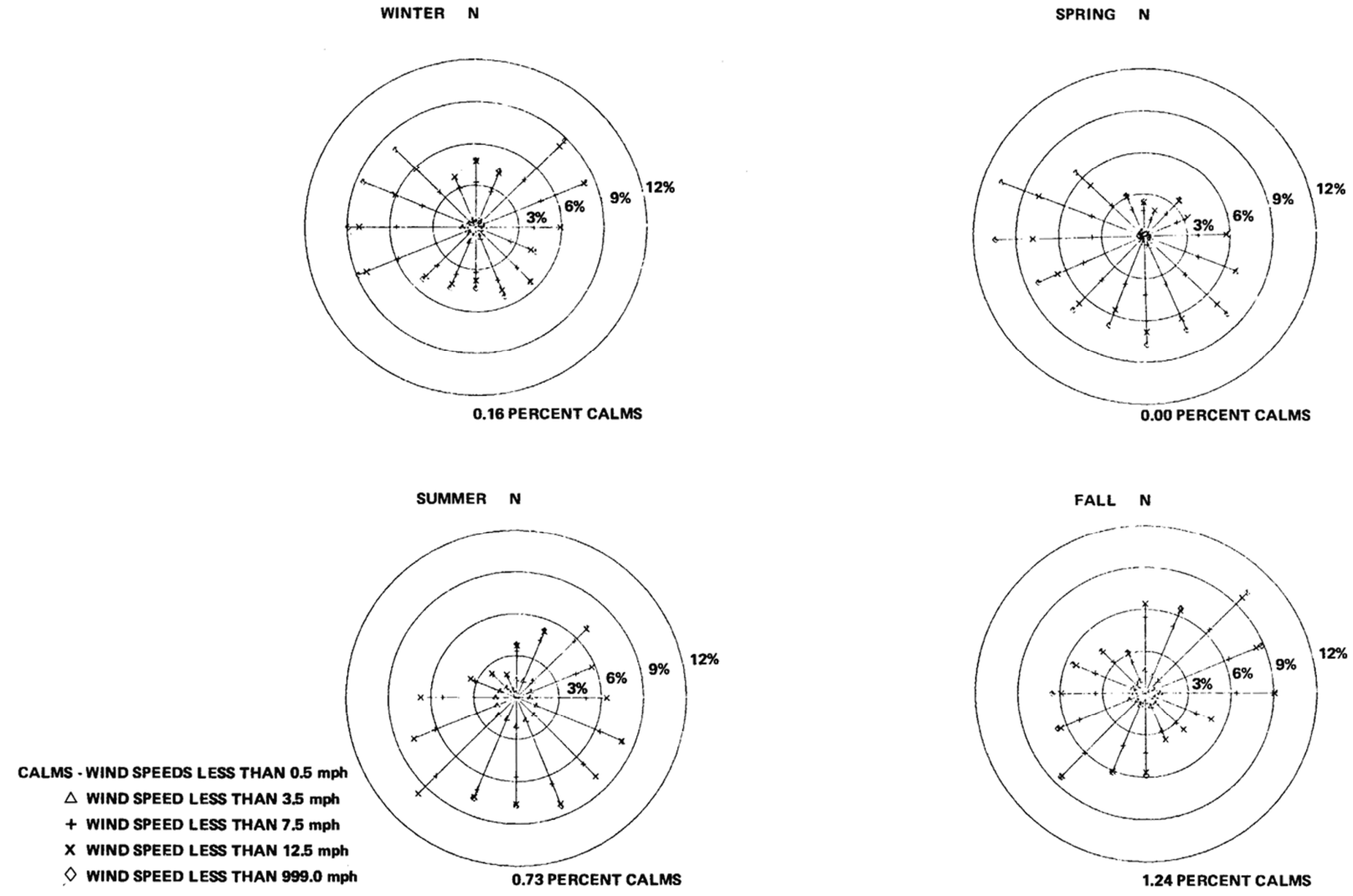
33–FT LEVEL MONTHLY WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–4 (SHEET 3 OF 3)



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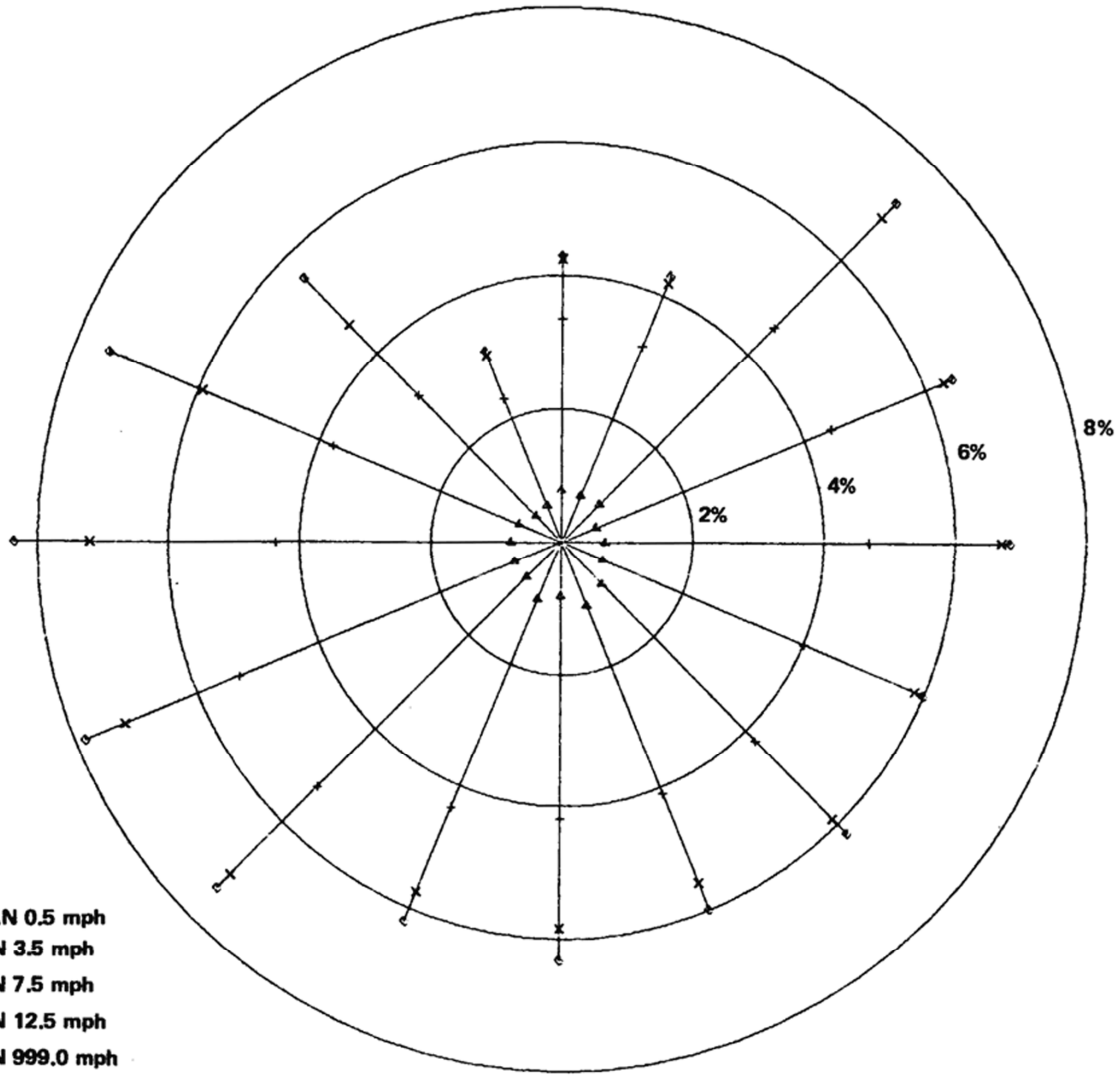
33–FT LEVEL SEASONAL WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–5



CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
◇ WIND SPEED LESS THAN 999.0 mph

0.49 PERCENT CALMS

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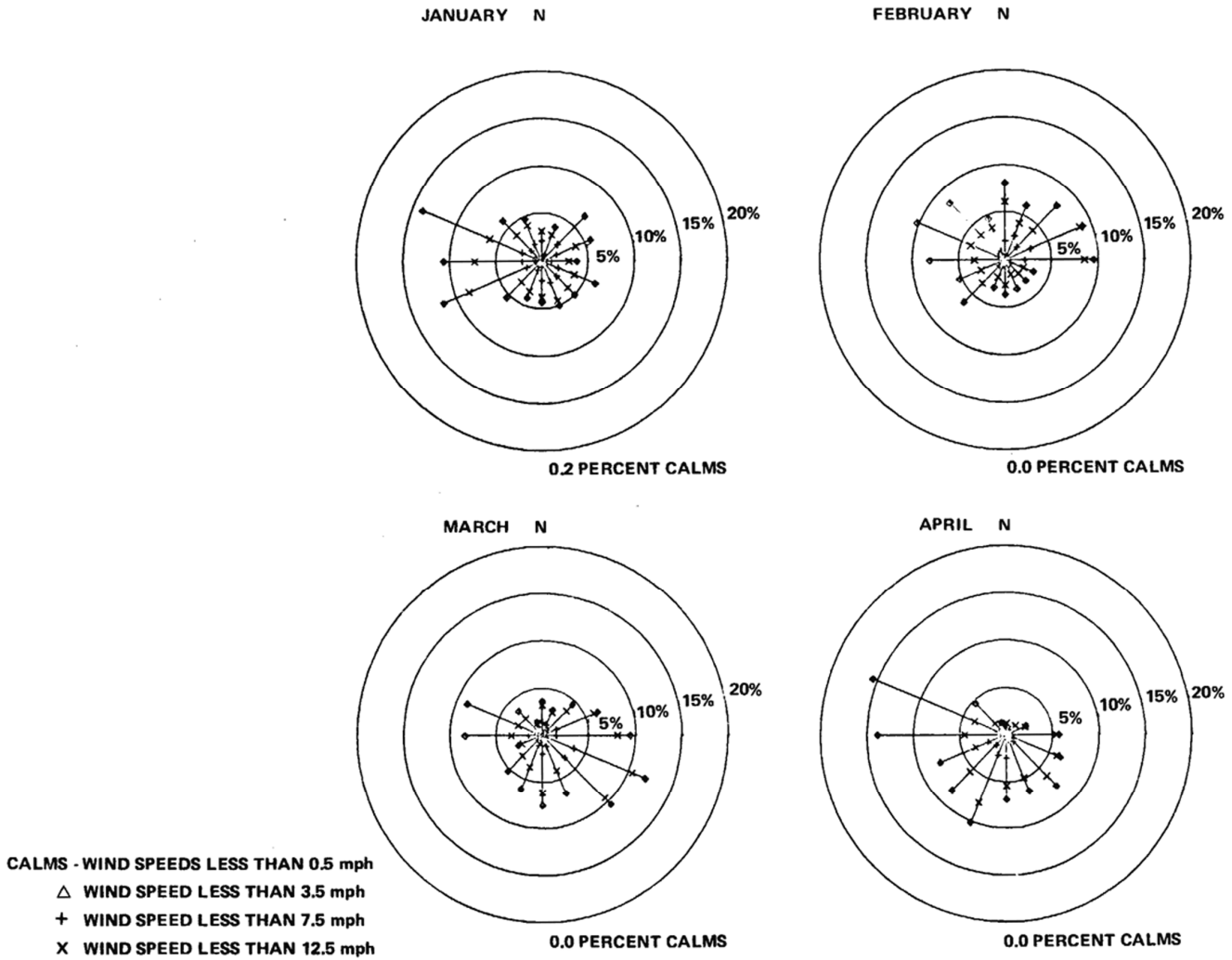
UNIT 1 AND UNIT 2

33–FT LEVEL ANNUAL WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–6



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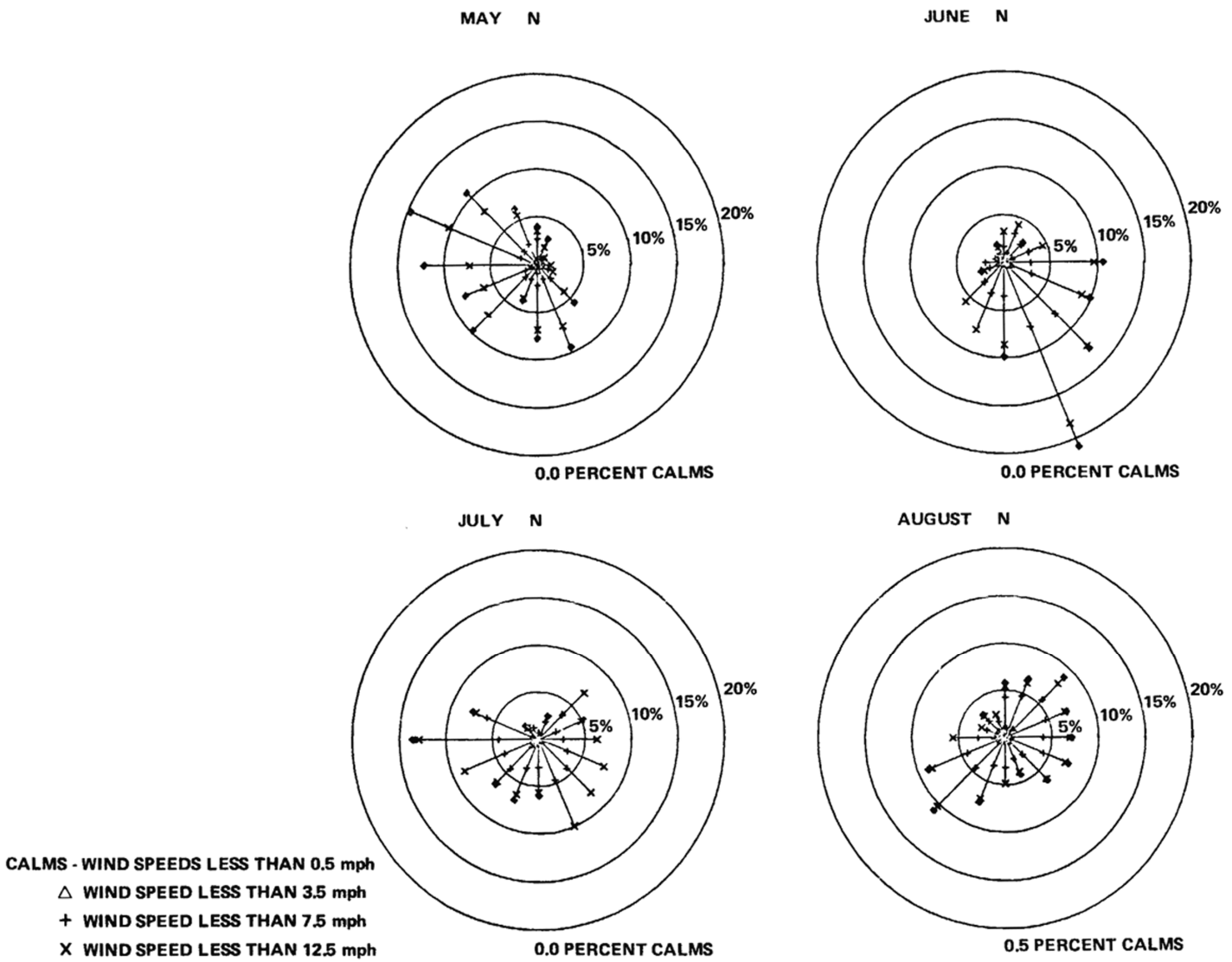
150–FT LEVEL MONTHLY WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–7 (SHEET 1 OF 3)



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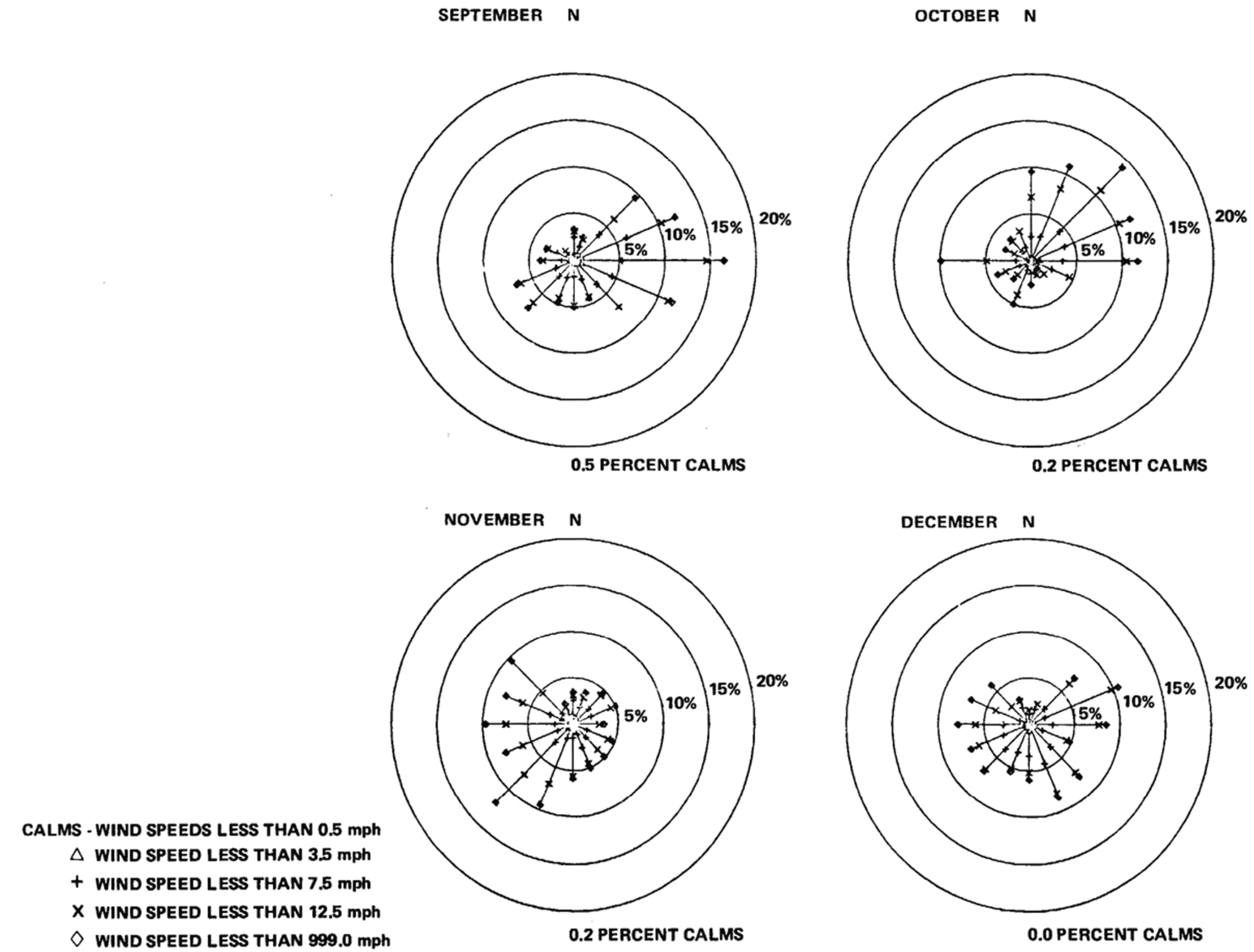
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLY WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–7 (SHEET 2 OF 3)



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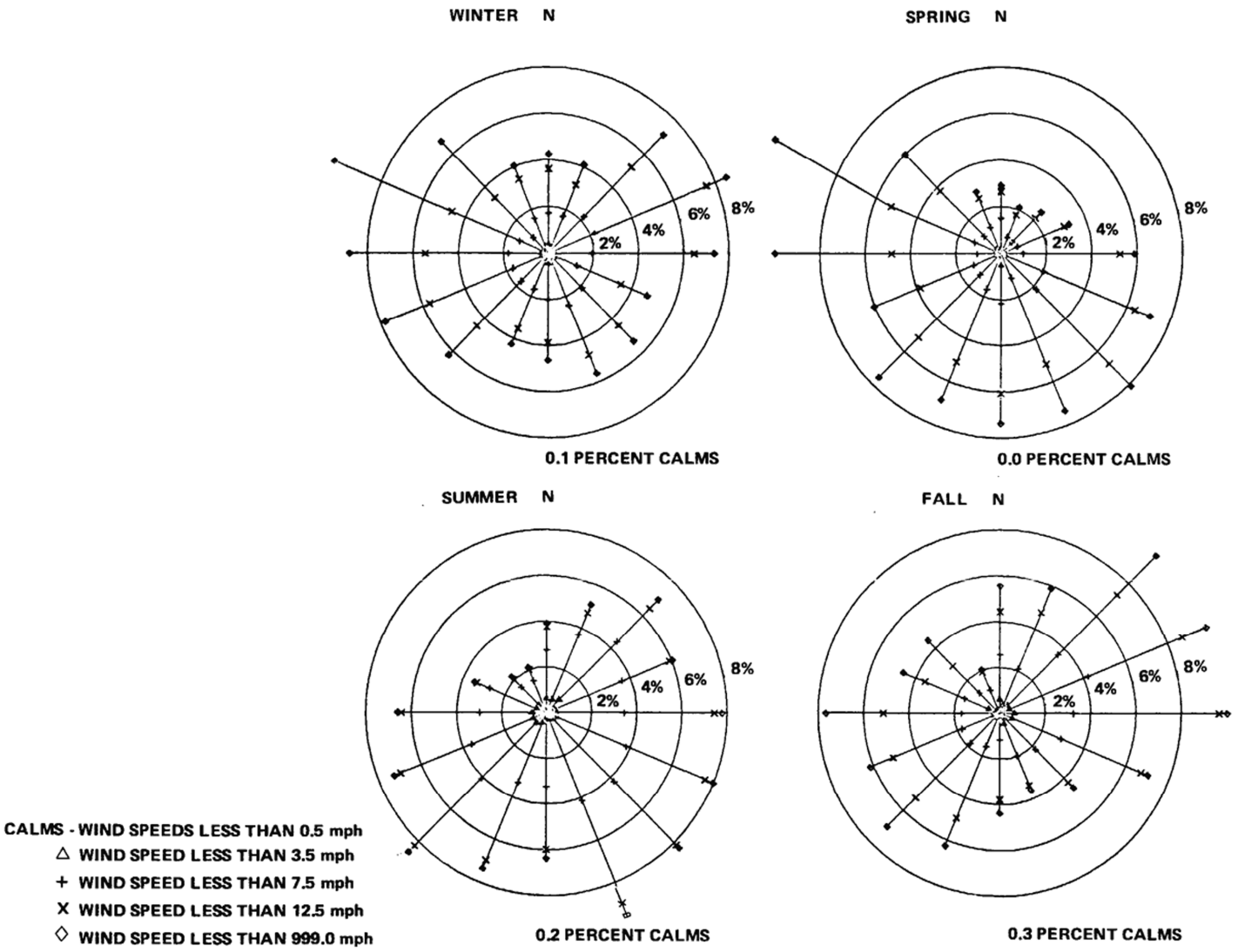
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLY WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–7 (SHEET 3 OF 3)



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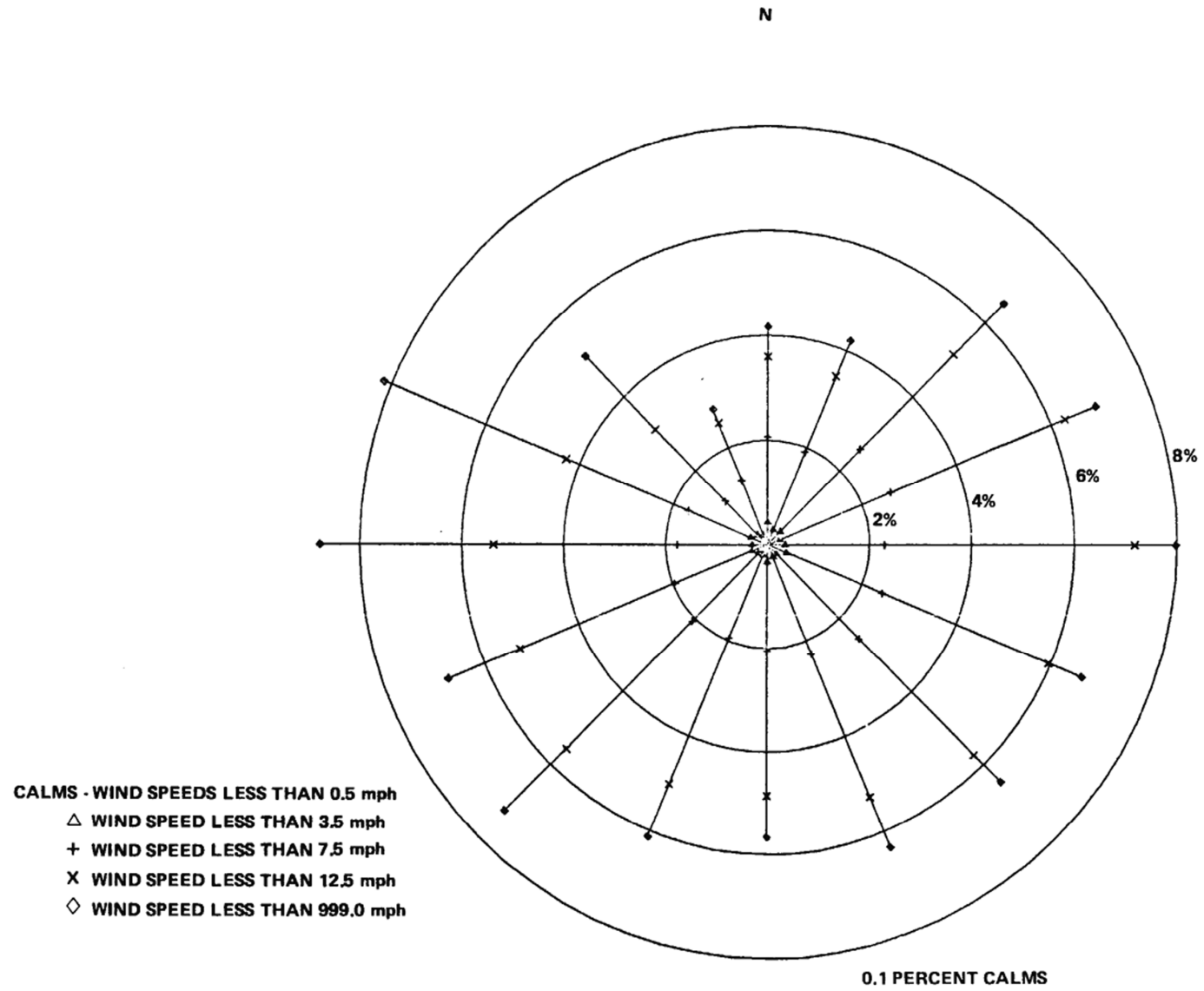
UNIT 1 AND UNIT 2

150–FT LEVEL SEASONAL WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–8



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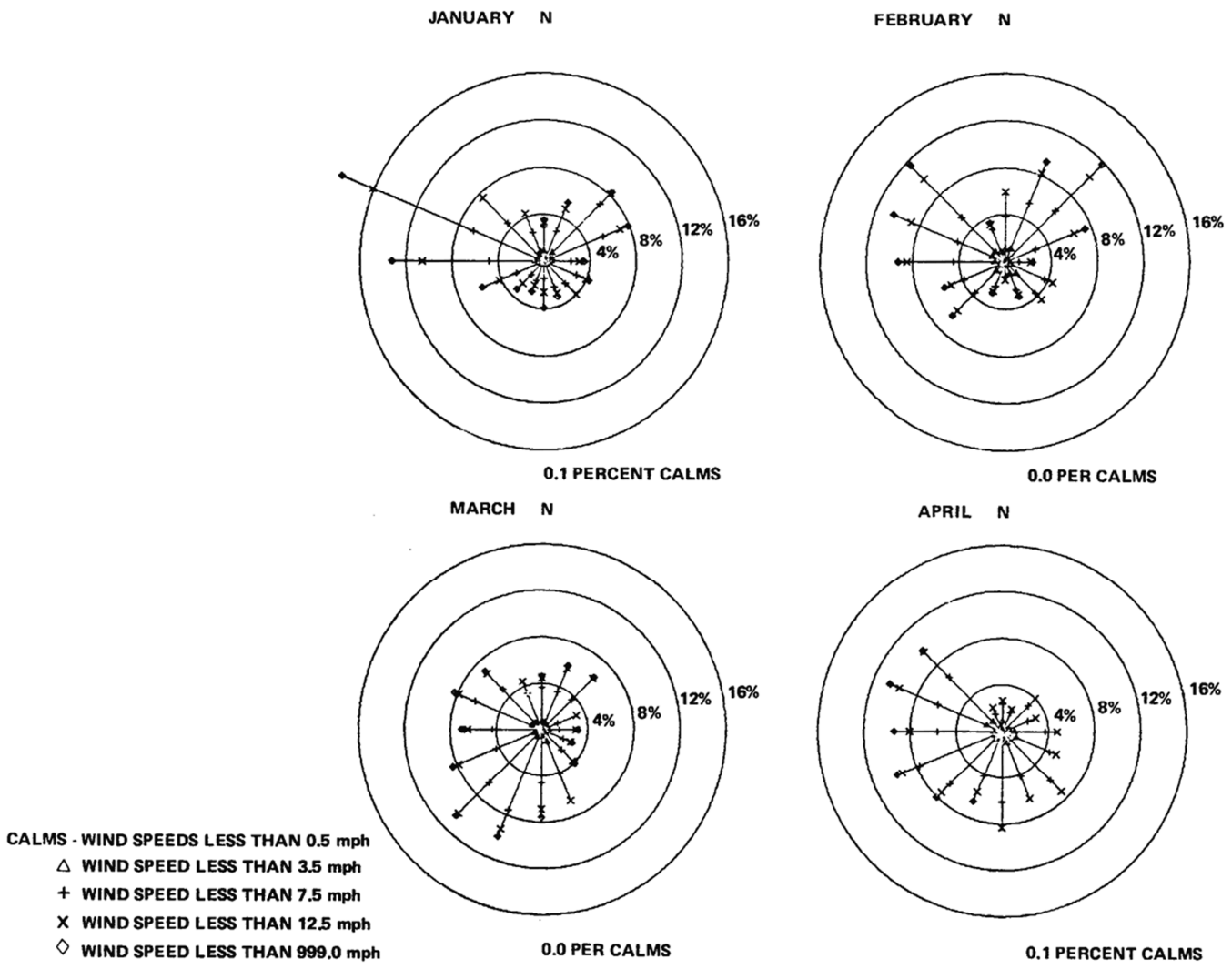
UNIT 1 AND UNIT 2

150–FT LEVEL ANNUAL WIND ROSE
1972 TO 1973 VEGP SITE DATA

FIGURE 2.3.2–9



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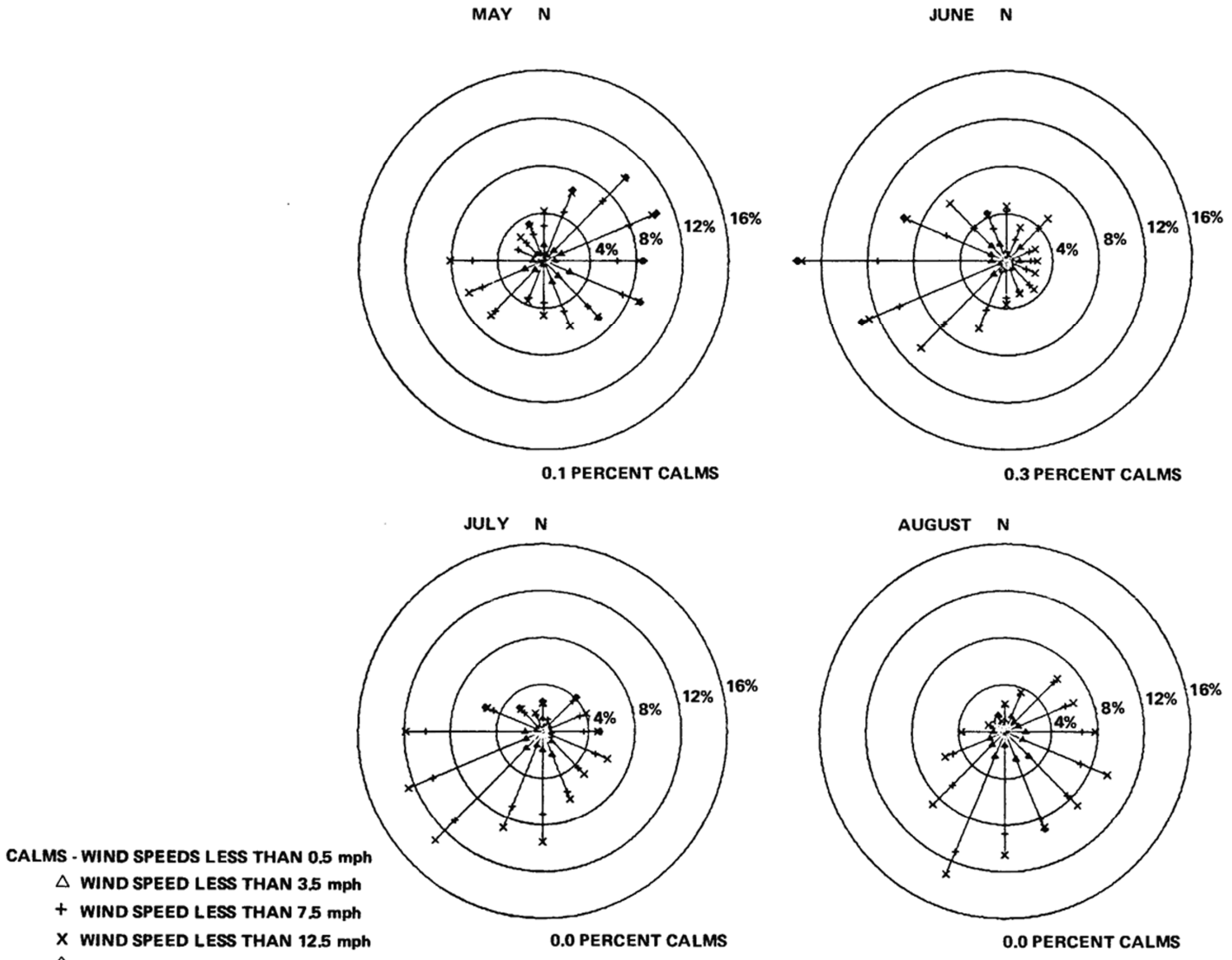
UNIT 1 AND UNIT 2

33–FT LEVEL MONTHLY WIND ROSE
1977 TO 1978 VEGP SITE DATA

FIGURE 2.3.2–10 (SHEET 1 OF 3)



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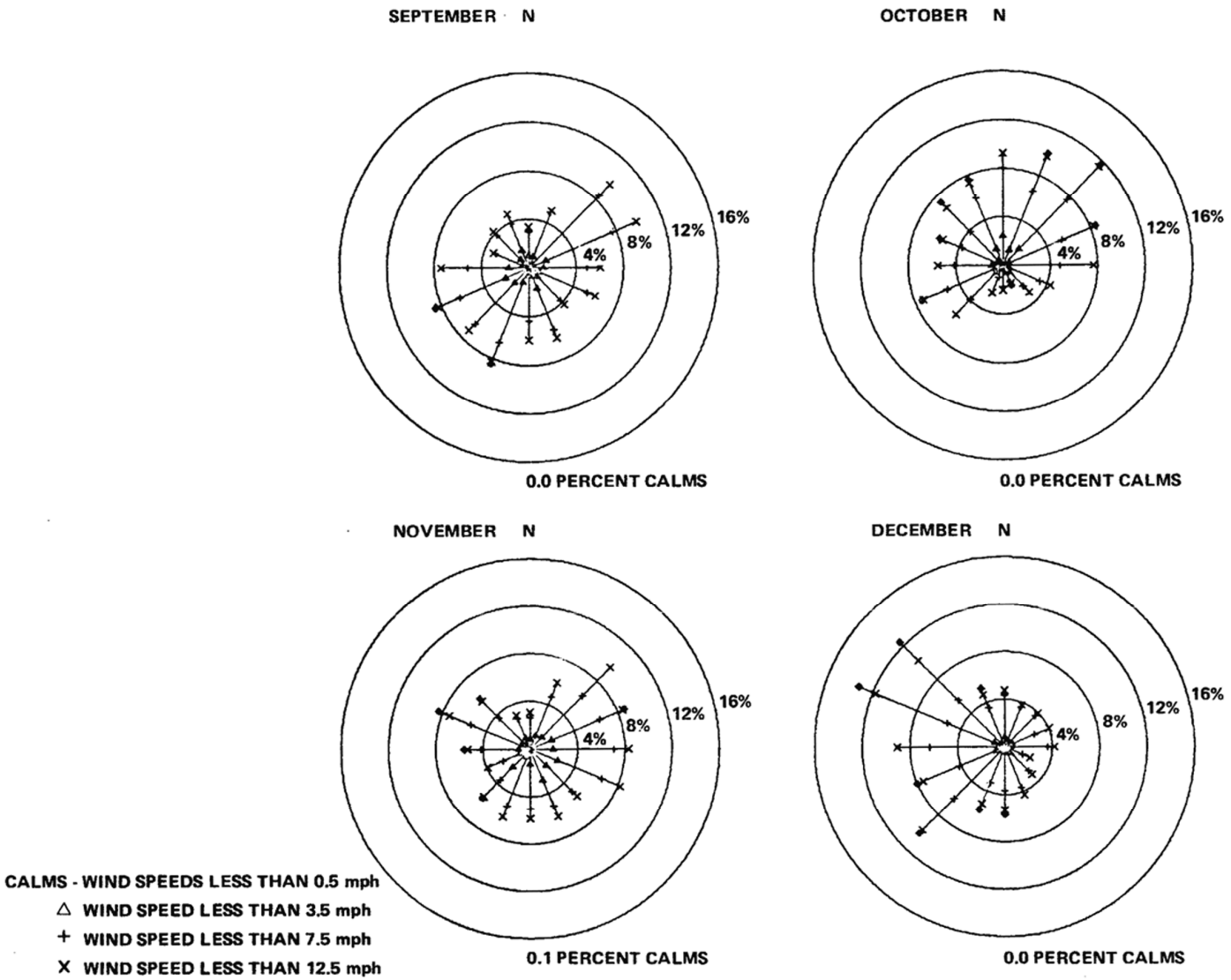
UNIT 1 AND UNIT 2

33–FT LEVEL MONTHLY WIND ROSE
1977 TO 1978 VEGP SITE DATA

FIGURE 2.3.2–10 (SHEET 2 OF 3)



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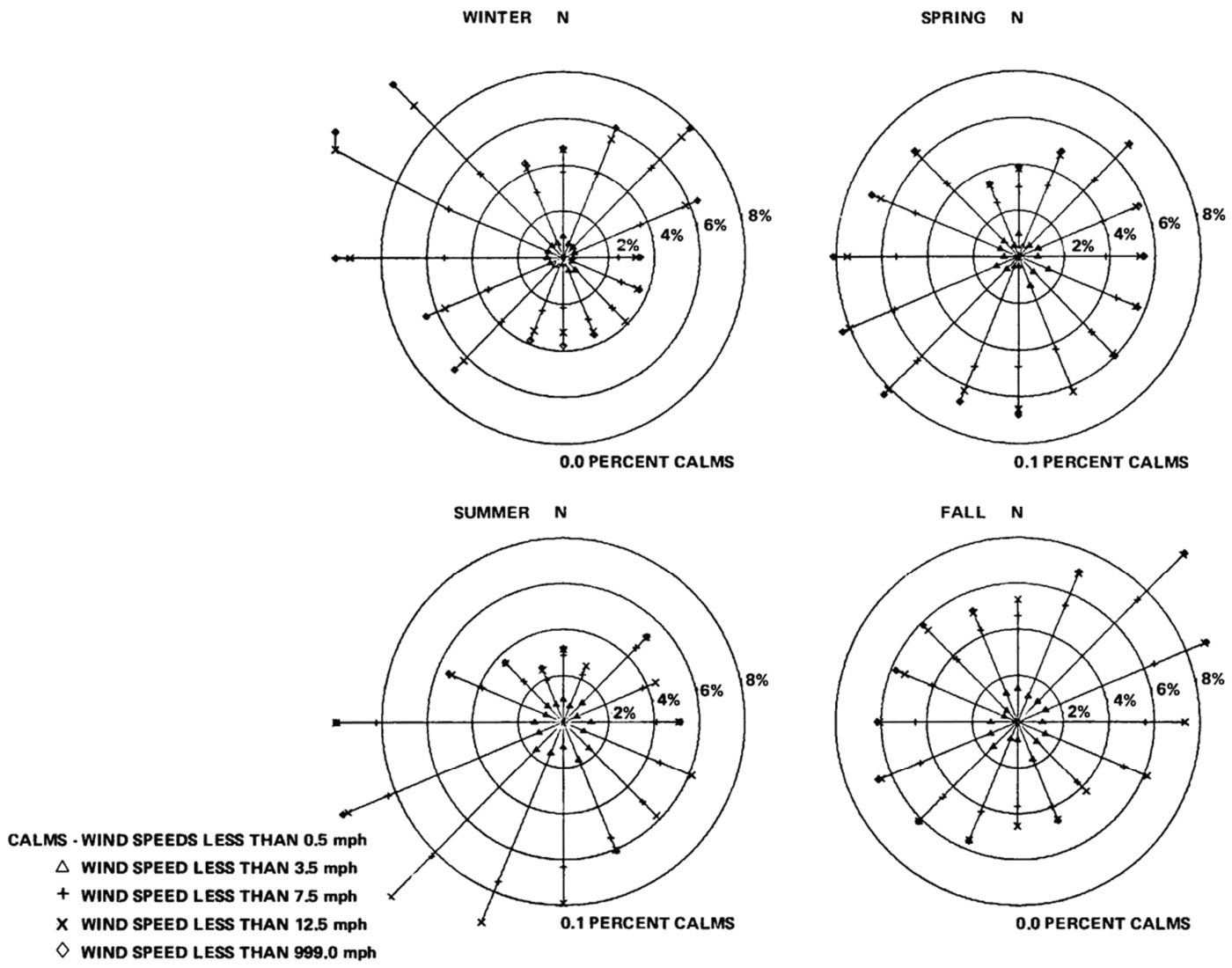
UNIT 1 AND UNIT 2

33–FT LEVEL MONTHLY WIND ROSE
1977 TO 1978 VEGP SITE DATA

FIGURE 2.3.2–10 (SHEET 3 OF 3)



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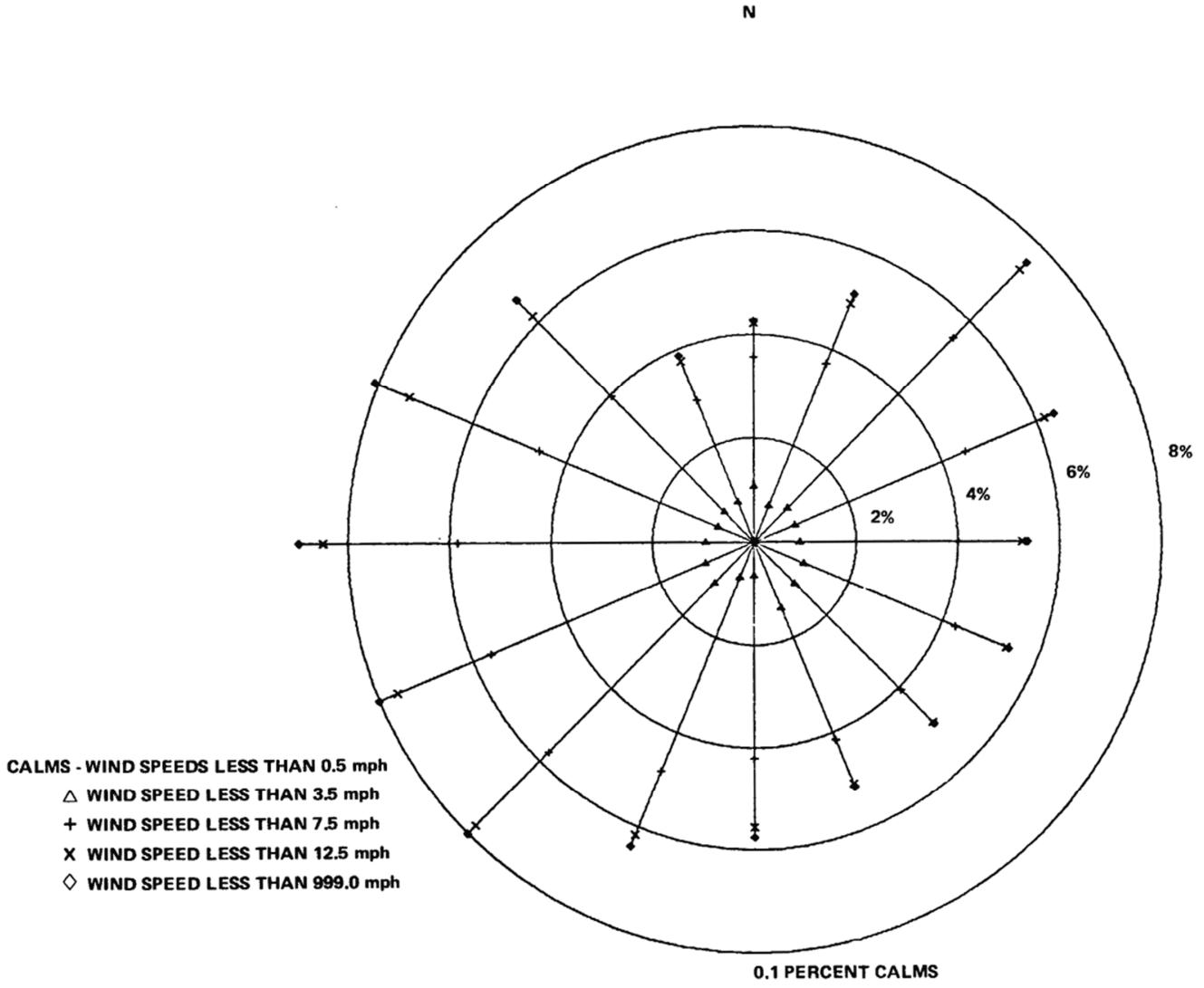
UNIT 1 AND UNIT 2

33–FT LEVEL SEASONAL WIND ROSE
1977 TO 1978 VEGP SITE DATA

FIGURE 2.3.2–11



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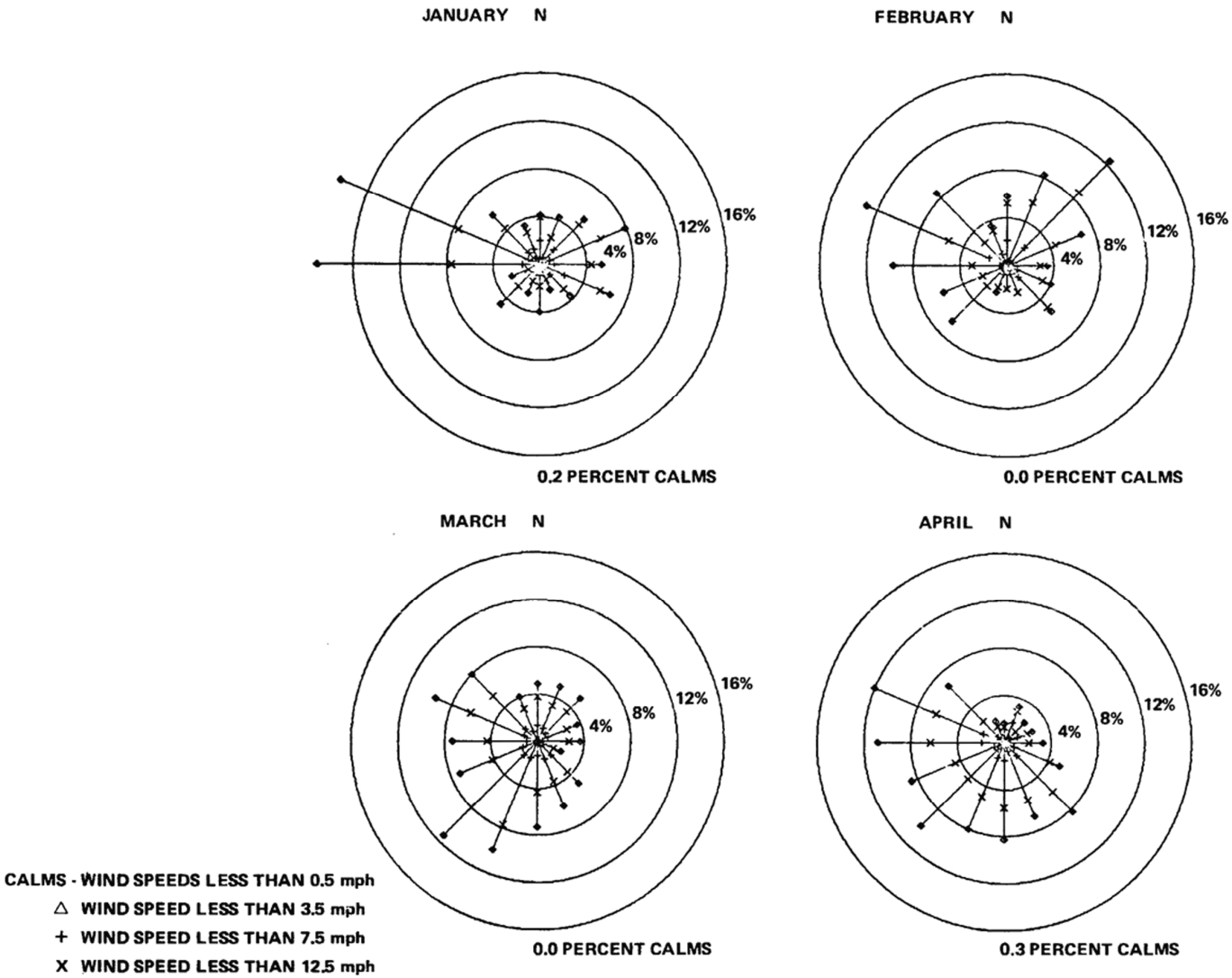
UNIT 1 AND UNIT 2

33–FT LEVEL ANNUAL WIND ROSE
1977 TO 1978 VEGP SITE DATA

FIGURE 2.3.2–12



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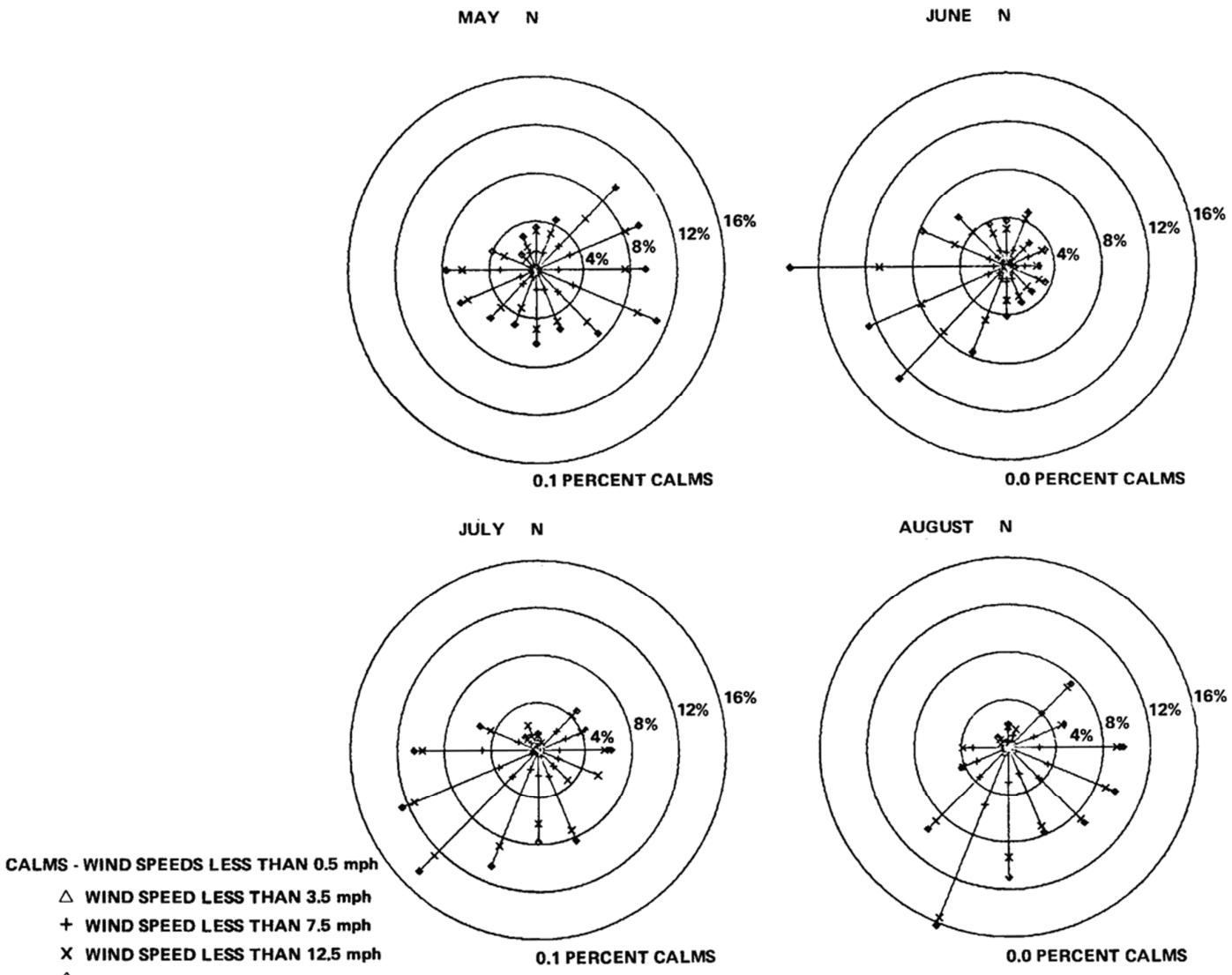
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLYWIND ROSE
1977 TO 1978 VEGP SITE DATA

FIGURE 2.3.2–13 (SHEET 1 OF 3)



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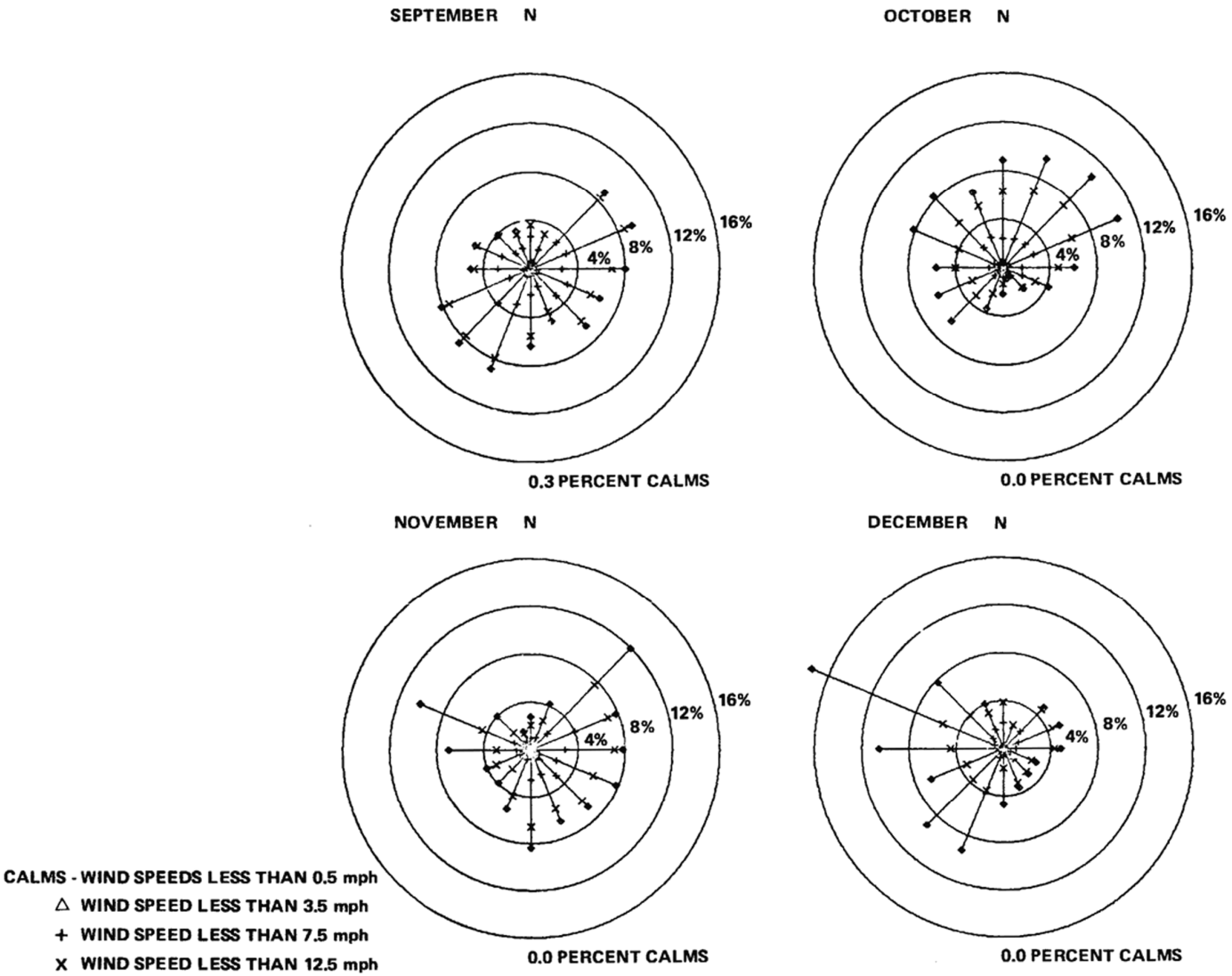
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLY WIND ROSE
1977 TO 1978 VEGP SITE DATA

FIGURE 2.3.2–13 (SHEET 2 OF 3)



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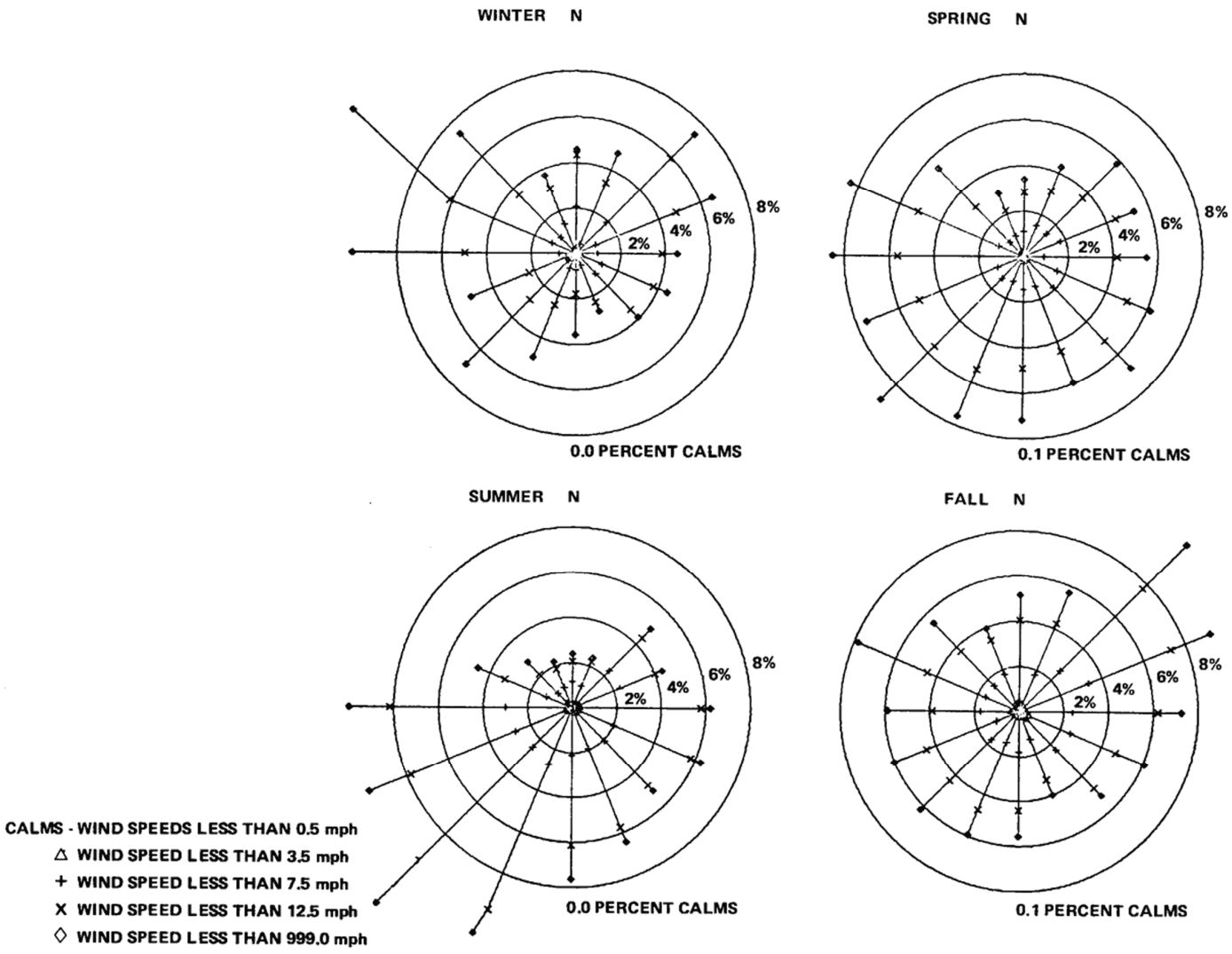
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLY WIND ROSE
1977 TO 1978 VEGP SITE DATA

FIGURE 2.3.2–13 (SHEET 3 OF 3)



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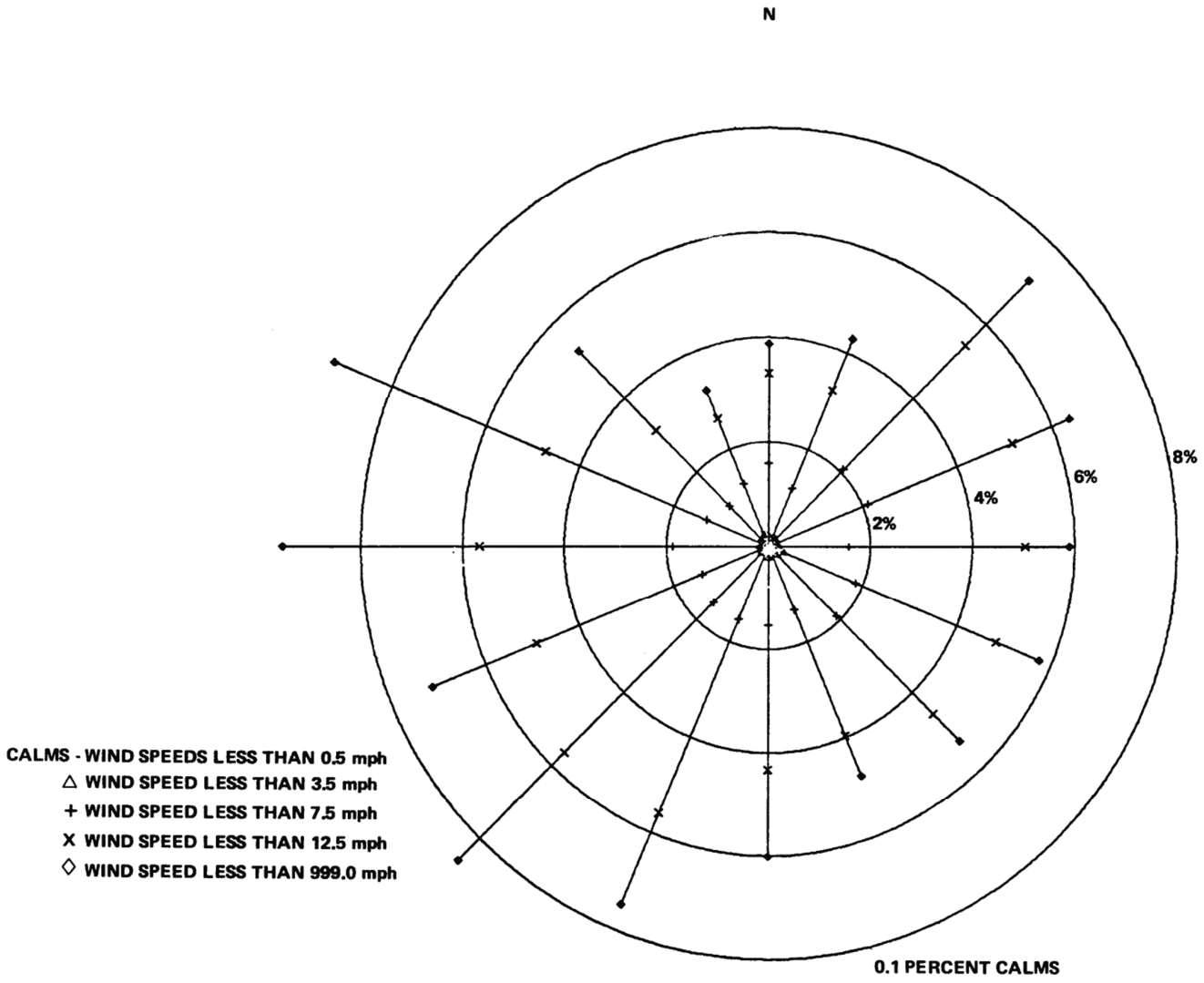
UNIT 1 AND UNIT 2

150–FT LEVEL SEASONAL WIND ROSE
1977 TO 1978 VEGP SITE DATA

FIGURE 2.3.2–14



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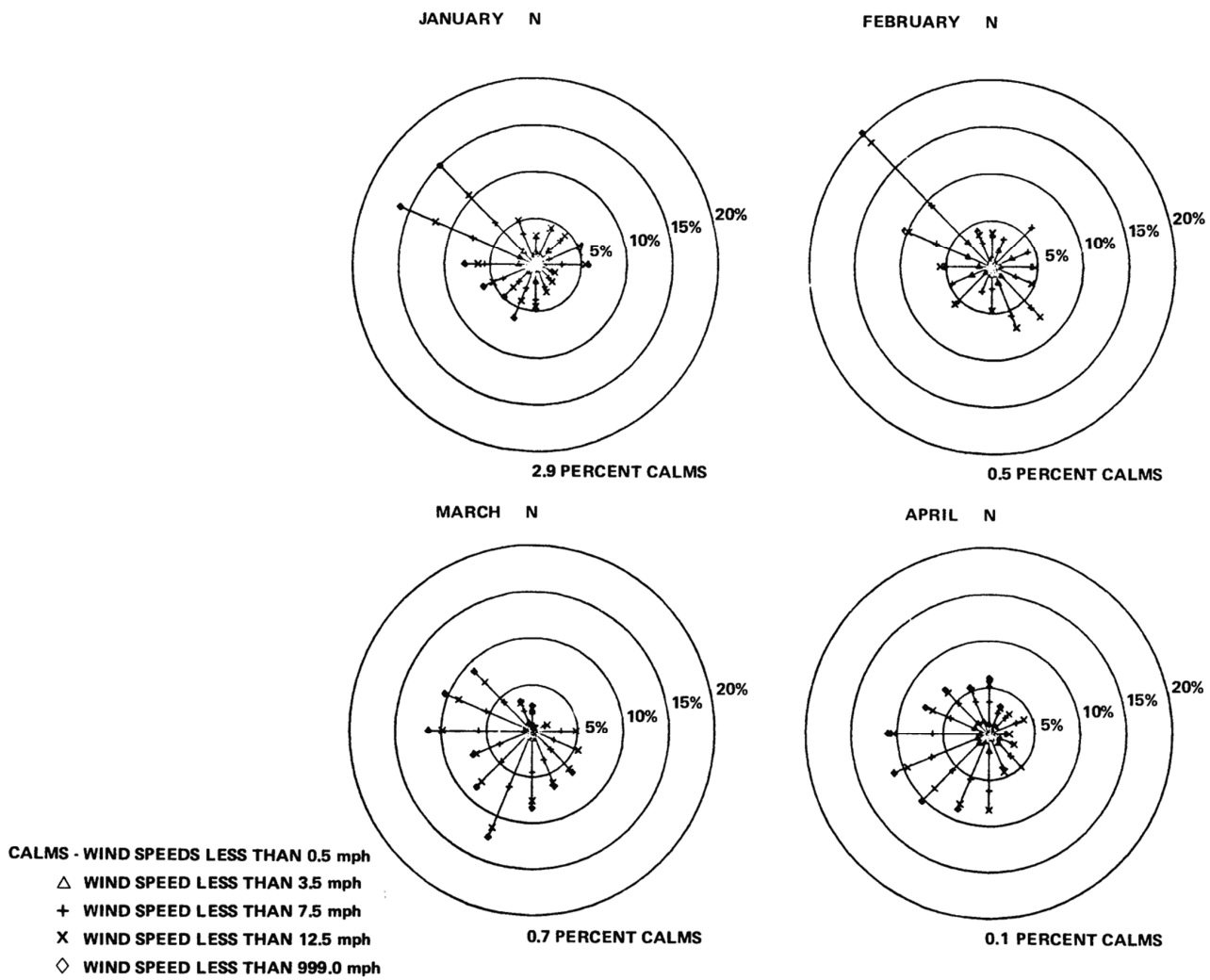
UNIT 1 AND UNIT 2

150–FT LEVEL ANNUAL WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–15



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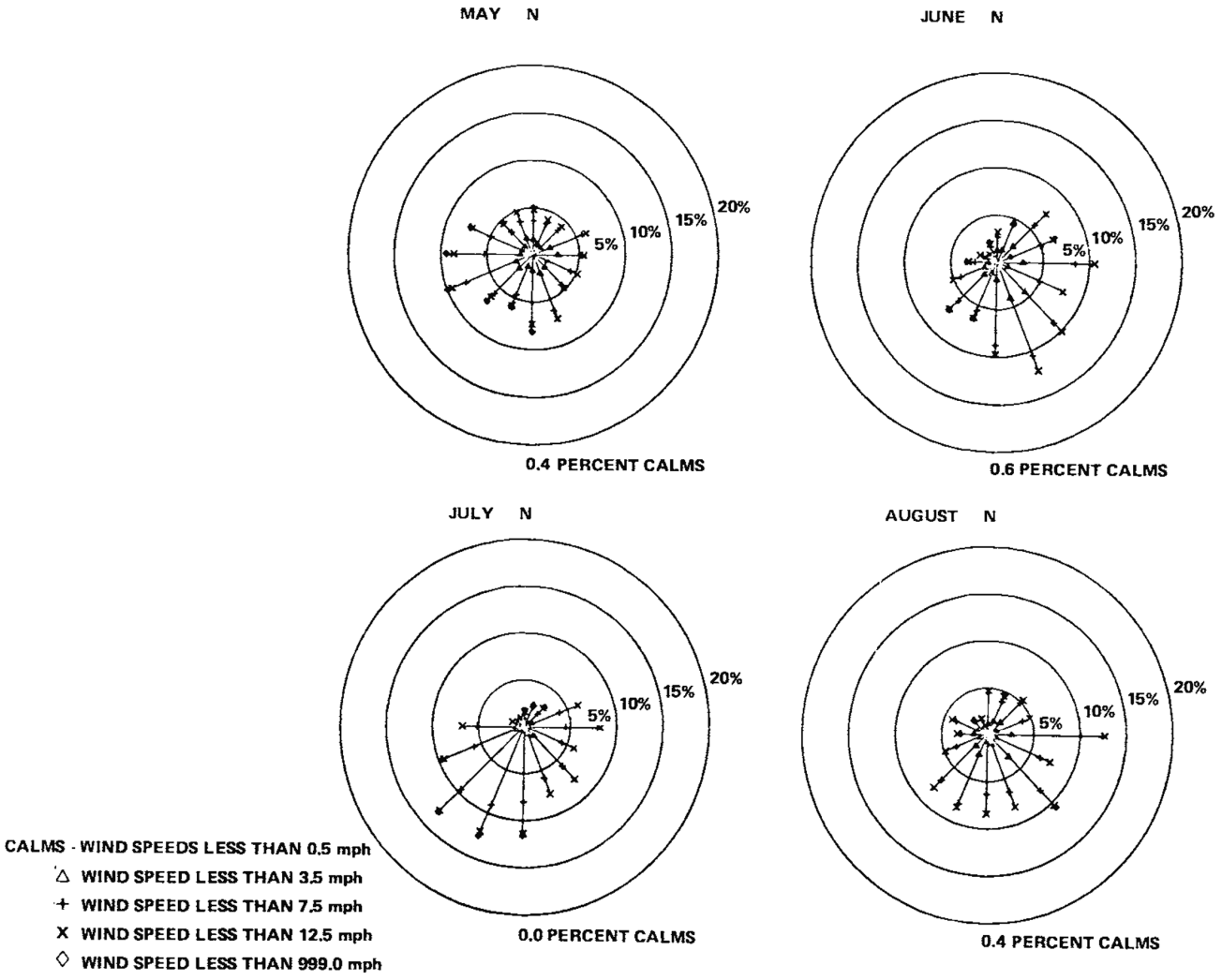
UNIT 1 AND UNIT 2

33–FT LEVEL MONTHLY WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–16 (SHEET 1 OF 3)



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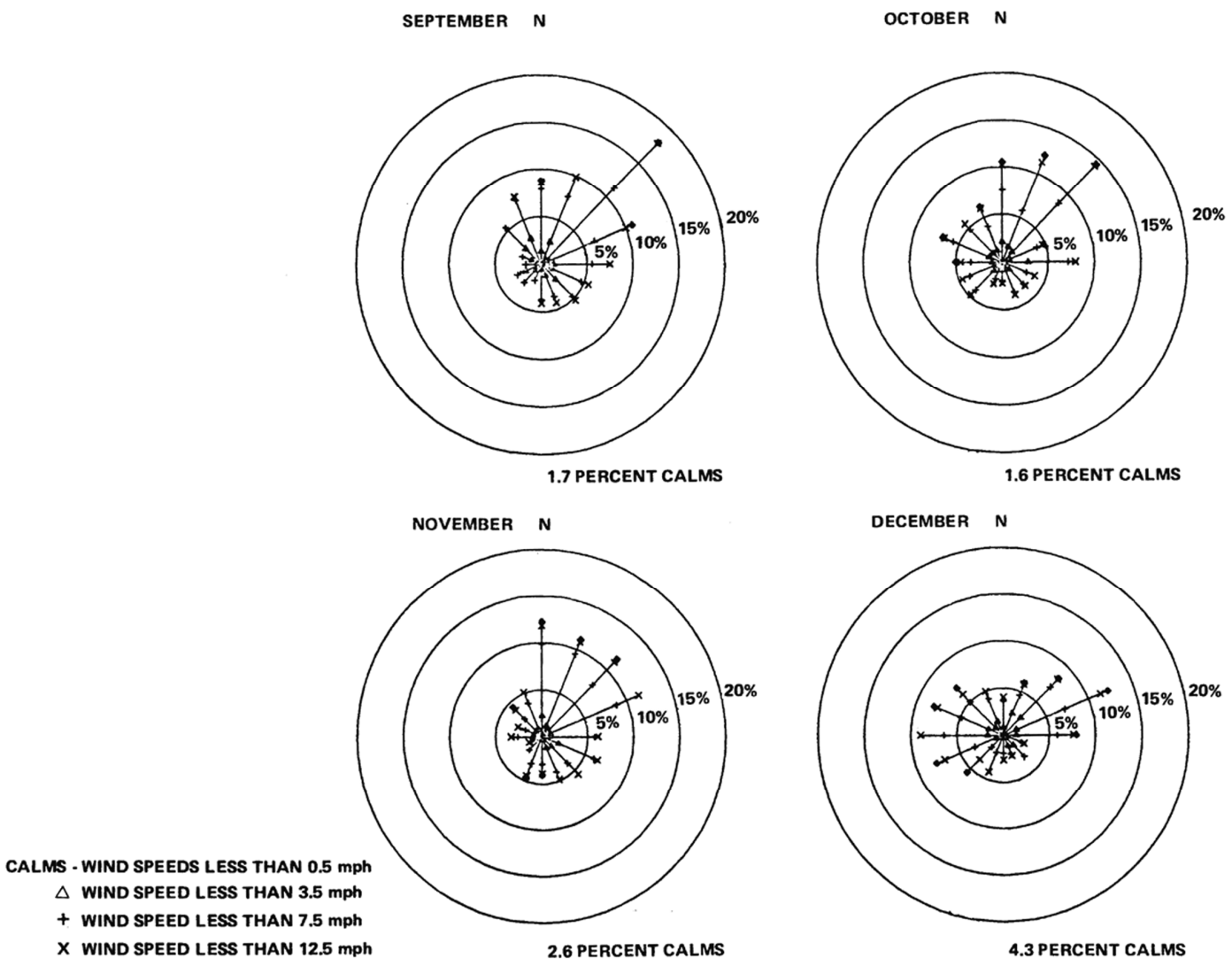
33–FT LEVEL MONTHLY WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–16 (SHEET 2 OF 3)



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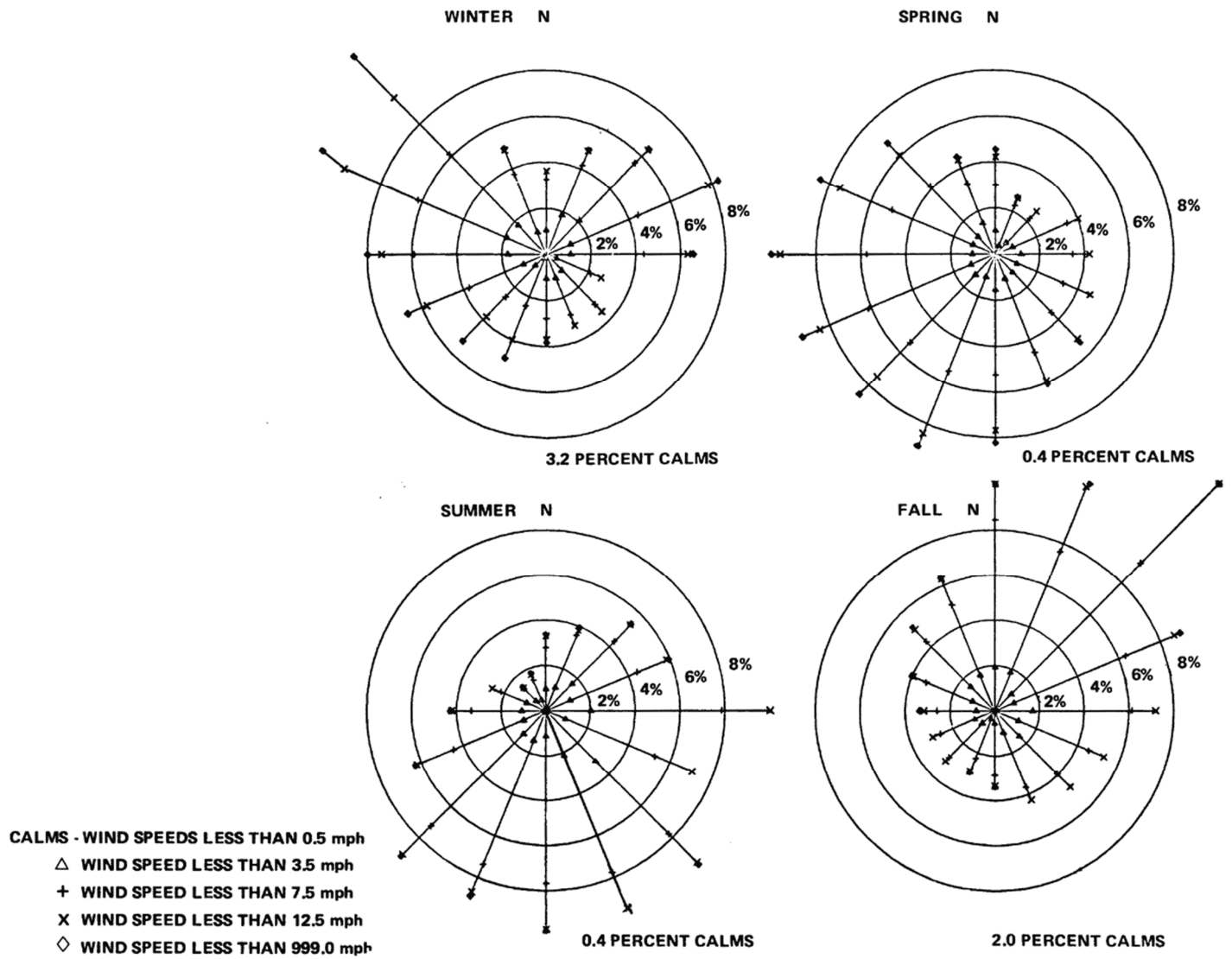
UNIT 1 AND UNIT 2

33–FT LEVEL MONTHLY WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–16 (SHEET 3 OF 3)



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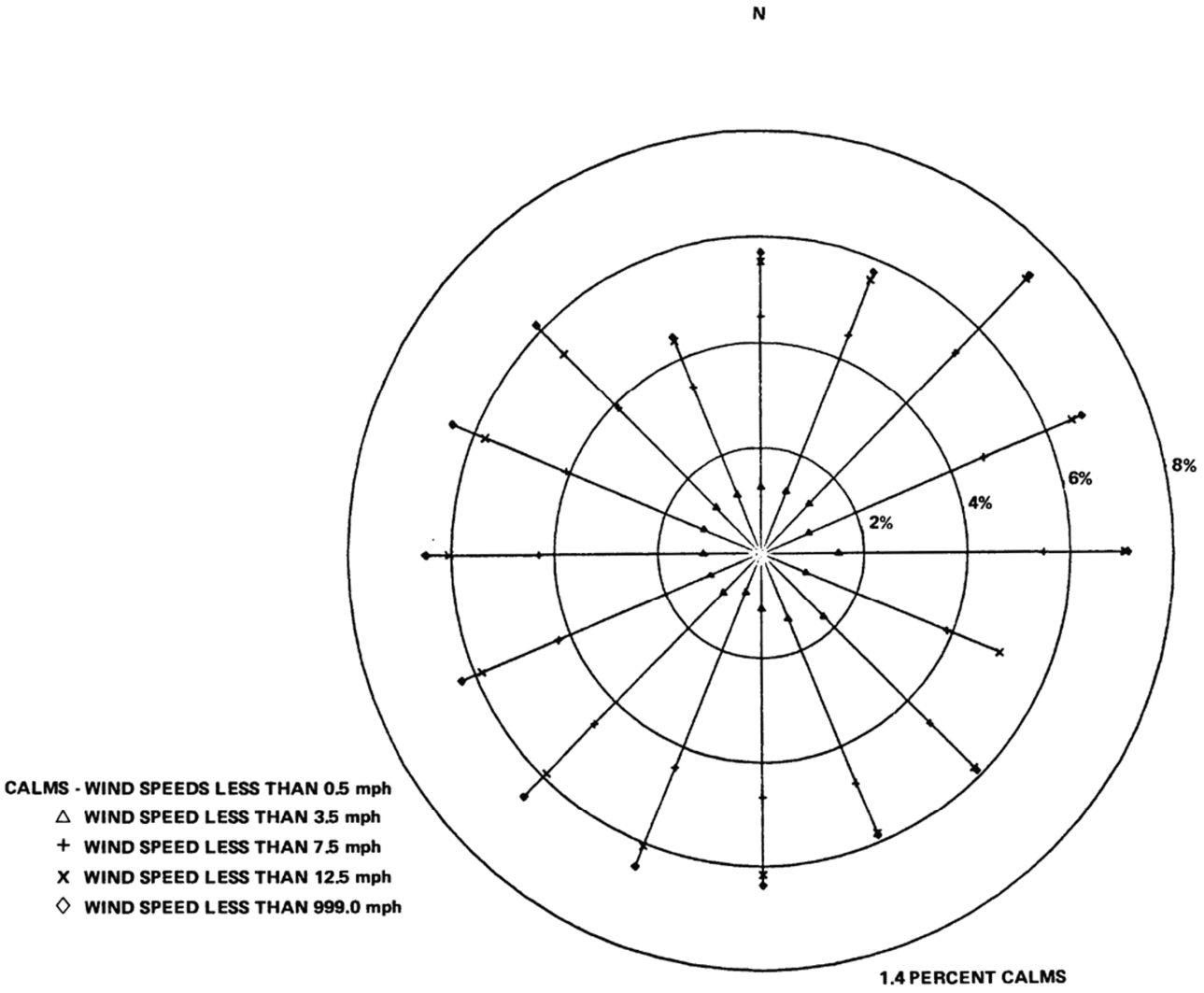
UNIT 1 AND UNIT 2

33–FT LEVEL SEASONAL WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–17



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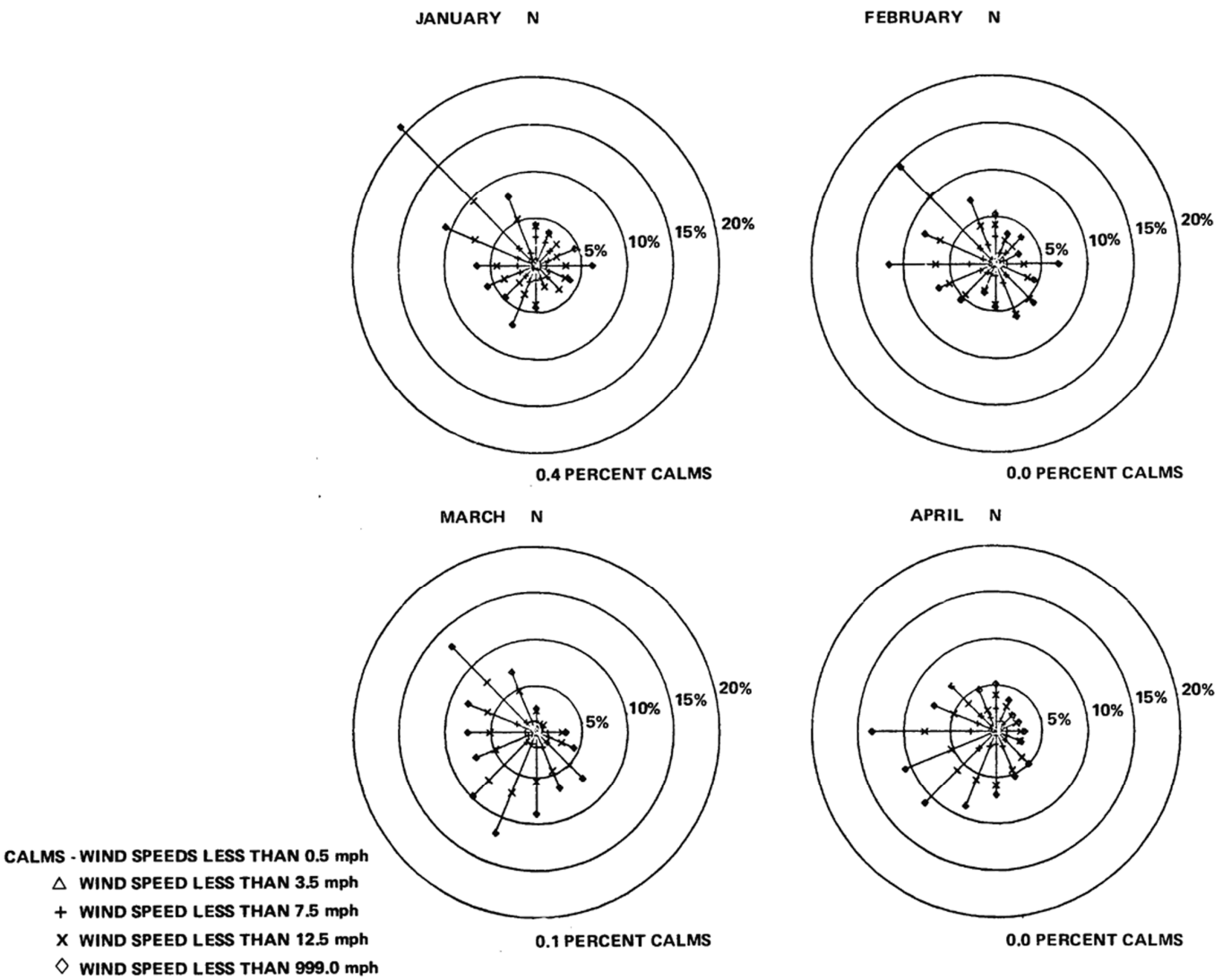
UNIT 1 AND UNIT 2

33–FT LEVEL ANNUAL WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–18



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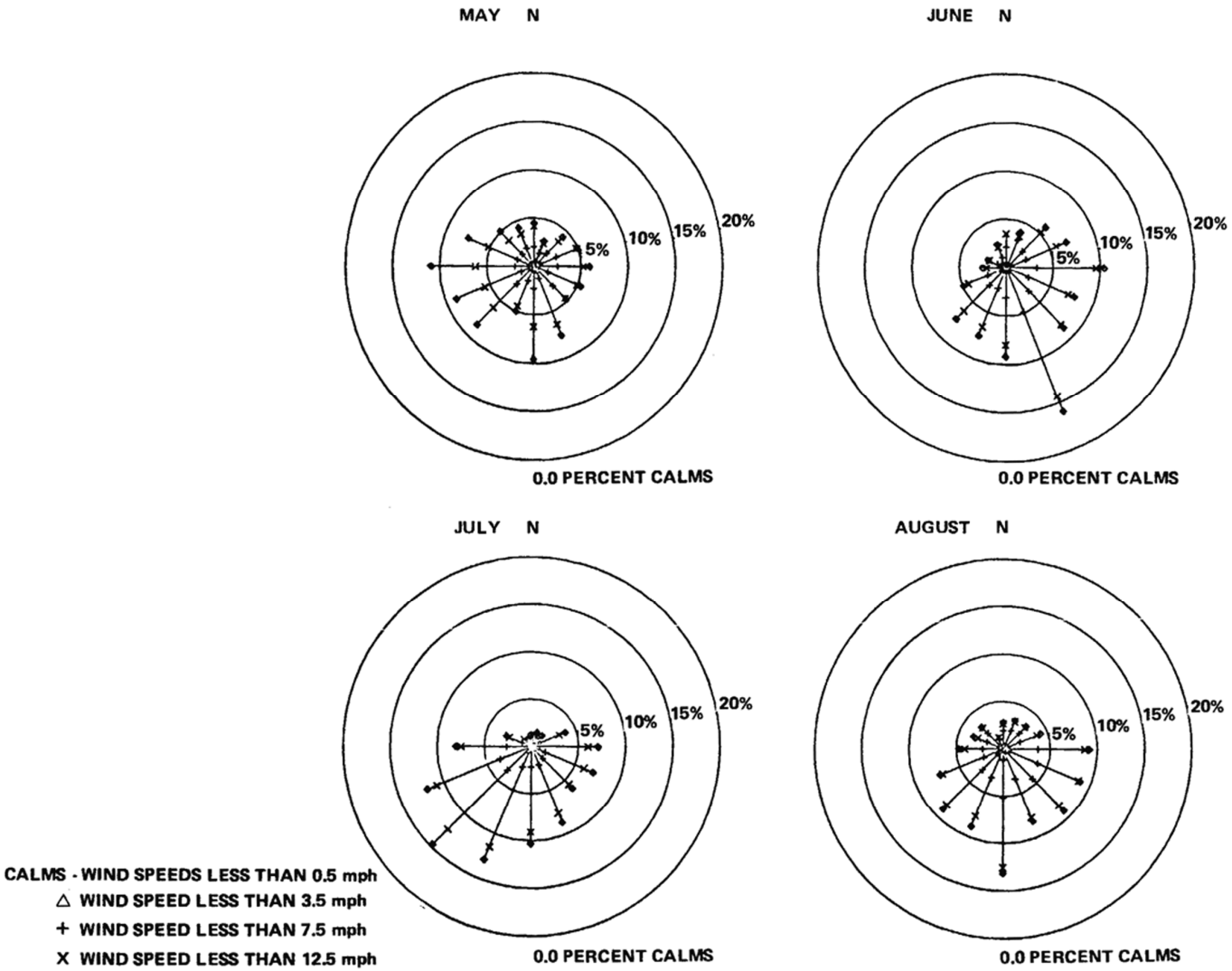
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLY WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–19 (SHEET 1 OF 3)



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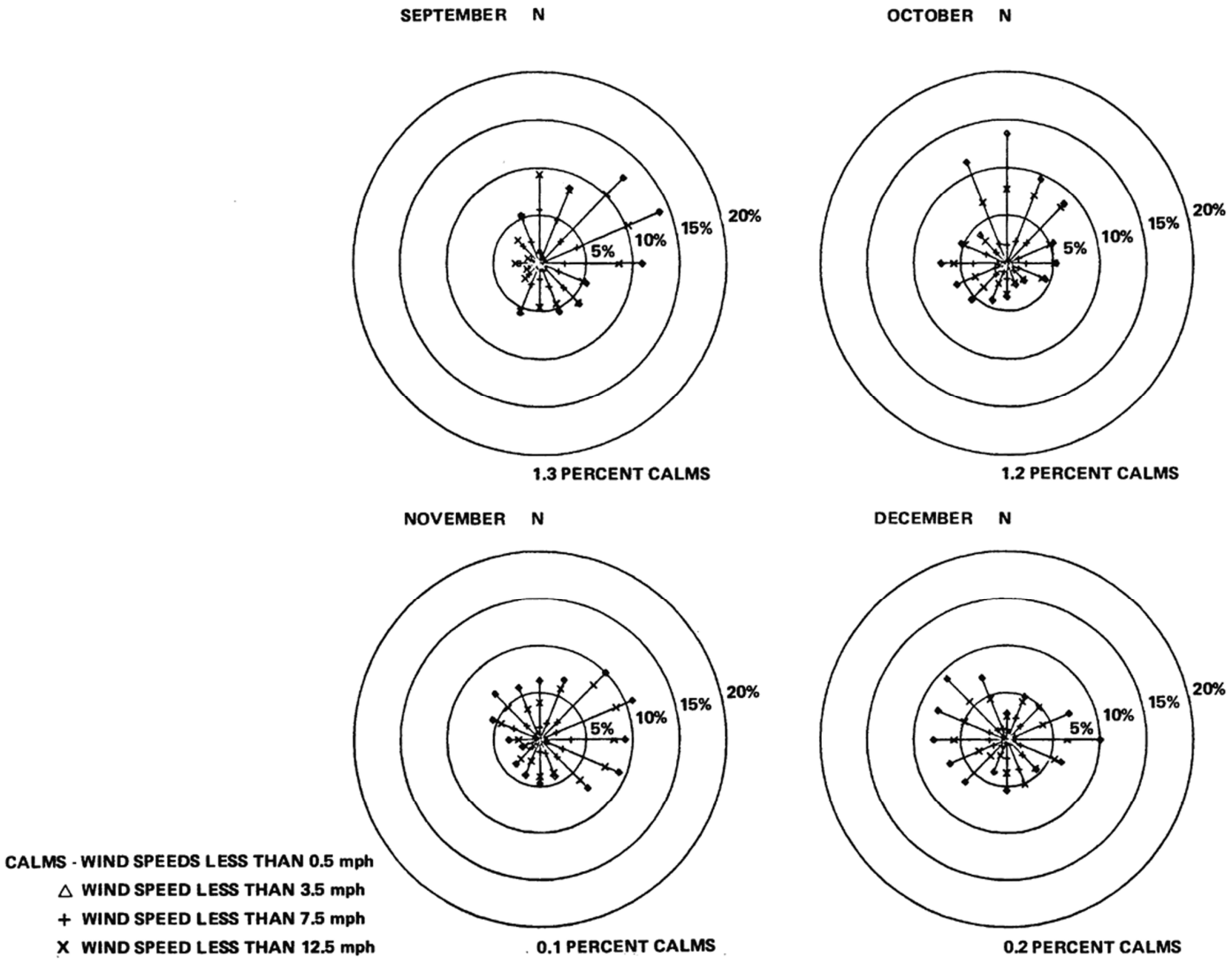
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLY WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–19 (SHEET 2 OF 3)



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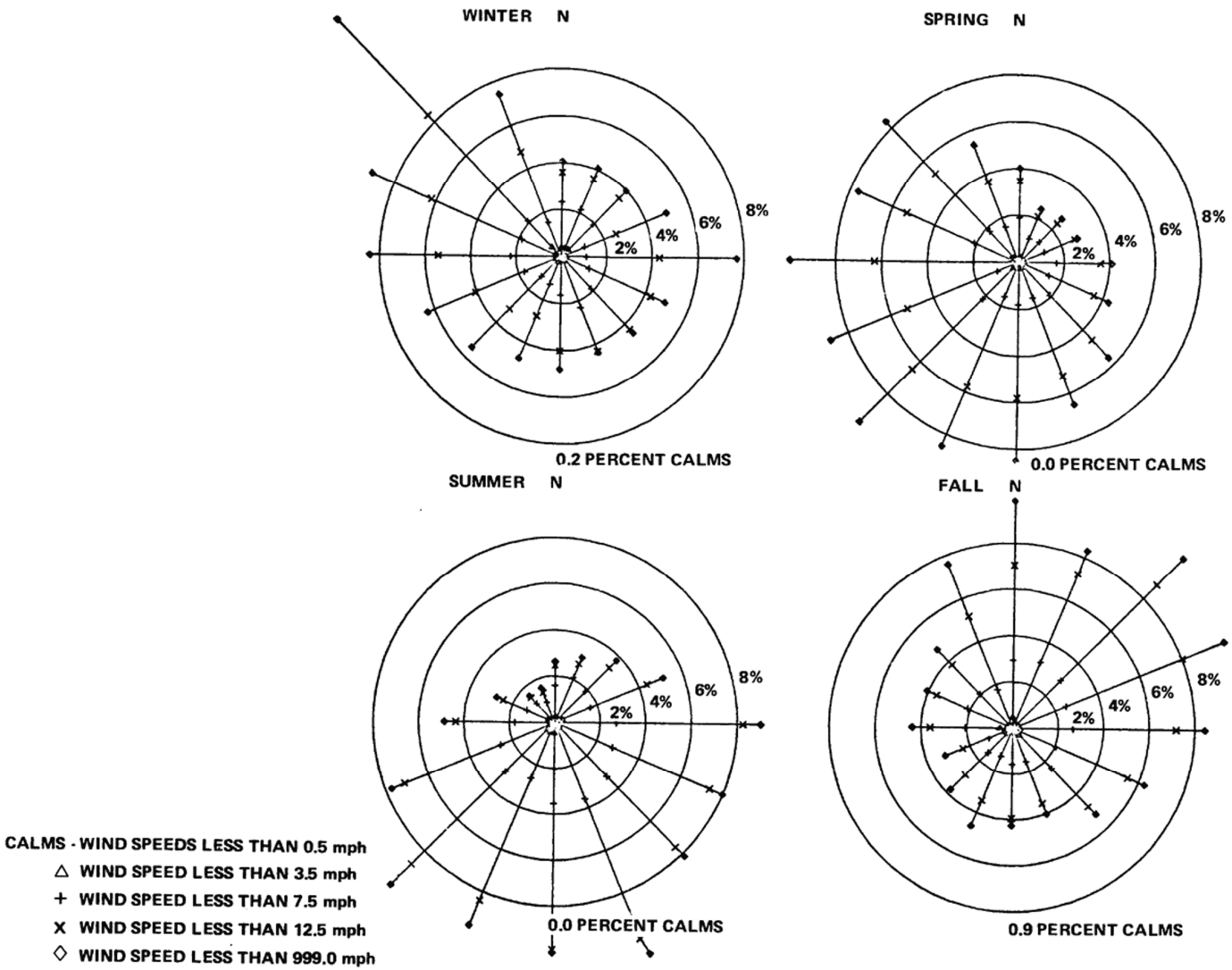
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLY WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–19 (SHEET 3 OF 3)



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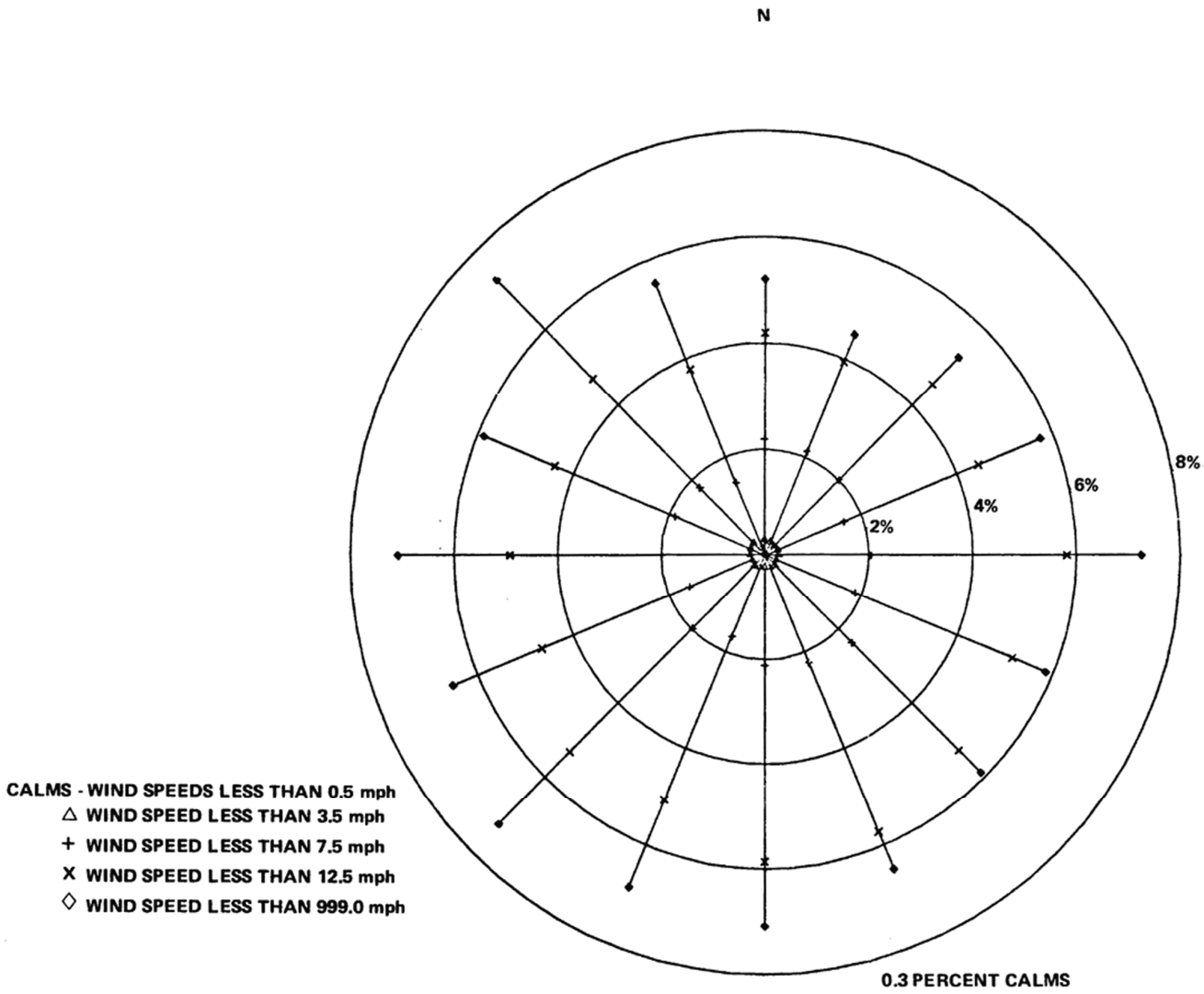
UNIT 1 AND UNIT 2

150–FT LEVEL SEASONAL WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–20



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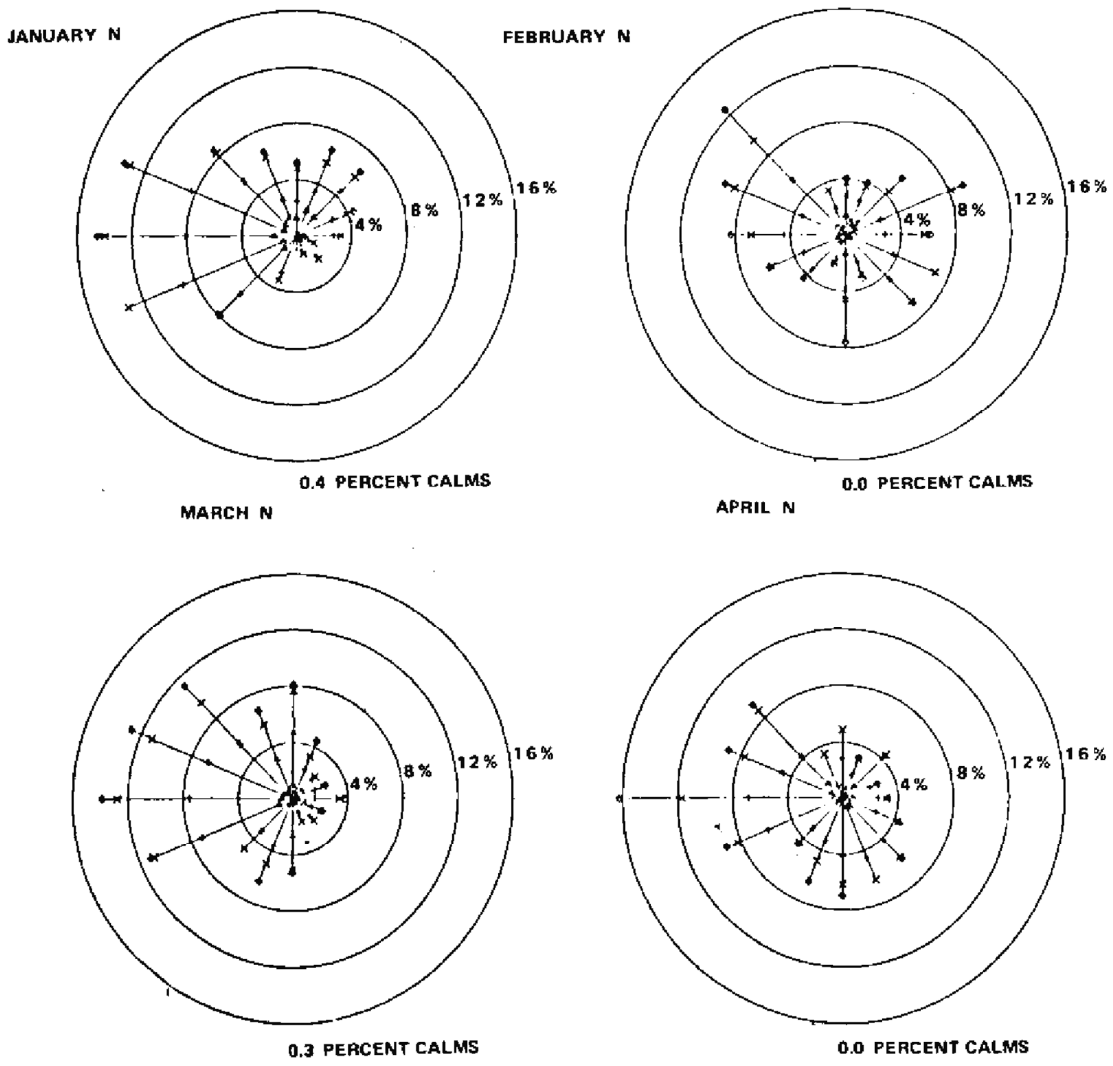
UNIT 1 AND UNIT 2

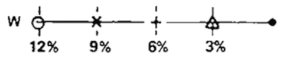
150–FT LEVEL ANNUAL WIND ROSE
1978 TO 1979 VEGP SITE DATA

FIGURE 2.3.2–21



NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH [INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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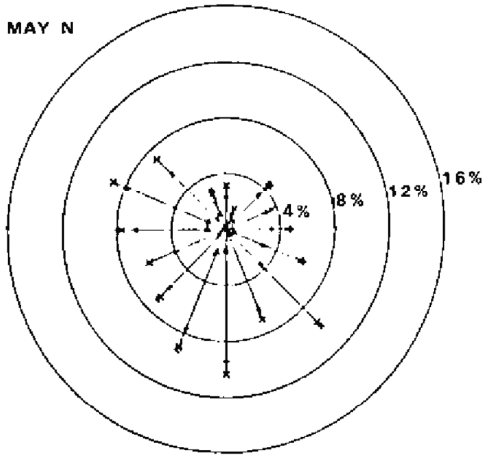
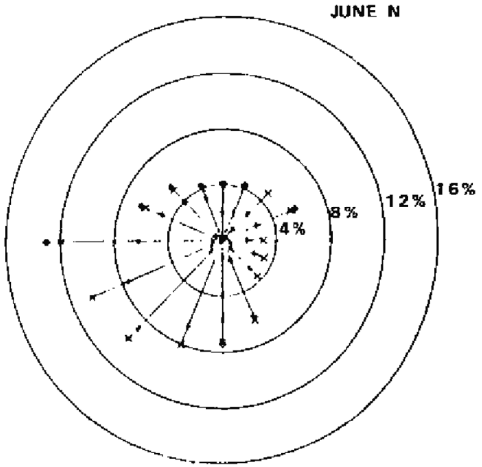
UNIT 1 AND UNIT 2

33–FT LEVEL MONTHLY WIND ROSE
1980 TO 1981 VEGP SITE DATA

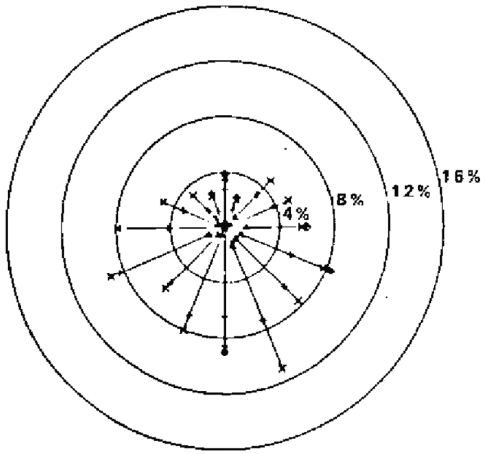
FIGURE 2.3.2–22 (SHEET 1 OF 3)



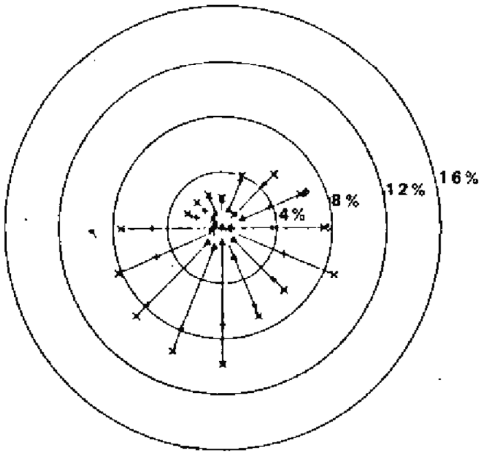
0.0 PERCENT CALMS



0.2 PERCENT CALMS



0.1 PERCENT CALMS



0.1 PERCENT CALMS

NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
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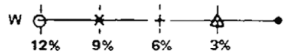
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH [INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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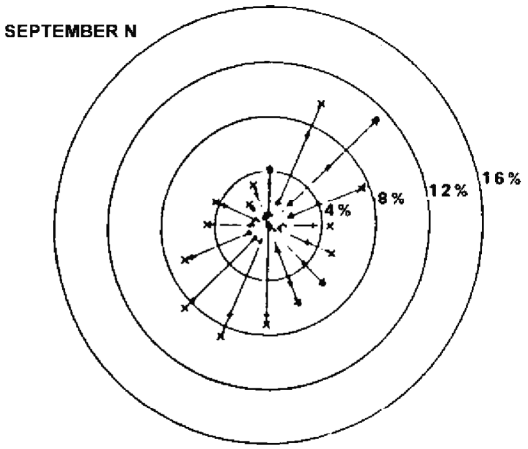
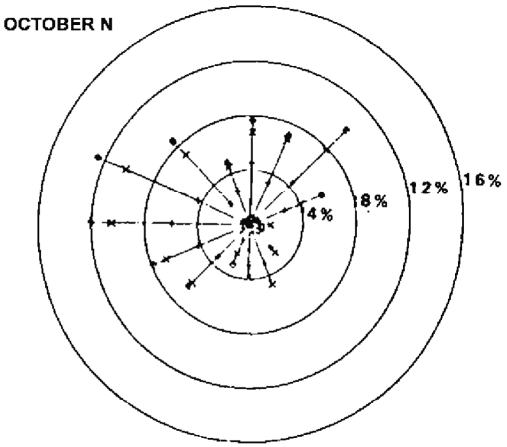
UNIT 1 AND UNIT 2

33–FT LEVEL MONTHLY WIND ROSE
1980 TO 1981 VEGP SITE DATA

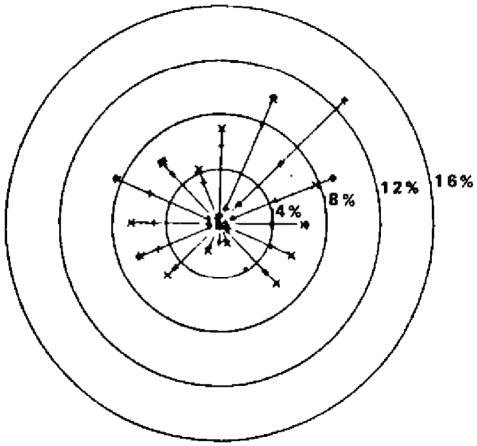
FIGURE 2.3.2–22 (SHEET 2 OF 3)



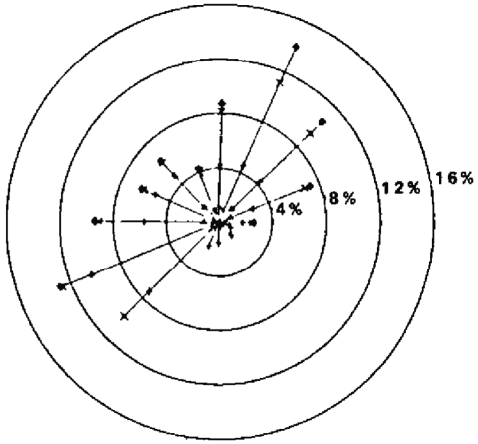
0.0 PERCENT CALMS



0.0 PERCENT CALMS



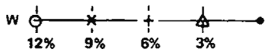
0.2 PERCENT CALMS



1.7 PERCENT CALMS

NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



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CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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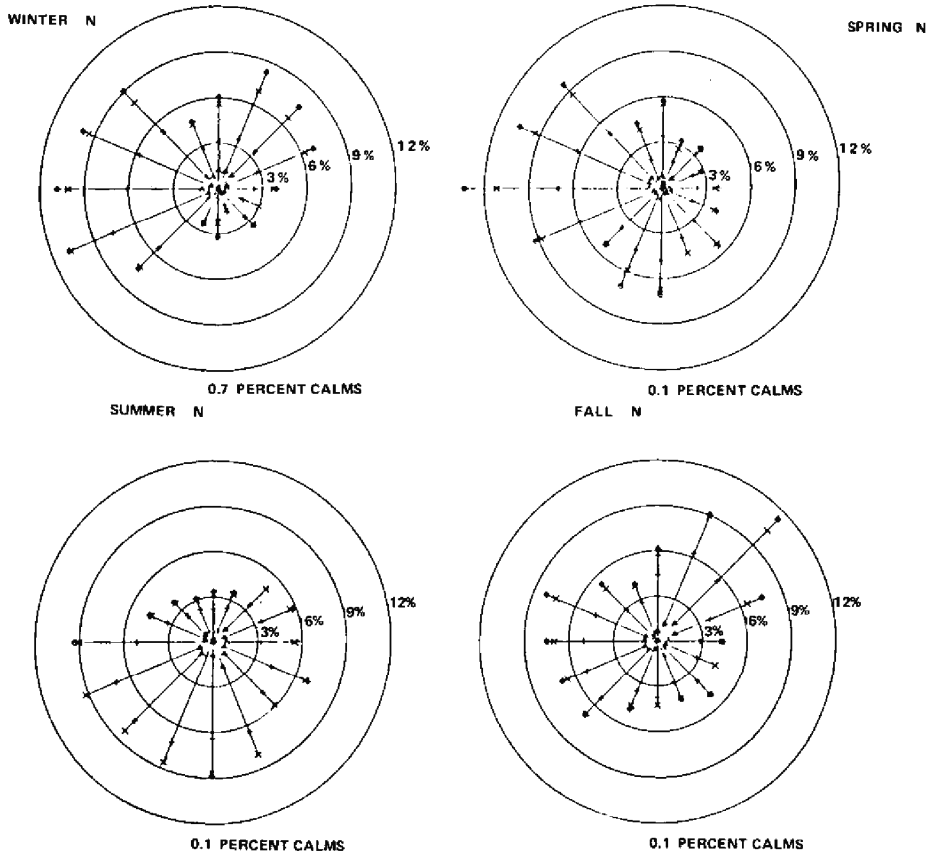
UNIT 1 AND UNIT 2

33–FT LEVEL MONTHLY WIND ROSE
1980 TO 1981 VEGP SITE DATA

FIGURE 2.3.2–22 (SHEET 3 OF 3)



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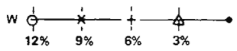
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.

FOR EXAMPLE.



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CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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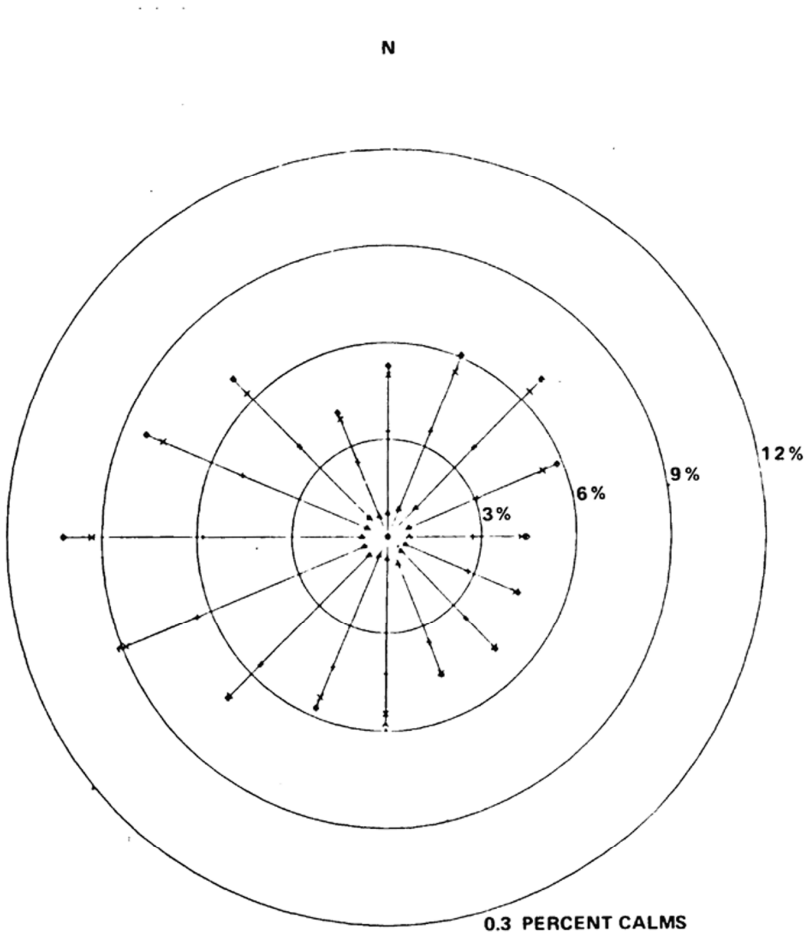
UNIT 1 AND UNIT 2

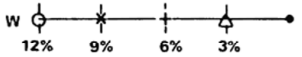
33–FT LEVEL SEASONAL WIND ROSE
1980 TO 1981 VEGP SITE DATA

FIGURE 2.3.2–23



NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △, +, X, O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH [INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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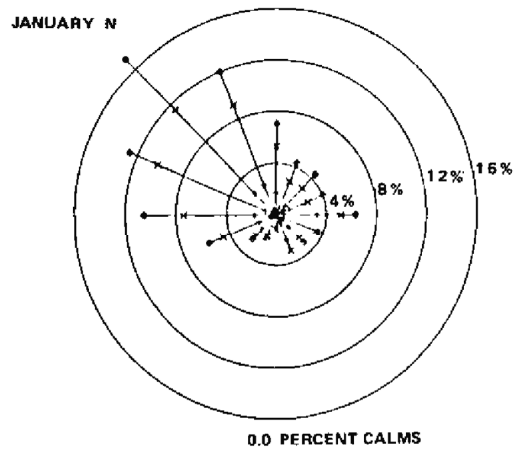
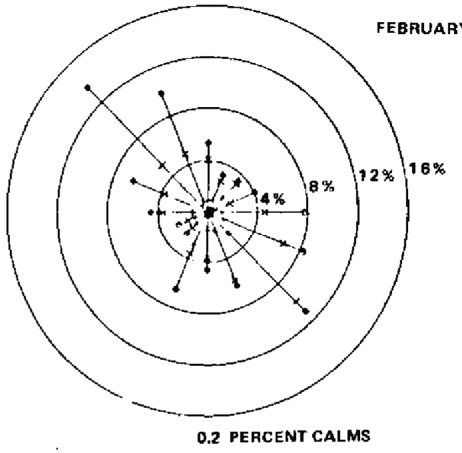
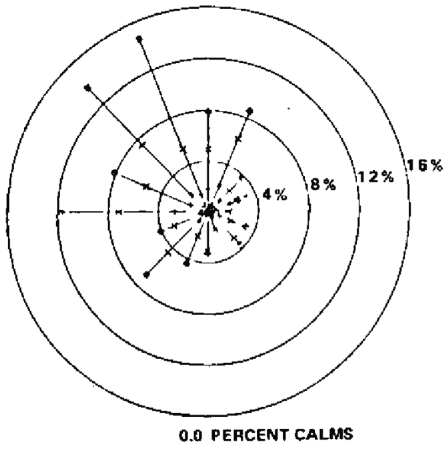
UNIT 1 AND UNIT 2

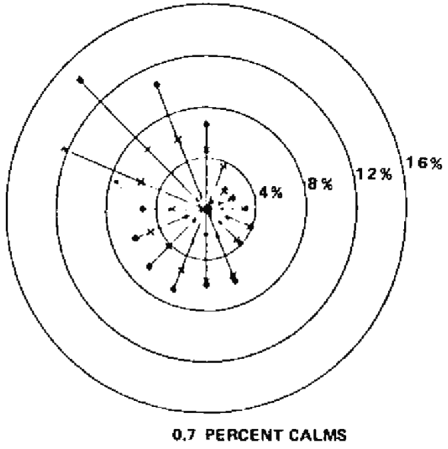
33–FT LEVEL ANNUAL WIND ROSE
1980 TO 1981 VEGP SITE DATA

FIGURE 2.3.2–24



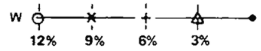






NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, *i.e.*, INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH [INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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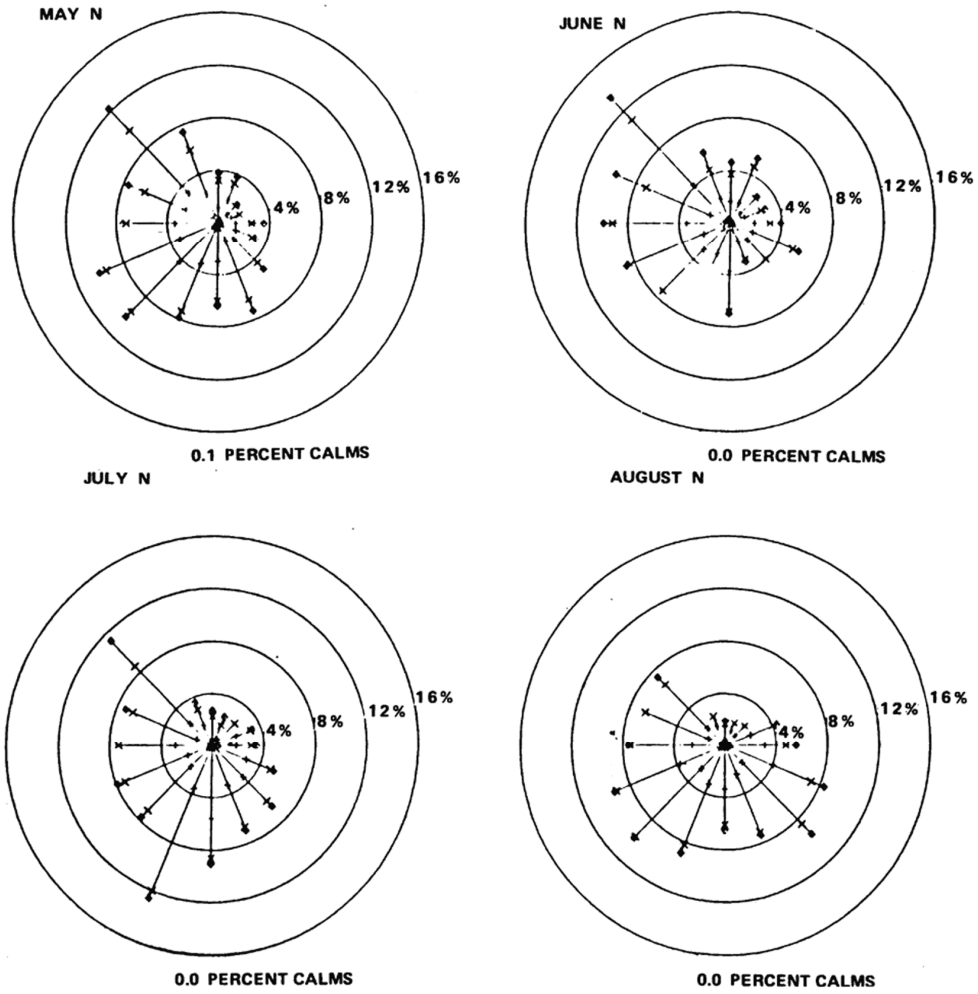
UNIT 1 AND UNIT 2

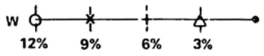
150–FT LEVEL MONTHLY WIND ROSE
1980 TO 1981 VEGP SITE DATA

FIGURE 2.3.2–25 (SHEET 1 OF 3)



NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST: OF THESE, 25 PERCENT HAD SPEEDS 0-3 MPH (INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE, 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph.
O WIND SPEED LESS THAN 999.0 mph

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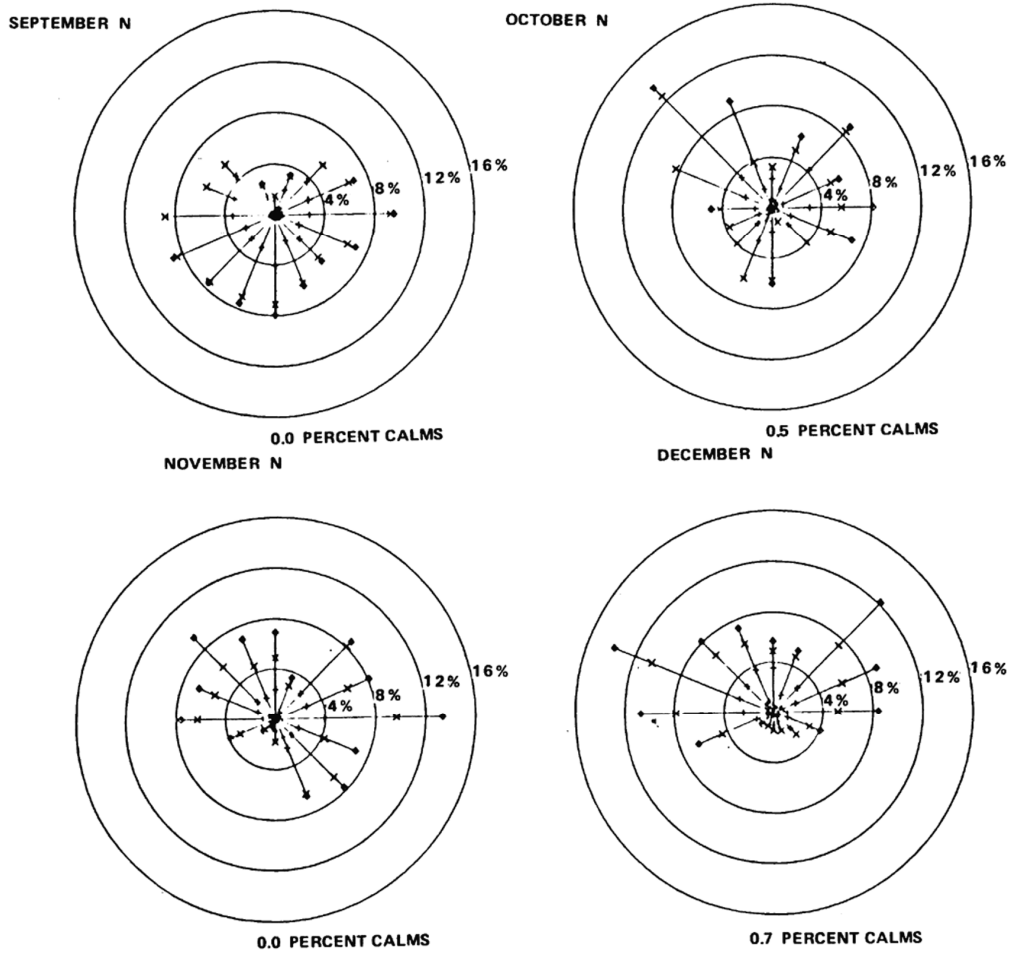
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLY WIND ROSE
1980 TO 1981 VEGP SITE DATA

FIGURE 2.3.2–25 (SHEET 2 OF 3)



NOTES:

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2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
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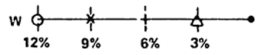
4. THE △ , + , X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH RANGE.
   FOR EXAMPLE.



THIS RADIAL, REPRESENTING WIND FROM THE WEST, INDICATES A TOTAL FREQUENCY OF 12 PERCENT WINDS FROM THE WEST; OF THESE 25 PERCENT HAD SPEEDS 0-3 MPH (INDICATED BY △ AT 3 PERCENT LINE (25 PERCENT OF 12 PERCENT)], 50 PERCENT HAD WINDS 0-7 MPH (INDICATED BY + AT 6 PERCENT LINE), 75 PERCENT HAD WINDS 0-12 MPH (INDICATED BY X AT 9 PERCENT LINE. 100 PERCENT HAD WINDS 0-999 MPH (INDICATED BY O AT END OF LINE; THIS WILL OBVIOUSLY BE TRUE OF EVERY RADIAL OF ALL WIND ROSES). WINDS IN CATEGORIES SUCH AS 7-12 MPH CAN BE OBTAINED BY DIFFERENCES OF CUMULATIVE PERCENTS.

CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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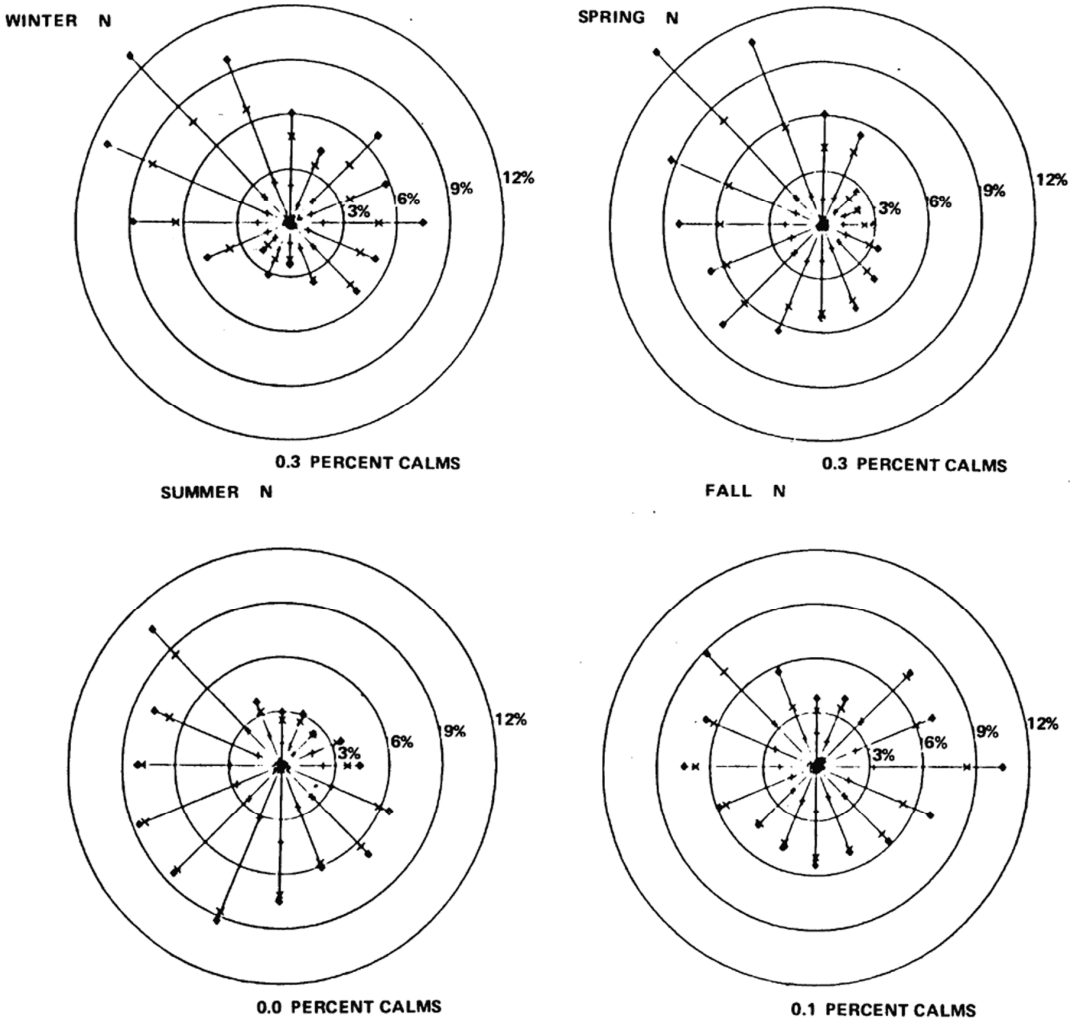
UNIT 1 AND UNIT 2

150–FT LEVEL MONTHLY WIND ROSE
1980 TO 1981 VEGP SITE DATA

FIGURE 2.3.2–25 (SHEET 3 OF 3)



NOTES:

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2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
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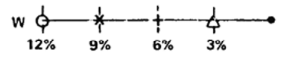
   FOR EXAMPLE.



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CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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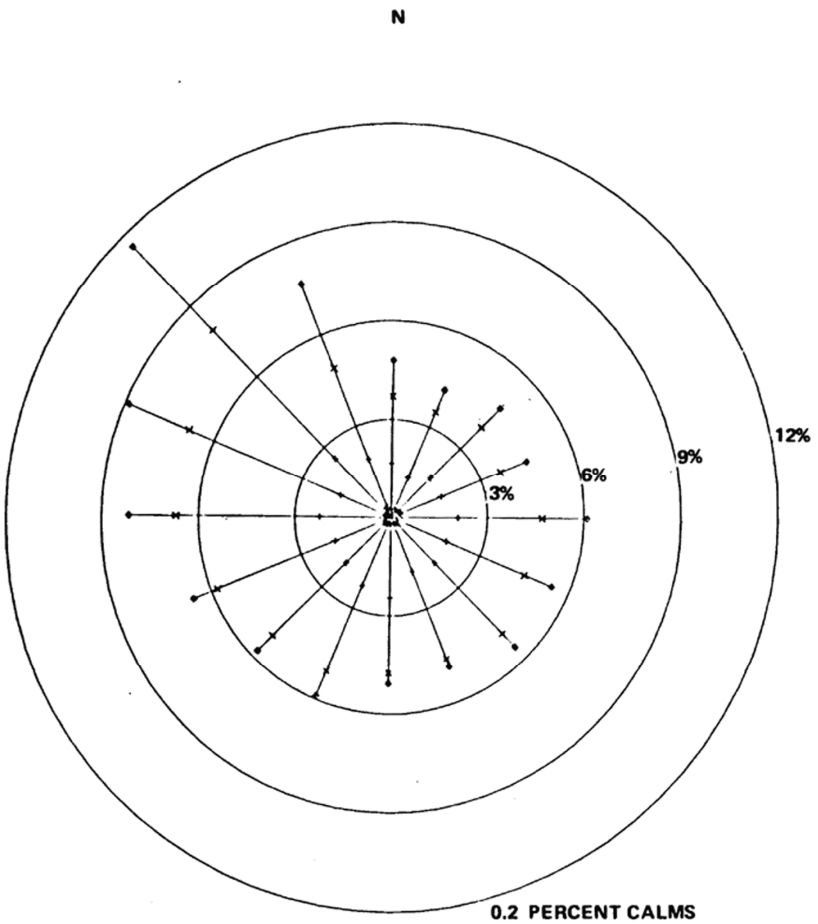
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

150–FT LEVEL SEASONAL WIND ROSE
1980 TO 1981 VEGP SITE DATA

FIGURE 2.3.2–26



NOTES:

1. WIND SPEEDS INDICATED ARE IN MILES PER HOUR.
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
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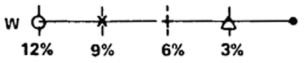
   FOR EXAMPLE.



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CALMS - WIND SPEEDS LESS THAN 0.5 mph
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O WIND SPEED LESS THAN 999.0 mph

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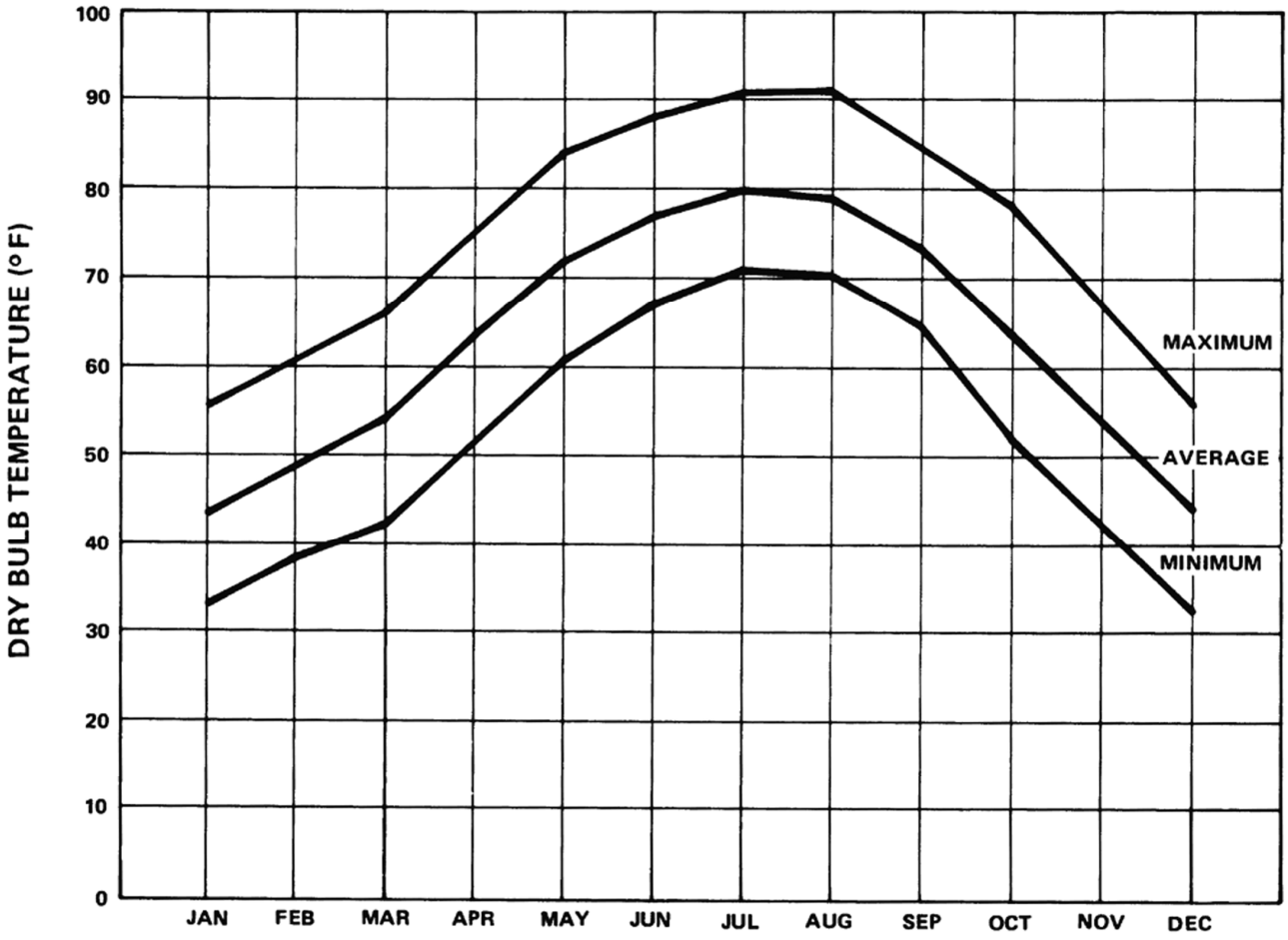
UNIT 1 AND UNIT 2

150–FT LEVEL ANNUAL WIND ROSE
1980 TO 1981 VEGP SITE DATA

FIGURE 2.3.2–27

AUGUSTA, GEORGIA
1959 TO 1963



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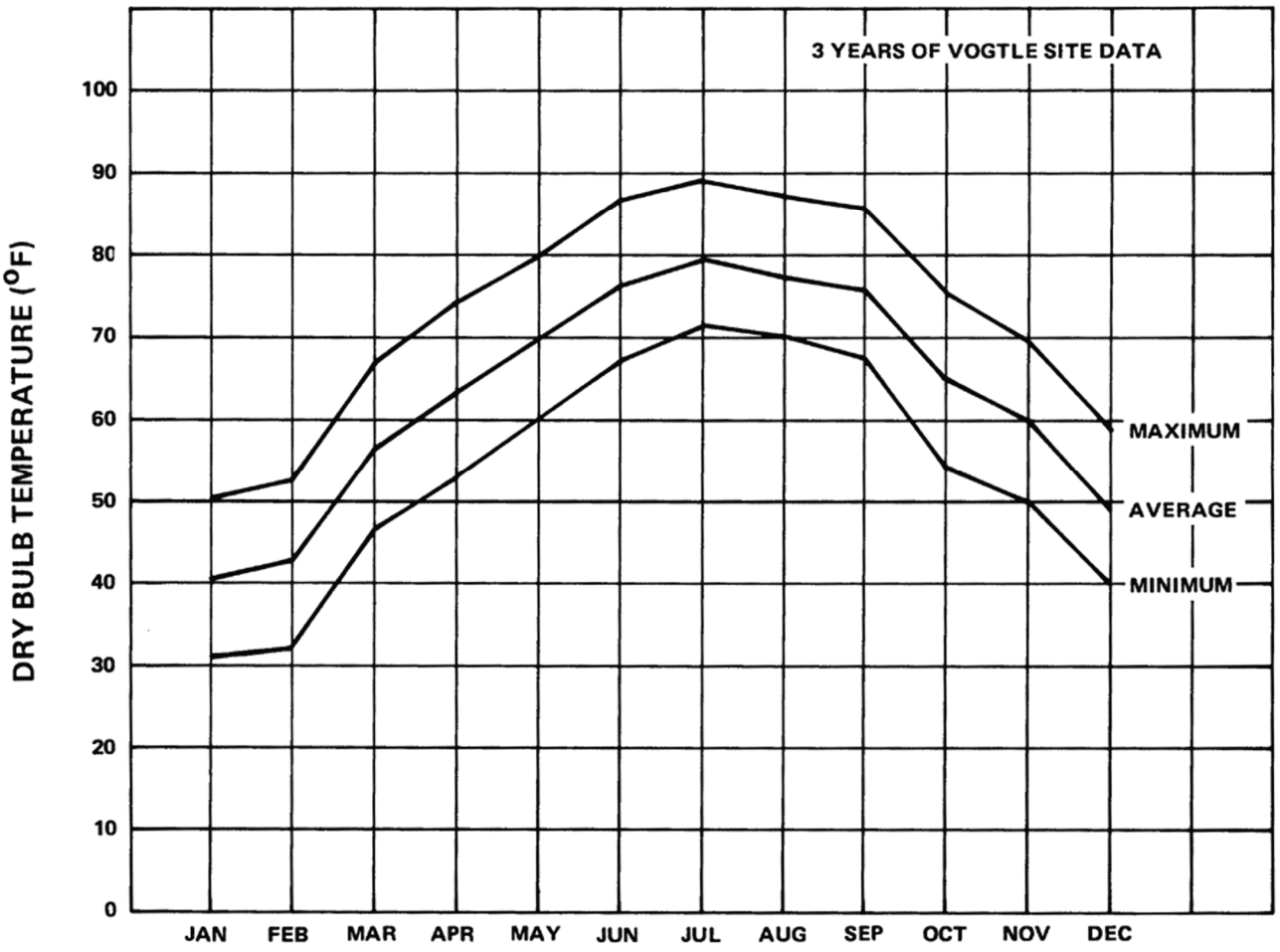
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	MONTHLY AVERAGE AND AVERAGE DAILY EXTREMES OF DRY BULB TEMPERATURES FIGURE 2.3.2–28



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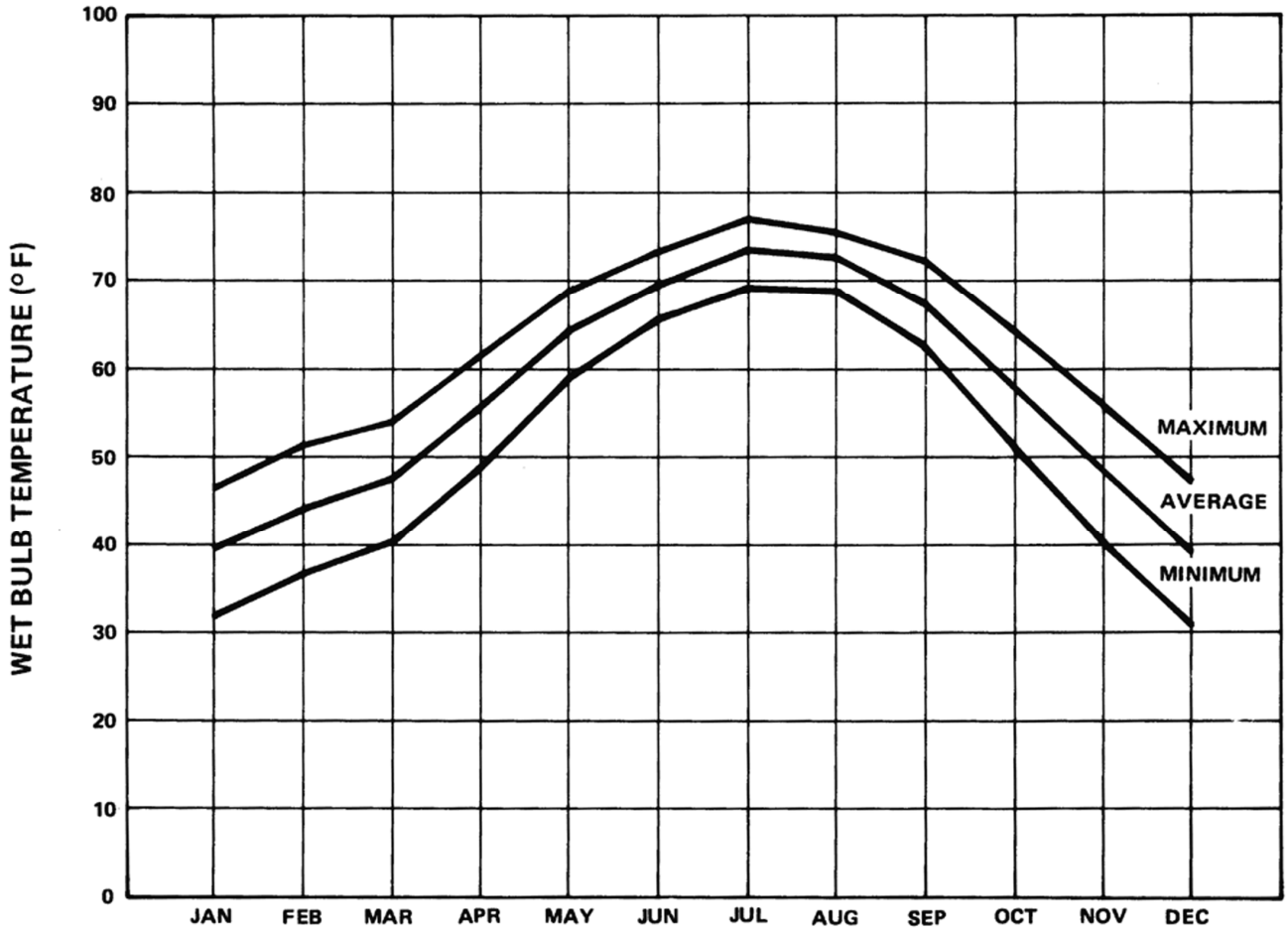
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND AVERAGE
OF DAILY EXTREMES
OF DRY BULB TEMPERATURE

FIGURE 2.3.2–29

AUGUSTA, GEORGIA
1959 TO 1963



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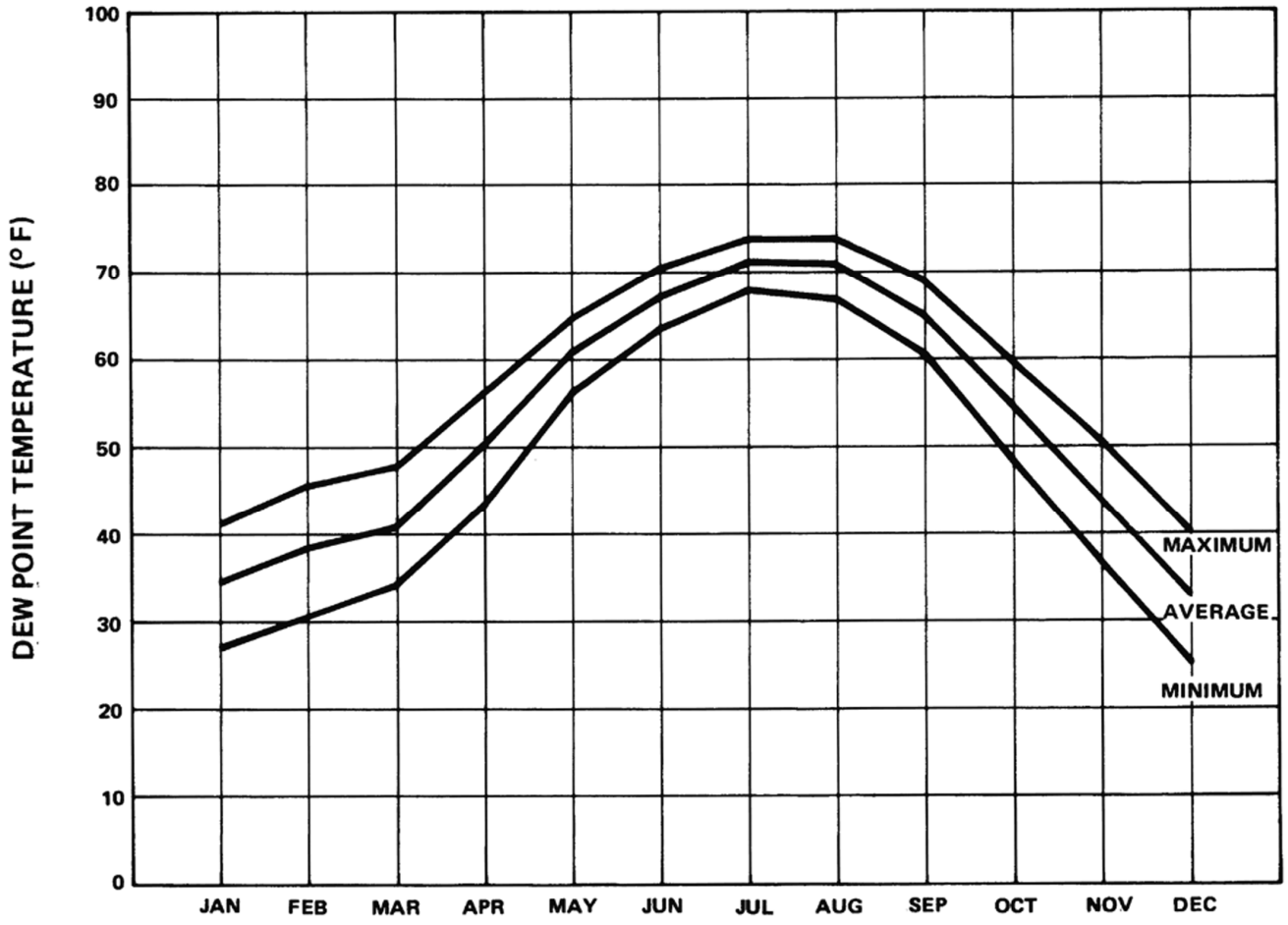
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE OF DAILY EXTREMES
OF WET BULB TEMPERATURES

FIGURE 2.3.2–30

AUGUSTA, GEORGIA
1959 TO 1963



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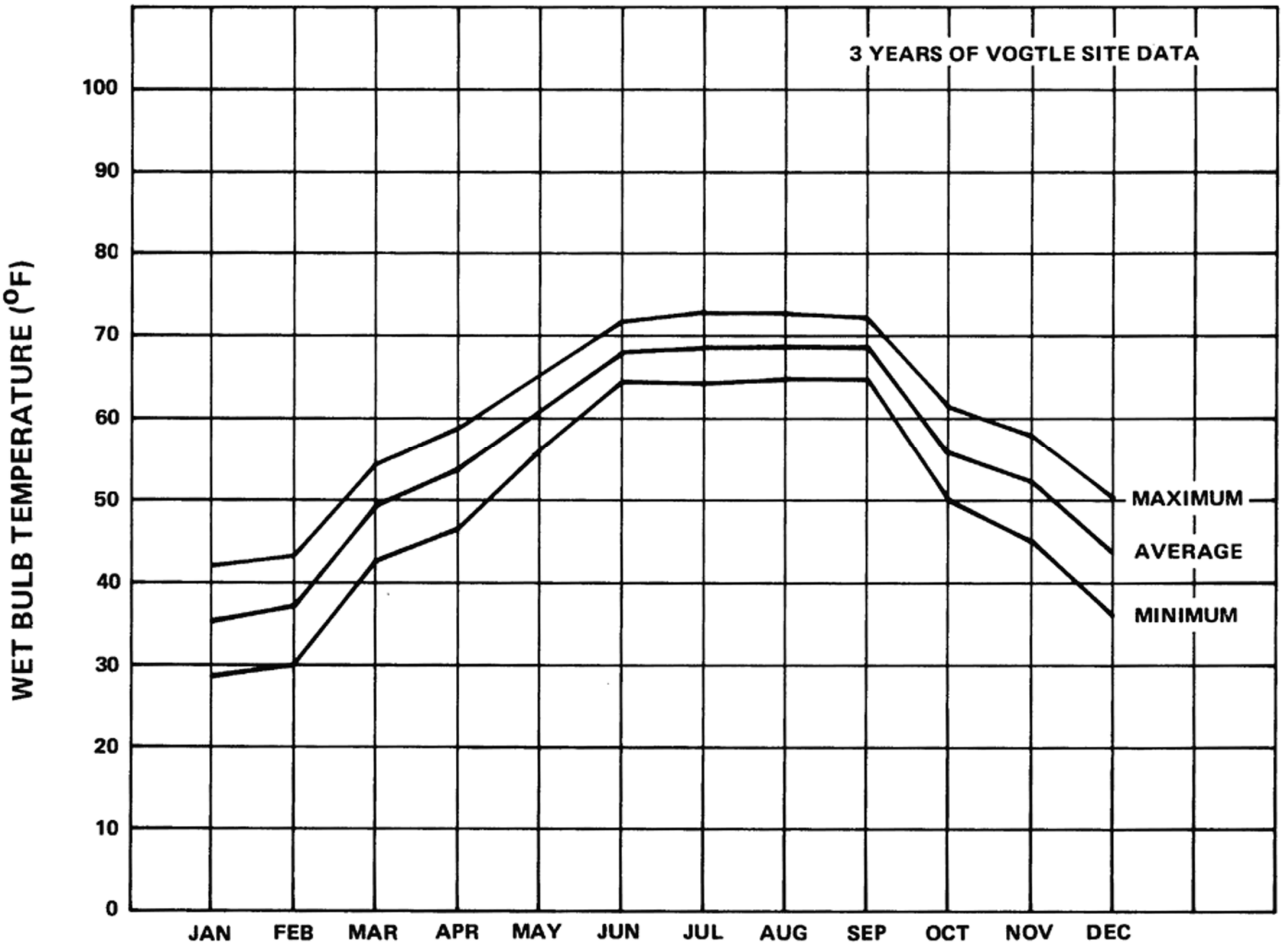
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE DAILY EXTREMES OF
DEWPOINT TEMPERATURE

FIGURE 2.3.2–31



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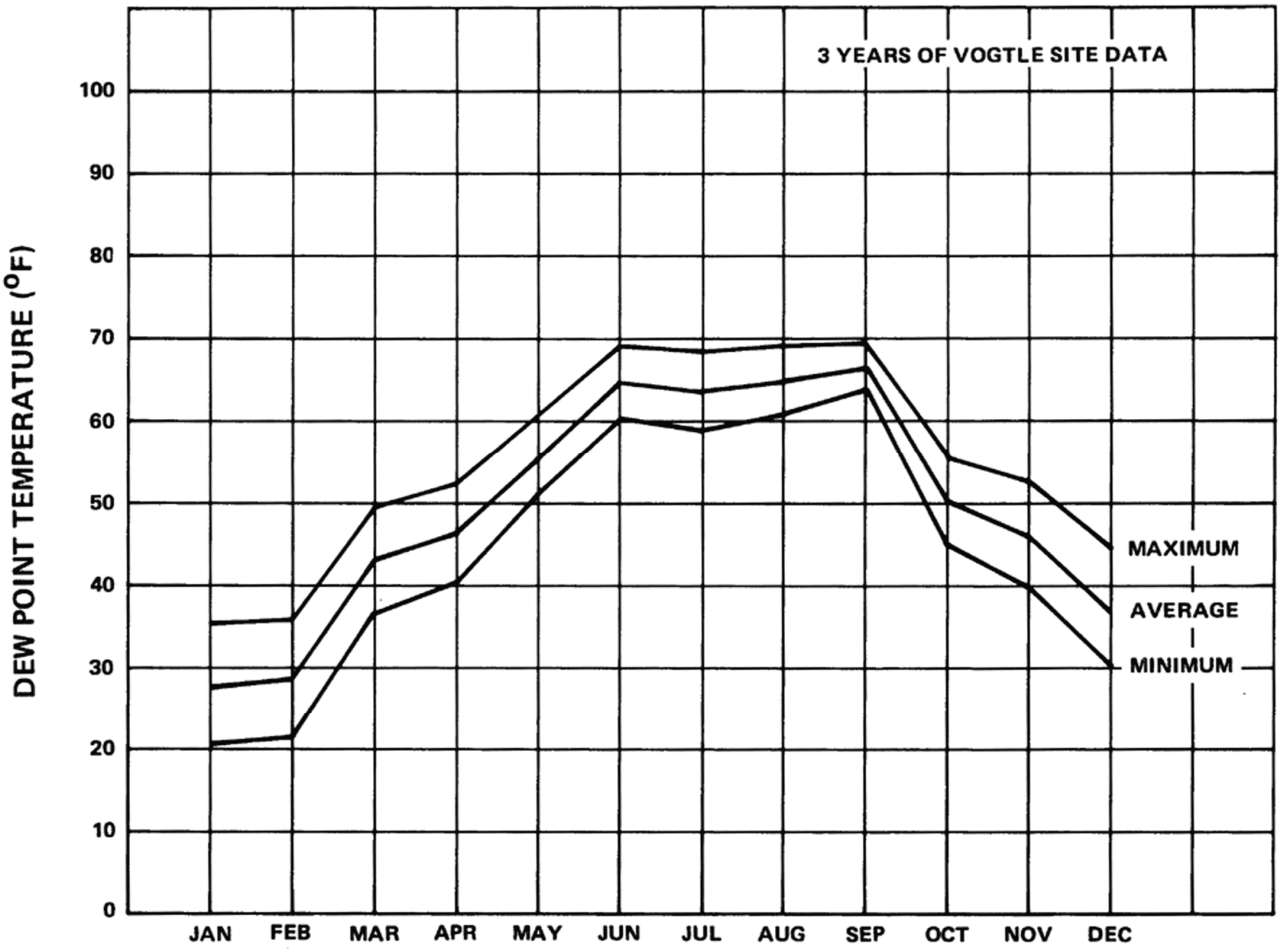
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND AVERAGE
OF DAILY EXTREMES
OF WET BULB TEMPERATURE

FIGURE 2.3.2–32



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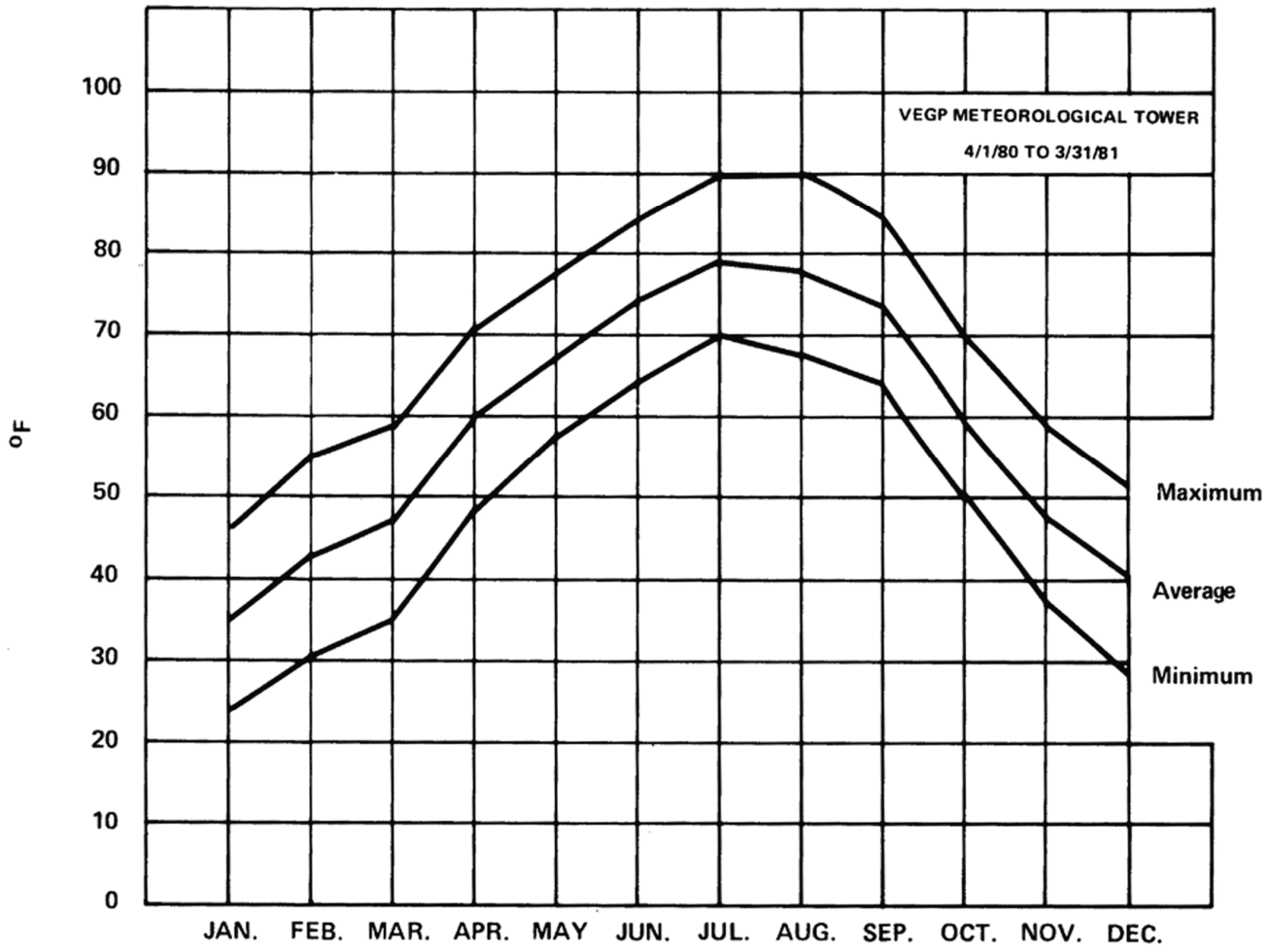
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE DAILY EXTREMES OF
DEWPOINT TEMPERATURE

FIGURE 2.3.2–33



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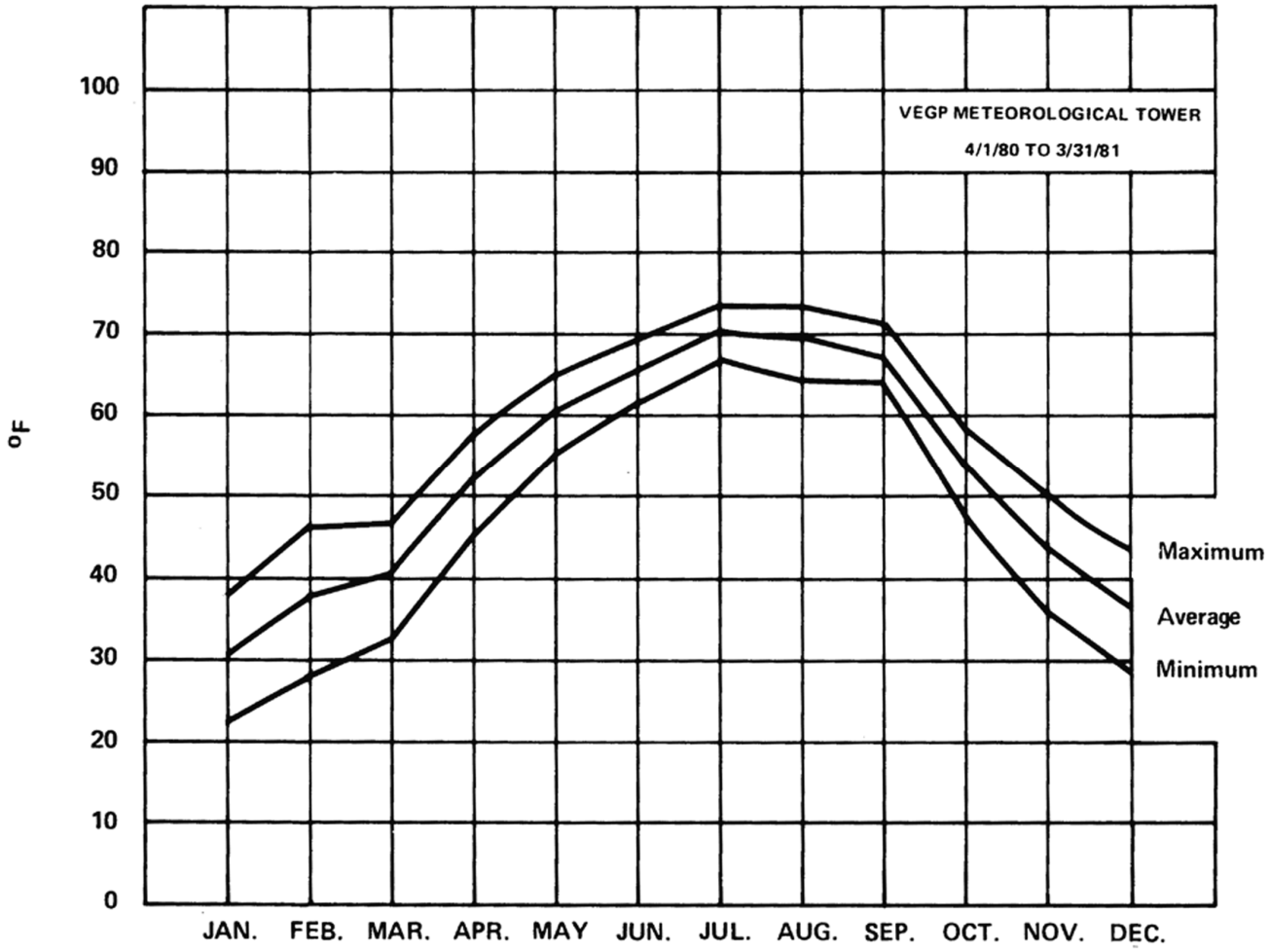
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE DAILY EXTREMES OF
DRY BULB TEMPERATURE

FIGURE 2.3.2–34



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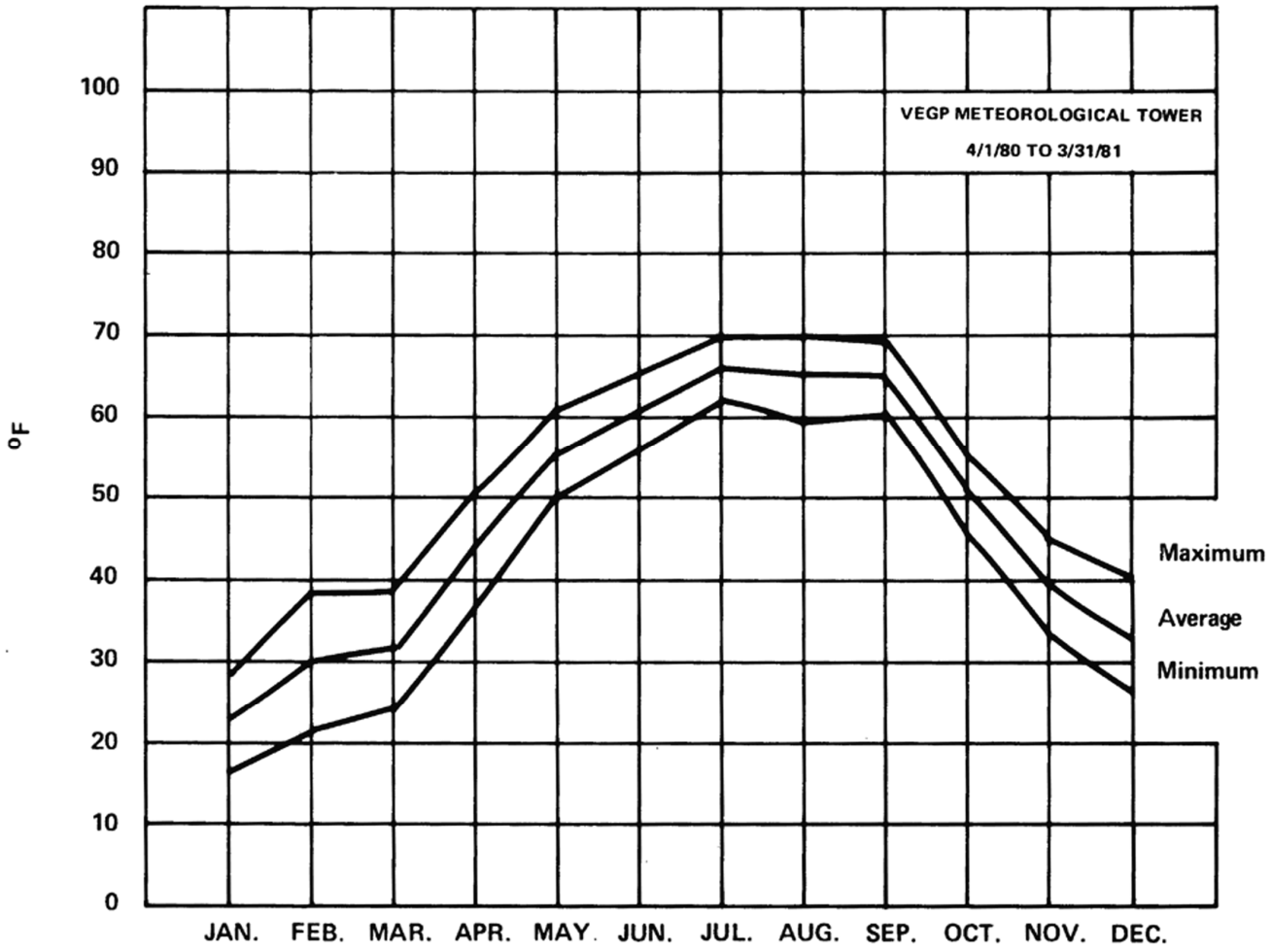
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE DAILY EXTREMES OF
WET BULB TEMPERATURE

FIGURE 2.3.2–35



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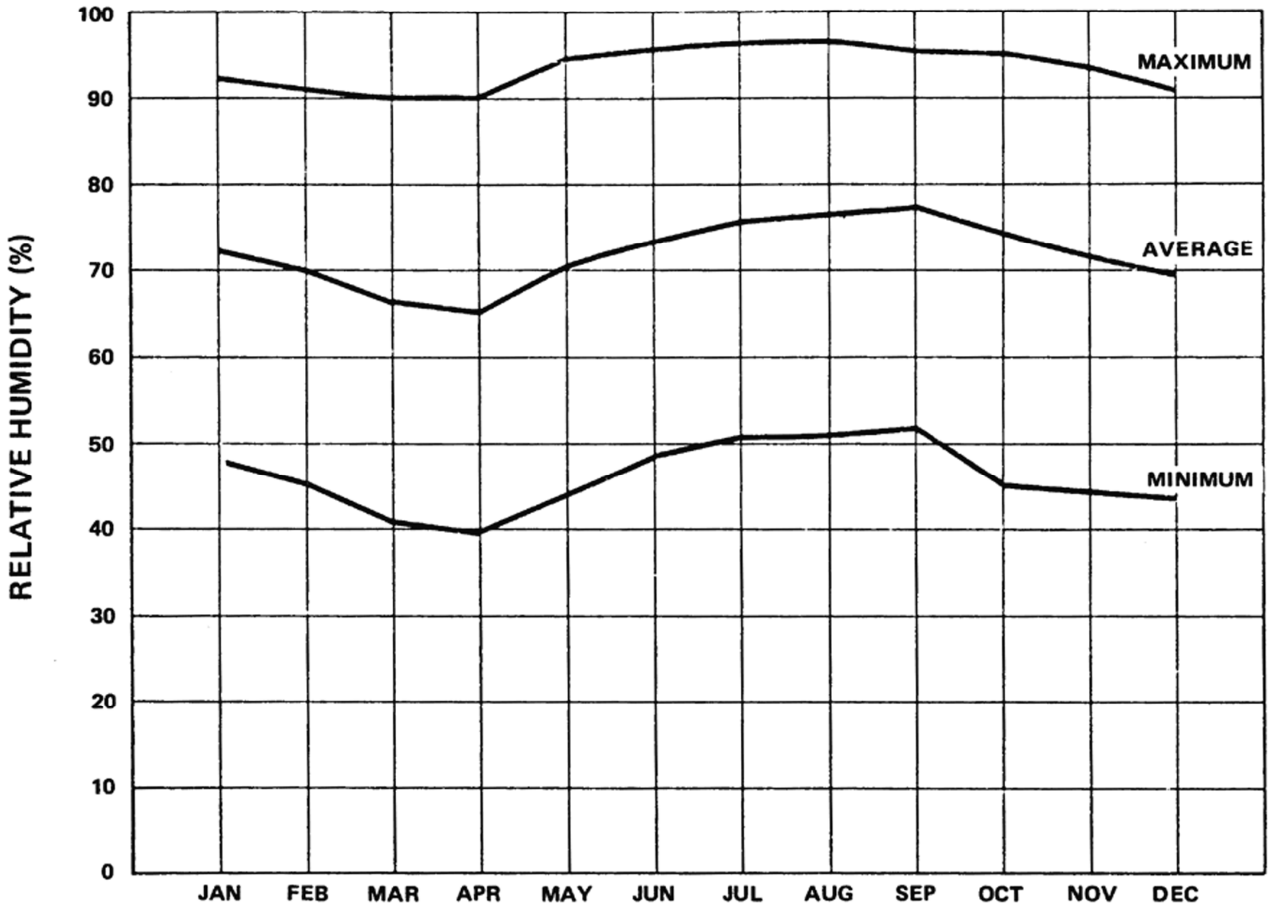
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE DAILY EXTREMES OF
DEWPOINT TEMPERATURE

FIGURE 2.3.2–36

AUGUSTA, GEORGIA
1959 TO 1963



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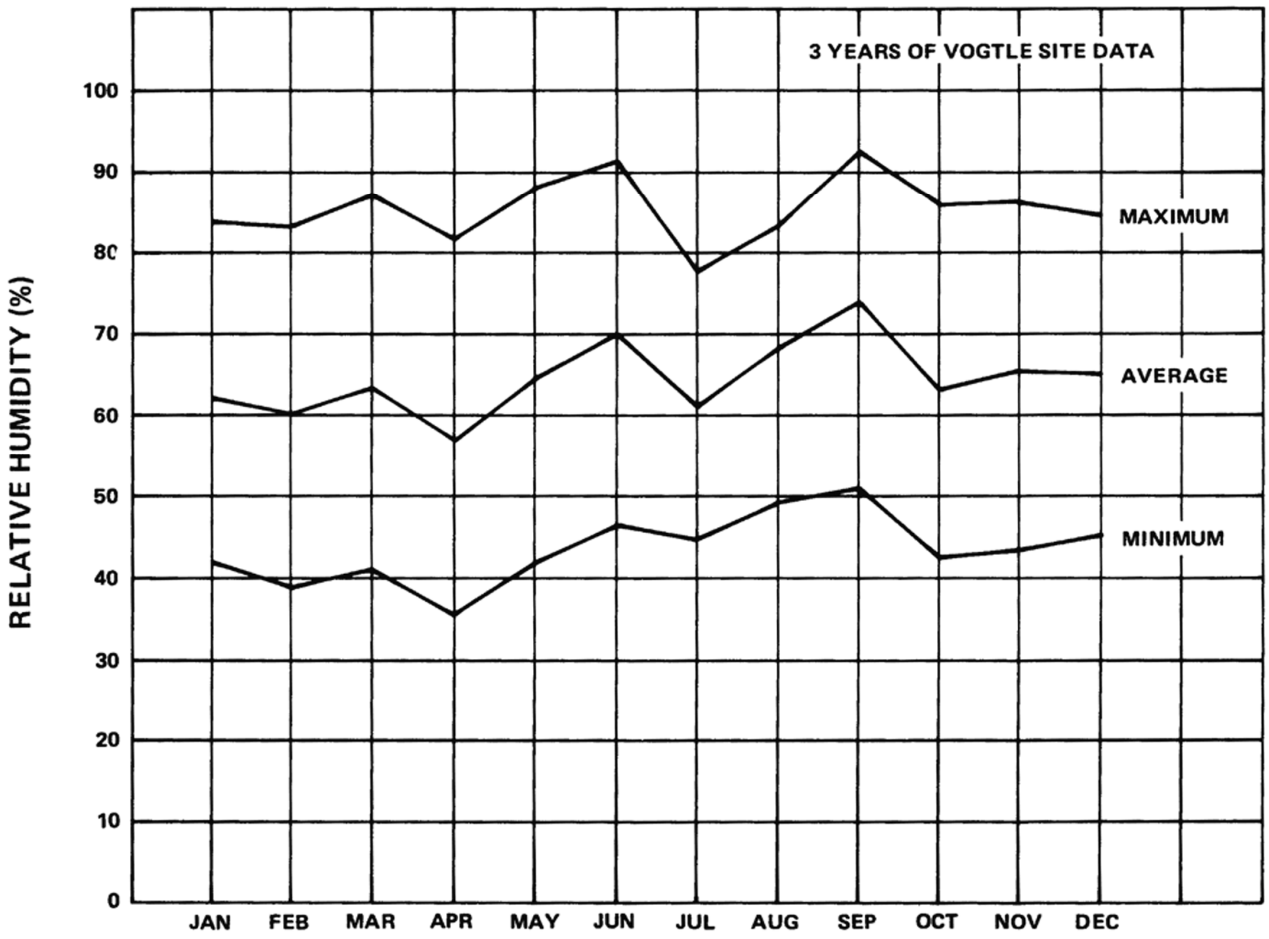
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE OF DAILY EXTREMES
OF RELATIVE HUMIDITY

FIGURE 2.3.2–37



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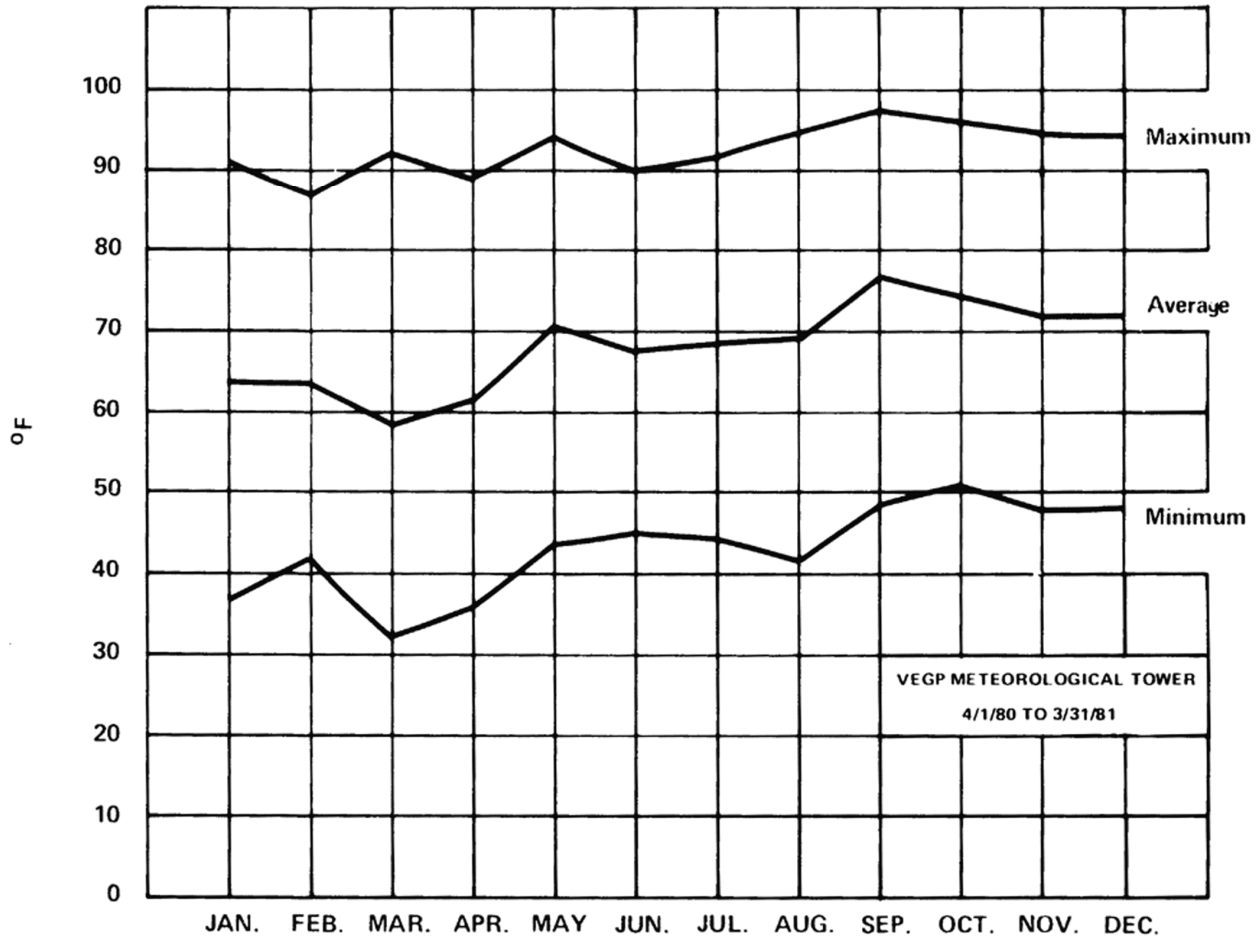
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND AVERAGE
OF DAILY EXTREMES
OF RELATIVE HUMIDITY

FIGURE 2.3.2–38



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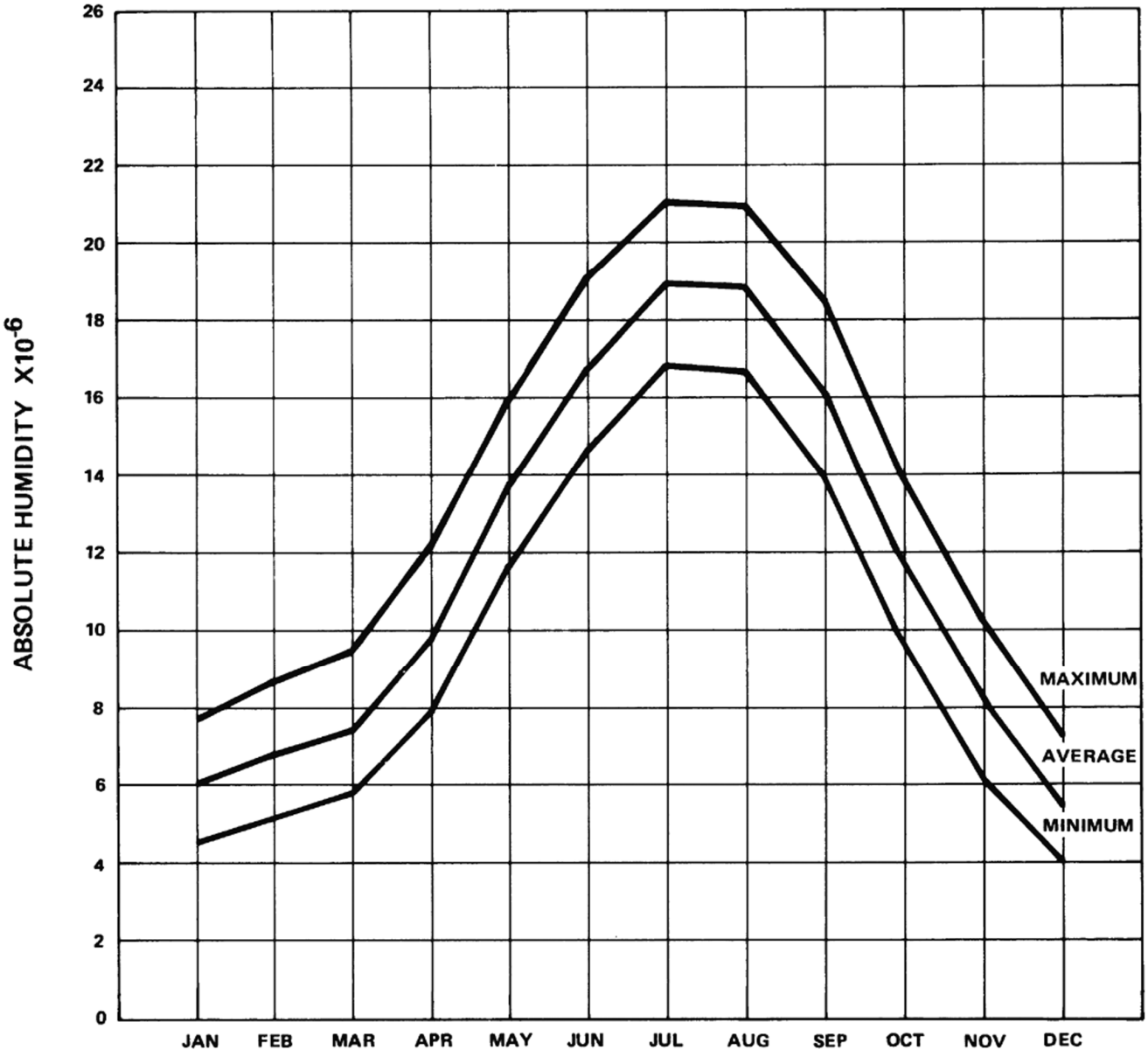
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE DAILY EXTREMES OF
RELATIVE HUMIDITY

FIGURE 2.3.2–39

AUGUSTA, GEORGIA
1959 TO 1963



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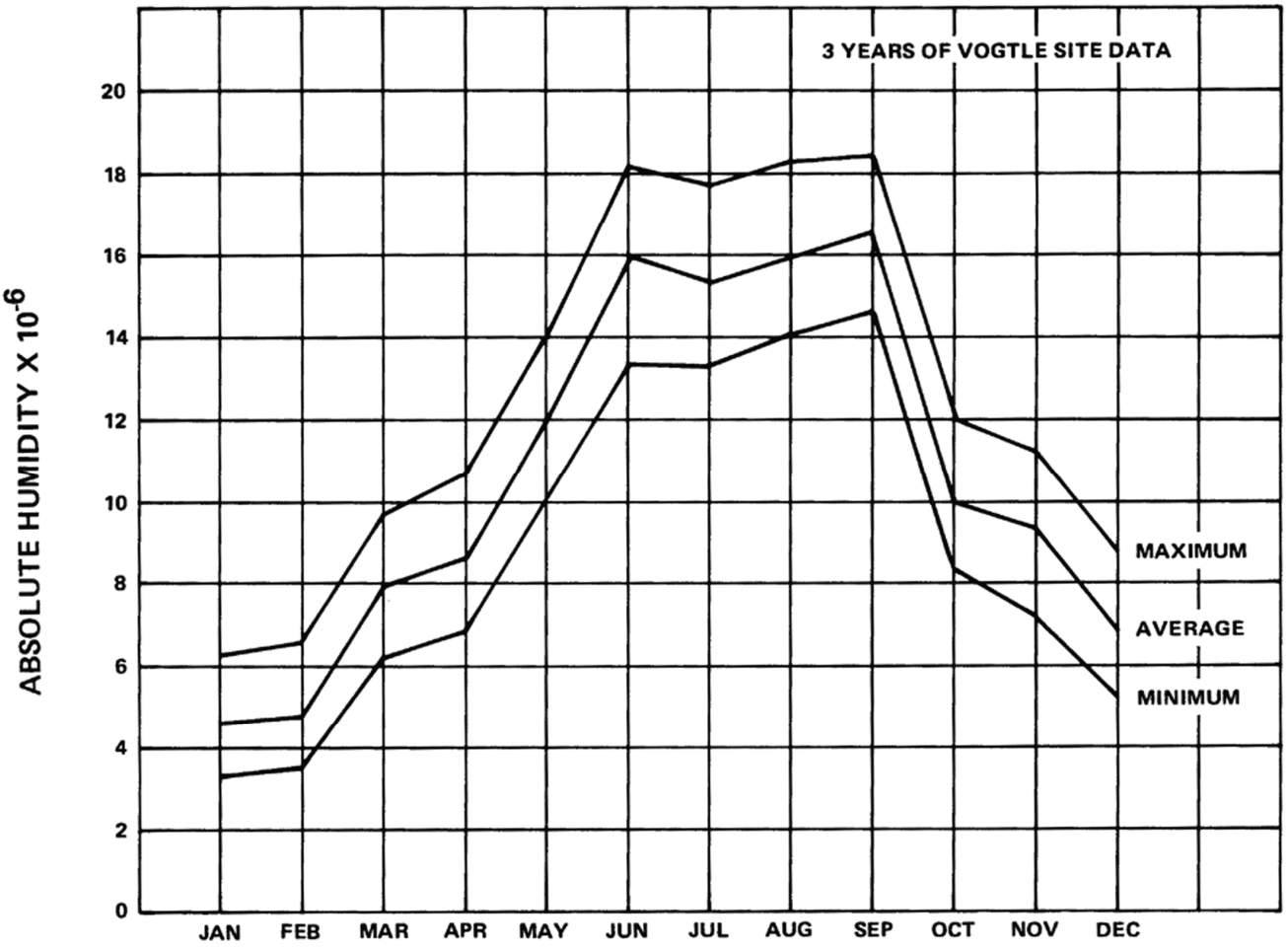
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE OF DAILY EXTREMES
OF ABSOLUTE HUMIDITY

FIGURE 2.3.2–40



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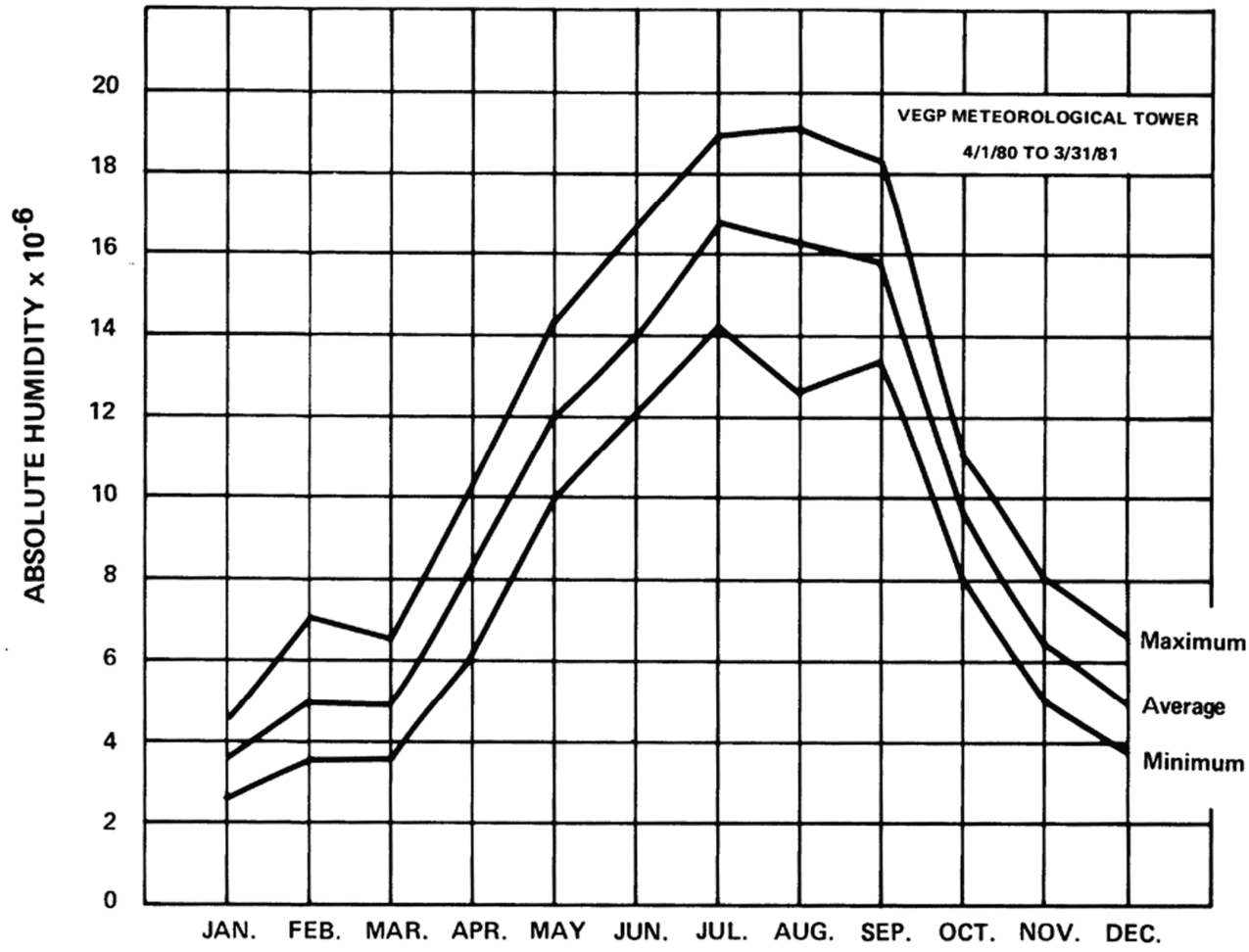
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND AVERAGE
OF DAILY EXTREMES
OF ABSOLUTE HUMIDITY

FIGURE 2.3.2–41



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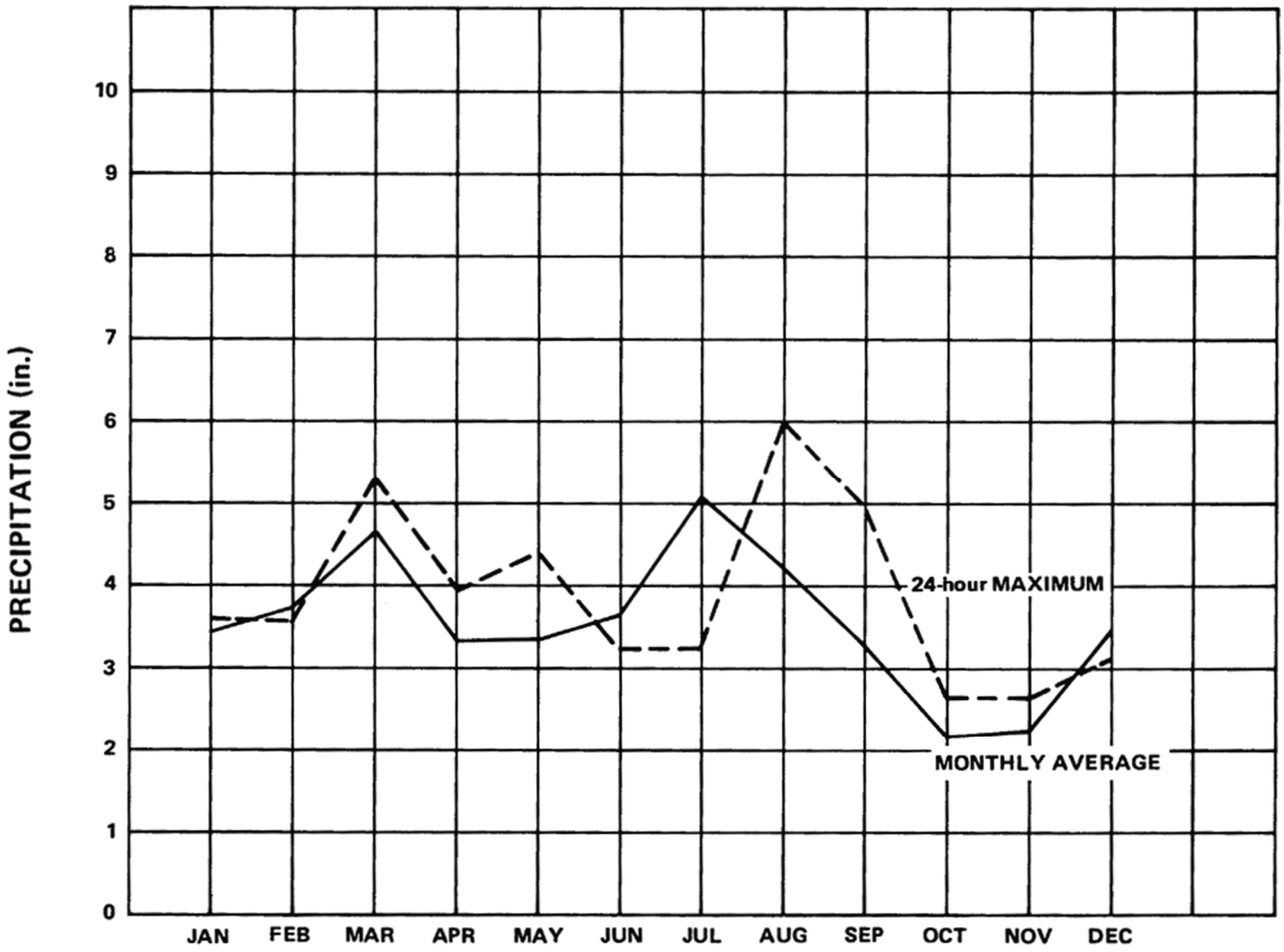
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
AVERAGE DAILY EXTREMES OF
ABSOLUTE HUMIDITY

FIGURE 2.3.2–42



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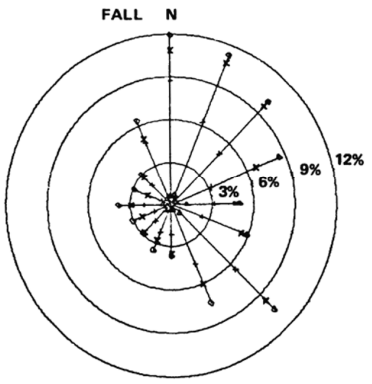
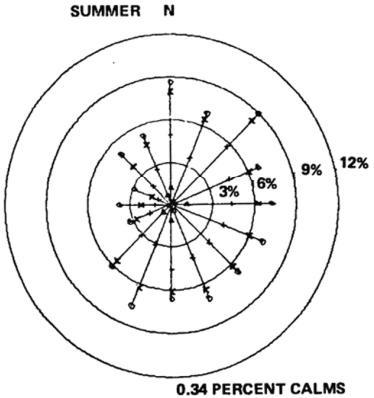
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND
24–HOUR MAXIMUM PRECIPITAION

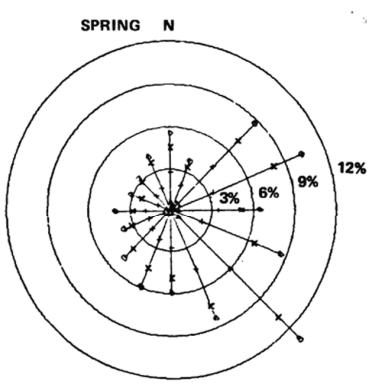
FIGURE 2.3.2–43



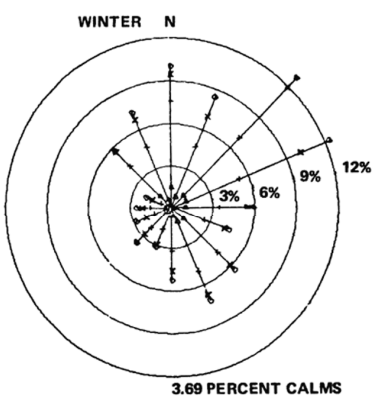
2.13 PERCENT CALMS



0.34 PERCENT CALMS



1.97 PERCENT CALMS



3.69 PERCENT CALMS

NOTES:

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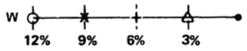
2. ALL WIND ROSES USE UPWIND SECTORS, i.e., INDICATED NORTH MEANS WIND FROM THE NORTH.
3. PRECIPITATION WIND ROSES ARE BASED ON WINDS WHICH OCCURRED SIMULTANEOUSLY WITH PRECIPITATION OF ANY TYPE OR INTENSITY.
4. THE △ , +, X , O ENTRIES ON WIND ROSE RADIALS ARE CUMULATIVE FRACTIONS OF WIND FROM THE PARTICULAR DIRECTION OF THAT RADIAL WHICH HAD SPEED IN GIVEN MPH-RANGE.
   FOR EXAMPLE.



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CALMS - WIND SPEEDS LESS THAN 0.5 mph
△ WIND SPEED LESS THAN 3.5 mph
\+ WIND SPEED LESS THAN 7.5 mph
X WIND SPEED LESS THAN 12.5 mph
O WIND SPEED LESS THAN 999.0 mph

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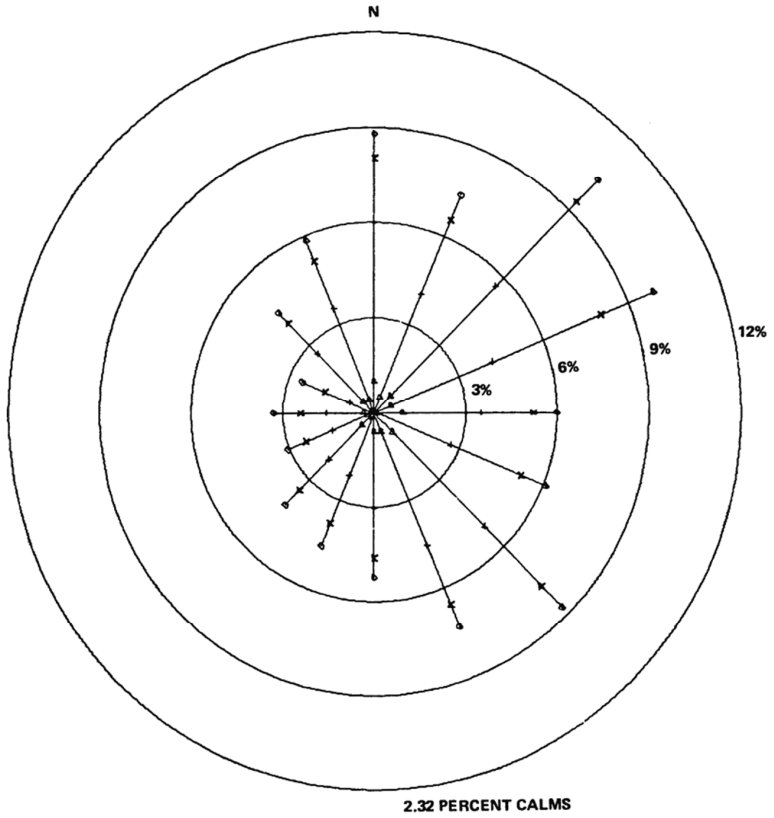
PRECIPITATION WIND ROSES FOR
AUGUSTA AIRPORT (1959 TO 1963)

FIGURE 2.3.2–44



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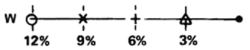
   FOR EXAMPLE.



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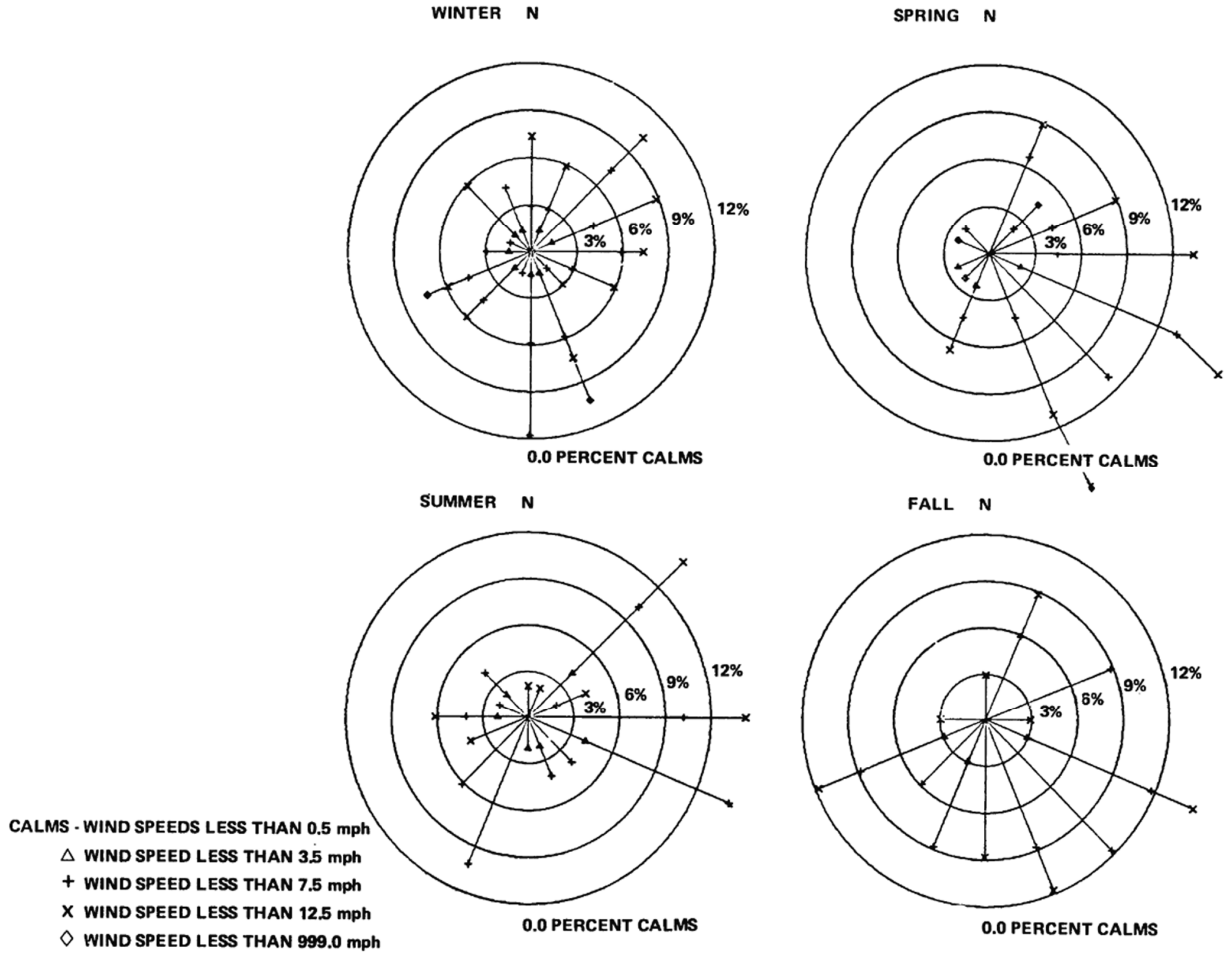
UNIT 1 AND UNIT 2

PRECIPITATION WIND ROSES FOR AUGUSTA AIRPORT (1959 TO 1963)

FIGURE 2.3.2–45



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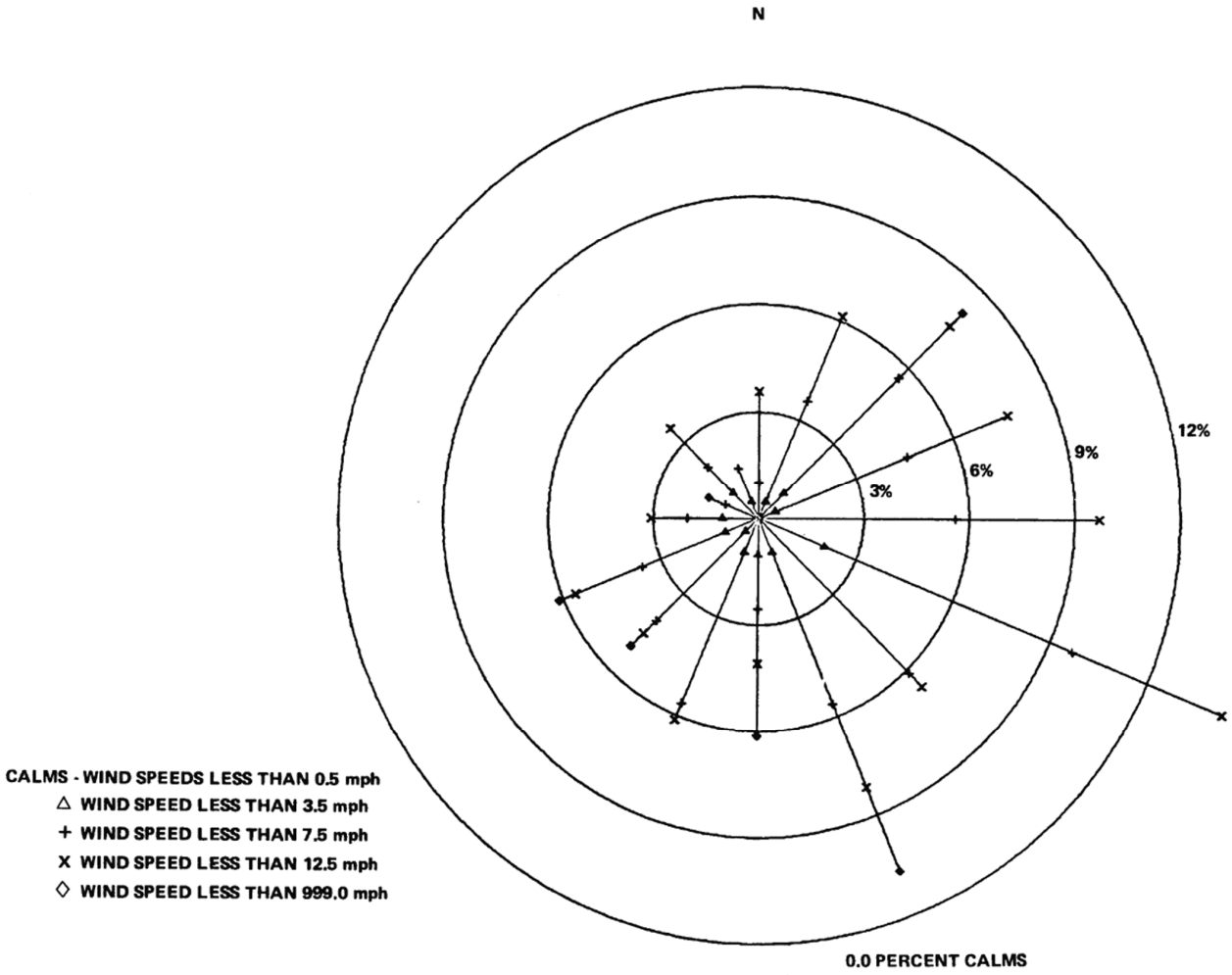
UNIT 1 AND UNIT 2

33–FT LEVEL PRECIPITATION
WIND ROSE 1972 TO 1973

FIGURE 2.3.2–46



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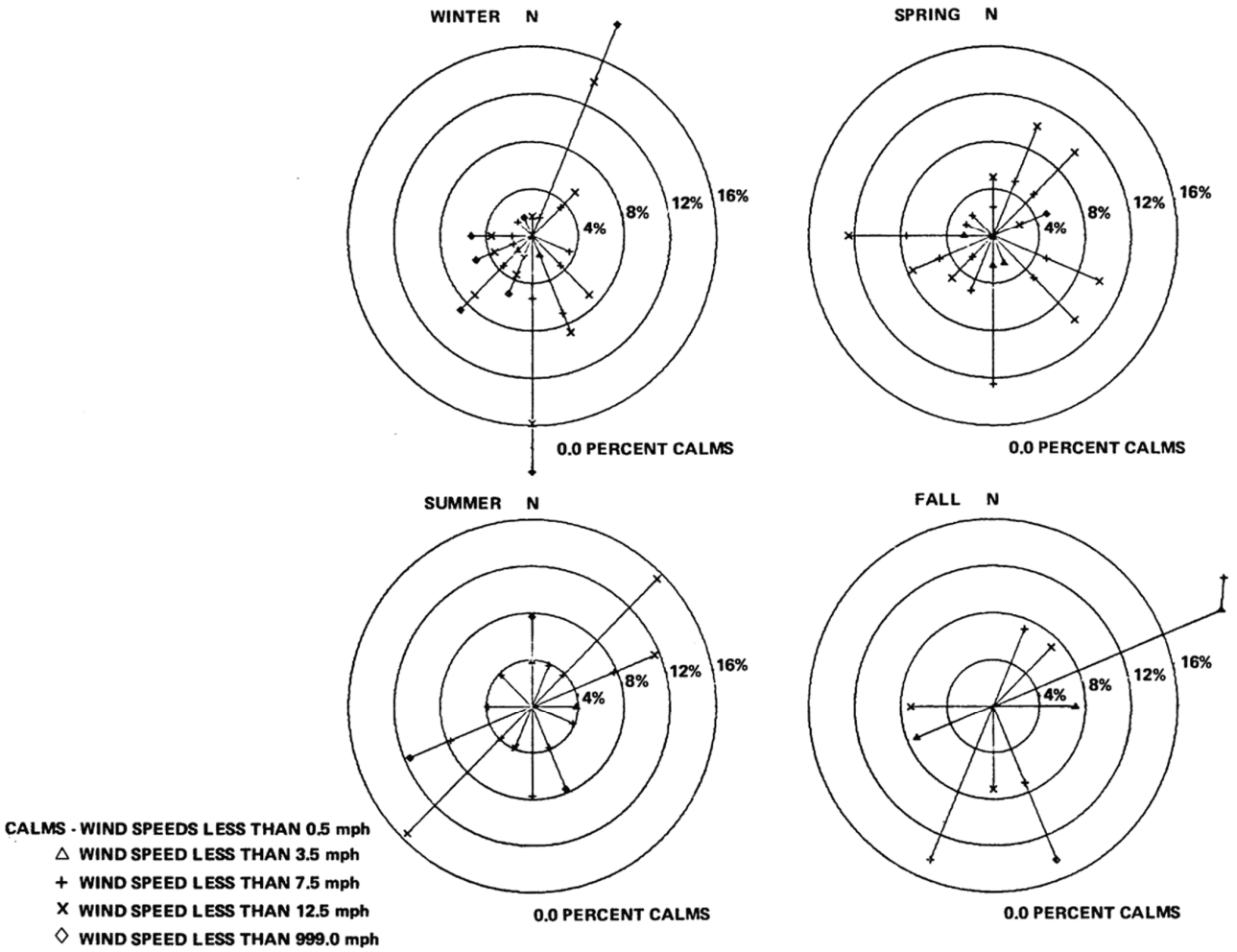
UNIT 1 AND UNIT 2

33–FT LEVEL PRECIPITATION
WIND ROSE 1972 TO 1973

FIGURE 2.3.2–47



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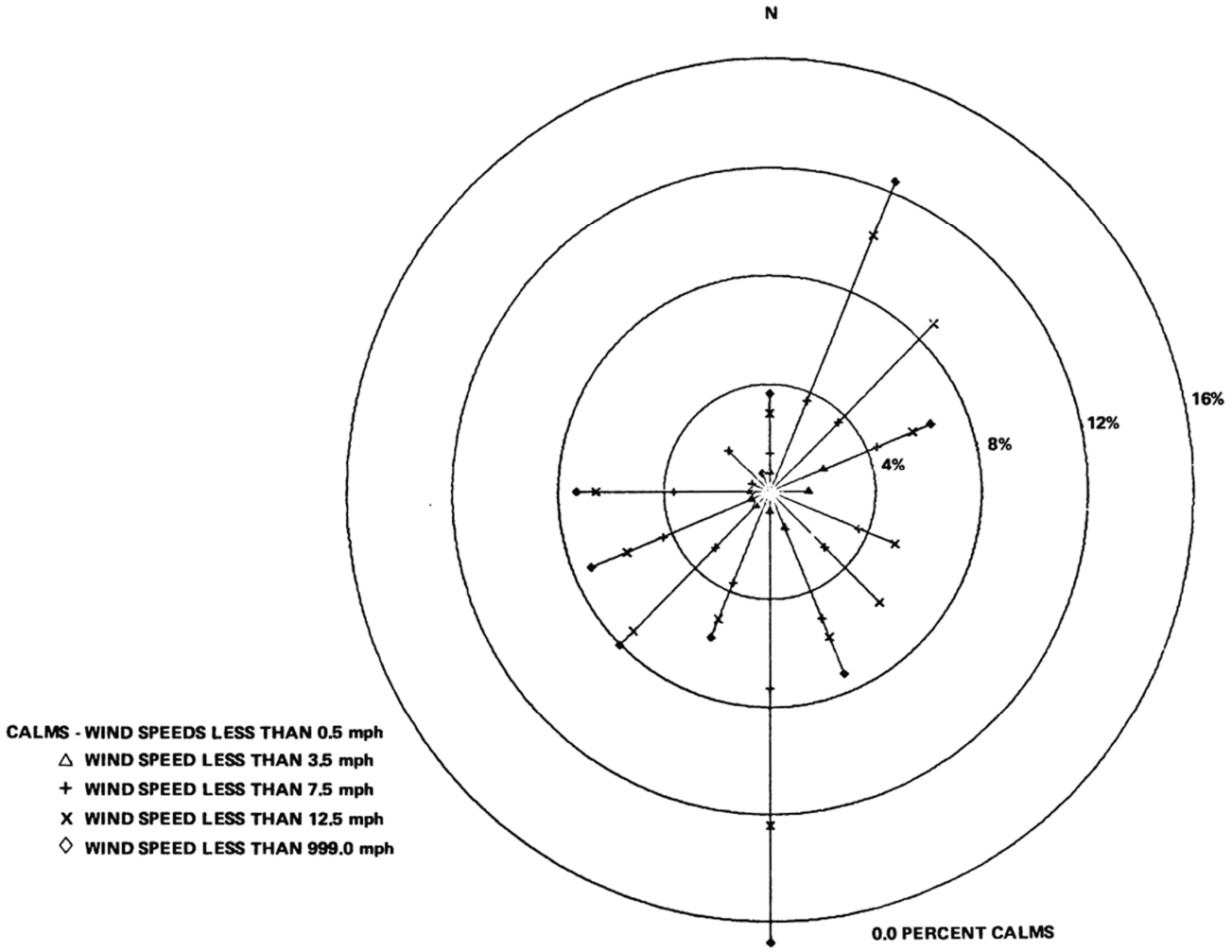
UNIT 1 AND UNIT 2

33–FT LEVEL PRECIPITATION
WIND ROSE 1977 TO 1978

FIGURE 2.3.2–48



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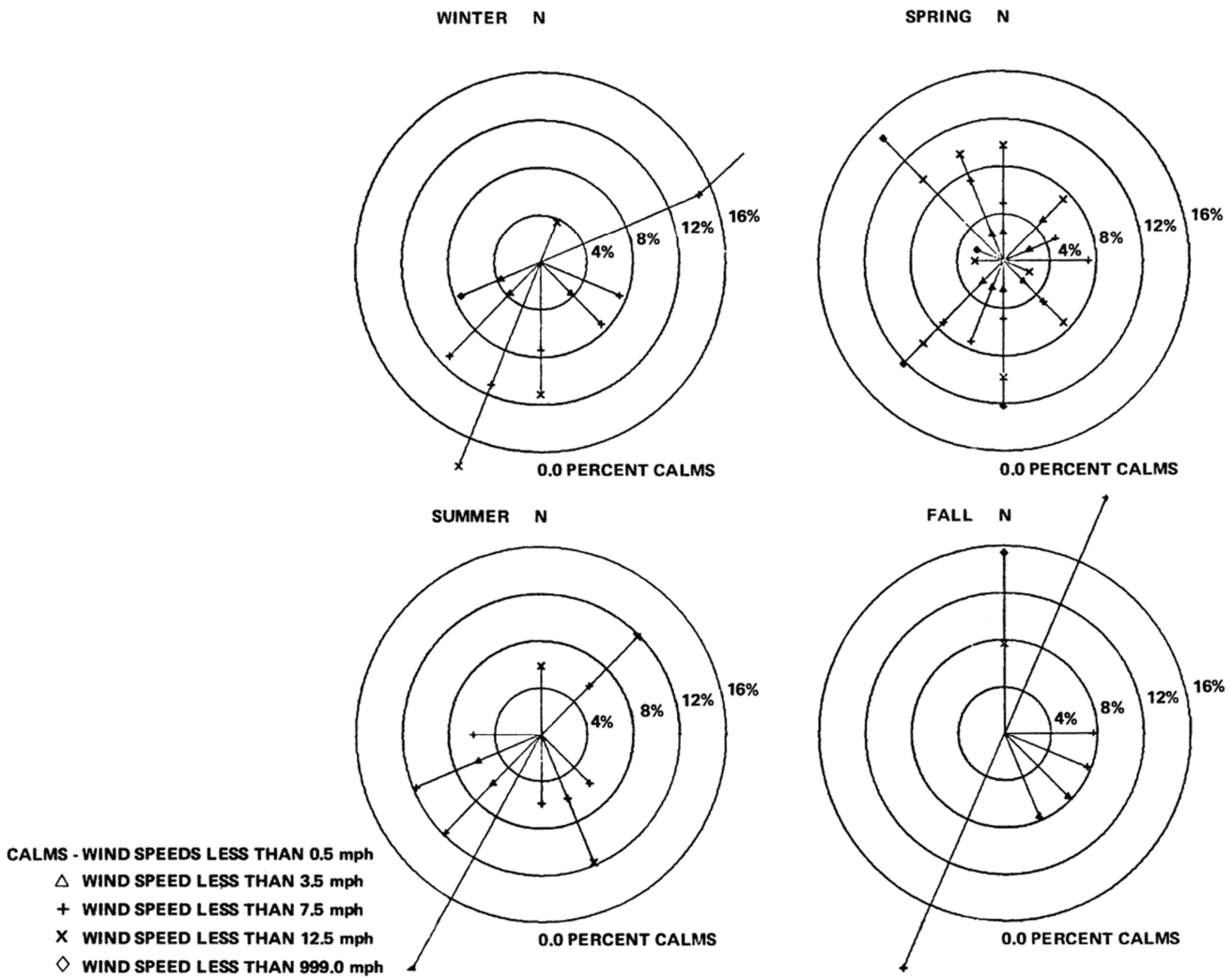
UNIT 1 AND UNIT 2

33–FT LEVEL PRECIPITATION
WIND ROSE 1977 TO 1978

FIGURE 2.3.2–49



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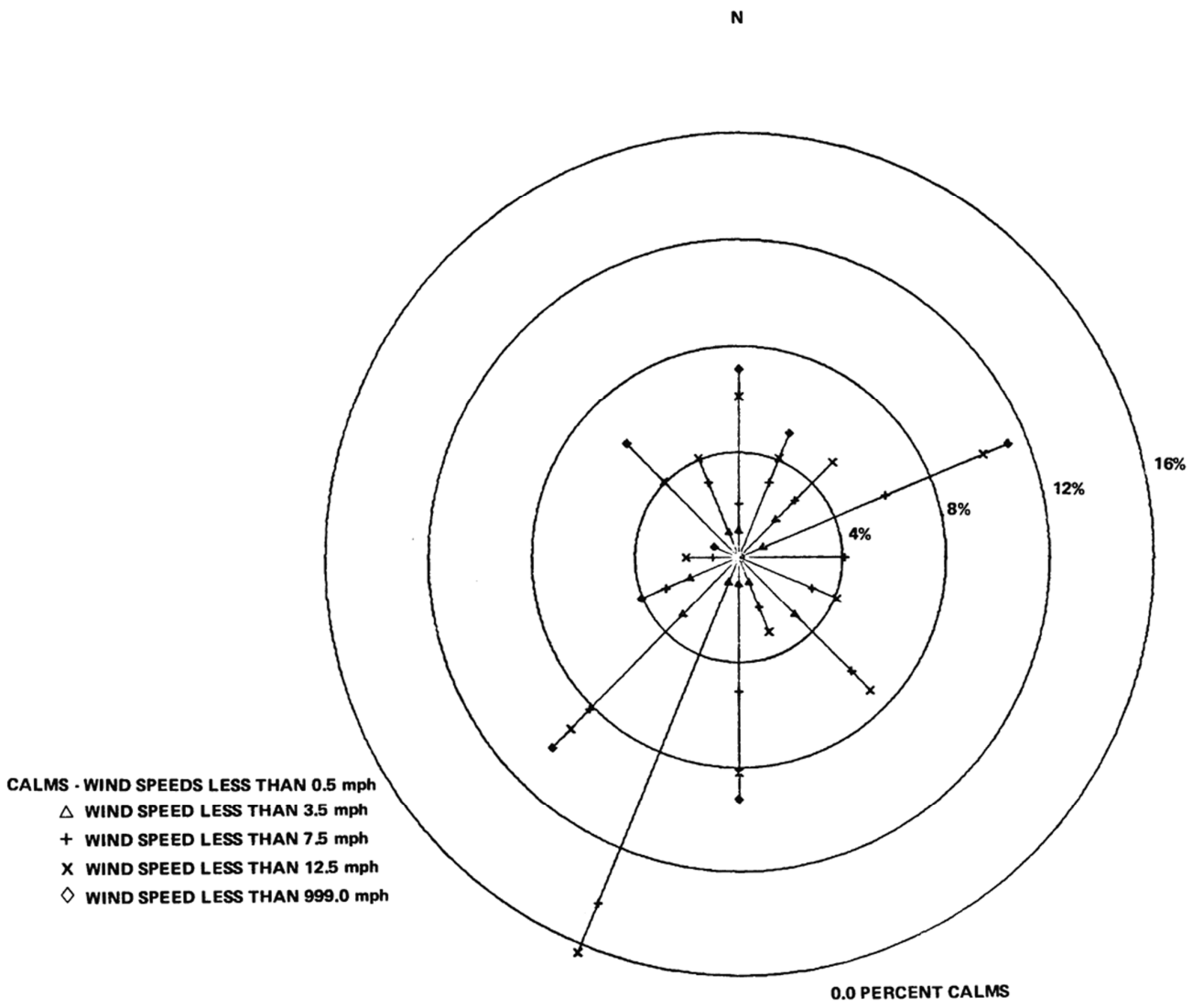
UNIT 1 AND UNIT 2

33–FT LEVEL PRECIPITATION
WIND ROSE 1978 TO 1979

FIGURE 2.3.2–50



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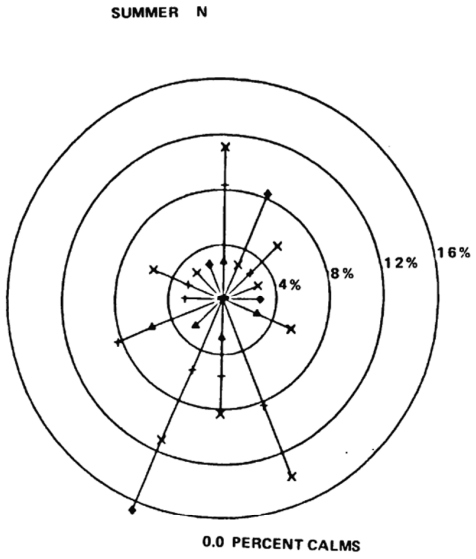
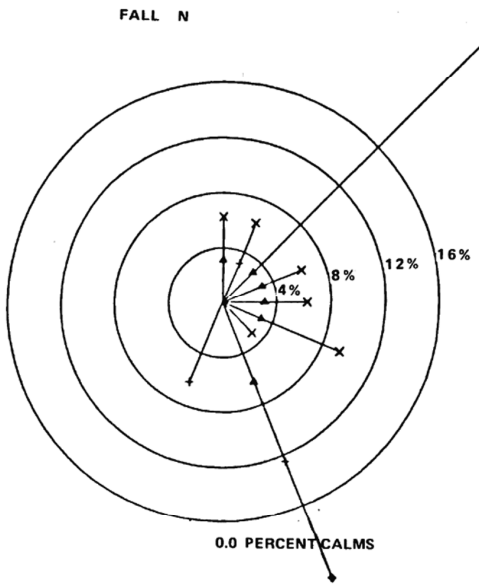
UNIT 1 AND UNIT 2

33–FT LEVEL PRECIPITATION
WIND ROSE 1978 TO 1979

FIGURE 2.3.2–51





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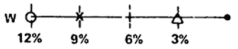

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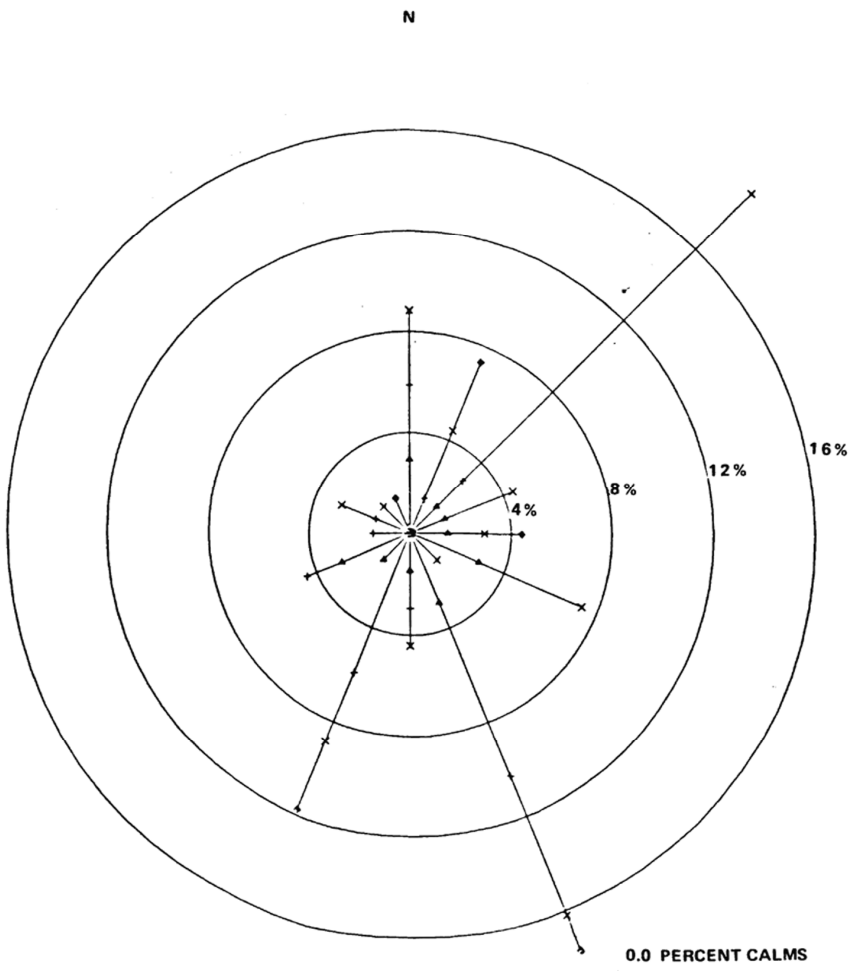
PRECIPITATION WIND ROSE
SEASONAL 1980 TO 1981

FIGURE 2.3.2–52



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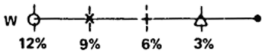
   FOR EXAMPLE.



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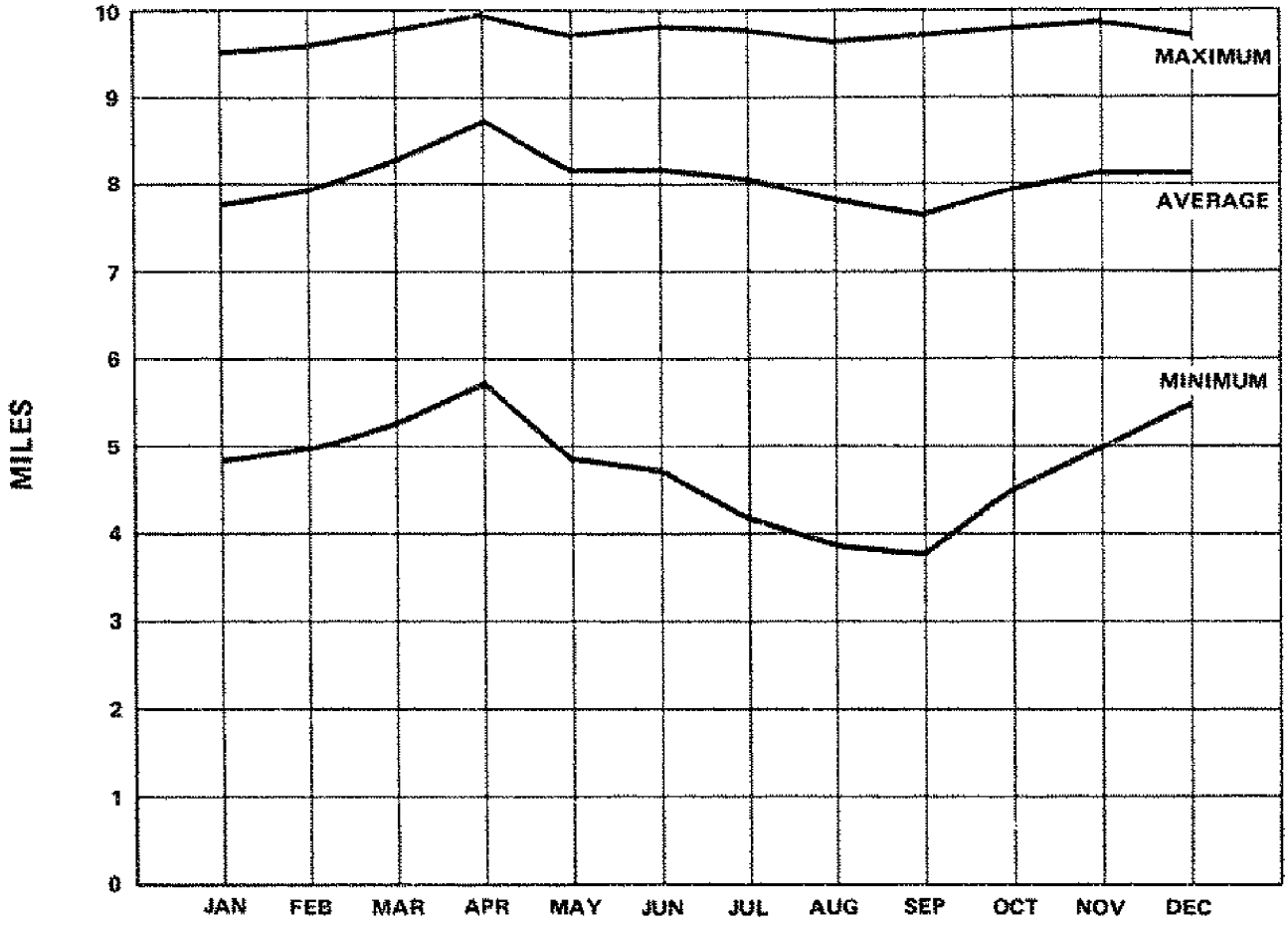
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRECIPITATION WIND ROSE
ANNUAL 1980 TO 1981

FIGURE 2.3.2–53



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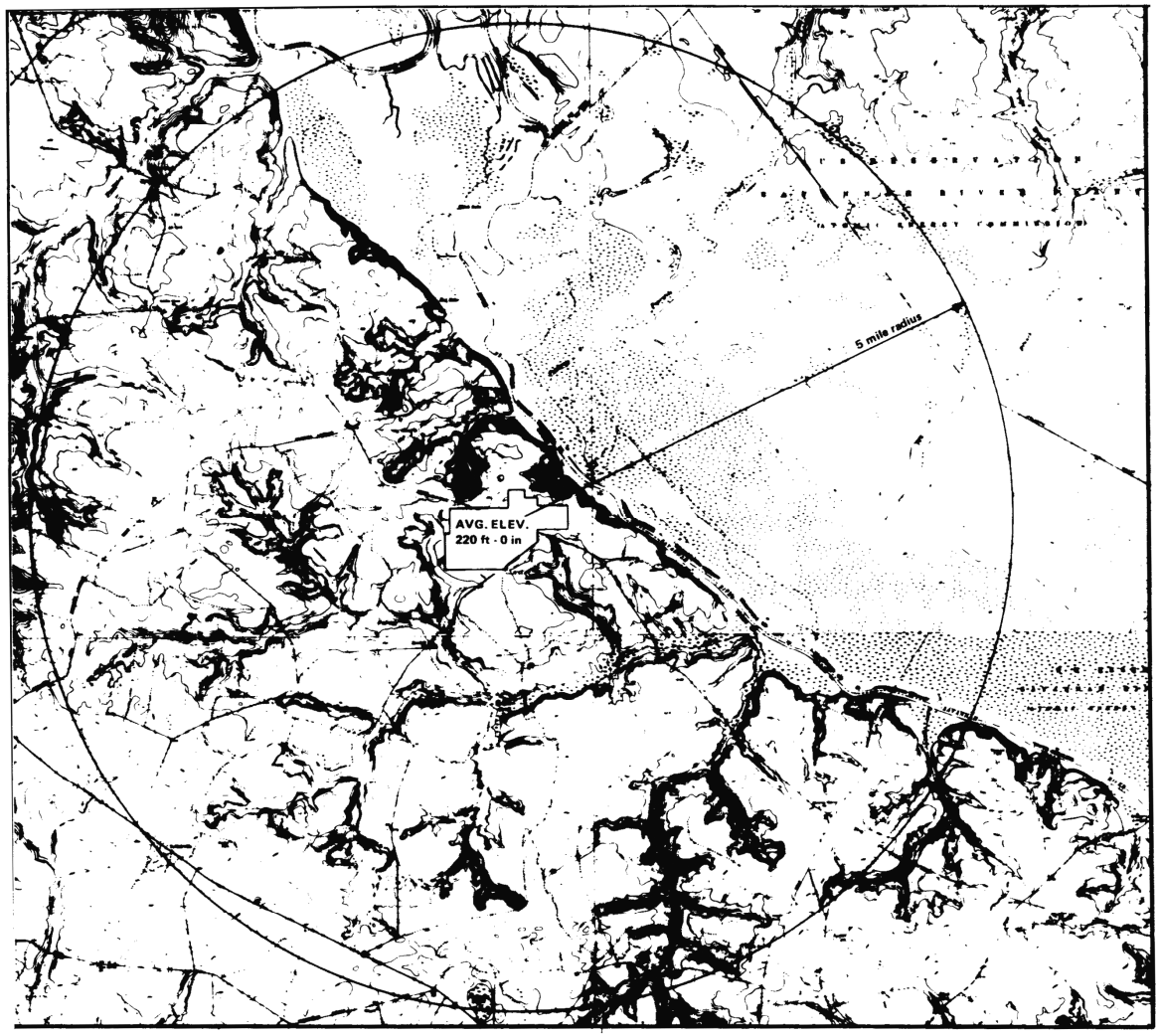
UNIT 1 AND UNIT 2

MONTHLY AVERAGE AND AVERAGES OF DAILY EXTREMES OF VISIBILITY

FIGURE 2.3.2–54



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UNIT 1 AND UNIT 2

TOPOGRAPHIC FEATURES
0– TO 5– MILE RADIUS FROM PLANT SITE

FIGURE 2.3.2–55

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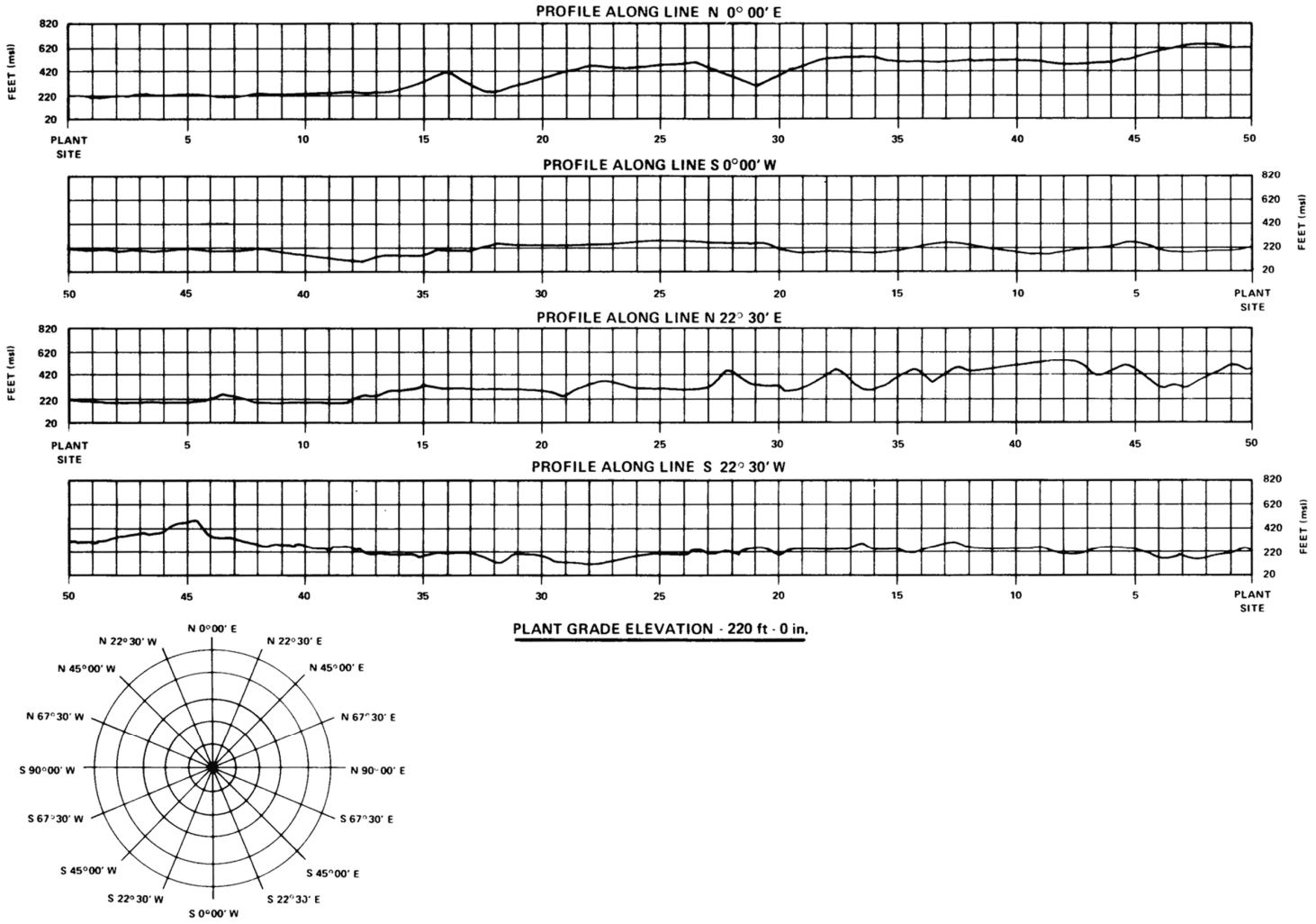
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TOPOGRAPHIC FEATURES
0– TO 5– MILE RADIUS FROM PLANT SITE

FIGURE 2.3.2–56



PLANT GRADE ELEVATION - 220 ft - 0 in.

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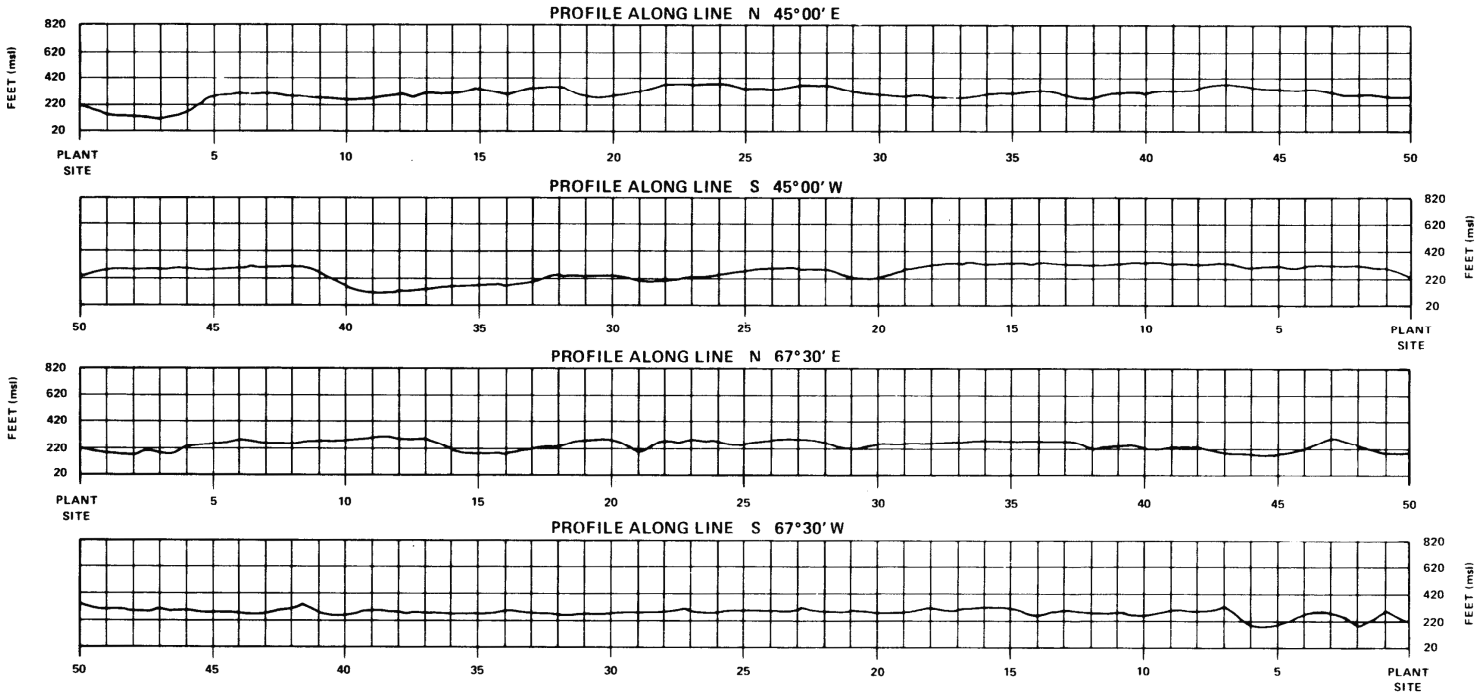
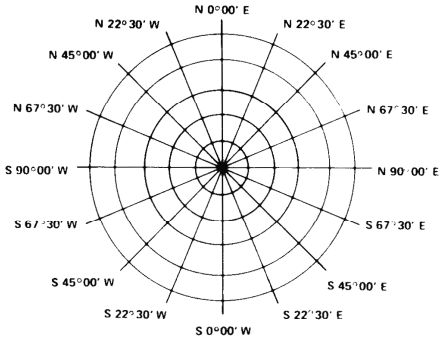
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TOPOGRAPHIC FEATURES

FIGURE 2.3.2–57 (SHEET 1 OF 4)



PLANT GRADE ELEVATION - 220 ft - 0 in.



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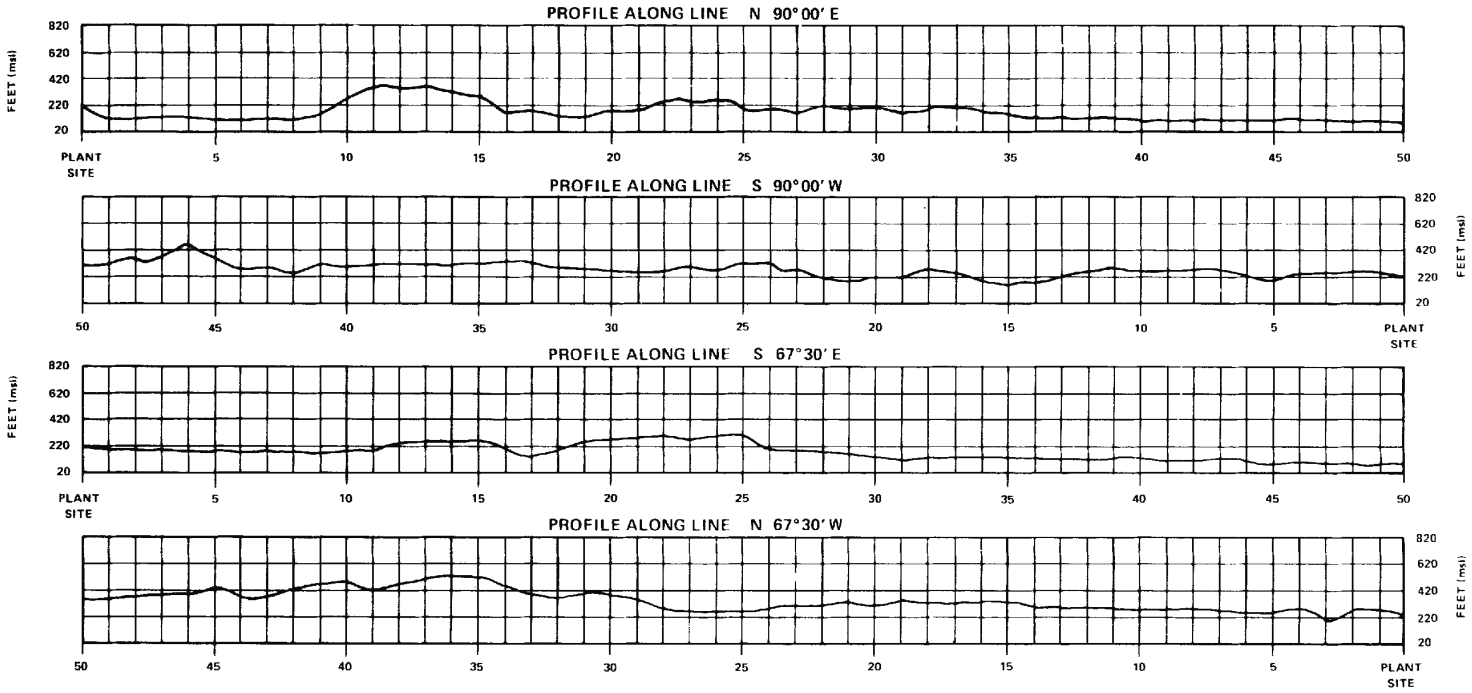
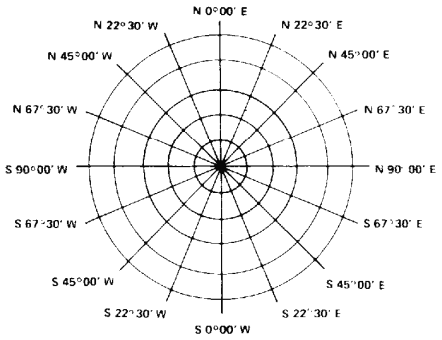
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TOPOGRAPHIC FEATURES

FIGURE 2.3.2–57 (SHEET 2 OF 4)



PLANT GRADE ELEVATION - 220 ft - 0 in.



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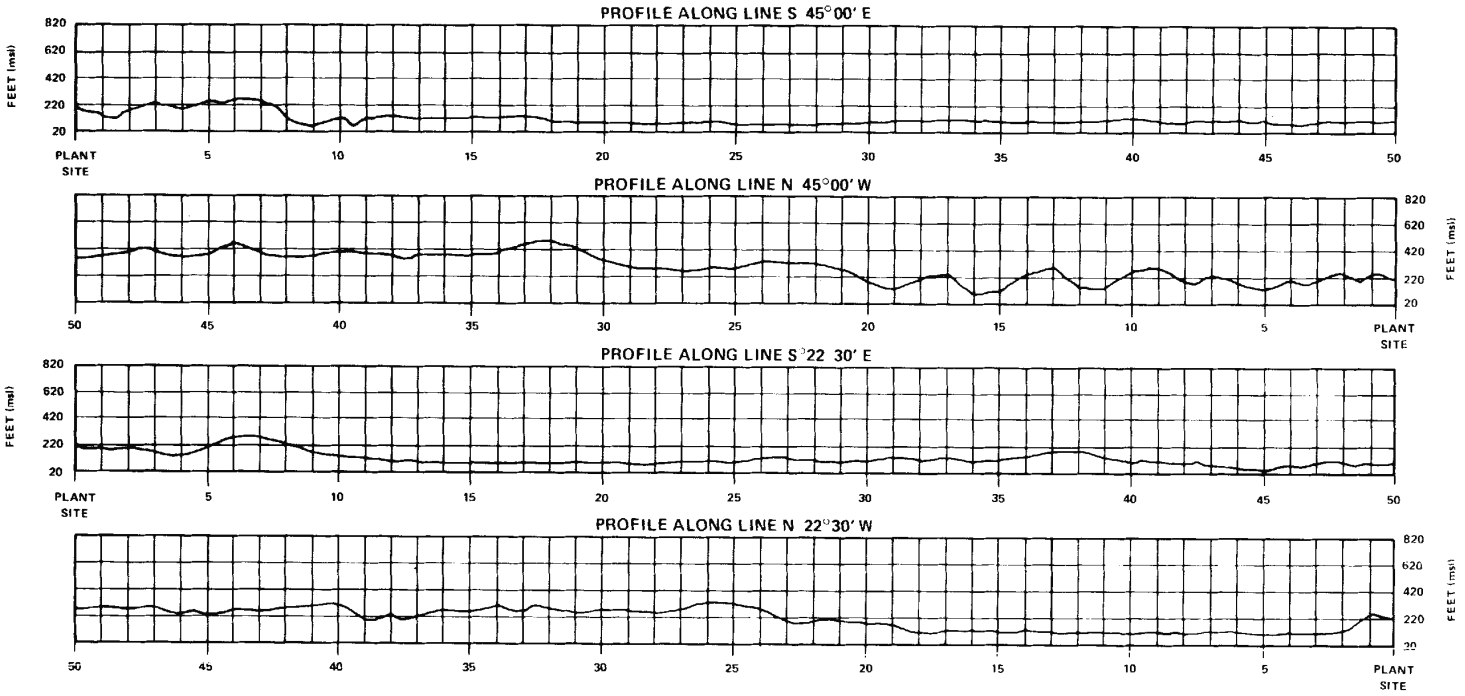
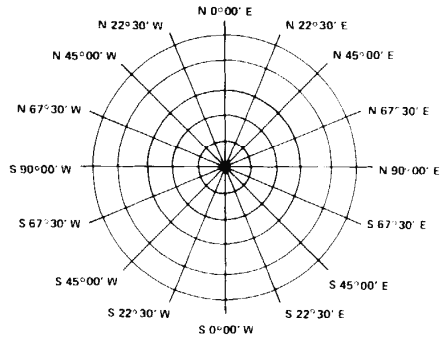
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TOPOGRAPHIC FEATURES

FIGURE 2.3.2–57 (SHEET 3 OF 4)





PLANT GRADE ELEVATION - 220 ft - 0 in.

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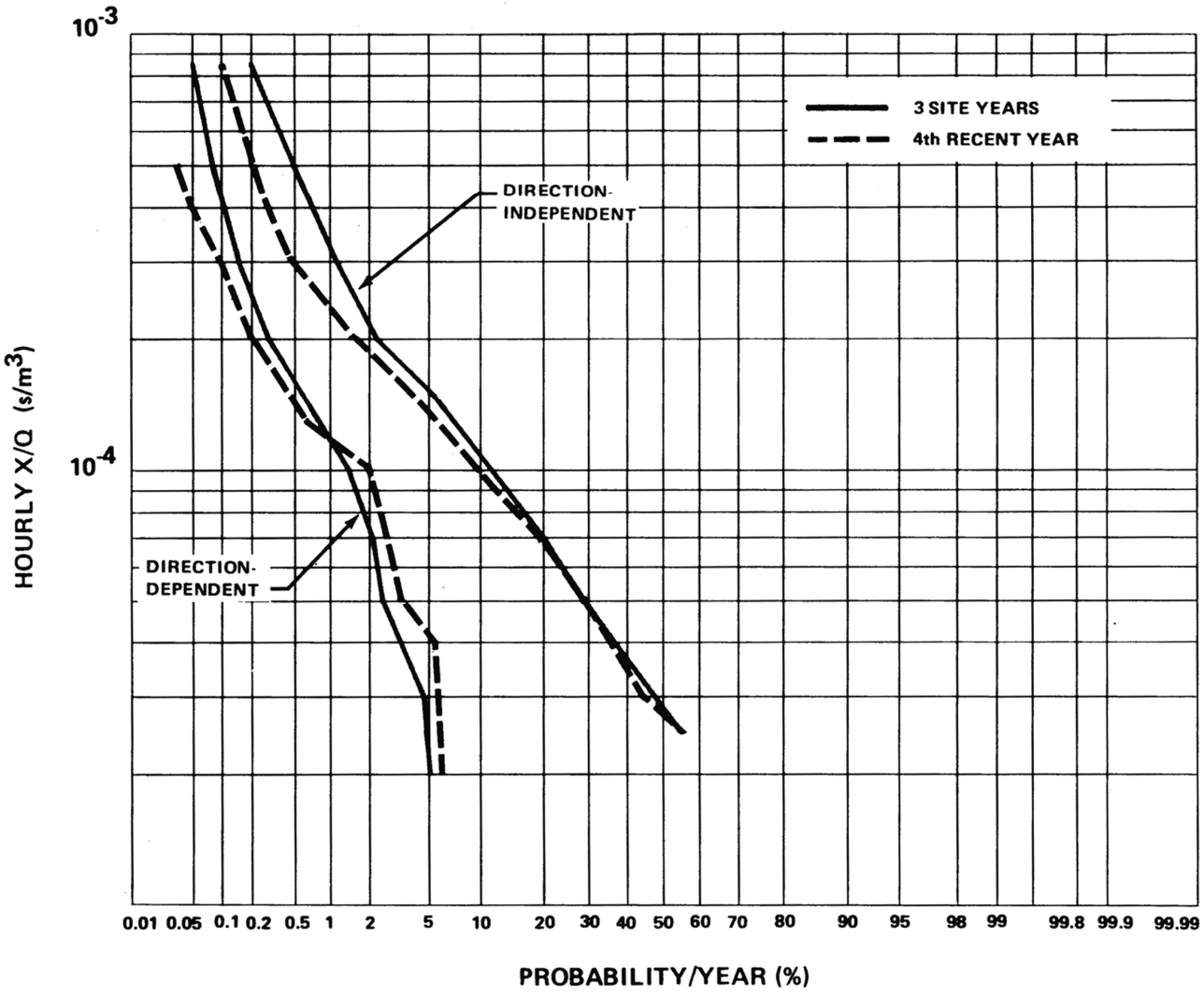
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TOPOGRAPHIC FEATURES

FIGURE 2.3.2–57 (SHEET 4 OF 4)



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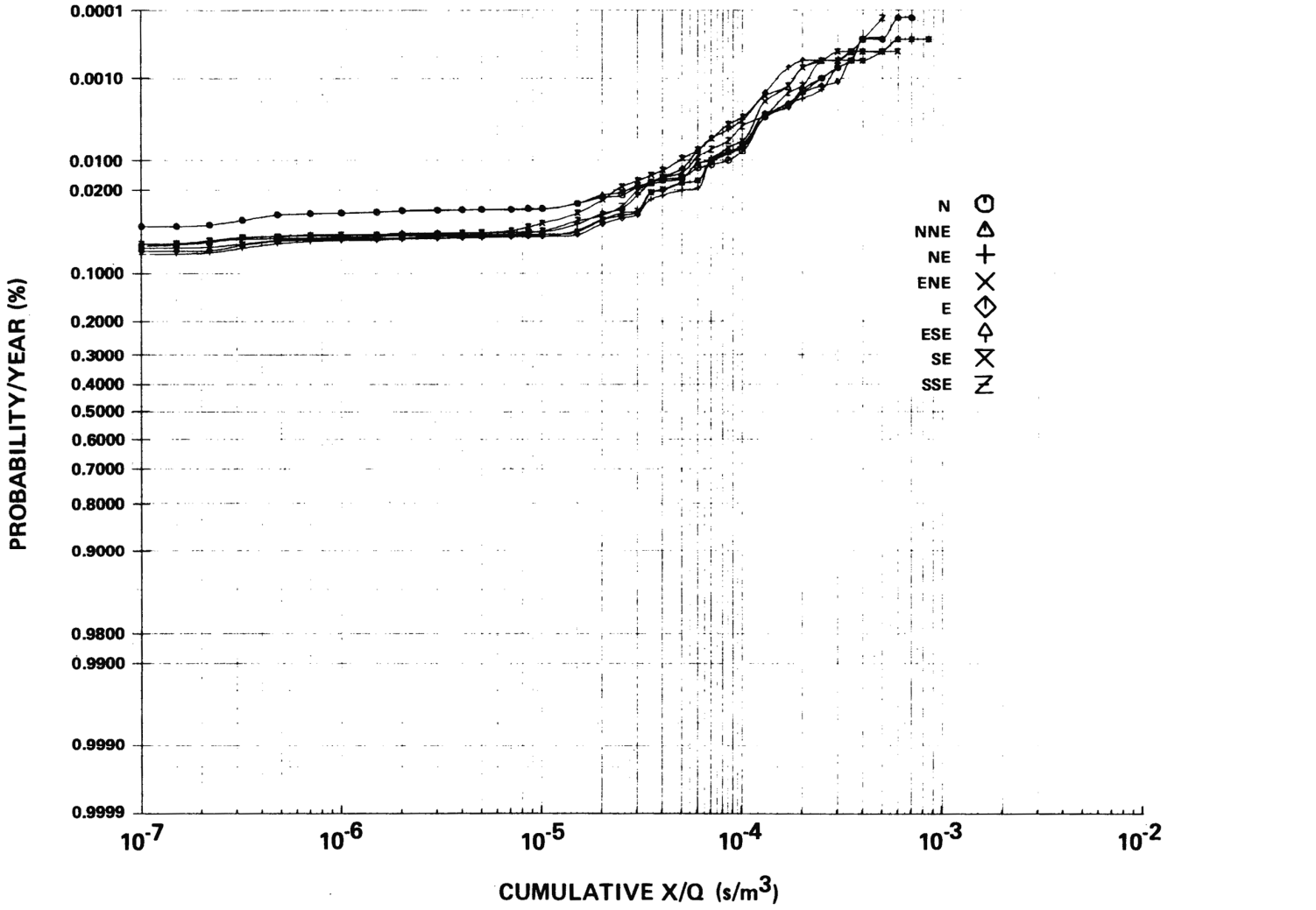
UNIT 1 AND UNIT 2

CUMULATIVE PROBABILITY OF
HOURLY X/Q VALUES

FIGURE 2.3.4–1



BASED ON VOGTLE SITE DATA (12/72 - 12/73)

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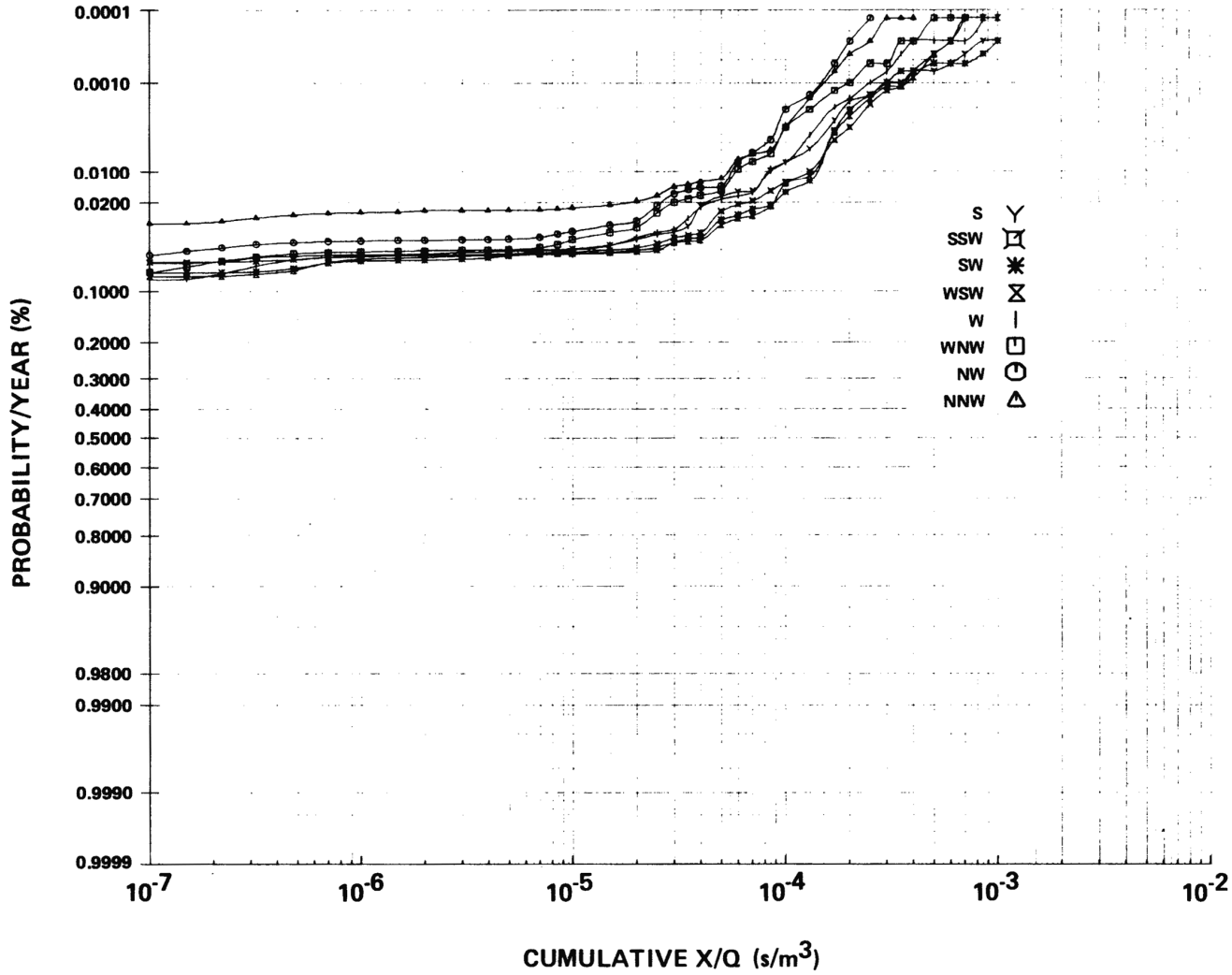
UNIT 1 AND UNIT 2

CUMULATIVE DISTRIBUTION OF X/Q
FOR DIRECTIONS (N THROUGH SSE)

FIGURE 2.3.4–2 (SHEET 1 OF 2)



BASED ON VOGTLE SITE DATA (12/72 - 12/73)

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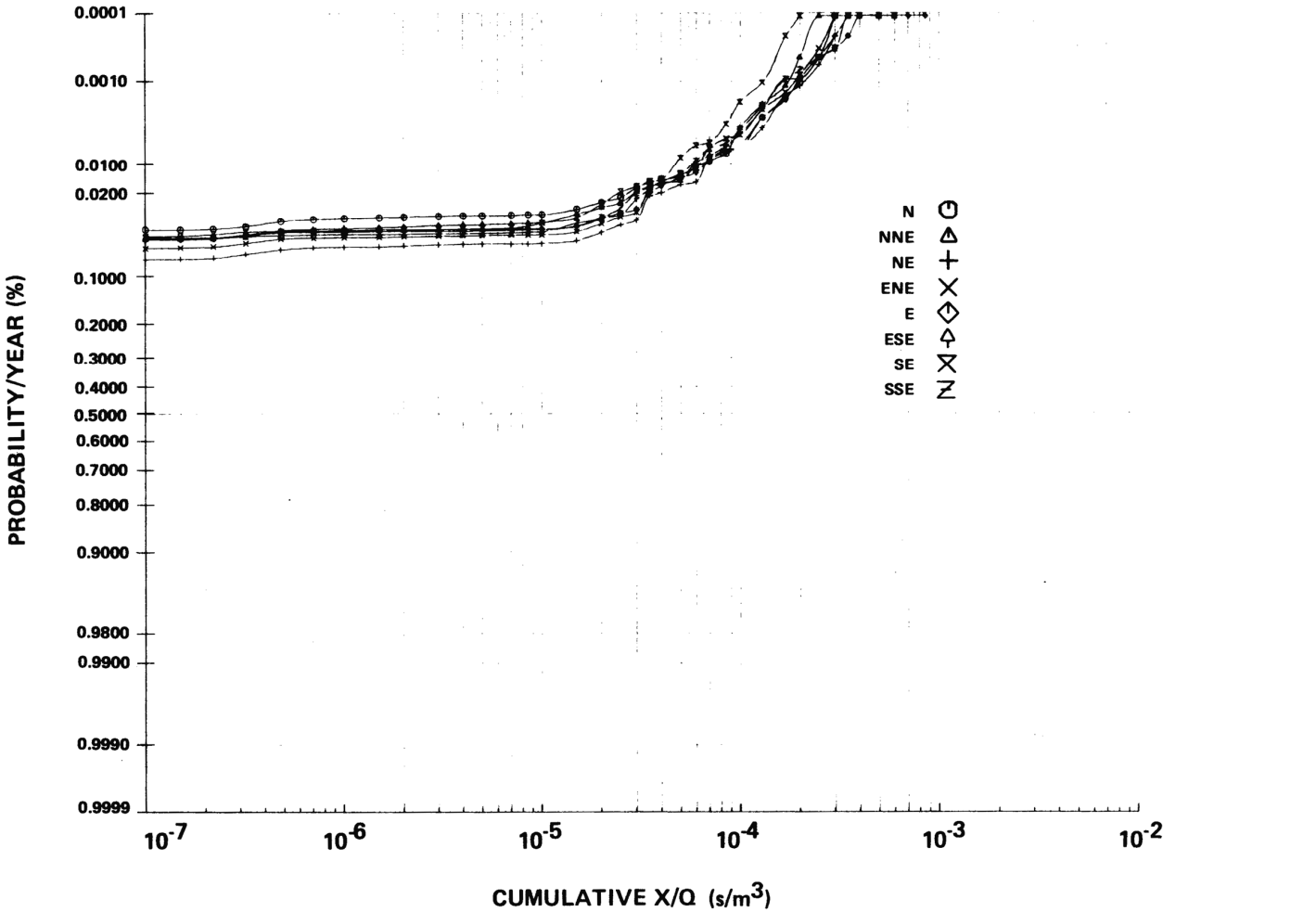
UNIT 1 AND UNIT 2

CUMULATIVE DISTRIBUTION OF X/Q
FOR DIRECTIONS (S THROUGH NNW)

FIGURE 2.3.4–2 (SHEET 2 OF 2)



BASED ON VOGTLE SITE DATA (4/77 - 4/78)

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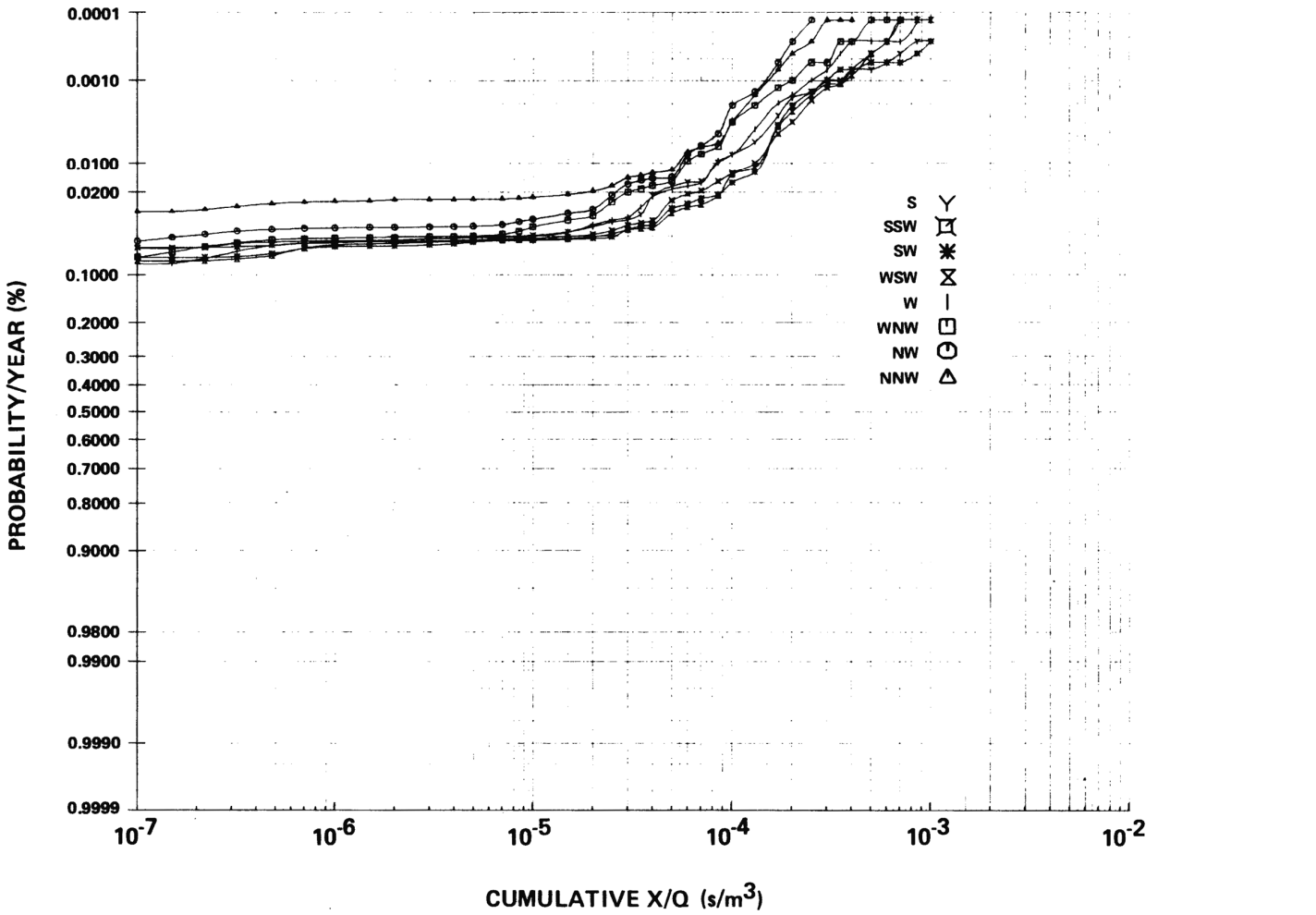
UNIT 1 AND UNIT 2

CUMULATIVE DISTRIBUTION OF X/Q
FOR DIRECTIONS (N THROUGH SSE)

FIGURE 2.3.4–3 (SHEET 1 OF 2)



BASED ON VOGTLE SITE DATA (12/72 - 12/73)

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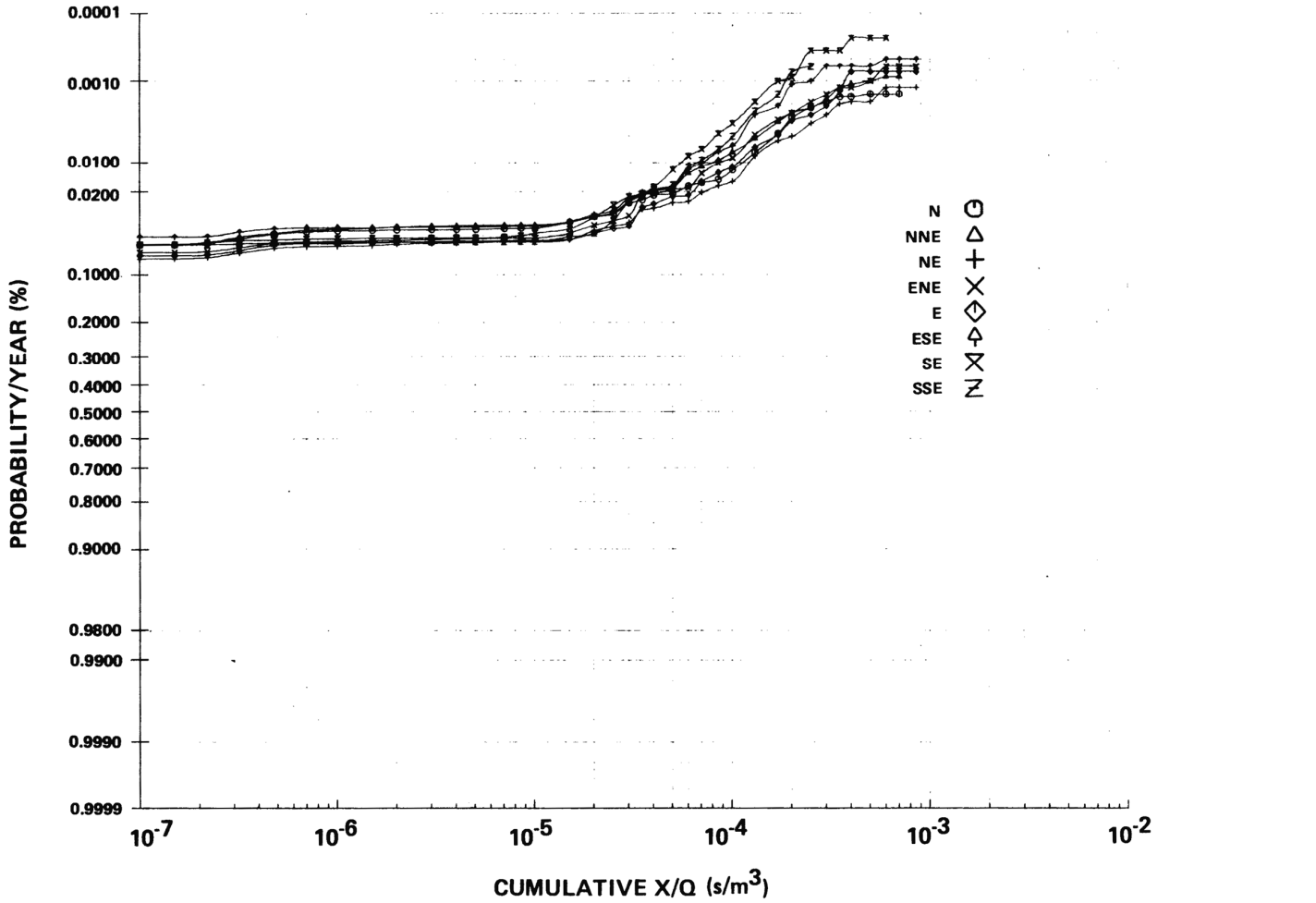
UNIT 1 AND UNIT 2

CUMULATIVE DISTRIBUTION OF X/Q
FOR DIRECTIONS (S THROUGH NNW)

FIGURE 2.3.4–3 (SHEET 2 OF 2)



BASED ON VOGTLE SITE DATA (4/78 - 4/79)

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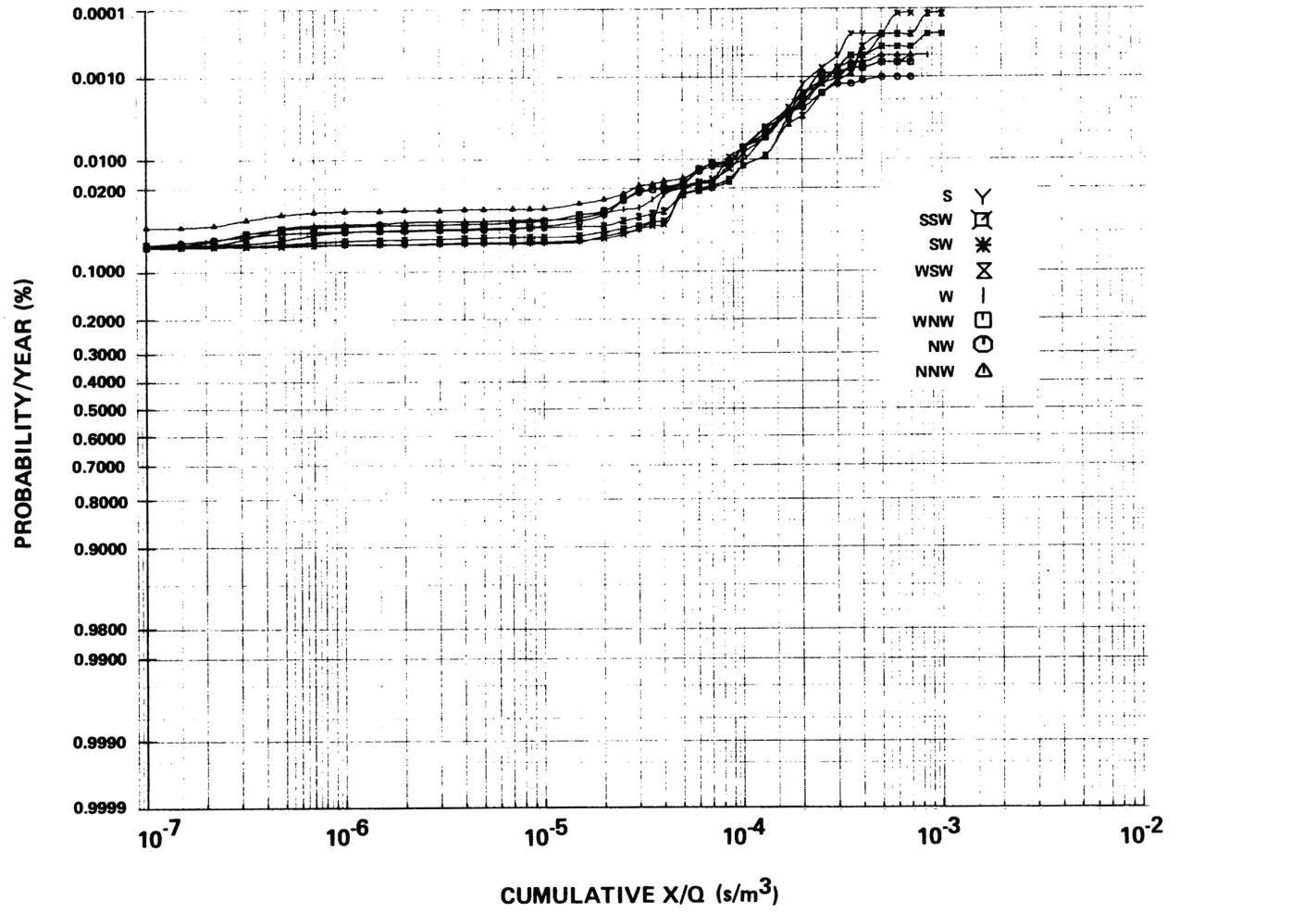
UNIT 1 AND UNIT 2

CUMULATIVE DISTRIBUTION OF X/Q
FOR DIRECTIONS (N THROUGH SSE)

FIGURE 2.3.4–4 (SHEET 1 OF 2)



BASED ON VOGTLE SITE DATA (4/78 - 4/79)

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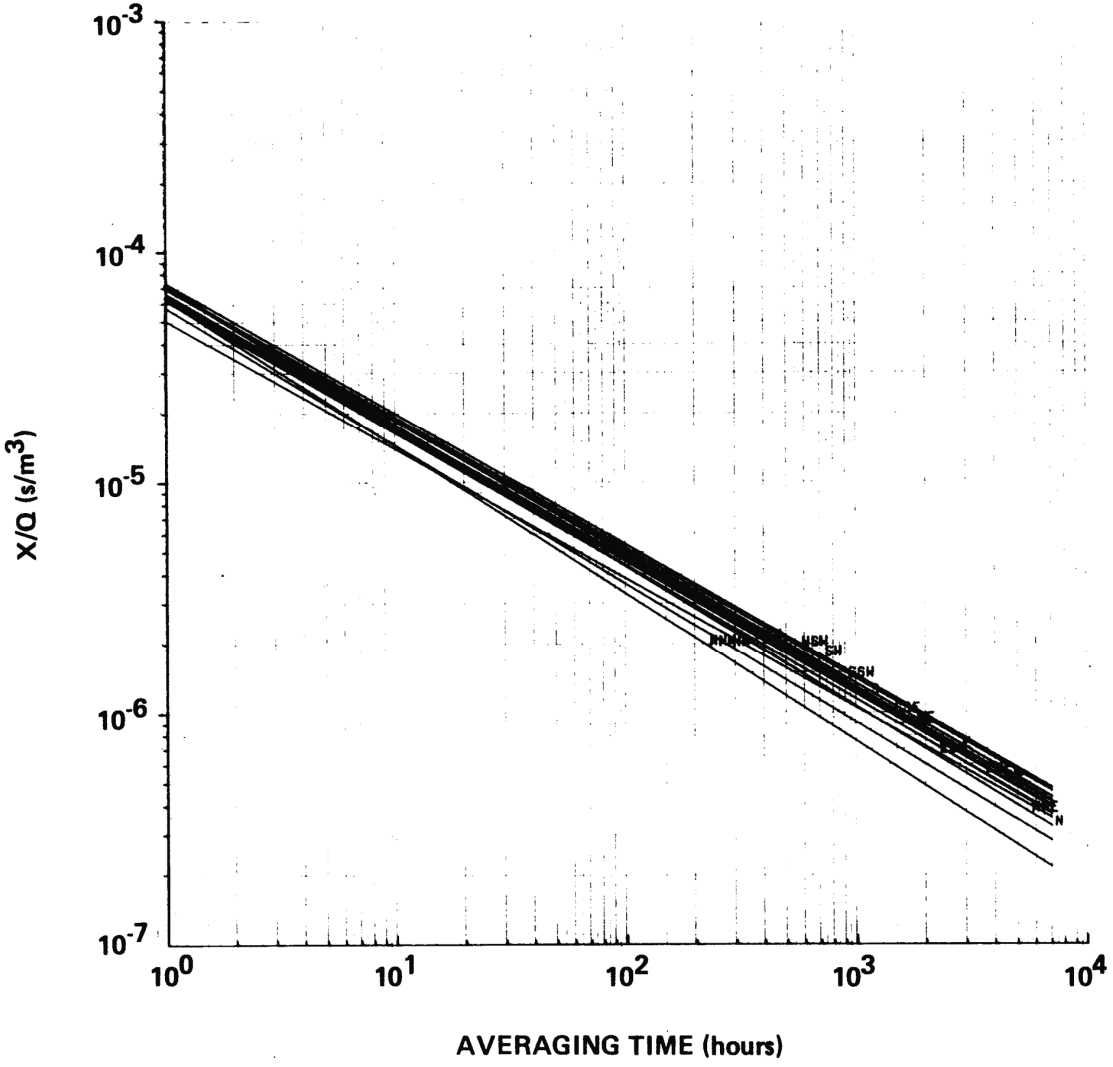
UNIT 1 AND UNIT 2

CUMULATIVE DISTRIBUTION OF X/Q
FOR DIRECTIONS (S THROUGH NNW)

FIGURE 2.3.4–4 (SHEET 2 OF 2)



BASED ON VOGTLE SITE DATA (12/72 - 12/73)

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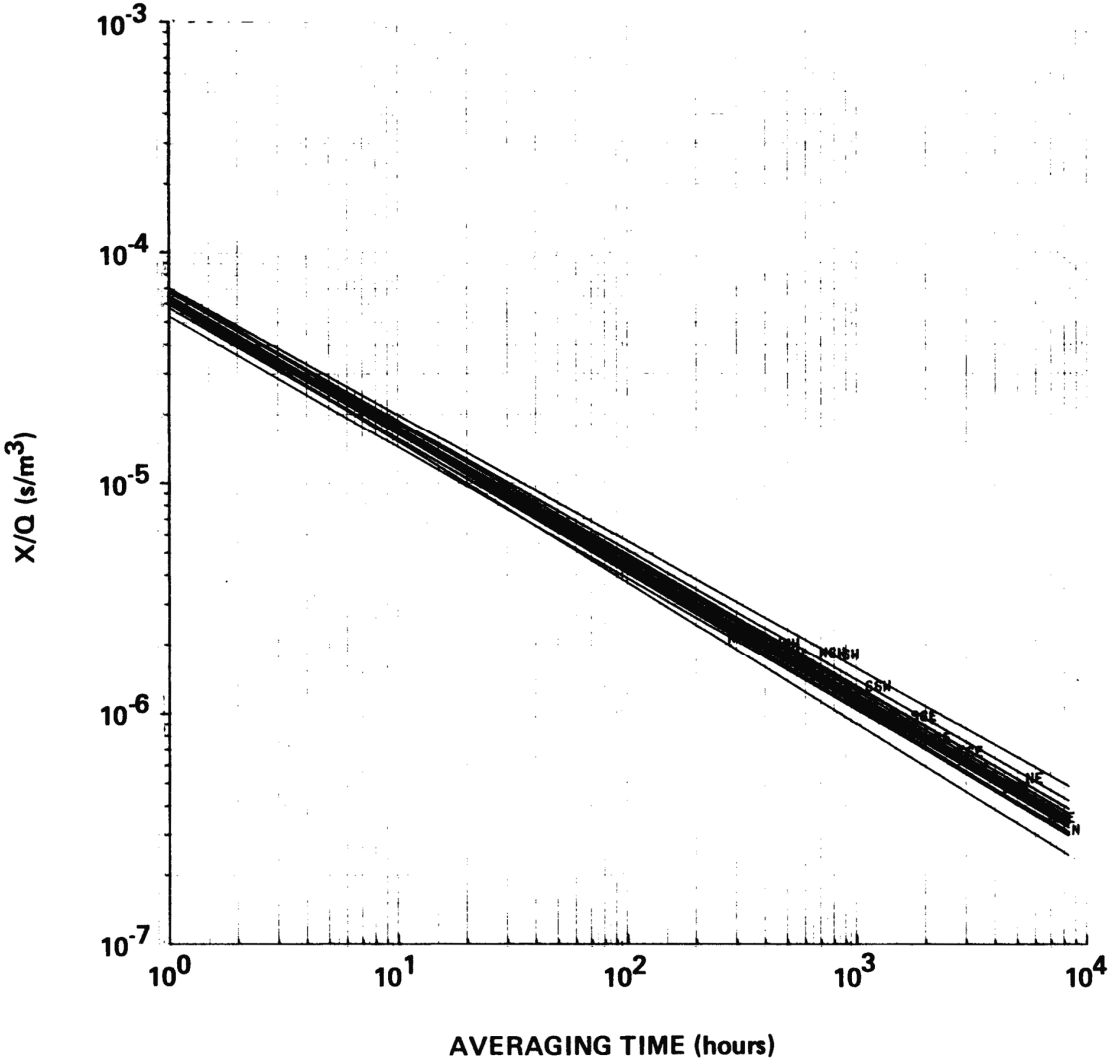
UNIT 1 AND UNIT 2

PLOT OF X/Q AT THE LPZ VS AVERAGING TIMES FOR ALL DIRECTION SECTORS

FIGURE 2.3.4–5



BASED ON VOGTLE SITE DATA (4/77 - 4/78)

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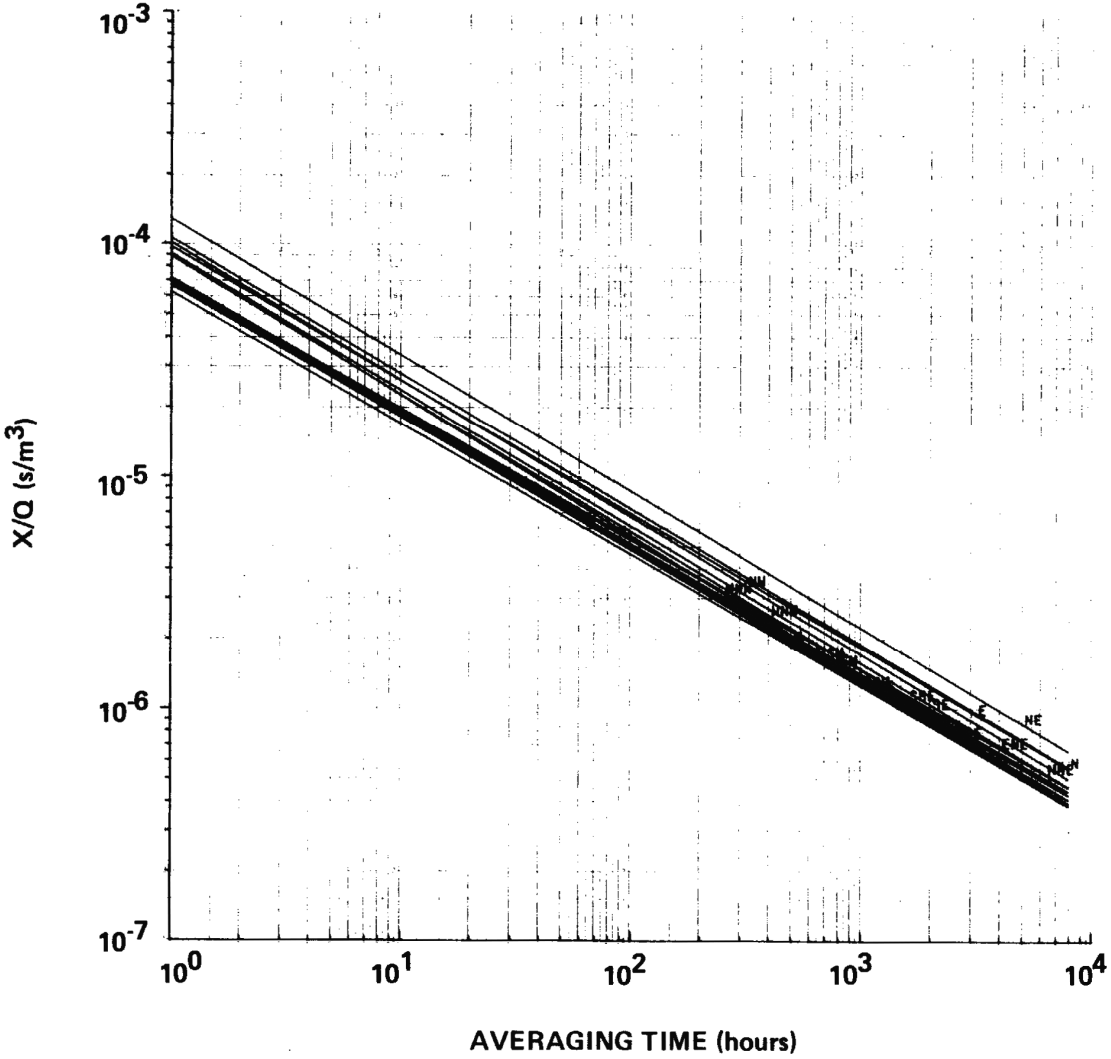
UNIT 1 AND UNIT 2

PLOT OF X/Q AT THE LPZ VS AVERAGING TIMES FOR ALL DIRECTION SECTORS

FIGURE 2.3.4–6



BASED ON VOGTLE SITE DATA (4/78 - 4/79)

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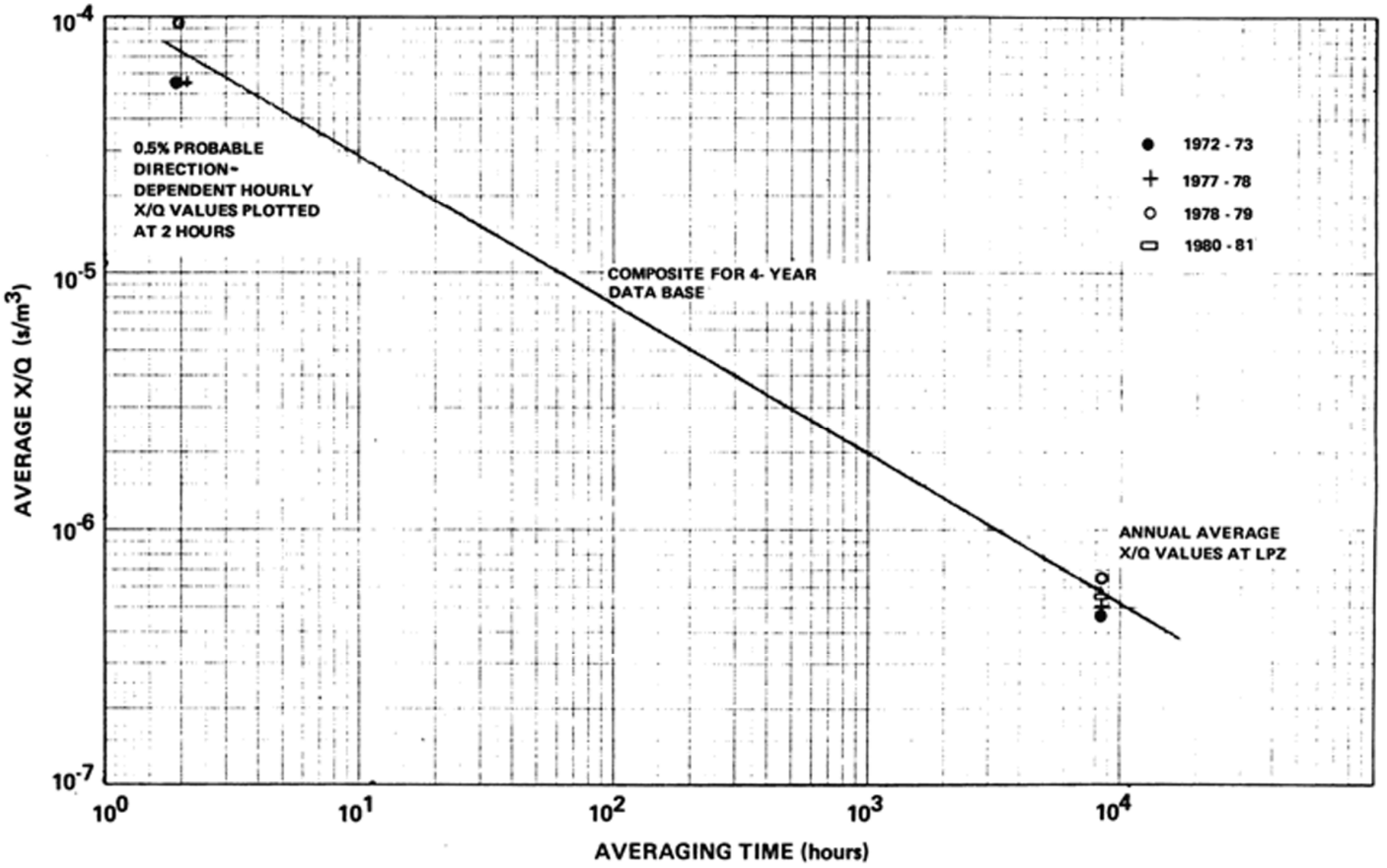
UNIT 1 AND UNIT 2

PLOT OF X/Q AT THE LPZ VS AVERAGING TIMES FOR ALL DIRECTION SECTORS

FIGURE 2.3.4–7



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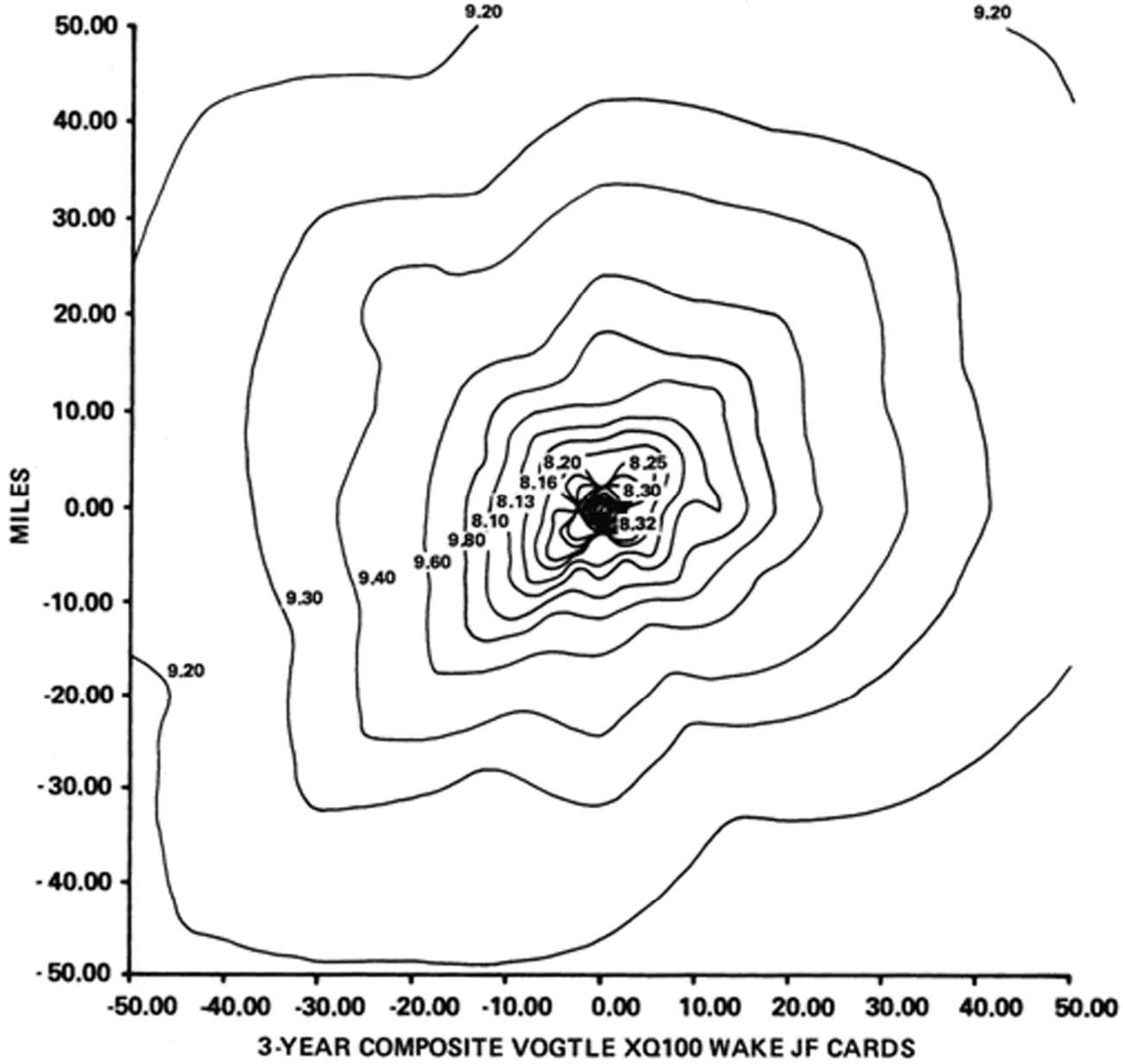
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PLOT OF X/Q AVERAGING
TIMES BASED ON VOGTLE SITE DATA
(4–YEAR COMPOSITE)

FIGURE 2.3.4–8

BASED ON 3 YEARS OF RECORD
AND JOINT FREQUENCY TABLES
(0 TO 50 MILES)



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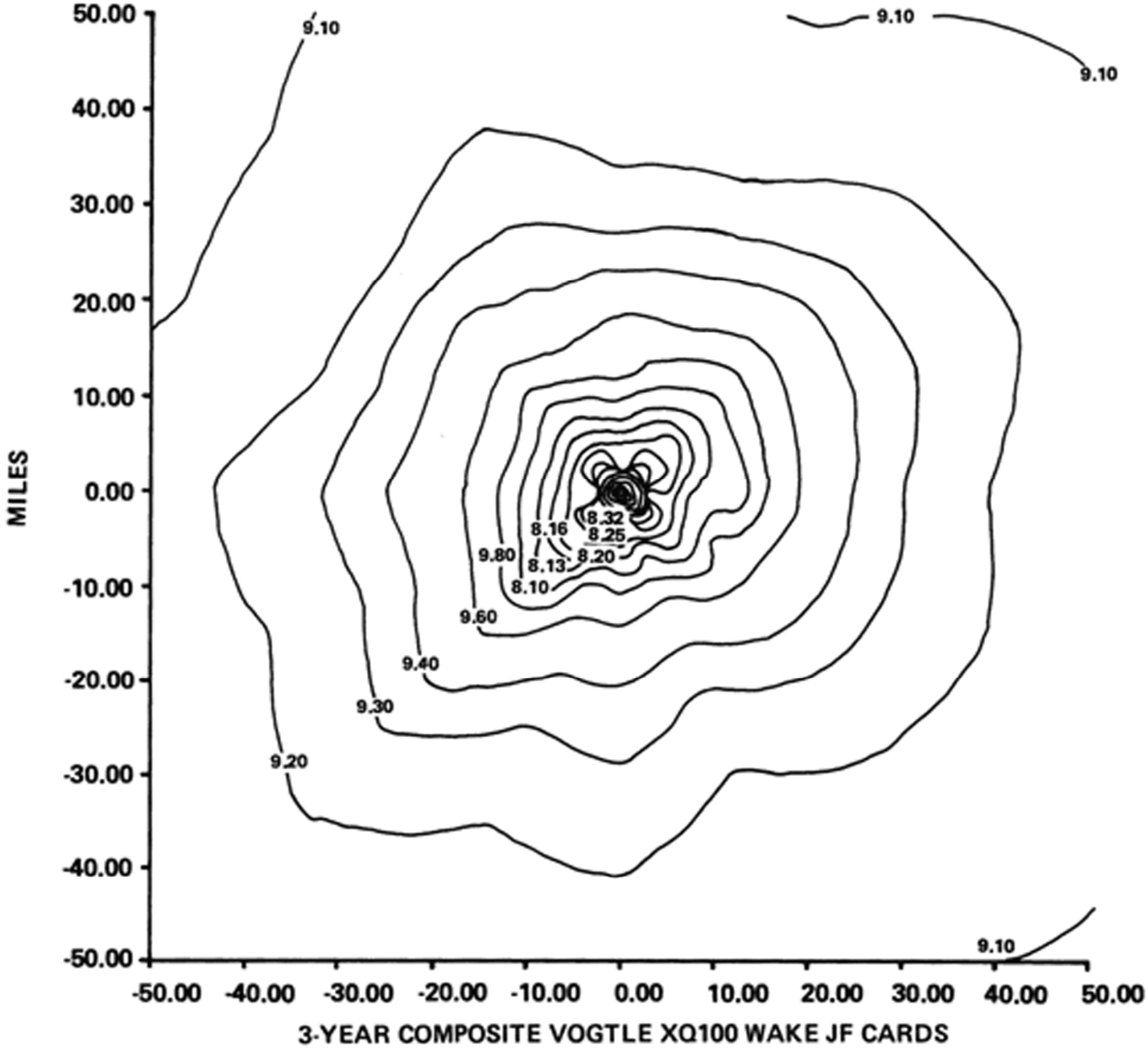
UNIT 1 AND UNIT 2

ISOPLETH OF ANNUAL X/Q

FIGURE 2.3.5–1

BASED ON 3 YEARS OF RECORD
AND JOINT FREQUENCY TABLES
(0 TO 50 MILES)



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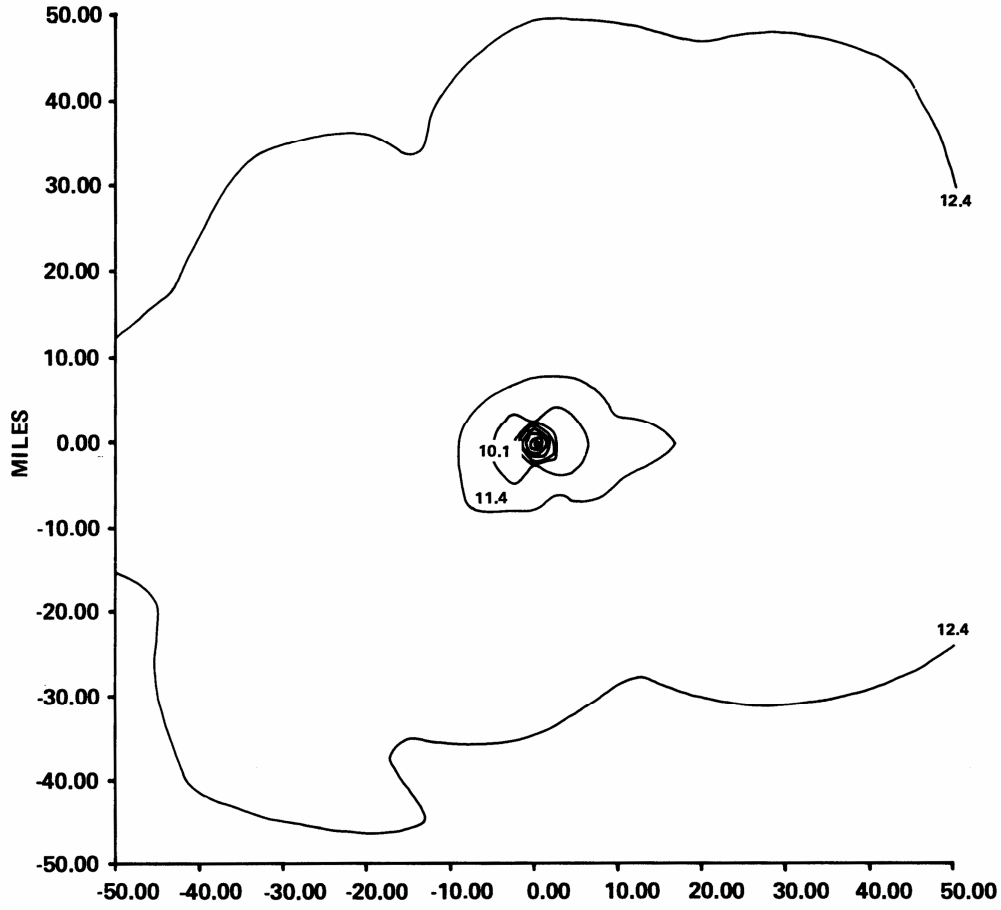
UNIT 1 AND UNIT 2

ISOPLETH OF ANNUAL AVERAGE
DEPLETED X/Q

FIGURE 2.3.5–2

BASED ON 3 YEARS OF RECORD
AND JOINT FREQUENCY TABLES
(0 TO 50 MILES)



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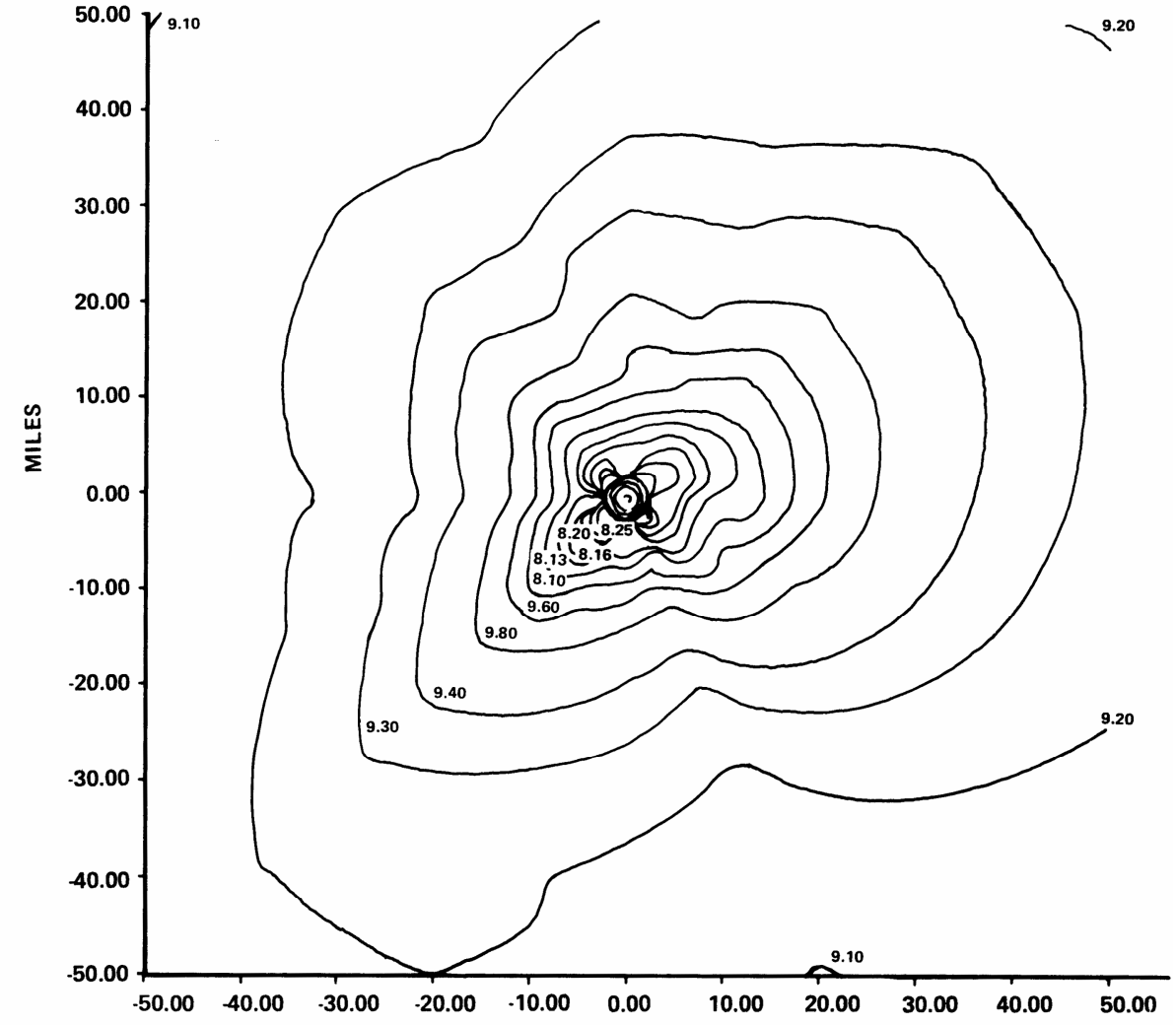
SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	ISOPLETH OF ANNUAL AVERAGE DEPOSITION (D/Q) FIGURE 2.3.5–3

BASED ON 4TH YEAR OF VEGP SITE
METEOROLOGICAL DATA 1980-81
(0 TO 50 MILES)



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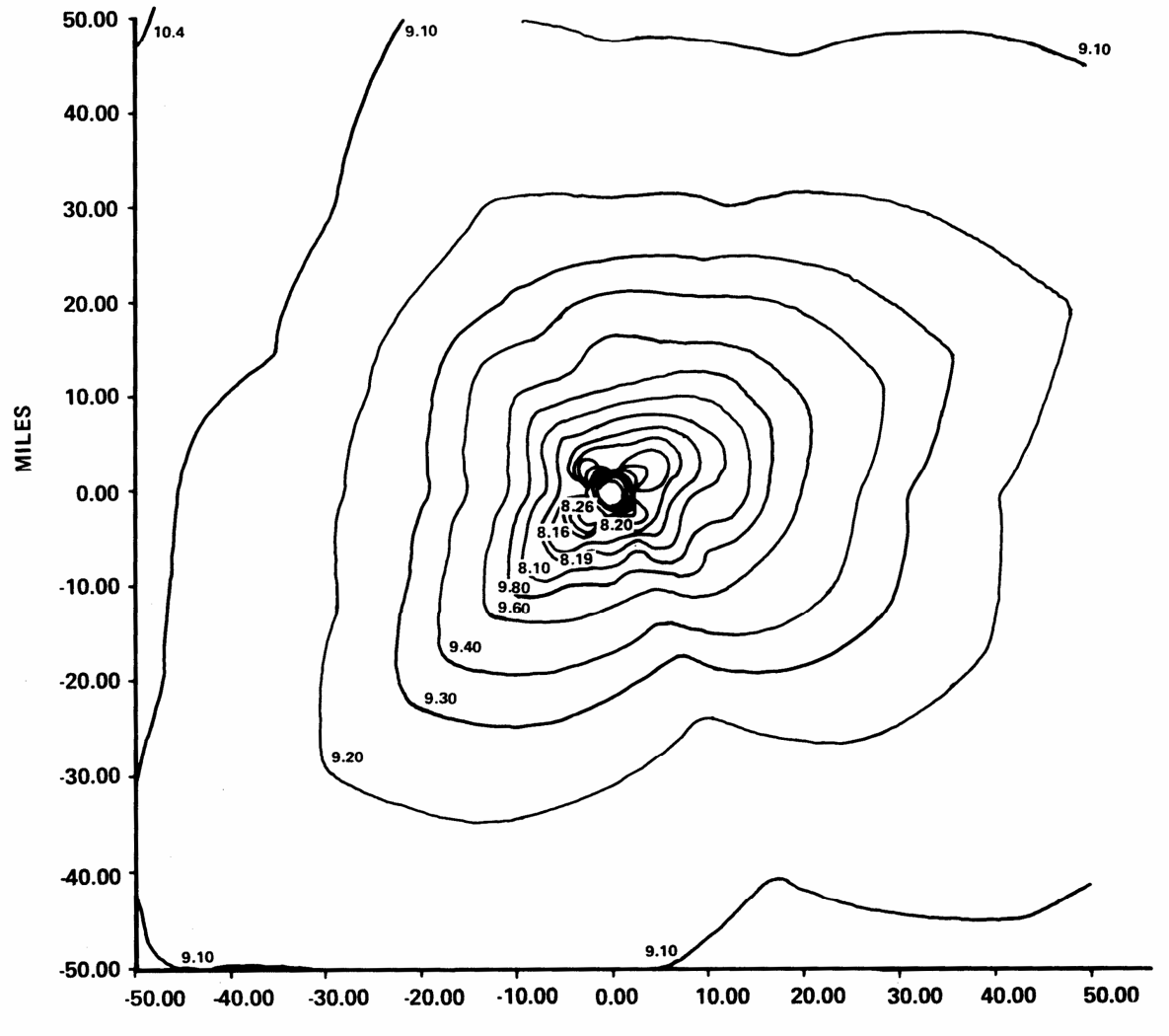
UNIT 1 AND UNIT 2

ISOPLETH OF ANNUAL AVERAGE DEPOSITION (D/Q)

FIGURE 2.3.5–4

BASED ON 4TH YEAR OF VEGP SITE
METEOROLOGICAL DATA 1980-81
(0 TO 50 MILES)



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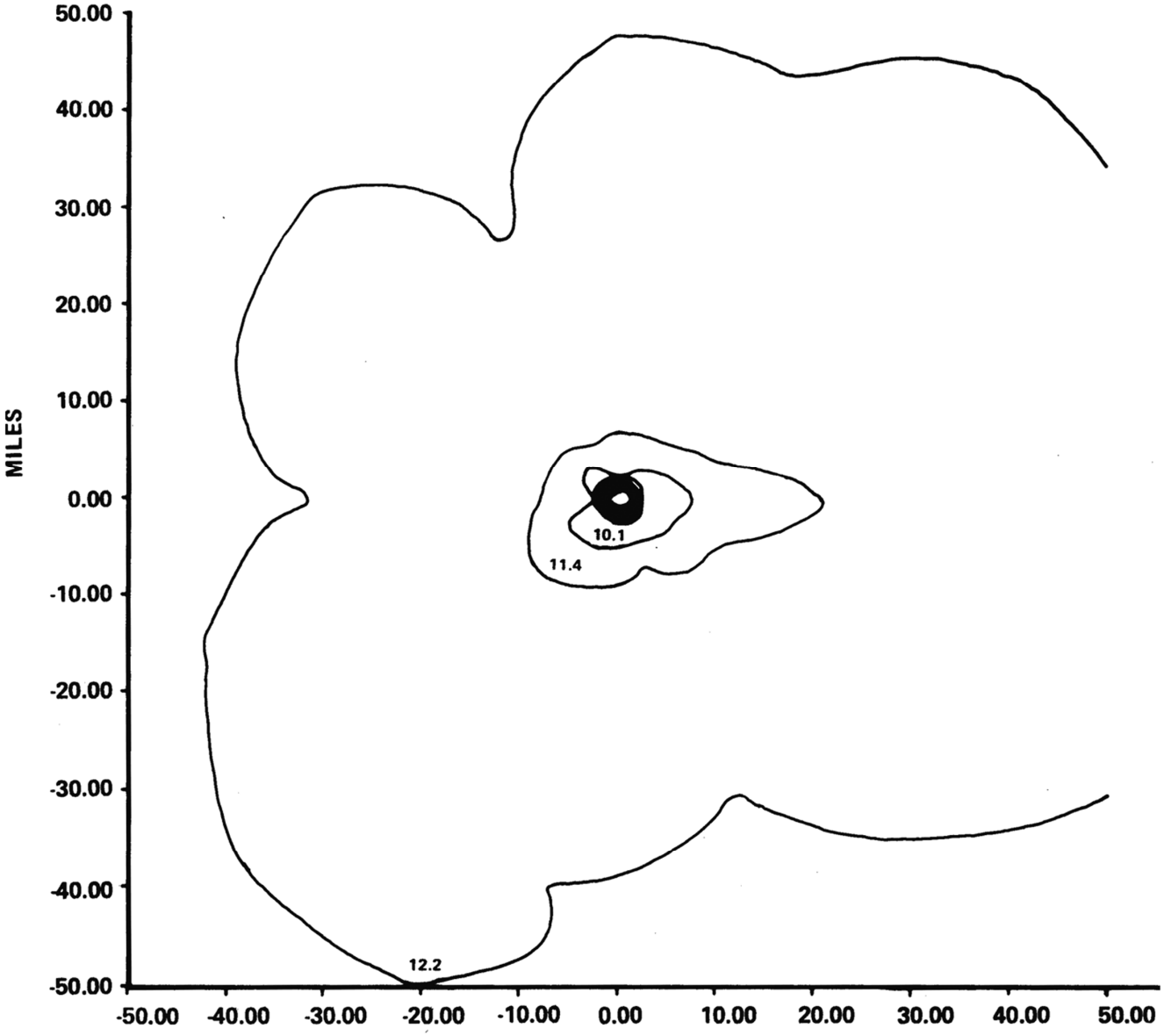
UNIT 1 AND UNIT 2

ISOPLETH OF ANNUAL AVERAGE DEPOSITION (D/Q)

FIGURE 2.3.5–5

BASED ON 4TH YEAR OF VEGP SITE
METEOROLOGICAL DATA 1980-81
(0 TO 50 MILES)



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SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	ISOPLETH OF ANNUAL AVERAGE DEPOSITION (D/Q) FIGURE 2.3.5–6

2.4 HYDROLOGIC ENGINEERING

2.4.1 HYDROLOGIC DESCRIPTION

2.4.1.1 Site and Facilities

The site of the VEGP, which encompasses an approximate area of 3169 acres, is owned by Georgia Power Company and the co-owners. The plant is located about 26 air miles south-southeast of Augusta, Georgia, along the west bank of the Savannah River, and 15 air miles east-northeast of Waynesboro, Georgia, in the eastern sector of Burke County at river mile 151.1. The drainage area above the plant site is about 8015 mi^2 including the Beaverdam Creek area (about 35 mi^2). The Savannah River Basin and its subbasins above the plant are shown in figure 2.4.1-1. The drainage areas of the subbasins are given in table 2.4.1-1. The plant site and the approximate boundary of the property line topography and site environs are given in drawings CX2D46V003, CX2D46V004, CX2D46V005, and CX2D46V006.

The plant is on high ground with the entrance to the power block buildings at grade el 220 ft msl, approximately 140 ft above minimum river level, and about 80 ft above probable maximum flood (PMF) level. The PMF level is based on all-weather probable maximum precipitation (PMP) of Hydromet No. 51[(1)] and Hydromet No. 52[(2)] by the National Weather Service. As shown in drawings CX2D46V003, CX2D46V004, CX2D46V005, and CX2D46V006, nominal drainage changes will be made to divert runoff away from the buildings and to improve the safety and drainage of the plant area. During the construction of Units 3 and 4, contours and grading in the Units 3 and 4 construction impact area will be controlled such that the flooding analysis for Units 1 and 2 is not adversely affected. The grade elevation at the river intake structure is approximately 125 ft msl, and the grade elevation at the control building, containment buildings, and diesel generator buildings is approximately 220 ft msl. The access to major safety-related structures is well above the PMF level. The safety-related structures, their access elevations, and types of access openings to these structures are given in table 2.4.1-2.

2.4.1.2 Hydrosphere

The VEGP site is along the Savannah River about 50 river miles below Augusta, Georgia. At a minimum flow of 5800 ft^3/s, the river at this location is about 340 ft wide and from 9 to 16 ft deep and has an average velocity of about 2 mph. The Savannah River Basin has a drainage area of 10,577 mi^2, of which 4581 mi^2 are in western South Carolina, 5821 mi^2 in Georgia, and 175 mi^2 in southwestern North Carolina. The Tallulah and Chattooga Rivers, which form the Tugaloo River on the Georgia-South Carolina State line, and Whitewater and Toxaway Rivers, which form Keowee River in South Carolina, start in the mountains of North Carolina. Keowee River and Twelve Mile Creek join near Clemson to form the Seneca River. The two principal headwater streams, the Seneca and Tugaloo Rivers, join near Hartwell, Georgia, to form the Savannah River.[(3)]

From this point, the Savannah River flows about 300 miles south-southeasterly to discharge into the Atlantic Ocean near Savannah, Georgia. Its major downstream tributaries include Broad

River in Georgia, the two Little Rivers in Georgia and South Carolina, and Brier Creek in Georgia, as shown in figure 2.4.1-2. The topography of the basin varies from el 5500 ft at the headwaters of the Tallulah River to about 1000 ft in the rolling and hilly Piedmont province, descending to around 200 ft at Augusta, Georgia, and from there, gently rolling to the nearby Coastal province from Augusta to the Atlantic Ocean. Rainfall is generally abundant and is about 43 in. annually.

Snow cover is rare except in the mountains. Runoff average is about 15 in. annually for the entire drainage area, while runoff at Augusta, Georgia, averages about 19 in. Total stream flow varies considerably from year to year. Streams in the basin are typically high in the winter and early spring. During the summer, flows recede and remain low through autumn. Industry has settled along the Savannah River at Augusta, Georgia, where there is an inland port, and at Savannah, Georgia, where there is a deep draft harbor. Two upstream reservoirs, Clark Hill and Hartwell, along with certain channel improvements, ensure minimum water requirements. River regulation has increased the minimum daily flow from a record of 1105 ft^3/s before construction of the dams to 6100 ft^3/s after their construction.(3)

The Savannah River Basin is wider at upper reaches and has about uniform width from Burton Dam to Clark Hill Dam. From Clark Hill Dam to Augusta the width gradually narrows. Below Augusta the basin further narrows to the outlet with mild land slope.(3)

There are three major Corps of Engineers dams in the Savannah River Basin: namely, Hartwell, Richard B. Russell, and Clark Hill. The Richard B. Russell Dam is under construction and will be completed in 1984. After completion the reservoir, along with the Clark Hill Reservoir to the south and the Hartwell Reservoir to the north, will form a chain of reservoirs about 120 miles along the shoreline. The Hartwell Dam is located 89 miles above Augusta and 7 miles below the confluence of the Tugaloo and Seneca Rivers, which form the Savannah River. It is a multipurpose project with 5 ft of storage above the maximum power pool (660 ft msl) reserved for flood control. This is equivalent to a flood control storage capacity of 293,000 acre-ft.

The reservoir covers 55,950 acres at full power pool (660 ft msl). Surface area at the top of flood control pool (665 ft msl) is 61,350 acres. Minimum power pool elevation is 625 ft msl.(3)

Richard B. Russell Lake and Dam is another multipurpose project in the Savannah River Basin and is located on the Savannah River in Georgia and South Carolina, 275.1 miles above its mouth, 63 miles above Augusta, and about 16 miles southeast of Elberton, Georgia. At maximum power pool (el 475 ft msl) the reservoir has an area of 26,650 acres and has a stable lake with only 5 ft of drawdown. This project is scheduled for completion in 1984.(3)

Clark Hill Dam project was begun in August 1946 and completed in July 1954. It is a multipurpose project designed to reduce floods in the Savannah River and to ensure a required minimum river flow. The Clark Hill project is credited with reducing the sediment load in the Savannah River carried into the Savannah Harbor by 22 percent. At full power pool (330 ft msl), Clark Hill provides a total storage of 2,900,000 acre-ft. The flood control storage at Hartwell combined with the flood control storage at Clark Hill Dam (390,000 acre-ft at el 335 ft msl) reduces flood peak downstream. Due to Clark Hill Dam being in place, the flood stage at Augusta in March 1964 was reduced from 38 ft to 25 ft. The flood stage at Augusta is 32 ft.(3) The reservoir elevation is expected to recede to 326 ft msl from September to mid-December.

Flow regulation at Hartwell Dam regulates dependable power at Clark Hill Dam which, in turn, guarantees minimum flow downstream of Clark Hill Dam.(3)

A minimum flow of 5800 ft^3/s (based on the period of record) is required for navigation below Augusta; however, a discharge of 6300 ft^3/s is normally provided 80 percent of the time. Clark Hill Dam is designed for a maximum drawdown of 18 ft from the top of the power pool at el 330 ft msl to a minimum pool at el 312 ft msl. However, it is not anticipated that the minimum pool will be reached more often than once in 150 years.[3]

Heavy inflows into the lake begin generally in mid-December and continue through April with a full power pool by the first of May. For these major reservoirs and other water-controlling structures, the drainage areas, ownership, seismic design criteria, spillway design criteria, location, and type of structure are tabulated in table 2.4.1-3.

There are two domestic water users of surface water (the Savannah River) downstream of the plant. These users are Beaufort/Jasper County, 112 river miles downstream, and the Cherokee Hill (Port Wentworth) water treatment plant, 122 river miles downstream. Subsection 2.4.13 discusses the consequences of effluent releases to surface waters.

2.4.1.3 References

1. Schreiner, Louis C., and Riedel, John T., "Probable Maximum Precipitation Estimates, United States East of the 105th Meridian," Hydrometeorological Report No. 51, National Weather Service, NOAA, U.S. Department of Commerce, Washington, D.C., 1978.

2. Hansen, E. M., and Schreiner, L. C., "Application of Probable Maximum Precipitation Estimates, United States East of the 105th Meridian," Hydrometeorological Report No. 52 (preliminary), National Weather Service, Silver Springs, Maryland, 1979.

3. U.S. Army Corps of Engineers, South Atlantic Division, Water Resources Development in South Carolina, Atlanta, Georgia, March 1975.

2.4.2 FLOODS

2.4.2.1 Flood History

A partial list of annual peaks at Augusta, Georgia, from 1796 through 1979, covering periods before and after Clark Hill Dam started operations in 1954, is shown in table 2.4.2-1. These flood events were natural flood events before the upstream controlling structures were in place. No major flood has occurred at Augusta, Georgia, since 1951. The record flood at the site occurred in 1796 with an estimated water surface elevation of 116 ft msl. There have been no floods due to surges, seiches, or tsunami because the site is not located near a large body of water.

2.4.2.2 Flood Design Considerations

There are no safety-related structures that could be affected by floods and flood waves. For the derivation of probable maximum flood (PMF), the most recent National Oceanic and Atmospheric Administration, formerly Weather Bureau, publication Hydromet No. 51,[1] in conjunction with Hydromet No. 52,[2] has been used for developing probable maximum

precipitation (PMP) envelopes for the Savannah River drainage basin above the VEGP. The PMP distribution is explained in paragraph 2.4.3.1. The PMF stage at the plant site is about 138 ft msl without wave runup, and with wave runup the water may reach as high as 165 ft msl. All safety-related structures have a grade elevation of 220 ft msl, which is well above the flood stage. If the valley storage effect between Clark Hill Dam and the VEGP site is taken into account, this results in lower flood peak and lower flood stage (drawing AX6DD319).

The makeup water river intake structure is nonsafety-related and has a deck elevation of 125 ft msl. This corresponds to the standard project flood stage elevation of 120 ft msl plus 5 ft freeboard allowance for waves. The river intake makeup water facilities are provided as a secondary backup source for the makeup water system. The makeup water for the nuclear service cooling water will be provided by wells drilled into the Tuscaloosa aquifer, which has enough water yield capacity to meet the makeup requirements for emergency conditions. In the unlikely event that the river intake structure is rendered inoperable, the safety of the plant operation or the plant safe shutdown will be unaffected, because the nuclear service cooling water tower basins are designed to provide 30-day inventory to permit safe shutdown (see subsection 9.2.5).

2.4.2.3 Effects of Local Intense Precipitation

The PMP, which is based on the world record envelope[3] and has a maximum intensity of 15 in. of rainfall in 1 h, is used to evaluate the effects of local intense precipitation. Runoff is directed away from the plant by the sloping ground surface to natural drainage channels and subsequently to the Savannah River. The site grading and drainage plan is shown in drawing CX2D46V003; pertinent elevations and additional slope arrows are shown in drawing CX2D46V006. It is assumed that all catch basins and roof drains are plugged for PMP flood analysis and that all flow is overland. The drainage system for the site precludes flooding of safety-related structures and equipment. The site drainage plan for local PMP storm is given in drawing CX2D46V004; the 100-year rainfall event is given in drawing CX2D46V005.

The normal yard drainage system, consisting of pipe culverts and open channels, is designed to drain the runoff from a 100-year rainfall event with a maximum intensity of 4 in. of rainfall in 1 h.

The rational formula is used to determine peak runoff rates from specified areas. Time of concentration is determined with the aid of the Kirpich nomograph.[4] Duration is assumed to be equal to time of concentration.

Corresponding intensity is taken from table 2.4.2-3 for 100-year storm flood analysis. Runoff coefficients for use with the rational method in determining 100-year peak rates of flow are as follows:

A.	Unpaved Areas	
	Power block area	0.65
	Adjacent to power block	0.50
	All other	0.40

B.	Paved Areas	
	Concrete	0.95
	Asphalt	0.90

For the PMP flood event, the analysis is based on the following: The general power block area and plant are bounded by peripheral railroads and roads typically at or near el 218.50 ft. The ground surface is sloped from the power block with threshold el 220 ft to the railroads and roads. All ditches, inlets, and storm drains are considered to be plugged, allowing local ponding due to the PMP storm until the flow discharges over the railroad. The rational formula is used to determine peak runoff rates from specified areas (power block to the railroad). The time of concentration used is T_c = 5.0 min with corresponding intensity of 55.0 in./h. The resulting flow is considered to discharge over the peripheral railroad acting as a broad-crested weir. The formula $Q = CLH^{3/2}$ is then used to determine the water surface el at the railroad, with C = 2.60. The flow then follows the natural ground surface overland to drainage channels.

Corresponding intensity is taken from table 2.4.2-2 for PMP flood analysis.

Runoff coefficients for use with the rational method in determining PMP peak rates of flow are as follows:

A.	Specific Areas	
	Paving	0.95
	Grass	0.85
	Gravel	0.75
	Forest/shrub	0.65
B.	Overall Areas	
	Power block	0.86
	Adjacent to power block	0.77

All runoff coefficients are exclusive of structures. The runoff coefficient for all power block structures is 1.0.

Depth of flow in channels is determined using Manning's formula for open channel flow, with n = 0.014 for concrete lined channels and 0.020 for air blown mortar lined channels. Manning's n for pipe flow is taken as 0.024 for corrugated steel and 0.012 for concrete pipe. There are no unpaved channels. The roof drain systems for all safety-related structures are designed to pass the runoff from the PMP. The design includes measures to guard against wind-induced seepage through roof and wall penetrations and doors where safety-related equipment could be damaged.

Icing normally does not occur. The combination of icing followed by heavy local precipitation is not considered for determining the effects of local intense precipitation for site drainage.

Seismic Category 1 structure roofs are typically flat and surrounded by 2-ft-high parapet walls. With the exception of the control building part of the equipment building, which contains no Seismic Category 1 equipment, all roofs are drained by scuppers in the parapet walls.

For the 100-year storm the scuppers are located at the roof elevation, and the storm water is dumped into downspouts which run down the outside faces of structures onto adjacent roofs or below ground level to the yard drainage system. Since downspouts are located external to structures, there are no internal drain lines which could flood safety-related equipment.

Additional scuppers are provided to drain the PMP. These are located a maximum of 6 in. above the roof line, and are a minimum of 6 in. deep and 12 in. wide. Their size minimizes their susceptibility to clogging, and flow is allowed to fall directly to adjacent roofs or to grade. Downspouts, which would have the potential for clogging, are not provided to drain the PMP. The site is graded away from Seismic Category 1 structures so that flooding of safety-related structures and equipment will not occur.

The roofs of Seismic Category 1 structures are designed for 18 in. of ponded water corresponding to a load of 93.6 lb/ft^2. Flow is routed between adjacent structures and a sufficient number of PMP scuppers are provided to ensure that the 18-in. maximum ponded depth is not exceeded. It is assumed that during the PMP all 100-year storm scuppers and one-half the PMP scuppers could be plugged. A 30-lb/ft^2 load is considered in the design of the roofs of all Seismic Category 1 structures to account for the snow or 100-year rain load.

2.4.2.4 References

1. Schreiner, Louis C., and Riedel, John T., "Probable Maximum Precipitation Estimates, United States East of the 105th Meridian," Hydrometeorological Report No. 51, National Weather Service, NOAA, U.S. Department of Commerce, Washington, D.C., 1978.

2. Hansen, E. M., and Schreiner, L. C., "Application of Probable Maximum Precipitation Estimates, United States East of the 105th Meridian," Hydrometeorological Report No. 52 (preliminary), National Weather Service, Silver Springs, Maryland, 1979.

3. Chow, Ven Te, Handbook of Applied Hydrology, McGraw-Hill Book Company, New York, pp 9-46 through 9-48, 1964.

4. Kirpich, P. Z., "Time of Concentration of Small Agricultural Watersheds," Civil Engineering, Vol 10, No. 6, p 352, June 1940.

2.4.3 PROBABLE MAXIMUM FLOODING ON STREAMS AND RIVERS

The probable maximum flooding for the Savannah River Basin above the VEGP site is determined from probable maximum precipitation (PMP). The 72-h PMP depth-area-duration (DAD) envelope is prepared from Hydromet No. 51.(1) The runoff is routed through the streams and reservoirs. The resulting maximum stage at the VEGP site is 165 ft msl including allowance for wave action. The procedure for the development of DAD envelope, temporal distribution, and spatial distribution is described in paragraph 2.4.3.1. The procedures for flood hydrograph development and subsequent routing through the reservoirs and streams are described in paragraphs 2.4.3.3 and 2.4.3.4, respectively.

As shown in drawings CX2D46V003, CX2D46V004, CX2D46V005, and CX2D46V006, the plant is located on a ridge line, and access to it follows the high ground. The potential for flooding is minimal.

2.4.3.1 Probable Maximum Precipitation

The PMP values for 10, 200, 1000, 5000, 10,000, and 20,000 mi^2 drainage areas for 6, 12, 24, 48, and 72 h are read from all- season PMP charts.(1) These values are given in table 2.4.3-1 and are used to prepare the DAD curve as shown in figure 2.4.3-1. From the DAD curve, a cumulative PMP envelope for the drainage area above the VEGP site is prepared as given in figure 2.4.3-2. The PMP values at 6-h intervals are obtained from figure 2.4.3-2 and are given in table 2.4.3-2. For the spatial distribution of the storm over the drainage area, the storm is positioned at the center of the drainage basin above the VEGP site. Also, the storm is positioned at noncentral locations to see the effect of storm centering on the resulting hydrograph.

The elliptical isohyets are drawn to cover various drainage areas and are assigned the precipitation depth as read from DAD curves corresponding to the area covered by the isohyet. The areas between two isohyets are measured, and the weighted arithmetic average of PMP is obtained for the two positions as explained below.

Position No. 1: The storm is positioned at the center of the drainage basin above the plant site. The major axis of the isohyet is aligned with the major axis of the drainage basin as shown on drawing AX6DD315.

Position No. 2: This is the same as position No. 1 except that the center of the isohyet is offset to the west about 6 miles and laid over the drainage basin in such a way that the major axis of the isohyet is approximately parallel to the major axis of the drainage basin at the center, as shown on drawing AX6DD316.

In addition, the PMP for a storm centered at position No. 1 is determined by an alternate method identified below. The storm is positioned as in No. 1, but for the storm distribution the procedure given in Hydromet No. 52(2) is closely followed.

For positions No. 1 and 2, a two-to-one major to minor axis elliptical storm isohyetal pattern is used, while for Hydromet No. 52, a three-to-one major to minor axis pattern is used. The storm isohyets, as positioned on the drainage basin, are shown on drawings AX6DD315, AX6DD316, and AX6DD317. For positions No. 1 and 2, all 6-, 12-, 24-, 48-, and 72-h values are averaged over the subbasins. In the case of the Hydromet No. 52 procedure, only the three greatest 6-h PMP values are spatially and temporally adjusted. No adjustment is applied to the remaining 6-h PMP values. These values are given in tables 2.4.3-3 through 2.4.3-5. The PMP for positions No. 1 and 2 and that based on Hydromet No. 52 are given in table 2.4.3-6 for all subbasins above the VEGP site. In the case of positions No. 1 and 2, no spatial or temporal adjustment has been made. The critical sequence arrangement of PMP in all three procedures is the same.

The PMP computations for positions No. 1 and 2 are straightforward. The weighted arithmetic average PMP over the subbasins for 6-,12-, 24-, 48-, and 72-h durations is determined.

The DAD curve for each subbasin is prepared. The unit hydrograph for each subbasin and its duration is known.(3) The incremental PMP is read from the DAD curve at a duration equal to unit hydrograph duration. The Corps of Engineers' computer program HEC-1(4) is used to develop a flood hydrograph for each subbasin. It is routed through the stream or reservoir to a predetermined location to be combined with other hydrograph(s) at that location. The procedure is continued until the flood hydrograph is routed past the VEGP site.

In the procedure identified here as Hydromet No. 52, only the three greatest 6-h PMPs are spatially and temporally adjusted. The adjustment factor depends upon the basin orientation.[2] The isohyet adjustment factors for the three greatest 6-h PMPs are given in table 2.4.3-3, while the three greatest 6-h PMPs for isohyets enclosing various areas are given in table 2.4.3-4. The details are available in reference 2.

The spatially adjusted, idealized elliptical isohyets are laid over the drainage basin as in position No. 1. The first, second, and third greatest 6-h PMP values from table 2.4.3-4 are assigned to each isohyet enclosing the area given in table 2.4.3-4. The average PMP for the first, second, and third greatest 6-h period for each subbasin is computed by the weighted arithmetic average of the isohyets enclosing the subbasin area.

For each subbasin, as given in figure 2.4.1-1, the 72-h PMP at 6-h intervals is arranged in critical sequence and is given in table 2.4.3-5. The flood hydrograph for each subbasin is developed and routed as mentioned earlier. Identical initial rainfall loss and uniform infiltration loss are used in all the aforementioned cases (procedures) for the development of the flood hydrograph. Also, identical parameters and routing procedures are used to route the flood hydrograph resulting from the three procedures and are then used to compare them objectively.

2.4.3.2 Precipitation Losses

The Savannah River is predominantly forested. Rainfall is generally abundant, up to 80 in. annually. Snow cover is rare except in the mountains. Runoff averages about 15 in. annually for the entire drainage area.[5] The infiltration and retention losses are very high.

An initial loss of 0.5 in. and a nominal uniform rate of 0.05 in./h is used. These values are well below the losses that would be used in an ordinary situation.

2.4.3.3 Runoff and Stream Course Models

The River Forecasting Center in Atlanta has divided the Savannah River Basin into subbasins and has established gauging points for its forecasting purpose.[3] The center has developed unit hydrographs for each subbasin. For the purpose of this study the River Forecasting Center unit hydrographs and subbasin drainage areas are used. The subbasins are numbered as shown on drawing AX6DD315. The basins and their drainage areas along with unit hydrograph duration are shown in table 2.4.1-1. In the Savannah River Basin three major multipurpose dams, namely, the Hartwell Dam, the Richard B. Russell Dam, and the Clark Hill Dam are modeled.

The Corps of Engineers' HEC-1 computer program[4] is used to develop the inflow hydrograph and to route it to the desired location to be combined with other local or routed hydrograph(s). The unit hydrograph and lag time for each subbasin are provided by the River Forecasting Center in Atlanta. The lag time, as shown on drawing AX6DD320, is utilized to delay the peak. The Tatum method[4][6] with three steps is used to route the hydrograph through the streams, and the modified pulse method[4] is used to route through the reservoirs (Hartwell, Richard B. Russell, and Clark Hill). The reservoirs are assumed at maximum power pool levels to release more water and obtain a conservative hydrograph at the VEGP site. Channel storage was omitted in the HEC-1 routing procedure.[4] For comparison purposes, the National Weather Service (NWS) dam break model[7] is also used to route the probable maximum flood (PMF)

outflow hydrograph from Clark Hill Dam to the VEGP site. The hydrographs on two schemes of routing are shown on drawings AX6DD318 and AX6DD319.

The following steps are followed to obtain the PMF hydrograph at the VEGP site:

1. Compute inflow into Burton Dam using unit hydrograph for subbasin 1.
2. Determine outflow from Burton Dam.
3. Route Burton Dam outflow to Hartwell Dam.
4. Compute local area hydrograph from Burton Dam to Hartwell Dam using unit hydrograph for subbasin 2.
5. Add results from steps 3 and 4 to obtain Hartwell inflow forecast.
6. Determine outflow from Hartwell Dam.
7. Route Hartwell Dam outflow to Calhoun Falls.
8. Compute local area hydrograph from Hartwell Dam to Calhoun Falls using unit hydrograph for subbasin 3.
9. Add results from steps 7 and 8 to obtain Calhoun Falls inflow forecast.
10. Route Calhoun Falls to Clark Hill Dam.
11. Compute Carlton Bridge forecast using unit hydrograph for subbasin 4.
12. a. Lag Carlton Bridge to Bell.

    b. Route lagged Carlton Bridge to Bell.
13. Compute local area hydrograph from Carlton Bridge to Bell using unit hydrograph for subbasin 5.
14. Add results from steps 12b and 13 to obtain Bell forecast.
15. a. Lag Bell to Clark Hill Dam.

    b. Route lagged Bell to Clark Hill Dam.
16. Compute local area hydrograph from Bell-Calhoun Falls to Clark Hill Dam using hydrograph for subbasin 6.
17. Add results from steps 10, 15b, and 16 to obtain Clark Hill inflow forecast.
18. Determine outflow from Clark Hill Dam.
19. Route Clark Hill outflow to Stevens Creek Dam.
20. Compute Modoc forecast using unit hydrograph for subbasin 7.

21. a. Lag Modoc to Stevens Creek Dam.

    b. Route lagged Stevens Creek outflow to Butler Creek.

22. Compute local area hydrograph from Clark Hill-Modoc to Stevens Creek Dam using unit hydrograph for subbasin 8.

23. Add results from steps 19, 21b, and 22 to obtain Stevens Creek inflow forecast.

24. Determine outflow from Stevens Creek Dam.

25. a. Lag Stevens Creek outflow to Butler Creek.

    b. Route lagged Stevens Creek outflow to Butler Creek.

26. Compute local area hydrograph from Stevens Creek to Butler Creek using unit hydrograph for subbasin 9.

27. Add results from steps 25b and 26 to obtain Butler Creek forecast.

28. a. Lag Butler Creek to plant site.

    b. Route lagged Butler Creek to plant site.

29. Compute local area hydrograph from Butler Creek to plant site using unit hydrograph for subbasin 10.

30. Route results obtained in step 29 to plant site.

31. Add results from steps 28b and 30 to obtain PMF hydrograph at plant site.

The hydrograph development scheme used in the HEC-1[4] computer program is given on drawing AX6DD320. The routing scheme used in the NWS[7] computer program is similar to the one shown on drawing AX6DD320, except that the outflow hydrograph from Clark Hill Dam is input to the NWS[7] computer model and routed through the downstream reach taking into consideration the downstream valley storage. The river reach between Clark Hill Dam and Burton's Ferry Bridge is divided into 42 subreaches, and the river reach between Clark Hill Dam and the VEGP site is divided into 27 subreaches, as shown on drawing AX6DD320. The runoff hydrographs from subbasins 7, 8, 9, and 10 are input as lateral inflows. The resulting hydrographs from both schemes are shown on drawings AX6DD318 and AX6DD319. It is obvious that the peaks are substantially reduced when the valley storage is considered in routing. The location of the river cross-sections used are shown on drawing AX6DD320.

2.4.3.4 Probable Maximum Flood Flow

The PMF is developed and routed through the stream course to the VEGP site as explained in paragraph 2.4.3.3. The routed hydrographs at the plant site are given on drawings AX6DD318 and AX6DD319. The three major dams upstream (Clark Hill, Richard B. Russell, and Hartwell) are all designed to withstand PMF combined with wind setup and wave runup from the wave action. It is unlikely that the dams will be overtopped. The dominotype upstream dam failure is discussed in paragraph 2.4.4.1. The Savannah River Basin embraces three distinct areas: the

Blue Ridge Mountains at the headwater; the Piedmont belt which ends around Augusta, Georgia; and the Coastal Plain from there to the Atlantic Ocean. The river reach between Clark Hill Dam and Augusta is narrow and has little valley storage. Below Augusta the river channel has wide flood plains which store flood flows and modulate the peak discharge as the flood progresses downstream. This causes the peak discharge by the VEGP site to be less than at Augusta.

The PMF peak discharge at the VEGP site, ignoring the valley storage effect, is 895,000 ft^3/s, which corresponds to a river stage of 136 ft msl read from a steady-state rating curve (figure 2.4.3-3). The PMF peak discharge at the same location, if valley storage of the upstream reach is taken into consideration, is 540,000 ft^3/s, and maximum flood wave elevation is 126 ft msl.

The Corps of Engineers has a PMF outflow hydrograph at Clark Hill Dam.[8] The NWS computer model[7] is used to route the hydrograph to the VEGP site. The river cross-sections used are those given on drawing AX6DD320.

The hydrographs from subbasins 7, 8, 9, and 10 are routed and combined with the main hydrograph at proper location as lateral hydrographs and are based on PMP from Hydromet No. 51 in conjunction with Hydromet No. 52. The PMF discharge based on this routing is 710,000 ft^3/s at the VEGP site, and the maximum flood wave elevation at the site is 138 ft msl. The routed hydrograph is shown in figure 2.4.3-4.

In a previous study done for the Preliminary Safety Analysis Report (PSAR), the floods of 1929, 1940, and 1948 were routed using the unsteady, nonuniform flow TVA computer program Simulated Open Channel Hydraulics (SOCH)[9] and assumed Manning's n values to produce observed stages. The Manning coefficient values, which are shown on drawing AX6DD406, are for left over bank, main channel, and right over bank, respectively, and are assumed the same at all elevations at the same cross-section. The river cross-sections used in this routing are shown on drawing AX6DD406. These Manning values reproduced the flood's stages close to the observed ones and therefore were assumed to be adequate and representative for use in other flood routings.

2.4.3.5 Water Level Determinations

A stage discharge relationship is developed for the VEGP site (river mile 151.1 as shown in figure 2.4.3-3), by using a computer program obtained from the Corps of Engineers, designated HEC-2 Water Surface Profiles.[10] In this study 29 cross-sections are taken from United States Geological Survey (USGS) topographical maps. The locations of these sections are shown on drawing AX6DD407. Three of these sections at the VEGP site are field checked and compared with the USGS maps as shown on drawing AX6DD407. In order to obtain reasonable n values, the floods of 1929, 1940, and 1948 are run to compare high-water marks with computer results. Drawing AX6DD407 shows the n values used in the comparison. Using these values are the steady flow conditions for various discharges, and the results are shown on drawing AX6DD407. These data do not include wind wave setup or wave runup effects.

2.4.3.6 Coincident Wind Wave Activity

A wave height analysis is made in accordance with standard procedure. (See references 11, 12, 13, 14, and 15.)

A reasonable sustained wind speed for this site is considered to be 50 mph.[13][16][17] This is about 40 percent more than the recorded maximum wind speed for a 1-h duration at Greenville, South Carolina. The longest fetch is for a location in the middle of the river opposite the plant site with the wind blowing in an upstream direction. The fetch diagrams[11][14][15] used to determine this longest fetch are shown on drawing AX6DD321. The effective fetch of 9.5 miles is used in the wind wave estimate for all conditions, because the water surface configuration would be about the same for each condition considered.

To generate the maximum wave the 50-mph wind must be from a southerly direction and must exist for at least 78 min. The wave study results are given below.

On a three-to-one riprapped slope:

A.	Significant wave height		8.4 ft
B.	Significant wave runup		9.1 ft
C.	Wind setup		1.9 ft
		Total runup (B + C)	11.0 ft
D.	Maximum wave height		14.0 ft
E.	Maximum wave runup		12.8 ft
F.	Wind setup		1.9 ft
		Total runup (E + F)	14.7 ft

On a three-to-one smooth slope:

G.	Maximum wave height		14.0 ft
H.	Maximum wave runup		25.3 ft
I.	Wind setup		1.9 ft
		Total runup (H + I)	27.2 ft

These total runups are added to standard project flood (SPF) pool el 119.9 ft and to SPF plus dam failure surge el 141 ft to obtain the maximum elevation the water would reach shown earlier in this section. When the total runup is added to the PMF level of 138 ft msl, the water level may reach 165 ft msl. The safety-related equipment is well above the flooding stage.

2.4.3.7 References

1. Schreiner, Louis C., and Riedel, John T., "Probable Maximum Precipitation Estimates, United States East of the 105th Meridian," Hydrometeorological Report No. 51, National Weather Service, NOAA, U.S. Department of Commerce Washington D.C., 1978

2. Hansen, E. M., and Schreiner, L. C., "Application of Probable Maximum Precipitation Estimates, United States East of the 105th Meridian," Hydrometeorological Report No. 52 (preliminary), National Weather Service, Silver Springs, Maryland, 1979.

3. Fox, William E., Hydrologist in Charge, River Forecast Center, Atlanta, Georgia, Letter of Correspondence, January 2, 1980.

4. U.S. Army Corps of Engineers, HEC-1 Computer Program User's Manual, 1975.

5. U.S. Army Corps of Engineers, South Atlantic Division, Water Resources Development in South Carolina, Atlanta, Georgia, March 1975.

6. Chow, Ven Te, Handbook of Applied Hydrology, McGraw-Hill Book Company, New York, New York, pp 9-46 through 9-48, 1964.

7. National Weather Service, The New Dam Break Flood Forecasting Model System User's Manual, October 1980.

8. U.S. Army Corps of Engineers, Reservoir Regulation Manual, Savannah, Georgia, December 1962 (Revised January 1968).

9. Tennessee Valley Authority, Computer Code Simulated Open Channel Hydraulics (SOCH), Transient Flow Mathematical Model, January 1972.

10. U.S. Army Corps of Engineers, HEC-2 Water Surface Profiles, February 1972.

11. Saville, T., Jr., McClendon, E. W., and Cochran, A. L., "Freeboard Allowances for Water in Inland Reservoirs," J. Waterways and Harbors Division, American Society of Civil Engineers, May 1962.

12. Saville, T., Jr., McClendon, E. W., and Cochran, A. L., "Freeboard Allowances for Water in Inland Reservoirs," J. Waterways and Harbors Division, American Society of Civil Engineers, p 81, May 1963.

13. McCartney, B. L., "Wave Runup and Wind Setup on Reservoir Embankments," U.S. Army Corps of Engineers, ETL 1110-2-221, November 29, 1976.

14. U.S. Army Corps of Engineers, "Computation of Freeboard Allowances for Waves in Reservoirs," ETL 1110-2-8, December 16, 1966.

15. U.S. Army Corps of Engineers, Coastal Engineering Research Center, Shore Protection Manual, Vol 2, 3rd Edition, 1977.

16. U.S. Army Corps of Engineers, "Freeboard Criteria for Dams and Levees," Concepts for Surface Wind Analysis and Record Velocities, Technical Bulletin No. 1, March 1959.

17. U.S. Army Corps of Engineers, "Freeboard Criteria for Dams and Levees," Severe Windstorms of Record, Technical Bulletin No. 2, January 1960.

2.4.4 POTENTIAL DAM FAILURES

The major dams upstream of the VEGP site are designed such that seismic events will not cause failure of these structures. However, domino-type failure of the upstream dams is assumed to evaluate the effect of the flood wave at the plant site. All the safety-related structures and equipment are above the probable maximum flooding coincident wave height elevation of 165 ft mean sea level (msl). The guidance provided in Regulatory Guide 1.59 was considered in the evaluation of potential dam failures.

2.4.4.1 Dam Failure Permutations

There are 13 dams on the Savannah River and its tributaries above the VEGP site.(1) The Hartwell and Clark Hill Dams are major dams immediately upstream of the plant site. Between Clark Hill and Hartwell Dams another major dam, Richard B. Russell, is under construction and is scheduled for completion in 1984. The existing and proposed dams are shown in figure 2.4.4-1.

A profile of the river is shown in figure 2.4.4-2 along with a list of locations for various dams. Data on the dams are listed in table 2.4.1-3. Plan and sections of Clark Hill Dam, a large dam 71.6 miles above VEGP and the closest upstream dam, are shown in figure 2.4.4-3. Area-capacity curves for five large reservoirs above VEGP are shown in figure 2.4.4-4.

The seismic design criteria and spillway design criteria for Jocassee and Keowee Dams are given in the Preliminary Safety Analysis Report (PSAR) for Oconee project, subsections 2.4.2 and 2.4.3.(2) Design criteria for Corps of Engineers' dams are given in the project report for each dam.(2)(3)(4)

The VEGP site is considered a dry site. Even if Clark Hill Dam fails due to a seismic event coincident with probable maximum flooding, the flood wave dissipates substantially due to valley storage before it reaches the plant site. In that event the intake structure and the river makeup water facility may go out of service. But, since the river makeup water facility is a secondary source of makeup water, it will not affect the safety of the plant, as discussed in paragraphs 2.4.11.5 and 2.4.11.6.

A study of the locations of the dams above VEGP and the Savannah River stream profile, as shown in figures 2.4.4-1 and 2.4.4-2, respectively, to determine the basic assumptions for the worst possible dam failure to affect VEGP, indicates that the worst reasonable failure that can be postulated would be to have Jocassee Dam fail during a standard project flood (SPF) with an earthquake and the chain reaction that would follow. A seismic failure of Jocassee Dam is not reasonably possible, since it has been designed to include earthquake forces (design basis earthquake - 0.1 g) as stated in the PSAR for the Oconee project referred to in the previous paragraph.(2) However, it is more conservative to postulate this failure in determining the maximum water level at VEGP.

Jocassee Dam is a rockfill dam 400 ft high, and it would be unlikely that the dam would fail in its entirety. For this analysis it is assumed that a breach would cause a wave about 30 ft high below the dam. This breach could be a slide at least 50 ft deep across the entire top of the dam caused by an earthquake at the peak of SPF, or it could be a deep, narrow section that would go out triggering a breach. This wave would be 16 ft(5) high at Keowee Dam, 15 miles downstream. Keowee Dam, during its SPF, would have a freeboard of 15 ft, so that the surge wave would not be contained but would overtop and breach the earth dike section. This could cause a 15-ft surge wave at Hartwell Dam 51 miles downstream. Since Hartwell Dam would have a freeboard of 11 ft during its SPF, it would be overtopped by the surge wave, starting a breach in the earth dam section. Assuming Richard B. Russell Dam has been built, which could cause a worse wave at VEGP, it would be overtopped. It is anticipated that a 20-ft surge wave could reach Richard B. Russell Dam from a Hartwell Dam failure and overtop the dam by about 9 ft during its SPF. From this failure a 15-ft surge wave would reach Clark Hill Dam, overtopping it by 6 ft during its SPF.

At this instant the spillway discharge at Clark Hill Dam would increase suddenly. The additional water pressure against the concrete structures would increase the horizontal load by 140 klb/linear ft and move the resultant about 10 ft. Thus, the concrete structures would remain stable. The 6-ft flow over the earth sections would start to erode them. The breaching of these sections would take some time; however, to be conservative, we assume their instantaneous removal. Then the flow elevation in the breach sections would be about el 290 ft based on the standard equations,[5] and the discharge through the combined east and west breaches would be about 1,600,000 ft^3/s. At this time the flow through the spillway would cease, because the reservoir level against the dam would be el 290 ft and below the spillway crest of el 300 ft. If the discharge through the breached dam is based on the submerged broad crested weir equations, the flow is estimated to be 2,400,000 ft^3/s.

Tailwater will be controlled by a river section 2.3 miles downstream of Clark Hill Dam. The Reservoir Regulation Manual, (Savannah River Basin, December 1962, Revised January 1968) shows a tailwater elevation of 255 ft msl for a spillway design flood of 1,055,000 ft^3/s (figure 2.4.4-5). Using the above discharge and tailwater elevation values, a channel n value of 0.065 is computed,[6] which is reasonable but low. For this value of n, the required elevation at the section to pass 2,400,000 ft^3/s is found to be el 292 ft msl.[6][7] This value is higher than the stage at the breach and indicates that flow through the breach will be reduced by the tailwater effects. However, to be conservative, the flow of 2,400,000 ft^3/s is transferred to below Augusta undiminished in magnitude. At this point, the rise of 1,800,000 ft^3/s in discharge above the SPF is assumed to occur in 3 h and the peak of 2,400,000 ft^3/s lasts for 4 h. It should be noted that the SPF at Clark Hill is computed[3] by taking one-half of the ordinates of probable maximum flooding. The routing of the dam failure surge from below Augusta is described in paragraph 2.4.4.2.

It should be noted that stages at VEGP are not very sensitive to either peak flow magnitude or peak flow duration, probably due to the great width of flood plain in the vicinity of VEGP. Some initial studies were made, which show the relative insensitivity of VEGP stage to variation in peak flow and duration. At 2,400,000 ft^3/s an increase of peak duration from 4 h to 8 h increased the peak flow at VEGP about 30,000 ft^3/s. Increasing the flow from 2,400,000 ft^3/s to 3,400,000 ft^3/s, both at 4-h duration, increased the flow at VEGP about 100,000 ft^3/s. The corresponding steady flow stage changes at VEGP are about 0.5 ft and 2 ft, respectively.

The following is a description of conditions if an operating basis earthquake (OBE) is assumed during an SPF, which could affect both Clark Hill and Richard B. Russell Dams simultaneously. Richard B. Russell Dam is designed for a seismic factor of 0.1 g.[4] Since this closely corresponds to the OBE, the dam should not fail. At Clark Hill Dam the concrete structures were conservatively designed.[3] A review of the most critical section during maximum pool elevation of 346 ft with an OBE of 0.12 g shows the resultant falls within the base with a toe stress of 207 psi, which is within the allowable limit (figure 2.4.4-6). It is assumed that failure of the earth dike section will occur because of insufficient data for analysis. This is not the critical condition because the maximum water surface in the reservoir is 11 ft lower and the resulting flood wave is some 6 ft lower than that described above.

If a simultaneous earthquake at Hartwell and Clark Hill Dams is assumed, it is likely that the resulting conditions are less critical than those described initially in this section. If Hartwell Dam fails during an OBE, the resulting wave would be several feet less than that assumed from upstream domino-type failure. When this wave arrives at Clark Hill where the earth section had failed at the time of the OBE, the stage would be significantly lower than that assumed initially in this section.

The failure of War Woman Dam, shown in figure 2.4.4-2, has been considered, and it is found that the surge wave diminishes as it progresses downstream and the volume in War Woman Reservoir is contained in Hartwell Reservoir. Since the assumptions made for the failure of Clark Hill Dam and its routing to VEGP are conservative, no other mode of failure is assumed for determining effect at VEGP.

Since the flood plains at and below VEGP are wide, there is no likelihood that a landslide will occur to raise the flood level postulated. The effect of PMF concurrent with waves and the maximum stage reached is discussed in paragraph 2.4.3.6.

2.4.4.2 Unsteady Flow Analysis of Potential Dam Failures

The hydrograph for the failure of Clark Hill Dam, resulting from the failure of the dams further upstream as described in paragraph 2.4.4.1, is considered to be at New Savannah Bluff Lock and Dam, which is at the south end of Augusta, Georgia, and is routed to Burton's Ferry Bridge using the unsteady, nonuniform flow computer program SOCH[(7)] and roughness values stated in paragraph 2.4.3.4.

To determine the effect of the dam failure, the hydrograph for the SPF is routed past VEGP using the same computer program referred to above. The SPF hydrograph is computed by taking one-half of the ordinates of the Corps of Engineers' spillway design flood hydrograph (PMF hydrograph). The results at the plant site are shown in figure 2.4.4-7. The peak stage is el 119.9 ft msl. Then the dam failure hydrograph, as stated above, is routed using the same procedure. The results are shown in figure 2.4.4-8. This gives the peak stage an elevation of 141 ft msl and a discharge of 980,000 ft^3/s. Wind wave and runup effects are not shown, but the resultant water elevation discussed in paragraph 2.4.4.3 considers these phenomena.

2.4.4.3 Water Level at Plant Site

A wind velocity of 50 mph, as stated in paragraph 2.4.3.6, is also considered for this condition. The runup at the river pump station under the above conditions for the significant wave coincident with the dam failure surge wave by VEGP, as stated above, would reach el 152 ft msl. The wave runup would reach el 168 ft msl on the natural slope (3 to 1) along the plant side of the river.

2.4.4.4 References

1. U.S. Study Commission, Southeast River Basin, Plan for Development of the Land and Water Resources of the Southeast, Appendix 1, 1963.

2. Duke Power Company, Oconee Nuclear Plant Preliminary Safety Analysis Report, subsections 2.4.2 and 2.4.3.

3. U.S. Army Corps of Engineers, Reservoir Regulation Manual, Savannah River Basin, Savannah River, Clark Hill Lake.

4. U.S. Army Corps of Engineers, Design Memorandum II, Construction Procedure Diversion Plan, Hydraulic Data, Richard B. Russell Project, 1969.

5. Rouse, Hunter, ed, Engineering Hydraulics, John Wiley & Sons, Inc., New York, New York, 1950.

6. Chow, Ven Te, Open Channel Hydraulics, McGraw-Hill Book Company, Inc., New York, New York, 1959.

7. Tennessee Valley Authority, Computer Code Simulated Open Channel Hydraulics (SOCH), Transient Flow Mathematical Model, January 1972.

2.4.5 PROBABLE MAXIMUM SURGE AND SEICHE FLOODING

Since the VEGP is not near a large body of water, this section does not apply.

2.4.6 PROBABLE MAXIMUM TSUNAMI FLOODING

Since VEGP is neither near a large body of water nor in a volcanically active seismic region, this section does not apply.

2.4.7 ICE EFFECTS

Icing does not occur on the lower reaches of the Savannah River. Based on records of minimum temperature as shown in table 2.4.7-1, the temperature is higher than 5°C most of the time.(1)

With Hartwell, Richard B. Russell, and Clark Hill Dams upstream of Augusta the ice jam is unlikely to occur at the VEGP river intake, since these upstream dams will modulate the water temperature.

If surface icing did occur at VEGP, the design of the river intake and the normal water depth of 14 ft ensure that an ice sheet across the entire river will not interfere with flow of water into the intake.

2.4.7.1 REFERENCE

1. Department of the Interior, U.S. Geological Survey, Atlanta, statistical data on river flow, temperature, and chemical analyses for the Savannah River (data retrieved from USGS computer on July 8, 1981, for the Hydro Projects Department, Southern Company Services).

2.4.8 COOLING WATER CANALS AND RESERVOIRS

2.4.8.1 Canals

There is a short canal less than 400 ft long formed by vertical sheet piling walls connecting the river intake structure and the Savannah River. As discussed in paragraph 2.4.11.5, the river

makeup water is the secondary backup source for the nuclear service cooling water tower basins. The river intake structure as well as the intake canal have no safety function. Thus, failure of this river intake canal will not cause a threat to the safety of the plant operations (subsection 2.4.9).

2.4.8.2 Reservoirs

There are no reservoirs at VEGP.

2.4.9 CHANNEL DIVERSIONS

The river upstream from the site has bluffs and steep slopes along the west bank. If it is assumed that a bluff slid into the riverbed just upstream from the pump station and blocked the river flow for a few hours and if it occurred during low flow, the water would pile up above the obstruction and then divert over the flood plain. If this postulated event occurs, the plant can be safely shut down due to available storag3e onsite of adequate quantities of nuclear service cooling water (NSCW). The water supplies for the NSCW system and the mechanisms by which primary and secondary supplies are activated are discussed in subsection 9.2.5.

The Savannah River near the VEGP site has a relatively straight and stable channel, and also in this vicinity the river does not have an oxbow forming. It is very unlikely that the river will be diverted from the intake by natural causes. Any possible effect on water supply to the intake from channel changes should come from extremely slow changes, which can be remedied as they occur.

2.4.10 FLOOD PROTECTION REQUIREMENTS

None of the safety-related facilities at the VEGP is susceptible to flooding by the most severe flood at the site. One flood source is the river, and highest postulated flood level due to an upstream dam break is el 141 ft (paragraph 2.4.4.2). Waves resulting from such a flood and wind are postulated to reach maximum elevation of 168 ft, and since the plant structures are at finished grade elevation of 220 ft, flooding is not probable from this source.

Another source of flooding is severe rainfall (paragraph 2.4.3.1). The power block area for Units 1 and 2 is on a high plateau and is not in the path of any adjacent watershed. The topography is such that the runoff is directed away from the power block by a combined system of culverts and open ditches to natural drainage channels (drawings CX2D46V003, CX2D46V004, CX2D46V005, and CX2D46V006). The plant drainage system is designed for a maximum precipitation of 4 in./h and 4.8 in. of rain in 2 h (paragraph 2.4.2.3); the system has been checked to make sure that flooding of safety-related equipment does not occur as a result of the probable maximum precipitation (PMP).

The roofs of all safety-related structures are designed to pass the local PMP, corresponding to time of concentration of flow. The design includes measures to guard against wind-induced seepage through roof and wall penetrations or doors where safety-related equipment could be damaged. The design is based on PMP having an intensity of 15 in. of rainfall in 1 h and 22 in. of rainfall in 2 h (paragraph 2.4.2.3).

2.4.11 COOLING WATER SUPPLY

2.4.11.1 Low Flow in Streams

The low flow in the Savannah River by the VEGP site is regulated by Clark Hill Dam and the New Savannah Bluff Lock and Dam. A minimum flow of 5800 ft^3/s is required for navigation in the Savannah River downstream of Clark Hill Dam.(1) However, it should be noted that a discharge of 6300 ft^3/s is normally provided 80 percent of the time.(1) As shown in figure 2.4.11-1, a minimum required flow of 5800 ft^3/s is released from New Savannah Bluff Lock and Dam. The Clark Hill Dam project is designed for a maximum drawdown of 18 ft from the top of the power pool elevation of 330 ft msl to a minimum pool at 312 ft msl. However, it is not anticipated that the minimum pool will be reached more often than once in 150 years.(1) The Clark Hill Dam project also ensures a required minimum river flow at the Savannah River Plant of the former Atomic Energy Commission and helps to regulate the water temperature.(2)

A low flow stage at the VEGP site corresponding to minimum river flow of 5800 ft^3/s is el 80.4 ft msl.(3)

A low flow rating curve as given in figure 2.4.11-2 and an intermediate flow rating curve as given in figure 2.4.11-3 were developed at river mile 151.1 for the pumped intake structure. These curves are based on backwater profiles computed from actual river cross-sections near the site and from assumed sections based on river bottom profiles. The location of surveyed sections is shown in figure 2.4.11-4.

Echo soundings and probings were also made in this section of the river, and the results are shown in figure 2.4.11-5. For a distance of approximately 2 miles downstream, much of the river bottom is at el 70 ft. Studies have led to the conclusion that degradation of river, if any, will be slow and small. If a conservative allowance of a 2-ft decrease in stage were made for degradation, a dashed curve in figure 2.4.11-2 would result.

Flow records for Augusta, Georgia, for the periods 1884 through 1906 and 1926 through 1970 are examined. A hypothetical extreme drought flow of 957 ft^3/s is determined by a statistical analysis of the period 1926 through 1950 using a method developed by Gumbel.(4)(5) This period is used because it was a period during which no major dams were built on the river or its tributaries upstream of Augusta. It is then concluded that the hypothetical extreme drought would have a stage elevation of 74 ft msl.

From the flow records for the 62 years of examined data from the United States Geological Survey (USGS), it is concluded that a sustained minimum release of 5800 ft^3/s (the planned operation of Hartwell and Clark Hill Reservoirs) could have been maintained for this period. A flow of 500 ft^3/s is required under present conditions to provide water at el 73 ft at the pump intake. A flow of 1500 ft^3/s is required using the allowance for channel degradation. It is concluded that the regulated river flow is a highly dependable source of water for any makeup operation.

2.4.11.2 Low Water Resulting from Surges

This situation does not apply because VEGP does not withdraw water from a large body of water, nor is it located in a volcanic region.

2.4.11.3 Historical Low Water

The available flow records for Augusta, Georgia, for 62 years for the periods 1884 through 1906 and 1926 through 1970, as shown in Flow Characteristics of Georgia Streams,[6] are examined. The low flow of record for gauging station No. 2-1970, Savannah River at Augusta, Georgia, at New Savannah Bluff Lock and Dam (river mile 189.8) prior to construction of Clark Hill Dam, occurred on September 24, 1939. This was caused by the operation of gates at New Savannah River Lock and Dam. If the rating curve is extended below 1400 ft^3/s, an extreme minimum discharge of 648 ft^3/s is reached. This is an extrapolated instantaneous minimum. Water stage recorder graphs and discharge measurements were furnished by the Corps of Engineers. On the day this low flow was recorded, the average daily flow was 2940 ft^3/s. A copy of the recorded graph was obtained from the USGS, and figure 2.4.11-6 was prepared to show this hydrograph, which indicates that the lowest flow occurred for about 10 h, the daily flow being over 2000 ft^3/s. The lowest mean daily flow shown in the Augusta record was 1040 ft^3/s, which occurred on October 2, 1927.[6]

The minimum mean daily discharge for the period 1963 through 1970 (subsequent to the filling of both reservoirs) was 5130 ft^3/s in 1963. The storage for power and navigation releases (between normal and minimum pool levels) from Hartwell and Clark Hill Reservoirs is 2,445,000 acre-ft, which would provide an average release of 3350 ft^3/s for 1 year assuming no inflow.[1] The total storage (between top of gates and minimum pool level) from both reservoirs is 3,128,000 acre-ft, which would provide an average release of 4300 ft^3/s for 1 year assuming no inflow.[1]

2.4.11.4 Future Control

Since the minimum flow condition is controlled mainly by upstream dam releases and no large future users of water are known, the future minimum should be similar to past minimum.

2.4.11.5 Plant Requirements

The heat removal requirement is greater for Unit 2 than for Unit 1 due to the larger spent fuel storage capacity of the Unit 2 pool. This results in some differences in the system performance and available inventory for each unit. These differences, where they are significant, are outlined in the following paragraphs. When a single value is presented, it represents the upper bound for both units.

Evaporation losses from the safety-related nuclear service cooling water system (NSCWS) tower basins are presented in subsection 9.2.5. During normal plant operation, a flow of 435-gal/min makeup water is required to replace these evaporative losses and also losses due to drift, seepage, and cooling tower blowdown. During accident conditions, maximum evaporative losses are 1490 gal/min (for 2-train NSCWS operation) immediately after reactor shutdown, decreasing to 340 gal/min after approximately 1 day and 165 gal/min after 16 days. Tower blowdown is isolated during accident conditions, and therefore, total losses and makeup requirements during accident conditions comprise only evaporation, a seepage loss of 1 to 2 gal/min, and drift. However, as discussed in subsection 9.2.5, tower makeup is not required during the period subsequent to reactor shutdown under accident conditions concurrent with a loss of offsite power. The combined storage capacity of the two NSCWS cooling tower basins provided for each generating unit is sufficient to provide 30-day inventory for Unit 1 and 30-day

inventory for Unit 2 to permit safe reactor shutdown and maintenance of the plant in a safe shutdown condition in conformance with the short-term recommendations of Regulatory Guide 1.27. Total NSCWS water required and available storage capacity of the NSCWS cooling tower basins are discussed in subsection 9.2.5.

As discussed in subsections 9.2.1 and 9.2.5, the NSCWS pumps take suction from the NSCWS cooling tower basins and circulate cooling water through the component cooling water heat exchangers, the auxiliary component cooling water heat exchangers, the containment coolers, and miscellaneous pump coolers. To ensure that the inventory of both NSCW basins is available in the event that only one NSCWS train is available, one safety-grade transfer pump is installed in each NSCWS cooling tower basin, with the train A transfer pump installed in the train B NSCWS tower basin and vice versa. The rated capacity of the transfer pumps is 600 gal/min, which is well in excess of the NSCWS makeup requirements during accident conditions, except during the first few hours of the transient, thus ensuring adequate makeup to the operating tower.

To ensure adequate NSCWS pump and transfer pump suction conditions at all times, all pumps are installed in pits, which extend 7 ft below the floor of the tower basin at el 137 ft.

The suction of the NSCW transfer pumps is located at elevation 132 ft 6 in., and the pumps are designed to operate with a minimum water level at elevation 141 ft 3 in. to provide adequate NPSH.

The suction of the NSCW transfer pumps is located at elevation 131 ft 4 in., and the pumps are designed to operate with a minimum water level at elevation 137 ft or as long as there is water in the tower basin.

Thus, the entire basin inventory of both NSCWS cooling towers, except for the last 4 ft 3 in., is available to satisfy the short term recommendations of Regulatory Guide 1.27 using only safety-related components.

To provide NSCWS makeup during normal operation and to provide long-term (after 30 days) cooling in conformance with Regulatory Guide 1.27, water from nonsafety-related systems (deep water wells and the Savannah River) is available. Two makeup wells provide the primary makeup to the NSCWS cooling towers for both generating units. Wells MU-1 and MU-2A are equipped with a 2000 gal/min and 1000 gal/min pump, respectively. The wells are approximately 2100 ft apart, and each is located approximately 1000 ft and 400 ft, respectively, from the immediate plant site boundary. These wells provide water for the following uses:

- Fire protection.
- Construction demands.
- Potable and sanitary uses.
- Makeup to the well water storage tank.

Water in the well water storage tank is utilized for utility purposes, for NSCW tower makeup, and for the makeup demineralizer. Plant operating procedures will be utilized by plant personnel to ensure proper operation of these water makeup systems.

The makeup wells have been sited on the basis of extensive investigations on the characteristics of the groundwater at plant site, which show the existence of two distinct aquifers, a shallow water aquifer and a deep, confined Tuscaloosa aquifer.

Estimated recoverable water quantity in the Tuscaloosa aquifer is approximately 21 billion acre-ft, thus providing a safe yield of 5 billion gal/day.

In the event that the well water is not available, river water can also be utilized for NSCW cooling towers makeup. The Savannah River provides the makeup water for the natural draft cooling towers and dilution water for the plant effluent discharge. The Savannah River, bordering the plant site on the east side, is approximately 340 ft wide and 13 ft deep at the site, with minimum and maximum flows of 5800 ft^3/s and 71,700 ft^3/s, respectively. The river temperature, recorded at the Burton's Ferry Bridge from January 1960 through September 1970, ranged from a minimum of 41°F (5°C) to a maximum of 84°F (28.9°C).

Four 22,000-gal/min capacity nonsafety-related river makeup water pumps provide makeup to the natural draft cooling towers. Provision is also made for these pumps to supply additional dilution water as required for the plant radwaste liquid discharge to the Savannah River. Piping is provided from the supply header of the river makeup water pumps to the basins of the NSCW cooling towers as a secondary source of makeup water. The plant makeup-water well system serves as the primary makeup source for the NSCW cooling towers.

To ensure adequate river makeup water pump suction conditions, the pump suction is installed 4 ft 10 in. below the minimum low river water surface elevation of 73 ft. The river intake pumps are designed to operate with a minimum submergence of 4 ft 6 in.

2.4.11.6 Heat Sink Dependability Requirements

As discussed in paragraph 2.4.11.5 and subsection 9.2.5, the ultimate heat sink for VEGP is provided by the water inventory in the NSCW cooling tower basins. As part of the power uprate, an analysis was performed to calculate the basin inventory, basin temperature, and the time-dependent evaporative loss. The combined volume of water in the two cooling tower basins is sufficient to provide greater than a 30-day cooling capacity for Unit 2 with two trains operating the first day and one train operating for the remaining days. The tower basin capacity for Unit 1 was not re-evaluated for the power uprate. The calculation was performed for Unit 2 since the higher heat loads from the reracked spent fuel make it the more conservative choice; therefore, the results will be bounding for Unit 1. As discussed in subsection 9.2.5, tower basin capacity and maximum evaporation losses are based upon the worst combination of recorded dry and wet bulb temperatures, which results in the greatest water loss over the postulated 30-day period following an accident. The total losses include tower drift and seepage losses but do not consider cooling tower blowdown, since blowdown is isolated during accident conditions.

Seepage into or out of the NSCW tower basins is minimized by the thick shell walls (3 ft thick above el 155 ft 5 1/2 in. and 5 ft thick below) and basemats (9 ft thick). In addition, level alarms are provided for each NSCW tower which will indicate when the water level in a tower basin drops below el 217 ft 3 in. (normal water level during plant operation is elevation 217 ft 9 in.). An automatic makeup valve will operate to maintain basin water level above el 217 ft 6 in. When a level alarm is activated, additional makeup water may be added and the basins will be monitored for any abnormal losses.

During normal operation, level instrumentation in the basins and pressure and flow instrumentation in the NSCWS piping give the operators a constant readout of system status with alarms as required to warn of abnormal system operation. As described in paragraph 2.4.11.5, there are two sources of makeup water: deep water wells and river water. During accident conditions concurrent with loss of offsite power, the safety-grade transfer pumps described in paragraph 2.4.11.5 can be manually operated to ensure full use of the basin inventory.

The safety-related portions of the NSCWS are designed and constructed as Nuclear Safety Class 3, Seismic Category 1, and in accordance with the ASME Boiler and Pressure Vessel Code, Section III, Class 3. Therefore, the system will remain functional during and after the safe shutdown earthquake and after any single failure. Thus, the system provides a dependable heat sink for the plant. Subsection 9.2.5 provides additional detail regarding the design of the ultimate heat sink.

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2.4.12 GROUND WATER

2.4.12.1 Description and Onsite Use

2.4.12.1.1 Regional Aquifers, Formations, Sources, and Sinks

There are two major regional aquifer systems recognized in the coastal plains of Georgia and South Carolina. The lower regimen is referred to as the Cretaceous aquifer system and consists primarily of the sands, gravels, and clays of the Tuscaloosa Formation. The upper regimen is variously referred to as the Tertiary aquifer system, the principal artesian aquifer, and the limestone aquifer. It consists primarily of the limestones and permeable sands of the Lisbon Formation or stratigraphic equivalents. In their outcrop areas, both systems are hydraulically connected and are under water table conditions. Both systems are under artesian conditions elsewhere throughout most of the region. The relatively impermeable clays and silts of the Huber and Ellenton Formations (Paleocene) overlie and confine the Cretaceous system, while the clays and clayey sands of the Barnwell Group (late Eocene), and other younger fine-grained sediments, overlie and confine the Tertiary system.

In addition to the major artesian aquifer systems, ground water also exists under water table conditions in the Late Eocene and younger formations, alluvial deposits, and Terrace deposits that are present throughout the region. Ground water is also present in the crystalline basement complex and the overlying Triassic Newark Group. In the paragraphs that follow, each of the regional systems is more fully described.

2.4.12.1.1.1 Pre-Cretaceous Formations. Pre-Cretaceous stratigraphy in the study region consists of Precambrian through Paleozoic igneous and metamorphic basement rocks overlain by Triassic fine- to coarse-grained clastic sediments and conglomerates. The rocks in the basement complex are dense and essentially impermeable, but ground water exists in secondary openings such as joints and fissures. Injection, pumping, and packer testing were performed in the crystalline basement in the early 1960s as part of an exploratory program to determine the feasibility of storing radioactive wastes in these rocks. This program was performed at the Savannah River Plant near Aiken, South Carolina, and indicated an average permeability of 0.0003 gal/d/ft^2 and an overall transmissivity of 160 gal/d/ft.(1)

These relative low values and correspondingly low yields to wells during pumping tests indicate that the crystalline basement complex is not a significant hydraulic unit, at least in the area tested. Hydraulic characteristics of the basement rocks in other areas are not as well known, but, based on lithologic descriptions, there are little data to indicate significantly different hydrogeologic properties elsewhere in the region.

Triassic rocks of the Newark Group overlie the basement complex, but their total areal extent has not been defined.[2] Where they are known to exist, the Triassic rocks consist of a variety of sedimentary grain sizes in various degrees of induration. Rocks of this lithologic character could conceivably include local permeable zones, but the highly productive and more shallow Cretaceous and Tertiary aquifer systems, which overlie the Triassic rocks, are available for use throughout the region, and, consequently, there has been little impetus to explore the deeper horizons for ground water potential. The very few wells that have been opened to the Triassic zones have yielded only meager quantities of ground water and this has also done little to encourage exploration. Until additional data are produced, the overall hydrogeologic character of the Triassic rocks will be obscure. Based on the sparse data presently available, the Triassic sequence does not appear to be a significant regional hydraulic unit, but it may have localized productive zones.

2.4.12.1.1.2 Cretaceous Aquifer System. Within the region, the Cretaceous ground water system is represented by the Tuscaloosa Formation, which consists of fluvial and estuarine deposits of cross-bedded quartzitic sand and gravel interbedded with silt, and clay. The coarse-grained sediments are mostly unconsolidated and are generally permeable, while the fine-grained sediments are partially consolidated and are generally permeable. In some areas, the fine-grained layers effectively confine the ground water to produce local artesian conditions that may differ from the overall system. In addition to the varying lithology, the formation also exhibits lateral facies changes, on-lap and off-lap relationships, and discontinuous lenses. The combined effects of these conditions have given rise to a complex multiaquifer system whose overall hydrogeologic character has not yet been fully defined.

The Cretaceous ground water system has not been extensively developed, primarily because the shallow Tertiary system is adequate for most ground water needs and is available for use throughout the region. Quantitative data from the limited number of test and production wells producing from the Cretaceous strata, and inferred data from geologic and stratigraphic studies, indicate clearly that the Cretaceous system is a highly transmissive regimen that is capable of yielding large quantities of good quality ground water. Callahan[12] estimates that the Cretaceous system has a safe yield of 5 billion gal/d throughout its known extent.

Recharge to the Cretaceous system is primarily by direct infiltration of rainfall in its outcrop areas, located north of VEGP in a 10- to 30-mile-wide belt extending from Augusta, Georgia, northeastward across South Carolina to the state line separating North and South Carolina. In the outcrop areas, precipitation penetrates the Cretaceous sediments and migrates downdip under a hydraulic gradient of 6 to 20 ft/mi.[2] Ground water in the outcrop areas is under water table conditions, but as it moves progressively downdip, it becomes confined beneath the Blue Bluff marl member of the Lisbon Formation in the vicinity of VEGP and beneath the Ellenton and Huber Formations farther downdip to the south and southeast. Hence, the Cretaceous system is under artesian conditions for most of its areal extent.

Discharge of the Cretaceous system ground water is primarily from subaqueous exposures of the aquifer that are presumed to occur along the Continental Shelf. Other discharge sources are to the Savannah River as described in section 2.14.12.1.2.2 and by pumping. Some discharge may also occur by upward leaking through the confining layers. This condition is inferred from the known variable permeability of the Ellenton and Huber Formations, but an area where such discharge is actually occurring has not been identified. Thickness of the Cretaceous system ranges from zero at the fall line to approximately 700 ft at the VEGP and 800 ft in east-central Georgia.

The quality of the ground water in the Cretaceous system has been satisfactory to good in the somewhat limited areas in which tests have been conducted. Such areas are mostly within the outcrop regions of the aquifer where it is shallow enough to develop economically. Several municipalities use water from the Cretaceous system and tests conducted by the United States Geological Survey (USGS) have shown that, except for marginally high concentrations of iron and nitrate, water from the Cretaceous aquifer meets all requirements of the U.S. Public Health Department.(2)

2.4.12.1.1.3 Tertiary Aquifer System. Underlying the southeastern United States is a highly productive aquifer system of Tertiary age that is referred to as the principal artesian aquifer in Georgia, Alabama, and South Carolina and as the Floridan aquifer in Florida(9). It is the primary source of municipal, industrial and agricultural water supply on the coastal plain of Georgia. Within Georgia and South Carolina, the aquifer system includes the carbonate rocks (and associated interbedded sands) of Eocene and Oligocene age. Within a single stratigraphic time horizon, facies changes downdip result in a confining layer becoming an aquifer. For example, beneath VEGP the Tertiary aquifer is comprised only of the unnamed sands of the Lisbon Formation and the overlying Blue Bluff marl acts as a confining layer. However, downdip, the Blue Bluff interfingers with a permeable limestone that becomes part of the Tertiary aquifer system, and beds younger than the Lisbon Formation form the confining layer.

Recharge to the Tertiary aquifer is primarily by infiltration of rainfall in its outcrop area, which is a belt 20 to 60 miles wide extending northeastward across central Georgia and into portions of Alabama to the west and South Carolina to the east. The strata, which make up the aquifer, dip southeastward beneath younger confining sediments. They include nearly impermeable clays and semipermeable fine-grained sediments of Oligocene and younger formations.

Discharge from the Tertiary aquifer occurs from pumping, from natural springs in areas where topography is lower than the piezometric level of the aquifer, and from subaqueous outcrops that are presumed to occur offshore. Discharge also occurs to the Savannah River where the river has incised down through the marl confining layer allowing discharge from the aquifer to the riverbed, as described in section 2.4.12.1.2.2.

The regional direction of flow of ground water in the Tertiary aquifer is south by southeast. Local deviations from the general flow direction occur in response to structural features, cones of depression caused by pumping, and by the effects of the discharge to the Savannah River.

Transmissivities of up to 220,000 ft^2/day have been measured for the Tertiary aquifer in southern Georgia.(4) The overall transmissivity of the aquifer has not been fully defined, but known variations in both vertical and horizontal permeabilities suggest that the average transmissivity in Georgia is substantially less than 220,000 ft^2/day.

Thickness of the principal Tertiary aquifer ranges from near zero at its northwestern limit area to approximately 900 ft in southern Georgia, where it is represented by the Claiborne Group. In Florida the aquifer thickens to several thousand feet.

Generally, the water quality of the Tertiary aquifer is good, although high hydrogen sulfide content and other objectionable constituents have been reported in some areas.(4)

2.4.12.1.1.4 Water Table Aquifer. Ground water is present under water table conditions in the late Eocene and younger sediments throughout the region. These include the sands and

carbonate rocks of the Barnwell Group (late Eocene), the clayey sand and gravel of the Hawthorne Formation (Miocene), and extensive Pleistocene to Holocene alluvial and terrace deposits. These sedimentary units are characterized by a wide spectrum of vertical and horizontal permeabilities, which have resulted in hydrogeologic conditions that are not favorable for significant development. Also, the water-bearing materials have been incised to various depths by the many streams throughout the region and are, therefore, not laterally continuous over large areas. Most wells open to the water table aquifer produce quantities that are adequate only for domestic and livestock use. A brief description of each stratigraphic unit and its water-bearing characteristics is given in table 2.4.12-1 and in the paragraphs that follow.

The Barnwell Group (Eocene) consists of fine- to coarse-grained clayey sand and sandy clay with subordinate lenses of limestone and sandstone. These deposits are semiconsolidated to consolidated and are present throughout the region. Limestone lenses in the Barnwell Group are prevalant in Georgia and occur as interbeds with the clayey sands that dominate the formation. The sands of the Barnwell Formation are relatively fine-grained, and the presence of silt and clay lenses reduces the overall permeability of the formation. Ground water is usually available only in quantities suitable for domestic or livestock use.

The Hawthorne Formation (Miocene) consists of tan, red, and purple, dense, sandy clay with interbedded lenses of gravel and numerous clastic dikes. Exposure of the Hawthorne Formation is extensive in the East Gulf Coastal Plain province. The fine-grained nature of the formation and clastic dike materials (silty to sandy clay) precludes significant ground water development. Ground water is locally available in more permeable lenses but only as small bodies of perched water.

Alluvial deposits ranging in age from Pliocene to Holocene, are present throughout the region. The Pliocene deposits occur principally as cappings on interstream divides, while the Pleistocene and Holocene deposits occur mainly in the tributary stream channels and flood plain of the Savannah River. All of the alluvial deposits consist of poorly sorted clay, sand, and gravel. Thickness rarely exceeds 30 ft. Areal extent is highly variable due to stream downcutting and erosional processes. In general, these deposits contain only small quantities of ground water.

Terrace deposits of Pleistocene age are also present in the Coastal Plain province. Cooke recognized seven such structures in South Carolina and postulated that they are of marine origin.[5] The deposits consist of sand, silty clay, and gravel; their thickness is not more than a few tens of feet. The water-bearing characteristics of the terrace deposits are not well known due to lack of development. Their principal hydraulic function is probably limited to acting as a recharge conduit to the lower aquifers.[2]

2.4.12.1.2 Local Aquifers, Formations, Sources, and Sinks

The initial geologic exploration program for VEGP included ground water field studies consisting of aquifer and permeability testing, well canvassing, chemical analyses, and observation well installation and monitoring. Additional hydrogeologic data were also acquired during the Millett fault study of 1982.[3] These data are the basis for establishing ground water conditions of the site area (figure 2.4.12-1)[3]. In general, those conditions conform to the regional aquifer systems described in paragraph 2.4.12.1.1. The local characteristics are explained under the appropriate headings in the paragraphs that follow.

2.4.12.1.2.1 Pre-Cretaceous Formations. None of the exploratory borings for the VEGP penetrated the crystalline basement rocks, but an exploration program conducted in South Carolina to determine the feasibility of storing radioactive wastes in these rocks produced a considerable amount of hydrogeologic data. This program was conducted in the early 1960s by the USGS at the Savannah River Plant a few miles east of VEGP.(1) The data indicate that a nominal amount of ground water exists in the crystalline basement complex in secondary openings, i.e., joints and fissures. Although the igneous and metamorphic rocks making up the basement complex are essentially impermeable, those zones in which an adequate number of interlacing fractures exist behave as a hydraulic unit and have transmissivities of up to 165 gal/d/ft. Other tests, including measurement of recovery after swabbing and injection permeability tests confirmed the existence of water-bearing zones in the basement rocks. However, the test program indicated that these permeable zones are limited in areal extent. The significance of these findings with regard to the VEGP is that the basement complex does not transmit ground water over a large area and that basement rocks are not a potential source for ground water within the study area.

The Triassic sequence of sedimentary rocks which overlie the crystalline basement were not penetrated by any of the exploratory borings associated with VEGP investigations; hence, no hydrogeologic data are available for this zone. However, the presence of Triassic rocks in the study area have been demonstrated during investigations relating to the Savannah River Plant. Based on the limited amount of data available, the Triassic rocks do not appear capable of transmitting ground water over a large area nor of providing a large volume of water to wells. It is therefore not considered a significant hydrogeologic unit.

All of the units above the Triassic sequence have been penetrated by wells constructed at the VEGP site. The initial test well, TW-1, provided data with which a well field was designed (paragraph 2.4.12.1.3.3). The log of that well, shown on drawing AX6DD331, illustrates the sequence and thickness of the important hydrogeologic units beneath the main plant area and provides a reference for the following discussions of those units.

2.4.12.1.2.2 Cretaceous and Tertiary Aquifers. The Cretaceous ground water system is represented in the site area by the Tuscaloosa Formation. This unit is approximately 700 ft thick.

It consists primarily of cross-bedded sands and gravels with subordinate beds of silt and clay. It is a highly transmissive aquifer system (reference 3 and drawing AX6DD331). At the VEGP site, the Tertiary aquifer is represented by the unnamed sands in the lower part of the Lisbon Formation (drawing AX6DD331). It consists of fluvial sands and sandy clays for which formal stratigraphic nomenclature has not yet been established. These sediments are moderately permeable, as shown by field permeability tests and by the operation of the VEGP construction water supply wells. Total thickness of the Tertiary aquifer at the site is approximately 100 ft.

Recharge to the Cretaceous aquifer is primarily from infiltration of rainfall where the formation is exposed northwest of VEGP. In the same general area, the Tertiary aquifer is also exposed and off-laps the Cretaceous aquifer. In this outcrop area, the Cretaceous and Tertiary systems are in hydraulic contact and the ground water is under water table conditions. After the water infiltrates the sediments, it migrates downdip in a south by southeast direction.

Within a few miles downgradiant of the recharge/outcrop area, ground water of the Cretaceous and Tertiary aquifers is confined beneath the Blue Bluff marl, the upper member of the Lisbon

Formation. The marl consists of semiconsolidated glauconitic marl with subordinate lenses of dense, well-indurated, well-cemented limestone. The marl layer immediately overlies the unnamed sands which form the Tertiary aquifer. The marl is approximately 70 ft thick. The permeability of the marl layer is very low, and it effectively confines the aquifer underlying it. It is a barrier to ground water movement.

At the VEGP site both aquifers are confined beneath the marl, but are in apparent hydraulic contact with one another. At some distance downdip of the VEGP site the Cretaceous aquifer becomes hydraulically separated from the Tertiary aquifer by the intervening, relatively impermeable clays and silts of the Huber and Ellenton Formations (Paleocene). The point at which this occurs is not well defined, but it is probably a few miles downdip (south) of VEGP. The Huber Formation is present beneath the VEGP site but, based on the lithology, it probably is not a significant barrier to movement of ground water in this area. The Ellenton Formation was not recognized beneath the VEGP site, but was identified in exploratory borings 7 miles southeast of the site[3].

At some point more than 10 miles downdip of the VEGP, the Blue Bluff marl undergoes a facies change (transition) to permeable limestone. Combined with the moderately permeable underlying unnamed sands and permeable overlying sediments, this limestone/sand sequence is referred to as the principal artesian aquifer. Overlying clays and fine-grained sediments of the Barnwell and younger formations confine the aquifer in this area.

To summarize, the Tertiary aquifer overlies and off-laps the Cretaceous aquifer in the outcrop areas north of VEGP. Ground water is under water table conditons in these areas. Progressing downdip, the two acquifers become separated stratigraphically (but not hydraulically) by the Huber Formation. They also become confined beneath the Blue Bluff marl member of the Lisbon Formation. These conditions prevail beneath VEGP and to an unidentified point, south of the site. Beyond that point the Ellenton and Huber formations confine the Cretaceous aquifer, hydraulically separating it from the Tertiary aquifer.

The regional direction of ground water flow in the Cretaceous and Tertiary systems in south-by-southeast toward the sea. However, from the fall line to a point a few miles south of the VEGP, the Savannah River has downcut through the Blue Bluff marl confining layer and into the underlying strata. This allows both the Cretaceous and the Tertiary aquifers to discharge to the riverbed. This condition gives rise to a ground water sink, and flow directions in this limited area do not follow regional trends. (A more comprehensive description of the sink is given in reference 3).

2.4.12.1.2.3 Water Table Aquifer. Unconfined ground water beneath the VEGP site is present in deposits of the Barnwell Group overlying the Blue Bluff marl, and in Quaternary deposits along adjacent stream channels. Immediately beneath and adjacent to the VEGP power block structures, the water table aquifer is the backfill material placed in the power block excavation. The Barnwell formations are extensively dissected in the VEGP area. Pleistocene alluvial and terrace deposits, and Holocene flood plain deposits are present along the Savannah River.

In the general vicinity of VEGP, the basal unit of the Barnwell Group is the Utley limestone member of the Clinchfield Formation (drawing AX6DD331). This is a fossiliferous and cavernous limestone unit which is capable of transmitting ground water. However, the unit rarely exceeds a few tens of feet in thickness, and it is of limited areal extent.

The remaining Barnwell units overlying the marl that contain ground water consisting of unconsolidated clays, silts, and sands. Data from packer and permeameter tests performed during exploration and plant construction indicate that both lateral and vertical permeability of these sediments varies considerably. This variation is attributed mainly to the highly variable quantities of clay distributed throughout the aquifer.

Recharge to the water table aquifer is almost exclusively by infiltration of direct precipitation. The presence of porous surface sands and the moderate topographic relief in the site area indicate that there is no significant storm runoff; hence, virtually all precipitation infiltrates the ground. Lateral recharge from adjacent areas is insignificant because the plant area is situated on an interfluvial high; i.e., it is isolated by drainage channels which have down cut to or near the marl and act as interceptor drains to potential recharge sources moving laterally toward the plant site. Lateral recharge from the Savannah River would be possible only in the case of a very severe flood, one that raises the river level some 30 ft or more.

2.4.12.1.3 Onsite Use

2.4.12.1.3.1 Plant Operating Requirements. Ground water is the primary source of supply for reactor cooling water makeup, normal makeup to the nuclear service cooling towers, and fire protection. Two makeup wells (designated MU-1 and MU-2A on drawing AX6DD332) producing from the combined Cretaceous/ Tertiary aquifers supply water to storage tanks from which water is drawn as needed. Wells MU-1 and MU-2A are capable of supplying water at 2000 gal/min and 1000 gal/min, respectively. The two wells pumping simultaneously have sufficient capacity to supply expected process makeup requirements and completely fill the major plant tanks in one day. The plant water well system is not required for safe shutdown of the plant but is the normal source of water supplied as makeup to the ultimate heat sink. An independent, alternate source, the Savannah River, is also available for cooling water makeup. (See section 2.4.11.) Table 2.4.12-2 summarizes expected ground water use at the site.

There is one well located at the simulator building (SB on drawing AX6DD332) which supplies potable water for that facility. This well is producing from the Tertiary aquifer. Another well (PW on drawing AX6DD332) provides wash water for the combustion turbine plant (Plant Wilson) on site. It produces from the water table aquifer.

2.4.12.1.3.2 Construction Requirements. Initial construction requirements were approximately 350,000 gal/d or 240 gal/min of untreated water for concrete batch plant operation, dust suppression, and sanitary needs. These requirements gradually increased to 600,000 gal/d (420 gal/min) to accommodate peak construction activities. Makeup wells 1 and 2 originally supplied most of these needs. The wells were pumped intermittently to fill two temporary tanks, a 25,000-gal construction water tank and a 10,000-gal batch plant water storage tank. In August 1980, a new 400-gal/min well, CW-2, producing from the Tertiary aquifer, was installed west of the power block as a primary source of construction water. In addition, the two temporary tanks were replaced by a permanent 30,000-gal tank near the batch plant. During April 1990, well CW-2 was grouted from the bottom to the top of the well and then abandoned.

Potable water for the temporary construction office on site was provided by a well, CW-1, producing from the Tertiary aquifer. Yield of this well is 100 gal/min with a specific capacity of

25 gal/min/ft of drawdown. In July 1989, well CW-1 was filled with sand and concrete and then abandoned. A third well, CW-3, producing from the Tertiary aquifer provided potable water for the nuclear operations garage. The construction wells are shown on drawing AX6DD332. A summary of the well data are on table 2.4.12-7D.

2.4.12.1.3.3 Wells and Well Field Design. During normal operation of the plant, the ground water supply is provided by one makeup well, with other makeup wells on standby. There are two makeup wells (MU-1 and MU-2A) and one alternate makeup well (TW-1). Well MU-1 is capable of producing 2000 gal/min on a continuous basis for the life of the plant. Beginning in 2001, Well-2A, originally rated at 2000 gal/min, became limited to producing 1000 gal/min on a continuous basis for the life of the plant. The alternate well (TW-1) is capable of producing 1000 gal/min on a continuous basis for the life of the plant.

Wells TW-1 and MU-2 were constructed as a test well and a makeup well, respectively. Well MU-2 was utilized for construction water, but due to a design change, its location interfered with other facilities. Therefore, it was replaced by well MU-2A. The test well (TW-1) was originally drilled and tested to provide design data for the makeup well field. Wells TW-1 and MU-2 were capped and utilized as observation wells to monitor the Cretaceous aquifer until 1995. Beginning in 1999, well TW-1 became the alternate makeup water source, limited to 1000 gal/min. Initial use of this well, due to its location near Category 1 structures, includes reading of settlement markers for structures near the well to confirm that drawdown does not affect these structures' settlement.

The two plant makeup water wells (MU-1 and MU-2A) are constructed as gravel pack wells extending to depths of 830 ft and 865 ft, respectively, and are open to selected aquifer zones below a depth of 435 ft. Well casing to a depth of 250 ft in MU-1 is 16-in. nominal diameter; below that depth, the casing and screen are 10-in. nominal diameter. Well casing to a depth of 280 ft in MU-2A is 14-in. nominal diameter (the original 16-in. diameter casing has been sleeved with a 14 in. casing); below that depth the casing and screen are 10-in. nominal diameter. To prevent movement of water between the aquifers above and below the marl the well bores were enlarged to 34 in. diameter to a depth of 160 ft in MU-1 and 98 ft in MU-2A, and 26-in. nominal diameter well casings were set at these depths. The annular space between the hole wall and casing was filled with grout to provide the seal. The alternate plant makeup water well (TW-1) is constructed as a gravel pack well extending to a depth of 860 ft and is open to selected aquifer zones below a depth of 450 ft. Well casing to a depth of 91 ft in TW-1 is 28-in. nominal diameter. A second casing 16-in. nominal diameter extends to 250-ft depth. A third casing and screen is 10-in. nominal diameter, and extends to 850-ft depth. To prevent movement of water between the aquifers above and below the marl, the well bore was enlarged to 34-in. diameter to a depth of 91 ft. The annular space between the hole wall and upper casing was filled with grout to provide the seal.

Well design criteria, including the well screen openings, are based on data collected in the drilling and testing of the initial test well (TW-1). The basic design criteria for the wells are as follows:

A. Allowing for pumping well interference and fluctuations of water levels caused by increased use of ground water and climatic changes, wells MU-1 and MU-2A can be pumped continuously at their rated capacity for the life of the plant.

B. Minimum construction specification is the American Water Works Association standard for deep wells (standard AWWA A-100).

C. The wells are lined with steel casing of sufficient strength to carry the stress loads anticipated during operation of the plant.

D. The well screen is a wire-wound type fabricated of stainless steel stock and has sufficient strength to carry anticipated stress loads. Screen openings in MU-1 and MU-2A are 0.04 in. and 0.035 in., respectively, as determined from size analyses of the aquifer materials and design of the gravel filter pack. Length of screen provides sufficient open area to reduce entrance velocities to less than 4 ft/min and to avoid migration of fine material into the wells.

E. Gravel pack material is clean washed and graded sand and gravel consisting of sizes (selected with reference to the aquifer materials) which provide an efficient filter without restricting movement of water to the well.

F. Allowance is made for potential corrosion (electrolytic and chemical) to ensure sufficient strength and integrity of the casing and screen during the life of the wells.

G. A grout seal is placed in the annular space between the 34-in.-diameter hole wall and the 26-in.-diameter surface casing to a minimum depth of 160 ft in MU-1 and 98 ft in MU-2A. Grout was placed in one continuous operation by placing a tremie pipe inside the annulus and filling the annular space from the bottom up.

The two plant makeup wells provide a well field system that was designed to consider the factors of pumping well interference, fluctuations of water levels caused by climatic changes, and future increased use of ground water and to preclude possible subsidence of plant foundations from drawdown of piezometric levels caused by pumping cones of depression.

A well to supply water for irrigation purposes, IW-4, was added in 1989. This well was drilled to a depth of 370 ft and a 4-in. PVC screening was installed to a depth of 370 ft. The well produces water from the Tertiary aquifer with a maximum pump capacity of 120 gal/min. A well for potable water supply to the security tactical training facility, SW-5, was added in 1990. This well was drilled to a depth of 200 ft and a 2-in. PVC screening was installed to a depth of 200 ft. The well produces water from the Tertiary aquifer with a maximum pump capacity of 20 gal/min. The locations of the water supply wells are shown on drawing AX6DD332. A summary of the water supply well data is in table 2.4.12-7D.

2.4.12.1.3.4 Temporary Dewatering. Construction of the foundations at VEGP required excavation of the Eocene and younger sands, silts, and clays of the unconfined aquifer from about el 216 ft to el 130 ft. The portion of the excavation below the water table elevation of the unconfined aquifer (approximately el 160 ft) was dewatered during excavation by a series of ditches oriented in an east- west direction and connected by a north-south ditch, which drained to a sump in the southwest corner of the excavation. The sump was equipped with four pumps with a capacity of 500 gal/min each to remove inflows from ground water. Additional capacity was provided for the removal of inflows of storm water into the excavation.

The ditch and sump approach was successful when the invert elevation of the ditches was maintained at 15 to 20 ft below the adjacent grade. This permitted the use of conventional excavation procedures in reasonably dry materials.

Upon reaching the marl, the system of ditches and sump was replaced by a perimeter drainage system as shown on drawing AX6DD324. This consisted of a porous concrete pipe around the perimeter of the power block excavation feeding into three sumps at the toe of the south slope. Water pumped from the sumps was discharged to debris basin 1 southeast of the power block. The porous concrete pipe was encased in a granular filter material which was carried up the surface of the adjacent 2 to 1 slope to about el 160 ft. This filter blanket was placed so that there was a minimum of 4 ft of filter material measured horizontally from the face of the slope out to the face of the filter blanket as shown on drawing AX6DD324.

The construction dewatering system, i.e., the filter blanket and porous concrete perimeter drain, was designed to prevent ground water from entering the power block excavation during construction operations. In this regard, it performed properly throughout the construction period. However, the system was not adequate to drain excessive rainfall which accumulated in compacted fill. The Category 1 backfill program necessarily proceeded in steps and stages that were commensurate to the gradual rise of structure walls. Fill was placed and compacted to thicknesses ranging from a few feet to a few tens of feet above the marl and then left as a functional surface to await the next stage of structure wall construction. In the interim, rainfall penetrated the fill, percolated to the marl layer, and accumulated to a fairly constant level in the same manner that a perched water table develops over a clay layer. In some areas the water rose to a level very close to the top of the fill, precluding further backfilling on that surface. In addition to near-surface saturation at various locations, there was seepage from the face of the slopes of the compacted fill which rested on the marl bearing surface. This condition precluded backfilling against these slopes.

The conditions described above were relatively minor and caused no major delays or problems until November 2, 1979. On that date almost 6 in. of rain fell at the site, causing severe surface erosion of slopes and rapid saturation of the more shallow fill areas. As a result of the additional water, it was necessary to implement a supplemental dewatering system so that backfill operations could continue. Along with the dewatering system, slope repair and settlement monitoring plans were also implemented. The remedial program was completely successful and is described fully in reference 6.

2.4.12.1.3.5 Chemical Quality of Water. Several samples of ground water and surface water were taken for chemical analysis to determine the quality of water in the area, to identify characteristics of the different aquifer water, and to determine any correlation between ground water and surface water. Tables 2.4.12-3 through 2.4.12-6 list analyses of water samples from observation wells, domestic wells, plant water supply wells, springs, and surface waters. Locations of sample points are shown on drawings AX6DD323, AX6DD325, and AX6DD332.

Water samples were collected in October 1971 from both the water table aquifer and the confined aquifer from observation wells set in each system. Additional water samples were collected from the construction and makeup water wells at various times from 1977 to 1985.

Overall, the ground water of the area is calcium and calcium- sodium bicarbonate types, with total dissolved solids less than 200 ppm. Samples of ground water from the confined aquifer system contain from 110 to 194 ppm total dissolved solids, and sodium is the dominant cation.

Ground water from the water table aquifer system is more variable in total dissolved solids; analyzed samples for the system ranged from 20.0 ppm (spring 2, table 2.4.12-5) to 169.5 ppm (well 143, table 2.4.12-3). Sodium is again the dominant cation. The variation in total dissolved solids is probably related to the length of time the water has remained in the ground, since more time allows more leaching of solids.

The surface water is low in total dissolved solids (less than 100 ppm), with calcium and magnesium as the dominant cations. Bicarbonate is the dominant anion. The highest total dissolved solids were measured at Mathes Pond where solutioning of calcite in the shell zone is believed to be the major source of cations.

2.4.12.2 Sources

2.4.12.2.1 Present Ground Water Use

Large quantities of water are stored in the confined aquifers underlying the region of the plant site, and to date relatively small withdrawals have occurred. Although many small communities derive water from wells, the draft on the aquifers is low because of the low population density, limited industrial developments, only moderate need for crop irrigation (due to high rainfall), and abundant surface waters.

The largest withdrawal of ground water in the region is concentrated in the Savannah area, 80 miles southeast of the VEGP site, near the Atlantic Coast. Ground water is extracted principally from limestone beds of the Ocala Group of upper Eocene age. Heavy pumping from a relatively small area has caused a large, deep cone of depression to form. Recharge to the aquifer in the vicinity of Savannah may eventually stabilize the cone but the hydrograph of well CHA 84, drawing AX6DD326, indicates a continued decline through the year 1979.

Closer to the VEGP site, the principal withdrawal of ground water is from the Tuscaloosa aquifer (i.e., the Cretaceous system) at the Department of Energy Savannah River Project facilities. Pumping of ground water for these facilities began in the early 1950s to supply water for construction. Since then, ground water extractions have continued for operational purposes, although the amounts have been considerably less than was originally anticipated. As reported by Savannah River Project personnel, ground water extractions have remained relatively constant at about 5000 gal/min. These withdrawals will have no effect on ground water conditions at VEGP.

Sylvania is one of the few communities in the vicinity of the VEGP site that derives its water supply from ground water. Located approximately 30 miles south-southeast of the site, the Sylvania wells withdraw water from the Tertiary aquifer. Water level history is shown on the hydrograph for Screven well 3 (drawing AX6DD326). Although short-term trends are indicated, no aquifer dewatering or change in storage of ground water is evident over the period of record, except for a slight decline in the late 1970s.

The city of Augusta is the nearest large municipality. Small domestic wells are present there, and a few large industries have wells that extract water from the recharge area of the Tuscaloosa Formation. The main source of municipal supply, however, is drawn from the Savannah River.

The area of Richmond County south of Gordon Highway is served by the Richmond County Water and Sewer Authority. The service area for the authority is about 12 mi^2 and serves about 15,000 customers from 18 wells, most of which are in the southern part of the county. The wells average 200 ft in depth and can produce about 750 gal/min each from the Tuscaloosa aquifer. Approximately 9.4 Mgal/d are used throughout the service area.

The area around McBean (located about 13 miles northwest of the site) is served by the Pine Hill Water Authority. The authority has five wells, three of which are operable, that draw water from Cretaceous sands of the Tuscaloosa aquifer. Total depth of each of the wells is about 450 ft, and the wells are distributed throughout the service area, which has about 2200 customers. The USGS has begun monitoring one of the wells to provide a record of future trends of water levels in the area.

Ground water use in eastern Burke County was determined by an extensive well canvass performed during the site exploration phase and supplemented by the Millett fault study of 1982.[7]

The well survey data indicate ground water use is almost exclusively for domestic needs. Small amounts are used for stock, and there are a few small commercial buildings in the communities served by municipal wells. Except for VEGP there are no known industrial, irrigation, or similar activities requiring continuous withdrawals of large quantities of ground water.

The only incorporated community within 10 miles of the plant site is the town of Girard, with a population of about 250. Although 12 private wells were in use in the recent past, these were abandoned when city water service lines were installed. City water is supplied from two wells producing from the Tertiary aquifer.

Sardis is a community 12 miles due south of the site. It is larger than Girard (population of 830), and the community water supply is provided by three wells open to the Tertiary aquifer with pumping capacities from 300 to 500 gal/min. Two of the wells are for standby and fire protection purposes. The 1000 customers are predominantly domestic users.

The many private wells in eastern Burke County are small, with a maximum capacity of less than 10 gal/min. The average of each well is estimated to be less than 0.5 gal/min.

2.4.12.2.2 Projected Future Ground Water Use

In 1965 and 1966, the Central Savannah River Area Planning and Development Commission conducted a regional investigation of present and future water and waste disposal needs. According to this survey, the population of eastern Burke County is expected to reach approximately 2800 by the year 1986. That is when the Augusta-Savannah Industrial Corridor is scheduled to extend into Burke County with a concomitant increase in water requirements to 540,000 gal/d. A large portion of this increase will be for industrial uses, and the source will be primarily from the Savannah River.

There is little anticipation of any substantial change in the agricultural development of the area. Assuming that present water requirements are primarily for domestic use and are derived from ground water, future increases in ground water use can be assumed to be correlative to the population.

The Richmond County Water Authority currently has a capacity of about 13 Mgal/d. By the year 2000 they estimate that their capacity will be enlarged to 18.5 Mgal/d based on annual growth of about 500 customers per year.

2.4.12.2.3 Water Levels and Ground Water Movement

A ground water monitoring program to determine the direction of ground water movement and the piezometric levels of the aquifers was established at the VEGP site with the first exploration work in 1971. That program has included an array of observation wells open to the water table aquifer above the Blue Bluff marl, and an array of wells open to the confined Tertiary aquifer immediately below the marl (in the unnamed sands of the Lisbon Formation). Two wells open to the Cretaceous (Tuscaloosa) aquifer were added to the monitoring program in 1985.

Special observation wells were installed to provide data on the distribution of hydrostatic pressure across the marl. Other special wells include a series of short-lived construction "piezometers" that were installed in the backfill as it was placed around the power block complex. They were utilized to assure the water table in the backfill was far enough below the surface to achieve effective compaction.

The initial array of observation wells installed during the exploration period 1971 through 1972 included several wells located in areas of plant construction. These were destroyed and sealed as required by the construction schedule, and when possible, replaced after the construction was completed.

Additional wells have been installed to provide an effective system to monitor ground water during plant operation as needed. A complete list of all observation wells, both active and inactive, is on tables 2.4.12-7A through 7C.

2.4.12.2.3.1 Observation-Well Monitoring. The original observation wells installed in 1971-1972 included 16 open to the water table aquifer, 10 open to the confined aquifer, and two monitoring hydrostatic pressure in the marl. This array remained until July 1974 when site grading and excavation for the power block commenced. A majority of the wells were terminated at that time to make way for construction. All activity at the site was interrupted three months later, September 1974.

Resumption of construction, which began in 1976, required dewatering the power block excavation. The dewatering continued, uninterrupted, until March 1983. As construction progressed, more wells had to be terminated. Only 3 of the original 16 observation wells open to the water table aquifer remain intact November 1985. Of the original 10 wells open to the confined aquifer, 2 remain. Other wells have been installed periodically to replace those destroyed by construction.

During the period September 1971 through March 1972, water levels in observation wells at VEGP were monitored by Law Engineering Company at least biweekly and, commonly, more frequently. No water level measurements were made between April 1972 and April 1973. Georgia Power Company personnel commenced monitoring on a quarterly basis in April 1973. Monitoring was again stopped July 1974 when site grading and excavation for the power block began. Monitoring was not resumed until June 1979, at which time quarterly measurements of all existing wells was again initiated.

Dewatering of the power block excavation was in effect from June 1976 through March 1983. Hence, water levels in observation wells of the water table aquifer during this period were influenced by construction dewatering. Daily readings were made in observation wells 800 and 802 during the period December 11, 1980, through September 15, 1982, as part of the monitoring conducted for placement of the backfill around the power block structures. Temporary wells were installed to monitor the saturated level in the backfill as it was being placed to assure proper compaction.

In July 1985, a program of frequent measurement of water table wells was implemented. The purpose of this program was to provide more detailed information to support the basis for the hydrostatic loading design (see section 2.4.12.4). Locations of the wells in this program are shown on drawings AX6DD333, AX6DD334, and AX6DD335. Additional water-table observation wells were installed in the backfill adjacent to the power block structures and several previously destroyed wells were replaced. There were 17 wells monitoring the water-table aquifer. A series of 6 Casagrande-type piezometers have been set in the Blue Bluff marl to monitor the hydrostatic pressure. They were in two clusters of 3 piezometers each, located on the northwest and southeast side of the power block. The piezometers in each cluster monitor pore pressure in the upper, middle, and lower portions of the marl. In addition, monitoring of the 10 Tertiary aquifer wells immediately below the Blue Bluff marl and the two observation wells in the Cretaceous aquifer was also maintained.

Water-table aquifer wells were measured on a weekly interval for a minimum period of 6 months. With each water-level measurement, depth to the base of each well was also measured to determine if silting occurred.

A review of water-level and precipitation data was made after 6 months of monitoring to establish a correlation (amount and lag-time) of water table response to precipitation. The review established a correlation and indicated that frequency of measurements could be reduced to once a month. Measurements on a monthly basis continued to establish seasonable correlations with precipitation.

The six piezometers open to the Blue Bluff marl were also monitored weekly. These piezometers were measured the same day each week as the wells in the water table aquifer. The data from these piezometers was reviewed after 6 months to determine the degree of hydrostatic pore pressure fluctuations in the marl. Once equilibrium was accomplished and degree of fluctuation was determined, monitoring frequency was reduced.

The wells in both confined aquifers (Tertiary and Cretaceous) were monitored on a monthly basis for a period of 6 months. The measurements were made concurrently with measurements in the water table aquifer. After the 6-month review of the data, it was determined that quarterly measurements could be made.

In December 1988, the well monitoring program was modified. Quarterly measurements of the four water table wells within the area of backfill around the power block (LT1B, LT7A, LT-12, LT-13) are to be taken. The balance of wells and piezometers are to be measured semiannually.

The frequency of monitoring of the observation wells was evaluated in 1991. The evaluation concluded that the water table aquifer water levels consistently follow the same trends and fluctuations; therefore, the number of wells being read could be reduced. The evaluation also concluded that the Blue Marl readings can be represented by one cluster of piezometers.

In 1993, the well monitoring program was modified to reduce the number of wells being read quarterly and semiannually. The well monitoring program was modified in 1995 to only read levels in the surface water table wells. The current well program is shown on drawing AX6DD335 and in table 2.4.12-9. A summary of the observation well data is in tables 2.4.12-7A through 2.4.12-7C.

2.4.12.2.3.2 Tertiary Aquifer. Water levels measured in observation wells open to the Tertiary aquifer confined beneath the marl indicate the direction of ground water movement is toward the Savannah River in the vicinity of the site. Drawings AX6DD327 and AX6DD328 show the contours of the piezometric surface for October 1971 and for December 1984. A comparison of the two sets of contours indicates very little difference. These data demonstrate the marked gradient toward the river. Selected hydrographs of Tertiary aquifer wells are on drawing AX6DD336.

2.4.12.2.3.3 Water Table Aquifer. As previously noted, the site is on an interfluvial ridge. Ground water present in the materials overlying the marl is under water table conditions and isolated hydraulically from other aquifers by the marl. Replenishment is by infiltration of precipitation, and after precolating to the water table, ground water moves laterally to the bordering interceptor streams. Hydrographs of water-table aquifer observation wells are on drawing AX6DD337.

Contours of the water table for November 1971 and for December 1984 are shown on drawings AX6DD329 and AX6DD330. Contours for the 1971 data are based on measurements of water levels in observation wells and springs and seepage areas on the interceptor streams. The water table is, in general, a subdued reflection of the ground surface, and movement is from the central portions of the interfluve toward the bordering interceptor streams.

Construction dewatering at the site was completed in March 1983. Water levels and the flow pattern of the water table aquifer have returned to a preconstruction pattern. No permanent dewatering facilities have been provided nor are any necessary at the plant site.

2.4.12.2.4 Hydrogeologic Properties of Subsurface Materials

The hydrogeologic properties of the materials beneath the site were determined by extensive subsurface exploration at and near the site by in situ tests conducted in the field, and supplemented by laboratory tests of numerous samples collected throughout the drilling programs.

The hydraulic conductivity (permeability) of the subsurface materials was measured in the field by pumping tests, constant-head inflow tests (including open standpipe and permeameter tests), variable-head inflow tests, and slug tests.

2.4.12.2.4.1 Cretaceous and Tertiary Aquifers. As stated in paragraph 2.4.12.1.2.2, the Cretaceous and Tertiary aquifers are believed to be in hydraulic connection at the site. The hydrologeologic properties of the combined Cretaceous/Tertiary aquifer were determined at the plant site by pumping tests on test well TW-1 during early site investigations and by pumping tests on makeup wells MU-1 and MU-2 during plant construction. A brief description of the makeup well pumping test program is given in the following paragraphs. (A detailed description

of the testing program for TW-1 was included in the Preliminary Safety Analysis Report). The transmissivities and storage coefficients for the wells are included in table 2.4.12-8.

The pumping test for MU-1 began on August 21, 1977, and concluded 72 h later. Recovery level readings began immediately following pump shutdown and continued for 24 h. The pump intake was placed at approximately 232 ft below ground level at approximately el -35 ft mean sea level (msl). Depth to water in the well before pump startup was 27.7 ft (from top of casing). The step-discharge test is summarized below:

Well Discharge (gal/min)	Step Duration (min)	Total Drawdown (ft)	Specific Capacity (gal/min/ft)
1150	720	32.7	35.2
2232	720	72.3	30.0
3334	2880	123.9	26.9

The transmissivity (T) of the aquifer was determined from the measurements of drawdown taken during the initial step-discharge test and from the measurements of recovery following the test.

The drawdown data were analyzed using the Jacob method (modified Theis nonequillibrium equation), and the recovery data were analyzed by the Theis recovery formula. The analyses show that the transmissivity is in the range of 110,400 to 116,600 gal/d/ft.

The pumping test for MU-2 began on December 15, 1977, and concluded 72 h later. Recovery level readings began immediately following pump shutdown and continued for 24 h. The pump intake was placed at 231 ft below top of 16-in diameter casing at approximate el -16 ft msl. Depth to water in the well before pump startup was 42.1 ft (from top of casing). The step discharge test is summarized below:

Well Discharge (gal/min)	Step Duration (min)	Total Drawdown (ft)	Specific Capacity (gal/min/ft)
1200	720	22.0	54.5
2175	720	47.5	45.8
3316	2880	81.3	40.8

The transmissivity of the aquifer was determined from the measurements of drawdown taken during the initial step- discharge test and from the measurements of recovery following the test. The drawdown data were analyzed using the Jacob method (modified Theis nonequilibrium equation), and the recovery data were analyzed by the Theis recovery formula. The analyses show that the transmissivity is in the range of 128,700 to 130,900 gal/d/ft.

Table 2.4.12-8 summarizes aquifer characteristics of the Cretaceous aquifer determined with test well TW-1 and with the makeup water wells. The moderately wide range of transmissivities and storage coefficients shown on the table suggests that the aquifer is not uniform in character and that permeability varies from place to place. No particular significance is attached to this condition because (1) the range of differences is not especially large and (2) the lowest of the values (110,400 gal/d/ft) is still indicative of a highly productive aquifer. It is noteworthy that the 158,000-gal/d/ft value, determined from a distance-drawdown analysis and considered more indicative of the average, falls very close to the median value between the highest and lowest transmissivities obtained. This implies that the value of 150,000 gal/d/ft is a realistic and conservative value to be used in evaluating the capability of the aquifer in this area to yield water to wells. The values of storage coefficient determined from the pumping tests indicate the aquifer is effectively confined.

2.4.12.2.4.2 Blue Bluff Marl. The Blue Bluff marl is approximately 70 ft thick. It extends over an area well beyond the limits of the plant site and the interfluvial ridge on which the plant site is located. The comprehensive exploration and testing that has been conducted demonstrates that the marl is an extensive and persistent unit. In particular, the marl's integrity as a barrier to ground-water movement has been demonstrated by (1) field permeability testing, (2) visual inspection of cored samples, the marl surface exposed during site excavation, and marl outcrops along the Savannah River, and (3) comparison of water levels in observation wells open to the water table aquifer with those observed in wells open to the confined aquifer immediately below the marl.

The continuity of this nearly impermeable material (i.e., the lack of voids, open joints or fractures) has also been demonstrated. Since 1971, there have been over 10,000 ft of marl penetrated at VEGP by drilling, coring, Standard Penetration Testing, and undisturbed sampling. When coring, the most revealing evidence for the occurrence of voids or permeable fractures is a loss of all or part of the drilling fluid and/or a sudden or rapid advance of the core barrel. At no time throughout this extensive testing was there any unaccountable fluid loss or abnormal tool advance in the marl. None of the borings encountered significantly fractured zones, nor was there evidence of leaching (removal of calcareous material by solution activity) of the marl.

Visual inspections and detailed logging of the many extracted samples of marl have likewise produced no indications of voids or extensive fracture zones. Over 1000 ft of the marl penetrated in drill holes have been either cored or sampled, and have been closely inspected and described. Very few joints or fractures were observed and those identified were consistently found to be tight, and without void space. More than 940,000 ft^2 of marl beneath the plant site was exposed during the excavation for the foundation and was directly examined and carefully logged. The results are discussed in detail in subsection 2.5.1.2.2.2.1.1. Additionally, marl outcroups along the Savannah River in the vicinity of VEGP have also been examined and mapped. These extensive and detailed mapping investigations of the marl formation at VEGP have found no evidence of voids, solution cavities, or systematic or extensive fractures in the marl.

The large and consistent hydraulic head differential between the water table aquifer above the marl and the confined aquifers immediately below the marl confirms that the marl is a barrier to ground water movement. The hydraulic head (energy potential) of ground water in an aquifer is commonly exprssed as ft (elevation) above sea level, and is determined from measuring the elevation of water in an observation well. In the vicinity of the plant, the hydraulic head in the

water table aquifer is 45 to 55 ft greater than the hydraulic head in the aquifer immediately below the marl. To bring about such a marked difference in hydraulic head, the barrier must be extensive and without significant through-going openings (such as fractures or solution cavities).

This difference in hydraulic head can be seen by comparing the ground water (equipotential) contours shown on drawings AX6DD328 and AX6DD330. The contours are based on water levels measured in observation wells in December 1984. Similar conditions were observed prior to plant construction, as indicated by the contours of water levels measured in wells in October and November 1971 and shown on drawings AX6DD327 and AX6DD329.

A nest of observation wells constructed at the site of exploratory hole 42 provided a measure of this hydraulic head differential between the overlying water table aquifer and the confined aquifer sands beneath the marl. The observation wells were constructed in 1971 and included two, 42B, and 42C, open to the marl itself, one, 42A, open to the confined aquifer and one, 42D, open to the water table aquifer. At these wells, the marl is 65 ft thick. The wells were monitored for 4 years until construction of the plant required their closure, at which time they were sealed. Hydrographs of the measured levels are shown on drawing AX6DD408.

A general relationship between the water levels (head) of the observation wells and the zone each well monitors can be seen on drawing AX6DD408. That is, the differences in water levels (head) between the observation wells is generally proportionate to the thickness of marl between zones monitored by the wells. The zones monitored by each observation well are illustrated on drawing AX6DD408. For example, the difference in water levels of the two wells open to the two aquifers (42D and 42A) is about 55 ft, (the head in the water table aquifer is higher) and the thickness of marl between them is 65 ft. In comparison, well 42B is open to an interval of the marl that is near the bottom of the marl. The water levels measured in well 42B are from 15 to 20 ft different (higher) than those measured in well 42A, which is open to the underlying confined aquifer, and the thickness of marl between them is about 10 ft. Water levels measured in well 42C follows this general relationship. It is open to an interval of the marl that is within 3 ft of the top of the marl. The water levels measured in 42C are from 50 to 52 ft higher than those measured in 42A, and the thickness of marl between them is about 60 ft.

Two clusters of piezometers (A&B) were installed in the marl in June and July 1985. The clusters are located at opposite corners of the power block, as shown on drawings AX6DD333, AX6DD334, and AX6DD335. The piezometers provide a direct measurement of the hydraulic head over the full thickness of the marl. The differences in hydraulic head between the piezometers within a cluster show a progressive decline in head with depth as was observed in the 42 series (see drawing AX4DD408).

In situ permeability tests were performed in the marl during early site investigations, river facilities investigations, and during recent (1985) installation of observation wells. Results of the tests are summarized on table 2.4.12-10. Tests were conducted in 95 intervals at different depths in 28 exploratory holes. In 90 percent of the intervals tested, no measurable water inflow occurred. In only three holes was any measurable water inflow confirmed; in two of these the inflow occurred in near surface, weathered marl. (Water inflow measured in three other holes was due to leakage around the packers.)

Laboratory permeability tests were conducted on 10 samples of the core collected from the marl observation wells. The range of laboratory permeability measurements is from 5.0×10^{3} ft/yr to 8.5 ft/yr. The results of the laboratory tests are summarized on table 2.4.12-10. The permeability tests indicate that the marl is nearly impermeable.

Porosity of the Blue Bluff marl was calculated from laboratory analyses of undisturbed samples taken in exploratory holes during the initial site studies. The porosities range from 24 percent to 62 percent, with a mean value of 47.5 percent. The calculated porosities are listed on table 2.4.12-11.

2.4.12.2.4.3 Water Table Aquifer. The permeability of the water table aquifer was measured by field-test methods and laboratory testing. Measurements were made of Barnwell sands and clayey sands, Utley limestone, and of backfill material. Preliminary estimates were made of the Barnwell sands and clay by correlation to grain-size analyses. Porosities of undisturbed samples of Barnwell sands, silty sands, and clay, as well as of samples of backfill material, were measured in the laboratory.

In-situ permeability of Barnwell sands and clayey sands was measured at two exploratory holes (183 and 184) at the plant site, and in the laboratory on four samples taken from hole 107A. Three of those samples were undisturbed; the fourth was a disturbed sample for which permeability was measured at three densities. Similarly, permeabilities were measured in the laboratory at four densities of two "grab" samples of backfill material. The backfill samples were selected for different amounts of material finer than the No. 200 sieve: one sample with 5.9 percent and one with 11 percent. Variation of permeability with density was determined by measuring each sample at four different densities. The testing procedure followed was to saturate the samples by the back-pressure technique, confine them at the effective overburden pressure, and then maintain a constant hydraulic gradient across the sample.

Preliminary estimates of permeability of the Barnwell sands as backfill material were made by the approximate relationship to grain size found by A. Hazen for filter sands [(11)]. These estimates are reported in reference 10.

The permeability of the backfill material was measured in situ by slug tests performed in four observation wells in the power block area [(13)].

The transmissivity values determined from the results of the insertion and extraction cycles of each test were averaged to obtain the transmissivity in the vicinity of each well. The average permeability of the backfill at each test site was then determined. The results are summarized on table 2.4.12-15.

Two test wells, each with an array of four observation wells, were constructed in the vicinity of the power block excavation to conduct aquifer pumping tests to measure the permeability of the Utley limestone. Beneath the site, the Utley limestone is composed predominantly of shells in a matrix of silt and clay, but includes thin and discontinuous beds of limestone, sand, and coquina. It was thought in the initial plans for dewatering that the limestone, which is at the base of the water table aquifer, might be very permeable and could act as a drain for dewatering.

The first well, W-1, was pumped at a rate of 36 gal/min for 97 h, and response was sufficient in all four observation wells for analysis. The second well, W-2, was in a much less permeable zone of the limestone than the first well, and response in the observation wells was negligible to small, precluding effective analysis. The average yield of W-2 was 12 gal/min, but it fluctuated considerably and the test was terminated after 27 h. The two tests indicated that transmissivity of the Utley limestone is relatively low and varies considerably from place to place. It was concluded it would not be an effective drain for dewatering.

In addition to the pumping tests described above, both falling-head and constant-head tests were conducted in some of the observation wells. During two of these tests, the water-level rise in an adjacent observation well was measured. Only the response of observation well 2A to the constant inflow of 74 gal/min at well 2B was adequate for analysis. The results of the tests of the Utley limestone are summarized in table 2.4.12-13.

Fifteen samples of undisturbed Barnwell sands and silty sands and clay were analyzed for porosity. Values measured range from 34 percent to 52 percent for the sands and silty or clayey sands. The mean of all tests is 43.9 percent. The porosity data are listed on table 2.4.12-14. In addition, porosity in relation to density of two recompacted samples of Barnwell sands used for backfill material is provided.

2.4.12.2.5 Potential Reversibility of Ground Water Flow

2.4.12.2.5.1 Confined Aquifer. On a regional basis, the general direction of ground water movement in the Tuscaloosa aquifer system is structure controlled toward south by the southeast under a gradient of 6 to 20 ft/mi.[2] Locally, piezometric surfaces take on discrete configurations under the influence of pumping, topography, recharge and discharge environments, geologic patterns, and other factors. Local ground water movement is therefore determined by the net effects of regional and local influences.

At the VEGP site, a local piezometric surface has been defined via observation wells set below the marl. Drawings AX6DD327 and AX6DD328 show clearly that the piezometric surface slopes to the northeast toward the Savannah River. This suggests that the aquifer is discharging to the river. Further, the marl layer is absent beneath the flood plain of the river; hence, the underlying Cretaceous and Tertiary aquifers are in hydraulic contact with the river. Therefore, as long as the hydraulic head of the aquifer is at a higher elevation than the riverbed, the river will receive water from the aquifer. Data from the Millett study substantiates these conclusions.[3]

Reversal of ground water movement in the confined Cretaceous or Tertiary aquifer would entail a change in conditions that would cause the Savannah River to discharge to the aquifer. The river would then be an influent stream. For this to occur, immense quantities of ground water would have to be withdrawn upgradient from the river, either at or in the vicinity of the VEGP. No significant new enterprises are anticipated in the vicinity of the plant. Plant needs are outlined on table 2.4.12-2 and discussed in paragraph 2.4.12.1.3.1. It can be seen that plant requirements are relatively small when compared to the large capacity and productivity of the aquifer. Therefore, the potential for reversing ground water flow is slight.

2.4.12.2.5.2 Water Table Aquifer. Ground water under water table conditions in the study area is available in quantities sufficient only for domestic use. Water for operation of VEGP is obtained from the confined Cretaceous aquifer and has no effect on the water table aquifer. Hence, the potential for change in flow direction as a result of operation of VEGP is negligible.

The power block construction dewatering system had a local effect on the ground water flow direction, i.e., flow was towards and into excavation. This was a temporary condition and existed only during construction. Dewatering has been discontinued, and the water table has recovered to near the preconstruction configuration (compare drawings AX6DD329 and AX6DD330).

There has been modification to that configuration due to site grading and plant excavation. These activities have lowered the topographic highs on site, with the effects that the general flow direction of ground water remains unchanged, but the water table is more subdued, and is lower in elevation.

2.4.12.3 Monitoring of Safeguard Requirements

Contamination of usable ground water by normal operation of VEGP, or by accidental spills, is unlikely. The potential for such an occurrence is very low. The marl underlying the site is an effective barrier to migration of fluids. Construction of the makeup water wells and observation wells includes a cement grout seal to prevent vertical movement of fluids. The pumping wells and observation wells extending beneath the marl provide a direct and available means of monitoring the confined ground water aquifer, if it is considered desirable or should a question arise. Samples may be taken immediately and analyses performed for prompt determination of any change in water quality.

Observation wells are also placed in the water table aquifer and provide a means to monitor and sample the ground water in the materials overlying the marl. Monitoring program observation wells are listed on table 2.4.12-9. Their locations are shown on drawings AX6DD333, AX6DD334, and AX6DD335.

2.4.12.4 Design Basis for Subsurface Hydrostatic Loading

The design basis for subsurface hydrostatic loading is elevation 165 ft msl. It is the estimated maximum probable level to which ground water could reasonably be expected to rise within the power block area during the life of the plant. The level was originally determined from a study of levels and fluctuations in observation wells in the water table aquifer in the vicinity of the power block site. Levels were measured during site exploration in 1971-1972. The wells nearest the power block during that period include 42D, 140, 143, and 245 (drawing AX6DD323).

Hydrographs of these four water table wells, and others, are shown on drawings AX6DD337 and AX6DD408. The highest level measured in the four wells nearest the power block site during 1971-72 was elevation 162 at well 140, located west and somewhat upgradient of the power block. The water table configuration during that period is illustrated by drawing AX6DD329. The largest fluctuation of water levels was 3.6 ft. From these records it was concluded that annual fluctuations could be expected to be less than 10 ft, and the maximum water table elevation at the power block would be 165.

Since that time, water level measurements in observation wells have continued. Several additional wells have been constructed, either to replace those wells that interfered with construction activities, or to supplement the observation network (see section 2.4.12.2.3.1 for details). Hydrographs of the levels measured through June 1985 are shown on drawing AX6DD337. The period of record extends over more than 14 years. The record demonstrates that natural fluctuations (not considering the response to dewatering of the power block excavation during construction) have been no greater than was expected, and the design level was not exceeded in the power block area, although it was approached in 1973.

Dewatering for the power block excavation was terminated in March 1983 and the water table in that area has recovered. A comparison with water table contours prior to construction, drawing

AX6DD329, with contours of the levels following recovery from dewatering, drawing AX6DD330, indicates that the water table has recovered to essentially the preconstruction levels and configuration. Excavation and grading have significantly reduced the topographic relief of the site. Because the water table is a subdued reflection of that relief, it follows that the post-construction water table can be expected to have lower maximum elevations than prior to construction.

Measurement of the water level in observation wells, as described in section 2.4.12.3, will be continued to verify that the design basis level is not exceeded.

2.4.12.5 References

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2.4.13 ACCIDENTAL RELEASES OF LIQUID EFFLUENTS IN GROUND AND SURFACE WATER

2.4.13.1 Consideration of Accidental Spill of Radioactive Material in Ground Water

In the very unlikely event that an accidental spill of radioactive fluid occurred at Plant Vogtle and infiltrated the ground without interception, it could percolate downward until it reached the water table (unconfined) aquifer. Because the Blue Bluff marl prevents significant vertical movement of ground water across it, continued migration of contaminants from an accidental spill at VEGP would be predominantly lateral in the direction of decreasing head in the water table aquifer. Accordingly, a spill would flow northwestward and after considerable travel time, would discharge into Mathes Pond and stream.

The impact of a postulated spill is assessed by assuming rupture of the recycle holdup tank (RHT), which is considered the worst-case release of radioactivity from the waste system. This tank would contain fluid with the highest concentration of critical isotopes. The migration of the spill in ground water is analyzed by a simplified, one-dimensional flow model in which a number of conservative assumptions are made. Extreme assumptions are imposed concerning the manner in which radioactivity is released to ground water. The analysis not only postulates tank failure, but also failure of the auxiliary building in which the tank is located (the floor on which the tank is located is below ground surface). It is assumed these failures are total, and the release to ground water occurs instantly, with no dilution or decay of the spilled waste. Analysis of the impact of the spill along the principal flow path (laterally, in the water-table aquifer) is first described. This is followed by a calculation of the possible rate of flow, vertically, across the marl to demonstrate the impact of a spill on the Tertiary and Cretaceous aquifers would be negligible.

2.4.13.1.1 Water Table Aquifer

Following these postulated events, the spilled waste would migrate in the ground water along a flow path northwestward to Mathes Pond. The flow path considered is a straight line between the auxiliary building and the south side of Mathes Pond, a distance of 3400 ft (drawing AX6DD330). The spill slug would first migrate an estimated distance of 550 ft through backfill material, and then would move through the Barnwell sands, eventually percolating through the underlying Utley limestone to reach Mathes Pond. Based on best estimates of the controlling parameters (i.e.; effective permeability, gradient) along the flow path, the time for ground water to migrate between the power block and Mathes Pond has been estimated to be as much as 350 years. Radionuclide decay during this long period is more than sufficient to reduce all radionuclide concentrations in a spill to below 10 CFR 20.1 - 20.601 limits.

An even more conservative analysis is to not consider the time of travel through the major portion of the flow path to Mathes Pond. Only that portion within the backfill material, a distance of 550 ft, is evaluated. Outside the backfill material, it is assumed that travel is rapid in an undefined conduit within the Utley limestone. Applying even this extreme assumption, analysis demonstrates that before any spill is discharged off site, concentrations of radioactivity would meet 10 CFR 20.1 - 20.601 limits.

The time required for ground water to migrate through the backfill is determined by the permeability and porosity of the materials, and the hydraulic gradient. The backfill is sand and

silty sand compacted to an average of 97 percent of its maximum density (paragraph 2.5.4.5.2). The permeability assigned to the backfill is the maximum value measured in situ, 1220 ft/year (table 2.4.12-15). Total porosity measurements of backfill samples that meet the compaction criterion range from 31.6 to 38.8 percent (table 2.4.12-14) and average total porosity is 34 percent. For sand and silty sand, the total and effective porosity are essentially the same (1). The hydraulic gradient in the backfill along the Mathes Pond flow path is 3.5×10^{-3}, but, again for conservatism, is rounded off to 4.0×10^{-3}. The relationship between these parameters in determining ground water seepage velocity is expressed in a form of Darcy's Law(1):

$$v = \frac{K_i}{n} \qquad (1)$$

where

v = ground water seepage velocity (L/T).

K = coefficient of hydraulic conductivity (permeability) (L/T).

i = hydraulic gradient (ratio).

n = effective porosity (ratio).

Applying the parameter values described, the calculated ground water velocity in the backfill is 14.4 ft/year. With a flow path length of 550 ft, the ground water travel time (t) in the backfill is 38.2 years.

The concentrations of spilled radionuclides that are ultimately transmitted through a ground water system to a discharge point (i.e., through the water table aquifer to Mathes Pond and stream, and, subsequently, discharged offsite to the Savannah River) is determined by the following factors:

- The source (tank) radionuclide inventory released to the ground water.
- The attenuation which takes place during transport through the system, caused principally by dispersion, dilution, adsorption, and radioactive decay.

Of the several radionuclides present in the liquid waste holding tanks, three are critical because of long half-lives. These include tritium (H^3), strontium-90 (Sr^{90}), and cesium-137 (Cs^{137}). Because they are chemically active and susceptible to adsorption, migration of Sr^{90} and Cs^{137} in the ground water will be retarded; they will move at a markedly slower rate than the water. Tritium is not adsorbed significantly, and tends to travel at the same rate as the ground water.

The degree of retardation is governed by various physical properties of the aquifer including bulk density and porosity, and the equilibrium distribution coefficients of the radionuclides. The relationship between ground water travel time (ground-water velocity and travel-path length), radionuclide adsorption, and the radionuclide fraction resulting from decay that is ultimately transmitted to Mathes Pond is given by the following expression (2):

$$\ln\left(C / C_o\right) = -\lambda t = \frac{-0.693\, t}{t_{1/2}}\ \text{wA} \qquad (2)$$

where:

C/C_o = transmitted fraction of original concentration, C_o, (ratio),

λ = radioactive decay constant = $\frac{\ln 2}{t_{1/2}}\left(\frac{1}{T}\right)$

t = $t_w A$ = radionuclide travel time (T).

t_w = ground water travel time (T).

$t_{½}$ = radionuclide half-life (T).

A = adsorption retention factor (ratio).

The adsorption retention factor (also called retardation factor) is equal to (1 + p/n K_d)

where:

p = dry (bulk) density of the aquifer (M/L^3).

n = porosity of the aquifer (ratio).

and

K_d = equilibrium distribution coefficient which is defined as the mass of radionuclide adsorbed per gram of soil divided by the mass of radionuclide dissolved per milliliter of ground water.

The density of backfill at the required compaction was determined for 12 samples(3). The values range from 1.62 to 1.79 g/cm^3. Assuming a value of 1.6 g/cm^3, and a porosity of 34 percent, the ratio, p/n, is 4.71 g/cm^3.

The K_ds (equilibrium distribution coefficients) for Sr^{90} and Cs^{137} of 4 samples of backfill were measured by the batch method. The resulting values, shown on table 2.4.13-1, are all more than 5 times greater than conservative estimates of average values proposed by Isherwood(4). Therefore, using the estimated average values, 10 and 100 cm^3/g for Sr^{90} and Cs^{137}, respectively, will impose definite conservatism in this analysis. Because tritium is not adsorbed, the K_d is zero.

Using the values of the parameters above, and a ground water travel time of 38.26 years, the calculated reductions in concentration in the backfill along a northward flow path are summarized as follows:

Nuclides	Kd (cm^3/gm)	A	$t_{1/2}$(yr)	C/C_o
H-3	0	1	12.2	1.10×10^{-1}
Sr-90	10	48.1	28	1.7×10^{-20}

Cs-137	100	472	30	6.8×10^{-182}

The concentration of radionuclides in the ground water after travel through the backfill is equal to the transmitted fraction times the initial concentration. The following summarizes the initial concentrations assumed to be present in the postulated worst-case spill, the reduced concentration after travel through the backfill due to radioactive decay and adsorption, and the maximum permissible concentration (MPC) for normal releases from Appendix B, Table II, Column 2 of 10 CFR 20.1 - 20.601.

Postulated RHT Rupture

Nuclides	Initial Concentration ($\mu Ci/cm^3$)	Concentration After Travel In Backfill ($\mu Ci/cm^3$)	MPC ($\mu Ci/cm^3$)
H-3	1.0	1.2×10^{-1}	3.0×10^{-3}
Sr-90	1.0×10^{-5}	1.7×10^{-25}	3.0×10^{-7}
Cs-137	1.9×10^{-2}	1.3×10^{-183}	2.0×10^{-5}

It can be seen that under this very conservative scenario, the concentrations of both Sr^{90} and Cs^{137} in ground water would meet 10 CFR 20.1 - 20.601 limits after travel through the backfill. Parameters that would reduce the concentration further, such as dispersion and dilution, need not be considered. Because H^3 is not retarded and migrates with the ground water, the tritium concentration in ground water traveling through the backfill would exceed the maximum permissible concentrations limits (still ignoring any dilution or dispersion of the spill).

Contaminated ground water exiting the backfill would continue migration through the Barnwell Group before reaching Mathes Pond. The high permeability measurements in the Utley limestone (table 2.4.12-13) raise the remote possibility of a continuous conduit allowing very rapid flow to Mathes Pond (negligible decay in short travel time). However, even if this hypothesis were correct, the contaminated ground water subsequently reaching Mathes Pond would be further diluted in the pond and in the stream running from the pond to the Savannah River, reducing the concentration below 10 CFR 20.1 - 20.601 limits before it flows off site. Flow into Mathes Pond is continuous, and the pond level is held constant by a spillway. The rate of flow discharging from Mathes Pond has been measured to be at least 250 gal/min. Measurements of stream flow at points downstream of the pond indicate it progressively increases in magnitude before discharging to the Savannah River.

The ratio of Mathes Pond stream flow to the rate at which the postulated spill would discharge from the backfill (and into Mathes pond) (assumed to be the rate of discharge into Mathes Pond) is a measure of the potential for dilution of the spill within the stream. The discharge rate of the spill in the backfill is determined by the velocity of ground water (14.4 ft/year) and the assumed volume and dimensions of the spill slug.

The volume of the spill slug is assumed to be 80 percent of the total capacity (112,000 gal) of the RHT, and is assumed to be transferred instantly to the backfill. To achieve a large effective discharge rate from the backfill, the shape of the slug is assumed to be rectangular, and relatively equidimensional. The dimensions take into account the saturated thickness of the

water table aquifer (estimated at 25 ft) and the effective porosity of the materials. With these parameters, the maximum spill discharge rate into Mathes Pond (discharge from backfill) is calculated to be 0.07 gal/min.

Contaminated ground water would first discharge to the pond, and would then flow into and down the stream below the pond for a distance of 4,000 ft before discharging off site. Mixing with the flow of Mathes Pond stream (a flow of at least 250 gal/min) would reduce the contaminant concentration by a factor of 2.8×10^{-4}. The concentration of tritium discharging from the backfill (0.12 $\mu Ci/cm^3$) would be reduced to 3.3×10^{-5} $\mu Ci/cm^3$, which is below the maximum permissible concentration.

As previously stated the analysis above is extremely conservative. It assumes an instantaneous release to the ground water, and thereby ignores any initial dilution or decay. It ignores any decay during the time to travel beyond the backfill. It ignores dispersion of the spill during its migration to Mathes Pond; and it ignores dilution due to the percolation of precipitation during the travel time. The effect of these additional parameters will be to further reduce the concentration of contaminants.

2.4.13.1.2 Blue Bluff Marl

Owing to the extent and very low permeability of the Blue Bluff marl, the impact of an accidental spill on the Tertiary and Cretaceous aquifers will be negligible. A calculation of the possible rate of flow across the marl demonstrates this conclusion.

The rate of flow is determined by the hydraulic gradient across the marl, and by the permeability and porosity of the materials (equation 1). The gradient is determined by the hydraulic head dissipated (the difference in piezometric levels of the water table and the Tertiary aquifers) over the travel path (the thickness of the marl). The difference in head beneath the power block can be determined from a comparison of piezometric surfaces of the two aquifers measured in December 1984 (drawings AX6DD328 and AX6DD330). These indicate a difference of about 50 to 55 ft. This is similar to the difference observed in a comparison of levels measured prior to construction November 1971. The minimum thickness of the marl is 38 ft, a result of excavation for structures beneath the power block. The maximum hydraulic gradient, then, is 55 ft of head over a distance of 38 ft, or 1.447.

The effective permeability of very-low permeable materials, such as the marl, is difficult to quantify. The most representative testing is by in-situ methods. However, the equipment and field conditions of these methods (such as packer and permeameter tests) limit the accuracy for measuring very low permeabilities. Under the field conditions and equipment used in these tests, no measurable water take (table 2.4.12-10) indicates that actual permeability is less than 0.1 ft/year.

Laboratory testing is capable of measuring very low permeabilities. However, the samples are small and may not be representative of in-situ conditions. The disturbance that occurs from collecting and handling the samples tends to increase the permeability of very low permeable materials.

The vertical permeability of the marl is heterogeneous, as is evidenced by the differences in head decline observed between the piezometers of well clusters A and B and well series 42 (drawing AX6DD408). The marl is composed of a series of beds, and a material comprised of

such layers, each of different permeabilities, is described as having layered heterogeneity.[1] The downward migration of ground water across the marl, however minute in quantity, will dissipate more head traversing the layers of lowest permeability than in traversing those layers of relatively higher permeability. The average or effective permeability across such a material (vertical flow) has been found to be equal to the harmonic mean of the layer permeabilities.[5] Assuming the ten laboratory tests (table 2.4.12-10) are a representative sample of the layers present in the marl (each sample represents an equal proportion of the total marl thickness), the harmonic mean permeability would be 0.045 ft/year (4.3×10^{-8} cm/sec). Adopting an average, or effective vertical permeability for the marl of 0.1 ft/year is therefore reasonably conservative.

Total porosity of the marl has been calculated for 19 samples, and the average value of those samples is 47.5 percent. Recent studies at the University of Waterloo[6] show that for clays the effective porosity (the porosity affecting the rate of ground water movement) is essentially equal to total porosity.

Applying the values above for the three controlling parameters -- hydraulic gradient (1.447), average permeability (0.1 ft/year), and effective porosity (47.5 percent) -- the average vertical ground water velocity in the marl is calculated to be 0.31 ft/year, and the time required to traverse 38 ft of marl would be 123 years. Taking into account retardation (discussed previously in paragraph 2.4.13.1.1), this travel time is sufficient to reduce all radionuclides in a worst case spill below the maximum permissible concentrations set forth in Appendix B, Table II, Column 2 of 10 CFR 20.1 - 20.601 (which applies to routine, continuous releases).

2.4.13.2 Dispersion, Dilution, and Travel Times of Accidental Releases of Liquid Effluents in Surface Waters

The only potentially radioactive tanks above grade are the refueling water storage tank, the reactor makeup water storage tank, and the condensate storage tanks. These tanks are designed and constructed to meet Seismic Category 1 requirements. Each tank has an approximately 2-ft-thick wall and 21-in.-thick roofs constructed of reinforced concrete. The tanks are lined with stainless steel liner plates. High level alarms are provided in the control room to alert the operator of a potential overflow condition. The tanks are surrounded by trenches to collect potential overflow. Provisions are made for sampling water collected in the trenches, and temporary connections are utilized to transfer potentially contaminated water to the liquid radwaste system. Thus no liquid effluent will be released to surface water from these tanks.

The radwaste processing facility described in subsection 1.2.2 is designed with a berm sufficient to contain the maximum liquid radwaste flow into the building for a period of greater than 30 minutes. A postulated leak or rupture of piping or components in the radwaste processing facility could potentially spray liquid over the berm and through gaps in the building wall around the door(s), resulting in a release outside the building. Bounding airborne and ground water releases are analyzed in paragraphs 15.7.2.5 and 15.7.3.4, respectively. Surface water runoff via the storm drain system (described in subsection 2.4.1) was analyzed and found to result in an exclusion area whole body dose less than the consquences of a liquid radwaste tank failure shown in table 15.7.2-2. Subsequent releases through the debris basin shown in drawing CX2D45V002 to the Savannah River were found to result in concentrations which would not exceed 10 CFR 20.1 - 20.601 limits at the nearest potable water intake.

An analysis of a radioactive release due to a failure of the most critical radwaste storage tank is provided in subsection 15.7.3.

The only direct discharge of radioactive liquids to surface water is from the waste monitor tanks, as discussed in subsection 11.2.3. The release point is shown in drawings AX4DB152-2 and AX4DB152-3. Normally, the discharge from the liquid waste processing system is combined with blowdown from the circulating water cooling system, the nuclear service cooling water system, and the steam generators as well as other station liquid wastes. If this flow is not sufficient to meet the Offsite Dose Calculation Manual limitations, additional river water is added to the discharge line to further dilute the radioactive wastes. Dilution factors are discussed in paragraph 11.2.3.4. An inadvertent release via the waste processing system or the steam generator blowdown system is prevented by interlocks between radiation monitors RE-018 and RE-021 (section 11.5) and associated downstream, air-operated valves. If the radioactivity in the streams exceeds a predetermined setpoint, then the streams are automatically isolated. In addition, locked closed valves in the waste stream ensure that administrative control of radioactive discharges is maintained.

Should an inadvertent discharge from the waste processing system occur, the release of one entire tank volume results in a release less than the normal annual releases discussed in subsection 11.2.3 and is within the limits found in the Offsite Dose Calculation Manual.

Discharge from the steam generator blowdown system (subsection 10.4.8) initially flows to the waste water retention basin. From there, it is pumped to the blowdown sump where it is combined with circulating water dilution flow and other blowdown flows before being discharged to the river. Upstream of the waste water retention basin, the steam generator blowdown is monitored for radioactivity by radiation monitor RE-021. If the monitor detects radioactivity exceeding a predetermined setpoint, the discharges to the waste water retention basin are automatically terminated.

2.4.13.3 References

1. Freeze, R.A., J.A. Cherry, Groundwater, Prentice-Hall Inc., New Jersey, 1979.

2. Friedlander, G., J.W. Kennedy, J.M. Miller, Nuclear and Radiochemistry, John Wiley and Sons, 1964.

3. Bechtel Civil and Minerals, Inc., Report on Confirmatory Laboratory Testing Program for Category I Backfill, September 1984.

4. Isherwood, D., Geoscience Data Base Handbook for Modeling a Nuclear Waste Repository, Lawrence Livermore Laboratory for the U.S. Nuclear Regulatory Commission, NUREG/CR-0912, Vol. 1, 1981.

5. Bouwer, H., Groundwater Hydrology, McGraw-Hill, 1978.

6. Desaulniers, D.E., Cherry, J.A., Gillham, R. W., "Hydrogeologic Analysis of Long Term Solute Migration in Thick Clayey Quaternary Deposits", Symposium on Ground Water Resources, Utilization and Contaminant Hydrogeology, International Association of Hydrogeologists, Montreal, 1984.

2.4.14 TECHNICAL SPECIFICATIONS AND EMERGENCY OPERATING REQUIREMENTS

There are no requirements for emergency protective measures designed to minimize the impact of hydrology-related events on safety-related facilities, and none will be incorporated into the Technical Specifications and emergency procedures.

TABLE 2.4.1-1

SAVANNAH RIVER SUBBASINS AND DRAINAGE AREAS ABOVE VEGP

Subbasin Number	Drainage Subbasin Gauging Station Description	Subbasin Hydrograph Area (mi^2)	Unit Duration (h)
1	Tallulah River at Burton Dam, Ga.	115.0	6
2	Savannah River at Hartwell Dam, Ga.	1850.0	6
3	Savannah River at Calhoun Falls, S.C.	788.0	6
4	Broad River at Carlton Bridge, Ga.	760.0	6
5	Broad River at Bell, Ga.	670.0	6
6	Savannah River at Clark Hill Dam	1740.0	6
7	Stevens Creek near Modoc, S.C.	545.0	6
8	Savannah River at Stevens Creek Dam, S.C.	484.0	6
9	Savannah River at Butler Creek, Augusta, Ga.	328.0	6
10	Savannah River at VEGP	735.0	6
TOTAL		8015.0	

TABLE 2.4.1-2 (SHEET 1 OF 5)

SAFETY-RELATED STRUCTURES AND ACCESS TO THEM

Building	Level	Elevation (ft)	Access Opening	Material
Control	1	220	Rollup door	Steel slats
Control	1	220	Emergency exit	Hollow metal
Control	1	220	Personnel access	Steel plate door
Control	1	220	TSC personnel access	Hollow metal
Control	1	220	TSC emergency exit	Hollow metal
Control	1	220	TSC equipment door	Hollow metal
Control	2	240	TSC roof access	Hollow metal
Control	3	260	Roof access door	Hollow metal
Control	4	280	Stairwell door	Hollow metal
Control	4	280	Elevator machine room	Hollow metal
Control	4	280	Unit 2 HVAC room	Hollow metal
Control	4	280	Unit 1 HVAC room	Hollow metal
Control	5	300	Roof access door	Hollow metal
North MSIV Control (Unit 1)	1	220	Access door	Hollow metal
North MSIV Control (Unit 2)	1	220	Access door	Hollow metal
Auxiliary	1	220	Emergency exit	Hollow metal
Auxiliary	1	220	Personnel missile door	Steel plate door

TABLE 2.4.1-2 (SHEET 2 OF 5)

Building	Level	Elevation (ft)	Access Opening	Material
Auxiliary	1	220	Railroad missile door	Steel plate door
Auxiliary	1	220	Emergency exit	Hollow metal
Auxiliary	1	220	Emergency exit	Hollow metal
Auxiliary	1	220	Emergency exit	Hollow metal
Auxiliary (Unit 2 South MSIV)	1	220	Access gate	Hollow metal
Auxiliary (Unit 1 South MSIV)	1	220	Access gate	Hollow metal
Auxiliary	3	260	Roof access(Unit 2)	Hollow metal
Auxiliary	3	260	Elevator machine room (Unit 2)	Hollow metal
Auxiliary	3	260	Roof access(Unit 1)	Hollow metal
Auxiliary	3	260	Elevator machine room (Unit 1)	Hollow metal
Containment (Unit 1)	B	183 ft 10 1/2 in.	Emergency escape door	Hollow metal
Containment (Unit 1)	1	220	Equipment building access	Hollow metal
Containment (Unit 1)	1	220	Containment equipment hatch	Steel hatch
Containment (Unit 2)	B	183 ft 10 1/2 in.	Emergency escape door	Hollow metal
Containment (Unit 2)	1	220	Equipment building access	Hollow metal

TABLE 2.4.1-2 (SHEET 3 OF 5)

Building	Level	Elevation (ft)	Access Opening	Material
Containment (Unit 2)	1	220	Containment equipment hatch	Steel hatch
Diesel generator (Unit 1)	1	220	Access door	Steel plate
Diesel generator (Unit 1)	1	220	Access door	Steel plate
Diesel generator (Unit 1)	1	220	Access door	Steel plate
Diesel generator (Unit 1)	1	220	Access door	Steel plate
Diesel generator (Unit 1)	1	220	Access door	Steel plate
Diesel generator (Unit 2)	1	220	Access door	Steel plate
Diesel generator (Unit 2)	1	220	Access door	Steel plate
Diesel generator (Unit 2)	1	220	Access door	Steel plate
Diesel generator (Unit 2)	1	220	Access door	Steel plate
Auxiliary feedwater pumphouse (Unit 1)	1	220	Access door	Steel plate
Auxiliary feedwater pumphouse (Unit 1)	1	220	Access door	Steel plate

TABLE 2.4.1-2 (SHEET 4 OF 5)

Building	Level	Elevation (ft)	Access Opening	Material
Auxiliary feedwater pumphouse (Unit 1)	1	220	Access door	Steel plate
Auxiliary feedwater pumphouse (Unit 1)	1	220	Access door	Steel plate
Auxiliary feedwater pumphouse (Unit 2)	1	220	Access door	Steel plate
Auxiliary feedwater pumphouse (Unit 2)	1	220	Access door	Steel plate
Auxiliary feedwater pumphouse (Unit 2)	1	220	Access door	Steel plate
Auxiliary feedwater pumphouse (Unit 2)	1	220	Access door	Steel plate
Train A NSCW pumphouse (Unit 1)	1	220	Access door	Steel plate
Train B NSCW pumphouse (Unit 1)	1	220	Access door	Steel plate
Train A NSCW pumphouse (Unit 2)	1	220	Access door	Steel plate

TABLE 2.4.1-2 (SHEET 5 OF 5)

Building	Level	Elevation (ft)	Access Opening	Material
Train A NSCW pumphouse (Unit 2)	1	220	Access door	Steel plate

TABLE 2.4.1-3 (SHEET 1 OF 2)

WATER CONTROL STRUCTURES SAVANNAH RIVER BASIN

Dam and Owner	River Mile	Storage (acre-ft)	Drainage Area (mi^2)	Earth Dike Length (ft)	Concrete Structure (Powerhouse Nonoverflow Wall) (ft)	Crest el (ft msl)	No. of Gates	Spillway Crest Length (ft)	Normal Pool el (ft msl)	Normal Head (ft)	Top of Dam el	Seismic Design Criteria	Spillway Design Criteria
New Savannah Bluff Lock and Dam Corps of Engineers	187.7	-	7508	-	-	-	-	-	13.1	152.6	-	-	-
Stevens Creek S.C. Elec. & Gas	209.7	-	-	-	-	-	-	-	-	29	187	-	-
Clark Hill Corps of Engineers	222.7	2,510,000 @ el 335	6144	3398	1186	300	23	1096	335	136	351	Yes	PMF
Richard B. Russell Corps of Engineers	260.2	600,000 @ el 475	2900	2640	1884	436	10	590	475	175	495	Yes	PMF
Hartwell Corps of Engineers	290.0	2,549,600 @ el 660	2088	15,952	1332	630	12	568	660	185	679	Yes	PMF
Yonah Georgia Power Co.	340.0	10,200 @ el 744.2	470	-	530	742.25	2'FB[(a)]	450	744.2	70.25	757	-	–
Tugaloo Georgia Power Co.	343.1	43,200 @ el 891.5	464	-	493	885	8'FB	357	891.5	144	905	-	–
Tallulah Falls Georgia Power Co.	346.7	2460 @ el 1500	186	-	110	1493	7'FB	316	1500	603.4	514	-	–
Mathis Georgia Power Co.	353.4	31,400 @ el 1689.6	151	370	312	1681.25	8.5'FB	285	1689.6	189.6	1704	-	–
Nacoochee Georgia Power Co.	362.1	8200 @ el 1752.5	136	-	350	1752.5	open	140	1752.5	62.2	1765	-	–

TABLE 2.4.1-3 (SHEET 2 OF 2)

Dam and Owner	River Mile	Storage (acre-ft)	Drainage Area (mi^2)	Earth Dike Length (ft)	Concrete Structure (Powerhouse Nonoverflow Wall) (ft)	Crest el (ft msl)	No. of Gates	Spillway Crest Length (ft)	Normal Pool el (ft msl)	Normal Head (ft)	Top of Dam el	Seismic Design Criteria	Spillway Design Criteria
Burton Georgia Power Co.	366.4	108,000 @ el 1866.6	118	-	845.5	1860	6.5'FB	197	1866.6	114.1	1873	-	-
Keowee Duke Power Co.	341.0+	940,000 @ el 800;	439	3500	-	765	4	176	800	140	815	-	-
Little River Duke Power Co.	366.0+	see Keowee	see Keowee	1750	- -	see Keowee	-	-	see Keowee	see Keowee	see Keowee	-	-
Jocassee Duke Power Co.	357.0+	1,100,000 @ el 1110	148	1800	-	1077	2	-	1100	310	1125	Yes	PMF

a. FB = flash board.

TABLE 2.4.2-1 (SHEET 1 OF 4)

GAUGING STATION RECORDS SAVANNAH RIVER BASIN
SAVANNAH RIVER AT AUGUSTA, GA.
ANNUAL FLOOD PEAKS[(a)]

Calendar Year	Month	Day	Discharge (ft^3/s)
1796	-	-	360,000
1840	May	28	270,000
1852	Aug.	29	250,000
1865	Jan.	11	240,000
1875	Dec.	30	86,400
1877	Apr.	14	119,000
1877	Nov.	23	51,500
1879	Aug.	3	44,000
1879	Dec.	16	102,000
1881	Mar.	18	130,000
1882	Sept.	12	93,300
1883	Jan.	22	111,000
1884	Apr.	16	81,000
1885	Jan.	26	77,000
1886	May	21	135,000
1887	July	31	173,000
1888	Sept.	11	303,000
1889	Feb.	19	149,000
1890	Feb.	27	48,500
1891	Mar.	10	197,000
1892	Jan.	20	140,000
1893	Feb.	14	60,000
1894	Aug.	7	54,000
1895	Jan.	11	106,000
1896	Jan.	10	107,000
1897	Apr.	6	93,300
1898	Sept.	2	117,000
1899	Feb.	8	112,000
1900	Feb.	15	138,000
1901	Apr.	4	124,000
1902	Mar.	1	175,000

TABLE 2.4.2-1 (SHEET 2 OF 4)

Calendar Year	Month	Day	Discharge (ft^3/s)
1903	Feb.	9	147,000
1904	Aug.	10	63,000
1905	Feb.	14	64,800
1906	Jan.	5	96,600
1906	Oct.	5	52,000
1908	Aug.	27	307,000
1909	June	5	87,300
1910	Mar.	2	69,800
1911	Apr.	14	32,800
1912	Mar.	17	234,000
1913	Mar.	16	156,000
1913	Dec.	31	48,000
1915	Jan.	20	61,000
1916	Feb.	3	82,400
1917	Mar.	6	68,000
1918	Jan.	30	45,500
1918	Dec.	24	128,000
1919	Dec.	11	133,000
1921	Feb.	11	129,000
1922	Mar.	12	92,000
1923	Feb.	28	59,700
1924	Apr.	6	56,400
1925	Jan.	20	150,000
1926	Jan.	20	55,300
1926	Dec.	30	39,000
1928	Aug.	17	226,000
1929	Mar.	6	191,000
1929	Oct.	2	350,000
1930	Nov.	17	26,100
1932	Jan.	9	93,800
1933	Feb.	9	48,200
1934	Mar.	5	73,200
1935	Mar.	14	63,700
1936	Apr.	8	258,000
1937	Jan.	4	90,200
1938	July	26	65,300

TABLE 2.4.2-1 (SHEET 3 OF 4)

Calendar Year	Month	Day	Discharge (ft^3/s)
1939	Mar.	2	82,400
1940	Aug.	15	252,000
1941	July	8	52,200
1942	Mar.	23	115,000
1943	Jan.	20	132,000
1944	Mar.	22	141,000
1945	Apr.	26	62,100
1946	Jan.	9	109,000
1947	Jan.	22	90,200
1948	Feb.	10	76,100
1949	Nov.	30	172,000
1950	Oct.	9	32,500
1951	Oct.	22	41,400
1952	Mar.	6	39,300
1953	May	8	35,200
1954	Mar.	30	25,500
1955	Apr.	15	23,900
1956	Apr.	12	18,600
1957	May	7	18,000
1958	Apr.	18	66,300
1959	June	8	28,500
1960	Feb.	14	34,900
1961	Apr.	2	34,800
1962	Jan.	9	32,500
1963	Mar.	23	31,300
1964	Apr.	9	87,100
1965	Dec.	27	34,600
1966	Mar.	6	39,300
1967	Aug.	25	26,500
1968	Jan.	12	35,900
1969	Apr.	21	45,600
1970	Apr.	1	25,200
1971	Mar.	5	63,900
1972	Jan.	20	33,700
1973	Apr.	8	40,200

TABLE 2.4.2-1 (SHEET 4 OF 4)

Calendar Year	Month	Day	Discharge (ft^3/s)
1974	Feb.	23	32,900
1975	Mar.	25	45,600
1976	June	5	33,300
1977	Apr.	7	34,200
1978	Jan.	26	43,100
1979	Feb.	27	37,300

a. U.S. Geological Survey, Surface Water Supply of the United States, Part 2, Vol 1, and Water Resources Yearly Data for the State of Georgia.

TABLE 2.4.2-2

PROBABLE MAXIMUM PRECIPITATION

Time (min)	Maximum Rainfall (in.)	Corresponding Intensity (in./h)
1	2.0	120
2	2.8	84
4	4.0	60
6	4.9	49
8	5.5	41
10	6.2	37
20	9.0	27
40	12.5	19
60	15.0	15
120	22.0	11
240	31.0	8

TABLE 2.4.2-3

YARD DRAINAGE
RECOMMENDED 100-YEAR RAINFALL CRITERIA[a]

Duration (min)	Intensity (in./h)
5	12.8
6	12.2
8	11.1
10	10.1
12	9.4
14	8.8
15	8.6
16	8.4
18	7.9
20	7.6
22	7.2
24	6.9
26	6.6
28	6.4
30	6.1
35	5.6
40	5.2
60	4.0
120	2.4
180	1.7
360	1.0

a. Hershfield, D. M., "Rainfall-Frequency Atlas of the United States for Durations from 30 Minutes to 24 Hours and Return Periods from 1 to 100 Years," U.S. Weather Bureau Technical Paper No. 40, 1961.

TABLE 2.4.3-1

PROBABLE MAXIMUM PRECIPITATION

Drainage Area (mi^2)	Probable Maximum Precipitation (in.)				
	6 h	12 h	24 h	48 h	72 h
10	31.0	37.2	43.8	48.2	51.0
200	23.0	28.0	34.7	38.8	42.0
1000	16.7	22.5	29.0	33.0	35.0
5000	9.7	14.0	19.5	24.0	27.0
10,000	7.5	11.2	16.0	20.0	23.5
20,000	5.4	8.8	12.5	16.4	19.5
For Drainage Area above VEGP					
8015	8.1	12.0	16.9	21.0	24.4

Location (approximate): latitude N 33°03'; longitude W 81°55'

TABLE 2.4.3-2

6-h SEQUENTIAL INCREMENTAL PROBABLE MAXIMUM PRECIPITATION (in.)

	6 h	12 h	18 h	24 h	30 h	36 h	42 h	48 h	54 h	60 h	66 h	72 h
Cumulative PMP	8.0	12.4	15.0	16.8	18.2	19.4	20.6	21.8	22.3	23.0	23.7	24.4
Incremental	8.0	4.4	2.6	1.8	1.4	1.2	1.2	1.2	0.5	0.7	0.3	0.7
Sequential	8.0	4.4	2.6	1.8	1.4	1.2	1.2	1.2	0.7	0.7	0.5	0.3
6-h duration interval	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0	11.0	12.0

TABLE 2.4.3-3

SPATIAL ISOHYETAL ADJUSTMENTS FOR
THE THREE GREATEST 6-h PROBABLE MAXIMUM PRECIPITATION
(percent)

		Isohyet Designation						
		P	A	B	C	D	E	F
Drainage area enclosed (mi^2)	10	35	270	800	3200	8700	19000	
Isohyet values in percent of PMP, greatest 6-h increase		258	227	174	137	95	55	32
Isohyet values in percent of PMP, second greatest 6-h increase		165	152	130	116	97	86	73
Isohyet values in percent of PMP, third greatest 6-h increase		113	111	107	104	100	97	93

TABLE 2.4.3-4

SPATIALLY ADJUSTED THREE GREATEST 6-h PROBABLE MAXIMUM PRECIPITATION

	Isohyet Designation						
	P	A	B	C	D	E	F
Drainage area enclosed (mi^2)	10	35	270	800	3200	8700	19700
Greatest 6-h PMP increment isohyet value (in.)	17.54	15.44	11.83	9.32	6.46	3.74	2.18
Second greatest 6-h PMP increment isohyet value (in.)	6.11	5.62	4.81	4.29	3.59	3.18	2.70
Third greatest 6-h PMP increment isohyet value (in.)	2.49	2.44	2.35	2.29	2.20	2.13	2.05

TABLE 2.4.3-5

6-h INCREMENTAL MAXIMUM PRECIPITATION ARRANGED IN CRITICAL SEQUENCE

Time from beginning of storm (h)	6	12	18	24	30	36	42	48	54	60	66	72
For critical sequence rank of 6-h increase, 1=highest, 12=lowest	9	7	8	3	2	1	5	4	6	10	11	12
Subbasin drainage identification number	6-h incremental PMP arranged in critical sequence (in.)											
Tallulah River at Burton Dam, Ga., No. 1	0.7	1.2	1.2	2.17	3.39	5.10	1.4	1.8	1.2	0.7	0.5	0.3
Savannah River at Hartwell Dam, Ga., No. 2	0.7	1.2	1.2	2.17	3.45	5.44	1.4	1.8	1.2	0.7	0.5	0.3
Savannah River at Calhoun Falls, S.C., No. 3	0.7	1.2	1.2	2.25	4.17	8.94	1.4	1.8	1.2	0.7	0.5	0.3
Broad River at Carlton Bridge, Ga., No. 4	0.7	1.2	1.2	2.18	3.49	5.66	1.4	1.8	1.2	0.7	0.5	0.3
Broad River at Bell, Ga., No. 5	0.7	1.2	1.2	2.18	5.73	3.51	1.4	1.8	1.2	0.7	0.5	0.3
Savannah River at Clark Hill Dam, No. 6	0.7	1.2	1.2	2.22	7.19	3.82	1.4	1.8	1.2	0.7	0.5	0.3
Stevens Creek near Modoc, S.C., No. 7	0.7	1.2	1.2	2.19	6.0	5.36	1.4	1.8	1.2	0.7	0.5	0.3
Savannah River at Stevens Creek Dam, No. 8	0.7	1.2	1.2	2.20	6.35	3.63	1.4	1.8	1.2	0.7	0.5	0.3
Savannah River at Butler Creek (Augusta, Ga.), No. 9	0.7	1.2	1.2	2.17	5.14	3.39	1.4	1.8	1.2	0.7	0.5	0.3
Savannah River at VEGP, No. 10	0.7	1.2	1.2	2.16	4.95	3.35	1.4	1.8	1.2	0.7	0.5	0.3

TABLE 2.4.3-6

CUMULATIVE PROBABLE MAXIMUM PRECIPITATION FOR VARIOUS STORM POSITIONS AND CALCULATION PROCEDURES

Subbasin number and description	Cumulative PMP for Storm Position No. 1 Based on Hydromet No. 51 Procedure (in.)					Cumulative PMP for Storm Position No. 2 Based on Hydromet No. 51 Procedure (in.)					Cumulative PMP for Storm Position No. 1 Based on Hydromet No. 52 Procedure (in.)				
	6 h	12 h	24 h	48 h	72 h	6 h	12 h	24 h	48 h	72 h	6 h	12 h	24 h	48 h	72 h
1. Tallulah River at Burton Dam, Ga.	0.52	1.92	3.65	7.65	10.74	1.09	3.52	5.16	11.21	13.57	1.6	2.2	4.0	7.3	10.7
2. Savannah River at Hartwell Dam, Ga.	1.53	4.27	6.22	11.37	14.3	3.71	8.07	10.72	15.11	18.35	1.8	3.4	6.2	11.6	14.3
3. Savannah River at Calhoun Falls, S.C.	6.32	12.4	15.74	19.26	22.36	13.36	19.28	25.7	28.93	31.42	7.6	10.8	14.8	19.4	22.4
4. Broad River at Carlton Bridge, Ga.	1.38	3.9	5.91	11.1	13.95	3.42	7.83	10.26	14.66	18.13	2.0	3.9	5.9	10.5	14.0
5. Broad River at Bell, Ga.	2.82	6.73	8.98	13.75	17.06	4.28	9.03	11.94	16.04	19.39	3.4	5.9	9.4	13.8	17.0
6. Savannah River at Clark Hill Dam	8.6	13.04	17.33	21.03	24.02	5.45	10.65	13.92	17.84	21.0	9.0	12.4	16.6	21.2	24.0
7. Stevens Creek near Modoc, S.C.	6.53	12.25	15.63	19.05	22.29	2.91	7.01	9.37	13.86	17.56	7.7	10.8	14.8	19.4	22.2
8. Savannah River at Stevens Creek Dam, S.C.	5.39	11.25	14.07	17.55	21.23	2.15	5.39	7.94	12.7	16.1	6.2	9.2	13.4	18.0	
9. Savannah River at Butler Creek, Ga.	3.92	9.21	11.4	15.4	19.67	1.26	3.78	5.64	11.41	14.06	4.8	7.8	11.7	16.1	19.6
10. Savannah River at VEGP, Ga.	2.04	5.18	7.66	12.54	15.85	0.73	2.67	4.26	9.64	12.11	1.5	5.0	8.2	12.6	15.8

TABLE 2.4.7-1

MINIMUM WATER TEMPERATURE OF THE SAVANNAH RIVER

Gauging Station	Period of Record	Statistical Parameter	Value Exceeded P Percent of Time	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
Savannah River at Augusta, Ga., river mile 187.4	1974 - 1980	Minimum temperature (°C)	V95	5.2	5.2	7.0	12.0	14.0	17.0	19.0	21.0	20.0	18.0	13.0	8.7
			V90	6.0	5.5	7.9	12.0	15.0	17.0	20.0	21.0	21.0	18.0	13.0	9.7
			V75	7.5	6.5	9.3	13.0	16.0	18.0	20.0	22.0	22.0	19.0	15.0	11.0
			V70	7.8	6.8	9.8	13.0	16.0	18.0	20.0	22.0	22.0	20.0	16.0	11.0
			V50	9.2	8.1	11.0	14.0	17.0	19.0	21.0	22.0	23.0	20.0	17.0	12.0
			V25	10.0	10.0	12.0	15.0	18.0	20.0	22.0	23.0	23.0	21.0	18.0	14.0
			V10	12.0	11.0	13.0	17.0	19.0	21.0	23.0	24.0	24.0	22.0	19.0	15.0
Savannah River at Jackson, S.C., river mile 156.8	1972 - 1980	Minimum temperature (°C)	V95	5.1	5.3	8.3	13.0	16.0	18.0	21.0	22.0	21.0	17.0	13.0	8.8
			V90	6.0	5.8	9.4	13.0	16.0	18.0	21.0	22.0	21.0	18.0	14.0	9.4
			V75	7.5	7.2	11.0	14.0	17.0	19.0	22.0	22.0	22.0	19.0	15.0	10.0
			V70	7.9	7.5	11.0	14.0	17.0	20.0	22.0	23.0	22.0	19.0	15.0	11.0
			V50	9.2	8.7	12.0	15.0	18.0	21.0	23.0	23.0	23.0	20.0	17.0	12.0
			V25	11.0	11.0	13.0	16.0	19.0	22.0	23.0	24.0	24.0	21.0	18.0	13.0
			V10	12.0	12.0	14.0	18.0	20.0	22.0	24.0	24.0	24.0	22.0	19.0	14.0
Savannah River below Steel Creek near Millet, S.C., river mile 138.8	1972 - 1980	Minimum temperature (°C)	V95	5.5	5.4	8.4	13.0	16.0	18.0	21.0	22.0	21.0	17.0	12.0	8.6
			V90	6.4	6.0	9.7	14.0	17.0	19.0	22.0	23.0	22.0	18.0	13.0	9.4
			V75	8.0	7.2	11.0	14.0	18.0	20.0	23.0	23.0	23.0	19.0	15.0	10.0
			V70	8.3	7.6	11.0	15.0	18.0	20.0	23.0	24.0	23.0	19.0	15.0	11.0
			V50	9.6	8.9	13.0	15.0	19.0	21.0	24.0	24.0	24.0	20.0	16.0	12.0
			V25	11.0	11.0	14.0	17.0	20.0	22.0	24.0	25.0	25.0	21.0	18.0	13.0
			V10	13.0	12.0	15.0	18.0	21.0	23.0	25.0	26.0	25.0	23.0	19.0	15.0
Savannah River at Burton's Ferry near Millhaven, Ga., river mile 118.7	1961 - 1969, 1973	Minimum temperature (°C)	V95	5.9	7.0	8.6	13.0	16.0	19.0	21.0	23.0	21.0	17.0	12.0	7.0
			V90	6.3	7.2	9.0	14.0	17.0	20.0	22.0	23.0	21.0	18.0	13.0	8.0
			V75	7.7	8.0	11.0	15.0	18.0	21.0	23.0	24.0	22.0	19.0	14.0	9.0
			V70	7.9	8.2	11.0	15.0	18.0	21.0	24.0	24.0	23.0	20.0	15.0	10.0
			V50	8.7	9.0	13.0	16.0	20.0	22.0	24.0	25.0	24.0	21.0	16.0	11.0
			V25	10.0	11.0	14.0	18.0	21.0	24.0	25.0	26.0	25.0	22.0	17.0	13.0
			V10	12.0	12.0	15.0	19.0	23.0	25.0	26.0	27.0	26.0	23.0	18.0	14.0

TABLE 2.4.12-1

WATER-BEARING PROPERTIES OF MATERIALS UNDERLYING VEGP AND VICINITY

Stratigraphic Unit	Thickness (ft)	Physical Characteristics	Water-Bearing Properties
Alluvium (Recent)	0-30	Tan to gray sand, silt, clay, and gravel	Very little water
Terraces (Pleistocene)	0-30	Sand, silt, clay, and gravel	Ground water supply moderate to none
Alluvium (Pliocene)	0-20	Gravel and sandy clay	Little or no water
Hawthorne Formation (Miocene)	0-80	Sandy clay, interbedded lenses of gravel, and numerous clastic dikes	Small to moderate amounts of ground water
Barnwell Group (Eocene-Jackson Age)	0-90	Red, brown, yellow, and buff sand and clay	Moderate amounts of ground water sufficient for domestic use
Lisbon Formation (Eocene-Claiborne Age)	0-250	Fine to coarse glauconitic sand interbedded with clay, sandy marl, or limestone, and siliceous limestone lenses	Ground water supply moderate to large in sands only; sufficient for limited industrial use
Huber/Ellenton Formation (Paleocene)	0-100	Dark, sandy lignitic micaceous clay containing disseminated crystals of gypsum; dark coarse sand and white kaolin	Ground water supply small in some areas
Tuscaloosa (Cretaceous)	0-700	Cross-bedded micaceous quartzite and arkosic sand and gravel, interbedded with clay and white kaolin	Large supplies of good quality ground water; yields up to 2000 gal/min from gravel packed wells
Newark Group (Triassic)	Unknown at VEGP	Sandstone, siltstone, graywacke, and claystone with sections of flaglomerate or conglomerate	Low yields to wells
Basement rocks (Paleozoic and Precambrian)	Unknown at VEGP	Granite, gneiss, schist, slate, and volcanic rocks	Limited supplies, ground water in fractures

Table derived from references 3 and 8.

TABLE 2.4.12-2

GROUND WATER USE

Service	No. Required	Maximum Water Demands/Capacity
Fire protection storage tank	2	300,000 gal/each
Construction water tank	1	30,000 gal (primary)
Batch plant water storage tank (plant construction only)	1	10,000 gal (backup)
Construction water tank	1	25,000 gal (backup)
Potable and sanitary water tank	1	25,000 gal
Potable and sanitary use	-	350 gal/min
Well water storage tank	1	300,000 gal
Utility water use	-	330 gal/min
Makeup demineralizer	1	440 gal/min process flow
Demineralizer water storage tank	1	250,000 gal
Reactor makeup water storage tanks	2	165,000 gal/each
Condensate storage tanks	4	480,000 gal/each
Demineralized water use	-	550 gal/min
Nuclear service cooling water towers	4	268 gal/min per tower at 4 cycles

TABLE 2.4.12-3 (SHEET 1 OF 2)

WATER QUALITY ANALYSES - VEGP WELLS

Observation Wells

(Values in parts per million unless otherwise designated)

Observation well	34[a]	121[a]	135[b]	124[c]	129[c]	142[c]	143[c]	24[b]	24[b]
Date sampled	10/06/71	09/22/71	10/31/71	10/12/71	10/13/71	10/13/71	10/13/71	07/21/71	10/06/71
Laboratory No.	17947	17840	17980	17978	17979	17981	17982		17944
Date of analysis	10/11/71	09/27/71	10/14/71	10/14/71	10/14/71	10/14/71	10/14/71		10/14/71
Constituents									
Sodium (Na)	13.3	15.9	16.3	6.8	18.2	5.2	21.6	6.75	4.6
Magnesium (Mg)	5.4	7.5	8.3	8.3	8.3	6.8	5.8	1.7	11.2
Potassium (K)	3.2	3.2	2.8	0.7	2.3	1.2	1.9	1.49	1.5
Calcium (Ca)	15.2	12.8	28.8	24.8	23.6	25.6	23.0	23.0	4.0
Carbonate (CO_3)	0.0	0.0	14.4	0.0	9.6	0.0	14.4	---	16.8
Bicarbonate (HCO_3)	95.2	84.2	114.7	96.4	93.9	103.7	96.4	---	45.1
Hydroxide (OH)	0.0	---	0.0	0.0	0.0	0.0	0.0	---	0.0
Sulfate (SO_4)	7.7	3.6	17.4	3.5	20.5	7.1	25.8	36.6	7.6
Chloride (Cl)	3.0	1.0	4.0	4.0	3.0	2.0	2.0	4.0	2.0
Nitrate (NO_3)	0.07	0.0	1.4	0.0	0.85	0.22	1.14	0.0	0.09
Fluoride (F)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	---	0.09
Total phosphate (PO_4)	0.15	0.66	0.26	0.26	0.28	0.26	0.63	---	0.5
Iron (Fe)	0.57	0.31	0.12	0.5	0.15	0.34	0.3	0.21	0.15
Silica (SiO_2)	7.9	8.0	9.2	5.2	5.4	7.7	8.4	6.75	7.4
Total dissolved solids	127.9	112.5	106.0	126.4	155.8	134.2	169.5	234.9	89.6
Conductivity (μmho)	140.0	130.0	180.0	130.0	170.0	120.0	150.0	390.0	115.0
pH	7.9	6.1	8.2	8.2	8.3	7.9	8.6	11.3	9.1
Free carbon dioxide	2.0	110.0	0.0	0.0	0.0	2.0	0.0	0.0	0.0
Total alkalinity as $CaCO_3$	78.0	69.0	106.0	79.0	85.0	85.0	91.0	119.0	51.0
Total hardness as $CaCO_3$	60.0	63.0	106.0	96.0	66.0	92.0	62.0	94.0	56.0
Temperature (°F)	---	65.0	---	---	---	---	68.0	---	68.0

TABLE 2.4.12-3 (SHEET 2 OF 2)

WATER QUALITY ANALYSES - VEGP WELLS

Supply Wells

(Values in parts per million unless otherwise designated)

Supply well	MU-1[d]	MU-1[d]	MU-2[d]	MU-2[d]	MU-2[d]	CW-2[e]	CW-3[e]	SB[e]
Date sampled	12/19/77	5/2/85	9/12/78	11/27/84	12/3/84	5/19/82	7/29/81	11/8/82
Date analyzed		5/2/85	9/19/78	3/1/85		6/14/82	7/30/81	11/9/82
Constituents								
Sodium (Na)	35.5	--	35.6	6.52	44.3	9.8	--	--
Magnesium (Mg)	0.63	--	0.4	0.35	0.36	1.48	--	--
Potassium (K)	1.4	--	1.6	0.47	0.81	2.3	--	--
Calcium (Ca)	3.8	3.8	2.2	2.0	1.96	20.7	--	--
Sulfate (SO_4)	4.8	3.12	3.0	6.7	6.0	--	--	6.7
Chloride (Cl)	2.2	2.67	3.13	2.1	2.4	4.74	3.58	198.4
Nitrate (NO_3)	--	0.2	0.04	--	--	0.10	0.014	1.15
Total phosphate ($P0_4$)	--	--	0.031	--	--	--	--	--
Iron (Fe)	--	0.13	0.21	0.06	--	1.28	0.14	3.07
Silica (SiO_2)	12.4	12.6	12.4	4.85	--	--	--	--
Total dissolved solids	114.0	--	201.0	112.0	101.0	96.0	--	568.0
Conductivity (μmho)	187.0	171.0	155.0	76.0	160.0	--	--	--
pH	7.9	7.52	8.03	7.8	7.4	7.74	7.61	10.49
Free carbon dioxide	--	--	--	--	--	--	3.3	--
Total alkalinity as $CaCO_3$	81.5	83.7	81.7	79.4	--	64.7	127.0	185.4
Total hardness as $CaCO_3$	12.1	4.6	7.3	7.0	--	57.8	116.0	18.8

a. Open to the Tertiary aquifer; flowing at the surface.

b. Open to the Tertiary aquifer; not flowing at the surface.

c. Open to the water table aquifer system (above the marl).

d. Open to the Cretaceous (confined) aquifer.

e. Open to the Tertiary (confined) aquifer.

TABLE 2.4.12-4

WATER QUALITY ANALYSES - DOMESTIC WELLS

(Values in parts per million unless otherwise designated)

Sample identification	Well 1	Well 3	Well 8	Well 10	Well 9	Well 6	Well 14	Well 14	Well 7
Date sampled	09/17/71	09/17/71	09/18/71	09/18/71	09/18/71	09/18/71	09/20/71	09/22/71	09/18/71
Laboratory No.	17803	17804	17806	17808	17807	17809	17812	17839	17805
Date of analysis	09/21/71	09/21/71	09/21/71	09/21/71	09/21/71	09/21/71	09/21/71	09/21/71	09/21/71
Constituents									
Sodium (Na)	2.0	2.2	1.9	2.2	3.3	6.0	3.8	3.6	2.9
Magnesium (Mg)	0.2	1.0	1.9	1.7	3.2	2.7	2.4	1.7	1.7
Potassium (K)	0.29	0.33	0.42	0.46	1.41	4.48	2.41	2.0	0.5
Calcium (Ca)	26.8	28.8	28.0	39.6	46.8	55.2	50.4	51.2	50.8
Bicarbonate (HCO_3)	78.1	95.2	98.8	118.3	152.5	128.1	156.2	158.6	159.8
Sulfate (SO_4)	0.6	0.0	0.0	7.5	4.8	5.2	4.9	4.9	5.3
Chloride (Cl)	0.0	0.0	0.0	0.0	2.0	6.0	0.0	3.0	0.0
Nitrate (NO_3)	0.42	0.0	0.0	0.0	0.0	7.8	0.0	0.0	0.0
Fluoride (F)	0.0	0.0	0.0	0.0	0.0	0.0	0.17	0.0	0.0
Total phosphate (PO_4)	0.6	0.66	0.79	0.46	0.44	1.7	1.0	0.35	0.92
Iron (Fe)	0.18	0.06	0.07	0.1	0.07	0.06	0.27	0.24	0.11
Silica (SiO_2)	6.9	7.3	8.6	15.5	14.8	6.1	13.4	15.5	11.8
Total dissolved solids	96.6	111.75	115.8	156.3	191.2	191.3	195.9	201.4	193.8
Conductivity (μmho)	115.0	120.0	130.0	165.0	200.0	260.0	205.0	220.0	200.0
pH	7.5	7.7	7.6	7.6	7.5	7.3	7.4	6.2	6.9
Free carbon dioxide	5.5	3.0	4.0	4.5	10.0	11.0	10.0	150.0	30.0
Total alkalinity as $CaCO_3$	64.9	78.0	81.0	97.0	125.0	105.0	128.0	130.0	131.0
Total hardness as $CaCO_3$	68.0	76.0	78.0	106.0	130.0	149.0	136.0	135.0	134.0
Temperature (°F)	72.0	---	---	---	69.0	---	67.0	68.0	68.0

TABLE 2.4.12-5

WATER QUALITY ANALYSES - SPRINGS

(Values in parts per million unless otherwise designated)

Sample identification	Spring 1	Spring 2	Spring3	Spring 5	Spring 6	Spring 7	Spring 4
Date sampled	09/17/71	09/22/71	09/22/71	09/20/71	09/22/72	09/18/72	09/18/72
Laboratory No.	17802	17835	17837	17811	17836	17838	17810
Date of analysis	09/21/71	09/21/71	09/21/71	09/21/71	09/21/71	09/21/71	09/21/71
Constituents							
Sodium (Na)	1.6	1.6	2.0	3.0	10.8	2.1	2.3
Magnesium (Mg)	0.0	2.2	2.4	0.0	4.6	1.9	0.2
Potassium (K)	0.29	0.25	0.0	0.54	0.33	0.0	0.5
Calcium (Ca)	2.0	0.8	2.8	37.2	29.2	2.8	22.0
Bicarbonate (HCO_3)	9.8	7.3	18.3	125.7	102.5	11.0	63.4
Sulfate (SO_4)	2.5	3.0	1.0	1.0	0.5	5.0	1.0
Chloride (Cl)	2.0	2.0	1.0	1.0	5.0	5.0	0.0
Nitrate (NO_3)	0.0	0.0	0.0	0.21	0.56	0.0	1.7
Fluoride (F)	0.0	0.0	0.09	0.0	0.0	0.09	0.26
Total phosphate (PO_4)	1.07	0.59	0.55	1.07	0.19	0.29	0.89
Iron (Fe)	0.12	0.11	0.08	0.06	0.04	0.96	0.06
Silica (SiO_2)	4.4	4.2	4.4	8.0	8.7	5.4	7.3
Total dissolved solids	21.8	20.0	28.1	147.9	136.8	32.3	83.8
Conductivity (μmho)	17.0	30.0	34.0	170.0	150.0	28.0	110.0
pH	6.4	6.1	6.0	7.2	6.0	6.1	7.1
Free carbon dioxide	6.0	12.0	25.0	11.0	160.0	15.0	8.0
Total alkalinity as $CaCO_3$	8.0	6.0	15.0	103.0	84.0	9.0	52.0
Total hardness as $CaCO_3$	7.0	11.0	17.0	99.0	92.0	15.0	56.0
Temperature (°F)	68.0	65.0	66.0	66.0	66.0	66.0	66.0

TABLE 2.4.12-6

WATER QUALITY ANALYSES - SURFACE WATER

(Values in parts per million unless otherwise designated)

Sample location[a]	Location 1	Location 2	Location 3	Location 3	Location 4
Date sampled	09/20/71			10/14/71	10/14/71
Laboratory No.	17813			17985	17987
Date of analysis		05/07/71	06/03/71	10/14/71	10/14/71
Constituents					
Sodium (Na)	1.8	2.9	1.9	1.9	2.2
Magnesium (Mg)	2.2	9.1	2.9	8.8	5.4
Potassium (K)	0.48	1.2	0.17	0.5	0.3
Calcium (Ca)	9.2	9.6	8.4	8.0	16.8
Carbonate (CO_3)	0.0	0.0	0.0	0.0	0.0
Bicarbonate (HCO_3)	18.3	---	---	31.7	29.3
Sulfate (SO_4)	0.77	5.2	2.6	2.8	1.5
Chloride (Cl)	2.0	4.0	2.0	1.0	2.0
Nitrate (NO_3)	0.0	0.79	0.75	0.39	0.44
Fluoride (F)	0.0	---	---	0.0	0.0
Total phosphate (PO_4)	1.05	---	---	0.16	0.26
Iron (Fe)	0.66	0.36	0.12	0.24	0.09
Silica (SiO_2)	6.5	10.1	5.6	5.4	4.9
Total dissolved solids	38.4	72.9	50.6	53.0	96.7
Conductivity (μmho)	37.0	60.0	57.0	50.0	90.0
pH	6.6	7.4	7.2	6.9	9.4
Free carbon dioxide	8.0	3.0	3.5	6.0	0.0
Total alkalinity as $CaCO_3$	15.0	30.0	28.0	26.0	58.0
Total hardness as $CaCO_3$	32.0	68.0	35.0	56.0	64.0
Temperature (°F)	76.0	---	---	69.0	---

a. Location 1 - Tributary to Daniels Branch.
Location 2 - Daniels Branch at road culvert.
Location 3 - Beaverdam Creek at River Road bridge.
Location 4 - Mathes Pond.

TABLE 2.4.12-7A (SHEET 1 OF 2)

OBSERVATION WELLS

Observation Wells in Water Table Aquifer

Well No.	History: Installed (YR)	History: Current Status	Coordinates: N	Coordinates: E	Ground Surface El (ft)	Top of PVC El (ft)	Depth Top of Marl (ft)	Screen Interval (ft)
129	1971	Inactive 1993	8856	9576	215.9	215.3	77	92 - 97
142	1971	Inactive 1993	8283	8262	231.2	224.5	92	85 - 95
179	1971	Inactive 1993	9059	7779	274.8	275.9	130	111 - 131
800	1979	Inactive 1993	8850	11011	213.7	215.3	83	69 - 89
801	1979	Inactive 1993	7656	10733	212.8	215.8	82	62.5- 82.5
802A	1985	Active	7196	10194	216.9	218.9	87.5	77 - 87
803A	1979	Inactive 1993	7085	8898	218.3	220.3	82	57 - 77
804	1979	Inactive 1993	6597	8227	224.1	226.1	87	60 - 80
805A	1979	Active	6672	10403	232.7	236.95	124	95 - 115
806B	1980	Active	8821	9726	214.8	221.45	77[(c)]	55 - 65
807A	1980	Grouted 1988	9047	9835	213.6	218.0	77[(c)]	65 - 75
808	1985	Active	9625	9300	207.0	216.47	66.3	45.5 - 68
809	1985	Inactive 1993	8320	7860	222.8	224.23	89	69.3 - 90
LT-1B	1985	Active	8388	9304	(4)	221.75	83.3	65.2 - 84.7
LT-7A	1985	Active	8151	9317	(4)	222.24	87	65 - 87
LT-12	1985	Active	7775	9600	(4)	219.20	79	58 - 79
LT-13	1985	Active	8135	10110	219.0[(d)]	220.67	89	68 - 90
42D	1971	Grouted, 1974	8403	9571	209.7	212.7	72	60 - 70
42E	1971	Grouted, 1974	8408	9580	209.6	-	72	45 - 55
124	1971	Inactive, 1979 (buried)	6896	9527	260.2	259.9	128	160 - 170
138	1971	Grouted, 1985	8000	8500	225.2	225.1	87	5 - 82
140	1971	Grouted, 1985	7846	8702	222.4	223.5	89	81 - 96
141	1971	Grouted, 1985	7860	8293	230.4	223.6	97	90 - 100

TABLE 2.4.12-7A (SHEET 2 OF 2)

(Observation Wells in Water Table Aquifer)

Well No.	History: Installed (YR)	History: Current Status	Coordinates: N	Coordinates: E	Ground Surface El[a] (ft)	Top of PVC El[b] (ft)	Depth Top of Marl (ft)	Screen Interval (ft)
143	1971	Grouted, 1980	8283	8738	224.5	225.0	81	78.5 - 88.5
145G	1971	Inactive, 1974 (buried)	7792	7063	218.7	219.7	82	72 - 82
176	1971	Inactive, 1974 (buried)	7117	11423	196.4	196.9	77	65 - 75
177	1971	Grouted, 1980	8560	10865	213.0	213.0	79	60 - 80
178	1971	Grouted, 1978	9958	8994	240.4	240.5	89	71 - 91
243	1972	Grouted 1985	9154	8618	213.0	225.2	71	60 - 80
244	1972	Inactive, 1979 (buried)	8835	8859	212.6	213.7	72	51 - 71
245	1972	Grouted, 1978	8501	9917	207.6	209.0	71.5	52 - 92
247	1972	Inactive, 1972 (buried)	5750	5424	211.3	--	82	70 - 80
248	1972	Inactive, 1972 (buried)	7469	5111	166.8	--	70.3	60 - 70
249	1972	Inactive, 1979 (buried)	8826	10154	193.0	194.0	57.9	47 - 57
802	1979	Grouted, 1985	7201	10199	215.8	217.7	91	69 - 89
LT-1A	1979	Grouted, 1985	8388	9300	204.8	206.9	69	65.4 - 75.4
LT-7	1979	Grouted, 1985	8151	9323	197.2	200.4	63	58.2 - 68.2

a. Elevations determined at time of drilling.

b. Elevations are current (Nov 1985) or latest prior to well abandonment.

c. Depth based on log of well 129.

d. Observation wells in backfill; not completed to grade, November 1985.

TABLE 2.4.12-7B (SHEET 1 OF 2)

Observation Wells In Confined Aquifers

Well No.	History: Installed (YR)	History: Current Status	Coordinates: N	Coordinates: E	Ground Surface El[a] (ft)	Top of PVC El[b] (ft)	Depth Bottom Marl (ft)	Screen Interval (ft)
Unnamed sands of Lisbon Formation (Tertiary)								
27	1971	Inactive 1995	8622	13931	210.0	209.0	148	180 - 190
29	1971	Inactive 1995	9975	12392	193.0	193.4	126	200 - 210
34	1971	Inactive 1995	12180	10846	86.0	90.5	(3)	90 - 100
850A	1984	Inactive 1995	11723	10494	225.9	227.8	135	169 - 179
851A	1984	Inactive 1995	8868	7066	262.7	264.3	195	269 - 279
852	1984	Inactive 1995	5993	13380	200.7	202.1	153.5	199 - 209
853	1984	Inactive 1995	11020	9204	227.6	229.1	145	195 - 205
854	1984	Inactive 1995	9899	7917	236.8	238.3	153	197 - 207
855	1984	Inactive 1995	7159	13951	218.0	219.4	173	219 - 229
856	1984	Inactive 1995	4927	12558	186.7	188.1	155	176 - 186
24	1971	Grouted	7850	9092	216.0	216.4	145	210 - 220
26	1971	Grouted, 1984	5963	15197	203.0	203.8	158	190 - 200
31	1971	Grouted, 1984	8764	11237	211.0	216.8	151	200 - 210
32	1971	Grouted, 1984	9784	9572	214.0	217.4	139	200 - 210
33	1971	Grouted, 1984	11834	10864	238.0	238.6	157	210 - 220
42A	1971	Grouted, 1974	8380	9535	210.6	213.0	137	140 - 150

TABLE 2.4.12-7B (SHEET 2 OF 2)

Observation Wells In Confined Aquifers

Well No.	History: Installed (YR)	History: Current Status	Coordinates: N	Coordinates: E	Ground Surface El[a] (ft)	Top of PVC El[b] (ft)	Depth Bottom of Marl (ft)	Screen Interval (ft)
101A	1971	Grouted, 1974	7950	9515	210.6	211.7	138	190 - 200
121	1971	Grouted, 1985	10467	12195	88.8	--	(3)	78 - 88
135	1971	Grouted	8992	8742	200.5	201.3	124.8	160 - 170
144	1971	Grouted	10403	12124	103.2	103.2	38	38.5- 48.5
147	1971	Grouted, 1978	7965	8471	226.2	227.4	152	280 - 300
181	1971	Grouted, 1985	8744	6833	258.3	--	194.5	190 - 200
246	1972	Grouted, 1984	10532	6553	210.4	213.5	179.7	220 - 230
Tuscaloosa Formation (Cretaceous)								
TW-1	1972	Inactive 1995-1998	7738	9984	218.5	928	140	506 - 850
MU-2	1977	Inactive 1995	9500	9135	214.5	850	150	450 - 820

a. Elevations determined at time of drilling.

b. Elevations are current (Nov 1985) or latest prior to well abandonment.

c. Marl not present.

TABLE 2.4.12-7C

Observation Wells In Marl

Well No.	History: Installed (YR)	Current Status	Coordinates: N	Coordinates: E	Ground Surface El[a] (ft)	Top of PVC El[b] (ft)	Marl Interval (ft)	Screen Interval (ft)
900	1985	Inactive 1995	7538	10119.5	216.3	218.05	92.6 - 148[c]	113.8 - 140.7
901	1985	Inactive 1995	7538	10104.5	215.58	220.75	91.6 - 148[c]	122.0 - 128.0
902	1985	Inactive 1995	7543.5	10110.5	215.97	221.11	91.0 - 148[c]	101.5 - 108.0
903	1985	Inactive 1993	8480	8900	215.75	216.73	78.0 - 148[c]	127.0 - 133.0
904B	1985	Inactive 1993	8464	8885	215.75	216.31	78.8 - 148[c]	90.0 - 96.0
905	1985	Inactive 1993	8450	8900	215.75	216.71	77.3 - 148[c]	109.8 - 116.0
42B	1971	Grouted, 1974	8386	9544	210.4	--	72 - 137	120 - 130
42C	1971	Grouted, 1974	8398	9563	210.0	--	72 - 137	80 - 90

a. Elevations determined at time of drilling.

b. Elevations are current (Nov 1985) or latest prior to well abandonment.

c. Bottom depth is interpolated from drawing AX6DD378.

TABLE 2.4.12-7D

Water Supply Wells

Well No.	Installed (YR)	Current Status	Coordinates N	Coordinates E	Ground Surface El[(a)] (ft)	Drilled Depth (ft)	Aquifer	Remarks
MU-1	1977	Active	9425	10531	196.9	851	Cretaceous	Cooling water makeup
MU-2A	1983	Active	8820	8400	225	884	Cretaceous	Cooling water makeup
TW-1	1972	Active	7738	9984	218.5	860	Cretaceous	Alternate cooling water makeup
CW-1	1976	Abandoned	6913	8919	255[(b)]	251	Tertiary	Construction supply
CW-2	1980	Abandoned	7452	6525	221	378	Tertiary	Construction supply
CW-3	1974	Active	3079	6645	210	220	Tertiary	Nuclear operations garage
SB	1981	Active	1666	15562	115[(b)]	340	Tertiary	Simulator building supply
PW	1973	Active	6880	14159	210[(b)]	100	Water table	Plant Wilson supply
IW-4	1989	Active	2979	6645	215[(b)]	370	Tertiary	Irrigation supply
SW-5	1990	Active	10704	8120	232[(b)]	200	Tertiary	Security training supply

a. Unless otherwise indicated, elevations shown are determined at time of drilling.

b. Estimated from topography, drawing AX6DD343.

TABLE 2.4.12-8

SUMMARY OF AQUIFER CHARACTERISTICS CALCULATIONS

Method of Analysis	Observation Point (s)	Calculated Transmissivity (gal/d/ft)	Storage Coefficient
Test Well Data (TW-1)			
Straight-line, distance drawdown	Pumping well observation points	158,000	(a)
Type-curve, time-drawdown	1	196,000	6.6×10^{-4}
Type-curve, time-drawdown	2	160,000	3.3×10^{-4}
Type-curve, time-drawdown	3	163,700	3.5×10^{-4}
Type-curve, time-drawdown	4	153,000	2.1×10^{-5}
Type-curve, time-drawdown	5	229,200	3.9×10^{-4}
Makeup Well Data MU-1 and MU-2)			
Type-curve, time-drawdown, MU-1	None	110,400	(a)
Type-curve, time recovery, MU-1	None	116,600	(a)
Type-curve, time drawdown, MU-2	MU-1	130,900	1.07×10^{-4}
Type-curve, time recovery, MU-2	MU-2	128,700	(a)

a. Storage coefficient calculated only from observation well data.

TABLE 2.4.12-9

OBSERVATION WELLS MONITORING PROGRAM 1995-PRESENT

Reading Frequency	Water Table Aquifer	Marl Aquiclude	Tertiary Aquifer	Cretaceous Aquifer
Quarterly	LT-1B LT-7A LT-12 LT-13	None	None	None
Semiannually	802A 805A 806B 808	None	None	None

Note: Details of well construction are in tables 2.4.12-7A, 7B, 7C, and 7D.

TABLE 2.4.12-10 (SHEET 1 OF 4)

PERMEABILITY TESTS
BLUE BLUFF MARL

Hole Number	Interval Tested (ft)	Permeability[(a)] (ft/yr)	Marl Interval (ft)	Remarks
Constant Head (Packer) Tests				
157	100.0-110.0	3.0	92.0-153.1	Tests rejected because of packer leakage. See results of tests in Hole 508.
	100.0-120.0	3.9		
	110.0-120.0	18.6-54.2		
	128.0-138.0	0		
	120.0-140.0	0		
170	104.5-124.5	0	92.0-152.0	
	110.0-130.0	0		
	120.0-140.0	0		
	130.5-150.5	0		
180	77.5-99.5	0	72.0-142.0	
	85.0-105.0	0		
	95.0-115.0	3.9		
	105.0-125.0	11.7		
245	80.0-100.0	0	71.5-135.5	
	82.0-102.0	0		
	86.0-106.0	0		
	103.0-123.0	0		
	110.0-130.0	0		
249	67.5-87.5	0	57.9-122.0	Tests questionable because of possible packer leakage.
	80.0-100.0	48.4		
	92.5-112.5	29		
501	76.5-96.5	0	74.0-150.0	
	88.0-113.0	0		
	114.0-130.0	0		
	135.0-150.0	0		
502	86.0-114.5	0	82.5-146.0	
	114.5-139.5	0		
	137.5-150.0	0		
503	63.5-82.0	0	58.0-121.5	
	66.0-102.0	0		
	81.0-102.0	0		
	100.0-122.0	0		

TABLE 2.4.12-10 (SHEET 2 OF 4)

Hole Number	Interval Tested (ft)	Permeability[(a)] (ft/yr)	Marl Interval (ft)	Remarks
Constant Head (Packer) Tests				
504	87.0-99.0	0	84.0-134.0	
	97.0-109.0	0		
	107.0-119.0	0		
	118.0-130.0	0		
	122.0-135.0	0		
505	148.0-160.0	0	147.0-187.0	
	157.0-167.0	0		
	166.0-178.0	0		
	175.0-187.0	0		
506	93.0-105.0	0	92.0-162.0	
	103.0-115.0	0		
	113.0-125.0	0		
	123.0-135.0	0		
	133.0-145.0	0		
	143.0-155.0	0		
	153.0-165.0	0		
507	112.0-124.0	0	111.0-180.5	
	125.0-137.0	0		
	135.0-147.0	0		
	140.0-152.0	0		
	150.0-162.0	0		
	160.0-172.0	0		
	165.0-177.0	0		
508	97.0-109.0	0	95.0-150.8	Drilled adjacent to Hole 157 to determine validity of original tests
	104.0-116.0	0		
	114.0-126.0	0		
	125.0-137.0	0		
	135.0-147.0	0		
	142.0-154.0	0		
510	95.0-107.0	0	93.0-154.0	
	105.0-117.0	0		
	115.0-127.0	0		
	125.0-137.0	0		
	135.0-147.0	0		
	141.0-153.0	0		

TABLE 2.4.12-10 (SHEET 3 OF 4)

Hole Number	Interval Tested (ft)	Permeability[(a)] (ft/yr)	Marl Interval (ft)	Remarks
Constant Head (Packer) Tests				
513	90.0-102.0	0	86.0-147.5	
	100.0-112.0	0		
	110.0-122.0	0		
	120.0-132.0	0		
	130.0-142.0	0		
518	124.0-129.0	0	77.5-139.7	
900	104.6-112.6	0	92.6-148	
	112.6-122.6	0		
	122.6-132.6	0		
	132.6-142.6	0		
	122.6-142.6	0		
901	118-128	0	91.6-148	
902	100-108	0	91-148	
903	85-96	0	78-148	
	96-106	0		
	106-116	0		
	116-126	0		
	126-133	0		
904B	85-96.7	0	78.8-148	
905	88.5-102.5	0	77.4-148	
	102.5-116	0		
P-1	11.0-31.0	0	4.0-33.0	
P-2	5.0-30.0	51	0.5-29.5	Analysis suggests leakage around packer.
P-3	7.1-17.0	0	7.0-39.5	Weathered marl
	17.0-37.0	0		
P-5	12.0-27.0	0	11.0-25.8	

TABLE 2.4.12-10 (SHEET 4 OF 4)

Hole Number	Interval Tested (ft)	Permeability[a] (ft/yr)	Marl Interval (ft)	Remarks
Well Parameter Tests				
P-1A	0.0-6.0	16	0.0-6.0	Weathered marl
P-3A	0.0-6.5	33	0.0-6.5	Weathered marl
Laboratory Permeability Tests[b]				
901	119.0	5.2×10^{-3}	91.6-148	Limestone
902	104.2	2.0×10^{0}	91 -148	Marl
903	108.2	2.0×10^{-1}	76 -148	Marl
903	112.7	5.0×10^{-1}		Marl and limestone Nodules
903	128.4	2.1×10^{0}		Marl
904B	92.3	2.5×10^{0}	78.8-148	Marl
905	91.6	1.5×10^{0}	77.3-148	Marl and limestone Nodules
905	96.7	8.8×10^{0}		Marl
905	107.5	1.4×10^{-1}		Marl
905	114.0	8.0×10^{-2}		Marl

a. Zero indicates no measurable water take.
b. Tests were performed by Harding Lawson Associates.

TABLE 2.4.12-11
POROSITY
(Blue Bluff Marl)

Boring No.	Sample Depth (ft)	Porosity Percent
102	125.8	48
	140.3	48
111	80.8	55
114	105.3	56
	80.8	54
	100.8	51
138A	97.0	56
	126.0	24
	134.0	36
	148.6	41
202	93.8-95.8	39
	134-136	41
203	82-84	62
	114.0	60
204	94.5	48
	132-134	42
216	84.5	47
	132.0	47

Note: Data from Law Engineering Testing Company, November 12, 1971.

TABLE 2.4.12-12 (SHEET 1 OF 2)

PERMEABILITY TEST RESULTS
(Barnwell Sands, Silts, and Clays; River Alluvial Sands, Silts, and Clays;
Unnamed Sands of Lisbon Formation)

Hole Number	Interval Tested (ft)	Permeability (ft/yr)	Material Tested and/or Remarks
Well Permeameter Tests			
183	50.0-60.0	200	Sand (Barnwell)
184	53.0-63.0	267	Sand (SW), clayey sand (SC) and clay (CL) (Barnwell)
P-4A	0 - 7.5	130	Silts/clay (river alluvium)
P-6A	0 - 4	260	Silts/clay (river alluvium)
Packer Tests			
P-6B	10-20	36,000[(a)]	Sands (river alluvium)
P-6C	20-30	21,000[(a)]	Sands (river alluvium)
P-6D	30-40	27,000[(a)]	Sands (river alluvium)
P-1	33-48	240	Unnamed sands Lisbon Formation
P-2	30-50	190	Unnamed sands Lisbon Formation
P-3	40-54.6	250	Unnamed sands Lisbon Formation
P-4	21-36	60	Unnamed sands Lisbon Formation
P-5	29-54	340	Unnamed sands Lisbon Formation
Laboratory Tests			
107A	13.8-14.4	302	Sand (SP); undisturbed sample
	34.0-36.0	9.8	Sand (SW); undisturbed sample
	49.0-51.0	19,973	Sand (SW); dry density = 83.1 per ft^3
		6,833	dry density = 84.0 per ft^3
		1,682	dry density = 91.0 per ft^3
	62.5-63.0	27.4	Sand (SW): undisturbed sample
S No. 10	Backfill (Grab sample)	6,070	Percent compaction = 92.9
		4,580	Percent compaction = 93.9
		4,400	Percent compaction = 95.7
		2,260	Percent compaction = 99.8

TABLE 2.4.12-12 (SHEET 2 OF 2)

Hole Number	Interval Tested (ft)	Permeability (ft/yr)	Material Tested and/or Remarks
Laboratory Tests			
S No. 11	Backfill (Grab sample)	4,110	Percent compaction = 91.2
		1,820	Percent compaction = 94.0
		1,430	Percent compaction = 97.0
		430	Percent compaction = 98.8

TABLE 2.4.12-13 (SHEET 1 OF 2)

PERMEABILITY TESTS
(Utley Limestone)

Observation Well No.	Tested Interval (ft)	Transmissivity (gpd/ft)	Permeability (ft/yr)	Remarks
1. Pumping Tests				
a. Well No. 1 pumped out at an average of 30 gal/min for 97 h				
1A	56-78	6,350	14,100	Theis curve match, maximum drawdown 1.92 ft
1B	68-78	25,700	125,400	Theis curve match, maximum drawdown 1.05 ft
1C	56-80	9,830	20,000	Theis curve match, maximum drawdown 2.49 ft
1D	56-80	21,700	44,100	Theis curve match, maximum drawdown 1.31 ft
1A,1B,1C,1D	59-70 (avg)	8,090	19,700	Distance-drawdown
b. Well 2B, pumped in at an average of 74 gal/min for 14 min				
2A	62-85	1,530	3,250	Semilog plot of recovery, maximum drawdown-6.22 ft
2. Falling Head (Variable Head) Tests				
Well No. 1	65-80	NA	5,800	Starting head = 36.7 ft
1A	63-78	NA	600	Starting head = 36.5 ft
Well No. 2	69-85	NA	980	Starting head = 44.1 and 10.6 ft (2 tests)
2A	70-85	NA	96	Starting head = 3.2 ft
2B	69-84	NA	360	Starting head = 0.6 and 0.9 (2 tests)
2C	65-85	NA	140	Starting head = 3.0 ft
2D	70-85	NA	2,100	Starting head = 1.8 ft

Table 2.4.12-13 (SHEET 2 OF 2)

Observation Well No.	Tested Interval (ft)	Transmissivity (gpd/ft)	Permeability (ft/yr)	Remarks
3. Constant Head Tests				
1A	56-78	NA	160	Total head = 61 ft
2A	56-85	NA	3,200	Total head = 64 ft
2B	56-84	NA	1,790	Total head = 75 ft
2D	56-85	NA	1,190	Total head = 77 ft

TABLE 2.4.12-14

POROSITY
Barnwell Sands, Silts, and Clays

Boring No. or Sample No.	Depth Interval (ft) or Sample Source	Porosity Percent	Remarks
102A	15.0-16.5	34	Silty sand-undisturbed sample
	35.0-36.5	47	Sand-undisturbed sample
	58.0-60.0	61	Clay-undisturbed sample
107A	13.2-15.2	45	Sand-undisturbed sample
	34.0-36.0	47	Sand-undisturbed sample
	62.0-64.0	52	Sand-undisturbed sample
138A	9-11	39	Silty sand-undisturbed sample
	14-16	40	Silty sand-undisturbed sample
	29-31	43	Silty, clayey sand-undisturbed sample
	34-36	38	Clayey sand-undisturbed sample
	49-51	49	Clayey sand-undisturbed sample
	59-60.5	44	Sand-undisturbed sample
204	18-19.3	40	Silty sand-undisturbed sample
226A	61-63	43	Sand-undisturbed sample
235	8-10	37	Sand-undisturbed sample
Sample No. 10	Backfill	39.4	@ 92.9 percent compaction
		38.8	@ 93.9 percent compaction
		37.6	@ 95.7 percent compaction
		35.0	@ 99.8 percent compaction
Sample No. 11	Backfill	36.9	@ 91.2 percent compaction
	Burrow area	34.9	@ 94 percent compaction
		32.9	@ 97 percent compaction
		31.6	@ 98.8 percent compaction

Note: Data from Law Engineering Testing Company, November 12, 1971 and August 31, 1984.

TABLE 2.4.12-15

PERMEABILITY OF BACKFILL

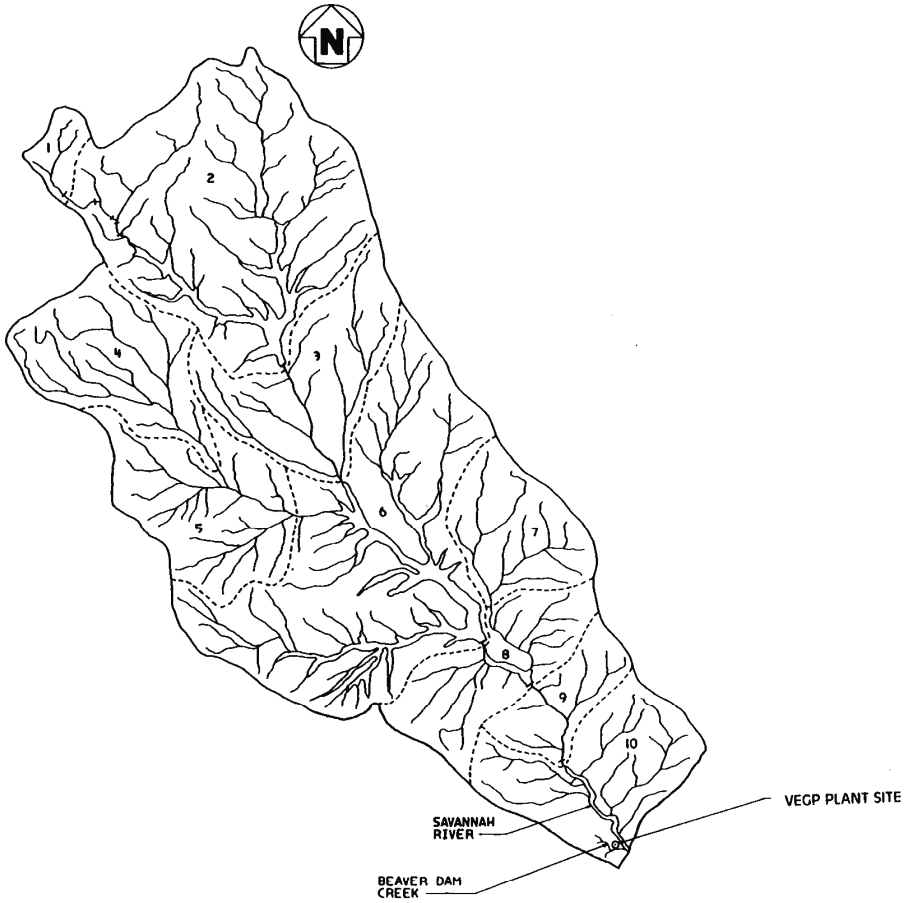
Well Number	Test Cycle	Transmissivity ft^2/d Test Cycle	Average	Open Interval ft	Permeability ft/yr
LT-1B	Insertion	59	65	19.5	1220
	Extraction	71			
LT-7A	Insertion	48	45	22	750
	Extraction	42			
LT-12	Insertion	29	28	21	480
	Extraction	26			
LT-13	Insertion	67	71	22	1180
	Extraction	75			

TABLE 2.4.13-1

Equilibrium Distribution Coefficient
Power Block Excavation Backfill
(ml/g)

Sample	$Cesium^{137}$	$Strontium^{90}$
1	385 ±138	40.8 ±6.0
2	1065 ±160	94.7 ±6.8
3	520 ±31	76.0 ±7.0
4	2134 ±589	56.2 ±20.3

NOTE: Values for each sample are the average of three replicates with the variability indicated.
Testing conducted by Battelle Pacific Northwest Laboratories.



LEGEND:
—— BASIN BOUNDARY
----- SUBBASIN BOUNDARY

SAVANNAH RIVER BASIN

SCALE:
10 5 0 10 20 30 40 MILES

REFERENCE:
U.S. DEPARTMENT OF COMMERCE
N.O.A.A. - NATIONAL WEATHER SERVICE,
ATLANTA RIVER FORCAST CENTER
AUGUST 1974

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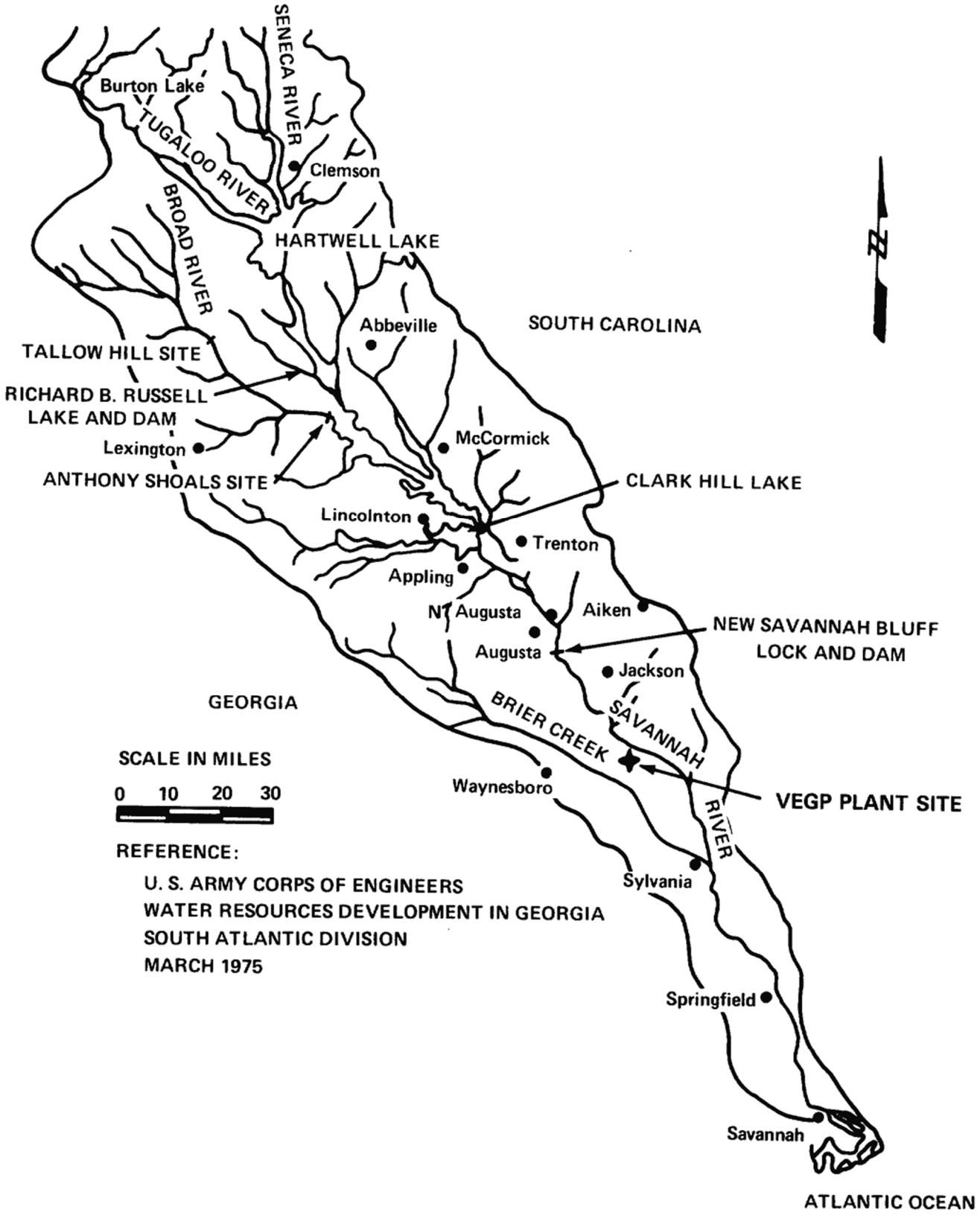
HYDROLOGIC MAP
SAVANNAH RIVER BASIN ABOVE SITE

FIGURE 2.4.1–1



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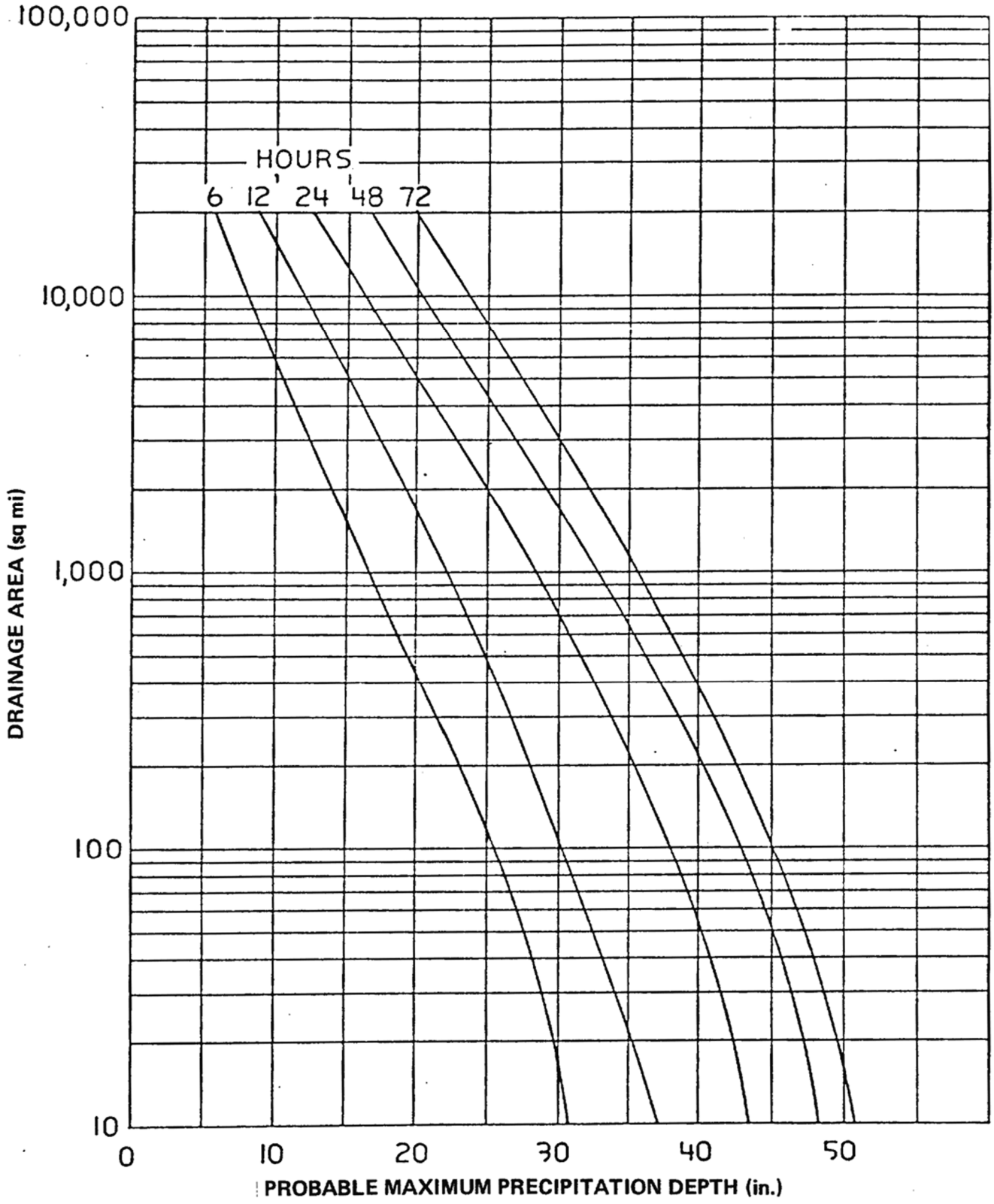
SAVANNAH RIVER
BASIN

FIGURE 2.4.1–2



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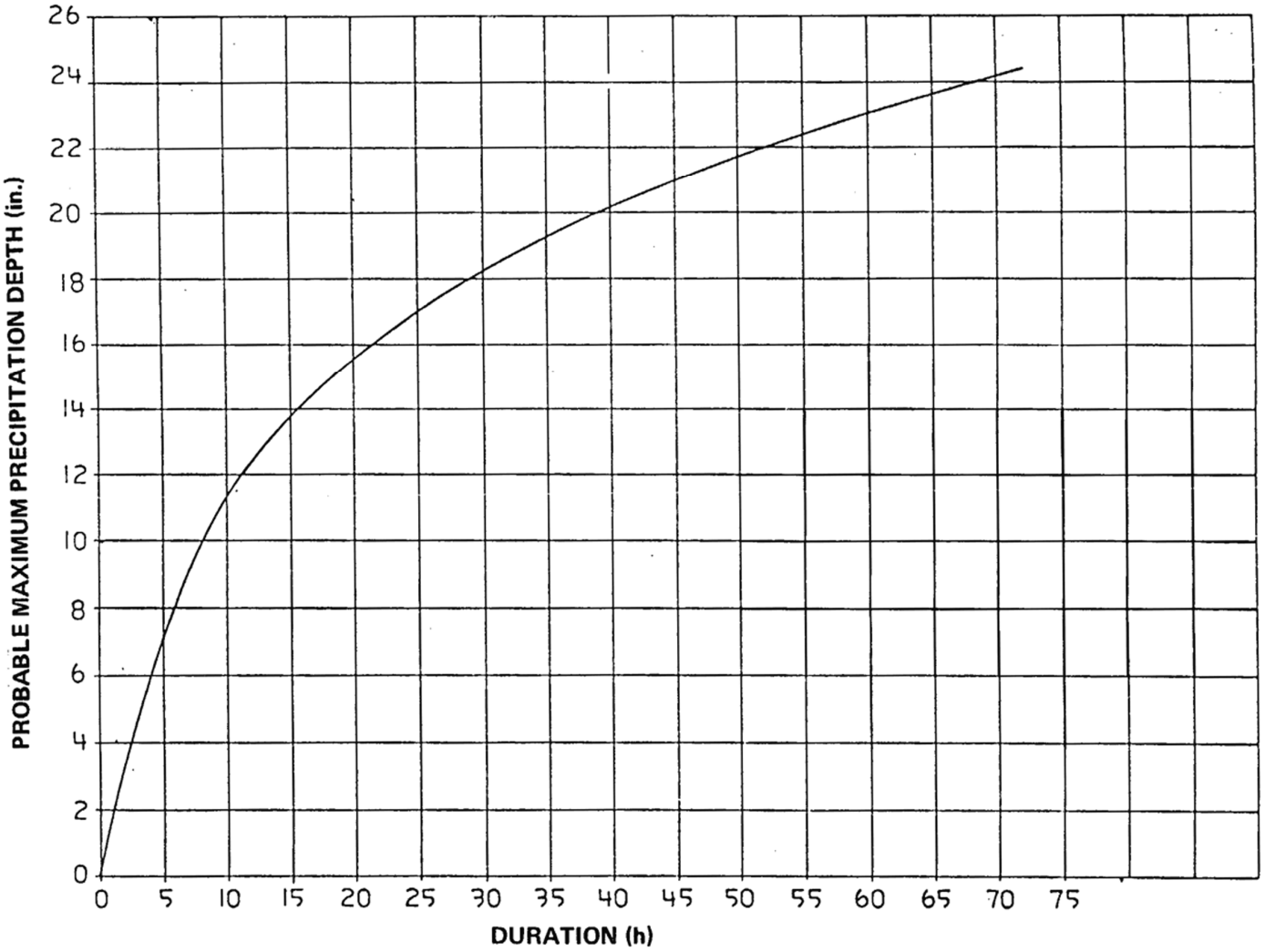
DEPTH–AREA–DURATION ENVELOPES

FIGURE 2.4.3–1



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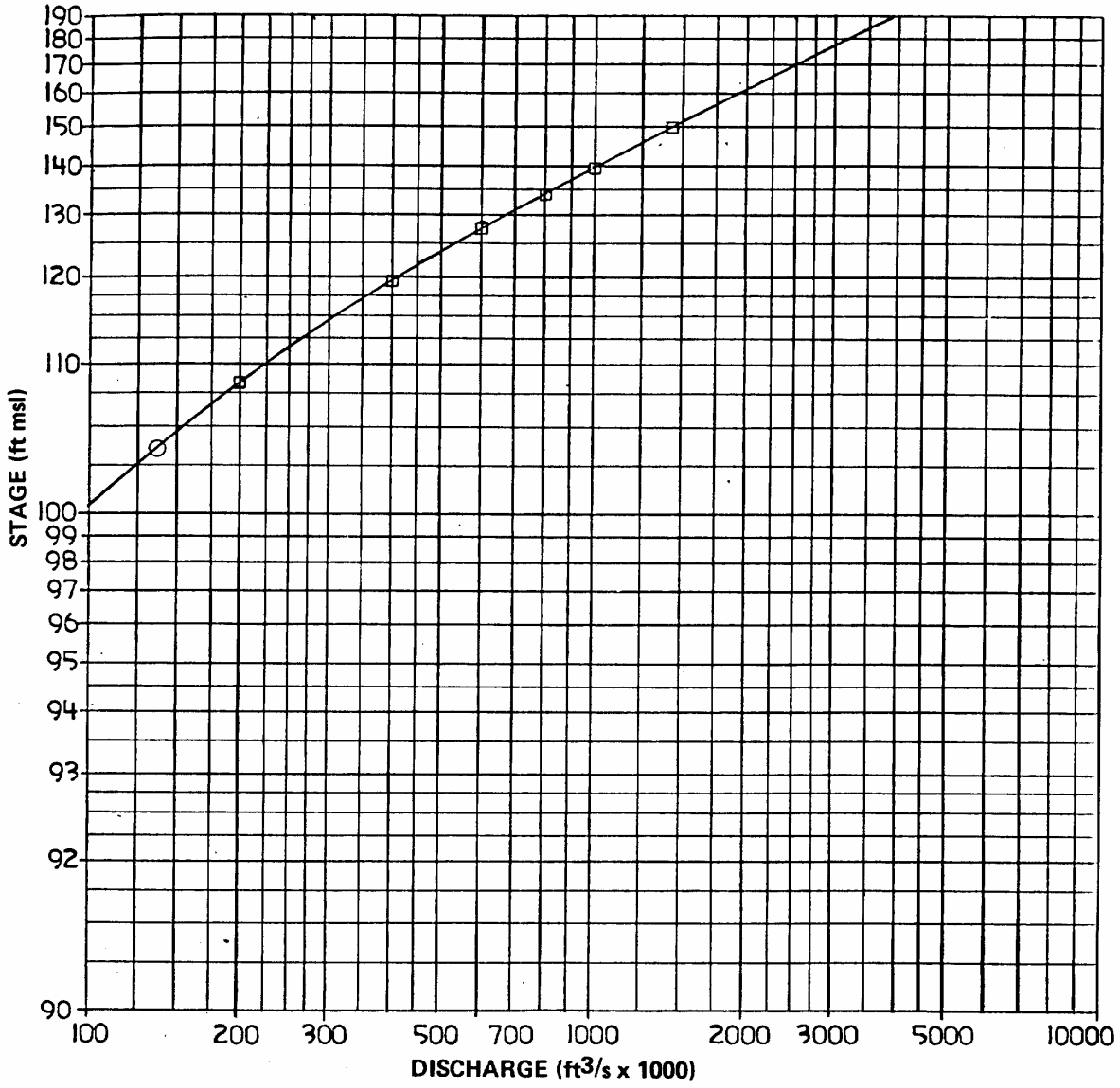
RMP DEPTH–DURATIN CURVE FOR
DRAINAGE (8015 MI^2) ABOVE THE PLANT

FIGURE 2.4.3–2



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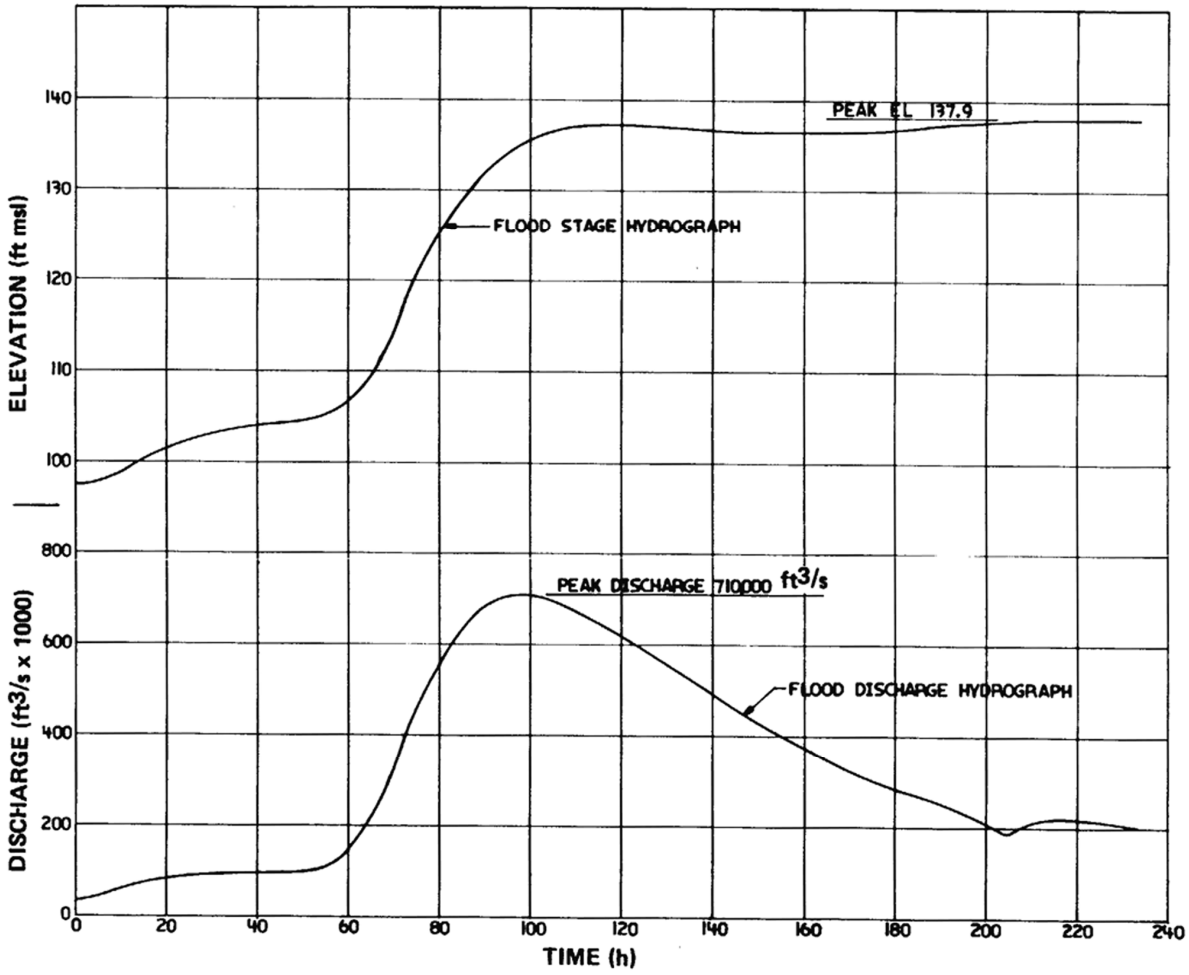
UNIT 1 AND UNIT 2

SAVANNAH RIVER STEADY FLOW CONDITION–DISCHARGE RELATION AT RIVER MILE 151.1

FIGURE 2.4.3–3



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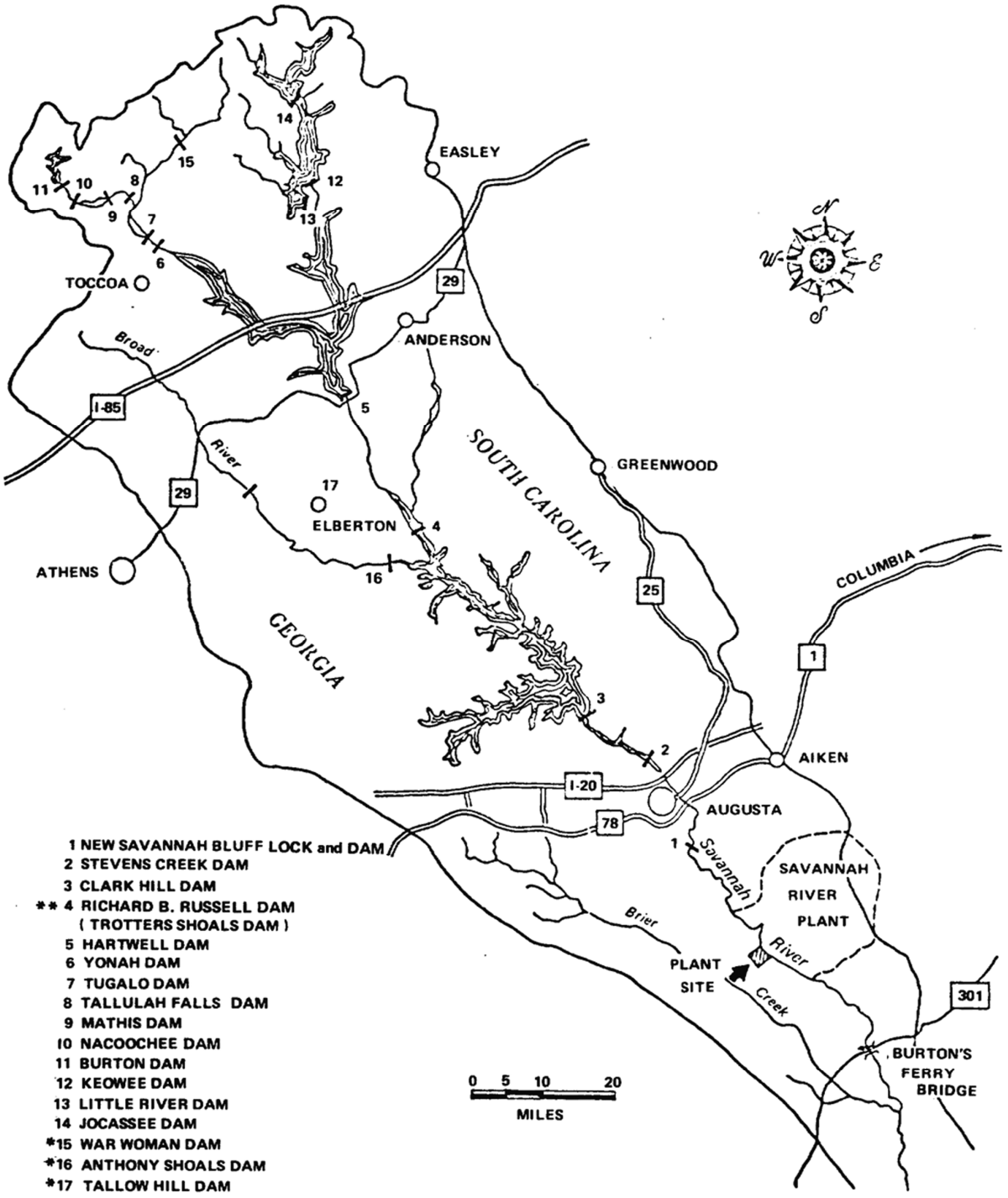
PROBABLE MAXIMUM FLOOD HYDROGRAPH AT RIVER MILE 151.1 BY ROUTING CORPS OF ENGINEERS PMF OUTFLOW HYDROGRAPH (VALLEY STORAGE TAKEN INTO ACCOUNT) AT CLARK HILL DAM THROUGH SAVANNAH RIVER

FIGURE 2.4.3–4



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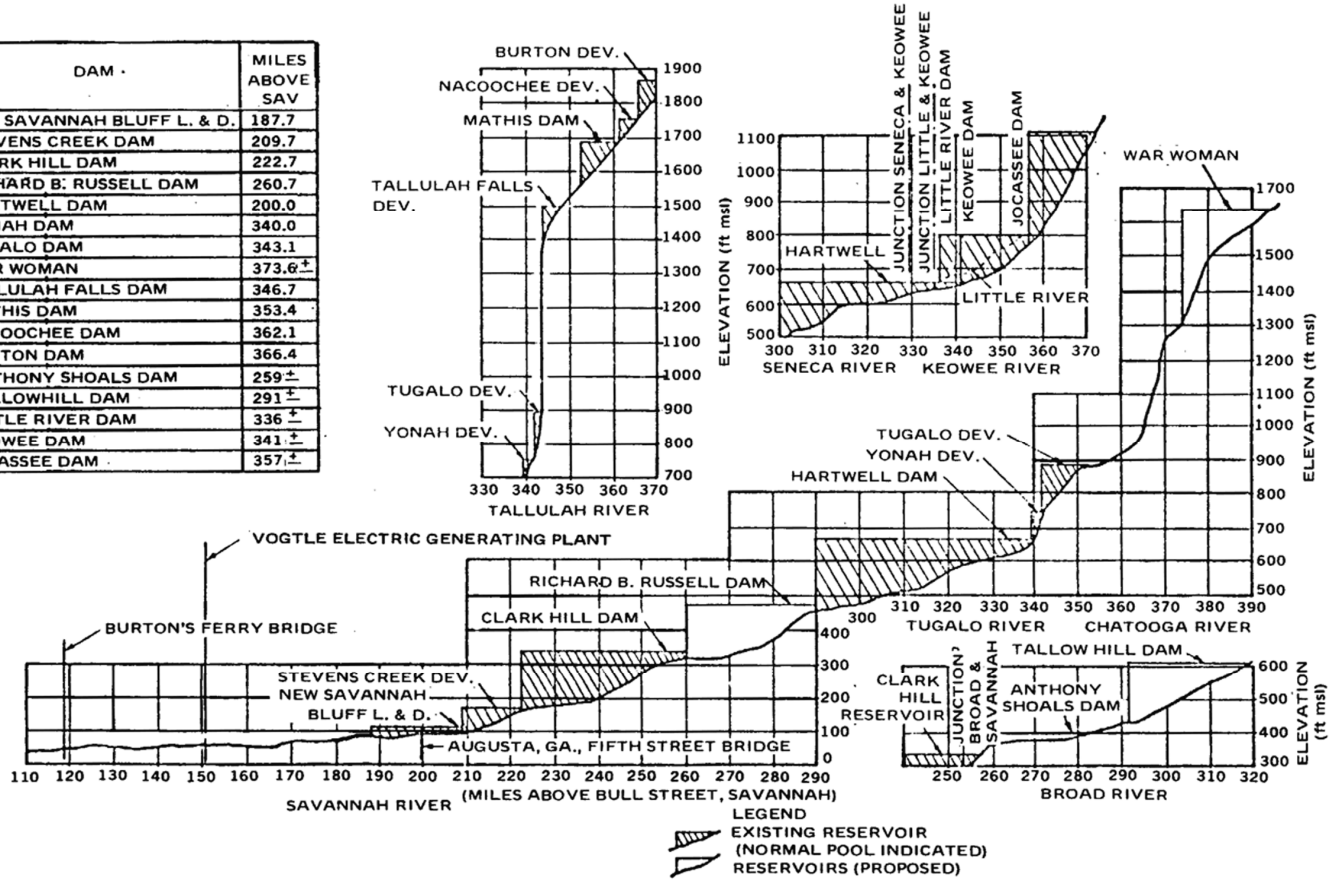
UNIT 1 AND UNIT 2

LOCATION OF DAMS
ON THE SAVANNAH RIVER

FIGURE 2.4.4–1

DAM	MILES ABOVE SAV
NEW SAVANNAH BLUFF L. & D.	187.7
STEVENS CREEK DAM	209.7
CLARK HILL DAM	222.7
RICHARD B. RUSSELL DAM	260.7
HARTWELL DAM	200.0
YONAH DAM	340.0
TUGALO DAM	343.1
WAR WOMAN	373.6±
TALLULAH FALLS DAM	346.7
MATHIS DAM	353.4
NACOOCHEE DAM	362.1
BURTON DAM	366.4
ANTHONY SHOALS DAM	259±
TALLOWHILL DAM	291±
LITTLE RIVER DAM	336±
KEOWEE DAM	341±
JOCASSEE DAM	357±



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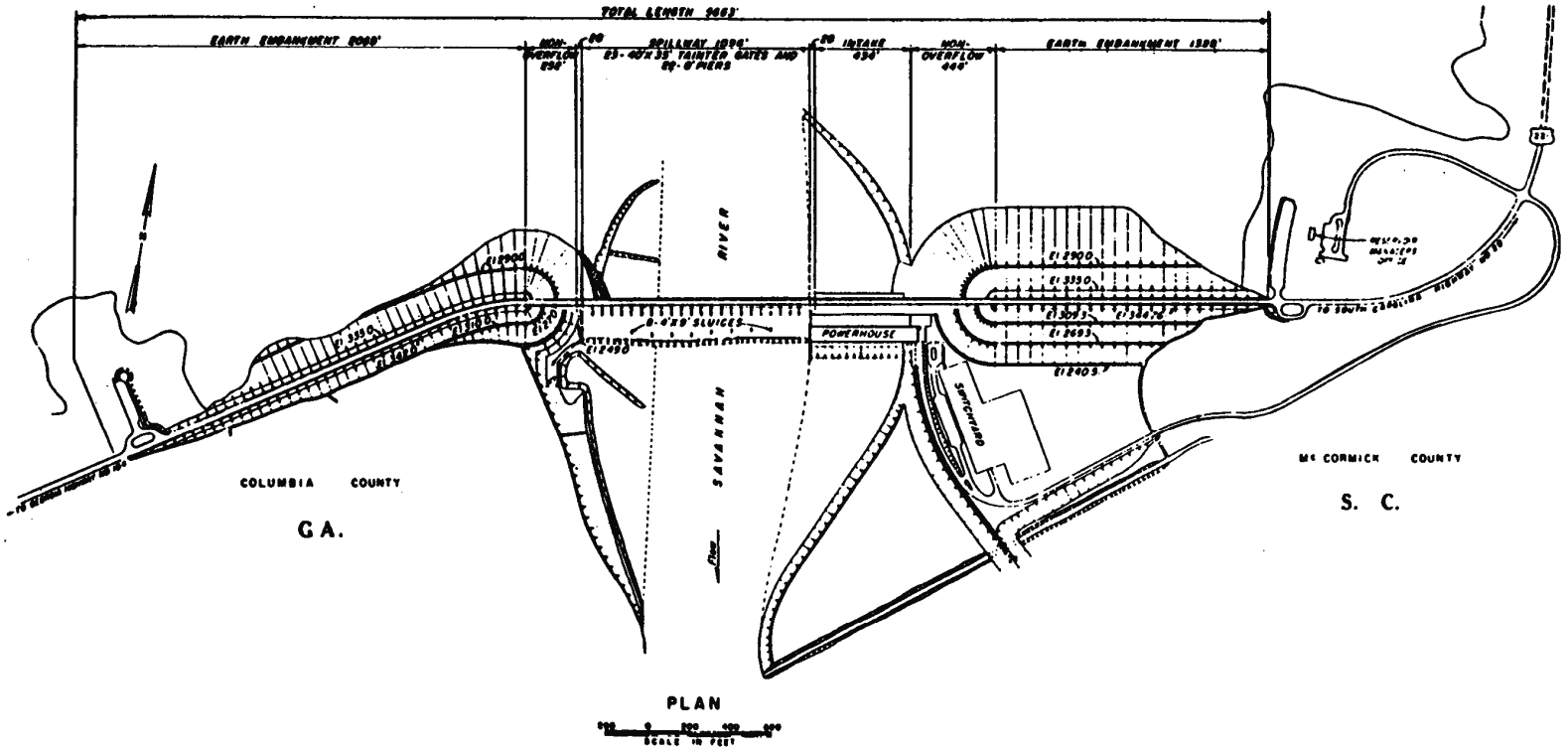
SAVANNAH RIVER
STREAM PROFILE

FIGURE 2.4.4–2



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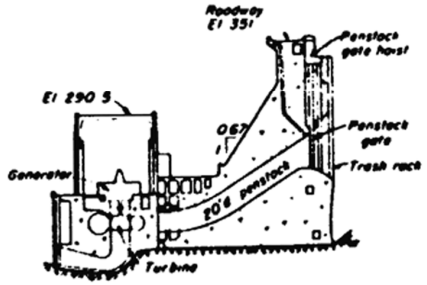
UNIT 1 AND UNIT 2

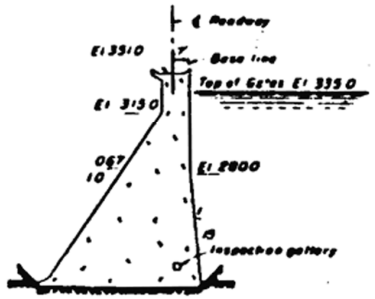
CLARK HILL DAM
PLAN AND SECTION

FIGURE 2.4.4–3 (SHEET 1 OF 2)



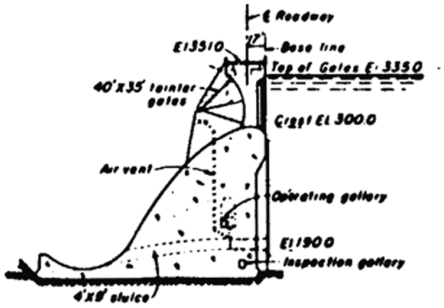
TYPICAL INTAKE SECTION

SCALE IN FEET



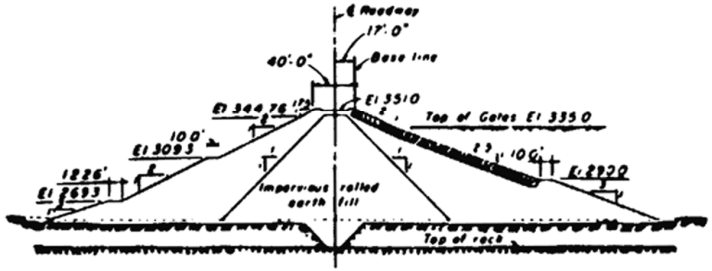
NON-OVERFLOW SECTION

SCALE IN FEET



SPILLWAY SECTION

SCALE IN FEET



EMBANKMENT SECTION

SCALE IN FEET

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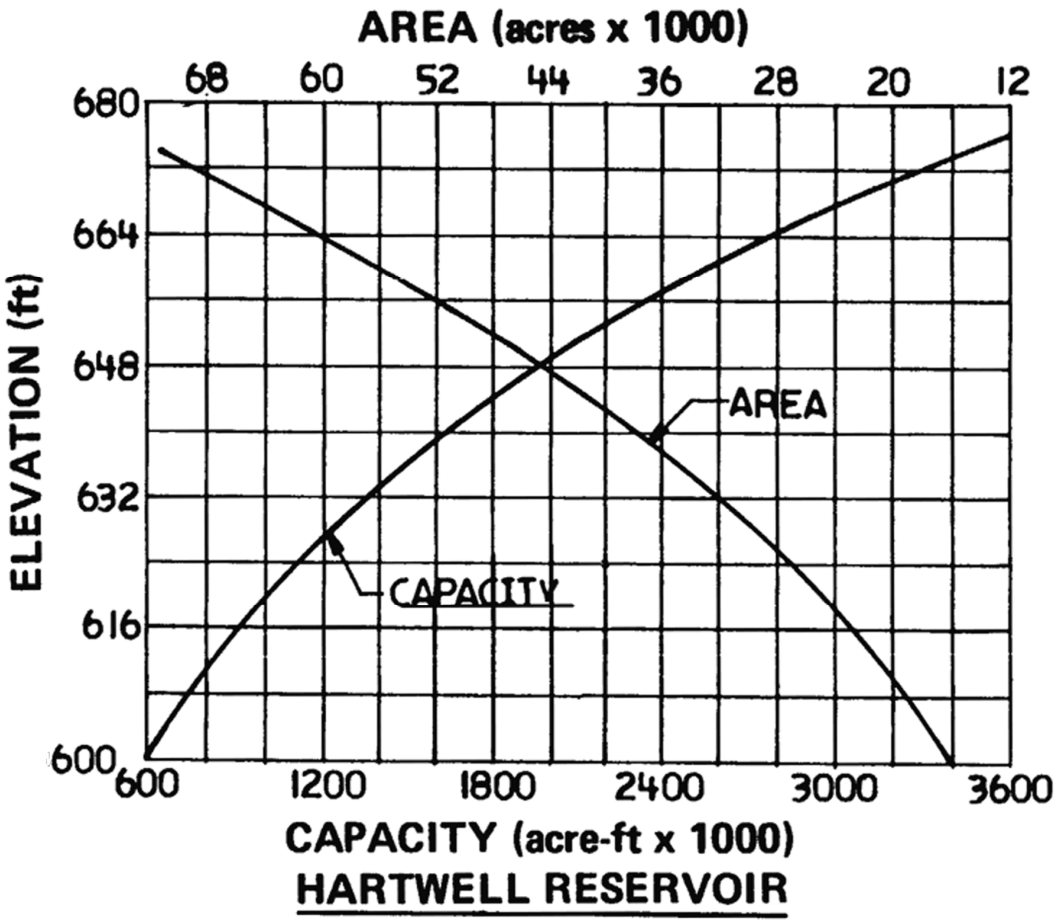
CLARK HILL DAM
PLAN AND SECTION

FIGURE 2.4.4–3 (SHEET 2 OF 2)



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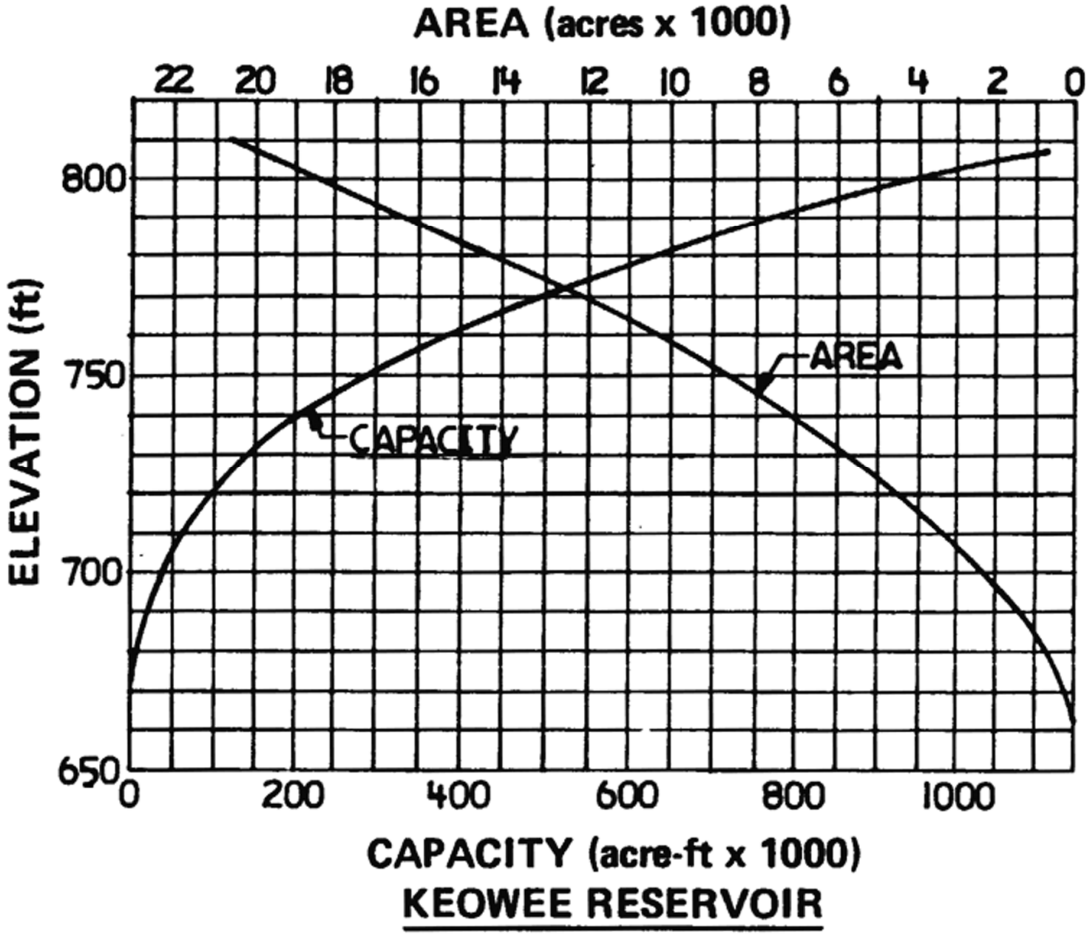
UNIT 1 AND UNIT 2

AREA AND CAPACITY CURVES FOR
MAJOR UPSTREAM DAMS

FIGURE 2.4.4–4 (SHEET 1 OF 5)



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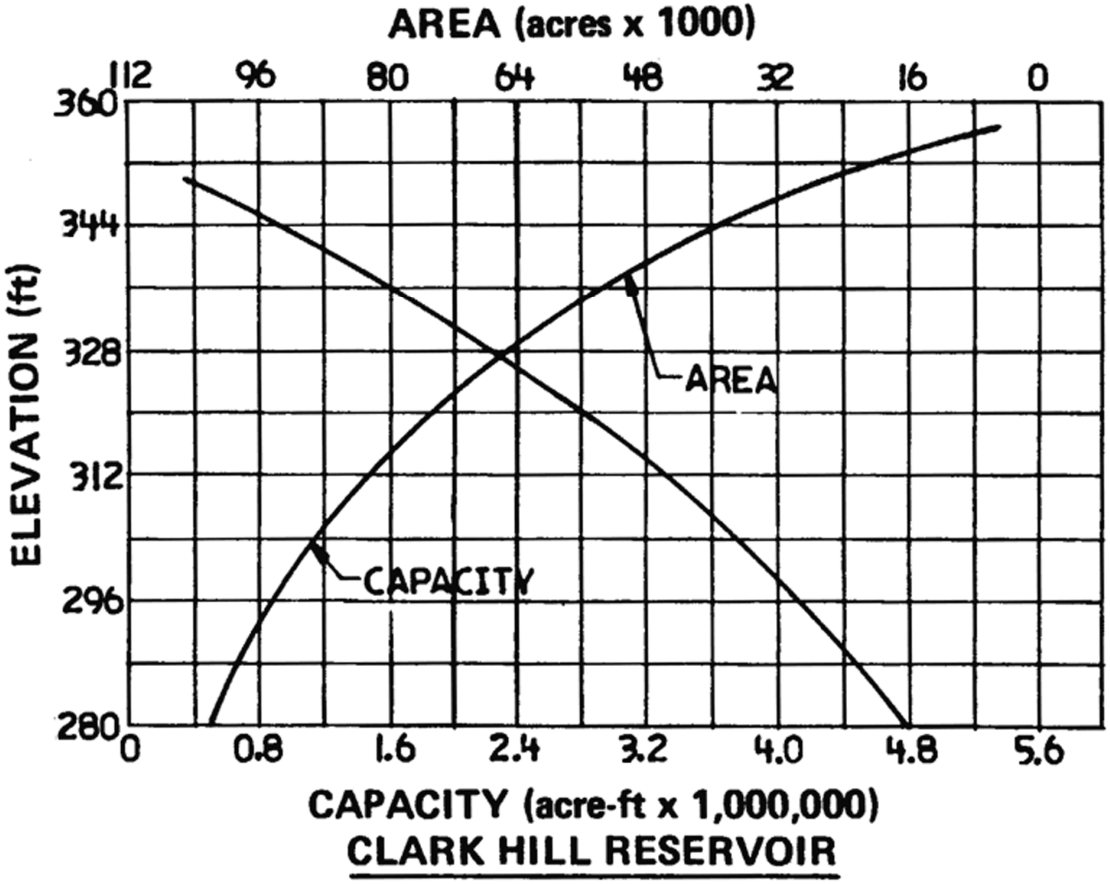
AREA AND CAPACITY CURVES FOR
MAJOR UPSTREAM DAMS

FIGURE 2.4.4–4 (SHEET 2 OF 5)



CLARK HILL RESERVOIR

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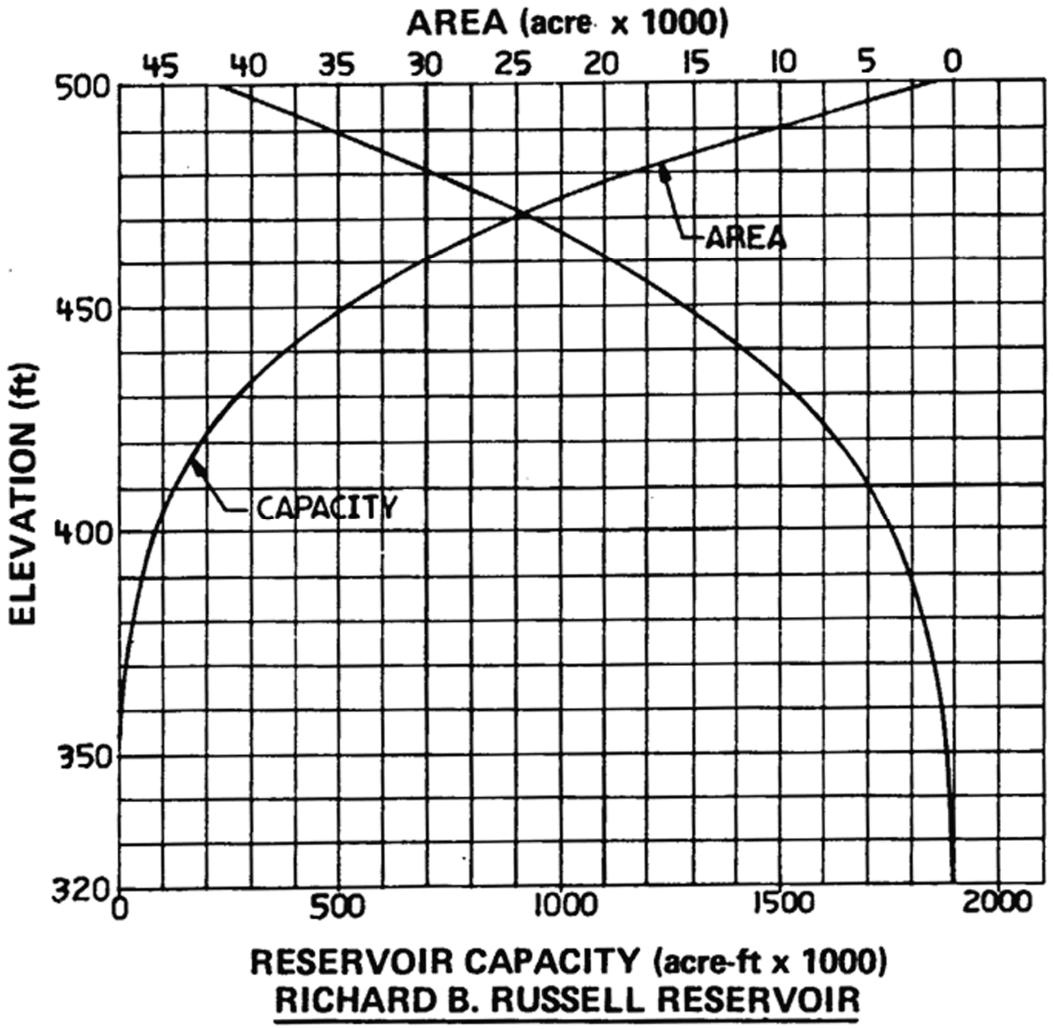
UNIT 1 AND UNIT 2

AREA AND CAPACITY CURVES FOR
MAJOR UPSTREAM DAMS

FIGURE 2.4.4–4 (SHEET 3 OF 5)



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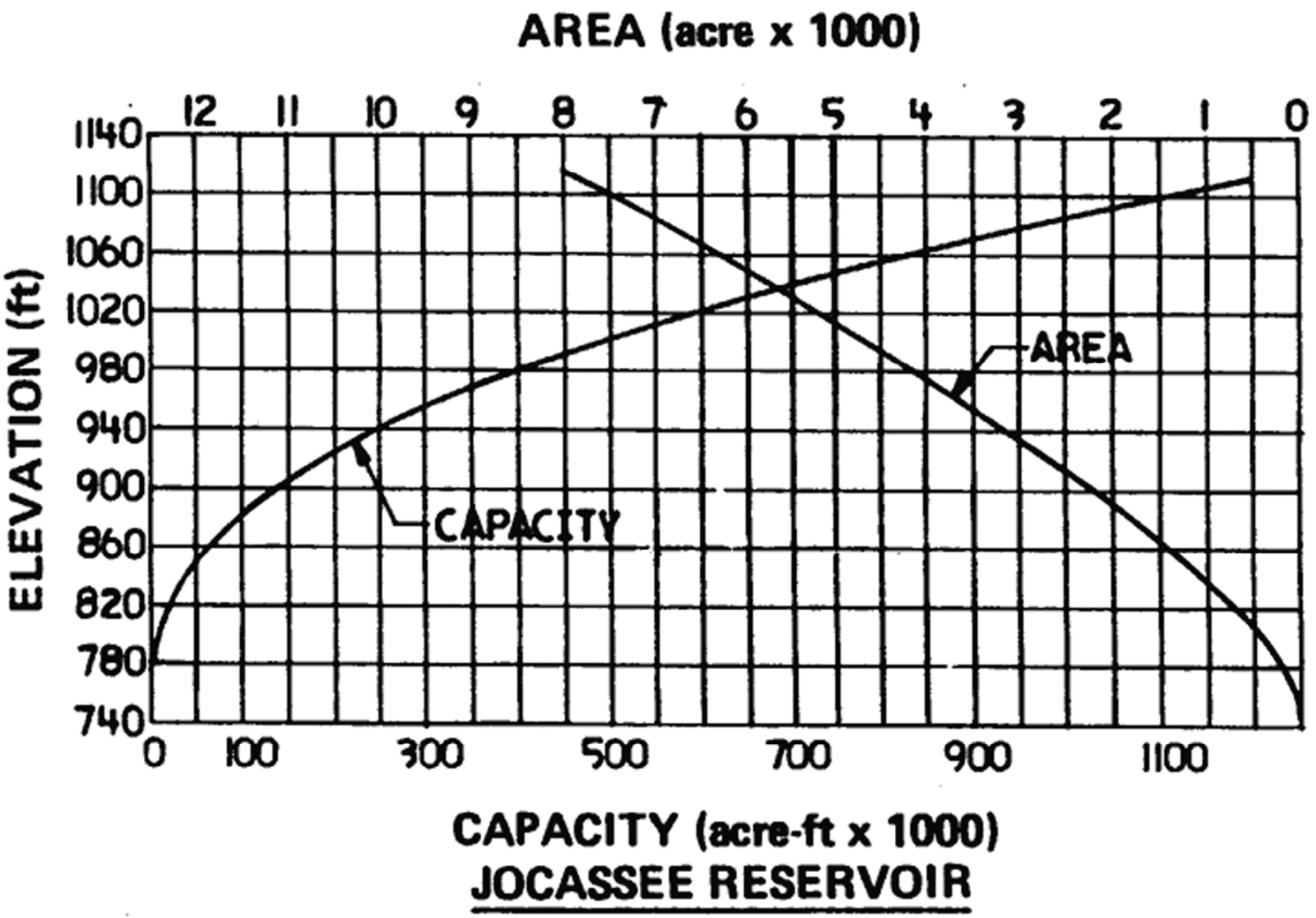
UNIT 1 AND UNIT 2

AREA AND CAPACITY CURVES FOR
MAJOR UPSTREAM DAMS

FIGURE 2.4.4–4 (SHEET 4 OF 5)



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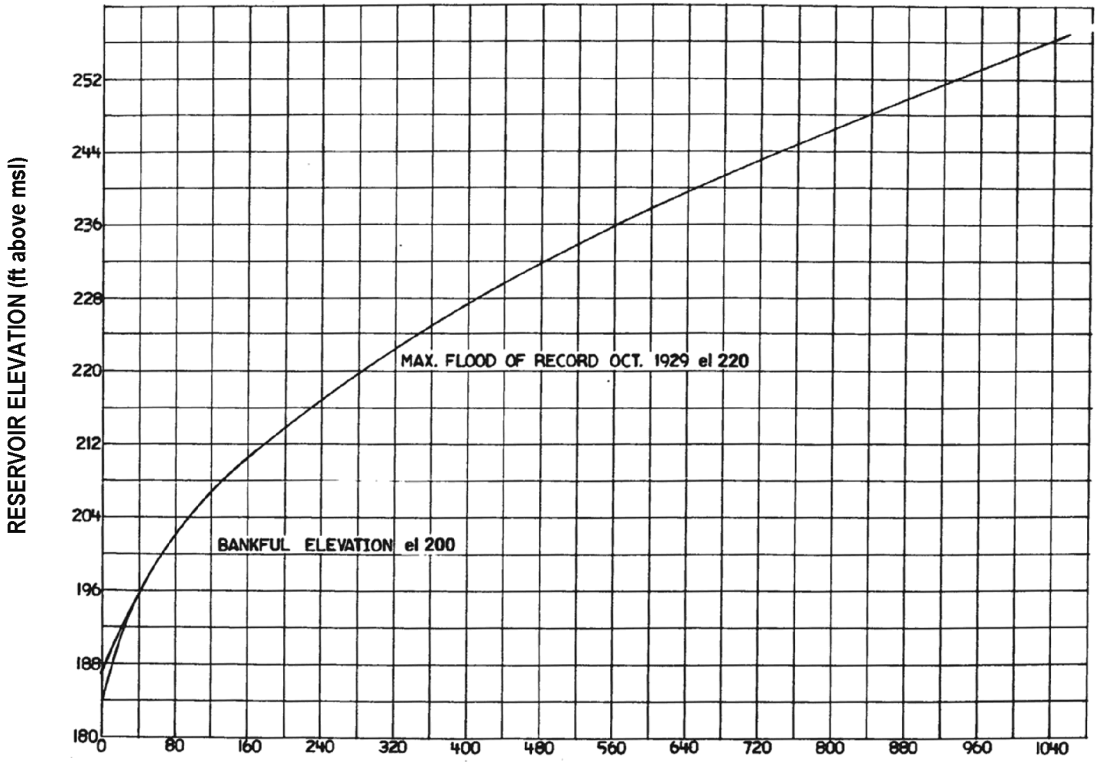
UNIT 1 AND UNIT 2

AREA AND CAPACITY CURVES FOR
MAJOR UPSTREAM DAMS

FIGURE 2.4.4–4 (SHEET 5 OF 5)



SAVANNAH RIVER BASIN
SAVANNAH RIVER GA. AND S. C.
RESERVOIR REGULATION MANUAL
CLARK HILL LAKE
U.S. ARMY ENGINEER DISTRICT, SAVANNAH
CORPS OF ENGINEERS
SAVANNAH, GEORGIA

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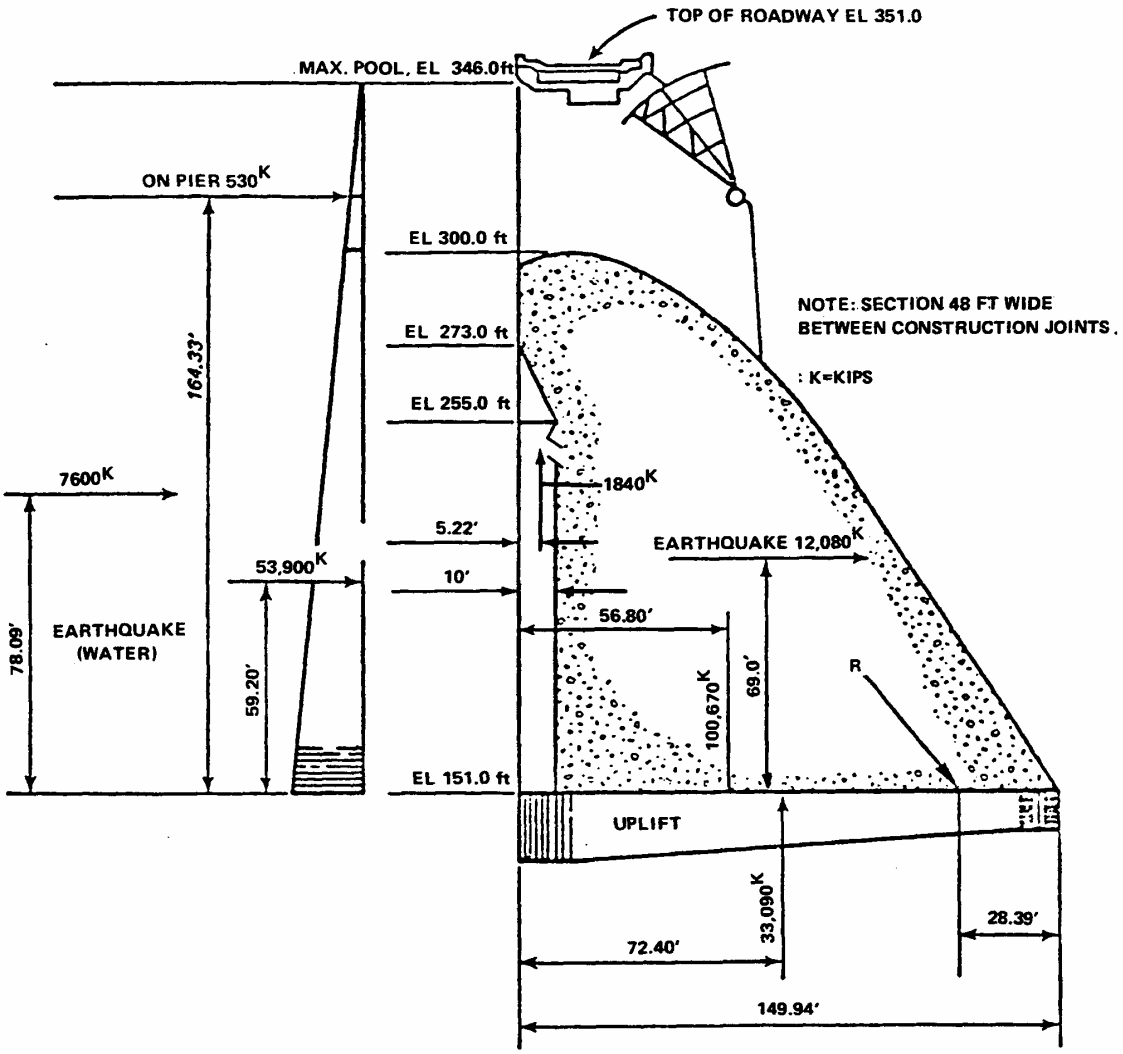
CLARK HILL TAILWATER
RATING CURVE

FIGURE 2.4.4–5



SPILLWAY SECTION

EARTHQUAKE LOADING
α = 0.12g

ELEV.	ΣV KIPS	ΣH KIPS	$\frac{\Sigma H}{\Sigma V}$	SHEAR FRICTION FACTOR	RESULTANT OUTSIDE KERN	CONCRETE STRESSES (lbs/in.2) HEEL	TOE
151	65,740	74,112	1.1	6.3	21.59'	-13	207

STRESS TABULATION

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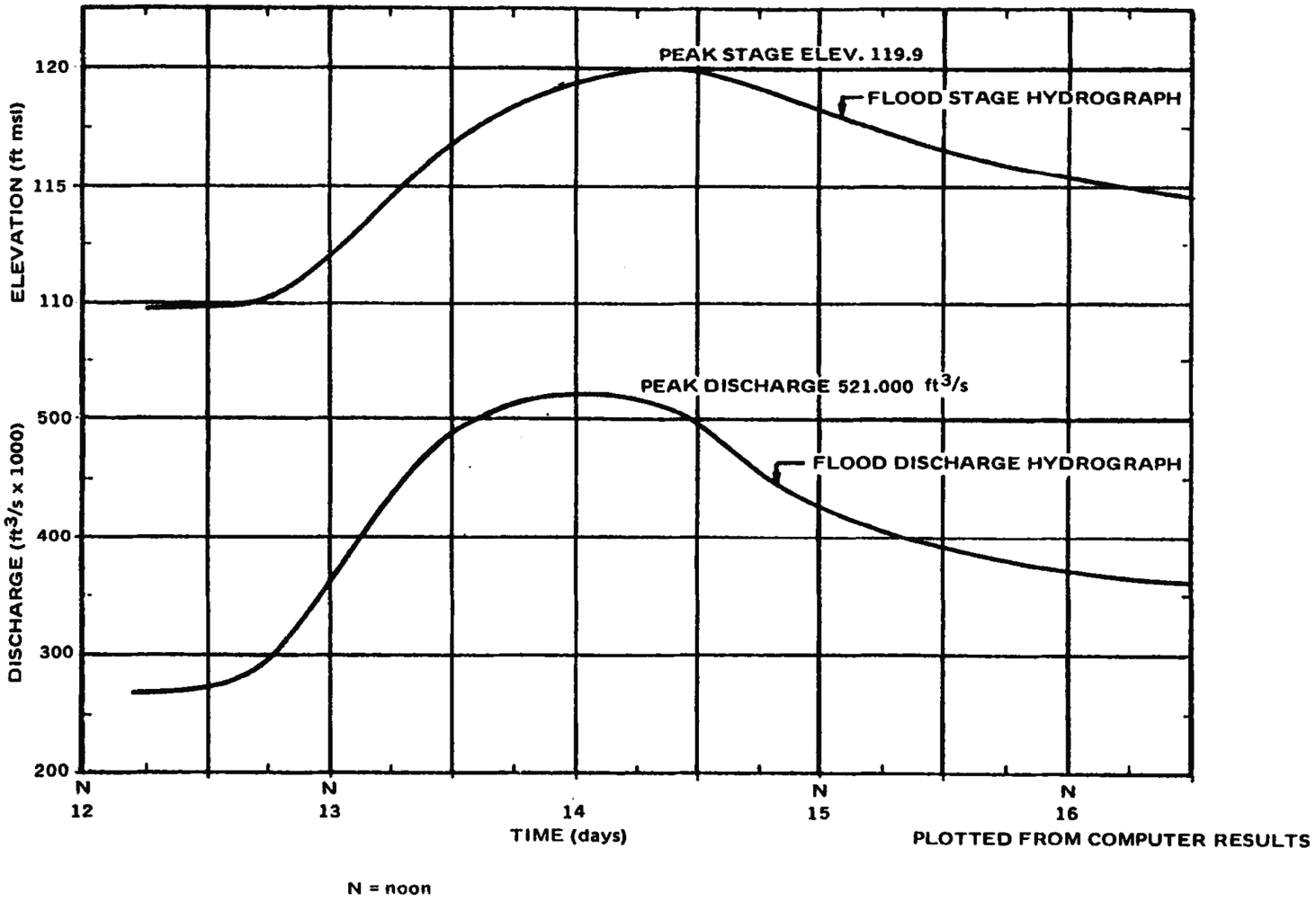
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STABILITY OF CLARK HILL DAM SPILLWAY SECTION

FIGURE 2.4.4–6



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UNIT 1 AND UNIT 2

STANDARD PROJECT FLOOD
AT RIVER MILE 151.1

FIGURE 2.4.4–7

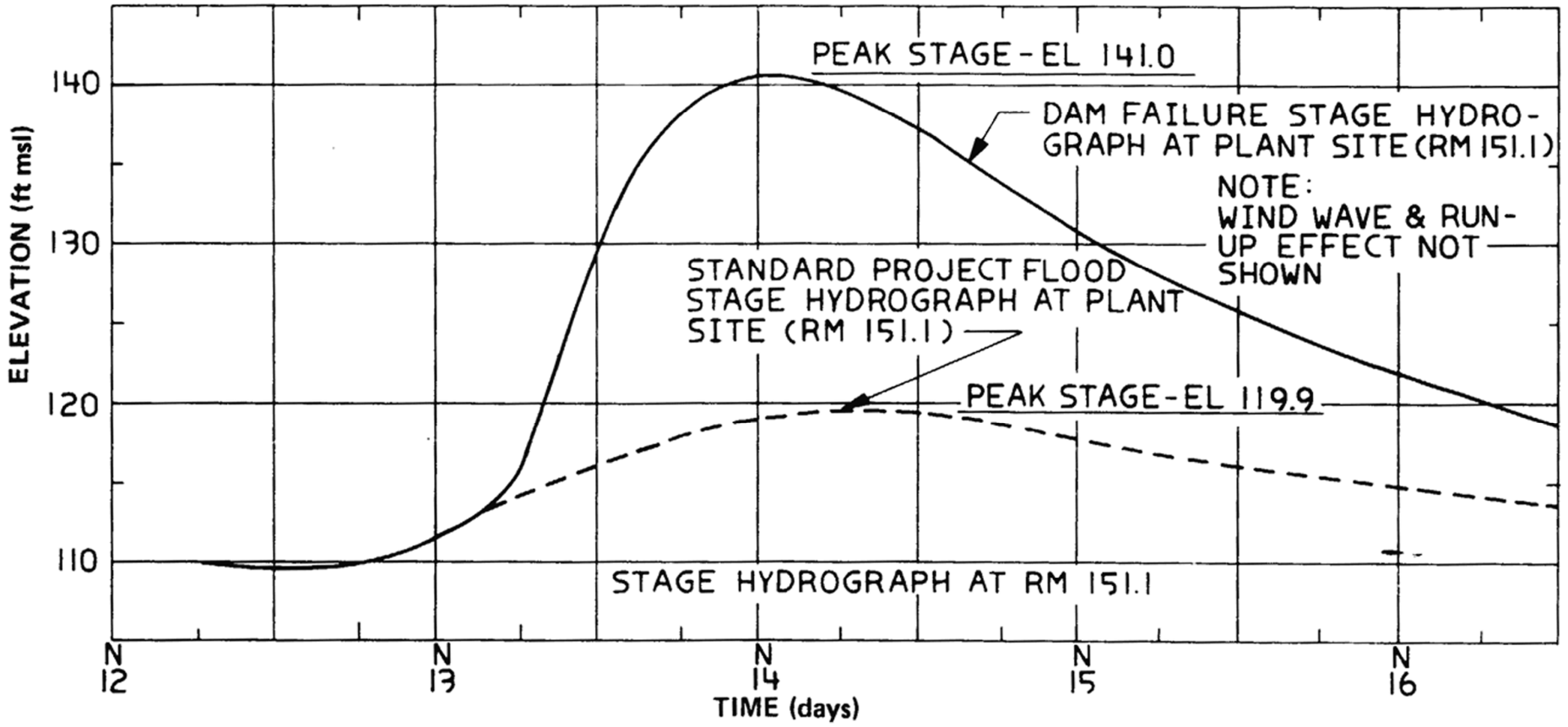

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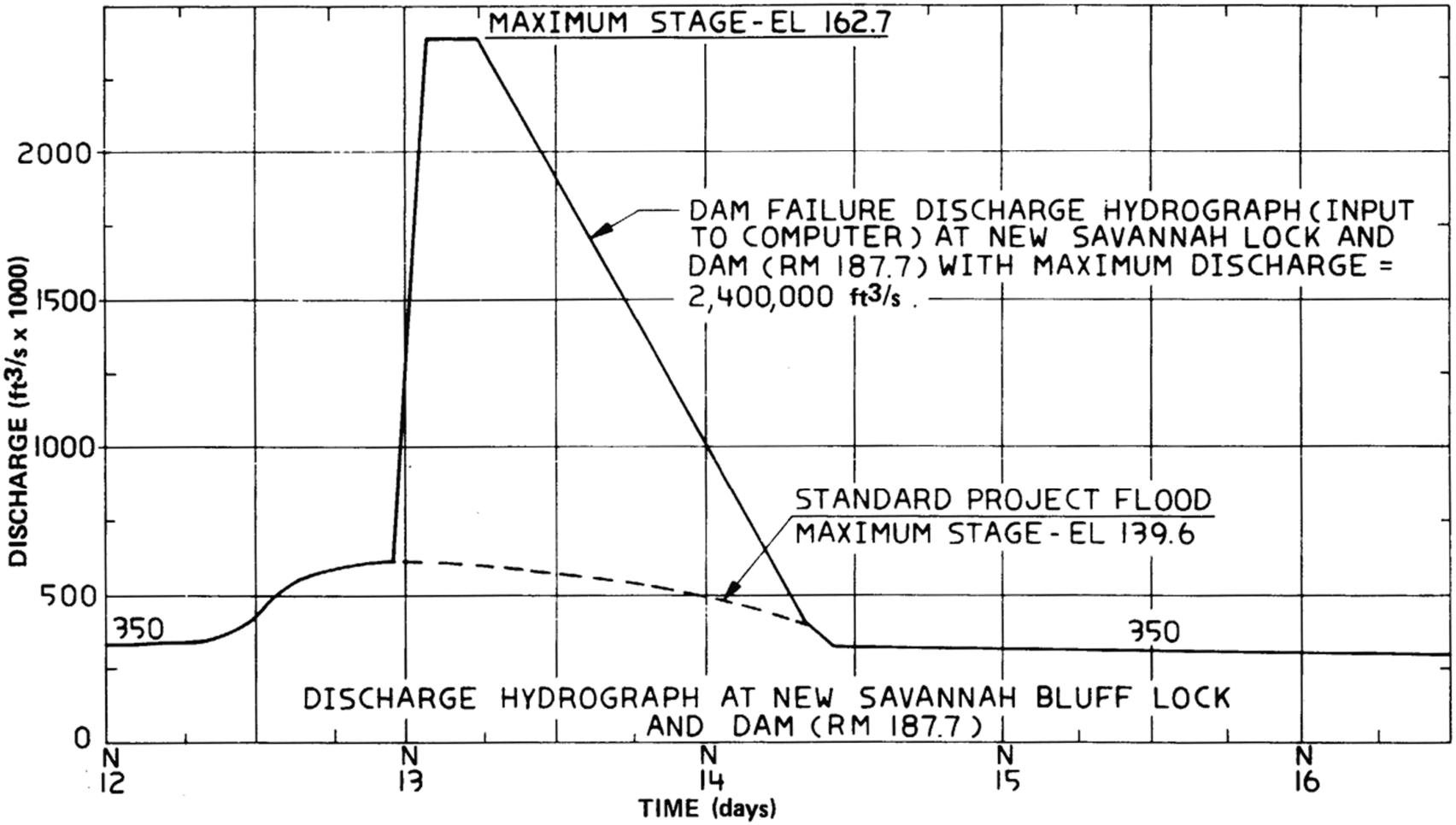
	FIGURE 2.4.4–8 (SHEET 1 OF 2)



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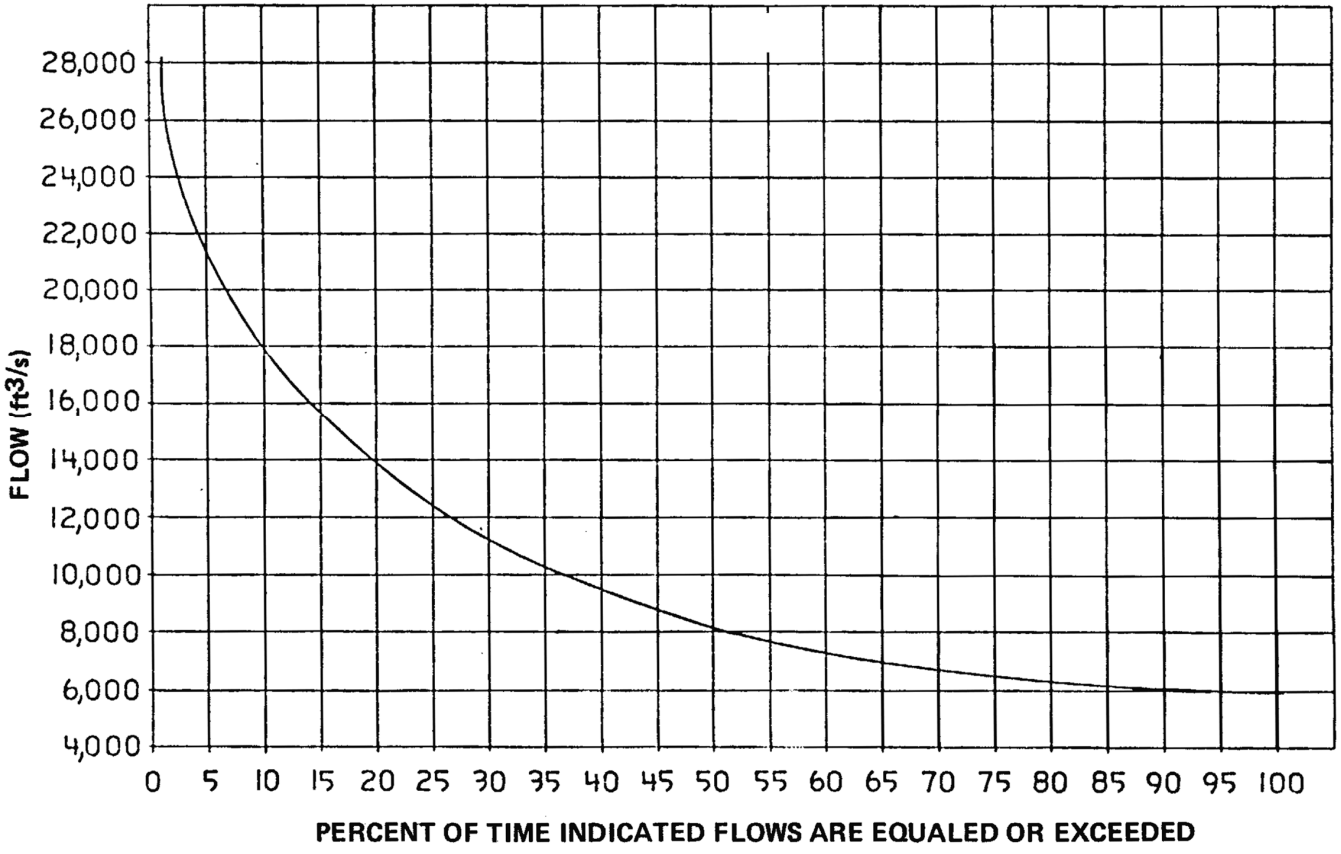
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	EFFECT OF DAM FAILURE AT VEGP SITE
	FIGURE 2.4.4–8 (SHEET 2 OF 2)

NOTES:
1. INFORMATION FURNISHED BY CORPS OF ENGINEERS, SAVANNAH DISTRICT.
2. FLOW IS CONTROLLED DUE TO CLARK HILL DAM UPSTREAM.



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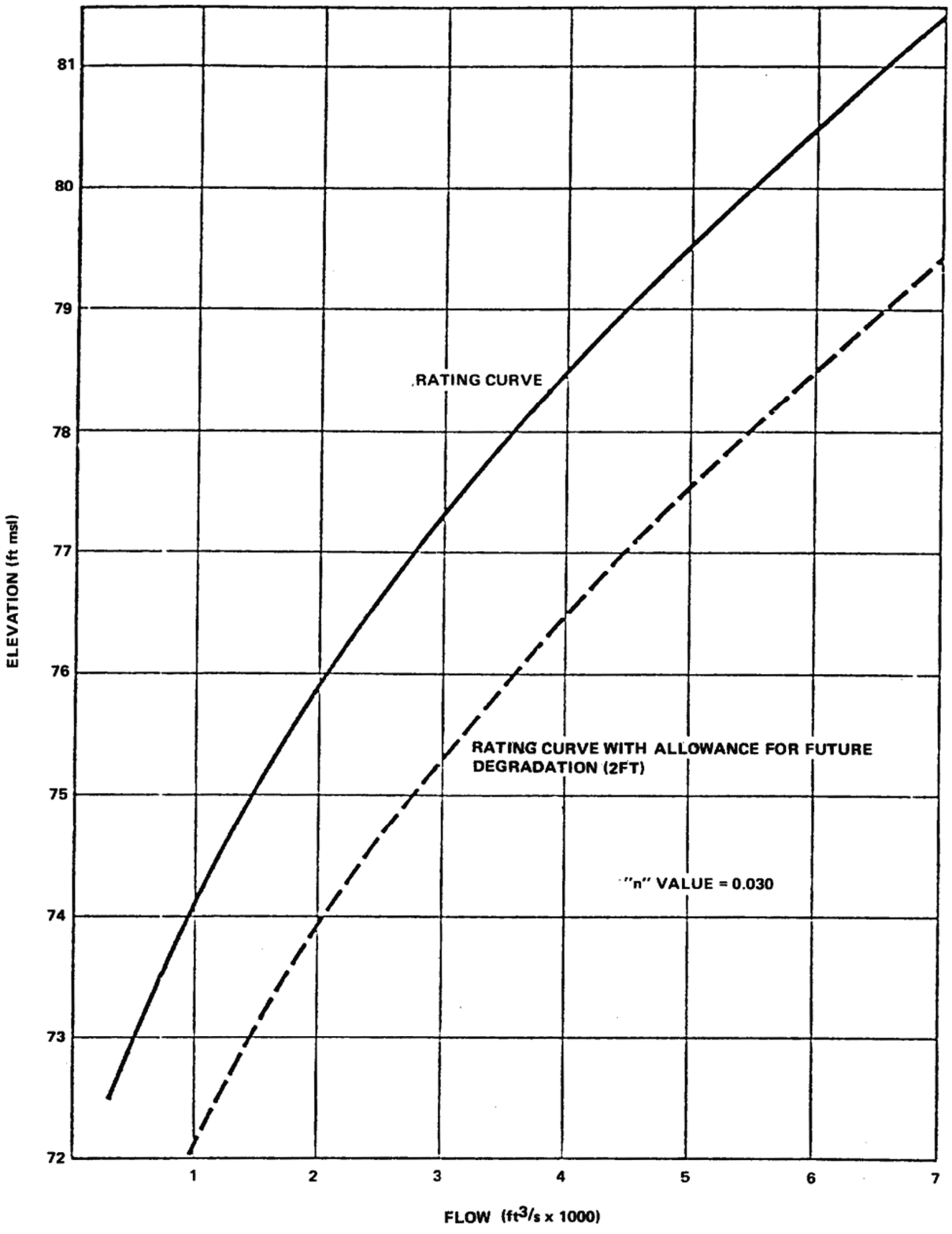
UNIT 1 AND UNIT 2

FLOW DURATION CURVE,
SAVANNAH RIVER AT BUTLER
CREEK–RIVER MILE 187.0

FIGURE 2.4.11–1



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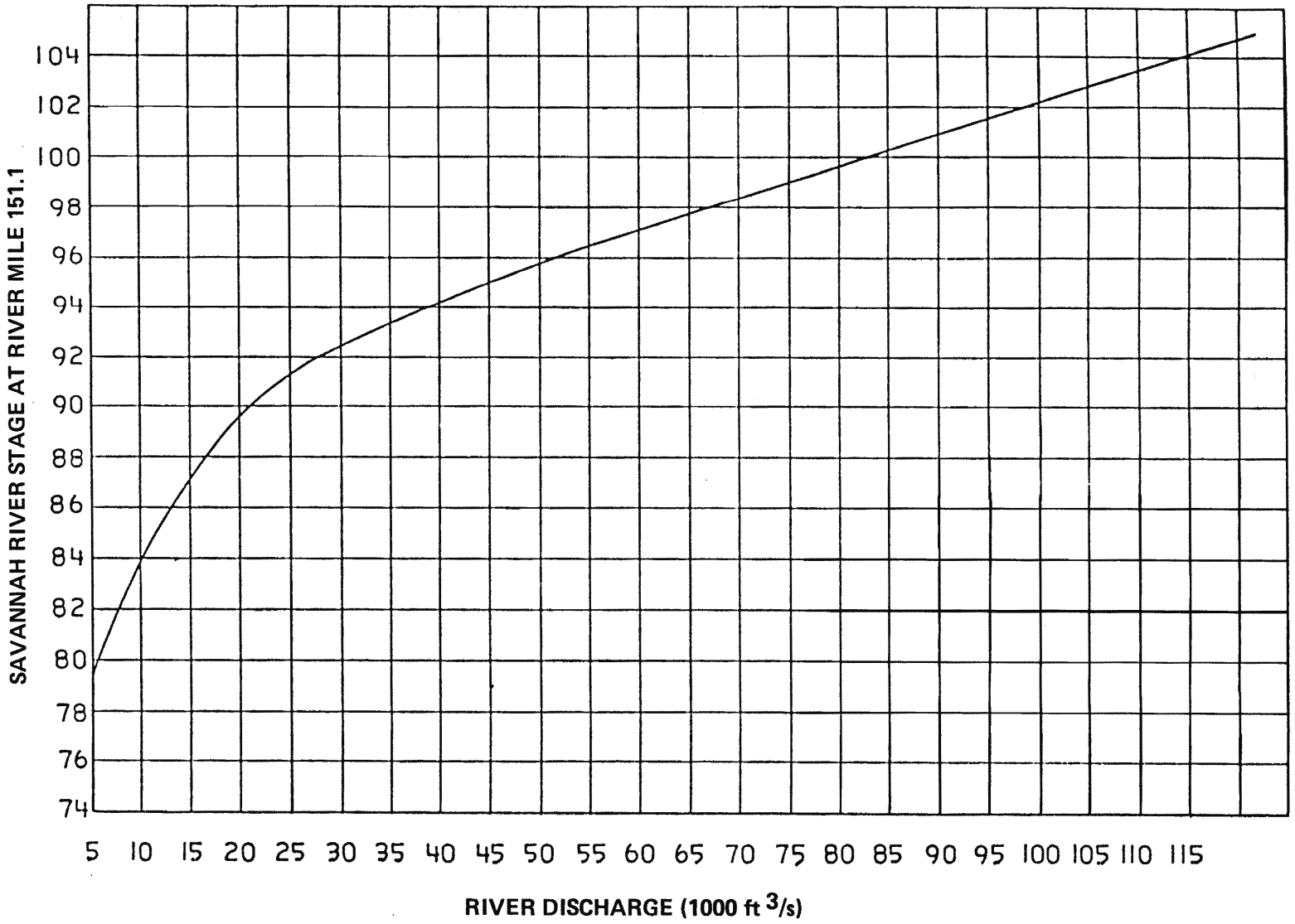
UNIT 1 AND UNIT 2

SAVANNAH RIVER LOW FLOW RATING CURVES AT INTAKE STRUCTURE RIVER MILE 151.1

FIGURE 2.4.11–2



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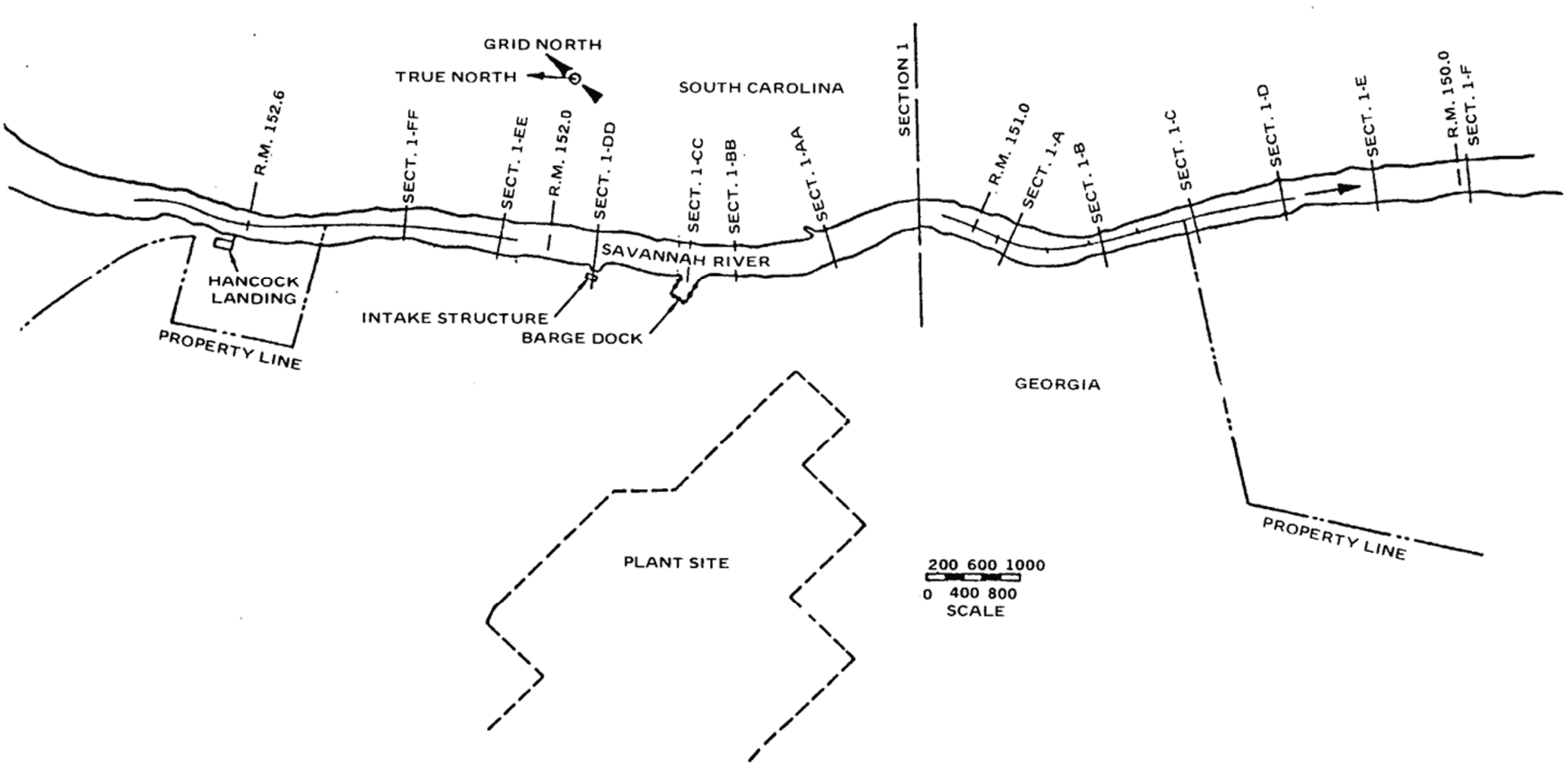
UNIT 1 AND UNIT 2

SAVANNAH RIVER INTERMEDIATE FLOW
RATING CURVE AT INTAKE STRUCTURE
RIVER MILE 151.1

FIGURE 2.4.11–3



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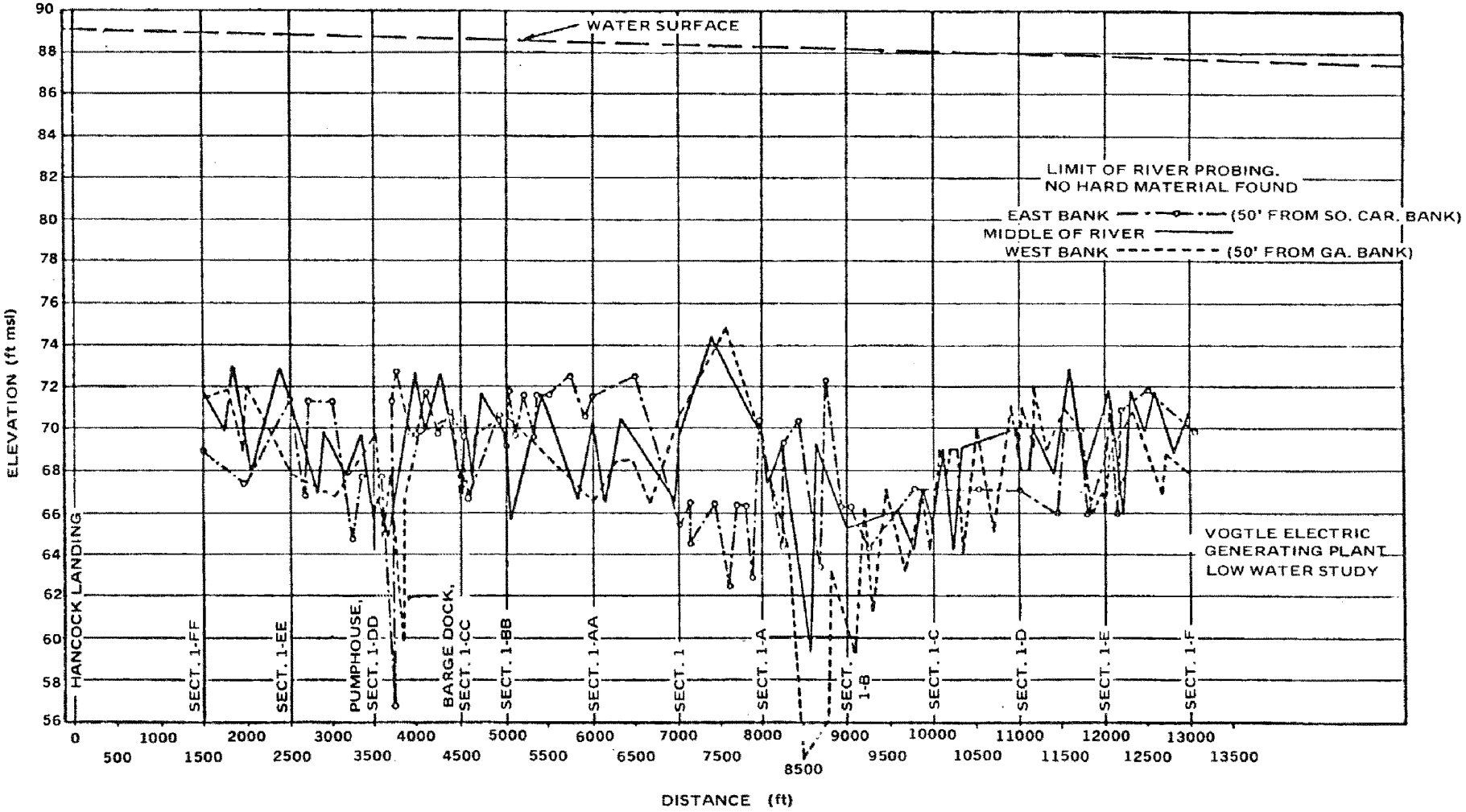
UNIT 1 AND UNIT 2

LOCATIONS OF SECTIONS
FOR LOW WATER STUDY

FIGURE 2.4.11–4



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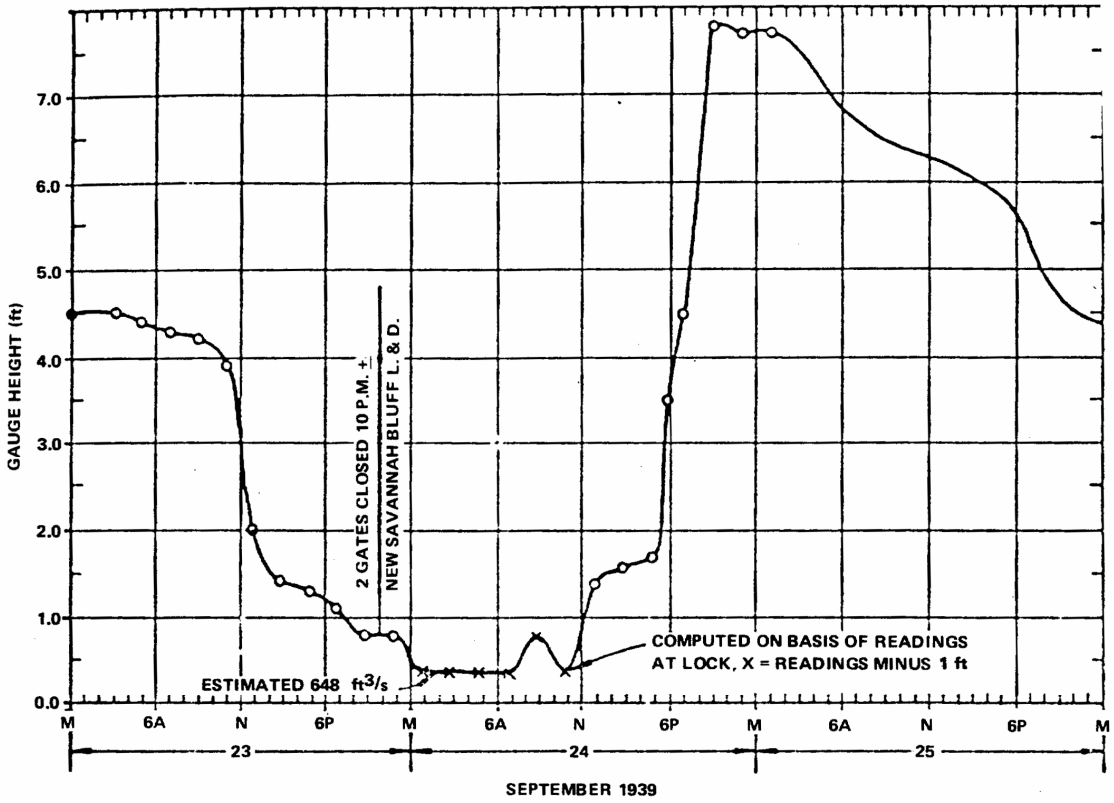
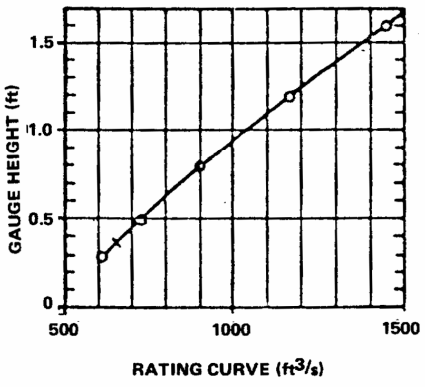
UNIT 1 AND UNIT 2

VEGP LOW WATER STUDY

FIGURE 2.4.11–5



TAKEN FROM COPY OF ORIGINAL RECORDER GRAPH AT BUTLER CREEK, 0.3 MILE DOWN-STREAM FROM NEW SAVANNAH BLUFF LOCK AND DAM. COPY FURNISHED BY U S G S



AVERAGE DAILY FLOW

DATE	FLOW ft3/s
9-23	2230
9-24	2940
9-25	5350

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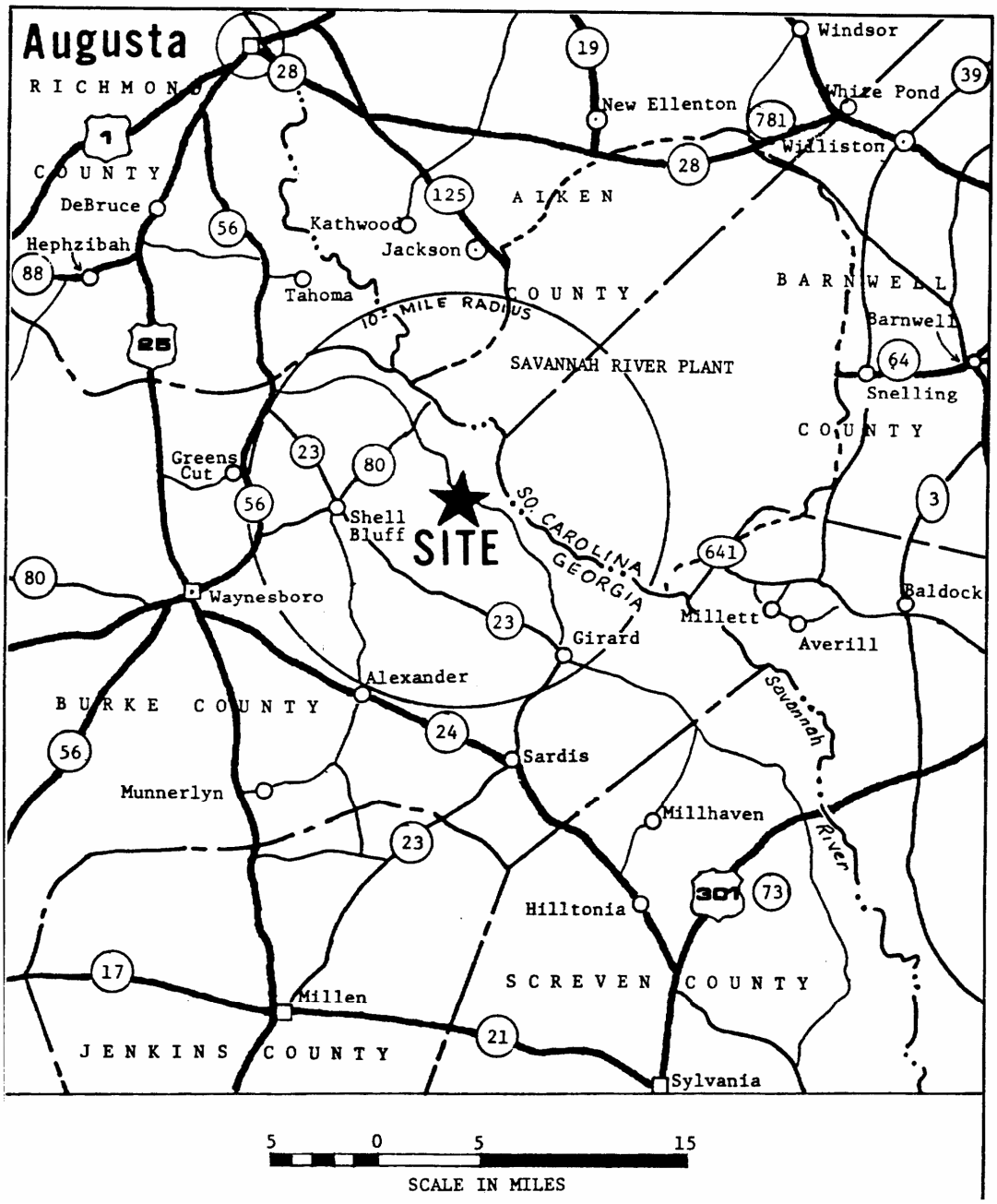
UNIT 1 AND UNIT 2

SAVANNAH RIVER AT AUGUSTA GAUGE READINGS FOR SEPTEMBER 23, 24 AND 25, 1939, MINIMUM FLOW OF RECORD

FIGURE 2.4.11–6



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	SITE AREA
		FIGURE 2.4.12–1

2.5 GEOLOGY, SEISMOLOGY, AND GEOTECHNICAL ENGINEERING

A summary of subsections 2.5.1 through 2.5.6 is provided below:

A. Summary of Investigation Program

The VEGP site is located approximately 26 miles south- southwest of Augusta, Georgia. The location of the site is shown on drawing AX6DD338. Comprehensive aerial geology and site-specific foundation investigations and examinations of the VEGP site were completed in accordance with the criteria outlined in Appendix A, Seismic and Geologic Siting Criteria for Nuclear Power Plants, of 10 CFR 100. The purpose of the investigation program and the subsequent evaluation of the geologic and soils data developed was to demonstrate the suitability of the site for a nuclear power generating facility.

Field investigations involved geologic mapping, drilling, geophysical survey, and ground water studies. During the Preliminary Safety Analysis Report (PSAR) phase of the investigations, 474 holes were drilled for a total of 60,000 ft of hole. A total of 111 holes were drilled subsequent to the PSAR investigations. The exploration program included electric logging, natural gamma, density, neutron, caliper, and three-dimensional velocity logs in selected drill holes. Water pressure tests and Menard pressure meter tests were performed to determine in situ properties of the marl stratum which provides bearing for plant structures and Seismic Category 1 backfill. Samples for fossil, mineral, or soluble carbonate analysis were taken in drill holes as required. The geophysical survey provided a total of 28,400 ft of shallow refraction seismic lines, 5000 ft of deep refraction lines, and cross-hole velocities in the upper 290 ft of materials.

Laboratory testing of selected soil samples consisted of the following tests:

1. Moisture content.
2. In situ unit weight.
3. Atterberg limits.
4. Static triaxial shear.
5. Direct shear.
6. Consolidation.
7. Relative density.
8. Unconfined compression.
9. Dynamic triaxial shear.
10. Resonant column.
11. Moisture density relationship.

Included in the investigation program was a thorough literature search, stereoscopic examination of color air photographs, evaluation of geologic conditions at and within 5 miles of the site, and geologic reconnaissance along 12 miles of the river bluff upstream and downstream of the site.

Drilling services were provided by Georgia Power Company, Girdler Drilling Company, and Law Engineering Testing Company. Geophysical seismic surveys were performed by Weston Engineers, Weston, Massachusetts. Down-hole geophysical surveys were performed by the Birdwell division of Seismograph Service, Incorporated. Laboratory testing services were provided by Law Engineering Testing Company, Atlanta, Georgia, and Geotechnical Engineers, Incorporated, Winchester, Massachusetts.

B. Summary of Geology

The site is located in the Atlantic Coastal Plain physiographic province in central Georgia. The portion of the Coastal Plain province in which the site occurs is known as the Tifton Upland which is characterized by rolling hills ranging in elevation from 80 to 280 ft in the site vicinity.

The geology within a 25-mile radius of the site consists of Precambrian and Paleozoic igneous and metamorphic basement rocks (gneisses and granites of the Kiokee Belt and phyllites and greenstones of the Belair Belt) overlain locally by Triassic basin sediments (Dunbarton Basin); these are, in turn, overlain by Cretaceous through Miocene Coastal Plain (shallow marine) sediments. Quaternary alluvial deposits occur along the Savannah River and its tributaries.

Virtually all tectonic activity occurred prior to the deposition of the Cretaceous sediments. The complex folding, faulting, and shear structures that developed in the basement rocks originated in the Precambrian and Paleozoic eras during orogenic episodes associated with the development of the southern Appalachians. Relatively undeformed Coastal Plain sediments indicate that this orogenic activity ceased prior to the Cretaceous period.

The geology within a 5-mile radius of the site reflects the geology of the region. The contact between the basement complex and Cretaceous sediments occurs more than 1000 ft below the surface. As a result of regional elevation fluctuations following the deposition of the basal Cretaceous sediments (Tuscaloosa formation), overlying Paleocene through Miocene sediments represent marine transgressive and regressive sequences. Strata include shallow marine sand, clay, gravel, limestone, and marl. Quaternary deposits of sand, gravel, silt, and clay occur as flood plain deposits in the Savannah River valley and the larger tributaries to the river. However, the Quaternary system is principally represented by erosion and weathering rather than deposition.

Cretaceous and post-Cretaceous formations underlying the site are essentially flat lying or gently dipping to the southeast, reflecting a regional dip of about 30 ft/mi. Localized solution occurs in a shallow formation stratigraphically above the marl. Therefore, the solution activity will have no effect on plant foundations.

No faults or lineaments have been found within 5 miles of the site, other than those associated with the Triassic Dunbarton Basin, and these structures do not extend into overlying Tertiary strata. There are no capable faults as defined by 10 CFR 100, Appendix A, anywhere within the site region. There are no other geologic hazards which could affect site safety or suitability for a nuclear power plant.

C. Summary of Seismology

The Atlantic Coastal Plain tectonic province, in which the site is located, is a large area with generally low seismic activity. The general seismicity of this province is expected to remain subdued, while high earthquake activity will be confined to the Charleston-Summerville seismic zone. In the adjacent southern Appalachian Mountains region, earthquakes are irregularly distributed, with concentrations in northeast Georgia, northwest South Carolina, eastern Tennessee, and Virginia, all at distances greater than 100 miles from the site.

The closest damaging earthquake to the site occurred on November 1, 1875. It was centered 60 miles to the northwest and had a maximum intensity of VI near the epicenter. This event may have been felt with low intensity at the site.

The New Madrid events of 1811-1812, which were centered 530 miles from the site, were probably felt at the site with an intensity less than VI. The Union County, South Carolina, earthquake of January 1, 1913, had an epicentral intensity of VII to VIII. However, the felt area does not include the VEGP site.

The source of seismicity most affecting the site, both in maximum historical intensity and number of earthquakes, is the Charleston-Summerville, South Carolina, area. The main shocks of August 31, 1886, probably produced an intensity of VI at the site. Recent studies have defined a linear north-northwest trending zone of epicenters in the Charleston- Summerville area. The closest approach of the Charleston-Summerville seismic zone to the site is 78 miles.

Detailed studies have revealed no seismological or geological evidence for capable faults within 200 miles of the site.

The maximum credible site intensity is VI-VII to VII. For conservatism a safe shutdown earthquake site intensity of VII-VIII is chosen. This intensity is associated with approximately 0.2 g peak horizontal acceleration.

Evidence indicates that the maximum historical intensity at the site was VI. For conservatism an operating basis earthquake of intensity VII is adopted. This intensity is associated with a peak horizontal ground surface acceleration of 0.12 g. A probabilistic analysis shows that the likelihood of this acceleration being exceeded during the 40-year[a] operating life of the plant is less than 8%.

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Seismic analyses are not related to aging and therefore are outside the scope of license renewal. NEI 98-03 guidance indicates that seismic data used to support original plant design bases are considered historical and do not need to be actively maintained. Therefore, this statement is not required to be updated as a result of license renewal.

D. Summary of Geotechnical Engineering

The surface soils at the VEGP site consist of three principal strata, i.e., the upper sand stratum, the marl stratum, and the lower sand stratum. The upper sand stratum is about 90 ft deep. The relative density of the upper sand stratum is very variable and ranges from very loose to dense. Based on the results of a liquefaction analysis, it was determined that the upper sand stratum would have a potential for liquefaction in the event of a seismic occurrence equivalent to the safe shutdown earthquake. The upper sand stratum was excavated down to the marl stratum and replaced with select sand and silty sand backfill compacted to an average of 97% of the maximum density determined by ASTM D 1557. This high degree of compaction will ensure an adequate factor of safety against liquefaction and reduce settlements to a tolerable level.

The auxiliary building, nuclear service cooling water towers, and instrumentation cavity of the containment are supported on the marl stratum. All the other power block structures are supported on compacted backfill. The marl stratum which directly or indirectly supports all the power block structures ranges in thickness from 60 to 100 ft. There is no evidence that the marl stratum has been subjected to or is potentially subject to subsidence, collapse, or uplift due to earthquake, solution processes, or other geological phenomena. The lower sand stratum underlying the marl stratum is estimated to be at least 750 ft thick and determined to be dense to very dense in relative density. The VEGP power block structures are therefore supported on competent bearing strata.

E. Conclusions

The studies described in subsections 2.5.1 through 2.5.6 have led to the following conclusions:

1. The site is suitable for design and construction of a multiple-unit nuclear power generating facility.

2. The geologic conditions exposed in the power block excavation confirm the conditions described in the PSAR.

3. No geologic features which could affect licensing aspects of the plant exist in the power block area.

4. The marl layer, which provides foundation for both Category 1 structures and Category 1 backfill, is as sound and as competent as anticipated and is free of solution cavities.

5. Results of laboratory testing and monitoring of the rebound of the marl confirm its competency.

2.5.1 BASIC GEOLOGIC AND SEISMIC INFORMATION

The following paragraphs contain the results and conclusions of the regional and site geologic and seismic investigations.

Information on regional and local ground water conditions is included in subsection 2.4.12 and is only summarized in the following geology subsections. The characteristics of the foundation materials with respect to their ability to support the major plant structures are discussed in subsection 2.5.4 and cross-referenced in this section. Sources used in the preparation of text, figures, and tables are contained in the references cited at the end of this subsection.

2.5.1.1 Regional Geology

This paragraph on regional geology describes the area surrounding the site and relates it to the Atlantic Coastal Plain and Piedmont regions in terms of physiography, geomorphology, geologic history, lithology, stratigraphy, structure, and tectonic conditions.

2.5.1.1.1 Regional Physiography and Geomorphology

The VEGP site is located approximately 26 miles south-southwest of Augusta, Georgia, in the Atlantic Coastal Plain province (drawing AX6DD338). The Atlantic Coastal Plain province covers approximately 60% of the surface area of the State of Georgia. Along its inner margin at the boundary with the Piedmont province is the Fall Line. This line marks the contact between the crystalline basement and the onlapped Cretaceous and Cenozoic sediments. The plant site is located near the eastern margin of the Tifton Upland topographic area, a major topographic division of the Atlantic Coastal Plain province (drawing AX6DD338). The Atlantic Coastal Plain province in Georgia also includes parts of three other major topographic areas:[1][2] from its western margin seaward, they are the Fall Line Hills, the Dougherty Plain, and the coastal terraces (drawing AX6DD338).

The Fall Line Hills, located northwest of the site, form a series of maturely dissected hills which adjoin the Fall Line to the northwest and the Tifton Upland to the southeast. The Fall Line Hills form a broad and continuous belt with the exception of some transverse divides. The Red Hills topographic subdivision, commonly considered part of the Fall Line Hills in Georgia, occupies the area immediately to the north of the Tipton Upland.[1] In Georgia, the Red Hills are equivalent to the tabular and incompletely dissected Louisville Plateau. The Fall Line Hills, including the Red Hills, form a zone between 20 and 40 miles wide.

The Dougherty Plain topographic division is widely exposed in west-central Georgia, situated between the Fall Line Hills on the north and the Tifton Upland on the south. The most noticeable features on its nearly flat plain are shallow, flat- bottomed or rounded depressions resulting from solution of the underlying rocks.

The plant site is located near the eastern margin of the Tifton Upland topographic division, a submaturely dissected area of the Coastal Plain just seaward of the Fall Line Hills in north-eastern Georgia. The dissection of the weak rocks has resulted in gently rolling, well-drained hills with broad, rounded summits and with relief of generally less than 50 and rarely more than 100 ft.[1]

In western South Carolina the trend of the Fall Line Hills is interrupted by the Aiken Plateau.[3][4] The trend is continued in west-central South Carolina as the Congaree Sand Hills.[3]

The remainder of the Coastal Plain province in both Georgia and South Carolina is comprised of belts of coastal terraces of Pleistocene age. The older terraces adjacent to the Tifton Upland show moderate erosion, whereas the younger terraces are only slightly eroded.

Immediately to the northwest of the Atlantic Coastal Plain province is the Piedmont province. Farther to the northwest, within a 200-mile radius of the site are the Blue Ridge and the Valley and Ridge provinces. These provinces are shown on drawing AX6DD338.

The Piedmont province is a northeast trending belt that adjoins the Atlantic Coastal Plain province to the south and the Blue Ridge province and, in part, the Valley and Ridge province to the north. Extending from central Alabama to southern New York, the Piedmont province is an intricately dissected upland plain. A few linear ridges are present in west-central Georgia. Approximately 20% of the State of Georgia lies in the Piedmont province.

The Fall Line forms the border between the Piedmont and Atlantic Coastal Plain provinces. In the vicinity of the Fall Line the upland plain of the Piedmont slopes down to meet the Coastal Plains. In this area erosion has caused large streams of the Piedmont to cut gorges and create waterfalls and rapids as they flow over this steeper gradient into the weaker Coastal Plain sediments.

The Blue Ridge province is situated to the northwest of the Piedmont province and extends for 550 miles from northeastern Georgia to southern Pennsylvania. The Blue Ridge province is relatively narrow north of North Carolina and is dominated by a central ridge, whereas the area in Georgia is a broad, mountainous upland.

The Valley and Ridge province lies to the northwest of the Blue Ridge province and extends from central Alabama to the St. Lawrence Lowland. The province can be divided along its length into two major parts. The eastern part consists of a broad valley; the western part consists of a series of ridges and valleys resulting in a pronounced northeast-southwest regional grain.

The Savannah River cuts a deep, transverse valley through the Atlantic Coastal Plain province along the eastern margin of the site. The river valley is of old age, with a broad flood plain at an elevation of approximately 85 ft adjacent to the site. The bordering upland has a general elevation of 250 to 300 ft above mean sea level and is dissected by the tributaries of the Savannah River. Drainage patterns are well developed consisting of deeply incised streams which have broad, gently sloping valleys.

Occasional small, shallow, closed surface depressions are present above el 200 ft in the site area. These are formed where leaching of a soluble carbonate horizon in the upper deposits has occurred, causing surface subsidence over the affected area. Site geologic conditions are discussed in paragraph 2.5.1.2 and appendix 2B. Stability of subsurface material at the site is discussed in subsection 2.5.4.

2.5.1.1.2 Regional Geologic History

Large-scale compressional and extensional tectonics characterize the history and evolution of the Appalachian Mountains and eastern margin of North America. Tectonic models initially proposed by Hatcher[5][6], Rodgers[7] and Rankin[8][9] provide the general tectonic framework for evolution of the central and southern Appalachians. The models generally describe a period of major extensional rifting during the late Precambrian, followed by the development of both eastward- and westward-dipping subduction zones at different times during the Paleozoic, followed by renewed extensional rifting during the Triassic and Jurassic.

Sequential tectonic development of the Appalachians has been confirmed and elaborated with respect to plate-tectonic interactions by more recent geological mapping and deep-crustal geophysical studies in the central and southern Appalachians.[10-37] These studies indicate that the Appalachians consist of a number of lithologically and structurally distinct terranes ("suspect terranes") sutured to the continent during Paleozoic plate collisions. Strike-slip displacement may have occurred along many of these terrane boundaries concurrent with or subsequent to the suturing process as indicated by paleomagnetic data from the New England and maritime Canada area.[38] Subsequent major lateral displacements of some or all of these terranes occurred in the late Paleozoic mainly along low angle detachment surfaces or decollements.

The Appalachians are classically divided into a series of northeast-southwest-trending physiographic provinces. From northwest to southeast these include the Valley and Ridge, Blue Ridge, Piedmont, and Coastal Plain (drawing AX6DD338). In the southern Appalachians, the Piedmont Province includes a number of Lithotectonic belts: the Inner Piedmont belt, the Kings Mountain belt, the Pine Mountain belt, the Charlotte belt, and the Carolina Slate belt.[5][39] Smaller local lithotectonic belts are also defined. Near the site, north of Augusta, the Kiokee belt parallels the southeastern margin of the Carolina Slate belt, separated from it by the Modoc fault. At the edge of Coastal Plain onlap, the Augusta fault is interpreted to juxtapose yet another belt (the "Belair belt") against the southeastern edge of the Kiokee belt (figures 2 and 10C of Cook, et al.[24]) Both the Brevard and the Augusta faults would represent, within the crystalline Piedmont rocks, major, east dipping ramp faults that are listric into the basal decollement at depth (figures 10C, 11, and 12 of Cook, et al.[24]). The regional distribution of these provinces and belts is shown in figure 1 of Hatcher and Odom[12] and figure 1 of Hatcher and Zietz[32].

Williams and Hatcher [20][21] group many of these various lithotectonic belts of the Appalachian Orogen into "suspect" Appalachian terranes because their internal consistency in structure, lithology, geologic history and provenance implies that they were once spatially separate crustal entities brought into mutual proximity by subsequent plate accretion. In the southern Appalachians, for example, the Charlotte and Carolina slate belts and part of the Kings Mountain belt comprise the Avalon terrane and the eastern Blue Ridge, Inner Piedmont, Chuaga belt, and much of the Kings Mountain belt comprise the Piedmont terrane (Williams and Hatcher [20][21]). To the southeast, magnetic and gravity data provide strong evidence for additional exotic terranes beneath the Coastal Plain sediments of Georgia and the Carolinas (Higgins and Zietz[35]). These include the Suwannee and Brunswick terranes of Williams and Hatcher[21].

The geologic history presented below is a synthesis of the current technical literature on Appalachian tectonics described above. Age constraints on the timing of tectonic events are determined primarily by the depositional history of sediments in the Appalachian Orogen and from radiometric dating of metamorphic and igneous rocks. This age control is summarized by Hatcher,[5-6] Hatcher and Odom,[12] and Glover.[18] A concise summary of the geologic history of the southern Appalachians is presented by Cook, et al.[24] following the work of Hatcher[6] and Hatcher and Odom[12].

2.5.1.1.2.1 Precambrian and Paleozoic Eras. In the late Precambrian (approximately 820 million years ago) extensional tectonics rifted the North American craton from the ancestral craton of Africa, Europe, and South America.[8][9] The present eastern and southern portions of the North American craton became a passive continental margin on which shallow marine shelf deposits accumulated (Chilhowee-Knox), transitional to a basinal facies farther east (Ocoee). Various continental fragments were distributed in the expanding proto-Atlantic or "Iapetus" Ocean as the major continents separated.

During the ensuing plate convergence or contraction of the Iapetus Ocean early in the Paleozoic, several island arcs developed offshore on these continental fragments or on oceanic crust. The Carolina Slate and Charlotte belts, for example, are believed to be one or more island arcs subsequently accreted to the margin of North America.[6][18][28] In the southern Appalachians, closure of the oceanic basin between North American and previously rifted continental and island arc fragments, and subsequent collision of these land masses during the Cambrian and Ordovician, produced the Taconic orogeny (480 to 450 million years ago). Fragments composing the Inner Piedmont-Blue Ridge terrane were accreted to North America by overthrusting.[6][9][23] Extreme deformation, metamorphism, and plutonism in the Inner Piedmont accompanied the orogeny. Sediment loading occurred on the cratonic lithosphere in past-Taconic time from detritus eroded from mountainous terrain to the east, representing a radical change in provenance compared to pre-Taconic paleogeography.

Continued closure of the Iapetus Ocean by subduction resulted in the collision of the Carolina Slate Belt with the Inner Piedmont-Blue Ridge terrane about 300 to 400 million years ago, approximately concurrent with the Acadian orogeny (Late Silurian through Late Devonian), and was accompanied by additional metamorphism and deformation. The intervening Kings Mountain belt and Charlotte belt include rocks caught in the collision. Rocks of the Charlotte belt represent high-grade metamorphic equivalents of the Carolina Slate belt and are of island arc affinity. Rocks of the Kings Mountain belt are predominantly a mixture of Charlotte belt and Inner Piedmont rocks. During the collision the allochthonous Inner Piedmont-Blue Ridge fragment was thrust further westward over the old continental margin.

The last phase of paleozoic plate convergence along the eastern margin of North America culminated in the continental collision between proto-North America and proto-Africa during the Alleghenian orogeny, from Middle Pennsylvanian through Early Permian time (300 to 250 million years ago). Most of the prominent Appalachian structures and provinces assumed their present configuration during this orogeny. Extensive igneous plutonism in the eastern Piedmont accompanied the orogeny, and large scale overthrusting deformed the sedimentary rocks in the Valley and Ridge province. Thrusting of the Piedmont over the Blue Ridge occurred along the Brevard zone at this time.

Over the past several years deep seismic reflection studies have suggested that these various thrust faults conveying the allochthonous slices westward are splays of a deeper, regionally extensive detachment surface or decollement. These studies suggest that the decollement extends eastward beneath the Piedmont province and may continue beneath the Coastal Plain and Atlantic shelf.[10][11][13][16][17][22 24][26][27][34][36] [40-43] The seismic profiles from Pennsylvania, Virginia, eastern Tennessee, South Carolina, and Georgia show laterally continuous subhorizontal reflectors ranging in depth from less than 6 km in the Valley and Ridge to more than 11 km beneath the Coastal Plain. Cook, et al.,[22, 24] and Harris and Bayer[27] interpret these subsurface reflectors to be elements of a regionally extensive decollement underlain by relatively undeformed, weakly unmetamorphosed, layered Paleozoic sedimentary rocks. The subhorizontal east dipping decollement presumably accommodated crustal foreshortening associated with continental collision in late Paleozoic time when metamorphic Precambrian and early Paleozoic rocks of the Blue Ridge and Piedmont were thrust to the west and northwest over relatively unmetamorphosed sedimentary rocks of the Valley and Ridge.

The southeastern extent of the decollement is still poorly known. The seismic reflection profiles confirm the presence of a decollement from the Valley and Ridge to at least as far as the inner Piedmont where the subsurface reflectors thicken, become discontinuous, and dip steeply to the southeast.[13][22- 24] In one interpretation, the steeply dipping reflectors mark the root zone of the decollement beneath the Kings Mountain Belt of the outer Piedmont.[12][21][32][36] In a second interpretation, Cook, et al., [22][24] suggest that the decollement may be rooted offshore where a complex structural configuration is evident on the seismic profiles.

Most investigators agree that major movement on the decollement occurred during the Alleghenian orogeny, although it must also have been active in some form during the earlier Taconic and Acadian orgenies. Such a model provides a rational kinematic mechanism for the thin-skinned fold and thrust deformation of the Valley and Ridge Province and the emplacement of the crystalline Blue Ridge and piedmont rocks as relatively thin, allochthonous sheets. Latest thrust movement is constrained to a time preceding the onset of Mesozoic extensional tectonics. Ellwood, Whitney, and Wenner[44] describe a 350-million-year-old intrusive complex (Elberton Granite) which was emplaced across the decollement; the intrusive complex would therefore provide a minimum age for thrusting.

2.5.1.1.2.2 Mesozoic Era.

2.5.1.1.2.2.1 Triassic and Jurassic Periods. The tectonic model which best explains the stratigraphic distribution of lower Mesozoic rocks on the eastern coast of North America includes the following sequence: (1) Permian to Late Triassic uplift and crustal thinning along the axis of the future Atlantic Ocean, (2) Middle to Late Triassic strike-slip faulting and volcanism along east-trending fracture zones followed by the advance of the Tethys Sea, and (3) Late Triassic rifting along the axis of the proto-Atlantic Ocean and shearing along east-west fracture zones. This action had the combined effect of decoupling segments of the African and North American plates and causing deposition of clastic sediments in the Triassic basins that formed. Late Triassic to Early Jurassic crustal extension and extrusion of basaltic lavas were followed by collapse of the continental margins and simultaneous deposition of marine carbonates.[45]

On the eastern seaboard of the United States many of the Triassic basins are exposed at the surface in the Piedmont province (see, for example, plate 1 of Wentworth and Mergner-Keefer[46]). Most, however, are covered by Coastal Plain sediments and have been delineated on the basis of core holes and wells (Plate 1 of Chowns and Williams[15]), aeromagnetic and gravity anomalies,[25][42][47] and seismic reflection and refraction studies.[11][24] These Triassic basins are shown on drawing AX6DD409. (For a general discussion of the development of Mesozoic rift basins in the southeastern United States, see Klitgord, Dillon, Popenoe[37] and Dillon, Klitgord, and Paull.[48])

The Triassic basins were filled with fanglomerates, landslide debris, and mudflow deposits originating along steep fault scarps. Streams draining the surrounding metamorphic highlands pumped most of the very coarse material close to the edge of the basin, but periodic increases in stream energy, due either to increased precipitation or to renewed uplift, carried coarser particles to the center of the basin.[49-51]

The Triassic sediments in the Dunbarton Basin beneath the plant site at one time may have reached a maximum thickness of 6000 to 8000 ft greater than at present. This estimate is based on the conversion of montmorillonite to illite, which varies with depth of burial.[49] Subsequent erosion reduced both mountains and valleys to a single peneplained surface.

2.5.1.1.2.2.2 Cretaceous Period. Both Cretaceous and Tertiary sediments of the Coastal Plain Province accumulated on the trailing eastern margin of the continent. The composition of these sediments and their gentle dip away from the Appalachian Mountains implies that the Appalachians have stood as an eroding structural high for over 200 million years.[52]

Following a period of uplift and erosion during the Late Jurassic and Early Cretaceous, there was a transgression of Late Cretaceous seas over part of the Coastal Plain.[53] The basal clastic formation in the vicinity of the plant site is the subaerial Tuscaloosa formation.

Deposition of this formation began sometime between 100 and 94 million years ago. A period of nondeposition occurred in the Upper Cretaceous series of South Carolina between 94 and 82 million years ago,[54][55] which may be correlated in part with an erosional surface within the Tuscaloosa formation downdip of the plant site.[56][57] Following this period of erosion the sea again transgressed onto the continent, and deposition of the Tuscaloosa continued in an estuarine environment near the plant site.

Rocks deposited during the close of the Cretaceous are not present in Georgia or South Carolina.[56-58] The Cretaceous-Tertiary boundary is marked by an erosional surface which would be due, in part, to a fall in sea level.[53]

During the Cretaceous Period and continuing into the Cenozoic Era structural deformations in the form of mild regional warping and faulting, or reactivation of older faults, occurred.

The Southeast Georgia Embayment of Toulmin[59] includes an area of downwarping and sediment thickening which formed during Cretaceous and Cenozoic time.[56][60] This feature has also been called the Okefenokee Embayment[61] and the Atlantic Embayment of Georgia.[62] A second sedimentary basin, the Appalachicola Embayment, is an area of thickened Tertiary sediments extending into the southwest corner of Georgia. This feature has also been called the Southwest Georgia Basin (LeGrand, 1961;[63] Murray, 1961[64]). Between these two embayments is a positive feature called the Central Georgia Uplift,[61] which is defined as a southeast-northwest striking upwarped feature between the two flanking downwarped areas. The southern extension of the Central Georgia Uplift is the Peninsula Arch[65] which also forms the spine of Florida.

The Yamacraw Ridge is a basement feature trending parallel to the Georgia and South Carolina coastlines.[60] Maps by Herrick and Vorhis[62] show that this feature may have had some influence on Upper Cretaceous sedimentation.[56]

Several small undulations appear within the confines of the Appalachicola Embayment. Most have been recognized from subsurface data, although a few are expressed as surface features. The folding in southwestern Georgia appears to be of Tertiary age, and some folding may have occurred as late as Miocene.[60][66]

Faults with minor displacement of Cretaceous and Cenozoic deposits are present in the southeastern United States.[67] Recent detailed work has indicated that northeast-trending faults with Later Cretaceous and Cenozoic displacements such as the Belair, Cooke, and Stafford fault zones exist in the Atlantic Coastal Plain and Piedmont.[40][68][69] Wentworth and Mergner-Keefer[70] propose that many of these faults may be reactivated Mesozoic and older high angle normal faults. Other isolated instances of Cretaceous and Cenozoic faulting in the coastal plain region have been listed by Prowell.[71]

Herrick and Vorhis [62] identified a feature they called the Gulf Trough within the Appalachicola Embayment from isopach and structure contour maps prepared from subsurface data. The structure is linear and more sharply defined than the surrounding folds, and for these reasons Cramer[60] concludes that the trough was formed by faulting. In an earlier work, Callahan[72] interpreted this feature as two parallel, down-to-the-southeast faults. Cramer and Arden[56] also suggested evidence for faulting within the trough. Various authors have suggested possible mechanisms for its formation.[56][62][73][74] Patterson and Herrick[74] have reviewed the proposals which include: normal faulting producing a graben; down-warping forming a syncline; and a Tertiary marine strait or valley.

2.5.1.1.2.3 Cenozoic Era.

2.5.1.1.2.3.1 Tertiary Period.

2.5.1.1.2.3.1.1 Paleocene Epoch. Sediments deposited during the early Paleocene are thickest in the southwest, indicating that seas transgressed from that direction. Following this period of deposition, uplift of the region resulted in the erosion and removal of most of these rocks in Georgia.[56][75]This uplift was accompanied by faulting in response to the tectonic forces resulting from the northwestward drift of a passive continental margin.[76]

A second transgression occurred in the late Paleocene.[56][75] Although this transgression is thought to have been extensive, no upper Paleocene sediments are interpreted to exist in the plant site area.

2.5.1.1.2.3.1.2 Eocene Epoch. Following a period of erosion during the early Eocene, the sea again transgressed over the Georgia Coastal Plain during the middle Eocene. The bulk of the middle Eocene sediments are carbonates, with up to 10% chert and evaporite. Toward the Fall Line all of the carbonate rocks become coarser and grade into calcareous sands, indicating a higher energy environment. Onlap of the marine sediments onto the Coastal Plain is evident and paleontological data indicate that the transgression was very slow.[56] Following the transgression of the middle Eocene seas, regression again occurred and erosion of the middle Eocene deposits began.

Late Eocene deposition is represented by a relatively thin, uniform blanket of shelf limestones and calcareous sands, which unconformably overlie deposits of middle Eocene age. Northeastward along the Fall Line the fluctuating strandline of the middle Eocene sea is apparent in the intertonguing of carbonate and clastic formations. A period of regression is apparent, and deposits of late Eocene age are overlain by upper Oligocene deposits.

2.5.1.1.2.3.1.3 Oligocene Epoch. At least two transgression/regression cycles occurred during the Oligocene. Only the late Oligocene transgression deposited material in the site area. The full extent of this overlap (Suwannee) is not known, since an undetermined quantity of updip rocks have been removed by erosion. Facies patterns indicate that the overlap was probably extensive. The Suwannee rocks that remain are shelf deposits, with none of the updip clastic facies preserved.

2.5.1.1.2.3.1.4 Miocene Epoch. The deposits of Miocene age appear to be a sequence of predominantly clastic sediments deposited during and following the regression of the coastline. In some places (VEGP site included) erosion has continued from the Miocene to the present.

2.5.1.1.2.3.2 Quaternary Period. During Pleistocene time the sea transgressed over the eastern part of the Coastal Plain several times. Each transgression/regression cycle left a distinct terrace as evidence of its occurrence. Surface uplift and subsidence of the Coastal Plain of Georgia and surrounding states continued through the Pleistocene.[77] Sediments have accumulated and related geomorphic features such as erosional scarps, and terraces have continued to develop over the last 1.8 million years.

2.5.1.1.3 Stratigraphy and Lithology

The stratigraphic nomenclature used to describe the formations in the Coastal Plain of Georgia has recently undergone a certain amount of reinterpretation and some of the formation names and intraformational boundaries are currently being changed.

The names of the stratigraphic units may change, but the formations are distinct, both lithologically and geophysically, and can be correlated. The stratigraphic nomenclature adopted for this report accepts, in general, the current thought of geologists working in the area. The terminology used was selected because, although not yet completely formalized, it is thought to accurately represent the best knowledge of the stratigraphic framework of the study area. A correlation chart (drawing AX6DD339) and a lithologic chart (drawing AX6DD340) are provided for reference.

2.5.1.1.3.1 Precambrian and Paleozoic Rocks. The crystalline basement rock ranges in age from late Precambrian through Paleozoic. The basement rocks exposed northwest of the VEGP site include the gneisses and granites of the Kiokee Belt and the phyllites and greenstones of the Belair Belt.[78][79] The upper surface of the basement rock has been eroded, tilted to the southeast, and buried. The general plane of this surface strikes approximately N62°E and dips southeast at 36 ft/mi.[80]

2.5.1.1.3.2 Mesozoic Rocks.

2.5.1.1.3.2.1 Triassic System. Triassic basins occur along the eastern seaboard from Connecticut south to Florida (drawing AX6DD409). Basins north of South Carolina are exposed in Piedmont crystalline rocks, while those south of North Carolina are overlain by Cretaceous and Cenozoic sediments.

The sediments within these basins have been tentatively correlated with the Newark Supergroup of Late Triassic through Early Jurassic age.[45][80-83] It is difficult to obtain an accurate age for the sedimentary rocks within these basins due to their time-transgressive nature.[45]

As shown on drawing AX6DD409 the plant site is underlain by the buried Dunbarton Triassic Basin. The sediments within this basin have been identified as Triassic, based on stratigraphic position and lithology. No microfossils[49][50][84] or igneous rocks[51][85] indicating Jurassic age have been found in the Dunbarton Basin.

Marine and Siple[84] have presented a complete lithologic description of the Triassic rocks of the Dunbarton Basin based on drill cores. In the central northwest portion of the basin, sediments consist of red-brown breccias in a matrix of claystone and siltstone. The central part of the basin is composed of alternating layers of sandstone and mudstone. Rocks from what may be the southeastern part of the basin include siltstones, claystones, and fine-grained sandstones which contain calcareous nodules.

2.5.1.1.3.2.2 Cretaceous System. The Upper Cretaceous Tuscaloosa Formation consists of fluvial and estuarine deposits of cross-bedded arkosic sand and minor gravel intercalated with lenses of variegated white, pink, red, brown, and purple silt and clay.[56][57][80] Coarse and fine sediments are interbedded in an irregular sequence and grade laterally into one another or pinch out within short distances. Abundant kaolin is present along with other clay minerals.

2.5.1.1.3.3 Cenozoic Deposits.

2.5.1.1.3.3.1 Tertiary System.

2.5.1.1.3.3.1.1 Paleocene Series. The lower Paleocene series in the vicinity of the site consists of the Ellenton and the Huber Formations (drawing AX6DD339).

Ellenton Formation

The Ellenton Formation is a dark-gray to black sandy lignitic micaceous clay interbedded with medium- to coarse-grained quartz sand. Authigenic gypsum is commonly associated. The lower part of the Ellenton is sandy lignitic clay with the sand portion becoming very coarse and gravelly.

The Ellenton is unconformable with the underlying Tuscaloosa Formation. The contact is characterized by a change in the color of the clay and in the composition of the sand. The Ellenton grades into the overlying Huber Formation in the vicinity of the plant site.(86)

Siple(80) originally assigned the Ellenton to the Late Cretaceous, but recent workers have assigned it an age of early Paleocene.(87)(88)

Huber Formation

The Huber Formation lies between the top of the Ellenton Formation and base of the overlying sands and limestones of middle Eocene age. The lithology of the Huber Formation is diverse, ranging from beds of multicolored clays, high-purity and sandy kaolin, to thick cross-bedded members of coarse, pebbly sand and conglomerate composed of boulders of pisolitic kaolin.(89) In drill cores the uppermost part of the Huber Formation shows signs of weathering and chemical reduction.

2.5.1.1.3.3.1.2 Eocene Series. The Eocene series consists of the middle Eocene Lisbon Formation and the upper Eocene Barnwell Group.

Lisbon Formation

The Lisbon Formation occurs between the top of the Huber Formation and an unconformity at the base of the Barnwell Group. In east-central Georgia the Lisbon Formation is subdivided into three members: an unnamed basal sand and limestone member, the Blue Bluff Member, and the McBean Limestone Member.

The lowermost portion consists of a quartz sand which grades both up section and downdip into a calcareous sand. Overlying these sands is a limestone. The Blue Bluff Member is a greenish- to bluish-gray, moderately hard calcareous siltstone of marl. In core holes recently drilled near the plant site the marl is thinly interbedded to laminated with isolated limestone nodules and shell fragments.(86) Updip, the McBean Limestone Member is composed of soft, gray limestone and calcareous sand. Downdip, the Blue Bluff Member interfingers with an unnamed gray calcareous sand and fossiliferous limestone.

At the plant site, the Blue Bluff Member is a bluish-gray marl. This marl forms the foundation for critical plant structures and structural backfill. This is discussed further in paragraph 2.5.1.2, subsection 2.5.4, and appendix 2B.

Barnwell Group

In east-central Georgia the Barnwell Group consists of the Clinchfield Formation which contains the Utley Limestone Member; the Dry Branch Formation which contains the Irwinton Sand, Griffins Landing, and Twiggs Clay Members; and the Tobacco Road Sand. Downdip the Barnwell Group grades into the carbonate facies of the Ocmulgee, Crystal River, and Williston Formations of the Ocala Group.

Clinchfield Formation

The Utley Limestone Member of the Clinchfield Formation is typically a sandy, glauconitic slightly argillaceous, and locally cavernous limestone of varying degrees of induration.(90)

Dry Branch Formation

The Dry Branch Formation consists of three distinct, interfingering lithofacies: a montmorillonite clay (Twiggs Clay); a distinctly bedded sand (Irwinton Sand); and an indistinctly to massively bedded, calcareous, fossiliferous sand (Griffins Landing).

The Twiggs Clay is a pale-greenish, olive-green, bluish-gray, dark-gray, or locally almost black silty clay with hackly, blocky, subconchoidal to conchoidal fracture and is found as interbeds in both the Irwinton Sand and the Griffins Landing Members of the Dry Branch Formation. The Irwinton Sand consists of fine- to medium-grained, well-sorted, deeply weathered, almost pure quartz sand that shows well developed horizontal and local cross-bedding in outcrop. Downdip, the Irwinton Sand interfingers with the Griffins Landing Member, a fairly well-sorted, massive to indistinctly bedded calcareous sand. The unit often contains lenses of Twiggs Clay associated with oyster shell beds. Downdip, the Griffins Landing grades into the Williston Formation, a nonfossiliferous, sandy equigranular limestone.

Tobacco Road Sand

The uppermost formation within the Barnwell Group is the Tobacco Road Sand, which is predominately quartz sand. The sand in the Tobacco Road varies from fine-grained and well-sorted to very coarse-grained, granular, pebbly and poorly sorted. The Tobacco Road is characteristically massively bedded and bioturbated, although locally the formation may be thinly and distinctly bedded, even laminated.(90)(91) In east-central Georgia the Tobacco Road Sand grades downdip into the limestone facies of the Ocmulgee Formation. Further downdip, the Ocmulgee grades into the Crystal River Formation.(86)

2.5.1.1.3.3.1.3 Oligocene Series.

Suwannee Limestone

Downdip of the plant site the Suwannee Limestone rests unconformably upon the Ocala Group which is the downdip equivalent of the Barnwell Group. (See drawing AX6DD339.) The basal part of the Suwannee consists of a sandy limestone that contains few fossils. Above this is a layer of predominately cream-colored, relatively soft, somewhat chalky, fossiliferous limestone. The upper part is a light-gray to cream color, dense nodular, cherty, and somewhat sandy limestone.(56)

2.5.1.1.3.3.1.4 Miocene Series. The Hawthorne Formation is the youngest Tertiary Formation in the vicinity of the plant site. The formation has been assigned to earliest Miocene (25 to 23 million years before present) or Altamaha age.(88) Hawthorne sediments include poorly sorted clayey sands and gravels, containing cross-bedded stringers of liminite-geothite pebbles. The sediments are variegated, orange through violet, with mottled or alligator-skin appearance due to weathering.

Exposures of Hawthorne and Barnwell Formation sediments in the region commonly contain patterned weathering structures. The weathering has produced an upper zone, commonly 2 to 3 ft thick, of mottled blotches and horizontal planes of offwhite bleached zones within the deep red sediments. Below this zone a series of vertical weathered fractures is found. The vertical features normally taper downward and pinch out within 10 ft of the upper Tertiary sediment surface. These features have been described as clastic dikes by various authors.

The occurrence of clastic dikes in Coastal Plain sediments has resulted primarily from alteration along near vertical fractures during a paleosol development. The material within the dikes consists of the same material as the host sediments, with some dikes containing high proportions of clay. The origin of the fractures is difficult to determine. Large exposures of dikes show polygon development associated with desiccation. Local small faults associated with solution collapse structures have dike alteration along them. In several locations the near vertical dike faults are offset by low angle reverse faults. These thrust faults are interpreted to result from later settlement and collapse, although the timing and exact relationship is unknown.

The geographic distribution of clastic dikes is the result of the paleoenvironment which caused the desiccation and alteration. The grain size and conduit geometry of the liquefaction feature studied by Cox[92] is very different from the clastic dikes found in the site area. It is concluded that clastic dikes near the site cannot be attributed to tectonic activity.

2.5.1.1.3.3.2 Quaternary System. The Quaternary system is represented by alluvial deposits consisting of coarse gravel and poorly sorted sand which occur irregularly and discontinuously in the tributary and main channels of the Savannah River.

2.5.1.1.4 Regional Structural Geology

Major structural and tectonic features in Georgia and South Carolina are shown on drawing AX6DD409. The major structural trend affecting the region is the pre-Mesozoic southern Appalachian Mountain system, exposed west of the Fall Line. Virtually all tectonic activity occurred prior to the deposition of the Cretaceous sediments east of the Fall Line. The complex folding, faulting, and shear structures that developed in the Piedmont, Blue Ridge, and Valley and Ridge Fold belts originated in the Precambrian and Paleozoic eras during one or more of the orogenic episodes associated with the development of the southern Appalachians. Relatively undeformed Coastal Plain sediments show that this orogenic activity ceased prior to the Cretaceous. Additional evidence for the pre-Cretaceous termination of orogenic activity is the lack of offset of the numerous Triassic diabase dikes which cross the earlier structural features.

The crystalline basement underlying the Georgia Coastal Plain dips toward the southeast at approximately 36 ft/mi. This regional dip is interrupted by several local structures.

2.5.1.1.4.1 Tectonic Framework of the Georgia Coastal Plain.

2.5.1.1.4.1.1 Triassic Features. The Dunbarton Basin (drawing AX6DD409) is one of several elongated basins filled with Triassic (and in some other cases Jurassic) rocks found buried beneath the Cretaceous and Cenezoic age sediments of the Georgia Coastal Plain.

The most probable origin of the Dunbarton Basin is the formation of a graben by normal faulting. Evidence has been presented for a northwest border fault of unknown displacement, and faulting has been hypothesized for the southeastern margin.[49] Substantial evidence for a southeastern border fault is lacking, however, and the nature and extent of this margin of the Dunbarton Basin is derived from gravity and aeromagnetic surveys.[49][84] The basin is oriented

northeast-southwest and is about 31 miles long and 6 miles wide (drawing AX6DD409) based on an aeromagnetic survey.

Recent geophysical studies have indicated the possibility of intrabasinal faulting, but an attempt to verify this by analyzing drill cores was inconclusive.[49][84]

Because of stratigraphic thickness and the nature of the gravity and magnetic data, faulting is a likely explanation for the southeastern boundary of the Dunbarton Basin.

2.5.1.1.4.1.2 Cretaceous and Cenozoic Features. The dominant structural features of the Georgia Coastal Plain are two large sedimentary basins separated by structural drawing AX6DD409). The southeast Georgia Embayment[59] includes an area of downwarping and sediment thickening which formed during Cretaceous and Cenozoic time.[56][60] A second sedimentary basin, the Appalachicola Embayment, is an area of thickened Tertiary sediments into the southwest corner of Georgia.[61] Between these two embayments is a positive feature called the Central Georgia Uplift,[61] which is defined as a southeast-northwest striking upwarped feature between the two flanking downwarped areas. The southern extension of the Central Georgia Uplift is the Peninsular Arch,[65] which also forms the spine of Florida. The Yamacraw Ridge is a basement feature trending parallel to the coastlines of Georgia and South Carolina which may have had some influence on Upper Cretaceous sedimentation.[60]

2.5.1.1.4.2 Faulting. Faults with minor displacement of Cretaceous and Cenozoic deposits are present in the southeastern United States.[67] The geology of the southeastern Atlantic Coastal Plain, however, is such that faulting is not easily recognized. Recent detailed work has demonstrated that northeast-trending faults with Late Cretaceous and Cenozoic reverse displacements do exist in the Atlantic Coastal Plain and Piedmont.[40][68][69][93]

2.5.1.1.4.2.1 Belair Fault Zone. The Belair fault zone is a structural feature extending along the inner margin of the Atlantic Coastal Plain (drawing AX6DD409). This fault is located a few miles west of Augusta and extends for about 29 miles from Fort Gordon Military Reservation on the south to a quarry just west of the Savannah River on the north.[78][69][94]

The Belair fault zone has been shown to consist of at least eight en echelon reverse faults trending from N23°E to N50°E and dipping 50° to the southeast.[69] The fault zone juxtaposes crystalline phyllite of the Little River Series of late Precambrian or Cambrian age with Coastal Plain kaolinitic sands and gravels, which are formally correlated with the Upper Cretaceous Tuscaloosa Formation.[46][93] Individual fault segments are from 1 to 3 miles in length, with gouge zones only a few feet wide at most. According to Prowell and others[94] the basal Tuscaloosa unconformity is vertically displaced from 15 to 100 ft. The most recent documentable movement along the Belair fault zone occurred about 40 million years ago.[46][70]

2.5.1.1.4.2.2 Gulf Trough. Much controversy surrounds the structure and origin of the Gulf Trough of Georgia (drawing AX6DD409). The structure is linear and more sharply defined than the surrounding folds, and for these reasons Cramer[60] concludes that the trough was formed by faulting. If the Gulf Trough is due to faulting, the available data indicate that this movement would have occurred prior to the beginning of the Miocene.[62][95] The evidence suggests that other geologic phenomena, such as erosion, local variations in regional tilt, local subsidence, or warping can also explain the trough.[95]

The origin of the Gulf Trough is not clear. Various authors have suggested possible mechanisms for its formation.[60][73][74][96] Patterson and Herrick[74] have reviewed the proposals which include: (1) normal faulting producing a graben, (2) downwarping forming a syncline, and (3) a Tertiary marine strait or valley.

2.5.1.2 Site Geology

2.5.1.2.1 Site Physiography and Geomorphology

The site is located near the boundary between two topographic subdivisions of the Atlantic Coastal Plain province (drawing AX6DD338). These are the Tifton Upland to the southwest, upon which the site is located, and the older terraces to the northeast. The nearly flat topography of the older terraces is separated from the moderately hilly Tifton Upland by an abrupt 70- to 100-ft-high bluff cut by the Savannah River, which flows along its base.

2.5.1.2.1.1 Tifton Upland. The plant is located on rolling hills at about el 300 ft. Elevations in the area range from 80 ft at the Savannah River to 280 ft at the crest of a knoll near the plant. Surface drainage is primarily northeastward toward the river via a dendritic stream pattern which surrounds the property. Rainfall is relatively evenly distributed on a monthly basis, and, except during heavy storms, rain tends to soak in rather than run off. The solution and removal of carbonates from shallow underlying beds of calcareous sands and shells have resulted in the formation of local depressions, creating areas of internal drainage. Since these soluble zones occur within nearly horizontal strata resting upon an essentially impervious, hard, clay marl, springs generally have emerged at the top of exposures of the marl, causing sapping and headward erosion of the overlying sands and clays and the formation of amphitheaters and eventually ravines. Where shell deposits are thick, small-scale cavernous conditions occur along preferred percolation paths. The coalescing of the solution depressions or collapse of these small subterranean channels on the top of the clay marl results in ravines with apparently small drainage areas and with amphitheaters at the head.

2.5.1.2.1.2 Older Terraces. The older terraces subdivision of the Atlantic Coastal Plain province (drawing AX6DD338) is represented principally by the Savannah River alluvial plain, which in the site area is broad and flat and at an elevation of 80 to 90 ft. The river valley is broad and mature and includes low, dissected, old marine terraces as well as various river plain features, such as cutoff oxbows and natural levees.

2.5.1.2.1.3 Site Geologic History. The site area is located upon a seaward-thickening wedge of sediments 950 ft thick at the plant, deposited upon the truncated and peneplained roots of the ancestral southern Appalachian Mountain system (paragraph 2.5.1.1.2). Igneous and metamorphic rocks of Precambrian through Paleozoic age and early Mesozoic Triassic sediments comprise the basement rock at the site. These deformed and faulted basement rocks reflect the complex geologic history of the Appalachian Mountain system, which has been essentially quiescent since late Mesozoic peneplanation. This period of tectonic stability during and following deposition of the sediments is evidenced by their nearly flat-lying and relatively undeformed nature. The seaward thickening of the sedimentary mantle indicates a progressive downwarping of the continental margin. Regional uplift of the Coastal Plain is the latest and current stage of the geologic history of the site area.

The stratigraphic and structural relationships of the lithologic units at the site reflect the geologic history of the region. The site area was relatively stable following the deposition of the nonmarine Cretaceous Tuscaloosa Formation and the overlying Ellenton Formation of Early Paleocene age. Unconformably overlying the Ellenton Formation in the site area are the Lisbon Formation and Barnwell Group of Eocene age, which are in turn unconformably overlain by the Hawthorne Formation of Miocene age. The Hawthorne Formation is the youngest deposit of formation status exposed in the site area.

The Tertiary shallow marine deposits represent periods of marine transgressions and regressions from Eocene through Miocene times, most likely the result of periods of minor regional uplift and subsidence. For example, the Barnwell Group includes lithologic units varying from coarse sand to clay and marl, zones of weathering and variations in fossil abundances indicative of variable near-shore and tidal conditions.

The current stage of regional uplift is evidenced in the site area by exposures of the Miocene marine Hawthorne Formation at elevations above 250 ft. The mature geomorphic expression and deep weathering of the Hawthorne Formation and exposures of the underlying Barnwell Group indicate an extended period of orderly erosion on a stable surface of emergence.

2.5.1.2.2 Site Lithology and Stratigraphy

The site lithology has been determined from the following:

- Geologic and foundation exploration borings.
- Seismic refraction surveys.
- Correlations between holes using spontaneous potential, resistivity, and gamma logs.
- Geological mapping of the surface and foundation excavations for plant structures.
- Millett fault study of 1982.

Surface distribution of geologic materials is shown on the 5-mile-radius local geologic map (drawing AX6DD345) and the site geologic map (drawing AX6DD351). Subsurface geological conditions are shown on the geologic sections (drawing AX6DD352). The drill logs of borings are discussed in appendix 2B. The stratigraphic succession of the lithologic units at the site and as used in this report is shown on drawings AX6DD339 and AX6DD340.

The geologic formations which were encountered during site exploration are shown on drawing AX6DD352 and table 2.5.1-1 and include materials ranging in age from Cretaceous to Eocene. Cretaceous sediments are known to underlie the site area and crop out a little more than 5 miles northeasterly from the site near the old town site of Ellenton. However, no materials identified as Cretaceous crop out within a radius of 5 miles of the site.

2.5.1.2.2.1 Pre-Tertiary. The pre-Tertiary rocks which underlie the site are described in paragraph 2.5.1.1. To summarize, approximately 600 ft of Cretaceous sediments rest unconformably upon a truncated and peneplained lithologic complex of Triassic, Paleozoic, and Precambrian age composed of indurated sediments, intrusive and extrusive igneous rocks, and metamorphic rocks.

2.5.1.2.2.2 Tertiary System. In the site area, all geologic exposures are sediments of Eocene through Miocene age, except for local alluvial cover. Most exploratory drill hole intercepts include sediments of Eocene age and, where drilling started at higher ground surface elevations, sediments of Miocene age. Deep borings, such as TW-1, encountered Paleocene and Cretaceous sediments. The regional stratigraphy is discussed in paragraph 2.5.1.1 and shown on drawings AX6DD339 and AX6DD340. The generalized lithology of the site, which is based in part on data obtained from exploratory drilling at the vicinity of the plant site, is presented in table 2.5.1-1.

2.5.1.2.2.2.1 Eocene Series. The Eocene series in the site area consists of two lithologic units. The older is the Lisbon Formation, which includes the bearing unit for the plant structures; the younger is the Barnwell Group. The local lithologic characteristics and

stratigraphy of these formations are summarized in table 2.5.1-1 and discussed in paragraphs 2.5.1.2.2.2.1.1 and 2.5.1.2.2.2.1.2.

2.5.1.2.2.2.1.1 Lisbon Formation. The Lisbon Formation of middle Eocene age is exposed only along the Georgia side of the Savannah River. In general, the exposed lithologic unit of this formation is an approximately 60-ft-thick, greenish-gray, fossiliferous clay marl with intercalated lenses of limestone. This clay marl unit, which is the bearing bed for the plant structures, is the Blue Bluff Member of the Lisbon Formation.

The lower portion of the Lisbon Formation, which is known in the site area only from exploration drilling, consists of an unnamed, approximately 100-ft-thick bed of fine-grained sand. The lower contact of the Lisbon Formation with the Paleocene Huber and Ellenton Formations is not exposed in the mapping area. Below the Lisbon Formation is the approximately 50-ft-thick lithologic unit comprised of interbedded clay, silty sand, and lignitic beds representing the Huber and Ellenton Formations. The upper contact of the Lisbon Formation with the Barnwell Group is well exposed in the power block excavation for the VEGP and along the Savannah River in the vicinity of the plant.

The best natural exposures of the Lisbon Formation within the mapping area are at Blue Bluff. They are described in detail in the report of investigation of the marl.[97] Excellent exposure of the Lisbon Formation in the auxiliary building excavation for the VEGP was mapped and described in the reports of the power block excavations.[98][99] Drawings AX6DD352, AX6DD360, AX6DD361, AX6DD362, AX6DD363, AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, and AX6DD370 show in detail a lithologic stratigraphic succession similar to that reported in the section at Blue Bluff.

Numerous black shark teeth were found in the interval immediately below the marl, and microfossil analysis of a sample taken from hole 152 just below the base of the Blue Bluff marl indicates an Eocene age for this material.

The Blue Bluff marl is a distinct unit that is relatively constant in thickness over many square miles, although variable in lithology. As may be seen from drawing AX6DD352, the marl has been eroded from much of the Savannah River flood plain and covered over in part by the higher river terraces. It is completely eroded from the section in hole 36. In hole 45, some 3 miles farther away, a facies change has occurred, with the marl becoming dense gray-green, silty sand and silty clay.

Parallel to the river, however, it is 50 ft thick at Shell Bluff, approximately 1.5 miles northwest of the site, and 65 ft thick at hole 156 on the Griffin Landing Road, nearly 5 miles to the southeast. Isopachs of the marl in the plant area are shown on drawing AX6DD371.

2.5.1.2.2.2.1.1.1 Lisbon Formation in the Power Block Excavation. The upper Eocene Lisbon Formation is represented in the site area by the Blue Bluff Member marl, which is the foundation for structures in the power block area. The marl has a total thickness of about 70 ft in the site area (on drawing AX6DD371). The upper approximately 25 ft of the marl were exposed in excavations and mapped in detail (drawings AX6DD360, AX6DD361, AX6DD362, AX6DD363, AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, AX6DD370, AX6DD372, AX6DD373, AX6DD374, and AX6DD375). A vertical section between el 108.6 ft (final excavated grade) and el 132 ft was exposed in the auxiliary building basement excavation. Ten subunits of the marl were recognized and mapped in this vertical section. The subunits, designated A through J are shown on drawings AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, and AX6DD370 and described in the following paragraphs.

Unit A, near the top of the excavation walls, is generally above el 128 ft to the upper contact of the marl with the Utley Limestone Member of the Barnwell Group. It consists of dark-gray, silty to clayey marl with very fine light-gray to white, fine, sandy laminations which are undulatory and discontinuous.

Scattered shell fragments and well-cemented lenses of sand up to 0.1 ft thick are present locally. The laminations are oriented parallel to the lower contact of the unit, and parting along the laminations is common. Unit A is dense and well consolidated. Surfaces exposed to the atmosphere tend to dessicate rapidly. Unit A interfingers with the underlying unit B. This was especially evident in the south wall of the vicinity of stations 0+70, 1+50, and 4+30 (drawings AX6DD365, AX6DD367, and AX6DD368). The contact with unit B is everywhere gradational.

Unit B, directly beneath unit A, was continuous around the auxiliary building basement excavation walls and varies from 1 to over 4 ft in thickness. It consists of massive to faintly laminated gray, sandy marl. It has a sugary texture and does not tend to dessicate as readily as does unit A. This property provides an easy means for differentiating the units after exposure to the atmosphere. Unit B is dense but poorly cemented and contains widely scattered shell fragments. A subunit of B, designated B_1, has been identified and is present locally within B. This subunit consists of laminated sandy marl which is locally fossiliferous. Subunit B_1 has been mapped at the base of B in the east wall and the easterly portions of the north and south walls. (For example, see drawing AX6DD366.) The contacts between B and B_1 are highly gradational.

Unit B is in turn underlain by a thin, relatively discontinuous but laterally extensive limestone, designated as unit C. This limestone is light gray and well indurated, and it exhibits conchoidal fracturing. It was continuous in the west end of the south wall but becomes discontinuous east of station 0+80. East of station 3+65, the limestone becomes a series of small, irregular, discontinuous pods at varying elevations (drawing AX6DD367). Where exposed in the north, east, and west walls, the limestone formed discontinuous lenses at a relatively consistent elevation. It averaged about 1 ft in thickness and dipped slightly to the east, being present at about el 127 to 128 ft at the west end of the auxiliary building and 125 ft at the east end.

During excavation of the auxiliary building basement, the irregularity of portions of unit C led to a special study to determine whether the irregularities could be related to fault offset. The concern was that lenses and pods of the limestone occurring at slightly different elevations might have been offset from one another. The study focused on an area of the south wall at station 2+80 and the north wall at station 1+70 (drawings AX6DD365 and AX6DD367). As both excavation and mapping of stratigraphically lower units progressed, it became very evident that the irregularities of unit C were due to processes other than faulting. The continuity of the lower units in the areas of interest precluded the possibility of fault offset. A report prepared by Bechtel[98] concluded that the only plausible explanation for the observed irregularities was a combination of erosional and depositional processes.

Underlying the limestone of unit C is medium-gray, highly fossiliferous, sandy to silty marl, designated as unit D. This zone, averaging 8 ft in thickness, was continuous around the walls of the auxiliary building excavation. The lithology of unit D is very uniform, and its upper and lower contacts are quite sharp. An abundance of pelecypods retaining both valves characterizes this unit. Near the base, a number of very hard, lime-cemented pods and lenses are present at roughly equivalent elevations and have highly gradational contacts with the surrounding marl. These pods and lenses are believed to represent accumulations of calcium carbonate cement leached from the surrounding fossiliferous marl. They are collectively considered to be a subunit of D, designated D_1.

Unit E underlies D and is thin, relatively continuous, impure limestone. It is light gray, very well indurated, and fossiliferous. It averages 1 ft in thickness and varies in elevation from 121 ft in

the northwest corner of the auxiliary building to 116 ft in the southeast corner. Locally, unit E is difficult to distinguish from D_1. This was seen in the north wall between stations 1+40 and 1+70 (drawing AX6DD365), where E is discontinuous and D_1 is represented by some fairly continuous lenses. In these cases, unit E is arbitrarily selected as the unit displaying the sharpest contacts with surrounding units and the one stratigraphically between the overlying unit D and underlying unit F. The similarity between portions of E and D_1 suggests that both may be cemented deposits resulting from leaching and redeposition of calcium carbonate from the overlying fossiliferous deposits. The relative continuity of E indicates a basic permeability change occurring at the horizon in the geologic past. This is a basis for differentiating the overlying unit D and underlying unit F.

Unit F, like D, is a fossiliferous marl which was seen to be continuous around the basement excavation walls. It is medium gray and sandy to silty; it varies in thickness from 1 to 4 ft. It is dense and well consolidated but poorly cemented and tends to dessicate upon exposure to the atmosphere. Unit F includes some cemented limey pods similar to D_1. These have gradational contacts with surrounding material and appear to be secondary in origin.

Unit G is light-to-dark gray laminated marl, which is present locally as lenses interfingering with units F and H. It was relatively continuous in the westerly portion of the south wall but pinches out at station 1+50. It reappeared between stations 1+85 and 2+25 (drawings AX6DD367 and AX6DD368) but then disappeared from the remainder of the south wall. It was present in portions of the west and north walls and was absent in the east wall. The unit is characterized by very fine, sinuous, and discontinuous sandy laminations; scattered shell fragments; and small, lenticular clay pods. It contains scattered carbonaceous lenses and is well consolidated.

Unit H underlies G and consists of massive gray marl which was continuous around the excavation. It is dense, well consolidated, and poorly cemented. Shell fragments are sparse in the upper part of the unit but become increasingly abundant toward the base. Unit H varies in thickness from 1 to 6 ft.

Unit I underlies H and is very similar to unit E. It is a thin, relatively continuous, light-gray, impure limestone which is generally less than 1 ft thick. It was continuous around the excavation walls, with the exception of the east wall between station 0+79 and the south end of the wall, where it was absent.

Unit J, the deepest marl unit exposed in the auxiliary building excavation, consists of medium gray, massive, fossiliferous marl similar to the stratigraphically higher units D and F. It was continuous around the excavation walls, with the exception of the east end of the excavation, where the upper contact of the unit dipped beneath the base of the excavation.

From the preceding descriptions, it is seen that the portion of the marl section exposed in the auxiliary building excavation represents cycles of fossil abundance and absence, interspersed with periods of formation of secondary limestone pods and lenses as a result of leaching of calcium carbonate from fossiliferous zones. Erosional and depositional processes have combined to create some of the interfingering of units as well as the irregularity of some of the limestone layers.

The upper contact of the Lisbon Formation was exposed around the perimeter of the power block excavation, because it exists at an elevation higher than the top of the more localized auxiliary building excavation. The top of the Lisbon Formation corresponds with the top of the Blue Bluff marl. This upper contact was examined in detail and surveyed. It varies in elevation from a high of 138.6 ft on the north side of the excavation to a low of 132.0 ft on the south side. The contact is erosional with very minor relief present. The uppermost few feet of the marl are locally weathered to a greenish color, and bioturbations (disturbance of the sediment due to the activity of organisms) were noted locally.

2.5.1.2.2.2.1.2 Barnwell Group. Late Eocene beds known as the Barnwell Group (paragraph 2.5.1.1.3.3.1.2) are present over much of the area within 5 miles of the site (drawing AX6DD345).

The formation is primarily comprised of tan, yellow, red, and white sands and clayey sands, although exposures of claystone, shelly limestone, and reef deposits are common. The Barnwell Group is comprised of four basic lithologic units, shown on drawing AX6DD352, which are listed below, from oldest to youngest:

- Utley Limestone.
- Twiggs Clay Member.
- Irwinton Sand Member.
- Tobacco Road Sand (Upper Sand).
- Upper Sand (Tobacco Road Sand).

The Barnwell Group rests unconformably upon an erosion surface at the top of the Lisbon Formation. The Barnwell Group is shown as a single lithologic unit on the 5-mile-radius geologic map (drawing AX6DD345).

The best exposures of the Barnwell Group within the mapping area include the power block excavation for the VEGP, a road cut just west of the intersection of River Road and Little Beaver Dam Creek, the intersection of River Road and Newberry Creek, and the intersection of the road just south of the railroad and Newberry Creek. Excellent exposures outside the mapping area which should be noted include a road cut just south of Brier Creek on Thomas Bridge Road and just south of McBean on State Highway 56. The Barnwell Formation was also encountered in numerous exploratory drill holes, which add to the data collected from surface exposure.

The oldest unit of the Barnwell Group is the Utley Limestone Member of the Clinchfield Formation. The Utley Limestone is a white to light-gray fossiliferous limestone, which has been referred to as the shell zone. The limestone was well exposed in the power block excavation for the VEGP and is locally exposed along the Georgia side of the Savannah River. This limestone layer, which is also thought to be of middle Eocene age, exhibits the effects of leaching. Surface topography, losses of drilling fluids during the exploratory drilling, and direct visual observation in the excavation and natural exposures all indicate the presence of solution cavities. The thickness of this unit varies from 0 to 100 ft.

Locally overlying the Utley Limestone of the Barnwell Group is the Late Eocene, Twiggs Clay Member. The Twiggs Clay was exposed only in the power block excavation where it was a medium-gray, moderately hard, sandy claystone. The upper 2 to 5 ft are weathered to greenish-gray, reflecting the unconformable relationship with the overlying sand units.

Unconformably overlying the Twiggs Clay is the Irwinton Sand Member. The Irwinton Sand is present through much of the mapped area. Although the Irwinton Sand was well exposed in the power block excavation, it apparently pinches out to the west.

The Irwinton Sand is typically represented by unconsolidated, tan, and white, medium-grained sand and clayey sand. The sands are typically massive, although some cross-bedding is present. Tan clay seams and clayey zones along with scattered shell fragments and carbonaceous zones are present. The upper few feet of this unit in the power block excavation

are comprised of shell fragments in a matrix of clay with manganese straining, providing a relatively sharp contact with the overlying sands.

Overlying the Irwinton Sand is the Tobacco Road Sand unit of the Barnwell Group. Tobacco Road Sand is typically red, although yellow, brown, tan, and mottled units are present. The sand is typically medium grained and locally cross-bedded. This sand unit is present throughout much of the mapping area and was particularly well exposed in the power block excavation and near the intersection of River Road and Little Beaver Dam Creek.

The upper portion of the Tobacco Road Sand locally contains lenses of limestone or relic features of limestone which have been leached. This limestone is well exposed near the intersection of Brier Creek and Thomas Bridge Road and near the intersection of the railroad and Newberry Creek. It should be noted that sands of the upper Barnwell Group are affected by surface weathering, forming mottled clayey sands and in many road cuts.

2.5.1.2.2.2.1.2.1 Barnwell Group in the Power Block Excavation. All of the sediments that were exposed in the sidewalls of the power block excavation are of Eocene age. Above the Blue Bluff marl of the Lisbon Formation, the exposures were comprised entirely of sedimentary beds of the Barnwell Group and include the units referred to in paragraph 2.5.1.2.2.2.1.1. The stratigraphic column shown on drawing AX6DD352 gives a summary of the lithologic characteristics of these sediments, as encountered in the power block excavation. Drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363, detailed geologic maps of the geological investigation of the excavation, provides comprehensive lithologic descriptions of the Barnwell Group sediments found in the excavation.

Although examined and described in detail, the deposits between the top of the Blue Bluff marl and approximately el 170 ft could not be mapped in detail. Consequently, the geologic map of the power block excavation (drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363) shows only the detailed lithology of the Tobacco Road Sand and the upper portion of the Irwinton Sand. This was due to extensive slumping of the slopes when excavation and dewatering were suspended during the period between September 1974 and June 1976. Extensive regrading obscured the contacts between units in this zone. Several portions of the slopes were covered with riprap in order to control seepage and improve stability, further obscuring contacts. Since seepage from the slopes was creating local stability problems, it was decided not to excavate back into the slopes to expose contacts and risk large slope stability problems. Detailed mapping of the units above and below this zone demonstrated the continuity of the strata and the absence of faulting.

The lowermost exposed unit within the Barnwell Group in the power block excavation is the Utley Limestone. The lower part of the limestone is grayish yellow, well indurated, and fossiliferous, grading locally into coquina. It was continuous around the power block excavation and varies in thickness from 0.5 to 3 ft. The upper part of the limestone is white to light gray and varies from 0 to 12 ft in thickness, present only in the north and northwest portions of the power block excavations. Although well indurated, this thicker limestone has been subjected to extensive leaching, producing a honeycomb network of cavities. Some individual cavities had mean diameters of several feet before being removed by excavation or filled in place. The location and extent of the cavities exposed in the slopes are shown on drawing AX6DD363. The filling of cavities in the limestone intersected by the excavation slopes is described in paragraph 2.5.4.5. Within the cavities, the limestone typically displayed a weathered and soft zone immediately adjacent to the cavity walls, which graded within a few inches to hard, unweathered limestone. Locally, extensive leaching of the limestone had left a residue of silt and clay impurities forming a soft mottled blackish material. Included in the Utley Limestone is a highly fossiliferous clay deposit which varies in color from tan to dark gray. The difference in colors appears to be due primarily to weathering effects. Prior to its removal, this clay was

present mainly in the northwest portion of the power block excavation. It contains abundant specimens of the oyster Crassostrea gigantissima, a key Eocene near-shore pelecypod. Lesser quantities of other pelecypods, gastropods, arthropod arts, and shark teeth have been identified in this clay.(78)

Unconformably overlying the Utley Limestone is the Twiggs Clay. This consists primarily of medium-gray, moderately indurated, laminated sandy claystone, which is quite similar to the underlying Blue Bluff marl of the Lisbon Formation. The Twiggs Clay was exposed only in the southeast portion of the power block excavation and varies in thickness from 0 to 13 ft. The upper 2 to 5 ft are weathered to a distinctive greenish-yellow color. The Twiggs Clay has alternating thin and thick beds (from less than 1 in. to greater than 1 ft), with gradational contacts between beds. No joints, fractures, or discontinuities were observed in the clay.

The Irwinton Sand of the Barnwell Group unconformably overlies the Twiggs Clay in the southeast portion of the power block and overlies the Utley Limestone elsewhere. The Irwinton Sand consists of an approximately 50-ft-thick vertical sequence of sands, clays, and reef deposits. At the base of the sequence is a massive, white, quartz-rich sand deposit. The presence of fossil shrimp burrows identifies this as an intertidal deposit. The upper surface of the sand is highly irregular, with reef-type accumulations of Crassostrea gigantissima present on the highs. These shell accumulations are well cemented and highly calcareous. This sand is fine to medium grained and very well sorted, and it exhibits extensive cross-bedding. It is extremely friable and tends to rapidly slump and ravel, assuming its angle of repose soon after excavation.

Above the white sand and reef deposits is a sequence of tan sand and clay. The sand is generally fine to medium and moderately sorted, and it contains thin seams of tan clay having high plasticity. Two continuous marker horizons are present within this sequence. The first, a zone of manganese-straining and shell debris, occurs generally between el 170 and 180 ft and was somewhat higher than this on the west side of the excavation. This zone, called the shell hash horizon, varies in thickness from less than 1 in. to almost 6 ft and could be traced continuously around the excavation slopes (drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363). A second shell hash horizon is locally present beneath the first one but is discontinuous. The second marker horizon is a zone of abundant tan clay seams, which varies from approximately 1 to almost 6 ft in thickness, and was found between el 180 and 200 ft. This clay zone marks the top of the Irwinton sand.

Both of the marker horizons undulate along the strike, with flexures in the bedding reflecting underlying reef highs as well as lows due to collapse of cavities in the stratigraphically lower Utley Limestone. These flexures are discussed further in paragraph 2.5.1.2.3.4.

Above the Irwinton Sand is the Tobacco Road Sand of the Barnwell Group. This sand extended up to the top of the excavation slopes and consisted of a thick (up to 40 ft) zone of predominately red sand with zones of lavender, purple, mustard yellow, and orange sand. The color changes are due to weathering effects and are not related to structure or lithology. The sand consists of fine- to medium-quartz grains which are moderately to well sorted and angular to subrounded. Colors are imparted by clay coatings on the individual grains. Differential weathering has produced mottled zones of bright colors which form an alligator-skin effect near the top of the unit. The sand is dense, well consolidated, and completely uncemented.

At the top of the excavation slopes, recent deposits of buff-colored, alluvial and windblown sand were present locally. These deposits form a thin veneer of fine- to medium-grained, angular to subangular, well-sorted quartz sand which is highly gradational with the underlying sand of the Barnwell Group.

2.5.1.2.2.2.2 Miocene Series. The Hawthorne Formation of Miocene age caps the ridge and hills above el 200 ft around the site area and lies unconformably upon the eroded surface of the upper sand member of the Eocene Barnwell Group. The Hawthorne Formation is typically red to brown mottled sandy clay and clayey sand. Lateral facies changes, however, result in significant lithologic variations including massive and cross-bedded lavender, purple, red, and brown medium- to coarse-grained sand. Channel deposits and localized lithologic changes are well illustrated in the railroad cut just east of Daniel Grove Baptist Church.

The contact between the Hawthorne Formation and the Barnwell Group is difficult to distinguish in the field. In gross terms, all lithologies present in the upper Barnwell Group occur in the Hawthorne Formation. The only unique distinguishing property of the Hawthorne Formation is the presence of siliceous gravel near the base of the formation. This gravel is present in road cuts near the intersection of Brier Creek and Thomas Bridge Road in the road cut across from DeLaigle Trailer Park, just west of the VEGP site entrance. In areas where the gravel is not present, it is difficult to differentiate the two formations.

The clastic dikes associated with the Hawthorne Formation are discussed in paragraph 2.5.1.1.3. No exposures of these clastic dikes have been found in the vicinity of the site.

2.5.1.2.2.3 Quaternary Deposits. The Quaternary sediments in the site area consist of sands, gravels, silts, and clays of Holocene and Pleistocene ages. The Quaternary is largely represented by flood plain deposits in the Savannah River Valley and alluvial trains along the courses of larger streams tributary to the Savannah River. At the plant site elevation, the Quaternary is principally represented by erosion and weathering rather than depositional processes, although deposits of buff-colored, windblown sand are seen on higher ground.

2.5.1.2.3 Site Structural Geology

The formations underlying the site area are essentially flat lying or gently dipping to the southeast, reflecting the regional dip. The site area structure is illustrated on drawings AX6DD377 and AX6DD378, which show subsurface contours on the top and bottom of the marl. The dip in the plant site area is about 30 ft/mi in a southeasterly direction. This gentle homoclinal structure is unbroken in the area except for a gentle dip reversal which is of depositional and differential compaction origin. This feature is discussed in paragraph 2.5.1.2.3.2.

Solution depressions are apparent on the geologic map of the area (drawing AX6DD351). These features, which have been investigated and found to be confined and related to lithologic units stratigraphically above the Blue Bluff marl, are discussed in paragraphs 2.5.1.2.3.3 and 2.5.1.2.2.

2.5.1.2.3.1 Faults and Lineaments. No faults or lineaments have been found within 5 miles of the site, other than those associated with the Triassic Basin, discussed in paragraph 2.5.1.2.3.2. These structures do not extend into the overlying Tertiary deposits. Examination of sediments exposed in the walls of the power block excavation has shown no evidence of faulting (paragraph 2.5.1.2.3.4).

In early 1982, U.S. Geological Survey Open-File Report 82-156 postulated the presence of two faults in the vicinity of VEGP.[100] The Millett fault was described as a northeast trending fault extending from northern Jenkins County, Georgia, to central Barnwell County, South Carolina. Its postulated trace passed close to the small community of Girard, Georgia, about 7 miles south of VEGP. The Statesboro fault was described as a northeast trending fault extending

from central Bulloch County, Georgia, to eastern Allendale County, South Carolina. Its postulated trace was about 32 miles south of VEGP.

In response to the open-file report, a detailed and comprehensive investigation was conducted to evaluate whether or not the postulated faults were capable, i.e., had moved once in the past 35,000 years or more than once in the past 500,000 years. Based on information in the open-file report, the postulated Millett fault was of primary interest and the postulated Statesboro fault was of secondary interest to the study. A review of the data and evaluations used in the open-file report indicated that additional data and evaluations would be needed to adequately determine the capability of the postulated faults. If a fault or faults were found to be present, the capability of such faults was to be determined.

The investigations encompassed several scientific fields which address the question of faulting. These include surface geology, subsurface geologic and geophysical characteristics, ground water aquifer characteristics, surface water hydrology, and the nature and distribution of historic seismicity in the area.

To provide guidance and review of the studies, a number of eminent consultants were retained. These were chosen because their fields of expertise were related to the planned studies: Dr. Bruce Bolt, director of the seismographic station at the University of California, Berkeley; Dr. R. D. Hatcher of the University of South Carolina; Dr. V. J. Henry of the University of Georgia; Dr. P. E. LaMoreaux, president of P. E. LaMoreaux and Associates; Mr. H. LeGrand, and independent consultant in geohydrology; Dr. R. Lyon of Stanford University; Dr. S. Papadopulos, president of S. Papadopulos and Associates; Mr. Carl Savit, senior vice president of Western Geophysical; Dr. Carl Stepp, affiliated with Woodward-Clyde Consultants, and Mr. L. Wood, ground water geology specialist with S. Papadopulos and Associates.

The results of the studies conclusively demonstrate the absence of a capable fault in the vicinity of the postulated Millet fault and strongly suggest that no capable fault exists near the location of the postulated Statesboro fault.

These conclusions are based on the following:

A. Core drilling and geophysical logging clearly demonstrate subsurface continuity of beds 40 to 80 million years before present across the trace of the postulated Millett fault.

B. Acoustic reflection surveys performed in the Savannah River demonstrate continuity of subsurface strata deposited across the strike of both the postulated Millett and Statesboro faults during the last 80 million years.

C. Geologic mapping and remote sensing studies reveal no surface expression of faulting.

D. Examination of recorded and reported seismic events indicate that there is no historic seismicity which can be associated with either of the postulated faults.

E. Surface and ground water hydrology studies do not support the presence of faults.

These studies are described in detail in Bechtel Power Corporation (1982).[86]

It is concluded that no capable faults exist in the vicinity of the postulated Millett and Statesboro faults; therefore, they can have no impact on the existing accepted seismic design bases of the VEGP.

2.5.1.2.3.2 Flexures and Folds. The foundation materials for the plant structures are comprised of essentially flat-lying Tertiary sediments, which include the Lisbon Formation of Eocene age. The bearing unit for the major plant structures is the Blue Bluff marl, the upper member of the Lisbon Formation. Contours of the upper and lower surfaces as well as an isopach map of the marl in the vicinity of the plant are shown on drawings AX6DD352, AX6DD371, and AX6DD372. The contours on these map figures were derived from outcrop and drill hole information and reflect the southeasterly regional dip of approximately 30 ft/mi throughout the plant siting area. This regional dip is interrupted approximately 1000 ft northwest of the plant siting area by a gentle dip reversal of a maximum of 3° (5%) northwesterly, along a northeast-southwest trend. The apparent dip reversal is shown with two-to-one, vertical-to-horizontal exaggeration on geologic section B-B' (drawing AX6DD352), where it resembles a monoclinal flexure.

This local anomaly in the regional dip has been the subject of detailed investigation and discussion. The hypotheses which have been considered concerning the origin of the anomaly include:

- Structural monocline.
- Solution collapse.
- Erosional-depositional feature.
- Stratigraphic facies change.

Numerous holes were drilled to determine the characteristics of the dip reversal, and water pressure tests were made to determine whether it affects the watertightness of the marl. No indication that it is a fault-controlled feature was found during the extensive investigations. It does not appear to be simply an erosional feature on the top of the unit, as it is reflected in both the top and bottom of the marl to an approximately equal extent. It dips in the wrong direction to reflect possible near-surface expression of the underlying Triassic Basin boundaries. No relationship to the assumed boundary fault contact at the northern edge of the Triassic basin could be found other than coincidence of location. As the assumed northern Triassic basin boundary fault would have to be down thrown towards the sea, the fact that the flexure in the bearing horizon slopes in the opposite direction (i.e., to the northwest) seems to negate any structural relationship.

The dip reversal seems to have been formed prior to and in part during the deposition of the Lisbon Formation. A local, well-developed, striated bedding plane was found in one hole (No. 246) at the base of the Blue Bluff marl. A few short, discontinuous fractures with slick surfaces have been noted. These appear to be similar to fractures found in clay during some compaction processes. These occur near the southwestern or lower side of the anomaly and well away from the plant area. Water losses related to jointing in the upper 15 ft of the marl were noted during the exploratory drilling investigation of this feature. These phenomena were not observed elsewhere throughout the plant site investigation. It is believed that the reversal represents deposition on an erosional irregularity on the underlying sands, with the possibility of some local differential compaction during or shortly after deposition of the overlying stratum.

A special study was made of the available information on the Triassic basin because of the proximity of the dip reversal in the bearing horizon to the best approximation of the basin's northern border. No evidence has been found that suggests a reflection of the subsurface basin boundaries in the overlying pre-Cretaceous and Cenozoic strata. The most recent and comprehensive subsurface investigations of this basin have been undertaken in the Savannah

River plant, some 5 to 11 miles east-northeast of the site. They have been reported by Marine;[84] some of the more pertinent portions are quoted below:

> Well P5R, drilled in 1962, was the discovery well for the buried Triassic sedimentary basin; it penetrated only about 95 feet of these sedimentary beds, consisting of maroon claystone and fine-grained sandstone. Well DRB 9 was drilled in 1969 at a location selected, on the basis of the short reflection seismic survey, to penetrate the edge of the Triassic basin and then pass into the crystalline metamorphic basement. At this location, the Triassic sedimentary rocks are 1593 feet thick and consist of maroon fanglomerate made up of clasts of pink weathered gneiss set in a maroon siltstone matrix. The Triassic is underlain by augen gneiss, with pink feldspar augen about 0.5 inches in diameter set in a matrix of fine-grained green hornblende. The well is about 0.4 mile from the edge of the basin; the contact of the Triassic with the augen gneiss dips about 35 degrees to the southeast, as do a few faint bedding planes in the fanglomerate. Well DRB 10 was drilled to a depth of 4206 feet and penetrated about 3035 feet of red siltstone and sandstone. Crystalline metamorphic rock was not expected in this well, and none was encountered. No well in the basin has penetrated any igneous rock or coal, even though they occur in other East Coast Triassic basins.
>
> Geophysical work, consisting of reflection seismic and gravity-magnetic surveys, was done or analyzed after the information from these three wells was available. The reflection seismic surveys showed a sharp northwest boundary but did not indicate termination of the Triassic rocks where the southeast border was inferred from the aeromagnetic map. The contact of the coastal-plain beds and the Triassic rocks could be followed as a reflection but with scattered apparent discontinuities (labeled inferred faults in figure 4). The contact of the basement with the Triassic rocks could not be detected at all.
>
> Gravity and magnetic surveys were made on the ground along many of the same traverse lines as the seismic survey in an effort to estimate the thickness of the Triassic rocks and to develop more information on the possible faults indicated by the reflection seismic surveys. One of the faults interpreted from the seismic data was selected for more intensive investigation by drilling. Two wells (DRB 11 and P12R) were drilled on either side of the fault. The elevation of the top of the Triassic was the same in both wells, showing that the last movement on the fault, if it exists, was before the development of the erosional surface. The age of the erosional surface is pre-Late Cretaceous or at least 100 million years. Thus, there has been no movement on the fault in the last 100 million years. This conclusion is substantiated by correlation of distinctive peaks on the gamma ray and electrical logs of the coastal-plain beds.
>
> Well DRB 11 was designed to deviate 15 degrees from vertical in order to intercept the fault that had been indicated by the seismic and gravity magnetic surveys. The well was drilled to a total depth of 3320 feet, which represented a true vertical depth of 3278 feet and a horizontal migration of 387 feet to the northwest. The well was cored completely, but the fault indicated by the geophysical surveys was not penetrated.

Wendell Marine noted in personal communication that, based on the investigations on the Savannah River plant site, he would conclude that all known faulting antedates the erosion surface between the coastal plain sediments and the Triassic sediments in the basin.

Early exploration drilling and water-pressure testing in the Blue Bluff marl resulted in water-loss data which seemed to show the marl to be permeable, possibly due to fracturing related to folding or faulting. Later, holes drilled within 15 feet of the early holes to further investigate the apparent water losses proved the marl to be quite tight. The previously indicated losses within the bearing stratum seem to have been caused by mechanical problems. The overlying shell zone (basal limestone) varies from moderately to extremely pervious. If the drill casing was not tightly seated to form a seal where it penetrated the marl, the vibrations caused by drilling may have loosened it, allowing the drilling fluid to escape beneath the base of the casing and into the pervious shell zone. This was reported by the driller as a water loss in the marl.

A similar situation exists with the pressure tests. The Blue Bluff marl, being nodular, causes drill hole diameter fluctuations, locally quite irregular (see caliper logs, drawings AX6DD379, AX6DD380, AX6DD381, AX6DD382, AX6DD383, and AX6DD384). This condition made difficult the proper leakage-free placement of packers during the pressure tests. Consequently, an apparent water loss into the marl would be indicated in the earlier pressure tests if either packer leakage occurred or its casing was not tightly seated. The holes showing possible water loss in the marl were checked out with the 500-series holes in which the casing was grouted to the surface of the marl before coring of the marl commenced. Packer settings were adjusted until a good seal was obtained before the interval was tested. The test results of the 500-series holes show that the marl is tight, and therefore the hypothesis of fold- and fault-related fracturing is unsupported.

Solution collapse, or apparent folding caused by subsidence of the marl into an underlying zone of solution, and removal of carbonates is a reasonable hypothesis. Slump and foldlike features have been ascribed to the leaching process in the Lisbon Formation in South Carolina. However, investigative drilling has found none of the associated features, which include tension fracturing, clastic dikes, and low-density, carbonate-impoverished materials or cavities within or beneath the marl.

Erosional and depositional processes, together with differential intraformational compaction of units below or within the marl, could produce apparent warping or folding in cohesive sediments. Such features have been described in the power block excavation. However, the magnitude of differential settlement, a net elevation change of 50 ft in 1000 ft of horizontal distance, seems to be large in consideration of the sandy nature and thickness of underlying strata.

Detailed mapping of the Lisbon Formation in excavations at the plant site has demonstrated rapid lithologic changes laterally and rapid changes in the thickness of mappable units within the formation. The upper boundary of the bearing stratum, the Blue Bluff marl, is basically established by the contact between it and an overlying shell bed, the Utley Limestone. The base is generally established by the presence of an underlying sand bed. In the excavation and in investigative borings, similar lithologic sequences are repeated vertically. The principal difference is one of scale. It can be shown within the excavation that thickness changes of 12 to 15 ft in a horizontal distance of less than 200 ft can occur in the Utley Limestone and in the Twiggs Clay, which is practically indistinguishable from the Blue Bluff marl. It seems that over a distance of 1000 ft, the magnitude of change of thickness, the pinching out of key units, and the appearance of similar key units at lower elevations could create the appearance of a flexure.

In conclusion, the origin of the dip reversal, or flexure, is probably due to erosional-depositional features possibly exaggerated by stratigraphic facies changes. There is no evidence to support either a structural monocline or solution collapse hypothesis.

2.5.1.2.3.3 Solution Depressions. Solution depressions are readily apparent on the geologic map (drawing AX6DD351) of the site and were the subject of considerable investigation using surface and down-hole geophysics, coring, and pressure-testing techniques.

The depressions at the site area originate from the solution and removal of calcareous shell material present in the sand above the marl horizon. Calcareous shell material is subjected to leaching by percolating meteoric water on its migration downward from the porous sandy soil to the water table, the base of which is formed by the top of the Blue Bluff marl. Subsequent settling of the overlying material results in the formation of the shallow depressions. To indicate the relationship of the surface depressions with the Utley Limestone and bearing horizon, the depressions have been accentuated by hachuring on drawings AX6DD371, AX6DD376, AX6DD377, and AX6DD378.

The calcareous shell material of the Utley Limestone above the Blue Bluff marl is of limited extent and varies considerably in thickness, as may be seen on drawing AX6DD376. The surface depressions are restricted to areas where the shell material is, or clearly has been, present. Where thick shell accumulations are present, solution features are common, including the depressions, cirque-like ravine heads, and, locally, a stream flowing from a small solution cave. The numerous drill holes which extend through the marl have intercepted no solution-related features in the marl or beneath it. The solution features are confined to the Utley Limestone zone above the marl.

The following data are submitted in support of the shallow origin of these depressions:

A. The depressions are broad and shallow rather than deep and steep sided, as would be the case in a near-surface cavern collapse. Solution and removal of calcium carbonate from a sand-shell mixture, however, produce the type of depression found here. That the shell materials above the marl are being dissolved is indicated by the nature of the core recovered and the voids and fluid losses commonly experienced while drilling the shell zone. Conversely, the impermeable nature of the marl determined from core and pressure tests data indicates that it acts as an aquiclude, essentially preventing the further vertical migration of the surface waters.

B. In the plant vicinity the depressions are confined to areas underlain by the Utley Limestone zone. (See drawing AX6DD376.)

C. There are no disappearing streams in which the flow reappears within or below the marl of the bearing horizon, as would be expected were the depressions caused by solution and removal of calcium carbonate from beneath the Utley Limestone. On the contrary, it appears that the marl forms the base of the streams over much of their courses until they start cutting down to the Savannah River base level. To the northwest, the stream drainage into Mathes Pond is via a small solution cave in the Utley Limestone in which the Blue Bluff marl forms the cave bottom.

D. Drawings AX6DD377, AX6DD378, and AX6DD371 present contour elevations on the top of the marl, the bottom of the marl, and the thickness of the marl, respectively. If the origin of the surface depressions were related to conditions beneath the shell zone, it would be reflected in these contours. A number of the drill logs in which the datum points for these contours were obtained were from holes in or adjacent to these surface depressions. This provides good contour control in these areas for these maps. Their data clearly show no

correlation between the surface depressions and contours of the isopach maps of the top of the bearing horizon.

Thus, the depressions are caused by the solution and removal of calcareous material above the Blue Bluff marl. Since the foundation excavation for the plant has removed all materials above the marl (drawings AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, and AX6DD370) from the overlying Utley Limestone, solution and removal of carbonate do not pose a geologic hazard for the plant.

2.5.1.2.3.4 Power Block Excavation. Two separate marker horizons in the Irwinton Sand have been mapped around the side slopes of the power block (drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363). Both horizons are continuous and unbroken, demonstrating the absence of faulting in these materials. In addition, the upper contact of the stratigraphically lower Lisbon Formation marl has been mapped with survey accuracy and has also been found to be uninterrupted by offsets (drawings AX6DD372 and AX6DD373). Subunits within the marl have been mapped around the walls of the auxiliary building basement excavation. These zones were likewise found to be undisturbed by faulting. Minor stratigraphic irregularities noted were shown to be related to erosional and depositional processes in the report of marl investigation.(97)

Only one joint was recognized in the entire power block excavation, and this was of limited extent. The joint was exposed in the southeast corner of the power block excavation, where it extended from the upper surface of the marl down to el 127 ft where it terminated at a depth of about 6 ft. The joint trended N81°E; it was approximately vertical and was tightly closed. Some secondary dark-green mineralization was noted as a fine coating on the joint faces. No other joints or fractures were identified in any other lithologic units.

The strata of the Irwinton Sand exhibited flexures in the power block excavation which are related to the following phenomena:

- Differential compaction of underlying sediments.
- Subsidence due to leaching of underlying calcareous materials and collapse of solution cavities.
- Deposition on uneven surfaces.

These flexures are particularly evident in the two marker horizons of the Irwinton sand.

Evidence for differential compaction includes the typical association of highs in the marker horizons with the occurrence of underlying reef deposits of shells. These cemented shell deposits form hard spots in comparison with the relatively more compressible sediments between them. As the load of the overburden increased during deposition, the sediments between the reefs were compressed relatively more than those immediately above them. This created downward flexures in the overlying marker horizons.

Evidence for subsidence due to leaching and collapse of cavities includes the presence of solution cavities in the limestone underlying zones of downwarping of the relatively incompetent sands and clays. Recent surface depressions in the area give added evidence for subsidence. Solution activity and cavity formation are confined to the limestone above the Blue Bluff marl. No cavities have been found in the marl, and the upper surface of the marl exposed in the power block excavation contains no depressions.

2.5.1.2.4 Site Geologic History

The geologic history of the site area reflects the geologic history of the region (paragraph 2.5.1.1.2), which was an active part of the ancestral Appalachian Mountain system. The complex geologic history of the Appalachian Mountain system is reflected by the igneous and metamorphic rocks of Precambrian through Paleozoic age and Triassic sediments underlying the site, which has been essentially quiescent since the deposition of the Cretaceous sediments in the coastal plains. Although rocks comprising the basement complex were not encountered in drill holes at the site, seismic refraction surveys at the site indicate that the basement complex Cretaceous contact occurs at a depth of over 950 ft.

Due to regional elevation fluctuations following deposition of the Cretaceous Tuscaloosa Formation and the overlying Paleocene Huber and Ellenton Formations, the time interval from Eocene through Miocene was one of marine transgressions and regressions. For example, the Lisbon Formation of Eocene age, overlying the Huber/Ellenton Formations in the site area, is representative of near-shore or tidal deposits. The presence of alternating, typically shallow water sediments, along with variations in fossil abundances in the Lisbon Formation and the Barnwell Group, suggests changing near-shore environments.

Variegated clays and sands, lithologically similar to the Miocene Hawthorne Formation, were encountered in the upper portion of some of the holes drilled at the site and are also the youngest Tertiary sediments underlying the soil horizon of the higher elevations in the site area.

2.5.1.2.5 Site Geologic Maps

The site geologic map is shown on drawing AX6DD361. The 5-mile-radius geologic map of the local area is shown on drawing AX6DD345. The 5-mile-radius map shows the location of the known and inferred contacts between surface materials at the site. A detailed geologic map of the power block area, drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363, was prepared from data obtained during the geologic mapping program conducted concurrent with excavation and grading. The geologic map of the power block area and geologic data concerning the area are discussed in paragraphs 2.5.1.2.2 and 2.5.1.2.3. Geologic sections through the power block excavations are shown on drawings AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, AX6DD370, AX6DD372, and AX6DD373. The regional geologic map is shown on drawings AX6DD341 and AX6DD342.

2.5.1.2.6 Plot Plan

Information concerning the locations of major structures of the plant, including all Seismic Category 1 structures, and exploration and test borings made at the site area is presented on drawings AX6DD343, AX6DD344, and AX6DD351. These figures are discussed in subsection 2.5.4. The geologic logs of the borings are discussed in appendix 2B.

2.5.1.2.7 Subsurface Profiles and Plant Foundations

Subsurface profiles showing lithologic correlations from drill hole data are given on drawing AX6DD352. Geologic profiles and geologic sections prepared from data obtained during the geologic mapping of the foundation excavations are presented on drawings AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, AX6DD372, and AX6DD373. They are discussed in paragraphs 2.5.1.2, 2.5.1.3, and 2.5.4.5. The site ground water conditions are discussed in detail in subsection 2.4.12 and summarized in paragraph 2.5.1.2.8.7. The significant engineering characteristics of the subsurface materials are

discussed in subsection 2.5.4. All Seismic Category 1 structures are founded on the Blue Bluff marl or upon compacted structural backfill placed upon the marl. The Blue Bluff marl is a gray-green, hard, sandy to silty, calcareous clay marl with thin limestone lenses, small calcareous nodules, and occasional macrofossils.

2.5.1.2.8 Engineering Geology Evaluation

2.5.1.2.8.1 Engineering Properties of Foundation Materials. The strength of foundation materials, static and dynamic properties, bearing capacities and settlement, rebound and heave, and foundation design criteria are discussed in subsection 2.5.4.

2.5.1.2.8.2 Prior Earthquake Effects. There is no evidence to suggest that surficial or subsurface materials have been affected by prior earthquake activity. No evidence of texture faults were found from any of the site exploration borings or in the power block excavations.

2.5.1.2.8.3 Deformational Zones. Examination of outcrops, excavation exposures, and subsurface samples have revealed that there are no deformational zones within the Blue Bluff marl, which is the foundation material for the major plant structures. Approximately 1000 ft northwest of the major structures, there is, however, a dip reversal of about 3° to the northwest. This gentle dip reversal in the otherwise very gently southeasterly dipping (approximately 30 ft/mi southeasterly) homocline of Tertiary sediments is of depositional origin and does not represent a structural (tectonic) deformation. Paragraph 2.5.1.2.3 contains a discussion of this anomaly.

During the construction phase at VEGP, a comprehensive inspection program was carried out to continuously monitor and assess the condition and character of all excavated marl throughout the power block area. A total of four joints was found in the uppermost strata of the marl. Two were found during routine inspection of the exposed marl surface prior to backfilling, and two were found during inspection of the radwaste solidification building caisson foundation. Each joint was independently investigated and found to be of limited depth and aerial extent and of nontectonic origin. Evidence produced by the investigations suggests that the joints were formed either during or immediately following late-stage diagenesis of the marl. Depositional loading from overlying sediments may have been a contributing factor. There is no evidence to suggest that these features are related to any processes that have occurred within recent geologic time.

With the exception of the joints described above, no other fractures, partings, or anomalous features were found in the marl.

2.5.1.2.8.4 Zones of Alternation or Weakness. The Blue Bluff marl is basically unweathered and unaltered, except for the uppermost section, which is up to 5 ft thick and slightly discolored by weathering from gray-green to green. This weathered zone was completely removed in foundation preparation.

2.5.1.2.8.5 Bedrock Stress. Over 1000 ft of unlithified to poorly lithified sediments overlie the pre-Cretaceous basement rock at the site. Rebound of the Blue Bluff marl, the bearing stratum, has been monitored by in situ instruments in the power block excavation. A total of nine heave points were installed between el 104 and 126 ft at the locations shown in figure 2.5.4-2. From 1974 to 1977, rebound ranged from less than 1 to 1.6 in., substantially less than predicted. A discussion of heave is contained in paragraph 2.5.4.10.

2.5.1.2.8.6 Effects of Man's Activity. There are no mining or underground mineral extraction activities occurring on or near the site. Ground water extraction is nominal in this area of low population. Therefore, there are no human activities which will affect site geologic conditions.

2.5.1.2.8.7 Site Ground Water. There are two confined aquifers beneath the site. The Cretaceous aquifer is the lowermost, and it consists primarily of the sands and gravels of the Tuscaloosa formation. It is often referred to as the Tuscaloosa aquifer. The overlying Tertiary aquifer is represented beneath the site by the "unnamed sands" member of the Lisbon Formation. These Tertiary sands are the local, minor equivalent of the regional principal artesian aquifer which consists primarily of permeable sands and limestones of several Tertiary formations extending throughout the Atlantic Coastal Plain. The Cretaceous aquifer and lesser Tertiary aquifer are believed to be hydraulically connected beneath the plant site. The beds that normally separate the Tertiary aquifer from the underlying Cretaceous aquifer are somewhat more permeable than they are elsewhere.

Overlying these aquifers is the Blue Bluff marl, the upper member of the Lisbon Formation. The marl layer, approximately 70 ft thick, is a near-impermeable layer that effectively confines the Tertiary and Cretaceous aquifers.

Ground water also exists in an unconfined water table aquifer in the Barnwell sands and limestone which overlie the marl. The water table aquifer a the site is on an interfluvial ridge, or a topographically high area in which the ground water in the water-table discharges along streams which nearly surround the high. The streams discharge to the Savannah River. A detailed discussion of site ground water conditions is contained in subsection 2.4.12.

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2.5.2 VIBRATORY GROUND MOTION

2.5.2.1 Seismicity

All significant historically reported earthquakes, that is, all earthquakes that could have reasonably affected the site region, are considered in this section. The site region as used here is the area within approximately 200 miles of the site. Within this site region all earthquakes of Modified Mercalli intensity IV or greater or magnitude 3.0 or greater are considered. Earthquakes that occurred outside the region but that were probably felt at the site are also treated.

All significant site region earthquakes are listed in table 2.5.2-1. To avoid crowding, only those Charleston Summerville area events of magnitude greater than or equal to magnitude 4.75 or intensity VI are shown on drawing AX6DD385. Elsewhere, all events in table 2.5.2-1 are shown on drawing AX6DD385. Several sources were used to compile the data given in table 2.5.2-1:

A. The Earthquake Data File of the National Geophysical and Solar-Terrestrial Data Center[1][2] is a collection of many published individual sources, including Earthquake History of the United States[3] (data source EQH), U.S. Earthquakes[4] (data source USE), and the U.S. Geological Survey (data source GS). Citations for these sources can be found in the Earthquake Data File records documentation.[1]

B. Data source CSC is from the earthquake catalog for the South Carolina seismic network of the University of South Carolina and is included in the Earthquake Data File.

C. Data source STR is a compilation of state earthquake catalogs by Stover and others[5][6] and Reagor and others.[7][8]

D. The Bollinger study[9] (data source BOL) is a catalog of southeastern United States from 1754 through 1974.

E. The Nuttli catalog (data source NUT) is an update of the Nuttli catalog of reference 10 and covers the central United States from 1811 through 1979.[11]

F. Data source EUS is attributed to Tarr of the U.S. Geological Survey. This catalog is a composite of many of the above catalogs, covering east and central United States. A number of events, usually of low intensity, listed in the EUS catalog are unique to that catalog and their existence could not be confirmed.[11] These events are generally small.

G. The Dewey and Gordon catalog[12] (data source D&G) covers the eastern continental United States and adjacent Canada, an area ranging between 52° to 104°W, 25° to 49°N. This is a compilation of previously cataloged events from 1925 through 1976, relocated with the benefit of newly developed location methods and improved regional travel-time curves, developed from data obtained from several dense seismic networks installed in the 1970s.

H. The Virginia Polytechnic Institute catalog (data source VPI) is from the Bulletin of the Southeastern United States Seismic Network, compiled at the Virginia Polytechnic Institute and State University (various publication dates since 1978). This catalog covers events primarily in the states of Alabama, Georgia, South Carolina, North Carolina, Tennessee, Kentucky, Virginia, and West Virginia, between July 1977 and June 1983. A recent effort by the author to remove blast- and reservoir-induced events from this catalog has been incorporated into this study's data set.

Before the early 1970s, most of what is known about site region seismicity was based on intensity data. These data are not derived from measurements made by instruments but are only chronicles of the sensible effects of earthquakes on people, structures, and landforms. The Modified Mercalli Scale of 1931[13] (an abridged version of this scale appears in table 2.5.2-2) is used throughout this subsection for the descriptive ranking of earthquake effects. The epicenters of earlier earthquakes were set at or near the center of the maximum intensity of shaking effects (I_o), which at times, especially in less populated areas, were sparsely reported. In the later years of this period, instrumental recordings from a few regional seismographic stations brought improvements to epicentral locations. The locations of site region earthquakes are given to an accuracy of 1/10° in table 2.5.2-1. This choice implies an uncertainty in epicentral position on the order of 5 to 10 miles or so. This is estimated to be a fair representation of the uncertainty in earthquake locations for site region events averaged over the chronological range of their occurrence. Roundoff at the 1/10° level is used in several of the sources,[3][9] from which the information in table 2.5.2-1 and drawing AX6DD385 is compiled, so that it is convenient and logical to follow this convention. However, it must be remembered that the 5- to 10-mile uncertainty is only an average estimate. For some of these events the actual uncertainties in location are realistically measured in several times this average.

Earthquake detection and location greatly improved in the 1970s with the installation of dense seismic networks in various areas in the eastern United States. A permanent seismic network was installed in 1974 in South Carolina between Charleston on the Atlantic Coast and Columbia in the central part of the state.

Other stations have been installed permanently or temporarily at sites of particular interest, especially at nearby reservoirs and at the Savannah River Plant just across the Savannah River from the project site. Recent events which have been recorded instrumentally with greater precision are given in table 2.5.2-1 with locations to an accuracy of 1/100°, implying an uncertainty in epicentral position on the order of 3 miles. This precision has also been given for events before the early 1970s that were relocated by Dewey and Gordon.

Few focal depths have been determined for events before the early 1970s. In their relocations of earthquakes Dewey and Gordon note that few of their calculated focal depths are estimated to be precise to within about 6 miles, although the data do indicate that small and moderate earthquakes in eastern North America typically have sources in the upper part of the earth's crust (about 20 miles). Site region earthquakes recorded since the mid-1970s frequently have focal depths, when determined, of improved precision to as fine as 1 mile. Asterisks (*) indicate that the depth has been constrained to enable a stable earthquake location solution.

The magnitude of an earthquake depends for its definition upon the amplitude of motion on a standard instrument normalized to take into account the separation of the earthquake location from the instrumental recording site. In the site region several magnitude scales are commonly employed which are not exactly equivalent. For the earthquakes of table 2.5.2-1 and drawing AX6DD385 the magnitudes are equivalent or roughly equivalent to body wave magnitudes, except where noted. The uncertainty in instrumentally determined magnitudes is roughly $\pm$0.3 magnitude units. Magnitudes estimated from various forms of intensity information, including intensity fall-off with distance and extent of felt area as well as by empirical conversions from site intensity, have uncertainties on the order of $\pm$0.3 to $\pm$0.6 magnitude units.

Distances between each site region epicenter and the site are tabulated in table 2.5.2-1. In view of the uncertainties in earthquake location noted above, the distances are given only to the nearest 5 miles for those events with locations given to an accuracy of 1/10° and to the nearest 1 mile for those events with locations given to an accuracy of 1/100°.

These estimates are based on available felt report data from several standard sources [3][4] supplemented by a number of specialized references.[14-20] The highest intensity at the site is VI, which is associated with the August 31, 1886, Charleston, South Carolina, earthquake. The earthquakes closest to the site are the October 28, 1974, intensity IV event about 45 miles north of the site and the July 26, 1945, intensity V event about 48 miles to the northwest. Although no specific felt reports are listed at the site for these events, U.S. Earthquakes indicates a site intensity of I-III in Augusta for the 1945 event.

The earthquake of epicentral intensity greater than or equal to VII occurring nearest the site, exclusive of those earthquakes in the Charleston-Summerville, South Carolina, area, was the Union County, South Carolina, event of January 1, 1913. A recent evaluation of the felt reports from this earthquake[16] indicates an epicentral intensity of VII-VIII and a felt area that does not include the site. Therefore, for this study it is concluded that this earthquake, occurring about 110 miles north, was not felt at the site.

Several other site region earthquakes, including some aftershocks of the 1886 Charleston earthquake, are estimated to have been felt with intensity less than or equal to IV at the site.

In addition to the earthquakes within the site region, several large events at greater distances are significant. For example, the New Madrid, Missouri, earthquake sequence of 1811 and 1812 occurred about 530 miles west-northwest but was probably felt at the VEGP site. The epicentral intensities of the largest earthquakes of this sequence have been variously reported as X-XI[21] or XII.[3] Only limited felt information in the site region is available. Fuller[22] cites evidence that may be interpreted as indicating a maximum intensity of VI anywhere in Georgia. This is in close agreement with a recent reevaluation of felt data from the New Madrid sequence

by Nuttli.[21] A generalized isoseismal map for the December 16, 1811, earthquake appearing in this reevaluation shows the site to lie in an area between the V and VI isoseismal lines. Therefore, the site intensity from this sequence is conservatively estimated to have been less than VI.

The Giles County, Virginia, earthquake of May 31, 1897, occurred about 280 miles north of the site. The epicentral intensity of this earthquake has been estimated to be as high as VIII.[16] A recent evaluation of the felt reports associated with this earthquake indicates that the southernmost extent of intensity III effects may have just included the site.[16] Therefore, it is possible that this earthquake may have been felt with low intensity at the site.

From the brief summary above, table 2.5.2-1, and drawing AX6DD385, it is clear that the Charleston-Summerville, South Carolina, area is the source of the seismicity most affecting the VEGP site. This is true both in terms of the maximum historical site intensity and the number of earthquakes felt at the site with lesser intensity.

Historic data indicate that this small area has been the location of earthquakes within South Carolina for as long as records have been kept. For example, a recent archival study[23] has indicated the occurrence of 18 probable earthquakes in South Carolina between 1698 and August 1886. Of these 18 events, 13 are believed to have originated near Charleston. This localization is in all likelihood due at least in part to demographic distribution. None of these 13 probable earthquakes had an intensity of greater than V. The seismicity of South Carolina as a whole was not anomalously high prior to the August 31, 1886, earthquake relative to that of the neighboring states of Tennessee, Georgia, and North Carolina.[23]

The Charleston earthquake of 1886 was associated with two main shocks 8 min apart and hundreds of aftershocks over the next several years. The center of the area of maximum intensity was located near Middleton Place, about 14 miles northwest of Charleston and 10 miles south-southeast of Summerville. There were two discrete zones of maximum damage: one near Middleton Place and Summerville and the other about 10 miles southwest near Ravenel and Rantowles. Both of these areas are included in the meizoseismal area of intensity X shown on drawing AX6DD385. It has been thought that this bimodal pattern resulted from each of the two main shocks having a different epicenter. It is possible, however, that local ground conditions and demography may have been responsible for the details of the high intensity pattern. The earthquakes were felt over an area of more than 2,000,000 mi^2; they were felt as far away as Boston, Milwaukee, New York, Cuba, and Bermuda. Minor damage was reported at Savannah, Georgia; Augusta, Georgia; and Columbia, South Carolina.

In the meizoseismal area more than 62 miles of railroad tracks were damaged. The damage included lateral and vertical displacement, formation of S-shaped curves, and the longitudinal movement of hundreds of meters of track.[15] This type of damage is first given at the intensity X level.[24] The formation of small sand craters and the ejection of sand were widespread effects. This is characteristic of intensity X if occurrence is on a large scale.[24]

Structures in the city of Charleston suffered extensive damage, but most masonry and frame buildings were not destroyed. Detailed descriptions of structural damage in Charleston may be found in Dutton.[14] This and the absence of rail damage and extensive ground effects indicate an intensity of IX in Charleston.[15]

A detailed isoseismal map throughout the State of South Carolina for the main shocks of the 1886 sequence is shown on drawing AX6DD385. This figure shows that the VEGP site was probably within an area of intensity VI, although pockets of higher intensity are present both closer to and farther from the epicenter.

Since the main earthquakes on August 31, 1886, both aftershocks and more recent seismic activity of epicentral intensity up to VII have occurred in the Charleston-Summerville zone. None of these earthquakes has been felt at the site with an intensity above IV.

2.5.2.2 Geologic Structures and Tectonic Activity

The site is located in the Atlantic Coastal Plain province at a point about 25 miles southeast of the boundary between this and the Piedmont province. Within 200 miles of the site are also the Blue Ridge province to the northwest of the Piedmont province and a small section of the Valley and Ridge province northwest of the Blue Ridge province. These physiographic provinces and the site location are shown on drawing AX6DD338.

Physiographic and geologic descriptions of these provinces may be found in paragraph 2.5.1.1.1. Regional geologic history is discussed in paragraph 2.5.1.1.2. This history consists of important episodes of Paleozoic orogeny, more modest tensional tectonism in the early Mesozoic resulting in Triassic basins and subsequent diabasic intrusions, and essential quiescence since the deposition of the Cretaceous sediments in the coastal plain. Some regional epeirogenic upwarping and arching during the late Eocene ended before Miocene deposition about 25,000,000 years ago.

The Pleistocene and recent history of the site region is largely represented by erosion of the southern Appalachian provinces and flood plain deposition and valley fill associated predominantly with the rivers and larger streams in the Atlantic Coastal Plain province.

In addition to the larger scale regional physiographic provinces and small scale Triassic basins, the Belair fault has been noted as a geologic structure of possible interest for evaluations of vibratory ground motion. This fault zone is discussed in paragraph 2.5.1.1.4.1. The most recent investigation of this fault zone[25] concludes that the last episode of movement occurred sometime within the last 60,000,000 years but prior to 23,000 to 2000 years ago. No intermediate age strata have been found that would provide a more definitive date of the last movement of the fault.

2.5.2.3 Correlation of Earthquake Activity with Geologic Structures or Tectonic Provinces

With the exception of the Charleston-Summerville area, seismicity of the site region is generally diffuse. There have been no definite correlations between earthquake epicenters and geologic structures. The evidence on the Belair fault zone is inconclusive as discussed in paragraphs 2.5.1.1.4.1 and 2.5.2.2. However, although lack of movement in the last 35,000 years has not been absolutely demonstrated, there is no correlation of any macroseismicity with this fault, and the general tectonic quiescence of the region argues against its likely significance. In a recent consideration,[26] it was concluded that the Belair fault zone is not a capable fault within the meaning of Appendix A to 10 CFR 100, section 3(g). This convention is followed in this study.

During the months of March through August 1982, further studies were made to determine the existence and capability of the postulated Millett Fault introduced in an open-file United States Geological Survey report.[27] No evidence was found in support of the existence of any fault in the region designated by the report. Details of the study can be found in a report entitled Studies of Postulated Millett Fault.[28]

Thus, for the purpose of vibratory ground motion at the VEGP site, historic earthquake activity is most logically correlated with tectonic provinces. Within 200 miles of the site, four distinct tectonic provinces are traditionally recognized. These are the Valley and Ridge, Blue Ridge,

and Piedmont provinces of the Southern Appalachian Mountains and the Atlantic Coastal Plain province. The boundaries of these provinces in the site region are shown on drawing AX6DD341.

In the last 10 years, several working hypotheses have been proposed in an attempt to associate geologic structure and tectonic activity in the Charleston region with the source of the Charleston earthquake. Seeber and Armbruster[29] suggest backslip on the Appalachian detachment (decollement) as a possible mechanism for the 1886 Charleston earthquake and cite potential normal displacement in the area of the Brevard fault zone and a number of coseismic phenomena in 1886 as supporting evidence for this hypothesis. Behrendt[30] proposes reverse slip on steeply dipping faults of small offset northwest of Charleston and associated listrically with the decollement. Wentworth and Mergner-Keefer[31][32] have proposed reactivation of faults bounding Triassic basins throughout the east coast as a model for recurrence of a Charleston type fault.

These working hypotheses both assume and require that the principal horizontal stress orientation in the Charleston region is essentially northwest-southwest as indicated by Zoback and Zoback.[33] Latest information[34] indicates that a principal compression almost at right angles to this direction is apparently now acting in this region. This, and the failure of the proposed types of structure to be correlated with well-located instrumental hypocenters, recent earthquake focal mechanisms, or any evidence of geologically young faulting, argue against the adoption of these models at this time.

In this report, the southern Appalachian Mountains provinces are treated as a single region. This usage is for convenience only and is not intended to imply that the distinct Valley and Ridge, Blue Ridge, and Piedmont tectonic provinces, in general, should be considered as a single unit. However, such a usage is adequate within the narrow context of the assumptions employed to determine design vibratory ground motion at VEGP. This point is discussed in paragraph 2.5.2.4. In this report, these three tectonic provinces are called, collectively, the Southern Appalachian Mountains Region.

The Southern Appalachian Mountains Region so defined is bounded on the east, southeast, and south by the fall line and the Atlantic Coastal Plain tectonic province. On the northwest, it is bounded by the Cumberland and Allegheny Plateaus. On the northeast, along structural trend, geologic and seismic discriminates are more tenuous. Here this boundary is chosen to include northern Pennsylvania, northern New Jersey, and southernmost New York State and to exclude the Appalachians north of this area. The outline of the Southern Appalachian Mountains Region in the site region is shown on drawing AX6DD385. This area is a region of consistent northeast-southwest structural trends. As may be seen on drawing AX6DD385, epicenters in this area are irregularly distributed, with concentrations in northeast Georgia, northwest South Carolina, and eastern Tennessee in the site region and in Virginia farther to the north. These concentrations are all at distances greater than 100 miles from the site.

The Atlantic Coastal Plain tectonic province extends from the fall line on the northwest to the edge of the Continental Shelf to the east and southeast. To the northeast the boundary is near the northeastward extent of the Southern Appalachian Mountains Region as defined above. To the southwest the Atlantic Coastal Plain is conveniently separated from the Gulf Coastal Plain by the Pickens-Gilbertown fault zone and a line extending down from the northwestern border of the Southern Appalachian Mountains Region. The Atlantic Coastal Plain tectonic province thus defined is a large area with generally low seismic activity. With the exception of the Charleston-Summerville area within the Atlantic Coastal Plain, seismicity is more infrequent and of smaller size than in the Southern Appalachian Mountains Region. As used in this study, the Charleston-Summerville seismic zone area includes the 1886 meizoseismal zone and the

epicenters of the cluster of earthquakes in this immediate vicinity that have occurred subsequently. This zone is sketched on drawing AX6DD385.

As discussed in paragraph 2.5.2.1, the intensity X earthquake of 1886 occurred in an area of previously low seismicity. It is the largest earthquake ever reported in the Southeastern United States. The seismic activity of the meizoseismal zone of this earthquake has remained anomalously high from that time. The occurrence of the 1886 earthquakes and the continued high activity of this area have prompted many recent studies aimed at better defining and understanding the source of this localized activity. Efforts to monitor local microearthquake activity [35-38] have revealed that a number of previously unrecordable small earthquakes do occur within the Charleston-Summerville seismic zone as shown on drawing AX6DD385. These epicenters near Middleton Place define a linear north-northwest trending zone about 15 miles long and 2 miles wide. Focal depths of earthquakes in this zone range from 1.25 to 7.5 miles, and hypocenter plots indicate that this zone is nearly vertical. The focal mechanism solution for the November 22, 1974, event (drawing AX6DD385) of intensity VI within this zone is consistent with dip-slip faulting on a nearly vertical fault striking northwest-southeast.[38] Studies of epicenters in this area are continuing.

The localized zone of seismicity is spatially associated with the northeast edge of a positive Bouguer gravity anomaly and the northeast lobe of a three-lobed positive magnetic anomaly. These anomalies have been interpreted to reflect the presence of mafic intrusives in the basement at depths of 1.5 to 2.8 miles.[39] This is within the range of depths established for the nearly vertical zone of seismicity. In turn, the presence of mafic intrusives with elastic rigidities differing from those of the surrounding shallow crustal rocks has been used to hypothesize several stress amplification mechanisms[40-43] that could help explain the localization of earthquakes in the Charleston-Summerville area. Scenarios of this type have not provided definitive conclusions to date. However, because of the clustering of historical earthquakes in the Charleston-Summerville area, because of the continuing microearthquakes there at levels higher than normal background seismicity for the Southeastern United States, and because of the geologic features peculiar to that area that may ultimately provide a mechanism for earthquake localization, it is accepted for the purposes of this analysis that near-future earthquakes in the Atlantic Coastal Plain tectonic province will continue to exhibit the features distinctive of the near past. That is, general seismicity will remain subdued, while high activity earthquake clustering in the site region will remain confined to the Charleston-Summerville seismic zone.

2.5.2.4 Maximum Earthquake Potential

The largest historical earthquakes within the Charleston-Summerville seismic zone are the two main events of August 31, 1886. As discussed in paragraph 2.5.2.1, recent reinterpretation of the damage reports associated with these earthquakes[15] indicates that the maximum intensity was X on the Modified Mercalli Scale.[13] The closest approach of the Charleston-Summerville seismic zone as drawn on drawing AX6DD385 and discussed in paragraph 2.5.2.3 is about 78 miles.

Exclusive of the Charleston-Summerville seismic zone, the maximum historical earthquake in the Atlantic Coastal Plain within the site region is of intensity VI. Several intensity VII earthquakes have been reported on or near the fall line between the Piedmont and Atlantic Coastal Plain tectonic provinces north of the site region; for example, the February 21, 1774, earthquake near Petersburg, Virginia; the November 11, 1840, and October 7, 1871, earthquakes near Philadelphia, Pennsylvania; and the December 23, 1875, event near Richmond, Virginia. As is discussed below, such events do not control the vibratory ground

motion design even if it is assumed that they or similar earthquakes should be considered, for conservatism, at the site.

The maximum intensity site region earthquake to occur in the Southern Appalachian Mountains Region is the VII-VIII[9] event of 1913 in Union County, South Carolina. The greatest historic earthquake in this region, as it has been defined in paragraph 2.5.2.3, is the intensity VIII Giles County, Virginia, earthquake of 1897. In both cases the currently accepted maximum intensities of these events have resulted from conservative reinterpretations of the previously accepted lower maximum intensities of VI-VIII and VII, respectively.[3] In addition, the Giles County earthquake occurred in the Valley and Ridge tectonic province that is typically separated from the Blue Ridge and Piedmont provinces for purposes of seismotectonic province characterization. However, as is discussed below, even if an intensity VIII earthquake is considered capable of occurring within the Southern Appalachian Mountain Region at its 26-mile closest approach to the site, it is not the controlling event for seismic design. This is the principal reason that the Southern Appalachian Mountains Region was defined as it was in paragraph 2.5.2.3.

In order to evaluate the effects at the site of the three maximum earthquakes discussed above, several intensity attenuation formulas were considered. Two of these[15][44] have been derived from studies of the intensity distribution for the 1886 Charleston earthquake, while the third[45] is derived from isoseismals from a number of more recent events in the Eastern United States.

McGuire[44] derives an intensity attenuation function from 783 intensity reports from the 1886 Charleston, South Carolina, earthquake as these reports were reevaluated in a recent study.[46] Bollinger[15] follows the same procedure but fits the data to a formula of slightly different form. These two attenuation relations are, respectively:

$$I_s = I_o + 3.08 - 1.34 \ \ell n\, R \text{ and}$$

$$I_s = I_o + 2.87 - 0.00052\, R - 2.88 \log R$$

Where ℓn is the logarithm to the base e (natural logarithm) and log is the base 10 (common) logarithm, I_s is near site intensity, I_o is epicentral intensity, and R is epicentral distance in kilometers. Substituting X for I_o and 78 miles x 1.6 km/mi = 125 km for R, the intensity at the site from a recurrence of the 1886 earthquake is 6.61 and 6.77, respectively. This is, as expected, entirely consistent with drawing AX6DD385, which shows the site at intensity VI to be in an area of lower than average intensity for its distance from the epicenter. Thus, best data indicates a recurrence of the 1886 Charleston earthquake would result in an intensity VI-VII to VII at the site.

To calculate the intensity at the site from the maximum earthquake in the Southern Appalachian Mountains Region, a recent formula developed by Anderson[45] is used. This formula is derived from the statistical consideration of isoseismal maps of 66 earthquakes in the Central and Eastern United States, including the 1897 Giles County earthquake. In this case the mean site intensity is given as:

$$I_s = I_o + 3.2 - 2.7\,(0.00039\, R + \log R)$$

For I_o = VII-VIII, R = 26 miles x 1.6 km/mi = 41.6 km, I_s = 6.28. For the more conservative I_o = VIII, I_s = 6.78. Thus, the effect at the site from the maximum, credible Southern Appalachian Mountains Region earthquake is very similar to the Charleston-Summerville event; that is, I_s = VI-VII to VII.

Finally, the site intensity from an earthquake in the Atlantic Coastal Plain province, exclusive of the Charleston-Summerville seismic zone, will be the same as the maximum historical event in this province. As discussed above, this is intensity VI.

Even if this event were assumed to be VII, it would not significantly supersede the site intensities from the other two pertinent earthquake sources in the site region.

Thus, the maximum potential earthquake intensity at the site is conservatively estimated to be VI-VII to VII.

2.5.2.5 Seismic Wave Transmission Characteristics of the Site

The site is underlain by Atlantic Coastal Plain sediments consisting of 800 to 1000 ft of predominantly clays, sands, limestone, and marl, ranging in age from Cretaceous to recent. Material properties for the sediments under the site and the methods used to determine these properties are discussed in paragraph 2.5.4.2.

No analysis to specifically take account of the site soil column is undertaken to translate the maximum potential site intensity into the safe shutdown earthquake (SSE) ground motion. This additional analysis is neglected because of the uncertainties involved in its performance and because the intensities of paragraph 2.5.2.4 are converted into peak accelerations in a conservative manner, using formulas relating peak acceleration to intensity on a wide variety of foundation conditions, including those similar to the foundation conditions at the site.

2.5.2.6 Safe Shutdown Earthquake

As discussed in paragraph 2.5.2.4, the maximum credible site intensity is VI-VII to VII. For additional conservatism a SSE site intensity of VII-VIII is chosen. According to the best available data relating peak horizontal ground surface acceleration to intensity on a variety of foundations,[47][48] intensity VII-VIII is associated with approximately 0.2 g peak horizontal acceleration. This relationship is appropriate for sites near the zone of energy release or for sites where I_s is not much less than I_o. This is the case for the Atlantic Coastal Plain maximum credible event and approximately the case for the Southern Appalachian Mountains maximum credible event. This condition is not well realized for the Charleston-Summerville seismic zone maximum credible event. Since acceleration attenuates more rapidly than intensity, the use of the empirical relationships noted above is probably very conservative in this case. However, the larger Charleston earthquake is likely to be richer in low frequency ground motion at the site than the other design earthquakes. Thus, for conservatism a single design response spectrum is proposed to define site SSE ground motion. This ground motion is defined in terms of Regulatory Guide 1.60 horizontal and vertical design response spectra[49] normalized to a peak ground acceleration of 0.20 g. These spectra are shown in section 3.7.

2.5.2.7 Operating Basis Earthquake (OBE)

As discussed in paragraph 2.5.2.1, the most recent best evidence indicates that the maximum historical intensity at the site was VI, associated with the Charleston earthquake of 1886. A less finely detailed study of attenuation of intensity from that earthquake would indicate a somewhat higher intensity should be expected at the site. For additional conservatism an OBE intensity of VII is adopted. Using the intensity/acceleration relationships noted in paragraph 2.5.2.6, this intensity is associated with a peak horizontal ground surface acceleration on average foundation condition and near the zone of energy release of approximately 0.12 g. This acceleration, and spectra of identical form as those characterized in paragraph 2.5.2.6, are used to define the OBE ground motion at the site and are shown in section 3.7.

A probabilistic estimate of the occurrence of the OBE acceleration at the site during its 40-year[a] operating life may be made based on the work of Algermissen and Perkins.[50] This study shows that the site acceleration with a 90% chance of nonexceedence in a 50-year interval is about 0.10 to 0.11 g.

Assuming, as is implicit in this characterization, that earthquakes occur as a Poisson point process, this is equivalent to estimating an 8% chance of occurrence of a site acceleration exceeding 0.10 to 0.11 g during the 40-year operating life of the plant. The chance of exceeding the 0.12 g OBE is, therefore, somewhat less than this.

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[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Seismic analyses are not related to aging and therefore are outside the scope of license renewal. NEI 98-03 guidance indicates that seismic data used to support original plant design bases are considered historical and do not need to be actively maintained. Therefore, this statement is not required to be updated as a result of license renewal.

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2.5.3 SURFACE FAULTING

No evidence of surface faulting has been uncovered in the site area. Detailed stratigraphic study and mapping of the excavations for Category 1 structures are discussed in paragraph 2.5.1.2.2. Mappable lithologic units may be traced unbroken around the perimeter of the excavations, demonstrating the absence of faulting. The 5-mile-radius site investigation showed no evidence of surface displacement that might localize earthquakes in the immediate vicinity of the site. This region has been relatively stable for a considerable length of time, and known faults in the Piedmont province to the west and the Triassic basin underlying the site are inactive. The geology of this area is discussed fully in paragraph 2.5.1.2.

2.5.3.1 Geologic Conditions of the Site

The lithologic, stratigraphic, and structural geologic conditions of the site are presented in paragraphs 2.5.1.2.2 and 2.5.1.2.3. Regional, local, site, and site excavation geologic maps are shown on drawings AX6DD341, AX6DD345, AX6DD351, AX6DD360, AX6DD361, AX6DD362, and AX6DD363, respectively. The regional geology and geologic history are discussed in paragraph 2.5.1.1.

2.5.3.2 Evidence of Fault Offset

The area within 5 miles of the site is not associated with known or suspected faulting. Geologic sections throughout the site and power block area, which are based on data obtained from field mapping, exploration borings, and mapping of the foundation excavation, reveal no evidence for the existence of any fault offset at the site (drawings AX6DD345, AX6DD351, AX6DD360, AX6DD361, AX6DD362, and AX6DD363).

2.5.3.3 Earthquakes Associated with Capable Faults

There are no known capable faults within 5 miles of the site.

2.5.3.4 Investigations of Capable Faults

Field investigations for this project, including exploratory drilling, field mapping, studies of aerial photography, and geophysical studies, show that no capable faults exist within 5 miles of the site. Reversal of the regional dip northwest of the site has been investigated and shown to be related to depositional and erosional processes, as discussed in paragraph 2.5.1.2.3. Stratigraphic irregularities discovered during excavation for power block foundation have been studied and shown to be related to depositional and erosional processes, as discussed in paragraph 2.5.1.2.2.2.

2.5.3.5 Correlation of Epicenters with Capable Faults

There are no known capable faults within 5 miles of the site. There is no correlation of epicenters with known or suspect faults within 5 miles of the site.

2.5.3.6 Description of Capable Faults

No capable faults are known to occur within 5 miles of the site.

2.5.3.7 Zone Requiring Detailed Faulting Investigation

In 1982, the U.S. Geological Survey released Open-File Report 82-156, which postulated the existence of two potentially capable faults within 32 miles of VEGP. According to the report, the Millett fault was located approximately 7 miles south of VEGP, while the Statesboro fault was located approximately 32 miles south of VEGP. The report did not assert that either of the faults were capable, but due to their proximity to the site, especially the Millett fault, a full-scale investigation was undertaken to determine exact location and date of last movement.

2.5.3.8 Results of Faulting Investigation

The investigation of the postulated Millett and Statesboro faults was completed with the conclusion that these faults did not exist within the depths to which the investigation extended and that, if they exist at some depth greater than the investigated depth, then they are not capable faults by virtue of the age of undisturbed overlying sediments. The investigative program is described completely in a separate report entitled, Studies of Postulated Millett Fault, prepared by Bechtel Power Corporation, dated October 1982.

2.5.4 STABILITY OF SUBSURFACE MATERIALS AND FOUNDATIONS

2.5.4.1 Geologic Features

There is no evidence that the Blue Bluff marl, which is the bearing stratum, has been subjected to or is potentially subject to subsidence, collapse or uplift due to earthquake, solution processes, or other geological phenomena (paragraph 2.5.1.2). Surface materials, comprised of strata which overlie the marl, have been subjected to and are potentially subject to subsidence due to solution processes (paragraph 2.5.1.2). These materials have been completely removed in the power block, and all Category 1 structures in the plant area are founded directly or indirectly on the marl (paragraphs 2.5.1.2.2 and 2.5.1.2.3.3).

The geologic history (paragraph 2.5.1.2.4) indicates that the plant site is located upon an area of regional uplift and has been subjected to subaerial erosion during Quaternary times. The stratigraphic sequence and investigative work (paragraph 2.5.1.2) indicate approximately 950 ft of unlithified to poorly lithified sediments resting upon pre-Cretaceous basement rock. Rebound in the marl, the bearing stratum has been monitored and is discussed in paragraph 2.5.4.10.

The surface of the shallow (unconfined) ground water table historically has been approximately el 160 ft. The marl, which is the bearing stratum, is essentially impermeable and is an effective aquiclude comprising the base of the ground water table and the cap of a confined aquifer. The hydrostatic surface elevation of the confined aquifer is approximately 115 ft.

There are no deformational zones, irregular weathering, jointing or fracturing systems, crushed zones, or other indications of structural weakness in the marl which is the bearing stratum (paragraphs 2.5.1.2.3 and 2.5.1.2.8).

There are no materials at the site that are hazardous or may become hazardous due to lack of induration or consolidation, variability, high water content, solubility, or undesirable response to natural or induced conditions.

2.5.4.2 Properties of Subsurface Materials

The subsurface conditions in the plant site may be subdivided into three principal strata. The top stratum consists of sands, silty sands, and clayey sands with occasional clay seams. This stratum, referred to hereinafter as the upper sand stratum (Barnwell Group), is about 90 ft thick. At the base of the upper sand stratum is a shelly limestone (Utley Limestone) which is about 5 ft thick on an average. Below the upper sand stratum is a stratum consisting of a very hard calcareous clay marl (Blue Bluff marl), ranging in thickness from 60 to 100 ft. This stratum is referred to as the marl bearing stratum. The stratum beneath the marl bearing stratum consists principally of dense, coarse to fine sand with minor interbedded silty clay and clayey silt. This unit (Ellenton Formation) is called the lower sand stratum. The thickness of this stratum is estimated to be at least 750 ft.

Based on the results of the site exploration, it was determined that the upper sand stratum would have a potential for liquefaction in the event of a seismic occurrence equivalent to the safe shutdown earthquake (SSE).[1] It was also determined that the shelly limestone layer is characterized by solution channels, cracks, and discontinuities within it. Consequently, it was concluded that the upper sand stratum materials and the shelly limestone layer should be excavated down to the marl bearing stratum and replaced with select sand and silty sand backfill compacted to a sufficient degree to preclude the possibility of liquefaction and to reduce settlement to a tolerable level. With the exception of the auxiliary building, nuclear service cooling water towers, and instrumentation cavity of the containment which are founded on the marl bearing stratum, all the power block structures including the containment basemat and the non-Category 1 turbine building are supported on Category 1 backfill. The location of these structures is shown in drawings CX2D45V003 and AX1D45A01. Compacted fill and marl foundations are indicated on drawing AX6DD386.

The static and dynamic engineering properties of the three principal soil strata and for compacted Category 1 backfill were determined by field investigation and laboratory testing. The results of all the field and laboratory work and data evaluation are covered in five separate reports.[1-5] A discussion and summary of the static and dynamic soil properties of the upper sand, marl, and lower sand strata are presented in paragraphs 2.5.4.2.1, 2.5.4.2.2, and 2.5.4.2.3, respectively. The static and dynamic soil properties of compacted Category 1 backfill are summarized and discussed separately in paragraph 2.5.4.5.2.

2.5.4.2.1 Properties of Upper Sand Stratum (Barnwell Group)

The static engineering properties of the upper sand stratum are summarized in table 2.5.4-1. A range of values is given for most properties. The standard penetration test data indicate that the relative density of the upper sand stratum is extremely variable and ranges from very loose to dense. The consistency of the clay lenses in this stratum ranges from soft to medium.

Unconsolidated undrained triaxial test results from samples in this stratum indicate that the Mohr strength envelope of total stresses may be defined by parameters ranging from about $c=2100\ lb/ft^2$, $\phi=6^\circ$ to $c=440\ lb/ft^2$, $\phi=32^\circ$ depending upon the predominance of clay or sand.

Similarly, consolidated undrained triaxial test results ranged from $c=1650\ lb/ft^2$, $\phi=17^\circ$ to $c=4000\ lb/ft^2$, $\phi=25^\circ$ for the Mohr strength envelope of total stresses and from $\phi=33^\circ$ to $\phi=34.5^\circ$ for the Mohr strength envelope of effective stresses. The design properties shown in table 2.5.4-2 were developed from the static engineering properties summarized in table 2.5.4-1.

The test data and the procedures used to obtain these data are included in reference 1 and its appendices.

A summary of the design dynamic shear modulus at strain levels of 10^{-4}% or lower for the upper sand stratum is given in table 2.5.4-3. The basic properties of the upper sand stratum to be used in dynamic analyses are summarized in table 2.5.4-4.

Values of the dynamic shear modulus are computed from in situ shear wave velocity measurements as follows:

$$G = \frac{\gamma}{g}(V_s)^2$$

where:

- G = shear modulus (lb/ft^2).
- γ = unit weight (lb/ft^3).
- g = acceleration due to gravity (ft/s^2).
- V_s = shear wave velocity (ft/s).

2.5.4.2.2 Properties of Marl Bearing Stratum (Blue Bluff Marl)

The marl bearing stratum is a zone of hard, slightly sandy, cemented, calcareous clay. It is the uppermost stratum capable of supporting heavy structural loads. Consistency of the marl varies from hard to very hard, moderately brittle material resembling a calcareous siltstone or claystone. Seismic explorations indicate a velocity interface about 15 ft below the top of the stratum. The material below that level has a compressional wave velocity approaching 7000 ft/s as compared to about 5000 ft/s for the upper portion of the stratum.

This is probably due to some degree of weathering of the upper 15 ft. The static engineering properties of the clay marl bearing stratum are summarized in table 2.5.4-1. Ranges of value are given for the most important properties.

The standard penetration test values range from 10 blows/ft in the weathered marl at the contact with the shell zone to well in excess of 100 blows/ft. The unconsolidated undrained shear strength based on one-point tests ranged from c=260 lb/ft^2 to c=500,000 lb/ft^2, with 10,000 lb/ft^2 being the value adopted for design. Samples that yield undrained strengths less than 10,000 lb/ft^2 exhibit large strains to failure which normally indicate sample disturbance in brittle materials of this type.

Laboratory tests indicate that the marl bearing stratum is highly preconsolidated. Atterberg limit tests indicate that the plasticity index is between 2 and 70%. Using an average of 25% this would yield a S_u/p ratio of about 0.2 based on work by Skempton,[6] where S_u is the undrained shear strength and p is the effective consolidation pressure at sample depth. This indicates that the preconsolidation pressure, would be 80 k/ft^2 for the average undrained shear strength of 16.0 k/ft^2. The average undrained strength is taken to be the average of all samples which failed at strengths less than 50 k/ft^2. With such a high preconsolidation pressure, it would be expected that settlements under structure loads would be small and would occur rapidly as load is applied. The design properties shown in table 2.5.4-2 were developed from the data summarized in table 2.5.4-1.

The basic test data and procedures used to obtain these data are contained in reference 1 and its appendixes.

The undrained shear strength of the marl bearing stratum was verified after completion of the power block excavations by testing representative cores. The results of these tests are included in reference 2. These test results verified that the recommended design strength

parameter of c=10,000 lb/ft^2, ϕ=0° is appropriately conservative. Actually, the average undrained shear strength of all core samples that failed was approximately 20 k/ft^2, and the lowest measured value was 11.7 k/ft^2. Therefore, all samples tested exceeded the design strength of 10 k/ft^2. During the excavation, the heave of the marl stratum was observed and recorded. Heave values were measured at nine different locations within the power block area. These data are included in reference 2. An average heave of approximately 1.25 in. was measured in the power block after the excavation was complete. Based on the heave data, the undrained Young's modulus of the clay marl stratum was computed to be 10,000 k/ft^2. This value was consistent with the range of values for the Young's modulus obtained from Menard pressuremeter and seismic velocity measurements but was significantly higher than the 4000 k/ft^2 used to evaluate elastic settlements in the Preliminary Safety Analysis Report (PSAR).

A summary of the design dynamic shear modulus at strain levels of 10^{-4}% or lower for the clay marl stratum is given in table 2.5.4-3. Reduction factors to be used for determining the shear modulus at strains larger than 10^{-4}% are given in figure 3.7.B.2-6. The properties used in dynamic analyses are summarized in table 2.5.4-4. Values of the dynamic shear modulus were computed from in situ shear wave velocity measurements as previously described in paragraph 2.5.4.2.1.

2.5.4.2.3 Properties of Lower Sand Stratum (Ellenton Formation)

Pertinent static engineering properties of the lower sand stratum are summarized in table 2.5.4-1. Design values for dry unit weight, in situ moisture content, and standard penetration test are summarized in table 2.5.4-2.

The standard penetration test values varied from 70 blows/ft to more than 100 blows/ft with the majority of values exceeding 100 blows/ft. Such high blow counts indicate the very dense state of the lower sand stratum. Based on seismic investigations, the lower sand stratum extends to basement rock which is located approximately 900 to 1000 ft below natural ground. Design dynamic properties of the lower sand stratum are summarized in tables 2.5.4-3 and 2.5.4-4.

2.5.4.3 Exploration

Geologic and soils investigation of the VEGP site has been completed and the results are described in subsection 2.5.1. This section summarizes the exploratory work performed.

The geologic and soils data and their evaluation were obtained in separate and individual investigations conducted by Law Engineering Testing Company (LETCO) and Bechtel Power Corporation. Included were a thorough search of the literature, stereoscopic examination of color air photographs, evaluation of geologic conditions at and within 5 miles of the site, and geologic reconnaissance along 12 miles of the river bluff upstream and downstream of the site.

2.5.4.3.1 Borings and Samplings

Field investigations started in January 1971 and involved drilling, geophysical survey, and ground water studies. Drilling was also carried out during construction excavation to verify and obtain further details concerning subsurface conditions in the power block area. In all, 474 holes have been drilled for a total of 60,000 ft of hole. The drilling logs of holes used for the primary geologic investigation and foundation investigation for the plant facilities for the PSAR are discussed in appendix 2B, and details of these borings are tabulated in table 2B-1. An inventory of cores retained and stored is contained in appendix 2B, table 2B-2. Drilling statistics of 111 holes drilled subsequent to PSAR investigations are as follows: 41 were drilled in the

river facilities area to define soil conditions and lateral extent of the marl and to obtain permeability data; 38 were drilled in the power block area to obtain samples of the marl for laboratory testing for a record of the properties of the marl; and 32 were drilled for natural draft cooling tower foundation information.

The locations of the borings are on drawings AX6DD343 and AX6DD344 with the exception of the 31 "CS" series holes drilled in the power block, which are shown on drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363.

The exploration program included electric logging, natural gamma, density, neutron, caliper, and "3D" (Birdwell) velocity logs in selected drill holes. Water pressure tests and Menard pressuremeter tests were conducted to determine in situ properties of the bearing horizon, the marl. Samples for fossil, mineral, or soluble carbonate analysis were taken in those drill holes where conditions were suitable. The geophysical survey conducted to supplement the drilling program is described in paragraph 2.5.4.4. It provided a total of 28,400 ft of shallow refraction seismic lines, 5000 ft of deep refraction lines, and cross-hole velocities in the upper 290 ft of materials. Seismic survey lines and profiles are shown on drawings AX6DD387, AX6DD388, AX6DD389, AX6DD390, AX6DD391, AX6DD392, and AX6DD393. Down-hole geophysical logs are discussed in appendix 2B.

Drilling services were provided by Georgia Power Company (GPC), Girdler Drilling Company, and LETCO. Geophysical seismic surveys were performed by Weston Engineers, Weston, Massachusetts. Down-hole geophysical surveys were performed by the Birdwell Division of Seismograph Service, Incorporated.

2.5.4.3.2 Backfill

For a summary of backfill exploration refer to paragraph 2.5.4.5.2.2.

2.5.4.4 Geophysical Surveys

Geophysical seismic refraction and cross-hole surveys were conducted at the site to evaluate the occurrence and characteristics of subsurface materials. The seismic refraction survey was used to determine depths to seismic discontinuities, based on measured compressional wave velocities. Shallow and deep refraction profiles were obtained throughout the site area, totaling 28,400 and 5000 linear ft, respectively. The cross-hole seismic survey was conducted in the power block area to determine in situ velocity data for both compression and shear waves to a depth of 290 ft (82 ft below sea level) in bore holes 136, 146G, 148, 149, 151, and 154. In this procedure, three-dimensional detectors were lowered into four of the bore holes to equal elevation levels. Energy was generated in a fifth bore hole, at the same elevation level, to determine cross-hole velocities.

The locations of the seismic survey lines, the borings used for cross-hole velocity measurements, and the seismic profiles are shown on drawings AX6DD387, AX6DD388, AX6DD389, AX6DD390, AX6DD391, AX6DD392,and AX6DD393. Table 2.5.4-5 is the compilation of the results of the cross-hole measurements. The seismic velocity zones are summarized and related to other data in table 2.5.4-6. These data were used in determining the elastic moduli, compiled in table 2.5.4-7.

2.5.4.5 Excavation and Backfill

2.5.4.5.1 Excavation

The natural ground surface in the plant area varied between el 200 and 230 ft. The power block area was excavated and graded to an elevation of approximately 130 to 135 ft near the top of the marl bearing stratum which is the clayey marl of the Blue Bluff Member of the Lisbon Formation. In the following and previous discussions, this is called the marl or the clay marl bearing stratum. The excavation for the power block structures at the VEGP site is roughly square in shape; there are three access ramps, one each in the northwest, southeast, and southwest corners of the excavation. It measures approximately 1400 ft on an edge at the top and 1000 ft on an edge at the toe. The side slopes were cut at a gradient of two horizontal to one vertical. The total excavated volume in the power block was approximately 5,000,000 yd^3 including the access ramp. Drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363 geologic maps of the excavation as of 1977, before access roads were completed.

Within the excavation, a deeper localized excavation was made for the auxiliary building basement (drawings AX6DD374 and AX6DD375). This consisted of a rectangular area measuring approximately 120 ft by 440 ft. The base of this excavation was at approximately el 108 ft, and the walls were cut vertically, with a horizontal bench at el 118 ft. The four nuclear service cooling water towers are founded directly on the marl just south of the auxiliary building. The other major power block structures are founded on structural backfill at elevations above the floor of the excavation.

Excavation work was started in May 1974 and postponed on September 12, 1974. The bottom elevation of the excavation averaged approximately 145 ft at this time and close to 900,000 yd^3 of excavation remained. The excavation work was resumed in February 1977 and the auxiliary building excavation was bottomed out in October 1977.

As excavation progressed, the exposed materials were geologically mapped (drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363), including the deeper localized excavation for the auxiliary building (drawings AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, AX6DD370, and AX6DD371). A discussion of the mapping is presented in paragraph 2.5.4.5.1.2.

2.5.4.5.1.1 Excavation Procedures. Excavation work started and progressed very rapidly using scrapers and bulldozers in the upper sands (above the water table) which are at about el 160 ft. Very little, if any, ripping was required because of the sandy nature of the deposits; a maximum rate of 120,000 yd^3/day was attained at the peak of activity. Upon reaching the water table, construction dewatering was begun. The site ground water conditions are discussed in detail in subsection 2.4.12. The procedures utilized during excavation for construction dewatering are discussed in paragraph 2.4.12.1.3.1.

When the excavation reached the zones of hard shell-rich limestone described earlier (Utley Limestone), limited blasting of the rock was utilized to facilitate its removal. Since the shell-rich limestone was immediately above the marl, it was necessary to control any required blasting in such a manner as to protect the underlying marl (marl bearing stratum) from damage. The major portion of the rock was removed by first breaking it with a hydraulic ram mounted on a backhoe, then loading it out with conventional equipment.

Excavation of the marl was accomplished by ripping, followed by conventional earth moving. The auxiliary building basement excavation was cut with bulldozers and front-end loaders. Trimming of the walls was accomplished with a backhoe. Some of the hard, indurated

limestone layers within the marl were first broken with the backhoe-mounted hydraulic ram, then removed by front-end loader. Fine grading of the floor of the power block was accomplished with motor graders in areas underlying future structural backfill and with Gradalls in the nuclear service cooling water tower foundation areas. In the foundation areas, shovels and air hoses were used for cleanup of loose material.

2.5.4.5.1.2 Geologic Mapping Procedures. The geologic mapping and recording of features exposed during excavation are described in the Bechtel Report of Geology and Foundation Conditions (appendix 2B.3). The mapping entailed these phases:

A. Detailed mapping of deposits above the marl; May 1974 to October 1974.

B. Detailed mapping of features within the marl and surveying of the upper contact of the marl; February 1977 to October 1977.

C. Detailed inspection and recording of areas in the marl approved for placement of concrete or backfill; June 1977 to January 1979.

The first phase of mapping was performed in conjunction with the excavation of the sediments above the marl. Features were located in the side slopes of the excavation as the bottom elevation was progressively lowered. The side slopes were cut at a gradient of two horizontal to one vertical, and survey stakes were installed on a grid pattern on the slopes. Locations of geologic features were measured by tape and hand-level methods using the slope stakes as reference points. Accuracy of these measurements is estimated to be within 0.5 ft. Mapping of the 2:1 slopes was recorded in plan on a base map compiled from the project excavation drawings and is shown on drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363.

The second phase of mapping was accomplished as the marl was exposed and prepared for placement of concrete and backfill. To demonstrate the absence of faulting in the marl, the contact between the marl and the overlying sediments was mapped and recorded with survey accuracy around the perimeter of the excavation. Five hundred seventy-five survey points were established by the geologists along this contact and these points were located instrumentally. The nature of the contact between the points was examined closely for continuity and absence of breaks. The contact is shown in plan view on drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363, and the details of the survey results are shown in both plan and section on drawings AX6DD372 and AX6DD373.

In addition to examining the upper contact, features within the marl were examined and recorded. The deep excavation for the auxiliary building basement, within the larger power block excavation, provided an excellent opportunity for this. The sides of the excavation exposed a vertical section of approximately 22 ft in height in the marl. A system of reference points was established on the walls of the excavation and stations were established for the purpose of describing locations of features. The stationing system adopted is shown on drawings AX6DD374 and AX6DD375. The mapping was recorded in the vertical plane and is presented as geologic sections on drawings AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, and AX6DD370. An explanation of geologic units used for mapping purposes is shown on drawing AX6DD370. By referring to drawings AX6DD374 and AX6DD375, the location of any section can be easily ascertained. Measurements were made by tape and hand-level methods on the excavation walls. Accuracy is generally within 0.1 ft.

The third phase of geologic mapping consisted of detailed inspection and photography of foundation areas rather than mapping in a strict sense. This effort was initiated in June 1977 when the first portion of the auxiliary building basemat excavation (vertical surface) was cleaned

off at final grade and prepared for application of a protective seal. Inspection and approval of final grade in the marl was documented and transmitted from the inspecting Bechtel geologists to GPC. This documentation has been transferred to Southern Nuclear Operating Company (SNC), as the exclusive operating licensee.

2.5.4.5.1.3 Construction Dewatering. A discussion of construction dewatering is contained in paragraph 2.4.12.1.3.3.1.

2.5.4.5.1.4 Slope Protection. During the early stages of excavation, intense rainfall caused erosion of the 2:1 side slopes of the power block excavation. The uncemented sands above the marl were eroded, resulting in deeply incised gullies in some areas. These gullies were backfilled with the native soil material, and local areas of the slope were regraded. One such area is seen on the geologic map (drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363) in the upper part of the east slope between stations N83+00 and N84+00. Another larger area exists in the south slope of the access ramp east of station E100+00. After regrading the eroded areas, berms were constructed around the tops of the slopes to control runoff. The surfaces of the slopes were sprayed with the chemical stabilizing agent Petroset, a colorless liquid which sets up and tends to bond the sand grains together. These measures proved to be generally successful in controlling further erosion.

After resumption of excavation work in 1977, erosion problems further down the slopes were encountered due to seepage of the perched ground water out of the slopes. Since stabilizing agents were expected to be ineffective under these conditions, the lower portions of the slopes were blanketed with a transition zone and covered with riprap to improve stability.

At the base of the upper sand stratum where the 2:1 slopes intersected a limestone shell bed (Utley Limestone on drawing AX6DD352), several cavities of varying size were exposed in the slopes. The location and extent of the cavities exposed in the slopes is shown on drawing AX6DD363. The largest of these existed in the northwest corner of the power block and had an opening measuring 10 ft by 10 ft. This cavity extended back into the slope some 30 ft before narrowing down to a small size. Other small cavities were encountered at varying intervals all along the north side of the power block excavation. It was necessary to fill these cavities so that an effective buttress would be formed against which the future structural backfill could be placed and compacted. The cavities were first cleaned of loose debris, then backfilled with crushed rock (Georgia State Standard No. 467). The crushed rock was packed into the cavities by means of a 20-ft-long ram attached to the blade of a bulldozer. The large cavity in the northwest corner was effectively filled in this manner to at least a distance of 25 ft back of the entrance.

To retard erosion of temporary slopes in Category 1 backfill placed in the power block excavation, these slopes were sprayed with a commercial compound known by the trade name Glassroot. It consists of a glass fiber material which is sprayed onto the slope, then coated with a film of asphalt emulsion. Other measures which also proved to be effective in controlling erosion of the compacted sandy backfill included the use of gunite, plastic sheeting, and sand bags.

2.5.4.5.1.5 Foundation Cleanup and Protection. As mentioned previously, the Blue Bluff marl (marl bearing stratum) at final grade in foundation areas was exposed using either a motor grader or Gradall. Loose material was then removed by shovel, broom, and air hose. On the vertical walls of the auxiliary building excavation, final trim to neat line was accomplished with a backhoe followed by pick and shovel and air hose techniques.

In all cases where final grade was exposed and cleaned off, the marl surface had to be covered in a manner approved by the geologist within 24 h of exposure. On horizontal surfaces the marl was covered either by structural backfill, a gunite protective layer, or a lean concrete mudmat depending on whether the particular area exposed was in a foundation or backfill area. The vertical walls of the auxiliary building basement excavation were coated with a 4-in.-thick layer of gunite reinforced with welded wire mesh.

In some cases, temporary covers such as loose soil or plastic sheeting were employed when the permanent cover material could not be applied within the 24-h limit. In all cases the temporary cover procedure was approved by either the geologist or the GPC inspector. Before placing the permanent cover material in any foundation area, the marl was inspected and approved by the geologist or soils engineer in accordance with prescribed procedures.

2.5.4.5.1.6 Foundation Inspection and Approval Procedures. All areas of marl (Blue Bluff marl) exposed and cleaned off in preparation for placement of concrete or backfill were examined closely for any evidence of loose or soft zones, geologic discontinuities, or unusual geologic features. After confirming the absence of such features, the inspecting geologist photographed and approved the excavated foundation area and documented the approval on a special form. The photographs and approval documents are part of the permanent project records.

2.5.4.5.1.7 Foundation Testing. During the general geologic mapping of the marl and other inspecting functions, a program of coring and testing samples of the marl was conducted to confirm the material properties used for design. The coring and sampling operation was performed under the direction of a Bechtel geologist and a GPC inspector, and the test assignments were made by the Bechtel foundation engineer.

A total of 38 core holes and offset replacement core holes were drilled by the rotary method in the floor of the power block excavation at 29 locations selected by the geologist. The hole locations are shown on drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363. The marl was cored to depths between 4 and 11 ft beneath the final excavated grade. Selected samples of 4-in.-diameter core were labeled and placed in wooden boxes for permanent storage at the site. Samples selected for laboratory testing were wrapped in cellophane, sealed with wax, and placed in special boxes for transportation to the laboratories of LETCO in Atlanta.

The results of the testing program are discussed in paragraph 2.5.4.2.2.

2.5.4.5.2 Backfill

Compacted backfill is placed in the power block area from the top of the marl stratum at approximately el 130 ft to the design elevation for each structure. The plant grade elevation is at 219 ft 6 in. or below. The auxiliary building and nuclear service cooling water towers, containment instrumentation cavity, and radwaste solidification building are supported directly on the marl stratum. The other safety-related power block structures are supported on compacted backfill. The foundation elevations of these structures are given in table 3.7.B.1-2. The radwaste solidification building foundation consists of large diameter drilled caissons extending into the marl stratum.

With the exception of an area north of the turbine building, an area over the in-situ slopes forming the west side of the power block excavation, and localized areas around nonsafety-related buried piping above the water table, all backfill in the power block area is compacted to an average of 97% of the maximum density determined by American Society of Testing

Materials (ASTM) D 1557, with no tests below 93% and not more than 10% of the tests between 95 and 93%. A procedure to achieve the required degree of compaction was developed in a test fill program. The results of the test fill program are discussed in paragraph 2.5.4.5.2.7 and presented in detail in reference 7.

The area north of the turbine building was compacted to an average of 95% of the maximum density determined by ASTM D 1557 with not more than 10% of tests between 93 and 95% and no test below 93%. The static stability and liquefaction analyses (paragraphs 2.5.4.8 and 2.5.4.10) were performed for the case where the power block backfill was assumed compacted to 97% relative compaction. A 95% relative compaction for the area north of the turbine building between el 185.5 to 219.5 ft has no effect on safety-related structures, since no Category 1 structures rely on this material for a load bearing foundation. The integrity of the turbine building design is not affected because the area does not project below the bottom of the building and does not provide foundation support for the turbine building. Since the area north of the turbine building is away from Category 1 structures and represents less than 10% of the total power block backfill, the factor of safety against liquefaction is not affected.

The area over the in-situ slopes forming the west side of the power block excavation was compacted with Category 2 material to an average of 95% of the maximum density determined by ASTM D 1557 with not more than 10% of tests between 93 and 95% and no test below 93%. The area is above elevation 206 and bounded by the coordinates N73+49, to N86+20 and E89+79 to E91+38. The use of Category 2 material compacted to 95% in this area has no effect on safety-related structures, since neither Category 1 structures nor the turbine building rely on this material for a load bearing foundation. The material is directly above the in-situ soils which are themselves not used to support Category 1 structures. Because the material is also over 40 ft above the water table and away from Category 1 structures the factor of safety against liquefaction is not affected.

Approximately 2.5 ft of undocumented fill was placed beneath a nonsafety-related concrete slab located above the in-situ excavation slopes on the south side of the power block. The slab is bounded by the coordinates N73+55 to N74+40 and E93+98 to E95+28, and is located at grade. The undocumented soils do not support, nor are they adjacent to, safety-related structures or equipment. They are located directly above in-situ soils which are themselves not used to support safety-related structures or equipment. Because the material is over 45 ft above the water table and located away from Category 1 structures, the factor of safety against liquefaction is not affected.

An approximate 6 inch lift of undocumented backfill was placed at grade level in a trench used for routing two separate 4-inch diameter PVC drain lines in the low voltage switchyard area. These additional pipe drains were installed to remove rainfall runoff that was accumulating in the bottom of the station service trenches and in two pull boxes. The undocumented soils do not support, nor are they adjacent to, safety-related structures or equipment. Because this material is at least 45 feet above the water table and located away from Category 1 structures, the safety factor against liquifaction was not affected.

Up to 24 in. of graded aggregate material was substituted for sand and silty sand backfill as the power block fill was brought to finished grade. This material was compacted to 97% of the maximum density determined in accordance with ASTM D 1557 and served to protect the sand and silty sand Category 1 backfill from erosion and vehicular traffic.

The localized area around nonsafety-related buried piping and similar conduits is compacted with concrete sand or other sands with similar properties to an average of 95% of maximum density determined by ASTM D 1557, with no tests below 93% and not more than 10% of the tests between 93 and 95%, unless compacted to an average of 97% using Category 1 backfill as defined below for safety-related piping. Typically, this localized area consists of backfill 3 ft

above, 1 ft below, and a maximum of 5 ft on either side of nonsafety-related buried piping or similar conduits. Only a few% of the total power block backfill utilizes this compaction criteria. All such areas are located above the water table so that the factor of safety against liquefaction is not affected. Sand compacted to an average of 95% in the limited areas around piping and similar conduits will not affect the structural integrity of any Category 1 structures. Sand compacted to an average of 95% will have static and dynamic properties consistent with those properties assumed for design of power block structures and piping. A static cone penetrometer reading of 200 is used to decide on the adequacy of concrete sand or other sands with similar properties between and below nonsafety-related piping in areas where constrained access prevents the use of the sand cone test.

Trenches containing safety-related piping or similar conduits are backfilled by placing lean concrete to the bottom of the pipe to provide continuous support and backfilling with Category 1 backfill, using wooden tampers, hand-held power tampers, or hand-held vibratory compactors as required. Use of these methods produces an average compaction of at least 97% of the maximum dry density determined in accordance with ASTM D 1557, with no tests below 93% and not more than 10% of the tests between 93 and 95%. Category 1 backfill material compacted between and immediately around pipes has a fines content below 10%. Static cone penetrometer readings developed from correlation with sand cone tests are used to decide on the adequacy of the compaction in areas where constrained access prevents the use of the sand cone test.

Post construction (after 1989) compaction may also be measured by a nuclear density test which has been correlated and calibrated with sand cone test per ASTM D2922-81 and FSAR paragraph 2.5.4.5.2.7.1D.

Lean concrete is used to backfill localized areas where placement of backfill material is impractical.

Undocumented backfill may be installed in areas designated for Category 1, Noncategory 1, and Category 2 backfill, on a case by case basis, and approved only when it is confirmed that there will be no effects on soil liquefaction, settlement, and load carrying capacities. The locations of the installment of undocumented backfill in a Category 1 area will be recorded on engineering drawings AX2D46T025 or AX2D46T001 to provide for proper consideration in regard to future installation or construction of structures, railroads, pipelines, utilities, etc.

2.5.4.5.2.1 Sources and Quantities of Backfill Material. An estimate of the total quantity of Category 1 backfill required in the power block was made by first determining the volume of the excavated area. The quantity obtained was increased by 10% to allow for compaction. The volume of all the structures below the grade level was then deducted from the total volume to obtain quantity of Category 1 backfill required. The quantity of Category 1 backfill required beneath and around the power block structures was estimated to be about 4.2 million yd^3.

Seven borrow sources within the plant site were investigated. These sources were designated as borrow areas 1, 2, 3, 4, 5, 1-A, and 1-B. In addition, two stockpiles, namely stockpiles A and B, were also investigated. These stockpiles contained material that had been obtained from excavations in the power block area. The approximate limits of all borrow areas and stockpiles are shown on drawing AX6DD394.

Detailed field exploration and laboratory testing programs were performed to identify suitable material in the borrow areas and stockpiles. The results of these investigations are presented in references 2, 3, and 4. Based on an evaluation of the field and laboratory data, it was determined that materials selected for Category 1 backfill should be sand and silty sand with not more than 25 weight% passing the U.S. No. 200 sieve size. This criterion was used to judge

the suitability of materials for backfill. It was concluded that borrow areas 1, 2, 3, 4, 5, 1-A, 1-B, and stockpile A were good sources of Category 1 backfill and that substantial quantities of sand and silty sand could be obtained by selective excavation. It was also concluded that stockpile B consisted of mainly clayey soils. From considerations of anticipated excessive structure settlement associated with clayey backfill and the difficulty of selective excavation to remove clayey soils, it was decided to exclude stockpile B as a suitable source for Category 1 backfill.

Estimated quantities of material suitable for Category 1 backfill were as follows:

Borrow Area	Estimated Quantity (yd^3)
1	1,503,000
2	251,000
3	1,007,000
4	1,197,000
5	190,000
1-A	1,700,000
1-B	1,700,000

In addition, the total quantity of material available in stockpile A was estimated to be approximately 600,000 yd^3. Thus, a total quantity of 8,148,000 yd^3 of Category 1 backfill was identified from the aforementioned sources, which was considered more than sufficient for backfill requirements in the power block.

An additional borrow area (3-A) was investigated through field exploration and laboratory testing during October and November 1992 to identify Category 1 backfill for future general use around the plant site. It was concluded that borrow area 3-A is a good source of Category 1 backfill, consisting of sand and silty sand meeting the above stated criteria, with the use of selective excavation.

The estimated quantity of Category 1 backfill, resulting from the 1992 investigation, which may be used for future plant requirements is as follows:

Borrow Area	Estimated Quantity (yd^3)
3-A	1,456,000

A portion of borrow area 3-A is a subset of old borrow area 3. This portion of borrow area 3 was never excavated. Future landfill trenches are planned due west of the existing landfill trench as shown on drawing AX6DD394. Therefore, it is likely that soil from borrow area 3-A will be excavated. Soil from borrow area 3-A which meets the criteria for Category 1 backfill (reference 28) may be stockpiled at the time of landfill excavation.

2.5.4.5.2.2 Exploration. Field exploration for borrow areas 1, 2, 3, 4, and 5 and stockpiles A and B was accomplished in early 1977. Subsurface exploration involved test pits excavated to a maximum depth of 25 ft by means of a backhoe. A total of 26, 8, 40, 12, and 3 test pits was excavated and logged in borrow areas 1, 2, 3, 4, and 5, respectively. Thirty-four test pits were excavated and logged in the two stockpiles. An appropriate number of jar and bulk samples were taken for laboratory testing.

Borrow area 1-A was investigated in the summer of 1978. Eighteen borings evenly spaced in a grid pattern covering the area were drilled and logged using a hollow stem auger. The borings extended to depths ranging from 13.5 to 66 ft below the existing grade and were terminated at

depths below the water table ranging from 0 to 14 ft. Representative soil samples were obtained at 5-ft intervals and whenever a change in soil type occurred.

Investigations in borrow area 1-B were performed in June and July 1979. Sixty borings were drilled and logged during this investigation. Holes were advanced using both rotary drilling and auger drilling techniques. The depth of borings ranged from 40.5 to 81.5 ft below existing grade and were terminated upon reaching the water table. In most of the borings representative split-spoon soil samples were obtained at 5-ft intervals. In some borings sampling was done at 2 1/2-ft intervals. In addition, bulk samples were obtained.

A field exploration of borrow area 3-A was conducted during October 1992. Fourteen soil borings were drilled using a 4-1/4 inch hollow stem auger and continuously sampled with a 3-1/2 inch I.D., 5-foot long continuous sampler. Samples were logged during the drilling operations. Three bulk samples were also obtained. The boring depths ranged from 23.8 feet to a maximum depth of 45 feet. Most of the borings were terminated upon encountering clay deposits. No borings were drilled beyond a depth of 45 feet due to the impracticality of excavating backfill to greater depths.

Logs of all test pits and borings drilled in the borrow areas and stockpiles are contained in references 2, 3, 4, and 28. Locations of all test pits and borings are shown on drawing AX6DD394.

2.5.4.5.2.3 Laboratory Testing. In order to classify the soils in borrow areas 1, 2, 3, 4, 5, 1-A, and 1-B; and stockpiles A and B and obtain the static and dynamic engineering properties of compacted backfill, many laboratory tests were performed on samples obtained from the field explorations.

These tests are listed below:

- Laboratory classification of soils.
- Grain size distribution.
- Atterberg limits.
- Moisture content of soil.
- Specific gravity.
- Moisture-density relation.
- Relative density.
- Static consolidated drained triaxial compression.
- Static consolidated undrained triaxial compression.
- Consolidation.
- Stress-controlled consolidated undrained cyclic triaxial.
- Strain-controlled consolidated undrained cyclic triaxial compression.

- Resonant column.
- Triaxial tests to determine volume changes due to cyclic loading.
  - Cyclically loaded without permitting drainage.
  - Cyclically loaded while permitting drainage.

In order to determine the suitability of soils in borrow area 3-A for use as Category 1 backfill, the following laboratory tests were performed:

- Grain size analysis, including hydrometer analysis on selected samples.
- Atterberg limits.
- Moisture content.
- Moisture-density relation.

All tests were performed in accordance with applicable ASTM test methods or recognized procedures where no ASTM was available.

Details of the test results and test procedures are included in references 2, 3, 4, 5, and 28.

2.5.4.5.2.4 Criteria for Category 1 Backfill Suitability. Soil classification test data obtained in accordance with ASTM D 2487, D 2488, D 1140, D 422, D 423, and D 424 were used to identify materials suitable for use as Category 1 backfill in the borrow areas and stockpiles. Cross-sections were developed based on the classification test data to facilitate selective excavation of acceptable material in the borrow sources. Summaries of classification test data and cross-sections for each borrow source are contained in references 2, 3, 4, and 28.

Select sand and silty sand materials are used for Category 1 backfill. In selecting soils for use as Category 1 backfill, only those soils with 25 weight% or less passing the U.S. No. 200 sieve size were considered suitable. Periodic tests in accordance with ASTM D 422 and D 1140 were used to ensure that the above criterion was being satisfied during backfilling operations.

2.5.4.5.2.5 Design Static Properties. For sand and silty sand backfill compacted to an average of 97% of the maximum dry density determined by ASTM D 1557, the static properties shown in table 2.5.4-8 were used for design. These properties were based on data obtained from the tests referred to in paragraph 2.5.4.5.2.3.

2.5.4.5.2.6 Design Dynamic Properties. For compacted sand and silty sand, the dynamic properties shown in table 2.5.4-9 are used for design.

For computing the dynamic shear modulus at higher strain levels, the attenuation factors shown in figure 3.7.B.2-5 were used for design.

The variation of damping ratio with shear strain (figure 3.7.B.1-8) was also used for design.

2.5.4.5.2.7 Test Fill Studies.

2.5.4.5.2.7.1 Test Fill for Heavy Equipment Compaction. A test fill program was performed to evaluate the performance of three different pieces of compactors: Ingersoll-Rand SPF 60,

Ingersoll-Rand SP 60, and Raygo 600A. The purpose of the program was to determine the appropriate lift thickness and number of passes required to achieve an average of 97% of the maximum density according to ASTM D 1557, with no tests below 93% and not more than 10% of tests between 95 and 93%. The material used for the test fill program consisted of sand and silty sand. Seven test fills were constructed.

The results of the test fill program are discussed in detail in reference 7.

A brief summary of the conclusions of the test fill program is given below:

A. The moisture content of the sand and silty sand material should be within ±2% of the optimum moisture content determined by ASTM D 1557. Upon initiation of the backfill program, the moisture content range was ±2%. This was later modified by program specifications to -3% to +2%. This modification was based on a review of the original test fill data and additional field tests performed during the backfill operation. Material too wet or too dry should not be compacted until brought within the required limits.

B. The Raygo 600A roller travelling at 1.5 mph can compact a 6-in. uncompacted lift, moisture conditioned within the range specified above, in four passes. Also, a combination of the Raygo 600A and Ingersoll-Rand SPF 60 rollers travelling at 1.5 mph can compact a 6-in. uncompacted lift with a total of four passes (two each). No test fills were made with the Ingersoll-Rand SPF 60 and SP 60 compactors for a 6-in. uncompacted lift thickness. These two rollers are capable of delivering a compaction effort comparable to or greater than the Raygo 600A compactor. Therefore the Ingersoll-Rand SPF 60 and SP 60 can be used for compacting a 6-in. uncompacted lift with four passes each separately.

C. The roller speed must be maintained at 1.5 mph.

D. The sand cone method (ASTM D 1556), modified to increase the minimum hole volume to 0.2 ft^3, provided consistent test results as compared to other test procedures. Therefore, it was used for all quality control testing during construction.

However, the nuclear density testing device may be used post construction (after 1989) in lieu of the sand cone method provided: 1) an acceptable laboratory calibration and field (soil specific) correlation between sand cone and nuclear test results can be achieved per the provisions of ASTM D2922-81, Calibration; 2) at least 10 test comparisons are made and the correlation is checked in an ongoing manner for every subsequent 10 nuclear density determinations; 3) the minimum 0.2 ft^3 sand cone test volume is employed; and 4) the correlation is established as per paragraphs 4.2.1, 4.2.2, and 4.2.3 of ASTM D2922-81 in order that minimal deviation is obtained.

Table 2.5.4-10 summarizes the results of the heavy equipment test fill program.

2.5.4.5.2.7.2 Test Fill for Hand Compaction Equipment. A test fill program was also performed to determine satisfactory compaction procedures involving three different types of hand compaction equipment. The test fill program was performed using the Wacker WS-74 Dual Drum, Wacker 100 (Jumping Jack), and Ingersoll-Rand SP 24 vibratory hand compactors. Based on the data obtained during the test fill operation, it was determined that the procedures outlined below will meet the Compaction requirements specified in paragraph 2.5.4.5.2.

Type of Equipment	Thickness of Lift (in.)	No. of Passes	Speed (ft/min)	Vibrations (per min)
Wacker WS-74 Dual Drum	6	4	60	3000
Wacker 100	6	2	20	630
Ingersoll-Rand SP 24	6	4	60	4000

Table 2.5.4-11 summarizes the results of the hand compaction equipment test fill program.

2.5.4.5.2.8 Nonsafety-Related Pipe Trench Backfill in Power Block Area. Trench backfill for nonsafety-related piping in Category 1 fill areas is compacted to an average of 95% relative compaction as defined in paragraph 2.5.4.5.2. The backfill material used is concrete sand with 2% or less fines. The sand is saturated and compacted by internal vibration using concrete vibrators.

A test fill program was implemented to determine whether the required degree of compaction could be achieved by the vibrated sand method. The resulting data demonstrate that the compaction above, between, and below the pipes meets the required compaction criteria. Results of the test fill program are summarized in reference 17.

2.5.4.5.2.9 Soil-Cement-Flyash Backfill. Plastic backfill consisting of cement, flyash, sand, and water is used as bedding material for Category 2 circulating water lines located in the Category 1 backfill zone north of the turbine building. Plastic backfill is being used in lieu of compacted sand and silty sand backfill because of the difficulty in obtaining the required compaction around the pipes.

Static and dynamic tests were performed on specimens consisting of different proportions of cement, flyash, sand, and water. The tests demonstrated that specimens of plastic backfill tested possess static and dynamic properties comparable to Category 1 backfill. The properties summarized below are of the plastic backfill that is used. The properties correspond to a plastic backfill mix of 65 lb of cement, 385 lb of flyash, 2586 lb of sand, and 469 lb of water per cubic yard of backfill.

Plastic unit weight	129.2 lb/ft^3
Slump	5 in.
Air content	3.5%
Unconfined compressive strength	
Average at 7 days	20.7 psi
Average at 28 days	30.5 psi
Average at 91 days	61.4 psi
Dry unit weight	
Average at 7 days	114.4 lb/ft^3
Average at 28 days	114.5 lb/ft^3
Average at 91 days	114.5 lb/ft^3
Moisture content	
Average at 7 days	14.0%

Average at 28 days	14.2%
Average at 91 days	14.6%
Cohesion	
Range at approximately 100 days	2100-5000 lb/ft^2
Angle of friction	
Range at approximately 100 days	36-48.5°
Range of shear modulus at approximately 100 days, under a confining pressure of 2 ksf for strain level of 10^{-4}%	4200-4400 lb/ft^2
Range of damping at approximately 100 days for strain level of 10^{-4}%	2.4-2.6%

2.5.4.6 Site Ground Water Conditions

The occurrence and movement of ground water beneath the site are described in detail in section 2.4.12. The first ground water body encountered beneath the VEGP site is a water table (unconfined) aquifer in the Barnwell sands and Utley limestone. It overlies the Blue Bluff marl. The site is on an interfluvial ridge that is nearly surrounded by streams that have cut down through the Barnwell sands and Utley limestone to the marl. This has isolated the water table aquifer beneath the site from adjacent areas. Ground water discharges from the water table aquifer to the surrounding streams. The streams discharge to the Savannah River.

Underlying the water table aquifer is the Blue Bluff marl, the upper member of the Lisbon Formation. The marl layer, approximately 70 ft thick, is a near-impermeable layer that effectively confines the underlying Tertiary and Cretaceous aquifers.

The Tertiary aquifer is represented beneath the site by the "unnamed sands" member of the Lisbon Formation. These sands are the local, minor equivalent of the regional Tertiary aquifer that is referred to as the principal artesian aquifer. The Cretaceous aquifer, the lowermost confined aquifer consists primarily of the sands and gravels of the Tuscaloosa Formation. It is often referred to as the Tuscaloosa aquifer. The Cretaceous aquifer and the lesser Tertiary aquifer are believed to be hydraulically connected beneath the plant site. Excavations for structures at the site do not extend through the marl; the marl remains as a hydrologic barrier beneath the site. Therefore, the confined aquifers will have no direct effect on structures.

Replenishment of the water table aquifer is by infiltration of precipitation, and after percolation to the water table, it moves laterally to the bordering interceptor streams. Contours of the water table for November 1971 and December 1984 are shown on drawings AX6DD329 and AX6DD330. The water table is, in general, subdued reflection of the ground surface, and movement is from the central portions of the interfluve toward the bordering interceptor streams.

Foundation design for the power-block facilities required excavation of the materials comprising the water table aquifer overlying the Blue Bluff marl. To construct and maintain the excavation the materials were dewatered by a series of ditches oriented in an east-west direction. They were connected by a north-south ditch, which drained to a sump in the southwest corner of the excavation. The sump was equipped with four pumps with a capacity of 500 gal/min each to remove inflows from ground water. Additional capacity was provided for the removal of inflows of storm water into the excavation. Dewatering for construction was terminated, in March 1983, and the water levels and flow pattern of the water table aquifer have returned near the

preconstruction pattern. Dewatering is discussed in more detail in paragraph 2.4.12.1.3.3.1, and in Appendix 2B.

Upon completion of construction, recharge is expected to be less in the plant area than prior to construction because of the structures, pavements, and surface drainage systems. Future recharge conditions will thus be such that the water table is not expected to rise as high as under preconstruction conditions. Power block structures are designed to accommodate ground water levels of el 165 ft; hence, no permanent dewatering system is required.

At the VEGP site the piezometric surface of the Tertiary aquifer, determined from observation wells set in the unnamed sands below the confining (marl) layer, slopes to the northeast toward the Savannah River. The river has cut through the marl in the vicinity of VEGP, and it is in hydraulic contact with the underlying Tertiary aquifer. This allows the aquifer to discharge to the river in this area. This is a relatively local condition, as downstream of the VEGP site, the confining layer is intact below the river, and the direction of ground water movement in the confined aquifers is to the southeast, the regional direction of migration of the aquifer.

Permeabilities of the aquifers and the confining layer were measured by field and laboratory methods. Details of the tests and the results are described in paragraph 2.4.12.2.4.

Permeability of Barnwell sands and clayey sands (water table aquifer) was measured in situ at two exploratory holes at the plant site and in the laboratory on three undisturbed samples. The results ranged from 10 to 302 ft/year. One disturbed sample of Barnwell sands (considered for use as backfill) and two "grab" samples of backfill material were measured at different densities. The results ranged from 430 to 20,000 ft/year. Two test wells, each with an array of 4 observation wells, were used to conduct field tests in the Utley limestone, which is at the base of water table aquifer. Data from the tests indicated that the permeability of the Utley limestone varies considerably from place to place. Calculated permeabilities range from 96 to 125,400 ft/year. The results of permeability tests of water table aquifer materials are summarized in table 2.4.12-12 and 2.4.12-13.
In-situ permeability tests in the Blue Bluff marl (the confining layer) were conducted in 95 intervals at different depths in 28 exploratory holes. In 90% of the intervals tested, no measurable water inflow occurred. In only three holes was any inflow confirmed: two of these were in near-surface, weathered marl. The range of laboratory permeability measurements is from 5.2×10^{-3} ft/year to 8.8 ft/year. Results of permeability tests in the confining layer are summarized in table 2.4.12-10.

Large quantities of ground water are stored in the confined aquifers underlying the region of the VEGP site, and relatively small withdrawals have occurred to date. Although many small communities derive water from wells, the draft on the aquifers is low because of the low population density, limited industrial development, abundant surface waters, and abundant rainfall (agricultural crops of the area do not require significant quantities of applied water). Future use of ground water for industrial and domestic use is expected to increase to some degree, but withdrawals from the confined aquifers are estimated to be small. This assessment takes into account the planned requirements of the VEGP project, which will draw from the Cretaceous aquifer for makeup water (paragraph 2.4.12.1.3.).

A comprehensive ground water monitoring program has been implemented at the VEGP. This program has been designed to monitor piezometric levels in the water table aquifer, the confined aquifers (Tertiary and Cretaceous), and hydrostatic pore pressure in the confining layer (marl). The program consists of various wells monitoring the unconfined aquifer, the Tertiary aquifer, the Cretaceous aquifer, and the confining layer. The ground water monitoring program is discussed in detail in paragraph 2.4.12.2.3.1.

2.5.4.7 Response of Soil and Rock to Dynamic Loading

This subject is addressed in subsections 3.7.1 and 3.7.2.

2.5.4.8 Liquefaction Potential

The liquefaction potential of the upper sand stratum was evaluated using the standard penetration test blow counts obtained during the investigation and the simplified procedure of Seed and Idriss.[8] This evaluation is described in detail in reference 1 and indicates that the upper sand below the ground water level is susceptible to liquefaction when subjected to the maximum SSE acceleration of 0.2 g. Based on this evaluation the upper sand stratum was removed to an approximate elevation of 130 to 135 ft in the power block area. Select sand and silty sand compacted to 97% of the maximum density determined by ASTM D 1557 is placed from the top of the marl stratum to the design elevation of the various power block structures with the exception of an area north of the turbine building as noted in paragraph 2.5.4.5.2. The liquefaction potential of compacted backfill in the power block area was evaluated for the PSAR and is discussed in detail in reference 1. The analysis indicated a factor of safety against liquefaction on the order of 1.9 to 2.0. The analysis was done utilizing cyclic strength data obtained from tests on specimens of compacted backfill.

During the investigations for borrow sources, additional dynamic data were obtained to supplement the cyclic strength data obtained previously and reported in reference 1. Cyclic triaxial tests were performed on compacted specimens of sands obtained from stockpile A and borrow area 1. The cyclic stress ratios versus the number of cycles to 2.5% total strain (initial liquefaction) are shown on drawings AX6DD395 and AX6DD396. The results show that the stress ratios for the cleaner sands are substantially lower than for silty sands. In the liquefaction analysis done previously (1) stress ratios for the cleaner sands were used to obtain the safety factor against liquefaction. Therefore, the cyclic stress ratios for the cleaner sands obtained during investigations for borrow material were compared with values obtained during the PSAR investigations. A comparison of the two test data is shown on drawings AX6DD397 and AX6DD398. The comparison indicates that the PSAR data represent a lower bound of test values. If the liquefaction analysis were performed using the upper bound values obtained during the borrow investigation, a factor of safety higher than 1.9 to 2.0 would have been obtained for the design SSE conditions.

From the discussion presented above, it is concluded that there exists an adequate factor of safety against liquefaction for backfill compacted to 97% of the maximum density obtained by ASTM D 1557.

2.5.4.9 Earthquake Design Basis

The design bases for the SSE and operating basis earthquake are addressed in paragraphs 2.5.2.6 and 2.5.2.7.

2.5.4.10 Static Stability

2.5.4.10.1 Bearing Capacity of Compacted Backfill and Marl Bearing Stratum Supporting Mat Foundations

The ultimate bearing capacity of the backfill is evaluated for the backfill consisting of sand and silty sand compacted to 97% of the maximum dry density (ASTM D 1557).

The ultimate bearing capacity of a soil is defined as the load at which shear failure will occur. A factor of safety of at least three is considered acceptable for the allowable bearing capacity for static loads. For dynamic loads, a minimum safety factor of two is required. The net ultimate bearing capacity of sand backfill supporting a rectangular foundation above the water table is given by the expression:(9)

$$q_{ult} = \gamma D n_q \left(1 + 0.2\frac{B}{L}\right) + 1/2\,\gamma B \left(1 - 0.3\frac{B}{L}\right) N\gamma - \gamma D$$

where:

q_{ult} = the net ultimate bearing capacity (k/ft^2).

γ = the total unit weight of the backfill (k/ft^3).

D = depth of embedment of the footing (ft).

B = width of the footing (ft).

L = length of the footing (ft).

$N_q, N\gamma$ = dimensionless bearing capacity factors.

For a circular foundation the expression is:

$$q_{ult} = 1/2\,\gamma B N\gamma\,(0.6) + \gamma D\,(N_q - 1)$$

If the water table is located at the bottom of a foundation supported by cohesionless material, the values obtained from the above expressions are approximately halved.

For a rectangular foundation supported entirely on the marl bearing stratum, the net ultimate bearing capacity is given by the expression:(10)

$$q_{ult} = cN_c \left(1 + 0.2\frac{D}{B}\right) \left(1 + 0.2\frac{B}{L}\right)$$

c = undrained shear strength of the marl bearing stratum (k/ft^2).

N_c = dimensionless bearing capacity factor.

For a circular foundation:

q_{ult} = $1.2\ cN_c$.

For sand and silty sand backfill compacted to 97% relative compaction (ASTM D 1557), strength parameters of c=0 and ϕ=34° derived from triaxial test data were used (paragraph 2.5.4.5.2). For the marl bearing stratum (Blue Bluff marl), strength parameters of c=10 k/ft^2 and ϕ=0 were used (paragraph 2.5.4.2.2).

A summary of power block structure loads and allowable bearing capacity is presented in table 2.5.4-12. The bearing capacity of compacted backfill was determined to be very high for the large structures under consideration. Consequently, the strength of the marl bearing stratum will govern the allowable bearing capacity of the plant structures. Since the net allowable bearing pressures in all cases far exceed the net static loads, bearing capacity of the supporting soils is not a problem. Settlement of structures will therefore govern the allowable bearing pressures.

2.5.4.10.2 Settlement of Power Block Structures on Mat Foundations

When a load of limited size is applied to a sand stratum, it will undergo shear deformation beneath the loaded area. The vertical component of this deformation is called the "initial" or elastic settlement which will occur immediately upon application of the load. Sand and silty sand drain relatively fast upon loading, and therefore, long-term volume changes with dissipation of pore water pressure do not occur in these soils.

Therefore, while estimating settlements in these soils, only elastic settlements based upon the Young's modulus of elasticity were considered.

When a load is applied to a column of saturated clay soil, the clay will deform and pore water pressures will be induced in it. Immediately after the application of the load, little, if any, pore water will be squeezed out and the clay will deform at constant volume. The vertical component of movement is called the initial or elastic settlement. In the course of time, pore water will be squeezed out of the clay and its volume will decrease. The vertical component of this volume decrease is known as "consolidation" settlement. Therefore, for estimating settlements in the marl bearing stratum, both elastic and consolidation settlements were taken into consideration.

Soil stresses and settlements were computed using the Settlement Problem Oriented Language (SEPOL) computer program developed at Massachusetts Institute of Technology.[11] The sand backfill, the marl bearing stratum, and the lower sand stratum were treated as layered systems and divided into layers of different thicknesses. The SEPOL program computes the stress and strain at the midpoint of each layer based on the theory of elasticity. The elastic settlement is computed by multiplying the calculated strain in the layer by the layer thickness. Consolidation settlement is obtained using the calculated vertical stress and the rebound portion of the laboratory consolidation test curve. The settlement of each layer is determined by taking the sum of elastic and consolidation settlement for the layer, and the total settlement is calculated as the sum of the settlements contributed by each layer.

The following soil parameters for sand and silty sand backfill, the marl bearing stratum, and the lower sand stratum are used in the calculation of elastic settlements:

Soil Parameter	Sand, Silty Sand Backfill	Marl Bearing Stratum	Lower Sand Stratum
Moist unit weight (lb/ft^3)	120	-	-
Saturated unit weight (lb/ft^3)	130	115	115
Submerged unit weight (lb/ft^3)	68	53	53
Poisson's ratio	0.4	0.5	0.4
Young's modulus (k/ft^2)	1500	10,000	See figure 2.5.4-3

For compacted backfill, the Young's modulus of 1500 k/ft^2 was obtained from static triaxial tests. For the marl bearing stratum, the value of Young's modulus of 10,000 k/ft^2 was derived from the heave data, in situ pressure meter, and seismic velocity test data. The values of Young's modulus for the lower sand stratum (figure 2.5.4-3) are based on shear wave velocity data and empirical correlations between shear wave (low-strain) Young's modulus and higher-strain static

Young's modulus. Using the available shear wave velocity data for the upper 140 ft of the lower sand stratum from table 2.5.4-5, a plot of low strain Young's modulus versus depth is obtained as shown on figure 2.5.4-3. Values of the low-strain Young's modulus below a depth of 140 ft in the lower sand stratum were obtained by extrapolation using the Seed-Idriss relationship: (18)

$$G_d = 1000K_2 (\sigma_m')^{1/2}$$

where

K_2 = a parameter that depends on the void ratio and the strain amplitude of motions.

σ_m' = mean effective stress in lb/ft^3

G_d = the low strain shear modulus in lb/ft^2.

The low strain Young's modulus (E_d) is obtained from the equation:

$$E_d = 2(1+\upsilon)G_d$$

Where υ is the Poisson's ratio of the lower sand stratum.

Based on studies made by Swiger(19) the static Young's modulus for the lower sand stratum may be obtained by taking it equal to about 1/3 the low strain value (E_d). Therefore, from figure 2.5.4-3, the static Young's moduli of the various layers in the lower sand stratum are as follows:

Depth Below Grade (ft)	E(k/ft^2)
160-260	10,800
260-460	13,500
460-760	17,500
760-1160	22,000

The lower sand stratum is approximately 1000-ft thick based on seismic investigations (paragraph 2.5.4.2.3). For the settlement analyses, the lower sand stratum was divided into four layers of thickness; 100, 200, 300, and 400 ft, respectively. Each layer was assigned an appropriate value of Young's modulus as shown above. Total elastic settlements were obtained by adding the elastic settlements obtained in the backfill, marl, and the various layers of the lower sand stratum. Elastic settlement of each layer was calculated by multiplying the strain (Σ_z) in the layer by the height of the layer. The strain (Σ_z) is given by the following expression: (9)

$$\Sigma_z = \frac{1}{E}\left\{\sigma - \upsilon(\sigma_x + \sigma_y)\right\}$$

where:

$\sigma_z' \sigma_x' \sigma_y$ = the principal stresses induced at center of the layer by the loads.

E = Young's modulus.

υ = Poisson's ratio.

The stresses σ_z, σ_x and σ_y were obtained at the midpoint of each layer from the SEPOL computer program. In calculating stresses and settlements, both the weight of the backfill and structure loadings were taken into consideration.

The consolidation settlement was computed using the following formula:[14, 15]

$$\rho_c = \frac{C_c h}{1+e_o} \log \frac{(\sigma_\upsilon^{'} + \Delta\sigma_\upsilon)}{\sigma_\upsilon}$$

where:

ρ_C = consolidation settlement.

C_C = compression index.

h = layer thickness.

e_0 = initial void ratio.

$\sigma_\upsilon^{'}$ = in situ effective vertical stress at midpoint of the marl stratum.

$\Delta\sigma_\upsilon$ = effective additional vertical stress at middepth of the marl stratum due to the surface load.

The thickness of the marl stratum was taken as 70 ft for purposes of computing consolidation settlement. Considering the highly preconsolidated nature of the marl stratum, the compression index (C_c) used in the above formula was taken equal to the rebound index (C_r) obtained from laboratory consolidated tests. [1] The value of $C_c/1 + e_0$ was determined for each consolidation test, and an average of 0.0046 was used for settlement calculations.

The results of the settlement analysis of the power block structures are presented in figure 2.5.4-1. The estimated total settlement at the center and corner of each structure are shown. Total settlements include both elastic and consolidation settlements of the marl stratum and elastic settlements of the sand backfill and the lower sand stratum. Due to the highly preconsolidated nature of the marl stratum, the consolidation settlements are small, compared to the elastic settlements. The total settlements do not include settlements in the marl and lower sand strata as a result of fill under foundations, since these settlements will occur prior to placement of building loads.

2.5.4.10.3 Caisson Foundation in Radwaste Solidification Building

2.5.4.10.3.1 Caisson Ultimate Downward Capacity. The ultimate downward capacity of a caisson supporting the radwaste solidification building is given by:

$$V_u = q_{nu} A_b + f_S A_S$$

where:

V_u = ultimate downward capacity of caisson (k).

q_{nu} = net ultimate bearing capacity of caisson (k/ft^2).

f_S = frictional resistance of caisson shaft (k/ft^2).

A_b = area of caisson base (ft^2).

A_S = peripheral area of caisson shaft (ft^2).

The net ultimate bearing capacity (q_{nu}) of a caisson foundation bearing on marl is given by:

q_{nu} = 9 S_u, where S_u = undrained shear strength of marl.

= 90 k/ft^2 for S_u = 10 k/ft^2.

For computing the frictional resistance along the caisson shaft embedded in marl, the adhesion between the marl and the concrete is taken as 1.5 k/ft^2. Shaft friction in the upper sand stratum is conservatively ignored.

The equation for the ultimate downward capacity of a caisson bearing on marl can therefore be rewritten as follows:

$$V_u = 90 A_b + 1.5 pH_m$$

where:

V_u = ultimate downward capacity of caisson (k).

A_b = area of caisson base (ft^2).

p = perimeter of caisson shaft (ft).

H_m = depth of caisson penetration in marl (ft)

The estimated total dead plus sustained live load transmitted to the radwaste building foundation is approximately 116,000 k. The foundation system consists of 54, 8-ft diameter caissons spaced 18 ft center to center. The required design capacity per caisson is approximately 2150 k. An 8-ft diameter caisson with 20 ft of penetration into the marl would develop an ultimate downward capacity of approximately 5280 k and is satisfactory for radwaste solidification building foundation support.

2.5.4.10.3.2 Caisson Settlement. A load caisson would undergo settlement as a result of three causes:

A. Elastic compression of the shaft of the caisson.

B. Elastic compression of the marl on which the caisson is supported.

C. Consolidation settlement of the marl due to dissipation of pore water pressure occurring over a period of time.

The total settlement that would result at the top of the caisson would be the sum of A, B, and C above.

The applied pressure at the top of an 8-ft diameter caisson is estimated to be approximately 40 k/ft^2. The total settlement of the caisson with an applied stress of 40 k/ft^2 at the top is not expected to exceed 1 in.

2.5.4.10.4 Foundation Heave

Prior to excavations, the soil conditions in the power block consisted of an upper sand stratum, followed by a 70-ft layer of the clay marl bearing stratum and a lower stratum of dense sand with clay to a 750-ft depth. All of the upper sand stratum was removed in the power block area. Mass excavations were carried out from the existing grade elevation of 210 ft to the top of the clay bearing stratum which is at an approximate elevation of 130 ft. The excavation commenced in May 1974 and continued through September 1974. Because of project suspension, no excavation was done from September 1974 to February 1977. Upon restart of the project in February 1977, further excavation was resumed and it was completed in August 1977. During this period of time the heave of the clay marl stratum resulting from the removal of the overburden was frequently observed and recorded by GPC. Heave values were measured at different locations within the power block area.

An average heave of approximately 1.25 in. was measured in the power block area. Field records indicate that practically the entire heave as a result of the excavations has probably occurred over the period of time of the excavations.

2.5.4.10.5 Lateral Earth Pressures

The lateral earth pressure on subterranean walls of the power block structures was computed for sand and silty sand backfill having the properties discussed in paragraph 2.5.4.5. The coefficient of earth pressure "at rest" is used and a value of 0.7 (appropriate for 97% relative compaction) are considered in the computations. Additionally, the walls are designed for surcharge loadings and dynamic soil pressures where appropriate.

2.5.4.10.6 Hydrostatic Ground Water Pressures

The maximum predicted ground water elevation is 35 ft above the top of the clay marl bearing stratum. The effect of hydrostatic ground water pressures was considered in evaluating bearing capacity and settlement of soils supporting the power block structures, as discussed in paragraph 2.5.4.10.1.

Based on information obtained from piezometers extending into the lower sand stratum, the piezometric surface of the water contained in the lower sand stratum is approximately 20 to 30 ft below the top of the clay marl bearing stratum. This pressure has no effect on safety-related structures supported on compacted backfill and the marl stratum.

2.5.4.11 Design Criteria

As discussed in paragraph 2.5.4.2, the upper sand stratum would have a potential for liquefaction from an occurrence of the SSE event at the site. Further, the shelly limestone layer is characterized by voids and discontinuities and is unsuitable for foundation support from the standpoint of bearing capacity and settlement. Therefore, all Category 1 structures are supported on either the clay marl bearing stratum or on sand and silty sand backfill compacted to an average of 97% of the maximum density determined by ASTM D 1557. The allowable bearing pressure under both static and dynamic conditions satisfies these requirements:

A. A minimum factor of safety of three against shear failure under sustained dead load plus live load.

B. A minimum factor of safety of two against shear failure under sustained dead load plus maximum live load.

C. Structure settlements within tolerable limits for sustained dead load plus live load.

Evaluation of the liquefaction potential of Category 1 backfill is based on a minimum factor of safety of 1.5 against liquefaction. Computed factors of safety against liquefaction and bearing capacity failure are identified in paragraphs 2.5.4.8 and 2.5.4.10, respectively. Methods of analyses, including assumptions made in the analyses, are also discussed in the referenced paragraphs.

2.5.4.12 Techniques to Improve Subsurface Conditions

2.5.4.12.1 Foundations in the Clay Marl Stratum

No special treatment was required to improve foundation conditions beneath Category 1 structures supported on the marl stratum.

2.5.4.12.2 Foundations in Soil

Category 1 foundations in soil are supported on sand and silty sand backfill compacted to an average of at least 97% of the maximum determined by ASTM D 1557. This subject has been addressed in paragraph 2.5.4.5.

2.5.4.13 Subsurface Instrumentation

2.5.4.13.1 Heave Instrumentation

The heave of the clay marl bearing stratum was monitored during the period of excavations in the power block area from 1974 to 1977.

The general procedure for measuring heave is outlined below:

A. Two permanent reference benchmarks were established far outside the excavation and away from all related construction activity.

B. Nine heave point tips with polyvinyl chloride protective sleeves were installed at selected locations approximately 5 ft below the eventual bottom of the excavation.

C. Invar steel reading rods were lowered through each protective sleeve to mate with the heave point tip. The rods were tensioned to alleviate any "snaking" and to rigidly clamp them into position.

D. The heave point tip elevation was determined by conducting a first order level survey from a benchmark to the top of the reading rods.

The locations of the heave points are shown in figure 2.5.4-2. Three of the heave points were damaged after installation; therefore, data from only six heave points are available. A report of

heave point measurements is presented in reference 16 and summarized in table 2.5.4-13. The data show that the measured heave of the marl stratum ranged from 0.6 to 1.7 in., with an average of 1.25 in. Results are also plotted on drawings AX6DD399 and AX6DD400.

2.5.4.13.2 Settlement Monitoring

2.5.4.13.2.1 Program Description. The foundation design parameters for all power-block structures were based on measured soil parameters obtained by field exploration and laboratory testing. The structures and the interconnecting piping are designed for building settlement. A settlement monitoring program was initiated to record settlements at various locations in the structures.

This monitoring program consists of two permanent benchmarks installed as reference points for measurements and monitoring points (i.e., settlement markers) as shown on drawing AX2D55V001. The settlement monitoring program is a separate engineering program implemented by plant procedures. Changes to this program will be made through the engineering design process.

2.5.4.13.2.2 Marker Reading Frequency. Markers were originally read at approximately 60-day intervals. Subsequent to October 1987, the 58 markers which have been demonstrated to be essentially stable have been read at maximum 6-month intervals and readings on two redundant markers (423-1 and 423-1B) have been discontinued (references 21, 22, and 24). Subsequent to September 1991, an additional 63 markers, which have been demonstrated to be essentially stable, have been read at maximum 6-month intervals, and readings on five markers associated with the radwaste solidification building and radwaste transfer tunnel (155, 158, 159-R, 160-R, and 161-R) have been discontinued. Subsequent to June 1, 1994, all markers except four (249, 252, 260, and 292) have been demonstrated to be essentially stable and were read annually. Subsequent to July 1, 1995, all markers have been demonstrated to be essentially stable and were read annually. Subsequent to July 1, 2002, the following markers have been demonstrated to be essentially stable and were read approximately every 24 months: 5, 14, 20, 26, 32, 40, 48, 54, 63, 72, 81, 87, 100-R, 101-R, 102-R, 103-R, 104-R, 105-R, 106, 119, 120, 121-R, 122, 123-R, 126, 129-R, 130-R, 131-R, 133-R, 135, 138, 139, 143, 146-R, 147, 149-R, 150-R, 157, 162, 163, 164-R, 165-R, 166-R, 167-R, 168-R, 169-R, 170, 171, 172-R, 173, 174, 175, 179, 184, 191, 197, 198, 199-R, 200-R, 201-R, 202-R, 203-R, 204-R, 205-R, 206, 219, 221, 222-R, 224, 225-R, 228, 229, 230, 234, 236, 237-R, 238-R, 239, 242, 247-R, 248-R, 250-R, 251-R, 254-R, 261, 262, 263-R, 264-R, 265-R, 266-R, 267-R, 268, 269, 270-R, 271, 272, 273, 276, 279, 280, 281, 282, 283-R, 284-R, 286, 287, 288, 290, 291, 293, 294, 295-R, 298, 420-2, 423-1A, 426, 427, 503, 506, 509, 512, 603, 606, 609, 612, 1000-R, 1002, 1003-R, 1004-R, 1005, 1006, 2000-R, 2001, and 2002. Subsequent to July 1, 2004, markers that are not required for differential settlement, Category 1 buried pipe, base mat, and building tilt evaluations and have been demonstrated to be essentially stable will be inactivated and no longer read. Most of the remaining active markers have been demonstrated to be essentially stable and will be read approximately every 5 years. However, some markers have indicated a higher percentage of allowable differential settlement or increasing differential settlement and these markers will be read annually. Active markers read annually or approximately every 5 years are identified in the plant procedures. In addition to the established reading frequencies described above, either of the two following conditions will result in the immediate monitoring of the settlement markers.

Settlement monitoring will be performed immediately after any earthquake event equal to or exceeding a free-field acceleration of 1/2 OBE (0.06g). A settlement survey will also be

performed if the groundwater level in the power block area drops more than 10 ft. below the reference groundwater level of 160 msl in more than one observation well monitoring the backfill.

2.5.4.13.2.3 Long Term Monitoring Commitments. Settlement monitoring for both units will be continued for important markers, as described in paragraph 2.5.4.13.2.4, through the first year following issuance of the operating license for Unit 2. At the end of this period, a brief technical report was provided to the NRC with supporting settlement data and graphical plots, and an evaluation of the settlement effects that justified a reduction in the frequency and number of markers monitored.

2.5.4.13.2.4 Marker Relocations. Throughout the life of the plant, important settlement markers that are destroyed or become inaccessible will be replaced with markers as near as possible to their original locations so that continuous readings can be provided and the total settlement at the original locations can be determined. Important settlement markers are defined as those markers located in safety-related structures whose total settlements have exceeded 1 in. and/or which are needed to continue the determination of differential settlements across a structure, or to establish differential settlements at piping penetrations.

In addition to markers which become inaccessible or are accidentally destroyed, a maximum of 96 markers are being moved from their original locations to new locations which provide more convenient access (references 21 and 22). The original markers will remain in place and will be available should any future correlations be required. Markers relocated to the outside of structures or to higher elevations for more convenient access, will remain as close as practicable to their original plan view locations (reference 23).

2.5.4.13.2.5 Total Structure Settlement. Actual total structure settlements have been compared with the predicted totals in FSAR figure 2.5.4-1 and were provided in a report to the NRC staff 3 months prior to fuel load of Unit 1 (reference 20). If total predicted settlements are exceeded, the settlement analysis will be reevaluated.

For additional discussion, see reference 25.

2.5.4.13.2.6 Differential Settlement Effect on Piping. Actual differential settlements between structures will be compared to those used in design. Piping has been installed as late in the construction schedule as practicable, and supports on either side of building interfaces typically have not been installed until structure construction is essentially complete. This has resulted in most structure settlements having taken place prior to permanent installation of piping. Subsequent to permanent installation of piping, differential settlement is monitored at building interfaces.

If future evaluations of the actual differential settlement indicate that 75% of the amount used in the design of piping has been reached, the situation will be reviewed. If it appears that the design differential settlement may be exceeded, the piping will be reanalyzed for an increased differential settlement and/or the supports will be adjusted or modified to satisfy the design requirements.

For additional discussion, see references 26 and 27.

2.5.4.13.2.7 Differential Settlement within Structures. The effect of differential settlements within structures on the structures themselves is addressed by reviewing the maximum net

slope of the deflection curve (δ/ℓ) relative to structure tilt. This review is performed for large power block structures, auxiliary building, control building, and fuel handling building. (Refer to reference 26.) Unless differential settlement within structures approach a value of δ/ℓ equal to 1/670, it is deemed unnecessary to evaluate such effects on the design of structures. Even at this net slope, no adverse effects are likely.

An update of differential settlements was provided in a report to NRC staff 3 months prior to fuel load of Unit 1 (reference 20). The report included an assessment of the net slope of the deflection curve (δ/ℓ) relative to structural tilt. The net slope was demonstrated to be well within the 1/670 limit. The review was performed for the control building, the auxiliary building and the fuel handling building. These large power block structures contain the number of markers required and the accessibility needed to perform such a review. The report, however, included the total and differential settlement for all other safety-related structures to demonstrate that the settlements which occurred were reasonable and within predicted limits.

The slope of the deflection curve relative to structure tilt for the above structures will be reviewed when settlement data warrants evaluation.

For additional discussion, see reference 26.

2.5.4.14 Construction Notes

There have been no significant construction problems, apart from the erosion of Category 1 backfill, that occurred as a result of heavy rainfall in early November 1979. Areas within the power block subjected to erosion are described in detail in a report submitted to the Nuclear Regulatory Commission.(14) The report outlined steps that had been initiated subsequent to the erosion to repair the affected and adjacent areas and to facilitate resumption of backfilling operations in the power block area. Also included in the report were recommended methods of repair and a description of future erosion and ground water control measures to prevent a recurrence of the problem.

All erosion in the power block backfill was satisfactorily repaired according to recommended procedures, with the exception of minor deviations that were necessitated by practical considerations.

Extensive field and laboratory tests were performed to verify the extent of disturbed material in the eroded areas. These tests were used to verify the competency of the backfill adjacent to the foundations of various Category 1 structures. The evaluation of the effect of erosion on Category 1 structure foundations was based on data developed during testing and visual observations made during the entire period of repair. The data and evaluation are contained in reference 15. The field testing and evaluations described in reference 15 provided adequate data which defined the disturbed zones in Category 1 backfill. All erosion was successfully repaired. This evaluation has established that there is no detrimental effect on the existing structures as a result of the heavy rainfall of early November 1979.

2.5.4.15 Standard Review Plan Evaluation

The Standard Review Plan calls for probabilistic as well as deterministic analyses of liquefaction potential at the site.

The liquefaction analyses performed for VEGP were of the deterministic type only.

The foundation properties for materials underlying Seismic Category 1 structures are known with much greater accuracy at VEGP than at most nuclear power plant sites. This is because

all potentially liquefiable foundation materials have been removed and replaced with homogeneous, well-compacted structural backfill. All Seismic Category 1 structures are founded either on this backfill or on the underlying very competent marl.

The deterministic evaluation of the liquefaction potential described in subsection 2.5.4 involved use of extensive laboratory test data that covered the upper and lower bound cyclic shear strengths of compacted Category 1 backfill. The deterministic analyses have demonstrated (paragraph 2.5.4.8) that an adequate factor of safety exists against liquefaction.

The backfill supporting Category 1 structure foundations has been placed under extremely well-controlled conditions and exceeds the minimum design compaction requirements (97% of the maximum density determined by ASTM D 1557). The auxiliary building and nuclear service cooling water towers are supported on the marl stratum and surrounded by compacted backfill. The remaining Category 1 structures, including containment, control, fuel, and diesel generating buildings, are supported on compacted sand and silty sand backfill. The inplace density of the sand backfill is so high that, when sheared, the backfill increases in volume and relieves all excess pore water pressures. Liquefaction will therefore not occur in compacted backfill under SSE conditions. The properties of the marl stratum have been well established based on extensive field and laboratory test data and show that it will adequately support Category 1 structures and that it is nonliquefiable. The backfill surrounding Category 1 structures is of sufficient width and length (paragraph 2.5.4.5) such that in the event of an SSE, none of the Category 1 structures will be affected by the potential liquefaction of the in situ upper sand stratum in the areas outside the backfill.

Because the foundation conditions are so well controlled and defined as described above and because liquefaction is shown to be clearly not possible under the design SSE, a probabilistic evaluation of liquefaction potential is not considered necessary for the VEGP site.

2.5.4.16 References

1. Bechtel Power Corporation, Report on Foundation Investigations, Alvin W. Vogtle Nuclear Project, July 1974.

2. Bechtel Power Corporation, Report on Backfill Material Investigations, Alvin W. Vogtle Nuclear Project, January 1978.

3. Bechtel Power Corporation, Report on Backfill Material Investigations, Alvin W. Vogtle Nuclear Project, Addendum No. 1, October 1978.

4. Bechtel Power Corporation, Report on Backfill Material Investigations, Alvin W. Vogtle Nuclear Project, Addendum No. 2, November 1979.

5. Bechtel Power Corporation, Report on Dynamic Properties for Compacted Backfill, Alvin W. Vogtle Nuclear Project, February 1978.

6. Skempton, "Discussion of the Planning and Design of the New Hong Kong Airport," Proceedings of the Institution of Civil Engineers, Vol 7, pp 305-307, 1957.

7. Bechtel Power Corporation, Test Fill Program, Phase II, Alvin W. Vogtle Nuclear Plant, October 1978.

8. Seed, H. B., and Idriss, I. M., "Simplified Potential for Evaluating Soil Liquefaction Potential," Journal of the Soil Mechanics and Foundations Division of the ASCE, Vol 97, No. SM 9, pp 1249-1273, September 1971.

9. Lambe, T. W., and Whitman, R. V., Soil Mechanics John Wiley and Sons, Inc., New York, 1969.

10. Terzaghi, K., and Peck, R. B., Soil Mechanics in Engineering Practice, 2nd Edition, John Wiley and Sons, Inc., New York, 1967.

11. Jordan, J. C., and Schiffman, R. L., ICES SEPOL-1—A Settlement Problem Oriented Language User's Manual, Publication No. 204, Civil Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts, December 1967.

12. Peck, R. B., Hansen, W. E., and Thornburn, T. H., Foundation Engineering, 2nd Edition, John Wiley and Sons, Inc., New York, 1973.

13. Winterkorn, H. F., and Fang, H. Y., ed, Foundation Engineering Handbook, Van Nostrand Reinhold Company, pp 181-185, New York, 1975.

14. Letter, with attachments, from D. E. Dutton (Georgia Power Company) to J. P. O'Reilly (Nuclear Regulatory Commission), dated January 8, 1980.

15. Bechtel Power Corporation and Georgia Power Company, Final Report on Dewatering and Repair of Erosion in Category 1 Backfill in Power Block Area, August 15, 1980.

16. Letter from C. J. Dunnicliff to J. D. Duffin, dated October 12, 1977.

17. Letter from D. O. Foster (Georgia Power Company) to E. G. Adensam (Nuclear Regulatory Commission), dated October 27, 1983.

18. Seed, H. B. and Idriss, I. M., "Soil Moduli and Damping Factors for Dynamic Response Analyses," EERC, University of California, Berkely, California, December 1970.

19. Swiger, F., "Evaluation of Soil Moduli for Soil Structure Interaction Analysis," presented at the Conference on Analysis and Design in Geotechnical Engineering, Austin, Texas, 1974.

20. Bechtel Power Corporation, VEGP Report on Settlement, August 1986.

21. Letter GN-1368, dated May 22, 1987.

22. Letter SL-3002, dated August 3, 1987.

23. Letter GN-1398, dated September 22, 1987.

24. Letter from Nuclear Regulatory Commission to Georgia Power Company, dated October 20, 1987.

25. Letter GN-350, dated May 9, 1984.

26. Letter GN-350, dated May 9, 1984.

27. Letter GN-350, dated May 9, 1984.

28. Category 1 Backfill Investigation, dated November 30, 1992 (Response to REA-VG-0672).

2.5.5 STABILITY OF SLOPES

2.5.5.1 Slope Characteristics

Category 1 slopes consisted of excavation cut slopes and temporary backfill slopes. The excavation slopes were cut in the upper sand stratum and shell zone at two horizontal to one vertical. The lower 5 ft of the cut was in the clay bearing stratum. Parameters for design of the excavation slopes were based on data developed for the upper sand and clay bearing strata (paragraph 2.5.4.2). A total stress design shear strength of c=0, ϕ=34° was used for the upper sand stratum and c=10,000 lb/ft^2, ϕ=0° for the clay bearing stratum (table 2.5.4-2).

Temporary fill slopes were constructed at a minimum of 1.5 horizontal to one vertical where the slope height exceeded 3 ft except for a few deviations which are addressed in references 1 and 2. Slopes or portions of slopes of heights less than 3 ft were placed with stable side slopes. Except for north of the turbine building, fill slopes consisted of sand and silty sand material compacted to an average of 97% of the maximum density by American Society of Testing Materials (ASTM) D 1557. Fill slopes north of the turbine building consisted of sand and silty sand backfill compacted to an average of 95% of the maximum density by ASTM D 1557. Parameters for design of temporary fill slopes were based on data developed for compacted Category 1 backfill (paragraph 2.5.4.5). Design effective stress parameters of C'=0, ϕ=34° were used in analyzing temporary fill slopes.

2.5.5.2 Design Criteria and Analysis

The stability of the excavation cut slopes in in situ soil was determined using a computer program based on a modification of the Swedish Slip Circle method of slices analysis.(3) The slopes were analyzed for stability by assuming the material below the water table to be dewatered. A peripheral dewatering system is being used to control ground water and will be continued until backfilling is completed above the ground water table. In a dewatered condition, the factor of safety against sliding for a slope of two horizontal to one vertical was determined to be 1.3. This was considered satisfactory for a temporary construction slope. Earthquake forces were not considered in the design of these slopes since they are temporary during the construction period only.

For temporary fill slopes (1.5 horizontal to one vertical), slope stability analysis was performed using the Integrated Civil Engineering Systems LEASE computer program.(4)

The analysis revealed that a deep seated sliding failure will not occur, and any instability in the fill will be manifested in the form of minor raveling of the fill surface if it is steeper than the effective angle of skin friction. Infinite slope analysis based on the design friction angle of 34° indicated that temporary fill slopes will have a minimum factor of safety against raveling of 1.01. This was considered satisfactory for temporary fill slopes in a dewatered condition.

Surcharge loadings, such as buildings, on the top of a slope will affect the slope stability. To prevent loss of bearing capacity for the structure foundation and to ensure slope stability, buildings were located a sufficient distance away from the top of the slope. When situations arose during construction that required a building to be placed near a temporary fill slope, each case was analyzed to determine the minimum setback distance.

2.5.5.3 Log of Borings

Log of borings is listed in references 5, 6, and 7.

2.5.5.4 Compacted Backfill

This subject is discussed in paragraph 2.5.4.5.2.

2.5.5.5 References

1. Letter, with attachments, from D. E. Dutton of GPC to J. P. O'Reilly of the NRC, dated January 8, 1980.

2. Bechtel Power Corporation, Final Report on Dewatering and Repair of Erosion in Category 1 Backfill in Power Block Area, August 1980.

3. U.S. Corps of Engineers, "The Method of Slices," Civil Works Engineering Manual, SM 1110-2-1902.

4. Berkley, W. A., and Christian, J. T., "ICES LEASE-1: A Problem Oriented Language for Slope Stability Analysis," User's Manual, Soil Mechanics Publication No. 235, Massachusetts Institute of Technology, April 1969.

5. Bechtel Power Corporation, Report on Backfill Material Investigations, Vogtle Electric Generating Plant, January 1978.

6. Bechtel Power Corporation, Report on Backfill Material Investigations, Addendum No. 1, Vogtle Electric Generating Plant, October 1978.

7. Bechtel Power Corporation, Report on Backfill Material Investigations, Addendum No. 2, Vogtle Electric Generating Plant, November 1979.

2.5.6 EMBANKMENTS AND DAMS

There are no earth, rock, or earth and rock fill embankments required for plant flood protection or for impounding cooling water required for the operation of the plant.

TABLE 2.5.1-1

STRATIGRAPHIC UNITS IN THE VICINITY OF VEGP

System		Series	Formation	Description
Quaternary	Recent to Pleistocene		Alluvium	Alluvial fill and terrace deposits in stream valleys, consisting of tan to gray sand, clay, slit, and gravel.
Tertiary	Miocene		Hawthorne Formation	Tan, red, and purple sandy clay, interbedded lenses of gravel, and numerous clastic dikes.
Tertiary	Eocene	Jackson Age	Barnwell Group	Red, brown, yellow, and buff, fine to course, massive to crossbedded sand and sandy clay.
		Claiborne Age	Lisbon Formation	Yellow-brown to green, fine to coarse, glauconitic quartz sand, interbedded with green, red, yellow, and tan clay, sandy marl or limestone, and lenses of siliceous limestone.
Tertiary	Paleocene		Huber/Ellenton Formation	Dark-gray to black, lignitic, micaceous clay containing disseminated crystals of gypsum. Medium- to dark-gray coarse sand and white kaolin.
Cretaceous	Upper		Tuscaloosa Formation	Tan, buff, red, and white cross- bedded micaceous quartzite and arkosic sand and gravel, interbedded with red, brown, and purple clay and white kaolin.
Triassic	Upper		Newark Group	Gray, dark-brown, and brick-red sandstone, siltstone, gray-wacke, and claystone with included sections of fanglomerate or conglomerate.
Paleozoic and Precambrian			Basement Rock of the Carolina Slate Belt and Charlotte Belt	Granite, gneiss, chlorite- hornblende, and chlorite- tremolite schist, slate, and volcanic rocks.

TABLE 2.5.2-1 (SHEET 1 OF 9)

SIGNIFICANT EARTHQUAKES WITHIN 200 MILES OF THE SITE

(Intensity ≥ 4 or Magnitude ≥ 3)

DATE			TIME			LOCATION		DEPTH	INTEN.	MAG.	SITE	DATA	
GMT						DEG N	DEG W	MI	MMI		DIST.	SOURCE	
YR	MO	DY	HR	MN	SEC						MI		
1776	11	5				35.3	83.2	-	IV	-	170	BOL	
1799	4	4				32.9	80.0	-	V	-	105	STR	
1799	4	11	8	20		32.9	80.0	-	V	-	105	STR	
1799	4	11	14	55		34.3	80.6	-	V	-	105	BOL	
1817	1	8	4			32.8	79.8	-	V	-	115	BOL	
1820	9	3	8	30		33.4	79.3	-	IV	-	145	STR	
1851	8	11	1	55		35.6	82.6	-	V	-	175	STR	
1853	5	20				34.0	81.2	-	VI	-	70	STR	
1857	12	19	9	4		32.8	79.8	-	V	-	115	BOL	
1860	1	16	18			32.8	79.8	-	V	-	115	BOL	
1869	0	0				32.9	80.0	-	IV	-	105	STR	
1872	6	17	20			33.1	83.3	-	V	-	90	EQH	
1874	2	10				35.7	82.1	-	VI	-	175	BOL	
1874	2	22				35.7	82.1	-	IV	-	175	EUS	
1874	3	17				35.7	82.1	-	IV	-	175	EUS	
1874	3	26				35.7	82.1	-	IV	-	175	EUS	
1874	4	14				35.7	82.1	-	IV	-	175	EUS	
1874	4	17				35.7	82.1	-	IV	-	175	EUS	
1875	11	2	2	55		33.8	82.5	-	VI	-	60	EUS	
1876	12	12				32.9	80.0	-	IV	-	105	STR	
1879	12	13				35.0	80.9	-	V	-	140	EUS	
1885	10	17	17	30		33.0	83.0	-	IV	-	70	BOL	
1886	8	27	8	30		33.0	80.2	-	V	-	90	BOL	
1886	8	28	8	45		32.9	80.0	-	VI	-	105	STR	

TABLE 2.5.2-1 (SHEET 2 OF 9)

DATE			TIME			LOCATION		DEPTH	INTEN.	MAG.	SITE	DATA	
GMT						DEG N	DEG W	MI	MMI		DIST.	SOURCE	
YR	MO	DY	HR	MN	SEC						MI		
1886	8	28	9	40	32.9	80.0	-	IV	-	105	STR		
1886	8	28	18	20	32.9	80.0	-	IV	-	105	STR		
1886	9	1	2	51	32.9	80.0	-	X	-	105	EQH		
1886	9	1	6	5	32.9	80.0	-	VI	-	105	STR		
1886	9	2	4	55	32.9	80.0	-	V	-	105	STR		
1886	9	3	21		30.4	81.7	-	IV	-	190	STR		
1886	9	4	4	1	32.9	80.0	-	VI	-	105	STR		
1886	9	4	9		30.4	81.7	-	IV	-	190	STR		
1886	9	5			30.4	81.7	-	IV	-	190	STR		
1886	9	6	4	6	32.9	80.0	-	VI	-	105	STR		
1886	9	6	16	35	32.9	80.0	-	IV	-	105	STR		
1886	9	8			30.4	81.7	-	IV	-	105	STR		
1886	9	9	18	47	30.4	81.7	-	IV	-	190	STR		
1886	9	17	6	29	32.9	80.0	-	VI	-	105	STR		
1886	9	21	10	15	32.9	80.0	-	VI	-	105	STR		
1886	9	21	10	30	32.9	80.0	-	V	-	190	STR		
1886	9	27	19	2	32.9	80.0	-	VI	-	105	STR		
1886	9	27	22	2	32.9	80.0	-	V	-	105	STR		
1886	10	9	3	40	32.9	80.0	-	IV	-	105	STR		
1886	10	9	5	40	32.9	80.0	-	IV	-	105	STR		
1886	10	9	6	48	32.9	80.0	-	VI	-	105	STR		
1886	10	22	10	20	32.9	80.0	-	VI	-	105	EQH		
1886	10	22	19	45	32.9	80.0	-	VII	-	105	EQH		
1886	10	23	1	7	32.9	80.0	-	IV	-	105	STR		
1886	11	5	17	20	32.9	80.0	-	VI	-	105	EQH		
1886	11	28	20	13	32.9	80.0	-	IV	-	105	STR		
1887	1	4	11	44	32.9	80.0	-	VI	-	105	STR		
1887	3	4	7		32.9	80.0	-	IV	-	105	STR		

TABLE 2.5.2-1 (SHEET 3 OF 9)

DATE			TIME			LOCATION		DEPTH	INTEN.	MAG.	SITE	DATA	
GMT						DEG N	DEG W	MI	MMI		DIST.	SOURCE
YR	MO	DY	HR	MN	SEC						MI		
1887	3	17	14	9		32.9	80.0	-	V	-	105	STR	
1887	3	18	23	10		32.9	80.0	-	IV	-	105	STR	
1887	3	19				32.9	80.0	-	IV	-	105	STR	
1887	3	24				32.9	80.0	-	IV	-	105	STR	
1887	3	24	4	5		32.9	80.0	-	IV	-	105	STR	
1887	3	28				32.9	80.0	-	IV	-	105	STR	
1887	4	7	4			32.9	80.0	-	IV	-	105	STR	
1887	4	8	9			32.9	80.0	-	IV	-	105	STR	
1887	4	10	11	30		32.9	80.0	-	IV	-	105	STR	
1887	4	14	7	25		32.9	80.0	-	IV	-	105	STR	
1887	4	26	10			32.9	80.0	-	IV	-	105	STR	
1887	4	28	8			32.9	80.0	-	V	-	105	STR	
1887	5	6				32.9	80.0	-	IV	-	105	STR	
1887	6	3	12			32.9	80.0	-	IV	-	105	STR	
1887	7	10	18			32.9	80.0	-	IV	-	105	STR	
1887	8	27	4	30		32.9	80.0	-	V	-	105	STR	
1887	8	27	9	20		32.9	80.0	-	IV	-	105	STR	
1888	1	12	15	54		32.9	80.0	-	VI	-	105	STR	
1888	1	16	17	52		32.9	80.0	-	IV	-	105	STR	
1888	2	29	11			32.9	80.0	-	V	-	105	STR	
1888	3	3				32.9	80.0	-	IV	-	105	STR	
1888	3	3	4	30		32.9	80.0	-	IV	-	105	STR	
1888	3	4				32.9	80.0	-	IV	-	105	STR	
1888	3	14	5			32.9	80.0	-	V	-	105	STR	
1888	3	20	5			32.9	80.0	-	IV	-	105	STR	
1888	3	25				32.9	80.0	-	IV	-	105	STR	
1888	4	16				32.9	80.0	-	IV	-	105	STR	
1888	4	16				32.9	80.0	-	IV	-	105	STR	

TABLE 2.5.2-1 (SHEET 4 OF 9)

DATE			TIME			LOCATION		DEPTH	INTEN.	MAG.	SITE	DATA	
GMT						DEG N	DEG W	MI	MMI		DIST.	SOURCE	
YR	MO	DY	HR	MN	SEC						MI		
1888	4	16				32.9	80.0	-	IV	-	105	STR	
1888	5	2				32.9	80.0	-	IV	-	105	STR	
1889	2	10		31		32.9	80.0	-	V	-	105	STR	
1889	7	12	2	54		32.9	80.0	-	IV	-	105	STR	
1891	10	13	5	55		32.9	80.0	-	IV	-	105	STR	
1893	6	21	4	5		30.4	81.7	-	IV	-	190	STR	
1893	6	21	7	7		30.4	81.7	-	IV	-	105	STR	
1893	7	5	8	10		32.9	80.0	-	IV	-	105	STR	
1893	7	6	9	5		32.9	80.0	-	IV	-	105	STR	
1893	7	8	7	48		32.9	80.0	-	IV	-	105	STR	
1893	7	8	15	25		32.9	80.0	-	IV	-	105	STR	
1893	9	19	7	5		32.9	80.0	-	IV	-	105	STR	
1893	9	19	7	40		32.9	80.0	-	IV	-	105	STR	
1893	9	19	8	55		32.9	80.0	-	IV	-	105	STR	
1893	11	8	4	40		32.9	80.0	-	IV	-	105	STR	
1893	11	8	6	5		32.9	80.0	-	IV	-	105	STR	
1893	12	27	6	51		32.9	80.0	-	IV	-	105	STR	
1893	12	27	7	17		32.9	80.0	-	IV	-	105	STR	
1893	12	27	9	9		32.9	80.0	-	IV	-	105	STR	
1893	12	27	9	56		32.9	80.0	-	IV	-	105	STR	
1893	12	28	2	20		32.9	80.0	-	IV	-	105	STR	
1894	1	10	8	5		32.9	80.0	-	IV	-	105	STR	
1894	1	10	8	49		32.9	80.0	-	IV	-	105	STR	
1894	1	10	9	15		32.9	80.0	-	IV	-	105	STR	
1894	1	30	4	5		32.9	80.0	-	IV	-	105	STR	
1894	2	1	5	21		32.9	80.0	-	IV	-	105	STR	
1894	6	16	2	16		32.9	80.0	-	IV	-	105	STR	
1894	12	11	5	27		32.9	80.0	-	IV	-	105	STR	

TABLE 2.5.2-1 (SHEET 5 OF 9)

DATE			TIME			LOCATION		DEPTH	INTEN.	MAG.	SITE	DATA	
GMT						DEG N	DEG W	MI	MMI		DIST.	SOURCE
YR	MO	DY	HR	MN	SEC						MI		
1895	1	8	5	40		32.9	80.0	-	IV	-	105	STR	
1895	1	8	5	58		32.9	80.0	-	IV	-	105	STR	
1895	1	8	7	29		32.9	80.0	-	IV	-	105	STR	
1895	4	27	7	40		32.9	80.0	-	IV	-	105	STR	
1895	7	25	4	1		32.9	80.0	-	IV	-	105	STR	
1895	10	6	6	25		32.9	80.0	-	IV	-	105	STR	
1895	10	20	17	8		32.9	80.0	-	IV	-	105	STR	
1895	11	12	23	33		32.9	80.0	-	IV	-	105	STR	
1896	3	19	8	22		32.9	80.0	-	IV	-	105	STR	
1896	8	11	5	58		32.9	80.0	-	IV	-	105	STR	
1896	8	11	6	14		32.9	80.0	-	IV	-	105	STR	
1896	8	11	8	15		32.9	80.0	-	IV	-	105	STR	
1896	8	11	9	24		32.9	80.0	-	IV	-	105	STR	
1896	8	12	7	42		32.9	80.0	-	IV	-	105	STR	
1896	8	14	5	43		32.9	80.0	-	IV	-	105	STR	
1896	8	30	3	24		32.9	80.0	-	IV	-	105	STR	
1896	9	8	18	16		32.9	80.0	-	IV	-	105	STR	
1896	11	14	8	15		32.9	80.0	-	IV	-	105	STR	
1899	3	10	5	45		32.9	80.0	-	IV	-	105	STR	
1899	12	4	12	48		32.9	80.0	-	IV	-	105	STR	
1900	10	31	16	15		30.4	81.7	-	V	-	190	EQH	
1901	12	2	0	26		32.9	80.0	-	IV	-	105	STR	
1903	1	24	1			32.9	80.0	-	IV	-	105	STR	
1903	1	24	1	15		32.1	81.1	-	VI	-	80	EQH	

TABLE 2.5.2-1 (SHEET 6 OF 9)

DATE			TIME			LOCATION		DEPTH	INTEN.	MAG.	SITE	DATA	
GMT						DEG N	DEG W	MI	MMI		DIST.	SOURCE	
YR	MO	DY	HR	MN	SEC						MI		
1903	1	31	10	54		32.9	80.0	-	IV	-	105	STR	
1903	2	3	10	6		32.9	80.0	-	IV	-	105	STR	
1907	4	19	8	30		32.9	80.0	-	V	-	105	EUS	
1911	4	20	22			35.2	82.7	-	V	-	150	BOL	
1911	4	21	3			35.2	82.7	-	V	-	150	EUS	
1911	6	12	10	30		32.9	80.0	-	VII	-	105	EUS	
1912	6	20				32.0	81.0	-	V	-	90	EUS	
1912	9	29	8	6		32.9	80.0	-	IV	-	105	STR	
1912	10	23	1	15		32.7	83.5	-	IV	-	105	STR	
1912	12	7	19	10		34.7	81.7	-	IV	-	105	STR	
1913	1	1	18	28		34.7	81.7	-	VII	-	105	EQH	
1914	3	5	20	5		33.5	83.5	-	VI	-	100	EQH	
1914	3	7	1	20		34.2	79.8	-	IV	-	135	STR	
1914	7	13	20	53		33.0	80.2	-	IV	-	90	BOL	
1914	9	22	2	4		33.0	80.2	-	V	-	90	BOL	
1915	10	29	6			35.8	82.7	-	V	-	190	EQH	
1916	2	21	17	39		35.5	82.5	-	VII	-	170	BOL	
1916	3	2	5	2		34.5	82.7	-	IV	-	105	EUS	
1923	12	31	20	6		34.8	82.5	-	IV	-	120	BOL	
1924	10	20	8	30		35.0	82.6	-	V	-	135	EQH	
1926	7	8	9	50		35.9	82.1	-	VI	-	190	EQH	
1928	11	20	3	45		35.8	82.3	-	IV	-	185	STR	
1928	12	23	2	30		35.3	80.3	-	IV	-	170	STR	
1929	1	3	12	5		33.9	80.3	-	IV	-	100	STR	
1929	10	28	2	15		34.3	82.4	-	IV	-	85	STR	
1930	12	10	0	2		34.3	82.4	-	IV	-	85	STR	
1930	12	26	3			34.5	80.3	-	IV	-	125	STR	
1931	5	6	12	18		34.3	82.4	-	IV	-	85	STR	

TABLE 2.5.2-1 (SHEET 7 OF 9)

DATE			TIME			LOCATION		DEPTH	INTEN.	MAG.	SITE	DATA	
GMT						DEG N	DEG W	MI	MMI		DIST.	SOURCE
YR	MO	DY	HR	MN	SEC						MI		
1933	6	9	11	30		33.3	83.5	-	IV	-	100	STR	
1933	12	19	14	12		33.0	80.2	-	IV	-	90	EUS	
1933	12	23	9	40		32.9	80.0	-	V	-	105	STR	
1933	12	23	9	55		32.9	80.0	-	IV	-	105	STR	
1934	12	9	5			33.0	80.2	-	IV	-	90	BOL	
1935	1	1	3	15		35.1	83.6	-	V	-	170	BOL	
1938	3	31	10	10		35.6	83.5	-	IV	3.8	195	NUT	
1940	12	25				35.9	82.9	-	V	-	200	USE	
1941	5	10	11	12		35.6	82.6	-	IV	-	175	STR	
1943	12	28	10	25		33.0	80.2	-	IV	-	90	BOL	
1944	1	28	17	30		32.9	80.0	-	IV	-	105	STR	
1945	1	30	20	20		32.9	80.0	-	IV	-	105	STR	
1945	7	26	10	32	16.4	33.75	81.38	3	V	4.4	48	D&G	
1947	11	2	4	30		32.9	80.0	-	IV	-	105	STR	
1949	2	2	10	52		32.9	80.0	-	IV	-	105	STR	
1949	6	27	6	53		32.9	80.0	-	IV	-	105	STR	
1951	3	4	2	55		32.9	80.0	-	IV	-	105	STR	
1951	12	30	7	55		32.9	80.0	-	IV	-	105	STR	
1952	11	19				32.8	80.0	-	V	-	105	BOL	
1956	1	5	3			34.3	82.4	-	IV	-	85	BOL	
1956	5	19	14			34.3	82.4	-	IV	-	85	BOL	
1956	5	27	18	25		34.3	82.4	-	IV	-	85	BOL	
1957	5	13	14	24	51.1	35.80	82.14	3*	VI	4.1	185	D&G	
1957	7	2	9	33	1.0	35.6	82.6	-	VI	4.6	175	EUS	
1957	11	24	20	6	17.0	35.0	83.5	-	VI	-	160	BOL	
1958	5	16	22	30		35.6	82.6	-	IV	-	175	EUS	
1958	10	20	1	16		34.5	82.8	-	V	-	110	BOL	
1959	8	3	6	8	36.8	33.05	80.13	1*	VI	4.4	96	D&G	

TABLE 2.5.2-1 (SHEET 8 OF 9)

DATE			TIME			LOCATION		DEPTH [1]	INTEN.	MAG. [2]	SITE	DATA	
GMT						DEG N	DEG W	MI	MMI		DIST.	SOURCE	
YR	MO	DY	HR	MN	SEC						MI		
1959	8	8	6	8	30.0	33.0	79.5	-	VI	-	135	BOL	
1959	10	27	2	7	28.0	34.5	80.2	-	VI	-	130	EUS	
1960	1	3	7	30		35.9	82.1	-	IV	-	190	STR	
1960	3	12	12	47	44.0	33.07	80.12	6	V	4.0	96	D&G	
1960	7	23				33.0	80.0	-	V	-	105	USE	
1960	7	28	3	37	30.0	32.8	82.7	-	V	-	60	EUS	
1963	4	11	17	45		34.9	82.4	-	IV	-	125	EUS	
1963	5	4	21	1	50.3	32.97	80.19	3*	IV	3.3	93	D&G	
1964	1	20	13	37	52.0	35.9	82.3	-	IV	-	190	EUS	
1964	3	7	18	2	58.6	33.72	82.39	3	-	3.3	53	D&G	
1964	3	13	1	20	17.5	33.19	83.31	1*	V	3.9	89	D&G	
1964	4	20	19	4	44.1	33.84	81.10	2	V	3.5	62	D&G	
1965	9	9	14	42	20.0	34.7	81.2	-	-	3.9	110	STR	
1967	10	23	9	4	2.5	32.80	80.22	12	V	3.4	94	D&G	
1968	7	12	1	12		32.8	79.7	-	IV	-	125	STR	
1968	9	22	21	41	18.2	34.11	81.48	1*	IV	3.5	69	D&G	
1969	5	18				34.0	82.6	-	-	3.5 M1	75	STR	
1969	12	13	10	19	29.7	35.04	82.85	4	V	3.7	145	D&G	
1971	5	19	12	54	3.6	33.36	80.65	1*	IV	3.7	67	D&G	
1971	7	13	6	42	26.0	34.8	83.0	-	IV	3.8	135	BOL	
1971	7	31	20	16	55.0	33.34	80.63	2	III	3.8	68	D&G	
1971	8	11	3	50		33.4	80.7	-	-	3.5	65	BOL	
1972	2	3	23	11	9.7	33.31	80.58	1	V	4.5	71	D&G	
1973	12	19	10	16	8.7	32.97	80.27	4	-	3.0	88	D&G	
1974	8	2	8	52	11.1	33.91	82.53	2	VI	4.1	68	D&G	
1974	10	28	11	33		33.79	81.92	-	IV	3.0	45	CSC	
1974	11	5	3			33.73	82.22	-	III	3.7	48	CSC	
1974	11	22	5	25	56.2	32.92	80.14	5	VI	4.3	96	D&G/BOL	

TABLE 2.5.2-1 (SHEET 9 OF 9)

DATE			TIME			LOCATION		DEPTH [1]	INTEN.	MAG. [2]	SITE	DATA	
GMT						DEG N	DEG W	MI	MMI		DIST.	SOURCE
YR	MO	DY	HR	MN	SEC						MI		
1974	12	3	8	25		33.95	82.50	-	III	3.6	69	CSC	
1975	4	1	21	9		33.20	83.20	-	-	3.9 M1	82	STR	
1975	4	28	5	46	52.6	33.00	80.22	6	IV	3.0 M1	91	STR	
1975	10	18	4	31		34.90	83.00	-	IV	-	140	STR	
1975	11	25	15	17	34.8	34.93	82.93	1	IV	3.2	140	D&G/GS	
1976	12	27	6	57	15.2	32.06	82.50	9	V	3.7	86	D&G/GS	
1977	1	18	18	29	14.2	33.04	80.21	7	VI	3.0 M1	91	STR	
1977	3	30	8	27	47.8	32.95	80.18	5	V	2.9 Ms	94	STR	
1977	8	25	4	20	7.0	33.39	80.69	-	V	3.1	65	GS	
1977	12	15	19	16	43.1	32.92	80.22	-	V	3.0	92	GS	
1979	9	6	20	38	16.3	35.30	83.24	6	-	3.2 M1	171	VPI	
1980	6	10	23	47	32.1	35.46	82.81	1*	-	3.0	170	VPI	
1981	4	9	7	10	31.2	35.51	82.05	-	-	3.0	164	VPI	
1981	5	5	21	21	56.7	35.33	82.42	6	-	3.5	155	VPI	
1982	10	31	3	12	12.2	32.64	84.89	0*	-	3.1	184	VPI	
1982	12	11	0	25	6.7	32.71	83.47	0	-	3.0 M1	103	VPI	

1 Asterisk (*) indicates that the depth has been constrained to enable stable earthquake location solution.

2 Magnitude is given as body wave magnitude (mb), unless otherwise specified as local magnitude (M1) or surface wave magnitude (Ms).

TABLE 2.5.2-2 (SHEET 1 OF 2)

MODIFIED MERCALLI INTENSITY SCALE OF 1931 (ABRIDGED)

I.	Not felt except by a very few under especially favorable circumstances (I Rossi-Forel Scale).
II.	Felt only by a few persons at rest, especially on upper floors of buildings. Delicately suspended objects may swing (I to II, Rossi-Forel Scale).
III.	Felt quite noticeably indoors, especially on upper floors of buildings, but many people do not recognize it as an earthquake. Standing motor cars may rock slightly. Vibration like passing truck. Duration estimated (III Rossi-Forel Scale).
IV.	During the day felt indoors by many; outdoors by few. At night some awakened. Dishes, windows, doors disturbed; walls made creaking sound. Sensation like heavy truck striking building. Standing motor cars rocked noticeably (IV to V Rossi-Forel Scale).
V.	Felt by nearly everyone; many awakened. Some dishes, windows, etc., broken, a few instances of cracked plaster, unstable objects overturned. Disturbance of trees, poles, and other tall objects sometimes noticed. Pendulum clocks may stop (V to VI Rossi-Forel Scale).
VI.	Felt by all; many are frightened and run outdoors. Some heavy furniture moved; a few instances of fallen plaster or damaged chimneys. Damage slight (VI to VII Rossi-Forel Scale).
VII.	Everybody runs outdoors. Damage negligible in buildings of good structures; considerable in poorly built or badly designed structures; some chimneys are broken. Noticed by persons driving motor cars (VIII Rossi-Forel Scale).
VIII.	Damage slight in specially designed structures; considerable in ordinary substantial buildings with partial collapse; great in poorly built structures. Panel walls thrown out of frame structures. Fall of chimneys, factory stacks, columns, monuments, walls. Heavy furniture overturned. Sand and mud ejected in small amounts. Changes in well water. Disturbs persons driving motor cars (VIII+ to IX Rossi-Forel Scale).

TABLE 2.5.2-2 (SHEET 2 OF 2)

IX.	Damage considerable in specially designed structures; well designed frame structures thrown out of plumb; great in substantial buildings, with partial collapse. Buildings shifted off foundations. Ground cracked conspicuously. Underground pipes broken (IX+ Rossi-Forel Scale).
X.	Some well-built wooden structures destroyed; most masonry and frame structures destroyed with foundations, ground badly cracked. Rails bent. Landslides considerable from river banks and steep slopes. Shifted sand and mud. Water splashed (slopped) over banks (X Rossi-Forel Scale).
XI.	Few, if any, masonry structures remain standing. Bridges destroyed. Broad fissures in ground. Underground pipe lines completely out of service. Earth slumps and land slips in soft ground. Rails bent greatly.
XII.	Damage total. Waves seen on ground surfaces. Lines of sight and level distorted. Objects thrown upward into the air.

TABLE 2.5.4-1

ENGINEERING PROPERTIES OF SITE SOILS

Static Properties	Upper Sand Stratum (Barnwell Group)	Marl Bearing Stratum (Blue Bluff Marl)	Lower Sand Stratum (Ellenton Formation)
In situ dry density (lb/ft^3)	41-120	51-155	69-118
In situ moisture content (percent)	5-86	3.2-60.6	15-45
Degree of saturation (percent)	18-100	100	-
ASTM D 1557 maximum dry density (lb/ft^3)	101-125	-	-
Optimum moisture content (percent)	7.5-17.4	-	-
Unconsolidated undrained shear strength c (lb/ft^2) and 0°	440-2100, 6°-32°	260-500,000, 0°	-
Consolidated undrained shear strength c (lb/ft^2) and 0°	1650-4000, 17°-25°	-	-
Consolidated drained shear strength c (lb/ft^2) and 0°	0, 33°-34.5°	-	-
Standard penetration test (blows/ft) range/average	2-60 / 30	10-100+ / 100+	70-100+ / 100+
Liquid limit (percent)	NP[(a)]	19-111	NP
Plastic limit (percent)	NP	15-55	NP
Plasticity index (percent)	NP	2-70	NP
Poisson's ratio	0.4-0.46	0.5	-
Porosity	-	0.403-0.619	-
Permeability (ft/year)	200-350	-	-
Specific gravity	2.67-2.74	2.37-2.84	
Modulus of elasticity (k/ft^2)	-	86-25,000	-

a. Majority of material nonplastic (NP): clay layers had liquid limit greater than 100.

TABLE 2.5.4-2

ENGINEERING PROPERTIES FOR DESIGN

Static Properties	Upper Sand Stratum (Barnwell Group)	Marl Bearing Stratum (Blue Bluff Marl)	Lower Sand Stratum (Ellenton Formation)
In situ dry density (lb/ft^3)	94	88	94
In situ moisture content (percent)	25	35	24
Degree of saturation (percent)	88	100	-
ASTM D 1557 maximum dry density (lb/ft^3)	115.2	-	-
Optimum moisture content (percent)	12.4	-	-
Unconsolidated undrained shear strength c (lb/ft^2) and 0°	2300, 6°	10,000, 0°	-
Consolidated undrained shear strength c (lb/ft^2) and 0°	1000, 18°	-	-
Consolidated drained shear strength c (lb/ft^2) and 0°	0, 34°	-	-
Standard penetration test (blows/ft)	30	100+	100+
Porosity	-	0.497	-
Poisson's ratio	0.4	0.5	-
Permeability upper sand stratum (ft/year)	350	-	-
Specific gravity	2.70	2.72	-
Modulus of elasticity (k/ft^2)	-	4000-10,000	-

TABLE 2.5.4-3

DESIGN VALUES OF SHEAR MODULUS[a]

In Situ Soils:

Elevation (ft)	Shear Modulus (lb/ft^2)
210 to 180	2.3×10^6
180 to -770	11.6×10^6

a. The values refer to shear modulus at strains of approximately 10^{-4} percent or lower.

TABLE 2.5.4-4

IN SITU SOILS - BASIC SOIL PROPERTIES FOR DYNAMIC DESIGN[a]

Stratum Designation	Elevation (ft)	Unit Weight Saturated Moist (lb/ft)	Submerged	Poisson's Ratio
Upper sand stratum	225 to 135	115	52.6	0.4 to 0.46
Marl bearing stratum	135 to 70	- 115	52.6	0.5
Lower sand stratum	70 to -770	- 115	52.6	0.4 to 0.46

a. In sandy soils the specific value of Poisson's ratio within the given range will be the most conservative value for the particular dynamic analysis being carried out.

Figures describing the variation of shear modulus and damping ratio with shear strain for clay marl bearing stratum and lower sand stratum are provided in subsections 3.7.B.1 and 3.7.B.2.

TABLE 2.5.4-5

COMPILATION OF SHEAR WAVE DATA[a]

Depth (ft)	Elevation[b] (ft)	Compression Wave Velocity (ft/s)	Shear Wave Velocity (ft/s)
0-15	208-193	1400	600[c]
20	188	2500	1000
30	178	2800	1000
40	168	2500	900
50	158	4600	1000
60	148	5200	1200
70	138	5100	1400
80	128	6600	1600
90	118	6700	1700
100	108	6900	1800
110	98	6600	1700
120	88	6400	1700
130	78	6600	1800
140	68	6500	1700
150	58	6800	1600
160	48	6600	1600
170	38	6800	1800
180	28	6600	--
190	18	6500	1800
200	8	6600	1800
210	-2	6600	1700
220	-12	6600	1800
230	-22	6700	1800
240	-32	6400	1700
250	-42	6500	1800
260	-52	6700	1800
270	-62	6800	1800
280	-72	6700	1800
290	-82	6700	1700

a. From cross-hole measurements or as noted.

b. Ground surface elevations average 208 ft above sea level in this area.

c. From surface data.

TABLE 2.5.4-6

SUMMARY OF DATA COMPILATION BASED ON CROSS-HOLE DATA[a]

Bore Holes	Depth (ft) From	Depth (ft) To	Elevation (ft) From	Elevation (ft) To	Compression Wave Velocity (ft/s)	Shear Wave Velocity (ft/s)	Generalized Material Correlations
B-136	0	15	208	193	1400	600 (from surface data)	Sands
B-146G	15	40+	193	168-	2500	1000	Sands
B-148	40+	50	168-	158	4600	1000	Saturated materials
B-149	50	80-	158	128+	5000	1300	Saturated materials
B-151	80-	90	128+	118	6650	1650	Compact materials, high blow-count values
B-154	90	290	118	-82	6800	1600-1800 (mostly 1800)	Compact materials, high blow-count values

a. Ground surface elevation averages 208 ft above sea level in this area.

TABLE 2.5.4-7

ELASTIC MODULI RESULTS

Depth (ft)		Elevation (ft)		P-Wave	S-Wave	Density	Poisson's	Young's Modulus	Shear Modulus
From	To	From	To	(ft/s)	(ft/s)	(lb/ft^3)	Ratio	($lb/in.^2$)	($lb/in.^2$)
0	15	208	193	1400	600	115	0.39	0.2×10^5	0.8×10^4
15	40+	193	168-	2500	1000	115	0.40	0.7×10^5	2.4×10^4
40+	50	168-	158	4600	1000	115	0.48	0.7×10^5	2.4×10^4
50	80-	158	128+	5000	1300	115	0.46	1.3×10^5	4.1×10^4
80-	90	128+	118	6650	1650	115	0.47	2.0×10^5	6.8×10^4
90	290	118	-82	6800	1800	115	0.46	2.3×10^5	8.0×10^4

TABLE 2.5.4-8

DESIGN STATIC PROPERTIES FOR BACKFILL COMPACTED TO 97-PERCENT RELATIVE COMPACTION (ASTM D 1557)

Soil Properties	Sand, Silty Sand
Unit weights (lb/ft^3)	
Moist	126
Saturated	132
Submerged	69.6
Effective shear strength parameters	
Cohesion or c (k/ft^2)	0
Angle of internal friction or Σ (degrees)	34
Undrained modulus of elasticity or E (k/ft^2)	1500
Poisson's ratio (υ)	0.4
Compression index	-

TABLE 2.5.4-9

DESIGN DYNAMIC PROPERTIES FOR BACKFILL COMPACTED TO 97-PERCENT RELATIVE COMPACTION (ASTM D 1557)

Soil Properties	Sand, Silty Sand
Unit weights (lb/ft^3)	
Moist	126
Saturated	132
Submerged	69.6
Poisson's ratio	0.33
Damping ratio	See figure 3.7.B.1-8.

Shear modulus at strain of 10^{-4} percent[(a)]

Elevation (ft)	Depth (ft)	Shear Modulus (lb/ft^2)
210	10	2.3×10^6
195	25	3.6×10^6
165	55	5.3×10^6
150	70	5.7×10^6
130	90	6.2×10^6

a. Shear modulus $G = 1000\ k_2(\sigma'_m)^{1/2}\ lb/ft^2$, where $k_2 \approx 79$.

Variation of shear modulus with strain is given in figure 3.7.B.2-5.

TABLE 2.5.4-10
SUMMARY OF TEST FILL RESULTS FOR HEAVY EQUIPMENT COMPACTION

Test Fill No.	Material From Stockpile[a]	Roller	Roller Speed (mph)	Number of Passes	Lift Thickness (in.)	Field Density Method	Depth of Test (in.)	Number of Field Density Tests[b]	Percent of Tests: Less Than 97% Compaction	Percent of Tests: 93 to 95% Compaction	Percent of Tests: Less than 93% Compaction	Remarks
I	C	SPF 60 and Raygo 600A	1.5	2 each	6	Sand cone	12	24	0	0	0	
							18	24	0	0	0	Acceptable
						Nuclear	12	24	8	0	0	
							18	24	8	0	4	
II	C	Raygo 600A	1.5	4	6	Sand cone	12	24	0	0	0	
							18	-	-	-	-	Acceptable
						Nuclear	12	24	0	0	0	
							18	-	-	-	-	
III	A	SPF 60 and Raygo 600A	1.5	2 each	6	Sand cone	12	24	33	4	0	
							18	-	-	-	-	Acceptable
						Nuclear	12	24	100	46	29	
							18	-	-	-	-	
IV	A	Raygo 600A	1.5	4	6	Sand cone	12	24	4	0	0	
							18	-	-	-	-	Acceptable
						Nuclear	12	24	96	42	46	
							18	-	-	-	-	
V	A	SP 60	3	3	12	Sand cone	12	24	42	8	0	
							18	6	83	17	0	Not acceptable
						Nuclear	12	24	100	4	96	
							18	6	100	50	50	
VI	C	SP 60	2, 1.5	3	12	Sand cone	12	24	0	0	0	
							18	6	0	0	0	Acceptable
						Nuclear	12	24	33	8	8	
							18	6	67	17	17	
VII	B	Raygo 600A	1.5	4	6	Sand cone	12	10	30	20	0	
							18	-	-	-	-	Inconclusive
						Nuclear	12	10	100	10	90	
							18	-	-	-	-	

a. Letters refer to designations on drawing AX6DD394.
b. A laboratory compaction test was performed for each field density test.

TABLE 2.5.4-11

SUMMARY OF TEST FILL RESULTS FOR HAND COMPACTION EQUIPMENT

Test Fill No.	Material From Stockpile[a]	Roller	Roller Speed (mph)	Number of Passes	Lift Thickness (in.)	Field Density Method	Depth of Test (in.)	Number of Field Density Tests[b]	Percent of Tests: 97% Compaction or Greater	Percent of Tests: 93 to 95% Compaction	Percent of Tests: 93% Compaction or Lower	Remarks
1	C	Wacker 74 Dual Drum	0.68	4	6	Sand cone	12	8	8	0	0	Acceptable
2	C	Wacker 100 Jumping Jack	0.23	2	6	Sand cone	12	8	8	0	0	Acceptable
3	C	Ingersoll - Rand- SP 24	0.68	4	6	Sand cone	12	16	15	1	0	Acceptable

a. Letters refer to designations on drawing AX6DD394.

b. A laboratory compaction test was performed for each field density test.

TABLE 2.5.4-12 (SHEET 1 OF 2)

SUMMARY OF RESULTS OF BEARING CAPACITY ANALYSIS

Structure	Supporting Stratum	El of Top of Foundation/El of Bottom of Foundation (ft)	Approx. Mat Size (ft)	Loading Pressure Gross Static (k/ft^2)	Loading Pressure NetLAM Static (k/ft^2)	Gross Dynamic (k/ft^2)	Net Dynamic (k/ft^2)	Ultimate[b] Net Bearing Capacity (k/ft^2)	Allowable[c] Net Bearing Capacity Static (k/ft^2)	Allowable[c] Net Bearing Capacity Dynamic (k/ft^2)	Factor of Safety = Ultimate Net Bearing Capacity ÷ Net Static or Dynamic Loading Pressure: Computed Static Factor of Safety	Factor of Safety = Ultimate Net Bearing Capacity ÷ Net Static or Dynamic Loading Pressure: Computed Dynamic Factor of Safety
Diesel generator building	Backfill	220/211	115x93	3.8	2.7	13.6	12.5	60.9	20.3	30.5	22.6	4.9
Turbine building	Backfill	195/186	184x604	3.6	-0.5	8.7	4.6	56.5	18.8	28.3	Very[d] high	12.3
Control building	Backfill	180/173	525x169	4.3	-1.3	13.4	7.8	57.8	19.3	28.9	Very[d] high	7.4
Containment building	Backfill	169/158.5	D=154.5	8.4	1.0	20.9	13.5	61.7	20.6	30.9	61.7	4.6
Fuel building	Backfill	160/154	198x75	8.1	0.1	23.4	15.4	64.0	21.3	32.0	640	4.2
Nuclear service cooling water tower	Marl	137/128	D=100	8.8	-2.4	34.2	23.1	61.7	20.6	30.9	Very[d] high	2.7
Auxiliary building	Marl	119.25/109.25	440x130	10.2	-3.3	28.7	15.2	63.7	21.2	31.9	Very[d] high	4.2
Nuclear service cooling water valve house	Backfill	204.5/198.5	110x33	3.2	0.6	12.6	10.0	133.5	44.5	66.8	222.5	13.4
Auxiliary feedwater pumphouse	Backfill	215/212	86x40	1.6	0.6	3.2	2.2	92.5	30.8	46.3	154.3	42.0

TABLE 2.5.4-12 (SHEET 2 OF 2)

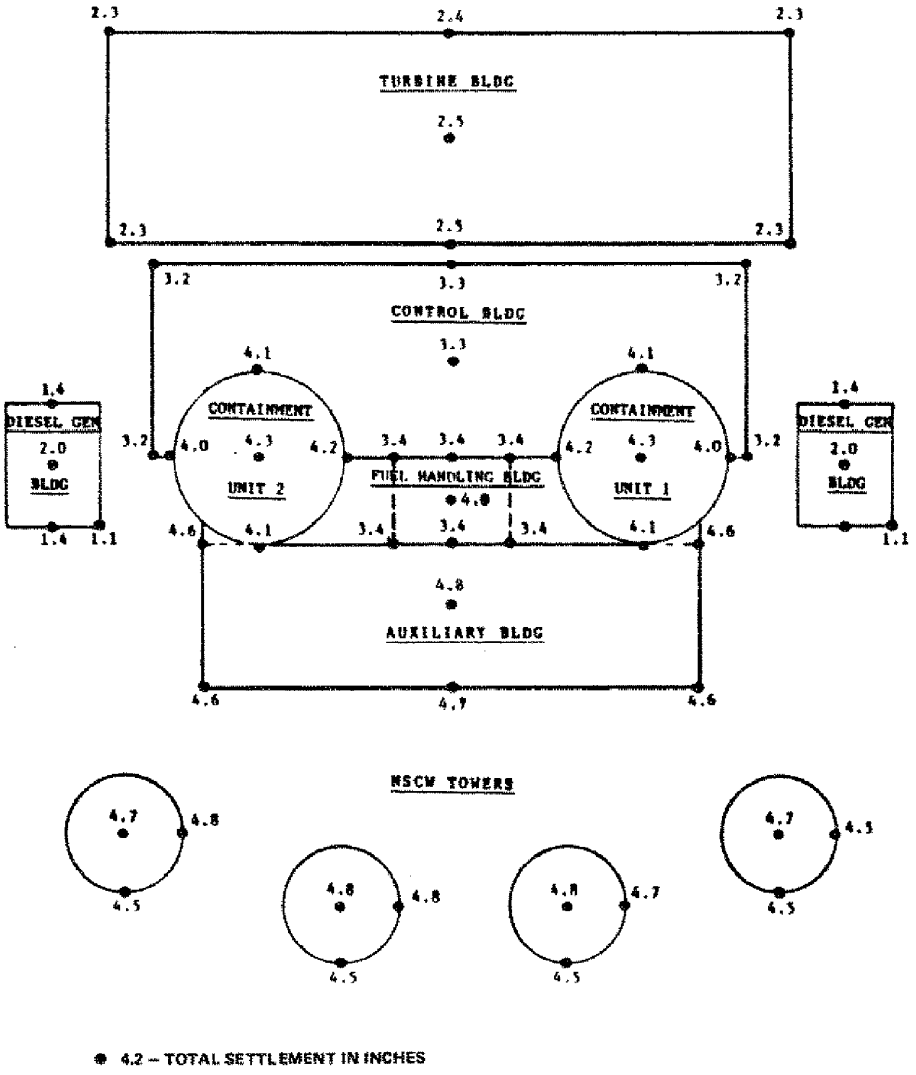
Structure	Supporting Stratum	El of Top of Foundation/El of Bottom of Foundation (ft)	Approx. Mat Size (ft)	Loading Pressure Gross Static (k/ft^2)	Loading Pressure NetLAM Static (k/ft^2)	Gross Dynamic (k/ft^2)	Net Dynamic (k/ft^2)	Ultimate[b] Net Bearing Capacity (k/ft^2)	Allowable[c] Net Bearing Capacity Static (k/ft^2)	Allowable[c] Net Bearing Capacity Dynamic (k/ft^2)	Factor of Safety = Ultimate Net Bearing Capacity ÷ Net Static or Dynamic Loading Pressure: Computed Static Factor of Safety	Factor of Safety = Ultimate Net Bearing Capacity ÷ Net Static or Dynamic Loading Pressure: Computed Dynamic Factor of Safety
Condensate storage tank	Backfill	220/212	115x63	3.1	2.1	5.3	4.3	115.3	38.4	57.7	54.9	26.8
Diesel fuel oil Storage tank Pumphouse	Backfill	211.5/209.5	114x26	1.7	0.4	2.5	1.2	81.9	27.3	41.0	204.8	68.3
Reactor makeup water storage tank	Backfill	220/212	51x51	2.3	1.3	5.3	4.3	95.7	31.9	47.9	73.6	22.3
Refueling water storage tank	Backfill	220/216	62x62	3.7	3.2	11.8	11.3	88.9	29.6	44.5	27.8	7.9
Radwaste transfer building	Backfill	220/216.5	93x45	3.6	3.2	7.7	7.3	86.5	28.8	43.2	27.0	12.0

a. The net static load is the load in excess of the overburden pressure at the base of the structure.
b. The ultimate net bearing capacity is the load in excess of the overburden pressure at the foundation level at which shear failure will occur in the foundation stratum.
c. The allowable net static and dynamic bearing capacities are obtained by dividing the net ultimate bearing capacity by factors of 3 and 2 respectively.
d. The net static bearing pressure is negative.

TABLE 2.5.4-13

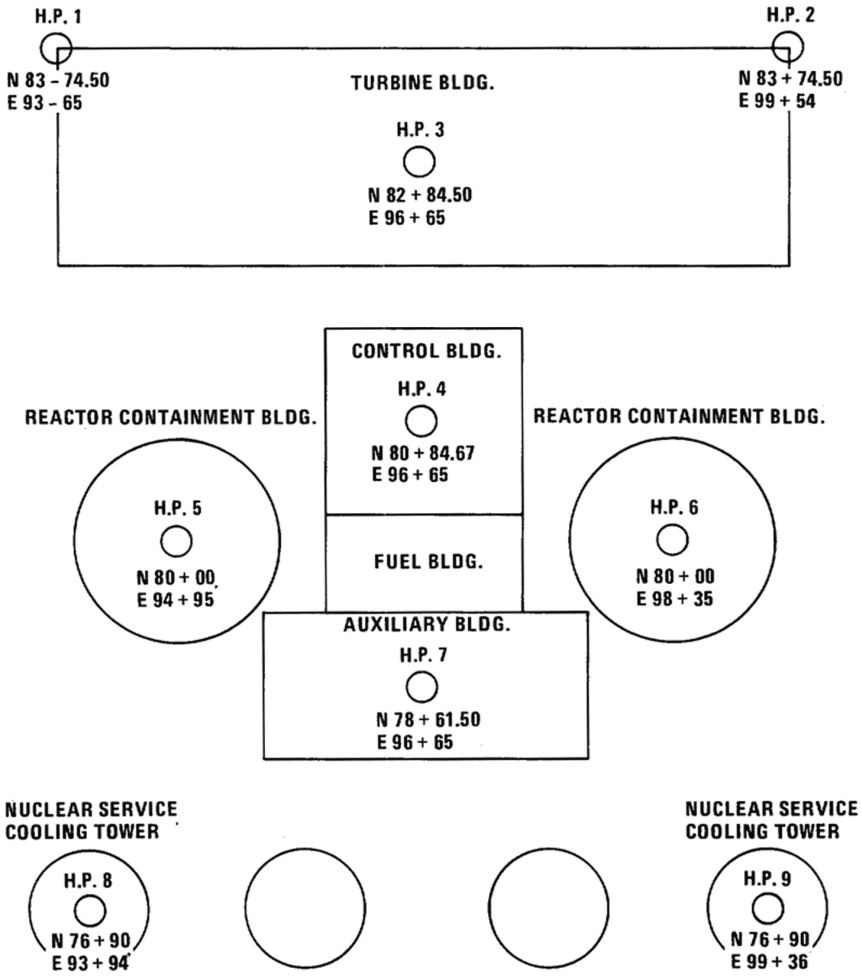
SUMMARY OF MEASURED HEAVE IN POWER BLOCK

Heave Point No.	Period		Measured Heave (in.)
	From	To	
1	06/22/74	08/07/77	1.1
2	06/16/74	06/22/76	1.4
3	06/16/74	10/02/74	0.6
5	06/16/74	02/26/77	1.7
7	06/16/74	06/05/77	1.2
9	06/30/74	08/07/77	1.5



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	SETTLEMENT OF UNITS 1 AND 2 POWER BLOCK STRUCTURE
		FIGURE 2.5.4–1



BENCHMARK NO. 1: N 80 + 00 E 139 + 00; TIP ELEV. = 105.0

BENCHMARK NO. 2: N 80 + 00 E 51 + 00; TIP ELEV. = 69.0

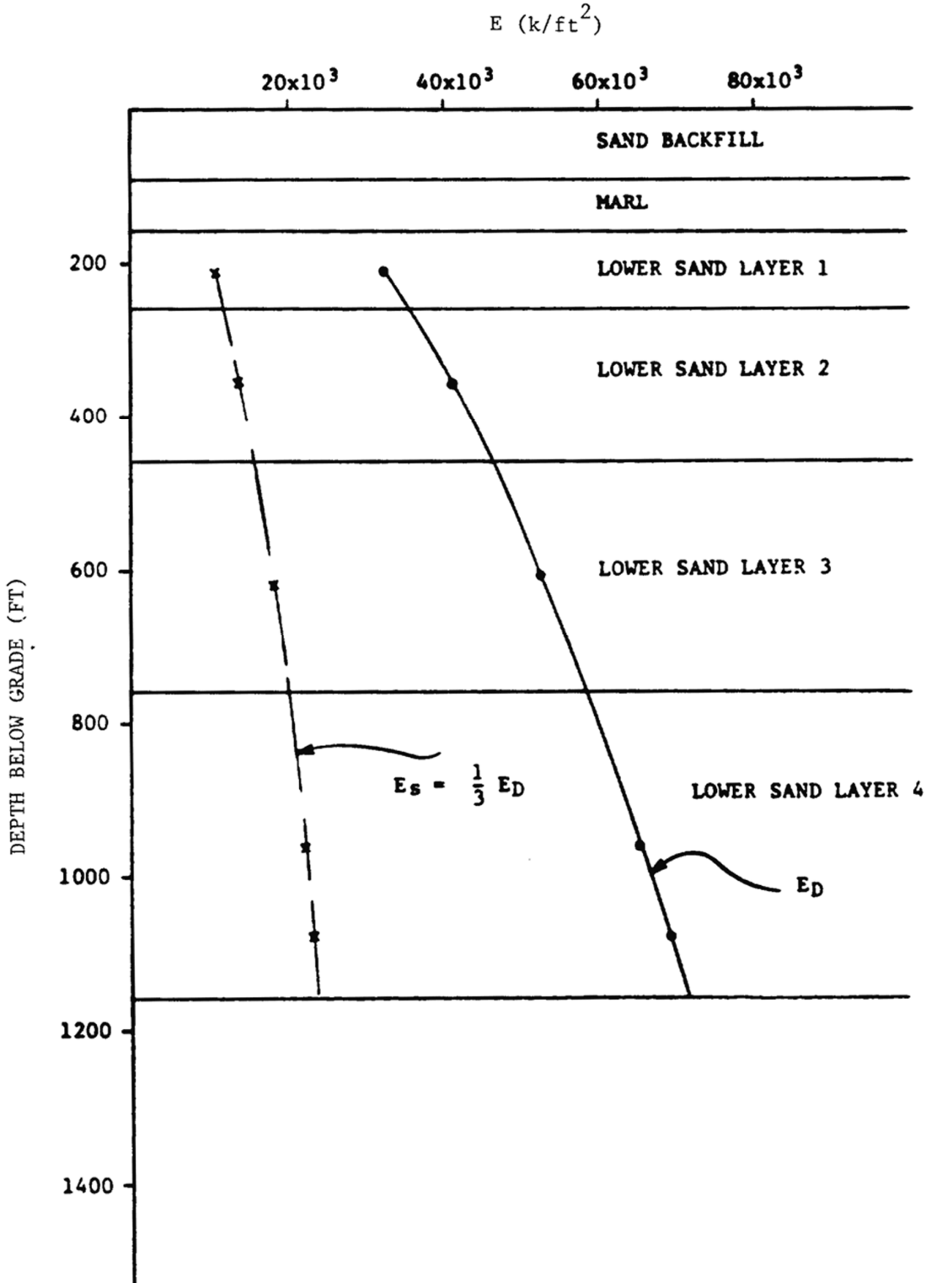
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

HEAVE POINTS LOCATION PLAN

FIGURE 2.5.4–2



REV 13 4/06

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ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

YOUNG'S MODULUS vs DEPTH

FIGURE 2.5.4–3

APPENDIX 2A (HISTORICAL)

POPULATION DISTRIBUTION METHODOLOGY

2A.1 INTRODUCTION (HISTORICAL)

This appendix documents the procedures followed in the preparation of paragraphs 2.1.3.1 and 2.1.3.2 of the Final Safety Analysis Report (0 to 50 miles) and paragraphs 2.1.2.1 and 2.1.2.2 of the Operating License Stage Environmental Report (50 to 500 miles). Included with a step-by-step documented review of the procedures is a presentation of the methodology used, a table of definitions, a review of the materials used, a discussion of the assumptions made, and a section addressing the procedures to follow should the submitted figures need updating.

2A.2 DEFINITIONS (HISTORICAL)

OPB: OPB is an acronym for the Office of Planning and Budget, State of Georgia.

SDC: SDC is an acronym for the State Data Center, State of South Carolina.

Sector: A sector is one of the 16 compass divisions comprising the area within any given circle centered on VEGP.

Segment: A segment is an area bounded by two sector divisions and two arcs.

USGS Quadrangle Maps: United States Geological Survey quadrangle maps are topographical maps bounded by parallels of latitude and meridians of longitude. Quadrangles covering 7 1/2 min of latitude and longitude are published at the scale of 1:24,000 (1 in. = 2000 ft.) Quadrangles covering 15 min of latitude and longitude are published at the scale of 1:62,500 (1 in. mile).

2A.3 ASSUMPTIONS (HISTORICAL)

The major assumptions taken to determine population distribution were as follows:

A. *The population density within the Savannah River Plant is zero.*

B. *The percentage of each county's population in a segment for 1980 will not change over the forecasted time span.*

C. *The curvature of the earth will not have a significant effect on the construction of rings and sectors for the 500-mile radius.*

D. *Due to possible human error in construction, rings are estimated to be accurate within ±1/2 mile.*

E. Population changes at Fort Gordon over the forecasted period are addressed in the OPB projections.

F. Population projections made for the area within a 500-mile radius of VEGP were based on 1980 census data and county population projections obtained from the OPB and the SDC.

G. The extrapolation method of population projections for county and subcounty areas is generally more accurate than the differential and share methods.

2A.4 MATERIALS (HISTORICAL)

The materials used to determine population distribution were:

A. The USGS quadrangle maps of the affected area.

B. The 1980 state maps depicting the affected counties.

C. U.S. Census Bureau figures indicating the population and number of housing units of the counties and cities in the affected area for 1980.

D. The OPB and the SDC county and city projections for the years 1990, 2000, 2010, and 2020.

E. The 1980 population figures for military installations in the area.

F. A house to house survey of the area within 5 miles of the VEGP site conducted in 1980.

The USGS quadrangle maps were used as the base maps for construction of rings and sectors and for the transcription of county and city boundaries; they were chosen as base maps because of their detailed representation of housing patterns. The OPB and SDC projections were chosen for future population estimates because of the high accuracy of their projections as demonstrated by a narrow margin of error between 1980 projected and actual population figures.

2A.5 METHODOLOGY (HISTORICAL)

Described in the steps below is the methodology used to determine population distribution:

A. Locate the center of VEGP on the USGS quadrangle maps. Construct concentric circles on the quadrangle maps at distances of 1 through 10, 20, 30, 40, 50, 60, 70, 85, 100, 150, 200, 350, and 500 miles.

B. Divide the constructed circles into sectors of 22 1/2° with each sector centered on one of the 16 compass points, e.g., true north, north-northeast.

C. Transcribe the city and county boundaries located within a 500-mile radius of VEGP to USGS quadrangle maps.

D. Estimate from the quadrangle maps the percentage of each city's and county's population that lies in each affected section.

E. Calculate from 1980 census statistics the percentage of each county's population that lies in each affected section.

F. Assume that the percentage of each county's population that resides in each affected segment remains the same over the forecasted time span.

G. Prepare population projections for the first year of plant operation (1987 for Unit 1) and beyond by first determining the margin of error for previous projections completed by the OPB and the SDC (the difference between 1980 population forecasts and 1980 census figures). Using the margin of error, adjust projections for the census years, the midpoint in the plant's operating life, and the endpoint in the plant's operating life.

H. Estimate a segment's population for any year by multiplying each affected county's adjusted population projection for that year by the percentage of that county's population which is in the segment.

To estimate a county's population for the anticipated first year of plant operation, the following methodology was used:

A. Subtract the county's previous census decade's count from the next projected census decade's estimate, divide by 10, and multiply this number times the number of years into the decade the first year of plant operation occurs.

B. Add the figure obtained in item A to the previous decade's estimate to obtain a county's estimated population for the anticipated initial year of plant operation. For example, assuming a starting date of 1987, the estimated population of county X for section A lying entirely within that county is determined as follows:

$$\frac{M - N}{10} \times 7 = 0.7M - 0.7N$$

$$0.7M - 0.7N + N = 0.7\ (M - N) + N$$

$$= \textit{estimated population for } 1987$$

where:

M = 1990 estimated population for county X.

N = 1980 population count for county X.

For segments within 50 to 500 miles, statewide projected growth rates were used to determine individual county forecasts. Population estimates within 5 miles of VEGP were based on a 1980 house to house survey of the area. Multifamily housing units related to construction worker demand were included in the population estimates for 1987. It was assumed that these housing units will continue to be in use through 1989, estimated completion year of Unit 2. However, projections for 1990 and beyond do not include construction-related housing.

2A.6 PROCEDURES (HISTORICAL)

Population distribution by sector for the area within a 500-mile radius of VEGP was determined in the following manner:

A. Base maps showing county boundaries within a 500-mile radius were overlain with annular rings and sectors.

B. Each segment's county composition was visually estimated. For example, sector 16-30 is composed of 4 percent Aiken County and 14 percent Richmond County.

C. All counties lying within the 500-mile radius were listed with their 1980 population. Cities over 25,000 were subtracted from county figures if they lay in more than one segment, or if the county lay in more than one segment. For example:

	1980 Population
Etowah County	*103,057*
Gadsden City	*-47,255*
Remaining population	*55,802*

Each city was listed with the segment or segments in which it lay.

D. Each segment's 1980 population was determined by multiplying the percentage of each county represented by its remaining 1980 population (U.S. Census Bureau).

APPENDIX 2B

GEOLOGY

2B.1 INTRODUCTION

Comprehensive aerial geology and site specific foundation investigations and examinations of the VEGP site have been completed. The results and conclusions are described and contained in section 2.5 and subsection 2.4.12. The geologic logs and geophysical logs are submitted under a separate cover.

A table of drilling statistics is presented in table 2B-1. A description of foundation conditions encountered during construction is contained in subsection 2B.3.1.

2B.2 FIELD INVESTIGATIONS

A total of 370 borings was drilled for the primary geologic and site specific foundation investigations for the plant facilities. The drill logs of 354 of these borings are submitted as described in section 2B.1. The remaining 16 borings were done for the revised locations of the cooling towers and the drill logs for these borings are included in the report Foundations Investigations for Natural Draft Hyperbolic Cooling Towers, Addendum, prepared by Bechtel Power Corporation, December 1978. Selected marl core samples from principal borings have been placed in protective storage; table 2B-2 provides an inventory of these core samples. An additional 12 borings were made for the studies of the postulated Millett Fault conducted in 1982. These borings are offsite and are described in detail in the report Studies of Postulated Millett Fault, dated October 1982. Logs of these borings are included in that report.

2B.3 REPORT OF GEOLOGY AND FOUNDATION CONDITIONS

2B.3.1 INTRODUCTION

This report presents the results of geologic work performed in conjunction with the excavation of the power block areas at the VEGP site. The purpose of the work was to identify, locate, and record details of the geologic structure, stratigraphy, and lithology of the soil and rock strata encountered in the excavation. In addition, samples of foundation rock were obtained for testing of physical properties.

The geologic work was performed by Bechtel geologists during the period May 1974 through October 1977 with certain tasks continuing on an as-needed basis. This time period included the initial startup of the construction work, the interim postponement of work between September 1974 and July 1976, and the subsequent restart of construction.

The work performed included detailed geologic mapping of the soil and rock strata exposed in the power block excavation, and coring and testing of the Blue Bluff marl, which forms the foundation for power block structures and structural backfill. This marl has sometimes been

referred to as the "clay bearing stratum." Geologic mapping was accomplished by recording the details of stratigraphy, structure, and lithology of the various soil and rock deposits on a base map prepared from excavation drawings. Mapping of the vertical surfaces of the auxiliary building excavation walls was recorded on geologic sections coinciding with the surfaces of the walls. All geologic mapping was performed using hand surveying techniques with the exception of the recording of the upper contact of the marl layer. This was recorded by instrumental survey of 575 points established by the geologists. Photography was employed as an aid to mapping and to provide a record of foundation geologic features.

As areas of the marl were cleaned off at final grade in the excavation, they were inspected and signed off by a qualified geologist or soil engineer. The documentation for the approved foundation areas was submitted to Georgia Power Company for permanent retention. This documentation has been transferred to Southern Nuclear Operating Company (SNC), as the exclusive operating licensee.

Subsection 2B.3.2 of this report presents a brief summary of conclusions from the studies performed. Subsection 2B.3.3 presents the details of the geologic structure, stratigraphy, and lithology of the various geologic materials encountered in the excavation. Ground water conditions encountered during the reference period are described. Geologic mapping procedures are discussed in detail. Subsection 2B.3.4 describes the excavation geometry along with the procedures utilized for advancing the excavation down to final grade. Temporary dewatering methods are described as are measures taken for protection of the side slopes from erosion. Foundation cleanup and protection procedures are discussed, and inspection and approval procedures are outlined. The marl testing program carried out to confirm the design physical properties of this material is discussed, and reference is made to the backfill report[(1)] in which the test results are compiled. The monitoring of rebound of the marl due to unloading by excavation of the overlying deposits is described. Subsection 2B.3.5 presents detailed conclusions drawn from the work described in the preceding sections.

The features described in the report are illustrated in maps, geologic sections, and miscellaneous figures which accompany the report.

This report is confined to geologic conditions exposed in the power block excavation at the VEGP site. For discussion of regional geology as well as other more general features of site geology, refer to section 2.5. The documents referenced at the end of this section present further details of topics mentioned briefly in this report.

The following organizations and individuals participated in the work covered by this report:

- Georgia Power Company--plant owner, construction manager, logistical support.
- Bechtel Power Corporation--geology, soil engineering.
- Law Engineering Testing Company--obtainment and laboratory testing of core samples.
- Goldberg, Zoino, Dunnicliff Associates--foundation instrumentation.
- R. Y. Bush--dewatering consultant.

2B.3.2 SUMMARY OF CONCLUSIONS

The studies described in this report have led to the following conclusions:

- The site is suitable for design and construction of a multiple-unit nuclear generating facility.
- The geologic conditions exposed in the power block excavation confirm the conditions described in the Preliminary Safety Analysis Report(PSAR).(2)
- No geologic features exist in the power block area, which could affect licensing aspects of the plant.
- The marl layer, which provides foundation for both Category 1 structures and Category 1 backfill, is as sound and as competent as anticipated and is free of solution cavities.
- Results of laboratory testing and monitoring of rebound of the marl confirm its competency.

2B.3.3 GEOLOGIC CONDITIONS

2B.3.3.1 General

The VEGP site is situated on the right (south) bank of the Savannah River in Burke County, Georgia, approximately 26 miles southeast of the city of Augusta. The area is a part of the Atlantic Coastal Plain physiographic province and is characterized by mature river valley topography. The elevation of the site varies from approximately 90 to about 255 ft above msl. The original ground surface elevation in the power block area varied from about 200 to 230 ft. The site topography is characterized by rolling hills and ridges dissected by tributaries of the Savannah River. Steep cliffs have formed along the south side of the Savannah River as a result of the steady southward migration of the main river channel. In contrast, the north bank is an extensive area of swampy lowland. This low area is outside the plant boundary.

The portion of the coastal plain in which the site lies consists of a thick sequence of Cretaceous and Tertiary shallow-water massive sediments overlying Triassic basement rocks. The sediments generally dip at a very slight angle towards the east and southeast. At the VEGP site, approximately 950 ft of these sediments cover the older Triassic rocks. The general stratigraphic sequence consists of a thin veneer of windblown sand on hills and ridges capping a sequence of shallow marine sands, clays, and isolated coquina and limestone deposits, which extend down to approximate el 135 ft. Below this, a layer of moderately to locally well-cemented marl having a thickness of about 70 ft overlies dense sands and clayey sands, which extend down to the Triassic basement.

The marl layer acts as an aquiclude separating the regional aquifer, confined beneath the marl, from the shallow water table existing in the deposits above the marl.

The excavation for the power block facilities cut through the upper sequence of sand, clay, limestone, and coquina and bottomed in the marl. Details of these deposits are discussed in the following paragraphs.

2B.3.3.2 Geologic Mapping Procedures

Geologic mapping and recording of the features exposed in the excavation was accomplished in three phases:

- Detailed mapping of deposits above the Blue Bluff marl, May 1974 to October 1974.
- Detailed mapping of features within the marl and surveying of the upper contact of the marl, February 1977 to October 1977.
- Detailed inspection and recording of areas in the marl approved for placement of concrete or backfill, June 1977 to January 1979.

The first phase of mapping was performed in conjunction with the excavation of the sediments above the marl. Features were located in the side slopes of the excavation as the bottom elevation was progressively lowered. The side slopes were cut at a gradient of two horizontal to one vertical (2:1), and survey stakes were installed on a grid pattern on the slopes. Locations of geologic features were measured by tape and hand-level methods using the slope stakes as reference points. Accuracy of these measurements is estimated to be within 0.5 ft. Mapping of the 2:1 slopes was recorded in plan view on a base map compiled from the project excavation drawings and is shown on drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363.

The second phase of mapping was accomplished as the marl layer was exposed and prepared for placement of concrete and backfill. Since it was desired to demonstrate the absence of faulting in the marl, the upper contact was measured and recorded with survey accuracy around the perimeter of the excavation. Five hundred and seventy-five survey points were established by the geologists along this contact, and these points were located instrumentally. The nature of the contact between the points was examined closely for continuity and absence of breaks. The contact is shown in plan view on drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363 and the details of the survey results are shown in both plan and section on drawing AX6DD372.

In addition to examining the upper contact, features within the marl were examined and recorded. The deep excavation for the auxiliary building basement, within the larger power block excavation, provided an excellent opportunity for this. The sides of the excavation exposed a vertical section approximately 22 ft in height in the marl. A system of reference points was established on the walls of the excavation, and stations were established for the purpose of describing locations of features. The stationing system adopted is shown on drawings AX6DD374 and AX6DD375. The mapping was recorded in the vertical plane and is presented as geologic sections on drawings AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, and AX6DD370. An explanation of geologic units used for mapping purposes is shown on drawings AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, and AX6DD370. By referring to drawings AX6DD374 and AX6DD375, the location of any section can be easily ascertained. Measurements were made by tape and hand-level methods on the excavation walls. Accuracy is estimated to be generally within 0.1 ft.

The third phase of geologic mapping consists of detailed inspection and photography of foundation areas rather than mapping in the strict sense. This effort was initiated in June 1977 when the first foundation area was cleaned off at final grade and prepared for the gunite protective seal. Inspection and approval of final grade in the marl continued on an intermittent basis as foundation areas were readied. Documentation of the geologic inspection and

approval of foundation areas have been transmitted from the inspecting Bechtel geologists to Georgia Power Company site personnel for permanent retention. The documentation described in this section has been transferred to SNC as the exclusive operating licensee.

Details of the methods used for protection of final grade areas in the marl are discussed in a later section.

2B.3.3.3 Stratigraphy

2B.3.3.3.1 General

The stratigraphy of the VEGP site has been described in the PSAR.[2] The subsequent detailed excavation mapping, as well as recent regional work by Georgia State Geological Survey, has resulted in a much clearer and more detailed understanding of the site stratigraphy.

The sequence of geologic strata exposed in the power block excavation is shown in the stratigraphic column presented in figure 2.5.1-5. Five distinct stratigraphic units are recognized in the excavation. These are, in sequence from the oldest to the youngest:

- The Lisbon Formation of middle Eocene age.
- The Utley Limestone Member of the Barnwell Group of late Eocene age.
- The Twiggs Clay Member of the Barnwell Group.
- The Irwinton Sand Member of the Barnwell Group.
- The Tobacco Road Sand unit of the Barnwell Group.

The stratigraphic nomenclature adopted in this report was chosen to conform with the latest regional stratigraphic interpretation adopted by the Georgia State Geological Survey.[3] Stratigraphic nomenclature in the Georgia-South Carolina area has long been disputed and has not yet been fully resolved.

The formational boundary between the Lisbon Formation and Barnwell Group coincides with the upper contact of the marl.

2B.3.3.3.2 Lisbon Formation

The middle Eocene Lisbon Formation is represented in the site area by the Blue Bluff marl which forms the foundation for structures and backfill in the power block area. The marl has a total thickness of about 70 ft in the site area. The upper approximate 25 ft of the marl were exposed in excavations and mapped in detail. A vertical section between el 108.6 ft (final excavated grade) and 132 ft was exposed in the auxiliary building basement excavation. Ten subunits were recognized and mapped in this vertical section. The subunits, designated A through J, are shown on drawings AX6DD364, AX6DD365, AX6DD366, AX6DD367, AX6DD368, AX6DD369, and AX6DD370.

Unit A, near the top of the excavation walls, is generally above el 128 ft and includes the marl from this point up to the upper contact of the marl with the Utley Limestone Member of the Barnwell Group. It consists of dark gray silty to clayey marl with very fine light gray to white fine

sandy laminations, which are undulatory and discontinuous. Scattered shell fragments and well-cemented lenses of sand up to 0.1 ft thick are present locally. The laminations are oriented parallel to the lower contact of the unit, and parting along the laminations is common. Unit A is dense and well consolidated. Surfaces exposed to the atmosphere tend to dessicate rapidly. Unit A interfingers with the underlying unit B. This is especially evident in the south wall in the vicinity of stations 0 + 70, 1 + 50, and 4 + 30. (See drawings AX6DD365, AX6DD367, and AX6DD368.) The contact with unit B is everywhere gradational.

Unit B, directly beneath unit A, is continuous around the auxiliary building basement walls and varies from 1 to over 4 ft in thickness. It consists of massive to faintly laminated gray sandy marl. It has a sugary texture and does not tend to dessicate as readily as unit A. This property provides an easy means for differentiating the units after exposure to the atmosphere. Unit B is dense but poorly cemented and contains widely scattered shell fragments.

A subunit of B, designated B_1, has been identified and is present locally within B. This subunit consists of laminated sandy marl, which is locally fossiliferous. Subunit B_1 has been mapped at the base of B in the easterly portions of the north and south wall and the east wall. (For example, see drawing AX6DD366.) The contacts between B and B_1 are highly gradational.

Unit B is in turn underlain by a thin, relatively discontinuous but laterally extensive limestone, designated unit C. This limestone is light gray, well indurated, and exhibits conchoidal fracture. It is continuous in the west end of the south wall but becomes discontinuous east of station 0 + 80. East of station 3 + 65, the limestone becomes a series of small, irregular discontinuous pods at varying elevations. (See drawing AX6DD367.) Where exposed in the north, east, and west walls, the limestone forms discontinuous lenses at a relatively consistent elevation. It averages about 1 ft in thickness and dips slightly to the east, being present at about el 127 to 128 ft at the west end of the auxiliary building and 125 ft at the east end.

During excavation of the auxiliary building basement, the irregularity of portions of unit C led to a special study to determine whether the irregularities could be related to fault offset. The concern was that lenses and pods of the limestone occurring at slightly different elevations might have been offset from one another. The study focused on an area of the south walls at station 2 + 80 and the north wall at station 1 + 70. (See drawings AX6DD365 and AX6DD367.) As both excavation and mapping of stratigraphically lower units progressed, it became very evident that the irregularities of unit C were due to processes other than faulting. The continuity of the lower units in the areas of interest precluded the possibility of fault offset. A report was prepared[(4)] which concluded that the only plausible explanation for the observed irregularities was a combination of erosional and depositional processes.

Underlying the limestone of unit C is medium gray, highly fossiliferous, sandy to silty marl, designated unit D. This zone, averaging 8 ft in thickness, is continuous around the walls of the auxiliary building excavation. The lithology of unit D is very uniform and its upper and lower contacts are quite sharp. An abundance of pelecypods retaining both valves characterizes this unit. Near the base, a number of very hard, lime-cemented pods and lenses are present at roughly equivalent elevations and have highly gradational contacts with the surrounding marl. These pods and lenses are believed to represent accumulations of calcium carbonate cement leached from the surrounding fossiliferous marl. They are collectively considered to be a subunit of D, designated D_1.

Unit E underlies D and is a thin, relatively continuous impure limestone. It is light gray, very well indurated, and fossiliferous. It averages 1 ft in thickness and varies in elevation from 121 ft in

the northwest corner of the auxiliary building to 116 ft in the southeast corner. Locally, unit E is difficult to distinguish from D_1. This is seen in the north wall between stations 1 + 40 and 1 + 70 (drawing AX6DD365) where E is discontinuous and D_1 is represented by some fairly continuous lenses. In these cases unit E is arbitrarily selected as the unit displaying the sharpest contacts with surrounding units, and the one stratigraphically in between the overlying unit D and underlying unit F. The similarity between portions of E and D_1 suggests that both may be cemented deposits resulting from leaching and redeposition of calcium carbonate from the overlying fossiliferous deposits. The relative continuity of E indicates a basic permeability change occurring at that horizon in the geologic past. This is a basis for differentiating the overlying unit D from the underlying unit F.

Unit F, like D, is a fossiliferous marl, which is continuous around the basement excavation walls. It is medium gray, sandy to silty, and varies in thickness from 1 to 4 ft. It is dense and well consolidated but poorly cemented and tends to dessicate upon exposure to the atmosphere. Unit F includes some cemented limy pods similar to D_1. These have gradational contacts with the surrounding material and appear to be secondary in origin.

Unit G is light to dark gray laminated marl, which is present locally as lenses interfingering with units F and H. It is relatively continuous in the westernly portion of the south wall but pinches out at station 1 + 50. It reappears between stations 1 + 85 and 2 + 25 (drawings AX6DD367 and AX6DD368) but then disappears for the remainder of the south wall. It is present in portions of the west and north walls and is absent in the east wall. The unit is characterized by very fine sinuous and discontinuous sandy laminations, scattered shell fragments, and small lenticular clay pods. It contains scattered carbonaceous lenses and is well consolidated.

Unit H underlies G and consists of massive gray marl, which is continuous around the excavation. It is dense, well consolidated, and poorly cemented. Shell fragments are sparse in the upper part of the unit but become increasingly abundant towards the base. Unit H varies in thickness from 1 to 6 ft.

Unit I underlies H and is similar to unit E. It is a thin, relatively continuous light gray impure limestone, which is generally less than 1 ft thick. It is continuous around the excavation walls with the exception of the east wall between station 0 + 79 and the south end of the wall where it is absent.

Unit J, the deepest marl unit exposed in the auxiliary building excavation, consists of medium gray, massive, fossiliferous marl similar to the stratigraphically higher units D and F. It is continuous around the excavation walls with the exception of the east end of the excavation where the upper contact of the unit dips beneath the base of the excavation.

From the preceding descriptions it is seen that the portion of the marl section exposed in the auxiliary building excavation represents cycles of fossil abundance and absence, interspersed with the formation of secondary limestone pods and lenses as a result of leaching of calcium carbonate from fossiliferous zones. Erosional and depositional processes have combined to create some of the interfingering of units as well as irregularity of some of the limestone layers.[4]

The upper contact of the Lisbon Formation was exposed around the perimeter of the power block excavation because it exists at an elevation higher than the top of the more localized auxiliary building excavation. The top of the Lisbon Formation corresponds with the top of the Blue Bluff marl. This upper contact was examined in detail and surveyed. It varies from a high

elevation of 138.6 ft on the north side of the excavation to a low of 132.0 ft on the south side. The contact is erosional with very minor relief present. The uppermost few feet of the marl is locally weathered to a greenish color, and bioturbations (disturbance of the sediment due to the activity of organisms) were noted locally.

2B.3.3.3.3 Barnwell Group

Deposits of the upper Eocene Barnwell Group overlie the Blue Bluff marl of the Lisbon Formation and include all of the sediments exposed in the side slopes of the power block excavation. The contact between Barnwell Group and Lisbon Formation deposits is a disconformity, representing a hiatus in the depositional history of the site.

As mentioned previously, four distinct units within the Barnwell deposits have been recognized and are described in this section. These units include, from oldest to youngest: The Utley Limestone Member, the Twiggs Clay Member, the Irwinton Sand Member, and the Tobacco Road Sand Member. These units are illustrated in the stratigraphic column in figure 2.5.1-5.

Although examined and described in detail, the deposits between the top of the Blue Bluff marl and approximate el 170 ft could not be mapped in detail. (See drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363.) Consequently, the geologic map of the power block excavation (drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363) shows only the detailed lithology of the Tobacco Road Sand and the upper portion of the Irwinton Sand. This was due to extensive slumping of the slopes when excavation and dewatering were suspended during the period between September 1974 and June 1976. Extensive regrading obscured the contacts between units in this zone. Several portions of the slopes were covered with riprap in order to control seepage and improve stability, thereby further obscuring contacts. Since seepage from the slopes was creating local stability problems, it was decided not to excavate back into the slopes to expose contacts and risk large slope stability problems. Detailed mapping of the units above and below this zone demonstrated the continuity of the strata and the absence of faulting.

The lowermost exposed unit within the Barnwell Group in the power block excavation is the Utley Limestone. The lower part of the limestone is grayish-yellow, well indurated, and fossiliferous, grading locally into coquina. It is continuous around the power block excavation and varies in thickness from 0.5 to 3 ft. The upper part of the limestone is white to light gray, varies from 0 to 12 ft in thickness, and is present only in the north and northwest portions of the power block excavations. Although well indurated, this thicker limestone has been subjected to extensive leaching, producing a honeycomb network of cavities. Some individual cavities had mean diameters of several feet before being removed by excavation or filled in place. The filling of cavities in the limestone intersected by the excavation slopes is described in paragraph 2.5.4.5. Within the cavities, the limestone typically displayed a weathered and soft zone immediately adjacent to the cavity walls, which graded within a few inches to hard, unweathered limestone. Locally, extensive leaching of the limestone left a residue of silt and clay impurities, forming a soft mottled blackish material. Included in the Utley Limestone is a highly fossiliferous clay deposit which varies in color from tan to dark gray. The difference in colors appears to be due primarily to weathering effects. Prior to its removal, this clay was present mainly in the northwest portion of the power block excavation. It contains abundant specimens of the oyster Crassostrea gigantissima, a key Eocene near-shore pelecypod. Lesser quantities of other pelecypods, gastropods, arthropod arts, and shark teeth have been identified in this clay.

Unconformably overlying the Utley Limestone is the Twiggs Clay. This consists primarily of medium gray, moderately indurated, laminated sandy claystone, which is quite similar to the underlying Blue Bluff marl of the Lisbon Formation. The Twiggs Clay is only present in the southeast portion of the power block excavation and varies in thickness from 0 to 13 ft. The upper 2 to 5 ft are weathered to a distinctive greenish-yellow color. The Twiggs Clay has alternating thin and thick beds (from less than 1 in. to greater than 1 ft), with gradational contacts between beds. No joints, fractures, or discontinuities were observed in the clay.

The Irwinton Sand of the Barnwell Group unconformably overlies the Twiggs Clay in the southeast portion of the power block and overlies the Utley Limestone elsewhere. The Irwinton Sand consists of an approximately 50 ft thick vertical sequence of sands, clays, and reef deposits. At the base of the sequence is a massive, white, quartz-rich sand deposit. The presence of fossil shrimp burrows identifies this as an intertidal deposit. The upper surface of the sand is highly irregular, with reef- type accumulations of Crassostrea gigantissima present on the highs. These shell accumulations are well cemented and highly calcareous. This sand is fine- to medium-grained and very well sorted, and it exhibits extensive crossbedding. It is extremely friable and tends to rapidly slump and ravel, assuming its angle of repose soon after excavation.

Above the white sand and reef deposits is a sequence of tan sand and clay. The sand is generally fine to medium and moderately sorted, and it contains thin seams of tan clay having high plasticity. Two continuous marker horizons are present within this sequence. The first, a zone of manganese-staining and shell debris, occurs generally between el 170 ft and 180 ft and is somewhat higher than this on the west side of the excavation. This zone, called the shell hash horizon, varies in thickness from less than 1 in. to almost 6 ft and can be traced continuously around the excavation slopes (drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363). A second shell hash horizon is locally present beneath the first one but is discontinuous. The second marker horizon is a zone of abundant tan clay seams, which varies from approximately 1 ft to almost 6 ft in thickness, and is found between el 180 ft and 200 ft. This clay zone marks the top of the Irwinton Sand.

Both of the marker horizons undulate along the strike, with flexures in the bedding reflecting underlying reef highs as well as lows due to collapse of cavities in the stratigraphically lower Utley Limestone. These flexures are discussed further in paragraph 2.5.1.2.3.4.

Above the Irwinton Sand is the Tobacco Road Sand of the Barnwell Group. This sand extends up to the top of the excavation slopes and consists of a thick (up to 40 ft) zone of predominately red sand with zones of lavender, purple, mustard yellow, and orange sand. The color changes are due to weathering effects and are not related to structure of lithology. The sand consists of fine to medium quartz grains which are moderately to well sorted and angular to subrounded. Colors are imparted by clay coatings on the individual grains. Differential weathering has produced mottled zones of bright colors which form an alligator-skin effect near the top of the unit. The sand is dense, well consolidated, and completely uncemented.

At the top of the excavation slopes, recent deposits of buff-colored, alluvial, and windblown sand are present locally. These deposits form a thin veneer of fine- to medium-grained, angular to subangular, well-sorted quartz sand which is highly gradational with the underlying sand of the Barnwell Group.

2B.3.3.4 Structure

The sedimentary sequence exposed in the power block excavation is flat lying, although many of the units within the Barnwell Group are discontinuous and highly variable in thickness. The regional dip is to the southeast at about 30 ft to the mile.

A complete absence of faulting of the sediments is demonstrated by the continuity of the upper surface of the Blue Bluff marl (Lisbon Formation), as well as of the two marker horizons in the Irwinton Sand discussed in the preceding section. Continuity of subunits of the marl mapped on the walls of the auxiliary building basement excavation give added proof of absence of faulting.

Only one joint was recognized in the entire power block excavation and this was only of limited extent. The joint was exposed in the southeast corner of the power block excavation where it extended from the upper surface of the marl down to el 127 ft, where it terminated at a depth of about 6 ft. The joint trended N 81° E, was approximately vertical, and was tightly closed. Some secondary dark green mineralization was noted as a fine coating on the joint faces. No other joints or fractures were identified in any other lithologic units.

The strata of the Irwinton Sand exhibit flexures that are related to the following phenomena:

- Differential compaction of the underlying sediments.
- Subsidence due to leaching of underlying calcerous materials and collapse of solution cavities.
- Deposition of an uneven surface.

These flexures are particularly evident in the two marker horizons of the Irwinton Sand.

Evidence for differential compaction includes the typical association of highs in the marker horizons, with the occurrence of underlying reef deposits of shells. These cemented shell deposits form hard spots compared to the relatively more compressible sediments between them. As the load of the overburden increased during deposition, the sediments between the reefs were compressed relatively more than those immediately above them. This created downward flexures in the overlying marker horizons between the shell deposits.

Evidence for subsidence due to leaching and collapse of cavities includes the presence of solution cavities in the limestone underlying zones of downwarping of the relatively incompetent sands and clays above the cavities. Recent surface depressions in the area give added evidence for subsidence. Leaching and cavity formation are confined to the limestone above the marl. No cavities have been found to exist in the marl, and the upper surface of the marl contains no depressions where exposed in the power block excavation.

2B.3.3.5 Ground Water Conditions

Ground water conditions at the VEGP site are discussed in subsection 2.4.12. The Blue Bluff marl of the Lisbon Formation forms an aquiclude between the shallow water table aquifer and the deeper confined regional aquifer. Since the excavation for the power block bottomed in the marl, only the shallow water table aquifer was encountered in the excavation.

The original water table elevation in the power block area prior to excavation was approximately 160 ft. When excavation had progressed to this level, seepage was encountered and temporary construction dewatering initiated. Dewatering methods employed are discussed in paragraph 2B.3.4.3. As excavation progressed downwards, seepage continued from the side slopes. The seepage was from water perched on clay seams in the Irwinton Sand. This water continued to seep out of the slopes even though the water surface was drawn down elsewhere in the excavation. Locations of some of the more prominent ground water seeps are shown on drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363.

In September 1974 the VEGP project was postponed indefinitely and the dewatering effort was terminated. Seepage continued into the excavation, which had bottomed generally above el 140 ft, and the water level rose in the excavation to about 155 ft where it remained for the next 2 years. In June 1976 the project was restarted and the excavation dewatered once more. As excavation progressed through the cavernous Utley Limestone above the marl, significant flows of ground water were encountered from cavities requiring special localized dewatering procedures. Slight artesian conditions were encountered locally within the limestone.

When excavation had extended into the marl and perimeter drainage had been provided, work progressed under dry conditions. The marl contains no free ground water, and no springs were observed which might indicate a hydraulic connection with the deeper artesian aquifer below the marl.

Seepage from the side slopes of the power block excavation continued during this period with gradual decline in the elevation of the top of the seepage zone. The zone of seepage was effectively obscured when the side slopes were lined with a blanket of riprap up to el 160 ft. Temporary construction dewatering was continued throughout a significant portion of the construction period.

2B.3.4 EXCAVATION AND FOUNDATION CONSIDERATIONS

2B.3.4.1 General

The excavation for the power block structures for Units 1 and 2 at the VEGP site is roughly square in shape, with two access ramps exiting from the southeast and southwest corners of the excavation. It measures approximately 1400 ft on an edge at the top and 1000 ft on an edge at the toe. The side slopes were cut a gradient of 2:1. The total excavated volume in the power block was approximately 5 million yd^3 including the access ramps.

The original ground surface in the power block area varied from an elevation of about 200 ft to slightly over 230 ft. The major portion of the excavation bottomed in the marl layer at approximately el 130 ft.

Within this larger excavation, a deeper localized excavation was made for the auxiliary building basemat. This consisted of a rectangular area measuring approximately 120 ft by 440 ft. The base of this excavation was at el 108.6 ft, and the walls were cut vertically with a horizontal bench at el 118 ft. The other major power block structures are founded primarily on structural backfill at elevations above the floor of the excavation.

The excavation is shown in plan view on drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363. These drawings show only the access road at the southeast corner since the one in the southwest corner was graded after geologic mapping had been performed on the slopes as shown in the drawings.

Excavation work was started in May 1974 and postponed on September 12, 1974. The bottom elevation of the excavation averaged approximately 145 ft at this time and close to 900,000 yd^3 of excavation remained. The excavation work resumed by February 1977, and the auxiliary building excavation was bottomed out in October 1977.

2B.3.4.2 Excavation Procedures

Excavation work started and progressed very rapidly in the upper sands above the water table at el 160 ft. A large fleet of bulldozers and scrapers was assembled for the job. Very little, if any, ripping was required because of the sandy nature of the deposits; and progress was extremely fast, attaining a maximum rate of 120,000 yd^3/day at the peak of activity. Upon reaching the water table, excavation progress was significantly slowed because of the tendency of the equipment to mire in the saturated sands. At this point construction dewatering was begun. The procedures utilized for dewatering are discussed in the following section, but the general approach consisted of trenching a system of parallel ditches to permit drainage from the area between the ditches. Once dry, these areas would be excavated by bulldozers and scrapers while the ditches were progressively deepened to maintain dry conditions between the ditches. Excavation below water in the ditches was accomplished by means of two draglines.

When the excavation reached the zones of hard Utley Limestone described earlier, limited blasting of the rock was utilized to facilitate its removal. Since the limestone to be removed directly overlaid the marl, which was to form the foundation for structural backfill, it was necessary to control the blasting in such a manner as to protect the underlying marl from damage.

First, the stipulation was made that the contractor not use explosives if conventional methods could be used, even if some difficulty resulted. Further, the use of explosives would be discontinued in any case, if, in the opinion of the engineer, the marl might be damaged as a result of blasting. Blast holes were not permitted to penetrate lower than el 135 ft, and a minimum stem of 18 in. was recommended below the charge in each hole. It was recommended that no blast holes exceed 3 in. in diameter and that the maximum charge weight should not exceed 30 lb per delay. The maximum allowable powder factor was set at 1 lb/yd^3.

Because of the concern for protecting the marl, only very limited blasting of the limestone was performed. The major portion of the rock was removed by first breaking it with a hydraulic ram mounted on a backhoe, then loading it out with conventional equipment.

Excavation of the marl was accomplished by ripping, followed by conventional earth moving. The auxiliary building basement excavation was cut with bulldozers and front-end loaders.

Trimming of the walls was accomplished with a backhoe. Some of the hard, indurated limestone layers within the marl described in paragraph 2B.3.3.3.2 were first broken with the backhoe-mounted hydraulic ram, then removed by front-end loader. Fine grading of the floor of the power block was accomplished with motor graders in areas underlying future structural

backfill and with Gradalls in the nuclear service cooling water tower foundation areas. In the foundation areas, shovels and air hoses were used for cleanup of loose material.

2B.3.4.3 Construction Dewatering

The construction dewatering system utilized in the power block excavation consisted of a system of east-west dewatering ditches connected by a north-south ditch leading to a sump and pumping plant in the southwest corner of the excavation. Because of the low permeability of the deposits, the dewatering consultant, Mr. R. Y. Bush, decided that a conventional well-point system would be ineffective, hence the ditch and sump approach.(5) This scheme proved to be successful when the invert elevation of the ditches was maintained 15 to 20 ft below the adjacent grade. This permitted conventional procedures in reasonably dry materials.

Upon reaching the marl, the system of ditches and sump was replaced by a perimeter drainage system as shown on drawing AX6DD324. This consisted of a buried porous concrete pipe around the perimeter of the power block excavation feeding into three small sumps at the toe of the south slope. Water pumped from the sumps was discharged to debris basin No. 1 southeast of the power block. The buried porous concrete pipe was encased in a granular filter material which was carried up the surface of the adjacent 2:1 slope to about el 160 ft. This filter blanket was placed so that there was a minimum of 4 ft of filter material measured horizontally from the face of the slope out to the face of the filter blanket. (See drawing AX6DD324.)

This dewatering scheme proved to be entirely successful and construction in the marl layer was able to proceed under totally dry conditions.

2B.3.4.4 Slope Protection

During the early stages of excavation, intense rainfall of short duration caused severe erosion of the 2:1 side slopes of the power block excavation. The uncemented sands rapidly washed out forming deeply incised gullies in some areas. These gullies were backfilled with the native soil material and local areas of the slope regraded. One such area is seen on the geologic map (drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363) in the upper part of the east slope between stations N83 + 00 and N84 + 00. Another larger area exists in the south slope of the access ramp each of station E100 + 00. After regrading the eroded areas, berms were constructed around the tops of the slopes to control runoff. The surfaces of the slopes were sprayed with the chemical stabilizing agent Petroset, a colorless liquid that sets up and tends to bond the sand grains together. These measures proved to be reasonably successful in controlling further erosion.

After the resumption of excavation work in 1977, erosion problems further down the slopes were encountered due to seepage of the perched ground water out of the slopes. Since stabilizing agents were expected to be ineffective under these conditions, the lower portions of the slopes were blanketed with riprap to improve stability. The riprap was subsequently covered with a finer grained filter transition material.

Where the 2:1 slopes intersected the cavernous limestone deposit, several cavities of varying sizes were exposed in the slopes. The largest of these existed in the northwest corner of the power block and had an opening measuring 10 ft by 10 ft. This cavity extended back into the slope some 30 ft before narrowing down to a small size. Other small cavities were encountered

at varying intervals all along the north side of the power block excavation. It was necessary to fill these cavities so that an effective buttress would be formed against which the future structural backfill could be placed and compacted. This consideration did not require complete filling of the cavities since a prism of fill material placed in the entrance and extending some distance into the cavity would provide an unyielding mass against which the structural backfill could be placed. The cavities were first cleaned of loose debris, then backfilled with crushed rock (Georgia State Standard No. 467). The crushed rock was packed into the cavities by means of a 20-ft-long ram attached to the blade of a bulldozer. This method proved to be very successful and actually resulted in the crushed rock being forced into small crevices, effecting an essentially complete filling of some of the cavities. The large cavity in the northwest corner was effectively filled in this manner. From the volume of crushed rock forced into the cavity, it was estimated that the cavity was completely filled to at least a distance of 25 ft back of the entrance.

To retard erosion of temporary slopes in Category 1 backfill placed in the power block excavation, these slopes were sprayed with a commercial compound known by the trade name Glassroot. It consists of a glass fiber material which was sprayed onto the slope and then coated with a film of asphalt emulsion. This proved to be effective in controlling erosion of the compacted sandy backfill but only for a limited period of time. By late 1979, the glassroot/asphalt coating began to show signs of excessive deterioration. Consequently, the slopes were stripped clean and recoated with gunite, which proved to be more durable and easier to maintain.

2B.3.4.5 Foundation Cleanup and Protection

As mentioned previously, the marl at final grade in foundation areas was exposed using either a motor grader or Gradall. Loose material was then removed by shovel, broom, and airhose. On the vertical walls of the auxiliary building excavation, final trim to neat line was accomplished with a backhoe followed by pick and shovel and airhose techniques.

In all cases where final grade was exposed and cleaned off, the marl surface had to be covered in a manner approved by the geologist within 24 h of exposure. On horizontal surfaces the marl was covered either by structural backfill or by mudmat concrete depending upon whether the particular area exposed was in a foundation or backfill area. The vertical walls of the auxiliary building basement excavation were coated with a 4-in.-thick layer of gunite reinforced with welded wire mesh.

In some cases temporary covers such as loose soil or plastic sheeting were employed when the permanent cover material could not be applied within the 24-h limit. In all cases the temporary cover procedure was approved by either the geologist or the Georgia Power Company inspector. Before placing the permanent cover material in any foundation area, the marl was inspected and approved by the geologist or soil engineer in accordance with the procedures described in the following section.

2B.3.4.6 Foundation Inspection and Approval Procedures

All areas of marl exposed and cleaned off in preparation for placement of concrete or backfill were examined closely for any evidence of loose or soft zones or geologic discontinuities. After confirming the absence of such features, the inspecting geologist documented the approval of

the area on field foundation approval forms. These field approval forms were transmitted to the Georgia Power Company site personnel for permanent retention. At intervals, the forms were countersigned by the supervising geologist and soil engineer for the project.

In addition, photographs of the foundation areas were taken. These were logged and transmitted to Georgia Power Company for permanent retention in the field office. The documentation described in this section has been transferred to SNC as the exclusive operating licensee.

2B.3.4.7 Foundation Testing

As a part of the general marl geologic mapping and inspecting functions, it was decided to carry out a program of coring and testing samples of the marl to confirm the material properties used for design. It was desired to obtain samples for record purposes. The coring and sampling operation was performed under the direction of the geologist and inspector, and the test assignments were made by the soils engineer.

A total of 38 core holes was drilled by rotary methods in the floor of the power block excavation at locations selected by the geologist. The hole locations are shown on drawings AX6DD360, AX6DD361, AX6DD362, and AX6DD363. The marl was cored to depths between 4 and 11 ft beneath the ground surface. Four-in.-diameter core samples were obtained, labeled, and placed in wooden boxes for permanent storage at the site. Samples for testing were selected by the geologist. These were then wrapped in cellophane, sealed with wax, and placed in special boxes for transportation to the laboratories of Law Engineering Testing Company in Atlanta.

A total of 31 core samples was tested for moisture content, bulk unit weight, unconfined compressive strength, and shear strength from one-point unconsolidated-undrained triaxial shear tests. The average wet unit was found to be 105.6 lb/ft^3, while the average moisture content was 36.2 percent. The average deviator stress at failure in the strength tests was 39.14 k/ft^2 (272 psi). A complete summary of test results is found in appendix 7 of reference 1. The results obtained are in the range anticipated and show that the marl is a competent foundation material.

2B.3.4.8 Foundation Rebound Monitoring

In order to monitor the rebound occurring in the Blue Bluff marl layer as a result of removal of approximately 100 ft of the overlying materials, the specialist firm of Goldberg, Zoino, Dunnicliff Associates was commissioned to provide in situ instrumentation. A total of nine heave points was installed at the bottom of drill holes made for this purpose. The heave points were installed at the locations shown in figure 2B-1, between approximate el 104 and 126 ft.

Throughout the excavation period, elevation changes of the heave points were surveyed. The measured heave is summarized in the table below:

	Period		
Measured Heave Point No.	From	To	(in.)
1	6/22/74	8/07/77	1.1
2	6/16/74	6/22/76	1.4
3	6/16/74	10/02/74	0.6

5	6/16/74	2/26/77	1.7
7	6/16/74	6/05/77	1.2
9	6/30/77	8/07/77	1.5

More complete data is presented in appendix 6 of reference 1.

The measured heave was substantially less than that predicted.

2B.3.5 CONCLUSIONS

The detailed geologic mapping of the strata exposed in the power block excavation at the VEGP site has better defined the structure and stratigraphy of this area. A much more comprehensive and detailed picture of the site geology has emerged as a result of this effort. The general conclusions of the PSAR[2] have been confirmed.

Two separate marker horizons in the Irwinton Sand have been mapped around the side slopes of the power block. Both horizons are continuous and unbroken, demonstrating the absence of faulting in these materials. The upper contact of the stratigraphically lower Blue Bluff marl of the Lisbon Formation has been mapped with survey accuracy and has also been found to be uninterrupted by offsets. Subunits within the marl have been mapped around the walls of the auxiliary building basement excavation. These zones were likewise found to be undisturbed by faulting. Minor stratigraphic irregularities noted were shown to be related to erosional and depositional processes.[4]

Surface depressions and subsidence features mapped in the upper sands were found to be related to collapse of solution cavities in the underlying limestone. Detailed examination of the exposed marl and surveying of its upper surface configuration has shown that the marl is free of solution cavities such as those present in the overlying limestone. The marl was found to contain no freely draining water and its function as an aquiclude was confirmed. Where the side slopes of the power block intersected solution cavities in the limestone layer, these cavities were backfilled with crushed rock to provide a firm buttress against which structural backfill could be placed and compacted. The back-filling of the cavities was inspected and found to be adequate.

Areas of the marl exposed at final grade were inspected, approved, and protected in an adequate manner as described in the report. All foundation areas inspected were found to expose sound competent marl suitable for supporting the backfill and plant structures.

The coring and testing of the marl at selected locations in the power block yielded results which confirm the design parameters used. Results of the rebound monitoring program showed that the measured rebound was less than that predicted, giving additional evidence of the competency of the marl.

The results of the geologic work described in this report lead to the conclusion that the VEGP site is suitable for design and construction of a multiple-unit nuclear generating plant. No geologic hazards were found to exist that might affect safety and licensing considerations.

2B.3.6 REFERENCES

1. Bechtel Power Corporation, Report on Backfill Material Investigations, two volumes, January 1978.

2. Georgia Power Company, Alvin W. Vogtle Nuclear Plant Preliminary Safety Analysis Report, chapter 2.

3. Huddleston, Paul, Georgia Geological Survey, Personal Communication, 1978.

4. Bechtel Power Corporation, Report on Stratigraphic Irregularities Exposed in the Auxiliary Building Excavation, February 1978.

5. Bush, R. Y., Consulting Engineer, Dewatering Study - Alvin W. Vogtle Nuclear Plant, January 12, 1973. (See also subsequent correspondence.)

2B.3.7 BIBLIOGRAPHY

Bechtel Power Corporation, Interim Report of Geologic Conditions –Power Block Excavating, October 23, 1974.

TABLE 2B-1 (SHEET 1 OF 36)

DRILL HOLE SUMMARY

Hole No.	Depth Location (ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
1[a]	91.5	98	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
2[a]	350	206	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
3[a]	220	230	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
5[a]	230	220	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
6[a]	210	198	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
7[a]	150	119	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
8[a]	225	218	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
9[a]	190		Preliminary investigation	Standard pen ASTM and rotary tricone	None	
10[a]	150	186	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
11[a]	190	215	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
13[a]	180		Preliminary investigation	Standard pen ASTM and rotary tricone	None	

TABLE 2B-1 (SHEET 2 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
14[a]		140		Preliminary investigation	Standard pen ASTM and rotary tricone	None	
15[a]		240		Preliminary investigation	Standard pen ASTM and rotary tricone	None	
12	N 1,150,535 E 620,169	240	263	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
16	N 1,148,930 E 617,946	200	233.7	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
17	N 1,144,080 E 620,200	280	264	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
18	N 1,143,403 E 621,922	225	222	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Carbonate solubility test
19	N 1,142,368 E 624,484	200	219	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Carbonate solubility test
19A		115	219	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
20	N 1,141,804 E 626,176			Preliminary investigation	Standard pen ASTM and rotary tricone	None	
21	N 1,139,810 E 621,875			Preliminary investigation	Standard pen ASTM and rotary tricone	None	Soil solubility test
22	N 1,141,735 E 622,001	260	253	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
23	N 1,142,357 E 620,273	220	215	Preliminary investigation	Standard pen ASTM and rotary tricone	None	

TABLE 2B-1 (SHEET 3 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
24	N 1,142,850 E 623,092	220	216	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Observe well/ gamma logged/ soil solubility test/ carbonate solubility test
25	N 1,142,600 E 627,657	220	230	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
26	N 1,140.963 E 629,199	200	203	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Observe well
27	N 1,143,622 E 627,931	190	210	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Observe well/ gamma logged
28	N 1,143,609 E 628,634	170	85	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
29	N 1,144,977 E 626,392	210	193	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Observe well/ gamma logged
30	N 1,145,072 E 626,534	85	91	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
31	N 1,143,764 E 625,237	210	211	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Observe well/ X-ray diffraction
32	N 1,144,784 E 623,572	210	214	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Observe well
33	N 1,146,834	220	238	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Observe well/ gamma logged
34	N 1,147,180 E 624,846	115	86	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Observe well
35A		70	94.4	Preliminary investigation	Standard pen ASTM and rotary tricone	None	

TABLE 2B-1 (SHEET 4 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
36A		70	98.3	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
36B		150	98.4	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
37	N 1,145,242 E 622,690	210	195	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Soil solubility test/ X-ray diffraction/ paleo analysis
38	N 1,143,474 E 619,772	270	257	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
39	N 1,149,703 E 622,835	90	118	Preliminary investigation	Standard pen ASTM and rotary tricone	None	X-ray diffraction
40	N 1,143,210 E 621,759	250	215	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
41	N 1,142,049 E 628,658	120	222.8	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
42	N 1,143,392 E 623,553	250	210	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Paleo analysis carbonate solubility test
42A	N 1,143,380 E 623,535	150	210.6	Hydrologic data	Standard pen ASTM and rotary tricone	None	Observe well – 150 ft
42B	N 1,143,386 E 623,544	130	210.4	Hydrologic data	Standard pen ASTM and rotary tricone	None	Gamma logged/ observe well - 130 ft
42C	N 1,143,398 E 623,563	90	210	Hydrologic data	Standard pen ASTM and rotary tricone	None	Gamma logged/ observe well - 90 ft
42D	N 1,143,403 E 623,571	70	209.7	Hydrologic data	Standard pen ASTM and rotary tricone	None	Observe well - 70 ft

TABLE 2B-1 (SHEET 5 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
42E	N 1,143,408 E 623,580	55	209.6	Hydrologic data	Standard pen ASTM and rotary tricone	None	Observe well – 55 ft
43	N 1,144,314 E 621,810	55	282.8	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
44	N 1,146,517 E 623,911	90	241.3	Preliminary investigation	Standard pen ASTM and rotary tricone	None	
45	18,300 NE of 36	370	273.52	Preliminary investigation	Standard pen ASTM and rotary tricone	None	Gamma logged/ paleo analysis
101	N 1,142,945 E 623,518	100	210.8	Plant foundation	Standard pen ASTM and rotary tricone	None	
101A	N 1,142,950 E 623,515	100	210.6	U.D. samples for reactor foundation	Rotary tricone, Denison, and Shelby	3-in. Denison-16 3-in. Shelby-12	Observe well to 200 ft
101B	20 ft North of 101	65	210.8	Bulk sample (100 lb)	24-ft auger	None	
102	N 1,142,796 E 623,727	200	211.5	Plant foundation	Standard pen ASTM and rotary tricone	None	
102A	Adjacent to No. 102	177	211.5	Plant foundation	Standard pen ASTM, Denison, and rotary tricone	Denison-15	
103	N 1,142,796 E 623,927	100	212.4	Plant foundation	Standard pen ASTM	None	
104	N 1,143,184 E 623,398	100	217.1	Plant foundation	Standard pen ASTM and rotary	None	
104A	Adjacent to No. 104	200	217.1	U.D. samples for reactor foundation	, ASTM, Shelby, and Denison	Shelby-5 Denison-6	Undisturbed samples from 100 ft to 200 ft standard pen ASTM with intermittent Denison samples

TABLE 2B-1 (SHEET 6 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
105	N 1,142,996 E 623,626	300	209.7	Reactor foundation	Standard pen ASTM and rotary tricone	None	
106	N 1,142,996 E 623,726	150	209.6	Plant foundation	Standard pen ASTM and rotary tricone	None	
107	N 1,142,996 E 623,876	300	209.4	Reactor foundation	Standard pen ASTM and rotary tricone	None	Gamma logged
107A	N 1,142,999 E 623,891	300	209.4	U.D. samples for reactor foundation	Rotary tricone, and Denison	Denison-20	
107B	20 North of107	65	209.4	Bulk sample (100 lb)	24-in. auger	None	24-in.-diameter bucket auger holes drilled adjacent to existing logged holes solely for the purpose of obtaining bulks oil samples from specific depths for testing
108	N 1,142,996 E 624,026	100	210.2	Plant foundation	Standard pen ASTM and rotary tricone	None	
109	N 1,143,405 E 623,357	200	216	Plant foundation	Standard pen ASTM and rotary tricone	None	
110	N 1,143,385 E 623,504	100	213.5	Plant foundation	Standard pen ASTM and rotary tricone	None	
111	N 1,143,256 E 623,726	200	207.2	Turbine foundation	Standard pen ASTM, Denison, Shelby, and rotary tricone	Denison-4 Shelby-10	
111A	Adjacent to111	142	207.2	Turbine foundation	Rotary tricone and Denison	4-in. Denison-3	
112	N 1,143,256 E 623,876	100	204.3	Turbine foundation	Standard pen ASTM and rotary tricone	None	

TABLE 2B-1 (SHEET 7 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
113	N 1,143,256 E 624,026	200	203.1	Turbine foundation	Standard pen ASTM and rotary tricone	None	
114	N 1,143,504 E 623,526	199	212	Plant foundation	Standard pen ASTM, Denison, Shelby, and rotary tricone	Denison-4 Shelby-10	Carbonate solubility test
114A	Adjacent to114	155	212	Plant foundation	Standard pen ASTM, Denison, and rotary tricone	4-in. Denison-3	
115	N 1,143,506 E 623,726	100	208.5	Plant foundation	Standard pen ASTM and rotary tricone	None	
116	N 1,143,503 E 623,928	200	208	Plant foundation	Standard pen ASTM and rotary tricone	None	Carbonate solubility test
117	N 1,143,940 E 624,343	100	197.8	Intake tunnel foundation	Standard pen ASTM and rotary tricone	None	
118	N 1,144,449 E 624,961	100	198	Intake tunnel foundation	Standard pen ASTM and rotary tricone	None	
119	N 1,144,966 E 625,639	100	117.76	Intake tunnel foundation	Standard pen ASTM and rotary tricone	None	
120	N 1,145,310 E 626,389	100	86.8	Intake tunnel foundation	Standard pen ASTM and rotary tricone	None	
121	N 1,145,467 E 626,195	200	88.8	Intake tunnel foundation	Standard pen ASTM and rotary tricone	None	Observe well to 88 ft
122	N 1,145,719 E 625,884	100	111.4	Intake tunnel foundation	Standard pen ASTM and rotary tricone	None	
123	N 1,146,101	200	89.3	Intake tunnel foundation	Standard pen ASTM and rotary tricone	None	

TABLE 2B-1 (SHEET 8 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
124	N 1,141,896 E 623,527	200	260.2	Cooling tower foundation	Standard pen ASTM and rotary tricone	None	Observe well to top of marl/gamma logged
125	N 1,142,156 E 624,027	100	248.1	Cooling tower foundation	Standard pen ASTM and rotary tricone	None	
126	N 1,142,997 E 625,306	100	241.4	Cooling tower	Standard pen ASTM and rotary tricone	None	
127	N 1,144,209 E 623,176	100	199.2	Switchyard foundation	Standard pen ASTM and rotary tricone	None	
128	N 1,144,206 E 623,876	100	198	Switchyard foundation	Standard pen ASTM and rotary tricone	None	
129	N 1,143,856 E 623,576	100	215.9	Switchyard	Standard pen ASTM and rotary tricone	None	Observe well to top of marl
130	N 1,142,796 E 623,527	100	209.6	Plant foundation	Standard pen ASTM and rotary tricone	None	
131	N 1,143,256 E 623,576	100	213.6	Plant foundation	Standard pen ASTM and rotary tricone	None	
132	N 1,144,988 E 626,154	150	169.5	Intake sructure bluff slope stability	Standard pen ASTM and rotary tricone	None	
133	N 1,145,145 E 626,089	150	155	Intake sructure foundation	Standard pen ASTM and rotary tricone	None	
134	N 1,146,750 E 621,024	200	191.3	Geologic fill-in section - between B16 and B37	Standard pen ASTM and rotary tricone	None	
135	N 1,143,992 E 622,742	200	200.5	Geologic-depression investigation	Cored with NWM barrel face discharge bit -	None	Observe well to below marl/carbonate solubility test

TABLE 2B-1 (SHEET 9 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
136	N 1,142,996 E 623,849	300	209.5	Geologic-center of reactor	Cored with4 x 5 ½ barrel F-D bit	None	Gamma logged/ carbonatesolublity
137	N 1,144,839 E 622,117	200	230.6	Unit 1 geologic depression investigation	Cored with NWM barrel, F-D bit	None	Observe well to below marl/gamma logged/carbonate solubility test
138	N 1,143,000 E 622,500	99.5	225.2	Reactor fan Units 3 and 4	Standard pen ASTM and rotary tricone	None	Observe well to top of marl
138A	N 1,142,966 E 622,509	200	224.9	U.D. samples for reactor foundation	Rotary tricone, Denison, and Shelby	4-in. Denison-17 3-in. Shelby-15	
139	N 1,142,996 E 623,526	300	210.9	Geologic hole edge of reactor Unit 2	Cored with4 x 5 1/2 barrel, F-D bit	None	Gamma logged/carbonate solubility test
140	N 1,142,845 E 622,702	96	222.4	Units 3 and 4	Standard pen ASTM and rotary tricone	None	Observe well to marl - 96 ft
141	N 1,142,860 E 622,292	105	230.4	Units 3 and 4	Standard pen ASTM and rotary tricone	None	Observe well to marl - 105 ft
142	N 1,143,283 E 622,262	105	231.2	Units 3 and 4	Standard pen ASTM and rotary tricone	None	Observe well to marl - 95 ft
143	N 1,143,283 E 622,738	88.5	224.5	Units 3 and 4	Standard pen ASTM and rotary tricone	None	Observe well to marl – 88.5 ft
144	N 1,145,411 E 626,127	48.5	103.2		Standard pen ASTM and rotary tricone	None	Observe well to marl – 48.5 ft
144A	N 1,145,406 E 626,133	51	103.9	Intake structure	Rotary tricone and Denison	3-in. Denison-14	

TABLE 2B-1 (SHEET 10 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
145	N 1,142,792 E 621,063	192	218.7	Geologic - in depression	Cored with NWM barrel, F-D bit		Gamma logged/observe well to marl – 82 ft
146	N 1,142,966 E 623,750	300	209.6	Seismic shot hole	Rotary tricone	None	
147	N 1,142,975 E 622,471	300	226.2	Geologic investigation Units3 and 4	Cored with4 x 5 1/2 barrel, F-D bit	None	Gamma logged/observe well to 300 ft /carbonate solubility test
148	N 1,142,996 E 623,814	300	209	Seismic shot hole	Rotary tricone	None	
149	N 1,142,996 E 623,779	300	209.2	Seismic shot hole	Rotary tricone	None	
150	N 1,142,996 E 623,556	170	210.3	Seismic shot hole	Rotary tricone	None	
151	N 1,142,946 E 623,849	300	210.4	Seismic shot hole	Rotary tricone	None	
152	N 1,133,831 E 633,344	200	152.7	Geologic hole to complete section betweenPlant site and Griffin Landing	Cored with NWM barrel, F-D bit	None	Paleo analysis
153	N 1,143,080 E 622,128	89.5	226.2	Determine depth of bearing horizon	Rotary tricone	None	
154	N 1,142,796 E 623,849	300	209.5	Seismic shot hole	Rotary tricone	None	Gamma logged
155	N 1,143,332 E 621,470	86.7	226	Determine depth of bearing horizon	Rotary tricone	None	

TABLE 2B-1 (SHEET 11 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
156	N 1,131,584 E 642,340	260	237.7	Geologic hole - Griffin Landing	Cored with NWM barrel, F-D bit	None	
157	N 1,145,605 E 621,598	184.1	207.6	Geologic hole	Cored with NWM barrel, F-D bit	None	Filled with gravel to 149.5 ft for Packer Test
158	N 1,143,838 E 622,866	72	213	Determine depth of bearing horizon	Rotary tricone	None	
159	N 1,143,931 E 622,401	80.2	222.2	Determine depth of bearing horizon	Rotary tricone	None	
160	N 1,144,157 E 622,625	78	213.7	Determine depth of bearing horizon	Rotary tricone	None	
161	N 1,144,102 E 622,899	65	201	Determine depth of bearing horizon	Rotary tricone	None	
162	N 1,144,977 E 622,318	90	235.5	Determine depth of bearing horizon	Rotary tricone	None	
163	N 1,144,748 E 621,985	95	238.6	Determine depth of bearing horizon	Rotary tricone	None	
164	N 1,145,401 E 626,120	145	103.2	Seismic shot hole	Rotary tricone	None	Gamma logged
165	N 1,145,354 E 626,138	155	112.2	Seismic shot hole	Rotary tricone	None	
166	N 1,145,215 E 626,194	185	143.1	Seismic shot hole	Rotary tricone	None	
167	N 1,145,388 E 626,087	145	104.6	Seismic shot hole	Rotary tricone	None	

TABLE 2B-1 (SHEET 12 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
168	N 1,145,375 E 626,055	147	105.8	Seismic shot hole	Rotary tricone	None	
169	N 1,145,364 E 626,027	147	106.5	Seismic shot hole	Rotary tricone	None	
170	N 1,142,988 E 622,440	180	228.3	Packer Test	Rotary tricone	None	Packer Test
171	N 1,143,420 E 621,944	90	223.1	Deep seismic shot hole	Rotary tricone	None	
172	N 1,143,452 E 621,959	90	224.1	Deep seismic shot hole	Rotary tricone	None	
173	N 1,141,664 E 626,629	80	188.6	Deep seismic shot hole	Rotary tricone	None	
174	N 1,141,691 E 626,642	89	189	Deep seismic shot hole	Rotary tricone	None	
175	N 1,143,386 E 621,363	165	233.1	Investigate geologic anomaly	Standard pen ASTM and rotary tricone	None	Gamma logged/ observe well set to 165 ft
176	N 1,142,117 E 625,423.05	80	196.4	Water observation well	Rotary tricone	None	Observe well to 75 ft
177	N 1,143,560 E 624,865	80	213	Water observation well	Rotary tricone	None	Observe well to 80 ft
178	N 1,144,958 E 622,994	93	240.4	Water observation well	Rotary tricone	None	Observe well to 91 ft
179	N 1,144,059 E 621,779	133	274.8	Water observation well	Rotary tricone	None	Observe well to 131 ft

TABLE 2B-1 (SHEET 13 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
180	N 1,142,965 E 623,724	162	210.1	Packer Test	Rotary tricone	None	Packer Test
181	N 1,143,744 E 620,833	200	258.3	Investigate geologic anomaly	Standard pen ASTM and rotary tricone	None	Observe hole 200 ft/gamma logged
182	N 1,144,232.04 E 620,820	220	260.4	Investigate geologic anomaly	Standard pen ASTM and rotary tricone	None	
183	N 1,143,026.04 E 623,526	60	210.8	Water observation well	Rotary tricone	None	Observe well to 60 ft
184	N 1,142,996 E 623,906	65	209.4	Water observation well	Rotary tricone	None	
200	N 1,142,860 E 623,560	100	209	Auxiliary building (Aux. bldg.) foundation	Standard pen ASTM and rotary tricone flight auger, and Shelby	Shelby-2	
201	N 1,142,860 E 623,740	100	211.4	Aux. bldg. foundation	Rotary tricone, standard pen ASTM, and Shelby	Shelby-1	
202	N 1,142,710 E 623,380	155.7	215.5	Emergency cooling tower foundation	Flight auger, standard pen ASTM, Shelby, and Denison	Shelby-8 Denison-8	
203	N 1,142,730 E 623,650	154.8	210.9	Railroad plant entrance foundation	Rotary tricone standard pen ASTM, Shelby, and Denison	Shelby-8 Denison-8	
204	N 1,142,710 E 623,910	156	212.8	Emergency cooling tower foundation	Rotary tricone standard pen ASTM, Shelby, and Denison	Shelby-8 Denison-8	
205	N 1,143,310 E 623,640	100	212	Turbine foundation	Rotary tricone standard pen ASTM	None	

TABLE 2B-1 (SHEET 14 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
206	N 1,143,310 E 623,900	99.5	204	Turbine foundation	Rotary tricone standard pen ASTM	None	
207	N 1,143,220 E 624,560	100.5	212.3	Cooling tower foundation	Rotary tricone standard pen ASTM	None	
208	N 1,143,220 E 625,070	90.5	218.1	Cooling tower foundation	Rotary tricone standard pen ASTM	None	
209	N 1,143,220 E 625,586	99.4	216.2	Cooling tower foundation	Rotary tricone standard pen ASTM	None	
210	N 1,142,680 E 624,560	101	216.9	Cooling tower foundation	Rotary tricone standard pen ASTM	None	
211	N 1,142,680 E 625,070	101.5	219	Cooling tower foundation	Rotary tricone standard pen ASTM	None	
212	N 1,142,680 E 625,580	96	211.1	Cooling tower foundation	Rotary tricone standard pen ASTM	None	
213	N 1,141,670 E 623,320	131	256.1	Cooling tower foundation	Rotary tricone standard pen ASTM	None	
214	N 1,141,670 E 623,830	126	248.6	Cooling tower foundation	Rotary tricone standard pen ASTM	None	
215	N 1,141,670 E 624,340	126	237.3	Cooling tower foundation	Rotary tricone standard pen ASTM	None	
216	N 1,142,930 E 623,650	142.5	210.6	Aux. bldg. foundation	Rotary tricone standard pen ASTM, Shelby, and Denison	Shelby-7 Denison-7	
217	N 1,143,130 E 623,650	141	207.5	Aux. bldg. and turbine foundation	Rotary tricone standard pen ASTM, Shelby, Denison, and Pitcher	Shelby-5 Denison-2 and Pitcher-7	

TABLE 2B-1 (SHEET 15 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
218	N 1,143,080 E 623,330	150.8	216.7	Diesel generator (gen.) bldg foundation	Rotary tricone standard pen ASTM, Shelby, Denison, and Pitcher	Shelby-4 Denison-6 and Pitcher-2	
219	N 1,143,080 E 623,950	140.3	207.8	Diesel gen. bldg foundation	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-5 Pitcher-8	
220	N 1,143,000 E 623,400	142.5	213.2	Reactor foundation	Rotary tricone standard pen ASTM, Shelby, Denison, and Pitcher	Shelby-4 Denison-8 Pitcher-4	
221	N 1,143,220 E 623,900	136	204.8	Turbine foundation	Rotary tricone standard pen ASTM, Shelby, and Denison,	Shelby-2 Denison-13	
222	N 1,143,310 E 623,370	150.5	216.5	Turbine foundation	Rotary tricone standard pen ASTM, Shelby, and Denison,	Shelby-5 Denison-8	
223	N 1,143,340 E 624,420	140.5	206.9	Intake structure foundation	Rotary tricone standard pen ASTM, Shelby, Denison, and Pitcher	Shelby-1 Denison-8 Pitcher-4	
224	N 1,142,300 E 623,180	186	250.2	Intake structure foundation	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-6 Pitcher-13	
225	N 1,142,940 E 624,560	148.7	211.6	Cooling tower foundation	Rotary tricone standard pen ASTM, Shelby, and Denison,	Shelby-4 Denison-11	
226	N 1,142,940 E 625,070	162	218.6	Cooling tower foundation	Rotary tricone standard pen ASTM, Shelby, and Denison,	Shelby-5 Denison-12	

TABLE 2B-1 (SHEET 16 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
227	N 1,142,940 E 625,580	150.7	209.5	Cooling tower foundation	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-6 Pitcher-12	
228	N 1,141,930 E 623,320	201	260.5	Cooling tower foundation	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-2 Pitcher-20	
229	N 1,141,930 E 623,830	203	255.5	Cooling tower foundation	Rotary tricone standard pen ASTM, Shelby, and Denison,	Shelby-8 Denison-12	
230	N 1,141,930 E 624,340	192	243	Cooling tower foundation	Rotary tricone standard pen ASTM, and Denison,	Denison-18	
231	N 1,142,200 E 623,320	124.7	254.2	Cooling tower foundation	Rotary tricone and standard pen ASTM	None	
232	N 1,142,200 E 623,830	125.3	250.1	Cooling tower foundation	Rotary tricone and standard pen ASTM	None	
233	N 1,142,200 E 624,340	105.5	229.8	Cooling tower foundation	Rotary tricone and standard pen ASTM	None	
234	N 1,143,760 E 621,970	105.5	245.5	Administration bldg. foundation	Rotary tricone, standard pen ASTM, and Shelby	Shelby-2	
235	N 1,143,650 E 624,450	135.5	206.2	Shops and warehouse foundation	Rotary tricone standard pen ASTM, Shelby, Pitcher, and Denison	Shelby-5 Pitcher-8 Denison-1	
236	N 1,145,550 E 626,040	81	90.5	Barge facility foundation	Rotary tricone and standard pen ASTM	None	

TABLE 2B-1 (SHEET 17 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
237	N 1,145,820 E 625,820	80	109.6	Intake structure foundation	Rotary tricone standard pen ASTM, and Denison,	Denison-6	
238	N 1,147,172 E 622,476	218	242.6	Geologic hole	Rotary tricone standard pen ASTM, and NX core,	None	
239	N 1,145,270 E 625,070	60.5	111.8	Water intake pipeline foundation	Rotary tricone and standard pen ASTM	None	
240	N 1,144,700 E 624,310	50	187.8	Makeup water pipeline foundation	Rotary tricone standard pen ASTM, Shelby, and Denison,	Shelby-3 Denison-1	
241	N 1,143,580 E 624,310	72.5	207.4	Warehouse and maintenance bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-3 Pitcher-4	
242	N 1,142,650 E 624,050	75.3	213.8	Water treatment area foundation	Rotary tricone and standard pen ASTM	None	
243	N 1,144,154 E 622,618	224	213	Geologic hole	NX core	None	
244	N 1,143,835 E 622,858	222.3	212.6	Geologic hole	5.5 in. x 4 in. core barrel	None	
245	N 1,143,491 E 623,924	150	207.6	Geologic hole	NX core	None	
246	N 1,145,532 E 620,553	400	210.4	Geologic hole	5.5 in. x 4 in. core barrel and 4.5 in. x 4 in. core barrel	None	
247	N 1,140,750 E 619,424	200	211.3	Geologic hole	5.5 in. x 4 in. core barrel	None	

TABLE 2B-1 (SHEET 18 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
248	N 1,142,466 E 619,111	200	166.8	Geologic hole	NX core	None	
249	N 1,143,826 E 624,154	183	193	Geologic hole	5.5 in. x 4 in. and 4.5 in. x 4 in. NX and core barrel	None	
301	N 1,142,692 E 622,264	161.3	227.3	Emergency cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-9 Pitcher-7	
302	N 1,142,690 E 622,390	165	229.3	Emergency cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-7 Pitcher-11	
303	N 1,142,690 E 622,605	161	229.2	Emergency cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-6 Pitcher-13	
304	N 1,142,690 E 622,776	173	231.4	Emergency cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-7 Pitcher-11	
305	N 1,142,870 E 622,505	161	227.7	Aux. bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-10 Pitcher-9	
306	N 1,143,000 E 622,335	159.9	229.8	Reactor cont.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-8 Pitcher-7	
307	N 1,143,000 E 622,675	154.5	218.7	Reactor cont.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-4 Pitcher-11	
308	N 1,143,106 E 622,740	151.3	225.9	Diesel gen. bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-9 Pitcher-7	

TABLE 2B-1 (SHEET 19 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
309	N 1,143,120 E 622,335	161.5	231.2	Control bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-6 Pitcher-14	
310	N 1,143,194 E 622,656	151.5	223.5	Control bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-6 Pitcher-9	
311	N 1,143,198 E 622,514	152.6	229.`	Control bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-4 Pitcher-5	
312	N 1,143,296 E 622,220	160.5	229.8	Turbine bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-1 Pitcher-7	
313	N 1,143,274 E 622,790	161.5	224.9	Turbine bldg	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-6 Pitcher-10	
314	N 1,143,335 E 622,494	160.5	229.5	Turbine bldg	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-5 Pitcher-7	
315	N 1,142,110 E 621,725	110.5	239.8	Unit No. 3 cooling tower	Rotary tricone and standard pen ASTM		
316	N 1,141,835 E 621,725	211.5	251.7	Unit No. 3 cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-1 Pitcher-25	
317	N 1,141,650 E 621,780	131	254.3	Unit No. 3 cooling tower	Rotary tricone and standard pen ASTM		
318	N 1,142,200 E 622,280	105.5	242.7	Unit No. 3 cooling tower	Rotary tricone and standard pen ASTM		

TABLE 2B-1 (SHEET 20 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
319	N 1,141,849 E 622,225	203	255.5	Unit No. 3 cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-6 Pitcher-14	
320	N 1,141,574	123.5	254.5	Unit No. 3 cooling tower	Rotary tricone and standard pen ASTM		
321	N 1,142,138 E 622,725	116	249.8	Unit No. 3 cooling tower	Rotary tricone and standard pen ASTM		
322	N 1,141,863 E 622,725	203	258.5	Unit No. 3 cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-4 Pitcher-16	
323	N 1,141,588 E 622,725	130.5	256.9	Unit No. 3 cooling tower	Rotary tricone and standard pen ASTM		
324	N 1,142,265 E 622,945	181.5	248.3	Unit No. 3 cooling tower structure	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-10 Pitcher-9	
325	N 1,143,276 E 620,548	105.5	236.5	Unit No. 4 cooling tower	Rotary tricone and standard pen ASTM		
326	N 1,143,000 E 620,548	206	230	Unit No. 4 cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-7 Pitcher-10	
327	N 1,142,726 E 620,548	99	228.2	Unit No. 4 cooling tower	Rotary tricone and standard pen ASTM		
328	N 1,143,262 E 621,048	104.5	234.1	Unit No. 4 cooling tower	Rotary tricone and standard pen ASTM		
329	N 1,142,987 E 621,048	165	222.4	Unit No. 4 cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-1 Pitcher-14	

TABLE 2B-1 (SHEET 21 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
330	N 1,143,248 E 621,548	85.5	221.8	Unit No. 4 cooling tower	Rotary tricone and standard pen ASTM		
331	N 1,142,873 E 621,548	161.5	220.9	Unit No. 4 cooling tower	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-5 Pitcher-9	
332	N 1,142,698 E 621,548	89.9	228.4	Unit No. 4 cooling tower	Rotary tricone and standard pen ASTM		
333	N 1,143,390 E 621,760	151.5	223.9	Unit No. 4 cooling water intake	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-6 Pitcher-9	
334	N 1,143,000 E 622,300	301	230	Reactor No. 4	Cored		
335	N 1,143,000 E 622,710	300	217.8	Reactor No. 3	Cored	Shelby-1 Pitcher-2	
336	N 1,143,535 E 622,320	170	226.8	Depression north of No. 4	Cored		
337	N 1,142,860 E 622,375	161.5	2311.2	Aux. bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-2 Pitcher-13	
338	N 1,142,860 E 622,625	153	223.5	Aux. bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-6 Pitcher-9	
339	N 1,143,070 E 622,270	160.5	229.8	Aux. bldg.	Rotary tricone standard pen ASTM, Shelby, and Pitcher	Shelby-1 Pitcher-13	
340	N 1,144,500 E 623,144	68	213.2	Groundwater monitoring	Rotary tricone and standard pen ASTM		

TABLE 2B-1 (SHEET 22 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
341	N 1,145,815 E 625,820	58	110.1	Groundwater monitoring	Rotary tricone and standard pen ASTM		
342	N 1,140,250 E 618,727	36.5	160	Dam site	Rotary tricone and standard pen ASTM		
343	N 1,140,232 E 619,064	31.5	152.2	Dam site	Rotary tricone and standard pen ASTM		
344	N 1,141,615 E 618,746	35.5	155.4	Dam site	Rotary tricone and standard pen ASTM		
345	N 1,137,248 E 628,175	31.5	118.5	Dam site	Rotary tricone and standard pen ASTM		
346	N 1,137,351 E 628,477	26.5	108	Dam site	Rotary tricone and standard pen ASTM		
347	N 1,138,000 E 627,800	25.5	114.3	Dam site	Rotary tricone and standard pen ASTM		
CT1	N 1,143,895 E 626,480	120	233	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT2	N 1,143,720 E 626,235	115	225.7	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT3	N 1,143,422 E 626,125	105	234	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT4	N 1,143,422 E 626,245	110	233.4	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT5	N 1,143,592 E 626,125	105	229.8	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT6	N 1,143,592 E 626,245	95	228.2	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	

TABLE 2B-1 (SHEET 23 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
CT7	N 1,143,848 E 626,125	95	220.4	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT8	N 1,143,848 E 626,245	100	225.2	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT9	N 1,143,935 E 626,515	100	233.4	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT10	N 1,143,935 E 626,645	100	231.4	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT11	N 1,143,820 E 626,580	95	230.8	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT12	N 1,143,620 E 626,580	101	231.8	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT13	N 1,143,505 E 626,515	102	234.6	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT14	N 1,143,505 E 626,645	110	234.2	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT15	N 1,141,077 E 627,940	120	208.3	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT16	N 1,141,247 E 627,940	100	208.5	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT17	N 1,141,417 E 627,940	95	206.3	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT18	N 1,141,672 E 627,940	115	207.7	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT19	N 1,141,077 E 628,060	100	207.7	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	

TABLE 2B-1 (SHEET 24 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
CT20	N 1,141,247 E 628,060	100	208.8	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT21	N 1,141,417 E 628,060	120	211.5	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT22	N 1,141,672 E 628,060	100	211.6	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT23	N 1,141,160 E 628,330	110	215.2	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT24	N 1,141,160 E 628,460	135	214.8	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT25	N 1,141,275 E 628,395	110	214.7	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT26	N 1,141,475 E 628,395	105	210.4	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT27	N 1,141,590 E 628,330	100	208	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
CT28	N 1,141,590 E 628,460	105	207.5	Combustion turbine foundation	Rotary tricone and standard pen ASTM	None	
RH-1	N 1,145,680 E 626,250	80	90.3	Haul road	Rotary tricone and standard pen ASTM	None	
RH-2	N 1,145,603 E 625,496	60	151.1	Haul road	Rotary tricone and standard pen ASTM	None	
RH-3	N 1,145,594 E 625,196	50	150.2	Haul road	Rotary tricone and standard pen ASTM	None	
RH-4	N 1,145,672 E 624,857	50	162.5	Haul road	Rotary tricone and standard pen ASTM	None	

TABLE 2B-1 (SHEET 25 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
OD-1	N 1,143,263 E 628,836	55	86.4	Barge dock	Rotary tricone and standard pen ASTM	None	
401	N 1,145,815 E 626,160	60	89.6	River intake	Rotary tricone and standard pen ASTM	None	
402	N 1,145,835 E 626,055	60	88.8	River intake	Rotary tricone and standard pen ASTM	None	
403	N 1,145,955 E 626,060	64	88.5	River intake	Rotary tricone and standard pen ASTM	None	
404	N 1,146,000 E 625,980	60	86.6	River intake	Rotary tricone and standard pen ASTM	None	
405	N 1,146,175 E 625,890	64	87.4	River intake	Rotary tricone and standard pen ASTM	None	
406	N 1,146,300 E 625,820	60	88.4	River intake	Rotary tricone and standard pen ASTM	None	
407	N 1,146,215 E 625,790	70	88.7	River intake	Rotary tricone and standard pen ASTM	6	
408	N 1,146,175 E 625,705	101.5	135.1	River intake	Rotary tricone and standard pen ASTM	10	
409	N 1,145,895 E 625,475	60	116.9	River intake	Rotary tricone and standard pen ASTM	None	
501	N 1,143,014 E 623,891	208	208.2	Check water loss in No. 701. Core for logging	Rotary tricone and NX split tube core barrel	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)

TABLE 2B-1 (SHEET 26 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
501A	N 1,143,109 E 623,896	141	208.2	Large core to check core loss in No. 501	Rotary ticone and 4-in. core	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
502	N 1,14,3,390 E 623,357	150	216	Check water loss in upper 5 ft of marl in No. 109	Rotary tricone and NX split tube core barrel	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
503	N 1,143,870 E 624,130	130	194.5	Check depression near No. 239. Water test for permeability	Rotary tricone and NX split tube core barrel	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
503A	N 1,143,877 E 624,126	121	194.5	Large core to check loss in No. 503	Rotary ticone and 4-in. core	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
504	N 1,144,139 E 622,611	186.5	214.6	Check water loss in No. 238. Water test for permeability. Core for logging.	Rotary tricone and NX split tube core barrel	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
505	N 1,147,040 E 622,299	201.5	241.9	Check water loss in No. 243. Water test for permeability. Check core recovery..	Rotary tricone and NX split tube core barrel	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
506	N 1,146,698 E 621,067	178	172.7	Check water loss in No. 134. Water test for permeability. Core for logging.	Rotary tricone and NX split tube core barrel	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)

TABLE 2B-1 (SHEET 27 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
506A	N 1,146,705 E 621,064	115	172.7	Core for logging to check core loss in No. 506.	Rotary tricone and 4 in. core	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
507	N 1,145,504.5 E 620,633.5	191	211.8	Check water loss in No. 246. Water test for permeability.	Rotary tricone and NX split tube core barrel	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
507A	N 1,145,503.5 E 620,628	134.5	211.8	Large core to check core loss in No. 507. Core for logging.	Rotary tricone and 4 in. core	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
508	N 1,145,605 E 621,613	163	190.5	Water test for permeability. Core for logging to check hole No. 157.	Rotary tricone and NX split tube core barrel	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
509	N 1,142,950 E 622,280	188	230.4	Menard pressure meter tests. Samples for laboratory testing.	Rotary tricone and NX split tube core barrel	14	3-D, neutron, density, and caliper geophysical logs (Birdwell)
510	N 1,143,047 E 622,353	185	230.7	Water tests for permeability. Menard tested. Samples for laboratory testing.	Rotary tricone and NX split tube core barrel	14	3-D, neutron, density, and caliper geophysical logs (Birdwell)
511	N 1,143,103 E 622,549	181	225.4	Core for logging. Samples for laboratory testing. Menard tested.	Rotary tricone and NX split tube core barrel	15	3-D, neutron, density, and caliper geophysical logs (Birdwell)

TABLE 2B-1 (SHEET 28 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
512	N 1,143,070 E 622,650	180	221.5	Core for logging. Samples for laboratory testing. Menard tested.	Rotary tricone and NX split tube core barrel	15	3-D, neutron, density, and caliper geophysical logs (Birdwell)
513	N 1,142,940 E 622,640	178	220.2	Core for logging and water tests for permeability. Menard tested. Samples for laboratory tests.	Rotary tricone and NX split tube core barrel	14	3-D, neutron, density, and caliper geophysical logs (Birdwell)
514	N 1,142,820 E 623,480	169.5	209.9	Core for logging. Menard tested. Samples for laboratory testing.	Rotary tricone and NX split tube core barrel	14	3-D, neutron, density, and caliper geophysical logs (Birdwell)
515	N 1,142,970 E 623,440	172	212.7	Core for logging. Menard tested. Samples for laboratory tests.	Rotary tricone and NX split tube core barrel	13	3-D, neutron, density, and caliper geophysical logs (Birdwell)
516	N 1,143,080 E 623,480	172	212.7	Core for logging. Menard tested. Samples for laboratory tests.	Rotary tricone and NX split tube core barrel	14	3-D, neutron, density, and caliper geophysical logs (Birdwell)
517	N 1,143,050 E 623,800	175	207.7	Core for logging. Menard tested. Samples for laboratory tests.	Rotary tricone and NX split tube core barrel	14	3-D, neutron, density, and caliper geophysical logs (Birdwell)
518	N 1,142,950 E 623,800	175	209.9	Core for logging. Menard tested. Samples for laboratory tests.	Rotary tricone and NX split tube core barrel	14	3-D, neutron, density, and caliper geophysical logs (Birdwell)

TABLE 2B-1 (SHEET 29 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
519	N 1,142,820 E 623,860	176	211.9	Core for logging. Menard tested. Samples for laboratory tests.	Rotary tricone and NX split tube core barrel	17	3-D, neutron, density, and caliper geophysical logs (Birdwell)
520	N 1,143,000 E 623,825	175	209.2	Experimental geophysical tests (high frequency sonic logging Holosonics, Inc.)	Rotary tricone	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
521	N 1,143,026.5 E 623,851.5	175	207.5	Experimental geophysical tests (high frequency sonic logging Holosonics, Inc.)	Rotary tricone	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
522	N 1,142,973 E 623,851	175	209	Experimental geophysical tests (high frequency sonic logging Holosonics, Inc.)	Rotary tricone	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
523	N 1,142,973 E 623,798	173	209.5	Experimental geophysical tests (high frequency sonic logging Holosonics, Inc.)	Rotary tricone	None	3-D, neutron, density, and caliper geophysical logs (Birdwell)
524	N 1,143,026 E 623,798	173	208.6	Experimental geophysical tests (high frequency sonic logging Holosonics, Inc.)	Rotary tricone	None	
601	N 1,143,120 E 625,035	165	218.6	Cooling tower No. 1	Standard pen ASTM, rotary tricone, and NX core		E-logged

TABLE 2B-1 (SHEET 30 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
602	N 1,143,056 E 625,191	115	217	Cooling tower No. 1	Standard pen ASTM, rotary tricone, and NX core		E-logged
603	N 1,142,900 E 625,255	170	216.7	Cooling tower No. 1	Standard pen ASTM, rotary tricone, and NX core		
604	N 1,142,744 E 625,191	119.5	217.1	Cooling tower No. 1	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
605	N 1,142,680 E 625,035	163	219	Cooling tower No. 1	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
606	N 1,142,744 E 624,879	115	216.5	Cooling tower No. 1	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
607	N 1,142,900 E 624,815	157	215.1	Cooling tower No. 1	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
608	N 1,143,056 E 624,879	115	217.9	Cooling tower No. 1	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	
609	N 1,143,320 E 625,915	160	232.1	Cooling tower No. 2	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
609A	N 1,143,320 E 625,885	187	232.6	Cooling tower No. 2	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
610	N 1,143,256 E 626,071	187.5	233.6	Cooling tower No. 2	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
611	N 1,143,100 E 626,135	185.5	235.1	Cooling tower No. 2	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
612	N 1,142,994 E 626,071	139	231.7	Cooling tower No. 2	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	
613	N 1,142,880 E 625,915	175.5	223.4	Cooling tower No. 2	Standard pen ASTM, rotary tricone, and NX core		E-logged

TABLE 2B-1 (SHEET 31 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
614	N 1,142,944 E 625,759	120	215.9	Cooling tower No. 2	Standard pen ASTM, rotary tricone, and NX core		
615	N 1,143,100 E 625,695	169	218.6	Cooling tower No. 2	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	
616A	N 1,143,256 E 625,759	135	222.4	Cooling tower No. 2	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	
617	N 1,143,320 E 620,255	203	244.4	Cooling tower No. 3	Standard pen ASTM, rotary tricone, and NX core		
618	N 1,143,256 E 620,411	138	239	Cooling tower No. 3	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	
619	N 1,143,100 E 620,475	211.5	231.9	Cooling tower No. 3	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
620	N 1,142,944 E 620,411	124.5	232.1	Cooling tower No. 3	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	
621	N 1,142,880 E 620,255	182	236.6	Cooling tower No. 3	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
622	N 1,142,944 E 620,099	153	247.3	Cooling tower No. 3	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	
623	N 1,143,100 E 620,035	219	250.3	Cooling tower No. 3	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
624	N 1,143,320 E 620,255	172	250.2	Cooling tower No. 3	Standard pen ASTM, rotary tricone, and NX core		
625	N 1,143,120 E 621,135	173.3	227.7	Cooling tower No. 4	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
626	N 1,143,056 E 621,291	114	227.4	Cooling tower No. 4	Standard pen ASTM, rotary tricone, and NX core		

TABLE 2B-1 (SHEET 32 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
627	N 1,142,900 E 621,355	168.3	221.8	Cooling tower No. 4	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
628	N 1,142,744 E 621,291	111.5	221.1	Cooling tower No. 4	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	
629	N 1,142,680 E 621,135	165.5	219.5	Cooling tower No. 4	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
630	N 1,142,744 E 620,979	110	219	Cooling tower No. 4	Standard pen ASTM, rotary tricone, and NX core		
631	N 1,142,900 E 620,915	172	223.2	Cooling tower No. 4	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	E-logged
632	N 1,143,056 E 620,979	118	225.3	Cooling tower No. 4	Standard pen ASTM, rotary tricone, and NX core	Menard Pressure Test	
633	N 1,142,363.4 E 629,540.9	165	211.2	Geology, marl at bluff	Standard pen ASTM, rotary tricone, and NX core		E-logged
701	N 1,143,245 E 625,343	110	213.88	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core		
702	N 1,143,328 E 625,500	165.2	213.27	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core	Menard Pressure Test	
703	N 1,135,127 E 625,658	125	219.13	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core		
704	N 1,142,846 E 625,500	162.1	210.73	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core	Menard Pressure Test	
705	N 1,144,032 E 625,500	172	203.83	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core	Menard Pressure Test	
705-A	10 north of boring 705	166.1	202.53	Cooling towers	Rotary tricone		

TABLE 2B-1 (SHEET 33 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
706	N 1,143,948 E 625,657	102	207.28	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core		
706-A	N 1,143,945 E 625,658	163.5	210.89	Cooling towers	Rotary tricone and NQ core		
707	N 1,143,791 E 625,741	180.5	214.43	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core	Menard Pressure Test	
708	N 1,143,634 E 625,657	116.5	214.93	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core		
709	N 1,143,550 E 625,500	165.3	213.35	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core	Menard Pressure Test	
710	N 1,143,634 E 625,343	107	213.23	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core		
711	N 1,143,791 E 625,259	161.5	208.66	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core	Menard Pressure Test	
712	N 1,143,949 E 625,343	110.5	207.3	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core		
712A	N 1,143,937 E 625,344	158.6	204.4	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core		
713	N 1,143,912 E 625,500	169	203.7	Cooling towers	Standard pen ASTM, rotary tricone, and NQ core		
800	N 1,143,851 E 625,012	94	213.7	Groundwater monitoring	Rotary tricone	None	
801	N 1,142,651 E 624,733	93.6	212.8	Groundwater monitoring	Rotary tricone	None	
802	N 1,142,196 E 624,198	94	215.8	Groundwater monitoring	Rotary tricone	None	

TABLE 2B-1 (SHEET 34 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
803	N 1,142,350 E 622,909	94	219.7	Groundwater monitoring	Rotary tricone	None	
803A	N 1,142,085 E 622,896	88	218.3	Groundwater monitoring	Rotary tricone	None	
804	N 1,141,597 E 622,227	96	224.1	Groundwater monitoring	Rotary tricone	None	
805	N 1,141,718 E 624,425	127	234.2	Groundwater monitoring	Rotary tricone	None	
805A	N 1,141,672 E 624,403	125	232.7	Groundwater monitoring	Rotary tricone	None	
VG-1	N 1,120,308.26 E 660,009.14	565	156.6	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper
VG-2	N 1,122,608.99 E 650,596.85	618	253.1	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper
VG-3	N 1,121,183.52 E 655,725.83	574.9	165.7	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper
VG-4	N 1,124,629.41 E 644,971.51	554.4	150.3	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper

TABLE 2B-1 (SHEET 35 OF 36)

Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
VG-5	N 1,116,669.12 E 665,818.68	502	94.5	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper
VG-6	N 1,110,896.34 E 669,643.15	620	217.1	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper
VG-7	N 1,127,245.60 E 640,322.37	392	250.6	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper
VG-8	N 1,104,446.34 E 678,744.09	355.4	103.7	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper
VSC-1	N 1,134,867.04 E 679,423,71	620	219	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper
VSC-2	N 1,141,512.71 E 673,492.62	600	201.7	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper
VSC-3	N 1,138,356.84 E 676,254.55	510	170.3	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper

TABLE 2B-1 (SHEET 36 OF 36)

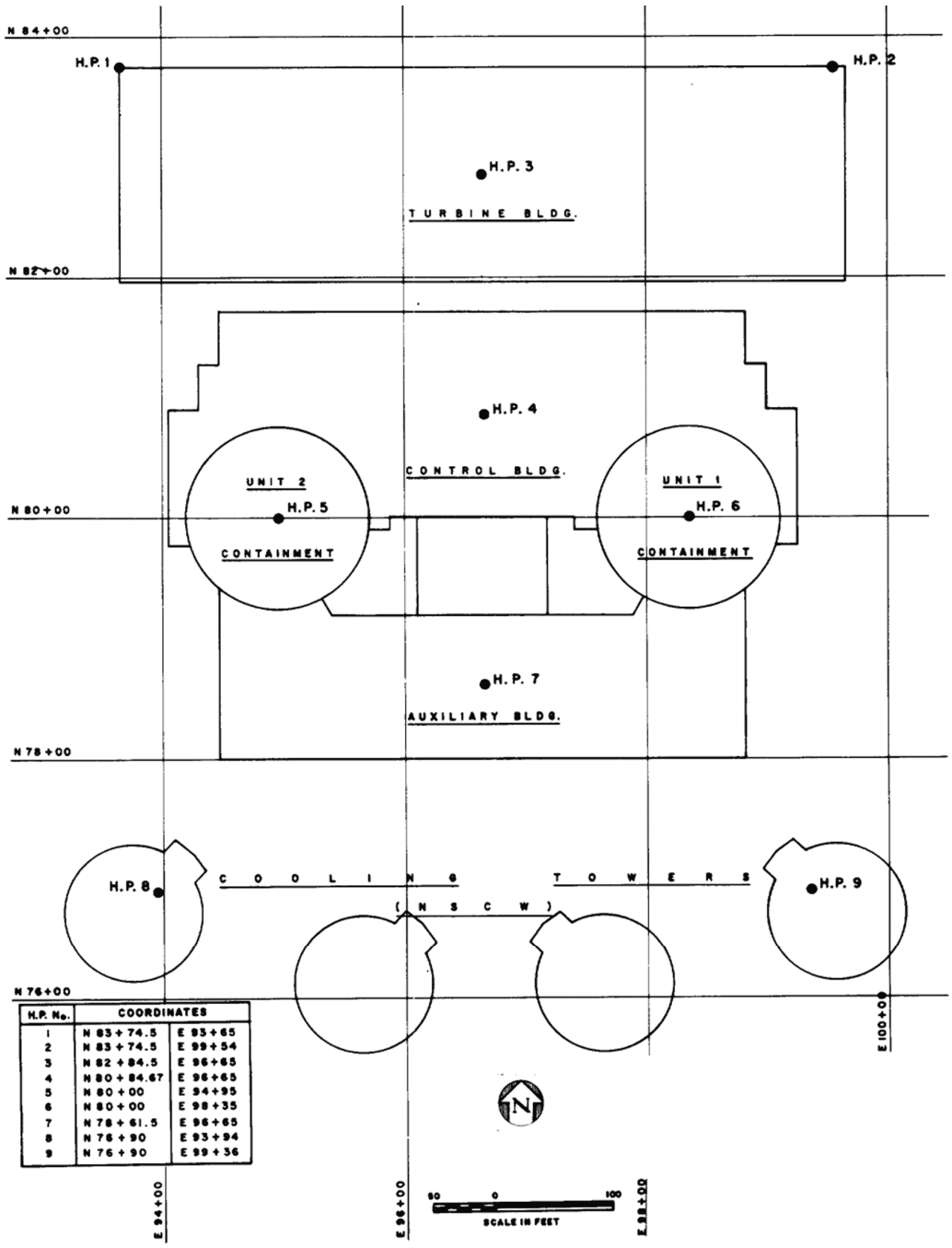
Hole No.	Depth Location	(ft)	Surface Elevation (ft)	Purpose	Type Drilling	Number and Type Undisturbed Samples Taken	Remarks
VSC-4	N 1,130,590.27 E 683,271.46	1024	156.7	Millett Fault investigation	Core	Continuous core samples taken triple-tube NQ wireline	Electric, gamma, neutron, caliper

a. Boring located outside site area, not shown on maps.

TABLE 2B-2

CORE SAMPLE INVENTORY

Hole No.	No. of Boxes	Hole No.	No. of Boxes
139	7	610	1
240	1	614	2
249	8	615	5
501	7	701	1
501B	1	703	1
502	3	704	3
503	3	705	4
503B	13	705A	3
513	2	706	2
514	4	707	5
515	2	708	2
516	4	709	3
517	4	710	2
518	4	711	3
519	2	712	3
602	2	721	1
603	5		



H.P. No.	COORDINATES	
1	N 83+74.5	E 93+65
2	N 83+74.5	E 99+54
3	N 82+84.5	E 96+65
4	N 80+84.67	E 96+65
5	N 80+00	E 94+95
6	N 80+00	E 98+35
7	N 78+61.5	E 96+65
8	N 76+90	E 93+94
9	N 76+90	E 99+36

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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

HEAVE POINT LOCATIONS

FIGURE 2B–1